

--- Début du chapitre: `_2d_offset` --- The 2D Offsetmodifier adds some two-dimensional offsets to the stroke backbone geometry. It has two sets of independent options/effects:

These two options add the given amount of offset to the start (or end) point of the stroke, along the (2D) normal at those points. The effect is blended over the whole stroke, if you for example, set only Start to 50, the start of the stroke is offset 50 pixels along its normal, the middle of the stroke, 25 pixels along its own normal, and the end point is not moved.

These two options simply add a constant horizontal and/or vertical offset to the whole stroke.

--- Fin du chapitre: `_2d_offset` ---

--- Début du chapitre: `_2d_transform` --- The 2D Transformmodifier applies two-dimensional scaling and/or rotation to the stroke backbone geometry. Scale is applied before rotation.

The center (pivot point) of these 2D transformations can be:

The median point of the stroke.

The beginning point of the stroke.

The end point of the stroke.

The Stroke Point Parameter factor controls where along the stroke the pivot point is (start point if set to 0.0; end point if set to 1.0).

The Pivot X and Y values define the position of the pivot point in the final render (from the bottom left corner).

Important

Currently, you have to take into account the real render size, i.e. resolution and resolution percentage.

The scaling factors, in their respective axes.

The rotation angle.

2D Transform modifier (blend-file).

--- Fin du chapitre: `_2d_transform` ---

--- Début du chapitre: `_3d_cursor` --- The 3D Cursor node outputs the position and orientation of the 3D cursor in the scene.

Note

This node can only be used in the Tool context.

This node has no inputs.

This node has no properties.

The position of the 3D cursor.

The orientation of the 3D cursor as a standard rotation value.

--- Fin du chapitre: `_3d_cursor` ---

--- Début du chapitre: `_3d_view` --- Sometimes when setting a large clipping range will allow you to see both near and far objects, but reduces the depth precision resulting in artifacts.

Model with no clipping artifacts.

Model with clipping artifacts.

Mesh with artifacts in Edit Mode.

To avoid this:

Increase the near clipping when working on large scenes.

Decrease the far clipping when objects are not viewed at a distance.

When perspective is disabled only the far Clip End is used, very high values can still result in artifacts. This is not specific to Blender, all graphics applications have these same limitations.

If you have a large scene, viewing it through Camera View may not display all of the objects in the scene. One possibility may be that the clipping distance of the camera is too low. The camera will only show objects that fall within the clipping range.

There are a couple of reasons why you may be experiencing a slow viewport.

Sometimes your hardware, mainly your graphics card, may be too slow to keep up with your model.

In some cases, slow selection is resolved by using updated drivers.

Blender uses OpenGL for selection, some graphics card drivers are slow at performing this operation.

This becomes especially problematic on dense geometry.

Possible Solutions:

See Preferences \xe2\x80\xa3 Viewport \xe2\x80\xa3 Selection.

This option is enabled by default, disabling it may give a better performance at the cost of selection accuracy.

In some cases, slow selection is resolved by using updated drivers. It is generally good to use recent drivers when using 3D software.

In Object Mode, holding Ctrl while selecting uses the object center point. While this can be useful on its own, it has the side effect of not relying on OpenGL selection.

Using Wireframe display mode can be used to more quickly select different objects.

Note

Obviously, the workarounds listed here are not long term solutions, but it is handy to know if you are stuck using a system with poor OpenGL support.

Ultimately, if none of these options work out it may be worth upgrading your hardware.

Having the viewport playback clamped to a maximum of 60 FPS is typically caused by the VSYNC setting on your GPU, for higher frame rates you may have to disable VSYNC functionality although this may be of limited use since frames rendered may be more than your GPU and monitor are able to display.

VSYNC is configured as part of your GPU driver options which vary depending on your system & GPU combination.

When navigating your scene, you may accidentally navigate away from your scene and find yourself with a blank viewport. There are two ways to fix this:

Select an object in the Outliner, then zoom to that object with View \xe2\x80\xa3 Frame Selected or Numpad Period.

Use Home to fit all objects into the 3D Viewport.

Sometimes when navigating you may be trying to zoom in but it seems that you have hit a limit to how far you can zoom. This is because Blender uses a central point to orbit around.

In practice this is good for modeling an object which you rotate about a lot to see from all sides (think of a potter using a wheel). However, this makes it awkward to explore a scene or model an object from the \xe2\x80\x99inside\xe2\x80\x99, for example.

Use View Dolly .

Use Walk/Fly Navigation .

Use Auto Depth and Zoom to Mouse Position .These tools will make sure the distance is always the value under the mouse cursor,
Use Zoom Region as it also resets the center point when zooming.
Center the view around the mouse cursor Alt- MMB .This will take the position under the cursor and make it your viewpoint center.
Use an NDOF, also known as a 3D mouse,see configuring peripherals for more information.

There are times when selection fails under some configurations,often this is noticeable in mesh Edit Mode,selecting vertices/edges/faces where random elements are selected.

Internally Blender usesOpenGLfor selection,so the graphics card driver relies on giving correct results.

Possible Solutions:

This is by far the most common cause of selection issues.

There are known problems with some graphics cards when using multisampling.

You can disable this option by turning multisampling off in your graphics card driver options.

Depending on your OpenGL configuration,some specific sample settings may work while others fail.

Unfortunately finding working configuration involves trial & error testing.

As with any OpenGL-related issues, using recent drivers can resolve problems.

However, it should be noted that this is a fairly common problem and remains unresolved with many drivers.

--- Fin du chapitre: _3d_view ---

--- Début du chapitre: _accumulate_field --- The Accumulate Fieldnode counts a running total of its input values, in the order definedby the geometry\xe2\x80\x99sindices. The node\xe2\x80\x99s essentialoperation is just addition, but instead of only outputting the final total, it outputs the currentvalue at every element.

The values to be accumulated.

Warning

When accumulating integer values, be careful to make sure that there are nottoo many large values. The maximum integer that Blender stores internally isaround 2 billion. After that, values may wrap around and become negative.See wikipediafor more information.

An index used to group values together for multiple separate accumulations.This can be thought of as a choice of the

\xe2\x80\x99bin\xe2\x80\x99d in which to place each value.This input has no effect when it is only a single value.

The node will accumulate a Floatfield.

The node will accumulate an Integerfield.

The node will accumulate a Vectorfield.

The node will accumulate a Matrixfield.

The attribute domain used for accumulationand for evaluation of the Valueinput.

The running total of values in the corresponding group, starting at the first value.

The running total of values in the corresponding group, starting at zero.

The total of all the values in the corresponding group.

Value

Group Index

Leading

Trailing

Total

A few examples of input values and the node's results. One important take-away from this table is that the specific values for the Group Input do not matter; it only matters that the values are shared between elements.

Here, the node is used in combination with the Random Value Node to create a stack of randomly scaled boxes. The Group Index input is not used, because all boxes are meant to be in the same stack.

A slightly more complicated version of the previous example, using the Group Index input to create three separate stacks.

--- Fin du chapitre: `_accumulate_field` ---

--- Début du chapitre: `_action` --- The Action constraint is powerful. It allows you to control an Action using the transformations of another object.

The underlying idea of the Action constraint is very similar to the one behind the Drivers, except that the former uses a whole action (i.e. multiple F-Curves of the same type), while the latter controls a single F-Curve of their owner's curve data.

Note that even if the constraint accepts the Mesh action type, only the Object, Pose and Constraint types are really working, as constraints can only affect objects or bones' transform properties, and not meshes' shapes. Also note that only the object transformation (location, rotation, scale) is affected by the action; if the action contains keyframes for other properties they are ignored, as constraints do not influence those.

As an example, let us assume you have defined an Object action (it can be assigned to any object, or even no object at all), and have mapped it on your owner through an Action constraint, so that moving the target in the (0.0 to 2.0) range along its X axis maps the action content on the owner in the (0 to 100) frame range. This will mean that when the target's X property is 0.0 the owner will be as if in frame 0 of the linked action; with the target's X property at 1.0 the owner will be as if in frame 50 of the linked action, etc.

Action panel. `xc2xb6`

Data ID used to select the constraints target, and is not functional (red state) when it has none. See common constraint properties for more information.

This property allows objects to be driven without a constraint target by interpolating between the Action Start and End frames. The relative position between the start and end frame can be controlled using the value slider. This is very helpful for more complex rigging and mechanical rigs, as it means the Action constraint can be controlled directly with a Driver or Custom Property.

Specifies how the keyframed transformation from the action is combined with the existing transformation. These modes are the same as in the Copy Transforms constraint.

The keyframed transformation is added before/after the existing transformation, as if it was applied to an imaginary parent/child of the constraint owner. Scale is handled like in the most basic Full Inherit Scale mode of bones, so combining non-uniform scale and rotation will create shear.

The keyframed transformation is added before/after the existing transformation, as if it was applied to an imaginary parent/child of the constraint owner. Scale is handled like in the Aligned Inherit Scale mode of bones to avoid creating shear.

This is equivalent to using the Split Channel option, but replacing the location component with the result of Full. If only uniform scale is used, the result is identical to Full.

Combines location, rotation and scale components of the transformation separately, similar to a sequence of three Copy Location, Copy Rotation and Copy Scale (with Offset) constraints bundled together in one operation; the result may be slightly different in case of sheared inputs. Unlike Aligned, in this mode location channels are simply added together, so rotation and scale components of the input transformations cannot affect the resulting location.

Warning

For technical reasons modes other than After Original (Full) and After Original (Aligned) may not work as expected for constraints on objects (not bones) without a parent.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

This selector controls which transform property (location, rotation or scale along/around one of its axes) from the target to use as the action driver.

This constraint allows you to choose in which space to evaluate its target's transform properties.

The lower and upper bounds of the driving transform property value.

Warning

Unfortunately, here again we find the constraint's limitations:

When using a rotation property as the driver, these values are mapped back to the $(-180.0 \text{ to } 180.0)$ range.

When using a scale property as the driver, these values are limited to null or positive values.

Select the name of the action you want to use.

Warning

Even though it might not be in red state (UI refresh problems), this constraint is obviously not functional when this field does not contain a valid action.

Bones only, when enabled, this option will make the constrained bone use the `object` part of the linked action, instead of the `same-named pose` part. This allows you to apply the action of an object to a bone.

The starting and ending frames of the action to be mapped.

Note

These values must be strictly positive.

By default, both values are set to 0, which disables the mapping (i.e. the owner just gets the properties defined at frame 0 of the linked action).

When the object or bone already has Action constraints, the next constraint using a newly keyframed action should be added before all others in order to get the same final combined transformation. This fact is not affected by the Mix mode.

Unlike usual, you can have a Startvalue higher than the Endone, or a Minone higher than a Maxone: this will reverse the mapping of the action (i.e. it will be \xe2\x80\x9cplayed\xe2\x80\x9d reversed\xe2\x80\xa6), unless you have both sets reversed, obviously! When using a Constraintaction, it is the constraint channel\xe2\x80\x99s names that are used to determine to which constraints of the owner apply the action. E.g. if you have a constraint channel named \xe2\x80\x9ctrackto_empty1\xe2\x80\x9d, its keyed Influence and/or Head/Tail values (the only ones you can key) will be mapped to the ones of the owner\xe2\x80\x99s constraint named \xe2\x80\x9ctrackto_empty1\xe2\x80\x9d. Similarly, when using a Poseaction (which is obviously only meaningful and working when constraining a bone!), it is the bone\xe2\x80\x99s name that is used to determine which bone channel\xe2\x80\x99s names from the action to use (e.g. if the constrained bone is named \xe2\x80\x9ccarm\xe2\x80\x9d, it will use and only use the action\xe2\x80\x99s bone channel named \xe2\x80\x9ccarm\xe2\x80\x9d\xe2\x80\xa6). Unfortunately, using a Poseaction on a whole armature object (to affect all the keyed bones in the action at once) will not work\xe2\x80\xa6 Actions can also be marked as Asset , but with certain limitations. For more info, see Pose Library .

--- Fin du chapitre: _action ---

--- Début du chapitre: _actions --- When animating objects and properties in Blender, Actions record and contain the data. As everything else in Blender, Actions are data-blocks.

Actions.\xc2\xba

So when you animate an object by changing its location with keyframes, the animation is saved to the Action.

Each property has a channel which it is recorded to, for example, Cube.location.x is recorded to Channel X Location. The X location and Y location properties can be shared across multiple objects, if all objects have X location and Y location properties beneath them.

Graph Editor. Each channel has an F-Curve represented by the lines between the keyframes.\xc2\xba

Record and contain animation data.

Are groups of channels.

Record properties.

F-Curves are used to interpolate the difference between the keyframes. Keyframes are used to set the values of properties bound to a point in time.

The Action data-block menu. \xc2\xba

When you first animate an object by adding keyframes, Blender creates an Action to record the data.

Actions can be managed with the Action data-block menu in the Dope Sheet Action Editor header, or the Sidebar region of the NLA Editor .

If you are making multiple actions for the same object, press the shield button for each action. This will give the actions a Fake User and will make Blender save the unlinked actions.

Objects can only use one Action at a time for editing. The NLA Editor is used to blend multiple actions together.

Actions with and without a Manual Frame Range in Dope Sheet. \xc2\xba

It is possible to manually specify the intended useful frame range of an action via a panel available in the Dope Sheet or the NLA Editor when a channel or NLA track is selected.

Manually specify the intended playback frame range for the action (this range is used by some tools, but does not affect animation evaluation). The manual frame range feature can be toggled with the checkbox.

When the range is set, it is used instead of the actual range occupied by key frames when adding a new track based on the action to NLA. It can also be used by exporters to determine the range of frames to export.

The range is displayed in the background of the editor as diagonal hash fill, to distinguish it from the solid fill of the current playback range. The frame values are most commonly expected to be integers, but can be fractional.

Specifies that the action is intended to be cyclic over the specified range. The first and last frames of the range should represent the same pose of the cycle one loop apart, i.e. the range should include the duplicated initial key of the loop.

Note

This option signifies intent and does not make the action cycle on its own. However, if Cycle-Aware Keying is enabled, it will automatically enable cyclic extrapolation and set up the loop period for curves newly added to the action.

Create and manage your own properties to store data in the action's data block. See the Custom Properties page for more information.

--- Fin du chapitre: _actions ---

--- Début du chapitre: _active_camera --- The Active Camera node outputs the scene's current active camera.

This node has no inputs.

This node has no properties.

Current active camera.

--- Fin du chapitre: _active_camera ---

--- Début du chapitre: _active_element --- The Active Element node outputs the index of the active vertex, edge, or face.

Note

This node can only be used in the Tool context.

This node has no inputs.

Which domain to return the index of.

Index of the active element in the specified domain.

True if an active element exists in the mesh, false otherwise

--- Fin du chapitre: _active_element ---

--- Début du chapitre: _active_spline --- Reference

Edit Mode

Properties \xa3 Curve \xa3 Active Spline

See also

Active Spline for curves.

The Active Spline panel is used in Edit Mode to control properties of the currently selected spline.

Active Spline panel.

Like curves, surfaces can be closed (cyclical) or open, independently in both directions, allowing you to easily create a tube, torus or sphere shape, and they can be viewed as \xe2\x80\x9csolids\xe2\x80\x9d in Edit Mode. This can be set per interpolation axis.

Makes the surface act like a B\xe3\x9zier curve. The control points act like Freehandles of a B\xe3\x9zier curve. Depending on the Order, 3 or 4 control points form one spline segment. This can be set per interpolation axis.

Makes the surface contact the end control points. This can be set per interpolation axis.

Endpoint U.\xc2\xb6

In the image below, the U interpolation axis is labeled as \xe2\x80\x9cU\xe2\x80\x9d and the V interpolation axis is labeled as \xe2\x80\x9cV\xe2\x80\x9d. The U\xe2\x80\x99s interpolation axis has been set to Endpoint and as such the surface now extends to the outer edges from E1 to E2 along the U interpolation axis.

To cause the surface to extend to all edges, Endpoint would be set for the V\xe2\x80\x99s axis as well.

This property is the same as with NURBS Curves; it specifies how much the control points are taken into account for calculating the curve of the surface shape. For high Orders 1 the surface pulls away from the control points, creating a smoother surface by assuming that the Resolution U/Vis high enough. For lowest Orders 2 the surface follows the control points, creating a surface that tends to follow the grid cage.

Order 2 and Order 4 surface.\xc2\xb6

For illustration purposes, in both Fig. Order 2 and Order 4 surface., the knot vectors were set to Endpoint, causing the surface to extend to all edges.

You can set independently the order for each interpolation axis, and like curves, it cannot be lower than 2, and higher than 6 or the number of control points on the relevant axis.

Alters the resolution of each segment by changing the number of subdivisions. This can be set per interpolation axis.

Use Smooth Shading for any 3D geometry.

--- Fin du chapitre: _active_spline ---

--- D but du chapitre: _adaptive --- In order for sculpting to give accurate and predictable results, Blender needs enough geometry. Instead of starting out with a highly subdivided mesh, add geometry dynamically by using either of the following adaptive sculpting methods.

\xe2\x80\x9cVoxel remeshing\xe2\x80\x9d rebuilds the geometry with a perfectly even distributed topology. Depending on the set voxel size, this will lead to a lower or higher resolution.

This technique is especially useful to block out the initial shape of an object. It also has the advantage of removing any overlapping geometry and creating a manifold volume as a result.

Any currently used mask, face sets and color attributes will be re-projected on the remeshed result. Reaching high vertex counts should still be achievable with this technique, depending on the used hardware.

Note

This technique will not work on objects that do not have an enclosed volume. Make sure to fill any holes in the mesh before remeshing. Or avoid any holes in the mesh/volume that are larger than the defined voxel size.

Tip

If in doubt, you can fill all holes in edit mode or by using the Mask Slice and Fill Holes operation to fill all holes in the mesh. If nothing is masked, it only fills any holes.

To more easily access this feature, use the shortcuts R to define the resolution, and Ctrl- R to execute the remeshing.

See also

More information at Remesh.

Dynamic topology (aka Dyntopo) is a dynamic tessellation sculpting method that automatically adds and removes topology under the brush.

Unlike the Voxel Remesher, this makes it possible to sculpt complex shapes without thinking about the resolution or topology. It also allows to define a different resolution wherever necessary. Much more complex base mesh sculpting is especially useful with this technique.

The disadvantages of this technique are a slower performance and limited support for some sculpt mode features. Custom attributes like Color Attributes, UV Maps and Face Sets are also lost or corrupted when using Dyntopo.

This feature shares the same shortcuts with voxel remeshing when enabled. Use R to define the resolution and Ctrl- R to flood fill the resolution (if Constant Detail is used).

Note

Because Dyntopo and the Voxel Remesher are mutually exclusive and cannot be used at the same time, both use the same shortcut to define the remeshing resolution.

Brushes like Density, Snake Hook and Clay Strips work especially well with this feature.

See also

More information at Dyntopo.

The Multiresolution Modifier can be used for subdivision based sculpting. This means the object will be subdivided, similar to the Subdivision Surface Modifier, only that the subdivisions can be freely sculpted for very high resolution detailing.

Note

For this technique it is highly recommended to use on a clean topology base mesh. This means the base mesh should be only made of quads and avoid non-manifold faces, as well as poles with two connected edges. More information at Quad Remeshing for an automatic retopology method.

This technique has the advantage of sculpting with multiple resolutions, meaning you have the ability to sculpt on any level of subdivision. This allows to add a much higher resolution of details for rendering and sculpting, while displaying lower resolutions for better viewport performance. It also allows sculpting on lower resolutions any time for broader changes.

As an example, you can sculpt general proportions in subdivision level 1, add high resolution details in level 4 and switch back to subdivision 1 to correct the shape further.

The disadvantages are that you may end up with some mesh distortions because the topology is not dynamic like voxel remeshing and dyntopo. The topology should also not be changed once already subdivided, since any edits to the base mesh will result in corrupted subdivision details.

Tip

Pay attention to the topology that you sculpt and how much it gets stretched. If more resolution is needed you can always subdivide another

time, but there will be worse performance and slower level switching once more than 5 subdivisions are used. Alternatively use the Slide Relax brush to slide topology to where it is needed.

Additional brushes like the Multires Eraser and Multires Smear are recommended for adjustments.

Here are general shortcuts to use the feature.

Step up one multires level Alt- 2

Step down one multires level Alt- 1

Set multires level / Create multires modifier Ctrl- 0 to Ctrl- 5

See also

More information at Multiresolution Modifier.

--- Fin du chapitre: _adaptive ---

--- Début du chapitre: _adaptive_domain ---Reference

Domain

Physics \xe2\x80\xa3 Fluid \xe2\x80\xa3 Adaptive Domain

When enabled, the domain will adaptively shrink to best fit the gas, saving computation time by leaving voxels without gas out of the simulation. Unless the Add Resolution is used, the adaptive domain will not exceed the bounds of the original domain.

Number of voxels to add around the outside of the domain.

Amount of extra space to leave around gas, measured in voxels. With very fast-moving gas larger margins may be required to prevent the gas from being cut off by the adaptive boundary, but note this will increase the number of voxels which need to be computed.

Smallest amount of gas a voxel can contain before it is considered empty and the adaptive domain is allowed to cut it out of the simulation.

--- Fin du chapitre: _adaptive_domain ---

--- Début du chapitre: _adaptive_subdiv ---Reference

Modifier \xe2\x80\xa3 Subdivision Surface

Note

Implementation not finished yet, marked as an Experimental Feature Set .

When using the Experimental Feature Set the Subdivision Surface Modifier gets changed to control the subdivision of a mesh at the time of rendering. For this, all the other settings are the same except the View and Render settings. These before mentioned settings get removed/renamed and the following settings are added:

Use OpenSubdiv to give different subdivision levels to near and far objects automatically. This allows nearer objects to get more subdivisions and far objects to get less.

Multiplier of the scene dicing rate to determine the final size of Micropolygons in pixels.

Subdivision off/on, Dicing Scale: 1.0 - 0.3 - 0.05 (monkeys look identical in viewport, no modifiers). \xc2\xbb6

Multi-user object data are currently made single users, leading to increased memory usage. For those it is better to use non-adaptive subdivision still.

Multi-view renders can have some inconsistencies between views.

Warning

Instances are not tessellated individually. Instead, the original object is tessellated and then duplicated on all instances. To take advantage of both adaptive subdivision and instancing you should place the original object at the position of the instance that is closest from the camera.

--- Fin du chapitre: _adaptive_subdiv ---

--- Début du chapitre: _add --- The Add effect strip adds the colors of two strips together. Use this effect with a base image strip, and a modifier strip. The modifier strip is either a solid color or a black-and-white mask, or another image entirely.

You can use this effect to increase the brightness of an image, or if you use a BW mask, selectively increase the brightness of certain areas of the image. The Mix node, in Add mode, does exactly the same thing as the Add SFX strip here, and is controlled the same way by feeding the Factor input.

The example shows what happens when you add gray to an image. The image gets bright because we are adding gray RGB(0.5, 0.5, 0.5) to say, a blue color RGB(0.1, 0.1, 0.5) resulting in RGB(0.6, 0.6, 1.0) which retains the original hue (relationship between the colors) but is much brighter (has a higher value). When applied to the whole image like this, it seems to flash.

This strip has no options.

Add Effect. \xc2\xba6

--- Fin du chapitre: _add ---

--- Début du chapitre: _addons --- Add-ons let you extend Blender's functionality using Python. Most of the time you can get add-ons as part of the Extension system.

Tip

If the Add-on does not activate when enabled, check the Console window for any errors that may have occurred.

If the add-on needs to use internet, it must check for the (read-only) property `bpy.app.online_access`. This option is controlled by Preferences, which can be overriding via command-line (`--offline-mode / --online-mode`).

For better error messages, you can check also for `bpy.app.online_access_overridden`, to determine whether users can turn Allow Online Access on the preferences, or not.

Note

Add-ons that follow this setting will only connect to the internet if enabled. However, Blender cannot prevent third-party add-ons from violating this rule.

For add-ons to be bundled as extensions, they must be self-contained. That means they must come with all its dependencies. In particular 3rd party Python modules.

There are a few options for this:

This is the recommended way to bundle dependencies.

This is recommended if an add-on depends on another add-on.

Make sure that both the individual and the combined add-on check for already registered types (Operators, Panels, \xc2\xa6). This avoids duplication of operators and panels on the interface if the add-ons are installed as a bundle and individually.

This can be used as a way to bundle a pure Python dependencies as a sub-module.

This has the advantage of avoiding version conflicts although it requires some work to setup each package.

Add-ons must not assume their own directory is user writable since this may not be the case for `\xc2\x80\x9cSystem\xc2\x80\x9d`

repositories. Writing files into the add-on directory also has the downside that upgrading the extension will remove all files. Add-ons which need their own user directory should use a utility function provided for this purpose:

If you wish create subdirectories, this can be done with the path argument.

This directory will be kept between upgrades but will be removed if the extension is uninstalled.

With the introduction of Extensions in Blender 4.2, the old way of creating add-ons is considered deprecated. While the changes are rather small they impact existing add-ons.

In order to allow a smooth transition process, the so-called legacy add-ons will continue to be supported by Blender. They can be installed via Install legacy Add-on button in the User Preferences.

All add-on maintainers are urged to convert the add-ons they want to share, so they are future proof and can support features like updating from the extensions platform.

- Create a manifest file.

- Remove the `bl_info` information (this is now in the manifest).

- Replace all references to the module name to `__package__`.

- Make all module imports to use relative import.

- Use `wheel` to pack your external Python dependencies.

- Remember to test it thoroughly.

Note

For testing it is important to install the extension from disk and check if everything is working well. This will get you as close to the final experience as possible.

The legacy add-ons would use their module name to access the preferences. This could lead to a name clash when extensions with the same name (from different repositories) would be installed. To prevent this conflict, the repository name is now part of the namespace.

For example, now instead of `kitsu` the module name would be `bl_ext.{repository}_module_name.kitsu` instead.

This has a few implications for preferences and module imports.

Add-ons can define their own preferences which use the package name as an identifier. This can be accessed using `__package__`.

This was already supported in the legacy add-ons, but not reinforced. As such this can break backward compatibility.

Before:

Now:

An add-on that defines sub-packages (sub-directories with their own `__init__.py` file) that needs to use this identifier will have to import the top-level package using a relative import.

Then `base_package` can be used instead of `__package__`. The `.. imports` relative to the packages parent, sub-sub-packages must use `...` and so on.

Note

The value of `__package__` will vary between systems so it should never be replaced with a literal string.

Extensions must not manipulate the value of `__package__` as this may cause unexpected behavior or errors.

```
from kitsu import utils
```

```
from importutils
```

Importing packages within the add-on module need to use relative paths. This is a standard Python feature and only applicable for add-ons that have multiple folders.

This was already supported in the legacy add-ons, but not reinforced. As such this can break backward compatibility.

--- Fin du chapitre: _addons ---

--- Début du chapitre: _addon_tutorial --- This tutorial is designed to help technical artists or developers learn to extend Blender. An understanding of the basics of Python is expected for those working through this tutorial.

Before going through the tutorial you should

Be familiar with the basics of working in Blender.

Know how to run a script in Blender's Text editor.

Have an understanding of Python primitive types (integer, Boolean, string, list, tuple, dictionary, and set).

Be familiar with the concept of Python modules.

Have a basic understanding of classes (object orientation) in Python.

Suggested reading before starting this tutorial.

Dive Into Python sections (1, 2, 3, 4, and 7).

Blender API Quickstart to help become familiar with Blender/Python basics.

To best troubleshoot any error message Python prints while writing scripts, you run Blender from a terminal. See Use The Terminal.

Tip

You can enable Developer Extras in the preferences to enable features that make developing add-ons easier.

While going through the tutorial, you may want to look into our reference documentation.

Blender API Overview: This document is rather detailed but helpful if you want to know more on a topic.

bpy.context API reference Handy to have a list of available items your script may operate on.

bpy.types.Operator The following add-ons define operators, these docs give details and more examples of operators.

An add-on is simply a Python module with some additional requirements so Blender can display it in a list with useful information.

To give an example, here is the simplest possible add-on:

is a dictionary containing add-on metadata such as the title, version and author to be displayed in the Preferences add-on list. It also specifies the minimum Blender version required to run the script; older versions won't display the add-on in the list.

is a function which only runs when enabling the add-on, this means the module can be loaded without activating the add-on.

is a function to unload anything setup by register, this is called when the add-on is disabled.

Notice this add-on does not do anything related to Blender (the blender_api: bpy module is not imported for example).

This is a contrived example of an add-on that serves to illustrate the point that the base requirements of an add-on are simple.

An add-on will typically register operators, panels, menu items, etc, but it's worth noting that any script can do this, when executed from the Text editor or even the interactive console there is nothing inherently different about an add-on that allows it to integrate

with Blender, such functionality is just provided by the `blender_api:bpy` module for any script to access. So an add-on is just a way to encapsulate a Python module in a way a user can easily utilize.

Note

Running this script within the Text editor won't print anything, to see the output it must be installed through the Preferences. Messages will be printed when enabling and disabling. The simplest possible add-on above is useful as an example but not much else. This next add-on is simple but shows how to integrate a script into Blender using an Operator which is the typical way to define a tool accessed from menus, buttons and keyboard shortcuts.

For the first example we will make a script that simply moves all objects in a scene.

Add the following script to the Text editor in Blender:

Click the Run Script button, all objects in the active scene are moved by 1.0 unit.

This add-on takes the body of the script above, and adds it to an operator's `execute()` function.

Note

`bl_info` is split across multiple lines, this is just a style convention used to more easily add items.

Note

Rather than using `bpy.context.scene`, we use the `context.scene` argument passed to `execute()`. In most cases these will be the same. However in some cases, operators will be passed a custom context so script authors should prefer the `context` argument passed to operators.

To test the script, you can copy and paste it into Blender's Text editor and run it. This will execute the script directly and call `register` immediately.

However running the script won't move any objects. For this, you need to execute the newly registered operator.

Operator Search menu.

Open the Operator Search menu and type in `Move X` by One (the `bl_label`), then Return.

The objects should move as before.

Keep this add-on open in Blender for the next step - Installing.

Once you have your add-on within Blender's Text editor, you will want to be able to install it so it can be enabled in the Preferences to load on startup.

Even though the add-on above is a test, let's go through the steps anyway so you know how to do it for later.

To install the Blender text as an add-on, you will first have to save it on drive. Take care to obey the naming restrictions that apply to Python modules and end with a `.py` extension.

Once the file is on drive, you can install it as you would for an add-on downloaded online.

Open the Preferences Add-ons Install and select the file.

Now the add-on will be listed and you can enable it by pressing the checkbox, if you want it to be enabled on restart, press Save as Default. The operator can be run in the same way as described in the previous section.

When the add-on is enabled, Blender executes the code and runs `register()` function. When the add-on is disabled, Blender runs `unregister()` function.

Note

The destination of the add-on depends on your Blender configuration. When installing an add-on the source and destination paths are printed in the console. You can also find add-on path locations by running this in the Python Console:

More is written on this topic here: [Directory Layout](#) .

For our second add-on, we will focus on object instancing \xe2\x80\x93 this is \xe2\x80\x93 to make linked copies of an object in a similar way to what you may have seen with the Array modifier.

As before, first we will start with a script, develop it, then convert it into an add-on.

Now try copying this script into Blender and run it on the default Cube. Make sure you click to move the 3D cursor before running as the duplicate will appear at the cursor \xe2\x80\x99s location.

After running, notice that when you go into Edit Mode to change the Cube \xe2\x80\x93 all of the copies change. In Blender, this is known as Linked Duplicates.

Next, we \xe2\x80\x99re going to do this in a loop, to make an array of objects between the active object and the cursor.

Try running this script with the active object and the cursor spaced apart to see the result.

With this script you \xe2\x80\x99ll notice we \xe2\x80\x99re doing some math with the object location and cursor, this works because both are `mathutils.Vector` instances, a convenient class provided by the `mathutils` module which allows vectors to be multiplied by numbers and matrices.

If you are interested in this area, read into `mathutils.Vector` \xe2\x80\x93 there are many handy utility functions such as getting the angle between vectors, cross product, dot products as well as more advanced functions in `mathutils.geometry` such as `Bzier` spline interpolation and ray-triangle intersection.

For now we will focus on making this script an add-on, but it \xe2\x80\x99s good to know that this 3D math module is available and can help you with more advanced functionality later on.

The first step is to convert the script as-is into an add-on:

Everything here has been covered in the previous steps, you may want to try run the add-on still and consider what could be done to make it more useful.

The two of the most obvious missing things are \xe2\x80\x93 having the total fixed at 10, and having to access the operator with `Operator Search` is not very convenient.

Both these additions are explained next, with the final script afterwards.

There are a variety of property types that are used for tool settings, common property types include: `int`, `float`, `vector`, `color`, `Boolean` and `string`.

These properties are handled differently to typical Python class attributes because Blender needs to display them in the interface, store their settings in keymaps and keep settings for reuse.

While this is handled in a fairly Pythonic way, be mindful that you are in fact defining tool settings that are loaded into Blender and accessed by other parts of Blender, outside of Python.

To get rid of the literal 10 for total , we use an operator property. Operator properties are defined via bpy.props module, this is added to the class body:

These properties from bpy.props are handled specially by Blender when the class is registered so they display as buttons in the user interface. There are many arguments you can pass to properties to set limits, change the default and display a tooltip.

See also

bpy.props.IntProperty()

This document doesn't go into details about using other property types. However, the link above includes examples of more advanced property usage.

Add-ons can add to the user interface of existing panels, headers and menus defined in Python.

For this example we add to an existing menu.

Menu Identifier. \xc2\xba6

To find the identifier of a menu, first enable Python Tooltips in the preferences. Then you can hover your mouse over the menu item and the identifier is displayed.

The method used for adding a menu item is to append a draw function into an existing class:

For docs on extending menus, see: bpy.types.Menu .

In Blender, add-ons have their own keymaps so as not to interfere with Blender's built-in keymaps.

In the example below, a new object mode bpy.types.KeyMap is added, then bpy.types.KeyMapItem is added to the keymap which references our newly added operator, using Shift- Ctrl- T as the key shortcut to activate it. Notice how the keymap item can have a total setting different than the default set by the operator, this allows you to have multiple keys accessing the same operator with different settings.

Note

While Shift- Ctrl- T is not a default Blender key shortcut, it is hard to make sure add-ons will not overwrite each other's keymaps. Thus at least take care when assigning keys that they do not conflict with important functionality of Blender (see also is key free add-on).

For API documentation on the functions listed above, see:

bpy.types.KeyMaps.new()

bpy.types.KeyMap

bpy.types.KeyMapItems.new()

bpy.types.KeyMapItem

In the menu. \xc2\xba6

Run the script (or save it and add it through the Preferences like before) and it will appear in the Object menu.

Operator Property. \xc2\xba6

After selecting it from the menu, you can choose how many instances of the cube you want create.

Note

Directly executing the script multiple times will add the menu each time too. While not useful behavior, there is nothing to worry about since add-ons will not register themselves multiple times when enabled through the Preferences.

Add-ons can encapsulate certain functionality neatly for writing tools to improve your workflow or for writing utilities for others to use.

While there are limits to what Python can do within Blender, there is certainly a lot that can be achieved without having to dive into Blender's C/C++ code.

The example given in the tutorial is limited, but shows the Blender API used for common tasks that you can expand on to write your own tools.

Blender comes with commented templates which are accessible from the Text editor's header. If you have specific areas you want to see example code for, this is a good place to start.

Here are some sites you might like to check on after completing this tutorial.

Blender/Python API Overview For more background details on Blender/Python integration.

How to Think Like a Computer Scientist Great info for those who are still learning Python.

Blender Development Blender Development, general information and helpful links.

Blender Developer Forum Forum where people ask Python development questions.

--- Fin du chapitre: _addon_tutorial ---

--- Début du chapitre: _add_cone ---Reference

Object Mode and Edit Mode

Toolbar Add Cone

Interactively add a cone mesh object.

First define the base of the object by dragging with LMB. Next, release LMB and move the mouse to define the height of the object. Finally, click LMB to confirm the shape of the object.

You can use the following hotkeys to temporarily change a setting (for as long as the key is held):

Ctrl

Toggles snapping.

Alt

Toggles the Origin setting.

Shift

Toggles the Aspect setting.

The initial depth (from the screen into the scene) used when placing the object.

Start placing on the surface under the mouse cursor. If there is no surface, this does the same as Cursor Plane.

Start placing on a plane that goes through the 3D Cursor and is aligned according to the Orientation and Plane Axis.

Start placing on a plane that goes through the 3D Cursor and is aligned to the view.

The new object's orientation is a set of three axes, out of which Plane Axis chooses one.

The object uses the normal orientation of the surface under the mouse cursor. If there is no surface, this does the same as Default.

The object uses the default Transform Orientation.

The target to use while Snapping.

Snap to all types of geometry (vertices, edges, and faces).

Snap to the target defined in the global snapping options.

Which of the three Orientation axes (X, Y or Z) is

used for the object. The object's base will be perpendicular to this axis.

Rather than using the Orientationaxis indicated by Plane Axis,use the one that\xe2\x80\x99s closest to the viewport\xe2\x80\x99s viewing direction(when not hovering over a surface).

Base

How the base is defined.

The base is defined from one corner to the opposing corner.

The base is defined from the centerpoint to a corner.

Whether the base has a free or fixed aspect ratio.

The width and depth of the base can be chosen independently.

The width and depth of the base are forced to be equal.

Height

How the height is defined.

The base becomes the bottom, after which you define the top.

The base becomes the center, after which you define the top.

Whether the side of the bounding box has a free or fixed aspect ratio.

The height can be chosen independently of the base.

The height is forced to be equal to the largest side of the base.

The number of vertices in the base.

Set how the circle at the base will be filled.

Fill with triangular faces which share a vertex in the middle.

Fill with a single N-gon .

Do not fill. Creates only the outer ring of vertices.

--- Fin du chapitre: _add_cone ---

--- Début du chapitre: _add_cube ---Reference

Object Mode and Edit Mode

Toolbar \xe2\x80\xa3 Add Cube

Interactively add acube mesh object.

First define the base of the object by dragging with LMB.Next, release

LMBand move the mouse to define the height of the object.Finally, click

LMBto confirm the shape of the object.

You can use the following hotkeys to temporarily change a setting(for as long as the key is held):

Ctrl

Toggles snapping.

Alt

Toggles the Originsetting.

Shift

Toggles the Aspectsetting.

The initial depth (from the screen into the scene) used when placing the object.

Start placing on the surface under the mouse cursor. If there is no surface, this does thesame as Cursor Plane.

Start placing on a plane that goes through the 3D Cursor and is aligned according to the Orientationand Plane Axis.

Start placing on a plane that goes through the 3D Cursor and is aligned to the view.

The new object\xe2\x80\x99s orientation \xe2\x80\x93 a set of three axes, out of which Plane Axischooses one.

The object uses the normal orientation of the surface under the mouse cursor.If there is no surface, this does the same as Default.

The object uses the default Transform Orientation .

The target to use while Snapping .

Snap to all types of geometry (vertices, edges, and faces).

Snap to the target defined in the global snapping options.
Which of the three Orientationaxes (X, Y or Z) is
\xe2\x80\x99s base
will be perpendicular to this axis.
Rather than using the Orientationaxis indicated by Plane Axis, use the one
that\xe2\x80\x99s closest to the viewport\xe2\x80\x99s viewing
direction(when not hovering over a surface).

Base

How the base is defined.

The base is defined from one corner to the opposing corner.

The base is defined from the centerpoint to a corner.

Whether the base has a free or fixed aspect ratio.

The width and depth of the base can be chosen independently.

The width and depth of the base are forced to be equal.

Height

How the height is defined.

The base becomes the bottom, after which you define the top.

The base becomes the center, after which you define the top.

Whether the object\xe2\x80\x99s side has a free or fixed aspect ratio.

The height can be chosen independently of the base.

The height is forced to be equal to the largest side of the base.

--- Fin du chapitre: _add_cube ---

--- Début du chapitre: _add_curves --- Used to distribute new curves on
the surface mesh. This tool requires the curve to have a surface object set.

The curves follow the surface normals. Using the interpolation options
allows the brush to take the characteristics of existing curves.

Number of curves added.

Note

Interpolation allows to add hair which are already combed. The new
curves are created following the previously created curves which are in
the vicinity.

Use the average length of the curves in close proximity.

Use the average radius of the curves in close proximity. If there is no
radius attribute, then the interpolation will skip.

Use the average shape of the curves in close proximity.

Use the average amount of control points of the curves in close
proximity.

Length of newly added curves when not interpolated.

Radius of newly added curves when not interpolated.

Number of Control Points for the new created curves when the point count
is not interpolated.

--- Fin du chapitre: _add_curves ---

--- Début du chapitre: _add_cylinder ---Reference

Object Mode and Edit Mode

Toolbar \xe2\x80\xa3 Add Cylinder

Interactively add a cylinder mesh object.

First define the base of the object by dragging with LMB. Next, release
LMB and move the mouse to define the height of the object. Finally, click
LMB to confirm the shape of the object.

You can use the following hotkeys to temporarily change a setting (for as
long as the key is held):

Ctrl

Toggles snapping.

Alt

Toggles the Originsetting.

Shift

Toggles the Aspectsetting.

The initial depth (from the screen into the scene) used when placing the object.

Start placing on the surface under the mouse cursor. If there is no surface, this does the same as Cursor Plane.

Start placing on a plane that goes through the 3D Cursor and is aligned according to the Orientation and Plane Axis.

Start placing on a plane that goes through the 3D Cursor and is aligned to the view.

The new object's orientation is a set of three axes, out of which Plane Axis chooses one.

The object uses the normal orientation of the surface under the mouse cursor. If there is no surface, this does the same as Default.

The object uses the default Transform Orientation.

The target to use while Snapping.

Snap to all types of geometry (vertices, edges, and faces).

Snap to the target defined in the global snapping options.

Which of the three Orientation axes (X, Y or Z) is

used for the object. The object's base will be perpendicular to this axis.

Rather than using the Orientation axis indicated by Plane Axis, use the one that's closest to the viewport's viewing direction (when not hovering over a surface).

Base

How the base is defined.

The base is defined from one corner to the opposing corner.

The base is defined from the centerpoint to a corner.

Whether the base has a free or fixed aspect ratio.

The width and depth of the base can be chosen independently.

The width and depth of the base are forced to be equal.

Height

How the height is defined.

The base becomes the bottom, after which you define the top.

The base becomes the center, after which you define the top.

Whether the side of the bounding box has a free or fixed aspect ratio.

The height can be chosen independently of the base.

The height is forced to be equal to the largest side of the base.

The number of vertices in the caps.

Set how the caps will be filled.

Fill with triangular faces which share a vertex in the middle.

Fill each ring with an N-gon.

Do not fill. Creates only the outer rings of vertices.

--- Fin du chapitre: `_add_cylinder` ---

--- Début du chapitre: `_add_icosphere` ---Reference

Object Mode and Edit Mode

Toolbar Add Icosphere

Interactively add an Icosphere mesh object.

First define the base of the object by dragging with LMB. Next, release LMB and move the mouse to define the height of the object. Finally, click LMB to confirm the shape of the object.

You can use the following hotkeys to temporarily change a setting (for as long as the key is held):

Ctrl

Toggles snapping.

Alt

Toggles the Origin setting.

Shift

Toggles the Aspect setting.

The initial depth (from the screen into the scene) used when placing the object.

Start placing on the surface under the mouse cursor. If there is no surface, this does the same as Cursor Plane.

Start placing on a plane that goes through the 3D Cursor and is aligned according to the Orientation and Plane Axis.

Start placing on a plane that goes through the 3D Cursor and is aligned to the view.

The new object's orientation is a set of three axes, out of which Plane Axis chooses one.

The object uses the normal orientation of the surface under the mouse cursor. If there is no surface, this does the same as Default.

The object uses the default Transform Orientation.

The target to use while Snapping.

Snap to all types of geometry (vertices, edges, and faces).

Snap to the target defined in the global snapping options.

Which of the three Orientation axes (X, Y or Z) is

the object's base will be perpendicular to this axis.

Rather than using the Orientation axis indicated by Plane Axis, use the one that's closest to the viewport's viewing direction (when not hovering over a surface).

Base

How the base is defined.

The base is defined from one corner to the opposing corner.

The base is defined from the center point to a corner.

Whether the base has a free or fixed aspect ratio.

The width and depth of the base can be chosen independently.

The width and depth of the base are forced to be equal.

Height

How the height is defined.

The base becomes the bottom, after which you define the top.

The base becomes the center, after which you define the top.

Whether the side of the bounding box has a free or fixed aspect ratio.

The height can be chosen independently of the base.

The height is forced to be equal to the largest side of the base.

Influences how many vertices are used to define the sphere. At level 1 the icosphere is an icosahedron, a solid with 20 equilateral triangular faces. Each increase in the number of subdivisions splits each triangular face into four.

Note

Subdividing an icosphere raises the vertex count very quickly even with few iterations(10 times creates 5,242,880 triangles).Adding such a dense mesh is a sure way to cause the program to crash.

--- Fin du chapitre: _add_icosphere ---

--- Début du chapitre: _add_uv sphere ---Reference

Object Mode and Edit Mode

Toolbar \xe2\x80\xa3 Add UV Sphere

Interactively add aUV sphere mesh object.

First define the base of the object by dragging with LMB.Next, release LMBand move the mouse to define the height of the object.Finally, click LMBto confirm the shape of the object.

You can use the following hotkeys to temporarily change a setting(for as long as the key is held):

Ctrl

Toggles snapping.

Alt

Toggles the Originsetting.

Shift

Toggles the Aspectsetting.

The initial depth (from the screen into the scene) used when placing the object.

Start placing on the surface under the mouse cursor. If there is no surface, this does thesame as Cursor Plane.

Start placing on a plane that goes through the 3D Cursor and is aligned according to the Orientationand Plane Axis.

Start placing on a plane that goes through the 3D Cursor and is aligned to the view.

The new object\xe2\x80\x99s orientation \xe2\x80\x93 a set of three axes, out of which Plane Axischooses one.

The object uses the normal orientation of the surface under the mouse cursor.If there is no surface, this does the same as Default.

The object uses the default Transform Orientation .

The target to use while Snapping .

Snap to all types of geometry (vertices, edges, and faces).

Snap to the target defined in the global snapping options.

Which of the three Orientationaxes (X, Y or Z) is

\xe2\x80\x93 for the object.The object\xe2\x80\x99s base will be perpendicular to this axis.

Rather than using the Orientationaxis indicated by Plane Axis,use the one that\xe2\x80\x99s closest to the viewport\xe2\x80\x99s viewing direction(when not hovering over a surface).

Base

How the base is defined.

The base is defined from one corner to the opposing corner.

The base is defined from the centerpoint to a corner.

Whether the base has a free or fixed aspect ratio.

The width and depth of the base can be chosen independently.

The width and depth of the base are forced to be equal.

Height

How the height is defined.

The base becomes the bottom, after which you define the top.

The base becomes the center, after which you define the top.

Whether the side of the bounding box has a free or fixed aspect ratio.

The height can be chosen independently of the base.
The height is forced to be equal to the largest side of the base.
Number of vertical segments. Like the Earth's meridians, going pole to pole.

Number of horizontal segments. These are like the Earth's parallels.

Note

Rings are face loops and not edge loops, which would be one less.

--- Fin du chapitre: `_add_uvsphere` ---

--- Début du chapitre: `_adjustment` --- The Adjustment Layer strip works like a regular input file strip except for the fact, that it considers all strips below it as its input.

Real-world use cases, you want to add some last finishing color correction on top of parts of your final sequence, timeline without messing with meta strips around. Just add an adjustment layer on top and activate the color balance.

Or you can stack a primary color correction and several secondary color corrections on top of each other (probably using the new mask input for area selection).

This strip has no options.

--- Fin du chapitre: `_adjustment` ---

--- Début du chapitre: `_alembic` --- From the Alembic home page:

Alembic is an open computer graphics interchange framework. Alembic distills complex, animated scenes into a non-procedural, application-independent set of baked geometric results. This distillation of scenes into baked geometry is exactly analogous to the distillation of lighting and rendering scenes into rendered image data.

Alembic is focused on efficiently storing the computed results of complex procedural geometric constructions. It is very specifically not concerned with storing the complex dependency graph of procedural tools used to create the computed results. For example, Alembic will efficiently store the animated vertex positions and animated transforms that result from an arbitrarily complex animation and simulation process which could involve enveloping, corrective shapes, volume-preserving simulations, cloth and flesh simulations, and so on. Alembic will not attempt to store a representation of the network of computations (rigs, basically) which are required to produce the final, animated vertex positions and animated transforms.

In brief, Alembic can be used to write an animated mesh to a drive, and read it back quickly and efficiently. This means that a mesh can be animated with a very CPU-intensive rig and then baked to an Alembic file. Finally it can be loaded into the shot file for shading and lighting with only moderate CPU usage.

Due to the Open Source nature of the Alembic standard as well as the C++ library implementing that standard, Blender can be used in a hybrid pipeline. For example, other software, such as Houdini or Maya, can export files to Alembic, which can then be loaded, shaded, and rendered in Blender. It is also possible to animate characters (or other models) in Blender, export to Alembic, and load those files into other software for further processing.

When importing an Alembic file, Mesh Sequence Cache modifiers are automatically added to time-varying meshes. For time-varying object transforms (so animation of rotation, location, or scale) the Transform Cache Constraint is used.

Value by which to enlarge or shrink the objects with respect to the world's origin

Select the file relative to the blend-file.

If checked, update scene's start and end frame to match those of the Alembic archive.

Set to true if the cache is split into separate files.

Check the imported mesh for corrupt data and fix it if necessary. When disabled, erroneous data may cause crashes displaying or editing the meshes. This option will make the importing slower but is recommended, as data errors are not always obvious.

Add cache modifiers and constraints to imported objects even if they are not animated so that they can be updated when reloading the Alembic archive.

This section describes the effect of the different export options.

General options.

This sets the global scale of the Alembic file. Keep it at the default value of 1.0 to use Blender's units.

When enabled, exports only the selected objects. When disabled, all objects are exported.

Limits the export to scene collections that are currently visible.

Scene options.

Sets the frame range to export to Alembic. This defaults to the current scene frame range.

These options control the sub-frame sampling of animations.

Transform Samples sets the number of times per frame at which animated transformations are sampled and written to Alembic.

Geometry Samples sets the same, but then for animated geometry.

Shutter Open/Close define the interval [open, close] over which those samples are taken. The valid range is -1 to 1, where -1 indicates the previous frame, 0 indicates the current frame, and 1 indicates the next frame.

For example, if information for detailed mesh motion blur is desired, some subframes around the current frame can be written to Alembic by using a sample count of 5, Shutter Open at -0.25 and Shutter Close at 0.25. This mimics a 180 degree shutter, opening 90 degrees before the frame and closing 90 degrees after the frame.

Exports data of duplicated or instanced objects as Alembic instances; speeds up the export and can be disabled for compatibility with other software.

When enabled (which it is by default), custom properties are exported to Alembic as well. The following custom property types are supported:

Numbers (int, float) and strings. These are exported as arrays of a single element, so 47 will be exported as [47] to Alembic, and "Agent" to ["Agent"]. This matches the behavior of many other DCCs.

Lists of numbers and strings. These are exported as-is, so [327, 47] is exported as [327, 47].

Matrices and nested arrays of numbers. These are flattened into one long list, so a 3x2 matrix of numbers will become a list of 6 numbers. Similarly, nested lists [[1, 2, 3], [4, 5], [6]] will be exported as [1, 2, 3, 4, 5, 6].

Numbers can be animated as well.

When disabled, parent/child relations between objects are exported too. Any parent object that is not exported itself, but with children that are exported, is replaced by an empty. When enabled, parent/child relations are not exported, and transformations are all written in world coordinates.

Determines visibility of objects, modifier settings, and other areas where there are different settings for viewport and rendering.

Use Render settings for object visibility, modifier settings, etc.

Use Viewport settings for object visibility, modifier settings, etc.

Geometry options.

When enabled, UV maps are exported. Although the Alembic standard only supports a single UV map, Blender exports all UV maps in a way that should be readable by other software.

When enabled, UVs sharing the same vertex and location will be merged into a single UV. The exported file can be slightly smaller.

When enabled, an object's Normals are exported. See Custom Split Normals of Meshes below for more information.

When enabled, exports Color Attributes.

Exports the material names per face. The material data is not exported but only material names.

Applies any Subdivision Surface modifiers before writing to Alembic.

Writes polygonal meshes using the 'SubD' Alembic schema, rather than the 'PolyMesh' schema. This sets an import option for the program, with which the file is opened, to apply its form of a non-destructive subdivision.

Triangulates the mesh before writing to Alembic. For more detail on the specific option see the Triangulate modifier.

Particle Systems options.

Alembic has no support for Particle Systems, in the same way that it does not support armatures.

Hair is exported as animated zero-width curves.

Particles are exported as animated points.

Blender supports the import and export of custom normals to Alembic files. As a basic rule of thumb, a completely smooth mesh will be exported without normals and thus produce the smallest Alembic file. This is reflected in the importer; an Alembic mesh without normals is loaded as a smooth mesh.

On export, for every mesh:

If it has Custom Loop Normals then the loop normals are exported.

If one or more polygons are marked flat then loop normals are also exported.

Otherwise, no normals are exported.

On import, when the Alembic mesh contains:

Loop normals (kFaceVaryingScope) are used as custom loop normals.

Vertex normals (kVertexScope or kVaryingScope) are converted to loop normals, and handled as above.

If there are no normals then the mesh is marked as smooth.

Unsupported normal types (kConstantScope , kUniformScope , kUnknownScope) are handled as no normals.

When an imported mesh does not contain normals, the final look can be controlled using the Normal's Shading .

Unlike Blender and many other applications and file formats, Alembic files don't have any concept of frames. Alembic works purely with time, and values that are sampled over time. For example, there is no

way to distinguish 30 FPS with 2 samples per frame, and 60 FPS with 1 sample per frame. This has caused many developers to just hard-code 24 FPS when reading Alembic files.

Blender uses the current scene frame rate to convert a frame number (in Blender) to a time in seconds (in Alembic). As a result, you can import an Alembic file that was produced at 120 FPS into a Blender scene that is 30 FPS and still not see any time stretching.

--- Fin du chapitre: `_alembic` ---

--- Début du chapitre: `_align` ---Reference

View `\xe2\x80\xa3 Align View`

These options allow you to align and orient the view.

Aligns the view to a certain local axis of the active object, bone, or (in Edit Mode) the normal of the active face. The view also becomes orthographic.

To return to the regular (untilted) perspective view, you can first press `Numpad3` to align to the global X axis, then orbit with `MMB`.

Moves and rotates the active camera so it matches the current viewpoint.

Moves the active camera (without changing its orientation) so that its view frames the selected objects.

Moves the 3D Cursor back to the world origin and changes the view so that you can see everything in your scene.

Centers the view on the 3D Cursor.

Centers the view on the active object and makes it the point of interest. The view will continue orbiting around the object even if you pan to a different location. In addition, it will follow the object if it moves.

Returns the view to how it was before using View Lock to Active.

--- Fin du chapitre: `_align` ---

--- Début du chapitre: `_align_euler_to_vector` --- The Align Euler to Vector node rotates an Euler rotation into the given direction.

Important

This node is deprecated, use the Align Rotation to Vector Node instead. The Euler rotation to align.

Important

This input has to be a rotation input. Be careful not to connect a direction vector like the normal.

Determines how much the points are rotated towards the vector. Zero effectively disables the node and one means that the points are aligned with the vector perfectly.

The direction vector that points should be rotated to. The vector is in the local space of the object that is being modified. When it is all zeros for a point, it is not rotated at all.

Local axis of the object that is to be rotated towards the vector input. The local axis to rotate around.

The best rotation angle is computed automatically. This minimizes the angle of rotation.

Rotate around a specific local axis.

The rotated Euler rotation.

--- Fin du chapitre: `_align_euler_to_vector` ---

--- Début du chapitre: `_align_objects` ---Reference
Object Mode

Object \xe2\x80\xa3 Transform \xe2\x80\xa3 Align Objects

The Align tool is used to align multiple selected objects so they line up on a specified axis.

Uses more precise math to better determine the locations for the objects. In case of positive or negative bounding box alignment, if one or more of the selected objects have any rotation transformations (or delta rotation transformations), it is recommended to check High Quality so that their bounding box is calculated with precision for all three global axes.

The Align Mode control will define what part of the objects will be aligned:

The objects centers.

The positive or negative sides (on the global axes) of their respective bounding boxes.

The Relative To control will let us choose to align the objects to:

The active object.

The median point of the selection.

The current position of the 3D Cursor.

The global origin.

Chooses which axis to align the selected objects on.

--- Fin du chapitre: _align_objects ---

--- Début du chapitre: _align_rotation_to_vector --- The Align Rotation to Vector node rotates an Euler rotation into the given direction. The Euler rotation to align.

Important

This input has to be a rotation input. Be careful not to connect a direction vector like the normal.

Determines how much the points are rotated towards the vector. Zero effectively disables the node and one means that the points are aligned with the vector perfectly.

The direction vector that points should be rotated to. The vector is in the local space of the object that is being modified. When it is all zeros for a point, it is not rotated at all.

Local axis of the object that is to be rotated towards the vector input.

The local axis to rotate around.

The best rotation angle is computed automatically. This minimizes the angle of rotation.

Rotate around a specific local axis.

The rotated Euler rotation.

--- Fin du chapitre: _align_rotation_to_vector ---

--- Début du chapitre: _align_transform_orientation --- Reference Object Mode and Edit Mode

Object \xe2\x80\xa3 Transform \xe2\x80\xa3 Align to Transform Orientation

Aligns (rotates) the selected objects so that their local orientation matches the active transform orientation in the Transform orientation panel or the Orientation selection in the Transform Adjust Last Operation panels.

--- Fin du chapitre: _align_transform_orientation ---

--- Début du chapitre: _all_by_trait --- Reference Edit Mode

Select \xe2\x80\xa3 Select All by Trait \xe2\x80\xa3 Non-Manifold
Selects the Non-manifold geometry of a mesh. This entry is available when editing a mesh, in Vertex and Edge selection modes only.

Lets you extend the current selection.

Selects all the edges that do not belong to any face.

Selects edges in boundaries and holes.

Selects edges that belong to three or more faces.

Selects edges that belong to exactly two faces with opposite normals.

Selects vertices that belong to wireand multiple faceedges, isolated vertices, and vertices that belong to non-adjoining faces.

Reference

Edit Mode

Select \xe2\x80\xa3 Select All by Trait \xe2\x80\xa3 Loose Geometry

This selection depends on the currently selected Selection Modes ; In vertex and edge selection mode it selects all vertices or edges that do not form part of a face. In face selection mode it selects all faces that do not share edges with other faces.

Reference

Edit Mode

Select \xe2\x80\xa3 Select All by Trait \xe2\x80\xa3 Interior Faces

Selects faces where all edges have more than two faces.

Reference

Edit Mode

Select \xe2\x80\xa3 Select All by Trait \xe2\x80\xa3 Faces by Sides

Selects all faces that have a specified number of edges.

Reference

Edit Mode

Select \xe2\x80\xa3 Select All by Trait \xe2\x80\xa3 Ungrouped Vertices

Selects all vertices which are not part of a vertex group .

--- Fin du chapitre: _all_by_trait ---

--- Début du chapitre: _along_stroke --- The Along Stroke modifier alters the base property with a new one from a given range mapped along each stroke \xe2\x80\x99s length. In other words, it applies a gradient along each stroke.

--- Fin du chapitre: _along_stroke ---

--- Début du chapitre: _alpha --- In this tab you control the alpha (transparency) of your strokes.

Line Style: Alpha. \xc2\xb6

The base alpha for this line style.

The modifier output can be mixed with the base property using the usual methods (see for example the Mix compositing node).

How much the result of this modifier affects the current property.

Either a linear progression (from 0.0 to 1.0), or a custom mapping curve.
Note

Note the linear non-inverted option is equivalent to \xe2\x80\x9cdo nothing \xe2\x80\x9d, as original values from materials are already in the (0.0 to 1.0) range. That is the case for: Crease Angle, Curvature 3D, Material, Noise, Tangent.

Inverts the Mapping.

Along Stroke

Crease Angle

Curvature 3D

Distance from Camera
Distance from Object
Material
Noise
Tangent

--- Fin du chapitre: `_alpha` ---

--- Début du chapitre: `_alpha_convert` --- The Alpha Convert Node converts the alpha channel format of an image.

For compositing and rendering, Premultiplied Alpha is the standard in Blender. Render layers will be premultiplied alpha, and images loaded into rendering or compositing will be converted to this.

If you want to do a compositing operation with straight alpha, the Alpha Convert node can be used. Typically this would be a color correction operation where it might give better results working on RGB channels without alpha. If the alpha is converted to straight in the Compositor, it should be converted back to premultiplied before the Composite Output node, otherwise some artifacts might occur.

Standard color input.

The direction of convert alpha. For details on the difference between both ways to store alpha values see Alpha Channel .

Converts from Straight Alpha to Premultiplied Alpha .

Converts from Premultiplied Alpha to Straight Alpha .

Standard color output.

--- Fin du chapitre: `_alpha_convert` ---

--- Début du chapitre: `_alpha_over` --- The Alpha Over node is used to layer an image on top of another with alpha blending.

The alpha of the foreground image, going from 0 (fully transparent) to 1 (fully opaque).

The background image.

The foreground image.

The Alpha Over node expects the foreground image to use Premultiplied Alpha . If it uses Straight Alpha instead, you can enable this checkbox to convert it.

Interpolate between Premultiplied Alpha and Straight Alpha .

When set to 1, the foreground color values will be multiplied by the alpha; this is equivalent to enabling Convert Premultiplied. When set to 0, the color values do not change.

If Premultiplied is not zero, Convert Premultiplied will be ignored.

Note

This is a legacy option.

The blended result.

In the node tree below, the Color Ramp Node is used to convert an opaque, grayscale swirl image to a red one with transparency. Then, the Alpha Over node is used to overlay it on top of another image.

Assembling a composite image using Alpha Over.

The example below uses the Time Curve Node to gradually increase the Alpha Over's Factor from 0 to 1 over the course of 30 frames.

This will result in the text fading in on top of the background image.

Animated fade in effect using Alpha Over.

--- Fin du chapitre: `_alpha_over` ---

--- Début du chapitre: _alpha_over_under_overdrop --- Using the alpha (transparency channel), this effect composites a result based on transparent areas of the dominant image. If you use a Scene strip, the areas of the image where there is not anything solid are transparent; they have an alpha value of 0. If you use a Movie strip, that movie has an alpha value of 1 (completely opaque).

So, you can use the Alpha Over/ Alpha Under effect to composite the CGI Scene on top of your movie. The result is your model doing whatever as if it was part of the movie. The Adjust α Compositing α Opacity controls how much the foreground is mixed over the background, fading in the foreground on top of the background. The colors of transparent foreground image areas are ignored and do not change the color of the background.

With Alpha Over, the strips are layered up in the order selected; the first strip selected is the background, and the second one goes over the first one selected. The Opacity controls the transparency of the foreground, i.e. Opacity of 0.0; will only show the background, and an Opacity of 1.0 will completely override the background with the foreground (except in the transparent areas of this one, of course!)

Warning

By clicking the Premultiply Alpha button in the Sidebar of the foreground strip, the alpha values of the two strips are not multiplied or added together. Use this effect when adding a foreground strip that has a variable alpha channel (some opaque areas, some transparent, some in between) over a strip that has a flat opaque (alpha=1.0 or greater) channel. If you notice a glow around your foreground objects, or strange transparent areas of your foreground object when using Alpha Over, enable Premultiply.

Alpha Over Effect. α

With Alpha Under, this is the contrary: The first strip selected is the foreground, and the second one, the background. Moreover, the Opacity controls the transparency of the background, i.e. an Opacity of 0.0; will only show the foreground (the background is completely transparent), and an Opacity of 1.0 will give the same results as with Alpha Over.

Over Drop is between the two others: as with Alpha Under, the first selected strip will be the foreground, but as with Alpha Over, the Opacity controls the transparency of this foreground.

The Over Drop effect is much like the Cross, but puts preference to the top or second image, giving more of a gradual overlay effect than a blend like the Cross does. Of course, all of the Alpha effects respect the alpha (transparency) channel, whereas Cross does not.

The degree of Alpha applied, and thus color mixing, can be controlled by an F-Curve. Creating a Sine wave could have the effect of the foreground fading in and out.

--- Fin du chapitre: _alpha_over_under_overdrop ---

--- Début du chapitre: _amd --- Blender uses of OpenGL for the 3D Viewport and user interface. The graphics card (GPU) and driver have a big impact on Blender's behavior and performance.

This section lists possible solutions for graphics glitches, problems with Eevee and Cycles, and crashes related to your GPU.

Upgrading to the latest graphics drivers often solves problems. Newer drivers have bug fixes that help Blender function correctly.

On macOS graphics drivers are built into the operating system and the only way to get newer drivers is to upgrade macOS as a whole to the latest version.

This means your graphics card and driver do not have the minimum required OpenGL 3.3 version needed by Blender.

Installing the latest driver can help upgrade the OpenGL version, though some graphics cards are simply too old to run the latest Blender. Using Blender 2.79 or earlier is the only option then.

Try running Blender from the command line, to see if any helpful error messages are printed.

On Windows, graphics drivers can sometimes get corrupted. In this case it can help to uninstall all graphics drivers (there may be multiple from Intel, AMD and Nvidia) and perform a clean installation with drivers from the manufacturer's website.

Update your graphics drivers (see above).

On laptops, make sure you are using a dedicated GPU (see above).

Try lowering quality settings in Preferences > System > Memory & Limits.

Try undoing settings in your graphics drivers, if you made any changes there.

See [Eevee](#) and [Cycles](#) documentation respectively.

See [Invalid Selection](#), [Disable Anti-Aliasing](#).

Running Blender inside a virtual machine is known to have problems when OpenGL drawing calls are forwarded to the host operating system.

To resolve this, configure the system to use PCI passthrough.

To find out which graphics card and driver Blender is using, use Help > Save System Info inside Blender. The OpenGL section will have information about your graphics card, vendor and driver version.

--- Fin du chapitre: _amd ---

--- Début du chapitre: _animation --- While rendering stills will allow you to view and save the image from the render buffer when it is complete, animations are a series of images, or frames, and are automatically saved directly out to a drive after being rendered.

After rendering the frames, you may need to edit the clips, or first use the Compositor to do green-screen masking, matting, color correction, DOF, and so on to the images. That result is then fed to the Sequencer where the strips are cut and mixed and a final overlay is done.

Finally you can render out from the Sequencer and compress the frames into a playable movie clip.

Generally, you do a lot of intermediate renders of different frames in your animation to check for timing, lighting, placement, materials, and so on. At some point, you are ready to make a final render of the complete animation for publication.

There are two approaches you can use when making a movie, or animation, with or without sound. The approach you should use depends on the amount of CPU time you will need to render the movie. You can render a typical frame at the desired resolution, and then multiply by the number of frames that will ultimately go into the movie, to arrive at a total render time.

If the total render time is an hour or more, you want to use the Frame Sequence approach. For example, if you are rendering a one-minute video clip for film, there will be (60 seconds per minute) X (24 frames per second) or 1440 frames per minute. If each frame

takes 30 seconds to render, then you will be able to render two frames per minute, or need 720 minutes (12 hours) of render time.

Rendering takes all available CPU time; you should render overnight, when the computer is not needed, or set Blender to a low priority while rendering, and work on other things (be careful with the RAM space!).

Direct Approach

The Direct Approach, which is highly not recommended and not a standard practice, is where you set your output format to an AVI or MOV format, and click Animation to render your scene directly out to a movie file. Blender creates one file that holds all the frames of your animation. You can then use Blender's Video Sequencer to add an audio track to the animation and render out to an MPEG format to complete your movie.

Frame Sequence

The Frame Sequence is a much more stable approach, where you set your output format to a still format (such as JPG, PNG or a multi-layer format). Click Animation to render your scene out to a set of images, where each image is a frame in the sequence.

Blender creates a file for each frame of the animation. You can then use Blender's Compositor to perform any frame manipulation (post-processing). You can then use Blender's Video Sequencer to load that final image sequence, add an audio track to the animation, and render out to an MPEG format to complete your movie. The Frame Sequence approach is a little more complicated and takes more drive space, but gives you more flexibility.

Here are some guidelines to help you choose an approach.

Direct Approach

Short segments with total render time under one hour.

Stable power supply.

Computer not needed for other uses.

Frame Sequence Approach

Total render time over one hour.

Post-production work needed:

Color/lighting adjustment

Green screen/matte replacement

Layering/compositing

Multiple formats and resolutions of the final product

Intermediate frames/adjustments needed for compression/codec.

Precise timing (e.g. lip-sync to audio track) needed in parts.

May need to interrupt rendering to use the computer, and want to be able to resume rendering where you left off.

First prepare your animation.

In the Format panel, choose the render size, Pixel Aspect Ratio, and the Range of Frames to use, as well as the frame rate, which should already be set.

In the Output panel set up your animation to be rendered out as images, generally using a format that does not compromise any quality. Choose the output path and file type in the Output panel as well, for example //render/my-anim- .

Confirm the range of your animation (frame Start and End).

Save your blend-file.

Press the Animation button and once the animation is finished, use your file manager to navigate to the output folder (render in this example). You will see lots of images that have a sequence number attached to. These are the single frames.

In Blender, open the Video Sequencer .

Note

The Video Sequencer does not support multi-layer EXR files. To render to a video format you will have to skip the next three steps and instead use an Image Input node in the Compositor.

Choose Add Image from the add menu. Select all the frames from your output folder that you want to include in your animation. They will be added as a strip in the Sequence editor.

Now you can edit the strip and add effects or leave it like it is. You can add other strips, like an audio strip.

Scrub through the animation to check if you have included all the frames.

In the Output panel, choose the container and codec you want (e.g.

MPEG-H.264) and configure them. The video codecs are described in Output Options .

Click the Animation render button and Blender will render out the Sequence editor output into a movie.

Unless your animation renders in a few minutes, it is best to render the animation as separate image files. Instead of rendering directly to a compressed movie file, use a lossless format (e.g. PNG).

This allows you an easy recovery if there is a problem and you have to re-start the rendering, since the frames you have already rendered will still be in the output directory.

Just disable the Overwrite option to start rendering where you left off.

You can then make a movie out of the separate frames with

Blender's Sequence editor or use 3rd party encoding software.

It can be useful to render a subset of the animated sequence, since only part of an animation may have an error.

Using an image format for output, you can use the Frame Step option to render every Nth frame. Then disable Overwrite and re-render with Frame Step set to 1.

--- Fin du chapitre: _animation ---

--- Début du chapitre: _animation_player ---Reference

Topbar \Render \View Animation

Ctrl- F11

The animation player is a utility typically used for previewing rendered animations, supporting all image and video formats also supported by Blender. This is a convenient way to play back image sequences at the correct frame rate.

Launching the animation player opens a new window, playing back images or a video located at the render output of the current scene. You can also drop images or movie files in a running animation player. It will then restart the player with the new data.

Tip

An external player can also be used instead of the one included in Blender. To do this, select it in the Preferences .

When enabled, playback loops forwards than backwards.

Flip the image horizontally or vertically.

Viewing the animation from a different perspective can help you see the animation with \fresh eyes\.

The following table shows the available hotkeys for the animation player.

Playback

Action

Hotkey

Start/Pause:
Spacebar
Start playback (when paused):
Return
Quit:
Esc
Timeline
Action
Hotkey
Scrub in time:
LMB
Step back one frame:
Left
Step forward one frame:
Right
Step back 10 frames:
Down
Step forward 10 frames:
Up
Manual frame stepping:
NumpadPeriod
Playback Options
Action
Hotkey
Backward playback:
Shift- Down
Forward playback
Shift- Up
Slow down playback:
NumpadMinus
Speed up playback:
NumpadPlus
Toggle looping:
Numpad0
Toggle frame skipping:
A
Toggle ping-pong:
P
Display
Action
Hotkey
Toggle Playhead (Indicator):
I
Flip image on the X axis:
F
Flip image on the Y axis:
Shift- F
Hold to show frame numbers:
Shift
Zoom in:
Ctrl- NumpadPlus
Zoom out:
Ctrl- NumpadMinus
Frame Rate

Action
Hotkey
60 fps
Numpad1
50 fps
Numpad2
30 fps
Numpad3
25 fps
Numpad4
24 fps
Shift- Numpad4
20 fps
Numpad5
15 fps
Numpad6
12 fps
Numpad7
10 fps
Numpad8
6 fps
Numpad9
5 fps
NumpadSlash

Image files are cached during playback for faster access.

While loading images is rarely a bottleneck, there are situations where high resolution images may slow down playback causing frame skipping.

See also

Memory Cache Limit preference to control this limit, which may be increased to cache more images during playback. Animation Playback Options to specify this value when launching from the command line.

--- Fin du chapitre: _animation_player ---

--- Début du chapitre: _anim_bvh ---Reference

Import-Export

File \xe2\x80\xa3 Import/Export \xe2\x80\xa3 Motion Capture (.bvh)

Imports or exports bvh-files or files with BioVision Hierarchical data of a skeleton (rig) including its animation. Useful for importing data from motion capture devices.

This add-on is enabled by default, in case it is not:

Open Blender and go to Add-ons section of the Preferences .

Search \xe2\x80\x9cBioVision Motion Capture (BVH) format\xe2\x80\x9d and check the Enable Extension checkbox.

The motion capture data type.

The bvh-file contains an animated rigged skeleton such as a walking motion capture.

The bvh-file contains a static (not animated) mesh object such as a character model.

Factor to increase the physical size of the BVH.

Rotation order of the BVH.

Since many applications use a different axis for pointing upwards, these are axis conversion for these settings, Forward and up axes \xe2\x80\x93

By mapping these to different axes you can convert rotations between applications default up and forward axes.

Blender uses Y forward, Z up (since the front view looks along the +Y direction). For example, its common for applications to use Y as the up axis, in that case -Z forward, Y up is needed.

The start frame, in Blender, to start playback of the BVH animation.

Scales the frame rate from the BVH file to the scene frame rate set in Blender, otherwise each BVH frame maps directly to a frame in Blender.

Cycles the animation playback.

Set the scene's frame rate to match the frame rate of the BVH file.

Extend the scene's duration to match the BVH's duration.

Factor to increase the physical size of the BVH.

Rotation order of the BVH.

Only write the translation animation channels for the root bone.

Sets the range of animation to export to the BVH file.

--- Fin du chapitre: _anim_bvh ---

--- Début du chapitre: _annotate_tool --- The annotation tool is available in multiple editors. It can be used to add notes to e.g. 3D objects or node setups. The arrow in the screenshot below is an annotation.

Annotations tool in a node editor.

The annotation tool can be activated in the Toolbar and has the following sub-tools:

Draw free-hand strokes in the main area.

Click and drag to create a line. Optionally, you can select the arrow style for the start and end of the line.

Click multiple times to create multiple connected lines, then press Return or Esc to confirm.

Click and drag to remove lines. The eraser has a Radius setting found in Tool Settings.

Adjust the color of existing and new strokes.

A pop-over menu, showing the name of the current layer, to access the Annotation Layers.

Determines where the annotations are drawn.

Only available in the 3D Viewport. The new annotations become part of the 3D scene; they're drawn on an imaginary plane that goes through the 3D Cursor and is aligned to your view.

Only available in the 3D Viewport. The new annotations become part of the 3D scene; they're drawn onto the surface of the object under the mouse. If there is no surface, you get the same behavior as 3D Cursor.

Only available in 2D editors such as the Image Editor. The annotations become part of the 2D space, meaning their position and size change as you pan and zoom in the editor.

The new annotations are 2D and get stuck to the screen. They keep the same position, rotation and size no matter how you pan, orbit or zoom in the editor.

Only use the first and last parts of the stroke for snapping.

Only project the strokes onto selected objects.

Helps to reduce jitter of the strokes while drawing by delaying and correcting the location of points.

Minimum distance from the last point before the stroke continues.

A smooth factor, where higher values result in smoother strokes but the drawing sensation feels like as if you were pulling the stroke.

The decoration to use at the beginning or end of the line segment. This can be used for example to create arrows to point out specific details in a scene.

When the annotation tool is enabled, the settings for managing multiple layers can be found in the Sidebar \xe2\x80\xa3 View \xe2\x80\xa3 Annotations panel.

Adjusts the opacity of existing and new strokes.

Adjusts the thickness of existing and new strokes.

Shows a ghosted image of strokes made in frames before and after the current frame. Onion skinning only works in the 3D Viewport and Sequencer. See the Grease Pencil documentation for an explanation of Onion Skinning.

Color to use before and after the current frame on ghost frames. The number defines how many frames to show before and after the current frame.

--- Fin du chapitre: _annotate_tool ---

--- Début du chapitre: _anti_aliasing --- The Anti-Aliasing node smooths away jagged edges.

Standard color input.

Controls edge detection sensitivity across the whole image.

Controls contrast level to consider when detecting edges.

The human eye does not perceive all edges equally. For instance, it tends to mask low contrast edges in the presence of much higher contrasts in the surrounding area. Therefore, applying anti-aliasing to unperceived edges will produce artifacts. This option quantifies the difference between low contrast and high contrast neighboring edges.

Detect corners to help preserve the original shape. Setting Corner Rounding to 0 means no corner detection and no corner rounding will take place. The higher the value the better corners will be preserved, i.e. resemble original image.

Standard color output.

Changing threshold affects all edges.\xc2\xbb

Changing contrast limit affects neighboring edges below contrast limit.\xc2\xbb

Corner detection and rounding off (set to 0). Notice how corners are smoothed because they are treated as artifacts.\xc2\xbb

Full corner detection and rounding preserves the sharp edges around the corner.\xc2\xbb

--- Fin du chapitre: _anti_aliasing ---

--- Début du chapitre: _ao --- The Ambient Occlusion shader computes how much the hemisphere above the shading point is occluded. This can be used for procedural texturing, for example to add weathering effects to corners only.

For Cycles, this is an expensive shader and can slow down render significantly. If render time is a concern, using Pointiness from the Geometry node or baking Ambient Occlusion will result in faster renders. Note

The object is either a Caustic caster or Caustic receiver while the scene contains an active Caustic caster, Caustic receiver, and Shadow Caustic Light.

Open Shading Language is active while using the OptiX rendering backend.
Tint for AO output color.

Distance up to which other objects are considered to occlude the shading point.

Normal used for ambient occlusion; if nothing is connected the default shading normal is used.

Number of samples to use for ray-traced ambient occlusion sampling. Keep as low as possible for optimal performance.

Detect convex rather than concave shapes, by computing occlusion inside mesh.

Only detect occlusion from the object itself, and not others.

Ambient occlusion with color tint.

Ambient occlusion factor without color tint.

White AO shader. \xc2\x6

--- Fin du chapitre: _ao ---

--- Début du chapitre: _aov --- Shader AOVs (Arbitrary Output Variables) provide custom render passes for arbitrary shader node components. As an artist this can be a good way to debug or tweak very fine details of a scene in post-processing. To use shader AOVs create the pass in the Shader AOV panel then reference that pass with the AOV Output shading node. Shader AOVs can be added or removed in the Shader AOV panel.

Tip

The AOV Output node can be used in Material and World shader nodes.

Output a color variable; as the name suggest can be used for a color but also a normal value.

Output a single numerical value.

The name of the render pass to assign the input value to. This property has the same Name that is specified in the Shader AOV panel.

This node has no outputs.

--- Fin du chapitre: _aov ---

--- Début du chapitre: _apply --- Reference

Pose Mode

Pose \xe2\x80\xa3 Apply

Ctrl- A

Conversely, you may define the current pose as the new rest pose (i.e. \xe2\x80\x9capply\xe2\x80\x9d current transformations to the Edit Mode).

When you do so, the skinned objects/geometry is also reset to its default, undeformed state, which generally means you will have to skin it again.

Same as Pose as Rest Pose but only applies to selected bones.

Applies the position of the bone after Constraints; allowing the constraints to be deleted and the bones will remain in their constrained positions.

Assign the current values of custom properties as their defaults, for use as part of the rest pose state in NLA track mixing.

--- Fin du chapitre: _apply ---

--- Début du chapitre: _applying_image --- Sooner or later, you may want to use an image texture on your model. The UV Editor allows you to map textures directly to the mesh faces. The 3D Viewport shows you the object being textured. If you set the 3D Viewport into Textured viewport shading, you will immediately see any changes made in the UV Editor, and

vice versa. This is because no real rendering is taking place; it is all just viewport shading. If you were to apply an image to UVs then render, the texture would not show up by default. So to render an image, you must: Create a Material for the object.

Tell Blender to use the UV textures on faces when rendering.

To create a Material, you have to click Add New Material in the Shading context.

There are two ways to tell Blender to use the UV texture when rendering; the quick way and the proper way.

The quick way is to use generated UV coordinates. Generated coordinates is used by default by all Texture nodes, with the exception of Image textures that use UV coordinates by default. To use generated coordinates for images as well use the Generated output of the Texture Coordinate node.

The proper way is to use UV Unwrapping to manually generate UV coordinates. To use UV mapping, use the Texture Coordinate node (UV output) or the UV Map node and select the UV map to use, \xe2\x80\x9cUV Map\xe2\x80\x9d by default.

Full details of using Image textures are on the Image Textures page.

Note

Material is Required for Rendering

You can perform UV texturing on a mesh within Blender without assigning a material, and you will even see it in your 3D Viewport in textured viewport mode. However, when you render, you will just get a default gray if the object does not have a Material assigned. You will get a black if you do not load an image. If you do not create a texture that uses the image, your object will render according to the procedural material settings.

If your image is a base uniform pattern and you want the application of that image to your model to look like cloth, you do not want any stretching (unless you want the cloth to look like spandex). You may also need to test your UV mapping with a test image:

The test grid applied to the UVs. \xc2\xb6

A preview of the texture on the geometry. \xc2\xb6

When you render, the mesh will have the test grid as its colors, and the UV texture will be the size image you specified. Note that Blender has a built-in test image. To use it, press the New button of the data-block menu in the Image editor header and change the Generated Type to UV Grid. See also

Render Bake

Texture Paint

The advantage to saving as a separate file is that you can easily switch textures just by copying other image files over it, and you can use external editing programs to work on it. The advantage of packing is that your whole project is kept in the blend-file, and that you only have to manage one file.

--- Fin du chapitre: _applying_image ---

--- Début du chapitre: _app_templates --- Application templates are a feature that allows you to define a re-usable configuration that can be selected to replace the default configuration, without requiring a separate Blender installation or overwriting your personal settings. Application templates can be selected from the splash screen or File \xe2\x80\x9c News submenu. When there are no templates found the menu will not be displayed on the splash screen.

New application templates can be installed from the Blender Menu .If you would like to keep the current application template active on restarting Blender, save your preferences.

In some cases it's not enough to write a single script or add-on, and expect someone to replace their preferences and startup file, install scripts and change their keymap.

The goal of application templates is to support switching to a customized configuration without disrupting your existing settings and installation. This means people can build their own applications on top of Blender that can be easily distributed.

An application template may define its own:

The default file to load with this template.

Only certain preferences from a template are used:

Themes.

Add-ons.

Keymaps.

Viewport lighting.

Templates may provide their own splash screen image.

While templates have access to the same functionality as any other scripts, typical operations include:

Modifying and replacing parts of the user interface.

Defining new menus, keymaps and tools.

Defining a custom add-on path for template specific add-ons.

Templates also have their own user configuration, so saving a startup file while using a template won't overwrite your default startup file.

Templates may be located in one of two locations within the scripts directory.

User configuration is stored in a subdirectory:

See Blender's Directory Layout for details on script and configuration locations.

Hint

Troubleshooting Paths

When creating an application template, you may run into issues where paths are not being found. To investigate this you can log output of all of Blender's path look-ups.

Example command line arguments that load Blender with a custom application template (replace my_app_template with the name of your own template):

You can then check the paths where attempts to access my_app_template are made.

Using the command-line arguments you can setup a launcher that opens Blender with a specific app template:

Each of the following files can be used for application templates but are optional.

Factory startup file to use for this template.

Factory preferences file to use for this template. When omitted preferences are shared with the default Blender configuration.

(As noted previously, this is only used for a subset of preferences).

Splash screen to override Blender's default artwork (not including header text). Note, this image must be a 1000x500 image.

A Python script which must contain register and unregister functions.

Note

Bundled `blend-files\startup.blend` and `userpref.blend` are considered Factory Settings and are never overwritten.

The user may save their own startup/preferences while using this template which will be stored in their user configuration, but only when the template includes its own `userpref.blend` file.

The original template settings can be loaded using: Load Template Factory Settings from the file menu in much the same way Load Factory Settings works.

While app templates can use Python scripts, they simply have access to the same APIs available for add-ons and any other scripts.

As noted above, you may optionally have an `__init__.py` in your app template. This has the following advantages:

Changes can be made to the startup or preferences, without having to distribute a blend-file.

Changes can be made dynamically.

You could for example configure the template to check the number of processors, operating system and memory, then set values based on this.

You may enable add-ons associated with your template.

On activation a register function is called, unregister is called when another template is selected.

As these only run once, any changes to defaults must be made via handler. Two handlers you are likely to use are:

```
bpy.app.handlers.load_factory_preferences_post
```

```
bpy.app.handlers.load_factory_startup_post
```

These allow you to define your own factory settings, which the user may change, just as Blender has its own defaults when first launched.

This is an example `__init__.py` file which defines defaults for an app template to use.

```
--- Fin du chapitre: _app_templates ---
```

```
--- Début du chapitre: _arc --- The Arcnode generates a poly spline arc.  
The node has two modes, Radius and Points.
```

```
Number of edges on the arc.
```

```
The radius of the arc. Radiusmode only.
```

```
Starting angle of the arc. Radiusmode only.
```

```
Length of the arc. Radiusmode only.
```

```
Connect the arc at the center.
```

```
Invert and draw opposite arc.
```

```
The three points on the arc. Pointsmode only. The order of the points  
determines the direction (clockwise or counterclockwise) of the arc. The  
arc will always draw from Start to End via the Middle point. This can be  
changed by using the Invert Arc option.
```

```
Offset angle of the arc. Pointsmode only.
```

```
Note
```

```
Because of the finite resolution, the middle point does not necessarily  
lie on the generated arc.
```

```
The position and radius of the arc are determined by three points. The  
center of the arc, radius and normal are also given as outputs.
```

```
The arc is determined by the radius, start angle and sweep angle.
```

```
Poly spline generated from the inputs.
```

```
The center of the arc described by the three points. Pointsmode only.
```

The normal direction of the plane described by the three points, pointing towards the positive Z axis. Pointsmode only.

The radius of the arc described by the three points. Pointsmode only.

--- Fin du chapitre: _arc ---

--- Début du chapitre: _areas --- Area boundaries are indicated by rounded corners (yellow highlights).

The Blender window is divided into a number of rectangles called Areas. Areas reserve screen space for Editors, such as the 3D Viewport or the Outliner. Each editor offers a specific piece of functionality.

Areas are grouped into Workspaces, which are geared towards particular tasks (modeling, animating and so on).

Note

While some keyboard shortcuts in Blender are global (such as Ctrl- S for saving), many depend on which editor the mouse cursor is hovering over. As an example, say you just selected two objects in the Outliner and want to join them. If you pressed the shortcut for this (Ctrl- J) while the cursor is still in the Outliner, nothing would happen as the shortcut isn't valid there; you first need to move your cursor to the 3D Viewport.

You can resize areas by dragging their borders with LMB. Move your mouse cursor over the border between two areas, so that the cursor changes to a double-headed arrow, and then click and drag.

Tip

Hold Ctrl to snap the size of areas to convenient sizes.

Docking describes several ways an area a user can interactively manipulate the size and location of areas along with splitting an area into new areas.

To start the interactive process, placing the mouse cursor in an area corner will change the cursor to a cross (+). Once the cursor is a cross, press and hold LMB to perform any of the following actions:

If you press Esc or RMB before releasing the mouse, the operation will be canceled.

Properties is being joined to the Outliner.

Dragging from an area corner into the space of a second area will join two areas. The areas that will be joined will be displayed brighter.

Splitting an area will create a new area. Dragging from an area corner left/right will split the area vertically, to split the area horizontally drag up/down.

You can split and join areas at once by dragging a split operation into a separate area.

Dragging an area into the middle of a second area will replace the second area with the first area.

RMB on the border opens the Area Options.

Shows an indicator line that lets you select the area and position where to split. Tab switches between vertical/horizontal.

Shows the join direction overlay.

Swaps this area with the adjacent one.

You can swap the contents of two areas by pressing Ctrl- LMB on one of the corners of the initial area, dragging towards the target area, and releasing the mouse there. The two areas do not need to be side-by-side, though they must be inside the same window.

Reference

View Area Duplicate Area into new Window

A new floating window containing an area can be created from View
\xe2\x80\xa3 Area \xe2\x80\xa3 Duplicate Area into new Window. (Not
available in some editors.)

The new window is a fully functional window, which is part of the same
instance of Blender. This can be useful, e.g. if you have multiple
monitors.

You can also create a new window from an existing area by pressing Shift-
LMB on an area corner, then dragging outward slightly.

Reference

View \xe2\x80\xa3 Area \xe2\x80\xa3 Toggle Maximize Area

Ctrl- Spacebar

Expands the Area so it fills the whole window (while keeping the Topbar
and Status Bar visible). To return to normal size, use the keyboard
shortcut again or click the Back to Previous button in the Topbar.

Reference

View \xe2\x80\xa3 Area \xe2\x80\xa3 Toggle Fullscreen Area

Ctrl- Alt- Spacebar

Expands the Area so it fills the whole window, hiding the Topbar, Status
Bar, and even these secondary regions (toolbars etc.) of the
Area\xe2\x80\x99s own editor. To return to normal size, use the keyboard
shortcut again or click the icon in the Area\xe2\x80\x99s top right
corner (only becomes visible when hovering).

--- Fin du chapitre: _areas ---

--- Début du chapitre: _arguments --- Run in background (often used for
UI-less rendering).

The audio device is disabled in background-mode by default and can be re-
enabled by passing in -setaudioDefault afterwards.

Render frames from start to end (inclusive).

Set the active scene <name> for rendering.

Render frame <frame> and save it.

+<frame> start frame relative, -<frame> end frame relative.

A comma separated list of frames can also be used (no spaces).

A range of frames can be expressed using .. separator between the first
and last frames (inclusive).

Set start to frame <frame> , supports +/- for relative frames too.

Set end to frame <frame> , supports +/- for relative frames too.

Set number of frames to step forward after each rendered frame.

Set the render path and file name. Use // at the start of the path to
render relative to the blend-file.

The # characters are replaced by the frame number, and used to define
zero padding.

animation_##_test.png becomes animation_01_test.png

test-#####.png becomes test-000001.png

When the filename does not contain # , the suffix #### is added to the
filename.

The frame number will be added at the end of the filename, eg:

//render_ becomes //render_#### , writing frames as //render_0001.png

Specify the render engine. Use -Ehelp to list available engines.

Use amount of <threads> for rendering and other operations [1-1024], 0 to
use the systems processor count.

Cycles add-on options must be specified following a double dash.

Set the device used for rendering. Valid options are:

CPUCUDAOPTIXHIPONEAPIMETAL .

Append +CPU to a GPU device to render on both CPU and GPU.

Example:

Log statistics about render memory and time usage.

Set the render format.Valid options are:

TGARAWTGAIPEGIRISAVIRAWAVIJPEGPNGBMPHRTIFF .

Formats that can be compiled into Blender, not available on all systems:

OPEN_EXROPEN_EXR_MULTILAYERFFMPEGCINEONDPXJP2WEBP .

Set option to add the file extension to the end of the file.

Instead of showing Blender's user interface, this runs Blender as an animation player,to view movies and image sequences rendered in Blender (ignored if -b is set).

Playback Arguments:

Open with lower left corner at <sx> , <sy> .

Read from disk (Do not buffer).

Specify FPS to start with.

Set frame step to <frame> .

Play from <frame> .

Play until <frame> .

Amount of memory in megabytes to allow for caching images during playback.Zero disables (clamping to a fixed number of frames instead).

Force opening with borders.

Force opening in full-screen mode.

Open with lower left corner at <sx> , <sy> and width and height as <w> , <h> .

Force opening maximized.

Start with the console window open (ignored if -b is set), (Windows only).

Do not use native pixel size, for high resolution displays (MacBook Retina).

Open behind other windows and without taking focus.

Enable automatic Python script execution.

Disable automatic Python script execution (Python-drivers & startup scripts), (default).

Run the given Python script file.

Run the given Python script text block.

Run the given expression as a Python script.

Run Blender with an interactive console.

Set the exit-code in [0..255] to exit if a Python exception is raised(only for scripts executed from the command line), zero disables.

Allow Python to use system environment variables such as PYTHONPATH and the user site-packages directory.

Comma separated list (no spaces) of add-ons to enable in addition to any default add-ons.

Allow internet access, overriding the preference.

Disallow internet access, overriding the preference.

Enable logging categories, taking a single comma separated

argument.Multiple categories can be matched using a .* suffix,so --

log"wm.*" logs every kind of window-manager message.Sub-string can be matched using a * prefix and suffix,so --log"*undo*" logs every kind of

undo-related message.Use \xe2\x80\x9c prefix to ignore, so -log"* , ^wm.operator.*" logs all except for wm.operators.* Use

\xe2\x80\x9c* to log everything.

Set the logging verbosity level (higher for more details) defaults to

1,use -1 to log all levels.

Only show file name in output (not the leading path).
Show a back trace for each log message (debug builds only).
Show a timestamp for each log message in seconds since start.
Set a file to output the log to.
Turn debugging on.
Enables memory error detection
Disables mouse grab (to interact with a debugger in some cases)
Keeps Python's sys.stdin rather than setting it to None
Set debug value of <value> on startup.
Enable debug messages for the event system.
Enable debug messages from FFMpeg library.
Enable debug messages for event handling.
Enable debug messages from libmv library.
Enable debug messages from Cycles.
Enable fully guarded memory allocation and debugging.
Enable time profiling for background jobs.
Enable debug messages for Python.
Enable all debug messages from dependency graph.
Enable debug messages from dependency graph related on evaluation.
Enable debug messages from dependency graph related on graph construction.
Enable debug messages from dependency graph related on tagging.
Switch dependency graph to a single threaded evaluation.
Enable debug messages from dependency graph related on timing.
Enable colors for dependency graph debug messages.
Verify validness of session-wide identifiers assigned to ID data-blocks.
Enable debug messages for Ghost (Linux only).
Enable debug messages for Wintab.
Enable GPU debug context and information for OpenGL 4.3+.
Enable workarounds for typical GPU issues and disable all GPU extensions.
Compile all statically defined shaders to test platform compatibility.
Capture the GPU commands issued inside the given scope name.
Enable Renderdoc integration for GPU frame grabbing and debugging.
Enable debug messages for the window manager, shows all operators in search, shows keymap errors.
Enable debug messages for virtual reality contexts. Enables the OpenXR API validation layer, (OpenXR) debug messages and general information prints.
Enable debug messages for virtual reality frame rendering times.
Enable all debug messages.
Enable debug messages for I/O (Collada, XML).
Enable floating-point exceptions.
Immediately exit when internal errors are detected.
Enable debug messages for Freestyle.
Disable the crash handler.
Disable the abort handler.
Set the logging verbosity level for debug messages that support it.
Suppress status printing (warnings & errors are still printed).
Force to use a specific GPU backend. Valid options: vulkan (experimental), metal, opengl.
Override the Max Compilation Subprocesses setting (OpenGL only).
Open the most recently opened blend file, instead of the default startup file.
Set the application template (matching the directory name), use default for none.

Skip reading the startup.blend in the users home directory.
 Enable event simulation testing feature bpy.types.Window.event_simulate .
 Set the BLENDER_SYSTEM_DATAFILES environment variable.
 Set the BLENDER_SYSTEM_SCRIPTS environment variable.
 Set the BLENDER_SYSTEM_EXTENSIONS environment variable.
 Set the BLENDER_SYSTEM_PYTHON environment variable.
 Force sound system to None.
 Force sound system to a specific device.
 NoneDefaultSDLOpenALCoreAudioJACKPulseAudioWASAPI .
 Run a command which consumes all remaining arguments.Use -chelp to list
 all other commands.Pass --help after the command to see its help text.
 This implies --background mode.
 Print this help text and exit.
 Print this help text and exit (Windows only).
 Register blend-file extension for current user, then exit (Windows &
 Linux only).
 Register blend-file extension for all users, then exit (Windows & Linux
 only).
 Unregister blend-file extension for current user, then exit (Windows &
 Linux only).
 Unregister blend-file extension for all users, then exit (Windows & Linux
 only).
 Print Blender version and exit.
 End option processing, following arguments passed unchanged. Access via
 Python\%s sys.argv .
 Arguments must be separated by white space, eg:
 %s will exit since -ba is an unknown argument.
 Arguments are executed in the order they are given. eg:
 %s will not render to /tmp because --render-frame1 renders
 before the output path is set.
 %s will not render to /tmp because loading the blend-file
 overwrites the render output that was set.
 %s works as expected.
 Replace default directory of all user files.Other BLENDER_USER_*
 variables override when set.
 Directory for user configuration files.
 Directory for user scripts.
 Directory for user extensions.
 Directory for user data files (icons, translations, ..).
 Replace default directory of all bundled resource files.
 Directory to add more bundled scripts.
 Directory for system extensions repository.
 Directory to replace bundled datafiles.
 Directory to replace bundled Python libraries.
 Path to override the OpenColorIO configuration file.
 Store temporary files here (MS-Windows).
 Store temporary files here (UNIX Systems).The path must reference an
 existing directory or it will be ignored.
 --- Fin du chapitre: _arguments ---

--- Début du chapitre: _armature --- Armature is the constraint version of
 the Armature Modifier, exactly reproducing the weight-blended bone
 transformations and applying it to its owner orientation. It can be used

like a variant of the `Child Of constraint` that can handle multiple parents at once, but requires all of them to be bones.

Note

Unlike the `Armature` modifier, the constraint does not take the `Deform` checkbox of bones into account, and can use any bone as target.

`Armature constraint`.

Like the matching option of the modifier, it enables the use of quaternions for preserving the volume of the object during deformation. To approximate envelope-only behavior of the modifier, add all relevant bones with weight 1.0 and enable this option.

Note

Unlike the modifier, the constraint always requires explicitly listing all of its target bones with associated weights. This option merely enables envelopes for all bones, as if they had `Envelope Multiply` enabled. Only for constraints on bones: Instead of using the rest location, use the current location of the owner bone to compute envelope weights or binding to B-Bone segments.

With envelope weights, this can be used to change the active parent bone of the owner bone dependent on its location. For non-bones this mode is always active, because they don't have a rest location.

This button adds a new empty entry at the end of the target list.

This button normalizes all weight values in the target list so that they add up to 1.0.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

This specifies the list of bones used by the constraint to deform its owner. Each target bone has the following input fields and controls:

Unlike the modifier, the constraint can use bones coming from different armatures at the same time. See common constraint properties for more information.

Name of the target bone.

Removes the entry from the target list.

Weight associated with the bone, equivalent to vertex groups in the modifier.

--- Fin du chapitre: `_armature` ---

--- Début du chapitre: `_arranging` --- The snapping options can be found on the rightmost side of the node editor's header.

Toggle snapping on or off. You can also do this temporarily by holding `Ctrl` after starting to drag one or more nodes around.

What to snap the selected nodes to:

Snap to the grid in the background.

Snap to the X coordinate of another node's vertical border.

Snap to the Y coordinate of another node's horizontal border.

Combination of the above.

Which part of the selected nodes to snap:

Snap closest point onto target.

Snap center of selected nodes onto target.

Snap median of selected nodes onto target.

Snap active node onto target.

When you drop a node with at least one input and one output socket onto an existing connection between two nodes, `Auto-offset` will, depending on the direction setting, automatically move the left or right node away to

make room for the new node. Auto-offset is a feature that helps organizing node layouts interactively without interrupting the user workflow. Auto-offset is enabled by default, but it can be disabled in the Preferences .

You can toggle the offset direction while you are moving the node by pressing T.

The offset margin can be changed using the Auto-offset Margin setting in the Editing section of the Preferences.

See also

Example Video:

Auto-Offset. A workflow enhancement for Blender's node editors.

--- Fin du chapitre: `_arranging` ---

--- Début du chapitre: `_array` --- The Array modifier creates an array of copies of the base object, with each copy being offset from the previous one in any of a number of possible ways.

Useful for creating complex repetitive drawings.

Multiple Array modifiers may be active for an object at the same time (e.g. to create complex three-dimensional constructs).

See also

This documentation refers to the Array Modifier specific to the Grease Pencil object. For uses with other object types refer to the general Array Modifier .

The Array modifier. `\xc2\x80`

Total number of copies.

Index of the material to use on duplicated strokes (0 use strokes original materials).

Adds a translation equal to the object's bounding box size along each axis, multiplied by a scaling factor, to the offset. X, Y and Z scaling factors can be specified.

Adds a constant translation component to the duplicate object's offset. X, Y and Z constant components can be specified.

Adds a transformation taken from an object (relative to the current object) to the offset. It is good practice to use an empty object centered or near to the initial object.

Add random offset values to the copies.

Add random rotation values to the copies.

Add random scale values to the copies.

Use the same random Seed for each scale axis in the copies for a uniform scale.

Seed used by the pseudo-random number generator.

Note

The Depth Order used in the Grease Pencil object has an influence on the visualization of the strokes when using the Array modifier. See Depth Order for more information.

See Influence Filters.

--- Fin du chapitre: `_array` ---

--- Début du chapitre: `_asset` --- Operations for managing the Asset status of an object.

See Creating an Asset .

See Removing Assets .

See Removing Assets .

--- Fin du chapitre: _asset ---

--- Début du chapitre: _asset_browser --- The Asset Browser is the main interface for organizing and using assets.

See also

For general information on Blender's asset library system, including how to create and edit assets, and design choices.

For organizing assets.

Built on top of the Asset Browser.

Asset Browser, showing materials in an asset library. \xc2\xb6

The center region of the Asset Browser lists the assets contained in the selected catalog.

Click LMB to select a single asset. Additionally hold Ctrl to add/remove that asset to/from the selection, or Shift to select a range of assets. You can also drag LMB to perform a box select.

The region has a context menu with the following operations:

Refreshes the list.

See Removing Assets.

See Removing Assets.

Opens the blend-file containing the asset.

Changes the size of the preview thumbnails.

The region on the left lets you select an asset library and shows its catalogs. You can show/hide this region by pressing T.

The asset library whose catalogs to show.

Show catalogs from all available libraries.

Show the catalogs in the current blend-file (even if that file is not yet part of an asset library). See The Current File Asset Library for more information.

Show the catalogs that come bundled with Blender.

Any libraries that you added in the File Path Preferences are listed here too.

Shown when Asset Library is set to Current File and the current blend-file is an asset bundle that's not yet part of an asset library.

Lets you select a target asset library, then opens a File Browser in that library's root folder so you can save the current blend-file there. Once saved, the assets in the blend-file become available as part of the library.

Tree view that shows the catalogs of the selected asset library. A catalog is a group of assets; when you select one, only the assets in that catalog and its child catalogs will be listed.

You can rename a catalog by double-clicking it, or assign it to a different parent catalog by dragging and dropping.

Add-ons and features like the Pose Library can show custom panels here.

The region on the right shows the metadata of the active asset. You can show/hide this region by pressing N or clicking the gear icon in the header.

Only metadata of assets contained in the current blend-file can be edited.

The asset's name. Unique for the asset data type within the same blend-file.

The full path of the blend-file that contains the asset.

Opens the blend-file that contains the asset in a new Blender instance. When this instance is closed, the Asset Browser will be automatically refreshed.

Optional name of the license under which this asset is distributed. Not used by Blender itself.

Optional copyright notice. Not used by Blender itself.

Optional asset description. Not used by Blender itself.

Optional field for the asset author. Not used by Blender itself.

Shows the preview image of the asset. See Asset Previews .

Opens a File Browser where you can select a new image for the asset preview.

Autogenerate a new preview for the asset.

Menu of additional preview operators.

Generates a preview based on the 3D Viewport's Active object. This is useful for node groups, which cannot automatically generate their own preview.

Panel for viewing and editing asset tags. These do not have any meaning to Blender and can be chosen freely. When using the search field to filter the assets, the assets whose tags (partially) match the search term will also be shown.

Note

Depending on the current mode of the object and the selected asset types, more panels may be shown. For example, see Pose Library.

As a general rule, an asset can be used by dragging it from the Asset Browser to the desired location. Objects and worlds can be dragged from the Asset Browser into the scene. Materials can be dragged onto the object that should use them. Geometry nodes can also be dragged onto objects to add a Geometry Nodes Modifier . The use of pose assets is different, and is described in Pose Library .

There are several things that can happen when an asset is used, depending on the following configuration of the Asset Browser:

Determines how data is managed when an asset is imported. This option can be found in the center of the Asset Browser header (when an asset library other than Current File or Essentials is selected):

Use the import method set in the File Path Preferences .

Same as File Link

The asset will be linked to the current blend-file, and thus be read-only. Later changes to the asset file will be reflected in all files that link it.

Same as File Append

The asset and all its dependencies will be copied into the current file. Dragging a material into the scene three times will result in three independent copies. Dragging an object into the scene three times will also result in three independent copies.

\xcDependencies\x in this case means everything the asset refers to. For an object, this can be its mesh and materials, but also other objects used by modifiers, constraints, or drivers.

Since the file now has its own copy of the asset, later changes to the asset file will not be reflected in the file it's appended to.

Specific to the Asset Browser.

The first time an asset is used, it will be appended, including its dependencies, just as described previously. However, Blender will keep track of where it originated, and the next time the asset is used, as much data as possible will be reused. Dragging a material into the scene three

times will only load it once, and just assign the same material three times. Dragging an object into the scene three times will create three copies of the object, but all copies will share their mesh data, materials, etc.

Since the file now has its own copy of the asset, later changes to the asset file will not be reflected in the file it's appended to.

Tip

Instantiating Collections

Mimics the Instance Collections option when appending from the file browser

Some asset types such as collections can be created as an instanced collection. This is done by enabling the Instance option after dragging collection assets into the 3D Viewport. By enabling this option, an empty object is added that uses an instance of the collection. If this option is disabled, the full collection hierarchy will be added to the scene.

Collection Assets from the current file will always be instanced.

Note that all regular Blender operations are available after the asset has been added to the current file. For example, you could choose to link an object to the scene; this will also link its mesh and its materials. Then you can make the object itself local (Object > Relations > Make Local > Selected Objects), while keeping the mesh and materials linked to the asset files. This will result in a local, and thus editable, object, and keep the mesh and materials automatically up to date with any changes in the asset library.

Preview panel in the Asset Browser.

Preview images are typically generated automatically when you mark a data-block as an asset. Objects are captured from their local -Y axis, while collections are captured from the global -Y axis (as these don't have a local axis).

If the auto-generated preview image isn't sufficient, you can replace it by a custom one.

For previews of pose assets, see Controlling the Look of Preview Images.

Asset bundles are blend-files that do not reference any other file, and whose name ends in `_bundle.blend`. Any textures and other external files need to be packed into the current blend-file.

Asset bundles can be copied to an asset library via the Asset Browser: Open the asset bundle blend-file.

Switch its Asset Browser to Current File (if it's not set to that already).

Click on Copy Bundle to Asset Library.

Choose the asset library to copy it to.

A File Browser will open at the root folder of the selected asset library. Choose the desired location of the blend-file, and click the Copy to Asset Library button.

The blend-file will be saved at the chosen location, and any catalogs of the asset bundle will be merged into the target asset library.

Note

Both the word `casset` and the word `bundle` are commonly used, and not necessarily with the same meaning as described here. Not everything that's presented as an `casset bundle` will have the Copy to Asset Library functionality available; for that, the bundle file needs to adhere to the definition above.

--- Fin du chapitre: _asset_browser ---

--- Début du chapitre: _assigning_vertex_group --- Empty Vertex Groups panel. \xc2\xba

Vertex groups are maintained within the Object Datatab (1) in the Properties.As long as no vertex groups are defined (the default for new mesh objects),the panel is empty (2).

You create a vertex group by LMBon the Addbutton + on the right panel border (3).Initially the group is named \xe2\x80\x9cGroup\xe2\x80\x9d (or \xe2\x80\x9cGroup.nnn\xe2\x80\x9d when the name already exists) andgets displayed in the panel (2) (see next image).

One vertex group.\xc2\xba

Once a new vertex group has been added, the new group appearsin the Vertex Groups panel. There you find three clickable elements:

The group name can be changed by double-clicking LMBon the name itself.Then you can edit the name as you like.

When the little arrow icon in the left lower corner is clicked, a new row opens upwhere you can enter a search term. This becomes handy when the number ofvertex groups gets big.

If you have a large number of vertex groups and you want to see morethan a few groups, you can LMBon the small drag handle to makethe vertex groups list larger or smaller.

When a vertex group is created,then it is also automatically marked as the Active Group.This is indicated by setting the background of the panel entryto a light gray color. If you have two or more groups in the list,then you can change the active group by LMBonthe corresponding entry in the Vertex Groups panel.

Delete a vertex group. \xc2\xba

You delete a vertex group by first making it the active group(select it in the panel) and then LMBthe Removebutton - at the right panel border. Deleting a vertex group only deletes the vertex assignments to the group.The vertices themselves are not deleted.

Lock a vertex group. \xc2\xba

Right after creation of a vertex group,an open padlock icon shows up on the right side of the list entry.This icon indicates that the vertex group can be edited.You can add vertex assignments to the group or remove assignments from the group.And you can change it with the weight paint brushes, etc.

When you click on the icon,it changes to a closed padlock icon and all vertex group modifications get disabled.You can only rename or delete the group, and unlock it again.No other operations are allowed on locked vertex groups,thus all corresponding buttons become disabled for locked vertex groups.

Assign weights to active group.\xc2\xba

You add vertices to a group as follows:

Select the group from the group list, thus making it the active group (1).

From the 3D Viewport selectShift- LMBall vertices that you want to add to the group.

Set the weight value that shall be assigned to all selected vertices (2).

LMBthe Assignbutton to assign the selected vertices to the active group using the given weight (3).

Note that weight assignment is not available for locked vertex groups. The Assign button is grayed out in that case.

Note

Assign is additive

The Assign button only adds the currently selected vertices to the active group. Vertices already assigned to the group are not removed from the group.

Also keep in mind that a vertex can be assigned to multiple groups.

To be sure the selected vertices are in the desired vertex group, you can try press the deselect button. If the vertices remain selected then they are not yet in the current vertex group.

At this point you may assign them, but take care since all selected vertices will have their weight set to the value in the Weight field.

You remove vertices from a group as follows:

Select the group from the group list (make it the active group).

Select all vertices that you want to remove from the group.

LMBclick the Remove button.

Note that removing weight assignments is not available for locked vertex groups. The Remove button is grayed out in that case.

You can quickly select all assigned vertices of a group:

(Optionally) press Alt- A to deselect all vertices.

Select the group from the group list (make it the active group).

When you now LMBclick the Select button, then the vertices assigned to the active group will be selected and highlighted in the 3D Viewport.

When you LMBclick the Deselect button instead, then the vertices assigned to the active group will be deselected in the 3D Viewport.

Note

Selecting/Deselecting is additive

If you already have vertices selected in the 3D Viewport, then selecting the vertices of a group will add the vertices but also keep the already selected vertices selected. Vice versa, deselecting the vertices of a vertex group will only deselect the vertices assigned to the group and keep all other vertices selected.

You can find ungrouped vertices as follows:

Press Alt- A to deselect all vertices.

In the header of the 3D Viewport, navigate to Select \xe2\x80\xa3 Select All by Trait \xe2\x80\xa3 Ungrouped Vertices.

--- Fin du chapitre: _assigning_vertex_group ---

--- Début du chapitre: _assignment ---Reference

Material \xe2\x80\xa3 Material Slots

Materials are data-blocks that can be created and then assigned to one or more objects. An object can also have multiple materials assigned in different material slots, which correspond to different parts of an object. If a smooth transition between materials is desired, then mixing shader nodes with a Mix shader is a better solution.

Material slots link materials to objects and meshes. By default objects only have a single material slot, which assigns a material to the entire object. If different parts of the mesh need different materials, multiple material slots can be created.

Material slots panel. \xc2\xb6

The object \xe2\x80\x99s material slots and active material displayed in a List View.

Add a new material slot on the object.

Remove a material slot from the object.

Copy material shader nodes and settings to clipboard.

Paste material shader nodes and settings from clipboard.

Copy the same material assignment from the active to other selected objects.

Removes all material slots not assigned to the object.

The MaterialData-Block Menu for the selected material slot. Here new materials can be created, or existing materials can be assigned to the material slot.

Specifies whether the material is to be linked to the object or to the object data.

The Link selector has two choices, Data and Object. These two menu choices determine whether the material is linked to the object or to the data (e.g. a mesh or curve). The Data menu item determines that this material will be linked to the mesh's data-block which is linked to the object's data-block. The Object menu item determines that the material will be linked to the object's data-block directly.

This has consequences of course. For example, different objects may share the same mesh data-block. Since this data-block defines the shape of the object any change in Edit Mode will be reflected on all of those objects. Moreover, anything linked to that mesh data-block will be shared by every object that shares that mesh. So, if the material is linked to the mesh, every object will share it.

On the other hand, if the material is linked directly to the object data-block, the objects can have different materials and still share the same mesh.

Short explanation: If connected to the object, you can have several instances of the same Object Data using different materials. If linked to mesh data, you cannot. See Data System for more information.

To assign materials to different parts of a mesh, enter Edit Mode on the mesh. Additional buttons will then appear in the material slots panel.

Material slots panel in Edit Mode.

Assign active material slot and material to the selected faces in the mesh, strokes in a Grease Pencil, and similar for other object types.

Select faces assigned to the active material slot.

Deselect faces assigned to the active material slot.

Blender is built to allow you to reuse anything, including material settings, between many objects. Instead of creating duplicate materials, you can simply reuse an existing material. There are several ways to do this using the Material's data-block menu:

Single Object With the object selected, click the sphere located to the left of the Material name. A pop-up appears showing all the materials available in the current blend-file. To use one, just click on it.

Tip

Searching for Materials

The search field at the bottom of the material list allows you to search the names in the list. For example, by entering `wood` all existent materials are filtered so that only materials containing `wood` are displayed in the list.

Multiple Objects In the 3D Viewport, with Ctrl-L you can quickly link all selected objects to the material (and other aspects) of the active object. Very useful if you need to set a large number of

objects to the same material; just select all of them, then the object that has the desired material, and Ctrl- L links them to that
parent.

To delete a material, select the material and click X in the Available Materials List entry.

Although the material will seem to disappear immediately, the Delete action can depend on how the material is used elsewhere.

If the material is linked to the object and there are other objects which use this material, then the material will be removed from that object (but remain on all its other objects).

If the Fake User button has been lit in the Available Materials list, then the material will be retained when the file is saved, even if it has no users.

Only if it has 0 real users, and no Fake user, will the material be permanently deleted. Note that it will still remain in the Materials list until the blend-file is saved, but will have disappeared when the file is reloaded. Normally, different colors or patterns on an object are achieved by adding textures to your materials. However, in some applications you can obtain multiple colors on an object by assigning different materials to the individual faces of the object.

To apply several materials to different faces of the same object, you use the Material Slots options in the Materials header panel.

The workflow for applying a second material to some faces of an object covered by a base material is as follows:

In Object Mode, create a base material.

Go into Edit Mode and Face Select (a new list will appear below the Active Material list with Assign, Select, Deselect buttons).

Select the faces to be colored with the second material.

In the Object Material Slots list, click the + button to create a new slot or select an existing material.

Click the Assign button, and the material will appear on the selected object faces.

--- Fin du chapitre: _assignment ---

--- Début du chapitre: _at --- Returns the color of a texture at the specified coordinates.

Standard color input.

The point at which to sample the color. For images, the space is between -1 and 1 for X and Y. If the coordinates are not spatially varying, the node will return a single color.

This node has no properties.

Standard color output.

--- Fin du chapitre: _at ---

--- Début du chapitre: _attach_hair_curves_to_surface --- Attaches hair curves to a surface mesh.

Note

This node/modifier will not function without the Surface Geometry/object and Surface UV Map inputs.

Input Geometry (only curves will be affected).

Surface Geometry to attach hair curves to. This input takes priority over the corresponding object input, if used.

Surface Object to attach to (needs to have matching transforms).

Surface UV map stored on the mesh used for finding curve attachment locations.

Set the surface mesh into its rest position before attachment.

Tip

In a typical hair generation setup, this node or modifier will be combined with the Deform Curves on Surface Node. If that operation comes after this one, it makes sense to turn this option on so the position used is the pre-deformed position consistent with the expectations for the deformation's input.

Sample the surface UV map at the attachment point.

Snap the root of each curve to the closest surface point.

Align the curve to the surface normal (need guide as reference).

Blend deformation along each curve from the root.

This node has no properties.

Geometry

Surface UV coordinate at the attachment point.

Surface normal at the attachment point.

--- Fin du chapitre: `_attach_hair_curves_to_surface` ---

--- Début du chapitre: `_attribute` --- The Attribute node allows you to retrieve attributes attached to an object or mesh.

This node has no inputs.

Name of the attribute.

Specifies the type of the attribute.

The attribute is associated with the geometry of the object, and its value varies from vertex to vertex, or within the volume of the object.

Most geometry attributes are directly accessible through the various input nodes, except for these:

Gives a scalar defining where foam might appear when using an Ocean Modifier. This depends on the name you give this property.

See also

For a full list of options see [This Thread](#) on the Blender Stack Exchange. The attribute name specifies a custom property name, or an RNA path to a built-in property (like the single property driver variables).

The values of attributes of this type are defined once per object. The name or path is looked up first in the object data-block, followed by the mesh data-block if not found. Custom properties have priority over built-in ones.

The property value must be an integer, float, boolean, or a vector of 1 to 4 floats or ints; properties of other types are ignored. If a suitable property is not found, all sockets of the node, including Alpha, output 0.

Tip

The color attribute will output the value of the Color field in the Viewport Display panel of the object, unless overridden by a custom property.

Similar to Object, but the attribute is looked up in the instancer particle system settings, followed by Geometry Node instance

attributes (searching from the innermost instancing layer to outer ones), and finally in the instancer object. If the current object is not instanced, or the property is not found, it falls back to the Object mode.

Warning

Currently only up to 4 layers of Geometry Node instancing are searched. The attribute is looked up in the current View Layer, Scene and World, using the same lookup logic as Object, and likewise producing all zero

outputs including Alpha if not found. Attributes of this type have the same uniform value throughout the whole Render Layer.

Tip

This gives access to a number of useful built-in properties, for example:

Outputs the value of the Color field in the Viewport Display panel of the World properties.

Outputs the current rendering resolution.

Outputs the effective field of view of the active Camera.

See also

An alternative method to access the same set of properties is to use driverContext Properties, possibly with a manually emulated lookup fallback chain.

RGB color interpolated from the attribute.

XYZ vector interpolated from the attribute.

Scalar value interpolated from the attribute.

Alpha channel of the attribute, when available. If the attribute has no alpha channel, generally defaults to 1.

Warning

Currently, only View Layer attributes are supported in shaders used for the World or Light Objects.

--- Fin du chapitre: _attribute ---

--- Début du chapitre: _attributes_reference --- An attribute is a generic term to describe data stored per-element in a geometry data-block. For example, every vertex can have an associated number or vector. Attributes can be altered by connecting a value to the Group Output node, but also many nodes can change the values of specific attributes.

Note

Attribute data types and domains are converted implicitly where possible, just like node sockets.

Named attributes are created and used in other areas of Blender like shaders, painting, and UV mapping. In the modifier panel, a named attribute can be used for input by clicking the icon to the right of the value button. The string input input allows you to search and choose existing attributes from the modifier's input geometry.

Attribute Search.

The attribute search gives a bit of context about each attribute. To the left of the menu, the attribute domain is shown followed by the attribute name. To the right of the menu, the attribute data type is shown.

The Normal and Rotation outputs are examples of attribute fields, which refer to an attribute stored on a geometry.

An anonymous attribute is a set of generic data stored on a geometry that doesn't have a name. Usually, attributes exposed in Blender's interface all have names. However, in geometry nodes, attributes can be passed around with node sockets. In these cases, an Attribute Field output is created, which is used by nodes to find attribute data in an input geometry.

Anonymous attributes are still stored on the geometry like other attributes, and they are even automatically interpolated when the geometry changes with other nodes, except for a few cases. So generally, if the node link is still accessible, the attribute it references will be available too. However, anonymous attributes cannot be connected to a completely separate geometry that was created from a different source. To

transfer attributes between separate geometries, the Sample Index Node or other similar nodes like the Sample Nearest Surface Node can be used. The type of an attribute is the kind of data stored at each element.

True or false value.

32-bit integer.

Smaller integer with a range from -128 to 127.

Floating-point value.

3D vector with floating-point values.

2D vector with floating-point values.

RGBA color with 32-bit floating-point values.

RGBA color with 8-bit positive integer values.

Floating point Quaternion rotation.

The above list is in the order of least to most

complex (An integer can contain more data than a boolean, so it is more complicated). When joining separate geometries together, the more complex datatype is preferred when there are matching names. This is particularly important when joining geometry with named attributes with the Join Geometry Node.

To store 2D Vectors (UV maps) and Byte Colors the Store Named Attribute Node must be used because there are no sockets for those types.

Through the use of Geometry Nodes, data can be converted between types.

Valid conversions:

Between color and vector: mapping between color channels and vector components.

Between color and float: the color data is converted to its grayscale equivalent.

Between float and integer: integers simply become floats, floats are truncated.

Between float and vector: when a float becomes a vector the value is used for each component. When a vector becomes a float the average of the components is taken.

Between float and boolean: values greater than 0 are true, 0 maps to false, and false maps to 0.

The domain of an attribute refers to what type of geometry element the attribute corresponds to. Knowing the domain of an attribute is important because it defines how it may be interpolated and used in nodes and shading. You can use the Spreadsheet Editor to determine the domains of attributes.

Vertices of a mesh

Points of a point cloud

Curve control points

Edge domain attributes are associated with the edges of a mesh.

Face domain attributes are associated with the faces of a mesh.

Face Corner domain attributes are associated with the corners of the faces of the mesh. An example is a UV map attribute.

Spline domain attributes are associated with a group of connected curve control points.

Instance domain attributes exist on the Instances in a geometry. They can be used to store different values on copies of geometry data. Instance domain attributes are only supported in geometry nodes.

Layer domain attributes are associated with a Grease Pencil Layer. Attributes are automatically interpolated to other domains. For example, when the Position Node is connected to the selection input of the Set Material Node, the values are interpolated from the Point domain to

the Facedomain. Normally, domain conversions use simple averages for values, but Boolean data type attributes have special rules for interpolation:

From

To

Conversion

Point

Edge

An edge is selected if both of its vertices were selected.

Point

Face

A face is selected if all of its vertices were selected too.

Point

Corner

Each corner's value is simply a copy of the value at its vertex.

Point

Spline

A spline is selected if all of its control points were selected.

Edge

Point

A vertex is selected if any connected edge was selected.

Edge

Face

A face is selected if all of its edges are selected.

Edge

Corner

A corner is selected if its two adjacent edges were selected.

Face

Point

A vertex is selected if any of the connected faces were selected.

Face

Edge

An edge is selected if any connected face was selected.

Face

Corner

Each corner's value is simply a copy of the value at its face.

Corner

Point

A vertex is selected if all connected face corners were selected and it is not a loose vertex.

Corner

Edge

An edge is selected if all corners on adjacent faces were selected.

Corner

Face

A face is selected if all of its corners were selected.

Spline

Point

Each point's value is simply a copy of the corresponding value of the spline.

Built-in attributes always exist, and cannot be removed. Their data type and domain cannot be changed.

Name

Type

Domain

Notes

position

Vector

Point

Built-in attribute describing vertex or point locations, in the local space of a geometry. Any node that changes the location of points will adjust this attribute, like the Transform Geometry Node and the Set Position Node.

radius

Float

Point

A built-in attribute on point clouds used to set the size for the points in the viewport. Also built-in on curves, where it controls the size of each curve control point when converted to a mesh, or for other operations.

id

Integer

Point

Created by the Distribute Points on Face node to provide stability when the shape of the input mesh changes, and used on instances to create motion blur. Values are expected to be large, with no order. This attribute is used by nodes that generate randomness, like the Random Value Node. Unlike other built-in attributes, this attribute is not required, and can be removed.

material_index

Integer

Face

Used to specify the material slot for every face in a mesh.

sharp_edge

Boolean

Edge

Attribute determining if an edge should have flat (rather than smooth) shading enabled in the viewport or a render.

sharp_face

Boolean

Face

Attribute determining if a face should have flat (rather than smooth) shading enabled in the viewport or a render.

resolution

Integer

Spline

Determines the number of evaluated points between two control points of a spline. Only NURBS and Bezier splines have this attribute, for poly splines, the value is always one.

cyclic

Boolean

Spline

Determines whether the spline has a segment that connects its first and last control points.

handle_left

Vector

Point

Describes the location of the left handle of a curve control point, on the side of the curve's start. Only exists when the curve contains a B-spline.

handle_right

Vector

Point

Describes the location of the right handle of a curve control point, on the side of the curve's end. Only exists when the curve contains a B-spline.

These attributes do not exist by default, but are used implicitly by certain parts of Blender. The data type of these attributes can be changed, just like any attribute besides the built-in attributes. However, the attributes might be expected by Blender to have a certain type.

Name

Type

Domain

Notes

velocity

Vector

Point

Used to create motion blur when rendering animations.

rest_position

Vector

Point

Holds the position of points or vertices from before a geometry is deformed procedurally. Can be created automatically before Shape Keys and Modifiers are evaluated with the Add Rest Position option.

surface_uv_coordinate

2D Vector

Curve

Used to describe curve attachment locations on a mesh surface, typically used for the hair system.

crease_vert

Float

Point

Vertex attribute used by the Subdivision Surface modifier. The values are expected to be in a range of 0 and 1.

crease_edge

Float

Edge

Edge attribute used by the Subdivision Surface modifier. The values are expected to be in a range of 0 and 1.

uv_seam

Boolean

Edge

True if an edge is considered a boundary between UV islands when unwrapping.

bevel_weight_vert

Float

Point

Used as vertex control for the bevel modifier.

bevel_weight_edge

Float

Edge

Used as edge control for the bevel modifier.

sculpt_face_set

Integer

Face

Used by theSculpt Face Sets Feature.

sculpt_mask

Float

Point

Used by theSculpt Masking Feature.

sculpt_face_set

Integer

Face

Used by theSculpt Face Sets Feature.

Vertex groups, UV maps and Color Attributes are available as attributes in geometry nodes.They are referred to by their name.Naming collisions (e.g. a vertex group and a UV map with the same name) should be avoided.If there is a naming collision, only one of the attributes is accessible in geometry nodes.

Attributes with any other name can also be created by nodes, when the name is used for the first time.

Note that geometry nodes does not always produce e.g. vertex groups if a node like Join Geometry is used.Similarly, if the data type of a vertex group attribute is changed from the initial \xe2\x80\x9cFloat\xe2\x80\x9d type,the attribute will no longer be a vertex group.

This operator found in the Attributespanel of the property editor can change thedomain or data type of an attribute.

Due to ongoing development in the area of attributes, many areas of Blender can not yet work withthe generic (identified with a name, stored on any domain with any data type) attributes used bygeometry nodes. That makes this operator an essential workaround in some cases where existingtools must be used with data generated from geometry nodes. Interpolate and convert the attribute between the domains and data types described on this page.

Create a Vertex Group from the attribute, which corresponds to a float attribute on the point domain.

Note

This operator only works on originalobject data, not including the results of modifiers,so any attributes added or changed by geometry nodes will not be affected. To change the typeof an attribute generated procedurally, modifiers must be applied.

--- Fin du chapitre: _attributes_reference ---

--- Début du chapitre: _attribute_statistic --- The Attribute

Statisticnode evaluates a field on a geometry and outputs a statistic about the entire data set.

Standard geometry input.

A boolean field input for each element indicating whether to include its value in the statistics resultIf the boolean is false, the corresponding value from the Attributeinput will be ignored.

The attribute field to query a statistic from.

The output will be a single floating-point value.

The output will be a vector of three floating-point values.All calculations are elementwise.

The attribute domain used for the statistics and to evaluate the input Attribute field on.

The average value of all data.

The median value of all data.

The sum value of all data.

The min value of all data.

The max value of all data.

The difference between the max and min value.

How much values differ from the mean. A low standard deviation indicates that the values are grouped tightly together at the mean. A high standard deviation indicates that the values are spread out over a large range.

The variance of all data, defined as the square of the standard deviation.

--- Fin du chapitre: `_attribute_statistic` ---

--- Début du chapitre: `_axes_to_rotation` --- Creates a rotation based on two axis directions.

Tip

In many cases, these directions are a normal and tangent on a mesh or curve.

The desired direction of the primary axis.

The desired direction of the secondary axis. Ideally, this is orthogonal to the primary direction.

The axis (X, Y or Z) that should be aligned exactly to the primary direction.

The axis that should be aligned as closely as possible to the secondary direction.

The rotation that results in the given axes being aligned to the given directions.

--- Fin du chapitre: `_axes_to_rotation` ---

--- Début du chapitre: `_axis_angle_to_rotation` --- The Axis Angle to Rotation node converts axis angle rotation to a standard rotation value.

Unit vector representing the axis to rotate around.

The rotation angle around the axis.

Standard rotation value.

--- Fin du chapitre: `_axis_angle_to_rotation` ---

--- Début du chapitre: `_axis_locking` --- Axis locking.

This option limits the transformation to the specified axis.

Transformations (translation/scale/rotation) in Object Mode and Edit Mode (as well as extrusions in Edit Mode) can be locked to a particular axis relative to the current transform orientation. By locking a transformation to a particular axis you are restricting transformations to a single dimension.

A locked axis will display in a brighter color than an unlocked axis. For example in the image to the right, the Z axis is shown in light blue as movement is constrained to this axis. This example, can be achieved in two ways:

The axis of movement can be changed at any time during transformation by typing X, Y, Z.

Axis constraint in action.

Holding MMB after starting a transformation lets you select an axis to constrain to. A visual option to constrain the translation will be

available, showing the three axes in the 3D Viewport space. A dotted white line is used as a pointer. The axis of choice to confirm the operation will depend on the highlighted axis about which the MMB is released.

When you already moved the mouse in the desired direction, pressing MMB will lock to the axis which was pointed at.

Reference

Object and Edit Modes (move, rotate, scale, extrude)

X, Y, Z or MMB after moving the mouse in the desired direction.

Axis locking limits the transformation to a single axis (or forbids transformations along two axes). An object, face, vertex or other selectable item will only be able to move, scale or rotate in a single dimension.

Reference

Object and Edit Modes (move, scale)

Shift- X, Shift- Y, Shift- Z or Shift- MMB after moving the mouse in the desired direction.

Plane locking.

Plane locking locks the transformation to two axes (or forbids transformations along one axis), thus creating a plane in which the element can be moved or scaled freely. Plane locking only affects translation and scaling.

Note that for rotation, both axis and plane locking have the same effect because a rotation is always constrained around one axis. Trackball type rotations cannot be locked at all.

A single key press constrains movement to the current transform orientation selection. A second key press of the same key constrains movement to the corresponding Global axis (except if the transform orientation is set to Global, in which case the Local orientation is used). A third key press of the same key removes constraints.

The orientation can be set in the Transform Orientation selector of the 3D Viewport header.

For example, if the current transform orientation is set to Normal, pressing G to start translation, followed by Z will lock translation in the Z direction relative to the Normal orientation, pressing Z again will lock translation to the Z axis relative to the Global orientation. Pressing Z again will remove all constraints. The current mode will be displayed in the left-hand side of the 3D Viewport header.

Z axis locking in Global orientation.

Z axis locking in Local orientation.

Z axis locking in Global orientation with vertex selection.

Z axis locking in Normal orientation with vertex selection.

As can be seen in the Axis locking modes image, the direction of the transform also takes into account the selection.

Note that using a locked axis does not prevent you from using the keyboard to enter numeric transformation values.

--- Fin du chapitre: _axis_locking ---

--- Début du chapitre: _backbone_stretcher --- The Backbone Stretcher modifier stretches (adds some length to) the beginning and end of the stroke.

Length to add to the strokes' ends.

--- Fin du chapitre: _backbone_stretcher ---

--- Début du chapitre: `_background` --- The Backgroundshader node is used to add background light emission. This node should only be used for the world surface output.

Color of the emitted light.

Strength of the emitted light.

This node has no properties.

Standard shader output.

--- Fin du chapitre: `_background` ---

--- Début du chapitre: `_bake` --- The Bakenode allows saving and loading intermediate geometries. This node bakes parts of the node tree for better performance.

The data format used to store geometry data is not considered to be an import/export format. Volume objects, however, are saved using the OpenVDB file format which can be used interoperably.

Important

It's not guaranteed that data written with one Blender version can be read by another Blender version.

Standard geometry input, which is used as the default bake item. More bake items can be added by dragging sockets into the blank socket or in the Bake Items panel. Items can be renamed by Ctrl-LMB on the socket name or in the nodes Properties panel.

Reference

Geometry Node Editor

Sidebar `Node` `Bake Items`

The Bake Items panel is used to manage the input sockets of the node thus also managing what data is baked.

Used to manage the inputs and outputs of the bake node. Items can be added, removed, renamed, and sorted.

The data (Sockets) of the input/output.

The attribute domain used to evaluate the input field on.

Bake item is an attribute stored on a geometry.

Note

Some properties can only be edited in the Properties panel (Sidebar `Node` `Properties`).

The Bakenode can calculate the geometry of a single frame or an animation.

Bakes the geometry data for multiple frames. By default the scene frame range is used, however, a Custom Range can also be defined.

Bakes the geometry data of the current frame.

Performs all necessary geometry calculations and saves the data on disk or packed into the .blend file.

Packs or unpacks already baked data. When unpacking, there are different options to control where the data will be stored. The options are the same as for the Unpack Resources operator.

Deletes the bake data.

Specifies where the baked data should be stored.

The setting is copied from the modifier that contains the bake node. This is the default.

The baked data is packed into the .blend file. So no separate file is necessary.

The baked data is stored in a separate directory on disk.

Specify a path where the baked data should be stored manually.

Location on disk where the baked data is stored. Note, this path is also used for simulation zones .

Override the simulation frame range from the scene.

The start and end frame numbers for the custom range.

For each input, the same output is added to act as a pass through.

Standard geometry output, which is used as the default bake item. More

bake items can be added by dragging sockets into the blank socket or in the Bake Items panel. Items can be renamed by Ctrl- LMB on the socket name or in the nodes Properties panel.

--- Fin du chapitre: _bake ---

--- Début du chapitre: _baking --- Baking allows saving and loading intermediate geometries. Baking parts of the node tree can be used for better performance.

The data format used to store geometry data is not considered to be an import/export format. Volume objects, however, are saved using the OpenVDB file format which can be used interoperably.

Data can be baked using two methods:

Bake Node \xe2\x80\x93 used to bake any portion of the node tree.

Simulation Zone Baking \xe2\x80\x93 used to bake animations where the result of one geometry state can influence the next state.

Data is not stored to disk until the Bake operation is ran.

Important

Blend-files must be saved to a disk before data can be baked.

It \xe2\x80\x93s not guaranteed that data written with one Blender version can be read by another Blender version.

Bakes the simulations and bake nodes in all modifiers for the selected objects.

Tip

When data is baked the number of baked frames is displayed above the simulation zone or bake nodes.

Reference

Geometry Node Editor

Sidebar \xe2\x80\x93 Node \xe2\x80\x93 Data-Block References

Baked geometries that reference other data-blocks such as materials are listed here.

This panel allows changing these references after the data has been baked.

Note

Currently only material data-blocks are supported.

--- Fin du chapitre: _baking ---

--- Début du chapitre: _basic --- These rig types are used to generate simple single-bone features, and for custom rigging done directly in the meta-rig.

The single-bone rig types must be applied separately to every bone even within a connected chain, and can have connected children controlled by a different rig type. This is unlike chain-based rig types that usually consume the whole connected chain.

Copies the bone chain keeping all the parent relations within the chain untouched. Useful as a utility rig type for custom rigs.

Requirement: A chain of at least two connected bones.

When enabled control bones and widgets will be created.

When enabled deform bones will be created.

Single-bone rig type that creates a custom pivot control for rotating and scaling its child sub-rigs.

This type of control transforms its children when rotated or scaled, while moving it merely changes the pivot point used by rotation or scaling.

When enabled an extra parent control bone with a box widget is created to allow moving the rig. It is also required by all other options besides Deform Bone.

Allows selecting one of the predefined widgets to generate for the master control instead of the default cube.

Generates a mechanism for switching the effective parent of the rig based on the value of a custom property.

Registers the rig as a potential parent scope for its child sub-rigs' parent switches.

Specifies additional comma-separated tag keywords for the registered parent scope. They can be used by other rigs to filter parent choices, or for selecting the default parent.

Some of the existing tags that are useful here:

The parent scope will be made available for all children of the parent sub-rig, rather than just this rig's children.

A control for the object held in the character's hand.

Preferred by finger IK.

The injected, held_object combination is perfect for such a control.

Disabling this avoids generating the actual custom pivot control, effectively turning this rig type into a version of basic.super_copy with parent switching support and a different widget.

When enabled a deform bone will be created.

Single-bone rig type that copies the bone without the ORG- name prefix.

Normally all bones copied from the meta-rig are prefixed with ORG- and placed on an invisible layer. This precludes their use as controls or deforming bones, which makes it difficult to transfer complex custom rigging verbatim from the meta-rig.

This rig type does not add the automatic prefix, thus allowing an appropriate ORG- , MCH- or DEF- prefix to be manually included in the meta-rig bone name, or alternatively using no prefix to create a control bone.

Allows retargeting constraints belonging to the bone to point at bones created in the process of generating the rig, thus allowing custom rigging to integrate with generated bones.

To use this feature, add @ and the intended target bone name to the constraint name, resulting in the ...@bone_name syntax. After all bones of the rig are generated, the constraint target bone will be replaced. If the new bone name is just CTRL , MCH or DEF , this will just replace the ORG prefix in the existing target bone name. For the Armature constraint you can add a @ suffix for each target, or just one @CTRL , @MCH or @DEF suffix to update all.

If the field is not empty, applies the same name substitution logic to the parent of the bone.

When this feature is enabled, the bone will not be automatically parented to the root bone even if it has no parent; enter root in the Parent field if that is necessary.

Single-bone rig type that simply copies the bone. Useful as utility rig type for adding custom features or specific deform bones to your rigs.

When enabled a control bone and widget will be created.

When enabled a widget will be created in replacement to the standard. Allows selecting one of the predefined widget types to generate instead of the default circle.

When enabled a deform bone will be created.

Works the same as in the basic.raw_copyrig. In addition, when enabled any constraints that havenames prefixed with CTRL: are moved to the control, and with DEF: to the deform bone.

--- Fin du chapitre: _basic ---

--- Début du chapitre: _basics ---Add a meta-rig structure from the Add \xe2\x80\xa3 Armaturemenu.

Edit the bone positions to match the character geometry.

In the armature properties click on the Generate Rigbutton to generate the rig.

Reference

Object Mode

Add \xe2\x80\xa3 Armature

Shift- A

Rigify stores all the information required to generate complex rig controls and mechanism inmore simple armatures called \xe2\x80\x9cmeta-rigs\xe2\x80\x9d.

The predefined meta-rigs can be found in the Addmenu.Currently available meta-rig types are:

Basic Human (doesn\xe2\x80\x99t include face and fingers)

Basic Quadruped

Human

Cat

Wolf

Horse

Shark

To correctly match your character, meta-rig bones must be moved to correct positions.This can be achieved in two different ways: Pose Mode or Edit Mode.

Note

Rigify assumes that 1 unit corresponds to 1 meter. So a human is about 2 units tall.If your character is in a different scale and you are more familiar with modeling rather than rigging,it is suggested to scale it to Rigify dimensions before positioning the meta-rig bones.If you want to scale the character\xe2\x80\x99s geometry, we suggest you to first scale up the character in Object Mode,then apply the geometry scale with the Apply Scaletool.

Limbs: Keep the legs as straight as possible in the front view (Rigify human works better in predictable cases).Give the knee and the elbow a slight bend angle (Rigify needs to know where your knee/elbow is pointing).

Torso: Keep the spine as straight as possible in the front view (Rigify human works better in predictable cases).The last bone of the spine is the head. By default the next two bones (top to bottom)are considered the neck bones. It is suggested to keep the neck bones as aligned as possible while editing.

Face: Positioning face bones can be tricky if you are not an expert in bone editing andthey are almost useless if you plan to make facial animation through shape keys.Consider removing face features from your character if they aren\xe2\x80\x99t really needed.If you don\xe2\x80\x99t

need the face all the face bones can be deleted. All the face bones are in the Facearmature bone collection by default. You can select them by displaying only that collection, selecting all of its content and then deleting the bones in Edit Mode to correctly remove the face.

If you want to scale all the face bones at once, consider scaling the face master bone in Pose Mode (see Pose Mode matching method). The face master bone is placed in the same position of the head bone. To select it easily, hide all other bone collections.

For more tips, see the Positioning Guide .

Enter the meta-rig Pose Mode. Rotate, scale, and translate the bones in the correct position. When bones are in correct positions (always staying in Pose Mode) use Apply \xe2\x80\xa3 Apply Pose As Rest Pose.

Note

Connected bones cannot be translated in Pose Mode. You can scale the parent bones to match the general length and then refine child bones scale. For more detailed information on armature modes please refer to the armatures section .

Some basic armature display setup is suggested before entering bone Edit Mode.

With the meta-rig selected, go in the Properties and click on the Object tab. Scroll down to the display panel and enable X-ray and under Maximum Draw Type selector select Wire. This way the bones will always be drawn in wireframe on top of your geometry.

Then, always in the Properties click on the Armatures tab and under display check the Axis checkbox. This way you the bones rotation axes will be displayed during the edit process.

For more detailed information on armature display modes please refer to the Display panel page.

With the bones in the correct positions, jump back in Object Mode, go to the Armature tab, scroll down to the bottom and click on the Generate Rig button to finalize the rig creation. The generation process will take from few seconds to one minute depending on rig complexity and hardware specifications of your machine. If the generated rig needs tweaking, you can modify the meta-rig accordingly and then click again on the generate button. If the rig already exists, Rigify will simply overwrite it retaining all your modifiers and constraints and \xe2\x80\x93 where possible \xe2\x80\x93 all the previously generated features.

For information about additional generation options, see the Advanced Rig Generation section.

Tip

If the meta-rig uses the legacy face rig , you can use the Upgrade Face Rig button that appears above Generate Rig to automatically upgrade to the new modular face system.

The upgrade will preserve compatibility with existing skinning, but existing poses and animations will likely not be compatible due to subtle changes in control behavior.

Note

To make the rig overwriting work as expected, you need to have both the rig and the meta-rig visible before generating again. Rigify will try to unhide them in simple cases, but will abort generation if that fails.

Warning

As with all Python add-ons, Blender interface cannot be updated until the Python script execution is over. Wait until the rig appears to see the results.

Warning

Rigify is designed assuming a workflow where the meta-rig is kept available to allow re-generating the main rig whenever it is necessary to make changes to it. Removing the meta-rig after generating the main rig, or significantly modifying the generated rig is not advised: it will make it impossible to introduce features added in later versions of Rigify, or easily adapt it to breaking changes in later Blender versions. In general, automatic version update scripts will be provided for meta-rigs when necessary, but not generated rigs.

To bind the geometry to the rig you can use your preferred tools. Just a few things you have to know:

- All the deforming bones are in the DEFbone collection.

- Eyes and Teeth bones of the legacy face are not deforming. You are supposed to bind the eyes and teeth geometry through Child Of constraints.

- Usually armature deform with automatic weights do a really good job out of the box if you correctly place your bones (and there is enough topology to work with!).

- For more detailed information on bone collections, Armature modifier and weight painting refer to the Blender manual.

- By using options in the Advanced sub-panel, it is possible to:

 - Generate more than one rig per scene.

 - Update/Override a specific rig.

 - Force previously generated widget objects to be overwritten.

 - Choose whether to use linked duplicates for left and right side widgets.

 - Execute a script data-block after generation.

 - Advanced rig generation options are by default hidden in a sub-panel.

Click on the Advanced line to open it.

Some of the options will be automatically set by Rigify if they have no value when a rig is generated, while others are fully controlled by the user.

When a brand new rig is generated, as opposed to overwriting an existing one, the value of this option is used to name it.

If this field is empty, the new object will be named based on the name of the meta-rig according to the following rules:

- If the name contains META, it is replaced with RIG.

- If the name contains meta-rig, it is replaced with rig.

- Otherwise, RIG- is prepended to the name.

When overwriting an existing rig object specified by the Target Rig option, its name is not changed, allowing it to be freely renamed without having to keep the value of this option in sync.

This option specifies the generated rig to overwrite when re-generating from this meta-rig.

If the option is not set, Rigify will generate a new rig object and store it in this option.

Note

When the option isn't set, Rigify will create a brand new rig object even if an object with a matching name already exists.

This option specifies the generated script data-block to overwrite when re-generating, and works in the same manner as Target Rig.

The script controls the UI in the 3D Viewport that allows conveniently switching visible bone collections, changing custom properties, converting between IK and FK and so on.

This reference specifies the collection containing generated widgets, and works in the same manner as Target Rig.

If enabled, Rigify will generate new widgets every time the rig is re-generated. By default, it tries to reuse the already generated widget objects that exist in the widget collection, allowing them to be manually edited to fit the character better.

When enabled, Rigify generates widgets for left and right side bones as linked duplicates, using negative X scale to flip the right side version. This enforces symmetry and reduces the number of meshes to adjust to fit the character.

When reusing an already generated widget, Rigify detects if it was originally generated mirrored by checking object scale to avoid flipping existing controls. Therefore switching to mirrored widgets for an existing character requires deleting the right side widgets, or Force Widget Update.

It is possible to configure Rigify to execute a Python script contained in a text data-block after generation in order to apply user-defined customizations. The script is executed with the generated rig active and selected in Object Mode.

The simplest use of this may be adjusting properties of generated constraints when Rigify rig types don't have any relevant meta-rig settings. That can be done by using the Copy Full Data Path context menu option on the property, pasting it into the script and making an assignment, e.g.:

Doing such changes via a script ensures they aren't lost if the rig is re-generated.

Users familiar with Rigify scripting can import Rigify utility modules, and access the generator instance through `rigify.get_generator()`. Yet note that, since generation is already finished, the only use of that is reading data created in the generation process.

When linking a rig into another file, you generally want to create a collection that includes the generated rig and the character mesh. You do not need to include the meta-rig or the widget object collection. You then link in the collection and run Make Library Override.

The `rig_ui_template.py` text data-block responsible for the rig UI will be automatically linked along with the rig, you don't need to link it separately. However, the script will not run until you run it manually from the Text editor or save and restart Blender.

--- Fin du chapitre: _basics ---

--- Début du chapitre: _beautify_faces ---Reference

Edit Mode

Face \xe2\x80\xa3 Beautify Faces

Beautify Faces works only on selected existing faces. It rearrange selected triangles to obtain more \xe2\x80\x9cbalanced\xe2\x80\x9d ones (i.e. less long thin triangles).

An angle delimiter option to limit edge rotation to flat surfaces.

Text converted to a mesh. \xc2\xb6

Result of Beautify Faces. \xc2\xb6

--- Fin du chapitre: _beautify_faces ---

--- Début du chapitre: _bend ---Reference

Object and Edit Modes

Object/Mesh/Curve/Surface \xe2\x80\xa3 Transform \xe2\x80\xa3 Bend

Shift- W

Before. \xc2\xb6

Clamp on. \xc2\xbb

Clamp off. \xc2\xbb

This tool rotates a line of selected elements forming an arc between the mouse cursor and the 3D cursor.

The Bendtool can be used in any case where you might want to bend a shape in two with a gradual transition between both sides.

This may take a little getting used to, the basics are listed below controls are noted here:

The initial position of the cursors define the axis to bend on.

The distance of the mouse cursor to the 3D cursor controls how sharp the bend will be.

The relative angle of the mouse cursor to the initial axis defines the bend angle.

If this seems overly complicated, it's probably best to try the tool where it becomes quickly apparent how the tool reacts to your input.

The amount of rotation.

The sharpness of the bend.

Normally the arc turns through a clamped rotation angle with the selected elements extended along a tangent line beyond that (see above left). When the clamp is deactivated, the arc continues around aligning the selected elements into a circle (right).

When off Alt all selected elements follow a circle, even when outside the segment between the 3D cursor and the mouse.

Note

Unlike most other transform modes, Bend is not effected by Pivot Point or Transform Orientation, always using the View Plane instead.

Hint

You can turn the bend angle through multiple rotations potentially forming a spiral shape.

Bend Transform example. \xc2\xbb

Since the bend tool relies on cursor input, it does not support adjusting the last bend operation.

--- Fin du chapitre: _bend ---

--- Début du chapitre: _bendy_bones ---Reference

All Modes

Bone \xe2\x80\xa3 Bendy Bones

Bendy Bones (B-Bones) are an easy way to replace long chains of many small rigid bones. A common use case for curved bones is to model spine columns or facial bones.

Blender treats the bone as a section of a B-spline curve passing through the bones' joints. Each of the segments will bend and roll to follow this invisible curve representing a tessellated point of the B-spline curve. The control points at each end of the curve are the endpoints of the bone. The shape of the B-Bones can be controlled using a series of properties or indirectly through the neighboring bones (i.e. first child and parent). The properties construct handles on either end of the bone to control the curvature.

When using the B-bone as a constraint target Data ID offers an option to follow the curvature.

Note

However, if the bone is used as a target rather than to deform geometry, only Armature and Copy Transforms constraints will use the full transformation including roll and scale. You can see these segments only if bones are visualized as B-bones. When not visualized as B-Bones, bones are always shown as rigid sticks, even though the bone segments are still present and effective. This means that even in e.g. Octahedron visualization, if some bones in a chain have several segments, they will nonetheless smoothly deform their geometry.

The initial shape of a B-Bone can be defined in Edit Mode as a rest pose of that bone. This is useful for curved facial features like curved eyebrows or mouths.

B-Bones have two sets of the Bendy Bone properties \xe2\x80\x93 one for Edit Mode (i.e. the Rest Pose/Base Rig) and another for Pose Mode \xe2\x80\x93 adding or multiplying together their values to get the final transforms.

Bones with just one segment in Edit Mode. \xc2\xbb

The B\xca9zier curve superposed to the chain, with its handles placed at bones\xe2\x80\x99 joints. \xc2\xbb

The same armature in Object Mode. \xc2\xbb

In Fig. Bones with just one segment in Edit Mode. we connected three bones, each one made of five segments.

Look at Fig. The same armature in Object Mode. , we can see how the bones\xe2\x80\x99 segments smoothly \xe2\x80\x93blend\xe2\x80\x9d into each other, even for roll.

An armature in Pose Mode, B-Bone visualization: Bone.003 has one segment, Bone.004 has four, and Bone.005 has sixteen. \xc2\xbb

Bendy Bones panel. \xc2\xbb

The number of segments, which the given bone is subdivided into. Segments are small, rigid linked child bones that interpolate between the root and the tip. The higher this setting, the smoother \xe2\x80\x93bends\xe2\x80\x9d the bone, but the heavier the pose calculations.

Controls the visible thickness of the bone segments when the armature is rendered in the B-Bones mode.

Controls how vertices are weighted to the individual segments of a B-Bone for deformations:

A fast mapping that works well for B-Bones with a straight or gently curved rest pose.

A slower mapping that improves deformations for B-Bones with a strongly curved rest pose. This should be used selectively when needed.

Straight vs Curved vertex mapping on a B-Bone with a strongly curved rest pose. \xc2\xbb

Applies offsets to the curve handle positions on the plane perpendicular to the bone\xe2\x80\x99s primary (Y) axis. As a result, the handle moves per axis (XZ) further from its original location, causing the curve to bend.

The roll value (or twisting around the main Y axis of the bone) is interpolated per segment, between the start and end roll values. It is applied as a rotational offset on top of the rotation defined by the handle bones.

If enabled, the Roll Out value of the Start Handlebone (connected parent by default) will be implicitly added to the Roll In setting of the current bone.

Scaling factors that adjust the thickness of each segment for the X and Z axes, or introduce non-uniform spacing along the Y axis. Similar to Roll it is interpolated per segment.

Since all segments are still uniformly scaled in the Y direction to fit the actual length of the curve, only the ratio between Scale In and Scale Out actually matters.

The Ease In/Out number fields, change the `length` of the `B\Bezier` handle to control the `root` handle and `tip` handle of the bone, respectively. These values are proportional to the default length, which of course automatically varies depending on bone length, angle with the reference handle, and so on. Although easing is a scale-like value, the Edit Mode and Pose Mode versions of the values are added, so they get corresponding start values of 1 and 0 by default.

Bone.004 with default In and Out (1.0).

Bone.004 with In at 2.0, and Out at 0.0.

If enabled, the final easing values are implicitly multiplied by the corresponding Scale Y values.

B-Bones can use custom bones as their reference bone handles, instead of only using the connected parent/child bones.

Specifies the type of the handle from the following choices:

The connected parent (or first connected child) of the bone is chosen as the handle. Calculations are done according to the Absolute handle type below.

The `B\Bezier` handle is controlled by the position of the head (tail) of the handle bone relative to the head (tail) of the current bone. Note that for this to work, there must be a nonzero distance between these bones. If the handle is also a B-Bone, additional processing is applied to further smooth the transition, assuming that the bones in effect form a chain.

The `B\Bezier` handle is controlled by the offset of the head (tail) of the handle bone from its rest pose. The use of this type is not recommended due to numerical stability issues near zero offset.

The `B\Bezier` handle is controlled by the orientation of the handle bone, independent of its location.

For types other than Automatic, a bone to use as handle has to be manually selected. Switching to a custom handle type without selecting a bone can be used to effectively disable the handle.

It is valid for two bones to refer to each other as handles in this correlation is applied in connected chains with Automatic handles.

If enabled, the final Scale and/or Ease values are multiplied by the corresponding local scale channels of the handle bone. This step is applied independently of Scale Easing and doesn't interact with it, i.e. enabling Y and Scale Easing doesn't replace the Ease toggle. These toggles are a more efficient replacement for up to eight trivial drivers passing segment scale data from the handle bones into the B-Bone option properties.

Tip

Keying Set

The `B\Bone Shape` Keying Set includes all Bendy Bones properties.

Visualization of the Bendy Bones properties.

From Left: 1) Curve X/Y offsets, 2) Scale In/Out, 3) Roll In/Out

--- Fin du chapitre: _bendy_bones ---

--- Début du chapitre: _bevel --- Cycles Only

The Bevel shader node can be used for rendering rounded corners. Like bump mapping, this does not modify the actual geometry, only the shading is affected. Slight rounding on edges helps to capture specular highlights that you would also see in the real world.

Note that this is a very expensive shader, and may slow down renders by 20% even if there is a lot of other complexity in the scene. For that reason, we suggest to mainly use this for baking or still frame renders where render time is not as much of an issue. The Bevel Modifier is a faster option when it works, but sometimes fails on complex or messy geometry.

Note

The Bevel node will not produce a valid result when:
or Caustic receiver while the scene contains an active Caustic caster ,
Caustic receiver , and Shadow Caustic Light .

Open Shading Language is active while using the OptiX rendering backend.
Width of the bevel effect on edges.

Normal to apply bevel on top of, to be combined with a Bump Node for example. When not connected, uses the surface normal.

Number of samples to take for each shader evaluation. More samples give more accurate results, but are also slower to render. The default value of 4 works well for most cases, with any noise resolved by using more AA samples.

Standard normal output.

A minimal node setup for working with the Bevel node. \xc2\xbb6

Bevel shader bringing out specular highlights on the edges. \xc2\xbb6

--- Fin du chapitre: _bevel ---

--- Début du chapitre: _bevel_vertices --- Reference

Edit Mode

Edge \xe2\x80\xa3 Bevel Edges

Ctrl- B (Bevel Edges)

Vertex \xe2\x80\xa3 Bevel Vertices

Shift- Ctrl- B (Bevel Vertices)

The Bevel tool rounds off edges or corners of a mesh at the point of the selected vertices. In \xe2\x80\x9cvertex only\xe2\x80\x9d mode, the Bevel Vertices tool works on selected vertices but the option to switch to Bevel Edges is available. By doing so, more vertices are added in order to smooth out profiles with a specified number of segments (see the options below for details about the bevel algorithm).

Cubes with and without bevel. \xc2\xbb6

Note

With the Vertex Only option active, some of the other options available will not work. However, they will work with Bevel Edges .

Only the areas near vertices are beveled, the edges remain unchanged.

Bevel the edges, creating intersections at vertices.

You can change the bevel amount by moving the mouse towards and away from the object, like with transform tools. The exact meaning of the value depends on the Amount Type option (see below).

Selects how the Amount value controls the size of the bevel. According to the selection, the amount is:

The relative distance from the new edge to the original.

The distance between the two new edges formed by the bevel (or the edges on either side of the bevel if there is more than one segment).
The perpendicular distance from the original edge to the bevel face.
The percentage of the length of adjacent edges that the new edges are slid along.

The exact distance along edges adjacent to the beveled edge. A difference from Offset is visible when the unbeveled edges attached to beveled edges meet at an angle besides a right angle.

For vertex-only bevels, the Offset and Depth types measure from the original vertex, and the Width type is measured from a new vertex to the center of the new face (as half the amount).

The number of segments in the bevel can be defined by scrolling the mouse Wheel to increase or decrease this value. The greater the number of segments, the smoother the bevel. Or press S to change the number with mouse movements, as well as numeric input.

Alternatively, you can manually enter a segment number value while using the tool, or in the Mesh Tool options panel after using the tool.

Bevel with four segments. \xc2\xba6

This is a number between 0 and 1 that controls the shape of the profile (side view of a beveled edge). The default value, 0.5, gives a circular arc (if the faces meet at right angles). Values less than that give a flatter profile, with 0.25 being exactly flat, and values less than that giving a concave bevel. Values more than 0.5 give a more convex profile. Similarly as Segments it can be set with mouse movements and numeric input after toggling P.

The Material number specifies which material should be assigned to the new faces created by the Bevel tool. With the default, -1, the material is inherited from the closest existing face (\xe2\x80\x99closest\xe2\x80\x99d can be a bit ambiguous). Otherwise, the number is the slot index of the material to use for all newly created faces.

When enabled, the per-vertex face normals of the bevel faces are adjusted to match the surrounding faces, and the normals of the surrounding faces are not affected. This will keep the surrounding faces flat (if they were before), with the bevel faces shading smoothly into them. For this effect to work, a mesh must have custom split normals. As a convenience, that option will be enabled for you if it is not already when you enable Harden Normals here.

Limits the width of each beveled edge so that vertices do not overlap with other geometry.

If there are un-beveled edges along with beveled edges into a vertex, the bevel tries to slide along those edges when possible. Turning the option off can lead to more even bevel widths.

Set Face Strength on the faces involved in the bevel, according to the specified mode. This can be used in conjunction with a Weight Normals Modifier (with the Face Influence option checked).

Do not set face strength.

Set the face strength of new faces along edges to Medium, and the face strength of new faces at vertices to Weak.

In addition to those set for the New case, also set the faces adjacent to new faces to have strength Strong.

In addition to those set for the Affected option, also set all the rest of the faces of the model to have strength Strong.

Creates a bevel with a uniform concave or convex curve.

The custom profile widget. \xc2\xba6

This widget allows the creation of a user-defined profile with more complexity than with the single profile parameter. The modal tool allows toggling the custom profile, but the shape of the profile is only editable in the options panel after the operation is confirmed.

The profile starts at the bottom right of the widget and ends at the top left, as if it were between two edges intersecting at a right angle.

Control points are created in the widget and then the path is sampled with the number of segments from the Bevel modifier.

The Support Loops and Steps presets are built dynamically depending on the number of segments in the bevel. If the number of segments is changed, the preset will have to be re-applied.

Samples will first be added to each control point, then if there are enough samples, they will be divided evenly between the edges. The Sample Straight Edges option toggles whether the samples are added to edges with sharp control points on either side. If there aren't enough samples to give each edge the same number of samples, they will just be added to the most curved edges. So it is recommended to use at least as many segments as there are control points.

Edge Bevel.

Vertex Bevel.

See also

The Bevel Modifier is a non-destructive alternative to the Bevel tool.

--- Fin du chapitre: `_bevel_vertices` ---

--- Début du chapitre: `_bezier_curve` --- The Bezier Curve modifier replaces the stroke by a Bezier approximation of it.

The maximum distance allowed between the new Bezier curve and the original stroke.

Bezier Curve modifier demo by T.K. (blend-file).

--- Fin du chapitre: `_bezier_curve` ---

--- Début du chapitre: `_bezier_segment` --- The Bezier Segment node generates a 2D Bezier spline from the given control points and handles.

The number of edges on the curve.

Positions of the start and end control point of the curve.

Positions of the handles used to define the shape of the curve.

The handle inputs are the absolute positions of the handles in 3D space.

The handle positions are relative to the control point on the curve. The

handle inputs give the offset from the control points.

Bezier spline generated from the inputs.

--- Fin du chapitre: `_bezier_segment` ---

--- Début du chapitre: `_bilateral_blur` --- The Bilateral Blur node performs a high-quality adaptive blur, blurring the image while retaining sharp edges.

It can be used for various purposes like: smoothing noisy render passes to avoid longer computation times in example ray-traced ambient occlusion, blurry refractions/reflections, soft shadows, or to make non-photorealistic compositing effects.

Standard color input. If only the image input is connected, the node blurs the image depending on the edges present in the source image.

Which is non-obligatory and if the Determinator is connected, it serves as the source for defining edges/borders for the blur in the image. This has

great advantage in case the source image is too noisy, but normals in combination with Z-buffer can still define exact borders/edges of objects.

Defines how many times the filter should perform the operation on the image. It practically defines the radius of blur.

Defines the threshold for which color differences in the image should be taken as edges.

A fine-tuning variable for blur radius.

Standard color output.

Bilateral smoothed Ambient Occlusion. `blend-file example\image\render`

Render result. `\image\render`

Composite. `\image\render`

--- Fin du chapitre: `_bilateral_blur` ---

--- Début du chapitre: `_bisect` ---Reference

Edit Mode

Toolbar `\image\render` Knife `\image\render` Bisect

Mesh `\image\render` Bisect

The Bisect tool is a quick way to cut a mesh in two along a custom plane.

Use LMBclick and drag to draw cut line. Once the cut is done the Adjust Last Operation panel gives a few options:

The plane can be numerically adjusted for precise values.

Cuts can optionally fill in the holes created, with materials, UV maps, and Color Attributes based on the surrounding geometry.

Cuts may remove geometry on one side.

Cut along the straight plane or along the existing geometry below the distance from the plane.

Changes the location of the line.

Constrains the rotation of the line to 15 degree intervals.

Changes the side of the line that is the inner/outer side; this option is useful when using Clear Inner, Clear Outer and/or Fill.

Example of a common use of bisect. `\image\render`

Example of bisect with the fill option enabled. `\image\render`

--- Fin du chapitre: `_bisect` ---

--- Début du chapitre: `_blackbody` --- The Blackbody node converts a blackbody temperature to RGB value. This can be useful for materials that emit light at natural occurring frequencies.

The temperature in Kelvin.

This node has no properties.

RGB color output.

Example of the color ranges of the Blackbody node. `\image\render`

--- Fin du chapitre: `_blackbody` ---

--- Début du chapitre: `_blade` ---Reference

Sequencer Mode

Toolbar `\image\render` Blade

K, Shift- K

Cuts a strip in two. Specifically, it first shortens the strip so it only shows the content up to the cut point, then adds a second strip that shows the content after the cut point.

Splitting be done in two different ways:

Select the tool in the Toolbar and click a strip at the time point where you want to split it.

Alternatively, select one or more strips, place the Playhead at the time point where you want to split them, and press one of the keyboard shortcuts below.

You can choose between the following split types:

After splitting, it's still possible to restore the cut content in the new strips by dragging their handles.

After splitting, it's not possible to restore the cut content by dragging handles. However, you can still restore it by changing the Hold Offset in the Sidebar.

--- Fin du chapitre: `_blade` ---

--- Début du chapitre: `_blend` --- The Blend texture generates a smoothly interpolated progression. This is one of the most frequently used procedural textures. You can use blend textures to blend other textures together (with Stencil), or to create nice effects (especially with the Mapping: Normal trick).

Note

Remember that if you use a ramp to create a custom blending, you may have to use No RGB, if the Mapping value needs an intensity input.

Blend Texture panels.

Profile of blend.

A linear progression.

A quadratic progression.

A flowing, nonlinear progression.

A diagonal progression.

A progression with the shape of a three-dimensional ball.

A quadratic progression with the shape of a three-dimensional ball.

A radial progression: Horizontal/ Vertical. The direction of the progression is flipped a quarter turn.

--- Fin du chapitre: `_blend` ---

--- Début du chapitre: `_blender_default` --- While this isn't a comprehensive list, this page shows common keys used in Blender's default keymap.

Ctrl- O

Open file.

Ctrl- S

Save file.

Ctrl- N

New file.

Ctrl- Z

Undo.

Shift- Ctrl- Z

Redo.

Ctrl- Q

Quit.

F1

Help (context sensitive).

F2

Rename active item.

F3

Menu Search.

F4
File context menu.
F5- F8
Reserved for user actions.
F9
Adjust last operation.
F10
Reserved for user actions.
F11
Show render window.
F12
Render the current frame.
Q
Quick access (favorites).
Ctrl- Spacebar
Toggle Maximize Area.
Ctrl- PageUp/Ctrl- PageDown
Next/previous Workspace.
Spacebar
User configurable.
Toggle animation playback.
Tool switching with hotkeys (Shift- Spacebarfor play).
Search for actions (Shift- Spacebarfor play).
Shift- Ctrl- Spacebar
Playback animation (reverse).
X
Deletes the selected item, requires a confirmation dialog.
Delete
Deletes the selected item, does not require a confirmation dialog.
These keys are shared across editors such as the 3D Viewport, UV and Graph editor.
A
Select all.
Alt- A
Select none.
Ctrl- I
Invert selection.
H
Hide selection.
Alt- H
Reveal hidden items.
T
Toggle Toolbar.
N
Toggle Sidebar.
Tab
Toggle Edit mode.
Ctrl- Tab
Toggle Pose mode for armatures, or show a mode switching pie menu for others.
In Edit Mode, switch between editing vertices (1), edges (2), or faces (3).
Hold Shiftto toggle one of these without disabling the others.

Hold Ctrl to alter how the selection is transformed from the old mode to the new.

See Mesh Selection Modes for details.

Accent Grave

Show 3D Viewport navigation pie menu.

Ctrl- Accent Grave

Toggle gizmos.

Shift- Accent Grave

Start Walk/Fly Navigation.

The Cmd key can be used instead of Ctrl on macOS for all but a few exceptions which conflict with the operating system.

List of additional macOS specific keys:

Cmd- Comma

Preferences.

Controls which mouse button, either left or right, is used to select items in Blender. If Left is selected, the RMB will be a context sensitive menu. If Right is selected, the LMB will place the 3D Cursor.

Controls the action of Spacebar. These and other shortcuts can be modified in the keymap preferences .

Starts playing through the Timeline . This option is good for animation or video editing work.

Opens the Toolbar underneath the cursor to quickly change the active tool. This option is good if you are doing a lot of modeling or rigging work.

Opens up the Menu Search . This option is good for someone who is new to Blender and is unfamiliar with the menus and shortcuts.

The activation event for gizmos that support drag motion. This option is only available when Left click Select with Mouse Button is chosen.

The gizmo's operation gets initiated (and additional options become available in the Status Bar) the moment you press down the mouse button on the gizmo.

The operation only gets initiated once you start dragging the gizmo.

The method of keys to activate tools such as move, rotate, and scale.

Activate actions immediately.

Activate the tool for editors that support tools.

Tapping Alt shows a prompt in the status bar prompting a second keystroke to activate the tool. Note that this option is not available when using Emulate 3 Button Mouse .

Emulate 3 Button Mouse .

Hold Alt to use the Active Tool when the gizmo would normally be required. (For example, with the Move tool selected, you can hold Alt and drag the mouse anywhere in the viewport to move the selected object, rather than having to drag its gizmo.) This option is only available when Select with Mouse Button is set to Left click and Emulate 3 Button Mouse is disabled.

Causes selection shortcut A to deselect all when any selection exists.

Opens a pie menu to toggle Regions , rather than toggling a single region.

Navigation pie menu, useful on systems without a numeric keypad.

Transform gizmos pie menu, useful for quickly switching between transform gizmos. Note that this doesn't apply to tools that force a certain gizmo (Move, Rotate, Scale and Transform); if you have such a tool selected, the gizmo will stay the same no matter what you choose in the pie menu.

The action when MMBdragging in the viewport, this also applies to trackpads.

Rotates the view around a pivot point, Shift- MMB is used for panning the view.

Shifts the view towards the mouse, Shift- MMB is used for orbiting the view.

Set the view axis where each mouse direction maps to an axis relative to the current orientation.

Set the view axis where each mouse direction always maps to the same axis.

Causes Tab to open a pie menu (swaps Tab and Ctrl- Tab).

This allows keys to have a secondary drag action.

Toggles Edit Mode.

Object Mode pie menu.

Toggles wireframe view.

Display mode pie menu.

First person Fly/walk Navigation.

View axis pie menu.

Show additional items in the shading menu (Zkey).

During transformations, use Alt to navigate the 3D Viewport.

When disabled, hotkeys for proportional editing, automatic constraints, and auto IK chain length will require holding Alt.

Navigate into folders by clicking on them once instead of twice.

--- Fin du chapitre: _blender_default ---

--- Début du chapitre: _blender_directory_layout --- This page documents the different directories used by Blender.

This can be helpful for troubleshooting, automation and customization. User directories store preferences, startup file, installed extensions, presets and more. By default these use the standard configuration folders for each operating system.

If the \$XDG_CONFIG_HOME environment variable is set:

When running Blender from a portable drive, it's possible to keep the configuration files on the same drive to take with you.

To enable this, create a folder named portable at the following locations:

Windows: Next to the Blender executable, in the unzipped folder

Linux: Next to the Blender executable, in the unzipped folder

macOS: Inside the application bundle at Blender.app/Contents/Resources

This folder will then store preferences, startup file, installed extensions and presets.

The BLENDER_USER_RESOURCES environment variable can be set to a custom directory to replace the default user directory.

System directories store files that come bundled with Blender and are required for it to function. This includes scripts, presets, essential assets and more.

Archive downloaded from blender.org:

Linux distribution packages:

Zip file downloaded from blender.org:

Installer downloaded from blender.org:

Microsoft Store installation:

BLENDER_SYSTEM_SCRIPTS and BLENDER_SYSTEM_EXTENSIONS environment variables can be used to bundle additional scripts and extensions, that are not part of the regular Blender installation.

Other `BLENDER_SYSTEM` environment variables can override other system paths, though are not commonly used in practice.

Autosave blend-file location. (Windows only, temp directory used for other systems.)

Located in user directories.

User configuration and session info.

Located in user directories.

Blend file to load on startup.

User preferences.

File Browser bookmarks.

Recent file menu list.

Startup file for an application template.

User preferences file for an application template.

Data files loaded at runtime.

Located in both user and system directories. User data files either override or add to system data files.

Default `OpenColorIO` configuration.

User interface fonts.

Studio light images for 3D viewport.

Extension repositories.

Located in both user and system directories. Repositories are loaded from both directories.

Add-ons, presets, templates, user interface, startup scripts.

Located in both user and system directories. Scripts are loaded from both directories.

Python add-ons which may be enabled in the Preferences include import/export format support, render engine integration and many handy utilities.

Modules for add-ons to use (added to `Python\sys.path`).

The add-ons directory which is used for bundled add-ons.

Modules for `addons_core` to use (added to `Python\sys.path` when it found).

Python modules containing our core API and utility functions for other scripts to import (added to `Python\sys.path`).

Scripts which are automatically imported on startup.

Application templates installed in user directories.

Application templates automatically loaded from system directories.

Presets used for storing user-defined settings for cloth, render formats, etc.

Example scripts which can be accessed from Text Editor \x03

Templates \x03 Python.

Example OSL shaders which can be accessed from Text Editor \x03

Templates \x03 Open Shading Language.

Bundled Python distribution.

Located in system directories.

The cache directory is used to store persistent caches locally. Currently it is only used for the indexing of Asset Libraries. The operating system is not expected to clear this automatically.

The following path will be used:

`$XDG_CACHE_HOME/blender/` if `$XDG_CACHE_HOME` is set, otherwise `$HOME/.cache/blender/`

`/Library/Caches/Blender/`

`%USERPROFILE%\AppData\Local\BlenderFoundation\Blender\Cache\`

The temporary directory is used to store various files at run-time(including render layers, physics cache, copy-paste buffer and crash logs).

The temporary directory is selected based on the following priority:
User Preference (see File Paths).

Environment variables (TEMP on Windows, TMP & TMP_DIR on other platforms).

The /tmp/ directory.

--- Fin du chapitre: _blender_directory_layout ---

--- Début du chapitre: _blend_hair_curves --- Blends shape between multiple hair curves in a certain radius.

Input Geometry (only curves will be affected).

Factor to blend overall effect.

Radius to select neighbors for blending.

Amount of neighbors used for blending.

Preserve each curve's length during deformation.

This node has no properties.

Geometry

--- Fin du chapitre: _blend_hair_curves ---

--- Début du chapitre: _blend_shape ---Reference

Edit Mode

Vertex Blend from Shape

Blend in the shape from a shape key.

--- Fin du chapitre: _blend_shape ---

--- Début du chapitre: _blob ---Reference

Sculpt Mode

Sidebar Tool Brush Settings

Advanced Brush Type

Similar to Draw, but vertices are pushed outwards like an inverted pinching effect. This will lead to a more consistent spherical curvature and thickening of strokes.

By default at 0.5 to push out the mesh during the stroke. More info at Pinch/Magnify

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _blob ---

--- Début du chapitre: _blueprint --- The Blueprint modifier produces blueprint-like strokes using either circular, elliptical, or square contours. A blueprint here refers to those lines drawn at the beginning of free-hand drawing to capture the silhouette of objects with a simple shape such as circles, ellipses and squares.

Which base shapes to use for this blueprint: Circles, Ellipses or Squares.

How many rounds are generated, as if the pen draws the same stroke several times (i.e. how many times the process is repeated).

For the Circles and Ellipses shapes. Adds some randomness to each round in the relevant aspect. Using more than one round with no randomness would be meaningless, as they would draw over each other exactly.

For the Squares shapes. The first adds some extra length to each edge of the generated squares (also affected by the second parameter). The second adds some randomness to the squares.

Note that the Min 2D Length feature from the Strokes settings is quite handy here, to avoid the noise generated by small strokes.

--- Fin du chapitre: _blueprint ---

--- Début du chapitre: _blur --- The Gaussian Blur strip is used to blur the input strip in a defined direction. This can be used to blur a background or to blur a transition strip.

Distance of the blur effect on the X axis.

Distance of the blur effect on the Y axis.

Gaussian Blur Effect.

--- Fin du chapitre: _blur ---

--- Début du chapitre: _blur_attribute --- The Blur Attribute node smooths attribute values between neighboring geometry elements.

The goal of each step is mixing values of each element with its neighbors. The weight for element is factor for multiplying all neighbor's values before accumulating them as new primitive value.

Blurring will only work with certain geometry types and attribute domains. Therefore, the attribute can only be affected on the Meshes and Curves components.

The domains this node works on is based on the field context of the node's evaluation. Only domains with explicit relations with their neighbors will work with this node. Explicit relations for correct blurring are vertices, edges, and faces of meshes, and curve control points.

Note

Blurring of face corner attributes is not handled by this node, because the ideal behavior for mixing face corner values is not clear.

All attribute data types are supported except for boolean attributes. The immediate value of each primitive to blur.

Number of repetitions of mixing value with neighbors. Each iteration is independent. Until one iteration is completed, its results are not used as a source for next blurring.

Weight of each primitive.

The data type used for the evaluated data.

Values, mixed with neighbors defined number of times.

Input is Mesh Plane. First Subdivide Mesh Node add some faces for capture color with Random Value Node used as hue in Combine Color Node on this. Now second Subdivide Mesh Node split each face on a lot of new. Each one new duplicates original attribute. Blur Attribute node mixes all attributes for each face. Due to this, the result is smoothed.

--- Fin du chapitre: _blur_attribute ---

--- Début du chapitre: _boid --- Reference

Physics Force Fields

Boid

The Boid force fields do not affect physics, and are used together with the Boids Particles to define boid predators and goals for the Boid Brain rules.

--- Fin du chapitre: _boid ---

--- Début du chapitre: _boids ---Reference

Particle System \xe2\x80\xa3 Physics

Boids

Boids particle systems are controlled by a limited artificial intelligence, which can be programmed to follow basic rules and behaviors. They are ideal for simulating flocks, swarms, herds and schools of various kind of animals, insects and fishes or predators vs. preys simulations. They can react on the presence of other objects and on the members of their own system. Boids can handle only a certain amount of information, therefore the sequence of the Boid Brainrules is very important. In certain situations only the first three parameter are evaluated.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Movement

Boids try to avoid objects with activated Collision. They try to reach goal objects, and fly from \xe2\x80\x9cpredators\xe2\x80\x9d according to the Boid Brainsettings.

Boids can have different physics depending on whether they are in the air, or on land (on collision object).

Allow boids to move in the air.

Allow boids to move on land.

Allow boids to climb goal objects.

Set the Maximum velocity in the air.

Set the Minimum velocity in the air.

Lateral acceleration in air, percentage of the max velocity (turn). Defines how fast a boid is able to change direction.

Tangential acceleration in air, percent 180 degrees. Defines how much the boid can suddenly accelerate in order to fulfill a rule.

Radius of boids personal space in air. Percentage of particle size.

How smoothly the boids land.

Set the Maximum velocity on land.

Maximum speed for jumping.

Lateral acceleration on land, percent of max velocity (turn). Defines how fast a boid is able to change direction.

Tangential acceleration on land, percent 180 degrees. Defines how much the boid can suddenly accelerate in order to fulfill a rule.

Radius of boids personal space on land. Percentage of particle size.

How strong a force must be to start effecting a boid on land.

Only collide with objects in this collection.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Battle

Initial boid health when born.

Maximum caused damage per second on attack.

Boid will fight this time stronger than enemy.

Accuracy of attack.

Maximum distance of which a boid can attack.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Misc

Amount of rotation around velocity vector on turns. Banking of 1.0 gives a natural banking effect.

Amount of rotation around side vector.

Boid height relative to particle size.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Relations

This list view allows you to set up other particle systems to react with the boids.

A data ID to select an object with a particle system set on.

Index of the Object\xe2\x80\x99s particle system as set in the list view in the particle panel.

Setting the type to Enemy will cause the systems to fight with each other.

Will make the systems work together.

Will not cause them to align or fight with each other.

Boids will try to avoid deflector objects according to the Collision rule\xe2\x80\x99s weight. It works best for convex surfaces (some work needed for concave surfaces).

As other physics types, Boids is also influenced by external force fields.

In addition, special Boid force fields can be used with the Boids physics. These effectors could be predators (positive Strength) that boids try to avoid, or targets (negative Strength) that boids try to reach according to the (respectively) Avoid and Goal rules of the Boid Brain.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Boid Brain

The Boid Brain panel controls how the boids particles will react with each other. The boids\xe2\x80\x99 behavior is controlled by a list of rules. Only a certain amount of information in the list can be evaluated. If the memory capacity is exceeded, the remaining rules are ignored.

The rules are by default parsed from top-list to bottom-list (thus giving explicit priorities), and the order can be modified using the little arrows buttons on the right side.

There are three ways to control how rules are evaluated:

All rules are averaged.

A random rule is selected for each boid.

Uses fuzzy logic to evaluate rules. Rules are gone through top to bottom. Only the first rule that affect above the Rule Fuzziness threshold is evaluated. The value should be considered how hard the boid will try to respect a given rule (a value of 1 means the Boid will always stick to it, a value of 0 means it will never). If the boid meets more than one conflicting condition at the same time, it will try to fulfill all the rules according to the respective weight of each.

Note

A given boid will try as much as it can to comply to each of the rules it is given, but it is more than likely that some rule will take precedence on other in some cases. For example, in order to avoid a predator, a boid could probably \xe2\x80\x9cforget\xe2\x80\x9d about Collision, Separate and Flock rules, meaning that \xe2\x80\x9cwhile panicked\xe2\x80\x9d it could well run into obstacles, e.g. even if instructed not to, most of the time.

The current rule affects boids while they are flying.

The current rule affects boids while they are not flying.

Seek goal.

Specifies the goal object. If not specified, Boid force fields with negative Strength are used as goals.

Predict target\xe2\x80\x99s movements.

Avoid \xe2\x80\x9cpredators\xe2\x80\x9d.
 Specifies the object to avoid.If not specified, Boid force fields with positive Strength are used as predators.
 Predict target\xe2\x80\x99s movements.
 Avoid object if danger from it is above this threshold.
 Avoid objects with activated Deflection.
 Avoid collision with other boids.
 Avoid collision with deflector objects.
 Time to look ahead in seconds.
 Boids move away from each other.
 Copy movements of neighboring boids, but avoid each other.
 Follows a leader object instead of a boid.
 Distance behind leader to follow.
 Follow the leader in a line.
 How many boids that are allowed to follow in a line.
 Maintain average velocity.
 Percentage of maximum speed.
 How fast velocity\xe2\x80\x99s direction is randomized.
 How much velocity\xe2\x80\x99s Z component is kept constant.
 Move toward nearby boids.
 Attack boids at a maximum of this distance.
 Flee to this distance.
 --- Fin du chapitre: _boids ---

--- Début du chapitre: _bokeh_blur --- The Bokeh Blur node generates a bokeh type blur similar to Defocus.Unlike defocus an in-focus region is defined in the Compositor.There is also more flexibility in the type of blur applied throughtheBokeh Imagenode.
 Several performance optimizations are also available such calculation area restriction and masking.
 Standard color input.
 This is an input for the Bokeh Image node.
 Size controls the amount of blur. Size can either be a single value across the entire image or a variable valuecontrolled by an input image. In order to use the latter, the Variable Size option must be selected.See the examples section below for more on how to use this.
 This can be used with a Box Mask matte node or with a Mask input node to restrict the area of the image the blur is applied to. This could be helpful, for example,when developing a node system by allowing only a small area of the image to be filteredthus saving composite time each time adjustments are made.
 Allows a variable blur radius, if the Size input is an image.
 Max Blur is intended to act as an optimization tool bylimiting the number of pixels across which the blur is calculated.
 Standard color output.
 Three examples of how the size input may be used follow.
 An ID masked alpha image can be used so that a background is blurred while foreground objects remain in focus.To prevent strange edges the Dilate Node should be used.
 The Z pass can be visualized using a Map Value nodeand a Color Ramp nodeas described in Render Layers .A multiply Math node can be used following the color rampso that a blur value greater than one is used for objects outside the focal range.
 Z pass used. \xc2\xb6

A manually created grayscale image can be used to define the sharp and blurry areas of a preexisting image. Again, a Multiply Node can be used so that a blur value greater than one is used.

Image used. \xc2\xba6

Z pass used. \xc2\xba6

Image used. \xc2\xba6

--- Fin du chapitre: _bokeh_blur ---

--- Début du chapitre: _bokeh_image --- The Bokeh Imagenode generates a special input image for use with the Bokeh Blurfilter node.

The Bokeh Imagenode is designed to create a reference image which simulates optical parameters such as aperture shape and lens distortions which have important impacts on bokeh in real cameras.

This node has no input sockets.

The first three settings simulate the aperture of the camera.

Sets an integer number of blades for the camera's iris diaphragm.

Gives these blades an angular offset relative to the image plane.

Sets the curvature of the blades with (0 to 1) from straight to bringing them to a perfect circle.

Provides a type of distortion found in mirror lenses and some telescopes. This can be useful to produce a visual complex bokeh.

Introduces chromatic aberration into the blur such as would be caused by a tilt-shift lens.

The generated bokeh image.

In the example below the Bokeh Image is used to define the shape of the bokeh for the Bokeh Blur node.

Example of Bokeh Imagenode. \xc2\xba6

--- Fin du chapitre: _bokeh_image ---

--- Début du chapitre: _bone_collections --- Note

Bone Collections were introduced in Blender 4.0 as replacement of Armature Layers and Bone Groups. Bone colors are now managed directly on the bone.

Reference

Pose & Armature Edit Modes

Properties \xe2\x80\xa3 Armature \xe2\x80\xa3 Bone Collections

Pose \xe2\x80\xa3 Bone Collections \xe2\x80\xa3 \xe2\x80\xa6

The Bone Collections panel in the Armature properties. \xc2\xba6

This panel contains a Tree View to manage Bone Collection. From this panel, Bone Collections can be created, deleted, re-arranged, and more.

Collections can be renamed by double clicking on the name, or right clicking and selecting Rename. To nest a collection inside an existing collection, click and drag the name onto another collection \xe2\x80\x99s name. Child collection can also be made by RMB and selecting

\xe2\x80\x9cAdd Child Collection\xe2\x80\x9d.

To the right of the name gives a few controls of the collection:

Bones in this collection will be visible in the 3D Viewport.

Show only this bone collection, and others also marked as \xe2\x80\x9csolo\xe2\x80\x9d.

Furthermore, collection that are not empty will have a dot to indicate the collection has bones assigned to it.

Tip

The Bone Properties panel gives a slightly different view on the bone \xe2\x80\x99s collections. See Bone Relations .

Unhides any hidden bone collections.

Clear the `\xe2\x80\x98solo\xe2\x80\x99` setting on all bone collections

Remove all bone collections that have neither bones nor children. This is done recursively, so bone collections that only have unused children are also removed.

Assigns the selected bones to the active bone collection.

Removes the selected bones from the active bone collection.

Selects the bones in the active bone collection.

Deselects the bones in the active bone collection.

Note

Individual bones can also be unassigned from their collections via the Bone Relations panel.

Tip

For setting up custom selection sets of bones, take a look at the SelectionSets add-on. It is bundled with Blender.

Blender should be in Edit Mode or Pose Mode to move bones between collections. Note that as with objects, bones can be assigned to in several collections at once.

Shows a list of the Armature's editable bone collections.

Choosing a bone collection unassigns the selected bones from all other bone collections, then assigns them to the chosen one.

Available as Pose `\xe2\x80\x93 Move to Collection ( Pose Mode)` Armature `\xe2\x80\x93 Move to Collection ( Edit Mode)`, and M (either mode).

Shows a list of the Armature's editable bone collections. The collections that the active bone is assigned to are prefixed with a - , and choosing those will unassign all selected bones from that collection. Similarly, choosing a bone collection prefixed with a + will assign all selected bones to that collection.

Available as Pose `\xe2\x80\x93 Bone Collections ( Pose Mode)` Armature `\xe2\x80\x93 Bone Collections ( Edit Mode)`, and Shift- M (either mode).

Note

The above operators will only show the editable bone collections. When the Armature is linked, its bone collections will be read-only. New bone collections can still be added via library overrides; only those will be editable.

See Library Overrides of Bone Collections.

Create and manage your own properties to store data in the Bone Collection's data-block. See the Custom Properties page for more information.

--- Fin du chapitre: `_bone_collections` ---

--- Début du chapitre: `_bone_positioning` --- Start by identifying basic face landmarks to follow as guide for bones placement.

Basic Face Landmarks. 

Orange lines represent bones that should be placed in closed loops.

Yellow lines represent bones whose position depends on surrounding bone loops.

Red lines represent outer edge bones.

Purple lines represent bridging bones used to cover deforming flesh.

Eyes-Nose Landmarks. 

The eyes-nose loop area is split in different parts identified by bone names. Follow the image to place the bones.

Eyes-Nose Bone Positions. 

Tip

Brow Placement

Keeping aligned the mid bones in \xe2\x80\x99cbrow\xe2\x80\x99d, \xe2\x80\x99cbrow.b\xe2\x80\x99d, \xe2\x80\x99clid.t\xe2\x80\x99d, \xe2\x80\x99clid.t\xe2\x80\x99d andcheek will give better results after rig generation.

Jaw-Ear Bone Positions. \xc2\xb6

Also the jaw-ear area is split in different parts identified by bone names. Follow the image to place the bones.

Tip

Jaw Placement

Try to place \xe2\x80\x99cear.L\xe2\x80\x99d bone covering the part of the ear attached to the mandible (lower jaw).Do the same with temple bone trying to cover the part you don\xe2\x80\x99t want to move with the jaw,this way you will also determine the jaw pivot position.

Lips Merge Point. \xc2\xb6

Warning

While moving the face bones it is necessary to preserve merge points, i.e. whenever headsor tails of two or more bones overlap at the same point, they should still do so afterrepositioning. Tearing a merge point apart may result in multiple controls being createdinstead of one, or even the generation of errors.

Face Stretcher Bones. \xc2\xb6

After the main face bones are placed use the cheek bone to connect the eye-nose area to the jaw mouth area.Then do the same with the brow area. This process will automatically define face muscles compression areas. Position the eye bones in the eye pivot point facing right towardthe face on the Y axis.The length of the eye bones should correspond to the radius of the eye.

Eyes Pivot Position. \xc2\xb6

Tip

Eye Pivot

If your eye has a spherical shape you can define its pivot by entering Edit Mode.Select two opposite vertices on the center meridian \xe2\x80\x93 or the opposite poles \xe2\x80\x93 andsnapping the cursor to selection by pressing Snap \xe2\x80\x93 Cursor To Selected.If your eye is a complete sphere and its location it\xe2\x80\x99s not applied, then you can just use its center of mass.

Finally position the teeth bones on your teeth geometry and the tongue bone chain as described in the figure.

Mouth and Teeth Positions. \xc2\xb6

Tip

Tongue

The tongue will work better if the bones are aligned at the symmetry line.

Before generating the rig ensure the face master bone is facing upward. Start by identifying on your character basic torso zones to follow as guide for bones placement.

Head, chest and pelvis are rigid zones, so they require less bones.Having a good edge loop placement around zone boundaries on your modelwill help in having correct deformation after armature binding.

Torso Landmarks. \xc2\xb6

Starting from the side view, place the main spine bones trying to useone bone for the rigid areas and two for the flexible ones.In addition to the

main spine, the torso is provided with additional pelvis bones (to oppose the leg bending), two breast controls and two shoulder bones. Even if the pelvis bones will not appear in the final rig as controls, they will contribute to deformation.

Torso Bones Positioning. \xc2\xba

Tip

Bone Placement

Try to keep the spine as centered as possible inside the mesh bounding volume, just apply a slight offset toward the back. In a similar way, consider the shoulder bones as general deformer; placing it too forward \xe2\x80\x93 where the collar bone should be \xe2\x80\x93 could cause undesired deformations.

While placing the arm bones try to start having a straight line that goes from the shoulder to the hand in both front and top view. After this is done just add a slight bend to the elbow. This can be easily done by going in the top view, entering armature Edit Mode and sliding the bone junction between forearm and upper_arm slightly toward the world \xe2\x80\x93 Y axis.

Arm Bones Positioning. \xc2\xba

For the leg you can follow a similar process. Start by aligning the leg bones creating a straight line from the hips to the ankle, then place the foot and the toe accordingly. Remember to add a slight bend to the knee. This can be easily done by going in the side view, entering armature Edit Mode and sliding the bone junction between thigh and shin slightly toward the world \xe2\x80\x93 Y axis.

Leg Bones Positioning. \xc2\xba

Finally align the heel bone by going in the front view and placing the head and tail to fill the foot size from side to side. Then, in the side view, align the bone at the point where the heel just touches the ground floor.

Note

From version 0.5 and above there is no more need of manual bone rolls alignment. The generate function will take care of that for you by evaluating it from bend axis; just insert a slight bend in your limb and it \xe2\x80\x93 done! If you need more control on the orientation, follow the guidelines described in Advanced Usage.

Start by placing, finger by finger, all the knuckles in place.

Tip

Fingers Placement

An easy and effective method to do this operation is to select on the mesh the corresponding edge loop in Edit Mode and use the Cursor to Selectionsnap. Then you can snap the bone to the corresponding loop using the Selection to Cursorsnap.

Knuckles Edge Loops and Cursor Snapping. \xc2\xba

Finalize the positioning by taking care of bone rolls (the X axis is set as bend axis).

Tip

Bone Roll

Finger axis alignment can be easily be made consistent by selecting all the finger bones and recalculating the bone rolls Recalculate Roll \xe2\x80\x93 Global -Z Axis.

Thumb may require more tweaking depending on your character \xe2\x80\x93 mesh topology, usually Recalculate Roll \xe2\x80\x93 Global +Y Axis is a good starting point.

Once your bone rolls are consistent, try generating the rig and scaling the finger master controls. This should cause the fingers to curl. If they are rotating on the wrong axis, change the Bend Rotation Axis parameter on the first finger's parameters under Rigify Type.

Fingers Bend Axis.

When the fingers are in place proceed placing the palm bones.

Palm Alignment.

Tip

Palm Placement

Try to keep palm bones' heads at a little distance between each other. This distance is required for Rigify to define the palm controls hierarchy. Palm axis alignment can be easily done by selecting all the palm bones and recalculating the bone rolls Recalculate Roll \xa3 Global -Z Axis.

See also

For more detailed information on bones and rolls refer to the Bone Structure and Bone Roll.

--- Fin du chapitre: `_bone_positioning` ---

--- Début du chapitre: `_bone_roll` --- In Edit Mode, you can control the bone roll (i.e. the rotation around the Y axis of the bone).

However, after editing the armature, or when using Euler Rotation, you may want to set the bone roll.

Reference

Edit Mode

Armature \xa3 Bone Roll \xa3 Recalculate Roll

Shift- N

Align roll relative to the axis defined by the bone and its parent.

X, Z

Align roll to global X, Y, Z axis.

X, Y, Z

Follow the rotation of the active bone.

Set the roll to align with the viewport.

Set the roll towards the 3D cursor.

Reverse the axis direction.

Avoids rolling the bone over 90 degrees from its current value.

Reference

Edit Mode

Armature \xa3 Bone Roll \xa3 Set Roll

Ctrl- R

This is a transform mode where you can edit the roll of all selected bones.

--- Fin du chapitre: `_bone_roll` ---

--- Début du chapitre: `_boolean` --- The Boolean node provides a Boolean value.

This node has no input sockets.

Single Boolean value (true/false).

Standard Boolean output.

--- Fin du chapitre: `_boolean` ---

--- Début du chapitre: `_booleans` --- The Boolean modifier performs operations on meshes that are otherwise too complex to achieve with as few

steps by editing meshes manually. It uses one of the three available Boolean operations to create a single mesh out of two mesh objects:

The Union, Intersection and Difference between a Cube and a UV Sphere, with the modifier applied to the sphere and using the cube as target.

This modifier needs a second mesh object, or collection of mesh objects, to be the target (the second operand) of the operation.

Warning

Only Manifold meshes are guaranteed to give proper results, other cases (especially open meshes, Non-manifold but without any self-intersections) will usually work well, but might give odd glitches and artifacts in some cases.

Tip

If you have marked your objects to show the edges (in Properties > Object Properties > Viewport Display, enable Wireframe), you will see the edge creation process while you are moving your objects around. Depending on your mesh topology, you can also enable X-Ray and Transparency and see the topology being created in real-time.

The Boolean modifier.

Everything inside both the target mesh and the modified mesh is kept. If the target is a collection, then only the inside of all meshes is kept. The target mesh or collection is added to the modified mesh, removing any interior faces.

The target mesh, or collection of meshes, is subtracted from the modified mesh (everything outside of the target mesh or collection is kept).

Choose the type of the operand (target).

The target is a mesh object.

The target is a collection. When the target is a collection and the Solver is Fast, the Intersect operation is not allowed.

The name of the target mesh object.

The name of the target collection (may be empty if Solver is Exact, which can be useful in combination with the Self option).

Algorithm used to calculate the Boolean intersections.

Uses a mathematically simple solver which offers the best

performance; however, this solver lacks support for overlapping geometry.

Uses a mathematically complex solver which offers the best results and has full support for overlapping geometry; however, this solver is much slower than the Fast Solver.

Method for setting materials on the new faces.

Set the material on new faces based on the order of the material slot lists. If a material doesn't exist on the modifier object, the face will use the same material slot or the first if the object doesn't have enough slots.

Transfer materials from non-empty slots to the result mesh, adding new materials as necessary. For empty slots, fall back to using the same material index as the operand mesh.

Correctly calculates cases when one or both operands have self-intersections, this involves more calculations making it slower.

Optimizes the Boolean output for Non-manifold geometry at the cost of increased computational time. Because of the performance impact, this option should only be enabled when the Exact solver demonstrates errors with non-manifold geometry.

Maximum distance between two faces to consider them as overlapping. This helps solve the limitation of this solver, if the Boolean result seems unexpected try using the exact solver.

--- Fin du chapitre: `_booleans` ---

--- Début du chapitre: `_boolean_math` --- The Boolean Math node performs a basic logical operation on its inputs.

Two standard Boolean inputs.

True when both inputs are true. (AND)

True when at least one input is true. (OR)

Opposite of the input. (NOT)

(True when at least one input is false. NAND)

True when both inputs are false. (NOR)

True when both inputs are equal. Also known as `exclusive nor`. (XNOR)

(XOR) True when both inputs are different. Also known as `exclusive or`.

True unless the first input is true and the second is false. (IMPLY)

True when the first input is true and the second is false. Also known as `not imply`. (NIMPLY)

Standard Boolean output.

--- Fin du chapitre: `_boolean_math` ---

--- Début du chapitre: `_boundary` --- Reference

Sculpt Mode

Sidebar `Tool` `Brush Settings` `Advanced` `Brush Type`

Similar to the Pose brush but deforms the open boundaries of a mesh. The tool detects the mesh boundary closest to the active vertex and propagates the deformation using the brush falloff into the mesh.

The main use cases of this brush are the Bend and Expand geometry, which leads to the best results on evenly distributed quad based topology. Use the Inflate, Grab, Twist, and Smooth deformation modes, to further adjustments and tweaks to the result (which do not depend that much on a clean topology).

Tip

Boundaries to hidden geometry will also be counted as an open boundary.

The boundary origin is displayed via a white line, which indicates the reach of the deformation. The targeted boundary that will be deformed is highlighted in the brush cursor color.

If the Deformation Target is changed, the brush can also be used for cloth sculpting.

Note

Evenly distributed and quad based topology will lead to much better results. Triangles and N-gons are also supported but may lead to unpredictable outcomes.

Note

More info at General brush settings and on Advanced brush settings.

Deformation type that is used by the brush.

Rotates the boundary around the local Y axis. Useful for creating folding shapes, like sleeves.

Moves/extends the mesh boundary in the local X direction. Useful for extending the boundaries along the surface.

Works similar to theInflatetool but,the vertices that are inflated are constrained to the mesh boundary.

Works similar to theGrabtool but,the vertices that are grabbed are constrained to the mesh boundary.

Rotates the active boundary around the local Z axis.Useful for creating folds like on a skirt.

Works similar to theGrabtool but,the vertices that are smoothed are constrained to the mesh boundary.

How the brushFalloffis applied across the boundary.

Applies the same deformation in the entire boundary.

Applies the deformation only within the brush radius.

Applies the brush falloff in a loop pattern along the boundary.

Applies the falloff radius in a loop pattern,inverting the direction back & forth.

Offset of the boundary origin in relation to the brush radius.

--- Fin du chapitre: _boundary ---

--- Début du chapitre: _bounding_box --- The Bounding Boxnode creates a box mesh with the minimum volume that encapsulatesthe geometry of the input. The node also can output the vector positions of the bounding dimensions.

The mesh output and the Minand Maxoutputs do not take instances into account. Instead,for instanced geometry, a bounding box is computed for each instance rather than the whole geometry.To compute the bounding box including the instances,aRealize Instances Nodecan be used.

Standard geometry input.

This node has no properties.

The resulting box that encapsulate the input geometry.

The coordinates corresponding to the box\xe2\x80\x99s -X, -Y, -Z position values,i.e. how far the box extends in each of the negative axes directions.

The coordinates corresponding to the box\xe2\x80\x99s +X, +Y, +Z position values,i.e. how far the box extends in each of the positive axes directions.

Bounding Box node used to create a box that encapsulates the geometry of the monkey mesh. \xc2\xb6

--- Fin du chapitre: _bounding_box ---

--- Début du chapitre: _bounding_box_center ---Reference

Object Mode and Edit Mode

Pivot Point \xe2\x80\xa3 Bounding Box Center

Period

In this mode, the pivot point lies at the center of the bounding box, which is a box that\xe2\x80\x99swrapped as tightly as possible around the selection while still being aligned to the world axes.

The pivot point becomes the center of the bounding box around the selected objects\xe2\x80\x99origin points, not their geometry.

This means that, if you have a single object selected, the pivot point is the sameas the object\xe2\x80\x99s origin point \xe2\x80\x93 which can be customizedand doesn\xe2\x80\x99t have to be in the center. In the example below, the orange rectanglehas it in a corner instead.

Single object rotation. \xc2\xb6

If you have multiple objects selected, the pivot point becomes the centerof an imaginary box around their origins.

The image below shows the difference between Bounding Box Center and Median Point. The latter calculates the average position of the origins, meaning that the pivot point shifts towards the area with the most objects.

Difference between \xe2\x80\x9cBounding Box Center\xe2\x80\x9d (left) and \xe2\x80\x9cMedian Point\xe2\x80\x9d (right). \xc2\xbf6
The pivot point becomes the center of the bounding box around the selected mesh elements.

The effects of rotation in different mesh selection modes. The pivot point is shown by a yellow circle. \xc2\xbf6
Median Point may again give a different result.

Difference between \xe2\x80\x9cBounding Box Center\xe2\x80\x9d (left) and \xe2\x80\x9cMedian Point\xe2\x80\x9d (right). \xc2\xbf6

--- Fin du chapitre: _bounding_box_center ---

--- Début du chapitre: _box ---Reference

Draw Mode

Toolbar \xe2\x80\x9a3 Box

The Box tool create rectangular shapes.

You can configure the brush main settings exposed on the Tool Settings for convenience. For the draw brushes configuration and settings see: Draw Brush .

The number of stroke points between each stroke edge.

Use a curve widget to define the stroke thickness from the start (left) to end (right) of the stroke.

When enabled, the stroke use a curve profile to control the thickness along the line.

In the Tool Settings select the brush, material and color type to use with the tool. The Box tool uses Draw Brush types. See Brush Settings for more information.

Click (LMB or the Pentip) and drag the start point.

Release on the desired end point.

After releasing you can move the start and end point by clicking and dragging on the yellow manipulators.

Then confirm (Return/ MMB) or cancel (Esc/ RMB).

While dragging you can use Shift to make a perfect square or use Alt to create the box from a center point.

Numpad Plus and Numpad Minus or using the mouse Wheel will increase or decrease the amount of points in the final box.

click and dragging the start point. \xc2\xbf6

Moving start and end points with manipulators. \xc2\xbf6

The box after confirming. \xc2\xbf6

--- Fin du chapitre: _box ---

--- Début du chapitre: _box_mask --- The Box Mask node creates an image suitable for use as a simple matte.

An optional mask to use as the base for mask operations.

Intensity of the generated mask.

Position of the center of the box as a fraction of the total width or height. (0.5, 0.5 creates a centered box; 0.0, 0.0 creates a box in the lower left.)

Width of the box as a fraction of the total image width.

Height of the box as a fraction of the total image width, not height.

Rotation of the box around its center point.

Operation to use against the input mask.

This yields the union of the input mask and the generated mask: Areas covered by the generated mask are set to the specified Value. Other parts of the input mask are passed through unchanged, or set to black if there is no input mask.

Values of the input mask have the specified Value subtracted from them. This yields the intersection of this generated mask and the input mask: Values of the input mask are multiplied by the specified Value for the area covered by the generated mask. All other areas become black. Any area covered by both the input mask and the generated mask becomes black. Areas covered by the generated mask that are black on the input mask become the specified Value. Areas uncovered by the generated mask remain unchanged.

A generated rectangular mask merged with the input mask. The created mask is the size of the current scene render dimensions.

Tip

For soft edges, pass the output mask through a slightBlur node.

--- Fin du chapitre: `_box_mask` ---

--- Début du chapitre: `_braid_hair_curves` --- Deforms existing hair curves into braids using guide curves.

Geometry

Guide index map which describes which curve to use as the center of each braid group. If this input is provided, it has priority over an existing map in the `guide_curve_index` attribute, and the `Guide Distance` and `Guide Mask` attributes will be unused.

Minimum distance between two guides for new guide map.

Mask for which curve are eligible to be selected as guides.

Use the existing guide map attribute if available. If this is false, and the `Guide Index` input isn't provided, the `Guide Distance` and `Guide Mask` inputs will be used to generate a new guide map for this node. Creating the guide map in a separate node or modifier gives more complete control over its creation.

Factor to blend overall effect.

Subdivision level applied before deformation.

Percentage along each curve to blend deformation from the root.

Overall radius of the braids.

Shape of the braid radius along each curve.

Factor of the minimum radius of the braids.

Factor of the maximum radius of the braids.

Frequency factor of the braids. This input can vary for different points of the same curve.

Thickness of each strand of hair.

Shape adjustment of the strand thickness for the braids.

Asymmetry of the shape adjustment of the strand thickness.

Length of the flare at the end of the braid.

Opening radius of the flare at the tip of the braid.

Geometry used for the hair tie instance (priority).

Object used for the hair tie instance.

Scale of the hair tie instance.

This node has no properties.

Geometry

Guide index map that was used for the operation. If a new guide map is created by this node, it will be stored for this output.

Parameter from 0 to 1 along the flare.
Index of the group of hair in the braid that each hair curve belongs to.
--- Fin du chapitre: `_braid_hair_curves` ---

--- Début du chapitre: `_brick` ---Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

Tip

Texture nodes can produce details at a higher frequency than geometry can show. This may cause artifacts such as Moiré type patterns or a lack of detail due to insufficient sampling points.

The Brick Texture is used to add a procedural texture producing bricks. Color of the bricks.

The color of the area between bricks.

Overall texture scale.

The size of the filling between the bricks known as `mortar`; 0 means no mortar.

Blurs/softens the edge between the mortar and the bricks. This can be useful with a texture and displacement textures.

The color variation between Color 1/2. Values of -1 and 1 only use one of the two colors; values in between mix the colors.

The width of the bricks.

The height of the brick rows.

Determines the brick offset of the various rows.

Determines the offset frequency. A value of 2 gives an even/uneven pattern of rows.

Amount of brick squashing.

Brick squashing frequency.

Texture color output.

Mortar mask (1 = mortar).

Brick texture: Colors changed, Squash 0.62, Squash Frequency 3. `brick`

--- Fin du chapitre: `_brick` ---

--- Début du chapitre: `_bricks` --- The Bricks node creates a brick-like pattern.

Sets the color range of the bricks. Brick colors are chosen randomly between these two colors.

Sets the mortar color, in between the bricks.

Sets the thickness of the mortar.

The bias of randomly chosen colors, between (-1 to 1). -1 Makes all bricks Color 1, and a value of 1 makes them all Color 2.

Sets the horizontal size of all the bricks.

Sets the vertical size of all the bricks.

The relative offset of the next row of bricks.

Offset every N rows. The brick pattern offset repeats every N rows.

Scales the bricks in every N rows by this amount.

Squash every N rows.

Standard color output.

--- Fin du chapitre: `_bricks` ---

--- Début du chapitre: `_bridge_edge_loops` ---Reference

Edit Mode

Edge \xe2\x80\xa3 Bridge Edge Loops
 Bridge Edge Loops connects multiple edge loops with faces.
 Loops connected with open ends.
 Tries to connect to a circular loop (where the start and end are merged).
 Connects each even count of loops individually.
 Merges edge loops rather than creating a new face.
 Which edge loop the edges are merged to, a value of 0.5 will merge at a half-way point.
 Determines which vertices in both loops are connected to each other.
 The number of intermediate edge loops used to bridge the distance between two loops.
 Linear, Blend Path, Blend Surface
 Smoothness of the Blend Path and Blend Surface.
 How much intermediary new edges are shrunk/expanded.
 The shape of the new edges. See the Proportional Editing page for a description of each option.
 Simple example showing two closed edge loops.
 Input.\xc2\xbb
 Bridge result.\xc2\xbb
 Example of the Bridge tool between edge loops with different numbers of vertices.
 Input.\xc2\xbb
 Bridge result.\xc2\xbb
 Example using the Bridge tool to cut holes in face selections and connect them.
 Input.\xc2\xbb
 Bridge result.\xc2\xbb
 Example showing how Bridge tool can detect multiple loops and connect them in one step.
 Input.\xc2\xbb
 Bridge result.\xc2\xbb
 Example of the subdivision option and surface blending with UVs.
 Input.\xc2\xbb
 Bridge result.\xc2\xbb
 --- Fin du chapitre: _bridge_edge_loops ---

--- Début du chapitre: _bright_contrast --- Standard color input.
 An additive-type factor by which to increase the overall brightness of the image. Use a negative number to darken an image.
 A scaling type factor by which to make brighter pixels brighter, but keeping the darker pixels dark. Higher values make details stand out. Use a negative number to decrease the overall contrast in the image.
 By default, it is supposed to work in premultiplied alpha. If the Convert Premul checkbox is not enabled, it is supposed to work in straight alpha. See Alpha Channel .
 Standard color output.
 It is possible that this node will put out a value set that has values beyond the normal range, i.e. values greater than one and less than zero. If you will be using the output to mix with other images in the normal range, you should clamp the values using the Map Value node (with the Min and Max enabled), or put through a Color Ramp node (with all normal defaults).
 Clamp the values to normal range. \xc2\xbb

Either of these nodes will scale the values back to normal range. In the example image, we want to intensify the specular pass. The bottom thread shows what happens if we do not clamp the values; the specular pass has a value much less than one in the dark areas; when added to the medium gray, it makes black. Passing the brightened image through either the Map Value or the Color Ramp node produces the desired effect.

A basic example. \xc2\xba6

--- Fin du chapitre: _bright_contrast ---

--- Début du chapitre: _brush ---Reference

Physics \xe2\x80\xa3 Dynamic Paint

Brush

The Brush type makes object apply paint on the canvas.

Brush main panel. \xc2\xba6

From the first brush panel you can define how brush affects canvas color surfaces.

Color of the paint.

Defines brush alpha or visibility. Final wetness is also affected by alpha.

Defines how \xe2\x80\x9cwet\xe2\x80\x9d new paint is. Wetness is visible on \xe2\x80\x9cPaint\xe2\x80\x9d surface

\xe2\x80\x9cwetmap\xe2\x80\x9d. Speed of \xe2\x80\x9cDrip\xe2\x80\x9d and \xe2\x80\x9cSpread\xe2\x80\x9d effects also depends on how wet the paint is.

This setting limits brush alpha influence. Without it, brush is \xe2\x80\x9ccadded\xe2\x80\x9d on surface over and over again each frame, increasing alpha and therefore influence of brush on canvas. In many cases however, it is preferred to not increase brush alpha if it already is on brushes level.

Makes brush dissolve existing paint instead of adding it.

Reference

Brush

Physics \xe2\x80\xa3 Dynamic Paint \xe2\x80\xa3 Source

Paintsource setting lets you define how brush influence/intersection is defined.

Mesh Volume

The Brush affects all surface point inside the mesh volume.

Source: Mesh Volume. \xc2\xba6

Proximity

Only uses defined distance to the closest point on brush mesh surface. Note that inside of the volume is not necessarily affected because it is not close to the surface.

Source: Proximity. Brush affects all canvas pixels around it. \xc2\xba6

Mesh Volume + Proximity

Same as volume type, but also has influence over defined distance.

Applies proximity inside the mesh volume.

Negates brush alpha within mesh volume.

The Volume + Proximity brush with no additional settings. \xc2\xba6

Inner Proximity. Proximity falloff is now visible inside the volume. \xc2\xba6

Negate Volume. Inner side of the volume has become completely transparent. \xc2\xba6

Inner Proximity and Negate Volume enabled together. \xc2\xba6

Object Center

Instead of calculating proximity to the brush object mesh, which can be quite slow in some cases, only distance to only center is calculated. This is much faster and often good enough.

Source: Object Center. \xc2\xba6

Particle System

Brush influence is defined by particles from a selected particle system. Defines the distance, inside which paint is solid color.

Uses the settings in the particle panel to determine solid radius size. Solid Radius size disabled while Particle Radius enabled.

An additional radius outside Solid Radius to add a smooth falloff.

If you set \xe2\x80\x9cSmooth Radius\xe2\x80\x9d to zero, particle will be painted as a solid sphere. If you set \xe2\x80\x9cSolid

Radius\xe2\x80\x9d to zero, it gets painted as a smooth halo.

Source: Particle System. \xc2\xba6

The maximum distance to mesh surface to affect paint.

Projects brush to the canvas from a defined direction. Basically this can be considered as \xe2\x80\x9cdirection aligned\xe2\x80\x9d proximity.

The Project option enabled. See how brush only affects canvas in normal direction. \xc2\xba6

Paints solid paint within the defined distance.

Makes paint to linearly fade out until becoming completely invisible when it reaches the maximum distance.

Allows you to manually make a custom falloff behavior.

Reference

Brush

Physics \xe2\x80\x93 Dynamic Paint \xe2\x80\x93 Velocity

This panel shows brush options that are based on object velocity.

On top you have a color ramp and several related settings. Basically the color ramp represents brush velocity values: left side being zero velocity and right side being the \xe2\x80\x9cMax velocity\xe2\x80\x9d. Speed is measured in \xe2\x80\x9cunits per frame\xe2\x80\x9d.

Checkboxes above can be used to define color ramp influence.

Uses color ramp\xe2\x80\x99s alpha value depending on current velocity and multiplies brush alpha with it.

Replaces the brush color with the values from the Color Ramp Widget .

Multiplies brushes \xe2\x80\x9cdepth intersection\xe2\x80\x9d

effect. Basically you can adjust displace and wave strength depending on brush speed.

Enabling Smudge makes the brush \xe2\x80\x9csmudge\xe2\x80\x9d (or \xe2\x80\x9csmear\xe2\x80\x9d) existing colors on the surface as it moves. The strength of this effect can be defined from the Smudge

Strength property.

Even when smudge is enabled brush still does its normal paint effect. If you want a purely smudging brush use zero alpha. It is also possible to have Erase option enabled together with smudge.

Reference

Brush

Physics \xe2\x80\x93 Dynamic Paint \xe2\x80\x93 Waves

This panel is used to adjust brush influence to

\xe2\x80\x9cWave\xe2\x80\x9d surfaces.

Select what effect the brush creates in the wave simulation.

The brush create waves when the intersection depth with the surface is changed on that point. If the brush is not moved, it will have no effect.

Using a negative `\xcFactor\x` with this type can create a nice looking `\cwake\x` for moving objects like ships.

Constantly affects surface whenever intersecting. Waves are also reflected off this brush type. However, due the nature of wave simulation algorithm this type creates an unnatural `\xcdent\x` in the surface if the brush is not moved.

Directly affects the velocity of wave motion. Therefore the effect is not one-to-one with brush intersection depth, yet the force strength depends on it.

This type has no visible effect on the surface alone but reflects waves that are already on the surface.

Adjusts how strongly brush `\xcddepth\x` affects the simulation. You can also use negative values to make brush pull water up instead of down.

In some cases the brush goes very deep inside the surface messing whole simulation up. You can use this setting to `\xclimit\x` influence to only certain depth.

--- Fin du chapitre: `_brush` ---

--- Début du chapitre: `_brushes` --- Brushes for Grease Pencil Weight Paintmode bundled in the Essentials library.

Paints a specified weight over the strokes.

Smooths out the weighting of adjacent points. In this mode the `WeightValue` is ignored. The strength defines how much the smoothing is applied.

Smooths weights by painting the average resulting weight from all weights under the brush.

Smudges weights by grabbing the weights under the brush and `\xcdragging\x` them. This can be imagined as a finger painting tool.

--- Fin du chapitre: `_brushes` ---

--- Début du chapitre: `_brush_management` --- Brush assets are stored in asset libraries to make them accessible from any Blender session. There are two ways of managing brush assets:

Using asset operators: Create and update brush assets using utility operators from any Blender file. Storage is managed by Blender. Convenient for simple `\xcon the fly\x` management of personal brush asset libraries.

Using manual storage: Create and update brush assets by opening blend files within asset libraries, and managing brush asset data-blocks in there. Useful for careful curation of asset libraries, especially to prepare them for sharing with others.

Brushes can be managed through a few operators that let Blender handle the act of saving and updating the brushes in asset libraries for you.

Assets managed this way will be saved in special asset system files using a `.asset.blendfile` extension.

Note

Note that only brush assets created via `Duplicate Asset` can be edited further using these asset operators. For others, these operations will be grayed out, and manual management is necessary.

Brushes from the Essentials asset library cannot be edited.

Reference

All Paint Modes

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush AssetProperties

\xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Asset

Asset Shelf \xe2\x80\xa3 Context Menu

Brush Asset panel in the Sidebar showing asset operators. \xc2\xb6

Creates a copy of the currently active brush as asset, and activates it.

A popup is spawned to input some settings to use:

A custom name to use for the new brush.

Choose an Asset Library to store the new brush asset in. The available

asset libraries are configured in the Preferences .

Choose an Asset Catalog to assign the brush asset to. Entering an on-

existent name/path will create a new catalog accordingly.

Permanently remove this brush asset from the Asset Library it is stored

in. This cannot be undone, so a popup will ask for confirmation.

Spawns a popup to change some of the available asset metadata fields:

Choose an Asset Catalog to assign the brush asset to. Entering an on-

existent name/path will create a new catalog accordingly.

See Asset Author .

See Asset Description .

Opens a window with the File Browser to select an image for the asset preview.

Saves any changes made to the active brush to the asset library.

Discards any unsaved changes made to the brush asset.

See also

Complete description of the manual asset create, edit, share and use workflow.

It is also possible to manually manage brushes in blend-files like any other asset data-block. By marking brushes as assets and saving the file in an asset library, they become available from any Blender session. This gives full control over how assets are stored, and is particularly useful for curating asset libraries that can be shared with others.

The Mark as Asset operator used on a brush in the Outliner. \xc2\xb6

Brushes can be imported as normal data-blocks from other files (including

from .asset.blend files from an asset library) through appending . In the

Blender File mode of the Outliner, the brush will be listed under Brushes.

Right-click the brush and select Mark as Asset. By saving the file inside

of an asset library directory, the asset becomes available from all

Blender sessions. If necessary, configure an asset library directory in

the Preferences.

--- Fin du chapitre: _brush_management ---

--- Début du chapitre: _brush_settings --- Data-block selector for the material. Except for the Erase tool of course.

Pin the material to the brush.

The final appearance of the strokes is a combination of the brush and material used, binding the material to the brush gives more control and avoids a lack of coordination between the two.

--- Fin du chapitre: _brush_settings ---

--- Début du chapitre: _brush_tool --- Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Brush

Tool to use for any of the Sculptmode brushes. Activating a brush from an asset shelf or brushselector will also activate this tool for convenience.

See the list of Essentials Brushes (based on available Brush Types) for more details.

--- Fin du chapitre: `_brush_tool` ---

--- Début du chapitre: `_build` --- Converting the RST-files into pretty HTML pages is simple.

Open a terminal or Command Prompt in the `~/blender-manual` directory and simply run:

Tip

On Microsoft Windows you can simply open the `make.bat` file to easily run the command without having to open the Command Prompt and typing commands.

This is the command you should use when building the docs, however, other commands are available by typing `make help`. This command will convert the RST-files into HTML pages and automatically open your default web browser to view the result. The command will continue to run and watch for changes made to the RST-files and refresh the HTML pages as necessary.

Note

The converted pages can also be viewed manually by browsing the build directory: `~/blender-manual/build/html`. For example to open the home page, open `build/html/index.html` to read the manual.

The building process may take several minutes the first time (or after any major changes), but for subsequent changes it should only take a few seconds.

--- Fin du chapitre: `_build` ---

--- Début du chapitre: `_bump` --- The Bump node generates a perturbed normal from a height texture, for bump mapping. The height value will be sampled at the shading point and two nearby points on the surface to determine the local direction of the normal.

Strength of the bump mapping effect, interpolating between no bump mapping and full bump mapping.

Multiplier for the height value to control the overall distance for bump mapping.

Scalar value giving the height offset from the surface at the shading point; this is where you plug in textures.

Standard normal input.

Invert the bump mapping, to displace into the surface instead of out.

Standard normal output.

Tip

If the Height input is not connected, the node becomes a no-op that outputs its Normal input as is, or defaults to the geometry normal if not connected. Routing a nodegroup input via a no-op Bump node before doing math effectively makes it default to normal.

The above node setup will only bump the diffuse part of the shader, simulating a bumpy diffuse surface coated with a smooth glossy `\xc2\x80\x9cglaze\xc2\x80\x9d` layer.

--- Fin du chapitre: `_bump` ---

--- Début du chapitre: `_buttons` --- Operator button. `\xc2\xb6`

Operator buttons execute an Operator which in summary execute an action when clicked with LMB. Operator buttons may be an icon, text, or text with an icon.

Checkboxes and Toggle buttons. \xc2\xba6

These controls are used to activate or deactivate options. Use LMB to change their state. A tick is shown on checkboxes when the option is activated. Active status on toggle buttons is indicated either by color on the icon background, or a change in icon graphics.

To change many values at once on or off, you can press down LMB and drag over multiple buttons. This works for checkboxes, toggles and to select a radio button value.

Radio buttons. \xc2\xba6

Radio buttons are used to choose one option from a selection of options. The active button is indicated by a colored background.

Use Ctrl-Wheel while hovering the mouse over radio buttons to cycle between the options.

Direction buttons. \xc2\xba6

Clicking with LMB in the sphere and dragging the mouse cursor lets the user change the direction by rotating the sphere.

LMB (drag) rotates the direction.

Ctrl (while dragging) snaps to vertical & diagonal directions.

--- Fin du chapitre: _buttons ---

--- Début du chapitre: _by_attribute ---Reference

Edit Mode

Select \xe2\x80\xa3 By Attribute

Selects vertices, edges, or faces based on the Active Attribute.

Note

The active attribute must have a boolean type.

The active attribute must be on the vertex, edge, or face domain.

Select the desired attribute from the Attribute List .

Execute the By Attribute operator.

--- Fin du chapitre: _by_attribute ---

--- Début du chapitre: _cache ---Reference

Particle System \xe2\x80\xa3 Cache

In order to improve real-time response and avoid unnecessary recalculation of particles, the particle data can be cached in memory or stored on a drive.

The Emitter particle system uses a unified system for caching and baking (together with Soft Body and Cloth).

See also

See the General Baking docs for more information.

The simulation is only calculated for positive frames in between the Start and End frames of the Cache panel, whether you bake or not. So if you want a simulation that is longer than the default frame range, you have to change the End frame.

When an animation is played, each physics system writes each frame to the cache. Note that for the cache to fill up, one has to start the playback before or on the frame that the simulation starts.

The cache is cleared automatically on changes. But not on all changes, so it may be necessary to delete it manually, e.g. if you change a force field.

The system is protected against changes after baking. If for example the mesh changes the simulation is not calculated anew.

The bake result can be cleared by clicking on the Delete Bake button in the simulation cache settings.

A simulation can only be edited in Particle Edit Mode when it has been baked in memory. And cannot be edited if the Disk Cache is used.

If you are not allowed to write to the required subdirectory caching will not take place, e.g. if your blend-file path is very long and your operating system has a limit on the path length that is supported.

Be careful with the sequence of modifiers in the modifier stack. You may have a different number of faces in the 3D Viewport and for rendering (e.g. when using subdivision surface), if so, the rendered result may be very different from what you see in the 3D Viewport.

--- Fin du chapitre: _cache ---

--- Début du chapitre: _calligraphy --- The Calligraphy modifier mimics some broad and flat pens for calligraphy. It generates different thickness based on the orientation of the stroke.

The angle (orientation) of the virtual drawing tool, from the vertical axis of the picture. For example, an angle of 0.0 mimics a pen aligned with the vertical axis. Hence, the thickest strokes will be the vertical ones i.e. stroke's direction is aligned with the angle, and the thinnest will be the horizontal ones i.e. stroke's direction is perpendicular to the angle.

Calligraphy modifier demo by T.K. (blend-file).

--- Fin du chapitre: _calligraphy ---

--- Début du chapitre: _camera --- This panel contains all settings of the camera used for filming the movie which is currently being edited in the Clip editor. Different predefined settings can be used here and can be chosen from the panel header. But such settings as distortion coefficients and principal point are not included in the presets and should be filled in even if camera presets are used.

Is the width of the CCD sensor in the camera. This value can be found in camera specifications.

Is the pixel aspect of the CCD sensor. This value can be found in camera specifications, but can also be guessed. For example, you know that the footage should be 1920x1080, but the images themselves are 1280x1080. In this case, the pixel aspect is: $1920 / 1280 = 1.5$. Is self-explanatory; it is the focal length with which the movie was shot. It can be set in millimeters or pixels.

Is the optical center of the lens used in the camera. In most cases it is equal to the image center, but it can be different in some special cases. Check camera/lens specifications in such cases.

Tip

Optical Center also known as the principal point in photogrammetry. See Set Principal to Center.

Mathematical function to convert distorted to undistorted coordinates.

Polynomial radial distortion. Uses three distortion coefficients: K1, K2, and K3.

It defines high distortions, which makes this model suitable much better for cameras with fisheye lenses. Use two distortion coefficients: K1, K2.

Distortion model used by the Nuke compositor. Use two distortion coefficients K1, K2.

Brown-Conrady is one of most advanced mathematical lens distortion models. Used to model both radial and tangential distortion. Can use up to four radial distortion coefficients: K1 - K4 and up to two tangential distortion coefficients: P1 and P2.

Coefficients are used to compensate for lens distortion when the movie was shot. Currently these values can be tweaked by hand only (there are no calibration tools yet) using tools available in Distortion mode. To do this tweak K1 until the solving is the closest to the known focal length (but also take grid and annotations into account to prevent impossible distortion).

The coefficients in lens distortion models work independent from each other. Positive values will give a barrel distortion while negative values give a pincushion distortion. With a mixture of both negative and positive coefficients you can define more complicated mustache distortions or other complex distortions, that are less common but not rare.

Example of radial distortion for positive and negative K coefficients.
\xc2\xba6

Works independent and allow to compensate for situations when the sensor is not perpendicular to a group of lens. The optical center (also called principal point) will be shifted (distorted) from the center of the sensor. P1 is used to compensate for sensor rotation in Z (vertical) axes, while P2 is for compensating sensor rotation in X (horizontal) axes. Such distortions can be found in sources from cameras with a sensor stabilization system.

Example of tangential distortion for P coefficients. \xc2\xba6
--- Fin du chapitre: _camera ---

--- Début du chapitre: _cameras --- A camera is an object that provides a means of rendering images from Blender. It defines which portion of a scene is visible in the rendered image.

Cameras are invisible in renders, so they do not have any material or texture settings. However, they do have Object and Editing setting panels available which are displayed when a camera is the active object.

See also

3D Viewport Camera Navigation for documentation about managing cameras in the viewport.

Reference

Object Mode

Properties \xe2\x80\xa3 Camera

Type

The camera lens options control the way 3D objects are represented in a 2D image.

This matches how you view things in the real world. Objects in the distance will appear smaller than objects in the foreground, and parallel lines (such as the rails on a railroad) will appear to converge as they get farther away.

The Focal Length controls the amount of zoom, i.e. the amount of the scene which is visible all at once. Longer focal lengths result in a smaller FOV (more zoom), while short focal lengths allow you to see more of the scene at once (larger FOV, less zoom).

Perspective camera with 35 mm focal length. \xc2\xba6

Perspective camera with 210 mm focal length instead of 35 mm. \xc2\xba6

The focal length can be set either in terms of millimeters or the actual Field of View as an angle.

Hint

While the camera is moving towards an object the Focal Length property can be decreased to produce a Dolly Zoom camera effect, or vice versa.

This video demonstrates the Dolly Zoom camera effect.

With Orthographic perspective objects always appear at their actual size, regardless of distance. This means that parallel lines appear parallel, and do not converge like they do with Perspective.

Render from the same camera angle as the previous examples, but with orthographic perspective.

This controls the apparent size of objects projected on the image.

Note that this is effectively the only setting which applies to orthographic perspective. Since parallel lines do not converge in orthographic mode (no vanishing points), the lens shift settings are equivalent to translating the camera in the 3D Viewport.

Panoramic cameras only work in Cycles. See the Cycles panoramic camera settings for more information.

Allows for the adjustment of vanishing points. Vanishing points refer to the positions to which parallel lines converge. In these render examples, the most obvious vanishing point is at the end of the railroad.

Horizontal lens shift of 0.330.

Rotation of the camera object instead of a lens shift.

Notice how the horizontal lines remain perfectly horizontal when using the lens shift, but do get skewed when rotating the camera object.

Note

Using lens shift is equivalent to rendering an image with a larger FOV and cropping it off-center.

The interval in which objects are directly visible. Any objects outside this range still influence the image indirectly, as further light bounces are not clipped.

Note

For viewport rendering, setting clipping distances to limited values is important to ensure sufficient rasterization precision. Ray tracing renders do not suffer from this issue so much, and as such more extreme values can safely be set.

Tip

When Limits in the Viewport Display panel is enabled, the clip bounds will be visible as two yellow connected dots on the camera's line of sight.

See also

3D Viewport clipping

Real-world cameras transmit light through a lens that bends and focuses it onto the sensor. Because of this, objects that are a certain distance away are in focus, but objects in front and behind that are blurred.

Example of DOF bokeh effect.

The area in focus is called the focal point and can be set using either an exact value, or by using the distance between the camera and a chosen object:

Choose an object which will determine the focal point. Linking an object will deactivate the distance parameter.

Sets the distance to the focal point when no Focus Object is specified. This distance can be visualized in the 3D Viewport by enabling Limits in the camera's Viewport Display panel.

Hint

Use the eyedropper icon or hover the mouse over the Focal Distance property and press E to enable the depth Depth Picker. Then LMB on a point in the 3D Viewport to sample the distance from that point to the camera.

F-Stop ratio that defines the amount of blurring. Lower values give a strong depth of field effect.

Total number of polygonal blades used to alter the shape of the blurred objects in the render, and render preview. As with the viewport, the minimum amount of blades to enable the bokeh effect is 3, resulting in a triangular-shaped blur.

Rotate the polygonal blades along the facing axis, and will rotate in a clockwise, and counter-clockwise fashion.

Change the amount of distortion to simulate the anamorphic bokeh effect. A setting of 1.0 shows no distortion, where a number below 1.0 will cause a horizontal distortion, and a higher number will cause a vertical distortion.

These settings adjust properties that relate to a physical camera body. Several Presets can be chosen to match real-world cameras.

Adjusts how the camera's sensor fits within the output dimension adjusting the angular field of view.

Calculates a square sensor size based on the larger of the resolution dimensions.

Manually adjust the Width of the sensor, the Height is calculated based on the aspect ratio of the output's Resolution.

Manually adjust the Height of the sensor, the Width is calculated based on the aspect ratio of the output's Resolution.

This setting is an alternative way to control the field of view, as opposed to modifying the focal length. It is useful to match a camera in Blender to a physical camera and lens combination, e.g. for motion tracking.

Safe areas are guides used to position elements to ensure that the most important parts of the content can be seen across all screens.

Different screens have varying amounts of Overscan (especially older TV sets). That means that not all content will be visible to all viewers, since parts of the image surrounding the edges are not shown. To work around this problem TV producers defined two areas where content is guaranteed to be shown: action safe and title safe.

Modern LCD/plasma screens with purely digital signals have no Overscan, yet safe areas are still considered best practice and may be legally required for broadcast.

In Blender, safe areas can be set from the Camera and Sequencer views.

Red line: Action safe. Green line: Title safe.

The Safe Areas can be customized by their outer margin, which is a percentage scale of the area between the center and the render size. Values are shared between the Video Sequence editor and camera view.

Also known as Graphics Safe. Place all important information (graphics or text) inside this area to ensure it can be seen by the majority of viewers.

Make sure any significant action or characters in the shot are inside this area. This zone also doubles as a sort of margin for the screen which can be used to keep elements from piling up against the edges.

Tip

Each country sets a legal standard for broadcasting. These include, among other things, specific values for safe areas. Blender defaults for safe areas follow the EBU (European Union) standard. Make sure you are using the correct values when working for broadcast to avoid any trouble. Center-cuts are a second set of safe areas to ensure content is seen correctly on screens with a different aspect ratio. Old TV sets receiving 16:9 or 21:9 video will cut off the sides. Position content inside the center-cut areas to make sure the most important elements of your composition can still be visible in these screens.

Blender defaults show a 4:3 (square) ratio inside 16:9 (widescreen).

Cyan line: action center safe. Blue line: title center safe. \xc2\xba6

A background picture in your camera can be very helpful in many situations: modeling is obviously one, but it is also useful when painting (e.g. you can have reference pictures of faces when painting textures directly on your model \xe2\x80\xa6), or animation (when using a video as background), etc.

The source of the background image.

Use an external image, image sequence, video file or generated texture.

Use one of the Movie Clip data-blocks.

Display a Movie Clip from the scene \xe2\x80\x99s Active Clip.

Display the background image using undistorted proxies when available.

Select between full (non-proxy) display or a proxy size to draw the background image.

See also

To build a proxy, the Movie Clip Editor Proxy settings have to be used. Otherwise the proxy settings here have no effect.

The color space the image or video file uses within Blender.

Apply the color management settings when displaying this image on the screen.

Controls the transparency of the background image.

Choose whether the image is shown behind all objects, or in front of everything.

Controls how the image is placed in the camera view.

Forces the image dimensions to match the camera bounds (may alter the aspect ratio).

Scales the image down to fit inside the camera view without altering the aspect ratio.

Scales the image up so that it fills the entire camera view, but without altering the aspect ratio (some of the image will be cropped).

Positions the background image using these offsets.

In orthographic views, this is measured in the normal scene units. In the camera view, this is measured relative to the camera bounds (0.1 will offset it by 10% of the view width/height).

Rotates the image around its center.

Scales the image up or down from its center.

Swaps the image around, such that the left side is now on the right, and the right now on the left.

Swaps the image around, such that the top side is now on the bottom, and the bottom now on the top.

Note

Movie Clips or images with view as render are only visible behind objects when film transparency is enabled or the scene world is disabled in the viewport.

Camera view displaying safe areas, sensor and name. \xc2\xba6

Size of the camera visualization in the 3D Viewport. This setting has no effect on the render output of a camera. The camera visualization can also be scaled using the standard Scale Transform key.

Shows an orange line indicating the Clip Start and End values, as well as a yellow cross indicating the Focus Distance. If the Focus Distance gizmo is enabled in the 3D Viewport's gizmo settings, this cross can also be dragged with the mouse to adjust the distance.

Toggles viewing of the mist limits on and off. The limits are shown as two connected white dots on the camera line of sight. The mist limits and other options are set in the World panel, in the Mist section.

Displays a dotted frame in camera view.

Toggle name display on and off in camera view.

Composition Guide enable overlays onto the camera display that can help when framing a shot.

Adds lines dividing the frame in thirds vertically and horizontally.

Adds lines dividing the frame in half vertically and horizontally.

Adds lines connecting opposite corners.

Divides the width and height into Golden proportions (about 0.618 of the size from all sides of the frame).

Displays a diagonal line from the lower left to upper right corners, then adds perpendicular lines that pass through the top left and bottom right corners.

Same as A, but with the opposite corners.

Displays a diagonal line from the lower left to upper right corners, then lines from the top left and bottom right corners to 0.618 the lengths of the opposite side.

Same as A, but with the opposite corners.

This option darkens the area outside of the camera's field of view. The opacity of the passepout can be adjusted using the value slider.

Tip

If the Passepout is fully opaque, Blender will make optimizations to speed up the rendering of areas inside the camera view.

--- Fin du chapitre: _cameras ---

--- Début du chapitre: _camera_data --- The Camera Data node returns information about the shading point relative to the camera. This could be used for example to change the shading of objects further away from the camera, or make custom fog effects.

This node has no inputs.

This node has no properties.

A normalized vector, in camera space, from the camera to the shading point.

The distance each pixel is away from the camera.

Distance from the camera to the shading point.

--- Fin du chapitre: _camera_data ---

--- Début du chapitre: _camera_solver --- The Camera Solver constraint gives the owner of this constraint, the location and rotation of the solved camera motion.

The solved camera motion is where Blender reconstructs the position of the physical, real-world camera, when it filmed the video footage, relative to the thing being tracked.

Note

This constraint only works after you have set up a minimum of eight markers and pressed Solve Camera Motion (Movie Clip Editor \xe2\x80\xa3 Toolbar \xe2\x80\xa3 Solve \xe2\x80\xa3 Solve Camera Motion).

Camera Solver Constraint panel. \xc2\xb6

Receive tracking data from the scene\xe2\x80\x99s Active Clip .If unchecked, an option appears to choose from the other clips.

Applies the constraint, creating Keyframes for the transforms.

Controls the percentage of affect the constraint has on the object.See common constraint properties for more information.

--- Fin du chapitre: _camera_solver ---

--- Début du chapitre: _camera_view --- Demonstration of camera view.\xc2\xb6

The Camera view shows the current scene from the active camera\xe2\x80\x99s viewpoint.

The Camera view can be used to virtually compose shots and preview how the scene will look when rendered.The rendered image will contain everything within the dashed frame.

See also

Camera Settings for details on how camera settings are used for display and rendering.

Hint

While in camera view, you can select the camera by clicking the dashed frame(assuming the camera object isn\xe2\x80\x99t hidden).

Reference

All Modes

View \xe2\x80\xa3 Cameras \xe2\x80\xa3 Active Camera, View \xe2\x80\xa3 Viewpoint \xe2\x80\xa3 Camera

Numpad0

This switches the view to the active camera.

Reference

Object Mode

View \xe2\x80\xa3 Cameras \xe2\x80\xa3 Set Active Object as Camera

Ctrl- Numpad0

Active camera (left) displayed with a solid triangle above it. \xc2\xb6 This sets the current active object as the active camera and switches to the camera view.

The active camera is the one that will be used for rendering,and which you\xe2\x80\x99ll look through when choosing camera view.

Another way of setting the active camera is through the Scenetab of the Properties .

Note

The active camera is normally defined on the scene level, so that it\xe2\x80\x99s the sameacross all 3D Viewports. However, it\xe2\x80\x99s also possible to make a camerathe active one within one Viewport only.SeeLocal Camera.

While a scene contains only one camera by default, it\xe2\x80\x99s possible to have multiple.You can then bind the cameras to specific time points in your animationto create jump cuts showing different viewpoints.SeeAnimating Cameras.

Reference

All Modes

View \xe2\x80\xa3 Cameras \xe2\x80\xa3 Frame Camera Bounds

Home

Centers the camera view inside the 3D Viewport's screen area and resizes the view to fit within the area's bounds.

Reference

All Modes

View Navigation Zoom Camera 1:1

Zooms the view so that the camera frame has the exact same size as the output resolution. This allows you to preview exactly how large objects will be in the rendered image/animation.

There are several different ways to position the camera in your scene. Some of them are explained below.

Hint

The active camera might be any kind of object, meaning these actions can also be used to position and aim a light for example.

Reference

Object Mode

View Navigation Align View Align Active Camera to View

Ctrl- Alt- Numpad0

Moves and rotates the camera so it perfectly matches your current viewport view.

By enabling Lock Camera to View in Sidebar View and switching to camera view or toggle the lock navigation gizmo button when in camera view, the camera will become 'glued' to the view and follow it around as you navigate.

See also

Fly/Walk Navigation for first person navigation that moves the active camera too.

To perform these camera moves, the camera must first be selected so transform operations apply to it. The following actions also assume that you are in camera view. Having done so, you can now manipulate the camera using the same tools that are used to transform any object:

Press R to enter object rotation mode. The default will be to rotate the camera along its local Z axis (the axis orthogonal to the camera view), which is the definition of a camera 'roll'.

This is just a rotation along the local X axis. Press R to enter object rotation mode, then X twice. (The first press selects the global axis, the second the local axis. This works with any axis; see Axis Locking).

This corresponds to a rotation around the camera's local Y axis. Press R, then Y twice.

To dolly the camera, press G then MMB (or Z twice).

Press G and move the mouse (you can use X or Y twice to get purely horizontal or vertical tracking).

--- Fin du chapitre: _camera_view ---

--- Début du chapitre: _canvas ---Reference

Physics Navigation Dynamic Paint

Canvas

The Canvas type makes object receive paint from Dynamic Paint brushes.

Canvas main panel. \xc2\xb6

A list of Dynamic Paint surfaces. These surfaces are basically layers of paint, that work independently from each other. You can define individual settings for them and bake them separately.

The checkbox toggles whether surface is active at all. If not selected no calculations are done.

Below you can set surface type and adjust quality and timing settings. Each surface has a certain format and type. Format determines how data is stored and outputted.

Dynamic Paint operates directly on mesh vertex data. Results are stored by point cache and can be displayed in viewports. However, using vertex level also requires a highly subdivided mesh to work.

Dynamic Paint generates UV wrapped image files of defined resolution as output.

You can adjust the output image dimensions for the Image Sequence surface type. For example using 256 will lead to 256x256 image output. Doubling the resolution will likely quadruple the baking time and vice versa.

Anti-Aliasing to smooth paint edges using a 5x multisampling method.

Defines surface processing start and end frame.

Extra samples between frames. They are usually required when there is a very fast brush.

Reference

Canvas

Physics \xe2\x80\xa3 Dynamic Paint \xe2\x80\xa3 Surface

From Surface panel you can adjust surface type and related settings.

Each surface has a \xe2\x80\x9ctype\xe2\x80\x9d that defines what surface is used for.

Paint Surface. \xc2\xb6

Paint is the basic surface type that outputs color and wetness values. In case of vertex surfaces, results are outputted as Color Attributes.

A wetmap is a black-and-white output that visualizes paint wetness. White being maximum wetness, black being completely dry. It is usually used as mask for rendering. Some \xe2\x80\x9cpaint effects\xe2\x80\x9d affect wet paint only.

Completely disable drying is useful for indefinitely spreading paint.

It can be used to define wetness level when paint colors start to shift to surface \xe2\x80\x9cbackground\xe2\x80\x9d. Lower values can be useful to prevent spreading paint from becoming transparent as it dries, while higher values give better results in general.

Displace Surface. \xc2\xb6

This type of surface outputs intersection depth from brush objects.

A new displace is added cumulatively on top of an existing displace.

The maximum level of intersection depth, larger values will be clamped to this value.

The multiplier for the intersection depth. You can use it to adjust final displace strength or use negative values to paint bumps.

Tip

If the displace output seems too rough it usually helps to add a Smooth Modifier after Dynamic Paint in the modifier stack.

Waves Surface. \xc2\xb6

This surface type produces simulated wave motion. Like displace, wave surface also uses brush intersection depth to define brush strength.

Allows waves to pass through mesh \xe2\x80\x9cedges\xe2\x80\x9d instead of reflecting from them.

Directly adjusts simulation speed without affecting simulation outcome. Lower values make simulation go slower and otherwise.

Affects how fast waves travel on the surface. This setting is also corresponds to the size of the simulation. Half the speed equals surface double as large.

Reduces the wave strength over time. Basically adjusts how fast wave disappears.

Adjusts the force that pulls water back to \xe2\x80\x9czero level\xe2\x80\x9d.

Limits the maximum steepness of the wave slope between simulation points. This greatly helps getting rid of the \xe2\x80\x9cnoise\xe2\x80\x9d that occurs when using objects with sharp edges (like cubes) as a brush. The default value should be enough to only get rid of the sharpest spikes, in order to get even smoother waves use higher values at the expense of reduced detail.

Tip

In some cases the wave motion gets very unstable around brush. It usually helps to reduce wave speed, brush \xe2\x80\x9cwave factor\xe2\x80\x9d or even the resolution of mesh/surface.

Weight Surface. \xc2\xb6

This is a special surface type only available for vertex format. It outputs vertex weight groups that can be used by other Blender modifiers and tools.

Tip

It is usually preferred to use \xe2\x80\x9cproximity\xe2\x80\x9d based brushes for weight surfaces to allow smooth falloff between weight values.

For each surface type there are special settings to adjust. Most types have the settings Dissolve and Brush:

Used to make the surface smoothly return to its original state during a defined Time period.

Used to define a specific collection to pick brush objects from.

For tweaking brush settings individually for each surface.

Reference

Canvas

Physics \xe2\x80\xa3 Dynamic Paint \xe2\x80\xa3 Cache

This panel is currently only visible for Vertex format surfaces. You can use it to adjust and bake point cache.

Reference

Canvas

Physics \xe2\x80\xa3 Dynamic Paint \xe2\x80\xa3 Effects

This is a special feature for \xe2\x80\x9cPaint\xe2\x80\x9d type surface. It generates animated movement on canvas surface.

Paint slowly spreads to surrounding points eventually filling all connected areas.

Paint moves in specific direction specified by Blender force fields, gravity and velocity with user-defined influences.

Painted area slowly shrinks until disappears completely.

For spread and drip effects, only \xe2\x80\x9cwet paint\xe2\x80\x9d is affected, so as the paint dries, movement becomes slower until it stops.

Reference

Canvas

Physics \xe2\x80\xa3 Dynamic Paint \xe2\x80\xa3 Initial Color

Allows you to define the initial color of the canvas. (Todo 2.62)

None

Color

UV Texture

Vertex Color

Reference

Canvas

Physics \xe2\x80\xa3 Dynamic Paint \xe2\x80\xa3 Output

From Output panel you can adjust how surface outputs its results.

For Vertexformat surfaces, you can select a mesh data layer(color/weight depending on surface type) to generate results to.You can use the \xe2\x80\x9c+\xe2\x80\x9d/\xe2\x80\x9d-\xe2\x80\x9d icons to add/remove a data layers of given name.If layer with given name is not found, it is shown as red.

For Image Sequencesurfaces,you can define used UV maps and output file saving directory, filenames and image format.

--- Fin du chapitre: _canvas ---

--- Début du chapitre: _capture_attribute --- The Capture Attributenode stores the result of a field on a geometry,and outputs the data as a node socket so it can be used by other nodes.

The result is stored on the geometry just like a regular attribute with a name, but instead of referencing it with a name, it is retrieved whenever the socket is connected to the input of a node. Later on when evaluating the node tree,the attribute will be removed automatically if it is no longer used.

This node is essential because field input nodes like theRadius Nodework in the context of the node they are connected to. Meaning that in order to pass data likeradiusto a geometry that doesn\xe2\x80\x99t have radius, an explicit node link with the output of this node must be used.

Note

Because this node stores an anonymous attribute in the geometry,it\xe2\x80\x99s essential to use the geometry output for further operations in the node tree.The anonymous attribute will not exist for any other geometry besides the output.

Standard geometry input.

The input field to evaluate.More fields can be added by dragging sockets into the blank socket or in the Capture Itemspanel.Items can be renamed by Ctrl- LMB on the socket name or in the nodes Propertiespanel.

Which attribute domain to store the evaluated data on.

Reference

Sidebar \xe2\x80\xa3 Node \xe2\x80\xa3 Properties \xe2\x80\xa3 Capture Items

Manages the input fields sockets of the node.Field sockets can be added, removed, reorganized, and renamed from theList View.

The data typeused for the evaluated data.

Standard geometry output.

The result of the evaluated field, stored on the geometry.

Here, a noise texture is evaluated in along the path of the curve in one dimensionand rendered with a shader. The capture node is required because the output ofthe Curve to Mesh Node does not have a \xe2\x80\x9ccurve parameter\xe2\x80\x9d,since it is a mesh and not a curve. So, the Spline Parameter Node must be evaluated while the geometry is still a curve.

Internally, after the noise texture is evaluated on the curve,it is automatically copied to the mesh result of the Curve to Mesh node.This means that anywhere Attributeoutput of this node can be connected alongthe same stream of geometry nodes, the internal attribute will be available.

--- Fin du chapitre: `_capture_attribute` ---

--- Début du chapitre: `_cast` --- The Castmodifier shifts the shape of a mesh, curve, surface or lattice, towards any of a few predefined shapes (sphere, cylinder, cuboid).

It is equivalent to the To Spheretool in Edit Mode, and what other programs call `Spherify` or `Spherize`, but, as written above, it is not limited to casting to a sphere.

Tip

The Smooth Modifier is a good companion to Cast, since the cast shape sometimes needs smoothing to look nicer or even to fix shading artifacts.

Note

For performance reasons, this modifier only works with local coordinates. If the modified object looks wrong, you may need to apply its transformations, especially when casting to a cylinder.

The Cast modifier. 

Menu to choose target shape of the projection: Sphere, Cylinder or Cuboid. Toggle buttons to enable/disable the modifier in the X, Y, Z axes directions (X and Y only for Cylindercast type, since the Z axis remains unaffected).

The factor to control blending between original and cast vertex positions.

It is a linear interpolation: 0.0 gives original coordinates (i.e. modifier has no effect), 1.0 casts to the target shape.

Values below 0.0 or above 1.0 exaggerate the deformation, sometimes in interesting ways.

If nonzero, this radius defines a sphere of influence. Vertices outside it are not affected by the modifier.

Alternative size for the projected shape. If zero, it is defined by the initial shape and the control object, if any.

If activated, calculate Size from Radius, for smoother results.

If set, restrict the effect to the only vertices in that vertex group. This allows selective, real-time casting, by painting vertex weights.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier.

The setting reverses the weight values of the group.

The name of an object to control the effect. The location of this object's origin defines the center of the projection. Also, its size and rotation transform the projected vertices.

Hint

Animating (keyframing) this control object also animates the modified object's casting deformation.

Top: Suzanne without modifiers. Middle: Suzanne with each type of Cast Modifier (Sphere, Cylinder and Cuboid). Bottom: Same as above, but now only X axis is enabled. Sample blend-file. 

--- Fin du chapitre: `_cast` ---

--- Début du chapitre: `_catalogs` --- Asset Catalogs help you to organize your assets. They look a little bit like file directories, but they are completely independent of the location of your blend-files. Assign each asset in a blend-file to its own catalog, or have one big catalog with all

the assets of all the blend-files combined. It's all up to you.

Similar to Collections, catalogs can be nested i.e. you can have a main catalog that contains several nested catalogs. For example, this allows you to have a catalog of assets for Furniture with sub-catalogs of Tables, Chairs, Lamps, etc.

For more technical information, see Asset Catalogs on the Blender Developer Documentation.

Example file system and catalog structures.

Example file system and catalog structures.

There can be as many catalogs as you want, but an asset can be assigned to a single catalog at a time. This is similar to a file system, where a file is only in one directory (ignoring advanced things like symbolic links).

Catalogs themselves can be nested and moved by dragging and dropping. Moving a catalog will not modify the assets it contains; they will simply move along to the new location of the catalog.

Selecting a catalog in the Asset Browser will show all assets in that catalog and in child catalogs. So, in the preceding example, selecting Characters/Ellie/Poses will also show assets from Characters/Ellie/Poses/Head and Characters/Ellie/Poses/Hands.

New catalogs can be created in the Asset Browser through Header Catalog New Asset Catalog. Once the catalog is created you can double LMB on it's name in the Source List region of the editor to give the catalog a more descriptive name. Catalogs can also be created in this region by clicking the plus icon found at the top of the tree view.

Assigning a selection of Scale material assets to a catalog.

To assign assets to a catalog, just select and drag the assets on top of the catalog.

Tip

You can assign an asset to the Unassigned catalog, this will remove it from any existing catalogs.

Saving catalogs makes any edits to any catalogs permanent by writing the current set up to the asset library. Catalogs can be saved in the Asset Browser through Header Catalog Save Asset Catalog. Once the catalog is created you can double LMB on it's name. Catalogs can also be saved in the Source List region of the editor by clicking the save icon found at the top of the tree view.

Each catalog consists of a catalog path, a UUID, and a simple name. Normally you would only deal with the catalog path; the rest is for internal Blender use and/or for emergency situations.

The path of a catalog determines where in the catalog hierarchy the catalog is shown. Examples are Characters/Ellie/Poses/Hand or Kitbash/City/Skyscrapers, which would result in the following catalog tree. The highlighted catalog has path Characters/Ellie/Poses/Hand.

Example tree of asset catalogs.

Each catalog has a UUID, which is normally hidden from the user interface (enable Developer Extras and the experimental Asset Debug Info option to see them). This is what is stored in the asset, and what

determines the identity of the catalog. As a result, a catalog can be renamed or moved around (i.e. you can change its path), and all assets that are contained in it will move along with it. This only requires a change to the catalog itself, and not to any asset blend-file.

Each catalog has an optional simple name. This name is stored along with the UUID in each asset. The purpose is to make it possible for humans to recognize the catalog the asset was assigned to, even when the catalog definition file (see below) is lost.

Like the UUID, the simple name is normally hidden from the user interface. Enable Developer Extras in the interface preferences to make it visible in the Asset Browser.

Asset catalogs are stored in Catalog Definition Files (CDFs). Blender 3.0 supports a single CDF per asset library. It is stored in `blender_assets.cats.txt` in the root directory of the asset library. If the file does not exist, Blender will create it when the catalogs are saved. When catalogs are changed, Blender updates that file, but also creates a backup of the previous state to a file named `blender_assets.cats.txt~`.

Asset catalogs can be saved independently of the blend-file; the catalog editor has its own `Save` button.

Catalog Definition Files (CDFs) are relatively simple text files, encoded in UTF-8. Each CDF consists of a version indicator, and a line of text per catalog. Each catalog line is colon-separated, of the form `{UUID}:{path}:{simple name}`.

This is an example of a valid catalog definition file:

Catalog paths follow the following rules:

All paths are absolute; there is no difference between `/a/b` and `a/b`.

Only `/` as separator (no `\\`; think less filesystem path and more URL).

Not empty (it's required for a valid catalog).

No empty components (so not `/b/a/bis` fine).

Invalid characters: `., \`.

Paths are always interpreted as UTF-8.

--- Fin du chapitre: `_catalogs` ---

--- Début du chapitre: `_channels` --- A channel is a horizontal track that's similar to a layer in an image editing program: higher channels are displayed in front of lower ones.

Within each channel, you can create one or more strips, which contain either a segment of video content (a rendered scene, an external video file) or an effect (color blending, blurring). The X axis represents time, so the further a strip is placed to the right, the later it will play in the final video.

While a channel can contain multiple strips, they can't overlap each other. If you want two strips to play at the same time, you need to place them in different channels.

The Channel region sits on the left side of the editor and contains the channel properties listed below. Its visibility can be toggled with View `Channels`.

The name of the channel. Double-click to change.

Disable the entire channel so that none of its strips can be seen (or heard) in the final video. Note that you can also mute individual strips.

Lock the entire channel to protect all its strips against accidental changes. Note that you can also lock individual strips.

--- Fin du chapitre: `_channels` ---

--- Début du chapitre: `_channel_key` --- The Channel Keynode determines background objects from foreground objects by the difference in the selected channel's levels.

For example in the YUV Color Model, this node is useful when compositing stock footage of explosions (very bright) which are normally shot against a solid, dark background.

Standard color input.

This button selects what color model the channels will represent.

RGB, HSV, YUV, YCbCr

This button selects the channel, defined by the Color Space, to use to determine the matte.

Method to calculate the difference between levels.

Limit by the maximum of the other two channels other than the Key Channel.

Limit by the maximum of the selected Limiting Channel.

The channel to use when computing the maximum, the options are defined by the Color Space.

Determines the lowest values that are considered foreground. (Which is supposed to be relatively low values: from this value to 1.0.)

Determines the highest values that are considered to be background objects. (Which is supposed to be relatively high values: from 0.0 to this value.)

Tip

It is possible to have a separation between the High and Low values to allow for a gradient of transparency between foreground and background objects.

Image with an alpha channel adjusted for the keyed selection.

A black-and-white alpha mask of the key.

--- Fin du chapitre: `_channel_key` ---

--- Début du chapitre: `_charge` --- Reference

Physics Force Fields

Charge

A Charge force field is similar to a Force field except it changes its behavior (attract/repulse) based on the effected particles charge field (negative/positive), like real particles with a charge. Which means that this field has only an effect on particles that have also a Charge field (else, they have no charge, and hence are unaffected)!

--- Fin du chapitre: `_charge` ---

--- Début du chapitre: `_checker` --- Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

Tip

Texture nodes can produce details at a higher frequency than geometry can show. This may cause artifacts such as Moiré type patterns or a lack of detail due to insufficient sampling points.

The Checker Texture is used to add a checkerboard texture.

Texture coordinate to sample texture at; defaults to Generated texture coordinates if the socket is left unconnected.

Warning

This node can have precision issues with some vector inputs. See the notes for the White Noise Texture for ways to mitigate this issue.

Color of the checkers.

Overall texture scale. The scale is a factor of the bounding box of the face divided by the scale. For example, a scale of 15 will result in 15 alternate patterns over the overall UV bounding box. Different patterns could be achieved using other nodes to give different input patterns to this socket. For example, using the Math node.

This node has no properties.

Texture color output.

Checker 1 mask (1 = Checker 1).

Default Checker texture. \xc2\xbb6

--- Fin du chapitre: _checker ---

--- Début du chapitre: _checker_deselect ---Reference

Edit Mode

Select \xe2\x80\xa3 Checker Deselect

This tool applies an alternating selected/deselected checker pattern. This only works if you already have more than one mesh element selected.

Changes the current selection so that only every Nth elements (vertices, edges or faces, depending on the active selection mode) will remain selected, starting from the active one.

In case of islands of selected elements, this tool will affect only the island of the active element (if there is one), or the island of the first element in the order of internal storage (if there is no active element).

The number of deselected elements in each pattern repetition.

The number of selected elements in each pattern repetition.

Offset from the starting point.

--- Fin du chapitre: _checker_deselect ---

--- Début du chapitre: _children ---Reference

Particle System \xe2\x80\xa3 Children

See Children.

--- Fin du chapitre: _children ---

--- Début du chapitre: _child_of --- Child Of is the constraint version of the standard parent/children relationship between objects (the one established through the Ctrl-P shortcut, in the 3D Viewport).

Parenting with a constraint has several advantages and enhancements, compared to the traditional method:

You can have several different parents for the same object (weighting their respective influence with the Influence slider).

As with any constraint, you can key (i.e. animate) its Influence setting. This allows the object which has a Child Of constraint upon it to change over time which target object will be considered the parent, and therefore have influence over it.

Important

Do not confuse this \xe2\x80\x93basic\xe2\x80\x9d object parenting with the one that defines the chains of bones inside of an armature. This

constraint is used to parent an object to a bone(the so-called object skinning),or even bones to bones. But do not try to use it to define chains of bones.

Child Of panel. \xc2\xbb

The target object that this object will act as a child of. Data ID used to select the constraint\xe2\x80\x99s target, and is not functional (red state) when it has none.See common constraint properties for more information.

Each of these buttons will make the parent affect or not affect the location along the corresponding axis.

Each of these buttons will make the parent affect or not affect the rotation around the corresponding axis.

Each of these buttons will make the parent affect or not affect the scale along the corresponding axis.

By default, when you parent your owner to your target, the target becomes the origin of the owner\xe2\x80\x99s space.This means that the location, rotation and scale of the owner are offset by the same properties of the target.In other words, the owner is transformed when you parent it to your target.This might not be desired!So, if you want to restore your owner to its before-parenting state, click on the Set Inversebutton.

This button reverses (cancels) the effects of the above one,restoring the owner/child to its default state regarding its target/parent.

Controls the percentage of affect the constraint has on the object.See common constraint properties for more information.

When creating a new parent relationship using this constraint, it is usually necessary toclick on the Set Inversebutton after assigning the parent. As noted above,this cancels out any unwanted transform from the parent, so that the owner returns tothe location/rotation/scale it was in before the constraint was applied.Note that you should apply Set Inversewith all other constraints disabled(their Influenceset to 0.0) for a particular Child Ofconstraint,and before transforming the target/parent (see example below).

About the toggle buttons that control which target\xe2\x80\x99s (i.e. parent\xe2\x80\x99s)individual transform properties affect the owner,it is usually best to leave them all enabled, or to disable all three of the given Location,Rotation and Scale transforms.

If you use this constraint with all channels on,it will use a straight matrix multiplication for the parent relationship,not decomposing the parent matrix into loc/rot/size.This ensures any transformation correctly gets applied,also for combinations of rotated and non-uniform scaled parents.

No constraint.\xc2\xbb

Note the position of Owner empty 1.0 unit along the X and Y axes.

Child Of just added.\xc2\xbb

Here you can see that Owner empty is now 1.0 unit awayfrom Target_1 empty along X and Y axes.

Offset set.\xc2\xbb

Set Inverse has been clicked, and Owner is back to its original position.

Target/parent transformed.\xc2\xbb

Target_1 has been moved along the XY plane, rotated around the Z axis,and scaled along its local X axis.

Offset cleared.\xc2\xbb

Clear Inverse has been clicked. Owner is fully again controlled by Target_1.

Offset set again.\xc2\xba6

Set Offset has been clicked again.As you can see, it does not gives the same result as in (Target/parent transformed).As noted above, use Set Inverse only once, before transforming your target/parent.

--- Fin du chapitre: _child_of ---

--- Début du chapitre: _chroma_key --- The Chroma Keynode determines if a pixel is a foreground or background(and thereby should be transparent) based on its chroma values.

Use this, for example, to composite images that have been shot in front of a green or blue screen.

Standard color input.

The background color usually selected using the color picker and the original image.

An angle on the color wheel that represents how tolerant the keying color is. Larger angles allow for largervariation in the keying color to be considered background pixels.

Controls the level that is considered the pure background. Higher cutoff levels mean more pixels will be100% transparent if they are within the angle tolerance.

Increase to make nearby pixels partially transparent producing a smoother blend along the edges.

Image with its alpha channel adjusted for the keyed selection.

A black-and-white alpha mask of the key.

--- Fin du chapitre: _chroma_key ---

--- Début du chapitre: _circle ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Circle

The Circle tool create oval shapes using any of the Drawtype brushes.

You can configure the brush main settings exposed on the Tool Settings for convenience.For the draw brushes configuration and settings see: Draw Brush .

The number of stroke points between each stroke edge.

Use a curve widget to define the stroke thicknessfrom the start (left) to end (right) of the stroke.

When enabled, the stroke use a curve profile to control the thickness along the line.

Picks the brush asset used by the tool.

SeeBrush Assetfor more information.

SeeDraw Brushesfor a detailed list of all draw brushes and their options.

Parameters to control to look of the stroke.

SeeDraw Brushesfor details.

Settings to determine the color of strokes.

SeeColor

In the Tool Settings select the brush, material and color type to use with the tool.The Circle tool uses Draw Brushtypes.SeeBrush Settingsfor more information.

Click (LMBor the Pentip) and drag the start point.

Release on the desired end point.

After releasing you can move the start and end point by clicking and dragging on the yellow manipulators.

Then confirm (Return/ MMB) or cancel (Esc/ RMB).

While dragging you can use Shift to make a perfect circle or use Alt to create the circle from a center point.

Numpad Plus and Numpad Minus or using the mouse Wheel will increase or decrease the amount of points in the final circle.

Click and dragging the start point. \xc2\xba

Moving start and end points with manipulators. \xc2\xba

The circle after confirming. \xc2\xba

--- Fin du chapitre: _circle ---

--- Début du chapitre: _clamp --- The Clamp node clamps a value between a minimum and a maximum.

The input value to be clamped.

The minimum value.

The maximum value.

Method to clamp.

Constrain values between Min and Max.

Constrain values between Min and Max. When Min is greater than

Max, constrain between Max and Min instead.

The input value after clamping.

The Voronoi Texture node outputs a value whose minimum is zero. We can use the Clamp node to clamp this value such that the minimum is 0.2.

Example of Clamp node. \xc2\xba

--- Fin du chapitre: _clamp ---

--- Début du chapitre: _clamping --- Reference

Render \xe2\x80\xa3 Clamping

This option limits the maximum light intensity a surface can reflect. It reduces Aliasing noise and Fireflies at the cost of accuracy. Setting this option to 0.0 disables clamping altogether. Lower values have a greater effect on the resulting image than higher values.

Similar to Direct Light but limits the maximum light intensity reflected using ray-tracing and light-probes.

Note

These options provide a way to limit Fireflies and Aliasing of highly reflective surfaces and dense volumes. However, note that as you clamp out such values, other bright lights will be dimmed as well.

Care must be taken when using this setting to find a balance between mitigating fireflies and losing intentionally bright parts.

The same as Surface Direct Light but for volume direct lighting.

The same as Surface Direct Light but for volume indirect lighting.

--- Fin du chapitre: _clamping ---

--- Début du chapitre: _clamp_to --- The Clamp To constraint clamps an object to a curve. The Clamp To constraint is very similar to the Follow Path constraint, but instead of using the evaluation time of the target curve, Clamp To will get the actual location properties of its owner (those shown in the Transform panel), and judge where to put it by

\xe2\x80\x9cmapping\xe2\x80\x9d this location along the target curve.

One benefit is that when you are working with Clamp To, it is easier to see what your owner will be doing; since you are working in the 3D Viewport, it will just be a lot more precise than sliding keys around on an F-Curve and playing the animation over and over.

A downside is that unlike in the Follow Path constraint, Clamp To does not have any option to track your owner\xe2\x80\x99s rotation (pitch, roll,

yaw)to the banking of the targeted curve, but you do not always need rotation on,so in cases like this it\’s usually a lot handier to fire up a Clamp To,and get the bits of rotation you do need some other way.

The mapping from the object\’s original position to its position on the curve is not perfect,but uses the following simplified algorithm:

A main axis is chosen, either by the user, or as the longest axis of the curve\’s bounding box (the default).

The position of the object is compared to the bounding box of the curve in the direction of the main axis.So for example if X is the main axis, and the object is aligned with the curve bounding box\’s left side,the result is 0; if it is aligned with the right side, the result is 1.

If the cyclic option is unchecked, this value is clamped in the range 0-1.

This number is used as the curve time, to find the final position along the curve that the object is clamped to.

This algorithm does not produce exactly the desired result because curve time does not map exactly to the main axis position. For example an object directly in the center of a curve will be clamped to a curve time of 0.5 regardless of the shape of the curve,because it is halfway along the curve\’s bounding box.However, the 0.5 curve time position can actually be anywhere within the bounding box!

Clamp To panel. ¶

The target Data ID indicates which curve object the Clamp To constraint will track along.The target must be a curve object type.See common constraint properties for more information.

This button group controls which global axis (X, Y or Z) is the main direction of the path.When clamping the object to the target curve, it will not be moved significantly on this axis.It may move a small amount on that axis because of the inexact way this constraint functions.

For example if you are animating a rocket launch,it will be the Z axis because the main direction of the launch path is up.The default Autooption chooses the axis which the curve is longest in (or X if they are equal).This is usually the best option.

By default, once the object has reached one end of its target curve, it will be constrained there.When the Cyclicoption is enabled, as soon as it reaches one end of the curve,it is instantaneously moved to its other end.This is of course primarily designed for closed curves (e.g. circles),as this allows your owner to go around it over and over.

Controls the percentage of affect the constraint has on the object.See common constraint properties for more information.

--- Fin du chapitre: _clamp_to ---

--- Début du chapitre: _clay ---Reference

Sculpt Mode

Sidebar ¶ Tool ¶ Brush Settings ¶ Advanced ¶ Brush Type

Similar to theDrawbrush,but includes settings to adjust thesculpt planeon which the brush acts. That\’s because it behaves like a combination of theFlattenand Drawbrushes.

This brush is useful for building and removing volumes and shapes like real clay, because it flattens details as you add/subtract from the surfaces.

If used together with Dyntopoit's easy to continuously build shapes, even in a single stroke.

Slightly higher by default. This makes the profile of the brush more noticeable. More info at Hardness

Enabled by default for a consistent smoothing effect. With lower brush strength (for example with lower pen pressure) the smoothing effect will be more noticeable and can be used to create and then blend/polish shapes in a single stroke. Enable Pressure to modulate the use of auto-smooth even more with pen inputs. More info at Auto-Smooth

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _clay ---

--- Début du chapitre: _clay_strips ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Similar to the Claybrush, but it uses a square tip shape instead of a round one.

Just like the Claybrush, it's useful for building and removing volumes and shapes like real clay, because it flattens details as you add/subtract from the surfaces.

Clay Strips is very commonly used for aggressive building of volumes and deliberate control over shapes on the surface. This brush alone can be used for a fast rough pass over the entire sculpt, with additional smoothing or polishing often required afterwards. This brush can be very versatile with varying stroke directions, repeated strokes and pen pressure to achieve various results.

If used together with Dyntopoit's easy to continuously build shapes, even in a single stroke.

Higher by default. This ensures that the brush does not change directions to sporadically during a stroke. More info at Normal Radius

Very low by default for a square shape for more deliberate shaping. More info at Tip Roundness

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _clay_strips ---

--- Début du chapitre: _clay_thumb ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Similar to the Claybrush. It imitates the effect of deforming clay with the finger, accumulating material during the stroke. The sculpt plane tilts during the stroke in the front part of the brush to achieve this effect.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _clay_thumb ---

--- Début du chapitre: _cleanup --- These tools are to help cleanup degenerate geometry and fill in missing areas of a mesh.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Decimate Geometry

The Decimate Geometry tool allows you to reduce the vertex/face count of a mesh with minimal shape changes.

Ratio of triangles to reduce to.

Use the active vertex group as an influence.

Strength of the vertex group.

Inverts the vertex group.

Maintain symmetry on either the X, Y, or Z axis.

See also

This tool works similar to the Decimate Modifier.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Fill Holes

This tool can take a large selection and detect the holes in the mesh, filling them in.

This is different from the face creation operator in three important respects:

Holes are detected, so there is no need to manually find and select the edges around the holes.

Holes can have a limit for the number of sides (so only quads or tris are filled in for example).

Mesh data is copied from surrounding geometry (UVs, Color Attributes, multi-res, all layers), since manually creating this data is very time-consuming.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Make Planar Faces

The Make Planar Faces iteratively flattens faces. This can happen with faces over three vertices and it is a common convention that faces should be kept planar.

Distance to move the vertices each iteration.

Number of times to repeat the operation.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Split Non-Planar Faces

This tool avoids ambiguous areas of geometry by splitting non-flat faces when they are bent beyond a given limit.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Split Concave Faces

This tool can be used to convert any Concave Face to convex by splitting the concave into two or more convex faces.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Delete Loose

This tool removes disconnected vertices and edges (optionally faces).

Reference

Edit Mode

Mesh \xe2\x80\xa3 Clean up \xe2\x80\xa3 Degenerate Dissolve

This tool collapses / removes geometry which you typically will not want.

Edges with no length.

Faces with no areas (faces on a point or thin faces).

Face corners with no area.

Reference

Edit Mode

Vertex \xe2\x80\xa3 Merge by Distance, Context Menu \xe2\x80\xa3 Merge by Distance

Merge by Distance is a useful tool to simplify a mesh by merging the selected vertices that are closer than a specified distance to each other. An alternative way to simplify a mesh is to use the Decimate Modifier .

Sets the distance threshold for merging vertices.

Allows vertices in the selection to be merged with unselected vertices. When disabled, selected vertices will only be merged with other selected ones.

--- Fin du chapitre: _cleanup ---

--- Début du chapitre: _clear ---Reference

Pose Mode

Pose \xe2\x80\xa3 Clear Transform

Once you have transformed some bones, if you want to return to their rest position, just clear their transformations.

Resets location, rotation, and scaling of selected bones to their default values.

Clears individual transforms.

Note that in Envelope visualization, Alt-S does not clear the scale, but rather scales the Distance influence area of the selected bones. (This is also available through the Pose \xe2\x80\xa3 Scale Envelope Distance menu entry, which is only effective in Envelope visualization, even though it is always available \xe2\x80\xa6)

Clears the transforms to their keyframe state.

Operate on just the selected or all bones.

--- Fin du chapitre: _clear ---

--- Début du chapitre: _clip --- Clip can be modified within the Movie Clip Editor.

Reference

Sidebar region \xe2\x80\xa3 Strip \xe2\x80\xa3 Movie Clip

Used to select the movie clip. For controls see Data-Block Menu .

Use the 2D stabilized version of the clip.

Use the undistorted version of the clip.

Below the properties, shows the frame range of the movie clip (before strip adjustments).

--- Fin du chapitre: _clip ---

--- Début du chapitre: _clip_display --- This pop-over contains various display settings for both Tracking mode and Mask mode.

Controls the color channels used for the frame preview. The tracking algorithm works with grayscale images, and with these options, you can check which combination of enabled and disabled channels will yield the best contrast and the least noise.

Note that this only affects the preview. To select which channels to use for the actual tracking, use the Track tab in the Toolbar to set a

default for newly created markers, or the Track tab in the Sidebar to configure existing markers.

Shows the whole frame as a grayscale image.

Hides the movie clip and displays a black image instead. This helps to find markers that are tracked inaccurately or not at all.

Applies the Lens settings to the video preview to undo lens distortion. Does not change the footage itself.

Applies the 2D stabilization settings to the video preview. Does not change the footage itself.

Displays a grid which is originally orthographic, but is distorted by the Lens settings. This can be used for manual calibration: the distorted grid lines should match lines in the footage that are meant to be straight.

Applies the Lens settings to annotation strokes. Like the Grid, this option also helps to perform manual calibration.

Changes the aspect ratio for displaying only. It does not affect the tracking or solving process.

Determines how markers are displayed in the editor.

Whether to show the pattern areas of tracks. Can be used to reduce clutter and check how good tracking is.

Whether to show the search areas of selected tracks. Can be used to reduce clutter and check how good tracking is.

Shows past (red) and future (blue) positions of tracks relative to the current frame, visualizing how they move. This makes it easier to spot irregularities.

Length (in frames) of the Path.

When unchecked, hides the tracks that are disabled on the current frame (except for the active track, i.e. the one that was selected last). This helps to make the view more clear and see if the tracking is accurate enough.

Displays the name and status of each selected track. The status can be `keyframed`, `tracked`, `disabled` and so on.

Shows the result of solving the markers' 3D locations based on their 2D movement. Each 3D location is projected back to the movie clip and displayed as a small point, which is colored green if it's close to the original 2D marker (meaning a good solve) or red if it's far away (meaning it needs to be tweaked).

By default, marker areas are displayed as bright boxes with a black outline. This option displays them using thin dashed lines instead.

--- Fin du chapitre: `_clip_display` ---

--- Début du chapitre: `_cloth` ---

Reference
Sculpt Mode

Sidebar `Tool` `Brush Settings` `Brush Type`
Advanced `Brush Type`

This brush simulates cloth physics on the mesh under the brush cursor. There are various deformation types and settings to customize the brush.

It's also easy to sculpt the mesh with other brushes and tools in between using the cloth brushes.

Note

Using a relatively small brush size makes the calculations much faster, while larger brush sizes might be too slow to get a usable brush.

Note

More info at General brush settings and on Advanced brush settings.

Allows the cloth brush to not accumulate deformation after each stroke. This is convenient to always simulate the cloth based on the same initial shape, but applying different forces to it.

When disabled, deformations accumulate after each stroke.

Resets the base mesh so that you can add another layer of deformations.

Selects the part of the mesh that is going to be simulated when the stroke is active. This can greatly affect performance depending on the complexity of the mesh.

Simulates only a specific area around the brush limited by a fixed radius.

Simulates the entire mesh.

The active simulation area moves with the brush while still being limited by a fixed radius.

The Factor added relative to the size of the radius to limit the cloth simulation effects.

The area to apply deformation falloff to the effects of the simulation. This setting is a factor of the Simulation Limit and is shown as a dashed line around the cursor.

Lock the position of the vertices in the simulation falloff area to avoid artifacts and create a softer transition with unaffected areas.

The type of cloth deformation that is used by the brush.

Simulates pulling the cloth to the cursor, similar to placing a finger on a table cloth and pulling.

Simulates pushing the cloth away from the cursor, similar to placing a finger on a table cloth and pushing.

Simulates pulling the cloth into a point.

Simulates pulling the brush into a line.

Simulates air being blown under the cloth so that the cloth lifts up.

Simulates picking up and moving the cloth.

Simulates stretching the cloth out.

Simulates moving the cloth without producing any artifacts in the surface and creates more natural looking folds than any of the other deformation modes. This is accomplished by adjusting the strength of the deformation constraints per brush step to avoid affecting the results of the simulation as much as possible.

Shape used in the brush to apply force to the cloth.

Applies the force as a sphere.

Applies the force as a plane.

Mass of each simulation particle.

How much the applied forces are propagated through the cloth.

The amount the cloth preserves its original shape, acting as a Soft Body.

Enables the detection of collisions with other objects during the simulation. In order for the sculpt object to collide with objects, the collision object must have Collision Physics activated.

--- Fin du chapitre: _cloth ---

--- Début du chapitre: _cloth_filter ---Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Cloth Filter

This tool works similar to the Cloth Brush, however, it applies a cloth simulation to all vertices in the mesh at the same time. Click and drag

away from the object for a positive effect and towards for a negative effect.

Tip

Vertices can be pinned by masking vertices that should remain stationary, or by using Face Sets .

Operation that is going to be applied to the mesh.

Applies gravity to the simulation.

Inflates the cloth.

Expands the cloth's dimensions.

Pinches the cloth to the point where the cursor was when the filter started.

Scales the mesh as a Soft Body using the distance to the origin of the object as scale. This creates filter produces folds in the surface. The orientation of the folds can be controlled using the Force Axis and Orientation.

The amount of effect the filter has on the mesh.

Apply the force along the selected axis.

Orientation of the axis to limit the filter force.

Use the local axis to limit the force and set the gravity direction.

Use the world axis to limit the force and set the gravity direction.

Use the view axis to limit the force and set the gravity direction.

Mass of each simulation particle.

How much the applied forces are propagated through the cloth.

Only applies the cloth forces to the vertices assigned to the Face Set that are under the mouse.

Enables the detection of collisions with other objects during the simulation. In order for the sculpt object to collide with object, the collision object must have Collision Physics activated.

--- Fin du chapitre: _cloth_filter ---

--- Début du chapitre: _cloth_sculpting --- Instead of sculpting cloth manually or creating complex physics simulation setups, there are various tools directly in sculpt mode that offer a simplified Cloth Physics Simulation.

This has various advantages but is especially useful for base mesh creation and larger clothing folds and draping. Detailing is possible, but the slower performance on high resolution meshes and simplified cloth physics might not lead to desirable results.

The resolution of the topology is mainly responsible for the size of the folds and detail level of the simulation. So an optimal and evenly distributed topology is important.

Many sculpting features are supported, so for example masked vertices are pinned in the simulation. Another example is with auto-masked face set boundaries. The sculpt mode gravity factor is also applied on the cloth physics.

The main brushes and tools for this feature are the Cloth Brush and Cloth Filter, but other transform brushes like Pose and Boundary also support cloth sculpting in the brush settings.

A demo file for trying out the various brushes and tools is available here.

--- Fin du chapitre: _cloth_sculpting ---

--- Début du chapitre: _clouds --- Clouds represent Perlin noise. In addition, each noise-based Blender texture (with the exception of Voronoi

and simple noise) has a Noise Basis setting that allows the user to select which algorithm is used to generate the texture. This is often used for Clouds, Fire, Smoke. Well-suited to be used as a Bump map, giving an overall irregularity to the material.

Clouds Texture panels.

The standard noise, gives an intensity.

The noise gives an RGB value.

Soft or Hard, changes contrast and sharpness.

The dimension of the Noise table.

The depth of the Clouds calculation. A higher number results in a long calculation time, but also in finer details.

--- Fin du chapitre: _clouds ---

--- Début du chapitre: _clump_hair_curves --- Clumps together existing hair curves using guide curves.

Input Geometry (only curves will be affected).

Guide index map which describes which curve to use as the center of each braid group. If this input is provided, it has priority over an existing map in the guide_curve_index attribute, and the Guide Distance and Guide Mask attribute will be unused.

Minimum distance between two guides for new guide map.

Mask for which curve are eligible to be selected as guides.

Use the existing guide map attribute if available. If this is false, and the Guide Index input isn't provided, the Guide Distance and Guide Mask input will be used to generate a new guide map for this node. Creating the guide map in a separate node or modifier gives more complete control over its creation.

Factor to blend overall effect.

Shape of the influence along curves (0=constant, 0.5=linear).

Distance of random spread at the curve tips.

Offset of each clump in a random direction.

Falloff distance for the clumping effect (0 means no falloff).

Distance threshold for the falloff around the guide.

Random seed for the operation.

Preserve each curve's length during deformation.

This node has no properties.

Geometry

Guide index map that was used for the operation. If a new guide map is created by this node, it will be stored for this output.

--- Fin du chapitre: _clump_hair_curves ---

--- Début du chapitre: _collada --- Important

COLLADA I/O support is now considered as a legacy feature in Blender, and will be removed in a future release. Please see the official announcement for more insight on this topic.

The COLLADA module has been implemented as a flexible tool for exporting and importing .dae files. A design goal is to provide a set of parameters which should make it possible to export/import Collada files from/to a variety of tools. But please be aware that the Collada module is still a work in progress. So it may be possible that your particular usage scenario is not yet supported.

Reference

File \Export Collada (.dae) (Legacy)

Operator Presets

There are two operator presets (see top of the Sidebar) for Second Life (SL) users:

Second Life Static is good for exporting static meshes.

Second Life Rigged is good for exporting the SL default character.

Note

Special Notes for Second Life users:

Please use the Operator presets. All other export settings will not work for Second Life.

The character orientation needs to be such that the character looks towards positive X.

Scale and Rotation must be applied before the export!

When Selection Only is enabled, then only the selected objects will be exported. Otherwise the entire scene is exported with all visible and all invisible objects.

When this option is enabled then all children of the selected objects will also be exported regardless of their selection state.

Hint

You can select only an armature, then using this option, all rigged meshes attached to the armature will also be exported.

When this option is enabled, then all armatures related to the selected objects will also be exported regardless of their selection state.

Hint

You can select only the objects, then in the exporter enable this option to export the armature data also.

Includes the application of shape keys by exporting meshes with the current shape key configuration baked in.

Rotate all root objects to match the global orientation settings otherwise set the global orientation per Collada asset.

Since many applications use a different axis for pointing upwards, these are axis conversion for these settings. Forward and up axes By mapping these to different axes you can convert rotations between applications default up and forward axes.

Blender uses Y forward, Z up (since the front view looks along the +Y direction). For example, it is common for applications to use Y as the up axis, in that case -Z forward, Y up is needed.

When you export images either material based image textures, then the exporter creates absolute file references in the export file.

But if the Copy option is enabled, the exporter will create copies of the images instead and place the copies besides the export file. In that case the file references are made relative.

When your mesh contains multiple UV layers, then Blender exports all layers by default. This option allows you to only export the active (selected) UV layer.

The mesh can be triangulated on-the-fly. The triangulation is based on the same function which is used in the Triangulate Faces tool for triangulating the current selection of faces. For full control over the triangulation you can do this manually before exporting. However, this option allows to apply the triangulation only on the exported data; the mesh itself is not affected.

Export objects using the evaluated mesh, meaning the resulting mesh after all Modifiers have been calculated.

Controls whether to apply the 3D Viewport resolution or the render resolution for modifiers that provide a preview mode and a render mode.

Collada supports two types of transformation matrix specifications. Either as <Matrix> or as a set of transformation decompositions (for move, rotate and scale). Note that the exporter will not strictly follow this option setting, but will rather take it as a hint to use the option if ever possible. This is so because some of the exported data types have specific rules about how the transformation matrix has to be exported. This is ongoing development and a less ambiguous method may be provided in the future.

When this option is enabled, then the exporter strips all non-deforming bones from the exported armatures. This option is useful when your armatures contain control bones which are not actually part of the character skeleton. For example you can export the Avastar rig with this option enabled. The resulting exported rig is compatible with Second Life. But please note the restrictions further below.

When this option is enabled, some issues with bone orientation are calculated differently and is designed to be used to export to Second Life or OpenSim.

This is only relevant for rigged meshes, for static meshes it just does nothing at all.

In Blender you can reuse the same mesh for multiple objects. This is named \xe2\x80\x9cobject instantiation\xe2\x80\x9d. When you enable this option, then Blender will propagate object instantiation to the Collada file.

Collada can be extended with tool specific data (profiles). Blender has its own (unofficial) profile that allows to export rig information into the Collada file. Later it can be used to reconstruct the rig when it should ever be necessary to import a dae file back into Blender.

The export order of data is bound to internal object order and it can not be influenced in a reliable way. This option ensures that the Geometry nodes and the Object nodes are both exported in alphabetical order.

When a rig is imported to Blender, the rig\xe2\x80\x99s bind pose will be used as Blender\xe2\x80\x99s rest pose. So all Matrix information of the original rest pose is lost. But in some cases you may want to preserve the original rig information. This option checks each bone for having two arrays:

rest_mat \xe2\x80\x93 an array of 16 floats which represent the bone\xe2\x80\x99s original rest-pose matrix.

bind_mat \xe2\x80\x93 an array of 16 floats which represent the bone\xe2\x80\x99s original bind-pose matrix.

If the arrays are present, then those arrays will be used instead of the current rest pose/bind pose. Those two arrays are either created by a previous Collada import (see Collada Importer below), or they can be created manually, or by an add-on (script based).

Reference

File \xe2\x80\x93 Export Collada (.dae) (Legacy)

The Collada importer is mostly driven by the imported data. There is one option for controlling the import units:

Use the mesh normals defined in the collada file, if they are defined, otherwise Blender will recompute them during the import process.

If not enabled the imported data will be rescaled according to the currently used unit system. If this option is enabled, then Blender will adjust itself to the unit system as provided by the Collada file.

Collada only records \xe2\x80\x9cjoints\xe2\x80\x9d which is mostly similar to Blender\xe2\x80\x99s bone heads. But when you import a Collada

file then the bone head/tail are not defined. This does not matter for connected bones where the bone parent only has one child. In that case the parent bone's end location is adjusted to the child's joint position. But especially for unconnected bones and for bones with more than one child a problem arises.

When the Fix Leaf Bone option is enabled then Blender tries to guess where the bone head/tail of unconnected bones would best be placed. If the option is disabled, then the bone head/tail are placed at an offset along the Y axis. That is why bones often point towards the Y axis.

When a bone has multiple children, then it is not defined which (if any) of the children should be connected to the bone. When the Find Bone Chain option is enabled, then Blender determines the longest bone chain (of children) for each bone. All bones along this chain will then be auto connected.

If the option is disabled, then children will only be connected to parents, if the parent has only one child. But see the Auto Connect option below.

When this option is enabled, then children will automatically be connected to their parents, if the parent has only one child.

When this option is enabled, then the importer creates two custom properties for each bone:

rest_mat an array of 16 floats which represent the bone's original rest-pose matrix.

bind_mat an array of 16 floats which represent the bone's original bind-pose matrix.

Those two arrays can later be used when you want to export the rig again and be sure the original rest pose/bind pose combination must be used.

Supported geometry types are:

Tris (not tested)

Polylist

Polygons

N-gons

Tri-fans (not tested)

Lines

Mesh data is exported as <polylist> , <lines> and <vertices> .

Blender does a best effort on importing lights from a dae-file. If a Blender profile is detected for lights, all values from these will be used instead. This ensures full re-import from a Blender exported dae-file. <extra> support has been added in Blender 2.57.

A Blender profile for lights has been added through the <extra> tag. The entire Light struct from Blender will be exported through this profile, with the exception of light curve falloff.

Support for object (mesh, camera, light) transform animations. Only Euler rotations, which is the default option for Objects, can be exported. For armature bone animations, Euler and quaternion rotation types are supported.

Import and export of animations for the following parameters are supported:

Light

Camera

Material effects

Non-skin controlling armature bone animation.

Animations of armatures with skin deforming bones.

Animations of armatures in Object Mode.

Fully rigified armature animations (referring to the Rigify add-on). For export of rigified armature animations:

Run the Bake Action operator.

If you have only the deform bones selected check Only Selected. This will give smaller dae. Otherwise uncheck Only Selected.

Check Clear Constraints.

Bake Action.

Select the mesh and the deform bones. Then export to Collada while checking only selected option. (Selecting only the Mesh and bones is not strictly necessary. Selecting and export only selected will give smaller dae.)

Demonstration video

For bone nodes which are leaf nodes in the armature tree, or if a bone has more than one child, a Blender profile for tip with an <extra> tag, is added for those joint nodes. To correctly derive the bone-to-tail location on re-import.

Note

Important Things to Remember

Object and data-block names are constrained to 21 characters (bytes).

UV layer names are constrained to 32 characters (bytes).

Only armature animation on mesh, single skin controller.

No support for modifiers yet.

When importing a dae-file that has <instance_node> on exporting this information is essentially lost and these nodes will be <node>s.

--- Fin du chapitre: _collada ---

--- Début du chapitre: _collection ---Reference

Object Mode

Properties \xe2\x80\xa3 Object Properties \xe2\x80\xa3 Instancing
\xe2\x80\xa3 Collection

Instance Collections allows you to create an instance of a collection for each instance of another object. Collections may contain animations, objects with physics simulations and even other nested collections. Create a new collection (this can be done via the Outliner). Link the objects that need to be instanced as part of the newly created collection.

Add \xe2\x80\xa3 Collection Instance

At this point, an instance of the collection and an empty object will appear. You can duplicate the empty, and the Instance Collections settings will be preserved for each empty. This way, you can get multiple copies of linked data very easily.

See Appending and Linking to understand how to dynamically link data from another blend-file into the current file. You can dynamically link collections from one blend-file to another. When you do so, the linked collection does not appear anywhere in your scene until you create an object controlling where the collection instance appears.

If you want to make further edits on an instanced collection select the Instance Collection. Then call Make Instances Real to convert the collection into regular objects that can be transformed and animated normally.

Note

Note that if the instanced collection was linked from an external file, the Object Data (mesh, materials, textures, transforms) will also still be

linked from the original collection. However, the various object's parent-child relationships do not carry over.
--- Fin du chapitre: _collection ---

--- Début du chapitre: _collections ---Reference

Domain

Properties \xa3 Physics \xa3 Fluid \xa3

Collections

If set, only objects in the specifiedCollection will be allowed to act as Flow objects in this domain.

If set, only objects in the specifiedCollection will be allowed to act as Effector objects in this domain.

--- Fin du chapitre: _collections ---

--- Début du chapitre: _collection_info --- The Collection Info node gets information from collections. This can be useful to control parameters in the geometry node tree with an external collection.

Tip

A Collection Info node can be added quickly by dragging a collection into the node editor.

Collection to get the properties from.

Output each child of the collection as a separate instance. The list of instances will be sorted alphabetically with the objects and child collections sorted together. This can be used with the Pick Instance option in the Instance on Points Node to choose between collection children at each point.

Remove the transform of each of the collection's children when converting them to instances. This is useful in order to keep child objects visually separate in the viewport, while keeping every instance located directly at the location of the point it was added for.

The transformation of the output instances. The instances are transformed, but not the geometry of the collection in them.

Output the instances relative to the collection offset.

Join the input collection instances with the modified object as geometry, maintaining the relative position between the objects in the scene.

Instances of the collection in world space with all modifiers applied and represented as geometry in instances.

--- Fin du chapitre: _collection_info ---

--- Début du chapitre: _collision ---Reference

Object Mode

Physics \xa3 Collision

Particles, Soft Bodies and Cloth objects may collide with mesh objects. Boidstry to avoid Collision objects.

You may limit the effect on particles to a group of objects (in the Field Weights panel).

Deflection for soft body objects is difficult, they often penetrate the colliding objects.

Hair particles ignore deflecting objects (but you can animate them as soft bodies which take deflection into account).

If you change the deflection settings for an object you have to recalculate the particle, soft body or cloth system by Delete Bake, this is not done automatically.

A collider object can be temporarily disabled via an animatable toggle to the right of the button that permanently activates or deactivates it. Collision panel.

A deflector can also deflect effectors. You can specify some collision/deflector objects which deflect a specific portion of the effector force using the Field Absorption value. 100% absorption results in no force getting through the collision/deflector object at all. If you have three collision objects behind each other with e.g. 10%, 43% and 3%, the absorption ends up at around 50% $((100 \times (1 - 0.1) \times (1 - 0.43)) \times (1 - 0.03))$.

Fraction of particles passing through the mesh.

How much particles stick to the object.

Deletes Particles upon impact.

Damping during a collision (independent of the velocity of the particles).

Random variation of damping.

Friction during movements along the surface.

Random variation of friction.

It is also important to note that this collision panel is used to tell all simulations that this object is to participate in colliding/deflecting other objects on a shared layer (particles, soft bodies, and cloth).

Note

The object's shape deforms the cloth, so the cloth simulation must be inputted the true shape of that mesh object at that frame. This true shape is the basis shape as modified by shape keys or armatures. Therefore, the Collision Modifier must be after any of those. The image to the right shows the Modifiers panel for the Character mesh object (not the cloth object).

Damping during a collision. The amount of bounce that the surfaces will have.

0.0 - No damping, soft bodies will have a maximum bounciness.

1.0 - Maximum damping, soft bodies will not bounce at all.

A padding distance is added to the inside and outside of each face, to help to prevent intersections. The soft body will come to rest at this distance away from the face of the colliding object. Outside and inside is defined by the face normal, depicted as blue arrow in Fig. A soft body vertex colliding with a plane.

Size of the outer collision zone.

Size of the inner collision zone (padding distance).

A soft body vertex colliding with a plane.

A coefficient for how slippery the cloth is when it collides with itself. For example, silk has a lower coefficient of friction than cotton.

When enabled, the collider is considered to represent the boundary of a solid object rather than a thin surface, and ejects intersecting cloth in the direction of its normal.

When enabled, cloth collision impulses act in the direction of the collider normals.

Note

Soft body collisions are difficult to get perfect. If one of the objects move too fast, the soft body will penetrate the mesh. See also the section about Soft Bodies.

Deflected particles.

Here is a Metaobject, using Instancing Vertices to a particle system emitting downwards, and deflected by a mesh cube. Make sure that the normals of the mesh surface are facing towards the particles/points for correct deflection. Negative scales on the object can have a similar effect. Make sure to recalculate the normals after applying the scale.

Hair particles react directly to force fields, so if you use a force field with a short range you do not need necessarily collision.

Hair particles avoid their emitting mesh if you edit them in Particle Edit Mode. So you can at least model the hair with collision.

--- Fin du chapitre: _collision ---

--- Début du chapitre: _collisions ---Reference

Physics \xe2\x80\xa3 Cloth \xe2\x80\xa3 Collision

In most cases, a piece of cloth does not just hang there in 3D space, it collides with other objects in the environment. To ensure proper simulation, there are several items that have to be set up and working together:

The Clothobject must be told to participate in collisions.

Optionally (but recommended) tell the cloth to collide with itself.

Other objects must be visible to the Clothobject via shared layers.

The other objects must be mesh objects.

The other objects may move or be themselves deformed by other objects (like an armature or shape key).

The other mesh objects must be told to deflect the cloth object.

The blend-file must be saved in a directory so that simulation results can be saved.

You then Bake the simulation. The simulator computes the shape of the cloth for a frame range.

You can then edit the simulation results, or make adjustments to the cloth mesh, at specific frames.

You can make adjustments to the environment or deforming objects, and then re-run the cloth simulation from the current frame forward.

Cloth Collisions panel.\xc2\xb6

A general setting for how fine and good a simulation you wish. Higher numbers take more time but ensure less tears and penetrations through the cloth.

If the cloth object needs to be deflected by some other object. To deflect a cloth, the object must be enabled as an object that collides with the cloth object. To enable objects to collide with cloth objects enable collision physics for the collider object (not on the cloth object).

Note

If your colliding object is not a mesh object, such as a NURBS surface, or a text object, you must convert it to a mesh object using Convert. The distance another object must get to the cloth for the simulation to repel the cloth out of the way. Smaller values might give errors but gives some speed-up while larger will give unrealistic results if too large and can be slow. It is best to find a good in between value.

Prevents explosions in tight and complicated collision situations by restricting the amount of movement after a collision.

Faces that have all vertices assigned to this Vertex Group are excluded from collision with objects.

Only objects that are a part of this Collection can collide with the cloth. Note that these objects must also have Collision physics enabled. Real cloth cannot penetrate itself, so you normally want the cloth to self-collide. Enable this to tell the cloth object that it should not penetrate itself. This adds to the simulation's compute time, but provides more realistic results.

Tip

A flag, viewed from a distance does not need this enabled, but a close-up of a cape or blouse on a character should have this enabled.

A coefficient for how slippery the cloth is when it collides with itself. For example, silk has a lower coefficient of friction than cotton. As cloth at this distance begins to repel away from itself. Smaller values might give errors but gives some speed-up while larger will give unrealistic results if too large and can be slow. It is best to find a good in between value.

Prevents explosions in tight and complicated collision situations by restricting the amount of movement after a collision.

Faces that have all vertices assigned to this Vertex Group are excluded from self-collision.

See also

Example blend-file: Cloth self-collisions.

If you encounter some problems with collision detection, there are a few ways to fix them:

The fastest solution is to increase the Distance for Object/Self Collisions. This will be the fastest way to fix the clipping; however, it will be less accurate and will not look as good. Using this method tends to make it look like the cloth is resting on air, and gives it a very rounded look.

A second method is to increase the Quality (in the Cloth panel). This results in smaller steps for the simulator and therefore to a higher probability that fast-moving collisions get caught. You can also increase the Collisions Quality to perform more iterations to get collisions solved.

If none of the methods help, you can easily edit the cached/baked result in Edit Mode afterwards.

If the Cloth is torn by the deforming mesh; increase the stiffness settings.

--- Fin du chapitre: _collisions ---

--- Début du chapitre: _color --- The Color node outputs the color value chosen with the color picker widget.

This node has no inputs.

This node has no properties.

Color value indicated by the color picker widget.

--- Fin du chapitre: _color ---

--- Début du chapitre: _colorize --- The Colorize Visual Effect applies different preset colorizing effects to the object.

Blur Visual Effect. \xc2\xb6

Converts to a grayscale image.

Converts to a sepia tone image.

Converts to a posterize image with high contrast and brightness.

Primary color.

Secondary color.

Add color transparency.
Allows to define a tint custom color.
Sets the tint color.
Control the mix value.

Mode: Grayscale.\xc2\x66
Mode: Sepia.\xc2\x66
Mode: Duotone.\xc2\x66
Mode: Transparent.\xc2\x66

--- Fin du chapitre: _colorize ---

--- D but du chapitre: _colors --- The color of a texture can be modified with the Brightness, Contrast, and Saturation buttons. All textures with RGB values, including Images and Environment Maps, may be modified with the RGB sliders.

Set negative texture RGB and intensity values to zero, for some uses like displacement this option can be disabled to get the full range.

Tint the color of a texture by brightening each red, green and blue channel.

Change the overall brightness/intensity of the texture.

Change the contrast of the texture.

Change the saturation of the texture.

Activates a color ramp which allows you to remap the colors of a texture to new ones.

--- Fin du chapitre: _colors ---

--- D but du chapitre: _color_balance --- The Color Balance node adjusts the color and values of an image.

Controls the amount of influence the node exerts on the output image.

Standard color input.

The mathematical method to adjust the image's colors.

Adjusts the colors and tonal range of an image by controlling the shadows, midtones, and highlights separately.

Adjusts the darkest areas of the image (the shadows).

Primarily affects the midtones, the middle range of brightness in the image.

Controls the brightest parts of the image (the highlights).

A standardized model for adjusting the colors and tonal range of an image. This allows the same values to be used across different applications to yield the same result. See Advanced for more details on the underlying implementation.

Adjusts the darkest areas of the image (the shadows).

Additional offset, allows to specify a negative offset value.

Primarily affects the midtones, the middle range of brightness in the image.

Controls the brightest parts of the image (the highlights).

Adjusts the color that should be considered white. The white point is specified as setting the input's color temperature and then the desired output temperature.

The blackbody temperature of the primary illuminant. By default a D65 white point is used.

The amount of green/magenta shift of the blackbody curve.

Standard output image.

$$I_{out} = (I_{in} \times s + o)^p$$

where:

out: The color graded pixel code value.
i: The input pixel code value (0 to 1) (black to white).
s: Slope (any number 0 or greater, nominal value is 1.0).
o: Offset (any number, the nominal value is 0).
p: Power (any number greater than 0, nominal value is 1.0).
--- Fin du chapitre: _color_balance ---

--- Début du chapitre: _color_correction --- The Color Correction node adjusts the color of an image, separately in several tonal ranges (highlights, midtones and shadows). Specifies which RGB channels will be affected by the correction.
Adjusts the image's saturation.
Adjust image contrast.
Exponential gamma correction, affecting the midtones of the image. (Works like Power in the Color Balance node.)
Multiplier, stronger influence on the highlights. (Works like Slope in the Color Balance node.)
This value (can be negative) will be added (+), linear lightens or darkens the image. (Works like Offset in the Color Balance node.)
These sliders affect the entire tonal range.
These sliders only affect the highlights.
These sliders only affect the midtones.
Affects the dark tones of an image often affecting the shadows.
Defines the start and the end of midtones range, i.e. values where the whole tonal range is divided into the highlights, midtones and shadows (there is also a smooth transition between the ranges of width 0.2 units).
Standard color input.
Controls the amount of influence the node exerts on the output image.
Standard color output.
--- Fin du chapitre: _color_correction ---

--- Début du chapitre: _color_filter --- Reference
Sculpt Mode
Toolbar \ Color Filter
Apply color corrections or effects on the active color attribute on all vertices in the mesh at the same time.
To use this tool, click and drag away from left to right or from right to left for a negative effect.
Fills in a single color.
Shifts the Hue of each color.
Increases or decreases the saturation.
Increases or decreases the values.
Increases or decreases the brightness.
Increases or decreases the contrast.
Blurs or sharpens the colors.
Increases or decreases the red channel.
Increases or decreases the green channel.
Increases or decreases the blue channel.
Set a color that will be used for the fill filter type.
The amount of effect the filter has on the color attribute.
--- Fin du chapitre: _color_filter ---

--- Début du chapitre: `_color_key` --- The Color Key node creates a matte based on a specified color of the input image.

Standard color input.

The sliders represent threshold values. Higher values in this node's context mean a wider range of colors from the specified will be added to the matte.

Hue, Saturation, Value

Image with its alpha channel adjusted for the keyed selection.

A black-and-white alpha mask of the key.

--- Fin du chapitre: `_color_key` ---

--- Début du chapitre: `_color_management` --- Color management is important to create renders and assets that are physically accurate and look great on multiple display devices. It is used both to ensure all parts of the pipeline interpret colors correctly, and to make artistic changes like exposure and color grading.

Different views and exposures of the same render.

Blender's color management is based on the OpenColorIO library. By using the same OpenColorIO configuration in multiple applications, the same color spaces and transforms will be available for consistent results.

For correct results, different Color Spaces are needed for rendering, display and storage of images. Rendering and compositing is best done in scene linear color space, which corresponds more closely to nature, and makes computations more physically accurate.

An example of a linear workflow.

If the colors are linear, it means that if in reality, we double the number of photons, the color values are also doubled. Put another way, if we have two photos/renders each with one of two lights on, and add those images together, the result would be the same as a render/photo with both lights on. It follows that such a radiometrically linear space is best for photorealistic rendering and compositing.

However, these values do not directly correspond to human perception or the way display devices work. and image files are often stored in different color spaces. So we have to take care to do the right conversion into and out of this scene linear color space.

Transforming scene linear colors to display involves both technical and artistic choices.

Correct display of renders requires a conversion to the display device color space. A computer monitor works differently from a digital cinema projector or HDTV, and so needs a different conversion.

There is also an artistic choice to be made. Partially that is because display devices cannot display the full spectrum of colors and only have limited brightness, so we can squeeze the colors to fit in the gamut of the device. Besides that, it can also be useful to give the renders a particular look, e.g. as if they have been printed on real camera film. The default Filmic transform does this.

Conversion from linear to display device space.

When loading and saving media formats it is important to have color management in mind. File formats such as PNG or JPEG will typically store colors in a color space ready for display, not in a linear space. When they are used as textures in renders, they need to be converted to linear first, and when saving renders for display on the web, they also need to be converted to a display space.

For intermediate files in production, it is recommended to use OpenEXR files. These are always stored in scene linear color spaces, without any data loss. That makes them suitable to store renders that can later be composited, color graded and converted to different output formats.

Images can also contain data that is not actually a color. For example normal or displacement maps merely contain vectors and offsets. Such images should be marked as Non-Color Data so that no color space conversion happens on them.

Reference

Properties

Render Properties \xe2\x80\xa3 Color Management

Color Management properties. \xc2\xb6

These are color management settings that are used across Blender. These color management settings are Scene specific so settings can be customized per Scene. Color management can also be overridden when saving images; this behavior can be set in the Output Color Management Properties.

The color space for the display that Blender is being viewed on.

Most displays are sRGB by default with some newer displays having the option to use Rec. 2020. These displays have a wider color gamut and can display high dynamic range content. If you have an Apple display you probably will want to use Display P3.

It is important to check your OS and display setting to make sure they all match the display in use to view the most accurate image.

Used by most displays.

Used by most Apple devices.

Used by many older TVs.

Used for newer wide gamut HDR displays.

These are different ways to view the image on the same display device.

Does no extra conversion besides the conversion for the display device.

Often used for non-photorealistic results or video editing where a specific look is already baked into the input video.

A tone mapping transform designed specifically for PBR color accuracy, to get sRGB colors in the output render that match as faithfully as possible the input sRGB base color in materials, under gray-scale lighting. This is aimed toward product photography use cases, where the scene is well-exposed and HDR color values are mostly restricted to small specular highlights.

A tone mapping transform that improves on Filmic, giving more photorealistic results. AgX offers 16.5 stops of dynamic range and unsaturates highly exposed colors to mimic film's natural response to light.

A tone mapping transform designed to handle high dynamic range colors. Filmic is deprecated and is superseded by AgX which improves handling of saturated colors.

Converts to Filmic log color space. This can be used for export to color grading applications, or to inspect the image by flattening out very dark and light areas.

Shows a heat map of image intensities, to visualize the dynamic range, and help properly expose an image.

Below is a table that represents how normalized linear color data is represented with False Color.

Luminance Value

Color
Low Clip
Black
0.0001% to 0.05%
Blue
0.05% to 0.5%
Blue-Cyan
0.5% to 5%
Cyan
5% to 16%
Green-Cyan
16% to 22%
Grey
22% to 35%
Green-Yellow
35% to 55%
Yellow
55% to 80%
Orange
80% to 97%
Red
High Clip
White

Intended for inspecting the image but not for final export.Raw gives the image without any color space conversion.

Choose an artistic effect from a set of measured film response data which roughly emulates the look of certain film types. Applied before color space conversion.

Used to control the image brightness (in stops) applied before color space conversion.It is calculated as follows: $\text{render}(_value \times 2^{(\text{exposure})})$

Extra gamma correction applied after color space conversion.Note that the default display transforms already perform the appropriate conversion,so this mainly acts as an additional effect for artistic tweaks.

The color space that the Sequencer operates in.By default, the Sequencer operates in sRGB space,but it can also be set to work in Linear space like the Compositing nodes, or another color space.Different color spaces will give different results for color correction, crossfades, and other operations.

The list of color spaces depends on the active OCIO config .The default supported color spaces are described in detail here: Default OpenColorIO Configuration

Enable high dynamic range display in rendered viewport, uncapping display brightness.This requires a monitor with HDR support and a view transform designed for HDR(Filmic does not generate HDR colors).

This feature is currently only supported on macOS.

Adjust RGB Curves to control the image colors before the color space conversion.Read more about using theCurve Widget.

Adjusts colors so that a given white point (expressed in color temperature and tint) ends up as white on the display.

As an alternative to manually specifying the values, there's also a color picker.When a color is selected, temperature and tint are set such that this color ends up being balanced to white.This only works if the color is close enough to a blackbody emitter.

The blackbody temperature of the primary illuminant. By default a D65 white point is used.

The amount of green/magenta shift of the blackbody curve.

Blackbody temperature curve.

Tip

White balancing can also be accomplished as part of the compositing pipeline by using the Color Balance Node

When working with image files, the default color space is usually the right one. If this is not the case, the color space of the image file can be configured in the image settings. A common situation where manual changes are needed is when working with or baking normal maps or displacement maps, for example. Such maps do not actually store colors, just data encoded as colors. Those images should be marked as Non-Color Data.

Image data-blocks will always store float buffers in memory in the scene linear color space, while a byte buffer in memory and files in a drive are stored in the specified color space setting.

By default, only renders are displayed and saved with the render View Transformation applied. These images are the Render

Result and the files saved directly to a drive with the Render Animation operator. However, when loading a render saved to an intermediate OpenEXR file, Blender cannot detect automatically that this is a render (it could be e.g. an image texture or displacement map). We need to specify that this is a render and that we want the transformations applied, with these two settings:

Display the image data-block (not only renders) with view transform, exposure, gamma, RGB curves applied. Useful for viewing rendered frames in linear OpenEXR files the same as when rendering them directly.

Option in the image save operator to apply the view transform, exposure, gamma, RGB curves. This is useful for saving linear OpenEXR to e.g. PNG or JPEG files in display space.

Blender comes with a standard OpenColorIO configuration that contains a number of useful display devices and view transforms. The reference linear Color Space used is the linear color space with Rec. 709 chromaticities and D65 white point.

However, OpenColorIO is also designed to give a consistent user experience across multiple applications, and for this, a single shared configuration file can be used. Blender will use the standard OCIO environment variable to read an OpenColorIO configuration other than the default Blender one. More information about how to set up such a workflow can be found on the OpenColorIO website.

Blender currently uses the following color space rules:

Color space used for rendering, compositing, and storing all float precision images in memory.

Color space for non-color data.

ACES2065-1 color space. Used to derive chromaticities of the scene linear color space, for effects such as blackbody emission.

Defines the distribution of colors in color pickers. It is expected to be approximately perceptually linear, have the same gamut as the scene linear color space, map 0..1 values to 0..1 values in the scene linear color space for predictable editing of materials' albedo.

Default color space for the Sequencer, scene linear if not specified.

Default color space for byte precision images and files, texture_paint if not specified.

Default color space for float precision images and files, scene_linear if not specified.

The standard Blender configuration includes support for saving and loading images in ACES (code and documentation) color spaces. However, the ACES gamut is larger than the Rec. 709 gamut, so for best results, an ACES specific configuration file should be used. OpenColorIO provides an ACES configuration file, though it may need a few more tweaks to be usable in production.

Blender's OCIO configuration file is equipped by default to read/write files in these color spaces:

Standard RGB display space using Rec. 709 chromaticities and a D65 white point.

BT.2020 2.4 Exponent EOTF Display.

BT.1886 2.4 Exponent EOTF Display, commonly used for TVs.

Generic data that is not color, will not apply any color transform (e.g. normal maps).

Linear BT.709 chromaticities with illuminant D65 white point.

Linear BT.2020 with illuminant D65 white point.

Linear E-Gamut with illuminant D65 white point.

Linear DCI-P3 with illuminant D65 white point.

1931 CIE XYZ standard with assumed illuminant E white point.

1931 CIE XYZ with adapted illuminant D65 white point.

Similar to sRGB but uses the Filmic view transform.

Intermediate log color space of Filmic view transform.

Apple's Display P3 with sRGB compound (piece-wise) encoding transfer function, common on Mac devices.

An ACES color space that is designed to be used for rendering and compositing. It uses the AP1 color primaries, a D60 white point, and a linear transfer function. While similar to ACES2065-1, this color space has a smaller color gamut. The smaller gamut allows it to better represent the colors that fit inside the CIE 1931 chromaticities diagram. Colors that lie outside the CIE 1931 chromaticities are generally not important to rendering and compositing because the human stimulus cannot represent these colors.

An ACES color space using the AP0 color primaries, a D60 white point and a linear transfer function. This color space is meant to store and transfer data with the most amount of possible color information.

--- Fin du chapitre: _color_management ---

--- Début du chapitre: _color_mix --- The Color Mix effect strip mixes two strips by working on the individual and corresponding pixels of the two input strips.

This effect can do the exact same operation as the Add, Subtract, or Multiply effect strips but also other color blending modes.

The Blend modes can be selected in the select menu. See Color Blend Modes for details on each blending mode.

Add, Subtract, Multiply, Screen, Divide, Difference, Darken, Lighten, Overlay, Color Dodge, Color Burn, Hue, Saturation, Value, Color, Soft Light, Linear Light

The amount of the blend of the second image gets composed onto the first.

--- Fin du chapitre: _color_mix ---

--- Début du chapitre: `_color_palette` --- Color Palette.\xc2\xbf

Color Palettes are a way of storing a brush's color so that it can be used at a later time. This is useful when working with several colors at once.

- Click to select a palette.
- Adds the current brush's primary color to the palette.
- Removes the currently selected color from the palette.
- Moves the selected color up/down one position.
- Sort Colors by Hue, Saturation, Value, Luminance.

Each color that belongs to the palette is presented in a list. Clicking on a color will change the brush's primary color to that color.

- Ctrl-LMB open the color picker to change color. See Color Picker .
- Backspace reset the value.

--- Fin du chapitre: `_color_palette` ---

--- Début du chapitre: `_color_picker` --- Circle HSV.\xc2\xbf

The color picker is a pop-up that lets you define a color value. Holding Ctrl while dragging snaps the hue to make it quick to select primary colors.

- Click to pick the first and second color component. The shape can be changed; see Types.
- The slider with a gradient in the background defines the value/lightness of the color mixing. Fine control can be inputted with Wheel.
- Selects the Color Model for the number value fields.
- Create the final color by mixing red, green, and blue colors.
- Create the final color by adjusting hue, saturation, and value/lightness.

Note

In Blender, the RGB and HSV/HSB values are in Scene Linear color space, and are therefore not Gamma corrected. On the contrary, Hex are automatically Gamma corrected for the sRGB Color Space. For more information, see Color Management .

Blender uses values from 0 to 1.0 to express the color mixing for RGB and HSV/HSB colors.

- For color inputs with an Alpha Channel, another slider is added.
- The hexadecimal (hex) equivalent value to the mixed color. Shorthand hex colors can be typed in, e.g. dark yellow `#FFCC00` can be written as `#FC0`.
- Samples a color from inside the Blender window using the Eyedropper. Note, colors sampled from the eyedropper are in linear color space and do not account for view transform adjustments.
- Ctrl-LMB (drag) snaps the hue to 30° intervals.
- Shift-LMB (drag) precision motion.
- Wheel adjust the value/lightness.
- Backspace reset the value to the default value.

The default color picker type can be selected in the Preferences, see: Interface .

- Circle HSV.\xc2\xbf
- Circle HSB.\xc2\xbf
- Square (SV + H).\xc2\xbf
- Square (HS + V).\xc2\xbf
- Square (HV + S).\xc2\xbf

--- Fin du chapitre: `_color_picker` ---

--- Début du chapitre: `_color_ramp` --- The Color Ramp Node is used for mapping values to colors using a gradient. The value to map. 0.0 results in the leftmost color, while 1.0 results in the rightmost.

See Color Ramp Widget .

Standard color output.

Standard alpha output.

An often overlooked use case of the Color Ramp is to turn a black-and-white image into a colored image with transparency.

In the example above, a black-and-white swirl image, which is lacking an alpha channel, is fed into the Color Ramp node as a Factor.

The Color Ramp node is set to a purely transparent color on the left end of the gradient, and a fully red color on the right. As you can see in the Viewer node, the Color Ramp node outputs an image that is transparent where the input is black, and opaque where the input is white.

In this example, multiple colors are added to the color gradient, converting a black-and-white image into a flaming swirl.

The shades of gray in the input image are mapped to three colors: blue, yellow, and red, all fully opaque. Where the image is black, the Color Ramp substitutes blue (the first color stop). Where it is some shade of gray, the Color Ramp outputs a corresponding color from the gradient (bluish, yellow, to reddish). Where the image is fully white, the Color Ramp outputs red.

--- Fin du chapitre: `_color_ramp` ---

--- Début du chapitre: `_color_spill` --- The Color Spill node reduces one of the RGB channels so that it is not greater than any of the others.

This is common when compositing images that were shot in front of a green or blue screen. In some cases, if the foreground object is reflective, it will show the green or blue color; that color has spilled onto the foreground object. If there is light from the side or back, and the foreground actor is wearing white, it is possible to get green (or blue) light from the background onto the foreground objects, coloring them with a tinge of green or blue. To remove the green (or blue) light, you use this fancy node.

Standard color input.

Standard Factor.

R, G, B

Simple, Average

R, G, B

Scale limit by value.

Allows you to reduce the selected channel's input to the image greater than the color spill algorithm normally allows. This is useful for exceptionally high amounts of the color spill.

R, G, B

The image with the corrected channels.

Results with the nodes applied to an image from the Mango Open Movie.

Before: green border and green reflections.

After: no unwanted green.

--- Fin du chapitre: `_color_spill` ---

--- Début du chapitre: `_combine_color` --- Combines four grayscale channels into one color image, based on a particular Color Model.

The inputs of this node depend on the Mode property (see below).

The opacity of the output color.

The color model to use.

Red, Green, Blue.

Hue, Saturation, Value.

Hue, Saturation, Lightness.

Standard color output.

--- Fin du chapitre: `_combine_color` ---

--- Début du chapitre: `_combine_matrix` --- The Combine Matrixnode constructs a 4x4 matrix from its individual values.

The inputs of this node are split into panels for each column of the matrix. Each panel, has four value inputs for the four rows of the matrix.

This node has no properties.

The constructed matrix.

--- Fin du chapitre: `_combine_matrix` ---

--- Début du chapitre: `_combine_transform` --- The Combine Transformnode combines a translation vector, a rotation vector, and a scale vector into a Transformation Matrix.

The translation vector.

The rotation vector.

The scale vector.

This node has no properties.

The combined transformation matrix.

--- Fin du chapitre: `_combine_transform` ---

--- Début du chapitre: `_combine_xyz` --- The Combine XYZ Node combines a vector from its individual components.

X

Y

Z

This node has no properties.

Standard vector output.

Note

The vector is not normalized.

--- Fin du chapitre: `_combine_xyz` ---

--- Début du chapitre: `_comb_curves` --- Shape the curves by moving their control points while preserving the initial length of every curve segment.

Falloff that is applied from the tip to the root of each curve.

--- Fin du chapitre: `_comb_curves` ---

--- Début du chapitre: `_commit_guide` --- Access to directly submit changes is limited to people with commit access to the repository. Once you are provided with commit access you can start committing directly instead of creating a patch file.

You can make commits from your Git client or using the Git command line tool. The following command will create a commit and send it to the central repository:

If you leave out `-m "message"`, you will be prompted to type the message in a text editor.

Tip

You should make sure you are always on the latest revision before committing. You may not be able to commit directly if there are conflicting changes in the latest revision.

To avoid this update your local repository before committing (run `makeupdate`).

See also

Blender's Git usage guide

See also

See release branch documentation on how to make commits to a specific release branch and how to create merge commits.

When making changes to the manual that directly relate to a specific commit (change) in Blender, it is helpful to make the title of the commit the same as the commit made to Blender. It is requested that you include the commit hash of the commit made to the Blender source code.

For example, the commit `rBM8473` includes a descriptive indicative of the changes made along with the hash `rBa71d2b260170`. The hash can be extracted from the URL provided in the Documentation task for a specific upcoming release.

Other more general changes do not have to follow the above policy however, it is still important to make the description clear about what changes you made and why. It can be helpful to prefix the commit title with a prefix word such as `Cleanup:` or `Fix:` when you are making general cleanups or fixes respectively.

Writing good commit messages helps administrators keep track of changes made and ensures all new features are properly documented.

--- Fin du chapitre: `_commit_guide` ---

--- Début du chapitre: `_common` --- The Target Data ID field lets you link the constraint to a Target object of your choosing. This link provides data to the constraint so that it can begin to function. For example, the Copy Location Constraint needs location data to function. Fill in the Target field, and the Copy Location constraint will begin to use location data from the Target object.

The Target field must be filled in for the constraint to function.
\xc2\x80

By default, the Target will use the Object Origin as the target point. If the Target field links to a Mesh or Lattice object, a Vertex Group field will appear. Enter the name of a vertex group and the constraint will target the median point of this vertex group instead of the object's origin.

If the Target field links to an Armature, a Bone field will appear along with a Head/Tail slider. Enter the name of a bone and the constraint will target the bone instead of the entire armature object origin.

The slider moves the precise position of the target between the Head and Tail of the bone. Some constraints have a button next to the slider that enables using the curved shape of Bendy Bones.

Constraints need a frame of reference in order to function. This frame of reference is called the `space` of the constraint. Choosing one space vs. another will change this frame of reference and substantially alter the behavior of a constraint.

To understand how changing the space will change the behavior of the constraint, consider experimenting with two empties. Make sure they display as arrows so that you can see the local axes for each empty. Make sure to size one empty a little larger than the other so that they are both

always visible even if directly on top of each other. Then add a constraint to one empty that targets the other and experiment thoroughly by moving, rotating and scaling the target in many different ways.

This constraint is set to use World Space as the frame of reference for both its Target space and its Ownerspace.

The space used to evaluate the target of the constraint is called the Target space. The space used to evaluate the constrained object (the object that owns the constraint) is called the Ownerspace. Hover over the space select menu(s) to learn whether it affects the space of the target or the space of the owner.

When the constraints use a Target and/or an Ownerspace there will be no, one or two selector(s). The Copy Location constraint in example uses both Target and Ownerspace.

When a constraint uses both Target and Ownerspace, the Target and Owner can be any combination of space types.

In this space type the world is the frame of reference for the object (or bone). Location is relative to the world origin. Rotation and Scale are oriented to the world axes. Transformations to the object, the object's parent and any other constraints higher up in the constraint stack are all taken into account.

This space excludes all effects of the parent objects or bones, as well as the rest position and orientation of the bone itself. Only transformations applied to the object or bone itself are taken into account.

Warning

For objects without a parent Local Space has a special meaning, different from the normal behavior of local space for bones or objects that have a parent. This behavior is kept for backwards compatibility, but may be removed in the future and shouldn't be used.

The bone position and orientation is evaluated relative to its rest pose location and orientation, thus including both its own transformations and those caused by a possible parent relationship (i.e. the chain's transformations above the bone).

The bone position and orientation is evaluated in the armature object local space (i.e. independently from the armature transformations in Object Mode). Hence, if the armature object has null transformations, Pose Space will have the same effect as World Space.

The position and orientation is evaluated relative to the current position and orientation of an arbitrary object or bone that is specified via additional input fields that appear when this option is selected. This can be used to evaluate the constraint using an arbitrary coordinate system.

This space works like Local Space, with an additional coordinate space transformation that compensates for the difference in the rest pose orientations of the owner and target bones. If applied as the Local Space of the owner, this will produce the same global space movement as the target, provided parents are still at rest pose.

This option replaces the following setup with two additional bones:

An extra child bone of the target, rotated the same as the owner in rest pose.

An extra sibling bone of the target, positioned same as the child in rest pose and using Copy Transforms in World Space from the child.

The constraint uses Local Space of the sibling instead of the original target.

This video demonstrates the difference from ordinary Local Space:
The influence slider determines how much the constraint will affect the constrained object (target).
An influence of 0.0 will have no effect. An influence of 1.0 will have the full effect.
Values between (0.0 and 1.0) will have a partial effect, but be careful. These partial effects can be difficult to control, especially as the constraint stack grows in complexity.
The influence value is animatable, allowing constraints to be turned off, or partially on as needed.
Disables the constraint while trying to preserve the current object position. This may not work perfectly if other constraints remain active.
--- Fin du chapitre: _common ---

--- Début du chapitre: _common_options --- Some options are commonly used by many modifiers, and share the same behavior across all of those. In particular, many offer ways to precisely mask and weight their effect on a vertex basis (using either vertex groups and/or textures).

Typical modifier Vertex Group options. \xc2\xbf

Vertex Groups are an easy way to control which vertices are affected by a modifier, and to which extent (using their weights). They are available when modifying meshes or lattices.

Tip

Vertex groups can also be edited and even animated using the Vertex Weight modifiers.

The vertex group name.

Warning

The group is referenced by its name. That means that if you rename it, the link to the renamed vertex group will be lost by all modifiers using it (their field will turn red), and you'll have to select the proper group again in all of them.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group. Only available in some modifiers.

Typical modifier Texture options. \xc2\xbf

Those options allow to use any kind of image (including parametric ones) to control the modifier's effect. Most of the time, only the value (grayscale) of the texture is used, but in some cases (like with some modes of the Displace modifier), the whole RGB color components might be exploited.

Tip

Textures can be animated (either using videos, or by animating the mapping coordinates).

The texture data-block to use.

Tip

By clicking on the right-most button of this field (with the settings icon), you can go directly to the selected texture's settings in the Texture Properties tab.

The texture's coordinates to get each vertex's value:

Take texture coordinates from face UV coordinates.

The UV Map from which to take texture coordinates. If the object has no UV coordinates, it falls back to the Local coordinate system. If this field

is blank, but there is a UV map available (e.g. just after adding the first UV map to the mesh), the currently active UV map will be used.

Note

Since UV coordinates are specified per face, the UV texture coordinate system currently determines the UV coordinate for each vertex from the first face encountered which uses that vertex. Any other faces using that vertex are ignored.

This may lead to artifacts if the mesh has non-contiguous UV coordinates.

Take the texture coordinates from another object's coordinate system.

The object from which to take texture coordinates. Moving the object will therefore alter the coordinates of the texture mapping.

If this field is blank, it falls back to the Local coordinate system.

Note

Moving the original object will also result in a texture coordinate update. As such, if you need to maintain a displacement coordinate system while moving the modified object, consider parenting the coordinate object to the modified object.

Take the texture coordinates from the global coordinate system.

Take the texture coordinates from the object's local coordinate system.

Which channel to use as value source (only available with a few modifiers currently, others follow the Intensity behavior, unless otherwise specified).

The average of the RGB channels (if RGB(1.0, 0.0, 0.0) value is 0.33).

One of the color channels' values.

The hue from the HSV color model (i.e.; the color in the standard wheel, e.g. blue has a higher hue value than yellow).

The saturation from the HSV color model (e.g. the value for pure red is 1.0, for gray is 0.0).

The value from the HSV color model.

Note

All of the channels above are gamma corrected, except for Intensity.

--- Fin du chapitre: `_common_options` ---

--- Début du chapitre: `_community` --- Being freely available from the start, even while it was closed source, considerably helped

Blender's adoption by the community. A large, stable, and active community of users has gathered around Blender since 1998. The community showed its support for Blender in 2002 when they helped raise \$100,000 in seven weeks to enable Blender to go Open Source under the GNU GPL License.

There are several independent websites such as forums, blogs, news, and tutorial sites dedicated to Blender.

One of the largest community forums is Blender Artists, where Blender users gather to show off their creations, get feedback, ask and offer help and, in general, discuss Blender.

Blender's community is one of its greatest features, so apart from this user manual, there are many different ways to get support from other users, such as Chat, Stack Exchange, and Reddit.

For studios and organizations there is Enterprise support, and for studios looking to add Blender to their pipeline, Blender Studio contains

documentation and training material around this topic.If you think you have found an issue with Blender, please report a bug. More details about support can be found on the support page. Being open source, Blender welcomes development from volunteers.Communication between developers is done mostly through three platforms:

The projects.blender.orgsystem

Developer Forum

Online Chat (see below)

If you are interested in helping develop Blender,see the page.

For real-time discussion, we have chat.blender.orgwhich uses Blender IDfor authentication.

You can join these channels:

#generalFor general chat with the community.

#blender-codersFor developers to discuss Blender development.

#pythonFor support for developers using the Python API.

#docsFor discussion related to Blender's documentation.

Blender FAQ(Can I use Blender commercially? What is GPL/GNU?

)

Demo and benchmark files

Developer's Ask Us Anything!

--- Fin du chapitre: _community ---

--- Début du chapitre: _compare --- The Comparenode takes two inputs and does an operation to determine whether they are similar.The node can work on all generic data types, and has modes for vectors that contain more complexcomparisons, which can help to reduce the number of necessary nodes, and make a node tree more readable.

Standard value inputs of the selected type.

Compared against the dot product of two input vectors in when the Modeproperty is set to Dot Product.

This value is used as a threshold for still considering the two inputs as equalfor the Equal and Not Equal operations.

Compare each axis of the input vectors separately, and output true only when the result is truefor each axis.

Compare the length of the two input vectors.

Compare the average of the elements of the input vectors. This is the same as the implicitconversion used when setting the node's data type to Float.

Compare the dot product of the two vectors with the separate Cinput, using the selected operation.The dot product outputs a single value that says how much the two vectors agree.

Compare the angle between the two vectors with the separate Angleinput, using the selected operation.The vectors are normalized, so their length does not matter.

True when the first input is smaller than second input.

True when the first input is smaller than the second input or equal.

True when the first input is greater than the second input.

True when the first input is greater than the second input or equal.

True when both the difference between the two inputs is smaller than the Epsiloninput.

True when both the difference between the two inputs is larger than the Epsiloninput.

True when the first color input is brighter than the second.

True when the first color input is darker than the second.

Standard Boolean output.

Here, the compare node is used with the Directionmode to compare the direction of the sphere's face normals to the direction of the cube object's location. Anywhere that the directions are less than 32.9 degrees apart, the faces will be selected, and deleted.

--- Fin du chapitre: `_compare` ---

--- Début du chapitre: `_compatibility` --- Blender can open blend-files saved with both older versions of the software (backward compatibility), and newer ones (forward compatibility). This comes with some limitations though.

Tip

When having issues with opening much older (or newer) blend-files, it can help to use a few intermediary Blender releases to perform conversions by smaller steps.

Note

Here is a more exhaustive documentation about compatibility handling, in the developer's documentation.

Opening older files and converting them for the current version of Blender is usually straight-forward. It is expected to give very good and usable results.

There can be major feature changes, for which the backward compatibility will only be ensured for a limited amount of time. For example the changes to the animation system that happened during the Blender 2.5x project. This will never be less than a full major release cycle (i.e. two years at least).

Forward compatibility is inherently harder to ensure, and loss of feature should always be expected when opening a blend-file saved with a more recent version of Blender.

A warning is shown in the UI when editing a more recent blend-file. Trying to overwrite it (with a simple `Save` operation) will also show a confirmation popup, as this could make that loss of data permanent.

When Blender switches to a new major version release (e.g. from 3.x to 4.0), there can also be major changes that will make the blend-file fully incompatible with older versions of Blender.

In such cases, older Blender will fail opening (or appending/linking from) the newer blend-file, with a message stating which minimal version is needed to open it.

In such cases, the last LTS release of the previous release cycle will be kept compatible with the newer file format version, and will be usable as converter between both versions.

For example, Blender 3.6 LTS can open files from Blender 4.x, and will perform the necessary conversions such that when re-saved from 3.6, the files become compatible with all 3.x Blender versions.

--- Fin du chapitre: `_compatibility` ---

--- Début du chapitre: `_composite` --- The Composite node is where the actual output from the Compositor is connected to the renderer. This node is updated after each render, but also reflects changes in the node tree (provided at least one finished input node is connected).

Connecting a node to the Composite node will output the result of the priortree of that node to the Compositor.
RGB image. The default is black, so leaving this node unconnected will result in a black image.

Alpha channel.

Used alpha channel, colors are treated alpha premultiplied. If disabled, alpha channel gets set to 1, and colors are treated as alpha straight, i.e. color channels does not change.

This node has no output sockets.

Note

If multiple Composite nodes are added, only the active one (last selected, indicated by a red header) will be used.

--- Fin du chapitre: _composite ---

--- Début du chapitre: _compositor --- The Compositor lets you manage nodes for compositing.

Nodes in the Compositor.

The use of the Compositor is explained in Compositing.

--- Fin du chapitre: _compositor ---

--- Début du chapitre: _cone --- Generates a cone mesh that is optionally truncated.

Number of vertices in the top and/or bottom circle of the cone. No geometry is generated if the number is below three.

Number of vertically stacked face loops that make up the cone's sides. Increasing this will add horizontal cuts. No geometry is generated if the number is below one.

Number of concentric rings in the top and/or bottom. No geometry is generated if the number is below one.

The radius of the cone's top circle. If this is zero, the circle is reduced to a single vertex.

Same as Radius Top but for the bottom circle.

Height of the generated cone.

Note

If the top and bottom radii are both zero, this node will output a single line.

How the circles at the top and bottom are filled with faces when their radius is larger than zero.

Do not fill the circles.

Fill the innermost circles with a single face.

Fill the innermost circles with triangles connected to a vertex in the center.

Standard geometry output.

A boolean field with a selection of the faces on the top of the cone. If the Fill Type is set to None, this will be a selection of the top edges instead. If Radius Top is zero, this will be a selection of the top vertex.

A boolean field with a selection of the faces on the side of the cone.

A boolean field with a selection of the faces on the bottom of the cone.

If the Fill Type is set to None, this will be a selection of the bottom edges instead. If Radius Bottom is zero, this will be a selection of the bottom vertex.

The default UV coordinate of each face corner. This can be connected to the Store Named Attribute Node for populating a UV Map .

--- Fin du chapitre: _cone ---

--- Début du chapitre: _connect_vertex_pairs ---Reference

Edit Mode

Vertex \xe2\x80\xa3 Connect Vertex Pairs

This operator connects selected vertices by creating edges between them and splitting the face. It can be used on many faces at once.

Vertices before connecting. \xc2\xbf

After connecting vertices. \xc2\xbf

Resulting face pair. \xc2\xbf

The main difference between this operator and Connect Vertex Path is that this operator ignores the selection order and connects all selected vertices that share a face.

--- Fin du chapitre: _connect_vertex_pairs ---

--- Début du chapitre: _connect_vertex_path ---Reference

Edit Mode

Vertex \xe2\x80\xa3 Connect Vertex Path

J

This tool connects vertices in the order they are selected, splitting the faces between them. When there are only two vertices selected, a cut will be made across unselected faces, a bit like the Knife tool; but this is limited to straight cuts across connected faces.

Two disconnected vertices. \xc2\xbf

Result of connecting. \xc2\xbf

Running a second time will connect the first/last endpoints. When many vertices are selected, faces will be split by their selected vertices.

Before. \xc2\xbf

After. \xc2\xbf

Vertices not connected to any faces will create edges, so this can be used as a way to quickly connect isolated vertices too.

--- Fin du chapitre: _connect_vertex_path ---

--- Début du chapitre: _constraints --- Operators for working with an object \xe2\x80\x99s Constraints.

Reference

Object Mode and Pose Mode

Object \xe2\x80\xa3 Constraint \xe2\x80\xa3 Add Constraint (with Targets)

Adds a constraint to the active object. The type of constraint must be chosen from a pop-up menu, though it can be changed later from the Add Constraint (with Targets) Adjust Last Operation panel. If there is another object selected besides the active one, that object will be the constraint target (if the chosen constraint accepts targets).

When using a bone from another armature as the target for a constraint, the tool will look inside the non-active armature and use its active bone, provided that armature is in Pose Mode.

Reference

Object Mode and Pose Mode

Object \xe2\x80\xa3 Constraint \xe2\x80\xa3 Copy Constraints to Selected Objects

Copies the active object Constraints to the rest of the selected objects.

Reference

Object Mode and Pose Mode

Object \xe2\x80\xa3 Constraint \xe2\x80\xa3 Clear Object Constraints
Removes all Constraints of the selected object(s).
--- Fin du chapitre: _constraints ---

--- Début du chapitre: _contribute --- On this page French (fr) is used for examples. However, it can be replaced with other languages codes. So, be sure to change the/frs suffixes in this guide to the language you are translating!

To see which languages are currently available, you can check the online interface, or browse the underlying git repository.

The preferred way to contribute to the translation effort is the use the web-based interface, currently a Weblate instance.

Simple enhancement suggestions can be contributed by any user, even without logging in. Suggestions will be reviewed by the translating team before they get published.

Weblate also comes with new helping tools to improve coherence of translations, like the glossary.

If for some reasons the web-based translation interface does not work well for you, you can still download PO file from it, and upload back it later.

Warning

You will have to deal with potential conflicts yourself if some update happened in the mean time. Direct commit to the git repository for translations is not possible anymore.

Note

There is a known issue with the current tool behind the web interface, which will make heavy processing like upload and integration of a PO file take several minutes, with the web page staying in refresh mode for the whole time. If it takes more than ten minutes, it will even apparently fail with a server timeout error message. There is usually no actual problem though, so no need to re-try uploading the PO file then, refreshing the page after a few minutes should be enough to see the contribution in the web interface.

Note

First of all, it is assumed that you have the manual already building. If you have not done this already go back to this section.

From the directory containing your checkout of the manual run:

You will be prompted to type in the language folder you want to download. In the case of this example we will use fr. Pressing Return will confirm this selection.

It will take a few minutes to download but once complete it will create a locale/fr subdirectory.

You should have a directory layout like this:

Note

When running Git from the command line (such as updating), you will need to change directory to locale first rather than the blender-manual directory. The PO language files themselves can also be downloaded from the web interface, Files menu, on each dedicated language page of the Manual component.

To make edit the PO files you will need to install a PO editor. We recommend that you use Poedit, however any PO editor will do.

Note

For Linux users, you will have to check with your distribution's software center for a version of Poedit. This

editor is only a recommendation. There are others, such as Kate and Kwrite, that could offer syntax highlighting and basic tools for text editing, e.g. letter case transposes. Other platforms can use some text editors supporting the syntax highlighting for PO files, or allowing you to create a custom one (such as Notepad++ on Windows).

Now you can build the manual with the translation applied:

On Linux and macOS run:

On Windows run:

Now you will have a build of the manual with translations applied.

Now you can edit the PO translation files, in the `LC_MESSAGES` folder you have two files:

`blender_manual.po` This is the main translation file that you will be editing.

`sphinx.po` This translation file is much smaller and contains translations for the website theme.

To edit these files open them up in your translation editor, i.e.

Poedit. Once in your editor you will see a list of texts, each of these items represent some part of the user manual. You may need to adjust your editor to sort the list in a way that makes sense for example

by source.

You can now select an untranslated string and your editor will have an input box to add the translation. The modified .po files can now be submitted back to the web-based interface.

Tip

Make sure that you Building with Translations to catch any syntax errors you may make while translating. These errors will be displayed as warnings while building the manual.

When the manual is updated, those translations which are outdated will be marked as fuzzy. To keep track with that, you can use a tool we created for that task.

You can do this by running:

This will only give a quick summary however, you can get more information by running:

You should get a list of all the files with information about the number of empty and fuzzy strings. For more options see:

As the original manual changes, the templates will need updating. Note, doing this is not required, as administrator usually update the files for all languages at once. This allows all languages to be on the same version of the manual. However, if you need to update the files yourself, it can be done as follows:

The updated templates can then be committed to the repository.

See also

A guide how to add a new language can be found in the [Adding a Language](#).

--- Fin du chapitre: `_contribute` ---

--- Début du chapitre: `_controls` --- Reference

Sculpt Mode

Header `Auto-Masking`

Alt- A

These properties automatically mask geometry based on geometric features of the mesh. It's a quick alternative to frequent manual masking. These masks are initialized on every new stroke or tool usage.

They are also never visible as an overlay.

Note, these properties are applied across all sculpt brushes, however, they can also be configured per brush in the Advanced Brush Settings . These properties can be accessed via a Pie Menu by pressing Alt- A . All auto-masking modes can be combined, which makes the generated auto-mask more specific. For example it's possible to auto-mask a specific face set, while excluding disconnected topology and face set boundaries, and only affect faces that are oriented towards the view via View Normal.

Only vertices that are topologically connected to where you started the stroke/tool on are affected. So loose geometry islands will be auto-masked.

Additionally for the Grab and Thumb brushes, anything that is not connected within the brush radius will be auto-masked. So even if geometry is connected somewhere, it is considered separate if the connection is not within the radius.

Only vertices that are part of the same face set that you started the stroke/tool on are affected.

Tip

If no topology or face set is visible under the cursor at the start of the stroke, the previously auto-masked area will be targeted. This is especially useful with the Projected falloff shape in the Falloff Settings.

Vertices that are part of open boundary edges are not affected. This also includes boundary edges to hidden faces.

Increases the soft gradient towards the auto-masked boundary edges. Each step iterates the distance one edge further. This setting is used for both Mesh Boundary and Face Sets Boundary.

This will execute the Mask From Mesh Boundary operator with the current auto-masking settings. This is very useful to visualize the current auto-mask, or to edit the mask further manually.

Vertices that are part of a boundary between face sets are not affected. This also includes boundary edges to hidden faces. Propagation Steps are shared with Mesh Boundary auto-masking.

This will execute the Mask From Face Sets Boundary operator with the current auto-masking settings. This is very useful to visualize the current auto-mask, or to edit the mask further manually.

Vertices that are the peaks of the surface curvature are not affected. While this auto-mask is primarily meant for painting, it can also be used for regular sculpting.

The overall contrast of how strong the cavity is applied. The value of 0.5 is the default, but better results can also be achieved on 0.2 if a Custom Curve is used as well.

The number of times the cavity mask is blurred. A value of 0 will give the pure cavity auto-mask. Anything higher than 6 will likely have a less visible effect and decrease performance. Even though the value is capped to 10, it can be increased up to 25 if typing in the value.

Use a custom curve to fine tune the cavity auto-mask. This is very useful if only small crevices or flat surfaces should be affected. Or for example if the contrast should be increased/decreased in a specific way.

This will execute the Mask From Cavity operator with the current auto-masking settings. This is very useful to visualize the current auto-mask, or to edit the mask further manually.

This is the same as Cavity, but inverted. This means the valleys/crevices of the surface curvature will not be

affected. It cannot be used at the same time as Cavity and shares all of its settings. Enable this to quickly invert the cavity auto-mask. Only vertices with a Normal that face the viewer are affected. This is similar to the "Front Faces Only" toggle in the Brush Setting, to only affect visible geometry. The advantage of this auto-mask is that it has more options and works on sculpt mode as a whole.

Change the View Normal behavior to only affect vertices that are not occluded by other faces. This setting is incompatible with the other Limit and Falloff sliders. It also causes a much slower performance.

Determines the range of angles that will be affected. 90 degrees encompasses all that is visible.

Extends the angular range of the Limit slider with a soft falloff gradient. This falloff will visually extend the limit range further.

Very similar to the View Normal, but uses the Normal of the surface that you started the stroke/tool on. This way any direction can be chosen for what vertices will be affected. It has the same Limit and Falloff sliders as the View Normal auto-mask.

--- Fin du chapitre: _controls ---

--- Début du chapitre: _control_points --- Reference

Edit Mode

Surface "Extrude Curve and Move

E

Unlike meshes or curves, you cannot generally directly add new control points to a surface, as you can only extend a surface by adding a whole U or V row at once. The only exception is when working on a NURBS surface curve, i.e. a surface with only one control point on each U or V row. In this special case, all works exactly as with curves.

Most of the time, only extrusion is available. As usual, once the tool is activated the extrusion happens immediately and you are placed into select mode, ready to drag the new extruded surface to its destination.

There are two things very important to understand:

Surfaces are 2D objects. So you cannot extrude anything inside a surface (e.g. "inner" row); it would not make any sense!

The control "grid" must remain

"square", which means that you can only extrude a whole row, not parts of rows here and there.

To summarize, the Extrude tool will only work, when one and only one whole border row is selected, otherwise nothing happens.

Note

As for curves, you cannot create a new surface in your object out of nowhere. However, unlike for curves, there is no

"cut" option allowing you to separate a surface into several parts, so you only can create a new surface by duplicating an existing one, or adding a new one with the Add menu.

Images Fig. Selecting control point. to Fig. Extruding. show a typical extrusion along the side of a surface.

In Fig. Selecting control point. and Select Control Point Row., a border row of control points were highlighted by selecting a single control point, and then using Select Control Point Row to select the rest of the control points.

Selecting control point. "b6

Select Control Point Row. \xc2\xbb6

The edge is then extruded as shown in Fig.Extruding..Notice how the mesh has bunched up next to the highlighted edge.That is because the newextruded surface section is bunched up there as well.

Extruding.\xc2\xbb6

By moving the new section away from the area, the surface begins to \xe2\x80\x99cunbunch\xe2\x80\x99d.

You can continue this process of extruding or adding new surface sectionsuntil you have reached the final shape for your model.

Reference

Edit Mode

Surface \xe2\x80\xa3 Make Segment

F

Just like curves ,merging two surfaces requires that a single edge, a border row of control points,from two separate surfaces is selected. This means that the surfaces must be part of the same object.For example, you cannot join two surfaces while in Object Mode\xe2\x80\x93 but you can of course,as with any objects of the same type, join two or more Surfaceobjects \xe2\x80\x93 they just will not be \xe2\x80\x99clinked\xe2\x80\x99d or merged in a single one.

This tool is equivalent to creating edges or faces for meshes (hence its shortcut).The selection must contain only border rows of the same resolution(with the same number of control points),else Blender will try to do its best to guess what to merge with what,or the merge will fail (either silently, or stating that Resolutiondoesnotmatch if rows with different number of points are selected, or that there is Toofewselectionstomerge if you only selected points in one surface\xe2\x80\xa6). To select control points of different surfaces,in the same object, you must use either box select or circle select; Ctrl-LMB will not work.

So to avoid problems, you should always only select border rows with the same number ofpoints\xe2\x80\xa6 Note that you can join a border U row of one surface with a border V row of anotherone, Blender will automatically \xe2\x80\x99cinvert\xe2\x80\x99d the axis of one surface for them to match correctly.

NURBS surface curves are often used to create objects like hulls,as they define cross sections all along the object,and you just have to \xe2\x80\x99cskin\xe2\x80\x99d them as described above to get a nice, smooth and harmonious shape.

Fig.Joining ready.is an example of two NURBS surface curves, notNURBS curves, in Edit Mode, ready to be joined.Fig.Joining complete.is the result of joining the two curves.

Joining ready. \xc2\xbb6

Joining complete. \xc2\xbb6

Reference

Edit Mode

Surface \xe2\x80\xa3 Control Points \xe2\x80\xa3 Smooth

Iteratively smooths the selected control pointsby reducing the distance between neighboring control points.

Reference

Edit Mode

Surface \xe2\x80\xa3 Control Points \xe2\x80\xa3 Hooks

Ctrl- H

Hooks can be added to control one or more points with other objects.

Reference

Edit Mode

Surface \xe2\x80\xa3 Control Points \xe2\x80\xa3 Make Vertex Parent

Ctrl- P

You can make other selected objects children of one or three control points, as with mesh objects.

To select a mesh (that is in view) while editing a surface, Ctrl- P click on it. Select either one or three control points, then Ctrl- LMB the object and use Ctrl- P to make a vertex parent. Selecting three control points will make the child follow the median point between the three vertices. An alternative would be to use a Child Of constraint. See also the Curve modifier.

--- Fin du chapitre: _control_points ---

--- Début du chapitre: _convert --- Reference

Object Mode

Object \xe2\x80\xa3 Convert \xe2\x80\xa3 Curve

Converts the selected mesh, or text object to a curve object. For mesh objects, only edges belonging to no faces will be taken into account. The resulting curve will be a poly curve type, but can be converted to have smooth segments as described in Set Spline Type.

Reference

Object Mode

Object \xe2\x80\xa3 Convert \xe2\x80\xa3 Mesh

Converts the selected curve, metaball, surface, or text object to a mesh object. The actual defined resolution of these objects will be taken into account for the conversion. Note that it also keeps the faces and volumes created by closed and extruded curves.

Reference

Object Mode

Object \xe2\x80\xa3 Convert \xe2\x80\xa3 Grease Pencil

Converts the selected curve, mesh or text object to a Grease Pencil object with strokes matching the curve, mesh, or text; basic materials are also added. When multiple curves, meshes, or texts are selected, they are all converted into the same Grease Pencil object.

Duplicates the original object before converting it.

Strokes thickness.

Threshold value that determines the strokes end.

Sets offset to separate strokes from filled strokes.

Convert only edges marked as seam.

Convert faces as filled strokes.

Reference

Object Mode

Object \xe2\x80\xa3 Convert \xe2\x80\xa3 Trace Image to Grease Pencil

See Trace Image to Grease Pencil.

Reference

Object Mode

Object \xe2\x80\xa3 Convert \xe2\x80\xa3 Convert to Mesh Plane

Converts the selected image empty to a textured mesh plane.

For a description of the options see Import Images as Planes.

--- Fin du chapitre: _convert ---

--- Début du chapitre: _convert_colorspace --- The Convert Colorspace node converts images between color spaces.

Note

Images are already automatically converted into linear color space unless specified in the image's Color Space option.

Standard color input.

The color space of the input image and the color space to convert it to.

The list of color spaces depends on the active OCIO config. The default supported color spaces are described in detail here: Default OpenColorIO Configuration

Standard color output.

--- Fin du chapitre: `_convert_colorspace` ---

--- Début du chapitre: `_convex_hull` --- The Convex Hull node outputs a convex mesh that is enclosing all points in the input geometry.

Note

When the node is used on a geometry with instances, the algorithm will run once per instance, resulting in many convex hull meshes in the instance geometries. The Realize Instances node can be used to get a convex hull of an entire geometry.

Important

Volumes are not supported by this node, and attributes are not automatically transferred to the result.

Standard geometry input.

This node has no properties.

Mesh that encloses all points in the input.

--- Fin du chapitre: `_convex_hull` ---

--- Début du chapitre: `_coordinates` --- This node has no inputs.

This node has no properties.

The Coordinates node outputs the local geometry coordinates, relative to the bounding box.

--- Fin du chapitre: `_coordinates` ---

--- Début du chapitre: `_copyright` --- Blender itself is released under the GNU General Public License. More info blender.org/about/license.

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--- Fin du chapitre: `_copyright` ---

--- Début du chapitre: `_copy_global_transform` --- Copy and paste object and bone transforms with ease.

When copying, the global (World Space) transform is placed on the clipboard. This can then be pasted onto any object or bone, at the current frame or at another one.

Since the transform is placed on the clipboard as text, you can even copy-paste it into an instant messenger and send it to someone else.

Open Blender and go to Add-ons section of the Preferences .

Search "Copy Global Transform" and check the Enable Extension checkbox.

Located in 3D Viewport "N-panel" "Animation" tab.

The figure on the right shows the main functionality of the Copy Global Transform panel. The collapsed panels are described each in their own section below.

Inspects the active Object (in Object mode) or Bone (in Pose mode) and places its current global transform onto the clipboard as a matrix.

Takes the copied global transform and applies it to the active Object or Bone. This is done by adjusting its location, rotation, and scale properties.

Same as "Paste" above, but then mirrored relative to some other object or bone. This can be useful, for example, to copy the foot position of one foot to the other. See Mirror Options below.

Paste as described above and additionally use auto-keying to update one or more frames. The key selection is used to tell Blender which frame this should happen on; it does not influence which parts of the transform are keyed. What is keyed is determined by the active keying set. Almost the same as Paste to Selected Keys. Instead of only pasting on the selected keys, Paste and Bake will paste & auto-key on every frame between the first and last selected keys.

The copied transform can be mirrored relative to an object or a Bone. This requires choosing that object or bone first.

Choosing an Armature object as mirror object will show the bone selector. You can use that to pick the bone to use as mirror. This will always use the named bone on that specific armature object.

When you choose no mirror object at all, you can still choose a bone name. This is used for mirroring against a bone in the active armature. This can be useful to mirror bone transforms relative to the "chest" bone of the active character.

This will just mirror relative to the chosen object.

After pasting with "Paste Mirrored", the mirror axes can be chosen in the "do" panel.

Also known as "bake to camera", this operator will ensure selected objects/bones remain static (relative to the camera) on unkeyed frames.

This is done by generating new keys. These keys will be of type "Generated" so that it remains clear which keys were manually created, and which were generated. This way the tool can be re-run to regenerate the keys.

Ensure your animation is keyed using constant interpolation. If this is not the case yet, bake your animation (at least the transform channels). This tool does not work with the "Stepped" F-Curve modifier

Choose which of the Location/Rotation/Scale channels you want to fix to the camera. When unsure, make sure they are all checked.

Press the "Fix to Camera" button.

To undo the effect of the `Fix to Camera` operator, click on the trash bin button. That will remove all the generated keys in either the scene range or the frame range.

The tool operates on the scene frame range, or on the preview range if that is active. Keys outside that range are ignored, both when fixing to the camera and when removing generated keys.

Warning

This tool assumes that all keys with type `Generated` are equal. It will overwrite them (or remove them, depending on which button you press).

The `Relative` panel has copy/paste buttons that work relative to a chosen object. When copying, the world-space transform is determined, and then adjusted to become relative to the world-space transform of the chosen object. When pasting, this happens in reverse.

If no object is chosen, the copy/paste will happen relative to the active scene camera. What is the active scene camera is determined for every action, so when you paste it can be different from when you copied. This can help to keep an object visually in the same place when switching cameras, or when switching between scenes.

Pasting a transform adjusts the Object/Bone's location, rotation, and scale. This means that when copying a skewed transform, this skew is lost.

If there are constraints on the Object/Bone, the resulting visual transformation may not be the same as the pasted one. To give a concrete example: if you have a constraint that adds a rotation, it will always add that rotation on top of the pasted transform.

See also

Pose Library for a way to manage and share entire poses.

Reference

Animation

Simple add-on for copying world-space transforms.

3D Viewport `N`-panel `Animation` tab

`copy_global_transform.py`

Sybren A. Sticvel

Sybren A. Sticvel

GPL 2+

Official

This add-on is bundled with Blender.

--- Fin du chapitre: `_copy_global_transform` ---

--- Début du chapitre: `_copy_location` --- The Copy Location constraint forces its owner to have the same location as its target.

Important

Note that if you use such a constraint on a connected bone, it will have no effect, as it is the parent's tip which controls the position of your owner bone's root.

Copy Location panel. `Copy`

Data ID used to select the constraint's target, and is not functional (red state) when it has none. See common constraint properties for more information.

These buttons control which axes are constrained.

Invert their respective corresponding axis coordinates.

When enabled, this control allows the owner to be moved (using its current transform properties), relative to its target's position.

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

Let us animate a solar system with the Copy Location constraint and its Offset option. You can make the owner, called 'moon', describe perfect circles centered on the world origin (using e.g. Location X/Y sine and cosine F-Curves, see Built-in Function Modifier). Then copy the location of a target 'earth' with the Offset checkbox enabled to model a satellite in a (simplified) orbit around its planet. Repeat these steps for more planets circling around its center star 'sun'.

Following video is a small animation of a solar system created using (among a few others) the previously described technique:

Note that, this 'solar' system is not realistic at all (the wrong scale, the earth is rotating in the wrong direction around the sun, 'a6').

You can download the blend-file used to create this animation.

Furthermore you can also animate a few properties of each constraint using animation curves: e.g. you can animate the Influence of a constraint. It is used to first let the camera follow the moon, then the earth, and finally using two Copy Location constraints with Offset set.

--- Fin du chapitre: _copy_location ---

--- Début du chapitre: _copy_paste --- Reference

Pose Mode

Pose ' Copy Pose, Pose ' Paste Pose, Pose ' Paste Pose Flipped

Ctrl- C , Ctrl- V , Shift- Ctrl- V

Blender allows you to copy and paste a pose, either through the Pose menu, or by using hotkeys.

Copy the current pose of selected bones into the pose buffer.

Paste the buffered pose to the currently posed armature.

Paste the X axis mirrored buffered pose to the currently posed armature.

Here are important points:

This tool works at the Blender session level, which means you can use it across armatures, scenes, and even files. However, the pose buffer is not saved, so you lose it when you close Blender.

There is only one pose buffer.

Only the selected bones are taken into account during copying (i.e. you copy only selected bones' pose).

During pasting, on the other hand, bone selection has no importance. The copied pose is applied on a per-name basis (i.e. if you had a forearm bone selected when you copied the pose, the forearm bone of the current posed armature will get its pose when you paste it ' and if there is no such named bone, nothing will happen 'a6).

What is copied and pasted is in fact the position, rotation or scale of each bone, in its own space. This means that the resulting pasted pose might be very different from the originally copied one, depending on:

The rest position of the bones.

And the current pose of their parents.

The rest position of the original armature. \xc2\xbf
The rest position of the destination armature. \xc2\xbf
The first copied pose (note that only two bones are selected and hence copied). \xc2\xbf
The pose pasted on the destination armature. \xc2\xbf
The pose mirror-pasted on the destination armature. \xc2\xbf
The same pose as above is copied, but this time with all bones selected. \xc2\xbf
The pose pasted on the destination armature. \xc2\xbf
The pose mirror-pasted on the destination armature. \xc2\xbf
--- Fin du chapitre: _copy_paste ---

--- Début du chapitre: _copy_rotation --- The Copy Rotation constraint forces its owner to match the rotation of its target.
Copy Rotation panel. \xc2\xbf
Data ID used to select the constraints target, and is not functional (red state) when it has none. See common constraint properties for more information.
Allows specifying which Euler order to use during the copy operation. Defaults to the order of the owner.
These buttons control which axes are constrained.
Invert their respective corresponding axis coordinates.
Specifies how the new rotation is combined with the existing rotation.
The new axis values replace existing values.
The new axis values are added to the existing values.
The new rotation is added before the existing rotation, as if it was applied to a parent of the constraint owner.
The new rotation is added after the existing rotation, as if it was applied to a child of the constraint owner.
This replicates the behavior of the original Offset checkbox. It was intended to be similar to the Before Original behavior, but does not work correctly with multiple axis rotations, and is thus deprecated.
Standard conversion between spaces. See common constraint properties for more information.
Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.
--- Fin du chapitre: _copy_rotation ---

--- Début du chapitre: _copy_scale --- The Copy Scale constraint forces its owner to have the same scale as its target.
Note
Here we talk of scale, not of size! Indeed, you can have two objects, one much bigger than the other, and yet both of them have the same scale. This is also true with bones: in Pose Mode, they all have a unitary scale when they are in rest position, represented by their visible length.
Copy Scale panel. \xc2\xbf
Data ID used to select the constraints target, and is not functional (red state) when it has none. See common constraint properties for more information.
These buttons control which axes of the target scale are copied.
Allows raising the copied scale to the specified arbitrary power.
Instead of copying scale for individual axes, apply a uniform scaling factor to all axes of the owner that achieves the same overall change in volume.

When enabled, the constraint combines the copied scale with the owner's scale, instead of overwriting it.
Uses addition instead of multiplication in the implementation of the Offset option.

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

Note

Since scale is a multiplicative quantity, it should be combined using multiplication, and split into fractions or inverted via power. Thus the use of Power is more mathematically correct than Influence, which uses linear interpolation. The use of the Additive option is also not recommended.

Tip

To copy scale from one axis of the target to all axes of the owner, disable other axes, enable Make Uniform, and set Power to 3.

--- Fin du chapitre: `_copy_scale` ---

--- Début du chapitre: `_copy_transforms` --- The Copy Transforms constraint forces its owner to have the same transforms as its target.

Copy Transforms panel. `_copy_transforms`

Data ID used to select the constraints target, and is not functional (red state) when it has none. See common constraint properties for more information.

Removes shearing from the target transformation after the target space conversion, ensuring it consists purely of translation, rotation and scale. Note that Copy Rotation always does this.

Specifies how the copied transformation is combined with the existing transformation.

The new transformation replaces the existing transformation.

The new transformation is added before/after the existing transformation, as if it was applied to an imaginary parent/child of the constraint owner. Scale is handled like in the most basic Full Inherit Scale mode of bones, so combining non-uniform scale and rotation will create shear.

The new transformation is added before/after the existing transformation, as if it was applied to an imaginary parent/child of the constraint owner. Scale is handled like in the Aligned Inherit Scale mode of bones to avoid creating shear.

This is equivalent to using the Split Channels option, but replacing the location component with the result of Full. If only uniform scale is used, the result is identical to Full.

Combines location, rotation and scale components of the transformation separately, similar to a sequence of three Copy Location, Copy Rotation and Copy Scale (with Offset) constraints bundled together in one operation; the result may be slightly different in case of sheared inputs. Unlike Aligned, in this mode location channels are simply added together, so rotation and scale components of the input transformations cannot affect the resulting location.

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

This video shows the difference between the mix modes. The right input is mixed after the left one:

A general demonstration of the constraint:

--- Fin du chapitre: `_copy_transforms` ---

--- Début du chapitre: `_copy_uvmaps` ---Reference
Object Mode

Object `\xe2\x80\xa3 Link/Transfer Data \xe2\x80\xa3 Copy UV Maps`

The active UV map of the selected objects will be replaced by a copy of the active UV map of the active object. If the selected object doesn't have any UV maps, it is created. Objects must be of type mesh and must have a matching topology.

--- Fin du chapitre: `_copy_uvmaps` ---

--- Début du chapitre: `_corners_of_edge` --- Selects a neighboring face corner of an edge and outputs its index.

This node is a bit special because it operates in two different domains. First, it evaluates a Weight for each corner in the geometry. Then, for each item in the context domain, it will:

Pick an edge from the geometry based on the Edge Index.

Find some (not all) face corners connected to this edge \xe2\x80\x93 see below.

Sort these corners by their associated weight.

Pick a corner from the above sorted list based on the Sort Index, where 0 means the corner with the lowest weight, 1 means the corner with the second-lowest weight and so on.

Output the geometry-wide index of this corner.

Warning

As illustrated below, the node only looks at one corner per connected face. Even though the edge has four neighboring corners, Corner Index can only return the indexes of two of them, and Total will similarly return 2. You can use the Offset Corner in Face Node to retrieve the indexes of the other corners.

A graphic for which corners are returned for a given edge \xc2\xb6

Red: selected edge.

Blue: the corners whose index can be retrieved using this node.

Purple: the corners that can be retrieved by offsetting the blue corner indices using the Offset Corner in Face Node.

The index of the edge for which to find connected corners.

Note

If this input is not connected, it uses the index of the context item, which means it's important that the node is evaluated in the Edge domain.

The weights of the corners in the geometry. Unlike the other inputs which follow the context domain, this one is always evaluated in the Face Corner domain.

The corners are sorted by their associated weight in ascending order. Corners with the same weight are sorted by their index.

The 0-based index of the corner to select from the edge's sorted corners. If this value is outside the range of valid indices, it wraps around.

This node has no properties.

The geometry-wide index of the selected corner. You can pass this to the Evaluate at Index Node or the Sample Index Node (with the domain set to Face Corner) to retrieve details about the corner.

If the edge has no connected corners, Corner Index will be zero.

The number of faces (not face corners!) connected to the edge.

See also

The page for the Edges of Vertex Node has an example of how to work with the different domains.

--- Fin du chapitre: `_corners_of_edge` ---

--- Début du chapitre: `_corners_of_face` --- Selects a corner of a face and outputs its index.

This node is a bit special because it operates in two different domains. First, it evaluates a Weight for each corner in the geometry. Then, for each item in the context domain, it will:

Pick a face from the geometry based on the Face Index.

Find the corners of this face.

Sort these corners by their associated weight.

Pick a corner from the above sorted list based on the Sort Index, where 0 means the corner with the lowest weight, 1 means the corner with the second-lowest weight and so on.

Output the geometry-wide index of this corner.

The index of the face for which to find the corners.

Note

If this input is not connected, it uses the index of the context item, which means it's important that the node is evaluated in the Face domain.

The weights of the corners in the geometry. Unlike the other inputs which follow the context domain, this one is always evaluated in the Face Corner domain.

The corners are sorted by their associated weight in ascending order. Corners with the same weight are sorted by their index.

The 0-based index of the corner to select from the face's sorted corners. If this value is outside the range of valid indices, it wraps around.

This node has no properties.

The geometry-wide index of the selected corner. You can pass this to the Evaluate at Index Node or the Sample Index Node (with the domain set to Face Corner) to retrieve details about the corner.

The number of corners in the face, which is also its number of edges.

See also

The page for the Edges of Vertex Node has an example of how to work with the different domains.

--- Fin du chapitre: `_corners_of_face` ---

--- Début du chapitre: `_corners_of_vertex` --- Selects a neighboring face corner of a vertex and outputs its index.

This node is a bit special because it operates in two different domains. First, it evaluates a Weight for each corner in the geometry. Then, for each item in the context domain, it will:

Pick a vertex from the geometry based on the Vertex Index.

Find the face corners adjacent to this vertex.

Sort these corners by their associated weight.

Pick a corner from the above sorted list based on the Sort Index, where 0 means the corner with the lowest weight, 1 means the corner with the second-lowest weight and so on.

Output the geometry-wide index of this corner.

The index of the vertex for which to find the corners.

Note

If this input is not connected, it uses the index of the context item, which means it's important that the node is evaluated in the Point domain.

The weights of the corners in the geometry. Unlike the other inputs which follow the context domain, this one is always evaluated in the Face Corner domain.

The corners are sorted by their associated weight in ascending order. Corners with the same weight are sorted by their index.

The 0-based index of the corner to select from the vertex's sorted corners. If this value is outside the range of valid indices, it wraps around.

This node has no properties.

The geometry-wide index of the selected corner. You can pass this to the Evaluate at Index Node or the Sample Index Node (with the domain set to Face Corner) to retrieve details about the corner.

The number of adjacent corners, which is also the number of faces.

See also

The page for the Edges of Vertex Node has an example of how to work with the different domains.

--- Fin du chapitre: _corners_of_vertex ---

--- Début du chapitre: _corner_pin --- The Corner Pin node uses explicit corner values for a plane warp transformation. It works like the Plane Track Deform node, but without using the plane track data from the Movie Clip Editor.

Standard color input.

Four vector inputs to define the plane warping. (Z component of vector inputs is ignored.)

This node has no properties.

Standard color output. (The image after distorting.)

A black-and-white alpha mask of the plane.

An example of the Corner Pin node.

An example of the distorted image.

In the example above, the image of the bird is distorted by the vectors specified by the Corner Pin node.

--- Fin du chapitre: _corner_pin ---

--- Début du chapitre: _corrective_smooth --- The Smooth Corrective modifier is used to reduce highly distorted areas of a mesh by smoothing the deformations.

This is typically useful after an Armature modifier, where distortion around joints may be hard to avoid, even with careful weight painting.

To use this modifier effectively, it is important to understand the basics of how it works.

Used as a reference to detect highly distorted areas. The original vertex locations are used by default.

Many options for this modifier relate to smoothing which is used internally to correct the distorted regions.

The Smooth Corrective modifier. \xc2\xba6

The factor to control the smoothing amount. Higher values will increase the effect.

Values outside expected range (above 1.0 or below 0.0) will distort the mesh.

The number of smoothing iterations, equivalent to executing the Smooth tool multiple times.

Additional scaling factor to increase the size of the mesh. This is useful because sometimes the Smooth Corrective modifier will introduce volume loss, especially when used with a rig.

Select the smoothing method used.

This simply relaxes vertices to their connected edges.

Uses a method of relaxing that weights by the distance of surrounding vertices. This option can give higher quality smoothing in some cases, by better preserving the shape of the original form.

If set, restrict the effect to the only vertices in that vertex group. This allows for selective, real-time smoothing, by painting vertex weights.

This option is included to preview the smoothing used, before correction is applied.

Prevent boundary vertices from smoothing.

Select the source for reference vertex positions that defines the undeformed state.

Use the original input vertex positions. This relies on the original mesh having the same number of vertices as the input, modified mesh.

Optionally you may bind the modifier to a specific state. This is required when there are constructive modifiers such as Subdivision Surface or Mirror in the stack before this modifier.

Armature only. \xc2\xba6

Armature and Corrective Smooth. \xc2\xba6

--- Fin du chapitre: _corrective_smooth ---

--- Début du chapitre: _crash --- The most common causes of Blender crashes:

- Running out of memory.
- Issues with graphics hardware or drivers.
- Bugs in Blender.

Firstly, you may be able to recover your work with File \xe2\x80\xa3 Recover \xe2\x80\xa3 Auto Save \xe2\x80\xa6.

To prevent the problem from happening again, you can check that the graphics drivers are up to date (Troubleshooting Graphics Hardware), upgrade your machine \xe2\x80\x99s hardware (the RAM or graphics card), and disable some options that are more memory intensive:

- Reduce undo steps Preferences \xe2\x80\xa3 System \xe2\x80\xa3 Memory & Limits \xe2\x80\xa3 Undo Steps.
- Using multisample anti-aliasing also increases the memory usage and makes the display slower.

On Linux, the Window Manager (KDE and Gnome for example) may be using hardware accelerated effects (e.g. window shadows and transparency) that are using up the memory that Blender needs. Try disabling the desktop effects or switch to a lightweight Window Manager.

To check memory usage by Blender:

- On Windows, use Task Manager and sort by Memory.

On macOS, use Activity Monitor.app and open Memory tab. Alternatively, `runtop -o MEM`.

On Linux, `runtop -o %MEM`.

When Blender crashes, it writes out a text file which contains information that may help identify the cause of the crash. Usually, this file is written in the Temporary Directory directory.

This file contains a log of tools used up until the crash as well as some other debug information. When reporting bugs about crashes it can be helpful to attach this file to your reports, especially when others are unable to reproduce the crash.

On a crash, a file is written based on the name of the currently loaded blend-file, so `test.blend` will create a file called `test.crash.txt`.

Batch scripts are provided in Blender installation directory which may be run to obtain the Blender debug log and system info text files:

`blender_debug_log.cmd` is used in most cases.

`blender_debug_gpu.cmd` and `blender_debug_gpu_workaround.cmd` log GPU-related errors.

`blender_factory_startup.cmd` starts Blender with default settings which is recommended for debugging.

If the crash happens in Blender module, stack trace is also written to a file named `blender.crash.txt`. The path to that file can be found at the end of `blender_debug_log.txt` file.

After crash, macOS Crash Reporter shows a window with backtrace after some time, or when Blender is opened again. Copy the text in the crash report and save it in a text file. That file should be attached to the bug report while following other bug reporting guidelines.

Some crash files can also be found

in `~/Library/Logs/DiagnosticReports/` with the name of format: `Blender_YYYY-MM-DD-HHMMSS_MACNAME.crash`. If a report is present corresponding to the time of crash, that file can also provide hints about cause of the crash. Alternatively, Console.app can be used to navigate all `~/Library/Logs/DiagnosticReports/` (see sidebar in the app).

On a crash, a file named `blender.crash.txt` is written in `/tmp` directory.

Note

More logs can be obtained by running Blender from Command Line and using `--factory-startup--debug-all` flags. See Launching from the Command Line and Command Line Arguments .

--- Fin du chapitre: _crash ---

--- Début du chapitre: _crease ---Reference

Sculpt Mode

Sidebar `Tool` `Brush Settings` `Brush Type`

Create sharp indents or ridges by pushing or pulling the mesh, while pinching the vertices together.

Crease can also be used to sharpen and polish existing creases. Enable pressure sensitivity on Strength to regulate the add/subtract effect while pinching creases.

Adds a consistent pinching effect to your stroke. If set to 0 it the brush will behave like the Draw brush. If set to 1 and the brush strength set to 0, the brush will behave like a Pinch brush.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _crease ---

--- Début du chapitre: `_crease_angle` --- A modifier based on the Crease Angle (angle between two adjacent faces). If a stroke segment does not lie on a crease (i.e. the edge does not have the Crease Angle nature), its properties are not touched by the modifier.

The range of input values to the mapping. Out-of-range crease angle values will be clamped by the Min and Max angles and their corresponding property values.

Crease Angle modifier example by T.K. (blend-file).

--- Fin du chapitre: `_crease_angle` ---

--- Début du chapitre: `_create_guide_index_map` --- Creates an integer attribute named `guide_curve_index` that stores the nearest guide curve for every curve to its nearest guide via index.

Other nodes in the Hair Guides Nodes category can generate guide maps themselves for convenience, but the behavior is always the same as this node.

Geometry

Guide Curves or Points used for the selection of Guide Curves.

Minimum distance between two guides.

Mask for which curve are eligible to be selected as guides.

ID to group together curves for guide map creation. Curves will only choose a guide with the same ID value.

This node has no properties.

Output geometry including the new map attribute and the guide selection anonymous attribute as well. This geometry includes the guide curves, they are not separated.

Output geometry including only the selected guide curves.

The index of the closest curve with the same Group ID value.

A selection in the Geometry output set to true for only the curves that were chosen as guides.

--- Fin du chapitre: `_create_guide_index_map` ---

--- Début du chapitre: `_crop` --- The Crop node crops an input image to a selected region by either making the cropped area transparent or by resizing the input image.

Standard color input. If no image is selected, an image filled with the selected color is used. You can use and crop this image in combination with another background image.

When disabled, the image remains the same size, but the cropped areas become transparent pixels. When enabled, the image size is cropped to the specified region and gets a new width or height or both.

Note that this will probably reposition the image in the render output because the cropped image is automatically centered.

When enabled, crop dimensions are a percentage of the input image's width and height. When disabled, the range of the Crop Region Values are the width and height of the image in pixels.

Define borders of the crop region; Left or Right can vary between 0 and the width of the image. Up or Down can vary between 0 and the height of the image.

Note

The terminology (Left, Right, Up, Down) can be misunderstood easily. First, the numbers do not represent the amount of cropping, e.g. Left is set to 50 and Right to 50 does not mean that you will be cropping

the image for 50 pixels on both the left and right side. In fact, it will result in zero-sized image because you are cropping from pixel 50 to pixel 50. So, the numbers defines a position in the input image.

Secondly, depending on which one is bigger, Left should be interpreted as Right and vice versa. If Left is greater than Right then both values are switched and Left gets the value of Right and vice versa. The same operation is done for Up and Down, where you can think of them as the top and bottom of the image.

Thirdly, the terms Up and Down are ambiguous and suggest an action; e.g. \xe2\x80\x9cCrop down\xe2\x80\x9d. The values, however, are not amounts but positions. The term Down should be interpreted as \xe2\x80\x9cBottom\xe2\x80\x9d and Up as \xe2\x80\x9cTop\xe2\x80\x9d. Standard color output.

The following workflow removes some possible confusion:

Uncheck Crop Image Size for this step, so that you can see the borders of the input image. To see this border, you have to select the Viewer node. If you don't need pixel-perfect cropping, check Relative so that you do not have to consider the exact dimensions of the input image. Set Left and Down to zero. Set Right and Up to one, or to the width and height of the input image. You should see now the entire input image in the backdrop. Up is thus interpreted as the top of the image. The origin of the image (0, 0) is at the bottom (down) left corner. To crop from the left, change the Left value. To crop from the right, change the Right value. To crop from the top, change the Up value. To crop from the bottom, change the Down value.

--- Fin du chapitre: _crop ---

--- Début du chapitre: _cross --- The Cross transition fades from one strip to another, also known as a crossfade. Strips can be overlapping or have a gap between them, however, when strips contain a gap the last and first frame of each strip extend which can cause a pause if any of the strips are a sequence.

Automatically calculates a linear fade over the length of the strip. Allows you to manually keyframe a custom fade. This can be used with different easings to fine-tune the fade in/out.

Cross Effect. \xc2\xb6

--- Fin du chapitre: _cross ---

--- Début du chapitre: _cryptomatte --- The Cryptomatte node uses the Cryptomatte standard to efficiently create mattes for compositing. Cycles and EEVEE output the required render passes, which can then be used in the Compositor or another compositor with Cryptomatte support to create masks for specified objects.

Unlike the Material and Object Index passes, the objects to isolate are selected in compositing, and mattes will be anti-aliased and take into account effects like motion blur and transparency. Standard color input.

The source of the Cryptomatte data.

Use Cryptomatte data that are stored as part of the render.

Use Cryptomatte data that are stored inside a multilayered OpenEXR image.

Scene selector. Only available when Render Source is selected.

Image selector. Only available when Image Source is selected.

Selector of the Cryptomatte layer.

List of object and material crypto IDs to include in matte. This can be used for example to quickly clear all mattes by deleting the text used to copy-and-paste crypto IDs from other software.

A colored output of the input image with the matte applied to only include selected layers.

A black-and-white alpha mask of all the selected crypto layers.

A colored representation of the Cryptomatte pass which can be used as a higher contrast image for color picking.

Enable Cryptomatte Object render pass in the Passes panel, and render.

In the compositing nodes, create a Cryptomatte node and select the Cryptomatte layer.

Attach a Viewer node to the combined pass of the render layers.

Use the Cryptomatte Add/Remove button to sample objects from the Compositor backdrop.

Use the Matte output of the Cryptomatte node to get the alpha mask.

The Image editor, UV editor, node backdrop or Movie Clip editor can be used to pick a Cryptomatte sample. They don't need to show any Cryptomatte layer. The node will use the sample image coordinate to sample in the Cryptomatte layer that is selected in the node.

In the example below, you can see the pass output on the right side. On the left side you can see a couple of objects that were selected through the Cryptomatte node. Notice how the cube on the left has a sphere-shaped cut-out from a sphere that was not selected in the node.

Cryptomatte sidecars (metadata files) are not supported.

Cryptomatte node cannot be used in node groups.

Volume Objects are not supported.

--- Fin du chapitre: _cryptomatte ---

--- Début du chapitre: _cryptomatte_legacy --- The Cryptomatte node uses the Cryptomatte standard to efficiently create mattes for compositing. Cycles and Eevee output the required render passes, which can then be used in the Compositor or another compositor with Cryptomatte support to create masks for specified objects.

Unlike the Material and Object Index passes, the objects to isolate are selected in compositing, and mattes will be anti-aliased and take into account effects like motion blur and transparency.

Important

The Cryptomatte Legacy node is deprecated and replaced by Cryptomatte Node. The legacy node will be removed in a future Blender release.

Standard color input.

Each crypto layer will be given its own render pass; each of these render passes must be connected to one of these crypto layer inputs. By default there are only four layers, see Adding/Removing Layers to add more.

Adds/Removes an object or material from matte, by picking a color from the Pick output.

List of object and material crypto IDs to include in matte. This can be used for example to quickly clear all mattes by deleting the text used to copy-and-paste crypto IDs from other software.

A colored output of the input image with the matte applied to only include selected layers.

A black-and-white alpha mask of all the selected crypto layers.

A colored representation of the Cryptomatte pass which can be used with a Viewer node to select which crypto passes are used to create the matte image.

Enable Cryptomatte Object render pass in the Passes panel, and render.
In the compositing nodes, create a Cryptomatte node and link the Render Layer matching Image and Cryptomatte passes to it.

Attach a Viewer node to the Pick output of the Cryptomatte node.

Use the Cryptomatte Add/Remove button to sample objects in the Pick Viewer node.

Use the Matte output of the Cryptomatte node to get the alpha mask.

By default there are only four crypto layers available as inputs to the Cryptomatte node. You can add or remove layer inputs through Sidebar \xe2\x80\xa3 Item \xe2\x80\xa3 Properties \xe2\x80\xa3 Add/Remove Crypto Layer. These operators will add/remove layers from the bottom of the pass inputs.

In the example below, you can see the pass output on the right side. On the left side you can see a couple of objects that were selected through the Cryptomatte node. Notice how the cube on the left has a sphere-shaped cut-out from a sphere that was not selected in the node.

--- Fin du chapitre: _cryptomatte_legacy ---

--- Début du chapitre: _cube --- The Cube node generates a cuboid mesh with variable side lengths and subdivisions. The inside of the mesh is still hollow like a normal cube.

Side lengths along each of the main axes.

Number of vertices for each side of the cube. The number of vertices should be at least 1.

This node has no properties.

Standard geometry output.

A 2D vector representing the default X/Y coordinates of the UV Map for the primitive \xe2\x80\x99s shape. This can be connected to the Store Named Attribute Node, to be used once the Geometry Nodes Modifier gets applied. The UV map must be stored on the face corner in order to be accessed.

--- Fin du chapitre: _cube ---

--- Début du chapitre: _curl_hair_curves --- Deforms existing hair curves into curls using guide curves.

Geometry

Guide index map which describes which curve to use as the center of each braid group. If this input is provided, it has priority over an existing map in the guide_curve_index attribute, and the Guide Distance and Guide Mask attributes will be unused.

Minimum distance between two guides for new guide map.

Mask for which curve are eligible to be selected as guides.

Use the existing guide map attribute if available. If this is false, and the Guide Index input isn't provided, the Guide Distance and Guide Mask inputs will be used to generate a new guide map for this node. Creating the guide map in a separate node or modifier gives more complete control over its creation.

Factor to blend overall effect.

Subdivision level applied before deformation.

Percentage along each curve to blend deformation from the root.

Overall radius of the curls.

Factor for the radius at the curl start.

Factor for the radius at the curl end.

Frequency factor of the curls.This input can vary for different points of the same curve.

Amount of random offset per curve.

Random Seed for the operation.

This node has no properties.

Geometry

Guide index map that was used for the operation.If a new guide map is created by this node, it will be stored forthis output.

--- Fin du chapitre: `_curl_hair_curves` ---

--- Début du chapitre: `_cursor` ---Reference

All Paint Modes

Tool Settings \xe2\x80\xa3 Brush Settings \xe2\x80\xa3 Cursor

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Cursor

Cursor options.\xc2\xb6

While painting or sculpting a special cursor is shown to display information about the active brush.The cursor is shown as a circle in the 3D Viewport, the radius of the circle matches the size of the brush.

The cursor can be disabled by toggling the checkbox in the panel\xe2\x80\x99s header.

Set the color of the brush ring while performing an add/positive stroke.

In some paint/sculpt modes the brush can be negative and subtract information from the paint target;these brushes can be given a separate color.

Depending on the paint or sculpt mode different overlays are shown within the cursorto give information on how the brush is textured.This is most commonly used to show the brush falloff with a gradient from the circle center to the perimeter.

You can change the amount of transparency usedwhen showing the texture using the slider.

Allows you to turn off the viewport overlay during strokes.

Toggles whether to show or hide the given brush texture overlay.

--- Fin du chapitre: `_cursor` ---

--- Début du chapitre: `_curvature_3d` --- A modifier based on radial curvatures of the underlying 3D surface.The curvature of a 2D curve at a point is a measure of how quickly the curve turns at the point.The quicker the turn is, the larger the curvature is at the point.The curvature is zero if the curve is a straight line.Radial curvatures are those computed for a 2D curve that appears at the cross sectionbetween the 3D surface and a plane defined by the view point (camera location)and the normal direction of the surface at the point.

For radial curvatures to be calculated (and therefore for this modifier to have any effect),the Face Smoothnessoption has to be turned on and the object needs to have Smooth Shading.

The limits of the mapping.If the current point of the stroke is at Min Curvature or less from the target,it will take the start point of the mapping. And conversely,if it is at Max Curvature or more from the target, it will take the end-point value of the mapping.

Curvature 3D modifier demo by T.K.(blend-file).\xc2\xb6

--- Fin du chapitre: `_curvature_3d` ---

--- Début du chapitre: `_curve` ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Curve

The Curve tool create complex B\xc3\xa9zier style curves using any of the Drawtype brushes..

You can configure the brush main settings exposed on the Tool Settings for convenience. For the draw brushes configuration and settings see: Draw Brush .

The number of stroke points between each stroke edge.

Use a curve widget to define the stroke thickness from the start (left) to end (right) of the stroke.

When enabled, the stroke use a curve profile to control the thickness along the curve.

Picks the brush asset used by the tool.

See Brush Asset for more information.

See Draw Brushes for a detailed list of all draw brushes and their options.

Parameters to control to look of the stroke.

See Draw Brushes for details.

Settings to determine the color of strokes.

See Color

In the Tool Settings select the brush, material and color type to use with the tool. The Curve tool uses Draw Brush types. See Brush Settings for more information.

Click (LMB or the Pentip) and drag the start point.

Release on the desired end point.

After releasing you can tweak the curve using two cyan B\xc3\xa9zier like manipulators.

Then confirm (Return/ MMB) or cancel (Esc/ RMB).

While dragging you can hold Shift to use only one manipulator to tweak the curve (like the Arc tool), use Alt to create the arc from a center point.

Numpad Plus and Numpad Minus or using the mouse Wheel will increase or decrease the amount of points in the final curve.

click and dragging the start point. \xc2\xb6

Tweaking curve with the manipulators. \xc2\xb6

The curve after confirming. \xc2\xb6

Before confirming you can use E to extrude the end point of the curve to generate multiple connected curves.

End point extruding. \xc2\xb6

Tweaking the last curve with the manipulators. \xc2\xb6

The connected curves after confirming. \xc2\xb6

--- Fin du chapitre: _curve ---

--- Début du chapitre: _curves --- The Vector Curves node maps an input vector components to a curve.

Use this curve node to slow things down or speed them up from the original scene.

In the shader context the node also has an additional Factor property. Controls the amount of influence the node exerts on the output vector. Standard vector input.

X, Y, Z

For the curve controls see: Curve widget .

Standard vector output.

--- Fin du chapitre: _curves ---

--- Début du chapitre: `_curves_to_grease_pencil` --- The Curves to Grease Pencil node Converts top-level curve instances into Grease Pencil layers Either plain curves or curve instances.
Either a curve or instance selection.
Create a separate layer for each instance. If instances are used, realized curve geometry will be ignored. Layer names will use the instance geometry name. If real curve geometry is used, a single layer is created with the input geometry's name.
This node has no properties.
Standard Grease Pencil geometry.
--- Fin du chapitre: `_curves_to_grease_pencil` ---

--- Début du chapitre: `_curve_circle` --- The Curve Circle node generates a poly spline circle.
Number of edges on the circle.
The radius of the circle.
The three points on the circle. The order of the points determines the direction (clockwise or counterclockwise) of the circle.
Note
Because of the finite resolution, the three points do not necessarily lie on the generated curve.
The position and radius of the circle are determined by three points. The center of the circle is also given as an output. If the three points lie on one line, no geometry is generated.
The circle is determined by the radius.
Poly spline generated from the inputs.
The center of the circle defined by the three points.
--- Fin du chapitre: `_curve_circle` ---

--- Début du chapitre: `_curve_display` --- Reference
Edit Mode
3D Viewport \xe2\x80\xa3 Viewport Overlays \xe2\x80\xa3 Curve Edit Mode
When in Edit Mode, curves have special overlays to control how curves are displayed in the 3D Viewport.
Toggles the option to display the Bezier handles.
Toggles the display of the curve normals.
Length of the axis that points the direction of the normal.
--- Fin du chapitre: `_curve_display` ---

--- Début du chapitre: `_curve_guide` --- Reference
Physics \xe2\x80\xa3 Force Fields
Curve Guide
The Curve Guide is used to force particles to follow a certain path defined by a Curve Object. A typical scenario would be to move a red blood cell inside a vein, or to animate the particle flow in a motor. You can also use Curve Guide to shape certain hair strands.
Note
You can also use the Particle Edit Mode to define a path.
Since you can animate curves as a soft body or any other usual way, you may build very complex animations while keeping great control and keeping the simulation time to a minimum.
To make particles point in the direction of the curve, you need to set their Orientation Axis to Velocity / Hair, enable Dynamic, and set their

Angular Velocity Axis to Velocity, all in the theRotation settings of the particle system. The Follow Path Constraint and the curve\– legacyFollow option won\– work for this.

A Curve Guide force affects all particles on the same layer, independently from their distance to the curve. If you have several guides in a layer, their fields add up to each other (the way you may have learned it in your physics course). But you can limit their influence radius by changing the Minimum Distance (see below).

A particle follows a Curve Guide during its lifetime, the velocity depends on its lifetime and the length of the path.

Note

The Curve Guide does not affect soft bodies .

Fraction of particle life time, that is not used for the curve.

This setting governs the strength of the guide between Min Distance and Max Distance. A falloff of 1 means a linear progression.

If you use Additive, the speed of the particles is also evaluated depending on the falloff.

Use Curve weights to influence the particle influence along the curve. The particles come together at the end of the curve (1) or they drift apart (-1).

Defines the form in which the particles come together. +0.99: the particles meet at the end of the curve. 0: linear progression along the curve. -0.99: the particles meet at the beginning of the curve.

The distance from the curve, up to where the force field is effective with full strength. If you have a falloff of 0, this parameter will have no effect, because the field is effective with full strength up to Max Distance (or the infinity). Min Distance is shown with a circle at the endpoints of the curve in the 3D Viewport.

The maximum influence radius. Shown by an additional circle around the curve object.

Warning

This feature is broken in the current version, see Bug Report #46776.

Changes the shape that the particles can take.

Todo.

Braid.

The radius of the influence depends on the distance of the curve to the emitter.

A three-dimensional, standing wave.

A one-dimensional, standing wave.

Todo.

A two-dimensional, standing wave.

It is not so easy to describe the resulting shapes, so have a look at the example below.

Kink options of a curve guide. From left to right: Radial, Wave, Braid, Roll. Animation. \–

Which axis to use for the offset.

The frequency of the offset.

Adjust the offset to the beginning/end.

The Amplitude of the offset.

Curve Guide force field. \–

--- Fin du chapitre: _curve_guide ---

--- D  but du chapitre: _curve_handle_positions --- Gets the two handle positions of each control point in a B\\©zier spline.

You can use the `Set Handle Positions Node` to change these positions.
Output the handle positions relative to the control point instead of in the local space of the geometry.

This node has no properties.

The position of the control point's left handle.

The position of the control point's right handle.

--- Fin du chapitre: `_curve_handle_positions` ---

--- Début du chapitre: `_curve_info` --- Reads information about each curve.

This node has no inputs.

This node has no properties.

Index of each Curve.

ID of each Curve.

Length of each Curve.

Direction from root to tip of each Curve.

Random vector for each Curve.

Attachment surface UV coordinate of each Curve.

--- Fin du chapitre: `_curve_info` ---

--- Début du chapitre: `_curve_length` --- The Curve Length node outputs the length of all splines added together.

Standard geometry input.

This node has no properties.

Accumulated length of all splines of the curve.

--- Fin du chapitre: `_curve_length` ---

--- Début du chapitre: `_curve_line` --- The Curve Line node generates poly spline line.

Position of the first control point.

Position of the second control point. This is only available in the Points mode.

Direction the line is going in. The length of this vector does not matter. This is only available in the Direction mode.

Length of the line. This is only available in the Direction mode.

Define the spline with start and end points.

Define the spline with a start, direction and length.

Standard geometry output.

--- Fin du chapitre: `_curve_line` ---

--- Début du chapitre: `_curve_of_point` --- The Curve of Point node retrieves the index of the curve a control point is part of. This node is conceptually similar to the Face of Corner Node.

The index of the input control point.

Note

By default this uses the index from the field context, which makes it important that the node is evaluated on the point domain.

This node has no properties.

The index of the curve the control point is part of.

How far along the control point is along its curve, with a value of 0 for the first point in each curve.

--- Fin du chapitre: `_curve_of_point` ---

--- Début du chapitre: `_curve_root` --- Reads information about each curve's root point.
This node has no inputs.
This node has no properties.
Boolean selection of curve root points.
Position of the root point of a Curve.
Direction of the root segment of a Curve.
Index of the root point of a Curve.
--- Fin du chapitre: `_curve_root` ---

--- Début du chapitre: `_curve_segment` --- Reads information each point's previous curve segment.
This node has no inputs.
This node has no properties.
Distance to previous point on Curve.
Direction from previous neighboring point on segment.
Index of previous neighboring point on segment.
--- Fin du chapitre: `_curve_segment` ---

--- Début du chapitre: `_curve_spiral` --- The Curve Spiral node generates a poly spline in a spiral shape. It can be used to create springs or other similar objects. By default the spiral twists in a clockwise fashion.
Number of points in one rotation of the spiral.
Number of times the spiral makes a full rotation.
Radius of the start point and end point of the spiral. The radius of the spiral changes linearly between the two values over the whole spiral.
Height of the spiral.
Boolean value that changes the direction from clockwise to counterclockwise when it is enabled.
This node has no properties.
Poly spline generated from the inputs.
--- Fin du chapitre: `_curve_spiral` ---

--- Début du chapitre: `_curve_svg` --- Reference
Import-Export
File \xe2\x80\xa3 Import \xe2\x80\xa3 Scalable Vector Graphics (.svg)
Note
Currently the script allows only importing and is limited to path geometry only.
This add-on is enabled by default, in case it is not:
Open Blender and go to Add-ons section of the Preferences .
Search \xe2\x80\x9cScalable Vector Graphics (SVG)\xe2\x80\x9d and check the Enable Extension checkbox.
This add-on does not have any properties.
Todo.
--- Fin du chapitre: `_curve_svg` ---

--- Début du chapitre: `_curve_tangent` --- The Curve Tangent node outputs the direction that a curve points in at each control point, depending on the direction of the curve (which can be controlled with the Reverse Curve Node). The output values are normalized vectors.
Warning
For NURBS and B-spline curves, keep in mind that the value retrieved from this node is the value at every control point, which may

not correspond to the visible evaluated points. For example, a B-spline might have 48 evaluated points, but only four control points, if its resolution is 12. For NURBS splines the difference may be even more pronounced and the result may not be as expected. A Resample Curve Node can be used to create a poly spline, where there is a control point for every evaluated point.

This node has no inputs.

This node has no properties.

The direction of the curve at every control point.

--- Fin du chapitre: _curve_tangent ---

--- Début du chapitre: _curve_tilt --- The Curve Tilt node outputs the angle used to turn the curve normal around the direction of the curve tangent in its evaluated points. Keep in mind that the output is per control point, just like the values that can be controlled in curve Edit Mode. For NURBS and B-spline splines, the values will be interpolated to the final evaluated points.

The input node for this data is the Set Curve Tilt node.

This node has no inputs.

This node has no properties.

The tilt angle for the normal in radians.

--- Fin du chapitre: _curve_tilt ---

--- Début du chapitre: _curve_tip --- Reads information about each curve's tip point.

This node has no inputs.

This node has no properties.

Boolean selection of curve tip points.

Position of the tip point of a Curve.

Direction of the tip segment of a Curve.

Index of the tip point of a Curve.

--- Fin du chapitre: _curve_tip ---

--- Début du chapitre: _curve_to_mesh --- The Curve to Mesh node converts all splines of a curve to a mesh. Optionally, a profile curve can be provided to give the curve a custom shape. This profile curve is scaled by the radius at each control point; see the Set Curve Radius Node.

The node transfers attributes to the result. Attributes that are built-in on meshes but not curves, like sharp_face, will be transferred to the correct domain as well.

Tip

The output mesh has sharp edges set from the profile curve tagged automatically. If any splines in the profile curve are B-spline splines and any of the control points use Free or Vector handles, the corresponding edges will be shaded sharp.

Standard geometry input. All non-curve components are ignored.

If a profile curve is provided, it will be extruded along all splines. Otherwise the generated mesh will just be a chain of edges.

If the profile spline is cyclic, fill the ends of the generated mesh with n-gons. The resulting mesh is Manifold, the two new faces for each spline are simply connected to existing edges.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _curve_to_mesh ---

--- Début du chapitre: `_curve_to_points` --- The Curve to Points node generates a point cloud from a curve.

Standard curve input.

Number of points generated. This input is only available for Count mode.

Length of the curve. This input is only available for Length mode.

Creates points from the curve's evaluated points based on the resolution attribute for NURBS and B-spline splines. This mode will generally be the fastest, since the second step of sampling equal lengths is not necessary.

Sample each spline by evenly distributing the specified number of points.

Sample each spline by splitting it into segments with specified length. The length will be rounded down so that a whole number of samples will fit in each input spline. To counteract jumping when the length of the spline changes, the Trim Curve Node can be used with a multiple of this length.

Generated point cloud.

The normalized curve tangent at the sampled position, or the direct evaluated normal in Evaluated mode.

The normal value from the evaluated curve at each result point. This is the same value from the Normal Node at those positions.

The Euler rotation build from the Tangent and Normal outputs, for convenience.

--- Fin du chapitre: `_curve_to_points` ---

--- Début du chapitre: `_custom_data` --- This panel is used to manage any generic data attributes that a mesh could have.

Warning

Clearing any data will result in the data loss of these values.

Deletes the internal `sculpt_mask` attribute. This attribute is used by the Sculpt Masking Feature.

Used to manage the skin data which is used by the Skin Modifier. This operator can be needed in case a Skin modifier is created but no skin data exist.

Adds Custom Split Normals data, if none exists yet.

--- Fin du chapitre: `_custom_data` ---

--- Début du chapitre: `_custom_properties` --- Custom Properties panel.

Custom properties are a way to store your own data in Blender's data-blocks. It can be used for rigging (where bones and objects can have custom properties driving other properties), and Python scripts, where it's common to define new settings not available in Blender. It is also possible to access custom properties from materials via the Attribute Node.

Only certain data supports custom properties:

All data-blocks types.

Bones and pose bones.

Sequence strips.

To add a custom property, search for the Custom Properties panel, found at the bottom of most Properties or Sidebar region, and click New. Properties can be removed from the same location with the delete icon. Once properties are added they can be configured via the edit icon to work for a particular use case; see Editing Properties for more information.

Custom Properties edit pop-up.

Custom properties can be edited using the panel available for data types that support it. Editing the properties allows you to configure things such as default values, ranges, and even add a custom tooltip.

The data type of the property; different data types have can only have specific data properties.

A numeric value with decimals e.g. 3.141, 5.0, or 6.125.

A collection of multiple float data types e.g. [3.141, 5.0, 6.125]. This data type can also be used for data that can be represented as a float array such as colors. These special float arrays can be set in the Subtypeselector.

A numeric value without any decimals e.g. 1, 2, 3, or 4.

A collection of multiple integer data types e.g. [1, 2, 3, 4].

A data type that has two possible values e.g. True or False.

A collection of boolean values e.g. [True, False, True]

A sequence of characters such as "Some Text".

A reference to a Blender object, see Data-Blocks.

Edit a Python data type directly, used for unsupported data types.

The number of elements in the array. Note that if the array length is greater than 7 you cannot directly edit its elements, you must press Edit Value to edit the elements of the array.

The text that is displayed to the left of the value. This name is also used to access the property via Python.

This sets the default value of the property used by the Reset to Default Value operator.

Warning

Default values are used as the basis of NLA blending, and a nonsensical default (e.g. 0 for a property used for scaling) on a property intended for being keyframed is likely to cause issues.

The minimum/maximum value the custom property can take.

Allow the property to be overridden when the data-block is linked.

Enables limits that the Property Values slider can be adjusted to without having to input the value numerically.

The minimum/maximum value for the soft limit.

A multiplier to control how much the data type is incremented at a time. The internal step size for floats is 0.01, so a Step value of 5 will increment at a rate of 0.05 and a Step value of 100 will increment by 1.0. For integers the internal step size is 1.

The number of digits after the decimal to display in the user interface for float data types.

Specifies the type of data the property contains, which affects how it appears in the user interface. This option is only available for float properties and has different options for regular floats and float arrays. Note, the unit often depends on the Scene Units.

For regular floats:

Data values do not have any special behavior.

A measure digital image resolution.

The displayed value is a percentage, typically you will want the Min and Max values to be 0 and 100.

A percentage between an upper and lower bound which typically have a numerical significance.

A measure between intersecting lines.

Time specified in seconds.

Measure of space between items.

Work as a factor of time, measured in watts. This is used in Blender to measure light intensity.

Intensity of heat present.

The distance between cycles of a wave measured in millimeters (mm), micrometers (μm), nanometers (nm), or picometers (pm).

For float arrays:

Data values do not have any special behavior.

Color in linear color space.

Color in gamma corrected color space.

Euler Rotation angles.

Quaternion Rotation angles.

Note

For either of the color subtypes to work as expected the Property Value must be a vector with three or four values depending on the availability of an Alpha Channel .

The ID-block type. For example: Key, Image, Object, Material. See Data-Block Types for a full list.

Allows you to write a custom Tooltip for your property.

Custom properties can be accessed in a similar way to dictionaries, with the constraints that keys can only be strings, and values can only be strings, numbers, arrays of such, or nested properties.

See the API documentation for details.

--- Fin du chapitre: _custom_properties ---

--- Début du chapitre: _cylinder --- The Cylinder node generates a cylinder mesh. It is similar to the Cone node but always uses the same radius for the circles at the top and bottom.

Number of vertices on the circle at the top and bottom. No geometry is generated if the number is below three.

Number of edges running vertically along the side of the cone. No geometry is generated if the number is below one.

Number of concentric rings used to fill the round faces at the top and bottom. No geometry is generated if the number is below one.

Distance of the vertices from the Z axis. If this is zero, the output will be a single line.

Height of the cylinder.

How the circles at the top and bottom are filled with faces when their radius is larger than zero.

Do not fill the circles.

Fill the innermost segment of the circles with a single face.

Fill the innermost segment of the circles with triangles connected to a new vertex on the Z axis.

Standard geometry output.

A boolean attribute field with a selection of the faces on the top of the cylinder. If the Fill Type property is None, then this will be a selection of the top edges instead. If the Radius is zero, this will be a selection of the top point.

A boolean attribute field with a selection of the faces on the side of the cylinder.

This is the same as the Top selection output, but on the bottom side of the geometry instead.

A 2D vector representing the default X/Y coordinates of the UV Map for the primitive's shape. This can be connected to the Store Named Attribute Node, to be used once the Geometry Nodes Modifier get

applied. The UV map must be stored on the face corner in order to be accessed.

--- Fin du chapitre: `_cylinder` ---

--- Début du chapitre: `_damped_track` --- The Damped Track constraint constrains one local axis of the owner to always point towards Target. This constraint uses a pure Swing rotation, i.e. the shortest possible single axis rotation. In other 3D software you can find it with the name `Look at` constraint.

Although usually associated with bones, Damped Track can align objects to point to (and follow) other objects or bones. It is important to note that the constraint aligns the origin's axes to point to the target's origin point. This is illustrated in the following figure. In each case the objects are set as Damped Track to +X.

A: Object vertices aligned along axis origin. B: Object vertices aligned away from origin.

Damped Track panel.

Data ID used to select the constraint's target, and is not functional (red state) when it has none. See common constraint properties for more information.

Once the owner object has had a Damped Track constraint applied to it, you must then choose which axis of the object you want to point at the Target object. The negative axis direction causes the object to point away from the Target object along the selected axis direction.

-X, -Y, -Z, X, Y, Z

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_damped_track` ---

--- Début du chapitre: `_dash` --- The Dot Dash modifier generates dot dash segments from the original stroke.

The Dot Dash modifier.

Determines the starting offset of the pattern.

Makes up individual stroke of a dot dash pattern.

Use the plus/minus button on the side of the list to add/remove segments.

The number of consecutive points from the original stroke to include in this segment.

The number of points skipped after the segment ends.

The factor to apply to the original point's radius for the new points.

The factor to apply to the original point's opacity for the new points.

Use this index on generated segment, use -1 for existing material.

Close the segment.

See Influence Filters.

--- Fin du chapitre: `_dash` ---

--- Début du chapitre: `_data` --- Grease Pencil Object Data.

The Grease Pencil data-block menu can be used to link the data between objects.

Strokes can be grouped in 2D layers, a special Grease Pencil layer that help to organize the drawing order and visibility of the strokes. Layers can be organized into layer groups.

Onion skinning is used in animation to see several frames at once and make decisions based on how the previous/next frames are drawn. General settings for Grease Pencil strokes.

Layers can store Custom Attributes .The attributes are stored on the Layer domain.

For example, the Layer Adjustments are stored as layer attributes.

List view of all the attributes stored on the layers.

Name of the layer attribute.

The Data Type of the attribute.

Vertex groups can be used to assign a group or weighted group to some operator.An object can have several weight groups and can be assigned in Weight Paint Mode .

Create and manage your own properties to store data in the Grease Pencil's data-block.

--- Fin du chapitre: _data ---

--- Début du chapitre: _data_block --- Lets you select aData-Block(such as a material)in order to link it to something else (such as an object).

A data-block menu with a search field.

Shows an icon indicating the data-block type. Clicking the image or the down arrow opens the popup menu.Dragging the image lets you apply the data-block to something else. (For example, you can drag a material onto an object in the 3D Viewport to assign it.Dragging onto Data ID fields is also possible.)

A list of data-blocks available in the current blend-file, or a link to select an item from.The menu may show a preview besides the items and a search field to search the items in the list by name.

Note

Data-blocks with names that begin with . are hidden from the list, unless a string that also starts with . is entered into the search field, or the Show Hidden Files/Data-Blocks user preference is enabled.

Displays, and allows editing of, the name of the selected data-block.

Displays the number of users of the data (if there's more than one user).Clicking it will create a single-user copy.

As an example, if three separate objects referenced the same material, the material's user count would be 3.Changing the material would affect all three objects. If you now selected an object and clicked the user count,the object would receive its very own copy of the material, which can be modified independently of the original that's still used by the other two.

If a data-block has no real users, it'll normally be cleaned up(deleted) when saving the blend-file. To prevent this, you can give it a fake user;that way, it's guaranteed to survive.Data-blocks with a fake user have an 'F' prefix in the drop-down list.

The Outliner can show an overview of all data-blocks without real users in the blend-file. Simply change its Display Mode to Orphan Data.

Creates a new data-block (or duplicates the current one) and selects it.

Opens the File Browser, for importing an image for example.

Unpack the file packed into the current blend-file to an external one.

Clears the link.Shift-LMB to set the users to zero allowing the data to be fully deleted from the blend-file.

Sometimes there is a list of applied data-blocks(such as a list of materials used on the object).

See also

Data-blocks are discussed further in the Data System chapter .

A Data-Block menu with preview. \xc2\xba

Some data-block menus have large preview images in their drop-down instead of just icons and names.

A Data ID field. \xc2\xba

A Data ID field is similar to a Data Block Menu, but is only for selecting (and not for other features like creating new data or managing users).

It can show the following elements:

The icon on the left specifies the accepted data-block type.

The text field functions as a search field by matching elements in the list. Press Tab to auto-complete names up to the level where a match is found. If more than one match exists, you have to continue typing. If you type an invalid name, the value will remain unchanged.

Lets you select the data-block directly.

In some Data IDs there is an Eyedropper available through the pipette icon on the right side.

Click the button on the right to clear the reference.

Related types of ID sub-data may become available to select, depending on the data-block type and its intended usage.

Sub-data Example. \xc2\xba

If the selected Object in the Target field is a mesh or a lattice, an additional field may be displayed to select one of its vertex groups.

If the selected Object in the Target field is an armature, an additional field may be displayed to select one of its bones.

Once a bone is selected, a numeric field may become available for specifying a point along the bone. A value of 0.0 corresponds to the bone's head, while a value of 1.0 corresponds to its tail. Any values between these will result in linear interpolation (so a value of 0.5 matches the bone's center).

If the bone is a bendy bone, clicking on this button will make the point follow the curvature of the B-spline between head and tail, rather than simply going in a straight line.

--- Fin du chapitre: _data_block ---

--- Début du chapitre: _data_blocks --- The base unit for any Blender project is the data-block. Examples of data-blocks include: meshes, objects, materials, textures, node trees, scenes, texts, brushes, and even Workspaces.

Blender File view of the Outliner. \xc2\xba

A data-block is a generic abstraction of very different kinds of data, which features a common set of basic features, properties and behaviors.

Some common characteristics:

They are the primary contents of the blend-file.

They can reference each other, for reuse and instancing. (Child/parent, object/object-data, materials/images, in modifiers or constraints too \xe2\x80\xa6)

Their names are unique within a blend-file, for a given type.

They can be added/removed/edited/duplicated.

They can be linked between files (only enabled for a limited set of data-blocks).

They can have their own animation data.

They can haveCustom Properties.

User will typically interact with the higher level data types (objects, meshes, etc.).When doing more complex projects, managing data-blocks becomes more important,especially when inter-linking blend-files.The main editor for that is theOutliner.

Not all data in Blender is a data-block,bones, sequence strips or vertex groups e.g. are not,they belong to armature, scene and mesh types respectively.

Data-blocks types with their icon. \xc2\x9c\x93

For reference, here is a table of data-blocks types stored in blend-files.

Library Linking, supports being linked into other blend-files.

File Packing, supports file contents being packed into the blend-file(notapplicable for most data-blocks which have no file reference).

Type

Link

Pack

Description

Action

\xe2\x9c\x93

\xe2\x80\x94

Stores animation F-Curves.Used as data-block animation data, and the Nonlinear Animation editor.

Armature

\xe2\x9c\x93

\xe2\x80\x94

Skeleton used to deform meshes.Used as data of armature objects, and by the Armature Modifier.

Brush

\xe2\x9c\x93

\xe2\x80\x94

Used as brush assets in sculpt and paint modes.

Camera

\xe2\x9c\x93

\xe2\x80\x94

Used as data by camera objects.

Cache File

\xe2\x9c\x93

\xe2\x80\x94

Used by Mesh Cache modifiers.

Curve

\xe2\x9c\x93

\xe2\x80\x94

Used as data by curve, font & surface objects.

Font

\xe2\x9c\x93

\xe2\x9c\x93

References font files.Used by curve object-data of text objects.

Grease Pencil

\xe2\x9c\x93

\xe2\x80\x94

2D/3D sketch data used by Grease Pencil objects.Used as overlay helperinfo, by the 3D Viewport, Image, Sequencer & Movie Clip editors.

Collection

\xe2\x9c\x93
\xe2\x80\x94
Group and organize objects in scenes.Used to instance objects, and in library linking.

Image
\xe2\x9c\x93
\xe2\x9c\x93
Image files.Used by shader nodes and textures.

Keys (Shape Keys)
\xe2\x9c\x97
\xe2\x80\x94
Geometry shape storage, which can be animated.Used by mesh, curve, and lattice objects.

Light
\xe2\x9c\x93
\xe2\x80\x94
Used as object data by light objects.

Library
\xe2\x9c\x97
\xe2\x9c\x93
References to an external blend-file.Access from the Outliner\xe2\x80\x99s Blender Fileview.

Line Style
\xe2\x9c\x93
\xe2\x80\x94
Used by the Freestyle renderer.

Lattice
\xe2\x9c\x93
\xe2\x80\x94
Grid based lattice deformation.Used as data of lattice objects, and by the Lattice Modifier.

Mask
\xe2\x9c\x93
\xe2\x80\x94
2D animated mask curves.Used by compositing nodes & sequencer strip.

Material
\xe2\x9c\x93
\xe2\x80\x94
Set shading and texturing render properties.Used by objects, meshes & curves.

Metaball
\xe2\x9c\x93
\xe2\x80\x94
An isosurface in 3D space.Used as data of metaball objects.

Mesh
\xe2\x9c\x93
\xe2\x80\x94
Geometry made of vertices/edges/faces.Used as data of mesh objects.

Movie Clip
\xe2\x9c\x93
\xe2\x9c\x97
Reference to an image sequence or video file.Used in the Movie Clipeditor.

Node Tree

- \xe2\x9c\x93
- \xe2\x80\x94
- Groups of re-usable nodes.Used in the node editors.
- Object
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - An entity in the scene with location, scale, rotation.Used by scenes & collections.
- Paint Curve
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - Stores a paint or sculpt stroke.Access from the paint tools.
- Palette
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - Store color presets.Access from the paint tools.
- Particle
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - Particle settings.Used by particle systems.
- Light Probe
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - Help achieve complex real-time lighting in EEVEE.
- Scene
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - Primary store of all data displayed and animated.Used as top-level storage for objects & animation.
- Sounds
 - \xe2\x9c\x93
 - \xe2\x9c\x93
 - Reference to sound files.Used as data of speaker objects.
- Speaker
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - Sound sources for a 3D scene.Used as data of speaker object.
- Text
 - \xe2\x9c\x93
 - \xe2\x9c\x97
 - Text data.Used by Python scripts and OSL shaders.
- Texture
 - \xe2\x9c\x93
 - \xe2\x80\x94
 - 2D/3D textures.Used by brushes and modifiers.
- Window Manager
 - \xe2\x9c\x97
 - \xe2\x80\x94
 - The overarching manager for all of Blender\xe2\x80\x99s user interface.Includes Workspaces, notification system, operators, and keymaps.
- World
 - \xe2\x9c\x93
 - \xe2\x80\x94

Define global render environment settings.

Workspace

\xe2\x9c\x97

\xe2\x80\x94

UI layout.Used by each window, which has its own workspace.

Every data-block has its usage counted (reference count), when there is more than one,you can see the number of current users of a data-block to the right of its name in the interface.Blender follows the general rule that unused data is eventually removed.

Since it is common to add and remove a lot of data while working,this has the advantage of not having to manually manage every single data-block.This works by skipping zero user data-blocks when writing blend-files.

Since zero user data-blocks are not saved,there are times when you want to force the data to be kept irrespective of its users.

If you are building a blend-file to serve as a library of assets that you intend to link to and from other files,you will need to make sure that they do not accidentally get deleted from the library file.

To protect a data-block, use the button with the shield icon next to its name.The data-block will then never be silently deleted by Blender,but you can still manually remove it if needed.

Data-blocks names are unique within their namespace. A data-block namespace is defined by its type, and the blendfile it is stored in.

This means that there can be for example an Object and a Mesh named the same, but there cannot be two local objects named the same in a blendfile. However, it is possible to have one local and several linked Objects sharing the same name.

Data-block names have a fixed length of 63 bytes, i.e. 63 basic ASCII characters, or less when using diacritics or non-latin glyphs (the UTF8 encoding will then typically use more than a byte per character).

When Blender has to name a new data-block, or rename an existing one, it will check for name collisions. If a data-block with the same name already exists, the (re)named data-block will get a numeric extension added as a post-fix to its \xe2\x80\x98root name\xe2\x80\x99, like e.g. .001. The first available index is used (up to the 999value, after that the postfix index values are simply incremented until no collision happen anymore).

In case adding the numeric suffix would make the data-block name too long, the root name part will be shortened as needed.

Blender will never rename another data-block when doing automatic naming.

So e.g. when adding a new Cubeobject and there are already Cubeand Cube.001local objects, the new one will be named Cube.002.

Local data-blocks can be renamed by the user in several places in the UI (like the ID selection widget, or the Outliner view). When renamed from the UI, the behavior in case of name collision is as follow:

If the original root name is different than in the new requested name, the renamed data-block gets the first available numerical suffix.

E.g. assuming that there are three objects named Sphere, Cubeand Cube.001, renaming Sphereto Cubewill rename the data-block to Cube.002.

If the original root name is the same as in the new requested name, the renamed data-block gets the requested name, and the conflicting of data-block is renamed accordingly.

E.g. assuming that there are three objects named Sphere, Cube and Cube.001, renaming Cube.001 to Cube will rename the data-block to Cube, and the other data-block to Cube.001.

Data-blocks can be shared among other data-blocks.

Examples where sharing data is common:

Sharing textures among materials.

Sharing meshes between objects (instances).

Sharing animated actions between objects, for example to make all the lights dim together.

You can also share data-blocks between files, see linked libraries .

When a data-block is shared between several users, you can make a copy of it for a given user. To do so, click on the user count button to the right of its name. This will duplicate that data-block and assign the newly created copy to that usage only.

Note

Objects have a set of more advanced actions to become single-user, see their documentation.

As covered in Life Time, data-blocks are typically removed when they are no longer used. They can also be manually unlinked or deleted.

Unlinking a data-block means that its user won't use it anymore. This can be achieved by clicking on the 'X' icon next to a data-block's name. If you unlink a data-block from all of its users, it will eventually be deleted by Blender as described above (unless it is a protected one).

Deleting a data-block directly erases it from the blend-file, automatically unlinking it from all of its users. This can be achieved by Shift-LMB on the 'X' icon next to its name.

Warning

Deleting some data-blocks can lead to deletion of some of its users, which would become invalid without them. The main example is that object-data deletion (like mesh, curve, camera) will also delete all objects using it.

Those two operations are also available in the context menu when RMB-clicking on a data-block in the Outliner.

--- Fin du chapitre: _data_blocks ---

--- Début du chapitre: _data_transfer --- The Data Transfer modifier transfers several types of data from one mesh to another. Data types include vertex groups, UV maps, Color Attributes, custom normals

Transfer works by generating a mapping between source mesh's elements (vertices, edges, etc.) and destination ones, either on a one-to-one basis, or mapping several source elements to a single destination one, using interpolation.

Transferring normals between objects, see example blend-file. See also

Transfer Mesh Data Operator

Data Transfer Modifier.

Mesh object to copy data from.

If the button to the right of the field is unset, both the source and the destination geometry is considered in global space when generating the mapping, otherwise they are evaluated in local space (i.e. as if both object's origins were at the same place).

Controls how destination data are affected:

Replaces everything in destination (note that Mix Factor is still used). Only replaces destination value if it's above given threshold Mix Factor. How that threshold is interpreted depends on the data type, note that for Boolean values this option fakes a logical AND. Only replaces destination value if it's below given threshold Mix Factor. How that threshold is interpreted depends on the data type, note that for Boolean values this option fakes a logical OR. Apply that operation, using mix factor to control how much of source or destination value to use. Only available for a few types (vertex groups, Color Attributes).

How much of the transferred data gets mixed into existing one (not supported by all data types).

Allows per-element fine control of the mix factor. Vertex group influence can be reverted using the small  button to the right.

This modifier cannot generate needed data layers itself. Once the set of source data to transfer is selected, this button shall be used to generate matching destination layers, if needed.

To keep the size of the modifier reasonable, the kind of elements to be affected must be selected first (vertices, edges, face corners and/or faces).

How is generated the mapping between those source and destination elements. Each type has its own options, see Geometry Mapping below for details.

The left column of toggle buttons, to select which data types to transfer.

In those cases (vertex groups, Color Attributes, UVs), one can select which source layers to transfer (usually, either all of them, or a single specified one), and how to affect destination (either by matching names, matching order/position, or, if a single source is selected, by specifying manually the destination layer).

This setting only affects UV transfer currently. It allows to avoid a given destination face to get UV coordinates from different source UV islands. Keeping it at 0.0 means no island handling at all. Typically, small values like 0.02 are enough to get good results, but if you are mapping from a very high-poly source towards a very low-poly destination, you may have to raise it quite significantly.

First key thing to keep in mind when using this modifier is that it will not create destination data layers. Generate Data Layers button shall always be used for this purpose, once the set of source data to transfer has been selected. It should also be well understood that creating those data layers on destination mesh is not part of the modifier stack, which means e.g. that they will remain even once the modifier is deleted, or if the source data selection is modified.

Geometry mapping is how a given destination mesh relates to a source mesh. In this process a destination vertex/edge gets a part of the source mesh assigned with functions as its data source. It is crucial to understand this topic well to get good results with this modifier.

The simplest option, expects both meshes to have identical number of elements, and match them by order (indices). Useful e.g. between meshes that were identical copies, and got deformed differently.

Those always select only one source element for each destination one, often based on shortest distance.

Uses source's nearest vertex.

Uses source's nearest vertex of source's nearest edge.

Uses source's nearest vertex of source's nearest face.

Uses source's edge which vertices are nearest from destination edge's vertices.

Uses source's nearest edge (using edge's midpoints).

Uses source's nearest edge of source's nearest face (using edge's midpoints).

A face corner is not a real element by itself, it's some kind of split vertex attached to a specific face. Hence both vertex (location) and face (normal, ϕ) aspects are used to match them together.

Uses source's corner having the most similar splitnormal with destination one, from those sharing the nearest source's vertex.

Uses source's corner having the most similar facenormal with destination one, from those sharing the nearest source's vertex.

Uses source's nearest corner of source's nearest face.

Uses source's nearest face.

Uses source's face which normal is most similar with destination one.

Those use several source elements for each destination one, interpolating their data during the transfer.

Uses nearest point on nearest source's edge, interpolates data from both source edge's vertices.

Uses nearest point on nearest source's face, interpolates data from all that source face's vertices.

Uses point of face on source hit by projection of destination vertex along its own normal, interpolates data from all that source face's vertices.

This is a sampling process. Several rays are cast from along the destination's edge (interpolating both edge's vertex normals), and if enough of them hit a source's edge, all hit source edges' data are interpolated into destination one.

A face corner is not a real element by itself, it's some kind of split vertex attached to a specific face. Hence both vertex (location) and face (normal, ϕ) aspects are used to match them together.

Uses nearest point of nearest source's face, interpolates data from all that source face's corners.

Uses point of face on source hit by projection of destination corner along its own normal, interpolates data from all that source face's corners.

This is a sampling process. Several rays are cast from the whole destination's face (along its own normal), and if enough of them hit a source's face, all hit source faces' data are interpolated into destination one.

When the `pressure stylus` icon button to the right is enabled, this is the maximum distance between source and destination to get a successful mapping. If a destination element cannot find a source one within that range, then it will get no transferred data.

This allows to transfer a small sub-detailed mesh onto a more complete one (e.g. from a `chand` mesh towards a `full body` one).

The starting ray radius to use when Ray Casting against vertices or edges. When transferring data between meshes Blender performs a series of ray casts to generate mappings. Blender starts with a ray with the radius defined here, if that does not return a hit then the radius is progressively increased until a positive hit or a limit is reached.

This property acts as an accuracy/performance control; using a lower ray radius will be more accurate however, might take longer if Blender has to progressively increase the limit. Lower values will work better for dense meshes with lots of detail while larger values are probably better suited for simple meshes.

--- Fin du chapitre: `_data_transfer` ---

--- Début du chapitre: `_decimate` --- The Decimate modifier allows you to reduce the vertex/face count of a mesh with minimal shape changes.

This is not usually used on meshes which have been created by modeling carefully and economically (where all vertices and faces are necessary to correctly define the shape). But if the mesh is the result of complex modeling, sculpting and/or applied Subdivision Surface/Multiresolution modifiers, the Decimate one can be used to reduce the polygon count for a performance increase, or simply remove unnecessary vertices and edges.

Unlike the majority of existing modifiers, this one does not allow you to visualize your changes in Edit Mode.

The modifier displays the number of remaining faces as a result of the Decimate modifier.

The Decimate modifier in Collapse mode.

Merges vertices together progressively, taking the shape of the mesh into account.

The ratio of faces to keep after decimation.

On 1.0: the mesh is unchanged.

On 0.5: edges have been collapsed such that half the number of faces remain (see note below).

On 0.0: all faces have been removed.

Note

Although the Ratio is directly proportional to the number of remaining faces, triangles are used when calculating the ratio.

This means that if your mesh contains quads or other polygons, the number of remaining faces will be larger than expected, because those will remain unchanged if their edges are not collapsed.

This is only true if the Triangulate option is disabled.

Maintains symmetry on a single axis.

Keeps any resulting triangulated geometry from the decimation process.

A vertex group that controls what parts of the mesh are decimated.

The amount of influence the Vertex Group has on the decimation.

The Decimate modifier in Un-Subdivide mode.

It can be thought of as the reverse of subdivide. It attempts to remove edges that were the result of a subdivide operation. It is intended for meshes with a mainly grid-based topology (without giving uneven geometry). If additional editing has been done after the subdivide operation, the results may be unexpected.

The number of times to perform the un-subdivide operation. Two iterations is the same as one subdivided operation, so you will usually want to use even numbers.

The Decimate modifier in Planar mode.

It reduces details on forms comprised of mainly flat surfaces.

Dissolve geometry which forms angles (between surfaces) higher than this setting.

Prevent dissolving geometry in certain places.

Does not dissolve edges on the borders of areas where the face normals are reversed.

Does not dissolve edges on the borders of where different materials are assigned.

Does not dissolve edges marked as seams.

Does not dissolve edges marked as sharp.

Does not dissolve edges that are part of a UV map.

When enabled, all vertices along the boundaries of faces are dissolved. This can give better results when using a high Angle Limit.

--- Fin du chapitre: _decimate ---

--- Début du chapitre: _decorators --- Decorators are small buttons that appear to the right of other buttons and show the state of the property. Decorators may appear next to number fields, menus, and checkboxes to indicate the property can be animated.

Decorators indicating different property states.

Clicking on the decorator dot icon will add a keyframe to that property. Clicking the rhombus icon again will remove the keyframe. A solid rhombus icon indicates there is a keyframe on the current frame, while a non-solid rhombus icon indicates that the property has a keyframe on another frame. Clicking the non-solid rhombus icon will create a keyframe on the current frame with the current property value.

If a property is being driven by another, the decorator shows the driver icon.

Decorators make it quick and easy to glance over properties and see their state.

See also

State Colors

--- Fin du chapitre: _decorators ---

--- Début du chapitre: _defaults --- When you start Blender for the first time or update to a new version, the interactive region of the Splash Screen is replaced with a couple of initial preferences to configure how you interact with Blender.

Note

These options can always be changed later in the Preferences.

This is where you can copy preferences from an older version of Blender. Doing so will copy preferences and startup files from the previous version of Blender and then loads them.

The preferences need to be imported from previous versions because the configuration files of each Blender version are stored in separate folders. Refer to the Blender's Directory Layout page for the location of these folders.

If you would like to start fresh with the new version, continue to Create New Preferences.

The language used in the user interface. The list is broken up into categories determining how complete the translations are. More language preferences can be set in the Translation Preferences . Choose between a light or dark theme for Blender. Themes can be customized more in the Preferences .

Presets for the default keymap for Blender. Note that this manual assumes that you use the \xe2\x80\x9cBlender\xe2\x80\x9d keymap.

This is the default keymap. Read more about this keymap here .

This keymap is intended to match an older series of Blender versions and is designed for people upgrading who do not want to learn the updated keymap.

This keymap is intended to match common creation software and is intended for people who use many different such applications. Read more about this keymap here .

Controls which mouse button, either right or left, is used to select items in Blender.

Controls the action of Spacebar. These and other shortcuts can be modified in the keymap preferences .

Starts playing through the Timeline . This option is good for animation or video editing.

Opens the Toolbar underneath the cursor to quickly change the active tool. This option is good if doing a lot of modeling or rigging.

Opens up the Menu Search . This option is good for someone who is new to Blender and is unfamiliar with its menus and shortcuts.

Saves the preferences set above and opens the regular Splash Screen .

The preferences are automatically saved when changed.

Changing the default startup file can be done via File \xe2\x80\xa3 Defaults \xe2\x80\xa3 Save Startup File. See Startup File .

There are two areas where Blender\xe2\x80\x99s defaults are stored:

The Preferences file stores keymap, add-ons theme and other options.

The Startup File stores the scene and UI setup which are displayed at startup and when creating a new file (File \xe2\x80\xa3 New).

You can revert your customizations to Blender\xe2\x80\x99s defaults:

The Preferences Load Factory Settings.

File \xe2\x80\xa3 Defaults \xe2\x80\xa3 Load Factory Settings.

Note

After loading the factory settings, the preferences won\xe2\x80\x99t be auto-saved.

See Managing Preferences for details.

--- Fin du chapitre: _defaults ---

--- Début du chapitre: _defocus --- The Defocus Node blurs areas of an image based on a Z depth map or mask input.

It is typically used to emulate depth of field (DOF) using a post-processing method with a Z-buffer input. But also allows to blur images that are not based on Z depth too.

Standard color input.

Z-buffer input, but could also be a (grayscale) image used as a mask, or a single value input.

The number of iris blades of the virtual camera\xe2\x80\x99s diaphragm.

Disk (to emulate a perfect circle), Triangle (3 blades), Square (4 blades), Pentagon (5 blades), Hexagon (6 blades), Heptagon (7 blades) or Octagon (8 blades).

This slider is deactivated, if the Bokeh Type is set to Disk. It can be used to add a rotation offset to the Bokeh shape. The value is the angle in degrees.

Applies a gamma correction on the image before and after blurring it.

This option controls the amount of focal blur in the same way as a real camera. It simulates the aperture of a real lens's iris, without modifying the luminosity of the picture. The default value 128 is assumed to be infinity: everything is in perfect focus. Half the value will double the amount of blur. This slider is deactivated, if No Z-buffer is enabled.

This value limits the amount of blur by setting a maximum blur radius. Can be used to optimize the performance. The default value of 0 means no limit.

Some artifacts, like edge bleed, may occur, if the blur difference between pixels is large. This value controls how large that blur difference considered to be safe.

Tip

Only change this value, if there is an occurring problem with an in-focus object.

If enabled a limited amount of (quasi-)random samples are used to render the preview. This way of sampling introduces additional noise, which will not show up in the final render.

To select the linked scene.

Should be activated for a non Z-buffer in the Z input. No Z-buffer will be enabled automatically whenever a node that is not image based is connected to the Z input.

Only active when No Z-buffer is enabled. When No Z-buffer is used, the input is used directly to control the blur radius (similar to F-Stop when using the Z-buffer). This parameter can be used to scale the range of the Z input.

Standard color output.

In this blend-file example, the ball array image is blurred as if it was taken by a camera with an f-stop of 2.8 resulting in a fairly narrow depth of field centered on 7.5 units from the camera. As the balls recede into the distance, they get blurrier.

Sometimes might want to have more control to blur the image. For instance, you may want to only blur one object while leaving everything else alone (or the other way around), or you want to blur the whole image uniformly all at once. The node, therefore, allows you to use something other than an actual Z-buffer as the Z input. For instance, you could connect an Image node and use a grayscale image where the color designates how much to blur the image at that point, where white is the maximum blur and black is no blur. Or, you could use a Time node to uniformly blur the image, where the time value controls the maximum blur for that frame. It may also be used to obtain a possibly slightly better DoF blur, by using a fake depth-shaded image instead of a Z-buffer. (A typical method to create the fake depth-shaded image is by using a linear blend texture for all objects in the scene or by using the `fog/mist` fake depth shading method.) This also has the advantage that the fake depth image can have Anti-Aliasing, which is not possible with a real Z-buffer.

The parameter No Z-buffer, becomes then the main blur control. The input has to be scaled, because usually the value of a texture is only in the numeric range 0.0 to 1.0.

Distance setting in the Camera Depth of Field panel.

The Defocus node uses the actual camera data in your scene if supplied by a Render Layer node.

To set the point of focus, the camera now has a Distance parameter, which is shorthand for Depth of Field Distance. Use this camera parameter to set the focal plane of the camera (objects Depth of Field Distance away from the camera are in focus). Set Distance in the main Camera edit panel; the button is right below the Depth of Field.

To make the focal point visible, enable the camera Limit option, the focal point is then visible as a yellow cross along the view direction of the camera.

In general, use preview mode, change parameters to your liking, only then disable preview mode for the final render.

For minimum artifacts, try to setup your scene such that differences in distances between two objects that may visibly overlap at some point are not too large.

Keep in mind that this is not real DoF, only a post-processing simulation. Some things cannot be done which would be no problem for real DoF at all. A typical example is a scene with some object very close to the camera, and the camera focusing on some point far behind it. In the real world, using shallow depth of field, it is not impossible for nearby objects to become completely invisible, in effect allowing the camera to see behind them. Hollywood cinematographers use this visual characteristic to achieve the popular "focus pull" effect, where the focus shifts from a nearby to a distant object, such that the "other" object all but disappears. Well, this is simply not possible to do with the current post-processing method in a single pass. If you really want to achieve this effect, quite satisfactorily, here is how:

Split up your scene into "nearby" and "far" objects, and render them in two passes.

Now, combine the two results, each with their own "defocus" nodes driven by the same Time node, but with one of them inverted (e.g. using a Map Value node with a Size of -1). As the defocus of one increases, the defocus on the other decreases at the same rate, creating a smooth transition.

At very low values, less than 5, the node will start to remove any oversampling and bring the objects at DoF Distance very sharply into focus. If the object is against a contrasting background, this may lead to visible stair-stepping (aliasing) which OSA is designed to avoid. If you run into this problem:

Do your own OSA by rendering at twice the intended size and then scaling down, so that adjacent pixels are blurred together.

Use the Blur node with a setting of 2 for X and Y.

Set DoF Distance off by a little, so that the object in focus is blurred by the tiniest bit.

Use a higher f-stop, which will start the blur, and then use the Z socket to a Map Value to a Blur node to enhance the blur effect.

Rearrange the objects in your scene to use a lower-contrast background.

A final word of warning, since there is no way to detect if an actual Z-buffer is connected to the node, be very careful with the No Z-buffer switch. If the Z scale value happens to be large, and you forget to set it back to some low value, the values may suddenly be interpreted as huge blur radius values that will cause processing times to explode.

--- Fin du chapitre: _defocus ---

--- Début du chapitre: _deform ---Reference

All Modes

Bone \xe2\x80\xa3 Deform

The Deform panel.\xc2\xb6

In this panel you can set deformation options for each bone.

Toggling the checkbox in the panel header off, prevents the bone from deforming the geometry at all, overriding any weights that it might have been assigned before; it mutes its influence.

It also excludes the active bone in the automatic weight calculation when the mesh is parented to the armature using the Armature Deform tool with the With Automatic Weight option.

Bone influence areas for envelopes method. \xc2\xb6

Envelopes is the most general skinning method. It works with all available object types for skinning (meshes, lattices, curves, surfaces and texts). It is based on proximity between bones and their geometry, each bone having two different areas of influence, shown in the Envelope visualization:

The inside area, materialized by the \xe2\x80\x9csolid\xe2\x80\x9d part of the bone, and controlled by both root and tip radius.

The outside area, materialized by the lighter part around the bone, and controlled by the Distance setting.

See also

The editing pages for how to edit these properties.

The Distance defines a volume which is the range within the bone has an influence on vertices of the deformed object. The geometry is less and less affected by the bone as it goes away by following a quadratic decay.

Single bone with various envelope sizes. \xc2\xb6

A bone property, that controls the global influence of the bone over the deformed object, when using the envelopes method.

It is only useful for the parts of geometry that are

\xe2\x80\x9cshared\xe2\x80\x9d, influenced by more than one bone

(generally, at the joints \xe2\x80\xa6) \xe2\x80\x93 a bone with a high weight will have more influence on the result than one with a low weight \xe2\x80\xa6 Note that when set to 0.0, it has the same effect as disabling the Deform option.

This option controls how the two deforming methods interact, when they are both enabled. By default, when they are both active, all vertices belonging to at least one vertex group are only deformed through the vertex groups method. The other \xe2\x80\x9corphan\xe2\x80\x9d vertices being handled by the envelopes one. When you enable this option, the \xe2\x80\x9cdeformation influence\xe2\x80\x9d that this bone would have on a vertex (based from its envelope settings) is multiplied with this vertex \xe2\x80\x99s weight in the corresponding vertex group. In other words, the vertex groups method is further

\xe2\x80\x9cweighted\xe2\x80\x9d by the envelopes method.

Set the radius for the head and the tail of envelope bones. Inside this volume, the geometry is fully affected by the bone.

Three Armature Bones all using Envelope Weight. \xc2\xb6

The 1st with a default radius value, the two others with differing Tail and Head radius values.

--- Fin du chapitre: _deform ---

--- Début du chapitre: `_deform_curves_on_surface` --- The Deform Curves on Surface node translates and rotates each curve based on the difference in its root position. The root position is defined by UV coordinates stored on each curve and the UV Map selected for the purpose in the Curves surface settings.

The transformation is calculated based on the difference of the original mesh (before shape keys and modifiers are evaluated), and the final mesh.

Unlike other geometry nodes, this node has quite a few implicit inputs:

The original and evaluated mesh are retrieved from the modifier object's surface property. This means the node only works for curves objects.

The original and evaluated UV map are also retrieved from the object's surface property.

A 3D vector attribute named `rest_position`, used for calculating tangents for rotating curves that are consistent with the tangents calculated on the original mesh (the rotation needs to be calculated from the normal and tangent of the original and evaluated meshes).

A 2D vector attribute on the curve domain named `surface_uv_coordinate` to store the location of the root positions on the surface mesh's UV map.

In future development, this node will be generalized so the setup is more flexible.

Parts of the internal operation are similar to the Sample UV Surface Node.

Warning

In order to achieve consistent deformation after the Subdivision Surface Modifier, the UV Smooth option of the modifier should be set to None. Otherwise the surface UV map will be subdivided in a way that may invalidate the curve UV attachment points stored in the `surface_uv_coordinate` attribute.

Standard curves input.

This node has no properties.

Standard curves output.

--- Fin du chapitre: `_deform_curves_on_surface` ---

--- Début du chapitre: `_delete` --- Reference

Edit Mode

Armature `Delete` Bones

X

This tool delete selected bones, selected joints are ignored.

If you delete a bone in a chain, its child(ren) will be automatically re-parented to its own parent, but not connected, to avoid deforming the whole armature.

An armature with two selected bones, just before deletion.

The two bones have been deleted. Note that Bone.002, previously connected to the deleted Bone.001, is now parented but not connected to Bone.

Reference

Edit Mode

Armature `Delete` Dissolve

Ctrl- X

Todo 2.76.

--- Fin du chapitre: `_delete` ---

--- Début du chapitre: `_delete_curves` --- Remove existing curves. The tool deletes the entire curves, if any of its segments fall under the brush falloff radius.

--- Fin du chapitre: `_delete_curves` ---

--- Début du chapitre: `_delete_geometry` --- The Delete Geometry node removes the selected part of a geometry. It behaves similarly to the Delete tool in Edit Mode. The type of elements to be deleted can be specified with the domain and mode properties.

Standard geometry input.

Boolean field that is true for parts of the geometry to be deleted.

The domain on which the selection field is evaluated.

The selection is on the points, control points, and vertices of the geometry.

The selection is on the edges of the mesh component. The other components are not modified.

The selection is on the faces of the mesh component. The other components are not modified.

The selection is on the splines in the curve component. For each spline, it will either be deleted entirely or not at all. The other components are not modified.

The selection is on the top-level instances, and the entire instance is removed if it is selected. If the input also contains realized geometry, that will be unaffected.

The type of elements to be affected. This only applies to the mesh component.

Vertices, edges, and faces in the selection will be deleted.

Vertices won't be deleted, even if they are in the selection.

Only faces in the selection will be deleted.

Standard geometry output.

--- Fin du chapitre: `_delete_geometry` ---

--- Début du chapitre: `_denoise` --- The Denoise node is used to denoise renders from Cycles and other ray tracing renderers. This helps to significantly reduce render time by rendering with fewer samples.

It uses Open Image Denoise, which transforms noisy images into clean images with machine learning.

Noisy image input.

Optional normal render pass to better preserve detail. For Cycles, it is recommended to use the Denoising Normal render pass, which is available when enabling the Denoising Data passes.

Optional albedo render pass to better preserve detail. For Cycles, it is recommended to use the Denoising Albedo render pass, which is available when enabling the Denoising Data passes.

Does not apply any prefiltering to the input passes. This option retains the most detail and is the fastest, but assumes the input passes are noise free which may require a high sample count. If the input passes are not noise free, then noise will remain in the image after denoising.

Assumes the input passes are not noise free, yet does not apply prefiltering to the input passes. This option is faster than Accurate but produces a blurrier result.

Prefilters the input passes before denoising to reduce noise. This option usually produces more detailed results than Fast with increased processing time.

Preserve colors outside the 0 to 1 range.

Denoised image output.

Render before and after denoising, with a very low number of samples as input. As more samples are used, the denoiser will be able to better preserve detail.

--- Fin du chapitre: _denoise ---

--- Début du chapitre: _density_curves --- Create (or remove) curves based on a target distance. It generates a high number of points and then rejects the ones that are too close to existing points.

Determines whether the brush adds or removes curves.

Either add or remove curves depending on the distance between existing curve roots under the cursor.

Add new curves between existing curves, taking the minimum distance into account.

Remove curves whose root points are too close.

Goal distance between the curve roots.

Interactively sets the Distance Minvalue by displaying a graphic inside the brush cursor, giving a representation of the density.

The density can be adjusted by moving the mouse cursor closer or farther from the paint cursor. The Distance Min will be changed once the operator is confirmed.

The maximum amount of points that the brush tries to sample in the surface.

--- Fin du chapitre: _density_curves ---

--- Début du chapitre: _deploying_blender --- This page contains tips for setting up Blender in environments such as animation studios and schools.

These environments often have special requirements regarding security, automated deployment and customization.

Blender downloads can be extracted to any directory on the system, as a self-contained installation. Multiple Blender versions can co-exist on the same system, and deployment can be automated using standard file management tools.

New Blender versions may add, remove or change functionality that affects the results of production files. For a given project, it is advisable to use a single LTS version of Blender. LTS versions receive bug fixes for two years.

For security or other reasons, workstation may not have internet access.

By default Blender does not access the internet, however this can be enabled in the System preferences with the Online Access option.

Working offline can be enforced by running with the --offline-mode command line argument. Users will then be unable to enable online access in the preferences.

Note

Add-ons that follow this setting will only connect to the internet if enabled. However, Blender cannot prevent third-party add-ons from violating this rule.

When working offline or in a more controlled environment, it may be useful to provide a set of extensions to all users. For this there is a default read-only System repository. This repository can for example be located on a read-only network drive or in a system directory.

System repository

The `$BLENDER_SYSTEM_EXTENSIONS` environment variable controls the default location. This should point to a directory, within which a system directory should exist.

Extensions packages should be extracted in this system directory, with a resulting path like this:

In the Extensions preferences, it's possible to manually set a custom directory for the default System repository, or to create multiple repositories.

Besides extensions, it's possible to bundle scripts for presets, application templates, legacy add-ons, as well as scripts run on startup.

Script directories can be manually added in the File Paths preferences. The `$BLENDER_SYSTEM_SCRIPTS` can also be used to add a script directory without modifying the preferences.

These script directories are expected to contain specific directories like presets, addons and startup for different types of scripts. See Path Layout for a complete list.

The Blender Python API can be used to customize Blender. This includes changing preferences, changing the startup file and adding UI elements.

For example, a script can enable add-ons for every user.

Application Templates can be used to set up Blender for particular tasks or projects, separate from the default configuration. When creating a new file the user can choose the template.

The files are expected to be placed in the system script directories like this:

Add-ons that have not been converted to become an extension yet need to be placed in the addons script directory.

For example, an add-on could be located at:

Blender follows the VFX reference platform, which means it is able to run on the same systems as other VFX software and exchange image, volume and scene files with them.

Blender and the `bpy` module are only compatible with a single Python version. This makes it possible for add-ons and VFX software in general to only have to target a single Python version.

Blender bundles a complete Python installation and does not interact with the system Python by default. This can be changed with the `--python-use-system-env` command line argument, if care is taken to set up a compatible Python version.

--- Fin du chapitre: `_deploying_blender` ---

--- Début du chapitre: `_depth_of_field` --- To render a scene, `EEVEE` uses a pinhole camera model which produces a perfectly focused image of the scene. For an enhanced realism, `EEVEE` can simulate the optical Depth of Field using a post-process filter, and a sample-based method. The optical settings are located in the camera settings properties. Whereas the quality of the effect can be controlled by the settings found in the present section.

Note

In the 3D Viewport, depth of field only works while in Camera View.

The post-process method is computed in two passes. The first pass is using a blur that fails to produce quality bokeh for highlights but works for the general case. Followed by a second pass which is sprite-based and improves only the quality of very bright highlights. That is because it is

too slow to be applied on every part of the image. So it just includes very bright isolated parts of the image such that are different from their surroundings. Which pixels are being processed by second pass can be control with the Sprite Threshold and Neighbor Rejection options.

Secondly the sample-based method works by randomizing the camera position for every sample. It is more accurate but needs many samples to achieve a smooth result. Accordingly the post-process blurring radius is scaled down to remove undersampling. Yet some scenes might still need more post-process blur in order to remove the noticeable sample pattern. This is exactly what the Overblur option does, but it will also reduce the bokeh shape sharpness.

Reference

Render \xe2\x80\xa3 Depth of Field

Maximum size in pixels of the depth of field post-process effect (lower is faster). A value of 0 will disable the post-process effect but not the sample-based method.

Minimum brightness a pixel needs to have to be considered by the sprite-based depth of field. Higher values will improve the performance but will also reduce the quality of highlights. Brightness is in the scene\xe2\x80\x99s referred color space.

Maximum intensity to consider when doing sprite neighborhood rejection. This should be set to a brightness value above which there is small visual differences to be noticeable after color management. Lower values will improve the performance but will also reduce the quality of highlights. Brightness is in the scene\xe2\x80\x99s referred color space.

Randomize the camera position for every scene render sample to increase precision. Enabling this option can change the scene\xe2\x80\x99s actual sample count.

Note

Be aware that the actual sample count can grow quite rapidly.

Hint

The actual number of samples is computed by the following formula:

where ring_count is the number of ring in the hexa web pattern. The ring_count is chosen so that the entire pattern contains at least the number of samples set in the Render Settings .

Scales the post-process depth of field radius to reduce artifacts. Higher values will soften the bokeh shape.

See also

Limitations .

--- Fin du chapitre: `_depth_of_field` ---

--- Début du chapitre: `_despeckle` --- The Despeckle node is used to smooth areas of an image in which noise is noticeable, while leaving complex areas untouched.

This works by the standard deviation of each pixel and its neighbors is calculated to determine if the area is one of high complexity or low complexity. If the complexity is lower than the threshold then the area is smoothed using a simple mean filter.

Controls the amount the filter effects the image.

Standard color input.

The threshold to control high/low complexity.

The threshold to control the number of pixels that must match.

Standard color output.

--- Fin du chapitre: `_despeckle` ---

--- Début du chapitre: `_dial_gizmo` --- The Dial Gizmonode is ideal for creating gizmos that control angles.
Special gizmo value socket. Everything that linked into this socket will be modified when the gizmo is rotated.
Position of the gizmo in the local space of the object.
Up or normal direction of the gizmo in the viewport.
If enabled, the gizmo will always have the same size in the viewport independent of the zoom level. This affects the meaning of the radius input.
In screen space mode, this is a factor on top of the default radius. Otherwise, this is the radius of the gizmo in Blender units.
Controls which theme color is used for this gizmo.
Should be joined into the geometry that is controlled by this gizmo.
--- Fin du chapitre: `_dial_gizmo` ---

--- Début du chapitre: `_difference_key` --- This node produces a matte that isolates foreground content by comparing it with a reference background image.
Contains foreground content against the background that is to be removed.
The reference background image.
Where pixels match the reference background to within the specified threshold, the matte is made transparent.
Increase to make nearby pixels partially transparent producing a smoother blend along the edges.
Image with its alpha channel adjusted for the keyed selection.
A black-and-white alpha mask of the key.
--- Fin du chapitre: `_difference_key` ---

--- Début du chapitre: `_diffuse` --- The DiffuseBSDFnode is used to add Lambertian and Oren-Nayar diffuse reflection.
Color of the surface, or physically speaking, the probability that light is reflected or transmitted for each wavelength.
Surface roughness; 0.0 gives standard Lambertian reflection, higher values activate the Oren-Nayar BSDF.
Normal used for shading; if nothing is connected the default shading normal is used.
This node has no properties.
Standard shader output.
Lambertian reflection.\xc2\xbb
Diffuse shader behavior.\xc2\xbb
Oren-Nayar reflection.\xc2\xbb
--- Fin du chapitre: `_diffuse` ---

--- Début du chapitre: `_diffusion` --- Reference Domain
Physics \xe2\x80\xa3 Fluid \xe2\x80\xa3 Diffusion
Liquid diffusion defines the physical properties of a liquid and in turn define how a liquid interacts with its environment. The main factors of diffusion are the Viscosity and Surface Tension. These properties can be adjusted to create virtual liquids that behave like water, oil, honey, or any other liquid. A couple presets exist to change the diffusion for different substances are predefined and can be changed in the preset

menu.Fluid Diffusion settings can be enabled/disabled in the panel header.

The viscosity refers to the η of the fluid and actually the force needed to move an object of a certain surface area through it at a certain speed.

For manual entry, please note that real-world viscosity (the so-called dynamic viscosity) is normally measured in Pascal-seconds ($\text{Pa}\cdot\text{s}$), or in Poise units (P, equal to $0.1 \text{ Pa}\cdot\text{s}$), and commonly centiPoise units (cP, equal to $0.001 \text{ Pa}\cdot\text{s}$).

Blender, on the other hand, uses the kinematic viscosity which is the dynamic viscosity divided by the density, $\frac{\text{Pa}\cdot\text{s}}{\text{kg/m}^3}$, which is m^2/s . So for example, the viscosity of water at room temperature is 1.002 cP, or $0.001002 \text{ Pa}\cdot\text{s}$; the density of water is about 1000 kg/m^3 , which gives a kinematic viscosity of $0.000001002 \text{ m}^2/\text{s}$ so the entry would be 1.002 times 10 to the minus six (1.002×10^{-6} in scientific notation).

The table below gives some examples of fluids together with their dynamic and kinematic viscosities.

Fluid

Dynamic viscosity (in cP)

Kinematic viscosity (Blender, in m^2/s)

Water (20 °C)

1.002 (1.002)

1.002×10^{-6} (0.000001002)

Oil SAE 50

5.0 (500)

5.0×10^{-5} (0.00005)

Honey (20 °C)

1.0 (10,000)

2.0×10^{-3} (0.002)

Chocolate Syrup

3.0 (30,000)

3.0×10^{-3} (0.003)

Ketchup

1.0 (100,000)

1.0×10^{-1} (0.1)

Melting Glass

1.0 (15)

1.0×10^0 (1.0)

Tip

You can find the kinematic viscosity of more materials in the proper units by asking Wolfram Alpha, e.g. η kinematic viscosity of alcohol in m^2/s .

To simplify the input of these numbers, the viscosity is changed by entering values in scientific notation by entering a base value and the exponent of that number.

The base of the viscosity value (e.g. 1.002 in the case of water (20 °C)).

The exponent of the viscosity value that will be multiplied by 10 - 1 (e.g. 6 in the case of water (20 °C)).

Note

Viscosity Varies

The default values in Blender are considered typical for those types of fluids and look right when animated. However, actual viscosity of some fluids, especially sugar-laden fluids like chocolate syrup and honey, depend highly on temperature and concentration. Oil viscosity varies by SA rating. Glass at room temperature is basically a solid, but glass at 1500 °C flows (nearly) like water.

Warning

The simulator is not suitable for non-fluids, such as materials that do not flow. Simply setting the viscosity to very large values will not result in rigid body behavior, but might cause instabilities.

Surface tension in grid units. Higher value will produce liquids with greater surface tension.

The high viscosity liquid solver can be used to simulate fluids with increased viscosity, replicating the behavior of substances like honey or molasses. This specialized solver enhances the accuracy of slow-moving and thick liquid simulations.

The viscosity of the liquid. Higher values result in more viscous fluids.

Note

A strength value of 0 will still apply some viscosity. Uncheck the High Viscosity Solver to disable the high viscosity liquid solver simulation step completely.

Strength of 0.2 (at frame 65).

Strength of 0.4 (at frame 200).

--- Fin du chapitre: `_diffusion` ---

--- Début du chapitre: `_dilate_erosde` --- Expands or shrinks a mask using a morphological operator.

A grayscale image.

Sets each pixel to the maximum (for dilation) or minimum (for erosion) value that is found within a square surrounding it. This approach keeps the original gray levels and is best suited for masks that contain sharp corners; rounded shapes such as circles will look more square-like in the output.

Despite the name, this is not an iterative process; the dilation/erosion is only performed once regardless of the chosen Distance.

Makes all the pixels fully black or white depending on whether they are darker or brighter than 50% gray. Then, sets each pixel to the maximum (for dilation) or minimum (for erosion) value that is found within a circle surrounding it. This approach loses the original gray levels. Shape wise, it is well-suited for masks that contain rounded corners; sharp ones will be rounded off. Sets each pixel to the maximum (for dilation) or minimum (for erosion) value that is found within a circle surrounding it. This approach preserves the original gray levels and is well-suited for masks that contain rounded corners.

Blurs the image.

The size of the surrounding area to look at for each pixel; or in other words, how much to dilate (for positive values) or erode (for negative values) the mask.

For the Threshold mode, determines how much to blur the edges after dilation/erosion.

For the Feathermode, determines the brightness curve of the blurred edges.

The resulting mask.

In the image below, notice that:

The light gray disk has turned white and the dark gray rectangle has turned black because of the Thresholdmode.

The shapes have become thicker \xe2\x80\x93 dilated because of the positive Distance.

The shapes appear blurred because of the positive Edge.

--- Fin du chapitre: `_dilate_erode` ---

--- Début du chapitre: `_directional_blur` --- Blurs an image along a specified direction. Can be used to fake motion blur.

Standard color input.

Controls how many times the image is duplicated to create the blur effect. Higher values give smoother results.

Sets the position where the blur center is. This makes a difference if the angle, spin, and/or zoom are used.

How large the blur effect is.

Image is blurred at this angle from the center.

Rotates the image each iteration to create a spin effect, from the center point.

Scales the image each iteration, creating the effect of a zoom.

Standard color output.

--- Fin du chapitre: `_directional_blur` ---

--- Début du chapitre: `_directory_structure` --- A video project is most likely a combination of several different assets. You can separate them into three broad categories.

Video files: video clips (or movies in Blender-talk), photos, graphic files (charts, logos, \xe2\x80\xa6), and Visual Effects (VFX) such as masks, lens flares, animation.

Audio files: recorded dialog, voice-over, music, and Sound Effects (SFX) such as environmental sounds, swooshes, \xe2\x80\xa6

Project files: the blend files and backups, (partial) render results, documentation such as scripts and storyboards.

Together they can form rapidly an intangible heap of files. It's good practice to collect all those assets in one project directory with appropriate subdirectories. Why? It will lower the probability that you accidentally delete a file (which will result in a \xe2\x80\x93 file not found \xe2\x80\x93 error) or that you forget to include a necessary file when transferring the project (\xe2\x80\x93 file is missing \xe2\x80\x93 error). And most of all, an appropriate directory structure, will help you to keep a clear overview of your assets.

See also

Blender can incorporate some files within the Blend file (see Packed Data). However, this doesn't work with video files, which can have very huge file sizes. So, it's better to assure that your project directory contains all necessary files.

It's also good practice to use some kind of naming convention and add metadata. Figure 1 shows a possible example, based on our categorization of the assets above. The directories are numbered so that they mimic a normal workflow.

Figure 1: Organizing your project. \xc2\xb6

--- Fin du chapitre: `_directory_structure` ---

--- Début du chapitre: `_displace` --- The Displace Node displaces the pixel position based on an input vector.

This node could be used to model phenomena, like hot air distortion, refractions of uneven glass or for surreal video effects. Standard color input.

Input of the displacement map. If the a color output is implicitly converted in the vector input, the first channel (red) value determines displacement along X axis. The second channel (green) the displacement along Y axis. If the input is a grayscale image, where both the channel values are equal, the input image will be displaced equally in both X and Y directions.

Separate scaling of the vector input in X and Y direction. Acting as multipliers by increasing or decreasing the strength of the displacement along their respective axes.

This node has no properties.

Standard color output.

--- Fin du chapitre: `_displace` ---

--- Début du chapitre: `_displacement` --- The Displacement node is used to displace the surface along the surface normal, to add more detail to the geometry. Both procedural textures and baked displacement maps can be used.

By default, Blender only uses Bump Mapping to render displacement. However with true displacement, the rendered geometry will be physically displaced. To use true displacement the Displacement method must be set accordingly.

Tip

For best results when using true displacement, the mesh must be subdivided finely to bring out the detail in the displacement texture.

See also

Material Displacement for more details on displacement workflows.

Distance to displace the surface along the normal. This is where a texture node can be connected.

Neutral displacement value that causes no displacement. With the default 0.5, any lower values will cause the surfaces to be pushed inwards, and any higher values will push them outwards.

Increase or decrease the amount of displacement.

Standard normal input.

Object Space means the displacement scales along with the object. When using World Space the object scale is ignored.

Displacement offset to be connected into the Material Output.

Typical displacement node setup. \xc2\xbb6

Bump only, displacement only, and displacement and bump combined. \xc2\xbb6

--- Fin du chapitre: `_displacement` ---

--- Début du chapitre: `_displace_hair_curves` --- Displaces hair curves by a vector based on various options.

Input Geometry (only curves will be affected).

Factor to scale overall displacement.

Shape of the influence along curves (0=constant, 0.5=linear).

Object used to define the displacement space.

Vector for displacement.

Surface geometry used to sample the normal for displacement. This input takes priority over the corresponding object input, if used.

Surface object used to sample the normal for displacement.

Surface UV map used to sample the normal for displacement.

Amount of displacement along the surface normal.

This node has no properties.

Geometry

--- Fin du chapitre: `_displace_hair_curves` ---

--- Début du chapitre: `_display` ---Reference

Particle System \xe2\x80\xa3 Display

Display hair as curves.

Display just the end points of the hairs.

The number of segments (control points minus 1) of the hair strand. In between the control points the segments are interpolated. The number of control points is important:

For the soft body animation, because the control points are animated like vertices, so more control points mean longer calculation times.

For the interactive editing, because you can only move the control points (but you may recalculate the number of control points in Particle Edit Mode).

Hint

Segments

Ten Segments should be sufficient even for very long hair, five Segments are enough for shorter hair, and two or three segments should be enough for short fur.

--- Fin du chapitre: `_display` ---

--- Début du chapitre: `_display_mode` --- Using this pop-up, you can choose between displaying the preview image or a scope that visualizes its color distribution.

Previews what the final video will look like, and lets you change the image layout using various tools.

This scope visualizes the luminosity (brightness) distribution of the image, letting you see at a glance if there's enough contrast and if any areas are under- or overexposed.

The scope works by plotting a curve for each scanline in the current video frame. Another way of saying this is that each pixel column in the luma waveform is a brightness histogram of the corresponding pixel column in the frame. Specifically:

The horizontal position of a pixel in the waveform refers to a pixel column in the frame.

The vertical position of a pixel in the waveform refers to a brightness value, going from 0 at the bottom to 1 at the top.

The brightness of a pixel in the waveform indicates how many pixels in the above frame column have the above brightness. If no pixels in the frame column have this brightness, the waveform pixel is black. If at least three pixels in the frame column have this brightness, the waveform pixel is white.

When this scope is selected, you have the following option in Sidebar \xe2\x80\xa3 View \xe2\x80\xa3 View Settings:

The examples below show two images and their corresponding luma waveforms.

The various horizontal lines in the luma waveform match the uniform-colored lines of the picture. Note that the 20% one-pixel width line (inside the yellow strip) is represented in the Luma waveform by a gray line. The two lines drawing an X are from the two monochrome gradients. Finally, the broken line matches the colored gradient at the bottom.

The curves are quite visible. We found a luma of 80-100% for the sky, a luma around 40% for the sea, and a luma of 10-20% for the mountains, growing around 40% for the sunny part.

Shows three waveforms for the red, green, and blue color channels instead of just one for the overall image brightness.

This scope visualizes the color distribution of the image. Each point has:

- An angle indicating its hue.

- A distance-from-center indicating its saturation.

- A brightness indicating how many pixels in the video frame have the above hue and saturation.

Example image.

Corresponding Chroma Vectorscope.

Shows three overlapping graphs, one for each color channel. Within each graph:

The X axis corresponds to color intensity, going from 0 on the left (black) to 1 on the right (fully red/green/blue).

The Y axis corresponds to number of pixels.

Use this mode to balance out the tonal range in an image. A well-balanced image should have nice and smooth distribution of color values.

Example image.

Corresponding Histogram.

--- Fin du chapitre: _display_mode ---

--- Début du chapitre: _display_settings --- The Workbench engine does not work with shader trees. In various tabs of the Properties are Viewport Display panels where settings can be adjusted that the Workbench engine uses.

The Viewport Display panel in the Object Properties has several settings that are used by the Workbench Engine.

Reference

Properties Object Viewport Display

When the Shadow in the Options is enabled this object will cast a shadow.

When checked the object will be rendered in front of the other objects in the scene.

The color to render the object in when object color needs to be rendered. The alpha channel can be used to render the object transparent.

The Viewport Display panel in the Material Properties has several settings that are used by the Workbench Engine.

Reference

Properties Material Viewport Display

The color when rendering the material. The alpha channel can be used to render the object transparent.

Changes the amount of specular lighting. This is only available when Specular Lighting in the Options is enabled.

Changes the amount of roughness for specular lighting. This is only available when Specular Lighting in the Options is enabled.

The Viewport Display panel in the World Properties has several settings that are used by the Workbench Engine.

Reference

Properties \xe2\x80\xa3 World \xe2\x80\xa3 Viewport Display

The color of the world background. This color will be rendered in the background of the scene.

--- Fin du chapitre: _display_settings ---

--- Début du chapitre: _distance --- Computes the distance between two 3D coordinates.

First coordinate point.

Second coordinate point.

This node has no properties.

Calculated distance.

--- Fin du chapitre: _distance ---

--- Début du chapitre: _distance_from_camera --- The Distance from Camera modifier alters the base property with a new one from a given range using the distance to the active camera.

The limits of the mapping from \xe2\x80\x9cdistance to camera\xe2\x80\x9d to \xe2\x80\x9cproperty in mapping\xe2\x80\x9d. If the current point of the stroke is at Range Minor less from the active camera or the object, it will take the start value. And conversely, if it is at Range Max or more from the camera/object, it will take the end value. These values are in the current scene\xe2\x80\x99s units, not in pixels!

Set the min/max range values from the distances between the current selected mesh vertices and the camera or the target.

--- Fin du chapitre: _distance_from_camera ---

--- Début du chapitre: _distance_from_object --- The Distance from Object modifier alters the base property with a new one from a given range using the distance to the active camera or to a given object as the parameter.

The object to measure distance from.

The limits of the mapping from \xe2\x80\x9cdistance to camera\xe2\x80\x9d to \xe2\x80\x9cproperty in mapping\xe2\x80\x9d. If the current point of the stroke is at Range Minor less from the active camera or the object, it will take the start value. And conversely, if it is at Max or more from the camera/object, it will take the end value. These values are in the current scene\xe2\x80\x99s units, not in pixels!

Set the min/max range values from the distances between the current selected mesh vertices and the camera or the target.

--- Fin du chapitre: _distance_from_object ---

--- Début du chapitre: _distance_key --- The Distance Key node determines a pixel\xe2\x80\x99s alpha value based on the three-dimensional distance between the image pixel color and the key color in a 3D color space.

This key works well when trying to single out a specific color in a background (not necessarily green).

Standard color input.

The color that is to be keyed.

A threshold what the node considers a match between the key color and the foreground pixel. The tolerance affects how close a pixel needs to be to the background pixel to be considered an absolute match.

When the Falloff value is high, pixels that are close to the Key Color are more transparent than pixels that are not as close to the Key Color (but still considered close enough to be keyed). When the Falloff value is low, it does not matter how close the pixel color (Image) is to the Key Color, it is transparent.

It is also possible to work with YCbCr color space, but only the Cb and Cr channels are taken into consideration for determining the distance between the foreground and background pixels.

RGB, YCC

The image with an alpha channel adjusted for the keyed selection.

A black-and-white alpha mask of the key.

--- Fin du chapitre: `_distance_key` ---

--- Début du chapitre: `_distorted_noise` --- Distortion Noise takes the option that you pick from Noise Basis and filters it, to create hybrid pattern. It is often used for grunge but is also very complex and versatile.

Distorted Noise Texture panels.

The texture to be distorted.

The texture to use to distort another.

The amount that Distortion Noise affects Basis.

The size of the noise generated.

Almost all procedural textures in Blender use derivatives for calculating normals for texture mapping (except Blend and Magic). This is important for Normal and Displacement Maps. The strength of the effect is controlled with the `Nabla` number field.

--- Fin du chapitre: `_distorted_noise` ---

--- Début du chapitre: `_distribute_points_in_volume` --- The Distribute Points in Volume node creates points inside of volume grids. The node has two basic modes of operation: distributing points randomly, or in a regular grid. Both methods operate on all of the float grids in the volume.

Standard volume geometry input.

Number of points to sample per unit volume.

Spacing between grid points.

Minimum value of a volume cell to contain a grid point

Distribute points randomly inside of the volume. The local point count is implicitly defined as a product of the global from the Density input and the local voxel value. This method creates a distribution that is not stable as the input volume deforms.

Distribute the points in a grid pattern inside of the volume. At each grid point, the voxel value is used to determine whether to add a point.

Standard point cloud geometry output.

--- Fin du chapitre: `_distribute_points_in_volume` ---

--- Début du chapitre: `_distribute_points_on_faces` --- The Distribute Points on Faces node places points on the surface of the input geometry object. Point, corner, and polygon attributes of the input geometry are transferred to the generated points. That includes vertex weights and UV maps. Additionally, the node has Normal and Rotation outputs.

The node also generates a stable ID, stored in the built-in `id` attribute, used as a stable identifier for each point. When the mesh is deformed or the density changes the values will be consistent for each remaining

point. This attribute is used in the Random Value and Instance on Points nodes.

Standard geometry input.

Note

The input geometry must contain a mesh with faces.

The selection of which face corners should be considered for point distribution.

The minimal distance points can have to each other. This option is only available for the Poisson Disk distribution method. At its default value of zero, the node's behavior is the same as it is in Random mode, because none of the internally generated points are removed. The point density for the point distribution. The unit is in number of points per square meter. This value is multiplied by the values from the Density input. Only available in Poisson Disk mode.

Note

This will be capped on distributions by the Distance Min option. If the density is greater than what the minimal distance allows, no new points will be added after this threshold has been passed.

The number of points to distribute per square meter on each mesh face. This value is multiplied by the values from the Density Attribute. In Poisson Disk mode, this value is multiplied by the Density Max input for the final density.

The random Seed to use when generating points.

Distribute points randomly on the surface. This is the fastest distribution method.

Distribute points randomly on the surface while taking a minimum distance into account.

By default, the node uses smooth and custom normals for the Normal and Rotation values. An earlier version of this node only uses

smooth normals, this option brings back this behavior of only using smooth normals.

This option can only be available in the Sidebar.

Generated points. Named attributes are copied to the result mesh, along with the data in the other attribute field outputs.

The Normal of the triangle on which each point is scattered.

An XYZ Euler rotation built from the normal attribute for convenience.

Such a value can also be built from the normal with the Euler to Rotation Node. Keep in mind that the Z axis of the result rotation will be arbitrary, since the mesh normal used to create the rotation does not have enough information to set all three rotation axes.

--- Fin du chapitre: `_distribute_points_on_faces` ---

--- Début du chapitre: `_domain_size` --- The Domain Size outputs the size of an attribute domain on the selected geometry type, for example, the number of edges in a mesh, or the number of points in a point cloud.

For more information about attribute domains, see the geometry attributes page.

Standard geometry input.

Which geometry type to retrieve the domain sizes from.

The size of the Point domain on any supported geometry.

The size of the Edge domain on meshes.

The size of the Face domain on meshes.

The size of the Face Corner domain on meshes.

The size of the Spline domain on curves.

The number of top-level instances in the geometry. Nested instances are not considered.

--- Fin du chapitre: `_domain_size` ---

--- Début du chapitre: `_dope_sheet` --- Dope Sheet View.\xc2\xbb

The Dope Sheet View is used to visualize motion tracking data, it is implemented as separate view of the Movie Clip editor just like the Graph View.

It displays channels for selected tracks and each channel visualizes tracked segments of tracks as dark bars and keyframed positions of tracks as small diamonds.

The background is highlighted depending on the number of tracks in a frame. This means that if for a frame (or sequence of frames) there are less than eight tracks, the background will turn red; if there are from eight to sixteen tracks, the background will be yellow.

This is only a visual feedback, which doesn't mean that the camera motion will not be reconstructed with less than eight tracks. It only means that you should pay attention to those frames and check if all possible good feature points are tracked there. Remember, if there are no good feature points in the frame and there are less than 16 tracks in the frame, it doesn't mean the solution won't be accurate. Rather, adding more tracks on bad feature points will reduce the accuracy of solution.

Sort order of the channels. \xc2\xbb

Limits Dope Sheet channels to only information about selected tracks.

Includes information from hidden tracks.

Sort order of the tracks.

Sort selected tracks in alphabetical order based on their names.

Sort tracks by longest tracked segment length.

Sort tracks by overall amount of frames.

Sort tracks by their average reprojection error after solving camera or object motion.

Sort channels by first frame number.

Sort channels by last frame number.

To change the sort order from ascending to descending.

The Dope Sheet View is for visualization and does not have any tools to actually edit data.

--- Fin du chapitre: `_dope_sheet` ---

--- Début du chapitre: `_double_edge_mask` --- The Double Edge Mask node creates a gradient between two masks.

A mask representing the inside shape, which will be fully white.

A mask representing the outside shape, which will fade from black at its edge to white at the Inner Mask.

All shapes in the Inner Mask contribute to the gradient, even ones that do not touch the Outer Mask shape.

Only shapes in the Inner Mask that overlap with the Outer Mask contribute to the gradient.

All.\xc2\xbb

Adjacent Only.\xc2\xbb

Parts of the Outer Mask that touch the edge of the image are treated as if they stop at the edge.

Parts of the Outer Mask that touch the edge of the image are extended beyond the boundary of the image.

Keep In.\xc2\xba6
Bleed Out.\xc2\xba6
Standard mask output.
Double Edge Mask Example Video
--- Fin du chapitre: _double_edge_mask ---

--- Début du chapitre: _drag ---Reference
Physics \xe2\x80\xa3 Force Fields
Drag
A Dragforce field resists particle motion by slowing it down.
Drag component proportional to velocity.
Drag component proportional to the square of the velocity.
--- Fin du chapitre: _drag ---

--- Début du chapitre: _draw ---Reference
Draw Mode
Asset Shelf \xe2\x80\xa3 Draw
The Draw brush allows you to draw free-hand strokes.
Data-block selector for the material .
The radius of the brush in pixels.
Allows you to change the brush size interactively by dragging the
pointer orby typing a number then confirm.
Adjusts the radius based on the stylus pressure when using a Graphics
Tablet .The gradient of the pressure can be customized usingthe curve
widget .
Control the stroke transparency (alpha).From fully transparent (0.0) to
fully opaque (1.0).
You can change the brush strength interactively by pressing Shift- F in
the 3D Viewport and then moving the pointer and then LMB.You can also
enter the size numerically.
Adjusts the strength based on the stylus pressure when using a Graphics
Tablet .The gradient of the pressure can be customized usingthe curve
widget .
The shape of the start and end of the stroke.
Strokes start and stop with a curved shape.
Strokes start and stop with a straight cutoff.
Controls how often the input device is read to generate points on the
stroke.Higher values give a higher precision (more points) but produce an
irregular stroke,while lower values give a lower precision (fewer points)
but produce a soften stroke.(0 disabled extra input device samples.)
You have to set up this value according to your input device to
obtainthe right balance between accuracy and softness for your
strokes.SeeInput Devicefor more information.
The number of smoothing iterations to apply to the stroke while drawing.
Direction of the input device that gives the maximum thickness to the
stroke (0\xc2\xba0 for horizontal).
Amount of thickness reduction when the stroke is perpendicular to the
Anglevalue.
Amount of transparency (alpha) to apply from the border of the point to
the center.Works only when the brush is using stroke materials of Dotor
Boxstyle.
Controls the width and height of the alpha gradient.

Post-processing methods that are executed on the strokes when you finished drawing, right after releasing the LMB or Pen tip. You can toggle the use of post-processing using the checkbox in the section panel header.

Strength of smoothing process on the points location along the stroke.

The number of smoothing iterations to apply to the stroke.

Number of subdivisions to apply to newly created strokes.

Reduces final points numbers in the stroke with an adaptive algorithm.

Automatically trim intersection strokes ends.

Activate the conversion of the newly created stroke to its outline.

Material used for outline stroke.

Thickness used for outline stroke.

Adds randomness to the position of the points along the stroke. You can toggle the use of Randomize using the checkbox in the section panel header.

The amount of randomness to apply using the pressure of the input device.

The amount of randomness to apply to the stroke strength value (alpha).

The amount of randomness to apply to the UV rotation.

Randomizes the hue, saturation, and value of the stroke's Color.

The amount of jittering to add to the stroke.

Common Options

Use randomness only at stroke level.

Uses the stylus pressure to control how strong the effect is. The gradient of the pressure can be customized using the curve widget.

Stabilize Stroke helps to reduce jitter of the strokes while drawing by delaying and correcting the location of points. You can toggle the use of Stabilize Stroke using the checkbox in the section panel header.

Minimum distance from the last point before the stroke continues.

A smooth factor, where higher values result in smoother strokes but the drawing sensation feels like as if you were pulling the stroke.

The cursor can be disabled by toggling the checkbox in the Cursor header.

Shows the brush linked material color in the viewport.

--- Fin du chapitre: `_draw` ---

--- Début du chapitre: `_drawing_planes` ---Reference

Draw Mode and Sculpt Mode

Drawing Plane

Drawing Planes pop-over.

The Drawing Planes selector helps to select the plane in which strokes are drawn.

To see which plane you are using when drawing strokes, you can enable Canvas in Viewport Overlays. See Viewport Display to know more about Canvas settings.

Note

The Drawing Plane only affects new strokes and does not affect existing strokes.

Strokes are drawn with the current 3D Viewport orientation.

Strokes are drawn on the plane determined by the XZ axes (front view).

Strokes are drawn on the plane determined by the YZ axes (side view).

Strokes are drawn on the plane determined by the XY axes (top view).

Strokes are drawn with the current 3D cursor orientation.

Front.

Side.

Top.

View. \xc2\xbb

Cursor. \xc2\xbb

--- Fin du chapitre: _drawing_planes ---

--- Début du chapitre: _draw_facesets ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Draw new or extend existingFace Setswith each stroke.

Holding Ctrlwill continue drawing the same face set as the one under the cursor.Holding Shiftwill relax or smooth the edges of the face setsby modifying the underlying topology so edges flow along the perimeter of the face sets.This will remove the jagged lines visible after drawing or creating a face set.

Note

More information in the Face Set Introduction .

While a lot of the general brush settings are supported,it\xe2\x80\x99s not needed to change them from the default,as the brush purpose is very simple.

Note

More info at General brush settingsand on Advanced brush settings.

--- Fin du chapitre: _draw_facesets ---

--- Début du chapitre: _draw_sharp ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Similar to theDrawbrush,but it always deforms the mesh from the original coordinatesand uses the SharperFalloff by default.

Draw Sharp is useful on high density meshes for creating cloth wrinkles, stylized hair or hard surface edges.To further sharpen or polish sharp edges in the case that the mesh density is not enough,it\xe2\x80\x99s recommended to use thePinch,CreaseorMultiplane Scrapebrushes.

A limitation is that the brush does not remesh the sculpted surfaceswithDyntopoenabled.Because of that, a better brush to use with Dyntopo can beCrease.

On Subtractby default to carve in creases. More info atDirection

Note

More info at General brush settingsand on Advanced brush settings.

--- Fin du chapitre: _draw_sharp ---

--- Début du chapitre: _drivers_editor --- This editor lets you set

upDrivers, which calculate the value for a propertybased on other

properties. In other words, they make a set of source properties

\xe2\x80\x99drive\xe2\x80\x9d the target property,and can thus serve as an alternative to animating the property by hand.

The Drivers Editor, showing how you might drive a cube\xe2\x80\x99s rotation based on its position.\xc2\xbb

The user interface is largely the same as that of theGraph Editor,with two important differences:

The Sidebar has an additionalDrivers tab. This is wherethe source properties are brought together to calculate an intermediate value for the target property.

The curve doesn't represent the property's value over time, but a mapping from the above intermediate value (X axis) to the final value (Y axis).

--- Fin du chapitre: _drivers_editor ---

--- Début du chapitre: _drivers_panel --- Edit Driver popover.

Reference

Graph editor

Drivers

Sidebar region Drivers

N

Reference

Context menu Edit Driver

Ctrl- D

This panel is visible in Sidebar of the Drivers Editor as a popover when adding a driver to a property.

It shows the property that is being driven, followed by a series of settings that determine how the driver works.

There are two categories of drivers:

Built-in functions(Average, Sum, Min and Max)

The driven property will have the value of the average, sum, lowest or highest (respectively) of the values of the referenced Driver Variables. If there is only one driver variable, these functions will yield the same result.

Custom(Scripted Expression).

An arbitrary Python expression that can refer to the Driver Variables by name. See Expressions.

The current result of the driver setup. Useful for debug purposes.

See Driver Variables.

Forces an update for the Driver Value dependencies.

Opens the fully featured Drivers Editor. This button only appears in the popover version of the Drivers panel.

Variables are references to properties, transformation channels, or the result of a comparison between transformations of two objects.

Drivers should access object data via Driver Variables, rather than direct references in the Python expression, in order for dependencies to be correctly tracked.

Add, Copy, Paste buttons.

Adds a new Driver Variable.

Copies the current variable list so it can be pasted into another driver's variable list.

Name for use in scripted expressions. The name must start with a letter, and only contain letters, digits, or underscores.

The type of variable to use.

Retrieves the value of an RNA property, specified by a data-block reference and a path string.

In case of transform properties, this will return the exact value of the UI property, while Transform Channel will take parenting and/or constraints into account as needed.

See also Custom Properties .

The ID-block type. For example: Key, Image, Object, Material.

The ID of the ID-block type. For example:

Material.001.

The RNA name of the property, based on a subset of Python attribute access syntax. For example: `location.x` or `location[0]` for the X location animation channel value (before parenting or constraints), or `["prop_name"]` for a custom property.

If enabled, allows specifying a fallback value to use as the variable value if the RNA Path cannot be resolved, instead of causing a driver evaluation failure. For more info see Context Property below.

Tip

The easiest way to create a variable of this type is to use the Copy As New Driver context menu option of the input property, and paste the result into the driver via Paste Driver Variables.

Retrieves the value of a Transform channel from an object or bone. ID of the object. For example: `Cube`, `Armature`, `Camera`.

For armatures, the name of the Armature bone. For example:

`\bone\bone.002`, `\bone`,

`\bone.r`.

For example, X Location, X Rotation, X Scale.

The Average Scale option retrieves the combined scale value, computed as the cubic root of the total change in volume. Unlike X/Y/Z Scale, this value can be negative if the object is flipped by negative scaling.

For rotation channels, specifies the type of rotation data to use, including different explicit Euler orders. Defaults to using the Euler order of the target. See Rotation Channel Modes.

World Space, Transform Space, Local Space.

Provides the value of the rotational difference between two objects or bones, in radians.

For armatures, the name of the Armature bone. For example:

`\bone\bone.002`, `\bone`,

`\bone.r`.

Provides the value of the distance between two objects or bones.

For armatures, the name of the Armature bone. For example:

`\bone\bone.002`, `\bone`,

`\bone.r`.

World Space, Transform Space, Local Space.

Provides the value of a property that is implicitly referring to either a scene or a view layer of the currently evaluating animation system. This is a weak reference which does not lead to the scene or view layer referenced from the driver to be linked when linking animation data.

An example when such properties come in play is referring to a transformation of the active camera. It is possible to set up a driver in a character file, and make the driver use the set camera when the character is linked into a set.

Active Scene, Active View Layer.

The RNA name of the property, based on a subset of Python attribute access syntax. For example: `camera.location.x` or `camera.location[0]` for the camera X location animation channel value (before parenting or constraints), or `["prop_name"]` for a custom property.

If enabled, allows specifying a fallback value to use as the variable value if the RNA Path cannot be resolved, instead of causing a driver evaluation failure.

This feature can be very useful for making drivers more robust when implementing scene-global options using custom properties. When the object is linked into a different scene, these custom properties may not exist there, and the fallbacks can be used to provide sensible default values.

Fallbacks can also be used to emulate the lookup behavior of the View Layer mode of the material Attribute Node .

Tip

Although the values of the x/y/z animation channels for the camera location can be accessed via `camera.location[0/1/2]`, retrieving its world space location and orientation after parenting and constraints currently requires using `camera.matrix_world`. This property can be understood easily by viewing the matrix as an array of four vectors in Worldspace:

`matrix_world[0][0/1/2]` is the Screen Right direction vector (camera local X).

`matrix_world[1][0/1/2]` is the Screen Up direction vector (camera local Y).

`matrix_world[2][0/1/2]` is the opposite of the direction the camera is pointing.

`matrix_world[3][0/1/2]` is the location of the camera.

Shows the value of the variable.

Rotation Transform Channels support a number of operation modes, including:

- Uses the Euler order of the target to decompose rotation into channels.

- Explicitly specifies the Euler rotation order to use.

- Provides the Quaternion representation of the rotation.

- Decomposes the rotation into two parts: a Swing rotation that aims the specified axis in its final direction, followed by a Twist rotation around that axis. This is often necessary for driving corrective Shape Keys and bones for organic joint rotation.

This decomposition is often produced in rigs by using a helper bone with a Damped Track Constraint to extract the swing part, and its child with Copy Transforms to extract the twist component.

The channels values for Swing and Y Twist are:

- Falloff curves for weighted angles.

- True angle of the twist rotation.

- True angle of the swing rotation, independent of its direction.

- Weighted angles that represent the amount of swing around the X/Z axis.

The magnitude of the angle equals W Rotation when the rotation is purely around that axis, and fades out to zero as the direction changes toward the other axis, following the falloff curves from the graph on the right.

Mathematically, the swing angles are computed from quaternion components, using $\sqrt{2} \arccos(w)$ for W and $\sqrt{2} \arcsin(x)$ etc. for the others. The component of the swing rotation that corresponds to the twist axis is always 0, and is replaced by the twist angle.

A text field where you can enter an arbitrary Python expression that refers to Driver Variables by their names.

The expression has access to a set of standard constants and math functions from `math`, `bl_math` and other modules, provided in the Driver Namespace. For an example of adding a custom function to the namespace, see the driver namespace example .

For performance reasons it is best to use the Simple Expression subset as much as possible.

If this option is enabled, the variable `self` can be used for drivers to reference their own data. Useful for objects and bones to avoid having creating a Driver Variable pointing to itself.

Example: `self.location.x` applied to the Y rotation property of the same object will make the object tumble when moving.

Note that dependencies for properties accessed via `self` may not be fully tracked.

Blender can evaluate a useful subset of Python driver expressions directly, which significantly improves performance, especially on multi-core systems. To take advantage of this, the driver expression must only use the following features:

- Use only ASCII characters.

- Floating-point and decimal integer.

- frame

- pi, True, False

- +, -, *, /, ==, !=, <, <=, >, >=, and, or, not, conditional operator/ ternary if
- min, max, radians, degrees, abs, fabs, floor, ceil, trunc, round, int, sin, cos, tan, asin, acos, atan, atan2, exp, log, sqrt, pow, fmod
- lerp, clamp, smoothstep

Simple expressions are evaluated even when Python script execution is disabled.

When an expression outside of this subset is used, Blender displays a `Slow Python expression` warning. However, as long as the majority of drivers use simple expressions, using a complex expression in select few is OK.

See also

- Extending Blender with Python .

- Python and its documentation.

- functions.wolfram.com.

--- Fin du chapitre: `_drivers_panel` ---

--- Début du chapitre: `_dual_mesh` --- The Dual Mesh Node converts a mesh into its dual, i.e. faces are turned into vertices and vertices are turned into faces. This also means that attributes which were on the face domain are transferred to the point domain in the dual mesh.

Warning

The Dual Mesh node only works on manifold geometry. To work with non-manifold geometry it's best to remesh the geometry first.

Standard geometry input.

Keeps the non-manifold boundaries of the input mesh in place, by creating extra geometry, and avoiding the dual mesh transformation there.

This node has no properties.

Standard geometry output.

The Dual Mesh Node combines nicely with triangulated meshes. In this case an Ico Sphere is used, which is made up of nice and evenly spaced triangles.

--- Fin du chapitre: `_dual_mesh` ---

--- Début du chapitre: `_duplicate` --- Reference

Edit Mode

- Armature `3 Duplicate`

- Shift- D

Note

This tool works on selected bones; selected joints are ignored.

As in mesh editing, by pressing Shift- D the selected bones will be duplicated. The duplicates become the selected elements and they are placed in select mode, so you can move them wherever you like.

If you select part of a chain, by duplicating it you will get a copy of the selected chain, so the copied bones are interconnected exactly like the original ones.

The duplicate of a bone which is parented to another bone will also be parented to the same bone, even if the root bone is not selected for the duplication. Be aware, though, that if a bone is parented and connected to an unselected bone, its copy will be parented, but not connected to the unselected bone (see Fig. Duplication example.).

An armature with three selected bones and a selected single root.
\xc2\xbf

The three duplicated bones. Note that the selected chain is preserved in the copy, and that Bone.006 is parented but not connected to Bone.001, as indicated by the black dashed line. Similarly, Bone.007 is parented but not connected to Bone.003. \xc2\xbf

--- Fin du chapitre: `_duplicate` ---

--- Début du chapitre: `_duplicate_elements` --- The Duplicate Elements node creates a new geometry with the specified elements from the input duplicated an arbitrary number of times. The positions of elements are not changed, so all of the duplicates will be at the exact same location. Standard geometry input.

Boolean field that is true for parts of the geometry to be deleted.

Field indicating how many times each input element should be duplicated. If the value is zero for an element, it will not be included in the output at all.

The type of geometry element to duplicate

Duplicate the points of meshes, curves, or point clouds. Any other elements will not be included in the output.

Duplicate mesh edges. Faces will not be included in the output.

Duplicate mesh faces. Each duplicated face will be separate, in other words they will not share edges with other faces.

Individual curves from the input curves component will be duplicated.

Input top-level instances will be duplicated.

Standard geometry output.

An attribute field with a value for every output element describing which duplicate of the corresponding input. The value for every input element will start at 0 and increase to Amount-1.

Combined with the Geometry to Instance Node, this can be used to create a basic efficient `\xe2\x80\x9cArray\xe2\x80\x9d` operation. This should be more efficient because the duplicates are instances.

The `\xe2\x80\x9cDuplicate Index\xe2\x80\x9d` is used to move each instance in the result a different amount.

--- Fin du chapitre: `_duplicate_elements` ---

--- Début du chapitre: `_duplicate_hair_curves` --- Duplicates hair curves a certain number of times within a radius.

Input Geometry (only curves will be affected).

Amount of duplicates per curve.

Percentage of amount used for the viewport.

Radius in which the duplicate curves are offset from the guides.

Shape of distribution from center to the edge around the guide.

Offset of the curves to round the tip.

Keep an even thickness of the distribution of duplicates.

Random Seed for the operation.

This node has no properties.

Geometry

Guide index map that was used for the operation.

--- Fin du chapitre: _duplicate_hair_curves ---

--- Début du chapitre: _duplicate_linked ---Reference

Object Mode

Object \xe2\x80\xa3 Duplicate Linked

Alt- D

You also have the choice of creating a Linked Duplicate rather than a Duplicate; this is called a deep link. This will create a new object with all of its data linked to the original object. If you modify one of the linked objects in Edit Mode, all linked copies are modified. Transform properties (object data-blocks) still remain copies, not links, so you still can rotate, scale, and move freely without affecting the other copies. Reference the Examples for the discussions below.

If the original object was animated, the duplicate will link to the same Action. This means that, even though each object has separate transform properties, they will be set to the same values by the animation system. If this is not desired, make the action a single-user copy in the Action or NLA Editor.

In the Duplicate Objects Adjust Last Operation panel the Linked checkbox is checked unlike with Duplicate.

Hint

If you want to make changes to an object in the new linked duplicate independently of the original object, you will have to manually make the object a \xe2\x80\x9csingle-user\xe2\x80\x9d copy by LMB the number in the Object Data panel of the Properties. (See Data-Block Menu .)

See also

Make Single User for unlinking data-blocks.

The Cube object was linked duplicated. \xc2\xb6

The object Cube was linked duplicated, using Alt- D. Though both these cubes are separate objects with unique names: Cube and Cube.001, the single mesh named Cube, is shared by both.

As a mesh is edited in Edit Mode in one object, the same occurs in the other cube as well. The mesh data are links, not copies.

In contrast, if one of these two cubes is rotated or scaled in Object Mode, the other remains unchanged. The transform properties are copied, not linked.

As in the previous example, the newly created cube has inherited the material of the original cube. The material properties are linked, not copied.

A common table has a top and four legs. Model one leg, and then make linked duplicates three times for each of the remaining legs. If you later make a change to the mesh, all the legs will still match. Linked duplicates also apply to a set of drinking glasses, wheels on a car \xe2\x80\xa6 anywhere there is repetition or symmetry.

See also

Linked Library Duplication

Linked Libraries are also a form of duplication. Any object or data-block in other blend-files can be reused in the current file.

Hint

If you want transform properties (i.e. object data-blocks) to be \xe2\x80\x9clinked\xe2\x80\x9d, see the page on parenting.

--- Fin du chapitre: _duplicate_linked ---

--- Début du chapitre: _dynamics ---Reference

Particle System \xe2\x80\xa3 Hair Dynamics

Hair particles can have dynamic properties using physics. To enable hair physics, click the checkbox beside Hair Dynamics.

Quality of the simulation in steps per frame (higher is better quality but slower).

Spring stiffness of the vertex target position.

Warning

If you use motion blur in your animation, you will need to bake one extra frame past the last frame which you will be rendering.

A general setting for how fine and good a simulation you wish. Higher numbers take more time but ensure less tears and penetrations through the hair.

The distance another object must get to the hair for the simulation to repel the hair out of the way. Smaller values might cause errors but provide some speed-up while larger will give unrealistic results if too large and can be slow. It is best to find a good in between value.

Prevents explosions in tight and complicated collision situations by restricting the amount of movement after a collision.

Only objects that are a part of this Collection can collide with the hair. Note that these objects must also have Collision physics enabled.

Reference

Particle System \xe2\x80\xa3 Hair Dynamics \xe2\x80\xa3 Structure

Value for the mass of the hair.

Controls the bending stiffness of the hair strands.

Random stiffness of hair.

Damping of bending motion.

Reference

Particle System \xe2\x80\xa3 Hair Dynamics \xe2\x80\xa3 Volume

Some phenomena of real-world hair can be simulated more efficiently using a volumetric model instead of the basic geometric strand model. This means constructing a regular grid such as those used in fluid simulations and interpolating hair properties between the grid cells.

Controls how thick the air is around the hair causing the hair to flow slower.

Amount of friction between individual hairs.

Size of the voxel grid cells for interaction effects.

Maximum density of the hair.

The influence that the Density Target has on the simulation.

--- Fin du chapitre: _dynamics ---

--- Début du chapitre: _dynamic_paint --- The Dynamic Paint modifier is a container for a Dynamic Paint Physics simulation.

As the modifier is only a container its actual options are configured in the Physics Properties tab. See the Dynamic Paint Physics Properties for more information.

--- Fin du chapitre: _dynamic_paint ---

--- Début du chapitre: _dynamic_repository --- A dynamic repository allows you to serve a smaller JSON file with only the latest version of the extensions which are compatible with the query parameters. This is only relevant for repositories which contain multiple version of multiple extensions.

For small or personal repositories it is simpler and recommended to use static repositories instead.

To setup a dynamic extensions repository, follow the steps for static repositories ,since the format and the listing are the same. When Blender fetches the extensions listing it passes the following arguments to make sure only compatible extensions are listed:

```
platform
blender_version
```

These arguments are passed as parameters to the server via a query URL:
<https://extensions.blender.org/api/v1/extensions/>

https://extensions.blender.org/api/v1/extensions/?blender_version=4.2.0&platform=linux-x64

Some repositories may require authentication. The user can specify an accesstoken for a repository, which is passed along with the API request from Blender.

This is passed to the servers via an Authorization header:

--- Fin du chapitre: _dynamic_repository ---

--- Début du chapitre: _dyntopo ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Dyntopo

Dynamic Topology (aka Dyntopo) can be toggled with the checkbox. With dynamic topology active, most brushes will subdivide the mesh during the stroke.

For a general explanation of dynamic topology, visit the Introduction.

Each Detail Type \xe2\x80\x99s detail is set here. Depending on the Detail Type being used this property will rather show as a pixel count (px), or percentage.

Allows you to interactively set the detail, with a preview of the detail \xe2\x80\x99s density in the 3D Viewport.

When using Constant Detail, it is possible to sample the detail value of a certain mesh area by clicking the pipette icon next to the detail setting and then clicking on the area.

Setting the option will determine which of the methods will be used when altering the topology.

Just like the Subdivide tool, this method will only subdivide topology to match the detail given.

When topology is too dense, and is smaller than the detail given, edges will be collapsed to fit the detail size appropriately.

This method combines the two methods, subdividing edges smaller than the detail size, and collapsing topology.

Dyntopo uses the following different detail methods to create dynamic detail on an object.

This method uses a detail size based on the number of pixels, and in turn will create topology in that size. Zoom out big details, zoom in small fine details.

To keep detail uniform across the entire object, Constant Detail can be used. The detail is based on the percentage of a single unit.

Giving more control over the topology, with this method you can create topology based on the brush size. You can increase and lower topology by resizing the brush itself. The detail size is based on the size of the brush itself, where full detail will create topology the size of the brush radius itself.

Similar to constant detail, this value sets a percentage value uniform across the object, but only applies detailing changes when using Flood Fill.

When using Constant or Manual Detailing, this option is made available, allowing you to fill the entire object with a uniform detail, based on the detail size.

--- Fin du chapitre: _dyntopo ---

--- Début du chapitre: _edges ---Reference

Physics \xe2\x80\xa3 Soft Body \xe2\x80\xa3 Edges

Allow the edges in a mesh object to act like springs. See interior forces.

Use a specified vertex group for spring strength values.

The spring stiffness for edges (how much the edges are allowed to stretch). A low value means very weak springs (a very elastic material), a high value is a strong spring (a stiffer material) that resists being pulled apart.

A value of 0.5 is latex, 0.9 is like a sweater, 0.999 is a highly-starched napkin or leather. The soft body simulation tends to get unstable if you use a value of 0.999, so you should lower this value a bit if that happens.

How much the soft body resists being scrunched together, like a compression spring. Low values for fabric, high values for inflated objects and stiff material.

The friction for edge springs. High values (max of 50) dampen the Push/Pull effect and calm down the cloth.

Permanent deformation of the object after a collision. The vertices take a new position without applying the modifier.

This option creates virtual connections between a vertex and the vertices connected to its neighbors. This includes diagonal edges. Damping also applies to these connections.

The edges can shrink or be blown up. This value is given in percent, 0 disables this function. 100% means no change, the body keeps 100% of its size.

Checks for edges of the soft body mesh colliding.

Checks for any portion of the face of the soft body mesh colliding (which is computationally intensive). While Face enabled can solve collision errors, there does not seem to be any dampening settings for it. So parts of the soft body object near a collision mesh tend to \xe2\x80\x9cjitter\xe2\x80\x9d as they bounce off and fall back, even when there is no motion of any meshes. Edge collision has dampening, so that can be controlled, but Deflection dampening value on a collision object does not seem to affect the face collision.

Force from surrounding media. See exterior forces for details.

Edges receive a drag force from the surrounding media.

Edges receive a lift force when passing through the surrounding media.

How much aerodynamic force to use. Try a value of 30 at first.

For quad faces, the diagonal edges are used as springs. This stops quad faces to collapse completely on collisions (what they would do otherwise).

Stiffness of the virtual springs created for quad faces.

--- Fin du chapitre: _edges ---

--- Début du chapitre: _edges_of_corner --- The Edges of Corner node retrieves the edges on both sides of a face corner.

The index of the input face corner.

Note

By default this uses the index from the field context, which makes it important that the node is evaluated on the face corner domain.

This node has no properties.

The index of the neighboring edge in the face, in the direction of increasing face corner indices.

The index of the neighboring edge in the face, in the direction of decreasing face corner indices.

--- Fin du chapitre: `_edges_of_corner` ---

--- Début du chapitre: `_edges_of_vertex` --- Selects a neighboring edge of a vertex and outputs its index.

This node is a bit special because it operates in two different domains. First, it evaluates a Weight for each edge in the geometry. Then, for each item in the context domain, it will:

Pick a vertex from the geometry based on the Vertex Index.

Find the edges connected to this vertex.

Sort these edges by their associated weight.

Pick an edge from the above sorted list based on the Sort Index, where 0 means the edge with the lowest weight, 1 means the edge with the second-lowest weight and so on.

Output the geometry-wide index of this edge.

The index of the vertex for which to find the edges.

Note

If this input is not connected, it uses the index of the context item, which means it's important that the node is evaluated in the Point domain.

The weights of the edges in the geometry. Unlike the other inputs which follow the context domain, this one is always evaluated in the Edge domain.

The edges are sorted by their associated weight in ascending order. Edges with the same weight are sorted by their index.

The 0-based index of the edge to select from the vertex's sorted edges. If this value is outside the range of valid indices, it wraps around.

This node has no properties.

The geometry-wide index of the selected edge. You can pass this to the Evaluate at Index Node or the Sample Index Node (with the domain set to Edge) to retrieve details about the edge.

If the vertex has no connected edges, Edge Index will be zero.

The number of edges connected to the selected vertex.

The example below creates a cone at each vertex of a cube, aligned to the neighboring edge that's the most vertical.

First, we calculate a verticality score for each of the cube's edges. To do this, we subtract the positions of its vertices to get its direction vector, which we normalize and use to calculate the dot product with the Z axis. The absolute value of that gives us a number between 0 and 1, where 0 means fully horizontal and 1 means fully vertical.

Because the edges will be sorted by ascending weight, we set weight = 1 - verticality. This way, the most vertical connected edge of each vertex will have the lowest weight and come first in the list.

Next, in the point domain, we need to calculate the rotation of each cone. By using the Align Rotation to Vector Node, the problem gets simplified and we only need to calculate a direction vector. The direction vector of each cone is the centerpoint of the most vertical neighboring edge minus the position of the vertex. Finding that most vertical neighboring edge is where the Edges of Vertex node comes in: for each vertex, it sorts the connected edges by their weight and pick the first one (because the Sort Index is 0). Once we have the edge's index, we use the Evaluate at Index Node to retrieve its centerpoint.

With the rotations of the cones calculated, we use the Instance on Points Node to create them.

Example node setup. (Rightclick and choose "Open image in new tab" to see a larger version.)

The resulting geometry.

--- Fin du chapitre: `_edges_of_vertex` ---

--- Début du chapitre: `_edges_to_face_groups` --- The Edges to Face Groups node group faces into regions surrounded by the selected boundary edges.

Edges used to split faces into separate groups.

This node has no properties.

Index of the face group inside each boundary edge region.

--- Fin du chapitre: `_edges_to_face_groups` ---

--- Début du chapitre: `_edge_angle` --- The Edge Angle node calculates the angle in radians between two faces that meet at an edge. For the Face, Face Corner, and Point domains, the node uses simple domain interpolation to move values from the mesh's edges.

Note

The output of this node depends on the density of the mesh. If there are more edges closer together and the curvature of the mesh stays the same, the edge angle will be different.

This node has no inputs.

This node has no properties.

The shortest angle in radians between two faces where they meet at an edge. The range of the data is from zero to PI. Flat edges and Non-manifold edges have an angle of zero. An edge between two faces completely folded back on each other has an angle of PI, or 180 degrees.

Tip

Computing this value is slightly faster than the signed angle, so if there is no need to distinguish between convex and concave angles, using this value can provide a performance improvement.

The signed angle in radians between two faces where they meet at an edge. Flat edges and Non-manifold edges have an angle of zero. Concave angles are positive and convex angles are negative.

--- Fin du chapitre: `_edge_angle` ---

--- Début du chapitre: `_edge_data` --- Edges can have several different properties that affect how certain other tools affect the mesh.

Reference

Edit Mode

Edge "Edge Crease"

Shift-E

This operator interactively sets the Edge Crease amount by moving the mouse (or typing a value with the keyboard). Selecting more than one edge will adjust the mean (average) crease value. A negative value will subtract from the actual crease value, if present. To clear the crease edge property, enter a value of -1.

Reference

Edit Mode

Edge `\xe2\x80\xa3` Edge Bevel Weight

Sets the value for the `bevel_weight_edge` attribute, a value between (0.0 to 1.0).

This attribute is used by the Bevel Modifier to control the bevel intensity of the edges.

This operator enters an interactive mode (a bit like transform tools), where by moving the mouse (or typing a value with the keyboard) you can set the bevel weight of selected edges. If more than one edge is selected, this operator alters the average weight of the edges.

See also

Vertex Bevel Weight

Reference

Edit Mode

Edge `\xe2\x80\xa3` Mark Seam/Clear Seam

These operators set or unset this mark for selected edges. Seams are a way to create separations, `\xe2\x80\x99`islands`\xe2\x80\x9d`, in UV maps. See the UV Mapping section for more details.

Reference

Edit Mode

Edge `\xe2\x80\xa3` Mark Sharp/Clear Sharp

Signifies the selected edge(s) as being

`\xe2\x80\x99`sharp`\xe2\x80\x9d`. This edge attribute can either be set (mark) or unset (clear).

This influences the rendering of Normals to appear flat if smooth shading is enabled for the connecting face or object. This attribute can also be used by many modifiers or operators to mask their effect.

Internally, this uses the sharp edge attribute .

Reference

Edit Mode

Edge `\xe2\x80\xa3` Set Sharpness by Angle

Sets the sharp edge attribute based on the angle between neighboring faces.

Maximum angle between face normals that will be considered as smooth.

Add new sharp edges without clearing existing sharp edges.

--- Fin du chapitre: `_edge_data` ---

--- Début du chapitre: `_edge_neighbors` --- The Edge Neighbors node outputs topology information relating to each edge of a mesh.

This node has no inputs.

This node has no properties.

The number of faces that use the edge as one of their sides. When the value is one, the edge is a non-manifold boundary edge. Alternatively, when the value is zero, the edge is a loose edge, not used by any faces.

Using the Face Count output to create a curve on a mesh `\xe2\x80\x99`s boundary edges. `\xc2\xbb6`

--- Fin du chapitre: `_edge_neighbors` ---

--- Début du chapitre: `_edge_paths_to_curves` --- The Edge Paths to Curves node output curves that follow paths across mesh edges.
See also

This node is meant to use the output of the Shortest Edge Paths Node .It is similar to the Edge Paths to Selection Node , but it creates a curve that follow each path, rather than a selection of every visited edge.
Standard mesh input.

A selection of the vertices to start at when traveling along the next vertex indices.

Describes the path to follow at every vertex.

This node has no properties.

Standard curves output.

--- Fin du chapitre: `_edge_paths_to_curves` ---

--- Début du chapitre: `_edge_paths_to_selection` --- The Edge Paths to Selection node follows paths across mesh edges and outputs a selection of every visited edge.

See also

This node is meant to use the output of the Shortest Edge Paths Node .It can be combined with the Separate Geometry Node to remove any unused edges.

A selection of the vertices to start at when traveling along the next vertex indices.

Describes the path to follow at every vertex.

This node has no properties.

A boolean field indicating all edges visited when traversing the mesh.

--- Fin du chapitre: `_edge_paths_to_selection` ---

--- Début du chapitre: `_edge_slide` --- Reference

Edit Mode

Edge `\xe2\x80\xa3` Edge Slide

G, G

Slides one or more edges across adjacent faces with a few restrictions involving the selection of edges (i.e. the selection must define a valid loop, see below).

Forces the edge loop to match the shape of the adjacent edge loop. You can flip to the opposite vertex using F.

When Even mode is active, this flips between the two adjacent edge loops the active edge loop will match.

Toggle clamping the slide within the edge extents.

Determines the amount of slide performed. Negative values correspond to slides toward one face, while positive ones, refer to the other one. It is also displayed in the 3D Viewport footer.

Lets you propagate the operation to the symmetrical elements of the mesh (if present, in local X direction).

Corrects the corresponding UV coordinates, if these exist, to avoid image distortions.

By default, the position of vertices on the edge loop move as a percentage of the distance between their original position and the adjacent edge loop, regardless of the edges `\xe2\x80\x99` lengths.

Selected edge loop. `\xc2\xb6`

Repositioned edge loop. `\xc2\xb6`

Evenmode keeps the shape of the selected edge loop the same as one of the edge loops adjacent to it, rather than sliding a percentage along each perpendicular edge.

In Evenmode, the tool shows the position along the length of the currently selected edge which is marked in yellow, from the vertex that has an enlarged red marker. Movement of the sliding edge loop is restricted to this length. As you move the mouse the length indicator in the header changes showing where along the length of the edge you are.

Even Mode enabled. \xc2\xbb

Even Mode with Flip enabled. \xc2\xbb

Moving the mouse moves the selected edge loop towards or away from the start vertex, but the loop line will only move as far as the length of the currently selected edge, conforming to the shape of one of the bounding edge loops.

There are restrictions on the type of edge selections that can be operated upon. Invalid selections are:

This means that the tool could not find any suitable faces that were adjacent to the selected edge(s). An example that shows this is selecting two edges that share the same face. A face cannot be adjacent to itself.

The selected edges are not in the same edge loop, which means they do not have a common edge. You can minimize this error by always selecting edges end-to-end or in a \xe2\x80\x9cchain\xe2\x80\x9d. If you select multiple edges just make sure they are connected. This will decrease the possibility of getting looping errors.

When a single edge was selected in a single-sided object. An edge loop cannot be found because there is only one face. Remember, edge loops are loops that span two or more faces.

A general rule of thumb is that if multiple edges are selected they should be connected end-to-end such that they form a continuous chain. This is literally a general rule because you can still select edges in a chain that are invalid because some of the edges in the chain are in different edge loops.

--- Fin du chapitre: _edge_slide ---

--- Début du chapitre: _edge_split --- The Edge Split modifier splits, duplicates edges within a mesh, breaking \xe2\x80\x9dlinks\xe2\x80\x9d between faces around those split edges.

The edges to split can be determined from the edge angle (i.e. angle between faces forming that edge), and/or edges marked as sharp.

Splitting an edge affects vertex normal generation at that edge, making the edge appear sharp. It can also be used for manual control of the smoothing process, where the user defines which edges should appear smooth or sharp (see Mesh Smoothing for other ways to do this). If desired, both modes can be active at once.

Note

This modifier is kept mostly for historical/compatibility reasons. Everything it can do in shading, and much more, can now be achieved using custom normals.

Unless you really need the topology changes it generates, it is not advised to use it in new projects.

Note

Splitting edges can also be performed manually in Edit Mode.

The Edge Split modifier. \xc2\xbb

When enabled, an edge will be split if the angle between its two adjacent faces is greater than the Split Angle.

On 0: all edges are split. On 180: no edges are split.

When enabled, edges will be split if they were marked as sharp .

Note

Non-manifold edges will always be split.

Flat shading.\xc2\xba

Smooth shading.\xc2\xba

Smooth shading with Edge Split.\xc2\xba

Smooth shading with Edge Split and Subdivision Surface.\xc2\xba

--- Fin du chapitre: _edge_split ---

--- Début du chapitre: _edge_vertices --- The Edge Vertices node outputs the position and index of the two vertices of each of a mesh's edges.

Note

The order of the two vertices of an edge is arbitrary. In some cases it may be predictable based on the internals of the algorithm that created the mesh, but in general the order should not be relied upon.

This node has no inputs.

This node has no properties.

The index of the two vertices of the edge.

The position of each of the edge's vertices. This output is for convenience, it is the same as using index output to retrieve the position from the Evaluate at Index Node .

--- Fin du chapitre: _edge_vertices ---

--- Début du chapitre: _editing --- You can modify the manual by editing local text files. These files are kept in sync with those online via a repository, based on this the server will update the online manual.

The manual is written in the reStructuredText (RST) markup language and can be edited using a plain text editor. For a local preview, you convert (build) the manual source files from RST into HTML web pages.

Firstly, make sure that your local copy of the manual is up to date with the online repository using:

You can now edit the documentation files, which are the .rst files inside the manual folder using a text editor of your choice.

Be sure to check the Writing Style Guide for conventions and Markup Style Guide to learn how to write in the reStructuredText markup language.

Happy writing!

If you are going to add or overhaul a section, be sure to check carefully that it does not already exist. In some places, the docs are so disorganized that sections may be duplicated or in a strange location. In the case that you find a duplicate or out of place section, create an issue explaining the issue, and optionally include a revision (actual changes).

Before you make any edits that are not simple and plainly justified (for example, moving folders around), you should verify with a manual maintainer that your contribution is along the community's vision for the manual. This ensures the best use of your time and good will as it is otherwise possible that, for some reason, your changes will conflict and be rejected or need time-consuming review. For example, another person may be already working on the section you wish to

change, the section may be scheduled for deletion or to be updated according to a planned change to Blender. Communicating early and frequently is the key to have a productive environment, to not waste people's effort and to attain a better Blender manual as a result.

If you are in doubt about functionality that you wish to document, you should pose your questions to the Blender developers responsible for that area or ask at the unofficial usersupport channel in the #docs or #blender-coders channel in Chat.

Blender itself has a system of module owners with developer and artist members who are responsible for the design and maintenance of the assigned Blender areas. See the module pages for more information.

To view your changes, build the manual as instructed. Keep in mind that you can also build only the chapter you just edited to view it quickly. Open the generated .html files inside the build/html folder using your web browser, or refresh the page if you have it open already.

When you are happy with your changes, you can upload them, so that they will be in the online manual. At first, this is done by submitting patches so that someone can review your changes and give you feedback. After, you can commit your changes directly. This process is described in detail in the next section.

--- Fin du chapitre: _editing ---

--- Début du chapitre: _edit_face_set --- Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Edit Face Set

Grow/Shrink Face Sets

Edits the Face Set under the cursor.

The operation to apply to the face set.

Grows the face sets boundary by one face based on mesh topology. This is also available as a shortcut operator via Ctrl- W .

Shrinks the face sets boundary by one face based on mesh topology. This is also available as a shortcut operator via Ctrl- Alt- W .

Deletes the faces that are assigned to the face set.

Creates a perfectly flat and smooth geometry patch from the face set. This is the ideal way to trim parts of your mesh if the vertex count is too high for other operations, or the vertex IDs must not be altered (Like when using Multires sculpting).

Creates a smooth as possible geometry patch from the face set by minimizing changes in vertex tangents. This is ideal for creating smooth curved surfaces on complex topology, where just using the smooth brush will not lead to desired results

Before fairing. \xc2\xbb6

After using Fair Positions. \xc2\xbb6

After using Fair Tangency. \xc2\xbb6

The amount of effect the filter has on the mesh. This setting is only available for the fairing operations.

Apply the edit operation to hidden face sets.

--- Fin du chapitre: _edit_face_set ---

--- Début du chapitre: _effector --- Effector objects are used to deflect fluids and influence the fluid flow. To define any mesh object as an effector object, add fluid physics by clicking Fluid in Properties \xe2\x80\xa3 Physics. Then select Effector as the fluid Type.

Tip

Force Fields (such as wind or vortex) are supported, like in most physics systems. The influence individual force types have can be controlled per domain object.

Reference

Physics \xe2\x80\xa3 Fluid \xe2\x80\xa3 Settings

Effector

Objects of this type will collide with fluid.

The velocity of objects of this type will be used when baking the guiding. So fluid guiding objects should move and have some velocity. Multiply the guiding object velocities by this factor. This is useful when working with multiple guiding objects and some of them should have higher or smaller velocities.

The mode describes how guiding velocities should be written into the global guiding velocity field of the domain.

The guiding object will compare the existing velocity in the global velocity field with its own velocity. If its absolute value is greater than the absolute value in the velocity field the guiding velocity will be kept.

A guiding object will compare the existing velocity in the global velocity field with its own velocity. If its absolute value is smaller than the absolute value in the velocity field the guiding velocity will be kept.

The most intuitive option. A guiding object will always write its own current velocity into the global guiding velocity field. Values in the velocity field from a previous frame or guiding object will be overridden. A guiding object will write the average of its own current velocity and the existing guiding velocity at that cell into the global guiding velocity field.

Number of substeps used to reduce gaps in collision of fluid from fast-moving effectors.

Additional area around the effector that will be considered as an effector.

Enables or disables the effector object effect on the fluid, this property is useful for animations to selectively enable and disable when the effector affects the fluid.

Defines the effector as either a single dimension object i.e. a plane or the mesh is Non-manifold. This ensures that the fluid simulator will give the most accurate results for these types of meshes.

A Manifold mesh can also be declared as planar. The fluid solver will then ignore the volume inside the mesh and just emit fluid from the mesh sides.

--- Fin du chapitre: _effector ---

--- Début du chapitre: _elastic_deform ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3 Advanced \xe2\x80\xa3 Brush Type

Used to simulate realistic deformations such as grabbing or twisting of Elastic objects. For example, this tool works great for modeling the shape of organic objects such as humans or animals. When pressing Ctrl, the brush deforms vertices along the normal of the active vertex.

Note

More info at General brush settings and on Advanced brush settings.

The surface alteration that is used in the brush.
Used to drag a group of vertices around.
Like Grabbut the falloff is more localized to the center of the brush.
Like Bi-scale Grabbut the falloff is more localized to the center of the brush.
Displaces vertices away from the active vertex.
Vertices are rotated around the active vertex.
Higher values preserve volumes more, but also lead to more bulging. (This value determines the poisson ratio for elastic deformation)
--- Fin du chapitre: `_elastic_deform` ---

--- Début du chapitre: `_ellipse_mask` --- The Ellipse Masknode creates an image suitable for use as a simple matte or vignette mask.
An optional mask to use as the base for mask operations.
Intensity of the generated mask.
Position of the center of the ellipse as a fraction of the total width or height. (0.5, 0.5 creates a centered ellipse; 0.0, 0.0 creates an ellipse with its center in the lower left.)
Width of the ellipse as a fraction of the total image width.
Height of the ellipse as a fraction of the total image width, not height. Equal Width and Height values will produce a circle.
Rotation of the ellipse around its center point.
Operation to use against the input mask.
This yields the union of the input mask and the generated mask: Areas covered by the generated mask are set to the specified Value. Other parts of the input mask are passed through unchanged, or set to black if there is no input mask.
Values of the input mask have the specified Value subtracted from them.
This yields the intersection of this generated mask and the input mask: Values of the input mask are multiplied by the specified Value for the area covered by the generated mask. All other areas become black.
Any area covered by both the input mask and the generated mask becomes black. Areas covered by the generated mask that are black on the input mask become the specified Value. Areas uncovered by the generated mask remain unchanged.
A generated elliptical mask merged with the input mask. The created mask is the size of the current scene render dimensions.
Tip
For soft edges, pass the output mask through a slightBlur node. For a vignette, pass the output of this through a heavy blur.
--- Fin du chapitre: `_ellipse_mask` ---

--- Début du chapitre: `_emission` --- The Emission node is used to add Lambertian emission shader. This can for example, be used for material and light surface outputs.
Light strength for point, spot and area lights is specified in Watts.
Sun lights are specified in Watts/m², which require much smaller values like 1 W/m². This can be confusing, but specifying strength in Watts would not have been convenient; the real sun for example has strength 384.6 × 10²⁴ W. Emission shaders on meshes are also in Watts/m².
Color of the emitted light.
Strength of the emitted light. For point and area lights, the unit is Watts. For materials, a value of 1.0 will ensure that the object in the

image has the exact same color as the Color input, i.e. make it
unshaded.

This node has no properties.

The Emission shader output can both be plugged into the Surface Input as well as the Volume Input of the Material Output node.

- Emission shader, with strength at 1.0.

- Emission shader, with strength at 3.0.

--- Fin du chapitre: _emission ---

--- Début du chapitre: _empties --- The Empty node is a single coordinate point with no additional geometry. Because an empty has no volume and surface, it cannot be rendered. Still it can be used as a handle for many purposes.

- Empty Display Types.

- Displays as six lines, initially with one pointing in each of the +X, -X, +Y, -Y, +Z, and -Z axis directions.

- Displays as arrows, initially pointing in the positive X, Y, and Z axis directions, each with a label.

- Displays as a single arrow, initially pointing in the +Z axis direction.

- Displays as a circle initially in the XZ plane.

- Displays as a cube, initially aligned to the XYZ axes.

- Displays as an implied sphere defined by three circles. Initially, the circles are aligned, one each, to the X, Y, and Z axes.

- Displays as a cone, initially pointing in the +Y axis direction.

- Empties can display images. This can be used to create reference images, including blueprints or character sheets to model from. The image is displayed regardless of the 3D display mode.

- Empty Displays settings can be accessed from Properties > Object Data > Empty panel.

- Offset the image origin (where 1.0 represents the width/height of the image).

- Object origin at image center.

- Object origin at image bottom, left.

- Object origin at image top, right.

- Use normal depth behavior.

- Always display on top of other objects.

- Always display behind of other objects.

Tip

When using the image as a reference for modeling, it can be useful to set the depth to Front, with a low Opacity.

- Display both the front and back of the empty.

- Only display the front of the image.

- Only display the back of the image.

Tip

This is useful if you are using an image as a reference where you have photos from both the front and back, so two empty images can be set only to show when viewed from the correct side.

- Show in orthographic view.

- Show in perspective view.

Hint

It's often useful to disable this so reference images don't get in the way when viewing a model.

- Only displays the image contents when the view is aligned with the object's local axis.

Blends the image with the background. The value slider adjusts the opacity of the image, changing how much of the image is blended with the background.

An empty can only be edited in Object Mode, which includes its transformation and parenting properties. For other tools see the Object section.

While empties don't exactly have any object data attached to them which can be used for supporting true apply scale (i.e. with non-uniform scaling), they do have Display Size which controls how large the empties are displayed (before scaling). This works by taking the scale factor on the most-scaled axis, and combines this with the existing empty Display Size to maintain the correct dimensions on that axis.

The Primitive empty type to display in the 3D Viewport.

Controls the size of the empties visualization. This does not change its scale, but functions as an offset.

Empties can serve as transform handles. Some examples of ways to use them include:

Parent object for a group of objects

An empty can be parented to any number of other objects. This gives the user the ability to control a group of objects easily, and without affecting a render.

Target for constraints

An empty can also be used as a target for normal, or bone constraints. This gives the user far more control; for instance, a rig can easily be set up to enable a camera to point towards an empty using the Track to constraint.

Array offset

An empty can be used to offset an Array Modifier, meaning complex deformations can be achieved by only moving a single object.

An example of an empty being used to control an array.

An example of an empty being used to control the Track To constraint.

Other common uses:

Placeholders

Rigging controls

DOF distances

Reference Images

--- Fin du chapitre: _empties ---

--- Début du chapitre: _endpoint_selection --- The Endpoint Selection node provides a selection for an arbitrary number of endpoints in each spline in a curve.

Note

The selection operates for every control point. This may not correspond to the evaluated points displayed in the viewport for NURBS and B-spline splines, where one control point may correspond to many evaluated points.

Tip

To use this data after the curve has been converted to another data type like mesh or a point cloud, the Capture Attribute Node can be used.

The number of points to select from the start.

The number of points to select from the end.

This node has no properties.

Selection of the start and end of each spline of the curve.
Anywhere the geometry is a curve, this node can be used to generate a selection of only the first and last points of each spline. Here, the Points input of the Instance on Points Node is a curve consisting of the poly spline shown in Edit Mode.

--- Fin du chapitre: `_endpoint_selection` ---

--- Début du chapitre: `_envelope` --- The Envelope modifier creates a shape known as envelope over the existing strokes connecting all the points that have *n* points between them.

The Envelope modifier. `\xc2\xbb`
Replaces the original stroke with the envelope shape.
Add segments to create the envelope shape keeping the original stroke.
Add segments to create the envelope without the original stroke.
The number of points to skip when creating the straight segments that define the envelope.

The thickness of the generated stroke segments.
The Opacity of the generated stroke segments.
Defines the material to use on the generated stroke segments.
The Number of generated stroke segments to skip to reduce complexity.
See Influence Filters.

--- Fin du chapitre: `_envelope` ---

--- Début du chapitre: `_environment` --- The Node Environmental Texture is used to light your scene using an environment map image file as a texture.

Texture coordinate for texture look-up. If this socket is left unconnected, the image is mapped as environment with the Z axis as up.
Image data-block used as the image source. Additional settings can be found in Sidebar `\xe2\x80\xa3` Item `\xe2\x80\xa3` Properties: These include options to control the alpha channel along with addition options for the color space. These addition options are documented with the rest of Common Image Settings .

Type of data that the image contains, either Color or Non-Color Data. For most color textures the default of Color should be used, but in case of e.g. a bump or alpha map, the pixel values should be interpreted as Non-Color Data, to avoid doing any unwanted color space conversions.

The list of color spaces depends on the active OCIO config. The default supported color spaces are described in detail here: Default OpenColorIO Configuration

Interpolation method used for the environment texture. The following interpolations are available:

Regular quality interpolation.

No interpolation, use closest pixel.

Smoother, better quality interpolation.

Bicubic when magnifying, otherwise Bilinear is used. This is only available for OSL .

Allows you to use different types of environmental maps. The following methods are supported:

Projection from an Equirectangular photo.

Projection from an orthographic photo or mirror ball.

RGB color from the image.

HDR image from OpenFootage.net. `\xc2\xbb`

--- Fin du chapitre: `_environment` ---

--- Début du chapitre: _erase ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Brush

The Erase tool erases already drawn strokes.

The Erase tool uses any of the Grease Pencil Erasedraw mode brushes. Activating a brush from an asset shelf or brush selector will also activate this tool for convenience.

The asset selector can be used to open a pop-up asset browser to select the active brush asset for the tool.

SeeAsset Operatorsfor more information.

The radius of the brush in pixels.

Allows you to change the brush size interactively by dragging the pointer orby typing a number then confirm.

Adjusts the radius based on the stylus pressure when using aGraphics Tablet.

Control how much will affect the eraser has on the stroke transparency (alpha).

You can change the brush strength interactively by pressingShift- Fin the 3D Viewport and then moving the pointer and then LMB.You can also enter the size numerically.

Adjusts the strength based on the stylus pressure when using aGraphics Tablet.

Determines how the erase tool behaves.

To simulate a raster type eraser, this eraser typeaffects the strength and thickness of the strokes before actually delete a point.

Delete one point at a time.

Delete an entire stroke.

The cursor can be disabled by toggling the checkbox in the Cursorpop-over menu.

In the Tool Settings select the brush to use with the tool.The Erase tool usesErase Brushtypes (soft, point and stroke).

Select an erase brush of type Soft/Hard.

Adjust brush settings.

Click and hold LMBor use the Pentip to delete strokes on the viewport.

Original drawing. \xc2\xbb

The eraser affect the transparency of the strokes. \xc2\xbb

Final result. \xc2\xbb

Select an erase brush of type Point.

Adjust brush settings.

Click and hold LMBor use the Pentip to delete strokes on the viewport.

Original drawing. \xc2\xbb

The eraser delete one point at a time. \xc2\xbb

Final result. \xc2\xbb

Select an erase brush of type Stroke.

Adjust brush settings.

Click and hold LMBor use the Pentip to delete strokes on the viewport.

Original drawing. \xc2\xbb

The eraser delete one stroke at a time. \xc2\xbb

Final result. \xc2\xbb

--- Fin du chapitre: _erase ---

--- Début du chapitre: _euler_to_rotation --- The Euler to Rotationnode creates a rotation value from an Euler rotation.

The Euler rotation.

Standard rotation value.

--- Fin du chapitre: `_euler_to_rotation` ---

--- Début du chapitre: `_evaluate_at_index` --- The Evaluate at Index node allows accessing data of other elements in the context geometry. It is similar to the Sample Index Node. The main difference is that this node does not require a geometry input, because the geometry from the field context is used.

This node is also similar to the Evaluate on Domain Node, except that the value to retrieve from the specified domain is specified by an index rather than an automatic domain interpolation.

The index of the element in the selected domain to retrieve data from, i.e. the fourth face, or the first control point.

The field to retrieve data from.

The attribute domain used for evaluation of the Value input. This is useful because it can be a different domain than the domain from the field context, i.e. to choose a vertex for each face.

The value of the input field at the given index.

--- Fin du chapitre: `_evaluate_at_index` ---

--- Début du chapitre: `_evaluate_on_domain` --- The Evaluate on Domain allows evaluating a field for a different attribute domain than the domain from the field context. For example, the face index could be used instead of the face corner index, when setting the values of a UV Map.

Note
This node is not necessary to retrieve data from other attribute domains; that is done automatically. Its utility comes from the fact that it's possible to control when the domain interpolation happens. Normally, input nodes interpolate their data to the current context's domain as soon as they create their output.

Tip

It may be preferable to use this node over the Capture Attribute Node, since it allows using a specific attribute domain without requiring a geometry socket input, which allows creating more reusable node groups. See also

The method of retrieving data from another domain is somewhat similar to the Evaluate at Index Node.

The field to evaluate on the chosen attribute domain.

The attribute domain used for evaluation of the Value input.

The values from the input, evaluated on the chosen domain, then interpolated to the domain from the field context.

--- Fin du chapitre: `_evaluate_on_domain` ---

--- Début du chapitre: `_examples` --- Here are some simple examples showing the power of soft body physics.

First, change your start and end frames to 1 and 150.

Then, add a plane, and scale it five times. Next, go to the physics tab, and add a collision. The default settings are fine for this example.

Now add a cube, or use the default cube, then enter Edit Mode to subdivide it three times. Add a Bevel Modifier to it to smoothen the edges and then to add a little more, press R twice, and move your cursor a bit.

When finished, your scene should look like this:

The scene, ready for soft body physics.

Everything is ready to add the soft body physics. Go to Properties
Physics and choose Soft Body. Uncheck the Soft Body Goal, and
check Soft Body Self Collision. Also, under Soft Body Edges, increase the
Bending to 10.

Playing the animation will now give a slow animation of a bouncing
cube. To speed things up, we need to bake the soft body physics.

Under Soft Body Cache change the values of your start and end frames. In
this case 1 and 150. Now, to test if everything is working, you can take a
cache step of 5 or 10, but for the final animation it is better to reduce
it to 1, to cache everything.

You can now bake the simulation, give the cube materials and textures
and render the animation.

The rendered bouncing cube

--- Fin du chapitre: _examples ---

--- Début du chapitre: _expand --- This is a multi-purpose modal operator
to intuitively create and edit masks/face sets. When executed, it
uniformly expand outwards a pattern from the vertex under the cursor.

Note

These operators are meant to be used interactively through the shortcut.

There is also a full showcase of the Expand features and use cases.

A preview of Expand Mask by Topology

Reference

Sculpt Mode

Mask Expand Mask by Topology

Shift- A

Expands a mask from the active vertex.

Start the operator and expand a mask from an origin to your mouse cursor
distance. Then confirm with LMB or Return

By default the expansion will use a Geodesic falloff to create perfectly
accurate distances along the surfaces. Use other falloff types via 2, 3 &
4 to expand via triangles, whole faces or scene distances instead.

The typical result when using the Diagonal falloff to expand along the
quads of the face.

Start Expand while pointing at an open boundary to expand from the
entire boundary loop. This will always use the Topology falloff mode.

Move your cursor outside of the boundaries of the mesh to mask the
entire connected mesh. This can be done repeatedly to quickly mask
separate meshes.

Hold Ctrl to snap to face sets under your cursor and fill them. Any face
set that was already covered in the expansion will be filled as well.

While using any Transform tool, the pivot point will automatically snap
the border of an Expand result. This way areas (Like limbs) can be masked
and then immediately rotated or otherwise transformed.

The different falloff types can be used for circular, triangular and
square patterns. More loops can also be added/removed via W & Q to repeat
the pattern across the mesh.

An example of using expand with mirror options, loops and a recursion to
create wood carving patterns.

Tip

Mirror options can also be combined with the expansion.

Linear gradients or brush falloff gradients B will help to add slanted
surfaces to the patterns.

A Recursions with RorAlt- R will start a new expansion along the border of the current expansion. Doing this multiple times, can help for increasingly random patterns or advanced pattern creation.

An example of using loops and gradients with multiple expanded masks.

Tip

Remember that Expand only affects visible geometry. So if a pattern should only be created on a part of the mesh, hide the other geometry first.

Use the Mesh Filter to deform the geometry and the Color Filter to add colors, to apply the patterns on the sculpt.

Textures can be used to affect the shape and gradients of the expanded mask. This feature can be combined with loops and recursion to create unique results.

To use a texture, you need to assign it to your currently active brush in the Texture Brush Settings. The texture can be edited/created in the Texture Properties.

Note

This texture only works when the Mapping is set to 3D.

Use Y and T to increase or decrease the affect the texture has on the edge of the mask.

F Flips between expanding a positive mask (value of one) or a negative mask (value of zero). In the case of face sets, this option flips between including areas inside the masked area or areas outside the masked area... needs visual technical examples for both

E Accumulate the new mask on top of the previous one or replace it. For Face Sets, this will toggle between creating Face Sets boundaries or replacing the existing Face Sets.

Spacebar Moves the initial vertex used for calculating the falloff... needs visual technical example

1 Uses a falloff based on the Geodesic distances from the edge boundary to the active vertex.

2 Uses a falloff based on a flood fill using edges.

3 Uses a falloff based on a flood fill using polygon diagonals and edges.

4 Uses a falloff based on the Euclidean distances from the edge boundary to the active vertex... needs visual technical examples

G Enables linear gradient of values from the origin to the current active vertex.

B Similar to linear gradient but uses the current brush Falloff to define the shape of the falloff... needs visual technical examples

R Start a new expansion with a Geodesic falloff from the boundary of the current falloff.

Alt- R Start a new expansion with a topology falloff from the boundary of the current falloff... needs visual technical examples

Ctrl Isolates the expanded region to the boundary of the face set under the cursor.

W Increase the number of loops or iterations the operator is run; using four loops will split the mask into four parts.

Q Decrease the number of loops or iterations the operator is run; using four loops will split the mask into four parts... needs visual technical examples

Y Increases the falloff distance when using a texture to distort the mask shape.

TDecreases the falloff distance when using a texture to distort the mask shape...needs visual technical examples

Reference

Sculpt Mode

Mask \xe2\x80\xa3 Expand Mask by Normals

Shift- Alt- A

Expand a mask, following the curvature of the surface.This operator uses the same internal operator as Expand meaning all the shortcuts and functionality works the same as that tool.

This operator is especially useful for hard surface sculpting.

Tip

If one expansion does not properly fill the entire desired surface,use the operator repeatedly with a different starting point.

Note

Using any of the Falloff shortcuts1- 4will replace the curvature falloff of this operator.

Reference

Sculpt Mode

Face Sets \xe2\x80\xa3 Expand Face Set by Topology

Shift- W

Expands a face set from the active vertex.This operator uses the same internal operator as Expand meaning all the hotkeys and functionality works the same as that tool,with the gradient features as the exception.

Expanding Face Sets has all the same use cases as expanding masks.The advantage for this one is that they will be saved for repeated usage.Face sets can be filled any time with a mask,so assigning areas face sets will save you time.(And of course face sets can be used tohide face sets)

When using thePose Brushit is most predictable when using it withFace Sets to define the face set boundaries as pivot point locations.Face Sets can be expanded from a point or from a boundary between hidden face setsto create them quickly.AlternativelyGrow/Shrink Face Setsor use theExpand Active Face Setto dynamically grow/shrink them.

Tools like theCloth FilterandCloth Brushwork especially well when only simulating small areas at a time.Face Sets can very easily be created with Expand to assign areas of action.

Reference

Sculpt Mode

Face Sets \xe2\x80\xa3 Expand Face Set by Topology

Shift- Alt- W

Expands an existing face set with a geodesic falloff.This operator uses the same internal operator as Expand meaning all the hotkeys and functionality works the same as that tool.

Note

Using any of the Falloff shortcuts1- 4the operator to switch toExpand Face Set by Topology.

Resize a Face Set along the surface distances.It is an alternative toGrow/Shrink Face Setswhich follows the topology instead of geodesic distances.

When holding Ctrlthe expansion will instead snap to other Face Sets.This is a fast way of joining multiple face sets into one.

--- Fin du chapitre: _expand ---

--- Début du chapitre: _experimental --- These preferences are reserved for features that are currently being worked on and are not yet

complete. This category can be enabled by enabling Developer Extras. Most experimental features are only available in Daily Builds.

Blender Preferences Experimental section.

Use legacy undo (slower than the new default one, but may be more stable in some cases).

Enables library overrides automatic resync detection and process on file load. Disable when dealing with older blend-files that need manual Resync (Enforce) handling.

Show the Cycles rendering debug panel.

Enable some extra fields in the Asset Browser to aid debugging.

Disabling the asset indexer forces every asset library refresh to completely reread assets from disk.

Enable viewport debugging options for developers in the overlays popover.

Enable EEVEE debugging options for developers.

--- Fin du chapitre: _experimental ---

--- Début du chapitre: _explode --- The Explode modifier is used to alter the mesh geometry by moving/rotating its faces in a way that roughly tracks particles emitted by that object, making it look as if the mesh is being exploded (broken apart and pushed outward).

For this modifier to have any visible effect, there needs to be a particle system on its object. That particle system will control how the mesh is exploded.

Both the number of emitted particles and number of faces determine how granular the Explode modifier is. More of each faces and particles will mean more individual pieces.

Here is a demo video showing a cube with a particle system and Explode modifier. (blend-file).

Note

The Explode modifier must come after the Particle System one in the modifier stack, in order for the former to get required data from the later.

The Explode modifier, with a Particle System above it. If set, the U value of the coordinates in that UV Map will be overwritten with the age of the particle attached to the matching mesh face (in proportion, from 0 for not yet born particles, to 1 for dead ones).

The V value is set to a constant 0.5 value.

You can for example vary the color of a fragment (face) during the explosion phase, by using a texture with a color gradient along its U axis.

Show faces when their attached particles are unborn.

Show faces when their attached particles are alive.

Show faces when their attached particles are dead.

Split the mesh in pieces based on location of emitted particles, instead of using existing faces. This will typically results in a splitting that appears more random.

Scale each face using the size of its attached particle, once that particle is alive.

Vertices in this group may not be affected by the

Explode modifier. Vertices with full weight are not affected at all, while vertices with less weight have a higher chance of being affected.

Vertices with null weight will be treated like those which do not belong to the group at all, and explode normally.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group.

Clean vertex group edges. Depending on the weights assigned to that vertex group, either completely protect those faces from being affected by the Explode modifier (which would happen if the faces had a weight value of 1), or completely remove protection from those faces (which would happen if the faces had a weight value of 0).

Refresh data in the Explode modifier.

This modifier uses the initial vertex weights. Modifiers that dynamically change weights will not influence the explosion as these values are only used once.

--- Fin du chapitre: _explode ---

--- Début du chapitre: _exposure --- The Exposure Node adjusts the brightness of an image using a camera exposure parameter.

See also

The exposure can also be adjusted in the scene Color Management . Standard color input.

Scalar factor to adjust the exposure of the image.

This node has no properties.

Standard color output.

In the example below, the Exposure node is used to increase the brightness of the window area using a mask.

Example of an Exposure node. \xc2\xba6

--- Fin du chapitre: _exposure ---

--- Début du chapitre: _extensions --- The Get Extensions section lets you install and manage extensions preferences.

Blender Preferences Extensions section. \xc2\xba6

See also

To learn about extensions and how to create them, refer to the Extensions page.

There are different ways to install an extension:

Drag the installation URL into Blender.

Search for the extension name and click on Install.

Use the drop-down menu in the top right, or drag-and-drop an extension .zip package into Blender.

Note

Any installed extension can be removed. This is a permanent change, though. To stop an extension temporarily, it is better to Disable it instead.

You need to manually check for available updates. Once an update is found, Blender will let you update any of the available extensions.

The current available version of an extension on the repository will always be considered the latest version.

Once an extension is installed it can be disabled (or re-enabled) as part of the user preferences. Some extension types do not support this, and will always be shown as enabled.

Tip

If the Add-on does not activate when enabled, check the Console window for any errors that may have occurred.

Opens extensions.blender.org <<https://extensions.blender.org/>> in a web browser.

Manually check the online repositories for available updates.
 Scan extension & legacy add-ons for changes to modules & meta-data(similar to restarting). Any issues are reported as warnings
 Update all the extensions that have an update available.
 Install an extension from a .zip package.This is installed to a Local Repository and no updates will be available.
 This can also be used to install legacy Add-ons, for more information see: Installing Legacy Add-ons .
 Or show only extensions of a single type:
 Only show add-ons.
 Only show themes.
 By default Blender has a Remote Repository pointing towards the Official Blender Extensions Platform and two Local Repositories.
 In the cases where more repositories are needed (e.g., to access third party extension platforms), new repositories can be added.
 Repositories. \xc2\xbb6
 To add new repositories click on the +icon:
 Add a repository from a URL.
 Add a repository which will be managed by the user (to be used with Install from Disk).
 To remove repositories click on the -icon:
 Remove an extension repository.
 Remove a repository and delete all associated files when removing.
 These changes are permanent and cannot be reversed.
 Remote repository with support for listing and updating extensions.
 Options:
 Allows Blender to check for updates upon launch.When updates are available a notification will be visible on the status bar.
 Personal access token, may be required by some repositories.
 A repository managed manually by the users.
 There are two types of local repositories. By default new local repositories are added as User repositories.This is what you want most of the time.
 After creating a repository they can be changed in the Advanced options to have a source System.These repositories are intended to bundle extensions with Blender, to make it portable.
 --- Fin du chapitre: _extensions ---

 --- D but du chapitre: _extension_arguments --- Command for managing Blender extensions.
 show this help message and exit
 subcommands:
 List all packages.
 Synchronize with remote repositories.
 Upgrade any outdated packages.
 Install packages.
 Install package from file.
 Remove packages.
 List repositories.
 Add repository.
 Remove repository.
 Build a package.
 Validate a package.
 Create a listing from all packages.

```

usage:
List packages from all enabled repositories.
show this help message and exit
Sync the remote directory before performing the action.
usage:
Download package information for remote repositories.
show this help message and exit
usage:
Download and update any outdated packages.
show this help message and exit
Sync the remote directory before performing the action.
usage:
The packages to operate on (separated by,without spaces).
show this help message and exit
Sync the remote directory before performing the action.
Enable the extension after installation.
Treat the user-preferences as read-only,preventing updates for
operations that would otherwise modify them.This means removing
extensions or repositories for example, wont update the user-preferences.
usage:
Install a package file into a user repository.
The packages file.
show this help message and exit
The repository identifier.
Enable the extension after installation.
Treat the user-preferences as read-only,preventing updates for
operations that would otherwise modify them.This means removing
extensions or repositories for example, wont update the user-preferences.
usage:
Disable & remove package(s).
The packages to operate on (separated by,without spaces).
show this help message and exit
Treat the user-preferences as read-only,preventing updates for
operations that would otherwise modify them.This means removing
extensions or repositories for example, wont update the user-preferences.
usage:
List all repositories stored in Blender's preferences.
show this help message and exit
usage:
Add a new local or remote repository.
The repository identifier.
show this help message and exit
The name to display in the interface (optional).
The directory where the repository stores local files (optional).When
omitted a directory in the users directory is automatically selected.
The URL, for remote repositories (optional).When omitted the repository
is considered local as it is not connected to an
external repository,where packages may be installed by file or managed
manually.
The access token to use for remote repositories which require a token.
The type of source in (USER,
SYSTEM).System repositories are managed outside
of Blender and are considered read-only.
Use package cache (default=1).

```

Clear all repositories before adding, simplifies test setup.

Treat the user-preferences as read-only, preventing updates for operations that would otherwise modify them. This means removing extensions or repositories for example, wont update the user-preferences.

usage:

Remove a repository.

The repository identifier.

show this help message and exit

Treat the user-preferences as read-only, preventing updates for operations that would otherwise modify them. This means removing extensions or repositories for example, wont update the user-preferences.

usage:

Build a package in the current directory.

show this help message and exit

The package source directory containing ablender_manifest.tomlmanifest.

Default\ue2\x80\x99s to the current directory.

The package output directory.

Default\ue2\x80\x99s to the current directory.

The package output filepath (should include a.zipextension).

Defaults to{id}-{version}.zipusing values from the manifest.

Reference a file path containing valid tags lists.

If you wish to reference custom tags a.jsonfile can be used. The contents must be a dictionary of lists where thekeymatches the extension type.

{"add-ons": ["Example", "Another"], "theme": ["Other", "Tags"]}

To disable validating tags, pass in an empty path--valid-tags="".

Build a separate package for each platform. Adding the platform as a file name suffix (before the extension).

This can be useful to reduce the upload size of packages that bundle largeplatform-specific modules (*.whlfiles).

Include verbose output.

usage:

Validate the package meta-data in the current directory.

The package source path (either directory containing package files or the package archive). This path must containing ablender_manifest.tomlmanifest.

Defaults to the current directory.

show this help message and exit

Reference a file path containing valid tags lists.

If you wish to reference custom tags a.jsonfile can be used. The contents must be a dictionary of lists where thekeymatches the extension type.

{"add-ons": ["Example", "Another"], "theme": ["Other", "Tags"]}

To disable validating tags, pass in an empty path--valid-tags="".

usage:

Generate a listing of all packages stored in a directory. This can be used to host packages which only requires static-file hosting.

show this help message and exit

The remote repository directory.

An optional server configuration to include information which can\ue2\x80\x99t be detected. Defaults toblender_repo.toml(in the repository directory).

This can be used to defined blocked extensions, for example

Create a HTML file (index.html) as well as the repository JSONto support browsing extensions online with static-hosting.

An optional HTML file path to override the default HTML template with your own.

The following keys will be replaced with generated contents:

`{body}` is replaced the extensions contents.

`{date}` is replaced the creation date.

--- Fin du chapitre: `_extension_arguments` ---

--- Début du chapitre: `_exterior` --- Exterior forces are applied to the vertices (and nearly exclusively to the vertices) of soft body objects. This is done using Newton's Laws of Physics:

If there is no force on a vertex, it stays either unmoved or moves with constant speed in a straight line.

The acceleration of a vertex depends on its mass and the force. The heavier the mass of a vertex the slower the acceleration. The larger the force the greater the acceleration.

For every action there is an equal and opposite reaction.

Well, this is done only in the range of computing accurateness, there is always a little damping to avoid overshoot of the calculation.

We will begin with a very simple example: the default cube.

To judge the effect of the external forces you should at first turn off the Goal, so that the vertices are not retracted to their original position.

Start playback to run the simulation.

What happens? The cube moves in negative Z direction. Each of its eight vertices is affected by a global, constant force the gravitation. Gravitation without friction is independent from the weight of an object, so each object you would use as a soft body here would fall with the same acceleration. The object does not deform, because every vertex moves with the same speed in the same direction.

Soft body vertices interact with all the Force Fields applied (usually to particles) in the layer, such as wind, force fields, and whatever physics field effect is on a common layer.

Reference

Physics Soft Body Field Weights

The Soft Body Field Weightspanel allows you to control how much influence each type of external force field has on the soft body system.

Limit effectors to a specified group. Only effectors in this group will have an effect on the current system.

Control how much the Global Gravity has an effect on the system.

Scale all of the effector weights.

Edges can be affected by wind as they move, and sail or flutter in a breeze. A simple aerodynamic model of a flag sailing in the wind.

This special exterior force is not applied to the vertices but to the connecting edges. Technically, a force perpendicular to the edge is applied. The force scales with the projection of the relative speed on the edge (dot product). Note that the force is the same if wind is blowing or if you drag the edge through the air with the same speed. That means that an edge moving in its own direction subject to no force, and an edge moving perpendicular to its own direction is subjected to maximum force. The angle and the relative speed between medium and edge is used to calculate the force on the edge. This force results that vertices with few connecting edges (front of a plane) fall faster than vertices with more connecting edges (middle of a plane). If all vertices have the same amount of edges in a direction they fall with equal speed.

The Aerodynamics settings are set in the Soft Body Edges panel.

A `Goal` is a shape that a soft body object tries to conform to. It acts like a pin on a chosen set of vertices, controlling how much of an effect soft body has on them.

Enabling Soft Body Goal tells Blender to use the position (or animated position) of a vertex in the simulation. Animating the vertices can be done in all the usual ways (F-Curves, armatures, parents, lattices, etc.) before the soft body simulation is applied. The `Goal` is the desired end position for vertices. How a soft body tries to achieve this goal can be defined using stiffness forces and damping.

See the Goal Settings for details.

The Goal Strength defines how much motion from an animation system gets applied.

A Goal value of 1.0 means no soft body simulation, the object act like any regular animated object (i.e. the vertex is kept at its original position). When setting Goal to 0.0 (or no goal), the vertex is only influenced by physical laws according to soft body simulation.

By setting goal values between 0.0 and 1.0, you can blend between having the object affected only by the animation system, and having the object affected only by the soft body effect.

Goal also serves as a memory, to make sure soft objects don't deform too much, ending up in the non-soft animated shape. Using the Vertex Group weight system, you can define a Goal weight per vertex. To make this look more natural, spring forces can be defined to control how far vertices can move from their original position.

Often Weight Paint is used to adjust the weight comfortably. For non-mesh objects the Weight parameter of their vertices/control points is used instead; Use the Context menu in Edit Mode or the Transform panel in the Sidebar region. The weight of Hair particles can also be painted in Particle Edit Mode.

In the Soft Body world, vertices of meshes are treated as particles having a mass. Their movement is determined by the forces affecting them. Beside other forces the individual particles can interact with another along edges using a physical model which is very close to shock absorbers used in cars. The working parts are:

A spring trying to keep the particles at a certain distance. How hard the spring tries to do that is controlled by the soft body parameter Stiffness.

A damping element to calm the movement down. The resistance the element builds up against motion is controlled by the soft body parameter Damping.

--- Fin du chapitre: `_exterior` ---

--- Début du chapitre: `_extrude` ---

Reference
Edit Mode

Armature `Extrude`

E, Shift- E

When you press E, for each selected tip (either explicitly or implicitly), a new bone is created. This bone will be the child of `tip owner`, and connected to it. As usual, once extrusion is done, only the new bones' tips are selected, and in select mode, so you can place them to your liking. See Fig. Extrusion example..

An armature with three selected tips. \xc2\xba

The three extruded bones. \xc2\xba

You also can use the rotating/scaling extrusions, as with meshes, by pressing respectively E and S \xe2\x80\x93 as well as locked extrusion along a global or local axis.

A single selected bone \xe2\x80\x93 tip. \xc2\xba

The two mirror-extruded bones. \xc2\xba

Bones have an extra \xe2\x80\x93 mirror extruding \xe2\x80\x93 tool, called by pressing Shift- E. By default, it behaves exactly like the standard extrusion. But once you have enabled X-Axis Mirror editing option, each extruded tip will produce two new bones, having the same name except for the \xe2\x80\x93_L \xe2\x80\x93 / \xe2\x80\x93_R \xe2\x80\x93 suffix (for left/right, see the naming conventions). The \xe2\x80\x93_L \xe2\x80\x93 bone behaves like the single one produced by the default extrusion \xe2\x80\x93 you can move, rotate or scale it exactly the same way. The \xe2\x80\x93_R \xe2\x80\x93 bone is its mirror counterpart (along the armature \xe2\x80\x93 local X axis), see Fig. Mirror extrusion example..

Important

Canceling the extrude action causes the newly created bones to snap back to the source position, (creating zero length bones). These will be removed when exiting Edit Mode, however, they can cause confusion and it \xe2\x80\x93 unlikely you want to keep them. If you realize the problem immediately, undo the extrude action.

In case you are wondering, you cannot just press X to solve this as you would in mesh editing, because extrusion selects the newly created tips, and as explained below the Delete tool ignores bones \xe2\x80\x93 joints. To get rid of these extruded bones without undoing, you would have to move the tips, then select the bones and delete them.

Reference

Edit Mode

Ctrl- RMB

If at least one bone is selected, Ctrl- RMB -clicking adds a new bone.

About the new bone \xe2\x80\x93 tip:

After you Ctrl- RMB -clicked it becomes the active element in the armature, it appears to be right where you clicked, but (as in mesh editing) it will be on the plane parallel to the view and passing through the 3D cursor.

The position of the root and the parenting of the new bone depends on the active element:

Ctrl-clicking when the active element is a bone. \xc2\xba

If the active element is a bone:

The new bone \xe2\x80\x93 root is placed on the active bone \xe2\x80\x93 tip.

The new bone is parented and connected to the active bone (check the Outliner in Fig. Ctrl-clicking when the active element is a tip.).

Ctrl-clicking when the active element is a tip. \xc2\xba

If the active element is a tip:

The new bone \xe2\x80\x93 root is placed on the active tip.

The new bone is parented and connected to the bone owning the active tip (check the Outliner in Fig. Ctrl-clicking when the active element is a tip.).

Ctrl-clicking when the active element is a disconnected root. \xc2\xba

If the active element is a disconnected root:

The new bone's root is placed on the active root.
The new bone is not parented to the bone owning the active root (check the Outliner in Fig. Ctrl-clicking when the active element is a disconnected root.).

And hence the new bone will not be connected to any bone.

Ctrl-clicking when the active element is a connected root. \xc2\xba

If the active element is a connected root:

The new bone's root is placed on the active root.

The new bone is parented and connected to the parent of the bone owning the active root (check the Outliner in Fig. Ctrl-clicking when the active element is a connected root.).

This should be obvious because if the active element is a connected root then the active element will be also the tip of the parent bone, so it is the same as the second case.

As the tip of the new bone becomes the active element, you can repeat these Ctrl- RMB clicks several times, to consecutively add several bones to the end of the same chain.

--- Fin du chapitre: _extrude ---

--- Début du chapitre: _extrude_cursor ---Reference

Edit Mode

Ctrl- RMB

Interactively places new vertices with Ctrl- RMB at the mouse cursor position.

The most basic element, a vertex, can be added with a Ctrl- RMB click when no other vertices are selected. Because the camera space (computer screen) is two-dimensional, Blender cannot determine all three vertex coordinates from a single mouse click, so the new vertex is placed at the depth of the 3D cursor.

To create interconnected vertices, you can add a vertex and continuously make subsequent Ctrl- RMB operations with the last vertex selected. This will link the last selected vertex with the vertex created at the mouse position with an edge (see Fig. Adding vertices one by one.), and will continuously create and connect new vertices if you continue repeating this operation.

Adding vertices one by one. \xc2\xba

Quad from an Edge with source automatically rotated. \xc2\xba

If you have two vertices selected and already connected with an edge, Ctrl- RMB click will create a planar face, also known as a quad. Blender will follow your mouse cursor and will use the planar view from your viewport to create those quads.

For Ctrl- RMB, Blender will automatically rotate the last selected Edge (the source) for the subsequent operations if you have at least one face created, dividing the angles created between the newly created edge and the last two edges, creating a smooth angle between them. Blender will calculate this angle using the last positive and negative position of the last X and Y coordinates and the last connected unselected edge. If this angle exceeds a negative limit (following a quadrant rule) between the recently created edge and the last two, Blender will wrap the faces. But if you do not want Blender to rotate and smooth edges automatically when extruding from Ctrl- RMB, you can also inhibit Blender from rotating sources using the shortcut Shift- Ctrl- RMB. In this case, Blender will not rotate the source dividing the angle between those edges when creating a face.

If you have three or more vertices selected, and Ctrl- RMB click, you will also create planar faces, but along the vertices selected, following the direction of the cursor. This operation is similar to an extrude operation.

Tip

When adding objects with Ctrl- RMB, the extrusions of the selected elements, being vertices, edges and faces with the Ctrl- RMB, are viewport dependent. This means, once you change your viewport, for example, from top to left, bottom or right, the extrusion direction will also follow your viewport and align the extrusions with your planar view.

--- Fin du chapitre: _extrude_cursor ---

--- Début du chapitre: _extrude_edges ---Reference

Edit Mode

Edge \xe2\x80\xa3 Extrude Edges Mesh \xe2\x80\xa3 Extrude \xe2\x80\xa3

Extrude Edges

E

Extrude edges as individual edges.

Edge selected. \xc2\xb6

Edge extruded. \xc2\xb6

--- Fin du chapitre: _extrude_edges ---

--- Début du chapitre: _extrude_faces ---Reference

Edit Mode

Face \xe2\x80\xa3 Extrude Faces, Mesh \xe2\x80\xa3 Extrude \xe2\x80\xa3

Extrude Faces

E

Extrude Faces duplicate faces, while keeping the new geometry connected with the original vertices.

Single vertex extruded. \xc2\xb6

Single edge extruded. \xc2\xb6

This tool is of paramount importance for creating new geometry. It allows you to create parallelepipeds from rectangles and cylinders from circles, as well as easily creating such things as tree limbs.

The axis on which faces are extruded along can be set interactively. Faces are extruded by default along their averaged normal. The extrusion can be limited to a single axis by specifying an axis; see Axis Locking.

The extrude tools differentiate in how the new geometry is connected in itself. Only the border loop gets extruded. The inner region of the selection gets moved unchanged with the extrusion.

Selected face. \xc2\xb6

During extrude. \xc2\xb6

Set to Z axis. \xc2\xb6

Only the normals of the new faces created from the extrusion will be flipped.

Removes and connects edges whose faces form a flat surface and intersect new edges.

Aligns the transformation axes to a specified orientation constraint. See Transform Orientations for more information.

The extruded face will affect nearby geometry. See Proportional Editing for a full reference.

Note

Even with the Proportional Sizeset to it's minimum, it will extrude the selected face as well as the new geometry and they will be layered on top of each other.

--- Fin du chapitre: _extrude_faces ---

--- Début du chapitre: _extrude_faces_normal ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Extrude Along Normals

Face \xe2\x80\xa3 Extrude Faces Along Normals, Mesh \xe2\x80\xa3 Extrude \xe2\x80\xa3 Extrude Faces Along Normals

E

Extrusion and offset will be locked in to only move along the local normals of the selected mesh.

Only the normals of the new faces created from the extrusion will be flipped.

Removes and connects edges whose faces form a flat surface and intersect new edges.

Amount to move geometry along the normals.

The length of the new edges will be uniform.

The extruded face will affect nearby geometry. See Proportional Editing for a full reference.

--- Fin du chapitre: _extrude_faces_normal ---

--- Début du chapitre: _extrude_individual_faces ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Extrude Individual

Face \xe2\x80\xa3 Extrude Individual Faces, Mesh \xe2\x80\xa3 Extrude \xe2\x80\xa3 Extrude Individual Faces

This tool allows you to extrude a selection of multiple faces as individuals, instead of as a region. The faces are extruded along their own normals, rather than their average. This has several consequences: first, \xe2\x80\x9cinternal\xe2\x80\x9d edges (i.e. edges between two selected faces) are no longer deleted (the original faces are).

Selection of multiple faces. \xc2\xb6

Extruded using extrude region. \xc2\xb6

Extruded using Extrude Individual. \xc2\xb6

--- Fin du chapitre: _extrude_individual_faces ---

--- Début du chapitre: _extrude_manifold ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Extrude Manifold

Mesh \xe2\x80\xa3 Extrude \xe2\x80\xa3 Extrude Manifold

This tool is very similar to Extrude Faces but enables Dissolve Orthogonal Edges by default. This causes the tool to automatically split and remove adjacent faces when extruding inwards.

Extrude Manifold Example. \xc2\xb6

--- Fin du chapitre: _extrude_manifold ---

--- Début du chapitre: _extrude_mesh --- The Extrude Mesh Node generates new edges or faces on the selected geometry elements and moves them by a certain offset.

The operations are similar to the extrude tools in mesh edit mode, though there are some differences. Most importantly, the node never keeps the

back-faces of the extrusion in place, they are always removed. Attribute propagation rules may also be different.

Standard geometry input.

A boolean field indicating which elements should be extruded.

The translation vector for each extruded element. By default, this is the element's normal.

Tip

If all the elements are extruded in the same direction, you may be able to improve performance by connecting a Vector Node to this input, thereby skipping the normal calculation.

Scaling factor for the above translation vector.

Whether to extrude each face individually rather than extruding connected groups of faces together.

Attaches a new freestanding edge to each selected vertex.

Attaches a new quad face to each selected edges. Vertices shared by the original selected edges are also shared in the duplicated edges.

Note

Depending on the situation, the normals of the new faces may be arbitrary. If the selected edges each have only one connected face, then the node can pick a consistent orientation for the new faces, but if there is more than one connected face, or no connected faces, the normals may have to be adjusted afterwards.

Extrudes contiguous regions of selected faces, or each selected face individually, depending on the Individual boolean input.

When the Individual input is false, the node will find regions of connected faces and generate new faces on the boundaries of those regions. Any vertices, edges or faces on the inside of the regions simply are moved, not duplicated. If the whole mesh is selected and it is already a Manifold shape, then the result will just be that the whole mesh gets resized.

Standard geometry output.

A boolean field indicating the elements in the extrusion. In Vertex mode, these are the new vertices; in Edge mode, the new edges; and in Face mode, the moved faces.

A boolean field indicating the elements in the extrusion. In Vertex mode, these are the new edges; in Edge mode, the new faces; and in Face mode, too, the newly generated faces (as opposed to the moved ones).

Here, the selection outputs are used to set materials on certain faces of the mesh. A Random Value Node is used to limit the extrusion to a random set of faces.

Attributes are transferred to the new elements with specific rules. An attribute will never change domains on the resulting mesh. The id attribute does not have any special handling.

Generally boolean attributes are propagated with

connected, meaning any

connected value that is mixed in for other types will cause the new value to be as well.

The following sections describe:

The new edges created in vertex mode use the average value of all connected edges.

New vertices have copied values from their original vertices.

New edges have the average value of any connected original edges. For boolean attributes, edges are selected if any connected edges were selected.

Attribute propagation for new connecting edges (the vertical yellow edge). The final value is a mix of the values from the two middle blue edges. The darker maroon edges lower on the image are not used.

New vertices have copied values from their original vertices.

Vertical connecting edges get the average value from any connected extruded edges. For booleans, the edges are selected if any connected extruded edges were selected. (Propagation rules are shown in the figure above.)

Horizontal duplicate edges have copied values from their original edges.

New faces get the average values of all faces connected to the selected edge. For booleans, faces are selected if any connected original faces were selected.

New face corners get the averaged value of corresponding corners in all faces connected to selected edges. For booleans, corners are selected if one of those corners are selected.

Attribute propagation for new connecting edges (the vertical yellow edge). The final value is a mix of the values from the two middle blue edges. The values from the darker maroon edges between unselected faces and on top of the extruded region are not used.

New vertices have copied values from their original vertices.

Vertical connecting edges get the average value from any connected extruded edges, not including the edges on top of extruded regions. For booleans, the edges are selected if any of those connected edges were selected. (Propagation rules are shown in the figure above.)

Horizontal duplicate edges have copied values from their original edges.

New faces have copied values from the corresponding extruded faces.

New face corners have copied values from the corresponding corresponding corners of extruded faces.

Attribute propagation for new connecting edge. Each edge uses the average values of the two neighboring edges on its extruded face.

New vertices have copied values from their original vertices.

Vertical connecting edges get the average value of the two neighboring edges on each extruded face. For booleans, the edges are selected when at least one neighbor on the extruded face was selected.

Horizontal duplicate edges have copied values from their original edges.

New side faces have copied values from their corresponding selected face.

New face corners have copied values from the corresponding corners of selected faces.

--- Fin du chapitre: _extrude_mesh ---

--- Début du chapitre: _extrude_region ---Reference

Edit Mode

Toolbar \Extrude Region

E

Extrusion tools duplicate vertices, while keeping the new geometry connected with the original vertices. Vertices are turned into edges and edges will form faces.

Single vertex extruded.

Single edge extruded.

This tool is of paramount importance for creating new geometry. It allows you to create parallelepipeds from rectangles and cylinders from circles, as well as easily creating such things as tree limbs.

The axis on which vertices and edges are extruded along can be set interactively. Faces are extruded by default along their averaged normal. The extrusion can be limited to a single axis by specifying an axis; see Axis Locking.

The extrude tools differentiate in how the new geometry is connected in itself.

Only the border loop gets extruded. The inner region of the selection gets moved unchanged with the extrusion.

Selected face. \xc2\xbf

During extrude. \xc2\xbf

Set to Z axis. \xc2\xbf

Although the process is quite intuitive, the principles behind Extrude are fairly elaborate as discussed below:

First, the algorithm determines the outside edge loop of the extrude; that is, which among the selected edges will be changed into faces. By default (see below), the algorithm considers edges belonging to two or more selected faces as internal, and hence not part of the loop.

The edges in the edge loop are then changed into faces.

If the edges in the edge loop belong to only one face in the complete mesh, then all of the selected faces are duplicated and linked to the newly created faces. For example, rectangles will result in parallelepipeds during this stage.

In other cases, the selected faces are linked to the newly created faces but not duplicated. This prevents undesired faces from being retained inside the resulting mesh. This distinction is extremely important since it ensures the construction of consistently coherent, closed volumes at all times when using Extrude.

When extruding completely closed volumes (like e.g. a cube with all its six faces), extrusion results merely in a duplication, as the volume is duplicated, without any link to the original one.

Edges not belonging to selected faces, which form an open edge loop, are duplicated and a new face is created between the new edge and the original one.

Single selected vertices which do not belong to selected edges are duplicated and a new edge is created between the two.

--- Fin du chapitre: _extrude_region ---

--- Début du chapitre: _extrude_vertices ---Reference

Edit Mode

Vertex \xe2\x80\xa3 Extrude Vertices, Mesh \xe2\x80\xa3 Extrude
\xe2\x80\xa3 Extrude Vertices

E

Extrude vertices as individual vertices.

Vertex selected. \xc2\xbf

Vertices extrude. \xc2\xbf

--- Fin du chapitre: _extrude_vertices ---

--- Début du chapitre: _eyedropper ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Eyedropper

The Eyedropper tool is used to create materials or palette color based on sampled colors in the 3D Viewport.
Create a new material with the Stroke Base Color to be the sampled color.
The color transformation will be applied on the stroke and/or the fill color.

Only paint over strokes.

Only paint over fill areas.

Paint over strokes and fill areas

Add a new color to the color palette based on the sampled color.

Sets the brush color to the sampled color.

LMB Create a stroke material.

Shift- LMB Create a fill material.

Shift- Ctrl- LMB Create both a stroke and fill material.

Holding LMB and dragging accumulates the average color under the mouse cursor.

--- Fin du chapitre: _eyedropper ---

--- Début du chapitre: _face --- These rig types implement components of a modular face.

Generates a simple tongue, extracted from the original PitchiPoy super_face rig.

Defines the number of b-bone segments each tweak control will be split into.

Optionally specifies bone collections for the main control.

Implements a skin system parent controller that manages two skin chains for the top and bottom eyelids in addition to generating the eye rotation mechanism.

The rig must have two child skin chains with names tagged with .T and .B symmetry to mark the top and bottom eyelid, which are connected at their ends forming eye corners. The chains are rigged to follow the surface of the eye and twist to its normal.

In addition, it creates target controls for aiming the eye, including a master control shared by all eyes under the same parent rig. The eyelids are rigged to follow the movement of the eyeball with adjustable influence.

Generates deform bones for the eyeball and the iris, the latter copying XZ scale from the eye target control. The iris is located at the tail of the ORG bone.

Generates a slider to disable the mechanism that keeps eyelid controls stuck to the surface of the eye.

Generates two separate sliders for controlling the influence of the eye rotation on X and Z eyelid motion.

Depending on Split Eyelid Follow Slider, specifies the default values for the split follow sliders, or fixed factors to be multiplied with the single common follow influence slider value.

Implements a skin system parent controller that manages one or more loops of mouth skin chains in response to the movement of jaw and mouth controls.

The rig must have one or more child chain loops, each formed by four skin chains tagged with .T / .B and .L / .R symmetrical names.

The lip loops are sorted into layers based on the distance from corners to the common center and rigged with blended influence of the jaw and the master mouth control. Other child rigs become children of the jaw.

Specifies the influence of the jaw on the inner bottom lip with mouth lock disabled.

Specifies the influence of the jaw on both lips of locked mouth.

Specifies the factor by which influence fades away with each successive lip loop (for bottom lip loops the blend moves away from inner bottom lip to full jaw influence).

--- Fin du chapitre: `_face` ---

--- Début du chapitre: `_faces` --- Will create a face system based on the bones child to the parent that has the property set on it.

Requirement: All the face bones bundled in the `faces.super_face` sample had to be present and child of the master bone that has the `Rigify-type face` property set.

Note

This rig type is being deprecated in favor of a new modular skin and face rigging system.

--- Fin du chapitre: `_faces` ---

--- Début du chapitre: `_face_area` --- The Face Area node outputs the surface area of a mesh's faces. The units are in Blender units no matter the unit system, equivalent to meters-squared at the default unit scale.

Note

For quads and N-gons, when the face's vertices are not planar, the output is not necessarily the same as the sum of every one of the face's triangles visible in the viewport. In this case it should only be used as an approximation. In some cases, the Triangulate Node can be used to get an exact value.

This node has no inputs.

This node has no properties.

The surface area of each of the mesh's faces.

Combined with the Attribute Statistic Node, this node can be used to calculate the total surface area of a mesh.

--- Fin du chapitre: `_face_area` ---

--- Début du chapitre: `_face_data` --- Reference

Edit Mode

Face Data Rotate Colors

Rotates the Color Attribute's colors inside faces either clockwise or counterclockwise.

Reference

Edit Mode

Face Data Reverse Colors

Flips the direction of Color Attribute's colors inside the selected faces.

Reference

Edit Mode

Face Data Rotate UVs

See Rotate UVs .

Reference

Edit Mode

Face Data Reverse UVs

See Reverse UVs .

Reference

Edit Mode
Face \xe2\x80\xa3 Face Data \xe2\x80\xa3 Flip Quad Tessellation
Internally, all quads are Tessellated into 2 triangles, this operator swaps which way the quad is split into triangles.

Reference

Edit Mode

Face \xe2\x80\xa3 Face Data \xe2\x80\xa3 Mark Freestyle Face
Todo.

Reference

Edit Mode

Face \xe2\x80\xa3 Face Data \xe2\x80\xa3 Clear Freestyle Face
Todo.

--- Fin du chapitre: _face_data ---

--- Début du chapitre: _face_group_boundaries --- The Face Group Boundaries Node finds the edges which lie on the boundaries of specified regions. These edges could be used to mark seams for UV unwrapping, for example.

Identifier for which group of faces this face belongs to. All contiguous faces with the same value are in the same region.

This node has no properties.

Selection of the boundary edges of the different face sets. An edge is considered to be at the boundary if it lies on at least two faces with different identifiers.

Combined with the UV Unwrap Node, this node is used to turn the face sets (right cube) into a UV map for a texture (left cube).

--- Fin du chapitre: _face_group_boundaries ---

--- Début du chapitre: _face_is_planar --- The Is Face Planar node outputs whether every triangle of a quad set is on the same plane as all of the others, in other words, if they have the same normal.

For example, a non-planar face can be created by moving a single vertex in a face but not the others. Triangles will always be planar.

The distance a point can be from the surface before the face is no longer considered planar.

This node has no properties.

Whether each mesh face is planar.

Combined with the Set Material Node, this node is used to visualize all non-planar faces in a mesh.

--- Fin du chapitre: _face_is_planar ---

--- Début du chapitre: _face_neighbors --- The Face Neighbors node outputs topology information relating to each face of a mesh.

This node has no inputs.

This node has no properties.

This output is simply the number of sides of each face, or how many corners each face has.

The number of faces that connect to this face with at least one edge. On a regular manifold mesh with only quads and triangles, this will be the same as the vertex count, otherwise it might be completely different.

--- Fin du chapitre: _face_neighbors ---

--- Début du chapitre: _face_of_corner --- Retrieves the face that a face corner is part of.

The geometry-wide index of the corner.

Note

If this input is not connected, it uses the index of the context item, which means it's important that the node is evaluated in the Face Corner domain.

This node has no properties.

The geometry-wide index of the face which the corner belongs to.

The face-local index of the corner. This is 0 for the first corner of the face, 1 for the next corner, and so on up to (number of corners - 1) for the last corner.

--- Fin du chapitre: `_face_of_corner` ---

--- Début du chapitre: `_face_set` --- The Face Set Node outputs which face set a face is in, and whether or not face sets exist in the mesh at all.

The corresponding data flow node is the Set Face Set Node.

Note

This node can only be used in the Tool context .

This node has no inputs.

This node has no properties.

Integer indicating which face set a face is in, or 0 when the mesh does not have face sets. When evaluated in the edge or point domain, outputs an interpolated value based on the connected faces.

Boolean value that indicates whether the element's mesh has face sets.

--- Fin du chapitre: `_face_set` ---

--- Début du chapitre: `_face_sets` --- This page details the face set related hotkey operators and menu operators in sculpt mode.

Tip

There is a face set pie menu that can be accessed with Alt- W .

Reference

Sculpt Mode

Face Sets \xe2\x80\xa3 Face Set from Masked

Creates a new face set from Masked Geometry .

Reference

Sculpt Mode

Face Sets \xe2\x80\xa3 Face Set from Visible

Creates a new face set from all visible geometry.

Reference

Sculpt Mode

Face Sets \xe2\x80\xa3 Face Set from Edit Mode Selection

Creates a new face set corresponding to the Edit Mode face selection.

Reference

Sculpt Mode

Face Sets \xe2\x80\xa3 Initialize Face Sets

Initializes all face sets on the mesh at once based off one of several mesh attribute properties.

The mesh data attribute used to define the boundaries for the face sets.

Creates a new face set per discontinuous part of the mesh.

Creates a face set for each isolated face set. This mode is useful for splitting the patterns created by Face Set Expand into individual Face Sets for further editing.

Creates a face set per Material Slot .

Creates face sets for Faces that have similar Normals .

Creates face sets using UV Seams as boundaries.
 Creates face sets using Edge Creases as boundaries.
 Creates face sets using Bevel Weights as boundaries.
 Creates face sets using Sharp Edges as boundaries.
 The minimum value to consider a certain attribute a boundary when creating the face sets.

Reference
 Sculpt Mode
 Face Sets \xe2\x80\xa3 Grow/Shrink Face Sets

Edit Face Set
 Ctrl- W, Ctrl- Alt- W
 Expands or contracts the face set under the cursor by adding or removing surrounding faces.

Note
 More info on Face Set Expand at the Expand page.

Reference
 Sculpt Mode
 Face Sets \xe2\x80\xa3 Extract Face Set

Creates a new mesh based on the selected face set. Once the operator is initiated, hover over the face set and LMB to create the new mesh. After the operator is finished the new mesh will be selected in Object Mode.

Reference
 Sculpt Mode
 Face Sets \xe2\x80\xa3 Randomize Colors

Generates a new set of random colors to render the face sets in the 3D Viewport.

Reference
 Sculpt Mode
 Viewport Overlays \xe2\x80\x93 Sculpt \xe2\x80\xa3 Face Sets

The face sets display can be toggled as a viewport overlay . In the overlay popover, the opacity of the face sets overlay can be adjusted to make it more or less visible on the mesh.

--- Fin du chapitre: _face_sets ---

--- Début du chapitre: _face_set_tools ---

Reference
 Sculpt Mode

Face Set gesture tools apply a single new Face Set to all faces within the selection area.

All Face Set gesture tools can be activated in the Toolbar and are comprised of the following:

Reference
 Sculpt Mode
 Toolbar \xe2\x80\xa3 Box Face Set

Creates a new Face Set based on a box gesture .

Reference
 Sculpt Mode
 Toolbar \xe2\x80\xa3 Lasso Face Set

Creates a new Face Set based on a lasso gesture .

Reference
 Sculpt Mode
 Toolbar \xe2\x80\xa3 Line Face Set

Creates a new Face Set based on a line gesture .

Reference
 Sculpt Mode

Toolbar \xe2\x80\xa3 Polyline Face Set
Creates a new Face Set based on a polyline gesture .
Only creates a face set on the faces that face towards the view.
--- Fin du chapitre: _face_set_tools ---

--- Début du chapitre: _falloff --- The Falloff allows you to control the Strengthfalloff of the brush.The falloff is mapped from the center of the brush (left part of the curve)towards its borders (right part of the curve).Changing the shape of the curve will make the brush softer or harder.Read more about using theCurve Widget.

Brush curve example.\xc2\xbb

You can choose how the strength of the falloff is determined from the center of the brushto the borders by manually manipulating the control points within the curve widget.There are also a couple of preset custom curves displayed at the bottom of the curve widgetthat can be used on their own or as a starting point for tweaking.

Smooth. \xc2\xbb

Sphere. \xc2\xbb

Root. \xc2\xbb

Sharp. \xc2\xbb

Linear. \xc2\xbb

Constant. \xc2\xbb

The center strength, the border strength, and the falloff transition between them are evenly distributed.

Similar to Smoothbut produces a wider center point of the brush before tapering off.

The strength of the brush is predominately at its strongest pointwith a steep falloff near the border of the brush.

Similar to a Sphere but the center is a more concentrated point.

The center of the brush is the strongest pointthen exponentially tapers off to a lower strength, creating a fine point.

With the center being the strongest,the strength will consistently weaken as it reaches the border of the brush.

Similar to Sharpbut the center point is more condensed.

A hybrid between Smoothand Sphere.

The strength of the brush remains unified across the entire brush.This will create a sharp edge at the border of the brush.

(From Left to Right) Smooth, Smoother, Sphere, Root,Sharp, Linear, Sharper, Inverse square, Constant.\xc2\xbb

Use projected or spherical falloff.Note, this is not supported in Texture Paint Mode.

Applies brushes influence in a sphere, outwards from the center.

This turns the brush influence into a cylinder (the depth along the view is ignored) instead of a sphere.It can be used along the outline of a mesh to adjust its silhouette.

As faces point away from the view the brush strokes fade away to prevent harsh edges.

If disabled, the normal of the surface has no effect on the falloff.

The angle at which the falloff begins.

--- Fin du chapitre: _falloff ---

--- Début du chapitre: _features ---Reference
Render

Experimental features are disabled / hidden by default, but can be enabled by setting Feature Set to Experimental in the Render properties. Enabling the Experimental Feature Set will use experimental and incomplete features that might be broken or change in the future.

Adaptive subdivision is currently the only experimental feature.

--- Fin du chapitre: `_features` ---

--- Début du chapitre: `_feature_sets` --- Rigify allows third party developers to implement sub-addons, called Feature Sets, which can provide new Meta-Rigs and Rig Types. Similar to regular add-ons, they can be installed from zip-files through Rigify settings.

These are some examples of Feature Sets currently provided by past and current Rigify developers:

This feature set provides the original Rigify rigs by Nathan Vegdahl, minimally ported and repackaged to work without switching Rigify to legacy mode. Note that their names were changed, so meta-rigs designed for legacy mode aren't directly compatible.

Rig experiments, some of which might be included in Rigify in the future. Examples include limbs with an extra IK system based at knee/elbow, a spline based tentacle, and more.

You can install these packages by clicking Code > Download ZIP, and then install the downloaded file through Rigify settings.

Developer documentation is available on the Blender Developer Documentation.

--- Fin du chapitre: `_feature_sets` ---

--- Début du chapitre: `_fields` --- Fundamentally, a field is a function: a set of instructions that can transform an arbitrary number of inputs into a single output. A field's result can then be calculated many times with different input data. They are used all over geometry nodes to allow calculations that have different results for every element (mesh vertices, faces, etc.).

A field input to a node.

For example, in the figure above, the field connected to the `Set Position` node depends on two inputs, `Position` and `Index`, and transforms them into a vector using a single instruction.

Socket shapes are used to convey which sockets are fields and which are regular data. There are three possible socket shapes, each visualizing its field status:

The socket requires a single real value, it cannot accept a field input. For output sockets, this means the node always outputs a single value.

The socket can accept a field input, or it outputs a field. A constant single value can be connected to these sockets, but then the output will often not vary per element.

The socket can be a field, but it is currently a single value. This is helpful because it allows tracking where single values are calculated, instead of a field with many different results. It also means that Socket Inspection will show the value instead of field input names.

The socket shape is a diamond with a dot, meaning the field has the same value for every element. Every point will be moved up by 5 m.

The socket shape is a diamond and the field input now has a varying input. In other words, the value can be different for every element. In

this case, the position will be doubled, since the offset for every point is the point's position.

Tip

Often it is desired to extract a single value from a field. While it doesn't make sense conceptually to simply change a field into a single value, the Sample Index Node or the Attribute Statistic Node can be used to retrieve a single value from a field evaluated on a geometry.

When a connection is made between two node sockets that support fields the node connection will be drawn as a dashed line. If you make the mistake of connecting a non-field socket to a field socket, the connection will be drawn as a solid red line indicating that there is an error.

Nodes can be separated into two categories: data flow nodes that usually pass geometry, and field nodes that operate on data per-element. Field nodes can be input nodes that bring geometry data into the node tree, or function nodes that operate on that data.

Nodes with a geometry input and a geometry output will almost always be data flow nodes. Meaning that they actually change geometry data that will be outputted from the Geometry Nodes modifier.

Nodes with diamond socket inputs and outputs are field nodes, and resemble the instructions that will be evaluated by data flow nodes. Examples of function nodes are the math nodes and also more complex nodes like the Geometry Proximity Node.

Input nodes provide data to the field evaluation process. By themselves, they mean nothing; they must be evaluated within the context of a data flow node (geometry) to actually output a value. Examples of input nodes are the built-in attribute input nodes like Position and ID, but also selection nodes like Endpoint Selection.

Field inputs may also come from other nodes that process geometry like the Distribute Points on Faces, in the form of Anonymous Attributes. All field nodes work in the context of the data flow node they are connected to. The context usually consists of a geometry component type and an attribute domain, so it determines what data is retrieved from input nodes.

One common misunderstanding is that the same field node tree used in multiple places will output the same data. This is not necessarily true, because the field node tree will be evaluated for every data flow node, potentially retrieving data from a different or changed geometry. Here, the Set Position node's input field is evaluated once. To evaluate the field, the node traverses backwards to retrieve the inputs from the field input nodes.

When a second Set Position node is added, the same field node tree is evaluated twice, once for each data flow node. At the second Set Position node, the results will be different since its geometry input will already have the changed position from the first node.

However, often it's necessary to use the same field values even after changing the geometry. The Capture Attribute Node evaluates a field, copying the result to an anonymous attribute on the geometry.

Here, a Capture Attribute node stores a copy of the initial position. Notice that evaluating the field input of the Capture Attribute node is an entirely different step. Later on, the input fields to the Set Position nodes don't use the actual position, but the anonymous attribute copy of it.

--- Fin du chapitre: _fields ---

--- Début du chapitre: `_field_weights` ---Reference

Properties \xe2\x80\xa3 Physics \xe2\x80\xa3 Fluid \xe2\x80\xa3 Field Weights

Domain

These settings determine how much gravity andForce Fieldsaffect the fluid.

When set, fluid can only be influenced by force fields in the specified collection.

How much the fluid is affected by Gravity.

Overall influence of all force fields.

The other settings determine how much influence individual force field types have.

--- Fin du chapitre: `_field_weights` ---

--- Début du chapitre: `_file_browser` --- The File Browser is used in all file-related operations. These include:

Opening and saving blend-files.

Browsing the content of other blend-files when appending or linking data-blocks(seeLinked Libraries).

Importing from/exporting to other file formats.

Updating the locations of previously imported media (images, videos, fonts\xe2\x80\xa6).

The most common way to use this editor is through modal operators (like opening or saving a blend-file).The File Browser will appear in a new window, wait for you to select a file, and then close again.

You can also use the File Browser like a regular, permanently visible editor. In fact,the predefined Video Editingworkspaceuses it this way.This lets you drag-and-drop media from the browser straight into e.g. the3D Viewportor theVideo Sequencer, saving you some overhead.

The File Browser.\xc2\xb6

The main region lists files, folders, or blend-file contents.Hovering over an item will show a tooltip with extra information.

In its Thumbnaildisplay mode, the File Browser supports many types of previews. These include:

Image and video formats

Fonts

Blend-files

Internal Data-blocks

In order to get previews for data-blocks, these must first be generated.See Blend-Files Previews .

The File Browser in Thumbnailmode. \xc2\xb6

Above the file list, there\xe2\x80\x99s a textbox showing the current folder path, along with buttons for navigating.

Move to previous folder in navigation history.

Move to next folder in navigation history.

Move up to parent directory.

Refresh current folder.

Create a new directory inside the current one.

The current folder path. Tabwill auto-complete an existing path.If you type a nonexistent path, you will be prompted to create it.

Filter items by name.The wildcard*will match anything, e.g.bl*erwill match bothblenderandblogger.There is always an implicit wildcard at the start and end of the search text,soblenderwill also

matchtest_blender_file.blend. This field can also be used to filter some specific file extension (e.g. .png will list all PNG files).

Control how files are displayed.

Displays files and folders in a vertical list.

Displays files and folders in a horizontal list.

Shows previews.

The size of the thumbnails.

The number of directory levels to show at once in a flat way.

List only the current directory's content.

List the whole content of a blend-file (only available when linking or appending data-blocks).

List all subdirectories' content, one level of recursion.

List all subdirectories' content, two levels of recursion.

List all subdirectories' content, three levels of recursion.

Hint

Showing several levels of directories at once can be handy to e.g. see your whole collection of textures, even if you have arranged them in a nice set of directories to avoid having hundreds of files in a single place.

In the Append/Link case, showing the content of the whole blend-file lets you link different types of data-blocks in a single operation.

Warning

The more levels you show at once, the more time it will take to list them all.

Sorts items by one of the four methods:

Sort the file list alphabetically.

Sort the file list by extension/type.

Sort files by modification time.

Sort files by size.

The toggle with the funnel icon controls whether filtering is enabled or not. The dropdown button next to it shows the filtering options.

Filters files by categories, like folders, blend-files, images, etc.

When appending or linking, you can also filter by data-block categories, like scenes, animations, materials, etc.

Shows hidden files (starting with a .).

These controls are at the bottom of the editor.

Text field to edit the file name and extension. Turns red to warn you about overwriting an existing file. Tab will auto-complete to existing names in the current directory.

Adds/increases or removes/decreases a trailing number in your file name (used e.g. to store different versions of a file).

Closes the File Browser and cancels the operation.

Confirm the current directory and file name. You can also double-click a file or data-block in the main region.

The region on the left contains a few panels that let you quickly jump to certain directories with a single click.

A custom list of folders that you use often. You can use the buttons to the right of the list to add/remove/move items.

Common directories such as the home directory in Linux or the %Documents% folder in Windows.

Drives and network mounts.

Recently accessed folders.

Clicking the down arrow button to the right reveals Clear Recent Items to fully clear this list.

You can control how many folders appear in this list with the Recent Files number field of the Save & Load tab in the Preferences.

The right region shows the options of the calling operator. Besides the common actions listed below, many import/export add-ons will also expose their options there.

See Opening & Saving.

See Supported Graphics Formats.

See Linked libraries.

For the common option:

See Relative Paths.

The header only contains two menus, one with the standard editor View controls and the other to list a few Selecting operators for the sake of discoverability. These menus are not visible when the browser is in a modal window.

Double-click a directory to enter it.

Takes you up one level of directory.

You can also drag and drop a file or directory from your file manager into the Blender File Browser. This will navigate to the item and select it.

Click LMB to select a single item. Additionally hold Ctrl to add/remove that item to/from the selection, or Shift to select a range of items.

Dragging with LMB starts a box selection .

Note

You can always select several entries in the File Browser. The last selected one is considered the active one. If the calling operation expects a single path (like e.g. the main blend-file Open one), it will get that active item's path, and the other selected items will be ignored.

It is also possible to select/deselect files by

walking through them using the arrow keys:

Press an arrow key to select the next/previous file in the list and deselect all the others.

Hold Shift to keep the current selection (and add to it).

Hold Shift-Ctrl to invert the selection as you pass over it.

If no file is selected, the arrow key navigation selects the first or last file in the directory, depending on the arrow direction.

The following operations are available in the file list's context menu.

Use the operating system to perform an action on the file or directory. The options listed below might not be available on all operating systems.

Open the file.

Open the folder.

Edit the file.

Create a new file of this type.

Search for files of this type.

Show this file.

Play this file.

Browse this file.

Preview this file.

Print this file.

Install this file.

Run as specific user.

Show OS Properties for this item.

Search for items in this folder.

Open a command prompt here.

Delete the currently selected files or directories by moving them to the operating system's trash.

Note, on Linux deleting directories requires KDE or GNOME.

Change the name of the currently selected file or directory.

--- Fin du chapitre: `_file_browser` ---

--- Début du chapitre: `_file_output` --- This node writes out an image, for each frame range specified, to the filename entered, as part of a frameset sequence.

This node can be used as a way to automatically save the image after a render; In addition, since this node can be hooked in anywhere in the node tree, it can also save intermediate images automatically.

The image(s) will be saved on rendering, writing to the current frame. An entire sequence of images will be saved, when an animation is rendered.

Unlike the render output filepath, this node uses a base directory and an image name, by default the output path is composed of:

`{basepath}/{filename}{framenum}. {extension}` .

Besides being split into two settings, in all other respects, this setting is treated the same as the render output path .

Label that shows the selected file format.

Note

More options can be set in the Sidebar region.

This node has no output sockets.

--- Fin du chapitre: `_file_output` ---

--- Début du chapitre: `_file_paths` --- The File section in Preferences allows you to configure auto-save preferences and set default file paths for blend-files, rendered images, and more.

Locations for various external files can be set for the following options:

Preferences File Paths section.

Hint

The default path `//` refers to the folder of the currently open blend-file (see Relative Paths for details).

Default location to browse for text object font files.

Default location to browse for image textures.

Default location to browse for sound files.

The directory for storing temporary save files. The path must reference an existing directory or it will be ignored and the system's temporary directory will be used instead. When left blank, the system's temporary directory will be used (see Temporary Directory for details).

Where rendered images/videos are saved.

The location where cached render images are stored.

Name and on-disk directory paths of asset libraries. To make Blender aware of an asset library, add it to this list. The name is for your reference only, and will appear in asset library selectors. The path should point to the location of the asset library.

Name and Location of asset libraries in the Preferences.

To create a new asset library, just create an empty directory and add it to the List View. Any asset from any blend-file contained in that directory (or subdirectories thereof) will appear in the Asset Browser .

Determines how data is managed when an asset is imported, unless overridden by the Asset Browser .

Same as File \xe2\x80\xa3 Link\xe2\x80\xa6

The asset will be linked to the current blend-file, and thus be read-only. Later changes to the asset file will be reflected in all files that link it in.

Same as File \xe2\x80\xa3 Append\xe2\x80\xa6

All of the asset and all its dependencies will be appended to the current file. Dragging a material into the scene three times will result in three independent copies. Dragging an object into the scene three times will also result in three independent copies.

\xe2\x80\x9cDependencies\xe2\x80\x9d in this case means everything the asset refers to. For an object, this can be its mesh and materials, but also other objects used by modifiers, constraints, or drivers. Since the file now has its own copy of the asset, later changes to the asset file will not be reflected in the file it\xe2\x80\x99s appended to.

Specific to the Asset Browser.

The first time an asset is used, it will be appended, including its dependencies, just like described previously. However, Blender will keep track of where it originated, and the next time the asset is used, as much data as possible will be reused. Dragging a material into the scene three times will only load it once, and just assign the same material three times. Dragging an object into the scene three times will create three copies of the object, but all copies will share their mesh data, materials, etc.

Since the file now has its own copy of the asset, later changes to the asset file will not be reflected in the file it\xe2\x80\x99s appended to. Use relative path when linking assets from this asset library.

Additional locations to search for Python scripts.

Each path can be given a Name to signify to purpose of that script directory.

By default, Blender looks in several directories (platform dependent) for scripts. By adding a user script path in the preferences an additional directory is used. This can be used to store your own scripts and add-ons independently of the current Blender version.

You will need to create specific subfolders in this path which match the structure of the scripts folder found in Blender\xe2\x80\x99s installation directory.

The following subdirectories will be used when present:

Modules in this folder will be imported on startup.

Legacy add-ons located here will be listed in the add-ons preferences.

Modules in this folder can be imported by other scripts.

Presets in this folder will be added to existing presets.

Hint

For add-ons it is now recommended to use a local extension repository if you wish to define additional locations to install and manage them.

To make use of these you will need to define them as extensions.

Note

You have to restart Blender for all changes to the users scripts to take effect.

The path to an external program to use for image editing.

The program used for playing back rendered animations via View Animation

.

By default this is set to Internal which uses Blender's built-in animation player .

This has the advantage that all image formats supported by Blender can be played back and no 3rd party application needs to be installed.

Command to launch the text editor when using Edit Externally, either a full path or a command in \$PATH. Use the internal editor when left blank.

Defines the specific format of the arguments with which the text editor opens files.

The supported expansions are as follows:

\$filepath: The absolute path of the file.

\$line: The line to open at (Optional).

\$column: The column to open from the beginning of the line (Optional).

\$line0&\$column0 similar to the above but they start at zero.

Example: -f \$filepath -l \$line -c \$column

Only visible when Developer Extras are enabled.

The path to the /branches directory of your local SVN translation copy, to allow translating from the UI.

Be sure that you have the right privileges for running the executable accessing the path defined. On Windows for instance, if the option \xe2\x80\x9cRun this program as an administrator\xe2\x80\x9d is enabled for the executable, it will lead to a failure to open the editor due to a limitation within the OS User Account Control. Running a program with elevated privileges is potentially dangerous!

--- Fin du chapitre: _file_paths ---

--- Début du chapitre: _fill ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Fill

The Fill tool is used to automatically fill closed strokes areas.

The Fill tool uses any of the Grease Pencil Fill draw mode brushes. Activating a brush from an asset shelf or brush selector will also activate this tool for convenience.

The asset selector can be used to open a pop-up asset browser to select the active brush asset for the tool.

See Asset Operators for more information.

You can also configure the brush main settings exposed on the Tool Settings for convenience.

The portion of area to fill.

Fills the area inside the shape under the cursor.

Fills the area outside the shape under the cursor.

Multiplier for fill boundary accuracy. Higher values are more accurate but slower.

Size in pixels to expand or shrink the fill area from the strokes boundary.

The thickness radius of the boundary stroke in pixels.

Sets the type of fill boundary limits calculation to perform.

Use the thickness of the strokes and the editing lines together.

Use only the thickness of the strokes (ignore edit lines).

Use only the edit lines (ignore strokes).

Toggle show auxiliary lines to see the fill boundary.

Determines which Layers are used for boundary strokes.

Calculates boundaries based on all visible layers.

Calculates boundaries based on the active layer.

Calculates boundaries based on the layer above the active layer.

Calculates boundaries based on the layer below the active layer.
Calculates boundaries based on all layers above the active layer.
Calculates boundaries based on all layers below the active layer.
Number of simplify steps to apply to the boundary line. Higher values reduce the accuracy of the final filled area.
When enabled, strokes with transparency does not take into account on fill boundary calculations.
The value slider controls the threshold to consider a material transparent.

When enabled, fill only visible areas in the viewport.

Gap closure lines are automatic temporarily lines that help to close gaps on the strokes.

Control the Size of the line extension or the circumference to use to calculate the lines that will close the gaps.

Sets the type of Gap closure method to use.

Uses the Radius of circumference of opened nearest points to calculate a line that close the gap.

Extends the opened strokes to close gaps.

Toggle show closure lines helper.

Check if extend lines collide with strokes, stopping the extension if a collision is detected.

Use for closing gaps only if the extend strokes collide.

In the Tool Settings select the brush, material and color type to use with the tool. The Fill tool uses Fill Brush types. See Brush Settings for more information.

Click LMB in a closed stroke area. The tool will automatically calculate the boundary and create a new closed stroke filled with the material selected.

Original Drawing. \xc2\xba6

Use the fill tool to leak materials on closed areas. \xc2\xba6

Final filled drawing. \xc2\xba6

If you have a large gap in an area that you want fill, you can add boundary strokes manually, a temporary auxiliary lines for closing open shapes. To create a boundary stroke use Alt-LMB and draw a line to close the desired area.

Original drawing. \xc2\xba6

Add boundary strokes to close open areas (red lines). \xc2\xba6

Use the Fill tool to leak material on the new closed area. \xc2\xba6

When you are satisfied with the fill result you can delete the boundary strokes using the Clean Up tool in the Grease Pencil Menu in Edit Mode.

A more automatic way to close gaps in an area that you want fill is using temporarily helper lines. There are two method to use

\xe2\x80\x9cRadius\xe2\x80\x9d or \xe2\x80\x9cExtend\xe2\x80\x9d

Radius use temporary auxiliary lines calculated from the radius of nearby open points to close open shapes. Set the size more than zero to control the circle size over opened points (the circle will disappear when the line close the gap). Click over the area you want to be filled and change the length of the strokes using PageUp/PageDown or Wheel. When you are satisfied with the length and you are sure the temporarily strokes cross each other, click again to fill the area.

Original Drawing. \xc2\xba6

Use Radius mode to close open areas (Red circles and cyan lines).

\xc2\xba6

Use Fill Tool to leak material on the new closed area. \xc2\xba6

Extend use temporary auxiliary lines extending the actual strokes ends for closing open shapes. Set the size more than zero to use the extended lines, click over the area you want to be filled and change the length of the strokes using PageUp/ PageDown, Wheel or a pen's MMB. When you are satisfied with the length and you are sure the temporarily strokes cross each other, click again to fill the area.

Original Drawing. \xc2\xba

Use Extend mode to close open areas (cyan lines). \xc2\xba

Use Fill Tool to leak material on the new closed area. \xc2\xba

--- Fin du chapitre: _fill ---

--- Début du chapitre: _fillet_curve --- The Fillet Curve rounds corners on curve control points, similar to the effect of the Bevel Modifier on a 2D mesh. However, a key difference is that the rounded portions created by the Fillet Curve node are always portions of a circle.

Standard geometry input with a curve component.

The radius of the circle portion generated at each fillet.

Only in Polymode, the number of control points to add for each fillet.

Whether to limit the maximum value of the radius in order to avoid overlapping fillets.

Only two control points will be generated for every filleted control point. The shape generated by the aligned handles on the generated control points on either side of the fillet is used to create the circle portion shape, meaning that the number of segments in the fillet shape depends on the spline's resolution value.

The number of control points generated for each field input is controlled directly with an integer field input. This mode works better for poly and NURBS splines.

Standard geometry input with a curve component.

The node can be used to round the corners of simple 3D poly splines.

\xc2\xba

The node can be combined with the curve primitive nodes to make more interesting shapes. \xc2\xba

--- Fin du chapitre: _fillet_curve ---

--- Début du chapitre: _fill_between_joints --- Reference

Edit Mode

Armature \xe2\x80\xa3 Fill Between Joints

F

The main use of this tool is to create one bone between two selected joints by pressing F, similar to how in mesh editing you can

\xe2\x80\x9c create edges/faces \xe2\x80\x9d.

If you have one root and one tip selected, the new bone:

Will have the root placed on the selected tip.

Will have the tip placed on the selected root.

Will be parented and connected to the bone owning the selected tip.

Active tip on the left. \xc2\xba

Active tip on the right. \xc2\xba

If you have two tips selected, the new bone:

Will have the root placed on the selected tip closest to the 3D cursor.

Will have the tip placed on the other selected tip.

Will be parented and connected to the bone owning the tip used as the new bone's root.

3D cursor on the left. \xc2\xba

3D cursor on the right. \xc2\xbb

If you have two roots selected, you will face a small problem due to the event system in Blender not updating the interface in real-time.

When clicking F, similar to the previous case, you will see a new bone:

With the root placed on the selected root closest to the 3D cursor.

With the tip placed on the other selected root.

Parented and connected to the bone owning the root used as the new bone\xe2\x80\x99s root.

If you try to move the new bone, Blender will update the interface and you will see that the new bone\xe2\x80\x99s root moves to the tip of the parent bone.

Before UI update (3D cursor on the left). \xc2\xbb

After UI update, correct visualization. \xc2\xbb

Clicking F with only one bone joint selected will create a bone from the selected joint to the 3D cursor position, and it will not parent it to any bone in the armature.

Fill with only one tip selected. \xc2\xbb

Fill with only one root selected. \xc2\xbb

You will get an error when:

Trying to fill two joints of the same bone.

Trying to fill more than two bone joints.

--- Fin du chapitre: _fill_between_joints ---

--- Début du chapitre: _fill_curve --- The Fill Curvenode generates a mesh using the constrained Delaunay triangulation algorithm with the curves as boundaries. The mesh is only generated flat with a local Z of 0.

Standard geometry input with a curve component.

Value used to group curves together. Curves with different Group ID are treated separately.

The type of geometry the output consists of.

The output is made up of triangles.

The output is made up of n-gons.

The filled-in curves.

One or many \xe2\x80\x99single point spline\xe2\x80\x9d can be used to customize the triangulation of the filled-in curves.

This is the default behavior of the Fill Curvenode applied to the star primitive.\xc2\xbb

Here, a single curve point is joined to the star primitive to customize the triangulation.\xc2\xbb

Here, 300 single curve point are joined to the star primitive to customize the triangulation.\xc2\xbb

The following figures display diverse application of the Group ID.

Here, the 4 curves share the same Group ID, resulting in 1 mesh island (default behavior).\xc2\xbb

Here, the 4 curves have different Group ID, resulting in 4 mesh islands.\xc2\xbb

Here, the 4 curves are separated into two groups based on their positions, resulting in 2 mesh islands.\xc2\xbb

--- Fin du chapitre: _fill_curve ---

--- Début du chapitre: _film ---Reference

Render \xe2\x80\x93 Film

This can be used to change the brightness of an image. Different than the Exposure option found in the Color management panel, this exposure option works on the data while the Color management exposure is on the view transform.

Due to limited resolution of images and computer screens, pixel filters are needed to avoid Aliasing. This is achieved by slightly blurring the image to soften edges.

Pixel Filtering algorithm to use.

No filter.

Smooth filter.

Default filter with a better balance between smoothness and detail preservation.

Lower values give more crisp renders, higher values are softer and reduce aliasing.

Render the background transparent, for compositing the image over another background after rendering.

Render transmissive surfaces as transparent, for compositing glass over another background.

For transparent glass, keep surfaces with roughness above the threshold opaque.

--- Fin du chapitre: _film ---

--- Début du chapitre: _filter --- The Filter node implements various common image enhancement filters.

Controls the amount of influence the node exerts on the output image.

Standard color input.

The Soften, Laplace, Sobel, Prewitt and Kirsch all perform edge detection (in slightly different ways) based on vector calculus and set theory equations.

Slightly blurs the image.

Increases the contrast, especially at edges.

Less aggressive than box sharpen, reducing sharpening artifacts.

Edge highlighting filter susceptible to highlighting visual noise.

Creates a negative image that highlights edges.

Produces a similar results to Sobel.

Gives better blending than Sobel or Prewitt, when approaching an edge.

Performs a relief, emboss effect, darkening outside edges.

Standard color output.

Original image.\xc2\xbb

Soften.\xc2\xbb

Box Sharpen.\xc2\xbb

Diamond Sharpen.\xc2\xbb

Laplace.\xc2\xbb

Sobel.\xc2\xbb

Prewitt.\xc2\xbb

Kirsch.\xc2\xbb

Shadow.\xc2\xbb

--- Fin du chapitre: _filter ---

--- Début du chapitre: _filters --- Filters are tools which provide an alternative way of sculpting, because they do not rely on a brush radius. Instead they will affect any vertices that are visible and not masked.

The strength is controlled by click & dragging from left to right. The position of the cursor can be used to only affect specific areas, if auto-masking is used.

Many of the same brush types are also available as a filter type. This way much of the mesh can simultaneously be smoothed, colored or have some cloth simulation applied.

Tip

A common example for using the Mesh Filter is to smooth everything after increasing the resolution with the Voxel Remesher or Dynatopo .

See also

More information at Mesh Filter , Cloth Filter , Color Filter and Mask Filters .

--- Fin du chapitre: _filters ---

--- Début du chapitre: _fixed ---Reference

Physics \xe2\x80\xa3 Rigid Body Constraint

Fixed

This constraint cause the two objects to move as one. Since the physics system does have a tiny bit of slop in it, the objects do not move as rigidly as they would if they were part of the same mesh.

--- Fin du chapitre: _fixed ---

--- Début du chapitre: _flatten ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Flatten or contrast surfaces by pulling and pushing them towards (or away from) a defined medium height. This medium height is determined via an area plane within the brush radius.

This area plane height can be further defined in the brush settings.

Invert the direction to push surfaces away from the sculpt plane, creating more surface contrast as a result.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _flatten ---

--- Début du chapitre: _flip --- This node flips an image along a defined axis.

You can use this node to just flip or use it as a part of mirror setting. Mix half of the image to be mirrored with its flipped version to produce mirrored image.

Standard color input.

This can be either X or Y. Also, flipping can be done on both X and Y axis simultaneously.

Flip X, Flip Y, Flip X & Y

Standard color output.

--- Fin du chapitre: _flip ---

--- Début du chapitre: _flip_faces --- The Flip Faces Node reverses the order of the vertices and edges of each selected face. The most common use of this node is to flip the normals of a face. Any face corner domain attributes of selected faces are also reversed.

Though this node is usually used to affect normals, it is not called \xe2\x80\x9cFlip Normals\xe2\x80\x9d for an important reason. The node

does not actually interact with normals directly. Normals are defined by the right hand rule, so if a face's vertex list is reversed, then its normal will point in the opposite direction.

Standard geometry input.

Whether to flip the direction of each face. True values mean the face will be flipped, false means the face will be unaffected.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_flip_faces` ---

--- Début du chapitre: `_flip_quats` ---Reference

Pose Mode

Pose \xe2\x80\xa3 Flip Quats

Alt- F

Flip quaternion values to achieve desired rotations, while maintaining the same orientations.

--- Fin du chapitre: `_flip_quats` ---

--- Début du chapitre: `_float_curve` --- The Float Curve node maps an input float to a curve and outputs a float value.

Controls the amount of influence the node exerts on the output value.

Standard float input.

For the curve controls see: Curve widget .

Standard float output.

--- Fin du chapitre: `_float_curve` ---

--- Début du chapitre: `_float_to_integer` --- The Float To Integer node takes a single floating point number input and converts it to an integer with a choice of methods.

Standard float value input.

Outputs the closest integer to Float, rounding either up or down based on the value.

Outputs the closest integer less than Float, always rounding down.

Outputs the closest integer greater than Float, always rounding up.

Outputs the closest integer between Float and zero. For positive numbers, acts like Floor. For negative numbers, acts as Ceiling.

Standard integer output.

Input Value

Round

Floor

Ceiling

Truncate

--- Fin du chapitre: `_float_to_integer` ---

--- Début du chapitre: `_floor` --- The Floor constraint allows you to use its target position (and optionally rotation) to specify a plane with a

forbidden side, where the owner cannot go. This plane can have any orientation you like. In other words, it creates a

floor (or a ceiling, or a wall)! Note that it is only capable of

simulating entirely flat planes, even if you use the Vertex Group option. It cannot be used for uneven floors or walls.

Floor panel. \xc2\xb6

Data ID used to select the constraints target, and is not functional (red state) when it has none. See common constraint properties for more information.

Allows you to offset the floor plane from the target's origin, by the given number of units. Use it e.g. to account for the distance from a foot bone to the surface of the foot's mesh.

Controls which plane will be the floor plane. The names of the buttons correspond, indeed, to the normal to this plane (e.g. enabling Z means XY plane, etc.). By default, these normals are aligned with the global axes. However, if you enable Use Rotation (see above), they will be aligned with the local target's axes. As the constraint does not only define an uncrossable plane, but also a side of it which is forbidden to the owner, you can choose which side by enabling either the positive or negative normal axis (e.g. by default Z, the owner is stuck in the positive Z coordinates).

Forces the constraint to take the target's rotation into account. This allows you to have a floor plane of any orientation you like, not just the global XY, XZ and YZ ones.

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: _floor ---

--- Début du chapitre: _flow --- Fluid Flowtypes are used to add or remove fluid to a domain object. Flow objects should be contained within the domain's Bounding Box in order to work.

To define any mesh object as a Flow object, add Fluid physics by clicking Fluid in Properties > Physics. Then select Flow as the fluid Type. Now you should have a default fluid flow source object.

Reference

Physics > Fluid > Settings

Flow

Emit only smoke.

Emit both fire and smoke.

Emit only fire. Note that the domain will automatically create some smoke to simulate smoke left by burnt fuel.

Emit liquid.

Controls if the Flow object either adds (Inflow), removes (Outflow), or turns the mesh itself into fluid (Geometry).

This object will emit fluid into the simulation, like a water tap or base of a fire.

Any fluid that enters the Bounding Box of this object will be removed from the domain (think of a drain or a black hole). This can be useful in combination with an inflow to prevent the whole domain from filling up.

Outflow objects can be animated and the area where the fluid disappears will follow the object as it moves around.

All regions of this object that are inside the domain bounding box will be used as actual fluid in the simulation. You can place more than one fluid object inside the domain. Also make sure that the surface normals

are pointing outwards or else they will not simulate properly. In contrast to domain objects, the actual mesh geometry is used for fluid objects. Enables or disables the flow of fluid, this property is useful for animations to selectively enable and disable when fluid is being added to or removed from the domain.

Number of sub-steps used to reduce gaps in emission of fluid from fast-moving sources.

Sampling sub-steps: 0.

Sampling sub-steps: 3.

Note that these emission sub-steps occur at every simulation step and not per frame. The simulation step count is controlled by the adaptive time stepping.

The color of emitted smoke. When smoke of different colors are mixed they will blend together, eventually settling into a new combined color.

If this checkbox is enabled, the emitter will only produce more smoke or fire if there is space for it in the emitter region. Otherwise smoke or fire will always be produced and add up.

Difference between the temperature of emitted smoke and the domain's ambient temperature. This setting's effect on smoke depends on the domain's Heat Buoyancy.

Amount of smoke to emit at once. Larger values result in more density being produced.

Amount of fuel being burned per second. Larger values result in larger flames, smaller values result in smaller flames:

Fuel: 0.5.

Fuel: 1.0.

When set, use the specified Vertex Group to control where smoke is emitted.

This setting defines the method used to emit fluid.

Emit fluid directly from the object's mesh.

Defines the effector as either a single dimension object i.e. a plane or the mesh is non-manifold. This ensures that the fluid simulator will give the most accurate results for these types of meshes.

Maximum distance in voxels from the surface of the mesh in which fluid is emitted. Since this setting uses voxels to determine the distance, results will vary depending on the domain's resolution.

Amount of fluid to emit inside the emitter mesh, where 0 is none and 1 is the full amount. Note that emitting fluid based on volume can have unpredictable results if your mesh is non-manifold.

Create smoke or fire from a particle system on the flow object. The particle system can be selected with a Data ID.

Note that only emitter type particle systems can add smoke. See Particles for information on how to create a particle system.

When this setting is enabled, it allows the Size setting to define the maximum distance in voxels at which particles can emit smoke, similar to the Surface Emission setting for mesh sources.

When disabled, particles will fill the nearest voxel with smoke.

When enabled, the fluid will inherit the momentum of the flow source.

Factor for the inherited velocity. A value of 1 will emit fluid moving at the same speed as the source.

This option controls how much velocity fluid is given along a face normal. Note that, initial velocities will always be applied along all face normals. Thus with a closed flow source mesh, fluid will always be emitted in more than one direction. To set initial velocities along only

one direction all normals need to point in the same direction. This is can be achieved when using a plane as the flow object.

Initial velocity along X, Y, Z coordinates in world space. This can be used in addition to the initial velocity along the normal.

Reference

Flow

Physics \xe2\x80\xa3 Fluid \xe2\x80\xa3 Settings \xe2\x80\xa3 Texture

When enabled, use the specified texture and settings to control where on the mesh smoke or fire can be emitted from. These settings have no effect on Outflow Flow Behavior.

A data ID selector to choose the texture.

Controls whether to use generated UVs or manual UV mapping.

Overall texture scale.

Translates the texture along the Z axis.

--- Fin du chapitre: _flow ---

--- Début du chapitre: _fluid --- Reference

Particle System \xe2\x80\xa3 Physics

Fluid

Fluid particles are similar to Newtonian ones but this time particles are influenced by internal forces like pressure, surface tension, viscosity, springs, etc. From liquids to slime, goo to sand and wispy smoke the number of possible use cases is endless.

Blender particle fluids use the SPH techniques to solve the particles fluid equations. Smoothed-particle hydrodynamics (SPH) is a computational method used for simulating fluid flows. It has been used in many fields of research, including astrophysics, ballistics, vulcanology, and oceanography. It is a mesh-free Lagrangian method (where the coordinates move with the fluid), and the resolution of the method can easily be adjusted with respect to variables such as the density.

Fluid physics share options with Newtonian Physics. These are covered on that page.

How incompressible the fluid is.

Linear viscosity. Use lower viscosity for thicker fluids.

Artificial buoyancy force in negative gravity direction based on pressure differences inside the fluid.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Advanced

How strongly the fluid tries to keep from clustering (factor of stiffness). Checkbox sets repulsion as a factor of stiffness.

Creates viscosity for expanding fluid. Checkbox sets this to be a factor of normal viscosity.

Fluid \xe2\x80\x99s interaction radius. Checkbox sets this to be a factor of 4 \xc3\x97 particle size.

Density of fluid when at rest. Checkbox sets this to be a factor of default density.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Springs

Spring force.

Rest length of springs. Factor of particle radius. Checkbox sets this to be a factor of 2 \xc3\x97 particle size.

Use viscoelastic springs instead of Hooke \xe2\x80\x99s springs.

How much the spring has to be stretched/compressed in order to change its rest length.

How much the spring rest length can change after the elastic limit is crossed.

Use initial length as spring rest length instead of 2 particle size.

Create springs for this number of frames since particle's birth (0 is always).

--- Fin du chapitre: `_fluid` ---

--- Début du chapitre: `_fluid_flow` ---Reference

Physics Force Fields

Fluid Flow

The Fluid Flowforce field creates a force based on aFluid simulationair flow.It applies the smoke simulation air flow velocity as a force to other simulations that use force fields.To use it you need to add a Fluid Flowforce field and select a domain object for it.For example fire sparkles can realistically flow along the air turbulence near the simulated fire.

An object that is used as a domain for the smoke simulation.

Adjust the force strength based on the smoke density.

--- Fin du chapitre: `_fluid_flow` ---

--- Début du chapitre: `_follow_path` --- The Follow Pathconstraint places its owner onto a curvetarget object,and makes it move along this curve (or path).It can also affect its owner's rotation to follow the curve's bends,when the Follow Curveoption is enabled.

It could be used for complex camera traveling,a train on its rails and most other vehicles can also use invisible tracks,the links of a bicycle chain, etc.

The owner is always evaluated in the global (world) space:

Its location (as shown in the Transformpanel)is used as an offset from its normal position on the path. E.g.if you have an owner with the (1.0, 1.0, 0.0) location,it will be one unit away from its normal position on the curve, along the X and Y axis.Hence, if you want your owner onits target path, clear its locationAlt- G!

This location offset is also proportionally affected by the scale of the target curve.Taking the same (1.0, 1.0, 0.0) offset as above,if the curve has a scale of (2.0, 1.0, 1.0),the owner will be offset two units along the X axis (and one along the Y one).

When the Follow Curveoption is enabled, its rotation is also offset to the one given by the curve.E.g. if you want the Y axis of your object to be aligned with the curve's direction,it must be in rest, non-constrained state, aligned with the global Y axis.Here again, clearing your owner's rotationAlt- Rmight be useful

The movement of the owner along the target curve/path may be controlled in two different ways:

The most simple is to define the number of frames of the movement,in thePath Animation panelof the Curve tab,via the Framesnumber field, and its start frame via the constraint's Offset option(by default, start frame: 1 [= offset of 0], duration: 100).

The second way, much more precise and powerful,is to define an Evaluation Timeinterpolation curve for the Targetpath(in the Graph Editor). See theGraph Editor chapterto learn more about F-Curves.

If you do not want your owner to move along the path, you can give to the target curve a flat SpeedF-Curve(its value will control the position of the owner along the path).

Follow Path is another constraint that works well with the Locked Track one. One example is a flying camera on a path. To control the camera's roll angle, you can use a Locked Track and a target object to specify the up direction, as the camera flies along the path.

Note

Follow Path & Clamp To

Do not confuse these two constraints. Both of them constraint the location of their owner along a curve, but Follow Path is an animation-only constraint, inasmuch as the position of the owner along the curve is determined by the time (i.e. current frame), whereas the Clamp To constraint determines the position of its owner along the curve using one of its location properties values.

Note

Note that you also need to keyframe Evaluation Time for the Path. Select the path, go to the Path Animation panel in the curve properties, set the overall frame to the first frame of the path (e.g. frame 1), set the value of Evaluation time to the first frame of the path (e.g. 1), right-click on Evaluation time, select create keyframe, set the overall frame to the last frame of the path (e.g. frame 100), set the value of Evaluation time to the last frame of the path (e.g. 100), right-click on Evaluation time, select create keyframe.

Follow Path panel. `\xc2\xbb6`

Data ID used to select the constraint's target, which must be a curve object, and is not functional (red state) when it has none. See common constraint properties for more information.

The number of frames to offset from the animation defined by the path (by default, from frame 1).

The axis of the object that has to be aligned with the forward direction of the path (i.e. tangent to the curve at the owner's position).

The axis of the object that has to be aligned (as much as possible) with the world Z axis. In fact, with this option activated, the behavior of the owner shares some properties with the one caused by a Locked Track constraint, with the path as `\xc2\xbb9caxle`, and the world Z axis as `\xc2\xbb9cmagnet`.

Object will stay locked to a single point somewhere along the length of the curve regardless of time.

Objects scaled by the curve radius. See Curve Editing .

If this option is not activated, the owner's rotation is not modified by the curve; otherwise, it is affected depending on the Forward and Up Axes.

Adds an F-Curve with options for the start and end frame. To Do: from above.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_follow_path` ---

--- Début du chapitre: `_follow_track` --- By default the Follow Track constraint is making objects have the same position at a frame as the

track has. The motion of this object happens on a single plane defined by the camera and the original position of the object.

Follow Track Constraint panel. \xc2\xba6

Receive tracking data from the scene\ xe2\x80\x99s Active Clip .If unchecked, an option appears to choose from the other clips.

Use the 3D position of the track to parent to.

Parent to the undistorted position of the 2D track.

Defines how the footage is fitted in the camera frame.

Select the camera to which the motion is parented to (if empty, the active scene camera is used).

If this object is set, constrained objects will be projected onto the surface of this depth object which can be used to create facial makeup visual effects.

Creates F-Curves for the object that copies the movement caused by the constraint.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

Follow Track Example Video

--- Fin du chapitre: _follow_track ---

--- Début du chapitre: _force ---Reference

Physics \ xe2\x80\xa3 Force Fields

Force

Force force field visualization.\xc2\xba6

The Forcefield is the simplest of the fields. It gives a constant force towards (positive strength) or away from (negative strength) the object\ xe2\x80\x99s origin. Newtonian particles are attracted to a field with negative strength, and are blown away from a field with positive strength.

--- Fin du chapitre: _force ---

--- Début du chapitre: _force_field --- Reference

Particle System \ xe2\x80\xa3 Field Weights

The Field Weight panel allows you to control how much influence each type of external force field, or effector, has on the particle system. Force fields are external forces that give dynamic system\ xe2\x80\x99s motion. The force fields types are detailed on the Force Field Page . Limit effectors to a specified group. Only effectors in this group will have an effect on the current system.

Control how much the Global Gravity has an effect on the system.

Scale all of the effector weights.

Reference

Particle System \ xe2\x80\xa3 Force Fields Settings

The Force Field Settings panel allows you to make each individual act as a force field, allowing them to affect other dynamic systems, or even, each other.

Causes the particle force fields to have an effect on other particles within the same system.

Set how many of the particles act as force fields. 0 means all of them are effectors.

You can give particle systems up to two force fields. By default they do not have any. Choose an effector type from the selector to enable them. Settings are described in the Common Settings section .

--- Fin du chapitre: _force_field ---

--- Début du chapitre: `_format` --- Format panel.

Several render presets exist with common resolution and frame rates for TVs and screens can be selected in the panel header.

The number of pixels horizontally and vertically in the image.

Slider to reduce or increase the size of the rendered image relative to the Resolution values. This is useful for small test renders that have the same proportions as the final image.

Older televisions may have non-square pixels, so this can be used to control the shape of the pixels along the respective axis. This will pre-distort the images which will look stretched on a computer screen, but which will display correctly on a TV set. It is important that you use the correct pixel aspect ratio when rendering to prevent re-scaling, resulting in lowered image quality.

Renders just a portion of the view instead of the entire frame. See the Render Region documentation to see how to define the size of the render region.

Crops the rendered image to the size of the render region, instead of rendering a transparent background around it.

The number of frames that are displayed per second, relevant for Animation. The menu gives several common frame rates, custom frame rates can be used by selecting Custom which gives access to the following properties:

The frame rate, expressed in frames per second.

Some standards require a more precise frame rate, for example NTSC. These can be represented as a fraction where the Base value is used as the denominator and the FPS being the numerator:
$$\frac{\text{FPS}}{\text{Base}}$$

See also

Viewport Playback Frame Rate Limited

--- Fin du chapitre: `_format` ---

--- Début du chapitre: `_for_each_geometry_zone` --- This zone type allows executing nodes for each element of a geometry. For example, the nodes can process every face of a mesh, or every instance.

The For Each Element zone.

The zone is ideal for tasks that generate large or complex geometry for every element of an input geometry. For example, generating a unique tree for every input curve, or a unique building on every input face.

The zone makes less sense for processing small amounts of geometry. In that case (for example each of a character's hairs separately) it will likely always be slower than working on fewer larger geometries. The additional flexibility from processing each element separately comes at the cost that Blender can't optimize the operation as well. For node groups that need to handle lots of geometry elements, it's recommended to design the node setup so that iteration over tiny sub-geometries is not required.

Geometry whose elements are iterated over.

Which subset of the chosen Domain to process.

Index of the element in the source geometry. Note that the same index can occur more than once when iterating over multiple geometry component types at once.

The input geometry is split up into a separate geometry for each element. This is the single element geometry for the current iteration.

This is not available for the Face Corner domain, since face corners cannot exist without their face.

Note

It can be quite inefficient to split up large geometries into many small elements. Because this output isn't computed if it's not used in the node graph, not using it will typically improve performance.

Which attribute domain to process.

Geometry element index that is used by inspection features like the Viewer Node or socket inspection .

The Main Geometry outputs create attributes on the main output geometry (the first output). Every single value on the inside of the zone becomes a value of the attribute at the current index.

The outputs in the Generated panel, including the default Geometry output are joined together from the geometry generated from each element. Any non-geometry type below a specific geometry in this list will output as an anonymous attribute on that joined geometry (and not the others). Attributes from the zone's input geometry are also propagated to these geometry outputs.

--- Fin du chapitre: `_for_each_geometry_zone` ---

--- Début du chapitre: `_frame` --- The Frame node is useful for organizing nodes by collecting related nodes together in a common area. Frames are useful when a node setup becomes large and confusing yet the re-usability of a Node Group is not required.

Font size of the label. For example, for subordinate frames to have smaller titles.

Once a node is placed in the Frame, the Frame shrinks around it so as to remove wasted space. At this point it is no longer possible to select the edge of the Frame to resize it, instead resizing occurs automatically when nodes within the Frame are rearranged. This behavior can be changed by disabling this option.

When you need to display more comprehensive text, frame nodes can display the contents of a text data-block. This is read-only, so you will need to use the Text Editor to modify the contents.

Reference

Node `Join in new Frame`

Ctrl- J

Make a new frame including the selected nodes.

Reference

Ctrl- P

Once a frame node is placed, nodes can be added by dropping them onto the frame or by selecting the node(s) then the frame and using Ctrl- P. This can be thought of as Parenting the selection to the frame.

Reference

Node `Remove from Frame`

Alt- P

To remove nodes from a frame, select and use Alt- P. This can be thought of as unparenting the selection from the frame.

--- Fin du chapitre: `_frame` ---

--- Début du chapitre: `_frame_range` --- Frame Range panel.

This panel defines how long an animation will last in terms of frames. Frames can be divided by the scene's Frame Rate to get the animation duration in terms of time. For example, a 250 frame animation at a frame rate of 30 will last 8.3 seconds.

Set the Start and End frames for Rendering Animations.

Controls the number of frames to advance by for each frame in the timeline.

Use to remap the length of an animation; making it run slower or faster. The Old and New settings may either be used as absolute values or as a ratio: For example, setting Old to a value of 2 and New to 1 will run the animation twice as fast.

Warning

Using Time Stretching will not influence the Start or End frames set above, so make sure that your animation is not cut off or has extraneous still frames at the end.

The length in frames of the original animation.

The length in frames the new animation will last.

--- Fin du chapitre: `_frame_range` ---

--- Début du chapitre: `_freestyle` --- Reference

Properties \xe2\x80\xa3 View Layer \xe2\x80\xa3 Freestyle

There is only one view map per view layer. It controls the edge detection parameters. Freestyle can be enabled/disabled per View Layer by toggling the checkbox in the panel header.

View Layer: Freestyle panel. \xc2\xb6

Which detected edges are actually rendered, and how, can be controlled either through:

Lines are rendered via parameters defined in a user-friendly interface to define and control Line Sets and line styles.

A view map (hence a view layer) can have multiple Line Sets, and each line set is linked to one line style.

Lines are rendered via Python scripting, powerful but complex.

An option to reuse a previously computed view map for subsequent rendering. The cache is automatically updated when the mesh geometry of the input 3D scene has been changed.

This functionality offers a major performance boost for Freestyle animation rendering when the camera-space mesh geometry is static, as well as for repeated still renders with updates of line stylization options.

Although the View Map Cache checkbox is a view layer option, the cache memory is shared by all view layers and scenes. This means that if Freestyle is used for two or more view layers (possibly in different scenes through the Compositor), then the cached view map for one view layer is replaced by a new view map for another view layer and hence no performance gain is expected.

Freestyle lines will not immediately be visible on top of the render image. Instead, Freestyle lines are rendered as a Render Pass which can be composited with the rendered image with an Alpha Over node.

If two adjacent faces form an angle less than the defined Crease Angle, the edge between them will be rendered when using Crease edge type selection in a line set. The value also affects Silhouette edge type selection.

Ignore the edges that are out of view. (Saves some processing time and memory, but may reduce the quality of the result in some cases.)

Takes Smooth Shading into account for edges calculation.

Affects the calculation of curvatures for Ridge, Valley and Suggestive Contoured edge type selection in a line set. The curvature at each vertex is computed by averaging the shape of the surface within the specified radius. Increasing the value reduces noise and detail. Controls the threshold on the minimum rate of change of curvature used to filter the output of the Suggestive Contoured edge type selection. Increasing the value reduces the amount of rendered lines, starting from smoother areas of the object (further information in this pdf).
--- Fin du chapitre: `_freestyle` ---

--- Début du chapitre: `_fresnel` --- The Fresnel or Dielectric Fresnel node computes how much light is reflected off a layer, where the rest will be refracted through the layer. The resulting weight can be used for layering shaders with the Mix Shader node. It is dependent on the angle between the surface normal and the viewing direction. The most common use is to mix between two BSDFs using it as a blending factor in a Mix Shader node. For a simple glass material you would mix between a glossy refraction and glossy reflection. At grazing angles more light will be reflected than refracted as happens in reality. For a two-layered material with a diffuse base and a glossy coating, you can use the same setup, mixing between a diffuse and glossy BSDF. By using the Fresnel as the blending factor you are specifying that any light which is refracted through the glossy coating layer would hit the diffuse base and be reflected off that. Index of refraction (IOR) of the material being entered. Input meant for plugging in bump or normal maps which will affect the output. This node has no properties. Fresnel weight, indicating the probability with which light will reflect off the layer rather than passing through.
--- Fin du chapitre: `_fresnel` ---

--- Début du chapitre: `_frizz_hair_curves` --- Deforms hair curves using a random vector per point to frizz them. Input Geometry (only curves will be affected). Apply offset cumulatively (previous points affect points after). Factor to blend overall effect. Overall distance factor for the deformation. Shape of the influence along curves (0=constant, 0.5=linear). Random Seed for the operation. Preserve each curve's length during deformation. This node has no properties. Geometry Vector by which each point was offset during deformation.
--- Fin du chapitre: `_frizz_hair_curves` ---

--- Début du chapitre: `_gabor` --- The Gabor Texture node evaluates a Gabor noise at the input texture coordinates. Gabor noise is visually characterized by random interleaved bands whose direction and width can be controlled. Additionally, it can be used to create omnidirectional noise like the standard Noise Texture node, but since it is more expensive to compute, using the Noise Texture node is probably the better option in those cases. See the examples for more information. Note

This node is ported from shader nodes. The manual and images arereferencing the shader version of the node.This node accepts field inputs and outputs.When not connected the Vector input has an implicit position attribute value.

The coordinates at which Gabor noise will be evaluated. The Z component is ignored in the 2Dcase. Defaults to Generatedtexture coordinates if the socket is left unconnected.

Scale of the Gabor noise.

The rate at which the Gabor noise changes across space. This is different from the Scale inputin that it only scales perpendicular to the Gabor noise direction.

The directionality of Gabor noise. 1 means the noise is completely directional, while 0 meansthe noise is omnidirectional.

The direction of anisotropic Gabor noise. This is an angle for the 2D case, while it is a unitdirection vector for the 3D case.

Type of Gabor noise texture.

Evaluates the noise in 2D space. The Z component of the input vector is ignored.

Evaluates the noise in 3D space.

Note

Higher dimensions corresponds to higher render time, so lower dimensions should be usedunless higher dimensions are necessary.

The Gabor noise value with both random intensity and phase. This is equal to sine the phasemultiplied by the intensity.

The phase of the Gabor noise, which has no random intensity.

The intensity of the Gabor noise, which has no random phase.

The following table demonstrates different outputs of the node with different parameters. As can beseen, the noise is visually characterized by interleaved bands that are generally oriented in aspecific direction. But the Anisotropyparameter can be decreased below 1 to make the bands morerandom in directions. The Frequencyparameter determines the number of bands perpendicular to thedirection of the noise. However, the Scaleparameter can also be used to globally increase thenumber of bands, so consider increasing the scale first since high frequency noise can suffer fromlow contrast and limited interleaving of bands.

Value output. Frequency = 2. Anisotropy = 1.\xc2\xbb

Phase output. Frequency = 2. Anisotropy = 1.\xc2\xbb

Intensity output. Frequency = 2. Anisotropy = 1.\xc2\xbb

Value output. Frequency = 3. Anisotropy = 1.\xc2\xbb

Phase output. Frequency = 3. Anisotropy = 1.\xc2\xbb

Intensity output. Frequency = 3. Anisotropy = 1.\xc2\xbb

Value output. Frequency = 2. Anisotropy = 0.7.\xc2\xbb

Phase output. Frequency = 2. Anisotropy = 0.7.\xc2\xbb

Intensity output. Frequency = 2. Anisotropy = 0.7.\xc2\xbb

Gabor noise is decomposed into a Phaseand an Intensitycomponents, where the Gabor value iscomputed as sine the phase multiplied by the intensity, noting that the phase output is normalizedto the [0, 1] range.

Compute the value output from the phase and intensity outputs. \xc2\xbb
The advantage of the Phaseoutput is that it has no random intensities and no low contrast areasas in the value output, so it can be used as a base for textures that are more structured in nature,like sand dunes.

Sand dune-like structures creates using the phase output. \xc2\xbb
The main advantage and use of the Intensityoutput is that it provides information about thelocation of singularitiesin the Phaseoutput.

Singularities are those areas in the phase where the bands meet, which are shown in red in the following figure. Those areas will be close to zero in the Intensity output. So if those areas are undesirable, they can be hidden by multiplying by a variant of the Intensity output.

Visualization of the areas where singularities happen. \xc2\xba
Inputs can be varied across space to get more interesting patterns.

Varying the frequency and orientation across space. \xc2\xba

--- Fin du chapitre: `_gabor` ---

--- Début du chapitre: `_gamma` --- Use this node to apply a gamma correction.

Standard color input.

An exponential brightness factor.

This node has no properties.

Standard color output.

Example of a Gamma node. \xc2\xba

--- Fin du chapitre: `_gamma` ---

--- Début du chapitre: `_gamma_cross` --- The Gamma Cross transition is similar to the Cross Strip transition, however, the Gamma Cross strip transition uses color correction while transitioning between the two strips, resulting in a smoother transition that is easier on the eyes. Automatically calculates a linear fade over the length of the strip. Allows you to manually keyframe a custom fade. This can be used with different easings to fine-tune the fade in/out.

--- Fin du chapitre: `_gamma_cross` ---

--- Début du chapitre: `_general` --- Sculpt Mode is similar to Edit Mode in that it is used to alter the shape of a model, but Sculpt Mode uses a very different workflow: instead of dealing with individual elements (vertices, edges, and faces), an area of the model is primarily changed using brushes.

Sculpting Mode Example. \xc2\xba

Sculpt Mode is accessed from the mode menu of the 3D Viewport header or with the pie menu via `Ctrl-Tab`. Once inside Sculpt Mode, the Toolbar and Tool Settings of the 3D Viewport will change to Sculpt Mode specific panels. The cursor will change to a circle, to indicate the size of the brush.

Warning

To have predictable brush behavior, make sure to apply the scale of your mesh.

The following pages will briefly explain the fundamental features and concepts of Sculpt Mode, including various links to other pages for more details.

--- Fin du chapitre: `_general` ---

--- Début du chapitre: `_generate_hair_curves` --- Generates new hair curves on a surface mesh. The curves are generated from scratch at point locations; if creating curves that depend on existing curves is desired, the Interpolate Hair Curves is a better choice.

Note

This node/modifier will not function without the Surface Geometry/object and Surface UV Map inputs.

Surface geometry for generation. This input takes priority over the corresponding object input if both are provided.
Surface object for generation (The transforms of this object must match the modifier object).

Surface UV map stored on the mesh used for finding curve attachment locations.

Set the surface mesh into its rest position before attachment.

Tip

In a typical hair generation setup, this node or modifier will be combined with the Deform Curves on Surface Node. If that operation comes after this one, it makes sense to turn this option on so the position used is the pre-deformed position consistent with the expectations for the deformation's input.

Length of the generated hair curves.

Material of the generated hair curves.

Amount of control points of the generated hair curves.

Use poisson disk distribution method to keep a minimum distance. See the Distribute Points on Faces for more information.

Surface density of generated hair curves.

Factor applied on the density for curve distribution.

Discard points based on a mask texture after distribution. The image is sampled with the Surface UV Map input.

Tip

The accuracy of sampling the image doesn't depend on the density of the surface mesh's vertices because it is sampled after the curve root points are generated, the accuracy. However, using the Density Mask input instead can give better performance. Using them in combination can give the benefits of both methods.

Factor applied on the density for the viewport.

Random seed for the operation.

This node has no properties.

Geometry

Curves

Normal direction of the surface mesh at the attachment point.

--- Fin du chapitre: `_generate_hair_curves` ---

--- Début du chapitre: `_generic` --- Reference

Physics \xa3 Rigid Body Constraint

Generic

The generic constraint has a lot of available parameters.

The X, Y, and Z axis constraints can be used to limit the amount of translation between the objects. Clamping the min/max to zero has the same effect as the Point constraint.

Clamping the relative rotation to zero keeps the objects in alignment. Combining an absolute rotation and translation clamp would behave much like the Fixed constraint.

Using a nonzero spread on any parameter allows it to oscillate in that range throughout the course of the simulation.

Enables/disables limit rotation around X, Y or Z axis respectively.

Lower limit of rotation for X, Y or Z axis respectively.

Upper limit of rotation for X, Y or Z axis respectively.

Enables/disables limit translation on X, Y or Z axis respectively.

Lower limit of translation for X, Y or Z axis respectively.

Upper limit of translation for X, Y or Z axis respectively.

--- Fin du chapitre: _generic ---

--- Début du chapitre: _generic_spring ---Reference

Physics \xe2\x80\xa3 Rigid Body Constraint

Generic Spring

The generic spring constraint adds some spring parameters for the X/Y/Z axes to all the options available on the Generic constraint. Using the spring alone allows the objects to bounce around as if attached with a spring anchored at the constraint object. This is usually a little too much freedom, so most applications will benefit from enabling translation or rotation constraints.

If the damping on the springs is set to 1, then the spring forces are prevented from realigning the anchor points, leading to strange behavior. If your springs are acting weird, check the damping.

Enables/disables limit translation on X, Y or Z axis respectively.

Lower limit of translation for X, Y or Z axis respectively.

Upper limit of translation for X, Y or Z axis respectively.

Enables/disables limit rotation around the X, Y or Z axis respectively.

Lower limit of rotation for X, Y or Z axis respectively.

Upper limit of rotation for X, Y or Z axis respectively.

Enables/disables springs translation on X, Y or Z axis respectively.

Spring Stiffness of the translation on X, Y or Z axis respectively.

Specifies how \xe2\x80\x9cbendy\xe2\x80\x9d the spring is.

Spring Damping of the translation on X, Y or Z axis respectively. Amount of damping the spring has.

Enables/disables springs rotation around the X, Y or Z axis respectively.

Spring Stiffness of the rotation around the X, Y or Z axis

respectively. Specifies how \xe2\x80\x9cbendy\xe2\x80\x9d the spring is.

Spring Damping of the rotation around the X, Y or Z axis respectively.

Amount of damping the spring has.

--- Fin du chapitre: _generic_spring ---

--- Début du chapitre: _genindex ----- Fin du chapitre: _genindex ---

--- Début du chapitre: _geometry --- In this tab you control the geometry of your strokes. It contains only the option to add modifiers.

As they always completely apply to the strokes\xe2\x80\x99 geometry (like object modifiers do). They take the resulting two-dimensional strokes from the Freestyle line set and displace or deform them in various ways.

As with other modifier stacks in Blender, they are applied from top to bottom.

Line Style: Geometry.\xc2\xb6

2D Offset

2D Transform

Backbone Stretcher

B\xe3\x9zier Curve

Blueprint

Guiding Lines

Perlin Noise 1D

Perlin Noise 2D

Polygonization

Sampling

Simplification

Sinus Displacement

Spatial Noise

Tip Remover

--- Fin du chapitre: _geometry ---

--- Début du chapitre: _geometry_node --- The Geometry Node editor is used to edit Node Groups which are used by the Geometry Node Modifier. Such a node group can define many operations to modify an object's geometry.

A list of all Geometry Nodes is available in the modeling section.

Geometry Nodes can have multiple contexts depending on the intended function of the node group. Changing the context adjusts the user interface to best fit the needs of the selected context.

Used to create node groups that will be used by the Geometry Nodes Modifier.

Used to create node groups that will be used to create Node-Based Tools.

Standard view menu.

Menu for Selecting Nodes.

Menu for adding new Geometry Nodes.

Menu for Editing Nodes.

Data-Block Menu for creating and selecting node groups.

The pin button will keep the current node group selection fixed, instead of using the Active Modifier. When a node group is pinned, it will remain visible in the Geometry Node editor even when another object or modifier is selected elsewhere.

Jumps up a node group level. See Edit Group for details.

Snapping options. See Arranging Nodes for details.

See Overlays.

See Selecting Nodes.

See Annotations.

See Cut Links.

This tab gives access to the active node's properties.

This tab gives access to the active tool's settings.

This tab allows managing annotations.

This tab allows you to edit the current node group's inputs and outputs.

Tip

In the Geometry Node Modifier, you can specify values for the root node group's inputs, as well as select destination attributes for its outputs.

These popover menus are displayed in the header when the tool context is enabled. These properties determine where the tool is available in the user interface.

See Supported Modes & Data-Types for more information.

The Object Type the tool supports.

The node tree supports Mesh Objects.

The node tree supports Curve Objects.

The Object Mode the tool supports.

The node group can be used in Object mode.

The node group can be used in edit mode.

The node group can be used in Sculpt Mode.

Wait for a mouse click input (LMB) before running the operator from a menu. This is useful for the Mouse Position Node.

--- Fin du chapitre: _geometry_node ---

--- Début du chapitre: `_geometry_nodes` --- The Geometry Nodes modifier creates a modifier with a node group which defines its functionality.

A new Geometry Nodes modifier with a new node group.

This modifier is supported by mesh, curve, text, and volume objects. A Node Group with the geometry input and output. Those are respectively what is received and passed to the previous and next modifier in the stack. See Nodes for all available nodes.

A list of the node group's inputs which can have unique values even if the group is shared among multiple modifiers.

If the input is connected to a Field socket, there will be a toggle to switch between using a single value for the input or using an attribute on the input geometry. Using an attribute for input means the value can be different for every element.

The attribute name used by default when using the node group in a modifier for the first time is defined in the node group inputs panel.

Note

The attribute domain and the used to access the attribute is defined by the node the input is connected to.

Nodes that show a warning message in the node editor will also show that message here.

Custom warning messages can be created using the Warning Node.

By connecting a field socket to the group output node, you can create custom attributes from a field output of any node in the node tree. The domain of the attribute must be specified in the group node's output properties. Note, this does not work with Instanced Data.

The attribute name used by default when using the node group in a modifier for the first time is defined in the node group outputs panel.

This panel is hidden unless output node has attribute sockets.

Specifies where baked data should be stored. This can be overridden for individual bakes.

The baked data is packed into the .blend file. So no separate file is necessary.

The baked data is stored in a separate directory on disk.

Location on disk where the baked data for Simulation Zones and Bake Nodes are stored.

See also

Geometry Node Baking

This panel displays information about all custom named attributes used by the node group. More information is available in the geometry nodes inspection page.

Creates a new geometry node tree with the name of the current node tree with .wrapper appended to the name. This operation moves all inputs and outputs from the old modifier into a new node group. In order for this operator to function, there must be a Group Input and a Group Output each with a Geometry socket attached to the node group. This action causes all Output Attributes to become Internal Dependencies utilizing the Store Named Attribute Node. All modifier inputs will then also become inputs of the newly created node group.

This operator is useful to easily allow a node tree to be reused in other trees or to mark it as an Asset to be reused in other projects.

--- Fin du chapitre: `_geometry_nodes` ---

--- Début du chapitre: `_geometry_proximity` --- The Geometry Proximity node computes the closest location on the target geometry.

Tip

The Map Range Node is often helpful to use with the distance output of this node to create a falloff with a maximum distance.

Standard geometry input.

Splits the elements of the input geometry into groups which can be sampled individually.

The given position to calculate the closest location on the target.

Determines in which group the closest nearest element is detected.

Calculate the closest point anywhere on the faces of the target mesh geometry.

Calculate the closest point anywhere on the edges of the target mesh geometry.

Calculate the closest point or vertex on the target geometry. This mode is usually the fastest. This mode works for both point cloud and mesh geometry, the other modes only work for meshes.

Closest location on the surface of the target mesh, or the closest point in the target point cloud in Points mode.

Distance (as floating-point value) from the source position to the closest location in the target.

Whether the sampling was successful. It can fail when the sampled group is empty.

The different modes of the node: faces, edges and points. In this example the Geometry Nodes modifier is added on the target plane. Note that the larger plane is subdivided and the smaller plane is not.

The three target element modes: faces, edges, and points.

Points distributed on a sphere used as a target for a distance used in a shader.

--- Fin du chapitre: `_geometry_proximity` ---

--- Début du chapitre: `_geometry_to_instance` --- The Geometry to Instance node turns every connected input geometry into an instance. Visually, the node has a similar result as the Join Geometry Node, but it outputs the result as separate instances instead. The geometry data itself isn't actually joined.

The node can be used in combination with the Pick Instances option in the Instance on Points Node, as a way to pick between geometry generated in the node tree (as opposed to picking from separate instances from the Collection Info Node, for example).

Tip

This node can be much faster than the join geometry node when the inputs are large geometries. This is because the join geometry node must actually create a larger mesh, or a larger curve. Even though the operation is simple, just creating a large mesh can have a significant cost. This node can be better, because instead of merging large geometries, it just groups them together as instances.

Geometry that will be joined. Multiple inputs are allowed. When the node is muted, only the first link will be passed through.

This node has no properties.

Standard geometry output.

The node used in combination with the Instance on Points Node to choose between multiple primitives for instancing.

--- Fin du chapitre: `_geometry_to_instance` ---

--- Début du chapitre: `_gesture_tools` --- Separate from brushes and filters, Sculpt mode also has a set of tools that perform actions to a drawn selection area. These tools are similar to the selection tools (e.g. box selection and lasso selection in other areas of Blender).

These tools do not provide a selection of elements that are then modified, they directly modify the underlying mesh.

Dragging creates a rectangular area defined by where LMB was pressed and where LMB is released.

Hold to reposition the selection area.

Dragging creates a freeform area that follows the cursor defined by where LMB was pressed and where LMB is released.

Hold to reposition the selection area.

Helps to reduce jitter of the strokes while drawing by delaying and correcting the location of points.

Minimum distance from the last point before the stroke continues.

A smooth factor, where higher values result in smoother strokes but the drawing sensation feels as if you were pulling the stroke.

Dragging creates a line. The resulting action acts upon everything on the highlighted side of the line. The area acted upon is extended in both directions of the viewport.

Toggles the side of the line that the tool affects.

Hold to constrain the rotation of the line to user-specified intervals. Defaults to 5 degree increments, customizable via the Snapping menu indicated by the magnet icon in the header.

Hold to reposition the line.

The affected area will not extend the length of the drawn line. This helps defining a smaller area instead of extending the line infinitely long.

Clicking places a point in the viewport. Each time LMB is pressed, a new point of the polygon is created. Pressing LMB on the starting point, pressing LMB twice, or pressing Return closes the selection area.

Hold to reposition the selection area.

--- Fin du chapitre: `_gesture_tools` ---

--- Début du chapitre: `_getting_started` --- Creating an extension takes only a few steps:

Open the directory containing the add-on code or theme file.

Add a `blender_manifest.toml` file with all the required meta-data (name, maintainer, ...).

Use the Blender command-line tool to build the extension .zip file.

How to publish to the Blender Extensions Platform:

Install from Disk to test if everything is working well.

Upload the .zip file (this step requires Blender ID).

The extension will be held for review, and published once the moderation team approves it.

An extension is shared as a .zip archive containing a manifest file and other files. The expected files depend on the extension type.

Add-ons need at least the manifest and an `__init__.py` file, while more complex add-ons have a few different .py files or wheels together.

A theme extension only needs the manifest and the .xml theme file.

Note

Extensions can optionally have all its files inside a folder (inside the archive). This is a common behavior when saving a repository as ZIP from version-control platforms.

A manifest is a file with all the meta-data required for an extension to be processed. This example is a good starting point to the `blender_manifest.toml` that should be inside the `.zip`.

Required values:

Minimum supported Blender version - use at least 4.2.0 .

Unique identifier for the extension.

List of licenses ,use SPDX license identifier.

Maintainer of the extension.

Complete name of the extension.

Internal version of the file format - use 1.0.0 .

One-line short description, up to 64 characters - cannot end with punctuation.

`\xe2\x80\x9cadd-on\xe2\x80\x9d`, `\xe2\x80\x9ctheme\xe2\x80\x9d`.

Version of the extension - must follow semantic versioning.

Optional values:

Blender version that the extension does not support, earlier versions are supported.

Website for the extension.

Some licenses require a copyright, copyrights must be `\xe2\x80\x9cYear Name\xe2\x80\x9d` or `\xe2\x80\x9cYear-Year Name\xe2\x80\x9d`.

List of tags. See the list of available tags .

List of supported platforms. If omitted, the extension will be available in all operating systems. The available options are [`\xe2\x80\x9cwindows-x64\xe2\x80\x9d`, `\xe2\x80\x9cwindows-arm64\xe2\x80\x9d`, `\xe2\x80\x9cmacos-x64\xe2\x80\x9d`, `\xe2\x80\x9cmacos-arm64\xe2\x80\x9d`, `\xe2\x80\x9clinux-x64\xe2\x80\x9d`]

List of relative file-paths Python Wheels .

Add-ons can list which resources they require. The available options are files, network, clipboard, camera, microphone. Each permission should be followed by an explanation (short single-sentence, up to 64 characters, with no endpunctuation).

Optional values for `\xe2\x80\x9cbuild\xe2\x80\x9d`:

These values are only used by the build sub-command.

A list of file-paths relative to the manifest to include when building the package.

A list of file-path patterns to exclude include when building the package.

The pattern matching is compatible with gitignore.

Note that setting this value isn't supported when paths is also declared.

If the [build] table isn't declared the following default is used:

These values must not be declared in a TOML and are reserved for internal use.

[build.generated]

Note

All the values present in the manifest file must be filled (i.e., cannot be empty, nor `text""`, nor `list[]`).

If you don't want to set one of the optional values just exclude it from the manifest altogether.

Extensions can be built, validated & installed via command-line.

To build the package defined in the current directory use the following commands:

See build docs.

\xe2\x80\x94

To validate the manifest without building the package:

You may also validate a package without having to extract it first.

See validate docs.

See also

Extensions Command Line Arguments.

If you want to host the extensions yourself, see the Creating an

Extensions Repository docs.

--- Fin du chapitre: _getting_started ---

--- Début du chapitre: _gizmo ---Reference

All Modes

Gizmos

Clicking the icon toggles all the gizmos in the Movie Clip Editor. The drop-down button displays a popover with more detailed settings, which are described below.

Toggle the visibility of the zooming and panning gizmos in the upper right corner.

--- Fin du chapitre: _gizmo ---

--- Début du chapitre: _gizmos --- Grid with gizmo in the 3D

viewport.\xc2\xba

Gizmos allow changing inputs of Geometry Nodes directly in the 3D viewport. This is often more intuitive than controlling the inputs in the modifier or node editor.

Node inputs that can be controlled with a gizmo have an additional icon. The gizmos of that node are shown if it is selected. Clicking on the icon pins the gizmo, so that it is shown even if the node is not selected. Modifying the gizmo in the 3D viewport, modifies the value in the socket.

Example of a node group that has gizmos. \xc2\xba

Note

Built-in nodes do not have their own gizmos yet. It's possible to create node groups that have gizmos though.

Gizmos are often automatically propagated when an input socket with a gizmo is linked. The gizmo then controls the value that it is propagated to, instead of the input of the node group directly. Not all nodes support propagating gizmos, but many math operations do. A double link indicates that the propagation was successful.

The gizmo is propagated from the Size X input socket to the Value node. \xc2\xba

Gizmos can also be propagated to Group Inputs, in which case they are also available on the parent group node. If the current group is used by a modifier directly, the gizmo will also be available on the modifier. Gizmos that are propagated to the modifier always show when the modifier is active, independent of whether any node is visible or selected.

Adding custom gizmos to a node group that generates or modifies geometry can make it more convenient to use.

To add a gizmo to a node group, one has to use one of the gizmo nodes

. The main aspect that makes gizmos unintuitive at first is that there is

a bidirection dependency:changing the gizmo position changes the controlled value and vice versa.

The most simple custom gizmo setup is shown below.The Linear Gizmo node adds a gizmo that is drawn in the 3D viewport.The gizmo controls the value that is plugged into it.When trying this, you will notice that the gizmo always jumps back to the origin while the value is still changed.That is because the Position of the gizmo node does not depend on the value yet.

The simplest custom gizmo setup. \xc2\xbf

When the gizmo position is made dependent on the value, the gizmo works more than one would expect.It now also works in both directions: changing the value moves the gizmo and moving the gizmo changes the value.

Simple gizmo setup where the gizmo position depends on the controlled value. \xc2\xbf

Multiple values can be plugged into the Value input of gizmo nodes at once.In that case, all these values are modified at the same time when moving the gizmo.Multiply or divide nodes can be used if the values should change at different rates.

The Transform output of gizmo nodes should be joined into the geometry that the gizmo controls.This helps Blender to understand that the gizmos should be transformed together with the geometry later on.

Example showing how to add simple gizmos to the built-inGridnode.

\xc2\xbf

Note

Generally, it is possible to build the entire node group functionality first and to add gizmos afterwards.

--- Fin du chapitre: _gizmos ---

--- D but du chapitre: _glare --- The Glare node is used to add lens flares, fog,glows around bright parts of an image.

Standard color input.

Simulates the glow around bright objects caused by light scattering in eyes and cameras.

Scale of the glow relative to the size of the image. 9 means the glow can cover the entire image, 8 means it can only cover half the image, 7 means it can only cover quarter of the image, and so on.

Creates a haze over the image.

Creates bright streaks used to simulate lens flares.

Total number of streaks.

The rotation offset factor of the streaks.

Fade out factor for the streaks.

Simulates the glow around bright objects caused by light scattering in eyes and cameras.This is similar to the Bloom mode, but is more physically accurate, at the cost of much slower computation time.

Scale of the glow relative to the size of the image. 9 means the glow will cover the entire image, 8 means it will cover half the image, 7 means it will cover quarter of the image, and so on.

Works similar to Streaks but gives a simpler shape looking like a star.

Fade out factor for the streaks.

Rotate the streaks by 45 .

If not set to something other than High, then the glare effect will only be applied to a low resolution copy of the image.This can be helpful to save render times while only doing preview renders.

The number of times to run through the filter algorithm. Higher values will give more accurate results but will take longer to compute. Note that, this is not available for Fog Glow as it does not use an iterative-based algorithm.

Used for Streaks and Ghosts to create a special dispersion effect.

Johannes Itten describes this effect, Color Modulation, as subtle variations in tones and chroma.

Value to control how much of the effect is added on to the image. A value of -1 would give just the original image, 0 gives a 50/50 mix, and 1 gives just the effect.

Pixels brighter than this value will be affected by the glare filter.

Standard color output.

--- Fin du chapitre: _glare ---

--- Début du chapitre: _glass --- The GlassBSDF is used to add a Glass-like shader mixing refraction and reflection at grazing angles. Like the transparent shader, only pure white will make it transparent. The glass shader tends to cause noise due to caustics. Since the Cycles path tracing integrator is not very good at rendering caustics, it helps to combine this with a transparent shader for shadows; for more details see here. Color of the surface, or physically speaking, the probability that light is transmitted for each wavelength.

Influences sharpness of the refraction; perfectly sharp at 0.0 and smoother with higher values.

Index of refraction (IOR) defining how much the ray changes direction. At 1.0 rays pass straight through like transparent; higher values give more refraction.

Normal used for shading.

Microfacet distribution to use.

GGX microfacet distribution.

Takes multiple scattering events between microfacets into account. This gives more energy conserving results, which would otherwise be visible as excessive darkening.

Cycles Only Beckmann microfacet distribution.

Standard shader output.

Sharp Glass example. \xc2\xbb6

Sharp Glass behavior. \xc2\xbb6

Rough Glass example. \xc2\xbb6

Rough Glass behavior. \xc2\xbb6

--- Fin du chapitre: _glass ---

--- Début du chapitre: _glossy --- The GlossyBSDF node is used to add reflection with microfacet distribution, used for materials such as metal or mirrors.

Color of the surface, or physically speaking, the probability that light is reflected for each wavelength.

Sharpness of the reflection; perfectly sharp at 0.0 and smoother with higher values.

Controls the amount the reflection stretches the reflection along the surface of the material. A value of 0.0 results in no anisotropic reflections. Higher values give elongated highlights orthogonal to the tangent direction; negative values give highlights shaped along the tangent direction.

This is a phenomenon known as `\xe2\x80\x9cAnisotropic Reflections\xe2\x80\x9d` which is often seen in metallic materials. Rotation of the anisotropic tangent direction. Value 0.0 equals 0 rotation, 0.25 equals 90 and 1.0 equals 360 = 0. This can be used to texture the tangent direction.

Anisotropic rotation on 0.

Anisotropic rotation on 0.25 (90).

Normal used for shading; if nothing is connected the default shading normal is used.

Tangent used for shading; if nothing is connected the default shading tangent is used.

Microfacet distribution to use.

GGX microfacet distribution.

Takes multiple scattering events between microfacets into account. This gives more energy conserving results, which would otherwise be visible as excessive darkening.

Cycles Only Beckmann microfacet distribution.

Standard shader output.

Sharp Glossy example.

Sharp Glossy behavior.

Rough Glossy example.

Rough Glossy behavior.

Anisotropic shading with 0 rotation, 90 rotation and textured rotation of the tangent direction. Example blend-file.

--- Fin du chapitre: `_glossy` ---

--- Début du chapitre: `_glow` --- This effect makes parts of an image glow brighter by working on the luminance channel of an image. The Glow is the superposition of the base image and a modified version, where bright areas are blurred.

To `\xe2\x80\x9canimate\xe2\x80\x9d` the glow effect, mix it with the base image using the Gamma Cross effect, crossing from the base image to the glowing one.

Areas brighter than the Threshold are blurred.

The maximum luminosity that is added.

Multiplier of the brightness.

The size of the blur.

Improves the quality of the glow by giving smoother results but will be slower.

This checkbox allows you to only show/use the

`\xe2\x80\x9cmodified\xe2\x80\x9d` version of the image, without the base one.

Glow effect.

--- Fin du chapitre: `_glow` ---

--- Début du chapitre: `_goal` --- Reference

Physics `\xe2\x80\x93 Soft Body \xe2\x80\x93 Goal`

Enabling this tells Blender to use the motion from animations (F-Curves, armatures, parents, lattices, etc.) in the simulation. The `\xe2\x80\x9cgoal\xe2\x80\x9d` is the desired end position for vertices based on this animation.

See exterior forces for details.

Use a vertex group to allow per-vertex goal weights (multiplied by the Default goal).

The spring stiffness for Goal. A low value creates very weak springs (more flexible \xe2\x80\x9cattachment\xe2\x80\x9d to the goal), a high value creates a strong spring (a stiffer \xe2\x80\x9cattachment\xe2\x80\x9d to the goal).

The friction coefficient for Goal. Higher values give damping of the spring effect (little jiggle), and the movement will soon come to an end.

Goal weight/strength for all vertices when no Vertex Group is assigned. If you use a vertex group the weight of a vertex defines its goal.

When you use a vertex group, you can use the Minimum and Maximum to fine-tune (clamp) the weight values. The lowest vertex weight will become Minimum, the highest value becomes Maximum.

--- Fin du chapitre: _goal ---

--- Début du chapitre: _gpu_rendering --- GPU rendering makes it possible to use your graphics card for rendering, instead of the CPU. This can speed up rendering because modern GPUs are designed to do quite a lot of number crunching. On the other hand, they also have some limitations in rendering complex scenes, due to more limited memory, and issues with interactivity when using the same graphics card for display and rendering.

To enable GPU rendering, go into the Preferences \xe2\x80\x93 System \xe2\x80\x93 Cycles Render Devices, and select either CUDA, OptiX, HIP, oneAPI, or Metal. Next, you must configure each scene to use GPU rendering in Properties \xe2\x80\x93 Render \xe2\x80\x93 Device.

Blender supports different technologies to render on the GPU depending on the particular GPU manufacturer and operating system.

CUDA is supported on Windows and Linux and requires an Nvidia graphics cards with compute capability 3.0 and higher. To make sure your GPU is supported, see the list of Nvidia graphics cards with the compute capabilities and supported graphics cards.

OptiX is supported on Windows and Linux and requires an Nvidia graphics cards with compute capability 5.0 and higher and a driver version of at least 495.89. To make sure your GPU is supported, see the list of Nvidia graphics cards.

OptiX takes advantage of hardware ray-tracing acceleration in RTX graphics cards, for improved performance.

GPU acceleration for OpenImageDenoise is available for compute capability 7.0 and higher, which includes all NVIDIA RTX cards.

HIP is supported on Windows and Linux and requires an AMD graphics card with the RDNA1 architecture or newer. Both discrete GPUs and APUs are supported.

Supported GPUs include:

Radeon RX 5000 Series

Radeon RX 6000 Series

Radeon RX 7000 Series

Radeon Pro W6000 Series

Minimum driver versions:

Windows: Radeon Software 21.12.1 or Radeon PRO Software 21.Q4

Linux: Radeon Software 22.10 or ROCm 5.3

Please refer to AMD's website for more information about AMD graphics cards and their architectures.

Experimental hardware ray-tracing support is available with the most recent drivers. This can be enabled in the preferences. However there are

currently known issues regarding motion blur, hair and point cloud rendering, and degenerate triangle shapes.

GPU accelerated denoising is available on discrete Radeon RX 6000 and Radeon RX 7000 GPUs.

oneAPI is a computation library that is supported on Windows and Linux and requires an Intel Arc graphics card with the Xe HPG architecture. Hardware acceleration for ray-tracing and denoising is supported.

Supported GPUs include:

Intel Arc A-Series

Minimum driver versions:

Windows: Intel Graphics Driver XX.X.101.5518

Linux: intel-level-zero-gpu package 1.3.27642, typically available through the intel-compute-runtime package XX.XX.27642.38

Please refer to Intel's website for more information about Intel graphics cards and their architectures.

GPU accelerated denoising is available on all supported GPUs.

Metal is supported on Apple computers with Apple Silicon. macOS 13.0 or newer is required to support all features.

GPU accelerated denoising is available on Apple Silicon.

Path Guiding is not supported on any GPU.

Open Shading Language is only supported for OptiX, with some limitations listed in the documentation.

On older GPU generations, graphics cards can only either render or draw the user interface. This can make Blender unresponsive while it is rendering. Heavy scenes can also make Blender unresponsive on newer GPUs, when using a lot of memory or executing expensive shaders, however this is generally less of a problem.

The only complete solution for this is to use a dedicated GPU for rendering, and another for display.

There may be multiple causes, but the most common one is that there is not enough memory on your graphics card. Typically, the GPU can only use the amount of memory that is on the GPU (see Would multiple GPUs increase available memory? for more information). This is usually much smaller than the amount of system memory the CPU can access. With CUDA, OptiX, HIP and Metal devices, if the GPU memory is full Blender will automatically try to use system memory. This has a performance impact, but will usually still result in a faster render than using CPU rendering.

Yes, go to Preferences > System > Compute Device Panel, and configure it as you desire.

Typically, no, each GPU can only access its own memory.

The exception is NVIDIA GPUs connected with NVLink, where multiple GPUs can share memory at a small performance cost. This can be enabled with Distributed Memory Across Devices in the preferences.

This varies depending on the hardware used. Different technologies also have different compute times depending on the scene tested. For the most up to date information on the performance of different devices, browse the Blender Open Data source.

In case of problems, be sure to install the official graphics drivers from the GPU manufacturers website, or through the package manager on Linux. The graphics drivers provided by the computer manufacturer can sometimes be outdated or incomplete.

This usually means there is not enough memory to store the scene for use by the GPU.

Note

One way to reduce memory usage is by using smaller resolution textures. For example, 8k, 4k, 2k, and 1k image textures take up respectively 256MB, 64MB, 16MB and 4MB of memory.

If a GPU is used for both display and rendering, Windows has a limit on the time the GPU can do render computations. If you have a particularly heavy scene, Cycles can take up too much GPU time. Reducing Tile Size in the Performance panel may alleviate the issue, but the only real solution is to use separate graphics cards for display and rendering.

Another solution can be to increase the time-out, although this will make the user interface less responsive when rendering heavy scenes. Learn More Here.

On Linux, depending on your GCC version you might get this error. See the Nvidia CUDA Installation Guide for Linux for a list of supported GCC versions. There are two possible solutions to this error:

If you have an older GCC installed that is compatible with the installed CUDA toolkit version, then you can use it instead of the default compiler. This is done by setting the `CYCLES_CUDA_EXTRA_CFLAGS` environment variable when starting Blender.

Launch Blender from the command line as follows:

(Substitute the name or path of the compatible GCC compiler).

If the above is unsuccessful, delete the following line in `/usr/local/cuda/include/host_config.h`:

This will allow Cycles to successfully compile the CUDA rendering kernel the first time it attempts to use your GPU for rendering. Once the kernel is built successfully, you can launch Blender as you normally would and the CUDA kernel will still be used for rendering.

This error may happen if you have a new NVIDIA graphics card that is not yet supported by the Blender version and CUDA toolkit you have installed. In this case Blender may try to dynamically build a kernel for your graphics card and fail.

In this case you can:

Check if the latest Blender version (official or experimental builds) supports your graphics card.

If you build Blender yourself, try to download and install a newer CUDA developer toolkit.

Normally users do not need to install the CUDA toolkit as Blender comes with precompiled kernels.

--- Fin du chapitre: `_gpu_rendering` ---

--- Début du chapitre: `_grab` --- Reference

Sculpt Mode

Sidebar `\xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3 Advanced \xe2\x80\xa3 Brush Type`

Drag geometry across the screen, following the cursor. Grab only moves the vertices that are under the brush radius at the start of the stroke. This is an essential sculpting brush to be used frequently to build shapes and adjust proportions.

Note

The effect is similar to moving geometry in Edit Mode with Proportional Editing enabled, except that Grab can make use of other Sculpt Mode options and brush settings.

Pressure Sensitivity is not supported for this brush type. More info at Radius.

Pressure Sensitivity is not supported for this brush type. More info atStrength.

For this brush, this setting is a purely visual change.It does not alter the brush behavior. More info atNormal Radius.

This setting is not supported. More info atAuto-Smooth.

Note

More info at General brush settingsand on Advanced brush settings.

Applies the maximum strength of the brush to the highlighted active vertex,making it easier to manipulate low-poly models or meshes with modifiers.

Enabling this option also enables a white wireframe overlay within the brush radius.This helps to visualize the real base geometry that is being manipulatedwhile sculpting withModifiers.

Preserves the object's silhouette shape by only grabbing vertices on one side of the mesh curvature.The shape of the silhouette is determined by the orientation of the 3D Viewport and the start of the stroke.

Note how in the image only the bottom side of the leg is pulled down, despite the size of the brush.

Tip

This setting is also useful for grabbing a single side of a creaseand pushing it further inwards, creating a more pinched crease.

If the Falloff Shape is set to Projected,the brush can grab infinitely deep into the viewport.This is especially useful for much broader changes to a sculpt.

The stroke can also be started outside of the mesh (like in empty 3D space)and grab the vertices within the brush radius.This can be useful for sculpting flat and tube-like meshes.

--- Fin du chapitre: _grab ---

--- Début du chapitre: _gradient ---Note

This node is ported from shader nodes. The manual and images arereferencing the shader version of the node.This node accepts field inputs and outputs.When not connected the Vector input has an implicit position attribute value.

The Gradient Texturenode generates interpolated color and intensity values based on the input vector.

Texture coordinate to sample texture at;defaults to Generated texture coordinates if the socket is left unconnected.

Controls the type of gradient generated.

Directly outputs the input X coordinate.

Interpolates the input X coordinate quadratically.

Uses a combination of quadratic and linear interpolationto generate a smooth gradient from the input X coordinate.

Averages the input X and Y coordinates.

Creates an inverse gradient using the length of the input vector; the maximum value is at (0, 0, 0).

The same as Spherical, except interpolated quadratically.

Outputs a value based on the angle of the input around the Z axis.

Texture color output.

Texture intensity output.

Gradient texture using object coordinates. \xc2\xb6

--- Fin du chapitre: _gradient ---

--- Début du chapitre: `_graph` --- Graph View.

The graph or curves view has numerous purposes based on the color of the lines. The red and green lines on the graph show you the speed of the trackers at a given frame. Green is vertical movement, Red is horizontal. Therefore the first frames will always be at zero. The blue line is the line that comes out when you click on the film strip is the average per-frame error. This curve is available only after pressing camera solve and is not editable. This is the one line that you want to be as flat as possible and as closer to zero as you can. The high points will show you where in your shot you are having inaccurate tracking.

Frames outside of scene frame range are darkened.

Displays the graph for only selected trackers.

Displays channels from objects that are hidden.

Display options, defines what curves are visible.

Visualizes per-frame average reprojection error of all tracks in the active tracking object.

Shows curves for X and Y speed of tracks.

Per-frame reprojection error of tracks.

The curves are useful to see if particular trackers are moving differently than the average. A line that spikes from the rest of the curve usually means a tracking error.

You can manually edit the curve by selecting a point in the curve and dragging it or deleting, that will affect the corresponding tracker on that particular frame.

Locks the view to selected markers during playback.

--- Fin du chapitre: `_graph` ---

--- Début du chapitre: `_gravity` ---Reference

Scene `\xe2\x80\xa3 Gravity`

Gravity is a global setting that is applied to all physics systems in a scene. It can be found in the scene tab. This value is generally fine left at its default, -9.810 on the Z axis, which is the force of gravity in the real world. Changing this value would simulate a lower or higher force of gravity. Gravity denoted g , measurement $m \cdot s^{-2}$.

Gravity is applied in the same way to all physics systems.

Gravity is practically the same around the entirety of planet Earth. For rendering scenes on The Moon, use $-1.622 m \cdot s^{-2}$ on the Z axis. Another popular gravity value might be for Mars which has a gravitation acceleration of $-3.69 m \cdot s^{-2}$ on the Z Axis.

Note

The gravity value per physics system can be scaled down in the Field Weightstab.

--- Fin du chapitre: `_gravity` ---

--- Début du chapitre: `_grease_pencil` ---Reference

Render `\xe2\x80\xa3 Grease Pencil`

This panel is comprised of settings to control the rendering of Grease Pencil Lines.

Threshold for the edge detection algorithm used to correct aliasing, higher values might over blur some part of the image.

--- Fin du chapitre: `_grease_pencil` ---

--- Début du chapitre: _grease_pencil_menu ---Strokes can be edited by transforming the locations of points.

Reference

Edit Mode

Toolbar \xe2\x80\xa3 Move, Rotate, Scale

Grease Pencil \xe2\x80\xa3 Transform \xe2\x80\xa3 Move, Rotate, Scale
G, R, S

Like other elements in Blender, points and strokes can be moved G, rotated R or scaled S as described in the Basic Transformations section. When in Edit Mode, Proportional Editing is also available for the transformation actions.

Basic move, rotate and scale transformations for selected points/strokes. See Move, Rotate, Scale Basics for more information.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Transform

Toolbar \xe2\x80\xa3 Bend/Shear

The Bend, Shear, To Sphere, Extrude and Shrink Fat transform tools are described in the Editing tools section.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Mirror

Ctrl- M

The Mirror tool is also available, behaving exactly the same as with mesh vertices .

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Snap

Shift- S

Mesh snapping also works with Grease Pencil components.

Reference

Edit Mode, Draw Mode

Grease Pencil \xe2\x80\xa3 Active Layer

Y

Select the active layer.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Animation

I

The stroke animation operations are described in the Animation section.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Interpolate Sequence

See Interpolate Sequence .

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Duplicate

Shift- D

Duplicates the selected elements, without creating any connections with the rest of the strokes (unlike Extrude, for example), and places the duplicate at the location of the original elements.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Copy

Ctrl- C

Copy the selected points/strokes to the clipboard.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Paste

Ctrl- V

Paste Grease Pencil points or strokes from the internal clipboard to the active layer.

Add pasted strokes behind all strokes.

Keep the world transform of strokes from the clipboard unchanged.

Reference

Edit Mode, Draw Mode

Grease Pencil \xe2\x80\xa3 Weights

The stroke weight operations are described in the Weights Menu section.

Contains operators to adjust the visibility of points and strokes in the viewport.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Show/Hide \xe2\x80\xa3 Show All Layers

Alt- H

Shows all Grease Pencillayers.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Show/Hide \xe2\x80\xa3 Hide Active Layer

H

Hides the active Grease Pencillayers.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Show/Hide \xe2\x80\xa3 Hide Active Layer

Shift- H

Hides the all Grease Pencillayersexcept the active layer.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Separate

P

Separate the selected geometry into a new Grease Pencil object.

Separate the selected geometry into a new object.

Separates the geometry by each material.

Separate all the strokes in the active layer into a new object. See 2D

Layers for more information.

These tools help to cleanup degenerate geometry on the strokes.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Clean Up \xe2\x80\xa3 Delete Loose Points

Removes strokes with only a few points.

The number of points to consider a stroke as loose.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Clean Up \xe2\x80\xa3 Delete Duplicate Frames

Removes any duplicate keyframes.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Clean Up \xe2\x80\xa3 Merge by Distance

Simplifies a stroke by merging the selected points that are closer than a specified distance to each other. Note, unless using Unselected, selected points must be contiguous, else they will not be merged.

Sets the distance threshold for merging points.

Allows points in selection to be merged with unselected points. When disabled, selected points will only be merged with other selected ones.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Clean Up \xe2\x80\xa3 Reproject

Sometimes you may have drawn strokes unintentionally in different locations in the 3D space but they look right from a certain plane or from the camera view. You can use Reproject to flatten all the selected strokes from a certain viewpoint.

Reproject selected strokes onto the front plane (XZ).

Reproject selected strokes onto the side plane (YZ).

Reproject selected strokes onto the top plane (XY).

Reproject selected strokes onto the current view.

Reproject selected strokes onto the mesh surfaces.

When Surface Mode is activated controls the stroke offset from the object.

Reproject selected strokes onto 3D cursor rotation.

Maintains the original strokes after applying the tool.

Original drawing from the front view. \xc2\xbb6

Original drawing in the 3D Viewport. \xc2\xbb6

Strokes reprojected onto the front plane to fix strokes misalignment.

\xc2\xbb6

Drawing after reprojection operation from the front view. \xc2\xbb6

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Delete

X, Delete

Opens a pop-up menu with operators to remove geometry from the grease pencil object.

Deletes all the strokes at the current frame and in the current layer/channel.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Delete \xe2\x80\xa3 Delete

Deletes the selected points. When only one point remains, there is no more visible stroke, and when all points are deleted, the stroke itself is deleted.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Delete \xe2\x80\xa3 Dissolve

Ctrl- X

Dissolving removes points between other points and connect the remaining points.

Ctrl- X Opens a pop-up to choose the dissolve type.

Deletes the selected points without splitting the stroke. The remaining points in the strokes stay connected.

Deletes all the points between the selected points without splitting the stroke. The remaining points in the strokes stay connected.

Deletes all the points that are not selected in the stroke without splitting the stroke. The remaining points in the strokes stay connected.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Delete \xe2\x80\xa3 Delete Active Keyframe
(Active Layer)

Deletes all the strokes at the current frame in the active layer.

Reference

Edit Mode

Grease Pencil \xe2\x80\xa3 Delete \xe2\x80\xa3 Delete Active Keyframes
(All Layers)

Shift- Delete

Deletes all the strokes at the current frame in all layer.

--- Fin du chapitre: _grease_pencil_menu ---

--- Début du chapitre: _grease_pencil_pdf --- The Portable Document
Format (PDF) is use for interchanging PDFs between applications,it
support the export of Grease Pencil animation creating one page in the
PDF document for each keyframe selected.

Warning

The exporter only works in Object Mode.

Determine which objects will be included in the export.

Export only the active Grease Pencil object.

Export all selected Grease Pencil objects.

Export all visible Grease Pencil object in the scene.

Determine which frames will be included in the export.

Export only the active keyframe.

Export all selected keyframes as different PDF pages.

Export all frames as different PDF pages.

Note

To enable multi-keyframe selection you must enable Multiframe
Edition.SeeMultiframe Editionfor more information.

Precision for the stroke sampling. Low values mean a more accurate
result.

When enabled, Export the Grease Pencil strokes fill.

When enabled, Export strokes with constant thickness.

Note

The export of the Grease Pencil strokes is always from camera view.

--- Fin du chapitre: _grease_pencil_pdf ---

--- Début du chapitre: _grease_pencil_svg --- The Scalable Vector
Graphics (SVG) format is use for interchanging vector based illustrations
between applicationsand is supported by vector graphics editors such as
Inkscape, and modern browsers among others.

Warning

The exporter only works in Object Mode.

Reference

File \xe2\x80\xa3 Import \xe2\x80\xa3 SVG as Grease Pencil
Resolution for generated strokes.

Generated strokes scale.

Reference

File \xe2\x80\xa3 Export \xe2\x80\xa3 Grease Pencil as SVG
Determine which objects include in the export.

Export only the active Grease Pencil object.

Export all selected Grease Pencil objects.

Export all visible Grease Pencil object in the scene.

Precision for the stroke sampling. Low values mean a more accurate result.

When enabled, Export the Grease Pencil strokes fill.

When enabled, Export strokes with constant thickness.

When enabled, and camera view is active export only the strokes clipped from camera view.

--- Fin du chapitre: `_grease_pencil_svg` ---

--- Début du chapitre: `_grease_pencil_to_curves` --- The Grease Pencil to Curves node converts each Grease Pencil layer into an instance that contains curves.

Standard Grease Pencil geometry.

Selects the layers to convert.

Create a separate curve instance for every layer.

Tip

This is equivalent to a Realize Instances operation after creating the curve geometry instances. This can simplify some logic later on, but typically keeping this operation off will be much better for performance.

This node has no properties.

Standard curves geometry.

--- Fin du chapitre: `_grease_pencil_to_curves` ---

--- Début du chapitre: `_grid` --- The Grid node generates a planar mesh on the XY plane.

Side length of the plane in the X direction.

Side length of the plane in the Y direction.

Number of vertices in the X direction. If this is smaller than two, no mesh is generated.

Number of vertices in the Y direction. If this is smaller than two, no mesh is generated.

This node has no properties.

Standard geometry output.

A 2D vector representing the default X/Y coordinates of the UV Map for the primitive's shape. This can be connected to the Store Named Attribute Node, to be used once the Geometry Nodes Modifier gets applied. The UV map must be stored on the face corner in order to be accessed.

--- Fin du chapitre: `_grid` ---

--- Début du chapitre: `_grid_fill` --- Reference

Edit Mode

Face's Grid Fill

Grid Fill uses a pair of connected edge loops or a single, closed edge loop to fill in a grid that follows the surrounding geometry.

The best predictable result can be achieved if you select two opposite edge loops with an equal number of vertices. When a single, closed edge loop is selected, the Span/Offset options allow you to adjust the way two opposite edge loops are detected from one closed edge loop.

Specifies the number of columns in the grid.

Defines the vertex that is considered to be the corner of the grid, by default, it's the active vertex. The Offset allows you to rotate the grid lines.

Use a simple interpolation algorithm to generate grid vertices from boundary loops, which does not try to maintain the shape, useful for flat surfaces or times when keeping the shape gives odd results.

Input. \xc2\xba

Grid Fill result. \xc2\xba

--- Fin du chapitre: `_grid_fill` ---

--- Début du chapitre: `_group` --- AGroup Node combines a set of nodes into a single one, and selectively exposes inputs and outputs of those nodes.

Group nodes can simplify a node tree by hiding away complexity and reusing functionality.

Exposes the inputs of the node group. You can have multiple of these nodes in your tree to keep it clean, bringing in each input right where you need it (rather than dragging long links all across your graph).

The input slots can be edited in the Grouptab of the Sidebar.

Receives the outputs of the node group. You can have multiple of these nodes in your tree to keep it clean, outputting each result right where it's produced (rather than dragging long links all across your graph).

The output slots can be edited in the Grouptab of the Sidebar.

This section lists all the node groups, both those in the current blend-file and those Linked or Appended from another blend-file.

--- Fin du chapitre: `_group` ---

--- Début du chapitre: `_groups` --- AGroup Node combines a set of nodes into a single one, and selectively exposes inputs and outputs of those nodes.

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The output slots can be edited in the Grouptab of the Sidebar.

This section lists all the node groups, both those in the current blend-file and those Linked or Appended from another blend-file.

--- Fin du chapitre: `_groups` ---

--- Début du chapitre: `_grow_shrink_curves` --- Change the length of existing curves preserving the amount of control points and resampling the curve to preserve the original shape.

Determines whether to grow or shrink the curves. It can be toggled holding Ctrl while sculpting.

Grow or shrink curves by changing their size uniformly instead of using trimming or extrapolation.

Avoid shrinking curves shorter than this length.

--- Fin du chapitre: `_grow_shrink_curves` ---

--- Début du chapitre: `_guides` --- Reference

Physics \xe2\x80\xa3 Fluid \xe2\x80\xa3 Guides

Domain

Fluid guides are used to apply forces onto the simulation. They are like simple external forces but also seek to preserve the physically accurate flow of the fluid. The Guides panel allows you to adjust guiding forces globally, i.e. for the entire domain. Enabling the guides hints the fluid solver to use the more accurate, but also computationally more expensive pressure solving step.

Even when there are no guiding objects baked or there is no guiding domain attached, the fluid solver will still perform the more expensive pressure guiding algorithm if guiding is enabled. It is therefore recommended to only enable Guides when there is a clear intention to use guiding in the simulation.

See also

Fluid guiding is an implementation of Primal-Dual Optimization for Fluids.

Controls the lag of the guiding. A larger value (also known as the α guiding value) results in a greater lag.

This setting determines the size of the vortices that the guiding produces. A greater guiding size (also known as the blur radius or β guiding value) results in larger vortices.

All guiding velocities are multiplied by this factor. That is, every cell of the guiding grid, which has the same size as the domain object, is multiplied by this factor.

Guiding velocities can either come from objects that move inside the domain or from other fluid domains.

All effector objects inside the domain will be considered for the global guiding velocity grid. Once effector objects have been baked it is not possible to change the fluid domain resolution anymore.

When using another fluid domain as the guiding velocity source this domain may have a different resolution and may also be of a different type (e.g. the guiding domain is of type Gas while the actual domain with the guiding effect in it is of type Liquid).

In order to use a domain as the velocity source, this domain needs to be baked already.

When using Domains as the velocity source, this field serves to select the guiding domain object.

This option is only available when using the Modular cache type and when using Effectors as the Velocity Source. Bake Guides writes vertex velocities of effector objects to drive. It is meant to be used before baking the fluid simulation.

The progress will be displayed in the status bar. Pressing Esc will pause the simulation.

Once the simulation has been baked, the cache can be deleted by pressing Free Guides. It is possible to pause or resume a Bake Guides process.

--- Fin du chapitre: _guides ---

--- Début du chapitre: _guiding_lines --- The Guiding Lines modifier replaces a stroke by a straight line connecting both of its ends.

Offset the start and end points along the original stroke, before generating the new straight one.

This modifier will produce reasonable results when strokes are short enough, because shorter strokes are more likely to be well approximated by straight lines. Therefore, it is recommended to use this modifier together

with one of the splitting options (by 2D angle or by 2D length) from the Strokespanel.

Guiding Lines modifier Demo by T.K. (blend-file).

--- Fin du chapitre: _guiding_lines ---

--- Début du chapitre: _hair --- Cycles Only

The HairBSDFnode is used to add shading forHair.

Color of the hair.

Controls the way the light is rotated (angular shift) for the reflection/transmission.

Reflection Offset.

Transmission Offset.

Controls the roughness in the direction light is skewed, and perpendicular to it.

Roughness when using the Reflection Component.

Roughness when using the Transmission Component.

Input tangent.

There are two components that can be used to control the look of the hair. Usually you are going to want each of these and use a Mix Node.

The light that bounces off the surface of the hair.

The light that passes through the hair and exits on the other side.

With Mix node: 0 is full Reflection, 1 is full Transmission.

Standard shader output.

--- Fin du chapitre: _hair ---

--- Début du chapitre: _hair_attachment_info --- Reads attachment information regarding a surface mesh.

Surface geometry of the curve attachment.

Surface UV map stored on the mesh used for finding curve attachment locations.

This node has no properties.

Surface attachment UV coordinate stored on each curve.

Whether the stored attachment UV coordinate is valid.

Normal direction of the surface mesh at the attachment point.

--- Fin du chapitre: _hair_attachment_info ---

--- Début du chapitre: _hair_curves_noise --- Deforms hair curves using a noise texture.

Geometry

Apply offset cumulatively (previous points affect points after).

Overall factor for the deformation.

Overall distance factor for the deformation.

Shape of amount along each curve (0=constant, 0.5=linear).

Scale of the noise texture by root position.

Scale of noise texture along each Curve.

Random offset of noise texture for each Curve.

Seed value for randomization.

Preserve the length of the Curves on a segment basis.

This node has no properties.

Geometry

Offset Vector

--- Fin du chapitre: _hair_curves_noise ---

--- Début du chapitre: `_hair_info` --- The Curves Info node gives access to Hair information.
 This node has no inputs.
 This node has no properties.
 Outputs 1 when the shader is acting on a strand, otherwise 0.
 The point along the strand where the ray hits the strand (1 at the tip and 0 at the root).
 The total measurement from the root to the tip of the strand, interpreted as a grayscale value from 0 to infinity.
 The thickness of the strand at the point where the ray hits the strand.
 Tangent normal of the strand.
 A random per-curve value in the range from 0 to 1. It can for example be used in combination with a color ramp, to randomize the curve's color.
 --- Fin du chapitre: `_hair_info` ---

--- Début du chapitre: `_hair_principled` --- Cycles Only
 The Principled Hair BSDF is a physically-based, easy-to-use shader for rendering hair and fur.
 Tip
 Realistic hair should have a minimum of variance between each strand. The shader allows for this by specifying two values, Random Color and Random Roughness, which remap the specified Melanin/Roughness values to the range $[(Color/Roughness) \cdot Randomization, 1]$.
 The RGB color of the strand. Only used in Direct coloring.

Hint
 The chosen color is converted to an absorption coefficient with the following formula (section 4.2 of [CBTB16]):

$$\mu_a = \frac{1}{\beta} \ln \left(\frac{1}{1 - \beta} \right)$$
 where β is the radial roughness of the hair after applying randomization (if specified).
 Coloring hair using the Direct coloring parametrization. (The numbers on top are the RGB values.)
 Absolute quantity of pigment. Range $[0, 1]$ equivalent to $[0\%, 100\%]$.

Hint
 This is a linear mapping to the underlying exponential function:

$$Melanin = \frac{1}{\ln(1.0 - MelaninRedness)} \cdot \ln(1.0 - MelaninRedness)$$
 Ratio of pheomelanin to eumelanin. Range $[0, 1]$ equivalent to $[0\%, 100\%]$.
 Hint
 The ratio formula is:
$$eumelanin = \frac{1.0 - MelaninRedness}{1.0 - MelaninRedness}$$
 The resulting quantities are converted (after randomization, if specified) to absorption concentration via the following formula (section 6.1 of [EFHLA11], adjusted for the range $[0, 1]$):

$$MelaninRedness = \frac{1.0 - e^{-\mu_a}}{1.0}$$
 Color used for dyeing the hair after applying the melanin pigment. It is not subject to randomization. It can be disabled by setting the color to white.
 Hint
 This is converted via the Color mapping above and added to the absorption coefficient of the melanin concentration.
 Tint, using Melanin 0.1 and the corresponding RGB values.
 Attenuation coefficient σ .

Index of refraction (IOR) defining how much the ray changes direction. At 1.0 rays pass straight through like in a transparent material; higher values give more refraction. Default value is 1.55 .

Tilts the glint of the hair by increasing the angle of the scales of the hair's cuticle with respect to the hair shaft. Human hair usually has low values.

For each strand, vary the melanin concentration by $\text{RandomFactor} \cdot \text{Range}([0, 1])$ equivalent to $[0\%, 100\%]$ of the initial melanin concentration.

Hint

The melanin concentration is multiplied by randomFactor , where $\text{randomFactor} = 1.0 + 2.0 \cdot (\text{Random} - 0.5)$ RandomColor .

Random Color.

For each strand, vary both Roughness values by $\text{RandomFactor} \cdot \text{Range}([0, 1])$ equivalent to $[0\%, 100\%]$ of the initial roughness values.

Hint

The applied formula is the same one as for Random Color.

Random Roughness.

Random number source. If no node is connected here, it is automatically instantiated with the value obtained from Hair Info Random .

The Chiang model is based on a Gaussian distribution with separate roughness along and orthogonal to the hair.

Specify how much the glints are smoothed in the direction of the hair shaft. Too low values will smoothen the hair to the point of looking almost metallic, making glints look like fireflies; while setting it too high will result in a Lambertian look.

Roughness.

Specify how much the glints are smoothed in the direction of the hair normal. Too low values will concentrate the glint; while setting it too high will spread the light across the width of the strand.

Hint

Mathematically, this parameter is mapped to the logistic distribution's scale factor s (section 4.1 of [CBTB16]).

Radial Roughness.

Simulate a shiny coat of fur, by reducing the Roughness to the given factor only for the first light bounce (diffuse). $\text{Range}([0, 1])$ equivalent to a reduction of $[0\%, 100\%]$ of the original Roughness.

Coat.

The Huang model is based on microfacet based reflection and transmission, and supports elliptically shaped hair.

The ratio of the minor axis to the major axis of an elliptical cross-section. Recommended values are $0.8 \sim 1$ for Asian hair, $0.65 \sim 0.9$ for Caucasian hair, $0.5 \sim 0.65$ for African hair. The major axis is aligned with the curve normal, which can be created with geometry nodes, but is not supported in legacy particle hair.

Microfacet roughness for reflection and transmission.

Optional factor for modulating the first light bounce off the hair surface. The color of this component is always white. Keep this 1.0 for physical correctness.

Optional factor for modulating the transmission component. Picks up the color of the pigment inside the hair. Keep this 1.0 for physical correctness.

Optional factor for modulating the component which is transmitted into the hair, reflected off the backside of the hair and then transmitted out of the hair. This component is oriented approximately around the incoming direction, and picks up the color of the pigment inside the hair. Keep this 1.0 for physical correctness

The shader provides three different ways, or parametrizations, to color the hair strands.

Choose the desired RGB color and the shader will approximate the necessary absorption coefficient (below).

This mode defines the color as the quantity and ratio of the pigments which are commonly found in hair and fur, eumelanin (prevalent in brown-black hair) and pheomelanin (red hair). The quantity is specified in the Melanin input, and the ratio between them in Melanin Redness. Increasing concentrations darken the hair (the following are with Melanin Redness $\backslash\backslash(1\backslash\backslash)$):

White (Melanin $\backslash\backslash(0\backslash\backslash)$)

Blonde (Melanin $\backslash\backslash(0.25\backslash\backslash)$)

Reddish (Melanin $\backslash\backslash(0.5\backslash\backslash)$)

Brown (Melanin $\backslash\backslash(0.75\backslash\backslash)$)

Black (Melanin $\backslash\backslash(1\backslash\backslash)$)

Additionally, the Tint input allows to dye the hair with the desired color.

Specifies the attenuation coefficient $\backslash\backslash(\sigma_a\backslash\backslash)$, as applied by the Beer-Lambert law. This mode is intended mainly for technical users who want to use coefficients from the literature without any sort of conversion.

Standard shader output.

This shader is an implementation of the papers by Chiang et al. [CBTB16] and Huang et al. [HHH22].

Chiang, M. J. , Bitterli, B. , Tappan, C. and Burley, B. (2016), A Practical and Controllable Hair and Fur Model for Production Path Tracing. Computer Graphics Forum, 35: 275-283. doi:10.1111/cgf.12830
Eon, E. , Francois, G. , Hill, M. , Letteri, J. and Aubry, J. (2011), An Energy Conserving Hair Reflectance Model. Computer Graphics Forum, 30: 1181-1187. doi:10.1111/j.1467-8659.2011.01976.x

Huang W., Hullin M.B. Hanika J. (2022), A Microfacet-based Hair Scattering Model. Computer Graphics Forum, 41: 79-91. doi:10.1111/cgf.14588

--- Fin du chapitre: `_hair_principled` ---

--- Début du chapitre: `_handle_type_selection` --- Creates a selection based on the handle types of the control points.

The handle type of each control point can be changed with the Set Handle Type Node.

This node has no inputs.

Whether to consider left and/or right handles. When both are selected, the output value is true if either of the handles are of the chosen type.

Consider the left handle.

Consider the right handle.

Handle type for which the selection will be true. See the `Bzier` curves page for more detail on the different handle types.

Boolean field set to true wherever the handle type matches.

--- Fin du chapitre: `_handle_type_selection` ---

--- Début du chapitre: `_hardware` ---A full HD display or higher is recommended. Multi-monitor setups are supported, and workspaces can be configured to span multiple monitors.

Example of Blender's multi-monitor support.

Blender supports various types of input devices:

Keyboard (recommended: keyboard with numeric keypad, English layout works best)

Mouse (recommended: three button mouse with scroll wheel)

Graphic Tablet

Touchpad

NDOF Device (also known as 3D Mouse)

Note

If you don't have a middle mouse button or numpad, you can emulate these in the Input Preferences.

If you do not have a 3 button mouse, you will need to emulate it by checking the option in the Preferences.

The following table shows the combinations used:

3-button Mouse

LMB

MMB

RMB

2-button Mouse

LMB

Alt- LMB

RMB

If you do not have a numpad on the side of your keyboard, you may want to emulate one. You can then use the number row at the top of the keyboard instead, but will no longer have access to these keys' original functions (such as switching between vertex/edge/face selection in Edit Mode).

See also

Read more about Numpad Emulation in the Preferences.

If you use a keyboard with a non-English layout, you may still benefit from switching to the UK or US layout while working with Blender.

Note

You can also change the keymap from the Preferences. However, this manual assumes you are using the default keymap.

Graphics tablets can be used to provide a more traditional method of controlling the mouse cursor using a pen. This can help provide a more familiar experience for artists who are used to painting and drawing with similar tools, as well as provide additional controls such as pressure sensitivity.

Note

If you are using a graphic tablet instead of a mouse and pressure sensitivity does not work properly, try to place the mouse pointer in the Blender window and then unplug/replug your graphic tablet. This might help.

Touchpad controls are available on Windows, macOS and Linux with Wayland. If you are working from a laptop without a mouse, you can emulate controls using multi-touch gestures with the trackpad from Preferences.

Gesture

Effect

Pan

Hold the Shiftkey while dragging two fingers on the pad.

Zoom

Hold the Ctrlor OSKeykey while dragging two fingers on the pad.

Orbit

Drag two fingers on the pad.

Emulate right-click

Tap two fingers on the pad.

3D mice or NDOFdevices are hardware that you can use to navigate a scene in Blender.Currently only devices made by 3Dconnexion are supported.These devices allow you to explore a scene, and makeFly/Walk Navigationeasier to control. The NDOF device can be configured in thePreferences.These settings can also be accessed directly from the viewport using the NDOFMenubutton on the NDOF device.

See also

See Input Preference for more information on configuring peripherals.

HMDsmake it possible to place users in an interactive, virtual environment.Attached to the head, they track head movements to project a seemingly surrounding world onto smallscreens in front of the user\xe2\x80\x99s eyes. If the system works well, they experience the virtual environment asif they were really inside of it.

Virtual reality support in Blender is implemented through the multi-platform OpenXR standard.This standard is new and therefore support for it is still limited.

Platform

Operating System

Notes

HTC Vive Cosmos

Windows

Developer Preview

HTC Vive Focus 3

Windows

Developer Preview

Monado

GNU/Linux

Notrecommended for general use yet.

Meta (formerly Oculus) (Rift and Quest)

Windows

Requires Oculus v31 Software Update. Oculus Link required for Quest.

SteamVR

Windows, GNU/Linux

Requires SteamVR 1.16 or greater.

Varjo

Windows

\xe2\x80\x93

Windows Mixed Reality

Windows

Requires Windows 10 May 2019 Update (1903).

The following subsections describe how an HMD can be set up for usage with the supported platforms.If this is not done, Blender will report an error when trying to start a virtual reality session.

The dedicated platform forthe HTC Vive Cosmosis currently targeted at developers and may lack features found in other platforms.

Follow the steps from the Vive Developer Forums.

Enable the VR Scene Inspection add-on in Blender.

The dedicated platform for the HTC Vive Focus 3 is currently targeted at developers and may lack features found in other platforms.

Follow the steps from the Vive Developer Forums.

Enable the VR Scene Inspection add-on in Blender.

Monado is a free and open source XR platform for Linux. It is not yet ready for production usage and should only be used for testing purposes. Packages are available for the following distributions:

Ubuntu (Eoan, Focal)

Debian (bullseye, sid)

For other systems, it has to be compiled from source, which in this case is not recommended for people with little experience in compiling software. Follow the Guides from Monado to do so nevertheless.

Enable the VR Scene Inspection add-on in Blender.

Meta (formerly Oculus) provides full support for OpenXR as of the Oculus v31 Software Update.

Download and install the Oculus Rift/Oculus Link software.

Set Oculus as the active OpenXR runtime via the General tab in the Oculus App Settings.

Enable the VR Scene Inspection add-on in Blender.

Passthrough Support

Currently, passthrough support over OpenXR is disabled by default in the Quest Link app, and must be manually enabled in its settings to use this feature.

The performance of the passthrough render varies with the quality of the connection between the headset and the computer. For better results, connecting the headset directly through USB to the PC, or at least connecting the computer to the local network over ethernet, is recommended.

SteamVR provides full support for OpenXR as of SteamVR 1.16.

Set SteamVR as the active OpenXR runtime via the Developer tab in the SteamVR Settings.

Enable the VR Scene Inspection add-on in Blender.

Note

The SteamVR runtime can also be used for HTC Vive Cosmos, Oculus, and Windows Mixed Reality HMDs.

Varjo includes full OpenXR support with its required Varjo Base software. Enable the VR Scene Inspection add-on in Blender.

Windows Mixed Reality provides full support for OpenXR. To check if a PC meets the requirements to run the software, Microsoft offers the Windows Mixed Reality PC Check application.

Make sure the Windows 10 May 2019 Update (1903) is installed.

If the system meets all requirements, the Mixed Reality Portal should already be installed. It is also available in the Microsoft Store.

Launch the Mixed Reality Portal. Click the menu button ... in the lower left corner. In the menu it opens, select the Set up OpenXR.

Enable the VR Scene Inspection add-on in Blender.

Note

To switch to Windows Mixed Reality from another OpenXR runtime (e.g. SteamVR), download the OpenXR Developer Tools from the Microsoft Store and set Windows Mixed Reality as the active runtime.

--- Fin du chapitre: _hardware ---

--- Début du chapitre: _harmonic ---Reference

Physics \xe2\x80\xa3 Force Fields

Harmonic

In a Harmonicforce field,the source of the force field is the zero point of a harmonic oscillator (spring, pendulum).If you set the Dampingparameter to 1,the movement is stopped in the moment the object is reached.This force field is really special if you assign it to particles. Controls the rest length of the harmonic force. Causes every point to be affected by multiple springs. Normally every particle of the field system influences every particle of the target system.Not with Harmonic! Here every target particle is assigned to a field particle.So particles will move to the place of other particles, thus forming shapes.

--- Fin du chapitre: _harmonic ---

--- Début du chapitre: _hash_value --- The Hash Valuenode takes a value input and hashes this to an integer.

Important

Hashes cannot be relied upon to be used as unique identifiers because they are not guaranteed to be unique.It can be used to generate somewhat stable randomnessespecially in cases where White Noise does not offer enough flexibility.

Value input determined by Data Typeproperty.

Integer input used to generate different Hashes.

The data type that is used for the Valueinput.The node supports float, integer, vector, color,boolean, rotation, matrix and string data types. Standard integer output.

--- Fin du chapitre: _hash_value ---

--- Début du chapitre: _header --- Every constraint has a header.The interface elements of the header are explained below using a Copy Location constraint as an example.

A Header sits at the top of every constraint.\xc2\xb6

Show or Hide the settings of the constraint.Tidy up the constraint stack by hiding constraints that do not currently need attention.Constraints will continue to affect the scene even when hidden. The constraint type icon.

Give the constraint a meaningful name in this text field, which describes its purpose.Meaningful names help you and your team members understand what each constraint is supposed to do.

The red background is a warning that the constraint is not yet functional.The background will turn gray when the constraint is functioning.When this Copy Location constraint has a valid target in the target field it will turn gray and begin to function.

Enable or Disable the constraint. Disabling a constraint will stop its affect on the scene.

Disabling a constraint is useful for turning off a constraint without losing all of its settings.Disabling means you can enable the constraint at a later time with the settings intact.Disabling is similar to setting the Influence to 0.0.

Makes the constraint \xe2\x80\x99real\xe2\x80\x9d by applying any transformations caused by the constraint to make the original object to match the results of the constraint and deletes the constraint.

Warning

Applying a constraint that is not first in the stack will ignore the stack order(it will be applied as if it was the first one), and may produce undesired results.

Creates a duplicate of the constraint just below current one in the stack.

Copies the constraint from theActiveobject to all selected objects.

Moves the constraint to the first or last position in the constraint stack.

Delete the constraint from the stack.The settings will be lost.The constraint will no longer affect the final outcome of the stack.

Move a constraint up or down in theconstraint stack.Since the stack is evaluated from top to bottom,moving a constraint in the stack can significantly affect the final outcome of the stack.

If there is only one constraint in the stack, the arrows will not be displayed.

If the constraint is at the top of the stack, only the down arrow will be displayed.

If the constraint is at the bottom of the stack, only the up arrow will be displayed.

--- Fin du chapitre: _header ---

--- Début du chapitre: _help --- Blender has a range of built-in and web-based help options.

Tooltip of the Renderer selector in the Info Editor. \xc2\xb6
When hovering the mouse cursor over a button or setting, after a few instants a tooltip appears.

The context-sensitive Tooltip might contain some of these elements:
Related details depending on the control.

A keyboard or mouse shortcut associated to the tool.

The value of the property.

Hovering over a color property will display a large swatch preview of the colorand the color\xe2\x80\x99s hexadecimal, RGBA, and HSVA values.

Source file of the active object. See alsoLinked Libraries.

The reason why the value is not editable.

WhenPython Tooltipsare enabled,a Python expression is displayed
forscripting(usually an operator or property).

Reference

All modes

Context menu \xe2\x80\xa3 Online Manual

F1

You may want to access help for a tool or area from within Blender.

To do so; hover the cursor over the tool or button you need help with anduse the keyboard shortcut or context menu item to visit pages of this reference manual from within Blender.This opens a web page relating to the button under the cursor, supporting both tool and value buttons.

Note

We do not currently have 100% coverage.You may see an alert in the info header if a tool does not have a link to the manual.

Other times, buttons may link to more general sections of the documentation.

The first options of this menu provide direct links to Blender-related websites.The same links can also be found in theSplash Screen.

This is a link to the Official Blender Manual (which you are now reading).

Link to the release notes for the current Blender version.
 Multiple tutorials to help you learn to use Blender.
 Links to various sites, providing both community and professional support.
 Lists of many different community sites and support venues.
 The Blender Bug Tracker (registration needed).
 This extracts system information which can be useful for including in bug reports, inspecting the configuration, or diagnosing problems.
 You will be prompted to save a text file called `system-info.txt`.
 It contains the following sections:
 This section shows you the Blender version, details about the build configuration, and the path in which Blender is running.
 The version and path of your Python installation.
 Paths used for scripts, data files, presets and temporary files.
 Those directories are configured using the `PreferencesEditor`.
 Shows the GPU vendor, version and the capabilities of your hardware and driver.
 Lists add-ons currently in use.
 --- Fin du chapitre: `_help` ---

--- Début du chapitre: `_hide_tools` ---Reference

Sculpt Mode

Hide gesture tools hide all selected vertices within the selection area and any of their connected edges and faces. Holding `Ctrl` while performing the selection reveals the vertices, edges, and faces.

Pressing `LMB` with any of these tools without also dragging reveals all elements of a mesh.

All hide gesture tools can be activated in the Toolbar and are comprised of the following:

Reference

Sculpt Mode

Toolbar `\xe2\x80\xa3` Box Hide

Hides vertices and connected edges and faces based on a box gesture .

Reference

Sculpt Mode

Toolbar `\xe2\x80\xa3` Lasso Hide

Hides vertices and connected edges and faces based on a lasso gesture .

Reference

Sculpt Mode

Toolbar `\xe2\x80\xa3` Line Mask

Hides vertices and connected edges and faces based on a line gesture .

Reference

Sculpt Mode

Toolbar `\xe2\x80\xa3` Polyline Mask

Hides vertices and connected edges and faces based on a polyline gesture .

.

Note

The Polyline Hide tool does not support showing all vertices via pressing `LMB`.

Determines whether all vertices inside or outside the selected area should be affected.

All vertices and connected elements inside the selection area will be hidden.

All vertices and connected elements outside the selection area will be hidden.

--- Fin du chapitre: `_hide_tools` ---

--- Début du chapitre: `_hinge` ---Reference

Physics \xe2\x80\xa3 Rigid Body Constraint
Hinge

The hinge permits one degree of freedom between two objects. Translation is completely constrained. Rotation is permitted about the Z axis of the object hosting the Physics constraint (usually an Empty, distinct from the two objects that are being linked). Adjusting the position and rotation of the object hosting the constraint allows you to control the anchor and axis of the hinge.

The Hinge is the only single-axis rotational constraint that uses the Z axis instead of the X axis. If something is wrong with your hinge, check your other constraints to see if this might be the problem.

Enables/disables limit rotation around Z axis.

Lower limit of Z axis rotation.

Upper limit of Z axis rotation.

--- Fin du chapitre: `_hinge` ---

--- Début du chapitre: `_history` --- In 1988, Ton Roosendaal co-founded the Dutch animation studio NeoGeo. NeoGeo quickly became the largest 3D animation studio in the Netherlands and one of the leading animation houses in Europe. NeoGeo created award-winning productions (European Corporate Video Awards 1993 and 1995) for large corporate clients such as the multinational electronics company Philips. Within NeoGeo, Ton was responsible for both art direction and internal software development. After careful deliberation, Ton decided that the current in-house 3D tool set for NeoGeo was too old and cumbersome to maintain, and needed to be rewritten from scratch. In 1995 this rewrite began and was destined to become the 3D software creation we all know as Blender. As NeoGeo continued to refine and improve Blender, it became apparent to Ton that Blender could be used as a tool for other artists outside of NeoGeo.

In 1998, Ton decided to found a new company called Not a Number (NaN) as a spin-off of NeoGeo to further market and develop Blender. At the core of NaN was a desire to create and distribute a compact, cross-platform 3D application for free. At the time, this was a revolutionary concept as most commercial 3D applications cost thousands of dollars. NaN hoped to bring professional level 3D modeling and animation tools within the reach of the general computing public. NaN's business model involved providing commercial products and services around Blender. In 1999 NaN attended its first SIGGRAPH conference in an effort to more widely promote Blender. Blender's first SIGGRAPH convention was a huge success and gathered a tremendous amount of interest from both the press and attendees. Blender was a hit and its huge potential confirmed!

Following the success of the SIGGRAPH conference in early 2000, NaN secured financing of \$4.5M from venture capitalists. This large inflow of cash enabled NaN to rapidly expand its operations. Soon NaN boasted as many as 50 employees working around the world trying to improve and promote Blender. In the summer of 2000, Blender 2.0 was released. This version of Blender added the integration of a game engine

to the 3D application. By the end of 2000, the number of users registered on the NaN website exceeded 250,000.

Unfortunately, NaN's ambitions and opportunities did not match the company's capabilities and the market realities of the time. This over-extension resulted in restarting NaN with new investor funding and a smaller company in April 2001. Six months later NaN's first commercial software product, Blender Publisher was launched. This product was targeted at the emerging market of interactive web-based 3D media. Due to disappointing sales and the ongoing difficult economic climate, the new investors decided to shut down all NaN operations. The shutdown also included discontinuing the development of Blender. Although there were clearly shortcomings in the then current version of Blender, such as a complex internal software architecture, unfinished features and a non-standard way of providing the GUI, the enthusiastic support from the user community and customers who had purchased Blender Publisher in the past, meant that Ton could not justify leaving Blender to fade into insignificance. Since restarting a company with a sufficiently large team of developers was not feasible, Ton Roosendaal founded the non-profit organization, Blender Foundation, in March 2002.

The Blender Foundation's primary goal was to find a way to continue developing and promoting Blender as a community-based open source project. In July 2002, Ton managed to get the NaN investors to agree to a unique Blender Foundation plan to attempt to release Blender as open source. The 'Free Blender' campaign sought to raise \$100,000 so that the Foundation could buy the rights to the Blender source code and intellectual property rights from the NaN investors and subsequently release Blender to the open source community. With an enthusiastic group of volunteers, among them several ex-NaN employees, a fundraising campaign was launched to 'Free Blender'. To everyone's surprise and delight the campaign reached the \$100,000 goal in only seven short weeks. On Sunday, October 13, 2002, Blender was released to the world under the terms of the GNU GPL. Blender development continues to this day, driven by a team of dedicated volunteers from around the world led by Blender's original creator, Ton Roosendaal.

The start!

1.00 January 1994: Blender in development at animation studio NeoGeo.

1.23 January 1998: SGI version published on the web, IrisGL.

1.30 April 1998: Linux and FreeBSD version, port to OpenGL and X11.

1.3x June 1998: NaN founded.

1.4x September 1998: Sun and Linux Alpha version released.

1.50 November 1998: First Manual published.

1.60 April 1999: C-key (new features behind a lock, \$95), Windows version released.

1.6x June 1999: BeOS and PPC version released.

1.80 June 2000: End of C-key, Blender full freeware again.

2.00 August 2000: Interactive 3D and real-time engine.

2.10 December 2000: New engine, physics, and Python.

2.20 August 2001: Character animation system.

2.21 October 2001: Blender Publisher launch.

2.2x December 2001: macOS version.

Blender goes Open Source

Blender goes Open Source, 1st Blender Conference.

Blender Publisher becomes freely available, and the experimental tree of Blender is created, a coder's playground.

The first truly open source Blender release.

The second open source Blender release.

First of the 2.28x series.

Preview release of the 2.3x UI makeover presented at the 2nd Blender Conference.

Upgrade to stable 2.3x UI project.

A major overhaul of internal rendering capabilities.

Game Engine returns, ambient occlusion, new procedural textures.

Particle interactions, LSCM UV mapping, functional YafRay integration, weighted creases in subdivision surfaces, ramp shaders, full OSA, and many (many) more.

Another version full of improvements: object hooks, curve deforms and curve tapers, particle duplicators and much more.

A stabilization version, much work behind the scenes, normal and displacement mapping improvements.

Transformation tools and widgets, soft bodies, force fields, deflections, incremental subdivision surfaces, transparent shadows, and multi-threaded rendering.

Full rework of armature system, shape keys, fur with particles, fluids, and rigid bodies.

Lots of fixes, and some Game Engine features.

The nodes release, Array modifier, vector blur, new physics engine, rendering, lip sync, and many other features. This was the release following Project Orange.

Multiresolution meshes, multi-layer UV textures, multi-layer images and multi-pass rendering and baking, sculpting, retopology, multiple additional mattes, distort and filter nodes, modeling and animation improvements, better painting with multiple brushes, fluid particles, proxy objects, Sequencer rewrite, and post-production UV texturing.

The big news, in addition to two new modifiers and re-awakening the 64-bit OS support, was the addition of subsurface scattering, which simulates light scattering beneath the surface of organic and soft objects.

Serious bug fixes, with some performance issues addressed.

The Peach release was the result of a huge effort of over 70 developers providing enhancements to provide hair and fur, a new particle system, enhanced image browsing, cloth, a seamless and non-intrusive physics cache, rendering improvements in reflections, AO, and render baking, a Mesh Deform modifier for muscles and such, better animation support via armature tools and drawing, skinning, constraints and a colorful Action Editor, and much more. It contained the results of Project Peach.

Bugfix release.

The Apricot release, cool GLSL shaders, lights and GE improvements, snap, sky simulator, Shrinkwrap modifier, and Python editing improvements. This contained the results Project Apricot.

Node-based textures, armature sketching (called Etch-a-Ton), Boolean mesh operation improvements, JPEG2000 support, projection painting for direct transfer of images to models, and a significant Python script catalog. GE enhancements included video textures, where you can play movies in-

game, upgrades to the Bullet physics engine, dome (fisheye) rendering, and more API GE calls made available.

Blender 2.5x \xe2\x80\x93 The Recode!

This series released four pre-version (from Alpha 0 in November 2009 to Beta in July 2010) and three stable versions (from 2.57 - April 2011 to 2.59 - August 2011). It was one of the most important development projects, with a total refactor of the software with new functions, redesign of the internal window manager and event/tool/data handling system, and new Python API. The final version of this project was Blender 2.59 in August 2011.

Blender 2.6x to 2.7x \xe2\x80\x93 Improvements & Stabilizing

Internationalization of the UI, improvements in the animation system and the GE, vertex weight groups modifiers, 3D audio and video, and bug fixes. The Cycles renderer was added to the trunk, the camera tracker was added, dynamic paint for modifying textures with mesh contact/approximation, the Ocean modifier to simulate ocean and foam, new add-ons, bug fixes, and more extensions added for the Python API.

The Carve library was added to improve Boolean operations, support for object tracking was added, the Remesh modifier was added, many improvements in the GE, matrices and vectors in the Python API were improved, plus new add-ons, and many bug fixes.

Bmesh was merged with the trunk, with full support for n-sided polygons, sculpt hiding, a panoramic camera for Cycles, mirror ball environment textures and float precision textures, render layer mask layers, ambient occlusion and viewport display of background images and render layers. New import and export add-ons were added, and 150 bug fixes.

A mask editor was added, along with an improved motion tracker, OpenColorIO, Cycles improvements, Sequencer improvements, better mesh tools (Inset and Bevel were improved), new keying nodes, sculpt masking, Collada improvements, a new Skin modifier, a new compositing nodes backend, and the fixing of many bugs.

Fire and smoke improvements, anisotropic shader for Cycles, modifier improvements, the Bevel tool now includes rounding, new add-ons, and over 200 bug fixes.

Dynamic topology, rigid body simulation, improvements in UI and usability (including retina display support), Cycles now supports hair, the Bevel tool now supports individual vertex beveling, new Mesh Cache modifier and the new UV Warp modifier, new SPH particle fluid solver. More than 250 bug fixes.

Freestyle was added, paint system improvements, subsurface scattering for Cycles, Ceres library in the motion tracker, new custom Python nodes, new mesh modeling tools, better support for UTF-8 text and improvements in Text editors, new add-ons for 3D printing, over 260 bug fixes.

New and improved modeling tools, three new Cycles nodes, big improvements in the motion tracker, Python scripts and drivers are disabled by default when loading files for security reasons, and over 280 bug fixes.

Even more modeling tools, Cycles improved in many areas, plane tracking is added to the motion tracker, better support for FBX import/export, and over 270 bugs fixed.

Cycles gets basic volumetric support on the CPU, more improvements to the motion tracker, two new modeling modifiers, some UI consistency improvements, and more than 560 bug fixes.

Deformation motion blur and fire/smoke support is added to Cycles, UI pop-ups are now draggable. There are performance optimizations for

sculpting mode, new interpolation types for animation, many improvements to the GE, and over 400 bug fixes.

Cycles gets volume and SSS support on the GPU, pie menus are added and tooltips greatly improved, the Intersection modeling tool is added, new Sun Beam node for the Compositor, Freestyle now works with Cycles, texture painting workflow is improved, and more than 220 bug fixes.

Cycles gets improved volumetric support, major upgrade to Grease Pencil, Windows gets Input Method Editors (IMEs) and general improvements to painting, Freestyle, Sequencer and add-ons.

Support for custom normals, viewport compositing and improvements to hair dynamics.

Integrated stereo/multi-view pipeline, Smooth Corrective modifier and new developmental dependency graph.

Pixar OpenSubdiv support, Viewport and File Browser performance boost, node auto-offset, and a text effect strip for the Sequencer.

OpenVDB support for caching of smoke/volumetric simulations, improved Cycles subsurface scattering, Grease Pencil stroke sculpting and improved workflow, and reworked library handling to manage missing and deleted data-blocks.

Cycles support for spherical stereo images for VR, Grease Pencil works more similar to other 2D drawing software, Alembic import and export support, and improvements to Bendy Bones for easier and simpler rigging.

New Cycles features: Denoising, Shadow catcher, and new Principled shader. Other improvements were made to Grease Pencil and Alembic. Support was also added for application templates.

Blender 2.8 \xe2\x80\x93 Revamped UI

A totally redesigned UI for easier navigation; improved viewport, gizmos, and tools. With EEVEE a new physically based real-time render engine was created. The Grease Pencil got a big overhaul and is now a full 2D drawing and animation system. Replacing the old layers, collections are a powerful way to organize objects. Other improvements: Cycles, Modeling, Animation, Import/Export, Dependency Graph.

Revamped sculpting tools, Cycles OptiX accelerated rendering, denoising, many EEVEE improvements, library overrides, UI improvements and much more.

UDIM and USD support, Mantaflow for fluids and smoke simulation, AI denoising, Grease Pencil improvements, and much more.

3D Viewport virtual reality scene inspection, new volume object type, Cycles adaptive sampling, Cycles viewport denoising, sculpting improvements, and much more.

Blender 2.9 \xe2\x80\x93 Refining 2.8

Improved sky texture, EEVEE motion blur, sculpting improvements, revamped modifier UI, improved modeling tools, and faster motion blur in Cycles.

Outliner improvements, property search, improved mesh Boolean operations, animation curves, volume object and display improvements, and more refined sculpting tools.

Geometry nodes, primitive add tool, sculpting improvements, Grease Pencil curve editing, Cycles Color Attribute baking, APIC fluid simulations, Video Sequencer improvements, and much more.

New geometry nodes, sculpting improvements, Grease Pencil Line Art modifier along with other improvements, an improved DOF for the EEVEE render engine, redesigned Cryptomatte workflow, and more.

Blender 3.0 \xe2\x80\x93 Optimizing Performance

Asset Browser added, Cycles X, EEVEE Attributes, New geometry nodes, animation update, Grease Pencil line art improvements, pose library, Open Image Denoising 2-8x faster, additional support for AMD on linux.

Major point clouds improvements, Cycles Apple Metal GPU support, Subdivision GPU support, image editor handles larger images, Major performance gains for geometry nodes, context aware search for geometry nodes.

Light groups for Cycles, true Shadow caustics, volume motion blur, GLTF improvements, AMD GPU Rendering on Linux, painting in sculpt mode, WEBP image support.

New hair object, procedural UV nodes, line art shadow and contour, Intel GPU rendering support via oneAPI, and improvements to library overrides. Cycles path guiding, sculpting auto masking improvements, even more geometry nodes, UV Editing improvements and Wayland support on Linux. New generative hair assets, vector displacement maps for sculpting, viewport compositor, and Cycle's light trees.

--- Fin du chapitre: `_history` ---

--- Début du chapitre: `_holdout` --- The Holdout shader node is used to create a hole in the image with zero alphaness, which is useful for compositing (see Alpha Channel). This node has no inputs.

This node has no properties.

Standard shader output.

The checkered area is a region with zero alpha.

--- Fin du chapitre: `_holdout` ---

--- Début du chapitre: `_hook` --- The Hook Modifier is used to deform stroke points using another object (usually an empty or a bone but it can be any object).

As the hook moves, it pulls points from the strokes with it. You can think of it as animated Proportional Editing.

See also

This documentation refers to the Hook Modifier specific to the Grease Pencil object. For uses with other object types refer to the general Hook Modifier.

The Hook modifier.

The name of the object to hook points to.

Restricts the effect only to a vertex group.

Adjust this hook's influence on the stroke points, from (0.0 to 1.0) (no change to fully follow the hook).

This can be used to adjust the type of curve for the Strength attenuation. You can also define a custom curve to get a much higher level of control.

The size of the hook's influence.

This setting is useful when using hooks on scaled objects, especially in cases where non-uniform scale would stretch the result of the hook.

See Influence Filters.

Note

The Hook Modifier stores points indices from the original strokes to determine what to affect; this means that modifiers that generate geometry, like a Subdivision Surface Modifier, should always be applied

after the Hook Modifier; otherwise the generated geometry will be left out of the hook's influence.

Empty used as a hook to manipulate a vertex group (right eye of the monkey).

--- Fin du chapitre: hook ---

--- Début du chapitre: hooks --- The Hook modifier is used to deform a mesh, curve or lattice using another object (usually an empty or a bone but it can be any object). As an object specified as hook moves, it pulls vertices or control points from the geometry with it. You can think of it as animated Proportional Editing.

While hooks do not give you the fine control over vertices movement that shape keys do, they have the advantage that you can select vertices directly for manipulation.

To assign selected vertices (in Edit Mode) you can use the Assign button on the modifier panel or use the Add Hook menu.

The Hook modifier.

The name of the object to hook vertices to.

Allows you to define the influence per vertex.

Useful when you want something other than a spherical field of influence.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier.

The setting reverses the weight values of the group.

Adjust this hook's influence on the vertices, where (0.0 to 1.0) (no change to fully follows the hook).

Since multiple hooks can work on the same vertices, you can weight the influence of a hook using this property.

The following settings are only available in Edit Mode:

Recalculate and clear the offset transform of the hook.

Set the hook center to the 3D cursor position.

Select the vertices affected by this hook.

Assigns selected vertices to this hook.

Warning

The Hook Modifier stores vertex indices from the original mesh to determine what to affect. This means that modifiers that generate geometry, like Subdivision Surface, should always be put after the Hook one in the stack. Otherwise, the generated geometry will be left out of the hook's influence.

This can be used to adjust the kind of influence curve that the hook has on the mesh. You can also define a custom curve to get a much higher level of control.

The size of the hook's influence.

This setting is useful when using hooks on scaled objects, especially in cases where non-uniform scale would stretch the result of the hook.

This is especially useful for lattices, where it is common to use non-uniform scaling.

Empty used as Hook to deform a subdivided cube.

--- Fin du chapitre: hooks ---

--- Début du chapitre: hue_correct --- The Hue Correct Node adjusts the hue, saturation, and value of an image, with an input curve.

Controls the amount of influence the node exerts on the output image.

Standard color input.

H (Hue), S (Saturation), V (Value)

For the curve controls see: Curve widget .By default, the curve is a straight line, meaning there is no change.The spectrum allows you to raise or lower HSV levels for each range of pixel colors.To change an H, S, or V level, move the curve points up or down. Pixels with hue values eachpoint in the horizontal position of the graph will be changed depending on the shape of the curve.

Standard color output.

--- Fin du chapitre: _hue_correct ---

--- Début du chapitre: _hue_saturation --- The Hue/Saturation/Value Nodeapplies a color transformation in the HSVColor Model.

The amount of influence the node exerts on the image.

Standard color input.

The hue rotation offset, from 0 (-180\xc2\xba) to 1 (+180\xc2\xba). Note that0 and 1 have the same result.

A value of 0 removes color from the image, making it black-and-white.A value greater than 1.0 increases saturation.

The value shift. 0 makes the color black, 1 keeps it the same, and highervalues make it brighter.

Standard color output.

Some things to keep in mind that might help you use this node better:

If you apply a Hue offset of 1 (+180\xc2\xba) to a blue image, you get the diametrically oppositecolor, which is yellow. If you apply a Hue offset of 1 to that yellow image, you get blue again.

Trying to change the Hue or Saturation of a grayscale image has no effect. You can only brightenor darken it by adjusting the Value. To add color, use the Mix node instead.

The different values can be animated using a Time Curvenode or by setting keyframes.

A basic example. \xc2\xba6

An example of using the Factor input for masking. \xc2\xba6

--- Fin du chapitre: _hue_saturation ---

--- Début du chapitre: _icosphere --- The Icospherenode generates a spherical mesh that consists of equally sized triangles.

Distance of the vertices from the origin.

Number of subdivisions on top of the most basic icosphere.The number of faces quadruple with every subdivision.

This node has no properties.

Standard geometry output.

A 2D vector representing the default X/Y coordinates of the UV Map for the primitive\xe2\x80\x99s shape.This can be connected to the Store Named Attribute Node ,to be used once the Geometry Nodes Modifier get applied.The UV map must be stored on the face corner in order to be accessed.

--- Fin du chapitre: _icosphere ---

--- Début du chapitre: _id --- The IDnode gives an integer value indicating the stable random identifier of each element on the point domain,which is stored in theidattribute.

The node to set this data is theSet ID Nodenode.

Note

Unlike other built-in attributes, the id attribute does not always exist.In that case, this node will output the index .

This node has no inputs.
This node has no properties.
Integer value.
--- Fin du chapitre: `_id` ---

--- Début du chapitre: `_id_mask` --- The ID Mask Node can be used to access an alpha mask per object or per material.

See also

The ID Mask node is superseded by the Cryptomatte Node. Cryptomatte is more feature complete and supported by Cycles and EEVEE. It is recommended to use this feature moving forward.

Input for the Object Index or Material Index render pass. Which is an output of the Render Layers node or the Image node with a multi-layer format.

Selection of the previously specified index.

This post-processing filter smooths the mask edges. See Anti-Aliasing .

The mask is white where the object is and black where it is not. If the object is transparent, the alpha mask represents that with gray values.

An index can be specified for any object or material in the scene. The Object Index can be set in Properties > Object Properties > Relations > Pass Index and Material > Settings > Pass Index for the Material Index. To be accessible after rendering, the Render Engine must be Cycles, and Object Index or Material Index render pass has to be enabled.

Object Pass Index. \xc2\xbf

In this example, the left rear red cube is assigned Pass Index 1, and the right cube Pass Index 2. Where the two cubes intersect, there is going to be noticeable pixelation because they come together at a sharp angle and are different colors. Using the mask from object 1, which is smoothed (anti-aliased) at the edges, we use a Mix Node set on Multiply to multiply the smoothed edges of the image, thus removing those nasty lines, thus, being smoothed out.

ID Mask node example. \xc2\xbf

Volume Objects are not supported.

--- Fin du chapitre: `_id_mask` ---

--- Début du chapitre: `_ies` --- The IES Texture is used to match real world lights based on IES files (IES). IES files store the directional intensity distribution of light sources.

Texture coordinate for lookup in the light distribution. Defaults to the normal.

Light strength multiplier.

The location to load the IES file from.

Use IES profile from a file embedded in a text data-block in the blend-file, for easy distribution.

Load IES profile from a file on the drive.

Light intensity, typically plugged into the Strength input of an Emission node.

Lights with different IES profiles. \xc2\xbf

--- Fin du chapitre: `_ies` ---

--- Début du chapitre: `_ik_solver` --- The Inverse Kinematics constraint implements the inverse kinematics armature posing technique. Hence, it is only available for bones. To quickly create an IK constraint with a target, select a bone in Pose Mode, and press Shift-I.

This constraint is fully documented in the Inverse Kinematics page, part of the rigging chapter.

Note

The IK constraints are special in that they modify multiple bones. For this reason, they ignore their position in the stack and always run after all other constraints on the affected bones. To apply constraints after IK, it is necessary to first copy the final transformation to a new bone chain, e.g. using Copy Transforms .

Inverse Kinematics panel. \xc2\xbb6

Data ID used to select the an armature. See common constraint properties for more information.

Object for pole rotation.

Maximum number of solving iterations.

How many bones are included in the IK effect. Set to 0 to include all bones.

Include bone's tail as last element in chain.

Enable IK stretching.

For Tree-IK: Weight of position control for this target.

Chain follows rotation of target.

Disable for targetless IK.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

If the iTaSC IK Solver is used, the IK Solver Constraint changes to add these addition parameters.

Equivalent to the traditional end effector position and orientation constraint: the end effector is constrained to take the position, and optionally the orientation, of a given target, which is set in the target field.

Allows to obtain various effect by not constraining the coordinates along certain axis.

Specifies how to compute the axis coordinates.

The coordinates are the position and orientation of the target relative to the bone.

The opposite of Bone, the coordinates are the position and orientation of the tip of the bone relative to the target.

Specify that the end effector will stay inside, at, or outside a sphere centered on the target object.

The end effector will stay inside of the distance from the target object.

The end effector will stay outside of the distance from the target object.

The end effector will stay exactly at the distance from the target object.

The radius from the target object.

Note

The Influence parameter is not implemented if Pole Target is used.

--- Fin du chapitre: `_ik_solver` ---

--- Début du chapitre: `_image` --- Note

Unlike the other texture nodes, this node operates differently in geometry nodes compared to the equivalent shader node. When not connected the Vector input has an implicit position attribute value.

The Image Texture node is used to add an image file as a texture. The image data is sampled with the input Vector and outputs a Color and Alpha value.

The image socket can be used to connect to the Group Input node. If this is not connected the image can be opened or selected from the node.

Texture coordinate for texture look-up. If this socket is left unconnected, the Position attribute is used.

If the Image supports animation, the frame can be set here. This can be keyframed so that the image changes between frames.

Method to scale images up or down for sampling.

Regular quality interpolation.

Smoother, better quality interpolation. For bump maps this should be used to get best results.

No interpolation, use only closest pixel for rendering pixel art.

Extension defines how the image is extrapolated past the original bounds:

Will repeat the image horizontally and vertically giving tiled-looking result.

Will extend the image by repeating pixels on its edges.

Clip to the original image size and set all the exterior pixels values to transparent black.

Repeatedly flip the image horizontally and vertically.

RGBA color from the image.

Alpha channel from image.

Image Texture displacing a plane. \xc2\xb6

--- Fin du chapitre: _image ---

--- Début du chapitre: _image_formats --- This is the list of image file formats supported internally by Blender:

Format
Channel Depth
Alpha
Metadata
DPI
Extensions
BMP
8bit
\xe2\x9c\x97
\xe2\x9c\x97
\xe2\x9c\x93
.bmp
Iris
8, 16bit
\xe2\x9c\x93
\xe2\x9c\x97
\xe2\x9c\x97
.sgi .rgb .bw
PNG
8, 16bit
\xe2\x9c\x93
\xe2\x9c\x93
\xe2\x9c\x93
.png
JPEG
8bit

```

\xe2\x9c\x97
\xe2\x9c\x93
\xe2\x9c\x93
.jpg .jpeg
JPEG 2000
8, 12, 16bit
\xe2\x9c\x93
\xe2\x9c\x97
\xe2\x9c\x97
.jp2 .jp2 .j2c
Targa
8bit
\xe2\x9c\x93
\xe2\x9c\x97
\xe2\x9c\x97
.tga
Cineon & DPX
8, 10, 12, 16bit
\xe2\x9c\x93
\xe2\x9c\x97
\xe2\x9c\x97
.cin .dpx
OpenEXR
float 16, 32bit
\xe2\x9c\x93
\xe2\x9c\x93
\xe2\x9c\x93
.exr
Radiance HDR
float
\xe2\x9c\x93
\xe2\x9c\x97
\xe2\x9c\x97
.hdr
TIFF
8, 16bit
\xe2\x9c\x93
\xe2\x9c\x97
\xe2\x9c\x93
.tif .tiff
WebP
8bit
\xe2\x9c\x93
\xe2\x9c\x93
\xe2\x9c\x93
.webp

```

Hint

If you are not interested in technical details, a good rule of thumb for selecting output formats for your project is:

- if you intend to do compositing or color grading on these images.
- if you intend on-screen output or encoding into multiple video formats.
- for on-screen output where file size is a concern and quality loss is acceptable.

All these formats support compression which can be important when rendering out animations.

Hint

Bit depths for image formats represent the following numbers of tonal levels per channel:

256 levels

1024 levels

4096 levels

65536 levels

Sets the file path to be relative to the currently opened blend-file.

See Relative Paths .

Automatically looks for image sequences in the selected images (based on the file name). Disable this when you do want to get single images that are part of a sequence.

Automatically looks for UDIM tiles in the directory of the selected image; if matches are found they are loaded into Blender as UDIMs. This works by detecting if the filename has a .xxxx (four digit number) before the file extension.

To load image sequence in any of the supported image file formats, the filename of the images must contain a digit to indicate the frame order (e.g. *-0001.jpg, *-0002.jpg, *-0003.jpg, etc, of any image format), indicating the frame.

The sequence could be opened by the selection of the images with any of the following methods by the confirmation with the Open Image button or Return.

Navigate into the directory and LMB click and drag over a range of names to highlight multiple files. You can page down and continue Shift-LMB click-dragging to add more to the selection.

Shift-LMB click selected non-related stills for batch processing; each image will be one frame, in sort order, and can be a mix of file types (jpg, png, exr, etc.).

Press A to select/deselect all files in the directory.

Choose what format to save the image as.

Choose the color format to save the image (or video) to. Note that RGBA is not available for all image formats, check the list above for details.

BW, RGB, RGBA

Some image file formats support a varying number of bits per pixel. This affects the color quality and file size. Commonly used depths:

Most common for on-screen graphics and video.

Used for some formats focusing on photography and digital films (such as DPX and JPEG 2000).

Since full 32bit float is often more than enough precision, half float can save drive space while still providing a high dynamic range.

Highest quality color depth.

Note

Internally Blender's image system supports either:

8 bits per channel (4 \xc3\x97 8 bits).

32 bits float per channel (4 \xc3\x97 32 bits) \xe2\x80\x93 using 4 times as much memory.

Images higher than 8 bits per channel will be converted into a float on loading into Blender.

Used to reduce the size of the image file. How this is done may vary depending on the file format and settings used.

Similar to Compression but is used for JPEG based file formats. The quality is a percentage, 0% being the maximum amount of compression and 100% is no compression.

Save image with render color management .For display image formats like PNG, apply view and display transform. For intermediate image formats like OpenEXR, use the default render output color space.

The Copy checkbox will define if the data-block will reference the newly created file or the reference will be unchanged, maintaining it with the original one.

To specify the color space of the source file.

The list of color spaces depends on the active OCIO config .The default supported color spaces are described in detail here: Default OpenColorIO Configuration

Note

Note, Cineon, DPX, OpenEXR, and Radiance HDR image types default to being saved in a linear color space.

Cineon is Kodak's standard for film scanning, 10 bits per channel and logarithmic. DPX has been derived from Cineon as the ANSI/SMPTE industry standard. DPX supports 16-bit colors/channels, linear as well as logarithmic. DPX is currently a widely adopted standard used in the film hardware/software industry.

DPX as well as Cineon only stores and converts the visible color range of values between 0.0 and 1.0 (as a result of rendering or composite).

ILM's OpenEXR has become a software industry standard for HDR image files, especially because of its flexible and expandable structure.

An OpenEXR file can store multiple layers and passes. This means OpenEXR images can be loaded into a Compositor keeping render layers and passes intact.

Available options for OpenEXR render output are:

Half saves images in a custom 16 bits per channel floating-point format. This reduces the actual bit depth to 10-bit, with a 5-bit power value and 1-bit sign.

Float (Half), Float (Full)

Lossy algorithm from Pixar, converting 32-bit floats to 24-bit floats.

Standard lossless compression using Zlib, operating on 16 scanlines at a time.

Lossless wavelet compression. Compresses images with grain well.

Run-length encoded, lossless, works well when scanlines have same values.

Standard lossless compression using Zlib, operating on a single scanline at a time.

JPEG-like lossy algorithm from DreamWorks; compresses blocks 32 scanlines together.

Same as DWAA but compresses blocks of 256 scanlines.

On rendering animations (or single frames via command line), Blender saves the same image also as a JPEG, for quick preview or download.

Radiance is a suite of tools for lighting simulation. Since Radiance had the first (and for a long time the only) HDR image format, this format is supported by many other software packages.

Radiance.hdr files store colors still in 8 bits per component, but with an additional (shared) 8-bit exponent value, making it 32 bits per pixel.

--- Fin du chapitre: _image_formats ---

--- Début du chapitre: `_image_info` --- The Image Info node gets information from image and animation. This can be useful to generate parameters in the geometry node for arbitrary images. Image information can be either general or frame-specific.
Source image to get parameters from.
Frame index for frame-specific outputs.
This node has no properties.
The number of pixels along the X axis. Specific to each frame.
The number of pixels along the Y axis. Specific to each frame.
Whether the transparency channel be different from 1 for the pixels of this image frame. Specific to each frame.
The number of frames in an image or video frame sequence. For a static image, always 1.
The number of frames per second. For static image is always 0.
--- Fin du chapitre: `_image_info` ---

--- Début du chapitre: `_image_movie` --- The term Image Textures simply means that a graphic image, which is a pixel grid composed of R, G, B, and sometimes Alpha values. It is used as the input source to the texture. As with other types of textures, this information can be used in a number of ways, not only as a simple `texture` but also as a `video texture`.

Video textures are a some kind of Image textures and based on movie file or sequence of successive numbered separate images. They are added in the same way that image textures are.

When the Texture Type Image or Movie is selected, three new panels present themselves allowing to control most aspects of how image textures are applied: Image, Image Sampling, and Image Mapping. Texture images take up precious memory space, often being loaded into a special video memory bank that is very fast and very expensive, so it is often very small. So, keep the images as small as possible. A 64x9764 image takes up only one fourth the memory of a 128x97128 image. For photorealistic rendering of objects in animations, often larger image textures are used, because the object might be zoomed in on in camera moves. In general, you want to use a texture sized proportionally to the number of pixels that it will occupy in the final render. Ultimately, you only have a certain amount of physical RAM to hold an image texture and the model and to provide workspace when rendering your image. For the most efficient memory usage, image textures should be square, with dimensions as powers of 2, such as 32x9732, 64x9764, 128x97128, 256x97256, 1024x971024, 2048x972048, and 4096x974096.

If you can reuse images across different meshes, this greatly reduces memory requirements. You can reuse images if you map those areas of the meshes that `look alike` to a layout that uses the common image.

When using file textures, it is very important that you have Mapped the UVs of the mesh, and they are laid out appropriately. You do not have to UV map the entire mesh. The sphere above on the left has some faces mapped, but other faces use procedural materials and textures. Only use UV textures for those portions of your mesh where you want very graphic, precise detail. For example, a model of a vase only needs UV texture for the rim where decorative artwork is incorporated. A throw pillow does not need a different image for the back as the front; in fact many throw pillows have a fabric (procedural material) back.

As another example, you should UV map both eyes of a head to the same image(unless you want one bloodshot and the other clear).Mapping both sides of a face to the same image might not be advisable,because the location of freckles and skin defects are not symmetrical.You could of course change the UV map for one side of the face to slightly offset,but it might be noticeable.Ears are another example where images or section of an image can be mapped to similar faces.

The Image Data-Block Menu .

Use the alpha channel information stored in the image.Where the alpha value in the image is less than 1.0,the object will be partially transparent and things behind it will be visible.Works withimage formats that store transparency information.

Calculate an alpha based on the RGB values of the Image.Black (0, 0, 0) is transparent, white (1, 1, 1) opaque.Enable this option if the image texture is a mask.Note that mask images can use shades of gray that result in semi-transparency,like ghosts, flames, and smoke/fog.

Image with Usealpha. The alpha values of the pixels are evaluated.
\xc2\xbf

Image with Calculatealpha only, Use Alphain the Imagepanel is disabled.
\xc2\xbf

Reverses the alpha value.Use this option if the mask image has white where you want it transparent and vice versa.

Image Mapping panel.\xc2\xbf

In the Mappingpanel,you can control how the image is mapped or projected onto the 3D model.

Rotates the image 90 degrees counterclockwise when rendered.

How the image is extrapolated beyond its original bounds.

Outside the image the colors of the edges are extended.

Clip to image size and set exterior pixels as transparent.Outside the image, an alpha value of 0.0 is returned.This allows you to
\xe2\x80\x98paste\xe2\x80\x99 a small logo on a large object.

Clips to cubic-shaped area around the images and sets exterior pixels as transparent.The same as Clip, but now the \xe2\x80\x98Z\xe2\x80\x99 coordinate is calculated as well.An alpha value of 0.0 is returned outside a cube-shaped area around the image.

The image is repeated horizontally and vertically.

X/Y repetition multiplier.

Mirror on X/Y axes. These buttons allow you to map the texture as a mirror, or automatic flip of the image,in the corresponding X and/or Y direction.

Checkerboards quickly made.You can use the option sizeon the Mappingpanel as well to create the desired number of checkers.

Set even/odd tiles.

Governs the distance between the checkers in parts of the texture size. The offset and the size of the texture in relation to the texture space.Pixels outside this space are ignored.Use these to crop, or choose a portion of a larger image to use as the texture.

In the Samplingpanel you can control how the information is retrieved from the image.

Image Sampling panel.\xc2\xbf

This option interpolates the pixels of an image.This becomes visible when you enlarge the picture. By default, this option is on.Turn this option off to keep the individual pixels visible and if they are correctly anti-aliased.This last feature is useful for regular patterns,

such as lines and tiles;they remain \xe2\x80\x98sharp\xe2\x80\x99 even when enlarged considerably.Turn this image off if you are using digital photos to preserve crispness.

Enlarged Image texture without Interpolation. \xc2\xba

Enlarged Image texture with Interpolation. \xc2\xba

Mip-maps are precalculated, smaller, filtered textures for a certain size.A series of pictures is generated, each half the size of the former one.This optimizes the filtering process. By default, this option is enabled and speeds up rendering.When this option is off,you generally get a sharper image, but this can significantly increase calculation time if the filter dimension(see below) becomes large. Without mip-maps you may get varying pictures from slightly different camera angles,when the textures become very small. This would be noticeable in an animation.

Used in conjunction with mip-mapping, it enables the mip-map to be made smaller based on color similarities.In game engines, you want your textures, especially your mip-map textures,to be as small as possible to increase rendering speed and frame rate.

Texture filter to use for image sampling.Just like a pixel represents a picture element, a texel represents a texture element.When a texture (2D texture space) is mapped onto a 3D model (3D model space),different algorithms can be used to compute a value for each pixel based on samples from several texels.

A fast and simple nearest-neighbor interpolation known as Monte Carlo integration.

One of the most efficient direct convolution algorithms developed by Paul Heckbert and Ned Greene in the 1980s.For each texel, EWA samples, weights, and accumulates texels within an elliptical footprint and then divides the result by the sum of the weights.

Maximum Eccentricity. Higher values give less blur at distant/oblique angles, but is slower.

Uses several isotropic probes at several points along a line in texture space to produce an anisotropic filter to reduce aliasing artifacts without considerably increasing rendering time.

Number of probes to use. An integer between 1 and 256.Further reading: McCormack, J; Farkas, KI; Perry, R; Jouppe, NP (1999) Simple and Table Feline: Fast Elliptical Lines for Anisotropic Texture Mapping, WRL

Area filter to use for image sampling.

Maximum Eccentricity. Higher values give less blur at distant/oblique angles, but is slower.

The filter size used in rendering, and also by the options Mip Map and Interpolation.If you notice gray lines or outlines around the textured object, particularly where the image is transparent,turn this value down from 1.0 to 0.1 or so.

Use Filter Size as a minimal filter value in pixels.

--- Fin du chapitre: _image_movie ---

--- Début du chapitre: _image_settings --- Image tab.\xc2\xba

Select the type of image to use. For images that come from files,see Supported Graphics Formats .

A single, static image.

An animation where each frame is stored in a separate file.See Opening an Image Sequence.For options, see Movie below.

A video file. Note that if you want to do motion tracking and video compositing rather than simply using the video as a texture, you should load it into the Movie Clip Editor instead.

Note

The options below are for preview purposes only; they don't affect the 3D Viewport or the render. For that, see the Image Texture Node .

Note

Blender plays all videos at the scene framerate, not their original framerate, meaning they'll be faster or slower than intended if these framerates don't match up. To work around this, see the Offset field of the Image Texture Node linked above.

How many frames of the video to play. Past this point, the video will be paused (unless Cyclic is enabled).

Sets the Frame to the number of frames in the video file.

Scene frame at which the video should start playing.

Number of frames to offset the video to an earlier point in time. (Put differently: how many frames at the start of the video to skip.)

Start over after the last frame to create a continuous loop.

Play the video in the Image Editor when the scene animation is playing. (The mouse cursor should be in the Image Editor or the Timeline when starting playback for this to work.)

Apply deinterlacing to interlaced (analog) video.

Image generated by Blender.

The width and height of the image in pixels.

Creates a 32-bit image. This has a larger file size, but holds much more color information than the standard 8-bit image. For close-ups and large gradients, it may be better to use a 32-bit image.

Creates a blank image of a single specified color.

Creates a checkerboard pattern with a colored cross (+) in each square.

Creates a more complex colored grid with letters and numbers denoting locations. It could be used to check for stretching or distortion in the UV mapping.

The fill color when creating a Blank image.

Used for replacing or packing files.

Embed the resource into the current blend-file. See Packed Data .

Path to the linked file.

Opens the File Browser to select a file from a drive.

Reloads the file. Useful when it has been reworked in an external application.

See Multi-View .

The Color Space the image file was saved in. This is used for converting the image to linear color (which is the color space Blender works with).

Textures and final renders are often stored in sRGB, while OpenEXR images are stored in a linear color space. Some images such as normal, bump or stencil maps do not strictly contain colors and should never have a color conversion applied to them. For such images, the color space should be set to Non-Color.

The list of color spaces depends on the active OCIO config. The default supported color spaces are described in detail here: Default OpenColorIO Configuration .

How the image uses its Alpha Channel. This option is only available if the image format supports transparency.

Store RGB and alpha channels separately with alpha acting as a mask, also known as unassociated alpha. Commonly used by image editing applications and file formats like PNG. This preserves colors in parts of the image with zero alpha.

Store RGB channels with alpha multiplied in, also known as associated alpha. The natural format for renders and used by file formats like OpenEXR. This can represent purely emissive effects like fire correctly, unlike straight alpha.

Different images are packed in the RGB and alpha channels, and they should not affect each other. Channel packing is commonly used by game engines to save memory.

Ignore alpha channel from the file and make image fully opaque.

Load the image with a bit depth of only 16 bits per channel instead of 32, which saves memory.

Apply the color management settings when displaying this image on the screen.

The thickness of the margin around UV islands for texture painting to bleed into. This margin ensures that no unpainted pixels remain at the island border.

Painting a stroke across a seam in 3D space makes it extend past the UV island borders in the texture, until it gets cut off at the margin.

\xc2\x80

A higher value will result in a thicker margin, which can be useful if you intend to create mipmaps of the texture. However, this may also reduce painting performance.

Note

This setting only affects Sculpt Mode, where texture painting support is currently experimental. In Texture Paint Mode, a fixed margin is used instead.

--- Fin du chapitre: _image_settings ---

--- Début du chapitre: _import_images_as_planes ---Reference

3D Viewport \xe2\x80\xa3 Add \xe2\x80\xa3 Image \xe2\x80\xa3 Images as Planes

Imports images and creates planes with them as textures. It automates the process of creating a plane, resizing it to fit the dimensions of the image, and creating a material with the image texture to it. The name of the plane, material, and texture will be derived from the name of the image file.

You can import a single image, multiple images, or an image sequence/movie clip. If you choose a single image, it creates one plane; if you choose multiple images, it creates as many planes as the number of images you selected, either stacked on top of each other or spaced apart. Selecting a movie clip or an image sequence will create a single plane with an animation.

You can save the current import settings as an Operator Preset .

Set link to the image file using a relative file path.

Reload the image file if it already exists as an image data-block.

Import sequentially numbered images as an animated image sequence instead of separate planes. They will be imported as a Clip texture on a single plane. The frame range will be automatically set but can be changed later.

Images as Planes sets up a material to display the image. You can set the type of material and related settings before the import.

The material will have a Principled BSDF shader node with default settings as its main component. An Image Texture node linked to the imported image will be connected to the Base Color of the Principled BSDF node.

A shadeless material is a material that does not respond to light from other objects and always has the same color in any lighting environment. This option creates a material with a node group of a mix between a Diffuse and an Emission shader controlled by a Light Path node.

The material will have a Principled BSDF shader node as its main component, but the Color output from the Image Texture node will be linked to the Emission input instead of the Base Color.

Set the strength of the emission.

Note

Blend Mode and Shadow Mode options are specific to the EEVEE renderer. For a detailed explanation of each option, see Material Settings .

Set the alpha blend mode of the material.

Set the shadow mode of the material.

Show backside of the transparent part.

Hide backside of the plane.

By default, the name of the new material from the name of the imported image. However, if there is already a material with the same name, Blender will append a number to the name of the material to avoid conflict. This Override Material option makes it overwrite the existing material of the same name in that case.

Note

For a detailed explanation of each option, see Image Texture Node .

Set the method to scale the image.

Set how the image is extrapolated past the original bounds.

Use the alpha channel of the image for transparency.

Automatically refresh the images in the viewport on frame changes.

Images as Planes creates the plane at the 3D Cursor's location. With Offset Planes, multiple planes will be placed with distance intervals set in Offset, along the axis set in Local Axis, beginning at the 3D Cursor's location.

Set how the plane's size will be determined.

The size of the plane will be set based on the height value set in Height. The width will be set in direct ratio to the height value. For example, with the default height value of 1 m, an image of 800 × 600 pixels will have a width of 1 / 600 × 800 or 1.33 m.

Set the height of the plane.

The size of the plane will be set to fit or fill the camera frame. This will automatically set the Align option to Face Camera. Make sure to have an active camera in the scene before the import.

Scale the plane to fit inside the camera frame while preserving the aspect ratio.

Scale the plane so that it fills the entire camera view while preserving the aspect ratio, but some part of the image can spill outside the camera frame.

The size of the plane will be set based on the pixels per inch value set in Definition. With the Unit System set to Metric and the default definition of 600 DPI, an image of 800 × 600 pixels will have a size of 0.0339 × 0.0254 units since 600 pixels are defined as 1 inch (0.0254 m).

Set the number of pixels to fit in 1 inch.

The size of the plane will be set based on the pixels per Blender Unit set in Definition. With the default definition value of 600, an image of 800 \xc3\x97 600 pixels will have a size of 1.33 \xc3\x97 1 units.

Set the number of pixels to fit in 1 Blender Unit.

Set the rotation of the plane.

The plane will be aligned to a major axis that is best to face the camera\xe2\x80\x99s view direction. If there is no camera in the scene, the plane will face toward Z+ (Up) axis.

Similar to the Main Axis but the plane will be rotated to directly face the camera\xe2\x80\x99s view direction.

The plane will be rotated to face toward the selected axis.

Add a Locked Track constraint to make the plane always face the camera, even if the camera moves. This option is only available when Main Axis or Face Camera option is selected in the Align menu.

Place multiple planes with an offset. If disabled, all planes will be created at the same location.

Choose a local axis (not the global axis) to offset the planes. For example, if you choose X+, the planes will be placed along the positive direction of the plane\xe2\x80\x99s local X axis.

Set a distance between each plane.

--- Fin du chapitre: `_import_images_as_planes` ---

--- Début du chapitre: `_index` ----- Fin du chapitre: `_index` ---

--- Début du chapitre: `_index_of_nearest` --- The Index of Nearest node is a way to find other close elements in the same geometry. If needed you can use Group ID to determine the group of neighbors to be analyzed together.

This is an alternative to the Sample Nearest Node. The main difference is that this node does not require a geometry input, because the geometry from the field context is used.

Tip

This is often combined with the Evaluate at Index Node or the Sample Index Node.

The position for each element to search. By default, this is the same as if the Position Node was connected.

ID to group elements together.

The index of the closest element in the same geometry component.

This is true when the group of the element has at least two elements. This is only relevant when using Group ID.

--- Fin du chapitre: `_index_of_nearest` ---

--- Début du chapitre: `_index_switch` --- The Index Switch node outputs one of its inputs depending on an index value. Only the input that is passed through the node is computed.

See also

The Menu Switch Node is similar but it exposes the choices as a menu. Determines which of the input options below will be passed through.

One input is created for every menu entry. The input is used when the matching option is selected.

Determines the type of the data that is handled by the node.

One of the inputs without any modifications.

--- Fin du chapitre: `_index_switch` ---

--- Début du chapitre: `_individual_origins` --- Reference

Object Mode and Edit Mode

Pivot Point \xe2\x80\xa3 Individual Origins
Period

While the other pivot point modes transform the whole selection around one point, Individual Origins transforms each item around itself. Each object gets transformed around its origin, which is a point that can be chosen freely and doesn't have to be in the center. In the example below, the orange rectangle has it in a corner instead.

Rotation around individual origins. \xc2\xb6

The images below compare Individual Origins to Median Point .

Starting situation, rotation around Individual Origins, rotation around Median Point. \xc2\xb6

Starting situation, scaling using Individual Origins, scaling using Median Point. \xc2\xb6

Each selected element is transformed around its own centerpoint.

Starting situation, rotation around Individual Origins, rotation around Median Point. \xc2\xb6

Starting situation, scaling using Individual Origins, scaling using Median Point. \xc2\xb6

When you transform adjacent faces or edges, they are treated as a single element (meaning they don't become disconnected).

--- Fin du chapitre: _individual_origins ---

--- Début du chapitre: _industry_compatible --- While this is not a comprehensive list, this page shows common keys used in the industry compatible keymap.

Switch Selection mode

Object Mode

Modes Pie Menu

RMB

Context menu

Tab

Menu Search.

Shift- Tab

Quick access (favorites)

Return

Rename

Ctrl- Return

Render

Ctrl- [

Toggle Toolbar

Ctrl-]

Toggle Sidebar

Backspace

Deletes the selected item, requires a confirmation dialog.

Delete

Deletes the selected item, does not require a confirmation dialog.

Ctrl- D

Duplicate

P

Set Parent

B

Proportional Editing / Soft Selection

Alt- LMB

Orbit View
Alt- MMB
Pan View
Alt- RMB
Zoom View
F1- F4
Front/Side/Top/Camera Viewpoints
F
Frame Selected
Shift F
Center View to Mouse
A
Frame All
LMB
Select
Ctrl- A
Select All
Shift- Ctrl- A
Deselect All
Ctrl- I
Select Inverse
Up
Select More
Down
Select Less
Double LMB
Select Loop
DoubleAlt- LMB
Select Ring
Ctrl L
Select Linked
W, E, R
Transform Tools
Q
Select Tools
D
Annotate
C
Cursor Tool
Ctrl- E
Extrude
Ctrl- B
Bevel
I
Inset
K
Knife
Alt- C
Loop Cut
Spacebar
Play/Pause
S
Set Location + Rotation + Scale keyframe
Shift- S

Insert Keyframe menu

Shift- W

Set Location Key

Shift- E

Set Rotation Key

Shift- R

Set Scale Key

The Cmdkey can be used instead of Ctrl on macOS for all but a few exceptions which conflict with the operating system.

--- Fin du chapitre: `_industry_compatible` ---

--- Début du chapitre: `_inflate` ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Similar to Draw, except that vertices are moved in the direction of their own normals. Especially useful when sculpting meshes with a lot of curvature.

Also available as a Mesh Filter to inflate all unmasked areas at once.

Either Inflate or Deflate sculpted areas. This is different from the typical Add & Subtract.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: `_inflate` ---

--- Début du chapitre: `_info_editor` --- The Info editor logs the executed operators as well as errors, warnings, and informational messages. You can select an entry by clicking it, optionally holding Shift to add it to the existing selection.

Info Editor. \xc2\xb6

Area controls. See the user interfacedocumentation for more information.

Selects all entries.

Deselects all entries.

Selects non-selected entries and deselects selected ones.

Selects all entries if there are currently no selected ones, and deselects them otherwise.

Lets you drag a box and adds the entries that overlap it to the selection.

Removes the selected entries from the log.

Copies the selected entries to the clipboard.

--- Fin du chapitre: `_info_editor` ---

--- Début du chapitre: `_inpaint` --- The Inpaint node is used to extend borders of an image into transparent or masked regions. This can be useful to solve problems like \xe2\x80\x99wire removal\xe2\x80\x9d and holes created during chroma keying.

Standard color input.

The number of times to extend the image.

Standard color output.

The left image shows the \xe2\x80\x99wire\xe2\x80\x9d in place and after chroma keying has been applied. You will see you are left with a blank space \xe2\x80\x93 it\xe2\x80\x99s shown as a black line here but it will be alpha in your Blender output.

Inpaint Node example. \xc2\xb6

Inpainting fills in a couple of pixels using the surrounding image and voil\xc3\xa0\xe2\x80\xa6 your wire is removed.

Note

The wider the \xe2\x80\x9chole\xe2\x80\x9d is, the more noticeable this effect is! If you use more than a few pixels of infill, the effect is almost as irritating as the wire and your viewers won\xe2\x80\x99t be impressed.

Inpainting can also cover up a multitude of other minor sins such as control points for motion capture: use it sparingly and it will amaze.

--- Fin du chapitre: _inpaint ---

--- Début du chapitre: _input --- In the Input preferences, you can customize how Blender reacts to the mouse and keyboard as well as define your own keymap.

The Numpad keys are used quite often in Blender and are not assigned to the same action as the regular number keys. If you have a keyboard without a Numpad (e.g. on a laptop), you can tell Blender to treat the standard number keys as Numpad keys by checking Emulate Numpad.

For transform mode, default to Advanced Mode, otherwise Simple Mode is used.

Blender can be configured to work with pointing devices which do not have an MMB. The functionality of the three mouse buttons by holding Alt- LMB. Mouse/Keyboard combinations referenced in this manual can be expressed with the combinations shown in the table. For example:

MMB drag becomes Alt- LMB drag for example.

Warning

This option prevents certain features from being accessed, since Alt- LMB is used for some operations.

Modifying multiple items values at once (objects, bones\xe2\x80\xa6 etc).

Deselecting edge/face rings in Edit Mode.

Detaching node links.

Moving the Compositor background image.

Some touchpads support three-finger tap for middle mouse button, which may be an alternative to using this option.

The modifier key to press to emulate the middle mouse keybindings. This option is unsupported on Microsoft Windows.

Use the Alt key to emulate the middle mouse button.

Use the OS Key to emulate the middle mouse button.

This has the advantage that it doesn\xe2\x80\x99t conflict with existing Alt- MMB shortcuts, noted above.

This feature is used to prevent the problem where an action such as moving objects or panning a view, is limited by your screen bounds.

This is done by warping the mouse within the view.

Note

Cursor warping is only supported by relative input devices (mouse, trackball, trackpad).

Graphics tablets, however, typically use absolute positioning, this feature is disabled when a tablet is being used.

This is detected for each action, so the presence of a tablet will not disable Continuous Grab for mouse cursor input.

Dragging LMB on an object will move it. To confirm this (and other) transform, an LMB is necessary by default. When this option is activated, the release of LMB acts as confirmation of the transform.

The time in milliseconds to trigger a double click.

The number of pixels that a User Interface element has to be moved before it is recognized by Blender, values below this will be detected as click events.

The drag threshold for tablet events.

The drag threshold for non mouse/tablet events (keyboard or NDOF for example).

This affects Pie Menu on Drag keymap preference.

The number of pixels the cursor must be moved before the movement is registered. This is helpful for tablet pens that are a lot more difficult to keep still, then this could help to reduce stuttering of the cursor position.

Note

Unlike the click/drag distinction, this is used to detect small movements for example, picking selection cycles through elements near the cursor. Once the cursor moves past this threshold, selection stops cycling and picks the closest item.

Note

This panel is available on Windows, macOS, and Linux with Wayland. Use multi-touch gestures for navigation with touchpad, instead of scroll wheel emulation. For more detail on supported gestures, see Configuring Peripherals .

The direction scrolling responds to the scroll gestures.

Only available on Linux using Wayland.

Scrolls content down when gestures move up.

Scrolls content up when gestures move up.

Select the native Windows Ink or older Wintab system for pressure sensitivity. Blender automatically selects the API for your operating system and tablet, however in case of problems this can be set manually. You may need to restart Blender for changes to take affect.

Amount of pressure required to achieve full intensity.

Controls how the softness of the low pressure response onset using a gamma curve.

These preferences control how an NDOF device interacts with the 3D Viewport. These preferences can also be accessed using the NDOF Menu button on the NDOF device to open a pop-up menu to adjust the settings directly from the 3D Viewport.

The overall sensitivity for panning in the 3D Viewport.

The overall sensitivity for orbiting in the 3D Viewport.

The threshold for the amount of movement needed from the device's rest position for Blender to interrupt that movement.

Navigation style for the viewport.

Uses the full 6-degrees of freedom.

Orbit about the view center.

Rotation style for the viewport.

Rotates the view keeping the horizon horizontal.

Is less restrictive, allowing any orientation.

Display the pivot point and axis during rotation.

Zoom using opposite direction.

Pan/zoom the camera view instead of leaving the camera view when orbiting.

Pan using up/down on the NDOF devices instead of forward/backwards.

Reverses the panning axis on the selected axes.

Reverses the orbit axis on the selected axes.

Settings to control how the NDOF device is used while using Walk/Fly Navigation .
Keeps the horizontal axis level while flying.
Moves the 3D Viewport up or down when moving the NDOF device up/down.
--- Fin du chapitre: _input ---

--- Début du chapitre: _input_index --- The Indexnode gives an integer value indicating the position of each element in the list, starting at zero. This depends on the internal order of the data in the geometry, which is not necessarily visible in the 3D Viewport. However, the index value is visible in the left-most column in the Spreadsheet Editor.

Note

Indices in geometry data are often defined by the internals of complex algorithms that create it. If no inputs change, indices will be the same when the same node tree is executed multiple times. However, they may not be predictable when inputs to nodes that generate geometry or change its topology are adjusted. Additionally, updates to algorithms in newer versions of Blender may change the order of generated elements.

To avoid relying on consistent indices, it is recommended to calculate them locally, or to avoid operations that change topology when they must be consistent over time.

This node has no inputs.

This node has no properties.

Integer value which enumerates each point on the geometry.

--- Fin du chapitre: _input_index ---

--- Début du chapitre: _inset_faces --- Reference

Edit Mode

Toolbar \xe2\x80\xa3 Inset Faces

Face \xe2\x80\xa3 Inset Faces

I

This tool takes the currently selected faces and creates an inset of them, with adjustable thickness and depth. Think of it as like creating an edge loop, but relative to the selected edges, even in complex meshes.

The tool is modal, such that when you activate it, you may adjust the thickness with your mouse position. You may also adjust the depth of the inset during the modal operation by holding Ctrl.

Selection to inset. \xc2\xb6

Result of inset with Select Outer enabled. \xc2\xb6

Inset operator options. \xc2\xb6

Determines whether open edges will be inset or not.

Scale the offset to give a more even thickness.

Scale the offset by lengths of surrounding geometry.

Created vertices slide along the original edges of the inner geometry, instead of the normals.

Set the size of the offset.

Raise or lower the newly inset faces to add depth.

Create an outset rather than an inset. Causes the geometry to be created surrounding selection (instead of within).

Toggle which side of the inset is selected after the operation.

By default the Inset tool operates on the region around selected faces, but with this option each selected face can be inset on its own.

Interpolate mesh data: e.g. UVs, Color Attribute \xe2\x80\x99s colors, weights, etc.

--- Fin du chapitre: `_inset_faces` ---

--- Début du chapitre: `_inspection` --- Inspecting intermediate values in a geometry node tree is useful while building/understanding one or when trying to figure out why something is not working. Blender provides multiple tools to understand how a node tree is working or why it is not working.

Note

Generally, the inspection tools display data from the last time the node tree has been evaluated. If it has not been evaluated, no information is available.

Socket Inspection. `\xc2\xba`

Socket inspection shows information about the value in a socket during the last evaluation. For primitive data types such as integers, vectors, and strings the actual value is shown. For geometry sockets only some data about the geometry is stored, including the set of data types the geometry contains, and a count of their elements.

Socket values are only logged from when the node tree was executed, so a node must be connected to the Group Output to have a value for inspection. Values are not logged during rendering, to improve performance.

Attribute Search. `\xc2\xba`

The attribute search is shown when clicking on an attribute input in the modifier. It contains a list of all the attributes that were available at that point in the modifier or node execution.

The Viewer node is used to display intermediate geometry in the Spreadsheet Editor and the Viewport. For more information see Viewer Node

Node Warning. `\xc2\xba`

When the inputs to a node are invalid, it displays a warning in the title. Hovering over the warning icon shows the error message. These warnings are only generated when the node is executed, so a node must be connected to the Group Output to have a warning.

The node timings overlay. `\xc2\xba`

Node timings show how long a node took to execute the last time the node group was evaluated. They can be turned on in the overlays popover on the top right of the node editor. When a node group is used in multiple places, the timings depend on the context of the node editor, which is displayed in the path on the top left.

Frame nodes display the total time from all of the contained nodes and the Group Output node displays the total time for the entire node group.

The displayed timings should only be considered an approximation, since they can also take into account actions like copying or deleting a geometry input that aren't part of the node's operation. Also, when a node uses multiple CPU cores, the evaluation system might work on other nodes at the same time.

It's also important to remember that field nodes generally don't do work by themselves, so their execution time is only added to the data-flow nodes they are connected to.

The `Named Attributes` overlay allows displaying when a custom named attribute is used by a node or a node group. Named attributes can be used by the Capture Attribute Node, the Named Attribute Node, and the Remove Named Attribute Node, and can be written to, read, or removed.

Using named attributes (as opposed to Anonymous Attributes) can be problematic when the original geometry already has attributes with the specified names. In that case a geometry node group might mistakenly overwrite some essential data. The overlay helpsto make detecting that situation easy.

The same data is also available in the Named Attributes panel in the modifier's UI.

Many nodes don't guarantee the order of elements in which they output things. For example, the order of edges coming out of the Triangulate node is deterministic but not well defined. The order may change between Blender versions. Therefore, if node setups depend on a specific order, they may break when the Blender implementation changes. Changing the order can often be necessary in order to fix bugs or improve performance.

Geometry randomization can be temporarily enabled to see if a blend-file depends on the indices in unstable ways. When enabled, various internal algorithms shuffle the result geometry elements so that any dependence on it would not work anymore. When building setups that are supposed to last a long time, it is recommended to check if they still work with randomization enabled.

To enable it, first enable Developer Extras in the preferences. Then search for Set Geometry Randomization. The popup allows enabling and disabling the randomization.

--- Fin du chapitre: _inspection ---

--- Début du chapitre: _instances --- The three types of instances.

In addition to storing real data like a mesh or a curve, objects can store instances, which themselves can reference more geometry, an object, or a collection. The purpose of instancing is to allow duplicating geometry and storing it in an object, without duplicating the actual data. This optimization allows render engines like Cycles to handle the same geometry data in many different locations better than when the data is duplicated.

Each instance keeps track of which geometry it corresponds to and how the instanced is transformed compared to its source geometry. Instances can also store the id attribute, which is used to correct motion blur when instances move in an animation.

Instances can be created with geometry nodes using the Instance on Points Node.

Warning

Currently instancing from geometry nodes cannot be mixed with instancing from the Instancing panel in the property editor.

Since instances can store a geometry, and a geometry can contain instances, nested instancing is possible. In other words, it is possible to instance an instance, or even a collection of instances. For example, by default, the Instance on Points Node will create nested instances by instancing instances on the points of real geometry.

A node group that creates nested instancing by chaining Instance on Points nodes.

Here, nested instancing is used to distribute geometry that contains both a mesh and instances. The output geometry contains a

real mesh and a group of instances. Each instance contains a sphere mesh and many instances of a cone geometry.

The tree of instanced geometry for the example above. \xc2\xba6
What makes this method helpful is the output geometry only contains three unique meshes: the plane, the sphere, and the cone. This would make the performance much better if the meshes were more complicated.

Warning

Only eight levels of nested instancing are supported for rendering and viewing in the viewport. Though deeper trees of instances can be made inside geometry nodes, they must be realized at the end of the node tree. The term \xe2\x80\x9crealizing\xe2\x80\x9d instances refers to converting the instances into unique geometry. When instances are realized they will take up more memory and manipulation to geometry will have to be processed individually rather than once per instancing geometry. To realize instance use the Realize Instances Node .

Almost all nodes that process geometry do so by processing each unique geometry separately rather than realized geometry. For example, if a Subdivision Surface Node was placed at the end of the example above, it would only have to subdivide three meshes, rather than each instance of a mesh. Another important example is processing with the output of the String to Curves Node , where each unique character only has to be processed once.

This method can improve performance a lot, but it means that the result of an operation will be the same for every instance of a certain geometry. In order to have unique results for every instance, the Realize Instances Node can be used.

--- Fin du chapitre: _instances ---

--- Début du chapitre: _instances_to_points --- The Instances to Points node generates points at the origins of top-level instances. Attributes on the instance domain are moved to the point cloud points.

Note

Top-level instances are those that are owned by the node \xe2\x80\x99s input geometry. Instances owned by other instances, i.e. nested instances, are not considered by this node.

Standard geometry input.

The instances used to generate points. True values mean a point is created for the instance, false values mean the instance is skipped.

Overrides the default position of generated point.

Controls the radius of the result points.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _instances_to_points ---

--- Début du chapitre: _instance_on_points --- The Instance on Points node adds a reference to a geometry to each of the points present in the input geometry. Instances are a fast way to add the same geometry to a scene many times without duplicating the underlying data. The node works on any geometry type with a Point domain, including meshes, point clouds, and curve control points.

Any attributes on the points from the Geometry input will be available on the instanced domain of the generated instances.

Tip

The Make Instances Real operator can be used to create objects from instances generated with this node.

Note

To instance object types that do not contain geometry, like a light object, the Object Info Node can be used. Other objects like Metaball objects are not supported for instancing.

Standard geometry input. The position of the points of this geometry affect the transforms of each instance output.

Note

If the input geometry contains instances, the node will create more instances on the points inside the instances, creating nested instancing. In this case, each new instance will have the transform created by the node from the Rotation and Scale inputs, but it will also be transformed based on the parent instances.

Whether to instance on each point. True values mean an instance will be generated on the point, false values mean the point will be skipped.

The geometry to instance on each selected point. This can contain real geometry, or multiple instances, which can be useful when combined with the Pick Instance option.

If enabled, instead of adding the entire geometry from the Instance input on every point, choose an instance from the instance list of the geometry based on the Instance Index input. This option is intended to be used with the Collection Info Node.

The selection of index for every selected point, only used when Pick Instances is true. By default the point ID is used, or the index if that doesn't exist. Negative values or values that are too large are wrapped around to the other end of the instance list.

The Euler rotation for every instance. This can use the rotation output of nodes like Distribute Points on Faces and Curve to Points. An Euler rotation can also be created from a direction vector like the normal with the Align Rotation to Vector Node.

The size of each generated instance.

This node has no properties.

Standard geometry output. If the id attribute exists on the input geometry, it will be copied to the result instances.

--- Fin du chapitre: `_instance_on_points` ---

--- Début du chapitre: `_instance_rotation` --- The Instance Rotation outputs the XYZ Euler rotation of each top-level instance in the local space of the modifier object.

The Instances page contains more information about geometry instances.

Note

Though rotations are often displayed in units of degrees in the spreadsheet or node editor, they are stored internally in radians, so this node outputs radians.

This node has no inputs.

This node has no properties.

Vector that indicates the rotation of each top-level instance in radians.

--- Fin du chapitre: `_instance_rotation` ---

--- Début du chapitre: `_instance_scale` --- The Instance Scale outputs the size of top-level instances on each axis in the local space of the modifier object.

The Instances page contains more information about geometry instances.

This node has no inputs.

This node has no properties.

Vector that indicates the scale of each top-level instance.

--- Fin du chapitre: `_instance_scale` ---

--- Début du chapitre: `_instance_transform` --- The Instance Transform outputs the Transformation Matrix of each top-level instance in the local space of the modifier object.

The Instances page contains more information about geometry instances.

This node has no inputs.

This node has no properties.

Matrix that indicates the Transformation of each top-level instance.

--- Fin du chapitre: `_instance_transform` ---

--- Début du chapitre: `_integer` --- The Integer node provides an integer value.

This node has no input sockets.

Single integer value.

Standard integer output

--- Fin du chapitre: `_integer` ---

--- Début du chapitre: `_integer_math` --- The Integer Math node performs math operations.

Standard integer value input. Depending on the operation there will be more than one input.

The mathematical operator to be applied to the input values:

The sum of the two values.

The difference between the two values.

The product of the two values.

The division of the first value by the second value.

The sum of the product of the two values with Addend.

The input value is read without regard to its sign. This turns negative values into positive values.

Changes the sign of the input value.

The Base raised to the power of Exponent.

Outputs the smallest of the input values.

Outputs the largest of two input values.

Extracts the sign of the input value. All positive numbers will output 1.

All negative numbers will output -1. And 0 will output 0.

Divide the values and round the result toward zero.

Divide the values and floor the result down to the nearest integer.

Divide the values and ceil the result up to the nearest integer.

Returns the positive remainder of a division operation.

Outputs the remainder once the first value is divided by the second value.

The largest positive integer that divides into each of the values.

The smallest positive integer that is divisible by both values.

Standard integer output.

--- Fin du chapitre: `_integer_math` ---

--- Début du chapitre: `_intel` --- Blender uses OpenGL for the 3D Viewport and user interface. The graphics card (GPU) and driver have a big impact on Blender's behavior and performance.

This section lists possible solutions for graphics glitches, problems with Eevee and Cycles, and crashes related to your GPU.

Upgrading to the latest graphics drivers often solves problems. Newer drivers have bug fixes that help Blender function correctly. On macOS graphics drivers are built into the operating system and the only way to get newer drivers is to upgrade macOS as a whole to the latest version.

This means your graphics card and driver do not have the minimum required OpenGL 3.3 version needed by Blender.

Installing the latest driver can help upgrade the OpenGL version, though some graphics cards are simply too old to run the latest Blender. Using Blender 2.79 or earlier is the only option then.

Try running Blender from the command line, to see if any helpful error messages are printed.

On Windows, graphics drivers can sometimes get corrupted. In this case it can help to uninstall all graphics drivers (there may be multiple from Intel, AMD and Nvidia) and perform a clean installation with drivers from the manufacturer's website.

Update your graphics drivers (see above).

On laptops, make sure you are using a dedicated GPU (see above).

Try lowering quality settings in Preferences > System > Memory & Limits.

Try undoing settings in your graphics drivers, if you made any changes there.

See [Eevee](#) and [Cycles](#) documentation respectively.

See [Invalid Selection](#), [Disable Anti-Aliasing](#).

Running Blender inside a virtual machine is known to have problems when OpenGL drawing calls are forwarded to the host operating system.

To resolve this, configure the system to use PCI passthrough.

To find out which graphics card and driver Blender is using, use Help > Save System Info inside Blender. The OpenGL section will have information about your graphics card, vendor and driver version.

--- Fin du chapitre: `_intel` ---

--- Début du chapitre: `_interface` --- Interface configuration lets you change how UI elements are displayed and how they react.

Adjusts the size of fonts and buttons relative to the automatically detected DPI. During typical usage, you may prefer to use zoom which is available in many parts of Blender interface.

Scale of lines and points in the interface e.g. button outlines, edges and vertex points in the 3D Viewport.

Thin, Default, Thick

Display the Splash Screen when starting Blender.

Show settings and menu items which are intended to help developers, this includes:

- Operator Search

- Sequencer Cache Settings

To open the Python reference manual.

To copy the expression used when pressing the button.

To edit Python source code that defines the button.

The option to edit UI translations (only available when the Manage UI translations add-on is also enabled).

The option to show mesh vertex/edge/face indices in the overlay popover.

Work in progress features can be enabled here which are currently being tested.

When enabled, a tooltip will appear when your mouse pointer is over a control. This tip explains the function of what is under the pointer, shows the associated hotkey (if any).

Displays a property's Python information below the tooltip.

Show most recently selected items at the top of search results, otherwise search results are sorted alphabetically.

This makes regions overlap the viewport. It means that the Toolbar and Sidebar regions will be displayed overlapping the main area.

Show navigation controls at top right of the area. This impacts the 3D Viewport as well as image spaces.

Note

If you are familiar with navigation key shortcuts, this can be disabled. Choose which type of Color Space you prefer for the Color picker. It will show when clicking LMB on any color field.

Circle HSV.

Circle HSL.

Square (SV + H).

Square (HS + V).

Square (HV + S).

The default header position when opening a new editor.

Uses top for most editor types and the positions saved in the start-up file.

Always positions the header at the top or the bottom of the editor.

How factor value types are displayed in the user interface.

Values are displayed as float numbers between 0.0 and 1.0.

Values are expressed as a percentage between 0 and 100.

When performing certain operations, Blender will open a new window. The behavior of these operations can be configured here.

When rendering, the user interface can do any of:

The user interface does not change and the render is computed in the background.

A new Image editor is opened as a temporary window in full screen mode.

The area that is the largest on screen is replaced by a temporary Image editor.

A new Image editor is opened as a regularly sized temporary window.

When opening files from the computer, the user interface can do any of:

A new File Browser editor is opened as a temporary window in full screen mode.

A new File Browser editor is opened as a regularly sized temporary window.

Preferences that affect the Status Bar.

Show

Shows information about the data in the active scene.

Collection: The name of the active Collection.

Active Object: The name of the active selected object.

Geometry: Information about the current scene depending on the mode and object type. This can be the number of vertices, faces, triangles, or bones.

Objects: The number of selected objects and the total count of objects.

Shows the total amount of time of the playback along with the current frame number and total frame count. The format of the duration text is determined by the Timecode Style.

Shows an estimate of Blender's RAM consumption. On a single-instance single-machine scenario, this estimate provides a measurement against the hardware limit of the machine.

Shows the number of extensions with available updates.

Shows the version number of Blender that is currently running.

The language used for translating the user interface (UI). The list is broken up into categories determining how complete the translations are.

Translates the descriptions when hovering over UI elements.

Translates all labels in menus, buttons, and panels.

Translates the names of new data-blocks.

Enable interface text Anti-Aliasing. When disabled, texts are rendered using straight text rendering (filling only absolute pixels).

Render text for optimal horizontal placement.

Adjust font hinting, controls the spacing and crispness of text display.

Replacement for the default user interface font.

Replacement for the default mono-space interface font (used in the Text editor and Python Console).

Select this to have the menu open by placing the mouse pointer over the entry instead of clicking on it.

Time delay in 1/10 second before a menu opens (Open on Mouse Over needs to be enabled).

Same as above for sub menus (for example: File > Open Recent).

Length of animation when opening Pie Menus.

Keystrokes held longer than this will dismiss the menu on release (in 1/100ths of a second).

The window system tries to keep the pie menu within the window borders. Pie menus will use the initial mouse position as center for this amount of time, measured in 1/100ths of a second. This allows for fast dragged selections.

The size of the Pie Menu set with the distance (in pixels) of the menu items from the center of the pie menu.

Distance from center before a selection can be made.

Distance threshold after which selection is made (zero disables).

--- Fin du chapitre: _interface ---

--- Début du chapitre: _interior --- By default, the edges of a soft-body mesh act like springs. This means that, like a mechanical spring, they can stretch under tension and squeeze under pressure. Their initial length is also their ideal or crest length, which they try to return to.

Having edges act like springs is what holds the mesh together. If you were to disable this behavior (as well as the Goal), each vertex would be free to go anywhere independently of the others, which would stretch the mesh until it's no longer recognizable.

Having springs along edges alone typically isn't enough, however: vertices in quads are still free to move towards their diagonal opposite, potentially collapsing the quad into a line.

You could solve this by creating diagonal edges everywhere, but fortunately, you don't have to: simply enable the Stiffness option to have Blender create diagonal springs internally. This way, you don't have to change your mesh.

Base springs along edges. \xc2\xbb6

Additional springs when Stiffness is enabled. \xc2\xbb6

Another method of preventing mesh collapse is applying Bending Stiffness, which adds rotational resistance: making edges try to keep their relative angles.

Both of these methods are described in more detail below. You can configure them, as well as other settings, in the Soft Body Edges panel. To show the effect of the Stiffness setting, we will drop two cubes onto a plane (see Collisions). The blue cube uses quads, while the red one uses tris. Both cubes have their Goal setting disabled.

If Stiffness is disabled, the quad-only cube will collapse completely, while the tri cube only temporarily deforms from the impact:

Frame 1.
Frame 36.
Frame 401.

If Stiffness is enabled, the quad cube maintains its shape as well thanks to the extra springs:

Frame 1.
Frame 36.
Frame 401.

The second method to stop an object from collapsing is to give it Bending Stiffness. Just like the other settings, this can be combined with Stiffness to add bending resistance to the diagonal springs as well. We first do the same cube experiment as before, using only Bending Stiffness:

Frame 1.
Frame 36.
Frame 401.

Both cubes keep their shape. Now, we try the same thing with subdivided planes, again a quad-based one and a triangulated one:

Two planes falling.
No bending stiffness.
High bending stiffness (10).

Without any Bending Stiffness, the faces can rotate freely as though their edges were hinges. Enabling Stiffness to add diagonal springs would not change this (just as triangulating doesn't).

With a high Bending Stiffness, however, the edges resist this rotation, and the planes act more like planks than towels.

--- Fin du chapitre: _interior ---

--- Début du chapitre: _interpolate ---Reference

Draw Mode

Toolbar  Interpolate

The Interpolate tool interpolates strokes between the previous and next keyframe by adding a single keyframe. When you are on a frame between two keyframes and click and drag a new breakdown keyframe will be added. This way you define the final interpolation for the new stroke.

Set the Playhead on the Timeline between the two keyframes you want to interpolate. Click and drag from left to right to set the desired interpolation percentage and release to confirm, a new breakdown keyframe will be added.

Restrict the interpolation to Active or All layers.

When enabled, only selected strokes will be interpolated.

Exclude existing Breakdowns keyframes as interpolation extremes.

Invert strokes start and end. Automatic will try to find the right mode for every stroke.

Amount of smoothing to apply to interpolated strokes for reducing jitter/noise.

Number of time to smooth newly created strokes.

--- Fin du chapitre: `_interpolate` ---

--- Début du chapitre: `_interpolate_curves` --- Generate new curves on points by interpolating between existing curves. This is useful to have a smaller set of original curves to make editing easier and faster while still generating high-density curves for the viewport or a final render. Base curves that new curves are interpolated between.

Optional up vector that is typically a surface normal. Supplying an up vector will improve the quality of the interpolation, making it aware of the surface shape which it otherwise wouldn't have a way to know about.

Tip

In a typical child hair generation setup, this up direction is retrieved with a combination of the Sample UV Surface Node using the same geometry that the points were distributed on, and the Normal Node.

Splits guides into separate groups. New curves interpolate existing curves from a single group. This input can be useful for generating hair parts by stopping curves in different sections (with different group IDs) from affecting each other.

The positions of the first root control points of the newly generated interpolated curves.

Optional up vector that is typically a surface normal.

The curve group to interpolate in.

Maximum amount of close guide curves that are taken into account for interpolation.

This node has no properties.

The guide curves with interpolated curves.

Index of the closest guide curve for each generated curve.

Note

Internally this node mixes the data from multiple guide curves, with the maximum number of sources depending on the Max Neighbors input. This output is only the index of the curve with the largest weight.

Weight of the closest guide curve for each generated curve.

--- Fin du chapitre: `_interpolate_curves` ---

--- Début du chapitre: `_interpolate_hair_curves` --- Interpolates existing guide curves on a surface mesh. The Duplicate Hair Curves is a similar option with simpler behavior that may offer better performance.

Note

This node/modifier will not function without the Surface Geometry/object and Surface UV Map inputs.

Input Geometry (only curves will be affected).

Surface geometry for generation. This input takes priority over the corresponding object input if both are provided.

Surface object for generation (Needs matching transforms).

Surface UV map stored on the mesh used for finding curve attachment locations.

Set the surface mesh into its rest position before attachment.

Tip

In a typical hair generation setup, this node or modifier will be combined with the Deform Curves on Surface Node. If that operation comes after this one, it makes sense to turn this option on so the position used is the pre-deformed position consistent with the expectations for the deformation's input.

Align the interpolated curves to the surface normal.

Use mesh islands of the surface geometry for parting.

Amount of guides to be used for interpolation per curve.

Distance around each guide to spawn interpolated curves.

Use poisson disk distribution method to keep a minimum distance.

Surface density of generated hair curves.

Factor applied on the density for curve distribution.

Discard points based on a mask texture after distribution. The image is sampled with the Surface UV Map input.

Tip

The accuracy of sampling the image doesn't depend on the density of the surface mesh's vertices because it is sampled after the curve root points are generated, the accuracy. However, using the Density Mask input instead can give better performance. Using them in combination can give the benefits of both methods.

Factor applied on the density for the viewport.

Random seed for the operation.

This node has no properties.

Geometry

Index of the main guide curve per curve.

Normal direction of the surface mesh at the attachment point.

--- Fin du chapitre: _interpolate_hair_curves ---

--- Début du chapitre: _interpolation --- Reference

Draw and Edit Modes

Toolbar \xe2\x80\xa3 Interpolate

Ctrl- E

When you are animating simple shapes you can use the interpolate tool to automatically add new breakdown keyframes.

See Interpolate tool for more details.

Reference

Draw and Edit Modes

Header \xe2\x80\xa3 Interpolate

Shift- Ctrl- E

Interpolate strokes between the previous and next keyframe by adding multiple keyframes. When you are on a frame between two keyframes and click the sequence button a breakdown keyframe will be added on every frame between the previous and next keyframe.

The number of frames between generated interpolated frames.

Restrict the interpolation to Active or All layers.

When enabled, only selected strokes will be interpolated.

Exclude existing Breakdowns keyframes as interpolation extremes.

Invert strokes start and end. Automatic will try to find the right mode for every stroke.

Amount of smoothing to apply to interpolated strokes for reducing jitter/noise.

Number of time to smooth newly created strokes.

Interpolation method to use for the sequence.

--- Fin du chapitre: _interpolation ---

--- Début du chapitre: _intersect_boolean ---Reference

Edit Mode

Face \xe2\x80\xa3 Intersect (Boolean)

Performs Boolean operations with the selection on the unselected geometry. While the Boolean Modifier is useful for non-destructive edits, access to these operations with a tool in Edit Mode can be useful to quickly perform edits.

Opposite of Difference (everything inside of the target mesh is kept).

The target mesh is added to the modified mesh.

The target mesh is subtracted from the modified mesh (everything outside of the target mesh is kept).

Algorithm used to calculate the Boolean intersections.

Uses a mathematically simple solver which offers the best performance; however, this solver lacks support for overlapping geometry.

Tolerance for close faces to be considered touching. It may be useful to increase this when some intersections aren't detected that should be and when extra geometry is being created because edges aren't detected as overlapping.

Warning

A threshold approaching the size of faces may cause very slow calculation, in general keep this value small.

Uses a mathematically complex solver which offers the best results and has full support for overlapping geometry; however, this solver is much slower than the Fast Solver.

Changes the order of the operations when using Difference to determine which side is kept.

Correctly calculates cases when one or both operands have self-intersections, this involves more calculations making it slower.

--- Fin du chapitre: _intersect_boolean ---

--- Début du chapitre: _intersect_knife ---Reference

Edit Mode

Face \xe2\x80\xa3 Intersect (Knife)

The Intersect tool lets you cut intersections into geometry. It is a bit like the Boolean tool, but, does not calculate interior/exterior geometry. Faces are split along the intersections, leaving new edges selected.

Operate on the overlapping geometry of the mesh.

Operate between the selected and unselected geometry.

Splits the geometry at the new edge.

Keep each side of the intersection separate without splitting the faces in half.

Merge all the geometry from the intersection.

Algorithm used to calculate the intersections.

Uses a mathematically simple solver which offers the best performance; however, this solver lacks support for overlapping geometry.

Tolerance for close faces to be considered touching. It may be useful to increase this when some intersections aren't detected that should be and when extra geometry is being created because edges aren't detected as overlapping.

Warning

A threshold approaching the size of faces may cause very slow calculation, in general keep this value small.

Uses a mathematically complex solver which offers the best results and has full support for overlapping geometry; however, this solver is much slower than the Fast Solver.

--- Fin du chapitre: `_intersect_knife` ---

--- Début du chapitre: `_introduction` --- Geometry Nodes is a system for modifying the geometry of an object with node-based operations. It can be accessed by adding a Geometry Nodes Modifier.

The properties of a Geometry Nodes modifier in the modifier stack.

The geometry node tree connected to a modifier is a Node Group. The geometry from the state before the modifier (the original geometry or the result of the previous modifier) will be passed to the Group Input node. Then the node group can operate on the geometry and pass an output to the Group Output node, where it will be passed to the next modifier.

Geometry nodes can modify different types of geometry:

Meshes

Curves

Point Clouds

Volumes

Instances

The interface of the modifier is described in the Modifier page.

To expand Blender with node-group operators see the description in the Node-Based Toolspace.

--- Fin du chapitre: `_introduction` ---

--- Début du chapitre: `_inverse_kinematics` --- Reference Pose Mode

Bone Inverse Kinematics

The Inverse Kinematics panel.

This panel controls the way a bone or set of bones behave when linked in an inverse kinematic chain.

--- Fin du chapitre: `_inverse_kinematics` ---

--- Début du chapitre: `_invert_color` --- Inverts the colors in the input image, producing a negative.

The amount of influence the node exerts on the image.

Standard color input.

In the compositing context, this node has the following properties:

Invert the color channels.

Invert the alpha channel.

Standard color output.

The Invert node is used to invert the mask.

--- Fin du chapitre: `_invert_color` ---

--- Début du chapitre: `_invert_matrix` --- Returns the inverse of the given matrix.

The matrix to invert.

This node has no properties.

The inverted matrix.

Returns whether the matrix could be inverted. This can be false when a transformation matrix has a scale of zero, for example. See Invertible matrix for more information.

Important

When a matrix is not invertible, the identity matrix is returned.

--- Fin du chapitre: `_invert_matrix` ---

--- Début du chapitre: `_invert_rotation` --- The Invert Rotation node inverts a rotation.

Standard rotation value.

The inverted rotation.

--- Fin du chapitre: `_invert_rotation` ---

--- Début du chapitre: `_in_between` --- In-Between Tools.

There are several tools for editing poses in an animation.

There are also in Pose Mode a bunch of armature-specific editing options/tools, like auto-bones naming, properties switching/enabling/disabling, etc., that were already described in the armature editing pages. See the links above.

Reference

Pose Mode

Pose In-Between Push Pose from Rest Pose

Similar to Push Pose from Breakdown but interpolates the pose to the rest position instead. Only one keyframe is needed for this tool unlike two for the other.

Reference

Pose Mode

Pose In-Between Relax Pose to Rest Pose

Similar to Relax Pose to Breakdown but works to bring the pose back to the rest position instead. Only one keyframe is needed for this tool unlike two for the other.

Reference

Pose Mode

Toolbar In-Between Tools Push

Pose In-Between Push Pose from Breakdown

Ctrl- E

Push Pose interpolates the current pose by making it closer to the next keyframed position.

Reference

Pose Mode

Toolbar In-Between Tools Relax

Pose In-Between Relax Pose to Breakdown

Alt- E

Relax pose is somewhat related to the above topic, but it is only useful with keyframed bones. When you edit such a bone (and hence take it

away from its keyed

position), using this tool will progressively

bring it back to its keyed

position, with smaller and smaller steps as it comes near it.

Reference

Pose Mode

Toolbar region In-Between Tools Breakdown

Pose In-Between Pose Breakdown

LMB-drag

Creates a suitable breakdown pose on the current frame.

The Breakdown tool can be constrained to work on specific transforms and axes, by pressing the following keys while the tool is active:

G, R, S: move, rotate, scale

B: Bendy bones
 C: custom properties
 X, Y, Z: to the corresponding axes
 Reference
 Pose Mode
 Pose \xe2\x80\xa3 In-Between \xe2\x80\xa3 Blend to Neighbor
 Shift- Alt- E
 Transitions the current pose with the neighboring keyframes in the timeline. In order for this operator to work, there must be a keyframe before and after the current frame.
 --- Fin du chapitre: `_in_between` ---

--- Début du chapitre: `_is_edge_smooth` --- The Is Edge Smooth node outputs true for each edge of the mesh that is not marked as sharp. Otherwise, if the edge is marked as sharp, then the node outputs false.
 See also
 Mark Sharp & Clear Sharp
 This node has no inputs.
 This node has no properties.
 Boolean value that indicates whether the edges of the mesh are not marked as sharp.
 --- Fin du chapitre: `_is_edge_smooth` ---

--- Début du chapitre: `_is_face_smooth` --- The Is Face Smooth node outputs true for each face of the mesh if that face is marked to render smooth shaded. Otherwise, if the face is marked to render as flat shaded, then the node outputs false.
 This node has no inputs.
 This node has no properties.
 Boolean value that indicates whether the normals of each face corner on the final mesh are smoothed with normal of all adjacent faces or not.
 --- Fin du chapitre: `_is_face_smooth` ---

--- Début du chapitre: `_is_spline_cyclic` --- The Is Spline Cyclic node controls whether each of the curve splines start and endpoints form a connection. Its output corresponds to the built-in `cyclic` attribute on the curve spline domain.
 The node to set this data is the Set Spline Cyclic Node.
 This node has no inputs.
 This node has no properties.
 Whether the spline is cyclic.
 --- Fin du chapitre: `_is_spline_cyclic` ---

--- Début du chapitre: `_is_viewport` --- The Is Viewport node outputs true when geometry nodes are evaluated for the viewport. For the final render the node outputs false.
 This node has no inputs.
 This node has no properties.
 Boolean value that indicates whether geometry nodes are evaluated for preview.
 --- Fin du chapitre: `_is_viewport` ---

--- Début du chapitre: `_join` --- Reference
 Object Mode

Object \xe2\x80\xa3 Join

Ctrl- J

Join merges all selected objects into the last selected Activeobject. All object data is linked to the active object (which must be selected). All objects must be of the same type: mesh, curve, surface or armature. If several curves are joined, each one will keep its subtype (NURBS or B\xe3\x9zier).

Note

Object data has many attributes which may be handled when joining.

Materials, vertex groups, UV and Vertex layers will be merged.

Modifiers, constraints, groups and parent relationships are ignored when joining and will not be applied to the active object.

--- Fin du chapitre: _join ---

--- D\xe9but du chapitre: _join_geometry --- The Join Geometry node merges separately generated geometries into a single one. If the geometry inputs contain different types of data, the output will also contain different data types.

Note

The node cannot handle the case when more than one geometry input has a volume component.

When multiple mesh inputs contain different materials, the material slots from each mesh geometry are merged so that the output mesh will contain all the input materials.

When merging attributes from multiple geometry inputs, the highest complexity data type is chosen for the output attribute. In other words, if a weight attribute has a Boolean type on one geometry input and a vector data type on another geometry, the weight attribute on the output geometry will have a vector data type. The same heuristic is used for attribute domains, the domain with the most information will be used for the output.

Warning

Like other geometry nodes, this node always outputs generic typed attributes. So instead of a Vertex Group attribute, it will create a \xe2\x80\x9cFloat\xe2\x80\x9d attribute on the result, and it will create a generic 2D vector attribute instead of a special \xe2\x80\x9cUV Map\xe2\x80\x9d attribute. Some other areas of Blender don\xe2\x80\x99t properly handle generic attributes in version 3.0.

Custom face corner normals are also not transferred currently.

Geometry that will be joined. Multiple inputs are allowed. When the node is muted, only the first link will be passed through.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _join_geometry ---

--- D\xe9but du chapitre: _join_strings --- The Join Strings node combines any number of input strings into the output string. The order of the result depends on the vertical ordering of the inputs in the multi-input socket.

Tip

This node can be used to create a multi-line string for the String to Curves Node, when combined with the line break output from the Special Characters Node.

String value to place between each concatenated string.

Multiple string values to be combined in connection order.

This node has no properties.

String result of the concatenation.

--- Fin du chapitre: `_join_strings` ---

--- Début du chapitre: `_keyed` ---Reference

Particle System \xe2\x80\xa3 Physics

Keyed

The path of Keyed particles is determined between particles of any two (or more) particle systems. This allows the creation of a chains of systems to create long strands or groovy moving particles. Basically the particles have no dynamics but are interpolated from one system to the next each frame.

To setup Keyedparticles you need at least two particle systems in the Keyslist.

Sets the number of times the entire Keyslist is repeated. Disabled if Use Timingis enabled.

Enabling this option allows you to specify the timing for each key independently, using the Timeand Durationoptions. By default, the Use Timingoption is deactivated, and the particles will pass through all keysfor a time equal to its lifetime. A shorter lifetime means faster movement. The lifetime will be split equally between the keys, this may lead to varying particle speeds between the targets.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Relations

The list view of keys (target particle systems).

The name of a target object for the selected key. If empty it uses the current particle system.

Index of particle system on the target object.

The time (frame number) at which the particles will be at the position of the selected system. Note also that the Startframe of the Keyed system adds an offset to this time.

How long (in frames) the particles stay on this system before they start moving to the next one.

--- Fin du chapitre: `_keyed` ---

--- Début du chapitre: `_keying` --- The Keyingnode is a one-stop-shop for \xe2\x80\x99cgreen screen\xe2\x80\x9d / \xe2\x80\x99cblue

screen\xe2\x80\x9d removal. It performs both chroma keying to remove the backdrop and despill to correct color cast from the backdrop. Additionally, you can perform common operations used to tweak the resulting matte.

Standard color input.

The color of content to be removed. This may be a single color, or a reference image such as generated by the Keying Screen Node .

An optional mask of area(s) to always exclude from the output. This is removed from the chroma key generated matte.

An optional mask of area(s) to always include in the output. This is merged with the chroma key generated matte.

Reduce the effects of color noise in the image by blurring only color by the given amount, leaving luminosity intact. This will affect matte calculation only, not the result image.

This is the balance between color channels compared with the key color. 0.5 will average the other channels (red and blue in the case of a green screen).

This may be tweaked in tandem with Clip Black and Clip White while checking the Matte output to create a mask with optimal separation.

Controls how much color bleed from the key color is removed from the input image: 0 means no despill, 1 means all possible spilling will be removed. The underlying implementation is the same as adjusting the Unspill amount of the Color Spill Node.

This controls how the color channels are compared when computing spill, affecting the hue and shade of the corrected colors. It is similar to setting the Limiting Channel in the Color Spill Node.

Defines the radius in pixel used to detect an edge.

Defines threshold used to check if pixels in radius are the same as current pixel: if the difference between pixel colors is higher than this threshold then the point will be considered an edge.

This sets the threshold for what becomes fully transparent in the output (black in the matte). It should be set as low as possible. Uneven backdrops will require this value to be increased. Use of the Keying Screen Node can help keep this value low. You may also use a Garbage Matte to exclude problematic areas.

This value does not impact areas detected as edges to ensure edge detail is preserved.

This sets the threshold for what becomes fully opaque in the output (white in the matte). It should be set as high as possible. Colors close to green in the foreground may require reducing this value and/or adjusting the Screen Balance. Particularly problematic parts can be fixed with a Core Matte instead of a low Clip White.

This value does not impact areas detected as edges to ensure edge detail is preserved.

Enlarge (positive numbers) or shrink (negative numbers) the matte by the specified number of pixels. This is similar to using the Dilate/Erode Node on the matte.

This is a simple way to include more or less along the edges of the matte, particularly combined with Post Blur.

The rate of the falloff at the edges of the matte when feathering, to manage edge detail.

Controls how much the matte is feathered inwards (negative number) or outwards (positive number).

Make the matte less sharp, for smoother transitions to the background and noise reduction.

Processed image with the Matte applied to the image's Alpha Channel.

Output matte to use for checking the quality of the key, or to manually apply using a Set Alpha Node or Mix Node.

Shows what edges were detected on the matte. Useful for adjusting the Edge Kernel Radius and Edge Kernel Tolerance.

Tip

If there are problems with the edges of the matte, it may help to start with adjusting the Edge Kernel parameters before adjusting feathering. Detected edges are not subject to Clip Black/Clip White thresholds to preserve fine edge detail. You can check edge detection by connecting a Viewer Node to the Edges output.

Sharper detected edges (smaller Edge Kernel Radius, like 2 / larger Edge Kernel Tolerance, like 0.4) will create a sharper matte, but may lose some detail like stray hairs. A sharp matte is good, but disappearing or flickering hairs are distracting.

Fat edges (larger Edge Kernel Radius, like 8 / smaller Edge Kernel Tolerance, like 0.05) will capture more edge detail, but may also produce a halo around the subject. The halo can be adjusted with Feather controls along with Dilate/Erode.

--- Fin du chapitre: _keying ---

--- Début du chapitre: _keying_screen --- The Keying Screen node creates plates for use as a color reference for keying nodes. It generates gradients from sampled colors on motion tracking points on movie clips. It can be used to deal with uneven colors of green screens.

This node has no input sockets.

The selectable clip data-block used as input for the gradient colors. Tracking Object to generate the gradient. You will probably want to create new a tracking object in the Object panel, because tracks used for gradients can not actually be used for camera/object tracking. After this tracks might be placed in places where gradient colors should be sampled. These tracks could be tracked or moved manually, so gradients would be updating automatically along the movie. Tracks might have an offset for easier tracking of feature-less screens.

Gradient image output.

Consider a node setup for green screen removal, using a Color Key :

Often, lighting is uneven across the backdrop.

Example from the Mango Open Movie, Tears of Steel. \xc2\xba6

That can result in a bad matte.

Example of a poor mask: Some of the backdrop is opaque, and some parts of the gun in the foreground are transparent. \xc2\xba6

If you increase the tolerances on the Color Key node, it will accept more shades of green to mask out. But it may also incorrectly mask out more of the foreground.

Instead of increasing the range of accepted shades to be masked out, the Keying Screen node lets you change what shade of green (or other color) to use for different parts of the image.

Start in the Movie Clip Editor. Open the Sidebar region and Toolbar to show tracking configuration. Tracks used for gradients are not useful for camera solving, because they do not track well. So create a new object track in the Objects selector. Place tracking markers on the clip to sample different parts of the backdrop.

These tracks may be tracked or moved manually, so gradients can be updated over time. If the marker is not enabled for a frame, it will not be used creating the gradient. (Such as the red-colored marker on the arm in the screenshot above)

Once the tracks are created, add the node to your compositing setup, and select the tracking object used for the backdrop.

Node configuration with Keying Screen \xe2\x80\x99s generated gradient plate connected to the Color input of the Keying node. \xc2\xba6

Gradient plate generated by Keying Screen. \xc2\xba6

The resulting image now has a better matte.

--- Fin du chapitre: _keying_screen ---

--- Début du chapitre: _keying_sets --- The Active Keying Sets data ID in the Timeline.\xc2\xba6

Keying Sets are a collection of animated properties that are used to animate and keyframe multiple properties at the same time. For example, pressing K in the 3D Viewport will bring up the available Keying Sets. Blender will then add keyframes for whichever Keying Set is chosen. There are some built-in Keying Sets and also custom Keying Sets called \xe2\x80\x9cAbsolute Keying Sets\xe2\x80\x9d.

Reference

Properties

Scene \xe2\x80\x93 Keying Set

This panel is used to add, select, manage \xe2\x80\x9cAbsolute Keying Sets\xe2\x80\x9d.

The Keying Set panel. \xc2\xba6

A List View of Keying Sets in the active scene. Selecting a keying set makes it active

Adds an empty Keying Set.

Removes the active keying set.

A short description of the Keying Set.

Export Keying Set to a Python script File.py .To re-add the Keying Set from the File.py , open then run the File.py from the Text Editor.

Reference

Properties

Scene \xe2\x80\x93 Active Keying Set

This panel is used to add properties to the active Keying Set.

The Active Keying Set panel. \xc2\xba6

A collection of paths in a List View each with a Data Path to a property to add to the active Keying Set.

Adds an empty path.

Removes the selected path.

Set the ID Type and the Object ID's data path for the property.

Set the rest of the Data Path for the property.

Use All Items from the Data Path or select the array index for a specific property.

This controls what group to add the channels to.

Keying Set Name, None, Named Group

These options control all properties in the Keying Set. Note that the same settings in Preferences override these settings if enabled.

These options control individual properties in the Keying Set.

Only insert keyframes where they are needed in the relevant F-Curves.

Insert keyframes based on the visual transformation.

Reference

Context menu \xe2\x80\x93 Add All/Single to Keying Set

K

Some ways to add properties to Keying Sets.

RMB the property in the User Interface, then select Add Single to Keying Set or Add All to Keying Set. This will add the properties to the active Keying Set, or to a new Keying Set if none exist.

Hover the mouse over the properties, then press K, to add Add All to Keying Set.

Reference

Shift- K

There are several ways to designate the active keying set:

Press Shift- K in the 3D Viewport.

Select a keying set in the Keying Set panel.
Select a keying set in the Keying popover in the Timeline header,
The built-in Whole CharacterKeying Set is made to keyframe all
properties that are likely to get animated in a character rig. It was also
implicitly used by the Old Pose Library system .
This keying set ignores bones whose name starts with one of the following
prefixes, as it assumes these are technical bones that are not meant to be
animated directly. The built-in Rigify add-on generates such bones, for
example.

COR (Corrective)

DEF (Deformation)

GEO (Geometry)

MCH (Mechanism)

ORG (Original from meta rig)

VIS (Visualization)

--- Fin du chapitre: _keying_sets ---

--- Début du chapitre: _keymap --- The keymap editor lets you adjust your
keymap via:

Predefined keymaps which come with Blender and can be added to.

Keymaps may define their own preferences to change the functionality or
add additional key bindings.

You may add/remove/edit individual keymap entries.

Blender Preferences Keymap section.

Select the keymap from a list of predefined keymaps.

You add a custom keymap configuration to the preset list by LMB on the
Add button + .

You remove a custom keymap configuration from the preset list by LMB on
the Remove button - .

Importing opens a File Browser to select a .py file to add to the list of
keymap presets.

Saves the current keymap configuration as a preset others may use.

When disabled, only the keymaps and categories that have been modified by
the user will be exported. In addition, add-ons may register keymaps to
their respective functions, however, these keymaps are not exported unless
changed by the user. This exported file may be thought of as a
\xe2\x80\x9ckeymap delta\xe2\x80\x9d instead of a full keymap export.

When enabled, the entire keymap is written.

Search the keymap item by the operator name it runs.

Search the keymap item by the key used to activate it.

Hint

You could for example search with CtrlShiftC for keymap items that use
all these keys.

The text to search (leave blank to disable).

Keymaps may define their own preferences, these are predefined
adjustments to the keymap you can make without having to manually adjust
individual keymap items which can cause problems with newer Blender
Versions.

See the default keymap preferences for options available in the default
keymap.

The Keymap editor lets you change the default hotkeys. You can change
keymaps for each of Blender's editors.

Keymap editor.

Usage

Select the keymap you want to change and click on the white arrows to open up the keymap tree.

Select which Input will control the function.

Change hotkeys as you want. Just click on the shortcut input and enter the new shortcut.

Uncheck to disable this keymap item.

Single hotkey or key combination.

Actions from mouse buttons, tablet or touchpad input.

Movement or button from a 3D mouse (NDOF) device.

Mouse click and drag (optionally map drag direction to different actions).

Use this function by entering a text.

Used to control actions based on a time period.E.g. by default, Animation Stepuses \xe2\x80\x9cTimer 0\xe2\x80\x9d, Smooth Viewuses \xe2\x80\x9cTimer 1\xe2\x80\x9d.

The identifier for the operator to call.

Hint

Seebpy.ops for a list of operators (remove thebpy.prefix for the identifier).

The key or button that activates this keymap item (depending on the map type).

The action (such as press, release, click, drag, etc.), (depending on the map type).

Additional keys to hold (such as Ctrl, Shift, Alt).

Changes to the defaults properties this operator is activated with

See also

Keymap Customizationfor more information on keymap editing.

If you want to restore the default settings for a keymap,just click on the Restorebutton at the top right of this keymap.

Tip

Instead of deleting the default keymap to create your custom one,you can just add a new Presetfor both the mouse and keyboard.

A problem with modifying your own keymap is newer Blender versions key change the way tools are accessed,breaking your customized keymap.

While the keymap can be manually updated, the more customizations you make, the higher the chance of conflictsin newer Blender versions is.

--- Fin du chapitre: _keymap ---

--- Début du chapitre: _keymap_editing --- When customizing keymaps it\xe2\x80\x99s useful to use keys which won\xe2\x80\x99t conflict with Blender\xe2\x80\x99s default keymap.

Here are keys which aren\xe2\x80\x99t used and aren\xe2\x80\x99t likely to be used in the future.

These F-keys (including modifier combination)have been intentionally kept free for users to bind their own keys to.

Blender doesn\xe2\x80\x99t use this key for any bindings.

macOSis an exception, where Cmdreplaces Ctrlexcept in cases it would conflict with the system\xe2\x80\x99s key bindings.

Binding modifier keys as primary keys is supported,to avoid conflicts with regular usage you can bind them to double click.

It\xe2\x80\x99s possible to configure a single key to perform multiple operationsusing Clickevent instead of Press. Then you may bind Dragto a separate action.

This is useful for mixing actions where one uses a drag event, e.g:Toggle a setting using with Tab, drag to open a pie menu showing all options related to the setting.

This is used in the default keymap in the 3D Viewport, Alt- MMB dragging in different directions rotates the view.

This section lists useful generic operations which can be used.

Menus and panels can be assigned key shortcuts,even if they\xe2\x80\x99re only accessible from submenus elsewhere.

- Open any menu on key press.

- Open any pie menu on key press.

- Open a pop-up panel (also known as a pop-over).

To find the name of a menu,enable the preference Interface \xe2\x80\xa3 Display \xe2\x80\xa3 Python Tooltips.

Then hover the cursor over the popover button or menu item.For submenus you will need to use the back arrow to prevent the submenu from opening and gaining focus.

There are many properties you might want to bind a key with.To avoid having to define operators for each property,there are generic operators for this purpose:

- Operators for adjusting properties begin withwm.context_.

- Some of these include:

- wm.context_toggletoggle a Boolean property.

- wm.context_cycle_enumcycle an enumproperty forwards or backwards.

- wm.context_menu_enumshow a pop-up menu for an enum property.

- wm.context_pie_enumshow a pie menu for an enum property.

- wm.context_scale_floatscale a number (used for increasing / decreasing brush size for example).

- wm.context_toggle_enumtoggle between two options of an enum.

- wm.context_modal_mousemoving the cursor to interactively change a value.

- See bpy.ops.wmfor a complete list.

- Each of these operators has adata_pathsetting to reference the property to change.

- To find thedata_path, basic Python knowledge is needed.

- For example, you can use the Python Console to access a Boolean property you wish to map to a key:

- To bind this to a key, add a new keymap item using the operatorwm.context_togglewithdata_pathset toobject.show_name(notice thebpy.contextprefix is implicit).

- See bpy.contextfor other context attributes.

- The Python API documentation can be used to find propertiesor you may use the Python Console\xe2\x80\x99s auto-complete to inspect available properties.

--- Fin du chapitre: _keymap_editing ---

--- Début du chapitre: _knife_project ---Reference

Edit Mode

- Mesh \xe2\x80\xa3 Knife Project

- Knife Project is a non-interactive tool where you can use objects to cookie-cut intoone or more meshes rather than hand drawing the line. The outline of the selected objectsthat are notin Edit Mode is projected along the view axis onto the meshes that arein Edit Mode, and then cuts into the faces there. Afterwards, the resulting geometryinside the cut gets selected.

- Note

The cutting objects must be curves or non-manifold meshes (e.g. flat shapes, loose edges). Select Non-Manifold will highlight the cutting edges of mesh objects.

Keep in mind that Knife Project works from the current view perspective. For best results, make sure to rotate your view to exactly the position you require before using this tool. Orthographic views such as Right, Front, and Top are commonly used for this.

Hint

3D Viewport Alignment to adjust the projection axis.

To use Knife Project, select the objects to be cut, switch to Edit Mode, select the cutting objects in the Outliner (Ctrl-LMB), and choose Mesh → Knife Project.

If Blender switches back to Object Mode when selecting the cutting objects, make sure that Edit → Lock Object Mode is checked in the topbar. Alternatively, if you have only one cutting object, you can select it in the viewport with Ctrl-LMB.

Projects the cut through the entire mesh, including back faces not currently visible.

Before projecting from a text object.

Resulting knife projection.

Before projecting from a mesh object.

Resulting knife projection (extruded after).

Before projecting from a 3D curve object.

Resulting knife projection (extruded after).

When cutting multiple meshes in Edit Mode at once, geometry from these meshes does not occlude separate mesh objects behind them.

--- Fin du chapitre: _knife_project ---

--- Début du chapitre: _knife_topology_tool ---Reference

Edit Mode

Toolbar → Knife

Mesh → Knife Topology Tool

K

The Knife tool can be used to interactively subdivide (cut up) geometry by drawing lines or closed loops to create holes.

When using Knife, the cursor changes to an icon of a scalpel and the header changes to display options for the tool. You can draw connected straight lines by clicking LMB, marked with small green squares. Red squares are already defined cuts. Surrounding red squares mean that there is a cut already in that position, so no additional vertex will be created (besides the first one).

If multiple objects are selected before entering Edit Mode, then knife cuts will affect all of those objects.

Mesh before knife cut.

Knife cut active.

After confirming knife cut.

Only cut geometry visible on screen.

Only cuts through selected geometry.

Show cut points on non-visible geometry too, when Occlude Geometry is deactivated.

Which visible measurements to show.

Distance, Angle, None, Both

Whether or not dragged lines should be constrained to particular angles, and if so, which coordinate system the angle is relative to.

None, Screen, Relative

When angle snapping is on, the angle will be constrained to a multiple of this angle.

Confirms the cut. Return will leave selected every edge except the new edges created from the cut.

Cancels the cut.

So you can draw a free-hand line over a surface, points will be created at edge intersections.

This is a quick way to close the loop you are currently cutting.

Begins a new cut. This allows you to define multiple distinct cut lines. If multiple cuts have been defined, they are recognized as new snapping points.

Creating multiple cuts.

Result of starting new cuts while in the tool.

Hold to snap the cursor to the midpoint of edges, meaning that all cuts will be performed at the exact center of each cut edge.

Hold to make the tool ignore snapping, unlike the default where mouse cursor snaps to near edges.

Allow the Cut tool to cut through to occluded faces, instead of only the visible ones.

Constrains the cut line to certain degree increments. The increment can be specified in the Tool Settings (see above), or can be typed when angle constraining is active. The default angles are in the plane of the screen, but typing A again makes it relative to the last cut edge. If the last cut edge is ambiguous (because the cut was on a vertex), typing R cycles through the possible reference edges.

Constraining cut angle.

Result of constraining cut angle.

Constrains the cut line to one of the coordinate system axes. Initially it will be the global axis with the given name, but pressing the same key again switches to the object's local axis system. Additionally, if the scene transformation orientation is set to a custom orientation (e.g. from a face), the constraints will be in that coordinate system. Shows measurements of the cuts being made: angles with respect to a mesh edge, lengths, or both. Pressing S repeatedly cycles between what can be shown.

Only Distance, Only Angles, Both, None

Only Distance.

Only Angles.

Both Angles and Distance.

Undoes the previous cut segment. The starting point for the next cut is adjusted accordingly. If a cut is a drag cut, the entire drag cut is undone.

Toggles whether or not cuts to segments behind the visible geometry are shown.

If you experience problems where duplicate vertices are being created by cuts, this is often caused by too large a near/far clipping range.

Try increasing the Clip Start to avoid this problem, see Depth Troubleshooting for details.

Cuts that begin or end in the middle of a face, will be ignored.

This is constrained by the kinds of geometry Blender can represent.

--- Fin du chapitre: _knife_topology_tool ---

--- Début du chapitre: _kuwahara --- The Kuwahara node implements the Kuwahara filter as well as its anisotropic variant. The Kuwahara filter is a smoothing filter that tries to preserve the edges in the image. The smoothing effect of the anisotropic variant is similar to brush strokes, so the node can be used to create stylized painting effects.

Standard color input.

Controls the size of the smoothing neighborhood. Large values may introduce artifacts for highly detailed areas. For the anisotropic method, the larger the size, the slower the filter.

Original.

Size: 3.

Size: 6.

Size: 9.

A simple smoothing method that averages the local square neighborhood of the image while preserving edges. Produces blocky results due to the square neighborhood and provides no tuning parameters, but is faster to compute.

A complex smoothing method that averages the local neighborhood of the image in the direction of the flow of the edges, thus preserving the edges in the output. Produces painterly-like results and provides multiple turning parameters, while being slower to compute.

Uses a more precise but slower method. Use if the output contains undesirable noise.

Controls the uniformity of the directions of the edges of the image. Nonuniform directions are nearly never desirable, so this should typically be increased until the user notices the result is no longer changing in a significant way. Further increases would produce worst results and increase compute time.

Controls the sharpness of the edges of the image.

Original.

Sharpness: 0.

Sharpness: 0.5.

Sharpness: 1.

Controls how thin and directional the filter is. Low eccentricity corresponds to circular omnidirectional features while high eccentricity corresponds to thin directional features.

Original.

Eccentricity: 0.

Eccentricity: 1.

Eccentricity: 2.

Standard color output.

The filter can be applied multiple times by chaining the node multiple times. This chaining can produce more flat filtering.

Original.

Iterations: 1.

Iterations: 2.

Iterations: 3.

The filter can be expensive to compute for high size input and high resolution images. To improve performance, consider scaling down the image, applying the filter, then scaling it up again. This can work well because the filter already attenuates low frequency details.

--- Fin du chapitre: _kuwahara ---

--- Début du chapitre: `_laplacian_deform` --- The Laplacian Deformmodifier allows you to pose a mesh while preservinggeometric details of the surface.

The user defines a set of `\xe2\x80\x9canchor\xe2\x80\x9d` vertices, and then moves some of them around.The modifier keeps the rest of the anchor vertices in fixed positions andcalculates the optimal locations of all the remaining vertices to preserve the original geometric details.

This modifier captures the geometric details with the use of differential coordinates.The differential coordinates capture the local geometric information, the curvature anddirection of a vertex based on its neighbors.

Note

You must define an Anchors Vertex Group. Without it the modifier does nothing.

The Laplacian Deform modifier. \xc2\xbb6

How many iterations to do to improve the found solution.The objective is to find the rotation of the differential coordinatespreserving the best possible geometric details.Details are retained better if more iterations are used,however, it will take longer to calculate.

Original Model.\xc2\xbb6

Repeat: 1.\xc2\xbb6

Repeat: 2.\xc2\xbb6

Repeat: 5.\xc2\xbb6

Original Model.\xc2\xbb6

Repeat: 1.\xc2\xbb6

Repeat: 2.\xc2\xbb6

Repeat: 10.\xc2\xbb6

The group of vertices that the user will use to transform the model.The weight of each vertex does not affect the behavior of the modifier,the method only takes into account vertices with weight greater than 0.

Inverts the influence of the selected vertex group, meaning that the groupnow represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group.

The Bindbutton is what tells the Laplacian Deformmodifier to actually capture the geometry detailsof the object, so that altering the anchor vertices actually alters the shape of the deformed object.

After binding the modifier, you may later decide to make changes to the Anchors Vertex Group.To do so you will first need to Unbindthe modifier before binding it again.

This message is displayed when a user deletes the vertex group or changes its name.

This message is displayed when a user adds or deletes vertices to/from the mesh.

This message is displayed when a user adds or deletes edges to/from the mesh.

This message is displayed if the solver could not find a solution for the linear system.

Note

If the mesh is dense, with a number of vertices greater than 100,000,then it is possible that the nonlinear optimization system will fail.

Laplacian Surface Editingis a method developed by Olga Sorkine and others in 2004.This method preserves geometric details as much as possible while the user makes editing operations.This method uses

differential coordinatescorresponding to the difference between a vector and the weighted averageof its neighbors to represent the local geometric detail of the mesh.

Differential Coordinate. \xc2\xbb

See also

Laplacian Surface Editing (Original paper)

Differential Coordinates for Interactive Mesh Editing

--- Fin du chapitre: _laplacian_deform ---

--- D but du chapitre: _laplacian_smooth --- The Smooth Laplacianmodifier allows you to reduce noise on a mesh\xe2\x80\x99s surface with minimal changes to its shape.

It can also exaggerate the shape using a negative Factor.

The Smooth Laplacianis useful for objects that have been reconstructed fromthe real world and contain undesirable noise. It removes noise while stillpreserving desirable geometry as well as the shape of the original model.

This modifier is based on a curvature flow Laplace Beltramioperator in a diffusion equation.

Hint

Mesher with a great number of vertices, more than ten thousand (10,000),may take several minutes for processing. You can use small portions of the mesh for testingbefore executing the modifier on the entire model.

The Smooth Laplacian modifier. \xc2\xbb

Repetitions allow you to run the smoothing operation multiple times.Each repetition causes the flow curvature of the mesh to be recalculated again,and as a result it removes more noise with every new iteration using a small Factor< 1.0.

When on 0, no smoothing is done.

Note

More repetitions will take longer to calculate.So beware of doing so on meshes with a large number of vertices.

Repeat: 0.\xc2\xbb

Repeat: 1.\xc2\xbb

Repeat: 5.\xc2\xbb

Repeat: 10.\xc2\xbb

Repeat: 0.\xc2\xbb

Repeat: 1.\xc2\xbb

Repeat: 5.\xc2\xbb

Repeat: 10.\xc2\xbb

Repeat: 0.\xc2\xbb

Repeat: 1.\xc2\xbb

Repeat: 5.\xc2\xbb

Repeat: 10.\xc2\xbb

Toggle buttons to enable/disable deforming vertices in the X, Y and/or Z axes directions.

X, Y, Z: Unselected.\xc2\xbb

X, Y, Z: Selected.\xc2\xbb

X, Z: Selected.\xc2\xbb

X: Selected.\xc2\xbb

X, Y, Z: Unselected.\xc2\xbb

X, Y, Z: Selected.\xc2\xbb

X, Z: Selected.\xc2\xbb

X: Selected.\xc2\xba

Controls the amount of displacement of every vertex along the flow curvature.

Using a small Factor, you can remove noise from the shape without affecting desirable geometry.

Using a large Factor, you get smoothed versions of the shape at the cost of fine geometry details.

Using a negative Factor, you can enhance the shape, preserving desirable geometry.

When the Factor is negative, multiple iterations can magnify the noise.

Factor: 0.0.\xc2\xba

Factor: 0.5.\xc2\xba

Factor: 2.5.\xc2\xba

Factor: 5.0.\xc2\xba

Factor: 0.0.\xc2\xba

Factor: 1.0.\xc2\xba

Factor: 10.0.\xc2\xba

Factor: 50.0.\xc2\xba

Factor: 0.0.\xc2\xba

Factor: -20.0.\xc2\xba

Factor: -50.0.\xc2\xba

Factor: -300.0.\xc2\xba

Since there is no way to calculate the curvature flow on border edges, they must be controlled separately. Border edges are smoothed using a much simpler method, using this property to control the influence.

Positive values will smooth the vertex positions, while negative values will enhance them by transforming them in the opposite direction.

Border: 0.0.\xc2\xba

Border: 1.0.\xc2\xba

Border: 2.5.\xc2\xba

Border: 10.0.\xc2\xba

Border: 0.0.\xc2\xba

Border: 1.0.\xc2\xba

Border: 5.0.\xc2\xba

Border: 20.0.\xc2\xba

Border: 0.0.\xc2\xba

Border: -20.0.\xc2\xba

Border: -50.0.\xc2\xba

Border: -200.0.\xc2\xba

The smoothing process can produce shrinkage. That is significant for large Factor or large Repeat values. You can use that option to reduce that effect.

Off.\xc2\xba

On.\xc2\xba

Off.\xc2\xba

On.\xc2\xba

When enabled, the results will depend on face sizes. When disabled, geometry spikes may occur.

Original Geometry.\xc2\xba

On.\xc2\xba

Off.\xc2\xba

Off, High Factor.\xc2\xba

A vertex group name, to constrain the effect to a group of vertices only. Allows for selective, real-time smoothing or enhancing, by painting vertex weights.

Original Geometry

No Group Chosen

Vertex Weights

Result

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group.

Femme front view blend-file.\xc2\xba

Femme side view blend-file.\xc2\xba

Cube smooth blend-file.\xc2\xba

Shape enhanced blend-file.\xc2\xba

See also

Smooth Modifier.

--- Fin du chapitre: _laplacian_smooth ---

--- Début du chapitre: _lattice --- Lattice \xe2\x80\x93 or commonly called deformation cage outside of Blender. A lattice consists of a three-dimensional non-renderable grid of vertices. Its main use is to apply a deformation to the object it controls with a Lattice Modifier. If the object is parented with Lattice Deforma Lattice Modifier is automatically applied.

Mirrors the vertices displacement from their base position.

U, V, W

Resets the whole lattice to a regular grid, where the cells are scaled to one cubic unit.

Lattice properties. \xc2\xba

A Data-Block Menu .

Rate of subdivision in the axes:

U, V, W

Selector for each axis. See Different types of interpolation..

Linear, Cardinal, Catmull-Rom, B-Spline

Takes only the vertices on the surface of the lattice into account.

The strength of the influence assigned as a weight to the individual vertices in the selected vertex group.

The lattice should be scaled and moved to fit around your object in Object Mode. Any scaling applied to the object in Edit Mode will result in the object deforming. This includes applying its scale with Ctrl- A as this will achieve the same result as scaling the lattice in Edit Mode, and therefore the object.

Lattice around the cube object in Object Mode. \xc2\xba

--- Fin du chapitre: _lattice ---

--- Début du chapitre: _layer --- Reference

Sculpt Mode

Sidebar \xe2\x80\x93 Tool \xe2\x80\x93 Brush Settings \xe2\x80\x93

Advanced \xe2\x80\x93 Brush Type

This brush is similar to Draw, except that the height is capped. This creates the appearance of a flat layer.

It is recommended to use the Persistent setting and regularly Set Persistent Base, so that multiple strokes do not add on top of each other.

Higher by default to ensure the profile of layers is more noticeable. More info at [Hardness](#)

Note

More info at [General brush settings](#) and on [Advanced brush settings](#).

The fixed height of each stroke. This is measured using the scene scale, so it is consistent no matter the amount of zoom or object size.

This will ensure that multiple strokes use the same height, as if sculpting a single layer.

This button resets a new base so that you can sculpt new layer.

--- Fin du chapitre: _layer ---

--- Début du chapitre: _layers ---Reference

All Modes

Object Data tab \xe2\x80\xa3 Layers

Grease Pencil Layers panel.\xc2\xb6

Grease Pencil objects can be organized into a tree known as the layer tree for grouping and arranging strokes.

Any stroke can only belong to a single 2D layer. The selected layer is the active layer. Only one layer or group can be active at a time. When you draw, the new strokes are added to the active layer. By default the view order of the layers in the viewport is top to bottom.

Layers can be grouped using Layer Groups. A layer can only be in one group at a time. Layers can be moved into groups using drag-and-drop. Groups can be color coded with a color tag.

Every layer correspond to a channel in the Dope Sheet editor (in Grease Pencil mode). See [Dope Sheet](#) for more information.

Layers can also be used together with Modifiers to only affects part of your drawing. See [Modifiers](#) for more information.

Layers can mask other layers by enabling Use Mask (mask icon) or using the checkbox in the Masks panel header. See [Masks](#) for more information.

Tip

Sometimes the layers you are not working on can be a distraction in the 3D Viewport. Activate the Fade Inactive Layers overlay to control the opacity of the non-active layers.

Tree view of all layers and groups for the Grease Pencil object.

Next to the layer name there are four icons buttons that control common properties of the layer:

Toggle the affect of Mask on the layer.

Toggle using the layer for Onion Skinning.

Toggle layer visibility in the viewport and in render.

Toggle layer from being editable.

Adds a new layer to the active object.

Adds a new layer group to the active object. Note, layer groups cannot be added from the Dope sheet; they must be added from the Properties editor.

Removes the active layer or layer group.

Operators for working with layers.

Makes an exact copy of the selected layer appending a number to differentiate its name.

Makes a copy of the selected layer but with empty keyframes. Useful to easily have empty keyframes preset to work on the cleanup or filling process.

Turns on the visibility of every layer in the list.

Turns off the visibility of every layer in the list except the active one.

Locks edition of all the layers in the list.

Unlocks edition of all the layers in the list.

Locks automatically the edition of every layer in the list except the active one. This way you avoid to make unwanted changes in other layers without the need to lock them every time.

Avoids editing locked materials in the layer. When disabled, any material can be edited even if they are locked in the material list.

Combine the selected layer with the layer below, the new layer keeps the name of the lower layer.

Combine layers in the active layer group into a single layer.

Combine all layers into the active layer.

Copy the active layer to the selected Grease Pencil object.

Copy all layers to the selected Grease Pencil object.

Moves the active layer or layer group up/down in the tree.

Below the layers list there are additional settings:

The layer blending operation to perform. See Color Blend Modes.

Used to set the opacity of the layer.

When enabled, the layer is affected by lights.

Masks panel. \xc2\xbb

In Grease Pencil there are no special mask layers, any layer can act as a mask for other layers. The mask system is flexible enough to allow top-bottom and bottom-top masking.

Layers used as masks can use all the blend modes and different opacity values like any other layer.

Tip

If you want to make a full transparent masking you will have to set the mask layer's opacity to 0.

The layer/s that will act as mask of the current layer could be added to the Mask list view .

In the Masks list next to the layers name there are two icons buttons that control common properties of the layer mask:

Inverts the mask.

Toggle layer visibility in the viewport and in render.

Original image (Blend: Regular, Opacity: 1). \xc2\xbb

Blend: Hard Light, Opacity: 1. \xc2\xbb

Blend: Regular, Opacity: 1. \xc2\xbb

Allows per-layer location, rotation and scale transformations.

Layers adjustment panel. \xc2\xbb

Color that tint any material colors used in the layer.

Controls the amount of tint color to apply.

Thickness value that override the strokes thickness in the layer.

Select a Parent object to manipulate the layer. The layer will inherit the transformations of the parent, this is especially useful when rigging for cut-out animation.

The layer index number can be used with some modifiers to restrict changes to only certain areas.

See Modifiers for more information.

Defines the View Layer to use for the Grease Pencil layer. If empty, the layer will be included in all View Layers. This is useful to separate drawings parts for compositing .

If disabled, no masks on the layer are included in the view layer render.

--- Fin du chapitre: _layers ---

--- Début du chapitre: `_layer_weight` --- The Layer Weight node outputs a weight typically used for layering shaders with the Mix Shader node. Bias the output towards all 0 or all 1. Useful for uneven mixing of shaders.

Input meant for plugging in bump or normal maps which will affect the output.

This node has no properties.

Dielectric Fresnel weight, useful for example for layering diffuse and glossy shaders to create a plastic material. This is like the Fresnel node, except that the input of this node is in the often more convenient 0.0 to 1.0 range.

Weight that blends from the first to the second shader as the surface goes from facing the viewer to viewing it at a grazing angle.

--- Fin du chapitre: `_layer_weight` ---

--- Début du chapitre: `_layout` --- You can copy a UV map from one mesh to another mesh provided both meshes have the same geometry/vertex order. This is useful for example when you want to recreate a UV map from an earlier version of your model with intact UVs. This works by: Select the target mesh (to which you want to copy the UV map).

Shift-select the source mesh (that contains the intact UV map).

Object menu `Make Links` `Transfer UV Layouts` (Shortcut: `Ctrl-L`).

The target mesh will now have a UV map that matches the original mesh.

You are not limited to one UV map per mesh. You can have multiple UV maps for parts of the mesh by creating new UV maps. This can be done by clicking the Add button next to UV maps list and unwrapping a different part of the mesh. UV maps always include the whole mesh.

When you have unwrapped, possibly using seams, your UV layout may be quite disorganized and chaotic. You may need to proceed with the following tasks: Orientation of the UV mapping, arranging the UV maps, stitching several maps together.

The next step is to work with the UV layouts that you have created through the unwrap process. If you do add faces or subdivide existing faces when a model is already unwrapped, Blender will add those new faces for you. In this fashion, you can use the UV texture image to guide additional geometry changes.

When arranging, keep in mind that the entire view is your workspace, but only the UV coordinates within the grid are mapped to the image. So, you can put pieces off to the side while you arrange them. Also, each UV unwrap is its own linked set of coordinates.

You can lay them on top of one another, and they will onion skin (the bottom one will show through the top one). To move only one though, RMB-select one of the UV coordinates, and use Select `Linked UVs`, `Ctrl-L` to select connected UVs, not box select because UVs from both will be selected.

Bad unwrap, note ear and neck.

Very often you will unwrap an object, such as the face example we have been using, and get it mostly right but with parts of the mesh that did not unwrap properly, or are horribly confusing. The picture to the right shows an initial unwrap of the face using the Unwrap from sphere option. The issues are with the ear; it is just a mush of UVs, and the neck, it is stretched and folded under. Too much work to clean up.

Unwrap face only, without ear or neck.\xc2\xba

We can tell that the ear would unwrap nicely with just a straightforward projection from the side view, and the neck with a tubular unwrap. So, our general approach will be to unwrap different parts of the object (face, ears, and so on) using different unwrap tools, selecting each tool according to which one works best for that part. So let us begin: We select only the face faces, unwrap them using the Sphere projection, and scale and rotate them a bit to fit logically within the image area of the UV Editor.

Unwrap ear only, using Project from View.\xc2\xba

Once we are satisfied with the face, it is time to turn our attention to the ear. First, deselect the faces you were working with. Their UVs will disappear from the UV Editor, but they are still there, just not shown. (To verify this, you can select a few faces in 3D Viewport and it will show up in the UV Editor.)

To work on the ear, in the 3D Viewport, we now select only the ear faces. You can use vertex groups to select the ear faces. Selecting sub-meshes is easy too, since they are not connected to the rest of the mesh. Simply selecting linked vertices will select that entire submesh. Basically, since you are in Edit Mode, all the selecting/deselecting features are available to you.

Now unwrap the ear again using the Project tool from side view, and scale and rotate them a bit (discussed in the next section), and place them off to the side. You can do this repetitively, using different unwrapping algorithms; each re-apply just puts those UVs for the selected faces somewhere else. Choose the tool for each part that gives you the best fit and most logical layout for subsequent painting of that part.

When all parts of the mesh have been unwrapped using the various methods, you should end up with something that looks like the example to the right. All sections of the mesh have been mapped, and all those maps are laid out in the same UV texture map. Congratulations! From here, it is a simple matter of stitching (discussed in the next section) to construct the entire UV map as a single map.

UV maps arranged together and stitched.\xc2\xba

When you have completed arranging and stitching, you will end up with a consolidated UV map, like that shown to the right, arranged such that a single image will cover, or paint, all of the mesh that needs detailed painting. All the detailed instructions on how to do this are contained in the next section. The point of this paragraph is to show you the ultimate goal. Note that the mesh shown is Mirrored along the Z axis, so the right side of the face is virtual; it is an exact copy of the right, so only one set of UVs actually exist. (If more realism is desired, the Mirror Modifier would be applied, resulting in a physical mirror and a complete head. You could then make both sides physically different by editing one side and not the other. Unwrapping would produce a full set of UVs (for each side) and painting could thus be different for each side of the face, which is more realistic).

At least for common people, we just do not get it right the first time. It takes building on an idea and iterating our creative process until we reach that magical milestone called Done. In software development, this is called the spiral methodology.

Applied to computer graphics, we cycle between modeling, texturing, animating, and then back to making modifications to mesh, UV mapping,

tweaking the animation, adding a bone or two, finding out we need a few more faces, so back to modeling, etc. We continue going round and round like this until we either run out of time, money, or patience, or, in rare cases, are actually happy with our results.

Refinement comes into play when we finally look at our character, and realize that we need more detail in a particular spot. For example, areas around the eyes might need close-up detail, or we need to add a logo to the vest. As you start to edit the image, you realize that there just are not enough pixels available to paint the detail that you want. Your only choice is to expand the size (scale out) of that UV face. Using the Minimize Stretch or Scale tools, you expand the UV faces around the eyes or chest, allocating more pixels to those areas, but at the same time taking away pixels (detail) from somewhere else, like the back of the head. After refining the UV map, you then edit the image so that it looks right and contains the details you want.

Another consideration is the need to conserve resources. Each image file is loaded in memory. If you can reuse the same image on different meshes, it saves memory. So, for example, you might want to have a generic face painting, and use that on different characters, but alter the UV map and shape and props (sunglasses) to differentiate.

You might want to have a faded blue jeans texture, and unwrap just the legs of characters to use that image. It would be good to have a generic skin image, and use that for character's hands, feet, arms, legs, and neck. When modeling a fantasy sword, a small image for a piece of the sword blade would suffice, and you would Reset Unwrap the sword faces to reuse that image down the length of the blade.

--- Fin du chapitre: _layout ---

--- Début du chapitre: _length --- The Length modifier can shrink or extend strokes.

The Length modifier. 

Length is in geometry space.

Length is in ratio to the stroke's length.

Added length to the start of the stroke. Negative value will shrink the stroke.

Added length to the end of the stroke. Negative value will shrink the stroke.

Define what portion of the stroke is used to calculate the direction of the extension.

When enabled, the extension will follow the curvature of the stroke.

Multiplied by Start/End for the total point count.

Factor to determine how much the length of the individual segments should influence the final computed curvature. Higher factors make small segments influence the overall curvature less.

Ignore points on the stroke that deviate from their neighbors by more than this angle when determining the extrapolation shape.

Invert the curvature of the stroke's extension.

Size of random length added to the start/end of each stroke.

Smoothly offset each stroke's random value.

Number used to generate different noise patterns.

Re-randomizes values over time.

Number of frames before recalculate random values again.

See Influence Filters.

--- Fin du chapitre: `_length` ---

--- Début du chapitre: `_lennard_jones` ---Reference

Physics \xe2\x80\xa3 Force Fields

Lennard-Jones

The Lennard-Jones force field is a very short range force with a behavior determined by the sizes of the effector and effected particle. At a distance smaller than the combined sizes, the field is very repulsive and after that distance it is attractive. It tries to keep the particles at an equilibrium distance from each other. Particles need to be at a close proximity to each other to be effected by this field at all.

Particles can have for example both a charge and a Lennard-Jones potential, which is probably something for the nuclear physicists among us.

--- Fin du chapitre: `_lennard_jones` ---

--- Début du chapitre: `_lens_distortion` --- Use this node to simulate distortions that real camera lenses produce.

Standard color input.

This creates a bulging or pinching effect from the center of the image. This simulates chromatic aberrations, where different wavelengths of light refract slightly differently, creating a rainbow colored fringe. Enable or disable slider projection mode. When on, distortion is only applied horizontally. Disables Jitter and Fit.

Adds jitter to the distortion. Faster, but noisier.

Scales image so black areas are not visible. Only works for positive distortion.

Standard color output.

--- Fin du chapitre: `_lens_distortion` ---

--- Début du chapitre: `_levels` --- The Levels Node read the input color channels and outputs analytical values. The output is one-dimensional meaning the visualization will be a uniform gray color.

Standard color input.

Selects which color values are used to calculate analytics.

Calculate values based on the red, green, and blue channels.

Calculate values based on the red channel.

Calculate values based on the green channel.

Calculate values based on the blue channel.

Calculate values based on the Luminance of the image.

The mean is the average value of all image pixels in specified channel. It represents the overall brightness of the image and can be used as such for setups that depend on how \xe2\x80\x93bright\xe2\x80\x9d or \xe2\x80\x9cdark\xe2\x80\x9d the input is.

How much pixel values differ from the mean. A low standard deviation indicates that the pixel values tend to be very close to the mean. A high standard deviation indicates that the values are spread out over a large range of values.

--- Fin du chapitre: `_levels` ---

--- Début du chapitre: `_library_overrides` --- Library Overrides is a system designed to allow editing linked data, while keeping it in sync with the original library data. Most types of linked data-blocks can be overridden, and the properties of these overrides can then be edited. When

the library data changes, unmodified properties of the overridden one will be updated accordingly.

Note

The old proxy system has been deprecated in Blender 3.0, and fully removed in Blender 3.2. Automatic conversion from proxies to library overrides happens when loading a blend-file, but results on complex characters are not guaranteed and may need manual fixes.

Library overrides supports:

Multiple independent overrides of a same linked data (e.g. having the same character multiple times in the same scene).

Adding new modifiers and constraints, anywhere in the stack.

Recursively chaining overrides (i.e. link and override overrides from another library file, etc.).

Note

There are known issues that have to be addressed. See the main task of the project, for more details.

Warning

While in most cases library overrides data is preserved across a loss of reference linked data (if e.g. the library file becomes unavailable or is relocated), there are some exceptions.

The main one is probably posed (but not animated) armature objects, when their Armature obdata itself is not overridden. The Pose bones of an armature object are fully linked to the bones of its Armature obdata, if the later goes missing, the pose bones are definitively lost.

Note

Proper Collections Layout Matters

For library overrides to work well, it is much better if all the collections needed by the character are children of the root (linked and instantiated) one, such that there is a clear hierarchy. Otherwise, some data may not be properly automatically overridden, and other operations may be less reliable.

Hierarchy is a very important concept to understand when working with library overrides. In Blender, a real-life asset (a character, a prop, a set, etc.) is almost never made of a single data-block, but is rather a group of data-blocks with dependency relationships to each other. E.g. a character will typically have an armature object, several geometry objects, rig-controllers objects, the object data for all of these objects, materials, textures, etc.

These relationships can be represented as a tree, with a root data-block linking in all its dependencies, recursively. With library overrides, typically, the root of the hierarchy is also the data-block that is directly linked when importing the asset (usually a collection).

This concept of hierarchy can also be seen as some sort of super meta-data-block. It is critical when there are several overrides of the same linked data, since it allows to clearly identify a given data-block to one override, leaving no ambiguity to processes that affect the whole hierarchy (e.g. resyncing overrides with their linked data). It also allows to share relationships between data-blocks of different hierarchies, like a parenting relationships between two different overrides of a same character.

Due to current design of animation data in Blender, what is editable in overrides animations can change greatly depending on whether animation data was already defined in the linked referenced data-block.

Animation data is created for a datablock if it gets animated by keyframes, or through drivers.

In general, an override can do much more with its animation data if no animation data exists in its linked reference data-block.

Keyframes (a.k.a. F-Curves)

Keyframed animation belongs to another data-block (an Action one). So it is possible to assign a purely local Action data-block replacing the one linked from the library. This will completely replace the keyframed animation from the linked data though, and not override it in any way. Overridden Action data-blocks only support a very limited amount of editing. For example, an existing F-Curve can be muted, but its keyframes cannot be edited, and no new F-Curve can be added.

Drivers

If the linked reference data has animation data, then its overrides only have limited possibilities to edit the existing drivers. For example, it will be possible to change the existing target of a driver, but it won't be possible to add new drivers, or new targets to an existing driver.

If the linked reference data has no animation data, then its overrides will create a new one when they get some drivers defined. Drivers can then be fully edited, added or removed, just as with purely local data-blocks.

NLA

The NLA editor data also belongs to the animation data of a data-block. However, this data does support some greater level of edition in overrides, including moving or resizing existing strips from the linked data, and adding new local strips.

The relationships between linked data-blocks can change, resulting in outdated overrides. When this happens, overrides need to be resynced to match the new structure of their hierarchy. Overrides are automatically resynced if needed on blend-files opening. However, it may be needed to resync them manually sometimes, see Troubleshoot an Override Hierarchy.

Tip

Blender is also able to resync library overrides from external libraries, that are then linked into a working file. However, this is a costly process that needs to be fully redone every time the working file is loaded, since Blender cannot edit/modify the external library directly.

So users linking overrides (or creating recursive overrides) should ensure that their library files are regularly updated, to avoid this overhead on file load (typically, opening and saving those library files should be enough to update them).

Tip

Auto resyncing can be disabled in the Experimental Preferences. For technical reasons (how relationships between data-blocks are stored), Blender needs to create overrides of a lot of data-blocks, even when only one or two of them actually needs to be edited by the user. To reduce the amount of information and risk of potential unwanted editing, most of these data-blocks are now marked as non-editable by default. This can be changed once the override has been created.

Reference

3D Viewport, Outliner, Properties

Object Mode

3D Viewport \xe2\x80\xa3 Header \xe2\x80\xa3 Object \xe2\x80\xa3 Library
Override \xe2\x80\xa3 Make Outliner \xe2\x80\xa3 Context Menu \xe2\x80\xa3

Library Override \xe2\x80\xa3 MakeID Widget \xe2\x80\xa3 Context Menu
\xe2\x80\xa3 Library Override \xe2\x80\xa3 Make
Shift- LMBon the
\xe2\x80\x98linked\xe2\x80\x99/\xe2\x80\x99overridden\xe2\x80\x99 button
of an ID Widget.

Create overrides from the selected data-blocks.

Blender automatically create overrides for all required data-blocks to
ensure thatvalid override hierarchies are created.

Only overrides created from selected items will be user-editable.

Warning

The support for the creation of library overrides from the ID Widget
(mainly from withinthe Propertieseditor) is limited. While the most
common usages should be supported,especially with Objects, meshes, etc.,
much remains to be implemented.

Depending on where from the override is created, there are several ways
to\xe2\x80\x98select\xe2\x80\x99 items to be overridden and user-
editable.

Note

This also applies to the other common operations (Resetand Clear).

The Troubleshootadvanced operations only available from the

Outlineralways apply to a whole override hierarchy.

The selected objects will be considered as selected.

When a selected object is a local Emptyinstantiating a linked
collection,the following will happen:* The Emptyobject will be removed.*
Its linked collection will be overridden, and that override will be
instancedin the same collection in the current View Layer.* If the
collection contains Armatureobjects, they will be user-
editable.Otherwise, no created override will be defined as user-editable.
The operation can be applied on either the selected items only, their
content only, or both.

Tip

Using Selected & Contentis an easy way to get all newly created
overrides immediatelyuser-editable.

Only the linked data-block in the ID Widget is considered as selected,
and set as editableonce overridden.

That same operation can also be used to make existing overrides user-
editable,after they have been created, orcleared

Reference

3D Viewport, Outliner, Properties

Object Mode

3D Viewport \xe2\x80\xa3 Header \xe2\x80\xa3 Object \xe2\x80\xa3 Library
Override \xe2\x80\xa3 ResetOutliner \xe2\x80\xa3 Context Menu
\xe2\x80\xa3 Library Override \xe2\x80\xa3 ResetID Widget \xe2\x80\xa3
Context Menu \xe2\x80\xa3 Library Override \xe2\x80\xa3 Reset
Reset the selected overrides to their original values (from the linked
reference data).Unlike with the Clearoperation, the overrides remain
fully editable, and are never deleted.

Reference

3D Viewport, Outliner, Properties

Object Mode

3D Viewport \xe2\x80\xa3 Header \xe2\x80\xa3 Object \xe2\x80\xa3 Library
Override \xe2\x80\xa3 ClearOutliner \xe2\x80\xa3 Context Menu
\xe2\x80\xa3 Library Override \xe2\x80\xa3 ClearID Widget \xe2\x80\xa3
Context Menu \xe2\x80\xa3 Library Override \xe2\x80\xa3 Clear

Shift-LMB on the overridden button of an ID Widget.

Reset the selected overrides to their original values, and if possible without breaking the existing hierarchy, delete them and replace them by their linked reference data. Otherwise, keep the overrides but mark them as non-editable.

Essentially, an override is edited the same way as a regular local data-block. You can use operators on them, edit their properties from various editors, etc. There are some limitations however, most notably Edit Mode is not allowed for overrides. In most cases, as soon as you edit a property, you can see that it's overridden by its teal blue outline/background.

You can also animate overrides, animated properties just replace/supersede overrides then. Note that you cannot override/edit an existing animation, you'll have to create a new action. You can manually define or remove an override from the context menu of the relevant property. If an override is not editable, you have to make it editable first.

Reference

Any

Object Mode

Context Menu $\rightarrow$ Define Overrides Context Menu $\rightarrow$ Define Override

Mark a property to be overridden in the local blend file. For array properties all elements will be overridden.

Reference

Any

Object Mode

Context Menu $\rightarrow$ Define Single Override

Mark a property to be overridden in the local blend file. For array properties only the selected element will be overridden.

Reference

Any

Object Mode

Context Menu $\rightarrow$ Remove Overrides Context Menu $\rightarrow$ Remove Override

Remove the property from the overrides. The value of the linked in data-block will be used. For array properties all elements will be removed from the override.

Reference

Any

Object Mode

Context Menu $\rightarrow$ Remove Single Override

Remove the property from the overrides. The value of the linked in data-block will be used. For array properties only the selected elements will be removed from the override.

Reference

Outliner

Object Mode

Context Menu $\rightarrow$ Library override $\rightarrow$ Troubleshoot

These operations are only available from the Outliner contextual menu. They can help fixing a broken override hierarchy.

Reference

Outliner

Object Mode

Context Menu \xe2\x80\xa3 Library override \xe2\x80\xa3 Troubleshoot
\xe2\x80\xa3 Resync

The hierarchy of the linked data (the relationships between linked data-blocks) can change. Overrides need to be resynced to match the new hierarchy. This operator will resync the override to match the new hierarchy in the library.

Warning

While resyncing a library override it is possible that edited overrides get deleted if they are changed in the original library. If this is the case, a warning message will be displayed stating how many overrides were deleted, if the deletion is undesirable the resync can be undone before saving the blend-file.

Note

This Process is Automatic

Usually, this operation happens automatically when blender detects it is needed, on file load, unless it is disabled in the Experimental Preferences .

Reference

Outliner

Object Mode

Context Menu \xe2\x80\xa3 Library override \xe2\x80\xa3 Troubleshoot
\xe2\x80\xa3 Resync Enforce

In some cases, especially with older blend-files that were saved with \xe2\x80\x98broken\xe2\x80\x99 (non-hierarchy-matching) overrides, a regular resync itself cannot rebuild properly the override as expected (e.g. some objects might go missing). To solve this issue, this operator rebuilds the local override from its linked reference, as well as its hierarchy of dependencies, enforcing that hierarchy to match the linked data (i.e. ignoring existing overrides on data-blocks properties). This is similar to a regular resync, but is more forceful, aggressive, at the cost of a potential loss of some overrides on ID pointers properties.

Reference

Outliner

Object Mode

Context Menu \xe2\x80\xa3 Library override \xe2\x80\xa3 Troubleshoot
\xe2\x80\xa3 Delete

Remove the whole library override hierarchy, and replace all of these override data-blocks by their original linked data-blocks. This fully reverts the Make operation.

--- Fin du chapitre: `_library_overrides` ---

--- Début du chapitre: `_license` --- When one hears about \xe2\x80\x99free software\xe2\x80\x99, the first thing that comes to mind might be \xe2\x80\x99no cost\xe2\x80\x99. While this is often true, the term \xe2\x80\x99free software\xe2\x80\x99 as used by the Free Software Foundation (originators of the GNU Project and creators of the GNU General Public License) is intended to mean \xe2\x80\x99free as in freedom\xe2\x80\x99 rather than in the sense of \xe2\x80\x99no cost\xe2\x80\x99 (which is usually referred to as \xe2\x80\x99free as in free beer\xe2\x80\x99 or gratis). Free software in this sense is software which you are free to use, copy, modify, redistribute, with no limit. Contrast this with the licensing of most commercial software packages, where you are allowed to load the software on a single

computer, are allowed to make no copies, and never see the source code. Free software allows incredible freedom to the end user. Since the source code is universally available, there are also many more chances for bugs to be caught and fixed.

When a program is licensed under the GNU General Public License (the GPL):

- You have the right to use the program for any purpose.

- You have the right to modify the program and have access to the source codes.

- You have the right to copy and distribute the program.

- You have the right to improve the program, and release your own versions.

In return for these rights, you have some responsibilities if you distribute a GPL program. These responsibilities are designed to protect your freedoms and the freedoms of others:

- You must provide a copy of the GPL with the program, so that recipients are aware of their rights under the license.

- You must include the source code or make the source code freely available.

- If you modify the code and distribute the modified version, you must license your modifications available under the GPL (or a compatible license).

- You may not restrict the licensing of the program beyond the terms of the GPL (you may not turn a GPL program into a proprietary product).

For more on the GPL, check its page on the GNU Project website.

Note

The GPL only applies to the Blender application and not the artwork you create with it; for more info see the Blender License.

--- Fin du chapitre: `_license` ---

--- Début du chapitre: `_licenses` --- The Blender Extensions Platform only supports free and open source extensions compliant with the Blender's license:

- For add-ons, the required license is GNU General Public License v3.0 or later.

- For themes, the recommended license is GNU General Public License v3.0 or later, but any GPL-compatible license is supported.

- For assets used in add-ons, the required license is Public Domain (CC0).

--- Fin du chapitre: `_licenses` ---

--- Début du chapitre: `_light` --- The Light Output node is used to output light information to a light object.

Shading for the surface of the light object .

Render engine the input shaders are used for. By default shaders are shared between Cycles and Eevee, with multiple output nodes specialized shader setups can be created for each.

This node has no outputs.

--- Fin du chapitre: `_light` ---

--- Début du chapitre: `_lighting` --- The Workbench engine does not use the lights of the scene. The lighting conditions that will be used can be set in the Lighting panel.

Reference

Properties \xe2\x80\xa3 Render \xe2\x80\xa3 Lighting
Objects are \xe2\x80\x9cshaded\xe2\x80\x9d in a flat color, without any highlights or shadows.

Use a predefined studio lighting setup, such as a key light shining from the front and a rim light shining from the back. Click the sphere to choose a different setup.

The studio lights can be configured in the Preferences. By default, they follow the viewport camera around, but this can be changed:

Keep the lights fixed in place rather than following the viewport camera.

The rotation of the lights on the Z axis.

Use a Material Capture, which is an image with texturing, lighting and even reflections baked into it. Objects are shaded by simply picking colors from this image based on the direction of the normal in relation to the camera.

Click the sphere to choose a different MatCap, or the double arrow button to flip it horizontally.

Custom MatCaps can be loaded in the Preferences.

--- Fin du chapitre: _lighting ---

--- Début du chapitre: _lights --- Blender Preferences Lights section.\xc2\xbf

Studio Lights are used to illuminate the 3D Viewport during Solid View and will not be rendered. Unlike lights in the scene, the lighting direction follows the viewport orientation.

There are up to four virtual light sources.

The Light toggles allow you to enable or disable individual lights. At least one of the four lights must remain enabled for the 3D Viewport. The lights are equal, except for their direction and color. You can control the direction of the lights, as well as their diffuse and specular colors.

Toggles the specific light.

This is the constant color of the light.

This is the highlight color of the light.

Smooth the shading from this light.

This has the effect of lighting to be less direct.

The direction of the light, (see Direction Buttons).

The direction of the light will be the same as shown at the sphere surface.

The color of unlit areas.

This panel manages MatCap image files which can be used to light the view when MatCap shading is enabled.

Two kinds of images are supported for MatCaps. Regular image files and multilayered OpenEXR files. When using multilayered OpenEXR files, the layer named \xe2\x80\x9cdiffuse\xe2\x80\x9d will be used as a diffuse pass, the layer named \xe2\x80\x9cspecular\xe2\x80\x9d will be used as a specular pass. Regular images will be handled as \xe2\x80\x9cdiffuse\xe2\x80\x9d and will not support specular highlighting.

The diffuse pass is multiplied with the base color of the objects and the specular pass is added on top. MatCaps, that only have a diffuse pass tend to look very metallic, with a separate specular pass it is possible to simulate a wider variety of materials.

This panel manages HDRI image files which can be used to light the view when Material Preview or Rendered shading is enabled.

--- Fin du chapitre: `_lights` ---

--- Début du chapitre: `_light_falloff` --- Cycles Only

The Light Falloff node allows you to manipulate how light intensity decreases over distance. In reality light will always fall off quadratically; however, it can be useful to manipulate as a non-physically-based lighting trick. Note that using Linear or Constant falloff may cause more light to be introduced with every global illumination bounce, making the resulting image extremely bright if many bounces are used.

Light strength before applying falloff modification.

Smooth intensity of light near light sources. This can avoid harsh highlights, and reduce global illumination noise. 0.0 corresponds to no smoothing; higher values smooth more. The maximum light strength will be strength/smooth.

This node has no properties.

Quadratic light falloff; this will leave strength unmodified if smooth is 0.0 and corresponds to reality.

Linear light falloff, giving a slower decrease in intensity over distance.

Constant light falloff, where the distance to the light has no influence on its intensity.

Todo <2.8 add.

--- Fin du chapitre: `_light_falloff` ---

--- Début du chapitre: `_light_linking` --- With light linking, lights can be set to affect only specific objects in the scene. Shadow linking additionally gives control over which objects act as shadow blockers for a light.

This adds more artistic control for lighting by breaking the laws of physics. For example the environment and characters in a shot might have different light setups. A character could have a dedicated linked rim light to make it stand out, and shadow linking could be used to ensure no objects from the environment block it.

Select the light or emissive mesh object and go to the Cycles Shading panel or Eevee Shading panel.

Create a new light or shadowing linking collection.

Drag & drop objects or collection from the outliner.

Note

Light linking emissive mesh object is only available for Cycles.

Links can also be set up in the 3D viewport, with the Link Data operator. The active light object is linked to selected receiver or blocker object.

A light or shadow linking collection can be assigned to and shared between multiple light objects. While a scene collection can be directly assigned as a light or shadow linking collection, it is recommended to instead create a dedicated collection and link any scene collection inside it instead. This way it's easy to include or exclude additional objects without affecting the scene layout.

Light receiver objects can be set to be either included or excluded. The behavior is as follows:

If only included objects are specified, the light only affects those objects.

If only excluded objects are specified, the light affects all objects in the scene except those specified.

If both included and excluded objects are specified, the light affects only included objects minus the excluded objects. This can be used to for example set a character collection to be included, and then exclude specific objects part of the character.

Sampling for light linking is most efficient with the light tree enabled, where a specialized acceleration structure is built for light linking.

When using shadow linking, renders can be slower and trace additional rays, as direct and indirect lighting take different paths.

--- Fin du chapitre: _light_linking ---

--- Début du chapitre: _light_object ---Reference

Properties \xe2\x80\xa3 Light and Shader Editor \xe2\x80\xa3 Sidebar
\xe2\x80\xa3 Settings

Defines the type of light.

Color tint of the emitted light.

EEVEE specific settings

Cycles specific settings

Point light. \xc2\xbf

The point light is an omni-directional point of light, that is, a point radiating the same amount of light in all directions. It\xe2\x80\x99s visualized by a plain, circled dot. Being a point light source, the direction of the light hitting an object\xe2\x80\x99s surface is determined by the line joining the light and the point on the surface of the object itself. It can be used as simple model of e.g. a light bulb. Light intensity/energy decays based on (among other variables) distance from the point light to the object. In other words, surfaces that are further away will be rendered darker.

Power of the light in Watts. Higher values increase the intensity of the light. Negative values can be set, but should be avoided for predictable and physically based result.

Apply falloff to avoid sharp edges when the light geometry intersects with other objects.

When larger than zero, light will be emitted from a spherical surfaces with the specified radius. Lights with larger size have softer shadows and specular highlights, and they will also appear dimmer because their power is distributed over a larger area.

A spot light emits a cone-shaped beam of light from the tip of the cone, in a given direction.

Power of the light in Watts. Higher values increase the intensity of the light. Negative values can be set, but should be avoided for predictable and physically based result.

Apply falloff to avoid sharp edges when the light geometry intersects with other objects.

When larger than zero, light will be emitted from a spherical surfaces with the specified radius. Lights with larger size have softer shadows and specular highlights.

Changing the spot options also changes the appearance of the spotlight as displayed in the 3D Viewport. \xc2\xbf

The size of the outer cone of a spot, which largely controls the circular area a spot light covers. This slider in fact controls the angle at the top of the lighting cone, and can be between (1.0 to 180.0).

The Blend slider controls the inner cone of the spot. The Blend value can be between (0.0 to 1.0). The value is proportional and represents that amount of space that the inner cone should occupy inside the outer cone Size.

The inner cone boundary line indicates the point at which light from the spot will start to blur/soften; before this point its light will mostly be full strength. The larger the value of Blend, the more blurred/soft the edges of the spotlight will be, and the smaller the inner cone's circular area will be (as it starts to blur/soften earlier).

To make the spot have a sharper falloff rate and therefore less blurred/soft edges, decrease the value of Blend. Setting Blend to 0.0 results in very sharp spotlight edges, without any transition between light and shadow.

The falloff rate of the spot light is a ratio between the Blend and Size values; the larger the circular gap between the two, the more gradual the light fades between Blend and Size.

Blend and Size only control the spot light cone's aperture and softness (radial falloff); they do not control the shadow's softness as shown below.

Render showing the soft edge spotlighted area and the sharp/hard object shadow.

Notice in the picture above that the object's shadow is sharp as a result of the ray tracing, whereas the spotlight edges are soft. If you want other items to cast soft shadows within the spot area, you will need to alter other shadow settings.

Displays a transparent cone in 3D Viewport to visualize which objects are contained in it.

The area light simulates light originating from a surface (or surface-like) emitter. For example, a TV screen, office neon lights, a window, or a cloudy sky are just a few types of area light. The area light produces shadows with soft borders by sampling a light along a grid the size of which is defined by the user. This is in direct contrast to point-like artificial lights which produce sharp borders.

Power of the light in Watts. Higher values increase the intensity of the light. Negative values can be set, but should be avoided for predictable and physically based result.

Shape of the light.

The shape of the light can be represented as a rectangle and changed with the X and Y values.

The shape of the light can be represented as a square and changed with the Size property.

The shape of the light can be represented as a disk and changed with the Size property.

The shape of the light can be represented as an ellipse and changed with the X and Y values.

Tip

Choosing the appropriate shape for your area light will enhance the believability of your scene. For example, you may have an indoor scene and would like to simulate light entering through a window. You could place a Rectangular area light in a window (vertical) or from neon lights

(horizontal)with proper ratio for Size Xand Size Y. For the simulation of the light emitted bya TV screen, a vertical Squarearea light would be better in most cases.

Dimensions for the Squareor Rectangle.

A sun light provides light of constant intensity emitted in a single direction from infinitely far away.It can be very handy for a uniform clear daylight open-space illumination. In the 3D Viewport,the sun light is represented by an encircled black dot with rays emitting from it,plus a dashed line indicating the direction of the light.

Note

This direction can be changed by rotating the sun light, like any other object,but because the light is emitted from a location considered infinitely far away,the location of a sun light does not affect the rendered result.

Strength of the lights in Watts per square meter. Typical values arearound 250 for an overcast day and 1000 or more for direct sunlight.See more details at Power of Lights.

The size of the sun light according to its angular diameteras seen from earth.

The power of sun lights is specified in Watts per square meter.The power of point lights, spot lights, and area lights is specified in Watts.But this is not the electrical Watts that consumer light bulbs are rated at.It is Radiant Flux or Radiant Powerwhich is also measured in Watts.It is the energy radiated from the light in the form of visible light.

If you want to set the power to real world values, you have to convert the wattage of consumer bulbsor LEDlights to radiant flux, but it is not a straightforward process.The wattage of bulbs means the electrical power required to power them. LED lights havea \xe2\x80\x9cWatt equivalent\xe2\x80\x9d which is neither the electrical power they require nor the amount of light they put out.Some consumer lights specify lumens or luminous fluxwhich is the radiant flux weighted with the wavelengths perceived by the human eye.

To save you from doing the conversion, here is a table of typical power values for point, spot, and area lights:

Real world light

Power

Suggested Light Type

Candle

0.05 W

Point

800 lm LED bulb

2.1 W

Point

1000 lm light bulb

2.9 W

Point

1500 lm PAR38 floodlight

4 W

Area, Disk

2500 lm fluorescent tube

4.5 W

Area, Rectangle

5000 lm car headlight

22 W

Spot, size 125 degrees

And a table of typical Strength values for sun lights:

Sun type

Strength

Clear sky

1000 W/m²

Cloudy sky

500 W/m²

Overcast sky

200 W/m²

Moonlight

0.001 W/m²

These values will likely produce much brighter or dimmer lights than you would expect, because our eyes adapt while a render engine does not. So to compensate, adjust the Exposure in Render \xe2\x80\xa3 Film.

To get realistic results, remember to also set the light size and color to realistic values. The color of your lights will also influence how bright they appear to the human visual system. If you leave the power unchanged, a green light will seem the brightest, red darker and blue the darkest. Thus you might want to manually compensate for these perceived differences.

--- Fin du chapitre: `_light_object` ---

--- Début du chapitre: `_light_path` --- The Light Path node is used to find out for which kind of incoming ray the shader is being executed; particularly useful for non-physically-based tricks. More information about the meaning of each type is in the Light Paths documentation.

This node has no inputs.

This node has no properties.

1.0 if shading is executed for a camera ray, otherwise 0.0.

1.0 if shading is executed for a shadow ray, otherwise 0.0.

1.0 if shading is executed for a diffuse ray, otherwise 0.0.

1.0 if shading is executed for a glossy ray, otherwise 0.0.

1.0 if shading is executed for a singular ray, otherwise 0.0.

1.0 if shading is executed for a reflection ray, otherwise 0.0.

1.0 if shading is executed for a transmission ray, otherwise 0.0.

Distance traveled by the light ray from the last bounce or camera.

Number of times the ray has been reflected or transmitted on interaction with a surface.

Note

Passing through a transparent shader does not count as a normal \xe2\x80\x9cbounce\xe2\x80\x9d.

Number of times the ray has gone through diffuse reflection or transmission.

Number of times the ray has gone through glossy reflection or transmission.

Number of times the ray has gone through a transparent surface.

Number of times the ray has gone through a transmissive surface. A typical use case is to avoid black spots in the render (caused by rays hitting the bounce limit) by switching from a transmissive to a diffuse shader past a certain point. See Mix Shader .

EEVEE has no real concept of rays, but in order to ease the workflow between Cycles and EEVEE, some of the outputs are supported in particular cases.

Is Camera: Supported.

Is Shadow: Supported.

Is Diffuse: Supported.

Is Glossy: Supported.

Is Singular: Not supported. Same as Is Glossy.

Is Reflection: Not supported. Same as Is Glossy.

Is Transmission: Not supported. Same as Is Glossy.

Ray Length: Distance from the camera to the shading point.

Ray Depth: Indicates the current bounce when baking the light cache.

Diffuse Depth: Same as Ray Depth but only when baking diffuse light.

Glossy Depth: Same as Ray Depth but only when baking specular light.

Transparent Depth: Not supported. Defaults to 0.

Transmission Depth: Not supported. Same as Glossy Depth.

Note

Is Glossy does not work with Screen Space Reflections/Refractions but does work with reflection planes (whether used with SSR or not).

--- Fin du chapitre: `_light_path` ---

--- Début du chapitre: `_light_paths` ---Reference

Render \xe2\x80\xa3 Light Paths

Ray types can be divided into four categories:

Camera: the ray comes straight from the camera.

Reflection: the ray is generated by a reflection off a surface.

Transmission: the ray is generated by a transmission through a surface.

Shadow: the ray is used for (transparent) shadows.

Reflection and transmission rays can further have these properties:

Diffuse: the ray is generated by a diffuse reflection or transmission (translucency).

Glossy: the ray is generated by a glossy specular reflection or transmission.

Singular: the ray is generated by a perfectly sharp reflection or transmission.

The Light Path node can be used to find out the type of ray the shading is being computed for.

See also

The objectray visibility settings.

The maximum number of light bounces can be controlled manually. While ideally this should be infinite, in practice a smaller number of bounces may be sufficient, or some light interactions may be intentionally left out for faster convergence. The number of diffuse reflection, glossy reflection and transmission bounces can also be controlled individually. Light paths are terminated probabilistically when specifying a minimum number of light bounces lower than the maximum. In that case, paths longer than minimum will be randomly stopped when they are expected to contribute less light to the image. This will still converge to the same image, but renders faster while possibly being noisier.

The Transparent BSDF shader is given special treatment. Rays pass straight through it, changing neither direction nor type as if there were no geometry at all.

Alpha pass output is also different for the transparent BSDF. Other transmission BSDFs are considered opaque, because they change the light

direction. As such they cannot be used for alpha-over compositing, while this is possible with the transparent BSDF.

Note that, while semantically the ray passes through as if no geometry was hit, rendering performance is affected as each transparency step requires executing the shader and tracing a ray.

Maximum number of light bounces. For best quality, this should be set to the maximum. However, in practice, it may be good to set it to lower values for faster rendering. A value of 0 bounces results in direct lighting only.

Maximum number of diffuse bounces.

Maximum number of glossy bounces.

Maximum number of transmission bounces.

Maximum number of volume scattering bounces.

Maximum number of transparency bounces.

Note, the maximum number of transparent bounces is controlled separately from other bounces. It is also possible to use probabilistic termination of transparent bounces, which might help rendering many layers of transparency.

This option limits the maximum intensity a sample from rays which have not yet bounced can contribute to a pixel. It reduces noise at the cost of accuracy. Setting this option to 0.0 disables clamping altogether. Lower values have a greater affect (dimmer samples) on the resulting image than higher values.

Note

This option provides a way to limit Fireflies. However, note that as you clamp out such values, other bright lights/reflections will be dimmed as well.

Care must be taken when using this setting to find a balance between mitigating fireflies and losing intentionally bright parts. It is often useful to clamp indirect bounces separately, as they tend to cause more fireflies than direct bounces. See the Clamp Indirect setting.

The same as Direct Light, but for rays which have bounced multiple times.

A common source of noise is Caustics.

See also

See Reducing Noise for examples of the clamp settings in use.

When using a value higher than 0.0, this will blur glossy reflections after blurry bounces, to reduce noise at the cost of accuracy. 1.0 is a good starting value to tweak.

Some light paths have a low probability of being found while contributing much light to the pixel. As a result these light paths will be found in some pixels and not in others, causing Fireflies. An example of such a difficult path might be a small light that is causing a small specular highlight on a sharp glossy material, which is observed through a rough glossy material. In fact in such a case there practically occurs a caustic.

With path tracing it is difficult to find the specular highlight, but if you increase the roughness on the material, the highlight gets bigger and softer, and so easier to find. Often this blurring will hardly be noticeable, because it is blurred by the material anyway, but there are also cases where this will lead to a loss of detail in lighting.

While in principle path tracing supports rendering of caustics with a sufficient number of samples, in practice it may be inefficient to the

point that there is just too much noise. This option can be unchecked, to disable reflective caustics.

The same as above, but for refractive caustics.

Reference

Render \xe2\x80\xa3 Light Paths \xe2\x80\xa3 Fast GI Approximation
Approximate diffuse indirect light with background tinted ambient occlusion. This provides fast alternative to full global illumination (GI), for interactive viewport rendering or final renders with reduced quality.

Fast GI approximation method.

Replace global illumination with ambient occlusion after a specified number of bounces.

Add ambient occlusion to diffuse surfaces.

The strength of the ambient occlusion.

Distance from shading point to trace rays. A shorter distance emphasizes nearby features, while longer distances make it also take objects farther away into account.

This option can also be overridden per object in the Object Properties, which is useful when you have both small and large scale objects in the same scene.

Replace global illumination with ambient occlusion after the specified number of bounces when rendering in the 3D Viewport. This can reduce noise in interior scenes with little visual difference.

Number of bounces when rendering final renders.

--- Fin du chapitre: _light_paths ---

--- Début du chapitre: _light_settings ---Reference

Properties \xe2\x80\xa3 Light and Shader Editor \xe2\x80\xa3 Sidebar
\xe2\x80\xa3 Options \xe2\x80\xa3 Light

Next to lighting from the background and any object with an emission shader, lights are another way to add light into the scene. The difference is that they are not directly visible in the rendered image, and can be more easily managed as objects of their own type.

Light settings for all renderers.

Maximum number of times light from the light is allowed to Bounce. Limited by scene-wide bounce settings.

By disabling this option, light from lights will not be blocked by objects in between. This can speed up rendering by not having to trace rays to the light source.

By default lights use only direct light sampling. For area lights and sharp glossy reflections, however, this can be noisy, and enabling this option will enable indirect light sampling to be used in addition to reduce noise.

Mark a light as a refractive caustic caster. This setting can be used in conjunction with the Cast and Receive caustics object settings to selectively speed up refractive caustic rendering of select objects. Area lights can also function as light portals to help sample the environment light, and significantly reduce noise in interior scenes. Note that rendering with portals is usually slower, but as it converges more quickly, less samples are required.

Light portals work by enabling the Portal option, and placing areas lights in windows, door openings, and any place where light will enter the interior.

In outdoor scenes most rays do not bounce much and just fly off into the sky and therefore, light portals are not helpful for outdoor scenes.

White Room model by Jay Hardy. \xc2\xbb

How wide the emitted light fans out controlling how diffused the light is. Larger values create soft shadows while smaller values create sharper lightsimulating a gridded softbox.

Example of Spread at different angles.\xc2\xbb

--- Fin du chapitre: _light_settings ---

--- D but du chapitre: _limbs --- These rig types handle generation of different kind of limbs and their features, like fingers.

Will create a simple bendy and stretchy b-bones tentacle chain, which can optionally replicate local rotation from preceding bones to the subsequent ones for use in cases like fingers.

Requirement: A chain of at least two connected bones.

Enables the automation on the selected axis. Multiple axis or none can be selected holding Shift- LMB .When enabled the subsequent control bones will copy the local rotations from the previous ones. The option is accessible in the controls of the final rig as a Copy Rotation constraint and can be disabled even after rig is generated, or at animation time.

If enabled, allows placing the Tweak controls in different bone collections from the main controls.

Will create a bendy and stretchy finger chain with a master control bone that controls the rotation of all joints through its scale.

Requirement: A chain of at least two connected bones.

Defines the automatic rotation axis to be linked to the scale of the master bone.

Defines the number of b-bone segments each tweak control will be split into.

Generates a very simple IK mechanism with only one control.

IK starts its work with the shape of the finger defined by FK controls and adjusts it to make the fingertip touch the IK control. It is designed as a tool to temporarily keep the fingertip locked to a surface it touches, rather than a fully featured posing system.

To improve performance, the switchable parent for the IK control contains only one option beside None. Thus it is advised to add a \xe2\x80\x9dheld object \xe2\x80\x99 control using the basic.raw_copy rig to act as the common parent for the fingers with a fully functional parent switch.

Specifies the value of the Local Location option for IK controls, which decides if the location channels are aligned to the local control orientation or world.

If enabled, allows placing the Tweak controls in different bone collections from the main controls.

If enabled, allows placing the extra IK control in different bone collections from the main controls.

Note

Rotation Axis (Bend Rotation Axis in the case of limbs.super_finger) affects the roll of the generated bones. Automatic mode recalculates the generated bones roll while any of the Manual modes copy the roll of the meta-rig bones.

A backwards compatibility wrapper around limbs.arm, limbs.leg and limbs.paw.

Will create a fully featured bendy and stretchy arm depending on the user-defined options.

Requirement: A chain of three connected bones (upper_arm, forearm, hand).

Arm required bones. \xc2\xba6

Generates an extra child of the hand IK control that rotates around the tail of the hand bone.

Defines the bend axis for the IK chain. FK chains will have a totally free degree of rotation on all axes.

Defines the number of additional tweak controls each limb bone will have on the final rig.

Defines the number of b-bone segments each tweak control will be split into.

Generates an extra control for the end of the IK limb that allows rotating it around an arbitrarily placed pivot.

If enabled, allows placing the FK chain in different bone collections from the IK bones.

If enabled, allows placing the Tweak controls in different bone collections from the IK bones.

Will create a fully featured bendy and stretchy leg depending on the user-defined options.

Requirement: A chain of four connected bones (thigh, shin, foot, toe) with one unconnectedchild of the foot to be used as the heel pivot.

Leg required bones. \xc2\xba6

Specifies where to put the pivot location of the main IK control, or whether to generate an additionalpivot control at the base of the toe.

Specifies that two separate toe controls should be generated for IK and FK instead of sharing one bone.This is necessary to get fully correct IK-FK snapping in all possible poses.

Generates a slider to switch the heel control to pivot on the tip rather than the base of the toe(for roll this obviously only applies on forward roll).

Defines the bend axis for the IK chain. FK chains will have a totally free degree of rotation on all axes.

Defines the number of additional tweak controls each limb bone will have on the final rig.

Defines the number of b-bone segments each tweak control will be split into.

Generates an extra control for the end of the IK limb that allows rotating it around an arbitrarily placed pivot.

If enabled, allows placing the FK chain in different bone collections from the IK bones.

If enabled, allows placing the Tweak controls in different bone collections from the IK bones.

Will create a fully featured bendy and stretchy paw depending on the user-defined options.

Requirement: A chain of four or five connected bones (thigh, shin, paw, optionaldigit, toe).

Front/Rear paw required bones. \xc2\xba6

Defines the bend axis for the IK chain. FK chains will have a totally free degree of rotation on all axes.

Defines the number of additional tweak controls each limb bone will have on the final rig.

Defines the number of b-bone segments each tweak control will be split into.

Generates an extra control for the end of the IK limb that allows rotating it around an arbitrarily placed pivot.

If enabled, allows placing the FK chain in different bone collections from the IK bones.

If enabled, allows placing the Tweak controls in different bone collections from the IK bones.

Derivative of limbs.pawwith extended IK suitable for use in front paws. The additional IK limits the degree of change in the angle between shin and paw bones (2nd and 3rd) as the main IK control moves and rotates. For best results, the shin bone should not be parallel to either thigh or paw in rest pose, i.e. there should be some degree of bend in all joints of the paw.

Influence of the extended IK. At full rotating the main IK control or digit bone would not affect the rotation of the paw bone, while lower values provide some blending.

Derivative of limbs.pawwith extended IK suitable for use in rear paws. The additional IK tries to maintain thigh and paw bones (1st and 3rd) in a nearly parallel orientation as the main IK control moves and rotates. For best results, thigh and paw bones should start nearly parallel in the rest pose.

Will create a palm system based on the distance between palm bones.

Requirement: At least two bones child of the same parent. The property has to be set on the inner palm bones (think it as index\bone\metacarpus), the rig control will appear on the last palm bone (think it as pinky\bone\metacarpus).

Generates controls on both sides of the palm, with influence on inner bones blended between them.

Defines the automatic rotation axis to be used on the palm bones.

This rig type implements a flexible tentacle with an IK system using the Spline IK constraint. The control bones define control points of a Bezier curve, and the bone chain follows the curve.

The curve control points are sorted into three groups: start, middle and end. The middle controls are always visible and active, while the other two types can be shown and hidden dynamically using properties; when enabled they appear next to the corresponding permanent start/end control and can be moved from there.

Specifies the number of optional start controls to generate.

Specifies the number of middle controls to generate.

Specifies the number of optional end controls to generate.

Specifies how the curve stretching and the final control bone work:

Stretches the whole bone chain to fit the length of the curve defined by the controls.

An end twist control is generated to control the twist along the chain.

Generates an IK end control, which directly controls the final bone of the chain similar to how regular IK works for limbs, as well as controlling the end of the bezier curve. The middle bones of the chain stretch to follow the curve and cover the gap.

The rig automatically deduces twist of up to 180 degrees based on the orientation of the end control. Higher amounts of twist have to be dialled in through an End Twist Estimate slider to avoid flipping.

This mode allows full manual control over the chain scaling, while the chain covers as much of the curve as it can given its current length.

The start control of the chain manages its uniform squash & stretch scale, while the end twist control manages both the twist of the chain, as well as its scale at the tip (blended gradually along the length).

Allows scaling the controls to control the thickness of the chain through the curve.

Specifies the maximum scale allowed by the Radius Scaling feature.

Generates an FK control chain and IK-FK snapping.

If enabled, allows placing the FK chain in different bone collections from the IK bones.

--- Fin du chapitre: `_limbs` ---

--- Début du chapitre: `_limitations` --- EEVEE's goal is to be an interactive render engine. Some features may not be there yet or may be impossible to implement into EEVEE's architecture without compromising performance.

Here is a rather exhaustive list of all the limitations you can expect while working with EEVEE.

Only 14 attributes from Geometry Nodes are supported in a material

Only 8 custom object properties are supported in a material

Only perspective and orthographic projections are currently supported.

Lights can only have one color and do not support light node trees.

Unlike in Cycles, the Size of spot lights does not change the softness of the cone.

The area light Beam spread option is not supported.

EEVEE supports up to 128 active light probe spheres.

EEVEE supports up to 16 active light probe planes inside the view frustum.

Active light probe volumes must fit inside the Light Probes Volume Memory Pool .

Light probe capture does not support specular reflections. Specular energy is treated as diffuse.

Shadow Map Raytracing can produce light leaking because of overlapping shadow casters. This can be mitigated by using lower step count , enabling jitter , or reducing the light shape size.

Thin objects (e.g. walls without thickness) might have light leaking on the shadowed side. This can be mitigated by making the object have some thickness or lowering Resolution Limit .

Only single scattering is supported.

Volumetrics are rendered only for the camera

crays. They don't appear in reflections/refractions and probes.

Volumetric shadowing only work in volumetrics. They won't cast shadows onto solid objects in the scene.

Volumetric shadowing only work for volumes inside the view frustum.

Blended materials cannot be correctly handled by the post-processing blur, but will be correctly handled by the sample-based method. For this, you need to disable the post-process depth of field by setting the Max Size to 0.

Ray-triangle intersection is not currently supported. Instead of this, EEVEE uses the depth buffer as an approximated scene representation. This reduces the complexity of scene scale effects and enables a higher performance. However, only what is inside the view can be considered when computing these effects. Also, since it only uses one layer of depth, only the front-most pixel distance is known.

These limitations creates a few problems:

The screen space effects disappear when reaching the screen border. This can be partially fixed by using the overscan feature.

Screen space effects lack deep information (or the thickness of objects). This is why most effects have a thickness parameter to control how to consider potential intersected pixels.

Objects behind other objects (occluded) are not considered by these effects.

Blended surfaces are not considered by these effects. They are not part of the depth prepass and do not appear in the depth buffer.

Objects that are a part of Holdout Collections will not be rendered with screen space effects.

Blended materials and materials using raytrace refractions will not appear in dithered materials reflections.

Blended materials are not compatible with raytracing.

Only one refraction event is correctly modeled. An approximation of the second refraction event can be achieved using the Thickness workflow.

Only dithered materials not using Raytrace Refractions can be refracted. All BSDFs are using approximations to achieve realtime performances so there will always be small differences between Cycles and Eevee.

Some utility nodes are not yet compatible with Eevee.

Certain combinations of BSDFs will result in more noise than others. This is the case when mixing Diffuse BSDF and Refraction BSDF.

Displacement of flat shaded surfaces will split the mesh into triangles. See Displacement for a workaround.

See also

For a full list of unsupported nodes see Nodes Support.

In Eevee, GPU memory management is done by the GPU driver. In theory, only the needed textures and meshes (now referred as "the resources") for a single draw call (i.e. one object) needs to fit into the GPU memory.

So if the scene is really heavy, the driver will swap things in and out to make sure all objects are rendered correctly.

In practice, using too much GPU memory can make the GPU driver crash, freeze, or kill the application. So be careful of what you ask.

There is no standard way of estimating if the resources will fit into the GPU memory and/or if the GPU will render them successfully.

Being a rasterization engine, Eevee only uses the power of the GPU to render. There is no plan to support CPU (software) rendering as it would be very inefficient. CPU power is still needed to handle high scene complexity as the geometry must be prepared by the CPU before rendering each frame.

There is currently no support for multiple GPU systems.

Headless rendering is not supported on headless Windows systems.

--- Fin du chapitre: limitations ---

--- Début du chapitre: limits --- While object positions, vertex locations are not clamped, larger values become increasingly imprecise. To get an idea of the precision you can work with using different scales. Here is a table of scales and their associated accuracy:

1/1,048,576 th

1/131,072 th

1/16,384 th

1/1,024 th

1/128 th

1/16 th

Hint

For a rough rule of thumb, values within -5,000/+5,000 are typically reliable (range of 10,000). Internally single precision floating-point calculations are used.

The maximum number of frames for each scene is currently 1,048,574, and allows for continuous shots for durations of:

12 hours, 8 minutes.

11 hours, 39 minutes.

9 hours, 42 minutes.

4 hours, 51 minutes.

Note

In practice, a finished work is typically composed of output from many scenes. So this limit does not prevent you from creating longer works. Fixed strings are used internally, and while it is not useful to list all limits, here are some common limits. Used for data-block names, modifiers, vertex groups, UV layers\

Note

Multi-byte encoding means some Unicode characters use more than a single ASCII character.

--- Fin du chapitre: `_limits` ---

--- Début du chapitre: `_limit_distance` --- The Limit Distance constraint forces its owner to stay either further from, nearer to, or exactly at a given distance from its target. In other words, the owner's location is constrained either outside, inside, or at the surface of a sphere centered on its target.

When you specify a (new) target, the Distance value is automatically set to correspond to the distance between the owner and this target.

Important

Note that if you use such a constraint on a connected bone, it will have no effect, as it is the parent's tip which controls the position of your owner bone's root.

Limit Distance panel. \

Data ID used to select the constraint's target, and is not functional (red state) when it has none. See common constraint properties for more information.

This number field sets the limit distance, i.e. the radius of the constraining sphere.

Resets the Distance value, so that it corresponds to the actual distance between the owner and its target (i.e. the distance before this constraint is applied).

The Limit Mode select menu allows you to choose how to use the sphere defined by the Distance setting and target's origin:

The owner is constrained inside the sphere.

The owner is constrained outside the sphere.

The owner is constrained on the surface of the sphere.

Transform operators will take the constraint into account to immediately restrict the resulting transform property values.

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

Tip

Evaluating both owner and target in aCustom Spaceusing the root bone or any other suitable parent bone will automatically scale the effective distance withthe relevant part of the rig.

--- Fin du chapitre: `_limit_distance` ---

--- Début du chapitre: `_limit_location` --- An object or unconnectedbone can be moved around the scene along the X, Y and Z axes.This constraint restricts the amount of allowed translations along each axis,through lower and upper bounds.

The limits for an object are calculated from its origin, and the limits of a bone, from its root.

It is interesting to note that even though the constraint limits the visual andrendered location of its owner, its owner's datablock still allows (by default)the object or bone to have coordinates outside the minimum and maximum ranges.This can be seen in its Transformpanel.

When an owner is selected and attempted to be moved outside the limit boundaries,it will be constrained to those boundaries visually and when rendered, but internally,its coordinates will still be changed beyond the limits. If the constraint is removed,its ex-owner will seem to jump to its internally specified location.

Similarly, if its owner has an internal location that is beyond the limits, dragging it backinto the limit area will appear to do nothing until the internal coordinates are back withinthe limit threshold (unless you enabled the Affect Transformoption, see below).

Setting equal the min and max values of an axis,locks the owner's movement along that axis. Although this is possible,using the Transformation Propertiesaxis locking feature is probably easier!

Limit Location panel. \xc2\xb6

These buttons enable the lower boundary for the location of the owner's origin along,respectively, the X, Y and Z axes of the chosen Space.The number field below them controls the value of their limit.Note that if a min value is higher than its corresponding max value,the constraint behaves as if it had the same value as the max one. These buttons enable the upper boundary for the location of the owner's origin along,respectively, the X, Y and Z axes of the chosen Space.Same options as above.

As pointed out before by default, even though visually constrained,the owner can still have coordinates out of bounds (as shown by the Transformpanel).Well, when you enable this checkbox, this is no longer possible \xe2\x80\x93the owner's transform properties are also limited by the constraint.However, note that, the constraint does not directly modify the coordinates:you have to select its owner one way or another for this to take effect\xa6

This constraint allows you to choose in which space to evaluate its owner's transform properties.See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object.See common constraint properties for more information.

--- Fin du chapitre: `_limit_location` ---

--- Début du chapitre: `_limit_rotation` --- This constraint restricts the rotation of an object or bone to be insidespecified angular limits.The

limits are given as Euler rotation ranges (a min and max angle), and a separate range can be given for each of the three axes.

As with all constraints in Blender, this does not (by default) restrict the user-set rotation values of the object/bone as seen in the Transform panel. When the object/bone is rotated outside the limit range, it will be constrained to that range in its final displayed/rendered position, but the user-set rotation values will still be outside that range. If the constraint is removed, the object/bone will then jump back to match those user-set values.

Something unique about the Limit Rotation constraint (as compared to the Limit Location and Limit Scale constraints) is that rotations loop, and therefore the meaning of the limit range is subtly different. All constraints in Blender internally work on transform matrices, which can't distinguish between e.g. 180 and -180 degrees, or 0, 360, and 720 degrees. In other words, any angles that result in the same visual rotation are indistinguishable to the constraint system.

What this means for the Limit Rotation constraint is that when the user-set rotation is outside of the limit range, the final displayed rotation will snap to the closest visual rotation in that range, not the closest numerical angle. For example, if you have a limit range of 0 to 90 degrees then a user-set rotation of 340 degrees will actually snap to 0 degrees because that is the closer visual rotation, even though 340 is numerically closer to 90.

Note that this constraint does not constrain the bone if it is manipulated by the IK solver. For constraining the rotation of a bone for IK purposes, see Inverse Kinematics.

Limit Rotation panel. 

These buttons enable the rotation limit around respectively the X, Y and Z axes of the owner, in the chosen Owner space. The Min and Max number fields to their right control the value of their lower and upper boundaries, respectively.

Note

If a min value is higher than its corresponding max value, the constraint behaves as if it had the same value as the max one.

Unlike the Limit Location constraint, you cannot separately enable lower or upper limits.

The constraint can be used to simply remove shear from the owner transformation by leaving all limits disabled.

Allows specifying which Euler order to use when applying the limits. Defaults to the order of the owner, or XYZ if the owner uses non-Euler rotations.

The constraint is taken into account when the object is manually rotated using transformation tools in the editors. This prevents assigning transformation property values (as shown in the Transform panel) that exceed the specified limits.

For backwards compatibility: make the constraint behave in the semi-broken way it did prior to Blender 4.2. This old behavior does not properly account for the looping nature of rotations, and therefore causes unpredictable/erratic rotation snapping. However, this behavior can still be useful in some specific circumstances when Owner is set to local space, and some older rig setups utilize that. However, that behavior is better and more robustly accomplished with drivers directly on the object/bone's rotation properties, so new rigs should favor that approach over using this option.

This constraint allows you to choose in which space evaluate its owner's transform properties. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_limit_rotation` ---

--- Début du chapitre: `_limit_scale` --- An object or bone can be scaled along the X, Y and Z axes. This constraint restricts the amount of allowed scaling along each axis, through lower and upper bounds.

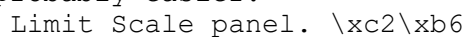
Important

This constraint does not tolerate negative scale values (those you might use to mirror an object): when you add it to an object or bone, even if no axis limit is enabled, nor the Affect Transform option, as soon as you scale your object, all negative scale values are instantaneously inverted to positive ones. And the boundary settings can only take strictly positive values.

It is interesting to note that even though the constraint limits the visual and rendered scale of its owner, its owner's data-block still allows (by default) the object or bone to have scale values outside the minimum and maximum ranges (as long as they remain positive!). This can be seen in its Transform panel. When an owner is scaled and attempted to be moved outside the limit boundaries, it will be constrained to those boundaries visually and when rendered, but internally, its coordinates will still be changed beyond the limits. If the constraint is removed, its ex-owner will seem to jump to its internally-specified scale.

Similarly, if its owner has an internal scale that is beyond the limits, scaling it back into the limit area will appear to do nothing until the internal scale values are back within the limit threshold (unless you enabled the Affect Transform option, see below, or your owner has some negative scale values).

Setting equal the min and max values of an axis locks the owner's scaling along that axis. Although this is possible, using the Transformation Properties axis locking feature is probably easier.

Limit Scale panel. 

These buttons enable the lower boundary for the scale of the owner along respectively the X, Y and Z axes of the chosen Space. The Min and Max number fields to their right control the value of their lower and upper boundaries, respectively.

Note

If a min value is higher than its corresponding max value, the constraint behaves as if it had the same value as the max one.

As pointed out before by default, even though visually constrained, and except for the negative values, the owner can still have scales out of bounds (as shown by the Transform panel). When you enable this checkbox, this is no longer possible, the owner transform properties are also limited by the constraint. However, note that, the constraint does not directly modify the scale values: you have to scale its owner one way or another for this to take effect.

This constraint allows you to choose in which space to evaluate its owner's transform properties. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_limit_scale` ---

--- Début du chapitre: `_line` ---Reference

Draw Mode

Toolbar `\xe2\x80\xa3` Line

The Line tool create straight lines using any of the Drawtype brushes. You can configure the brush main settings exposed on the Tool Settings for convenience. For the draw brushes configuration and settings see: Draw Brush .

The number of stroke points between each stroke edge.

Use a curve widget to define the stroke thickness from the start (left) to end (right) of the stroke.

When enabled, the stroke use a curve profile to control the thickness along the line.

Picks the brush asset used by the tool.

See Brush Asset for more information.

See Draw Brushes for a detailed list of all draw brushes and their options.

Parameters to control to look of the stroke.

See Draw Brushes for details.

Settings to determine the color of strokes.

See Color

In the Tool Settings select the brush, material and color type to use with the tool. The Line tool uses Draw Brush types. See Brush Settings for more information.

Click (LMB or the Pentip) and drag the start point.

Release on the desired end point.

After releasing you can move the start and end point by clicking and dragging on the yellow manipulators.

Then confirm (Return/ MMB) or cancel (Esc/ RMB).

While dragging you can use Shift to snapping the line to horizontal, vertical or 45° angle or use Alt to create the line from a center point.

Numpad Plus and Numpad Minus or using the mouse Wheel will increase or decrease the amount of points in the final line.

click and dragging the start point. `\xc2\xba`

Moving start and end points with manipulators. `\xc2\xba`

The line after confirming. `\xc2\xba`

Before confirming you can use E to extrude the end point of the line to generate multiple connected lines.

End point extruding. `\xc2\xba`

Moving the end point of the last line with the manipulator. `\xc2\xba`

The connected lines after confirming. `\xc2\xba`

--- Fin du chapitre: `_line` ---

--- Début du chapitre: `_linear_gizmo` --- The Linear Gizmo node provides the most widely applicable gizmo. It can e.g. be used to control the height of something.

Special gizmo value socket. Everything that linked into this socket will be modified when the gizmo is moved.

Position of the gizmo in the local space of the object.

Specifies the direction in with the gizmo points or is moved.

Controls which theme color is used for this gizmo.
Allows choosing between different styles of the gizmo.
Should be joined into the geometry that is controlled by this gizmo.
--- Fin du chapitre: _linear_gizmo ---

--- Début du chapitre: _line_art ---Reference
Material \xe2\x80\xa3 Line Art
Line Art material properties.\xc2\xb6
Material masks are a way to provide Line Art extra information about faces that caused the occlusion. So edges occluded by those faces can be selected to have different styles.
The layer to include faces of the current material.
Faces with this material will behave as if it has set number of layers in occlusion.
Assigns an intersection priority value for this material. Note that this priority takes precedent over Object or Collection priority values. The intersection line will be included into the object with the higher intersection priority value.
--- Fin du chapitre: _line_art ---

--- Début du chapitre: _line_project ---Reference
Sculpt Mode
Toolbar \xe2\x80\xa3 Line Project
This tool flattens the geometry along a plane determined by the camera view and a drawn line. The region of the mesh being flattened is visualized by the side of the line that is shaded.
Before Line Project. \xc2\xb6
After Line Project. \xc2\xb6
Use the tool by:
Orient the 3D Viewport to define the direction in depth.
LMB and hold while moving the cursor to define direction of the line projection.
Adjust the operation with extra Control shortcuts.
Release LMB to confirm.
Changes the side of the line that the tool projects geometry.
Constrains the rotation of the line to 15 degree intervals.
Reposition the line.
The affected area will not extend the length of the drawn line. This helps defining a smaller area instead of extending the line infinitely long
--- Fin du chapitre: _line_project ---

--- Début du chapitre: _line_set ---Reference
Properties \xe2\x80\xa3 View Layer \xe2\x80\xa3 Freestyle Line Set
A line set selects, among the lines (edges) detected by Freestyle, which ones will be rendered using its attached line style, through various methods.
Freestyle Line Set panel.\xc2\xb6
Causes Freestyle to only take geometry within the image border into consideration for line calculation. This reduces render times but increases continuity problems when geometry is moved out of and into camera view.
Determine how to use visibility for feature edge selection.
Only lines occluded by no surfaces are rendered.
Lines occluded by at least one surface are rendered.

Proof of concept of visible and hidden edges by LightBWK(blend-file).
\xc2\xba6

Lines occluded by a number of surfaces in the given range are rendered.
Min/max number of occluding surfaces for a line to be rendered.

QI Range proof of concept demo, Start: 3, End: 7, by LightBWK(blend-file). \xc2\xba6

Edge types are basic algorithms for the selection of lines from geometry. When using the parameter editor you have to choose at least one edge type in order to get a render output, but several edge types can be combined in one line set. Edge types can also be excluded from calculation by pressing the X next to them.

Examples of some basic edge types: Silhouette (green), Crease (black), Border (blue) and Edge Marks (red) (blend-file by LightBWK). \xc2\xba6

Draws silhouettes around your closed objects by rendering lines where the surface normal transitions between pointing toward and away from the camera. It is often good for organic objects (like Suzanne & Sphere), and bad for sharp edges, like a box. It cannot render open mesh objects like open cylinders and flat planes.

Shows only edges whose adjacent faces form an angle sharper than the defined view map's Crease Angle.

Crease Angle proof of concept for 121 \xc2\xba0 by LightBWK(blend-file).
\xc2\xba6

Border shows open mesh edges, i.e. edges that belong to only one face. An open cylinder has open edges at the top and bottom, and a plane is open all around. Suzanne's eye socket is an open edge.

Renders marked edges. See Edge Marks for details.

Draws lines around each object, separating it from other objects behind it, or the scene background.

Draws lines around all objects, separating them from the scene background, but not each other.

Left pair: Contour; Right pair: External Contour. \xc2\xba6

Draws lines where two materials meet on the same object.

Draws some lines which would form the Silhouette of the mesh if the view point was shifted. Depends on your view map settings for Kr Derivative Epsilon and Sphere Radius (further information: File:Manual-2.6-Render-Freestyle-Princeton Linestyle.pdf).

Draws lines marking crests of ridges and valleys, i.e. places where the surface curvature is at its minimum or maximum. Depends on your Sphere Radius view map settings.

In Edit Mode you can mark \xe2\x80\x9cFreestyle Edges\xe2\x80\x9d in the same manner you can mark \xe2\x80\x9cSeams\xe2\x80\x9d for UV unwrapping or \xe2\x80\x9cSharp\xe2\x80\x9d for edge split. These marked edges are available to render when you select Edge Mark.

This is done as follows:

Select the mesh object and enter Edit Mode.

Select the edges you want to be marked.

Press Ctrl-E and select Mark Freestyle Edge.

Edge marks are useful when you want to draw lines along particular mesh edges. The examples below explain the use of edge marks.

Marking Freestyle Edges in Edit Mode; the edge marks are highlighted in green. \xc2\xba6

With Edge Marks enabled, the previously-marked lines are always rendered. You can see the black contour lines and the blue lines that are made with edge marks.

Render without Edge Marks. \xc2\xba6

Render with Edge Marks enabled. \xc2\xba6

What are edge marks good for?

When you need to render marks on an almost-flat plane, when other edge types cannot detect any line.

When you want full control of edge rendering. Often used for edges of squarish shapes.

Mark the whole base mesh to be rendered for base mesh preview.

What are edge marks not good for?

Round outer edges (use instead Contour/External Contour/Silhouette).

Face marks are useful for removing lines from certain areas of a mesh.

To set a face mark:

Select a mesh object and enter Edit Mode.

Select the faces you want to be marked.

Press Ctrl- F and select Face Data \xe2\x80\xa3 Mark Freestyle Face.

In this example, two faces of the default cube are marked like the image on the left. On the right is a render without face marks activated.

Marked faces (Edit Mode).\xc2\xba6

Render output.\xc2\xba6

The line selection can be controlled via inclusion and faces options: Whether to include or exclude edges matching defined face mark conditions from the line set.

(De)select all edges which have one or both neighbor faces marked.

(De)select all edges which have both of their neighbor faces marked.

Inclusive, One Face.\xc2\xba6

Inclusive, Both Faces.\xc2\xba6

Exclusive, One Face.\xc2\xba6

Exclusive, Both Faces.\xc2\xba6

Include or exclude objects for line calculation, based on their belonging to a Collection .

The name of the object collection to use.

Whether to include or exclude lines from those objects in this line set.

--- Fin du chapitre: _line_set ---

--- Début du chapitre: _linked --- Reference

Edit Mode

Select \xe2\x80\xa3 Select Linked \xe2\x80\xa3 Linked

Ctrl- L

Select geometry connected to already selected elements. This is often useful when a mesh has disconnected, overlapping parts, where isolating it any other way would be tedious. To give more control, you can also enable delimiters in the Adjust Last Operation panel, so the selection is constrained by seams, sharp edges, materials or UV islands.

With Pick Linked you can also select connected geometry directly under the cursor, using the L shortcut to select or Shift- L to deselect linked. This works differently in that it uses the geometry under the cursor instead of the existing selection.

Reference

Edit Mode

Select \xe2\x80\xa3 Select Linked \xe2\x80\xa3 Shortest Path

Ctrl- LMB

Select a face or vertex path with Ctrl- LMB. \xc2\xba6

Selects all geometry along the shortest path from the active vertex, edge, or face to the one which was selected.

This select button indicates what should be done when selecting a vertex path with Ctrl- LMB :

Just selects all the edges in the path.

Marks all edges in the path as seams for UV unwrapping.

Marks all edges in the path as sharp for the Edge Split Modifier.

Marks all edges in the path as creases for the Subdivision Surface Modifier, with weight 1.0.

Gives bevel weight 1.0 (for the Bevel Modifier) to all edges in the path.

Marks all edges in the path as Freestyle edges.

Supports diagonal paths for vertices and faces, and selects edge rings with edges.

Only takes into account the number of edges of the path and not the length of the edges to calculate the distances.

Selects all elements in the shortest paths from the active selection to the clicked area.

Allows to quickly select alternate elements in a path.

The number of deselected elements in the repetitive sequence.

The number of selected elements in the repetitive sequence.

Offset from the starting point.

Reference

Edit Mode

Select \xe2\x80\xa3 Select Linked \xe2\x80\xa3 Linked Flat Faces

Selects all connected faces with a similar angle.

Select connected faces with a reference angle less than the value specified.

Looking at the image above, when at least one face is selected (as seen on the left), Linked Flat Faces will select all connecting faces that lie on the same or similar plane (as shown in the middle image). If the corners are smoothed, those faces are no longer lined up with the selected faces. At this point, increasing the Sharpness value in the tool options could include the smoothed faces.

--- Fin du chapitre: _linked ---

--- Début du chapitre: _link_append --- These functions help you reuse materials, objects and other data-blocks loaded from another blend-file. You can build libraries of common content and share them across multiple referencing files.

Newly added collections types are available in Add \xe2\x80\xa3 Collection Instance in the 3D Viewport.

Look in the Outliner, with display mode set to Blender File, to see all your linked and appended data-blocks.

Note

It is not possible to Append or Link data from much newer blend-files

Reference

Topbar

All Modes

File \xe2\x80\xa3 Link

Link creates a reference to the data in the source file such that changes made there will be reflected in the referencing file the next time it is reloaded. But linked data is usually not editable. Library Overrides can be created from linked data to allow some level of local editing, animation, etc.

In the File Browser, navigate to the external source blend-file and select the data-block you want to reuse.

When you link an object, it will be placed in your scene at the 3D cursor position. Many other data types, cameras, curves, and materials for example, must be linked to an object before they become visible.

See Relative Paths.

Makes the object Active after it is loaded.

The object will be added to the active collection of the active view layer. Otherwise, it will be added to a new collection in the active view layer.

This option instantiates the linked collection as an object, adding it to the active scene. Otherwise, the linked collection is directly added to the active view layer.

Create instances for object data which are not referenced by any objects.

Reference

Topbar

All Modes

File \xe2\x80\xa3 Link

Append makes a full copy of the data into your blend-file, without keeping any reference to the original one. You can make further edits to your local copy of the data, but changes in the external source file will not be reflected in the referencing file.

In the File Browser, navigate to the external source blend-file and select the data-block you want to reuse.

Tip

Blend-files can also be linked/appended by dragging and dropping blend-files into the Blender window.

Note

Appending data you already have linked will add objects/collections to the scene, but will keep them linked (and un-editable).

This is done so existing relationships with linked data remain intact.

Makes the object Active after it is loaded.

The object will be added to the active collection of the active view layer. Otherwise, it will be added to a new collection in the active view layer.

This option instantiates the linked collection as an object, adding it to the active scene. Otherwise, the linked collection is directly added to the active view layer.

Create instances for object data which are not referenced by any objects.

Defines the appended data-block as Protected.

Appends also all indirectly linked data, instead of linking them.

Reloading is useful if you changed something in the library blend-file and want to see those changes in your current blend-file without having to re-open it. You can reload and relocate a whole library from the context menu of the library items in the Outliner \xe2\x80\x99s Blender Fileview. Relocating allows you to reload the library from a new file path. This can be used to either fix a broken linked library (e.g. because the library file was moved or renamed after linking from it), or to switch between different variations of a same set of data, in different library files.

While loading a blend-file, if Blender cannot find a library, it will create placeholder data-blocks to replace missing linked ones. That way, references to the missing data is not lost, and by relocating the missing library, the lost data can be automatically restored.

Reference

3D Viewport
Object Mode
Object \xe2\x80\xa3 Relations \xe2\x80\xa3 Make Local\xe2\x80\xa6
Reference
Outliner

Context menu \xe2\x80\xa3 ID Data \xe2\x80\xa3 Make Local

Makes the selected or all external objects local to the current blend-file. Links to the original library file will be lost, but it will make those data-blocks fully editable, just like the ones directly created in that blend-file.

The operation available from the Outliner\xe2\x80\x99s context menu has no options, and only affects the selected data-block.

The operation available from the 3D Viewport only directly affects selected objects, but it can also make local the objects\xe2\x80\x99 dependencies:

Optionally unlinks the object\xe2\x80\x99s Object Data and Material Data.

Selected Objects, + Object Data, + Materials, All (i.e. including all scenes)

For the most part linking data will work as expected, however, there are some corner cases which are not supported.

In general, dependencies should not go in both directions. Attempting to link or append data which links back to the current file will likely result in missing links.

When linking objects directly into a blend-file, the Rigid Body settings will not be linked in since they are associated with their scene\xe2\x80\x99s world. As an alternative, you can link in the entire scene and set it as a Background Set.

Linking to blend-files with compression enabled may significantly increase memory usage while loading files.

Reading data on demand isn\xe2\x80\x99t supported with compression (this only impacts load time, once loaded there is no difference in memory use).

--- Fin du chapitre: `_link_append` ---

--- Début du chapitre: `_link_data` ---

Reference
Object Mode

Object \xe2\x80\xa3 Link/Transfer Data\xe2\x80\xa6

Links objects between scenes or data-blocks of the active object to all selected objects. In some case (i.e. Object Data, Modifier) the target objects must be of the same type as the active one or capable of receiving the data. If targets already have data linked to them, it will be unlinked first.

Data-block type to link.

Link Object Data

Link Materials

Link Animation Data

Link Collections

Link Instance Collection

Link Fonts to Text

Copy Modifiers

Copy Grease Pencil Effects

See also

Making Single User for unlinking data-blocks.

Replace assigned Object Data.
Replace assigned Materials.
Replace assigned Animation Data.
Replace assigned Collections.
Replace assigned Collection Instance.
Replace Text object Fonts.
Replace Modifiers.
Replace Grease Pencil Effects.
--- Fin du chapitre: `_link_data` ---

--- Début du chapitre: `_link_scene` ---Reference
Object Mode
Object `\xe2\x80\xa3 Link/Transfer Data \xe2\x80\xa3 Link Objects to Scene`
Links the selected objects into a different scene than the current one. The Link Objects to Scene in the Adjust Last Operation panel lets you choose between scenes.
This makes the same object exist in more than one scene at once, including its position and animation data. The object `\xe2\x80\x99s` origin will change its color to reflect that.
--- Fin du chapitre: `_link_scene` ---

--- Début du chapitre: `_linux` --- This guide covers the following topics:
Installing Dependencies
Downloading the Repository
Setting up the Build Environment
Below are listed the installation commands for popular Linux distributions.
For the appropriate system, run the command in a terminal:
Debian/Ubuntu:
Redhat/Fedora:
Arch Linux:
Simply check out the Blender Manual `\xe2\x80\x99s` repository using:
The repository will now be downloaded which may take a few minutes depending on your internet connection.

Tip
It is recommended to setup and activate a virtual Python environment where dependencies will be installed:
Repeat the `source .venv/bin/activate` command to re-activate the virtual environment, whenever you open a new terminal to build the documentation.
This step may be required on some distributions that enforce PEP 668.
Open a Terminal window.
Enter the blender-manual folder which was just added by git clone :
Inside that folder is a file called `requirements.txt` which contains a list of all the dependencies we need. To install these dependencies, we can use the `pip3` command:
Note
Every now and then you may want to make sure your dependencies are up to date using:
--- Fin du chapitre: `_linux` ---

--- Début du chapitre: `_linux_windowing_environment` --- On Linux Blender supports both X11 and Wayland for official releases.

When Wayland is detected, it is the preferred system, otherwise X11 will be used.

Hint

The current \xe2\x80\x9cWindowing Environment\xe2\x80\x9d is listed in File \xe2\x80\x93 About.

This is the windowing environment that has been used most widely on Linux & Unix systems.

There are no near-term plans to deprecate or remove X11 support.

Support for Wayland is a more recent addition, so there may be configurations that have not been tested yet. Please report a bug if you experience problems.

Blender has been tested with Gnome-Shell (mutter), KDE (plasma) & SWAY (wlroots) based compositors.

Under Gnome-Shell the libdecor library is required. This is available as a package on most Linux distribution.

If the library isn't found X11 will be used as a fallback.

Detailed Wayland output can help to track down problems. Launch Blender from the command-line with additional arguments:

Uninstall libdecor, then run Blender with an empty X11 display variable.

The cursor theme to use (must refer to a locally installed cursor).

The cursor size, defaults to 28, you may wish to increase the size on Hi-DPI displays.

Versions of Gnome-Shell prior to 44 don't fully support fractional scaling.

Using fractional under older versions of Gnome-Shell may result in glitches such as a small cursor size.

Currently NVidia drivers don't fully support features needed for Wayland. Graphical glitches and flickering are common problems. In some cases, there can be crashes on startup. This is not specific to Blender, so NVidia users may want to use X11 until driver support improves.

Feature

X11

Wayland

Notes

Smooth Scroll

\xe2\x9c\x97

\xe2\x9c\x93

Multi-Touch Gestures

\xe2\x9c\x97

\xe2\x9c\x93

Reliable Cursor Warping

\xe2\x9c\x97 *1

\xe2\x9c\x93

Window Positioning

\xe2\x9c\x93

\xe2\x9c\x97 *2

Other features which both systems support such as Hi-DPI, 3D-mouse, tablet input, \xe2\x80\x93 etc. have been left out of this list.

--- Fin du chapitre: _linux_windowing_environment ---

--- Début du chapitre: _list_view --- List view with expanded Filtering Options panel. \xc2\xb6

This control is useful for managing lists of items. In addition to the main list, there is a Filtering panel on the bottom (hidden by default) and modification buttons on the right.

To select an item, click LMB on it.

By double-clicking on an item, you can edit its name via a text field. This can also be achieved by clicking it with Ctrl-LMB.

The list view can be resized to show more or fewer items. Hover the mouse over the handle (:::), then click and drag to expand or shrink the list.

Click the Show filtering options button (triangle on bottom left) to show or hide the filter option panel.

Filters the list to only show items containing a certain term.

Toggle between including items that match the search term and those that do not contain the search term.

This button switches between alphabetical and non-alphabetical ordering.

Sort objects in ascending or descending order. This also applies to alphabetical sorting, if selected.

On the right of the list view are list modification buttons:

Adds a new item.

Removes the selected item.

A special menu with operators to edit list entries.

Moves the selected item up/down one position.

--- Fin du chapitre: _list_view ---

--- Début du chapitre: _local_view --- Reference

All modes

View \xe2\x80\xa3 Local View \xe2\x80\xa3 Toggle Local View

NumpadSlash, Slash

Global view shows all 3D objects in the scene. Local view isolates the selected object(s) so that they are the only ones visible in the viewport. This is useful for working on objects that are obscured by others, or to speed up the viewport performance in heavy scenes. Local view is contextual, meaning that it can be set per 3D Viewport.

You can toggle between Global and Local View by selecting the option from the View menu or using the shortcut NumpadSlash.

Global View.\xc2\xb6

Local View.\xc2\xb6

Note

In local view, the 3D cursor is not locked to the scene. Instead, each view has an independent cursor location.

Tip

Accidentally pressing NumpadSlash can happen rather often if you are new to Blender, so if a bunch of the objects in your scene seem to have mysteriously vanished, try pressing NumpadSlash again.

Reference

All modes

View \xe2\x80\xa3 Local View \xe2\x80\xa3 Remove from Local View

Alt-NumpadSlash, Alt-Slash

Objects can be removed from Local View by selecting them and using the Remove from Local View operator. This will move them back to the global view. If the last remaining object is removed, the local view will be left empty and you will have to exit it to see any objects.

Hint

This is useful when working with objects in dense scenes where painstakingly selecting objects to include in the local view

isn't practical, especially when they intersect or are obscured by objects you don't want to include. In this case it's simpler to select many objects in a region and enter local view, then remove the ones you don't need.

--- Fin du chapitre: `_local_view` ---

--- Début du chapitre: `_locked_track` --- The Locked Track constraint is a bit tricky to explain, both graphically and textual. Basically, it is a Track To constraint, but with a locked axis, i.e. an axis that cannot rotate (change its orientation). Hence, the owner can only track its target by rotating around this axis, and unless the target is in the plane perpendicular to the locked axis, and crossing the owner, this owner cannot really point at its target.

Let us take the best real-world equivalent: a compass. It can rotate to point in the general direction of its target (the magnetic North, or a neighbor magnet), but it cannot point directly at it, because it spins like a wheel on an axle. If a compass is sitting on a table and there is a magnet directly above it, the compass cannot point to it. If we move the magnet more to one side of the compass, it still cannot point at the target, but it can point in the general direction of the target, and still obey its restrictions of the axle.

When using a Locked Track constraint, you can think of the target as a magnet, and the owner as a compass. The Lock axis will function as the axle around which the owner spins, and the To axis will function as the compass's needle. Which axis does what is up to you!

If you have trouble understanding the buttons of this constraint, read the tooltips, they are pretty good. If you do not know where your object's axes are, enable Axis in Properties's Armature's Viewport Display. Or, if you are working with bones, turn on the Axes in the armature's Viewport Display panel.

This constraint was designed to work cooperatively with the Track To constraint. If you set the axes buttons right for these two constraints, Track To can be used to point the axle at a primary target, and Locked Track can spin the owner around that axle to a secondary target.

This constraint also works very well for 2D billboarding.

Locked track panel. 

Data ID used to select the constraint's target, and is not functional (red state) when it has none. See common constraint properties for more information.

The tracking local axis, i.e. the owner's axis to point at the target. The negative options force the relevant axis to point away from the target.

The locked local axis, i.e. the owner's axis which cannot be re-oriented to track the target.

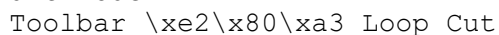
Important

If you choose the same axis for Tracking Axis and Locked Axis, the constraint will no longer be functional (red state). Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_locked_track` ---

--- Début du chapitre: `_loop` --- Reference

Edit Mode

Toolbar 

Ctrl- R

The Loop Cuttool is a modal tool version of the Loop Cut and Slide operator. This tool splits a loop of faces by inserting new edge loops intersecting the chosen edge.

The tool is interactive and has two steps:

Pre-Visualizing the Cut

After the tool is activated, move the cursor over a desired edge. The cut to be made is marked with a magenta colored line as you move the mouse over the various edges. The to be created edge loop stops at the poles (tris and n-gons) where the existing face loop terminates.

Perform the Cut

Once the desired location of the new edge loop is found, the edge loop can be created via LMB.

Mesh before inserting edge loop.

Preview of edge loop location.

Interactive placement of edge loop between adjacent loops.

Increases and decreases the number of cuts to create. These cuts are uniformly distributed in the original face loop, and you will not be able to control their positions.

Corrects the corresponding UV coordinates, if these exist, to avoid image distortions.

After the modal tool is run the Loop Cut and Slide Options are available in the Adjust Last Operation panel.

--- Fin du chapitre: _loop ---

--- Début du chapitre: _loopcut_slide ---Reference

Edit Mode

Edge \xe2\x80\xa3 Loop Cut and Slide

Ctrl- R

Loop Cut and Slide splits a loop of faces by inserting a new edge loop intersecting the chosen edge.

The tool is interactive and has two steps:

Pre-Visualizing the Cut

After the tool is activated, move the cursor over a desired edge. The cut to be made is marked with a magenta colored line as you move the mouse over the various edges. The to be created edge loop stops at the poles (tris and n-gons) where the existing face loop terminates.

Sliding the new Edge Loop

Once an edge is chosen via LMB, you can move the mouse along the edge to determine where the new edge loop will be placed. This is identical to the Edge Slide tool. Clicking LMB again confirms and makes the cut at the pre-visualized location, or clicking RMB forces the cut to exactly 50%. This step is skipped when using multiple edge loops (see below).

Mesh before inserting edge loop.

Preview of edge loop location.

Interactive placement of edge loop between adjacent loops.

These options are available while the tool is in use, and later in the Adjust Last Operation panel.

After activating the tool, but before confirming initial loop location, you can increase and decrease the number of cuts to create, by entering a number with the keyboard, scrolling Wheel or using PageUp and PageDown.

Note

When creating multiple loops, these cuts are uniformly distributed in the original face loop, and you will not be able to control their positions.

Preview of multiple edge loops.

Result of using multiple cuts.

Smoothing causes edge loops to be placed in an interpolated position, relative to the face it is added to, causing them to be shifted outwards or inwards by a given percentage, similar to the Subdivide Smooth tool. When not using smoothing, new vertices for the new edge loop are placed exactly on the preexisting edges. This keeps subdivided faces flat, but can distort geometry, particularly when using Subdivision Surfaces. Smoothing can help maintain the curvature of a surface once it is subdivided.

Added edge loops without smoothing.

Same edge loops, but with smoothing value.

Falloff type for Smoothness, changes the shape of the profile.

Position of the edge loop relative to the surrounding edge loops.

Only available for single edge loops. This matches the shape of the edge loop to one of the adjacent edge loops. (See Edge Slide tool for details.)

When Even is enabled, this flips the target edge loop to match. (See Edge Slide tool for details.)

Clamp within the edge extents.

Corrects the corresponding UV coordinates, if these exist, to avoid image distortions.

--- Fin du chapitre: _loopcut_slide ---

--- Début du chapitre: _loops --- Reference

Edit Mode (Vertex or Edge select mode)

Select  Select Loops  Edge Loops

Alt-LMB, or Shift-Alt-LMB for modifying existing selection.

Holding Alt while selecting an edge selects a loop of edges that are connected in a line end-to-end, passing through the edge under the mouse pointer. Holding Shift-Alt while clicking adds to the current selection.

Note

Vertex mode

In Vertex select mode, you can also select edge loops, by using the same shortcuts, and clicking on the edges (not on the vertices).

Longitudinal and latitudinal edge loops.

The left sphere shows an edge that was selected longitudinally. Notice how the loop is open. This is because the algorithm hit the vertices at the poles and is terminated because the vertices at the pole connect to more than four edges. However, the right sphere shows an edge that was selected latitudinally and has formed a closed loop. This is because the algorithm hit the first edge that it started with.

All boundary edges can be selected by performing a second loop select action on a boundary edge.

This can be useful for selecting boundaries for meshes that include triangles and n-gons, where loop select would not otherwise select the full boundary.

The second loop select action is shown on the right.

Reference

Edit Mode (Face or Vertex select modes)

Alt-LMB or Shift-Alt-LMB for modifying existing selection.

In face select mode, holding Alt while selecting an edge selects a loop of faces that are connected in a line end-to-end, along their opposite edges.

In vertex select mode, the same can be accomplished by using Ctrl-Alt to select an edge, which selects the face loop implicitly.

Face loop selection. \xc2\xba

This face loop was selected by clicking with Alt-LMB on an edge, in faceselect mode. The loop extends perpendicular from the edge that was selected.

Alt versus Ctrl-Alt in vertex select mode. \xc2\xba

A face loop can also be selected in Vertexselect mode. Technically Ctrl-Alt-LMB will select an Edge Ring, however, in Vertexselect mode, selecting an Edge Ring implicitly selects a Face Loop since selecting opposite edges of a face implicitly selects the entire face.

Reference

Edit Mode

Select \xe2\x80\xa3 Select Loops \xe2\x80\xa3 Edge Rings

Ctrl-Alt-LMB

In Edgeselect mode, holding Ctrl-Alt while selecting an edge (or two vertices) selects a sequence of edges that are not connected, but on opposite sides to each other continuing along a face loop.

As with edge loops, you can also select edge rings based on current selection, using either Select \xe2\x80\xa3 Select Loops \xe2\x80\xa3 Edge Rings.

Note

Vertexmode

In Vertexselect mode, you can use the same shortcuts when clicking on the edges (not on the vertices), but this will directly select the corresponding face loop \xe2\x80\xa6

A selected edge loop, and a selected edge ring. \xc2\xba

In Fig. A selected edge loop, and a selected edge ring. The same edge was clicked on, but two different \xe2\x80\x9cgroups of edges \xe2\x80\x9d were selected, based on the different tools. One is based on edges during computation and the other is based on faces.

Note

Convert Selection to Whole Faces

If the edge ring selection happened in Edge Select Mode, switching to Face Select Mode will erase the selection.

This is because none of those faces had all its (four) edges selected, just two of them.

Instead of selecting the missing edges manually or by using Shift-Alt-LMB twice, it is easier to first switch to Vertex Select Mode, which will kind of \xe2\x80\x9cflood \xe2\x80\x9d the selection. A subsequent switch to Face Select Mode will then properly select the faces.

Reference

Edit Mode (Edge select mode)

Select \xe2\x80\xa3 Select Loops \xe2\x80\xa3 Select Loop Inner-Region

Select Loop Inner-Region selects all faces that are inside a closed loop of edges. While it is possible to use this operator in Vertex and Faceselection modes, results may be unexpected. Note that if the selected loop of edges is not closed, then all connected edges on the mesh will be considered inside the loop.

Loop to Region. \xc2\xba

This tool handles multiple loops fine, as you can see. \xc2\xba

This tool handles \xe2\x80\x9choles\xe2\x80\x9d just fine as well.

\xc2\xb6

Reference

Edit Mode (Edge select mode)

Select \xe2\x80\xa3 Select Loops \xe2\x80\xa3 Select Boundary Loop

Select Boundary Loop does the opposite of Select Loop Inner-Region, based on all regions currently selected, it selects only the edges at the border (contour) of these islands. It can operate in any select mode, but when in Facemode it will switch to Edgeselect mode after running.

All this is much more simple to illustrate with examples:

Select Boundary Loop does the opposite and forces into Edge Select Mode.

\xc2\xb6

--- Fin du chapitre: _loops ---

--- Début du chapitre: _luminance_key --- The Luminance Key node determines background objects from foreground objects by the difference in the luminance (brightness) levels.

Stock footage of explosions, smoke or debris are normally shot against a solid, dark background rather than a green screen. This node can separate the foreground effect from the background. It can also be used for sky replacement for overexposed or gray skies that aren't suitable for chroma keying.

Tip

When compositing footage of something that emits light and has a dark background, like fire, a Mix Node using a Screen or Add operator will produce better results.

Standard color input.

Determines the lowest values that are considered foreground. (Which is supposed to be \xe2\x80\x93 relatively \xe2\x80\x93 light: from this value to 1.0.)

Determines the highest values that are considered to be background objects. (Which is supposed to be \xe2\x80\x93 relatively \xe2\x80\x93 dark: from 0.0 to this value.)

Note

Brightness levels between the two values form a gradient of transparency between foreground and background objects.

Image with an alpha channel adjusted for the keyed selection.

A black-and-white alpha mask of the key.

For this example the model was shot against a white background. Using the Luminance Key node, we get a matte out where the background is white, and the model is black; the opposite of what we want. If we wanted to use the matte, we have to switch the white and the black. How to do this? Color Ramp node to the rescue \xe2\x80\x93 we set the left color to White Alpha 1.0, and the right color to be Black Alpha 0.0. Thus, when the Color Ramp gets in black, it spits out white, and vice versa. The reversed mask is shown; its white outline is usable as an alpha mask now.

Using Luma Key with a twist. \xc2\xb6

Now to mix, we do not really need the Alpha Over node; we can just use the mask as our Factor input. In this kinda weird case, we can use the matte directly; we just switch the input nodes. As you can see, since the matte is white (1.0) where we do not want to use the model picture, we feed the background photo to the bottom socket (recall the Mix node uses the top socket where the factor is 0.0, and the bottom socket where the factor is 1.0). Feeding our original photo into the top socket means it will be used

where the Luminance Key node has spit out Black. Voil\,our model is teleported from Atlanta to aboard a cruise ship docked in Miami.

--- Fin du chapitre: _luminance_key ---

--- Début du chapitre: _macos --- This guide covers the following topics:

Installing Dependencies

Downloading the Repository

Setting up the Build Environment

Note

This guide relies heavily on command-line tools. It assumes you are the least familiar with the macOS Terminal application.

Install those packages or make sure you have them in your system.

PIP

Git

Git LFS

When using Homebrew, run the following commands in the terminal:

Simply check out the Blender Manual's repository using:

The repository will now be downloaded which may take a few minutes depending on your internet connection.

Tip

It is recommended to setup and activate a virtual Python environment where dependencies will be installed:

Repeat the `source .venv/bin/activate` command to re-activate the virtual environment, whenever you open a new terminal to build the documentation. Open a Terminal window.

Enter the blender-manual folder which was just added by git clone :

Inside that folder is a file called requirements.txt which contains a list of all the dependencies we need. To install these dependencies, we can use the pip command:

Note

Every now and then you may want to make sure your dependencies are up to date using:

--- Fin du chapitre: _macos ---

--- Début du chapitre: _magic --- Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

The Magic Texture node is used to add a psychedelic color texture.

Texture coordinate to sample texture at; defaults to Generated texture coordinates if the socket is left unconnected.

Scale of the texture.

Amount of distortion.

Number of iterations.

Texture color output.

Texture intensity output.

Magic texture: Depth 10, Distortion 2.0. \

--- Fin du chapitre: _magic ---

--- Début du chapitre: _magnetic --- Reference

Physics \ Force Fields

Magnetic

This field depends on the speed of the particles. It simulates the force of magnetism on magnetized objects.

--- Fin du chapitre: `_magnetic` ---

--- Début du chapitre: `_maintain_volume` --- The Maintain Volume constraint limits the volume of a mesh or a bone to a given ratio of its original volume.

See also

Harkyman on the development of the Maintain Volume constraint.

Maintain Volume Constraint. `\xc2\xba6`

Specifies how the constraint handles scaling of the non-free axes.

This mode overrides non-free axis scaling to strictly maintain the specified volume. Only the ratio between the scale of the non-free axes is passed through.

This mode maintains the volume as specified only when the pre-constraint scaling is uniform. Deviations from uniform scaling on non-free axes are passed through.

This mode maintains the volume only when the object is scaled just on its free axis. Any additional non-free axis scaling is passed through.

The free-scaling axis of the object.

The bone's rest volume.

This constraint allows you to choose in which space to evaluate its owner's transform properties. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_maintain_volume` ---

--- Début du chapitre: `_maintenance_guide` --- When RST-files are added or removed the corresponding locale files are added or removed automatically by the update script. However, if files need to be moved please use this Python script:

RST-files can then be freely moved and the remap script will move the locale file after:

It is best to avoid moving/renaming files as this breaks URLs and without this script translators will lose all their work in these files. Please ask an administrator if you think something should be renamed/moved.

Note

This script also works for image file names.

Create a release branch (`blender-3.2-release/`)

Update the splash image: `interface_splash_current.png` in the release branch.

Increase the `conf.py:blender_version` variable in the trunk version.

--- Fin du chapitre: `_maintenance_guide` ---

--- Début du chapitre: `_make_face_edge` --- Reference

Edit Mode

Vertex `\xc2\xba3` New Edge/Face from Vertices

F

This is a context-sensitive tool which creates geometry by filling in the selection. When only two vertices are selected it will create an edge, otherwise it will create faces.

The typical use case is to select vertices and press F, yet Blender also supports creating faces from different selections to help to quickly build up geometry.

The following methods are used automatically depending on the context.

Before. \xc2\xbb

After. \xc2\xbb

Before. \xc2\xbb

After. \xc2\xbb

When there are many edges Blender will make an n-gon. Note that, this does not support holes, to support holes you need to use the Fill Faces tool.

Before. \xc2\xbb

After. \xc2\xbb

Existing edges are used to make the face as well as an extra vertex.

Before. \xc2\xbb

After. \xc2\xbb

Sometimes you may have many connected edges without interior faces.

Before. \xc2\xbb

After. \xc2\xbb

When there are many isolated vertices, Blender will calculate the edges for an n-gon.

Before. \xc2\xbb

After. \xc2\xbb

With a single vertex selected on a boundary, the face will be created along the boundary, this saves manually selecting the other two vertices. Notice this tool can run multiple times to continue creating faces.

See also

For other ways to create faces see:

Fill

Grid Fill

Bridge Edge Loops

When you have a region of existing faces, creating a face on this selection will remove the shared vertices and edges, creating a single face.

This is simply a convenience for accessing Dissolve Faces.

--- Fin du chapitre: _make_face_edge ---

--- D but du chapitre: _make_single_user ---Reference

Object Mode

Object \xe2\x80\xa3 Relations \xe2\x80\xa3 Make Single User

Makes the selected or all object data-blocks single users, that is, not shared (linked) between other objects in the blend-file.

Additionally, it can also make single-user copies of its dependencies, like meshes, curves, materials, animations \xe2\x80\xa6

These actions work on the selected objects, or on all the objects of the scene.

All, Selected Objects

Lets you, in addition to the menu predefined selection, choose the type of data-blocks individually.

Make single user objects.

Make single user object data.

Make materials local to each data-block.

Make the animation of Object Properties data local to each object.

Make object data (mesh, curve etc.) animation data local to each object.
See also

Making Single User

--- Fin du chapitre: _make_single_user ---

--- Début du chapitre: _make_vertex_parent ---Reference
Edit Mode

Vertex \xe2\x80\xa3 Make Vertex Parent

Ctrl- P

This will parent the other selected object(s) to the
vertices/edges/faces selected,as described inparenting.

--- Fin du chapitre: _make_vertex_parent ---

--- Début du chapitre: _mapping --- The Mappingnode transforms the input
vector by applying translation, rotation, and scaling.

The inputs of the node are dynamic. In particular, the Locationinput is
only available inthe Textureand Pointvector types.

The vector to be transformed.

The amount of translation along each axis.

The amount of rotation along each axis. XYZ order.

The amount of scaling along each axis.

The node applies the transformation differently depending on the semantic
type of the input vector.

For this vector type, the node performs a straightforward transformation.

Transforming a texture coordinates is analogous to transforming a UV

map.For instance, translating the texture coordinates along the positiveX
axis would resultin the evaluated texture to move in the negativeX axis,

much like if one translated a UV map.Similarly, scaling the texture

coordinates upwould result in the evaluated texture to scale down.So

transforming the texture coordinates would appear to have the opposite
effect on the evaluated texture.

The order of transformation is: Scale \xe2\x80\x93 Rotate \xe2\x80\x93
Translate, which means:

Translation moves the input along the local rotation axis.

Rotation rotates the input around the origin of the space.

Scaling scales the input along the global axis.

For this vector type, the node performs an inverse transformation.

Inverse transforming a texture coordinates would, as opposed to the

Pointtype,transform the evaluated texture itself. For instance,

translating the texture coordinates alongthe positive X axis would result
in the evaluated texture to move in the positive X axis,as one would

expected. Similarly, scaling the texture coordinates up would result
inthe evaluated texture to scale up, as one would expect.

The order of transformation is: Translate \xe2\x80\x93 Rotate
\xe2\x80\x93 Scale, which means:

Translation moves the input along the global axis.

Rotation rotates the input around the translation vector.

Scaling scales the input along the local rotation axis.

For this vector type, a Pointttransformation is performed, but with zero
translation.

For this vector type, the node performs the inverse transpose of the

transformation and normalize the result.Such transformation ensures

correct normals after non-uniform scaling.So this type should be used

when transforming normals.

The input vector after transformation.

Mapping node example. \xc2\xba6

--- Fin du chapitre: _mapping ---

--- Début du chapitre: _map_range --- The Map Rangenode remaps a value from a range to a target range.

The input value or vector to be remapped.

The lower bound of the range to remap from.

The higher bound of the range to remap from.

The lower bound of the target range.

The higher bound of the target range.

The number of values allowed between To Min and To Max when using Stepped Linear interpolation. A higher value will give a smoother interpolation while lower values will progressively quantize the input.

Map Range supports both Float and Vector data types. Changing the data type will also update the sockets to reflect the data type chosen.

The mathematical method used to transition between gaps in the numerical inputs.

Linear interpolation between From Min and From Max values.

Stepped linear interpolation between From Min and From Max values.

Smooth Hermite edge interpolation between From Min and From Max values.

Smoother Hermite edge interpolation between From Min and From Max values.

If enabled, the output is clamped to the target range.

The input value after remapping.

The Noise Texture node outputs a value in the range [0, 1]. We can use the Map Rangenode to remap this value into the range [-1, 1].

Example of Map Range node. \xc2\xba6

--- Fin du chapitre: _map_range ---

--- Début du chapitre: _map_uv --- Map a texture using UV coordinates, to apply a texture to objects in compositing.

May be used in combination with Cryptomatte Node to apply the texture only to specific objects.

The new 2D texture.

The input for UV render pass. See Cycles render passes .

Hint

To store the UV pass a multi-layer OpenEXR format could be used.

Interpolation Methods.

Enhances the clarity of textures viewed at oblique angles, addressing issues like blurring and distortion.

No interpolation, uses nearest neighboring pixel.

Alpha threshold is used to fade out pixels on boundaries.

The resulting image is the input image texture distorted to match the UV coordinates. That image can then be overlay mixed with the original image to paint the texture on top of the original. Adjust alpha and the mix factor to control how much the new texture overlays the old.

Hint

When painting the new texture, it helps to have the UV maps for the original objects in the scene, it is recommended to keep those UV texture outlines around even, when shooting is done.

In the example below, we have overlaid a grid pattern on top of the two heads after they have been rendered. During rendering, we enabled the UV layer in the Properties Render Layer \xe2\x80\xa3 Passes. Using a Mix node (\xe2\x80\x9cOverlay\xe2\x80\x9d in figure), we mix that new UV

texture over the original face. We can use this grid texture to help in any motion tracking that we need to do.

Adding a grid UV textures for motion tracking. \xc2\xba

In the next example, we overlay a logo on top of a mesh composed of two intersecting cubes, and we ensure that we Enable the Alpha premultiply button on the Mix node. The logo is used as additional UV texture on top of the existing texture. Other examples include the possibility that there was used an unauthorized product box during the initial animation, and it is needed to substitute in a different product sponsor after rendering.

Hint

Due to limits of this node, it is not recommended to rush pre-production rendering under the guise of \xe2\x80\x9cfixing it later\xe2\x80\x9d.

Adding UV textures in post-production. \xc2\xba

--- Fin du chapitre: _map_uv ---

--- Début du chapitre: _map_value --- Map Value node is used to scale, offset and clamp values.

Standard Value input. (Value refers to each vector in the set.)

Factor added to the input value.

Scales (multiply) the input value.

Enable this to activate their related operation.

Defines a range between minimum and maximum to Clamp the input value to.

Standard value output.

This is particularly useful in achieving a depth of field effect, where the Map Value node is used to map a Z value (which can be 20 or 30 or even 500 depending on the scene) to the range between (0 to 1), suitable for connecting to a Blur node.

The Map Value node can also be used to multiply values to achieve a desired output value. In the mini-map to the right, the Time node outputs a value between 0.0 and 1.0 evenly scaled over 30 frames. The first Map Value node multiplies the input by 2, resulting in an output value that scales from 0.0 to 2.0 over 30 frames. The second Map Value node subtracts 1 from the input, giving working values between (-1.00 to 1.0), and multiplies that by 150, resulting in an output value between (-150 to 150) over a 30-frame sequence.

Using Map Value to multiply. \xc2\xba

--- Fin du chapitre: _map_value ---

--- Début du chapitre: _marble --- The marble texture is used to generate marble, fire, or noise with a structure. Bands are generated based on the sine, saw, or triangular formula and noise turbulence.

Marble Texture panels. \xc2\xba

Three settings for soft to more clearly defined Marble.

Soft, Sharp, Sharper

Shape of wave to produce bands.

Sine, Saw, Triangle

The noise function works with two methods.

Soft, Hard

The dimensions of the noise table.

The depth of the Marble calculation. A higher value results in greater calculation time, but also in finer details.

The turbulence of the sine bands.

--- Fin du chapitre: _marble ---

--- Début du chapitre: `_marker` --- Marker schematic.\xc2\xb6
This panel contains numerical settings for marker position, pattern and search area dimensions, and offset of anchor point from pattern center.
Toggles the markers affect on the current frame.
The X/Y positions of the marker at frame in screen coordinates.
The X/Y offset to parenting point.
The width/height of a marker pattern in screen coordinates.
The X/Y position of search at frame relative to the marker position.
The width/height of the markers search in screen coordinates.
--- Fin du chapitre: `_marker` ---

--- Début du chapitre: `_markers` --- Markers are used to denote frames with key points or significant events within an animation. E.g. it could be that a character's animation starts, the camera changes position, or a door opens. Markers can be given names to make them more meaningful at a quick glance. They are available in many of Blender's editors.

Note

Unlike keyframes, markers are always placed at a whole frame number, you cannot set a marker at frame 2.5.

Markers can be created and edited in the following editors:

Graph Editor

Dope Sheet

NLA Editor

Video Sequence Editor

Timeline

Note

A marker created in one of these editors will also appear in all others that support them.

Besides standard markers, pose markers are another type of markers, which are specific to armatures and shape keys. They are used to denote poses in the Action Editor mode and Shape Keys Editor of Dope Sheet.

In the supported editors, if at least one is created, markers are visualized in a separate row at their bottom. This area can be disabled per editor via the View \xe2\x80\xa3 Show Markers menu option.

Note

While the markers area is disabled, markers operators are not available in that editor, and in the header the Markers menu is hidden.

Regular markers are shown as small white triangles, empty if unselected or filled if selected, and with a dashed line that covers the editor height at the corresponding frame. If they have a name, this is shown to their right in white.

The 3D Viewport does not allow you to create, edit or remove markers, but it shows their name in the Object Info in the upper left corner, when on their frame.

Pose markers show a diamond-shaped icon in the Dope Sheet. In the NLA editor pose markers are shown as a red dashed line inside the relative action strip.

Reference

All modes

Marker \xe2\x80\xa3 Add Marker

M

The simplest way to add a marker is to move to the frame where you would like it to appear, and press M.

Hint

Markers can also be added during playback.

If Show Pose Markers is checked, a pose marker and a new pose in the Old Pose Library are added.

Reference

All modes

LMB

Click LMB on the marker's triangle to select it. Use Shift- LMB to select multiple markers.

In the Graph Editor, Dope Sheet, NLA Editor, Timeline, and Video Sequence Editor, you can also select all markers while hovering the mouse over the marker row, and apply selection tools on them like Box Select, etc. (as usual, LMB to select, RMB to deselect). The corresponding options are found in the Select menu of these editors.

Reference

All modes

Marker ▸ Duplicate Marker

Shift- D

You can duplicate the selected markers by pressing Shift- D. Once duplicated, the new ones are automatically placed in select mode, so you can move them to the desired location.

Note

Note that unlike most other duplications in Blender, the names of the duplicated markers are not altered at all (no .001 numeric counter append).

Reference

All modes

Marker ▸ Duplicate Marker to Scene ▸

Duplicates the selected markers into another scene.

Reference

All modes

Marker ▸ Delete Marker

X

To delete the selected markers simply press X, and confirm the pop-up message with LMB.

Reference

All modes

Marker ▸ Rename Marker

F2

Having dozens of markers scattered throughout your scene's time will not help you much unless you know what they stand for. You can name a marker by selecting it, pressing F2, typing the name, and press Return

Reference

All modes

Marker ▸ Move Marker

G

Once you have one or more markers selected, press G, while hovering with the mouse over the marker bar, to move them, and confirm the move with LMB or Return (as usual, cancel the move with RMB, or Esc). Or drag them with the LMB.

By default, you move the markers in one-frame steps, but if you hold Ctrl, the markers will move in steps corresponding to 1 second (according to the scene's FPS).

Reference

All modes

Marker `Select`

Convenient operators for selecting Marks; see [Selecting for more information on selecting Markers](#).

Selects all Markers.

Deselects any already selected Markers.

Select all unselected Markers and deselects all selected Markers.

Selects all Markers to the left of the current frame and the Marker on the current frame if it exists.

Selects all Markers to the Right of the current frame and the Marker on the current frame if it exists.

Reference

Dope Sheet

Action Editor or Shape Keys Editor mode

Marker `Show Pose Markers`

Shows markers belonging to the active action instead of scene markers.

Reference

All modes

Marker `Make Markers Local`

It is possible to convert standard markers into pose markers with Marker `Make Markers Local`. Note that the original marker will be gone. If you want to keep it, make a duplicate before you convert.

Reference

All modes

Marker `Jump to Next/Previous Marker`

Moves the Playhead to the next/previous marker relative to the current frame.

Reference

Timeline

Marker `Bind Camera to Markers`

Ctrl- B

Bind Camera to Markers is a special operator only available in the Timeline. The operator allows markers to be used to set the active object as the active camera.

To use this operator, select the object to become the active camera and select a marker to bind the active camera to. If no marker is selected when the operator is applied, a marker will be added. When an object is bound to a marker, the marker will be renamed to the name of the active object. These markers also have a camera icon next to the left of the name to easily distinguish them from other informative markers.

These markers can be moved to change the frame at which the active camera is changed to the object the marker is bound to.

--- Fin du chapitre: `_markers` ---

--- Début du chapitre: `_markup_guide` --- This page covers the conventions for writing and use of the reStructuredText (RST) markup syntax.

Three space indentation.

Lines should be less than 120 characters long.

Use italics for button/menu names.

Other loose conventions:

Avoid Unicode characters.

Avoid heavily wrapped text (i.e. sentences can have their own lines).

Note

Partss should only be used for contents or index pages.

Note

Each.rstfile should only have one chapter heading (*) per file.

See the overview on ReStructuredText for more information on how to style the various elements of the documentation and on how to add lists, tables, pictures and code blocks. The Sphinx reference provides more insight into additional constructs.

The following are useful markups for text styling:

:kbd:`LMB` \xe2\x80\x93 keyboard and mouse shortcuts.

Mirror \xe2\x80\x93 interface labels.

:menuselection:`3DViewport-->Add-->Mesh-->Monkey` \xe2\x80\x93 menus.

There is support for syntax highlighting if the programming language is provided, and line numbers can be optionally shown with the :linenos: option:

Figures should be used to place images:

For consistency, and since it would be good to ensure screenshots are all a similar size when floated next to text, writers should take screenshots in the following manner:

Prepare the area you would like to capture making sure to use the default theme and setting. (In some cases you may not want to use the default settings e.g. if some options are hidden behind a checkbox.)

Zoom to the maximum zoom level (hold Numpad Plus or Ctrl- MMB or similar).

Zoom out eight zoom levels (Numpad Minus \xe2\x80\x93 eight times).

In some cases you will want to leave a small margin around the thing you are trying to capture. This should be around 30px but does not have to be exact.

This can be applied to several parts of the interface but might not work for all cases.

Lower case filenames underscore between words.

Order naming with specific identifiers at the end.

Use.png for images that have solid colors such as screenshots of the Blender interface, and.jpg for images with a high amount of color variance, such as sample renders and photographs.

Do not use animated.gif files, these are hard to maintain, can be distracting and are usually large in file size. Instead use a video if needed (see Videos below).

Place the image in the manual/images folder. Use no other subfolders.

For naming files use underscores to separate chapters and sections, and use dashes to separate sections that are two or more words. So for image files should look like: chapter_subsection_sub-subsection_id.png, e.g:

interface_splash_current.png

interface_undo-redo_last.png

interface_undo-redo_repeat-history-menu.png

Do not use special characters or spaces!

Avoid specifying the resolution of the image, so that the theme can handle the images consistently and provide the best layout across different screen sizes.

When documenting a panel or section of the UI, it is better to use a single image that shows all of the relevant areas (rather than multiple images for each icon or button) placed at the top of the section you are

writing, and then explain the features in the order that they appear in the image.

Note

It is important that the manual can be maintained long term, UI and tool options change so try to avoid having a lot of images (when they are not especially necessary). Otherwise, this becomes too much of a maintenance burden.

Videos can be embedded from Blender's self-hosted PeerTube instance which can be found at video.blender.org. To embed a video using the following directive:

The ID is found in the video's URL, e.g:

The ID for <https://video.blender.org/videos/watch/47448bc1-0cc0-4bd1-b6c8-9115d8f7e08c> is 47448bc1-0cc0-4bd1-b6c8-9115d8f7e08c .

To get a new video uploaded, contact a Documentation Project Administrator or include the uploaded video in your Pull Request description.

Avoid adding videos that rely on voice or words, as this is difficult to translate.

Do not embed video tutorials as a means of explaining a feature, the writing itself should explain it adequately. (Though you may include a link to the video at the bottom of the page under the heading Tutorials).

`|BLENDER_VERSION|` \xe2\x80\x93 Resolves to the current Blender version.
:abbr:`SSAO` (ScreenSpaceAmbientOcclusion) \xe2\x80\x93 Abbreviations display the full text as a tooltip for the reader.

:term:`Manifold` \xe2\x80\x93 Links to an entry in the Glossary .

You can link to another document in the manual with:

To link to a specific section in another document (or the same one), explicit labels are available:

Linking to a title in the same file:

Linking to the outside world:

It is possible to link to a specific part of the manual from in Blender by opening the context menu (right click) of a property or operator and selecting Online Manual. In order for this to work, this needs to be accounted for in the documentation. To link a property or operator to a specific part of the manual you need to add an external reference link tag whose ID matches Blender's RNA tag. The easiest way to find out what the tag for a property is to open the context menu of the property/operator and select Online Python Reference to extract the tag from the URL. Some examples of how this looks in the RST document are given below:

For an operator:

Blender's icons can be included as inline text using:

To learn more about reStructuredText, see:

Good basic introduction.

Links to reference and user documentation.

--- Fin du chapitre: `_markup_guide` ---

--- Début du chapitre: `_mask` --- The Mask strip generates a mask image from the selected mask data-block generated in the Movie Clip Editor. This works similar to the Mask Node but without the options available for finer control. The mask image is always generated at the render resolution, scaling along with different proxy levels.

Data-block menu to select a mask.

--- Fin du chapitre: `_mask` ---

--- Début du chapitre: _mask_by_color ---Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Mask by Color

Click on any color on the mesh to create a new mask (based on the active color attribute).

How much changes in color affect the mask generation. A smaller threshold includes fewer similar colors. A larger threshold includes much more similar colors.

Mask only contiguous color areas. Colors that don't touch the one that you click on will not be masked.

Invert the generated mask.

Preserve previous mask and add or subtract the new one generated by the colors.

--- Fin du chapitre: _mask_by_color ---

--- Début du chapitre: _mask_display --- This popover controls how masks are displayed in Mask mode.

Toggles the display of the mask splines. Note that if they're hidden, you won't be able to edit them.

Line style of the splines.

Visualizes masks by shading the whole clip.

Displays just the masks as a grayscale image. Excluded areas are black, while included areas are white.

Displays the clip with excluded areas darkened.

How much excluded areas are darkened when using the \xe2\x80\x9cCombined\xe2\x80\x9d Overlay Mode.

--- Fin du chapitre: _mask_display ---

--- Début du chapitre: _mask_tools ---Reference

Sculpt Mode

Mask gesture tools apply a constant value to all selected vertices within the selection area. By default, these tools fully mask each vertex. Holding Ctrl while performing the selection clears the mask.

All mask gesture tools can be activated in the Toolbar and are comprised of the following:

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Box Mask

B

Creates a new Mask based on a box gesture .

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Lasso Mask

Creates a new Mask based on a lasso gesture .

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Line Mask

Creates a new Mask based on a line gesture .

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Polyline Mask

Creates a new Mask based on a polyline gesture .

Only creates a mask on the faces that face towards the view.

--- Fin du chapitre: `_mask_tools` ---

--- Début du chapitre: `_material` --- The Material input node outputs a single material. It can be connected to other material sockets to make using the same material name in multiple places more convenient.

Tip

The Material node can also be added by dragging and dropping a material data-block into the node editor. This will add the node and select the dropped material in the Data-Block Menu .

This node has no inputs.

Material

A reference to the selected material.

--- Fin du chapitre: `_material` ---

--- Début du chapitre: `_material_index` --- The Material Index node outputs which material in the list of materials of the geometry each element corresponds to. Currently the node supports mesh data, where `material_index` is a built-in attribute on faces.

The node to set this data is the Set Material Index node.

This node has no inputs.

This node has no properties.

Standard integer value, with a minimum value of zero.

--- Fin du chapitre: `_material_index` ---

--- Début du chapitre: `_material_selection` --- The Material Selection node provides a selection for meshes that use this material. Since the `material_index` is stored on each face, the output will be implicitly interpolated to a different domain when necessary. For example, every vertex connected to a selected face will be selected.

Standard material input.

This node has no properties.

Selection of faces that use the input material.

--- Fin du chapitre: `_material_selection` ---

--- Début du chapitre: `_material_settings` --- Reference

Material \xe2\x80\xa3 Settings and Shader Editor \xe2\x80\xa3 Sidebar
\xe2\x80\xa3 Settings

Method used to perform Displacement on materials.

Mesh vertices will be displaced before rendering, modifying the actual mesh. This gives the best quality results, if the mesh is finely subdivided. As a result, this method is also the most memory intensive.

When executing the surface shader, a modified surface normal is used instead of the true normal. This is a less memory intensive alternative to actual displacement, but only an approximation. Surface silhouettes will not be accurate and there will be no self-shadowing of the displacement. Both methods can be combined, to do displacement on a coarser mesh, and use bump mapping for the final detail.

The method used for sampling the emissive component of the material. This option will only have an influence if the material contains an emissive material node, otherwise it will be ignored.

Do not use this surface as a light for sampling.

Automatically determine if the surface should be treated as a light for sampling based on emission intensity.

Treat only the front side of the surface as a light, useful for closed meshes whose interior is not visible.

Treat only the back side of the surface as a light for sampling.

Treat surface as a light for sampling, emitting from both the front and back side.

Use transparent shadows if it contains a Transparent BSDF, disabling will render faster but will not give accurate shadows.

Applies corrections to solve shadow terminator artifacts caused by bump mapping.

For dense volumes lit from far away Distance sampling is usually more efficient.

If you have got a light inside or near the volume then equiangular sampling is better.

If you have a combination of both, then the multiple importance sampling will be better.

Interpolation method to use for the volume objects and smoke simulation grids.

Simple interpolation which gives good results for thin volumes.

Smoothed high-quality interpolation needed for more dense volumes, but slower.

Assume volume has the same density everywhere (not using any textures), for faster rendering. For example absorption in a glass object would typically not have any textures, and so the renderer can be set to avoid taking small steps to sample the volume shader. Usually this is automatically determined by the renderer. This setting provides a manual control for cases where it is not detected.

Adjust distance between volume shader samples for volume shaders. This is typically used to reduce the step size for procedural shaders that add more detail with procedural textures, when it is not captured by the default step size. See Volume Render Settings for more information.

--- Fin du chapitre: `_material_settings` ---

--- Début du chapitre: `_math` --- The Math Node performs math operations. The inputs of the node are dynamic. Some inputs are only available for certain operations. For instance, the Add input is only available for the Multiply Add operator.

Input Value. Trigonometric functions read this value as radians.

Input Addend.

Input Base.

Input Exponent.

Input Epsilon.

Input Distance.

Input Minimum.

Input Maximum.

Input Increment.

Input Scale.

Input Degrees.

Input Radians.

The mathematical operator to be applied to the input values:

The sum of the two values.

The difference between the two values.

The product of the two values.

The division of the first value by the second value.

The sum of the product of the two values with Addend.

The Base raised to the power of Exponent.
 The log of the value with a Base as its base.
 The square root of the value.
 One divided by the square root of the value.
 The input value is read without regard to its sign. This turns negative values into positive values.
 Raises Euler's number to the power of the value.
 Outputs the smallest of the input values.
 Outputs the largest of two input values.
 Outputs 1.0 if the first value is smaller than the second value. Otherwise the output is 0.0.
 Outputs 1.0 if the first value is larger than the second value. Otherwise the output is 0.0.
 Extracts the sign of the input value. All positive numbers will output 1.0. All negative numbers will output -1.0. And 0.0 will output 0.0.
 Outputs 1.0 if the difference between the two input values is less than or equal to Epsilon.
 Smooth Minimum.
 Smooth Maximum.
 Rounds the input value to the nearest integer.
 Rounds the input value down to the nearest integer.
 Rounds the input value up to the nearest integer.
 Outputs the integer part of the value.
 Returns the fractional part of the value.
 Outputs the remainder once the first value is divided by the second value.
 Returns the positive remainder of a division operation.
 Outputs a value between Min and Max based on the absolute difference between the input value and the nearest integer multiple of Max less than the value.
 Rounds the input value down to the nearest integer multiple of Increment.
 Bounces back and forth between 0.0 and the Scale as the input value increases.
 The Sine of the input value.
 The Cosine of the input value.
 The Tangent of the input value.
 The Arcsine of the input value.
 The Arccosine of the input value.
 The Arctangent of the input value.
 Outputs the Inverse Tangent of the first value divided by the second value measured in radians.
 The Hyperbolic Sine of the input value.
 The Hyperbolic Cosine of the input value.
 The Hyperbolic Tangent of the input value.
 Converts the input from degrees to radians.
 Converts the input from radians to degrees.
 Limits the output to the range (0.0 to 1.0). See Clamp .
 Numerical value output.
 Minimum and maximum function example.
 The top Render Layer's node has a cube that is about 10 units from the camera. The bottom Render Layer's node has a plane that covers the left half of the view and is 7 units from the camera. Both are fed through their respective Map Value nodes to multiply the depth value by 0.05 and clamp it

to [0.0, 1.0], bringing it into a suitable range for displaying it as a color.

The Minimum node selects the smallest of the two depth values at each pixel. In the left half, it chooses the plane (because it's closer than the cube), and in the right half, it chooses the cube (because it's closer than the background, which is infinitely far away).

The Maximum node selects the largest of the two depth values at each pixel. In the left half, it chooses the cube (because it's farther away than the plane), and in the right half, it chooses the background (because it's farther away than the cube).

Using sine function example.

This example has a Timenode putting out a linear sequence from 0 to 1 over the course of 101 frames. At frame 25, the output value is 0.25. That value is multiplied by 2π (6.28) and converted to 1.0 by the Sine function, since $\sin(2\pi \cdot 0.25) = \sin(\pi/2) = +1.0$.

Since the sine function can output values between (-1.0 to 1.0), the Map Valuenode scales that to 0.0 to 1.0 by taking the input (-1 to 1), adding 1 (making 0 to 2), and multiplying the result by 0.5 (thus scaling the output between 0 to 1). The default Color Ramp converts those values to a gray-scale. Thus, medium gray corresponds to a 0.0 output by the sine, black to -1.0, and white to 1.0. As you can see, $\sin(\pi/2) = 1.0$. Like having your own visual color calculator! Animating this node setup provides a smooth cyclic sequence through the range of grays.

Use this function to vary, for example, the alpha channel of an image to produce a fading in/out effect. Alter the Z channel to move a scene in/out of focus. Alter a color channel value to make a color pulse.

Scaling a channel example.

This example has a Math (Multiply) node increasing the luminance channel (Y) of the image to make it brighter. Note that you should use a Map Valuenode with min() and max() enabled to clamp the output to valid values. With this approach, you could use a logarithmic function to make a high dynamic range image. For this particular example, there is also a Brightness/Contrast node that might give simpler control over brightness.

Posterization example.

In this example, we restrict the color values to be one of the six values: 0, 0.2, 0.4, 0.6, 0.8, 1.

To split up a continuous range of values between 0 and 1 to certain set of values, the following function is used: $\text{round}(x \cdot n - 0.5) / (n - 1)$, where n is the number of possible output values, and x is the input pixel color. Read more about this function.

To implement this function in Blender, consider the node setup above. We string the Math nodes into a function that takes each color (values from 0 to 1), multiplies it up by six, the desired number of divisions (values become from 0 to 6), offsets it by 0.5 (-0.5 to 5.5), rounds the value to the nearest whole number (produces 0, 1, 2, 3, 4, 5), and then divides the image pixel color by five (0.0, 0.2, 0.4, 0.6, 0.8, 1.0).

In the case of a color image, you need to split it into separate RGB channels using Separate/Combine Color nodes and perform this operation on each channel independently.

--- Fin du chapitre: _math ---

--- Début du chapitre: `_matrix_determinant` --- The Matrix Determinant node computes the determinant of the passed in matrix. The matrix to compute the determinant of. This node has no properties. The compute determinant value.
--- Fin du chapitre: `_matrix_determinant` ---

--- Début du chapitre: `_measure` ---Reference
All Modes

Toolbar \xe2\x80\xa3 Measure

The Measure tool is an interactive tool where you can drag lines in the scene to measure distances or angles. Snapping to geometry could be activated for better accuracy or to measure wall thickness. The Measure tool can be accessed from the Toolbar.

Examples of the Measure tool. \xc2\xb6

Here are some common steps for using the Measure tool:

Activate the Measure tool from the Toolbar.

Click and drag in the viewport to define the initial start and end point for the ruler. You can add multiple rulers in the viewport.

Drag either end of the ruler to move it.

Holding Ctrl while moving enables snapping to edges and vertices.

Holding Shift while moving lets you measure the distance between faces. This only works well with parallel faces, e.g. walls.

You can always navigate (pan, zoom, \xe2\x80\xa6) or change the view (orthogonal, perspective) in the viewport to have better access to the ruler.

Click on the midpoint of a created ruler to convert it to a protractor. The midpoint can then be dragged just like the endpoints.

A selected ruler can be deleted with Delete or X. To delete all measurements, delete the \xe2\x80\x9cRulerData3D\xe2\x80\x9d layer in the Sidebar \xe2\x80\xa3 View \xe2\x80\xa3 Annotations panel (see image above).

All measurements are hidden when another tool is selected. They are shown when the Measure tool is selected again. However, you can do editing operations while the ruler is active. For example, you can edit the rotation or scale of the selected object in the Sidebar.

Measurements do not appear in the Render output.

Unit settings and scale from the scene are used for displaying dimensions. Changing the unit system (metric, imperial), or the units of length (cm, m, \xe2\x80\xa6) or angle (degrees, radians) will update the measurements.

Tip

In Edit Mode only, there is also a Measurement group in the Viewport Overlay popover. Using the settings in this group, you can have the viewport automatically display measurements for selected edges and faces, without the need to manually create a ruler.

--- Fin du chapitre: `_measure` ---

--- Début du chapitre: `_median_point` ---Reference

Object Mode and Edit Mode

Pivot Point \xe2\x80\xa3 Median Point

Period

Places the pivot point at the averaged-out position of the selected items.

In Object Mode, the Median Point is the averaged-out position of the origins of the selected objects. The shape and size of the objects is not taken into account.

Origins can be chosen freely and can even lie outside their object's geometry, so that the Median Point is not always what you might expect.

Median points in Object Mode. \xc2\xba

In Edit Mode, the Median Point is the averaged-out position of the selected vertices. This means that the pivot point will shift towards the area with the densest geometry.

In the example below, the pivot point lies perfectly in the middle if both cubes have the same number of vertices, but heavily leans towards the side if one cube is subdivided even though both cubes still have the same size.

Median points in Edit Mode. \xc2\xba

--- Fin du chapitre: _median_point ---

--- Début du chapitre: _menus --- Blender uses a variety of different menus for accessing options and Operators. Menus can be interacted with in the following ways:

LMB on the desired item.

You can use the number keys or numpad to input an item in the list to select. For example, Numpad 1 will select the first item and so on.

If a menu is too large to fit on the screen, a small scrolling triangle appears on the top or bottom. Scrolling is done by moving the mouse above or below this triangle.

Shortcuts

Use Wheel while hovering with the mouse.

Arrow keys can be used to navigate.

Each menu item has an underlined character which can be pressed to activate it.

Number keys or numpad can also be used to activate menu items. (1 for the first menu item, 2 for the second etc. For larger menus, Alt-1 activates the 11th and so on, up to Alt-0 for the 20th.)

Press Return to activate the selected menu item.

Press Esc to close the menu without activating any menu item. This can also be done by moving the mouse cursor far away from the menu, or by LMB clicking anywhere outside of it.

Image menu in the Header of the Image editor. \xc2\xba

Popup menus list Operators which can be executed by selecting with LMB or using the generated shortcut indicated by the underlined character of the operator name. All menu entries show any relevant shortcut keys, which can be executed without opening the menu.

All popup menus can be searched by pressing Spacebar and typing the name of the operator in the menu. If a popup menu has

\xe2\x80\x9cSearch\xe2\x80\x9d as one of the items, the menu can be searched without having to press Spacebar first.

All popup menus of an editor can be searched using the Menu Search feature.

Sometimes it's helpful to gain some extra horizontal space in the header by collapsing menus. This can be accessed from the header context menu: click RMB on the header and uncheck the Show Menus checkbox.

Right-click on any of the header menus. \xc2\xba

Access the menu from the collapsed icon. \xc2\xba

The 3D Viewport Mode Select menu. \xc2\xbf

A Select menu (or \xe2\x80\x9cselector\xe2\x80\x9d for short) lets you choose between a set of options. It appears as an icon and/or text with a down arrow on the right side. To use it, click the button with LMB to show the available options, then click the desired option (once selected, it\xe2\x80\x99ll appear inside the button). You can also use Ctrl- Wheel to cycle through the options without opening the menu.

The Transform Orientations popover menu. \xc2\xbf

Popover menus are similar to Select Menus, but can show more varied contents such as a title, buttons, sliders, etc.

Context menus are pop-ups that can be opened with RMB. In most editors, it\xe2\x80\x99s also possible to use the Menu key. The contents of the menu depend on the location of the mouse pointer.

When invoked in an editor, the menu contains a list of operators sensitive to the editor\xe2\x80\x99s mode. When invoked over buttons and properties, common options include:

Apply the change to a single value of a set (e.g. only the X coordinate of an object\xe2\x80\x99s Location).

Apply the change to all values in a set (e.g. all coordinates of an object\xe2\x80\x99s Location).

Replaces the current value by the default.

Copies the Python property data path, relative to the data-block. Useful for Python scripting.

Copies the full Python property data path including any needed context information.

Creates a new driver using this property as input, and copies it to the clipboard. Use Paste Driver to add the driver to a different property, or Paste Driver Variable to extend an existing driver with a new input variable.

Copies the property value to the selected object\xe2\x80\x99s corresponding property. A use case is if the Properties context is pinned. Lets you define a keyboard or mouse shortcut for an operation. To define the shortcut you must first move the mouse cursor over the button that pops up. When \xe2\x80\x9cPress a key\xe2\x80\x9d appears you must press and/or click the desired shortcut. Press Esc to cancel.

See also

Common Shortcuts.

Lets you redefine the shortcut.

Unlinks the existing shortcut.

Opens the containing folder using the operating system\xe2\x80\x99s file manager.

Opens an online page of the Blender Manual in a web browser.

Context-sensitive access to the Python API Reference.

For UI development \xe2\x80\x93 Creates a text data-block with the source code associated with the control, in case the control is based on a Python script. In the Text Editor it points at the code line where the element is defined.

For UI development \xe2\x80\x93 Points at the translation code line.

The Specials pop-up menu is similar to a context menu, but is opened using a button consisting of a down arrow on a dark background.

A pie menu is a menu whose items are spread radially around the mouse.

The 3D Viewport Mode Pie menu. \xc2\xbf

Tip

The fastest way to operate a Pie menu is to press down the key(s) that invoke the menu, move the mouse slightly towards a selection, and release the key(s) to activate the selection. Releasing the key without moving the mouse will keep the menu open so you can click the desired item. If you do move the mouse before releasing, the item closest to the mouse will be activated instantly. An open disc widget at the center of the pie menu shows the current direction of the pie menu. The selected item is also highlighted. A pie menu will only have a valid direction for item selection if the mouse is touching or extending beyond the disc widget at the center of the menu. Pie menu items support key accelerators, which are the letters underlined on each menu item. Number keys can also be used. If there are sub-pies available, it is indicated by a plus icon.

See also

Pie menu settings.

--- Fin du chapitre: _menus ---

--- Début du chapitre: _menu_switch --- The Menu Switch node outputs one of its inputs depending on a menu selection. Only the input that is passed through the node is computed.

The available menu entries are defined by the user. Menu items can be added and removed, as well as renamed and reordered in the editor sidebar. Renaming a menu entry keeps existing links of the matching input socket.

The menu can be used in node groups and the nodes modifier UI. Connecting the menu input with a Group Input node will expose the menu as a group input. A menu socket in a node group, reroute node, or other pass-through nodes needs to be connected to a Menu Switch node in order to work. An unconnected menu socket will show an empty menu by default.

Connecting multiple Menu Switch nodes to the same output socket creates a conflict (even when the menu entries are the same). To avoid this a menu switch can be wrapped in a node group. Multiple node groups of the same type can be connected to the same menu, since they contain the same menu switch node.

Conflict caused by connecting different menus. \xc2\xba6

Same node group can be connected without conflict. \xc2\xba6

See also

The Index Switch Node is similar but it exposes the choices as an integer index.

Determines which of the input options below will be passed through.

One input is created for every menu entry. The input is used when the matching option is selected. Items can be renamed by Ctrl-LMB on the socket name or in the nodes Properties panel.

Determines the type of the data that is handled by the node.

One of the inputs without any modifications.

--- Fin du chapitre: _menu_switch ---

--- Début du chapitre: _merge --- Reference

Edit Mode

Mesh \xe2\x80\xa3 Merge, Context Menu \xe2\x80\xa3 Merge

M

This tool allows you to merge all selected vertices to a unique one, dissolving all others. You can choose the location of the remaining vertex in the menu this tool pops up before executing:

It will place the remaining vertex at the center of the selection. Available in all select modes.

It will place the remaining vertex at the 3D Cursor. Available in all select modes.

Every island of selected vertices (connected by selected edges) will merge on its own median center, leaving one vertex per island.

It will place the remaining vertex at the location of the first one selected. Only available in Vertexselect mode.

It will place the remaining vertex at the location of the last one selected (the active one). Only available in Vertexselect mode.

Merging vertices of course also deletes some edges and faces. But Blender will do everything it can to preserve edges and faces only partly involved in the reunion.

Note

At First and At Last depend on that the selection order is saved: the order is lost, for instance, after changing selection mode.

If UVs is ticked in the Adjust Last Operation panel, the UV mapping coordinates, if existing, will be corrected to avoid image distortion. Todo.

--- Fin du chapitre: `_merge` ---

--- Début du chapitre: `_merge_by_distance` --- The Merge by Distance node merges selected mesh vertices or point cloud points within a given distance, merging surrounding geometry where necessary. This operation is similar to the Merge by Distance operator or the Weld Modifier.

Standard geometry input.

Boolean field that is true for parts of the geometry to be deleted. Unselected points will be completely unused for the operation \xe2\x80\x93 they will not be merged into other points, and no points will merge into them either.

Tip

When possible, using the selection input can be a simple way to speed up the node, since searching for nearby points is a relatively expensive operation that gets even more expensive when more points are involved. The distance to use for searching for nearby points.

Method for choosing which vertices are merged.

Merge includes all geometry including loose parts.

Merge only includes attached geometry i.e. the modifier will not merge loose parts together.

Standard geometry output.

Using the selection input to only merge some of the points in a point cloud. \xc2\x86

--- Fin du chapitre: `_merge_by_distance` ---

--- Début du chapitre: `_merge_layers` --- Combines multiple Grease Pencil Layers into a single layer.

See also

Merge Layers Operator

The grease pencil object with multiple layers to combine.

Operate of a subset of layers by setting the field layer index value to true. By default, all layers are selected.

The index numbers of the layer to be merged. This input is only visible when using By Group ID Mode.

Determines how to choose which layers are merged.

Combine all layers which have the same name.

Provide a custom group ID for each layer and all layers with the same ID will be merged into one.

The grease pencil object with combined layers.

--- Fin du chapitre: `_merge_layers` ---

--- Début du chapitre: `_mesh` --- The liquid mesh is, besides the liquid particles, another way to visualize the liquid simulation. It is generated directly from the liquid particles and often uses a higher resolution than the base Resolution Divisions.

Besides enabling parts of the interface, checking Meshlets the cache knows which simulation data to read. If, for example, Mesh is enabled but there is no mesh simulation data to read it will show the original domain. The checkbox does not reset the cache and can be used to switch the view between the original domain and the baked liquid mesh.

It is important to keep in mind that the shape of the mesh depends on a combination of all these parameters. E.g. changing the Particle Radius will lead to a different interpretation of the concavity values.

Factor by which to enhance the resolution of the mesh. The scaling factor is coupled to the Resolution Divisions (i.e. the mesh is this times bigger than the base simulation).

The radius of one liquid particle in grid cells units. This value describes how much area is covered by a particle and thus determines how much area around it can be considered as liquid. A greater radius will let particles cover more area. This will result in meshes covering more volume around liquid particles.

This property refers to the same Particle Radius described in the liquid domain settings. Yet for the mesh, it is useful to interpret the particle radius on its own. For one, the mesh can have a resolution different from the base resolution through the Upres Factor. For another, it is often desirable to be able to control the mesh size around a single liquid particle.

Creates a velocity Attribute which records the velocity of each vertex per frame. These will be used (automatically) when rendering with motion blur enabled.

Note

In order to render motion blur with Cycles, Deformation Motion Blur must be enabled.

Motion blur enabled. \xc2\xbf

Motion blur disabled. \xc2\xbf

The mesh generator method determines the accuracy of the mesh. The Final option produces a higher quality mesh and provides more configuration option than the Preview option which in turn is faster but not as smooth.

Positive mesh smoothing iterations. Higher values will make the mesh outline increasingly smooth. Yet higher values can prevent small details (e.g. smaller liquid drops) from getting meshed.

Negative mesh smoothing iterations. Higher values will make the mesh outline sharper. High values will preserve details, however, the mesh outline will become more ragged (e.g. a single mesh particle will become less rounded and have more flat sides).

Comparison of a liquid drop hitting a surface (viewed from top) with varying smoothing values. Left: 1, 1 (Smoothing Positive, Smoothing Negative). Middle: 10, 1. Right: 1, 10. Note the slightly sharper corners in the right splash (compared to the left one). \xc2\xbf

Upper mesh concavity bound. High values tend to smoothen and fill out concave regions.

Lower mesh concavity bound. High values tend to smoothen and fill out concave regions.

Using a lower concavity which is greater the upper concavity can result in distorted, non-manifold meshes. Unless the artist sees value in this kind of mesh, such concavity value combinations should be avoided.

Upper: 1.0, Lower: 0.0. \xc2\xba

Upper: 1.0, Lower: 0.5. \xc2\xba

Upper: 1.0, Lower: 1.0. \xc2\xba

Upper: 1.5, Lower: 0.0. \xc2\xba

Upper: 1.5, Lower: 0.5. \xc2\xba

Upper: 1.5, Lower: 1.0. \xc2\xba

Upper: 2.0, Lower: 0.0. \xc2\xba

Upper: 2.0, Lower: 0.5. \xc2\xba

Upper: 2.0, Lower: 1.0. \xc2\xba

This option is only available when using the Modular cache type.

The progress will be displayed in the status bar. Pressing Esc will abort the simulation.

Once the simulation has been baked, the cache can be deleted by pressing Free Mesh. It is possible to pause or resume a Bake Mesh process.

--- Fin du chapitre: _mesh ---

--- Début du chapitre: _mesh_analysis ---Reference

Edit Mode

Header \xe2\x80\xa3 Overlays \xe2\x80\xa3 Mesh Analysis

Mesh analysis is useful for displaying attributes of the mesh, that may impact certain use cases.

The mesh analysis works in Edit Mode and Solid Viewport shading. It shows areas with a high value in red, and areas with a low value in blue. Geometry outside the range is displayed gray.

Currently the different modes target 3D printing as their primary use. Extrusion 3D printers have a physical limit to the overhang that can be printed, this display mode shows the overhang with angle range and axis selection.

Minimum/Maximum angle to display.

Axis and direction to use as the bases to calculate the angle to visualize.

Overhang. \xc2\xba

Printers have a limited wall-thickness where very thin areas cannot be printed, this test uses ray casting and a distance range to the thickness of the geometry.

Minimum/Maximum thickness to display.

Number of samples to use to calculate the thickness.

Thickness. \xc2\xba

Another common cause of problems for printing are intersections between surfaces, where the inside/outside of a model cannot be reliably detected. Unlike other display modes, intersections have no variance and are either on or off.

Intersecting faces. \xc2\xba

Distorted geometry can cause problems since the triangulation of a distorted n-gon is undefined.

Distortion is measured by faces which are not flat, with parts of the face pointing in different directions.

Minimum/Maximum distortion to display.

Distorted Faces. \xc2\xba

Similar to wall-thickness, sharp edges can form shapes that are too thin to be able to print.

Minimum/Maximum angle to display.

Sharp edges. \xc2\xba

There are some known limitations with mesh analysis:

Currently only displayed with Deform Modifiers.

For high-poly meshes the performance is low while editing.

--- Fin du chapitre: _mesh_analysis ---

--- Début du chapitre: _mesh_boolean --- The Mesh Boolean Node allows you to cut, subtract, and join the geometry of two inputs. This node offers the same operations as the Boolean modifier.

Standard geometry input.

Correctly calculates cases when one or both operands have self-intersections. This involves more calculations making the node slower. Optimizes the Boolean output for Non-manifold geometry at the cost of increased computational time. Because of the performance impact, this option should only be enabled when the solver demonstrates errors with non-manifold geometry.

Produce a new geometry containing only the volume inside of both geometry 1 and geometry 2.

The two input meshes are joined, then any interior elements are removed. Geometry 2 is subtracted from geometry 1 (everything outside of geometry 2 is kept).

Algorithm used to calculate the Boolean intersections.

Uses a mathematically simple solver which offers the best

performance; however, this solver lacks support for overlapping geometry.

Uses a mathematically complex solver which offers the best results and has full support for overlapping geometry; however, this solver is much slower than the Float Solver.

Standard geometry output.

A boolean attribute field with a selection of the edges that were created where the two inputs meet.

--- Fin du chapitre: _mesh_boolean ---

--- Début du chapitre: _mesh_cache --- The Mesh Cache modifier main use is for animated mesh data to be applied to a mesh and played back, deforming the mesh.

This works in a similar way to shape keys, but is aimed at playing back external files and is often used for interchange between applications.

Tip

Both MDD and PC2 depend on the vertex order on the mesh remaining unchanged. This is a limitation of this method, so take care not to add, remove or reorder vertices once this modifier is used.

Mesh Cache Modifier. \xc2\xba

The input file format (currently .mdd and .pc2 are supported).

Path to the cache file.

Factor to adjust the influence of the modifier's deformation, useful for blending in/out from the cache data.

This setting defaults to Overwrite which will replace the vertex locations with those in the cache file. However, you may want to use shape keys, for

example, and mix them with the mesh cache. In this case you can select the Deform option which integrates deformations with the mesh cache result.

Note

This feature is limited to making smaller, isolated edits and will not work for larger changes such as re-posing limbs.

None, or Linear, which will blend between frames. Use linear when the frames in the cache file do not match up exactly with the frames in the blend-file.

If set, restrict the effect to the only vertices in that vertex group. Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group.

Select how time is calculated.

Allows you to control the frames, which will ignore timing data in the file but is often useful since it gives simple control.

Evaluates time in seconds, taking into account timing information from the file (offset and frame-times).

Evaluates the entire animation as a value in the [0, 1] range.

Select how playback operates.

Use the current frame from the scene to control playback.

Play the cache starting from this frame.

Scale time by this factor (applied after the start value).

Control animation timing manually.

Property used for animation time, this gives more control of timing (typically this value will be animated).

The axis for forward and up used in the source file.

In rare cases you may also need to flip the coordinates on an axis.

--- Fin du chapitre: _mesh_cache ---

--- Début du chapitre: _mesh_circle --- The Mesh Circle node generates a circular ring of edges that is optionally filled with faces. Number of vertices on the circle. No geometry is generated when the number is below three.

Distance of the vertices from the origin.

How the circle is filled with faces.

Output just the edge ring without any faces.

Fill the circle with a single face.

Fill the circle with triangles connected to a new vertex at the origin.

Standard geometry output.

--- Fin du chapitre: _mesh_circle ---

--- Début du chapitre: _mesh_deform --- The Mesh Deform modifier allows an arbitrary mesh (of any closed shape) to act as a deformation cage around another mesh.

Note

This modifier is reasonably easy to use, but it can be very slow to compute the binding (the mapping between the deform mesh cage to the deformed object geometry).

The Mesh Deform modifier. \xc2\xba6

The name of the mesh object to be used as the deforming cage.

An optional vertex group of the object \xe2\x80\x99s mesh to restrict the vertices that will be affected by this modifier. Vertices not in this group will not be deformed.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group. Controls the accuracy with which the deform mesh cage alters the deformed object, when the points on the cage are moved. Raising this value higher can greatly increase the time it takes to complete the binding calculations, but it will get more accurate cage mapping to the deformed object. This setting becomes unavailable once a cage has been bound. When activated, other mesh altering features (such as other modifiers and shape keys) are taken into account when binding, increasing deformation quality. The setting is deactivated by default to save memory and processing time when binding. Like with Precision, this setting is unavailable once a cage has been bound. Links the current vertex positions of both the modified geometry and the deforming Object chosen together. An unbound Mesh Deform modifier has no effect, it must be bound so that altering the shape of the deform mesh cage actually alters the shape of the modified object.

Warning

Depending on the settings of the modifier and complexity of the deform mesh cage and/or deformed object, it can take a long time for this operation to complete. This can result in Blender not responding to user's actions until it has completed.

It is also possible that Blender will run out of memory and crash.

To be safe, save your blend-file before proceeding!

When a deformed object has been associated to a deform mesh cage, it can later be disassociated by clicking the Unbind button which replaced the Bind one.

When Unbind is clicked, the deforming mesh cage will keep its current shape, it will not reset itself back to its initial shape. If you need this original shape, you will have to save a copy of it before you alter it. The deformed object will, however, reset back to its original shape that it had before it was bound to the deform mesh cage.

Warning

Significant changes to the entire cage mesh (such as rotating the cage upside down) can cause noticeable artifacts.

These can be reduced by binding with a higher Precision, however, it is a known limitation with this modifier and cannot be avoided entirely.

Ensure that the normals on the cage mesh point to the outside (they are used to determine the inside and outside of the cage).

Besides the outer cage, more faces within the cage, either loose or forming another smaller cage, can be used for extra control. Such smaller cages may also overlap with the main cage. For example, to get extra control over eyes, two small sphere cages could be added around them.

See also

The Lattice modifier.

Original paper

--- Fin du chapitre: `_mesh_deform` ---

--- Début du chapitre: `_mesh_filter` ---Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Mesh Filter

Applies a deformation to all vertices in the mesh at the same time.Masking, auto-masking and visibility will be taken into account.

To use this tool, click and drag away from left to right or from right to left for a negative effect.

These are all of the available filter deformations.

Smooths the positions of the vertices to either polish surfaces or remove volume from larger shapes.Especially useful to fix most of the artifacts of the voxel remesher.This filter works similar to the Smooth brush.

Increases the size of the mesh.This filter works similar to the Scale Transform .

Displaces vertices uniformly along their normal.This filter works similar to the Inflate brush.

Morphs the mesh progressively into a sphere.This filter works similar to the To Sphere Transform .

Randomly moves vertices along the vertex normal.This filter works similar to the Randomize Transform .

Tries to create an even distribution of quads without deforming the volume of the mesh.This filter works the same as holding Shift with the Slide Relax brush.

This will remove the jagged lines visible after drawing or creating a face set.This filter works the same as holding Shift with the Draw Face Set brush.

Eliminates irregularities of the mesh by making the positions of the vertices more uniform while preserving the volume of the object.This filter works similar to the Surface deformation type of the Smooth brush.

How much of the original shape is preserved when smoothing.

How much the position of each individual vertex influences the final result.

Sharpens and smooths the mesh based on its curvature,resulting in pinching hard edges and polishing flat surfaces.Especially useful when sculpting hard surfaces and stylized models with creasing and flattening brushes.

How much smoothing is applied to polished surfaces.

Increases the high frequency surface details of the mesh by intensifying the difference between creases and valleys.

The number of times the smoothing operation is applied per brush step.Controls how much smooth the resulting shape is, ignoring high-frequency details.

Increases the high frequency surface details of the mesh by intensifying the difference between creases and valleys.This filter works similar to the inverted direction of the Smooth brush.

Deletes displacement information of the Multires Modifier ,resetting the mesh to a regular subdivision surface result.This can be used to reset parts of the sculpt or to fix reprojection artifacts after applying a Shrinkwrap Modifier .

Negative strokes will intensify the displacement details,this method works similar to Enhance Details and can give better results in some circumstances.

The amount of effect the filter has on the mesh.At certain object scales it can be useful to change this value.

Apply the deformation only on the selected axis.

Orientation of the axis to limit the filter displacement.

Use the local axis to limit the displacement.

Use the global axis to limit the displacement.

Use the view axis to limit the displacement.

--- Fin du chapitre: `_mesh_filter` ---

--- Début du chapitre: `_mesh_island` --- The Mesh Island node outputs information about separate connected regions, or islands of a mesh. Whenever two vertices are connected together by an edge, they are considered as part of the same island, and will have the same Island Index output.

This node's behavior is similar to the Select Linked operator in edit mode, or the Random per Island output of the Geometry shader node.

This node has no inputs.

This node has no properties.

The index of each vertex's island. Indices are decided based on the lowest vertex index contained in each island.

The total number of mesh islands. This is a single value, and does not vary per element.

--- Fin du chapitre: `_mesh_island` ---

--- Début du chapitre: `_mesh_line` --- The Mesh Line node generates vertices in a line and connects them with edges.

Number of vertices on the line.

Length of individual edges. The node tries to fit as many vertices as possible between the start and end point. The exact end point might not be hit. This is only available when the mode is set to End Points and the count mode is set to Resolution.

Position of the first vertex.

Controls the direction of the line and distance between the vertices. This is only available when the mode is set to Offset.

Position of the last vertex. This is only available when the mode is set to End Points.

Inputs to use to control the line.

Specify the offset from one vertex to the next.

Specify the start and end point of the line.

Determines how the number of vertices is chosen. This is only available when the mode is set to End Points.

Specify the total number of vertices.

Specify the distance between vertices.

Standard geometry output.

--- Fin du chapitre: `_mesh_line` ---

--- Début du chapitre: `_mesh_sequence_cache` --- The Mesh Sequence Cache modifier loads data from Alembic and USD files. It supports static meshes, but is mostly used to load animated meshes. Despite its name, this modifier also supports curves. It also handles file sequences, as well as meshes and curves with varying topology (like the result of fluid simulations).

When importing an Alembic or USD file, Mesh Sequence Cache modifiers are automatically added to time-varying meshes. For time-varying object transforms (so animation of rotation, location, or scale), the Transform Cache Constraint is used. Files other than Alembic or USD, like MDD and PC2 files, can be loaded using the Mesh Cache modifier.

Data-block menu to select the Alembic or USD file.

Path to Alembic or USD file.

The path to the Alembic or USD object inside the archive or stage.

Type of data to read for a mesh object, respectively: vertices, polygons, UV maps, and Color Attributes.

Vertices, Faces, UV, Color

Whether or not the cache is separated in a series of files.

Whether to use a custom frame for looking up data in the cache file, instead of using the current scene frame.

The Framevalue is the time to use for looking up the data in the cache file, or to determine which to use in a file sequence.

Subtracted from the current frame to use for looking up the data in the cache file, or to determine which file to use in a file sequence.

The name of the Alembic attribute used for generating motion blur data; by default, this is .velocities which is standard for most Alembic files.

Note

The Velocity Attribute option is currently for Alembic files only.

Defines how the velocity vectors are interpreted with regard to time.

The velocity unit was encoded in frames and does not need to be scaled by scene FPS.

The velocity unit was encoded in seconds and needs to be scaled by the scene FPS (1 / FPS).

Note

The Velocity Unit option is currently for Alembic files only.

Multiplier used to control the magnitude of the velocity vector for time effects such as motion blur.

Note

The Velocity Scale option is currently for Alembic files only.

--- Fin du chapitre: _mesh_sequence_cache ---

--- Début du chapitre: _mesh_to_curve --- The Mesh to Curve node turns each string of connected mesh edges into a poly spline. Whenever two or more strings cross each other, the splines will be split.

Loose vertices are ignored \xe2\x80\x93 they will not be turned into single-point splines.

Attributes, both named and unnamed ones, are transferred to the resulting splines. If there is a radius attribute, it will be applied as such, although you may find it more convenient to use the Set Curve Radius Node for this.

Standard mesh input.

A field input evaluated on the edge domain to determine whether each edge will be included in the result.

Tip

Using this input is more efficient than deleting parts of the geometry before or after the conversion.

This node has no properties.

Generated curve.

--- Fin du chapitre: _mesh_to_curve ---

--- Début du chapitre: _mesh_to_points --- The Mesh to Points node generates a point cloud from a mesh.

Standard Mesh input.

The meshes used to generate a point cloud.

Positions of generated points. By default, this input is the same as if the Position Node was connected.

Radii of generated points.

Points are generated for each vertex.

Points are generated for each edge, at the middle of each edge, by default.

Points are generated for each face, at the average of all of each face's vertices, by default.

Points are generated for each corner. The points are all placed at the location of each corner's vertex, so they will overlap by default.

Generated point cloud.

--- Fin du chapitre: _mesh_to_points ---

--- Début du chapitre: _mesh_to_volume --- The Mesh to Volume node creates a fog volumes based on the shape of a mesh. The volume is created with a grid of the name "density".

Standard Mesh input.

Value of voxels inside the generated fog volume.

Specify the approximate number of voxels along the diagonal.

Specify the voxel side length.

The maximum distance of the included voxels to the surface on the inside of the mesh.

How the voxel size is specified.

Specify the approximate number of voxels along the diagonal.

Specify the voxel side length. It is recommended to be careful when tweaking this value, because small changes can have a large effect on the processing time.

The generated volume grid.

--- Fin du chapitre: _mesh_to_volume ---

--- Début du chapitre: _mesh_uv_layout --- Reference

Import-Export

UV Editor \xe2\x80\xa3 UV \xe2\x80\xa3 Export UV Layout

This add-on is enabled by default, in case it is not:

Open Blender and go to Add-ons section of the Preferences .

Search \xe2\x80\x9cUV Layout\xe2\x80\x9d and check the Enable Extension checkbox.

Using your favorite image painting program, you could use an exported UV layout to create a texture. Then save your changes, and back in Blender, use the Image \xe2\x80\xa3 Open to load it as your UV image for the mesh in Edit Mode for the desired (and active) UV map.

As a way of communicating to an artist who is painting your UV Texture for you, Blender has a tool called UV Layout (UV Editor \xe2\x80\xa3 UV \xe2\x80\xa3 Export UV Layout) that saves an image as a Targa (.tga), EPS , or SVG format for the object you have selected.

The image will be lines defining the UV edges that are within the image area of the UV mapping area. Edges outside the boundary, even if selected, will not be shown in the saved graphic. The artist will use this as a transparent layer in their paint program as a guide when painting your texture. The example below shows Blender in the background, and the Gimp working on the texture, using the saved layout as a guide. Note that targa format supports the Alpha channel, so you can paint transparent areas of the mesh. For using images as textures, see the page on Image Textures .

A UV layout in the UV Editor. \xc2\xbb

A UV layout in a paint program. \xc2\xbb

Export options. \xc2\xbb

if disabled, then only the UV faces selected will be outlined.

Export UVs from the modified mesh.

Select the type of image file to save (.png , .eps , .svg).

Select the size of the image in pixels.

Set the opacity of the fill.

--- Fin du chapitre: _mesh_uv_layout ---

--- Début du chapitre: _meta --- A Meta Strip is a strip which contain multiple strips treated as if it was one strip. It allows you to reduce the vertical space used in the Sequencer. You can edit it the same way as any other strips.

It is organization tool. For example, if you are using a lot of strips with complicated arrangement, you can group them together using Meta strips.

To create a Meta strip, select all the strips you want to group, and Ctrl-G to group them. The Meta strips will span from the beginning of the first strip to the end of the last one, and condenses all channels into a single strip.

Separating (ungrouping) the Meta strip restores the strips to their relative positions and channels. This can be used if you choose to delete a Meta strip and want to keep the strips inside.

Example of Meta strips.

You can edit the content inside a Meta strip by pressing Tab. It will expand the strip to the whole view and hide any other strips. To exit the Meta strip press Tab again. Meta strips can also be nested, which make editing them a little confusing. To exit out one level of Meta Strip make sure you do not have a Meta strips selected when you press Tab.

Note

The default blend mode for a Meta strip is Replace. There are many cases where this alters the results of the animation so be sure to check the results and adjust the blend mode if necessary.

One convenient use for Meta strips is when you want to apply the same effect to multiple strips. For example: if you have a video that was recorded in different files and want to add an effect strip. It is much more convenient to apply a single set of effects to one Meta strip than applying it to each individual strip.

See also

It is also possible to do the similar task described above with an Adjustment Layer effect strip.

--- Fin du chapitre: _meta ---

--- Début du chapitre: _metadata --- Metadata panel.

The Metadata panel includes options for writing metadata into render output.

Note

Only some image formats support metadata: See image formats .

Where to grab metadata from.

Use metadata from the current scene.

Use metadata from the strips in the Sequencer.

Includes the current date and time.

Includes the current scene time and render frame at HH:MM:SS.FF.

Includes the render time.

Includes the frame number.

Includes the start and end frame numbers.

Includes the peak memory usage.

Includes the rendering machine's hostname.
Includes the name of the active camera.
Includes the name of the active camera's lens value.
Includes the name of the active scene.
Includes the name of the last marker.
Includes the filename of the blend-file.
Includes the name of the foreground sequence strip.
Includes a custom note.

Hint

It can be useful to use the Notefield if you are setting up a render farm. Since you can script any information you like into it, such as an identifier for the render node or the job number. For details on stamping arbitrary values, see: [this page](#).

Add metadata as text to the render.

Set the size of the text.

Set the color and alpha of the stamp text.

Set the color and alpha of the color behind the text.

Displays the labels before the metadata text. For example, `Camera` in front of the camera name, etc.

--- Fin du chapitre: `_metadata` ---

--- Début du chapitre: `_metallic` --- The `MetallicBSDFnode` is used to recreate the appearance of metals.

Color of the material when viewed straight on.

Color of the material when viewed at a 82° angle.

Refractive index per color channel. This is the real part of a complex refractive index, scientifically denoted as n .

Extinction coefficients per color channel. This is the imaginary part of a complex refractive index, scientifically denoted as k .

Sharpness of the reflection; perfectly sharp at 0.0 and smoother with higher values.

Amount of anisotropy. Higher values give elongated highlights along the tangent direction; negative values give highlights shaped perpendicular to the tangent direction.

Rotates the direction of anisotropy, with 1.0 going full circle.

Compared to the Glossy BSDFnode, the direction of highlight elongation is rotated by 90°. Add 0.25 to the value to correct.

Normal used for shading; if nothing is connected the default shading normal is used.

Tangent used for shading; if nothing is connected the default shading tangent is used.

Microfacet distribution to use.

GGX microfacet distribution.

Takes multiple scattering events between microfacets into account. This gives more energy conserving results, which would otherwise be visible as excessive darkening.

Cycles Only Beckmann microfacet distribution.

Models for describing the metal's appearance, by specifying the apparent color or the physical IOR.

Uses the Adobe F82-Tint formula for the metallic fresnel. This allows for artist friendly control of the color near the edge of the material to simulate a complex IOR.

Accepts Complex IOR measurements from real world metals to replicate a more accurate rendering of metals than the F82 TintFresnel type.

Complex IOR values can be found from sources like the Physically Based database for CG artists and Refractive Index database.
Standard shader output.

Material

Titanium (Default)

Aluminum

Copper

Gold

Base Color

Edge Tint

Material

Titanium (Default)

Aluminum

Copper

Gold

IOR

Extinction

--- Fin du chapitre: _metallic ---

--- Début du chapitre: _metarigs --- Add a single bone from the Add
\xe2\x80\xa3 Armature menu.

Go in armature Edit Mode and build the meta rig by samples or Rigify-
types.

Define the Rigify bone collection UI, color sets, and selection sets.

In the armature properties click on the Generate button to generate the
rig.

A meta-rig is an assembly of bone chains. A bone chain is identified by
the Connected attribute. Bone chains can be further connected together by
parenting them without using the Connected attribute (i.e. using the Keep
Offset option while parenting).

Each first bone of a bone chain has a custom attribute on it which is a
Rigify custom property that identifies the sub-rig type. At rig generation
time Rigify will determine which controls and deform bones will be created
processing the meta-rig from the first bone to the last of each chain.

Human meta-rig split by samples. \xc2\xb6

Since a meta-rig is just a collection of sub-rigs, new meta-rigs can be
built assembling sub-rigs in different ways. This way an infinite number
of meta-rigs can be built from the same rigging blocks.

Cat meta-rig built by samples. \xc2\xb6

The meta-rig contains more information than the visualized bones. In fact
at generation time Rigify will identify each sub-rig type and depending
on the selected options will create all the sophisticated controls,
switches, and deforming bones with a single click.

Reference

Object Mode

Add \xe2\x80\xa3 Armature \xe2\x80\xa3 Single Bone

Shift- A

Building your own meta-rig from scratch requires an armature object to
work with. Just add a single bone from the Add menu.

Tip

At this stage naming the newly added armature meta-rig is a good idea. You
can do it at any time (or not at all) but it's suggested to do
it before going on so it will always be clear on which armature you have
to work when editing the meta-rig structure.

Now that there is an armature object to work with the armature selected enter armature Edit Mode. Building a meta-rig from scratch in Edit Mode can be done in two ways:

- Adding rig samples.

- Creating bone chains.

Adding pre-defined samples in Edit Mode is a good way to start building a meta-rig. This way you can become familiar with the available building blocks and how they are meant to be used. To add a rig sample:

- Go in the armature tab.

- Scroll down to Rigify panel.

- Select a sample from the list.

- Click on the Add sample button.

- Edit the bone positions to match your character.

For the list of available samples, see the Rig Types page.

For full control, you can use the Rigify Type panel of bone properties in Pose Mode to assign any Rigify sub-rig type to any bone, as well as change its options.

For the list of available sub-rig types and their options, see the Rig Types page.

At the top of the panel you can find a field specifying the rig type for the active bone. The drop-down list can be optionally filtered by the Feature Set it belongs to.

Below that you can change options relevant to the selected rig type, if it has any.

Some rig types that generate many control bones have options that reference Bone Collections. These reference lists have a standard UI with the following features:

- A checkbox controlling whether the reference should be used.

- A button to copy the reference list contents from the active to all selected bones.

- A plus button to add a new reference to the list.

- A list of references, each entry with a field to specify the target collection, and a button to remove the entry from the list.

Note

Each sub-rig has a required number of bones as input. If you are unsure on how to use rig-types properties, add a rig sample to your armature to see how it is supposed to be used.

Certain properties of the meta-rig bones are often copied to the generated rig control, deform and mechanism bones.

The exact set depends on the sub-rig and the specific generated bone, and the sub-rig may override some properties even when it preserves others from the same subset, but there are certain common patterns:

This subset consists of the parent ORG bone, Use Connect, Use Inherit Rotation, Use Local Location, and Inherit Scale.

It is usually copied to deform bones, FK controls, and in other cases where the sub-rig doesn't have a reason to completely override them.

Consist of the segment count, Vertex Mapping Mode, Ease In/Out, Roll In/Out, Curve In/Out and Scale In/Out.

The segment count is often overridden via a sub-rig option, but other settings are usually copied to deform bones as is.

Consist of the rotation mode, pose mode rotation values, and channel locks.

These settings are usually copied to FK controls.

Usually copied to one of the controls generated based on the metarig bone (mainly FK). Intra-armature drivers that access the property are retargeted to the copied instance.

Usually copied to one of the controls generated based on the metarig bone (mainly FK), and suppresses automatic generation of a widget for the bone if specified.

If the meta-rig contains a bone called root, it is used as the root control bone instead of creating a new one. This allows changing the rest position of the root bone, assigning a custom widget, or adding custom properties to the bone.

The custom root bone must have no parent, and use the basic.raw_copy sub-rig type or none.

The Color Sets panel is used to define the bone color scheme for the final rig. The colors from the list can be associated with bone collections from the relevant panel.

The top two rows of the Color Sets panel are used to define the general behavior of the bone colors. Usually color themes use a gradient of colors to define the different bone states: default, selected and active. When multiple color themes are used in the same rig, identifying which bone is selected or active can be tricky since each color will have its corresponding state.

To override this behavior Rigify unifies the active and selected states using the same color. This is defined by two values:

When this option is active adding a bone group in the list will always keep the colors consistent. When a color scheme is added from a theme, the color scheme is loaded as is. Click on the Apply button to force the system to unify selected and active colors.

These two color fields define respectively the Selected and Active colors. By default Rigify reads these colors from the theme defined by the user in the Blender preferences. This way the Selected/ Active colors can always have a predictable and consistent behavior in the UI. The colors can be customized by clicking on the relevant color field. To reset them to the Blender current theme value just click on the button with the update icon.

Color Sets can be added and deleted by clicking on the + or - buttons. All color sets can be deleted at once by clicking on the Specials menu.

To add the colors from the predefined Rigify default color scheme (as shown in the image) to the list click the Add Standard button.

To add a specific theme with its own color scheme, select it from the list and click on the Add From Theme button.

Bone Collections are used to group related bones together so that they can be hidden or revealed together.

Rigify can take advantage of collections to generate extra features and the user interface for the final rig. A panel named Rig Layers is generated with buttons for hiding the collections, arranged in an intuitive layout.

The Bone Collections UI panel allows configuring the layout of that generated panel, as well as specifying some other settings for bone collections, such as the color set to use.

The top of the panel is occupied by a list that duplicates the main bone collection list, but displays additional properties, such as the color set, whether the collection has a button, or whether it generates a selection set.

Some sub-rig types have references to bone collections in their properties. Rigify uses a referencing scheme that is robust to collection renames, but deleting collections or joining armatures can still lead to broken references.

This button runs a scan that validates and normalizes all collection references, reporting any errors, and reducing the chance of breakage being caused by subsequent user actions.

This scan is also performed automatically every time the rig is generated.

Warning

To avoid breakage this operation should be used both immediately before and after joining two metarig armatures. More specifically, it must be always done between the actions of renaming any collections and joining. Specifies the color set to use for bones in this collection. If a bone belongs to multiple collections, in general the collection located earlier in the list has priority.

Specifies whether a selection set should be generated for this collection.

If nonzero, specifies which row of the Rig Layers panel should contain the button controlling the visibility of this collection. When zero, no button is generated, and the collection is hidden.

This field can be used to override the title used on the UI button to be distinct from the true collection name. Unlike collection names, titles are not required to be unique, so this can be used to reduce clutter by relying on contextual cues within the panel.

The UI Layout sub-panel provides a WYSIWYG editor for the layout of the generated UI panel (as defined by the UI Row and UI Title settings above).

Each row contains three buttons at the end:

Moves the active collection button to this row.

Inserts a new row before the current one.

Removes the current row and shifts all buttons up.

To the left of the editing control buttons, rows display buttons corresponding to the collections, same as the final UI, except that rather than hiding or unhiding, clicking these buttons selects the collection.

For the active collection the selection button is replaced with an input field for editing the UI Title, and an X button to unassign the collection from the UI.

For any collections not assigned to the UI, their select buttons are displayed in a separate section at the bottom of the sub-panel.

The Root collection will be added and/or assigned a UI button automatically if necessary when the rig is generated. If desired, it is possible to manually assign UI buttons to the internal ORG, DEF and MCH collections.

Tip

Blank rows appear much thinner in the final interface, since they don't have to contain editing buttons, and can be used as logical separators.

The Action constraint allows applying poses defined by an action to bones based on the transformation of another bone. This requires adding the constraint to everyone affected by the action, which is very tedious. For this reason, Rigify includes a system to do this automatically through the Actions panel.

The panel defines a list of actions to be applied to the generated rig bones. Each action must be listed only once.

The list entries show the name of the action, the trigger (a bone or a corrective action driven by two others), and a checkbox that can be used to temporarily disable applying this action to the rig. The icon at the start of the entry is changed from an action icon to a link icon to highlight corrective actions that depend on the active normal one, or normal actions used by the active corrective action.

Note

The Action constraints are added to the bones in such an order as to exactly reproduce the intended deformation, assuming the actions were created (posed and keyframed) in the order listed.

Normal actions are applied based on the transformation of a specific control bone from the generated rig. They have the following properties:

- Specifies the bone that drives the action.

- If the control bone has a suffix that specifies that it belongs to the left or right side, this option can be enabled to automatically apply symmetry.

- When enabled, left-side bones keyframed in the action will be controlled by the left-side control, and right-side bones by the right side control. Bones that don't have a side suffix are assumed to belong to the center of the character. They are rigged with two Action constraints with influence 0.5 that are controlled by each of the control bones.

- Specifies the frame range of the action that will be used by the created constraints.

- Specifies the coordinate space and transformation channel of the target bone that should be used.

- Specifies the range of the transformation channel values that is mapped to the specified action frame range.

- Shows the frame within the action that maps to the neutral value (1 for scale and 0 otherwise) of the transformation channel, as computed from the specified range values.

Corrective actions are applied based on the progress of two other actions from the list, and are used to improve the pose when they are used together.

- Specifies the frame range of the action that will be used by the created constraints.

- Specifies the two actions that control the correction. The interface rows contain buttons to show the settings for that action, or jump to it in the list.

The progress of the corrective action from the start to the end frame is calculated as the product of the progress values of the two trigger actions. Thus, the start frame is applied when either of the triggers is at the start frame, and the end frame is used when both are at their end frame.

Corrective actions must be below their triggers in the list, which is enforced via an implicit reorder even if violated.

Tip

Corrective actions behave in the most intuitive way when both triggers have the Default Frame equal to Start Frame. To create a corrective action in such case:

Create the two trigger actions, add them to the panel and generate the rig.

Pose your controls so that both trigger actions are fully activated to the end frame.

Pose and keyframe the necessary corrections in the end frame of the new action, while keying the startframe to the neutral values.
Add the newly created action to the end of the list in the panel and configure its settings.

--- Fin du chapitre: `_metarigs` ---

--- Début du chapitre: `_mirror` --- The Mirror modifier mirrors the strokes along its local X, Y and/or Z axes, across the Object Origin. It can also use another object as the mirror center, then use that object's local axes instead of its own.

See also

This documentation refers to the Mirror Modifier specific to the Grease Pencil object. For uses with other object types refer to the general Mirror Modifier .

The Mirror modifier. \xc2\xbb6

The X, Y, Z axis along which to mirror, i.e. the axis perpendicular to the mirror plane of symmetry.

To understand how the axis applies to the mirror direction, if you were to mirror on the X axis, the positive X values of the original stroke would become the negative X values on the mirrored side.

You can select more than one of these axes. And will then get more mirrored copies. With one axis you get a single mirror, with two axes four mirrors, and with all three axes eight mirrors.

A Data ID to select an object (usually an empty), which position and rotation will be used to define mirror planes (instead of using the ones from the modified object).

See Influence Filters.

--- Fin du chapitre: `_mirror` ---

--- Début du chapitre: `_mix` --- The Mix Node mixes values, colors and vectors inputs using a factor to control the amount of interpolation. The Color mode has additional blending modes.

Controls the amount of mixing between the A and B inputs.

The two inputs that are mixed together.

The data type that is used for mixing. The node supports float, vector, color, and rotation data types.

The factor mode can be set to Uniform and Non-Uniform. In uniform mode, a single float controls the factor. In non-uniform mode, a vector controls the factor for each XYZ channel separately.

The Blend modes can be selected in the select menu. See Color Blend Modes for details on each blending mode.

Add, Subtract, Multiply, Screen, Divide, Difference, Darken, Lighten, Overlay, Color Dodge, Color Burn, Hue, Saturation, Value, Color, Soft Light, Linear Light

Limit the factor value between 0.0 and 1.0. If this option is unchecked then the node operates using Extrapolation.

Limit the Result to the range between 0.0 and 1.0.

Output the result of the mix using the data type selected.

See the Color > Mix page for additional examples: Mix Color Node

--- Fin du chapitre: `_mix` ---

--- Début du chapitre: `_mix_color` --- Blends two images together, much like how an image editing program blends two layers.

The opacity of the foreground image.

The background image. Determines the dimensions of the output.

The foreground image.

Keep in mind that, unlike image editing programs where the foreground layer is on top, the foreground slot in Blender is on the bottom.

The blending mode to use.

Regular alpha blending. Typically called Normal in image editing programs.

For each color component, takes the smallest of the two values being blended.

Multiplies the colors component by component. Blending with a white pixel (value 1.0) has no effect, while blending with a black one (0.0) always results in black.

Inverts the background color, divides it by the foreground color, and inverts the result.

For each color component, takes the largest of the two values being blended.

Inverts both colors, multiplies them, and inverts the result.

Divides the background color by the inverted foreground color.

Adds the two colors together.

Applies Multiply blending if the foreground color's lightness is below 0.5, or Screen blending if it's above.

Like Overlay, but more subtle.

Applies Linear Burn blending ($\text{background} + \text{foreground} - 1$) if the foreground color's lightness is below 0.5, or Linear Dodge ($\text{background} + \text{foreground}$) if it's above.

For each component, subtracts the lower value from the higher value.

Adds the two colors, then subtracts their multiple twice.

Subtracts the foreground color from the background color.

Divides the background color by the foreground color.

Combines the saturation and value of the background color with the hue of the foreground color.

Combines the hue and value of the background color with the saturation of the foreground color.

Combines the value of the background color with the hue and saturation of the foreground color.

Combines the hue and saturation of the background color with the value of the foreground color.

Whether to use the alpha channel of the foreground image during mixing. The alpha channel of the background image is always used.

Clamp the output value to the [0.0, 1.0] range.

The result of the mixing operation.

Below are examples of blending modes, as well as some practical use cases.

Blending a colored pattern with a flat color (top row) and a circular mask (bottom row).

The Compositing setup below shows how to fix an overexposed render by darkening it and increasing contrast.

Example node setup showing two RGB Curves nodes and a Mix node for composition.

The top RGB Curves Node darkens the image by linearly scaling each color value to a smaller one.

The bottom curve node increases contrast by making small values smaller and large values larger.

Finally, the Mix node blends the two together.

In the old days, a pattern was pressed into the paper much as it dried, creating a mark that identified who made the paper and where it came from. The mark was barely perceptible except in just the right light. Probably the first form of subliminal advertising.

Nowadays, people watermark their images to identify them as personal intellectual property, for subliminal advertising of the author or hosting service, or simply to track their image's proliferation throughout the web.

Blender provides a complete set of tools for you to both encode your watermark and to tell if an image has your watermark.

First, construct your own personal watermark. You can use your name, a word, or a shape or image not easily replicated. While neutral gray works best using the encoding method suggested, you are free to use other colors or patterns. It can be a single pixel or a whole gradient; it is up to you.

In the example below, we are encoding the watermark in a specific location in the image using the Translate node; this helps later because we only have to look at a specific location for the mark. We then use the RGB to BW node to convert the color image to grayscale numbers, which we then feed into the Map Range node to reduce the mark to one-tenth of its original intensity.

The Add node (Mix node with blending mode Add) adds the corresponding pixels, making the ones containing the mark ever-so-slightly brighter.

Embedding a watermark in an image. \xc2\xbf

Of course, if you want people to notice your mark, do not scale it so much, or make it a contrasting color. There are also many other ways, using other mix settings and fancier rigs. Feel free to experiment!

When you see an image that you think might be yours, use the node tree below to compare it to your stock image (pre-watermarked original). In this tree, the Mix node is set to Difference, and the Map Value node amplifies any difference. You can see how the original mark clearly stands out.

Checking an image for your watermark. \xc2\xbf

--- Fin du chapitre: _mix_color ---

--- Début du chapitre: _mix_rgb --- The Mix Node mixes values, colors and vectors inputs using a factor to control the amount of interpolation. The Color mode has additional blending modes.

Controls the amount of mixing between the A and B inputs.

The two inputs that are mixed together.

The data type that is used for mixing. The node supports float, vector, color, and rotation data types.

The factor mode can be set to Uniform and Non-Uniform. In uniform mode, a single float controls the factor. In non-uniform mode, a vector controls the factor for each XYZ channel separately.

The Blend modes can be selected in the select menu. See Color Blend Modes for details on each blending mode.

Add, Subtract, Multiply, Screen, Divide, Difference, Darken, Lighten, Overlay, Color Dodge, Color Burn, Hue, Saturation, Value, Color, Soft Light, Linear Light

Limit the factor value between 0.0 and 1.0. If this option is unchecked then the node operates using Extrapolation.

Limit the Result to the range between 0.0 and 1.0.

Output the result of the mix using the data type selected.

See the Color > Mix page for additional examples: Mix Color Node

--- Fin du chapitre: `_mix_rgb` ---

--- Début du chapitre: `_mix_vector` --- The Mix Node mixes values, colors and vectors inputs using a factor to control the amount of interpolation. The Color mode has additional blending modes. Controls the amount of mixing between the A and B inputs. The two inputs that are mixed together. The data type that is used for mixing. The node supports float, vector, color, and rotation data types. The factor mode can be set to Uniform and Non-Uniform. In uniform mode, a single float controls the factor. In non-uniform mode, a vector controls the factor for each XYZ channel separately. The Blend modes can be selected in the select menu. See Color Blend Modes for details on each blending mode. Add, Subtract, Multiply, Screen, Divide, Difference, Darken, Lighten, Overlay, Color Dodge, Color Burn, Hue, Saturation, Value, Color, Soft Light, Linear Light. Limit the factor value between 0.0 and 1.0. If this option is unchecked then the node operates using Extrapolation. Limit the Result to the range between 0.0 and 1.0. Output the result of the mix using the data type selected. See the Color > Mix page for additional examples: Mix Color Node

--- Fin du chapitre: `_mix_vector` ---

--- Début du chapitre: `_modal_map` --- During a transformation, some hotkeys can change the behavior of the operation.

You can check editing the keys of these modal modifiers in Blender Preferences > Keymap > Transform Modal Map (at the bottom of the keymap).

When moving, rotating or scaling, if you only want certain axes to be affected, you can restrict the transformation to those axes.

By default the constraint keys are X, Y and Z. This constraint can be restricted to a plane if Shift is pressed or automatically detected if MMB is pressed.

It is worth noting that if you press the same constraint hotkey a second time, you change the orientation from Local to Global or vice versa.

Pressing a third time disables the constraint.

Transform operations use the snapping settings set in the scene. However, some options can be changed during the transformation.

Even if the magnetic icon is disabled, you can still enable snapping during a transformation. The default hotkey in this case is Ctrl.

The Snap Base is automatically determined based on the Snap Base options. However, this automatic snap detection point of origin may not always align with the user's intentions. Therefore, transform operations include a utility to set a new snap origin point during the transformation. The new Snap Base will correspond to the snap point whose target is defined by the Snap Target.

By default the hotkey is B.

Note

If only Snap to Increment is enabled as Snap Target, the targets Vertex, Edge, Face and Edge Center will be used instead.

While you're transforming a selection with snapping enabled, you can press A whenever there's a highlighted snap

target to mark it. With multiple such targets marked, the selection will then be snapped to their average location.

Marking a target more than once will give it more weight.

Multiple snapping targets.

While performing a modal transformation, you can perform navigation actions such as zooming, panning, or rotating by holding Alt then perform the desired action.

--- Fin du chapitre: _modal_map ---

--- Début du chapitre: _mode --- Using Particle Edit Mode you can edit the keyed points (keyframes) and paths of Hair, Particle, Cloth, and Soft Body simulations. (You can also edit and style hair before baking.)

Since working in Particle Edit Mode is pretty easy and very similar to working with vertices in the 3D Viewport, we will show how to set up a particle system and then give a reference of the various functions.

Important

Particle Edit Mode, specifically for hair is deprecated; please use the new Empty Hair object with its associated Sculpt Mode instead.

Important

Editing a cached cloth simulation is not currently working, see: [blender/blender#77114](#) for details.

Tip

Only Frames Baked to Memory are Editable!

If you cannot edit the particles, check that you are not baking to a Disk Cache.

Create a Hair particle system.

Give it an initial velocity in the Normal direction.

Create a simulation.

Check the Hair Dynamics box.

Editing hair strands in Particle Edit Mode.

Use Emitter particles, or a cloth/soft body simulation.

Create a simulation by setting up objects and or emitters, set your time range (use a small range if you are just starting out and experimenting), set up the simulation how you want it, using Alt- A to preview it.

Once you are happy with the general simulation, bake the simulation from Object Mode. The simulation must be baked to enable editing.

Switch to Particle Edit from the Mode select menu in the header of the 3D Viewport to edit the particle's paths/Keyframes. You may need to press T from within the 3D Viewport to see the Particle Edit toolbox. Move to the frame you want to edit and use the various tools to edit your simulation.

Tip

Switch to the Point select mode (see below) in the header of the 3D Viewport to be able to see and select the keypoints.

Select single: LMB.

Add to/remove from selection: Shift- LMB .

All: A.

None: Alt- A .

Invert: Ctrl- I .

Box select: B.

Circle Select: C.

Lasso Select: Ctrl- Alt- LMB .

Select Linked: Move the mouse over a path and press L to add all its points to the selection.

Unselect Linked: Move the mouse over a path and press Shift- L to remove all its points from the selection.

Root/Tips: Select \xe2\x80\xa3 Roots / Tips.

- Randomly selects particles.

- Percent of particles to randomly select.

- Seed value to use for the selection.

- Select random can be either used to select or deselect particles.

Selects either hair or points. Here these terms can be confusing because hair/point does not refer to the particle type but the path/points of the hair/particle.

- Select Modes.\xc2\xb6

- No keypoint are visible, you can select/deselect only all particles.

- You see all of the keypoints.

You can see and edit (including the brushes) only the tip of the particles, i.e. the last keypoint.

- Reference

- Particle Edit Mode

- Toolbar

- Moves the keypoints (similar to the Proportional Editing tool).

Hair particles only \xe2\x80\x93 Do not move keypoints through the emitting mesh.

- The distance to keep from the Emitter.

- Parallels visually adjacent segments.

- Adds new particles.

- The number of new particles per step.

- Interpolate the shape of new hairs from existing ones.

- Amount of brush steps.

- How many keys to make new particles with.

Scales the segments, so it makes the hair longer with Grow or shorter with Shrink.

- Sets the brush to add the effect or reverse it.

Rotates the hair around its first keypoint (root). So it makes the hair stand up with Add or lay down with Sub.

Apply puff to unselected end points, (Helps to maintain the hair volume when puffing the root.)

- Scales the segments until the last keypoint reaches the brush.

This is especially useful for soft body animations, because the weight defines the soft body Goal. A keypoint with a weight of 1 will not move at all, a keypoint with a weight of 0 subjects fully to soft body animation. This value is scaled by the Strength Min to Max range of soft body goals \xe2\x80\xa6

- Below the brush types, their settings appear:

- Set the radius of the brush.

- Set the strength of the brush effect (not for Add brush).

- Reference

- Particle Edit Mode

- Tool Settings \xe2\x80\xa3 Options

Recalculate velocities of particles according to their edited paths. Otherwise, the original velocities values remains unchanged regardless of the actual distance that the particles moves.

Enable mirror editing across the local X axis.

Keep the length of the segments between the keypoints when combing or smoothing the hair. This is done by moving all the other keypoints. Keep first key unmodified, so you cannot transplant hair.

A mesh object which boundary is used by the Shape Cuttool.

This grooming tool trims hairs to a shape defined by the Shape Object. This is a quicker way of avoiding protruding hair sections from lengthening than using the Cutting tool. It works especially well for characters with extensive fur, where working in a single plane with the Cutting tool becomes tedious.

Before. \xc2\xbb

After. \xc2\xbb

The number of steps used to draw the path; improves the smoothness of the particle path.

Displays the children of the particles too. This allows to fine-tune the particles and see their effects on the result, but it may slow down your system if you have many children.

Displays the actual particles on top of the paths.

Fade out paths and keys further away from current time.

How many frames to fade.

To move selected keypoints press G, or use one of the various other methods to move vertices.

To move a particle root you have to turn off Keep Root in the Toolbar.

You can do many of the things like with vertices, including scaling, rotating and removing (complete particles or single keys).

You may not duplicate or extrude keys or particles, but you can subdivide particles which adds new keypoints Particle \xe2\x80\xa3 Subdivide.

Alternatively you can re-key a particle Particle \xe2\x80\xa3 Rekey.

How smoothly the hair and particle paths are displayed depends on the Path Steps setting in the Toolbar. Low settings produce blocky interpolation between points, while high settings produce a smooth curve.

Reference

Particle Edit Mode

Particle \xe2\x80\xa3 Mirror

If you want to create an X axis symmetrical haircut you have to do following steps:

Select all particles with A.

Mirror the particles with Particle \xe2\x80\xa3 Mirror.

Turn on X Mirror in Sidebar Region \xe2\x80\xa3 Tool \xe2\x80\xa3 Options.

It may happen that after mirroring two particles occupy nearly the same place. Since this would be a waste of memory and render time, you can use Merge by Distance from the Particle menu.

Reference

Particle Edit Mode

Particle \xe2\x80\xa3 Unify Length

This tool is used to make all selected hair uniform length by finding the average length.

Reference

Particle Edit Mode

Particle \xe2\x80\xa3 Show/Hide

Hiding and unhiding of particles works similar as with vertices in the 3D Viewport. Select one or more keypoints of the particle you want to hide and press H. The particle in fact does not vanish, only the key points.

Hidden particles (i.e. particles whose keypoints are hidden) do not react on the various brushes. But:

If you use Mirror Editing even particles with hidden keypoints may be moved, if their mirrored counterpart is moved.

To unhide all hidden particles press Alt-H.

--- Fin du chapitre: _mode ---

--- Début du chapitre: _modes --- The Mode select menu.

Modes allow editing different aspects of objects. While Object Mode allows you to position/rotate/scale them, Edit Mode allows changing their geometry, Pose Mode allows posing them, and so on.

You can change the current mode using the Modes selector in the 3D Viewport header. Which modes are available depends on the object's type. The complete list is shown below.

Apart from using the selector, you can also press Ctrl-Tab to bring up a pie menu around the cursor for faster access. (If the selected object is an Armature, this shortcut will instead switch between Object Mode and Pose Mode.)

Pressing Tab will toggle Edit Mode for objects that support it.

Modes can affect many things in Blender:

Each mode changes the header and Toolbar to show its own unique set of menus and tools. This also means it affects the available keyboard shortcuts.

Modes can completely change the look of the viewport. For example, Weight Paint mode will shade the object to show its vertex weights, which are not normally visible.

Modes can affect other editors. For example, the UV Editor can only be used if the 3D Viewport is in Edit Mode. In the Properties editor, too, certain buttons and panels can only be used in certain modes.

Icon

Name

Details

Object Mode

The default mode, available for all object types. Allows editing position, rotation and scale, duplicating objects, and so on.

Edit Mode

A mode for editing an object's shape (vertices/edges/faces for meshes, control points for curves/surfaces, points/strokes for Grease Pencil, etc.).

Sculpt Mode

Provides an alternative toolset for editing an object's shape (only for meshes).

Vertex Paint Mode

A mesh-only mode that allows you to set your mesh's vertex colors (i.e. to "paint" them).

Weight Paint Mode

A mesh-only mode, dedicated to vertex group weighting.

Texture Paint Mode

A mesh-only mode that allows you to paint a texture directly on the model, in the 3D Viewport.

Particle Edit Mode

A mesh-only mode dedicated to particle systems, useful for editable systems (hair).

Pose Mode

An armature-only mode, dedicated to posing.

Draw Mode

A Grease Pencil-only mode, dedicated to creating Grease Pencil strokes.

Sculpt Mode (Grease Pencil)

A Grease Pencil-only mode, used to deform and shape existing strokes more organically.

Edit Mode (Grease Pencil)

A Grease Pencil-only mode, dedicated to modifying individual strokes and points of Grease Pencil objects.

Vertex Paint Mode (Grease Pencil)

A Grease Pencil-only mode, dedicated to adding color the vertices of strokes directly.

Weight Paint Mode (Grease Pencil)

A Grease Pencil-only mode, dedicated to assigning vertex weights to stroke.

Note

The cursor becomes a brush in Paint and Sculpt Modes.

We will not go into any more detail on mode usages here, because they are dealt with in their own sections.

Hint

If you are reading this manual and some button or menu option is referenced that does not appear on your screen, it may be that you are not in the proper mode for that option to be valid.

Reference

All Modes

Alt- Q

If you enter a mode such as Weight Paint for an object and then select another object, Blender will typically switch back to Object Mode. This means that, if you want to weight paint the other object too, you have to enter the mode a second time.

There is a way of avoiding this, however. Once you enter a mode, the Outliner will show a dot next to other objects that also support it. By clicking such a dot, you can switch over to another object without leaving the mode.

Alternatively, you can hover over the other object in the 3D Viewport and press Alt- Q .

See also

Lock Object Modes for preventing accidental mode changes.

Edit Mode and Pose Mode let you work with multiple objects even more easily than described above, as they can have multiple objects in the mode at the same time.

There are two ways of accomplishing this:

If you're not yet in the mode, you can simply select all the objects and enter it.

If you're already in the mode, you can bring other objects into it by clicking Ctrl- LMB on the dot in the Outliner. Removing objects from the mode works in the same way.

Some points of note:

The Properties editor will only ever show the details (shape keys, UV maps) of the active object, not of all the selected ones. Selecting any element from an object will make it the active one.

There are limits to the edits you can make. For example, you can't create an edge that connects vertices from different objects.

--- Fin du chapitre: _modes ---

--- Début du chapitre: _modifiers --- Operators for working with an object\xe2\x80\x99sModifiers.

Reference

Object Mode

Object \xe2\x80\xa3 Modifiers \xe2\x80\xa3 Add Modifier

Opens a menu with a list of modifiers, selecting a modifier will add it to the bottom of The Modifier Stack .

Reference

Object Mode

Object \xe2\x80\xa3 Modifiers \xe2\x80\xa3 Copy Modifiers to Selected Objects

Copies all modifiers from the Active object to the other selected objects.

Reference

Object Mode

Object \xe2\x80\xa3 Modifiers \xe2\x80\xa3 Clear Object Modifiers

Deletes all modifiers from the selected objects.

--- Fin du chapitre: _modifiers ---

--- Début du chapitre: _more_less ---Reference

Edit Mode

Select \xe2\x80\xa3 Select More/Less \xe2\x80\xa3 More

Ctrl- NumpadPlus

Select \xe2\x80\xa3 Select More/Less \xe2\x80\xa3 Less

Ctrl- NumpadMinus

With at least one vertex, edge, or face selected, Select More/Less expands or shrinks the selection. However, if there is only one selection in any selection mode, Less will deselect it.

With Face Step on, each use of the tool will affect the size of the selection on a face by face basis. When deactivated, it will be based on either vertices or edges depending on which Selection Mode is active.

(From left to right) initial selection, without Face Step, with Face Step, and in edge selection mode.\xc2\xbb

Reference

Edit Mode

Select \xe2\x80\xa3 Select More/Less \xe2\x80\xa3 Next Active

Shift- Ctrl- NumpadPlus

Select \xe2\x80\xa3 Select More/Less \xe2\x80\xa3 Previous Active

Shift- Ctrl- NumpadMinus

This uses selection history to select the next vertex, edge, or face based on the surrounding topology. Which means that, it will derive the next selection from the previous two selections.

Initial selection.\xc2\xbb

Using Next Active once.\xc2\xbb

Using Next Active twice.\xc2\xbb

Only the last selected element will be removed.

--- Fin du chapitre: _more_less ---

--- Début du chapitre: _motion_blur ---Reference

Render \xe2\x80\xa3 Motion Blur

Blender\xe2\x80\x99s animations are by default rendered as a sequence of perfectly still images. While great for stop-motion and time-lapses, this

is unrealistic, since fast-moving objects do appear to be blurred in the direction of motion, both in a movie frame and in a photograph from a real-world camera.

Motion blur example. (blend-file) \xc2\xba6

Controls at what point the shutter opens in relation to the current frame.

Shutter is starting to open at the current frame.

Shutter is fully opened at the current frame.

Shutter is fully closed at the current frame.

Time (in frames) between when the shutter starts to open and fully closed. For example, shutter time 1.0 blurs over the length of 1 frame.

Creates a Rolling Shutter effect.

No rolling shutter effect.

Renders rolling shutter from the top of the image to the bottom.

Controls balance between pure rolling shutter effect (if the value is zero) and pure motion blur effect (if the value is one).

Note

If there are particles or other physics system in a scene, be sure to bake them before rendering, otherwise you might not get correct or consistent motion blur.

See also

Each object has its own settings to control motion blur. These options can be found in the Object tab of the Properties. See object setting for more information.

Curve defining how the shutter opens and closes. The X axis is time, Y values of 0 mean fully closed shutter, Y values of 1 mean fully opened shutter. The default mapping is set to when shutter opens and closes instantly.

Camera motion blur does not work for Orthographic Cameras .

Motion blur does not take into account the movement of Lights .

--- Fin du chapitre: _motion_blur ---

--- Début du chapitre: _motion_paths ---Reference

3D Viewport, Properties

Object Mode

Properties \xe2\x80\xa3 Object Properties \xe2\x80\xa3 Motion Paths

Reference

3D Viewport, Properties

Pose Mode

Properties \xe2\x80\xa3 Armature \xe2\x80\xa3 Motion Paths

Pose \xe2\x80\xa3 Motion Paths

An animated cube with its motion path displayed. \xc2\xba6

The Motion Paths tool allows you to visualize the motion of points as paths over a series of frames. These points can be object origins and bone joints.

To create or remove motion paths, it is necessary to first select the bones. Then:

To show the paths (or update them, if needed), click on the Calculate Path button.

To hide the paths, click on the Clear Paths button.

Note

Remember that only selected bones and their paths are affected by these actions!

The paths are shown in red for the section in the past and green for the section in the future. These colors follow the user preference options \xe2\x80\x9cBefore Current Frame\xe2\x80\x9d and \xe2\x80\x9cAfter Current Frame\xe2\x80\x9d, which can be found in the 3D Viewport section. Each frame is displayed by a small dot on the paths.

The paths are automatically updated when you edit your poses/keyframes, and they are also active during animation playback. Playing the animation affects the paths only when using the Around Frametype.

The Motion Paths panel in the Armature tab. \xc2\xb6
Type of range to show for Motion Paths.

Display paths of points within a fixed number of frames around the current frame. When you enable this button, you get paths for a given number of frames before and after the current one

Display paths of points within specified range.

The range of the motion path. Only active when Paths Type is set to In Range. Changing this option only takes effect when updating the path, via the Update Path or Update All Paths buttons.

Generate a motion path ranging from the first keyframe to the last.

Only the keys of the active object / bone are considered.

Same as All Keys except that it ranges from the first to the last selected keyframe.

Use the start & end frames of the scene, or the preview range if active.

Manually set the start and end frame.

Starting and Ending frame of range of paths to display/calculate (not for the Around Frametype).

Although the start and end frame are always editable, updating the motion path will change these according to the Calculation Range setting.

To ensure your chosen frame range is maintained, choose Manual Range there.

Number of frames to show before and after the current frame (only for the Around Frametype).

Allows displaying one point for every n frames on the path. Mostly useful when you enable the frame number display (see below), to avoid cluttering the 3D Viewport.

When enabled the motion path is calculated in screen space for the active scene camera. Note that the resulting motion path will only be useful for that single camera. Switching cameras using markers is not supported. It will only bake to the camera that is active when the bake was started.

These are the start/end frames of the range in which motion paths are shown. You cannot modify this range without deleting the motion path first.

If no paths have been calculated, Calculate Paths will create a new motion path in cache based on the options specified in the pop-up menu or the Adjust Last Operation panel. Note, if the current context is an Armature calculating the objects motion paths, and not the bones, this operator will calculate the motion paths for all the bones within the armature as well.

These are the start/end frames of the range in which motion paths are shown. The start frame is inclusive, so if you set Start to 1, you will really see the frame 1 as starting point of the paths\xe2\x80\xa6
Which point on the bones is used when calculating paths. Only available for bones while in Pose Mode.

Calculates the path position of the bone\xe2\x80\x99s heads.

Calculates the path position of the bone\xe2\x80\x99s heads.

In the case a path has already been calculated, this operator will update the path shape to the current animation. To change the frame range of the calculated path, you need to delete the path and calculate it again. Clears paths on all objects/bones or just the selected ones when holding Shift.

Recalculates the motion paths for all visible objects and poses.

When enabled, a small number appears next to each frame dot on the path, which is of course the number of the corresponding frame.

When enabled, big yellow square dots are displayed on motion paths, showing the keyframes of their bones (i.e. only the paths of keyed bones at a given frame get a yellow dot at this frame).

For bone motion paths, it searches the whole Action for keyframes instead of in groups with matching name only (this is slower).

When enabled, you will see the numbers of the displayed keyframes, so this option is obviously only valid when Show Keys is enabled.

Toggles whether the lines between the points are shown.

Line thickness for motion path.

Use custom color for this motion path. The custom color can be modified for time before and after the current frame.

An example of a motion path of an armature. \xc2\xba6

--- Fin du chapitre: _motion_paths ---

--- Début du chapitre: _motor ---Reference

Physics \xe2\x80\xa3 Rigid Body Constraint

Motor

The motor constraint causes translation and/or rotation between two entities. It can drive two objects apart or together. It can drive simple rotation, or rotation and translation (although it will not be constrained like a screw since the translation can be blocked by other physics without preventing rotation).

The rotation axis is the X axis of the object hosting the constraint. This is in contrast with the Hinge which uses the Z axis. Since the Motor is vulnerable to confusing perturbations without a matching Hinge constraint, special care must be taken to align the axes. Without proper alignment, the motor will appear to have no effect (because the hinge is preventing the motion of the motor).

Enable linear or angular motor respectively.

Target linear or angular motor velocity respectively.

Maximum linear or angular motor impulse respectively.

--- Fin du chapitre: _motor ---

--- Début du chapitre: _mouse_position --- The Mouse Position node returns information about the mouse cursor such as its position and the region's dimensions.

Tip

When using this node, enable Wait for Click .to wait for a mouse click input (LMB) before running the operator from a menu.

Note

This node can only be used in the Tool context .

This node has no inputs.

This node has no properties.

The region-space mouse X location, in pixels, increasing from 0 at the left.

The region-space mouse Y location, in pixels, increasing from 0 at the bottom.

The total X size of the region in pixels.

The total Y size of the region in pixels.

--- Fin du chapitre: `_mouse_position` ---

--- Début du chapitre: `_move` ---Reference

Object Mode, Edit Mode, and Pose Mode

Object/Mesh/Curve/Surface \xe2\x80\xa3 Transform \xe2\x80\xa3 Move

G

In Object Mode, the move option lets you move objects. Translation means changing location of objects. It also lets you move any elements that make up the object within the 3D space of the active 3D Viewport.

Pressing G activates \xe2\x80\x9cMove\xe2\x80\x9d transformation mode. The selected object or element then moves freely according to the mouse pointer\xe2\x80\x99s location and camera. To confirm the action, press LMB. While moving items, the amount of change along the X, Y, and Z axis is displayed in the header of the 3D Viewport.

Translation Display.\xc2\xb6

Tip

Moving an object in Object Mode changes the object\xe2\x80\x99s origin. Moving the object\xe2\x80\x99s vertices/edges/faces in Edit Mode does not change the object\xe2\x80\x99s origin.

See also

Using a combination of shortcuts gives you more control over your transformation. See Transform Control .

The amount to move the selection on the respected axis.

Aligns the transformation axes to a specified orientation constraint. See Transform Orientations for more information.

The extruded face will affect nearby geometry. See Proportional Editing for a full reference.

--- Fin du chapitre: `_move` ---

--- Début du chapitre: `_movie` --- To add a movie (with or without audio) select a movie file(s) in the File Browser. g. in the Audio-Video Interleaved format (*.avifile).

Note

Clips can be Huge

A three minute QuickTime .mov file can be 140MB. Loading it, even over a high-speed LAN can take some time. Do not assume your computer or Blender has locked up if nothing happens for awhile.

Tip

Movie strips can display thumbnails in the Sequencer overlaid on their strips by enabling the Thumbnails overlay.

Reference

Add \xe2\x80\xa3 Movie

Store the location of the image file relative to the blend-file.

The Start Frame to place the left handle of the strip.

The Channel to place the strip.

Replaces the currently selected strips with the new strip.

Determines how images with an aspect ratio different than the scene\xe2\x80\x99s Resolution are scaled to fit inside the render area.

Adjusts the strips Scale Transforms so the visual contents of the strip to fit exactly within the project's Resolution while maintaining the original aspect ratio.

This may mean that the transparent areas may be added along the content's border to fit the content in the rendered area.

Adjusts the strips Scale Transforms so the visual contents of the strip to span the project's Resolution while maintaining the original aspect ratio.

This may mean that portions of the original image no longer fit the content inside the rendered area.

Adjusts the strips Scale Transforms so the visual contents of the strip to fill the project's Resolution. Note, unlike the other two methods described above, Stretch to Fill does not maintain the original aspect ratio.

This may mean that the original image becomes distorted to fit the content inside the rendered area.

Automatically sets an appropriate View Transform based on the Color Space of the imported media. In most cases, the Standard should be used; using the wrong transform could result in inaccurate colors or degraded rendering performance.

Automatically adjusts the video's speed to playback at the original speed regardless of the scene's framerate.

Add a Sound Strip that contains the movie's audio track.

Sets the Scene Frame Rate to the frame rate encoded in the movie file.

Imported Movie strip with audio track underneath. \xc2\xb6

In the strip itself, you can see strip name, path to source file, and strip length.

--- Fin du chapitre: _movie ---

--- Début du chapitre: _movie_clip --- This node is a special node that uses some of the values taken from footage cameras and trackings and link them to the output. It is possible to load image sequences, but only Image and Alpha values will be available, because the other outputs will not have any values associated with them. When a tracked clip is chosen, Blender will fulfill the outputs using internal values taken from the tracking. So the controls for start and end frames will be defined in the Movie Clip editor.

This node has no input sockets.

Used to select the movie clip. For controls see Data-Block Menu.

The first two sockets are the minimum output.

Outputs the entire image in the specified color space.

The alpha value taken from the movie or image.

The X offset value from the footage camera or tracking.

The Y offset value from the footage camera or tracking.

The scale of the image taken from the footage camera or tracking.

The lens angle taken from the footage camera or tracking.

--- Fin du chapitre: _movie_clip ---

--- Début du chapitre: _movie_distortion --- In the real world, all camera lenses produce some or the other sort of lens distortion. But, whatever we render has got no distortion. So, this node helps in removing distortion from movies or adding distortion to render to make our render blend in with the movie clip.

Usually, it is used while motion tracking.

Before using this node, one has to calculate the lens distortion of the clip. This can be done by adjusting K1, K2 and K3 values in Movie Clip Editor \xe2\x80\xa3 Properties \xe2\x80\xa3 Lens. For more information on how to edit those values, check this out.

Standard color input.

Used to select the movie clip whose distortion is to be used. This can be useful if more than one movie clips are present, each having a different distortion setting. For controls see Data-Block Menu .

Used to undistort the image received, and is usually used for the raw distorted movie clip.

Used to distort the image received, and is usually used for rendered images.

The image after distorting/undistorting.

Although, both, distortion of render and undistortion of movie clip are possible, and produce similar results, there is a difference between these two methods.

There are two kinds of lens distortion possible and, in simple terms, they can be said as:

When the movie clip is bulging out.

When the movie clip is bulging in.

For the first case, it is recommended to distort the render and leave the movie clip as it is, because, undistorting the movie clip will require extra pixel information, which is not available to Blender. Similarly, in the second case, it is recommended to undistort the movie clip and leave the render as it is, because, distorting the render will require those extra unavailable pixels. Doing the wrong method in the wrong case can create weird results around the edges, such as in the image shown.

Problems (notice the edges?) \xc2\xbf

--- Fin du chapitre: _movie_distortion ---

--- Début du chapitre: _multicam --- The Multicam Selector strip is used for multi-camera editing. Multi-camera editing is when a scene is recorded using multiple cameras from different angles and then edited together afterwards.

The process of multi camera editing can be rather easy in the Video Sequencer if properly setup. The following guide shows the basic steps to setup a basic multi camera editing workflow.

First you are going to want to add in each of your video strips.

Next, you will want to sync all your cameras by either using Audio Waveforms or by the movement of objects.

Tip

To make syncing strips easier you can group cameras, their audio, and their effects together using Meta Strips.

Split the editor into many Previews , one for each input track. Then change the Display Channel of each of the previews to the channel number of the input track.

Add a Multicam Selector strip above all the video channel tracks.

After completing these steps you should get something similar to the following image:

Multi-camera editing setup. \xc2\xbf

Now select the Multicam strip, if you take a look at the strip options (in the Sidebar), you will notice, that Multicam is a rather simple effect strip: It just takes a selected channel as its input. That is all. The magic comes with the convenient keyboard layout.

When you select the Multicam strip, the keys 1 to 9 are mapped to the cut buttons. So, select the Multicam strip and start playback and press the keys for the correct input while watching the individual cameras. You will end up with a small Multicam Selector strip for every cut. In reality, it boils down to: watch a few seconds to see, what is coming, watch it again and do a rough cut using the number keys. Then fine-tune the placement by selecting the outer handles of two neighboring Multicam for A/B rolling.

Tip

To improve playback performance enable Proxies.

The channel which the Multicam Selector gets its input from.

Cuts the Multicam strip at the current frame and changes the Source Channel automatically to the selected channels.

First you are going to want to add in each of your video strips.

Next, you will want to sync all your cameras by either using Audio Waveforms or by the movement of objects.

Tip

To make syncing strips easier you can group cameras, their audio, and their effects together using Meta Strips.

Split the editor into many Previews, one for each input track. Then change the Display Channel of each of the previews to the channel number of the input track.

Add a Multicam Selector strip above all the video channel tracks.

After completing these steps you should get something similar to the following image:

Multi-camera editing setup. \xc2\xba6

Now select the Multicam strip, if you take a look at the strip options (in the Sidebar), you will notice, that Multicam is a rather simple effect strip: It just takes a selected channel as its input. That is all. The magic comes with the convenient keyboard layout.

When you select the Multicam strip, the keys 1 to 9 are mapped to the cut buttons. So, select the Multicam strip and start playback and press the keys for the correct input while watching the individual cameras.

You will end up with a small Multicam Selector strip for every cut.

In reality, it boils down to: watch a few seconds to see, what is coming, watch it again and do a rough cut using the number keys. Then fine-tune the placement by selecting the outer handles of two neighboring Multicam for A/B rolling.

Tip

To improve playback performance enable Proxies.

--- Fin du chapitre: _multicam ---

--- Début du chapitre: _multiframe --- Multiframe pop-over. \xc2\xba6

Multiframe allows you to draw, edit, sculpt, or weight painting on several frames at the same time. Extremely useful to avoid repeating a task one frame at a time when animating.

When enabled, the effects on the strokes start to fall off from the current frame as defined by a curve widget.

Select the desired keyframes to draw, edit or sculpt at the same time.

Activate the Multiframe tool in the 3D Viewport \xe2\x80\x99s header with the toggle button (faded lines icon).

Once activated you can:

Select the points in all the selected keyframes and make your editions.

Start sculpting. The sculpt brushes will affect all the strokes in the selected keyframes.

Start weight painting. The weight paint brush will affect all the strokes in the selected keyframes.

Start Drawing. The new strokes will be added in all the selected keyframes. If you are using the Fill tool then it will be applied in all the selected keyframes.

When interpolating you can select the stroke from the different frames in the right order. Interpolate tool will use the selection order to calculate the correct stroke pairs.

Note

Not all operators support Multiframe mode.

--- Fin du chapitre: _multiframe ---

--- Début du chapitre: _multiplane_scrape ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Scrapes the mesh with two angled planes at the same time, creating a sharp edge between them. This is useful for creating & polishing hard surface objects.

Note

More info at General brush settings and on Advanced brush settings.

The angle between the two planes of the brush, pressing Ctrl inverts the angle.

When enabled, the angle is dynamically updated based on the surface under the brush. The Plane Angle then controls how much the angle will increase when applying pen pressure. When pressing Ctrl, it locks the plane angle to 0 degrees.

Displays a preview of the two planes and the angle they form during the stroke.

--- Fin du chapitre: _multiplane_scrape ---

--- Début du chapitre: _multiple_objects --- Unlike Edit Mode, there is no multi-object editing supported for Sculpt Mode. Since sculpting often involves editing many separate objects, it is recommended to use the shortcut Alt Q while pointing at other objects, for switching Objects quickly.

The advantage of using multiple objects is that each can have its own origin and modifiers. Splitting the geometry among multiple objects can also improve the sculpt mode performance. Alternatively objects can also be joined so there is no need to switch objects.

In the case that Face Sets were already used, joining objects or creating new geometry in Edit Mode will automatically assign new Face Sets. This makes it immediately possible to target each new geometry, for example via auto-masking. If no Face Sets are created, use the Initialize Face Sets operator to create them.

Face Sets and Masked geometry can also be extracted via Expand Mask or sliced into a new object via Mask Slice.

--- Fin du chapitre: _multiple_objects ---

--- Début du chapitre: _multiple_strokes --- The Multiple Strokes modifier generate multiple parallel strokes around the original ones.

The Multiple Strokes modifier. \xc2\xb6

The number of additional strokes.

Distance between the original and the duplicate strokes.

Control the offset position (inner or outer) for duplicate strokes.

When activated, the duplicate strokes fades out using their opacity or thickness.

Control the initial position for the fading.

Fade influence on strokes thickness.

Fade influence on strokes opacity.

SeeInfluence Filters.

--- Fin du chapitre: `_multiple_strokes` ---

--- Début du chapitre: `_multiply` --- The Multiplyeffect multiplies two colors.Blender uses values between (0.0 to 1.0) for the colors.This operation does not have to be normalized, the multiplication of two termsbetween (0.0 to 1.0) always gives a result between (0.0 to 1.0).

(With the `\xe2\x80\x9ctraditional\xe2\x80\x9d` representation of three bytes, like RGB(124, 255, 56),the multiplications give far too high results, like RGB(7316, 46410, 1848),that have to be normalized (brought back) by dividing them by 256to fit in the range of (0 to 255)\xe2\x80\xa6)

This effect has two main usages:

With a Mask

A mask is a black-and-white picture which, after multiplication with a `\xe2\x80\x9cnormal\xe2\x80\x9d` image,only show this one in the white areas of the mask (everything else is black).

The opening title sequence to James Bond movies,where the camera is facing down the barrel of a gun at James, is a good example of this effect.

With Uniform Colors

Multiplying a color with a `\xe2\x80\x9cnormal\xe2\x80\x9d` image allows you to soften some hues of this one(and so `\xe2\x80\x93` symmetrically `\xe2\x80\x93` to enhance the others).

For example, if you have a brown pixel RGB(0.50, 0.29, 0.05), andyou multiply it with a cyan filter (uniform color RGB(0.0, 1.0, 1.0)), you will get a color RGB(0.0, 0.29, 0.05).Visually, the result is to zero the reds and bring up (by `\xe2\x80\x9csymmetry\xe2\x80\x9d` `\xe2\x80\x93` the real values remain unchanged!)the blues and greens. Physically, it is the same effect as shining a cyan light onto a chocolate bar.

Emotionally,vegetation becomes more lush, water becomes more Caribbean and inviting, skies become friendlier.

Note

This effect reduces the global luminosity of the picture(the result will always be smaller than the smallest operand).If one of the images is all white, the result is the other picture;if one of the images is all black, the result is all black!

This strip has no options.

Multiply Effect. `\xc2\xb6`

--- Fin du chapitre: `_multiply` ---

--- Début du chapitre: `_multiply_matrices` --- The Multiply Matricesnode performs a matrix multiplicationon two input matrices.

The first multiplication.

The second multiplication.

This node has no properties.

The resulting matrix multiplication.

--- Fin du chapitre: `_multiply_matrices` ---

--- Début du chapitre: `_multiresolution` --- The Multiresolutionmodifier (often shortened to `Multires`) gives you the ability to subdivide a mesh similarly to the Subdivision Surface modifier, but also allows you to edit the new subdivision levels in Sculpt Mode.

Note

Multiresolution is the only modifier that cannot be repositioned in the stack after any modifier that will change geometry or other object data (i.e. all Generate, some Modify and some Simulate modifiers cannot come before the Multiresolution).

Deform modifiers will be applied onto the Multires subdivision levels instead of the base mesh, if they come after the Multires.

Tip

This is especially useful for re-projecting details from another sculpt with a Shrinkwrap modifier. For the best result make sure to set the wrap method to Project, snap mode to Above Surface and enable Negative.

The Multiresolution modifier. `Multires`

Set the level of subdivisions to show in the viewport.

Set the level of subdivisions to use specifically in Sculpt Mode. While in Sculpt mode use Alt- 1 to decrease the level or Alt- 2 to increase.

Set the level of subdivisions to show when rendering.

Deform the unsubdivided base mesh instead of the higher levels. Meanwhile the set level will be previewed. This allows you to make much broader changes in visual context to higher sculpted details without creating surface noise and artifacts.

Only display the edges of the original geometry. So when rendering the wireframe of this object, the wires of the subdivided edges will be skipped.

Creates a smooth level of subdivision (using the default Catmull-Clark algorithm).

Creates a level of subdivision with un-smoothed base mesh edges (using a simple interpolation by subdividing edges without any smoothing).

Creates a completely un-smoothed level of subdivision (using linear interpolation of the current sculpted displacement).

Rebuild a lower subdivision level of the current base mesh.

Deletes all subdivision levels that are higher than the current one.

Copy the shape of another object onto the multires levels by copying its vertex coordinates.

To use it, first select a different mesh object with matching topology and vertex indices, then Shift-select the object you wish to copy vertex coordinates to, and click Reshape.

Modifies the original unsubdivided mesh to match the form of the subdivided mesh.

Rebuilds all possible subdivisions levels to generate a lower resolution base mesh. This is used to create an optimized multiresolution version of a preexisting sculpt. This option is only available when no subdivision level have been created through the modifier.

Saves displacements to an external .btx file.

How precisely the vertices are positioned (relatively to their theoretical position), can be lowered to get a better performance when working on high-poly meshes.

How to handle UVs during subdivision.
UVs remain unchanged.
UV islands are smoothed, but their boundary remain unchanged.
UVs are smoothed, corners on discontinuous boundary and junctions of three or more regions are kept sharp.
UVs are smoothed, corners on discontinuous boundary, junctions of three or more regions and darts and concave corners are kept sharp.
UVs are smoothed, boundaries are kept sharp.
UVs and their boundaries are smoothed.
Controls how open boundaries (and corners) are smoothed.
Smooth boundaries, including corners.
Smooth boundaries, but corners are kept sharp.
Use the Weighted Edge Creases values stored in edges to control how smooth they are made.
Interpolates existing Custom Split Normals of the resulting mesh.
--- Fin du chapitre: _multiresolution ---

--- Début du chapitre: _multires_displacement_eraser ---Reference
Sculpt Mode
Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3
Advanced \xe2\x80\xa3 Brush Type
This brush deletes displacement information of the Multires Modifier, resetting the mesh to a regular subdivision surface result.
This can be used to reset parts of the sculpt or to fix reprojection artifacts after applying a Shrinkwrap Modifier.
Tip
This brush works best after using Apply Base .
Note
More info at General brush settings and on Advanced brush settings.
--- Fin du chapitre: _multires_displacement_eraser ---

--- Début du chapitre: _multires_displacement_smear ---Reference
Sculpt Mode
Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3
Advanced \xe2\x80\xa3 Brush Type
This tool deforms displacement information of the Multires Modifier, moving the displaced vertices without affecting the base mesh.
Smearing effect can be used multiple times over the same area without generating any artifacts in the topology.
Tip
This brush works best after using Apply Base .
Note
More info at General brush settings and on Advanced brush settings.
Deformation type that is used by the brush.
Pulls the displacement values in the direction of the brush.
Pulls the displacement values towards the center of the brush, creating hard surface effects without pinching the topology.
Pushes the displacement values away from the brush center, smoothing the displacement.
--- Fin du chapitre: _multires_displacement_smear ---

--- Début du chapitre: _musgrave --- The Musgrave texture node was replaced by the Noise Texture node, which includes all the same functionality.

The Dimension input was replaced by a Roughness input, where
\\(Roughness = Lacunarity^{ -Dimension} \\).

The Detail input value must be subtracted by 1 compared to the old
Musgrave Texture node.

--- Fin du chapitre: _musgrave ---

--- Début du chapitre: _named_attribute --- The Named Attributenode
outputs the data of an attribute based on the context of where it is
connected (theField Context).

The name of the attribute to read.

The data type used for the retrieved data. Attribute Search can be used
to give a basic list of possible attribute names and data types. When a
value is chosen from the search menu, the datatype is set to
automatically choose the data type from the geometry nodes result.

The attribute data stored on the geometry.

True if the attribute accessed by the node is present in the connected
context.

--- Fin du chapitre: _named_attribute ---

--- Début du chapitre: _naming ---Reference

All Modes

Properties \xe2\x80\xa3 Bone Properties

You can rename your bones, either using the Namefield in the Bones
Properties. It is also possible to rename by double-clicking bones in the
Outliner.

Blender also provides you some tools that take advantage of bones named
in a left/rightsymmetry fashion, and others that automatically name the
bones of an armature.

Naming conventions in Blender are not only useful for you in finding the
right bone, but also to tell Blender when any two of them are
counterparts.

In case your armature can be mirrored in half (i.e. it is bilaterally
symmetrical), it is worthwhile to stick to a left/right naming
convention. This will enable you to use some tools that will probably save
you time and effort (like the X-Axis Mirrored editing tool).

An example of left/right bone naming in a simple rig. \xc2\xb6

First you should give your bones meaningful base-names, like

\xe2\x80\x9cleg\xe2\x80\x9d, \xe2\x80\x9carm\xe2\x80\x9d,
\xe2\x80\x9cfinger\xe2\x80\x9d, \xe2\x80\x9cback\xe2\x80\x9d,
\xe2\x80\x9cfoot\xe2\x80\x9d, etc.

If you have a bone that has a copy on the other side (a pair), like an
arm, give it one of the following separators:

Left/right separators can be either the second position \xe2\x80\x9cL
_calfbone\xe2\x80\x9d or last-but-one \xe2\x80\x9ccalfbone
.R\xe2\x80\x9d.

If there is a lower or upper case \xe2\x80\x9cL\xe2\x80\x9d,

\xe2\x80\x9cR\xe2\x80\x9d, \xe2\x80\x9cleft\xe2\x80\x9d or

\xe2\x80\x9cright\xe2\x80\x9d, Blender handles the counterpart

correctly. See below for a list of valid separators. Pick one and stick to
it as close as possible when rigging; it will pay off.

Examples of valid separators:

(nothing): handLeft \xe2\x80\x93 handRight

\xe2\x80\x9c_ \xe2\x80\x9d (underscore): hand _L \xe2\x80\x93 hand _R

\xe2\x80\x9c. \xe2\x80\x9d (dot): hand .l \xe2\x80\x93 hand .r

\xe2\x80\x9c- (dash): hand -l \xe2\x80\x93> hand -r
\xe2\x80\x9c (space): hand LEFT \xe2\x80\x93> hand RIGHT
Note

Note that all examples above are also valid with the left/right part placed before the name. You can only use the short

\xe2\x80\x9cL/ \xe2\x80\x9cR/ code if you use a separator (e.g. \xe2\x80\x9chandL/ \xe2\x80\x9chandR/ will not work!).

Before Blender handles an armature for mirroring or flipping, it first removes the number extension, e.g. \xe2\x80\x9c.001/.

You can copy a bone named \xe2\x80\x9cbla.L/ and flip it over using Flip Names. Blender will name the copy

\xe2\x80\x9cbla.L.001/ and flipping the name will give you \xe2\x80\x9cbla.R/.

Reference

Edit Mode

Armature \xe2\x80\x93 Names \xe2\x80\x93 Auto-Name Left/Right, Front/Back, Top/Bottom

The three AutoName entries of the Armature \xe2\x80\x93 Names menu allow you to automatically add a suffix to all selected bones, based on the position of their root relative to the armature's origin and its local coordinates:

Will add the \xe2\x80\x9cL/ suffix to all bones with a positive X coordinate root, and the \xe2\x80\x9cR/ suffix to all bones with a negative X coordinate root. If the root is exactly at 0.0 on the X axis, the X coordinate of the tip is used. If both joints are at 0.0 on the X axis, the bone will just get a period suffix, with no \xe2\x80\x9cL/ \xe2\x80\x9cR/ (as Blender cannot decide whether it is a left or right bone).

Will add the \xe2\x80\x9cBk/ suffix to all bones with a positive Y coordinate root, and the \xe2\x80\x9cFr/ suffix to all bones with a negative Y coordinate root. The same as with AutoName Left-Right goes for 0.0 Y coordinate bones.

Will add the \xe2\x80\x9cTop/ suffix to all bones with a positive Z coordinate root, and the \xe2\x80\x9cBot/ suffix to all bones with a negative Z coordinate root. The same as with AutoName Left-Right goes for 0.0 Z coordinate bones.

Reference

Edit Mode

Armature \xe2\x80\x93 Names \xe2\x80\x93 Flip Names

You can flip left/right markers (see above) in selected bone names. This can be useful if you have constructed half of a symmetrical rig (marked for a left or right side) and duplicated and mirrored it, and want to update the names for the new side. Blender will swap text in bone names according to the above naming conventions, and remove number extensions if possible.

--- Fin du chapitre: _naming ---

--- Début du chapitre: _navigating --- As with most editors, you can:

Pan the view vertically (channels) and horizontally (time) by dragging MMB.

Zoom in and out by rolling Wheel or dragging Ctrl- MMB.

You can also use the scrollbars for this.

Shows or hides the Sidebar Region .

Displays a pop-up panel to alter the properties of the last completed operation. See Adjust Last Operation .

Shows or hides the Channels region (the list of animated property names on the left).

Pans and zooms the view to focus on the selected keyframes.

Pans and zooms the view to show all keyframes.

Reset the horizontal view to the current scene frame range, taking the preview range into account if it is active.

Pans the view so the Playhead is in the center.

Lets you filter by multiple search terms instead of just one (in the search textbox above the channel list and in the Filters popover). The terms are space-separated, so you can for example type `clock rot` to find all channels that have `clock` or `rot` in their name. If this option were disabled, the list would only show channels containing the text `clock rot`, of which there are likely none.

Whether to update other views (such as the 3D Viewport) while you're moving keyframes around. If disabled, the other views only get updated once you finish the move.

Sliders. 

Shows a value slider next to each channel. Adjusting such a slider automatically creates a keyframe.

Handle types. 

Displays keyframes using shapes that represent their Bézier handle type. In addition, if a keyframe uses a non-default interpolation type for the curve segment that comes after it, this is indicated by a green line. See Handles & Interpolation Display .

Extreme markers. 

Detects keys where the curve changes direction, and marks them by displaying an arrow inside their shape. Local maxima (hills) are shown as up arrows, while local minima (valleys) are shown as down arrows. A keyframe may show both arrows, namely when it's part of a summary row containing a channel with a maximum and one with a minimum. Automatically merge keyframes that end up on the same frame after transformation.

Shows the marker region (provided any markers have been defined). When disabled, the Marker menu is also hidden and marker operators are not available in this editor.

Shows timing in seconds instead of frames.

Synchronizes the horizontal panning and scale of the editor with other time-based editors that also have this option enabled. That way, they always show the same section of time.

Lets you drag a box to define a time range for previewing. As long as this range is active, playback will be limited to it, letting you repeatedly view a segment of the animation without having to manually rewind each time.

You can change the start or end frame using the corresponding button in the Timeline editor's Playback popover. Alternatively, you can simply run Set Preview Range again.

Clears the preview range.

Applies a preview range that encompasses the selected keyframes.

Changes the area's editor to the Graph Editor .

Area controls. See the user interface documentation for more information. These filters are available in the funnel dropdown button in the header.

Toggles the Summary row at the top of the Channels region. This row shows the union of all keyframes across all channels.

Only show keyframes belonging to objects/bones that are selected.

Note

If this option is enabled, the Dope Sheet may not show all material keyframes of the selected objects. Instead, it only shows the keyframes belonging to the selected nodes in the Shader Editor.

Show keyframes from objects/bones that are hidden.

Only show channels that have errors (for example, because they try to animate a property that doesn't exist on the object).

Filters the channel list by a search term (or multiple search terms if Multi-Word Match Search is enabled).

Select a collection to only show keyframes from objects in that collection.

Filter curves by property type.

Sorts data-blocks alphabetically to make them easier to find.

If your playback speed suffers because of this (should only really be an issue when working with lots of objects), you can turn it off.

--- Fin du chapitre: _navigating ---

--- Début du chapitre: _navigation --- There are different preferences for navigating the 3D Viewport in Blender. For painting and sculpting specific workflows it is recommended to use any of the following methods.

Center the View on the surface directly under the mouse position. This way the rotation point of the viewport can be manually changed to any point you wish to orbit around.

Center the View on the average position of the last stroke.

Various preferences can also make navigation more convenient. These can be found in the Navigation tab of the preferences.

Tumble the view based on the mouse position in your 3D Viewport while rotating. This makes it very easy to tilt the viewport freely, instead of having the Z axis of the viewport locked.

Use the mouse position to zoom towards and rotate around the surface that is pointed at. This can be an alternative to the repeated manual use of the Center View to Mouse operator.

The disadvantage is that this navigation preference can lead to accidental navigation around backfacing geometry or very distant geometry.

--- Fin du chapitre: _navigation ---

--- Début du chapitre: _newtonian --- Reference

Particle System Physics

Newtonian

The particles will move according to classical (Newtonian) mechanics. Particles start their life with the specified initial velocities and angular velocities, and move according to external forces. The response to environment and to forces is computed differently, according to the given integrator chosen by the animator.

Reference

Particle System Physics Forces

Specify the amount of Brownian motion. Brownian motion adds random motion to the particles based on a Brownian noise field. This is nice to simulate small, random wind forces.

A force that reduces particle velocity in relation to its speed and size (useful in order to simulate air drag or water drag).

Reduces particle velocity (deceleration, friction, dampening).

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Integration

Integrators are a set of mathematical methods available to calculate the movement of particles. The following guidelines will help to choose a proper integrator, according to the behavior aimed at by the animator.

Also known as \xe2\x80\x9cForward Euler\xe2\x80\x9d. Simplest integrator. Very fast but also with less exact results. If no dampening is used, particles get more and more energy over time. For example, bouncing particles will bounce higher and higher each time. Should not be confused with \xe2\x80\x9cBackward Euler\xe2\x80\x9d (not implemented) which has the opposite feature, the energy decrease over time, even with no dampening. Use this integrator for short simulations or simulations with a lot of dampening where speedy calculations are more important than accuracy.

Very fast and stable integrator, energy is conserved over time with very little numerical dissipation.

Also known as \xe2\x80\x9c2nd order Runge-Kutta\xe2\x80\x9d. Slower than Euler but much more stable. If the acceleration is constant (no drag for example), it is energy conservative. It should be noted that in example of the bouncing particles, the particles might bounce higher than they started once in a while, but this is not a trend. This integrator is a generally good integrator for use in most cases.

Short for \xe2\x80\x9c4th order Runge-Kutta\xe2\x80\x9d. Similar to Midpoint but slower and in most cases more accurate. It is energy conservative even if the acceleration is not constant. Only needed in complex simulations where Midpoint is found not to be accurate enough. The amount of simulation time (in seconds) that passes during each frame. The number of simulation steps per frame. Subframes to simulate for improved stability and finer granularity in simulations. Use higher values for faster-moving particles.

The following options are only available for Fluid type physics:

Automatically set the number of subframes.

A tolerance value that allows the number of subframes to vary. It sets the relative distance a particle can move before requiring more subframes.

The number of steps per frame will be at least Subframes + 1. More subframes may be simulated if the fluid becomes turbulent, according to the Threshold.

Reference

Particle System \xe2\x80\xa3 Physics \xe2\x80\xa3 Deflection

Use the particle size in deflections.

Kill particle when it hits a deflector object.

If set, particles collide with objects from the collection.

--- Fin du chapitre: _newtonian ---

--- Début du chapitre: _nodes --- Cycles applies a number of shader node optimizations both at compile time and run-time. By exploiting them it is possible to design complicated \xe2\x80\x9cUber Shader\xe2\x80\x9d style node groups that incur minimal render time overhead for unused features.

As the first step in preparing a node shader for execution, Cycles expands all node groups, as if using the Ungroup tool, and discards UI only features like frames and reroute nodes.

After that, it applies some obvious transformations, for example, it can (the list is not exhaustive):

Replace the following nodes with the constant result of their evaluation, if all their inputs are determined to be constant:

RGB, Value, Mix RGB, Math, Vector Math, RGB to BW, Gamma, Bright Contrast, Invert, Separate/Combine RGB/XYZ/HSV, Blackbody, RGB Curves, Vector Curves, Color Ramps.

Detect Mix RGB, Math and Vector Math nodes that become no-op (without Clamp) or evaluate to 0 as a result of addition, subtraction, multiplication, division or dot/cross product with a known constant 0 or 1 input, and replace with the appropriate input link or constant result.

Eliminate Mix RGB Mix (without Clamp) and Mix Shader nodes when Factor is known to be 0 or 1 by replacing with the appropriate input value or link. Eliminate no-op Mix RGB (except Color Burn, Color Dodge, Lighten, or enabled Clamp), Invert, RGB Curves and Vector Curves nodes with known zero Factor.

Eliminate Emission and Background shader nodes that do not emit any light, and Add Shader nodes with one or both input arguments missing. Eliminate Bump with constant Height input, using its Normal input or Geometry Normal instead. This is useful for implementing node group inputs that default to normal via routing through a no-op Bump before doing math.

Replace Attribute nodes of the View Layer type with the evaluated attribute value (it is constant within the whole Render Layer).

Combine multiple copies of the same node with the same inputs into only one instance.

Finally, any nodes that end up not connected either directly or indirectly to the Output node are removed.

When executing shaders, a special optimization is applied to Mix Shader nodes. If Factor evaluates to 0 or 1, any nodes that are only reachable via the unused branch of the mix are not evaluated.

This can substantially reduce the performance cost of combining multiple materials in one shader with a Color Attribute, texture, or other input used as a switch.

If Open Shading Language is chosen as the rendering backend, node shaders are translated to OSL code and then compiled and executed by the OSL runtime. In the process it applies its own extensive set of optimizations, both at compile time and run-time.

Open Shading Language can optimize out Script nodes if their outputs are unused or constant, even if their OSL shaders have side effects like debug tracing and message passing, which may be confusing. For that reason message passing with setmessage and getmessage should generally not be used for passing information forward in the graph; explicitly passing information through sockets should be preferred.

--- Fin du chapitre: _nodes ---

--- Début du chapitre: _nodes_support --- Most nodes are taken from Cycles. However, some features are missing and may (or may not) be implemented in EEVEE in the future.

See also

Shader Nodes .

These nodes are only available if EEVEE is the active render engine. These nodes will not work in Cycles.

EEVEE supports the conversion of BSDF outputs into color inputs to make a wide variety of custom shading. This is supported using the Shader to RGB node. This node evaluates the lighting of the BSDFs connected to it just like a Blended material and inherits its limitation.

This node implements the specular workflow found in other render engines. If something is not listed here, it is supported.

In the general case, shader nodes should behave more or less like in Cycles. So be sure to check out the Cycles section of this manual for that.

See also

Materials .

Although most BSDFs are supported, many of them are approximations and are not feature complete.

Roughness is not supported. Only Lambertian diffusion is supported.

Only supports GGX and Multiple Scatter GGX distribution. See Raytracing limitations.

Only supports GGX and Multiple Scatter GGX distributions.

Random Walk sampling, IOR and Anisotropic are not supported.

Colored and additive transparency are only compatible with blended modes.

Does not diffuse the light inside the object. It only lights the object with reversed normals.

Cumulative limitations from Diffuse BSDF, Glossy BSDF, Refraction BSDF and Subsurface Scattering. Anisotropy is not supported. The Sheen layer is a crude approximation.

See Volume Limitation.

The anisotropy parameter will be mixed and averaged for all overlapping volumetric objects, which is not physically correct and differs from Cycles. Also see Volume Limitation.

Same as Volume Scatter. See Volume Limitation.

Partially supported, using dithered mode may give incorrect results.

Not supported.

Not supported.

Not supported.

Not supported.

Not supported.

The Only Local option is not supported.

Pointiness is not supported.

Random per Island is not supported.

Defaults to active UV layer. Only `density`, `color`, `flame` and `temperature` built-in Geometry attributes are supported. UVs and Color Attributes are supported. Only up to 8 Object or Instancer attributes per material (both types share the same limit), and 512 View Layer attributes per scene are supported.

Not supported.

The Random output uses a different RNG algorithm. Range and statistical distribution of the values should be the same but the values will be different.

EEVEE has no real concept of rays. But in order to ease the workflow between Cycles and EEVEE some of the outputs are only supported in

particular cases. This node makes it possible to tweak indirect lighting in the shader.

Is Camera: Supported.

Is Shadow: Supported.

Is Diffuse: Set to 1.0 when baking light probe volume. Otherwise is set to 0.0.

Is Glossy: Set to 1.0 when baking light probe sphere or plane. Otherwise is set to 0.0.

Is Singular: Not supported. Same as Is Glossy.

Is Reflection: Not supported. Same as Is Glossy.

Is Transmission: Not supported. Same as Is Glossy.

Ray Length: Not supported. Defaults to 1.0.

Ray Depth: Not supported. Defaults to 0.0.

Diffuse Depth: Partially supported. Set to 1.0 when baking light probe volume. Otherwise is set to 0.0.

Glossy Depth: Partially supported. Set to 1.0 when baking light probe sphere or plane. Otherwise is set to 0.0.

Transparent Depth: Not supported. Defaults to 0.

Transmission Depth: Not supported. Same as Glossy Depth.

Note

Is Glossy does not work with Screen Space Reflections/Refractions but does work with reflection planes (whether used with SSR or not).

Not supported.

From Instancer is not supported.

From Instancer is not supported.

Pixel size option does not give exactly the same output as Cycles. The width can be a bit different.

Most texture nodes are supported except for the exceptions listed below:

Not supported.

Smart Interpolation always uses Cubic interpolation. Artifact present using Tube or Sphere projection with linear interpolation. This is due to hardware mip-mapping and Anisotropic filtering. This kind of artifact will be also visible if the texture coordinates provided are not continuous. Using Box projection with Extend type set to Clip or Extend is not supported. Instead, it will always use Repeat.

Not supported.

In Nishita mode, the Sun Disc property is not supported.

Not supported.

--- Fin du chapitre: `_nodes_support` ---

--- Début du chapitre: `_node_wrangler` --- Node Wrangler provides various tools that help you to work with nodes quickly and efficiently.

While many of this add-on's functions work in all supported node editors (Compositor, Shader, Geometry Nodes, and Texture Nodes) some functions only work in specific node editors, and some functions work differently per editor. Functions that only work in specific editors are marked with labels (Compositor, Shader, Geometry Nodes, Texture Nodes). Functions without labels should work for all node editors.

Open Blender and go to Add-ons section of the Preferences.

Search `Node Wrangler` and check the Enable Extension checkbox.

Use the panel in Sidebar of the node editor or press Shift-W to bring up the quick access menu. You can also look up the shortcut list in the add-on preferences panel.

You can access most functions from the sidebar panel or quick access menu.

Reference

Alt- RMB -drag, Shift- Alt- RMB -drag

Connect two nodes without even clicking the sockets. Just drag the cursor from one node to another while holding Alt- RMB. It will select the nodes nearest the start and end points of the drag for connection, so you don't even have to click on the nodes.

Selection can be lazy.

It tries to connect the best-matched sockets possible, based on their names, types, and whether they are open or not.

For a more precise connection, you can alternatively use Shift- Alt- RMB. It brings up menus of available inputs and outputs before connection, so you can select the exact sockets to connect. It's especially useful when working with a large node tree since you can make connections without frequently zooming in and out.

Reference

Shift- Ctrl- RMB -drag

Connect the outputs of two nodes into an appropriate Mix type of node. This is the lazy way of selecting nodes and executing the Mix function from Merge with Automatic Type Detection.

Reference

Node Wrangler Merge Selected Nodes

Connect outputs of the selected nodes into a Mix type of node (Mix, Math, Z-Combine, Alpha Over, Mix Shader, Add Shader, Join Geometry).

Note

Merge currently does not support outputs of Integer, String, or Boolean types from Geometry Nodes.

There are automatic and manual ways of merging. The automatic ways let the add-on determine which Mix node to use based on the types of outputs to merge. The manual ways let you decide and force connections even if the types of outputs and the Mix node are not compatible.

Note

Generally, the modifier part of the shortcut signifies the type of Mix node you want to use (Ctrl for automatic detection, Ctrl- Alt for the Mix node, and Shift- Ctrl for the Math node), the non-modifier part signifies the mode of Mix node you want to set (Numpad Plus for add, Numpad Minus for subtract, Numpad Slash for divide, and Numpad Asterisk for multiply).

The automatic merge functions determine the type of Mix node to use based on the types of outputs to merge. If it has a Color output, it will use the Mix node. It will use the Math node if both outputs are of Value type. Add Shader, Mix Shader, and Join Geometry nodes will also be used for specific cases.

Merge into Mix or Math nodes, then set blend mode or math operation as Add. If the outputs are Shaders, it will use Add Shader node instead. Merge into Mix or Math nodes, then set blend mode or math operation as Multiply.

Merge into Mix or Math nodes, then set blend mode or math operation as Subtract.

Merge into Mix or Math nodes, then set blend mode or math operation as Divide.

Merge into Mix node, then set blend mode as Mix. If the outputs are Shaders, it will use Mix Shader node instead. If the outputs are Geometry, it will use Join Geometry node.

Reference

Node Wrangler \xe2\x80\xa3 Merge Selected Nodes \xe2\x80\xa3 Use Mix Nodes

Use the Mix nodes for merging, regardless of the selected nodes. You can choose the mode of the node via the menu. You can quickly set some operations by using corresponding shortcuts.

Add: Ctrl- Alt- = , Ctrl- Alt- =

Subtract: Ctrl- Alt- Minus , Ctrl- Alt- NumpadMinus

Multiply: Ctrl- Alt- 8 , Ctrl- Alt- NumpadAsterisk

Divide: Ctrl- Alt- Slash , Ctrl- Alt- NumpadSlash

Reference

Node Wrangler \xe2\x80\xa3 Merge Selected Nodes \xe2\x80\xa3 Use Math Nodes

Use the Math nodes for merging, regardless of the selected nodes. You can choose the mode of the node via the menu. You can quickly set some operations by using corresponding shortcuts.

Add: Shift- Ctrl- = , Shift- Ctrl- =

Subtract: Shift- Ctrl- Minus , Shift- Ctrl- NumpadMinus

Multiply: Shift- Ctrl- 8 , Shift- Ctrl- NumpadAsterisk

Divide: Shift- Ctrl- Slash , Shift- Ctrl- NumpadSlash

Greater than: Ctrl- Comma

Less than: Ctrl- Period

Compositor

Reference

Node Wrangler \xe2\x80\xa3 Merge Selected Nodes \xe2\x80\xa3 Use Z-Combine Nodes

Ctrl- NumpadPeriod

Use the Z-Combine nodes for merging. If possible, Image and Z-Depth outputs will be linked. If the current node editor is not Compositor, this will execute the Mix function from the automatic merge.

Compositor

Reference

Node Wrangler \xe2\x80\xa3 Merge Selected Nodes \xe2\x80\xa3 Use Alpha Over Nodes

Ctrl- Alt- 0

Use the Alpha Over nodes for merging. If the current node editor is not Compositor, this will execute the Mix function from the automatic merge.

Reference

Node Wrangler \xe2\x80\xa3 Batch Change

Change the blend mode or math operation of the selected Mix and Math nodes at once. You can use Alt- Up or Alt- Down to cycle through previous or next blend modes or math operations. You can also quickly set some operations by using corresponding shortcuts.

Add: Alt- = , Alt- =

Subtract: Alt- Minus , Alt- NumpadMinus

Multiply: Alt- 8 , Alt- NumpadAsterisk

Divide: Alt- Slash , Alt- NumpadSlash

Greater than: Alt- Comma

Less than: Alt- Period

Reference

Alt- Left , Shift- Alt- Left , Alt- Right , Shift- Alt- Right , Shift- Ctrl- Alt- Left , Shift- Ctrl- Alt- 0 , Shift- Ctrl- Alt- Right , Shift- Ctrl- Alt- 1

Change the Factor value of the selected Mix and Mix Shader nodes with shortcuts.

Increase Factor by 0.1:Alt- Right

Decrease Factor by 0.1:Alt- Left

Increase Factor by 0.01:Shift- Alt- Right

Decrease Factor by 0.01:Shift- Alt- Left

Set Factor to 0.0:Shift- Ctrl- Alt- Left,Shift- Ctrl- Alt- 0

Set Factor to 1.0:Shift- Ctrl- Alt- Right,Shift- Ctrl- Alt- 1

Reference

Node Wrangler \xe2\x80\xa3 Delete Unused Nodes

Alt- X

Clean up your node tree. Delete all nodes that don't contribute to the final result.

Reference

Node Wrangler \xe2\x80\xa3 Swap Links

Alt- S

When two nodes are selected, this swaps each other's output link. Note that some output connections can be lost if the two nodes have a different number of connected outputs.

With one node selected, if the node has one linked input, it cycles the link through the available input sockets. If the node has two linked inputs, it swaps those two links. If there are more than two inputs linked, it swaps the two inputs with matching types (the Mix node's two Color inputs, for example).

Swap works differently depending on the selected nodes and their links.

Compositor

Reference

Node Wrangler \xe2\x80\xa3 Reset Backdrop

Z

Reset the position and scale of the backdrop.

Shader

Reference

Header \xe2\x80\xa3 Add \xe2\x80\xa3 Input \xe2\x80\xa3 Attributes

Add an Attribute node with the selected attribute.

ShaderGeometry Nodes

Reference

Shift- Ctrl- LMB for Shader, Shift- Alt- LMB for Geometry Nodes

Connect an output of the selected node to the final output of the node tree (the Material Output or World Output for Shader, the final Group Output for Geometry Nodes) to preview its output in the viewport. You can cycle through the available outputs by clicking it again while holding the modifier keys.

See also

While in Shader, any output can be connected to the final output, in Geometry Nodes, only Geometry outputs can be connected to the final output. To preview other types of outputs in Geometry Nodes, use its own Viewer Node .

See also

Also check out Connect to Output. It is a similar function but has different behaviors. It also works in all node editors.

Reference

Node Wrangler \xe2\x80\xa3 Frame Selected

Shift- P

Insert the selected nodes into a Frame node.

CompositorShaderTexture Nodes

Reference

Node Wrangler \xe2\x80\xa3 Reload Images

Alt- R

Reload all of the images used in the node tree. This lets you reload the images without using the Image Editor.

Reference

Node Wrangler \xe2\x80\xa3 Copy to Selected \xe2\x80\xa3 Settings from Active

Shift- C

Copy the settings of the active node to all selected nodes of the same type.

Reference

Backspace

Revert the settings of the selected nodes to default while maintaining connections.

Reference

Node Wrangler \xe2\x80\xa3 Copy to Selected \xe2\x80\xa3 Copy Label

Shift- V , Shift- C

Copy custom labels to all of the selected nodes. You can copy them from the active node (Shift- V), from the nodes that are linked to the selected ones, or from the names of the sockets that the selected nodes are linked to. Shift- C will bring up a submenu with all available options.

Reference

Node Wrangler \xe2\x80\xa3 Clear Label

Alt- L

Clear the custom labels of selected nodes and revert them back to their default node names.

Reference

Node Wrangler \xe2\x80\xa3 Modify Labels

Shift- Alt- L

Batch rename the custom labels of selected nodes. You can add text to the beginning and the end and replace parts of the text.

Shader

Reference

Node Wrangler \xe2\x80\xa3 Add Texture Setup

Ctrl- T

Add a setup of a texture node, Texture Coordinate, and Mapping nodes to any shader node. If you select a texture node, it will only add the Texture Coordinate and Mapping nodes. For a background shader it will add an Environment Texture node.

Shader

Reference

Node Wrangler \xe2\x80\xa3 Add Principled Setup

Shift- Ctrl- T

Add a principled texture setup from the selected texture files. Select a Principled BSDF node, select Add Principled Setup from the quick access menu (or press Shift- Ctrl- T), and select texture files. It automates the

process of adding Image Texture nodes, loading images, selecting the appropriate Color Space, and connecting their outputs to the Principled BSDF node.

It detects the type of textures by looking at their file names. You can edit the tags used for this matching process in the add-on preferences.

Setting up these textures can take dozens of clicks, even with Node Wrangler's other tools. With Principled Texture Setup, you can reduce that to a few clicks.

Reference

Node Wrangler \xe2\x80\xa3 Add Reroutes
Slash

Add reroute nodes to each output of the selected nodes.

Reference

Node Wrangler \xe2\x80\xa3 Link Active to Selected
Backslash

Link the active node to the selected nodes based on various criteria.

Link the active node to all selected nodes. (K) You can force it to replace existing links. (Shift- K)

Link only to the selected nodes that have the same label as the active node. (\')

You can force it to replace existing links. (Shift- \')

Link only when the name of the outputs matches the name or label of the selected nodes. (;) You can force it to replace existing links. (Shift- ;)

This is handy for replacing sources at the same time. (For example, connecting outputs from Render Layer to image (multi-layer EXR) in Compositor.)

Reference

Node Wrangler \xe2\x80\xa3 Align Nodes
Shift- =

Align the selected nodes horizontally or vertically. The effect is similar to scaling nodes on an axis (S X 0 or S Y 0), but it places the nodes at an even distance.

] \xe2\x80\x93 Select all direct child nodes of the selected frame.

[\xe2\x80\x93 Select the direct parent frame node of the selected nodes.

Reference

Node Wrangler \xe2\x80\xa3 Detach Outputs
Shift- Alt- D

Detach the selected node's outputs while leaving linked inputs intact.

Reference

Node Wrangler \xe2\x80\xa3 Connect to Output
O

Connect the output of the selected node to the final output of the node tree (Composite in Compositor, Material Output or World Output in Shader, the final Group Output in Geometry Nodes, Output in Texture Nodes), or, if the node is inside a group, to the Group Output.

Compositor Shader

Reference

Add \xe2\x80\xa3 Input for Compositor, or Add \xe2\x80\xa3 Texture for Shader

Add an Image Sequence by only selecting one image from a sequence of image files. It will automatically detect the length of the sequence and set the node appropriately.

Compositor Shader

Reference

Add \xe2\x80\xa3 Input for Compositor, or Add \xe2\x80\xa3 Texture for Shader

Select multiple images and add a node for each image. (Useful for importing multiple render passes or renders for image stacking.)

See also

Please see the old Wiki for the archived original docs.

Reference

Node

Various tools to enhance and speed up node-based workflow.

Node editor \xe2\x80\xa3 Sidebar or see the shortcuts of individual tools.

node_wrangler.py

Bartek Skorupa, Greg Zaal, Sebastian Koenig, Christian Brinkmann, Florian Meyer

GPL

This add-on is bundled with Blender.

--- Fin du chapitre: _node_wrangler ---

--- Début du chapitre: _noise ---Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

The Noise Texture node evaluates a fractal Perlin noise at the input texture coordinates. It can be used for a single Perlin noise evaluation, or for combining multiple octaves (layers) with increasingly finer detail. The inputs are dynamic, they become available if needed depending on the node properties.

Texture coordinate to evaluate the noise at; defaults to Generated texture coordinates if the socket is left unconnected.

Texture coordinate to evaluate the noise at.

Scale of the base noise octave.

Number of noise octaves. The fractional part of the input is multiplied by the magnitude of the highest octave. Higher number of octaves corresponds to a higher render time.

Blend between a smoother noise pattern, and rougher with sharper peaks.

The difference between the scale of each two consecutive octaves. Larger values corresponds to larger scale for higher octaves.

An added offset to each octave, determines the level where the highest octave will appear.

An extra multiplier to tune the magnitude of octaves.

Amount of distortion.

The dimensions of the space to evaluate the noise in.

Evaluate the noise in 1D space at the input W.

Evaluate the noise in 2D space at the input Vector. The Z component is ignored.

Evaluate the noise in 3D space at the input Vector.

Evaluate the noise in 4D space at the input Vector and the input W as the fourth dimension.

Note

Higher dimensions corresponds to higher render time, so lower dimensions should be used unless higher dimensions are necessary.

If enabled, ensures that the output values stay in the range 0.0 to 1.0. When disabled, output values are in the range -1.0 to 1.0.

Type of Noise texture, with different ways to combine octaves.

Fractal Brownian motion, produces a homogeneous and isotropic result. Values from octaves are added together.

More uneven, varying by location similar to real terrain. Values from octaves are multiplied together.

Creates peaks and valleys with different roughness values, like real mountains rise out of flat plains. Combines octaves using both addition and multiplication.

Creates sharp peaks. Calculates the absolute value of the noise, creating canyons and ridges, and then flips the surface upside down. Similar to Hybrid Multifractal creates a heterogeneous terrain, but with the likeness of river channels.

Value of fractal noise.

Color with different fractal noise in each component.

Noise Texture with high detail. \xc2\xba

fBM (fractal Brownian Motion). \xc2\xba

Multifractal. \xc2\xba

Hybrid Multifractal. \xc2\xba

Heterogeneous Terrain. \xc2\xba

Ridged Multifractal. \xc2\xba

While the noise is random in nature, it follows a certain pattern that might not evaluate to random values in some configurations. For instance, consider the following configuration where a grid of objects have a material that evaluates a noise texture at their locations. One might expect the objects to have random values since they have different locations, but this is not the case.

An example configuration where the noise evaluates to a constant value. \xc2\xba

It seems all objects have a value of 0.5. To understand why this happens, let us look at the following plot of a 1D noise texture.

A plot of a 1D noise with zero details and zero distortion. \xc2\xba

The horizontal line denotes a value of 0.5 and the vertical lines denote whole numbers assuming a noise scale of 1. As can be seen, the noise always intersects the 0.5 line at whole numbers. Since the aforementioned objects were distributed on a grid and have whole number locations, they all evaluate to 0.5. Which explains the issue at hand.

Generally, any discrete evaluation of noise at integer multiples of the reciprocal of the noise scale will always evaluate to 0.5. It also follows that evaluations closer to that will have values close to 0.5. In such cases, it is almost always preferred to use the White Noise Texture.

Regardless, one can mitigate this issue in a number of ways:

Adjust the scale of the noise to avoid aligning the noise with the evaluation domain.

Add an arbitrary offset to the texture coordinates to break the alignment with the evaluation domain.

Evaluate the noise at a higher dimension and adjust the extra dimension until a satisfactory result is achieved.

Constant value issue. \xc2\xba

Mitigating the issue by adjusting the scale. \xc2\xba

Mitigating the issue by adding an arbitrary offset. \xc2\xba

Mitigating the issue by evaluating at a higher dimension. \xc2\xba

Similarly, in other configurations, one might experience some banding patterns in the noise, where there are bands of high contrast areas followed by banding of low contrast areas. For instance, planar surfaces

that are slightly tilted along one of the axis will have such a banding pattern.

An example configuration where the noise have a banding pattern.

\xc2\xbb

This happens because the slight tilt along one of the axis causes values along the perpendicular axis to change very slowly making the grid structure of the noise more apparent. The easiest way to mitigate this issue is to rotate the coordinates by an arbitrary amount.

Mitigating the issue by rotating the coordinates by an arbitrary amount.

\xc2\xbb

--- Fin du chapitre: `_noise` ---

--- Début du chapitre: `_normal` --- The Normal node returns a vector for each evaluated point indicating the normal direction. The output can depend on the attribute domain used in the node evaluating the field, but the output is always a normalized unit vector.

On the face domain, the normal is the direction of the face.

For mesh vertices, the normal is an average of the surrounding face normals. If the vertex does not have any connected faces, the output is simply the normalized position of that vertex.

The normal output for each edge is the average of the edge's two vertex normals.

The output for each face corner is the same as the face normal of the corresponding face.

The output of this node when used for curve geometry is the evaluated normal of the curve, which depends on the twist method. The normal vector is always perpendicular to the direction of the curve's path at every point.

Warning

For NURBS and Bezier spline curves, keep in mind that the value retrieved from this node is the value at every control point, which may not correspond to the visible evaluated points. For NURBS splines the difference may be even more pronounced and the result may not be as expected. A Resample Curve Node can be used to create a poly spline, where there is a control point for every evaluated point.

This node has no inputs.

This node has no properties.

Vector indicating the normal of each geometry element.

--- Fin du chapitre: `_normal` ---

--- Début du chapitre: `_normalize` --- Find the minimum and maximum values of a single channel. Then map the values to a range of 0 and 1. This is mostly useful for the Z-buffer.

Standard value input.

This node has no properties.

Standard value output.

--- Fin du chapitre: `_normalize` ---

--- Début du chapitre: `_normals` --- See also

The Normal Edit Modifier can be used to edit normals.

The Weighted Normal Modifier can be used to affect normals by various methods, including Face Strength (see below).

You can also copy normals from another mesh using Mesh Data Transfer (operator or modifier).

Reference

Edit Mode

Mesh \| Normals \| Flip

This will reverse the normals direction of all selected faces. Note that this allows you to precisely control the direction (not the orientation, which is always perpendicular to the face) of your normals, as only the selected faces are flipped.

Reference

Edit Mode

Mesh \| Normals \| Recalculate Outside and Mesh
\| Normals \| Recalculate Inside

Shift- N and Shift- Ctrl- N

These tools will recalculate the normals of selected faces so that they point outside (respectively inside) the volume that the face belongs to. The volume does not need to be closed; inside and outside are determined by the angles with adjacent faces. This means that the face of interest must be adjacent to at least one non-coplanar other face. For example, with a Grid primitive, recalculating normals does not have a meaningful result.

Reference

Edit Mode

Mesh \| Normals \| Set from Faces

Set the custom normals at corners to be the same as the face normal that the corner is part of.

Reference

Edit Mode

Mesh \| Normals \| Rotate

R N

This is an interactive tool. As you move the mouse around, the selected normals are rotated. You can also invoke the Rotate Normals tool by pressing the Rotate transform key R, followed by N.

Reference

Edit Mode

Mesh \| Normals \| Point to Target

Alt- L

All selected normals are set to point from their vertex to the target after confirmed by Return or LMB.

A target is set by the keys:

The mouse cursor M

The pivot L

The object origin O

The cursor (set at this click) Ctrl- LMB

A mesh item selection (set by this click) Ctrl- RMB

The tool operation can be modified; if one of the following keys has been previously pressed:

All normals will point in the same direction: from the center of selected points to the target.

Each normal will be an interpolation between its original value and the direction to the target.

The normal directions are reversed from what was specified above.

Will reset the custom normals back to what they were when the operation started.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Merge

Merge all of the normals at selected vertices, making one average normal for all of the faces.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Split

Split the normals at all selected vertices so that there are separate normals for each face, pointing in the same direction as those faces.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Average

Average all of the normals in each fan of faces between sharp edges at a vertex.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Copy Vectors

If a single normal is selected, copy it to an internal vector buffer.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Paste Vectors

Replace the selected normals with the one in the internal vector buffer.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Smooth Vectors

Adjust the normals to bring them closer to their adjacent vertex normals.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Reset Vectors

Put normals back to the default calculation of the normals.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Select by Face Strength

Another way to affect normals is to set a Face Strength on the faces of the model. The Face Strength can be either Weak, Medium, or Strong. The idea is that the Weighted Normal Modifier can be set to pay attention to the Face Strength as follows: When combining the normals that meet at a vertex, only the faces with the strongest Face Strength will contribute to the final value.

For example, if three faces meet at a vertex and have the face weights weak, medium, and strong, then only the normal associated with the strong face will be used to set the final result.

Use the submenu to pick one of Weak, Medium, or Strong. Then this tool selects those faces that have the chosen face strength.

Reference

Edit Mode

Mesh \xe2\x80\xa3 Normals \xe2\x80\xa3 Set Face Strength

Use the submenu to pick one of Weak, Medium, or Strong. Then this tool changes the Face Strength of currently selected faces to the chosen face strength.

--- Fin du chapitre: _normals ---

--- Début du chapitre: `_normal_edit` --- The Normal Edit modifier affects (or generates) custom normals. It uses a few simple parametric methods to compute them (quite useful in game development and architecture areas), and mixes back those generated normals with existing ones.

Normal Edit Modifier. 

Aligns normals with the (origin, vertex coordinates) vector, in other words all normals seem to radiate from the given center point, as if they were emitted from an ellipsoid surface.

Makes all normals point (converge) towards a given target object.

Uses this object's origin as reference point when generating normals.

Optional in Radial mode, mandatory in Directional mode.

Makes all normals parallel to the line between both objects' origins, instead of converging towards target's origin.

Only relevant in Directional mode.

How to affect existing normals with newly generated ones.

Note that the Multiply option is not a cross product, but a faster component-by-component multiplication.

How much of the generated normals get mixed into existing ones.

Allows per-item fine control of the mix factor. The vertex group influence can be inverted by using the arrow button to the right.

Forbids new generated normals to have an angle to the original normal above that given threshold. This is useful to prevent extreme changes, that can even lead to inverting the front/back sides of a face, and consequently to shading artifacts.

Prevents flipping (reversing front/back sides) of polygons which normal does not match anymore the side to which point its corners' custom normals. Can also help to avoid shading issues.

Gives modified object's origin an offset before using it to generate normals.

Only relevant in Radial mode if no Target Object is set, and in Directional mode when Parallel Normals is set.

This modifier can be used to quickly generate radial normals for low-poly tree foliage or to fix shading of toon-like rendering by partially bending default normals.

Tip

More complex normal manipulations can be achieved by copying normals from one mesh to another, see the Data Transfer Modifier. Some shading effects can also make use of the Weighted Normals modifier.

Editing custom normals to point towards a given direction (blend-file). 

The left tree mesh has unmodified normals, while on the right one a Normal Edit modifier is used to bend them towards the camera. This shading trick is often used in games to fake scattering in trees and other vegetation.

--- Fin du chapitre: `_normal_edit` ---

--- Début du chapitre: `_normal_map` --- The Normal Map node generates a perturbed normal from an RGB normal map image. This is usually chained with an Image Texture node in the color input, to specify the normal map image. For tangent space normal maps, the UV coordinates for the image must match, and the image texture should be set to Non-Color mode to give correct results.

Strength of the normal mapping effect.

Strength is set to 0, 0.5, 1, 2 (from left to right). \xc2\xba6
RGB color that encodes the normal map in the specified space.
The input RGB color can be in one of three spaces: Tangent, Object and World space. Tangent space normal maps are the most common, as they support object transformation and mesh deformations. Object space normal maps keep sticking to the surface under object transformations, while World normal maps do not.

Name of the UV map to derive normal mapping tangents from. When chained with an Image Texture node, this UV map should be the same as the UV map used to map the texture.

Normal that can be used as an input to BSDF nodes.

The Normal Map Strength is set to 1. \xc2\xba6

--- Fin du chapitre: _normal_map ---

--- Début du chapitre: _nudge ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Moves vertices in the direction of the brush stroke.

Similar to the Snake Hook brush, but instead only moves the geometry along the surface (using the area plane). This is useful for grabbing geometry along curved surfaces, without making too impactful changes to the overall object shape.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: _nudge ---

--- Début du chapitre: _numeric_input --- Using the mouse for transformations is convenient, but if you require more precise control, you can also enter numeric values. After pressing the shortcut type a number to indicate the magnitude of the transformation. Then confirm or cancel. E.g. pressing S 2, Return will double the scale of an object.

By default and with no other key presses, the translation will occur along the X axis.

The rotation is in clockwise direction for positive values.

Scaling works in almost identical fashion to translation. The primary difference is that by default, scaling applies equally to all three axes.

You can see the numbers you enter in the 3D Viewport footer.

Numeric input displayed in the footer. \xc2\xba6

Tip

Numeric input can also be inputted in the Properties region.

Blender has two \xe2\x80\x9cmodes\xe2\x80\x9d a simple and an advanced one. Simple mode only accepts simple numbers. You can use basic text editing except selection.

Decimals can be entered by pressing Period.

Negate the whole value by pressing Minus.

Hitting Slash during number entry switches the number being entered to its reciprocal, e.g. 2/ results in 0.5 (1/2); 20/ results in 0.05 (1/20).

Hitting Backspace after having deleted all leading chars will first reset the edited value to initial state, and on second press, the whole number editing will be canceled, going back to usual transform with mouse.

To enter numeric values for multiple axes, use Tab or Ctrl-Tab. E.g. To move an object, one unit on all three axes press: G1 and Tab1 and Tab1 .

You can also combine numeric input with Axis Locking to limit movement to a particular axis or tool specific shortcuts.

In advanced mode you can additionally enter expressions and units.

Use = or NumPadAsterisk to enable advanced mode, and Ctrl- = or Ctrl-NumPadAsterisk to switch back to simple mode.

It features:

Units (cm , " , deg , etc.). See unit system .

Basic operations from Python (+ , * , / , ** , etc.).

Math constants and functions (pi , sin , sqrt , etc.). See

Python's math module.

You can still use the negate and inverse shortcuts (Minus, Slash), as well as non-number inputs, but you have to hold Ctrl to activate them.

--- Fin du chapitre: _numeric_input ---

--- Début du chapitre: _nvidia --- Blender uses of OpenGL for the 3D Viewport and user interface. The graphics card (GPU) and driver have a big impact on Blender's behavior and performance.

This section lists possible solutions for graphics glitches, problems with Eevee and Cycles, and crashes related to your GPU.

Upgrading to the latest graphics drivers often solves problems. Newer drivers have bug fixes that help Blender function correctly.

On macOS graphics drivers are built into the operating system and the only way to get newer drivers is to upgrade macOS as a whole to the latest version.

This means your graphics card and driver do not have the minimum required OpenGL 3.3 version needed by Blender.

Installing the latest driver can help upgrade the OpenGL version, though some graphics cards are simply too old to run the latest Blender. Using Blender 2.79 or earlier is the only option then.

Try running Blender from the command line, to see if any helpful error messages are printed.

On Windows, graphics drivers can sometimes get corrupted. In this case it can help to uninstall all graphics drivers (there may be multiple from Intel, AMD and Nvidia) and perform a clean installation with drivers from the manufacturer's website.

Update your graphics drivers (see above).

On laptops, make sure you are using a dedicated GPU (see above).

Try lowering quality settings in Preferences > System > Memory & Limits.

Try undoing settings in your graphics drivers, if you made any changes there.

See Eevee and Cycles documentation respectively.

See Invalid Selection, Disable Anti-Aliasing.

Running Blender inside a virtual machine is known to have problems when OpenGL drawing calls are forwarded to the host operating system.

To resolve this, configure the system to use PCI passthrough.

To find out which graphics card and driver Blender is using, use Help > Save System Info inside Blender. The OpenGL section will have information about your graphics card, vendor and driver version.

--- Fin du chapitre: _nvidia ---

--- Début du chapitre: _obj --- Reference

File > Import/Export > Wavefront (.obj)

OBJ format is a popular plain text format, however, it has only basic geometry and material support.

Note

There is no support for armatures, lights, cameras, empty objects, parenting, or transformations. See Compatibility for more information.

Import geometry and curves to the OBJ format.

If there is a matching .MTL for the OBJ then its materials will be imported too.

Value by which to scale the imported objects in relation to the world's origin.

OBJ-files often vary greatly in scale, this setting clamps the imported file to a fixed size.

Since many applications use a different axis for \xe2\x80\x98Up\xe2\x80\x99, these are axis conversion for these settings, Forward and Up axes \xe2\x80\x93 By mapping these to different axes you can convert rotations between applications default up and forward axes.

Blender uses Y Forward, Z Up (since the front view looks along the +Y direction). For example, it's common for applications to use Y as the up axis, in that case -Z Forward, Y Up is needed.

Import each OBJ \xe2\x80\x93object name\xe2\x80\x9d group (o) as a separate object.

Import each OBJ \xe2\x80\x93object name\xe2\x80\x9d group (g) as a separate object.

Import OBJ groups as vertex groups.

Check the imported mesh for corrupt data and fix it if necessary. When disabled, erroneous data may cause crashes displaying or editing the meshes. This option will make the importing slower but is recommended, as data errors are not always obvious.

Character used to separate an object's name into a hierarchical structure using Collections.

Export geometry and curves to the OBJ format.

Only export the selected objects. Otherwise export all objects in the scene.

Global scale to use on export.

Since many applications use a different axis for \xe2\x80\x98Up\xe2\x80\x99, there are axis conversion settings, Forward and Up axis \xe2\x80\x93 By mapping these to different axis you can convert rotations between applications default up and forward axis.

Blender uses Y Forward, Z Up (since the front view looks along the +Y direction). For example, it's common for applications to use Y as the up axis, in that case -Z Forward, Y Up is needed.

Write out the active UV layers coordinates from Blender.

Write out Blender's face and vertex normals (depending on the faces smooth setting).

Mostly this isn't needed since most applications will calculate their own normals but to match Blender's normal map textures you will need to write these too.

Write out the active vertex colors attribute layer, if present. Colors are exported in \xe2\x80\x93xyzrgb\xe2\x80\x9d OBJ extension format.

Write out NURBS curves as OBJ NURBS rather than converting to geometry.

Write out quads as two triangles. Some programs only have very basic OBJ support and only support triangles.

Export objects using the evaluated mesh, meaning the resulting mesh after all Modifiers have been calculated.

For properties that have different settings for the viewport/final render pick which is used for output. One example where this is important is the Subdivision Surface Modifier .

Use viewport properties.

Use final render properties.

Write out each Blender object as an OBJ object.

Note

Note that as far as Blender is concerned there is no difference between OBJ Groups and Objects, this option is only included for applications that treat them differently.

Generate an OBJ group for each part of a geometry using a different material.

Export the name of the vertex group of a face. It is approximated by choosing the vertex group with the most members among the vertices of a face.

Write Blender's sharp edges as smooth groups.

Generate Bitflags for smooth Groups.

Write out the MTL-file along with the OBJ. Most importers that support OBJ will also read the MTL-file.

Export MTL library using PBR extensions (roughness, metallic, sheen, clearcoat, anisotropy, transmission).

When referencing paths in exported files you may want some control as to the method used since absolute paths may only be correct on your own system. Relative paths, on the other hand, are more portable but mean that you have to keep your files grouped when moving about on your local file system. In some cases, the path doesn't matter since the target application will search a set of predefined paths anyway so you have the option to strip the path too.

Uses relative paths for files which are in a subdirectory of the exported location, absolute for any directories outside that.

Uses full paths.

Uses relative paths in every case (except when on a different drive on Windows).

Uses relative / absolute paths based on the paths used in Blender.

Only write the filename and omit the path component.

Copy the file on exporting and reference it with a relative path.

Exports a numbered OBJ for each frame from the start to the end frame. Please be aware that this can take quite a long time.

The first and last frame to export, used to determine the range of exported frames.

NURBS surfaces, text3D and metaballs are converted to meshes at export time.

--- Fin du chapitre: _obj ---

--- Début du chapitre: _object --- The friction of the surrounding medium. Generally friction dampens a movement. The larger the friction, the more viscous is the medium. Friction always appears when a vertex moves relative to its surround medium.

Mass value for vertices. Larger mass slows down acceleration, except for gravity where the motion is constant regardless of mass. Larger mass means larger inertia, so also braking a soft body is more difficult.

You can paint weights and use a specified vertex group for mass values.

--- Fin du chapitre: _object ---

--- Début du chapitre: _objects --- Objects panel.\xc2\xba6

This panel contains a list view with all objects which can be used for tracking, camera or object solving. By default there is only one object in this list which is used for camera solving. It cannot be deleted and other objects cannot be used for camera solving; all added objects are used for object tracking and solving only. These objects can be referenced from Follow Track and Object Solver constraints. Follow Track uses the camera object by default.

If some tracks were added and tracked to the wrong object, they can be copied to another object using Track \xe2\x80\xa3 Copy Tracks and Track \xe2\x80\xa3 Paste Tracks.

The usage for all kind of objects (used for camera and object tracking) is the same: track features, set camera data, solve motion. Camera data is shared between all objects and refining of camera intrinsics happens when solving camera motion only.

--- Fin du chapitre: _objects ---

--- Début du chapitre: _object_data --- Reference

Object Properties \xe2\x80\xa3 Visibility

See also

There are several other general visibility properties.

Enables the object to only receive shadow rays. It is to be noted that, shadow catcher objects will interact with other CG objects via indirect light interaction. This simplifies compositing CG elements into real-world footage.

Note

The Shadow Catcher outputs different results depending on if the Shadow Catcher pass is enabled in Render Layer settings. With the Shadow Catcher pass enabled, all indirect light interactions are captured. With it disabled, a simple approximation is used instead. The simple approximation is used in viewport rendering.

Example of the shadow catcher. Note how the material of the plane can still be viewed in the spheres. \xc2\xba6

Objects can be set to be invisible to particular ray types. This can be used, for example, to make an emitting mesh invisible to camera rays. For instanced objects, visibility is inherited; if the parent object is hidden for some ray types, the children will be hidden for these too.

In terms of performance, using these options is more efficient than using a shader node setup that achieves the same effect. Objects invisible to a certain ray will be skipped in ray traversal already, leading to fewer ray casts and shader executions.

Makes the object visible to the Camera; this includes the viewport \xe2\x80\x99s perspective in viewport rendering.

Makes the object visible in diffuse rays.

Makes the object visible in glossy rays.

Makes the object visible in transmission rays.

Makes the object visible in volumetric scatter rays.

Enables the object to cast shadows.

In order to activate these options the respective camera cull options have to be enabled in the scene simplify panel.

Ignore and this way make objects invisible to rays outside of the camera frustum.

Will cull any objects further from the camera than a given distance. When used in combination with camera frustum culling, this can be used to avoid culling nearby objects that are outside the camera frustum, but still visible in reflections. It is also useful to cull small objects far from the camera.

Reference

Properties \xe2\x80\xa3 Object Properties \xe2\x80\xa3 Motion Blur
Each object has its own motion blur settings along with the Scene Level Motion Blur. These settings can be found in the Object Properties tab of the Properties.

Controls accuracy of deformation motion blur, more steps uses more memory. The actual number of time steps is $\lfloor 2^{\text{steps} - 1} \rfloor$.

Enables motion blur for deformed meshes such as animated characters, including hair.

Warning

An object modifier setup that changes mesh topology over time can not render deformation motion blur correctly. Deformation blur should be disabled for such objects. Common examples of this are animated Booleans, Deformation before Edge Split, Remesh, Skin or Decimate modifiers.

Reference

Properties \xe2\x80\xa3 Object Properties \xe2\x80\xa3 Shading
Offset rays from the surface to reduce shadow terminator artifacts on low-poly geometry. Higher values affect more triangles, a value of one affecting all triangles and zero having no affect. The default value only affects triangles at grazing angles to light and should eliminate most artifacts.

Unlike the Shading Offset, this option has little affect on the lighting making it the preferable method to handle shadow terminator artifacts.

Pushes the shadow terminator (the line that divides the light and dark) towards the light to hide artifacts on low-poly geometry such as the ones below:

Shadow Terminator Artifacts. \xc2\xbb

Result of using an offset of 0.15. \xc2\xbb

Note

This property artificially alters the scene's lighting and is not energy conserving and consequently not physically accurate (see Geometry Offset instead).

Override for the world's Distance, if the value is zero the world's distance is used.

Mark objects as caustic casters or receivers. This is used in conjunction with a Light or World Shader with Shadow Caustics enabled to selectively speed up caustic rendering of objects in your scene.

Note

The rendering technique used to speed up the rendering of caustics is based on MNEE. There are a number of limitations with this technique and its implementation in Cycles:

Only refractive caustics in the shadows of objects work. Caustics from reflections or caustics that fall outside shadows are not rendered with this technique.

MNEE Caustics are an approximation of caustics and will produce physically inaccurate results in many situations. Examples include incorrect brightnesses and the incorrect representation of caustics caused by rough or curved surfaces.

In complex materials with multiple refractive BSDFs, MNEE will only generate caustics for one of the BSDFs.

Filter Glossy settings are ignored when using MNEE for refractive caustics.

MNEE Caustic rays can pass through up to 6 Caustic Caster surfaces between a Caustic Receiver and a Shadow Caustic light before the ray is terminated and caustics are ignored.

The Ambient Occlusion and Bevel nodes will not produce a valid result on objects that are Caustic Caster or Caustic Receiver while the scene contains an active Caustic Caster, Caustic Receiver, and Shadow Caustic Light.

MNEE Caustics only work if the caustic Caster has smooth normals.

Volumetric materials are not considered when calculating MNEE caustics.

Bump and normal maps are ignored when calculating caustics.

Mark an object as a caustic Caster.

Mark an object as a caustic Receiver.

Rendering caustics inside an eye without MNEE at 32 samples per pixel.

\xc2\xbf

Rendering caustics inside an eye using MNEE at 32 samples per pixel.

\xc2\xbf

Select the Light Group to add the current Object's Light to.

If the name input into the Light Group field does not align with an existing Light Group, then pressing this button will create a Light Group with that name and assign the selected Object's Light to it.

Limit light influence to specified objects, with Light Linking.

Collection of objects that will receive light emitted from the object.

Limit shadows to specified objects, with Light Linking.

Collection of objects that will act as shadow blockers for light emitted from the object.

--- Fin du chapitre: _object_data ---

--- Début du chapitre: _object_info --- The Object Info node gets information from objects. This can be useful to control parameters in the geometry node tree with an external object, either directly by using its geometry, or via its transformation properties.

An Object Info node can be added quickly by dragging an object into the node editor.

Object to get the properties from.

Output the entire object as single instance instead of realized geometry. This allows instancing non-geometry object types, because the output will contain an instance of the object.

The transformation of the vector and geometry outputs.

Output the geometry relative to the input object transform, and the location, rotation and scale relative to the world origin.

Bring the input object geometry, location, rotation and scale into the modified object, maintaining the relative position between the two objects in the scene.

Transformation Matrix containing the location, rotation and scale of the object.

Location of the object in world space.

Rotation of the object in world space.

Scale of the object in world space.

Geometry of the object in world space with all its modifiers applied.

--- Fin du chapitre: _object_info ---

--- Début du chapitre: `_object_solver` --- The Object Solver constraint gives the owner of this constraint, the location and rotation of the solved object motion.

The solved object motion is where Blender thinks the physical, real-world (tracked) object was, relative to the camera that filmed it.

Can be used to add a mesh to video for example.

Note

This constraint only works after you have set up a minimum of eight markers and pressed Solve object Motion. Located at Movie Clip Editor > Toolbar > Solve > Solve Camera Motion. If it says Solve Camera Motion instead of Solve Object Motion then go into the Movie Clip Editor > Sidebar region > Objects and switch it from the camera, to an object.

Object Solver Constraint panel. >

Receive tracking data from the scene's Active Clip. If unchecked, an option appears to choose from the other clips.

Select a tracked object to receive transform data from.

Select the camera to which the motion is parented to (if left empty the active scene camera is used).

Moves the origin of the object to the origin of the camera.

Moves the origin of the object back to the spot set in the Movie Clip Editor > Toolbar > Solve > Orientation > Set Origin.

Applies the constraint, creating keyframes for the transforms.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_object_solver` ---

--- Début du chapitre: `_ocean` --- The Ocean modifier is a tool to simulate and generate a deforming ocean surface, and associated texture, used to render the simulation data. It is intended to simulate deep ocean waves and foam.

It is a port from the open source Houdini Ocean Toolkit.

The Ocean modifier. >

Creates a tiled mesh grid that exactly corresponds with the resolution of the simulation data.

When generating a mesh surface, the existing mesh object is completely overridden with the ocean grid; this also includes any data generated from previous modifiers in the stack. A UV channel is also added, mapping the (0.0 to 1.0) UV space to the simulation grid.

Controls the number of times the grid is tiled in X and Y directions. UVs for these tiled mesh areas continue outside of the (0.0 to 1.0) UV space.

Uses the existing geometry rather than replacing it. Vertices are displaced along the local Z axis.

The main control of quality vs speed in the simulation engine. This determines the resolution of the internal 2D grids generated by the simulation for the 3D Viewport or the final render.

The internal grids are powers of two of the resolution value, so a resolution value of 16, will create simulation data of size 256x256. The higher the resolution, the more details will be produced, but the slower it will be to calculate.

Note

When using the Generate modifier geometry option, this resolution value also determines the resolution of the generated mesh surface, equal to the resolution of the internal simulation data.

The time at which the ocean surface is being evaluated. To create an animated ocean, you will need to animate this value. The speed that the time value is changing will determine the speed of the wave animation. The constant depth of the ocean floor under the simulated area. Lower values simulate shallower waters by producing higher frequency details and smaller waves.

A simple scaling factor that does not affect the height of the waves or behavior of the simulation.

The width of the ocean surface area being simulated, in meters. This also determines the size of the generated mesh, or the displaced area. Of course, you can scale the object with the Ocean modifier in Object Mode to tweak the apparent size in your scene.

A different Seed will produce a different simulation result.

Simulates extra normal map data. This can be used by the Ocean texture, when mapped to Normals, as a bump map, and enables generating normal map image sequences when baking.

An overall scale control for the amplitude of the waves. It approximates the height or depth of the waves above or below zero. Rather than just scaling the ocean object in Z, it scales all aspects of the simulation, displacement in X and Y, and corresponding foam and normals too.

A minimum limit for the size of generated waves. Acts similarly to a low-pass filter, removing higher frequency wave detail.

The choppiness of the wave peaks. With a choppiness of 0, the ocean surface is only displaced up and down in the Z direction, but with higher choppiness, the waves are also displaced laterally in X and Y, to create sharper wave peaks.

Wind speed in meters/second. With a low velocity, waves are restricted to smaller surface waves.

The directionality of the wave shapes due to wind. At a value of 0, the wind and waves are randomly, uniformly oriented. With higher Alignment values, the wind is blowing in a more constant direction, making the waves appear more compressed and aligned to a single direction.

When using Alignment, the direction in degrees that the waves are aligned to (using local X axis as reference).

When using Alignment, this will define the amount that inter-reflected waves are damped out. This has the effect of making the wave motion more directional (not just the wave shape).

With a Damping of 0.0, waves are reflected off each other in every direction, with a Damping of 1.0, these inter-reflected waves are damped out, leaving only waves traveling in the direction of the wind.

Simulates extra foam data.

This can be retrieved by the Ocean texture for use in texturing (perhaps as a mask), and enables generating foam map image sequences when baking.

Optional name for the vertex data layer, used by the Ocean Modifier to store foam maps as a Color Attribute. This is required for accessing the foam data in the renderer.

Tweaks the amount of foam covering the waves, negative values will reduce the amount of foam (leaving only the topmost peaks), positive values will add to it. Typically ranges from (-1.0 to 1.0).

Using foam Color Attributes with a named data layer.

Generate map of spray direction as a Color Attribute. This map can be used to define the velocities for spray particles.

Name of the Color Attribute used for the spray direction map.

Inverts the spray direction map.

Used to choose the wave spectrum model to use. Wave spectra are used to describe how energy moves through the waves at different frequencies. Energy travels through waves differently depending on the depth of the water and the wind speed.

Use for turbulent seas with foam (Phillips).

Use for a large area, established ocean where the ocean would extend for miles with wind blowing for days allowing the waves to reach a point of equilibrium (Pierson-Moskowitz method).

Similar to regular Established Ocean however, waves will continue to grow with time creating sharper peaks (JONSWAP and Pierson-Moskowitz method). An extra parameter is used to define the sharpness of these peaks.

Use for shallow water with depths less than about 10 meters which makes it great for small lakes and ponds without heavy wind (JONSWAP and TMA \xe2\x80\x93 Texel-Marsen-Arsloe methods).

Turbulent Ocean. \xc2\xba

Established Ocean. \xc2\xba

Established Ocean (Sharp Peaks). \xc2\xba

Shallow Water. \xc2\xba

An artificial factor to control how sharp the peaks of the waves are in the Established Ocean (Sharp Peaks) and Shallow Water spectrum models.

Distance from a lee shore, called the fetch, or the distance over which the wind blows with constant velocity. Used by Established Ocean (Sharp Peaks) and Shallow Water spectrum models.

Rather than simulating the ocean data live, it can be baked to files in a given directory. When a simulation is baked, the simulator engine is completely bypassed, and all information for the modifier or texture is retrieved from the baked files.

Baking has the following advantages:

It is faster to use the stored data rather than recalculating it.

It allows rendering of ocean data in external renderers.

It enables more advanced foam maps.

Simulation data is stored as sequences of OpenEXR image maps, one for each of displacement, normals, and foam (if enabled to be generated). Upon loading the data from these baked files, when a frame of the bake sequence is read, it is cached in memory. This means that accessing loaded frames subsequent times is fast, not incurring the overhead of drive access.

Since these baked files are plain OpenEXR \xe2\x80\x93s, they can also be opened and rendered in any other application or renderer that supports them.

Folder to store the baked EXR files in. The sequences will be in the form disp_####.exr, normal_####.exr, and foam_####.exr, where #### is the four digit frame number. If the cache path folder does not exist, it will be created.

Frames of the simulation to bake (inclusive). The start and end frames of the bake are repeated when accessing frames outside of the baked range.

Baking also provides improved foam capabilities. When simulating live, the ocean simulator retrieves data for that current frame only. In the case of the foam map, this represents the tips of wave crests for that given frame. In reality, after foam is created by wave

interactions, it remains sitting on the top of the wave surface for a while, as it dissipates. With baking, it is possible to approximate that behavior, by accumulating foam from previous frames, leaving it remaining on the surface.

The following example was created and rendered in Blender, notice how the peaks of the waves are white; an effect generated from the foam data.

--- Fin du chapitre: _ocean ---

--- Début du chapitre: _offset --- The Offset Modifier changes the strokes location, rotation or scale starting from the object origin.

Offset Modifier. \xc2\xba6

Sets strokes location offset from its object origin.

Sets strokes rotation.

Sets strokes scale.

Add random values to the individual strokes offset.

Offset by layers.

Offset by strokes (based on the stroke draw order).

Offset by Materials.

Sets individual element location offset.

Sets individual element rotation.

Sets individual element scale.

Use the same random Seed for each scale axis in the strokes for a uniform scale.

Seed used by the pseudo-random number generator.

The number of elements to be grouped and offset together.

Offset the element starting point.

See Influence Filters.

--- Fin du chapitre: _offset ---

--- Début du chapitre: _offset_corner_in_face --- Retrieves another corner in the same face as the input corner. This is like

\xe2\x80\x99rotating\xe2\x80\x9d the input corner around in its face.

Conceptually the operation is similar to the Offset Point in Curve Node.

The index of the input face corner.

Note

If this input is not connected, it uses the index of the context item, which means it\xe2\x80\x99s important that the node is evaluated in the Face Corner domain.

The number of corners to move around the face before finding the result, circling back to the first corner if necessary.

This node has no properties.

The index of the offset face corner.

--- Fin du chapitre: _offset_corner_in_face ---

--- Début du chapitre: _offset_edge_slide --- Reference

Edit Mode

Edge \xe2\x80\x93 Offset Edge Slide

Shift- Ctrl- R

Add two edge loops on either side of selected loops.

Extends the loop by creating triangles at endpoints.

Location of the loop relative to the center loop and the outside edge loops.

Only available for single edge loops. This matches the shape of the edge loop to one of the adjacent edge loops. (See Edge Slide tool for details.)

When Even is enabled, this flips the target edge loop to match. (See Edge Slide tool for details.)

Clamp within the edge extents.

Correct UV coordinates when transforming.

--- Fin du chapitre: `_offset_edge_slide` ---

--- Début du chapitre: `_offset_point_in_curve` --- The Offset Point in Curve node retrieves other points in the same curve as the input control point. This is like starting at a specific control point and walking along neighboring points toward the start or end of the curve.

Conceptually the operation is similar to the Offset Corner in Face Node, but the point index doesn't wrap around to the other end of the curve unless it is cyclic.

The index of the input control point.

Note

By default this uses the index from the field context, which makes it important that the node is evaluated on the point domain.

The number of points to move around the curve before finding the result. If the curve is cyclic and the offset goes past the start or end point of the curve, it will wrap around to the other side.

This node has no properties.

Whether the input control point plus the offset is a valid index of the original curve. Any offset in a cyclic curve is always valid.

The index of the offset curve point.

--- Fin du chapitre: `_offset_point_in_curve` ---

--- Début du chapitre: `_onion_skinning` --- Onion Skinning shows ghosts of the keyframes before and after the current frame allowing animators to make decisions in the animation sequence.

The main switch to show/hide Onion Skinning is in the Viewport Overlays, but Grease Pencil Onion Skinning is per-layer and the visibility can be toggled in the layer list. See 2D Layers for more information.

Onion Skinning panel.

Shows Keyframes in the range determined by the Before/ After settings.

Shows Frames in the range determined by the Before/ After settings.

Shows only on the manually selected keyframes in the Dope Sheet.

Control the opacity of the ghost frames.

Filters what type of frames to show in the Onion Skinning range.

Sets how many frames or keyframes, depending on the Mode, to show before and after the current frame.

Color to use before and after the current frame on ghost frames.

Opacity of the ghost frames decreases the further away from the current frame.

Help working on loop animations showing the first keyframe/frame as ghost when you are on the last frame of your animation.

An example of Onion Skinning activated.

--- Fin du chapitre: `_onion_skinning` ---

--- Début du chapitre: `_opacity` --- The Opacity Modifier changes the opacity (alpha) value of the stroke points.

The alpha value in Grease Pencil is stored per-point. The modifier can alter these values to go from totally transparent points to totally opaque points.

Opacity Modifier.

The color transformation will be applied to the stroke/fill color or stroke Hardness. When Hardness is selected the opacity affects the stroke's transparency (alpha) from the center to the border. Stroke & Fill, Stroke, Fill, or Hardness

When enabled, makes the opacity equal for the entire strokes.

Absolute opacity for the stroke points.

Controls the opacity value of the stroke points. A value of 1.0 respects the original alpha value of the points, a shift less than 1.0 makes the points more transparent than originally, and a shift greater than 1.0 makes the points more opaque than originally.

Sets value to 2.0 makes the points alpha fully opaque.

See Influence Filters.

Opacity Factor: 0.3

Opacity Factor: 1.0 (original alpha)

Opacity Factor: 2.0 (fully opaque)

--- Fin du chapitre: _opacity ---

--- Début du chapitre: _open_save --- Opening and saving blend-files is usually done using the File Browser.

Tip

Blend-files can also be opened by dragging and dropping blend-files into the Blender window. This method also allows to link/append the file.

Note

Unsaved Changes

By default, when exiting Blender or loading a new blend-file, if you have unsaved changes, a pop-up will ask you to either confirm discarding those changes, or save them.

This behavior can be disabled with the Save Prompt option in the Save & Load section of the Preferences.

Reference

File > Open

Ctrl- O

The upper text field displays the current directory path, and the lower text field contains the selected filename.

The File Browser in open configuration.

When enabled, the screen layout saved inside each blend-file is used, replacing the current layout and Workspaces. Otherwise the file screen layout is ignored.

Tip

If you want to work on a blend-file using your own defaults, start a fresh Blender, open the File Browser and turn off the Load UI button, and open the desired file.

When enabled, Python scripts and drivers that may be included in the file will be run automatically. Enable this only if you created the file yourself, or you trust that the person who gave it to you did not include any malicious code with it. See Python Security to configure default trust options.

Reference

File > Open Recent

Shift- Ctrl- O

Displays a list of recently opened blend-files. Hovering over items will show a preview, and information about the blend-file. Select any of the file names in the list to open that blend-file. When RMB on a listed item, a context menu will appear; One of the available options is Open File

Location, which will open that location in an OS file explorer or Finder window.

- Removes items from the recent files list.

- Choose which type of items to remove.

- Removes all recent files.

- Removes files from the list that cannot be found i.e. that have been moved or deleted.

Reference

- File > Recover > Last Session

This will load the quit.blend file Blender automatically saved just before exiting. This option enables you to recover your last work session if, for example, you closed Blender by accident.

Reference

- File > Recover > Auto Save

This will allow you to open an automatically saved file to recover it.

See also

- Auto Save

- Reference

- File > Save

Ctrl- S

Save current blend-file over itself (if it was not saved yet, this will automatically switch to Save As).

The File Browser in save configuration.

Reference

- File > Save Incremental

Ctrl- Alt- S

Save the current Blender file with a numerically incremented name that does not overwrite any existing files.

Reference

- File > Save As

Shift- Ctrl- S

Choose a file path to save the blend-file to.

Warning

If a file with the same given name already exists, the text field will turn red as a warning that the file will be overwritten.

Tip

Use the plus or minus buttons to the right of the file name, or Numpad Plus, Numpad Minus to increase/decrease a number at the end of the file name (e.g. changing file_01.blend to file_02.blend).

Reduces the file size of the resulting blend-file but takes longer to save and load. This option is useful for distributing files online and saving drive space for large projects. But it can cause slowdowns when quitting Blender, or under normal operation when auto-saving backup files. See Compression & Memory Use for more information.

Hint

The used compression algorithm is Zstandard. It is not unique to Blender so files can be compressed/decompressed with external tools.

Changed in version 3.0: Prior to this version, the compression algorithm used was Gzip. This means to open newer blend-files in versions prior to 3.0, blend-files must first be saved without compression in a newer version of Blender or decompressed using an external Gzip tool.

This option remaps Relative Paths (such as linked libraries and images) when saving a file in a new location.

Saves a copy of the actual working state but does not make the saved file active.

Reference

File \xe2\x80\xa3 Save Copy\xe2\x80\xa6

Choose a file path to save the blend-file to, but return to editing the original file upon completion. This can be used to save backups of the current working state without modifying the original file.

For options see Save As .

Many blend-files reference external images or other linked blend-files. A path tells Blender where to look for these files. If the external files are moved, the blend-file that references them will not look right.

When you specify one of these external files, the default option is to make the path relative. Blender stores a partial path evaluated relative to the directory location of the referencing blend-file. This choice helps when you need to reorganize folders or move your files.

With a relative path, you can move the blend-file to a new location provided the externally linked files are moved along with it. For example, you could send someone a folder that contains a blend-file and a subfolder of external images that it references.

When relative paths are supported, the File Browser provides a Relative Path checkbox, when entering the path into a text field, use a double slash prefix (//) to make it so.

Relative paths are the default but this can be changed in the File tab of the Preferences.

Note

You cannot use relative paths into a new untitled blend-file. Save it before linking to external files.

Hint

If it is necessary to relocate a blend-file relative to its linked resources, use Blender\xe2\x80\x99s File Save As function which has an option to Remap Relative file links.

--- Fin du chapitre: _open_save ---

--- Début du chapitre: _operators --- Reference

Help \xe2\x80\xa3 Operator Cheat Sheet

Enable Developer Extras

Creates a text file in the Text Editor that gives a list of all operators and their default values in Python syntax, along with the generated docs. This is a good way to get an overview of all Blender\xe2\x80\x99s operators.

See also

Blender\xe2\x80\x99s API documentation

Reference

All Modes

Topbar \xe2\x80\xa3 Blender \xe2\x80\xa3 System \xe2\x80\xa3 Reload Scripts

Reloads all scripts found in the scripts data folder; any changes that have been made in the Text Editor will be lost!

Reference

--debug-memory

Topbar \xe2\x80\xa3 Blender \xe2\x80\xa3 System \xe2\x80\xa3 Memory Statistics

This operator which can be found by searching \xe2\x80\x9cMemory Statistics\xe2\x80\x9dwith theOperator Searchwill print useful information about memory objects, their size and user count.

Important

To fully use this operator run Blender from the console with --debug-memory .

Reference

All Modes

Topbar \xe2\x80\xa3 Blender \xe2\x80\xa3 System \xe2\x80\xa3 Debug Menu
This operator brings up a menu to set Blender into a certain debug mode.
See the source codefor a description of what each value does.

Tip

Developers can search the code for G.debug_value to find other possible uses for this operator.

Note

Additional debug options are available by launching Blender in debug mode or setting bpy.app.debug=True .

Reference

All Modes

Topbar \xe2\x80\xa3 Blender \xe2\x80\xa3 System \xe2\x80\xa3 Redraw Timer

This operator brings up a menu with a list of teststo benchmark UI render times along with undo/redo functions.

Reference

All Modes

Topbar \xe2\x80\xa3 Blender \xe2\x80\xa3 System \xe2\x80\xa3 Clean Up Space-data

Removes unused settings for invisible editors.

--- Fin du chapitre: _operators ---

--- Début du chapitre: _options ---Reference

Properties \xe2\x80\xa3 Render \xe2\x80\xa3 Optionspanel.

Use backface culling to hide backsides of faces.

Render the scene transparent. With the slider you can control howtransparent the scene should appear.

Renders a sharp shadow in the scene.

Defines how dark the shadow should be rendered. This slider can be adjustedbetween 0 (shadow not visible) and 1 (shadow is black).

Controls the direction of the light source that casts the shadows.

Controls the Shadow termination angle. It can be used to limit self shadowing artifacts.

Controls the falloff near the edge of the shadow.

Highlight ridges and valleys in the scene geometry.

How to calculate cavities.

More precise but is slower to calculate.

Fast but does not take the size of the ridges and valleys into account.

Use both methods.

Control the visibility of ridges.

Control the visibility of valleys.

Use the Depth of Field settings of the active camera in the viewport.Only visible when looking through the camera.

The settings are located on Properties \xe2\x80\xa3 Camera \xe2\x80\xa3 Depth of Fieldpanel.

Render the outline of objects in the viewport. The color of the outline can be adjusted.

Render specular highlights.

Note

Only available when Lighting is set to StudioLighting or when a MatCaphas been selected that contains a specular pass.

--- Fin du chapitre: _options ---

--- Début du chapitre: _orientation ---Reference

Object and Edit Modes

Header \xe2\x80\xa3 Transform Orientation

Comma

The Transform Orientation determines the orientation of the Object Gizmo. Changing this orientation can make it easier to perform transformations in the direction you want.

With the default Global transform orientation (left) it's tricky to move the plane in the direction it's facing, but with Local (right) it's easy.

The Transform Orientation can be changed using a selector in the 3D Viewport's header:

Transform Orientation selector.

The orientation can also be changed temporarily while performing a hotkey-based transformation with axis locking. For example, if you first press G to start moving an object, then X to lock to the orientation's X axis, and finally a second time, you'll get a lock to an alternative orientation: the Local orientation if it was Global previously, and the Global orientation otherwise.

In addition to the builtin orientations, you can also define your own (see Custom Orientations below).

Align the transformation axes to world space. The world axes are shown by the Navigation Gizmo in the top right corner of the viewport, as well as the Grid Floor.

Align the transformation axes to the active object's orientation.

In Edit Mode, orient the transformation axes so that the Z axis of the gizmo matches the average Normal of the selected elements.

In Object Mode, this is equivalent to Local orientation.

Orient the transformation axes to visualize the workings of the object's Rotation Mode. This is specifically useful for the Euler modes, where the object is rotated one axis at a time: the rotation axes don't stay perpendicular to each other and might even overlap, a phenomenon known as gimbal lock that complicates animation. Align the transformation axes to the view (meaning they change as you orbit around):

X: Left/Right

Y: Up/Down

Z: Towards/Away from the screen

Align the transformation axes to the 3D Cursor .

Align the transformation axes to the Parent .

Default cube with Global transform orientation selected.

Rotated cube with Global orientation, gizmo has not changed.

Local orientation, gizmo matches the object's rotation.

Normal orientation, in Edit Mode. \xc2\xbf
Gimbal transform orientation. \xc2\xbf
View transform orientation. \xc2\xbf
Parent transform orientation. Cube parented to rotated empty. \xc2\xbf

Reference

Object and Edit Modes

Header \xe2\x80\xa3 Transform Orientation

You can define custom transform orientations using objects or mesh elements. Custom orientations defined from an object use the Local orientation of that object, whereas those defined from mesh elements (vertices, edges, faces) use the average Normal orientation of those elements.

Transform Orientation panel. \xc2\xbf

The Transform Orientation panel, found in the header of the 3D Viewport, can be used to select, add, remove, and rename transform orientations.

The default name for these orientations is derived from the selection. If it's an object it will take that object's name, if it's an edge it will be titled 'Edge', and so on.

To create a custom orientation, select an object or mesh element(s) and click the '+' button in the Transform Orientation panel.

Create Orientation Adjust Last Operation panel. \xc2\xbf

Right after creating the orientation, the Create Orientation Adjust Last Operation panel gives a few options:

Text field for naming the new orientation.

The new orientation will be aligned to the view space.

The new orientation stays selected.

If the new orientation is given an existing name, a suffix will be added to it's name to avoid overwriting the existing orientation, unless Overwrite Previous is checked, in which case it will be overwritten.

To delete a custom orientation, simply select it and click the 'X' button.

--- Fin du chapitre: _orientation ---

--- Début du chapitre: _origin --- Each object has an origin point. The location of this point determines where the object is located in 3D space. When an object is selected, a small circle appears, denoting the origin point. The location of the origin point is important when translating, rotating or scaling an object. See Pivot Points for more.

The color of the origin changes based on the selection state of the object.

Object is active.

Object is selected, but not active.

Object is not linked and not selected.

Object is linked.

Object is selected, linked, but not active.

Note

Colors are themeable and might appear different. The colors described here are from the default Dark Theme.

Reference

Object Mode

Object \xe2\x80\xa3 Set Origin

The object origin and geometry can be moved relative to each other and to the 3D cursor.

Moves the model to the origin and this way the origin of the object will also be at the center of the object.

Moves the origin to the center of the object.

Moves the origin of the model to the position of the 3D cursor.

Moves the origin to the calculated center of mass of model (assuming the mesh has a uniform density).

Median Point Center, Bounding Box Center

Tip

To transform an object's origin directly, enable **Affect Only Origins** in the Tool Settings Options.

--- Fin du chapitre: _origin ---

--- Début du chapitre: _osl --- Cycles Only

It is also possible to create your own nodes using Open Shading Language (OSL). These nodes will only work with the CPU and OptiX rendering backend.

To enable it, select Open Shading Language as the shading system in the render settings.

Note

Some OSL features are not available when using the OptiX backend.

Examples include:

Memory usage reductions offered by features like on-demand texture loading and mip-mapping are not available.

Texture lookups require OSL to be able to determine a constant image file path for each texture call.

Some noise functions are not available. Examples include Cell, Simplex, and Gabor.

The trace function is not functional. As a result of this, the Ambient Occlusion and Bevel nodes do not work.

OSL was designed for node-based shading, and each OSL shader corresponds to one node in a node setup. To add an OSL shader, add a script node and link it to a text data-block or an external file. Input and output sockets will be created from the shader parameters on clicking the update button in the Node or the Text editor.

OSL shaders can be linked to the node in a few different ways. With the Internal mode, a text data-block is used to store the OSL shader, and the OSO bytecode is stored in the node itself. This is useful for distributing a blend-file with everything packed into it.

The External mode can be used to specify a .osl file from a drive, and this will then be automatically compiled into a .oso file in the same directory. It is also possible to specify a path to a .oso file, which will then be used directly, with compilation done manually by the user. The third option is to specify just the module name, which will be looked up in the shader search path.

The shader search path is located in the same place as the scripts or configuration path, under:

Tip

For use in production, we suggest to use a node group to wrap shader script nodes, and link that into other blend-files. This makes it easier to make changes to the node afterwards as sockets are added or removed, without having to update the script nodes in all files.

For more details on how to write shaders, see the OSL Documentation.

Here is a simple example:

OSL is different from, for example, RSL or GLSL, in that it does not have a light loop. There is no access to lights in the scene, and the material must be built from closures that are implemented in the renderer itself. This is more limited, but also makes it possible for the renderer to do optimizations and ensure all shaders can be importance sampled. The available closures in Cycles correspond to the shader nodes and their sockets; for more details on what they do and the meaning of the parameters, see the shader nodes manual.

See also

Documentation on OSL's built-in closures.

```
diffuse(N)
oren_nayar(N, roughness)
diffuse_ramp(N, colors[8])
phong_ramp(N, exponent, colors[8])
diffuse_toon(N, size, smooth)
glossy_toon(N, size, smooth)
translucent(N)
reflection(N)
refraction(N, ior)
transparent()
microfacet_ggx(N, roughness)
microfacet_ggx_aniso(N, T, ax, ay)
microfacet_ggx_refraction(N, roughness, ior)
microfacet_beckmann(N, roughness)
microfacet_beckmann_aniso(N, T, ax, ay)
microfacet_beckmann_refraction(N, roughness, ior)
ashikhmin_shirley(N, T, ax, ay)
ashikhmin_velvet(N, roughness)
hair_reflection(N, roughnessu, roughnessv, T, offset)
hair_transmission(N, roughnessu, roughnessv, T, offset)
principled_hair(N, absorption, roughness, radial_roughness, coat, offset, IOR)
```

Used to simulate subsurface scattering.

method(string) Rendering method to simulate subsurface scattering. **burley**: An approximation to physically-based volume scattering. This method is less accurate than **random_walk** however, in some situations this method will resolve noise faster. **random_walk_skin**: Provides accurate results for thin and curved objects. Random Walk uses true volumetric scattering inside the mesh, which means that it works best for closed meshes. Overlapping faces and holes in the mesh can cause problems. **random_walk**: Behaves similarly to **random_walk_skin** but modulates the Radius based on the Color, Anisotropy, and IOR. This method thereby attempts to retain greater surface detail and color than **random_walk_skin**.

Rendering method to simulate subsurface scattering.

burley: An approximation to physically-based volume scattering. This method is less accurate than **random_walk** however, in some situations this method will resolve noise faster.

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N(vector) \xe2\x80\x93 Normal vector of the surface point being shaded.
radius(vector) \xe2\x80\x93 Average distance that light scatters below the surface.Higher radius gives a softer appearance, as light bleeds into shadows and through the object.The scattering distance is specified separately for the RGB channels,to render materials such as skin where red light scatters deeper.The X, Y and Z values are mapped to the R, G and B values, respectively.

albedo(color) \xe2\x80\x93 Color of the surface, or physically speaking, the probability that light is reflected for each wavelength.

henyey_greenstein(g)

absorption()

emission()

ambient_occlusion()

holdout()

background()

Geometry attributes can be read through the getattribute() function.This includes UV maps, color attributes and any attributes output from geometry nodes.

The following built-in attributes are available through getattribute() as well.

Automatically generated texture coordinates, from undeformed mesh.

Default render UV map.

Default tangent vector along surface, in object space.

Position before displacement, in object space.

For instances, generated coordinate from instancer object.

For instances, UV coordinate from instancer object.

Three vertex coordinates of the triangle.

Number of vertices in the polygon (always returns three currently).

Vertex coordinates array of the polygon (always three vertices currently).

Name of the object.

Is mesh face smooth or flat shaded.

Is object a curve or not.

0..1 coordinate for point along the curve, from root to tip.

Thickness of the curve in object space.

Length of the curve in object space.

Tangent Normal of the strand.

Is point in a point cloud or not.

Radius of point in point cloud.

Center position of point in point cloud.

Random number, different for every point in point cloud.

Ray distance since last hit.

Random number, different for every object instance.

Object unique instance index.

Object location.

Material unique index number.

Particle unique instance number.

Particle age in frames.

Total lifespan of particle in frames.

Location of the particle.

Size of the particle.
Velocity of the particle.
Angular velocity of the particle.

CPU Only

We support the `trace(pointpos,vectordir,...)` function, to trace rays from the OSL shader. The `csshade` parameter is not supported currently, but attributes can be retrieved from the object that was hit using the `getmessage("trace",...)` function. See the OSL specification for details on how to use this.

This function cannot be used instead of lighting; the main purpose is to allow shaders to `cprobe` nearby geometry, for example to apply a projected texture that can be blocked by geometry, apply more `cwear` to exposed geometry, or make other ambient occlusion-like effects.

Metadata on parameters controls their display in the user interface. The following metadata is supported:

Name of parameter in the user interface

Hide parameter in the user interface.

Display integer parameter as a boolean checkbox.

--- Fin du chapitre: `_osl` ---

--- Début du chapitre: `_other` --- Blender uses OpenGL for the 3D Viewport and user interface. The graphics card (GPU) and driver have a big impact on Blender's behavior and performance.

This section lists possible solutions for graphics glitches, problems with Eevee and Cycles, and crashes related to your GPU.

Upgrading to the latest graphics drivers often solves problems. Newer drivers have bug fixes that help Blender function correctly.

On macOS graphics drivers are built into the operating system and the only way to get newer drivers is to upgrade macOS as a whole to the latest version.

This means your graphics card and driver do not have the minimum required OpenGL 3.3 version needed by Blender.

Installing the latest driver can help upgrade the OpenGL version, though some graphics cards are simply too old to run the latest Blender. Using Blender 2.79 or earlier is the only option then.

Try running Blender from the command line, to see if any helpful error messages are printed.

On Windows, graphics drivers can sometimes get corrupted. In this case it can help to uninstall all graphics drivers (there may be multiple from Intel, AMD and Nvidia) and perform a clean installation with drivers from the manufacturer's website.

Update your graphics drivers (see above).

On laptops, make sure you are using a dedicated GPU (see above).

Try lowering quality settings in Preferences `System` `Memory & Limits`.

Try undoing settings in your graphics drivers, if you made any changes there.

See [Eevee](#) and [Cycles](#) documentation respectively.

See [Invalid Selection](#), [Disable Anti-Aliasing](#).

Running Blender inside a virtual machine is known to have problems when OpenGL drawing calls are forwarded to the host operating system.

To resolve this, configure the system to use PCI passthrough.

To find out which graphics card and driver Blender is using, use Help
Save System Info inside Blender. The OpenGL section will have
information about your graphics card, vendor and driver version.

--- Fin du chapitre: _other ---

--- Début du chapitre: _outline --- The Outline modifier convert strokes
to outline tracing all strokes perimeter with new strokes.

The Outline modifier. \xc2\xbb6

The thickness of the generated strokes outline.

The perimeter strokes are maintaining inside the original stroke
perimeter trying to keep the original shape.

controls the number of subdivision of the generated strokes outline.

Controls the accuracy of the perimeter conversion.

Defines the material to use on the generated strokes outline.

Controls the origin of the cyclic strokes generated.

See Influence Filters.

--- Fin du chapitre: _outline ---

--- Début du chapitre: _output --- Output panel. \xc2\xbb6

This panel provides options for setting the location of rendered frames
for animations, and the quality of the saved images.

Choose the location to save rendered frames.

When rendering an animation, the frame number is appended at the end of
the file name with four padded zeros (e.g. image0001.png). You can set a
custom padding size by adding the appropriate number of # anywhere in the
file name (e.g. image_##_test.png translates to image_01_test.png).

This setting expands Relative Paths where a // prefix represents the
directory of the current blend-file.

Adds the correct file extensions per file type to the output files.

Saves the rendered view layers and their passes to a multi-layer OpenEXR
image. The Compositor can then use this file to improve performance,
especially for heavy compositing.

The image is stored in the Render Cache folder as specified in the File
Paths Preferences. You can also load it back into the Image
Editor's Render Result, even after closing and reopening
Blender; see Open Cached Render.

Choose the file format to save to. Based on which format is used, other
options such as channels, bit depth and compression level are available.

For rendering out to images see: saving images, for rendering to videos
see the Encoding panel.

Choose the color format to save the image to. Note that RGBA will not be
available for all image formats.

BW, RGB, RGBA

Overwrite existing files when rendering.

Create empty placeholder frames while rendering.

Hint

Primitive Render Farm

An easy way to get multiple machines to share the rendering workload is
to:

Set up a shared directory over a network file system.

Disable Overwrite, enable Placeholder in the Render Output panel.

Start as many machines as you wish rendering to that directory.

This panel controls how Color Management is applied when saving images.

Uses the same color management settings defined by the active Scene. These properties are defined in the Render Settings

Uses custom color management settings defined by the properties below in the panel; disregarding any color management settings set at the Scene level.

For a detailed description of color management properties, see the Color Management page.

Reference

Properties \xe2\x80\xa3 Output \xe2\x80\xa3 Encoding

Encoding panel. \xc2\xb6

Here you choose which video container, codec, and compression settings you want to use. With all of these compression choices, there is a trade-off between file size, compatibility across platforms, and playback quality. In the header, you can use the presets, which choose optimum settings for you for that type of output.

Tip

When you view the System Console, you can see some of the output of the encoding process. You will see even more output if you execute Blender as `blender -d`.

Video container or file type. For a list of all available options, see video formats .

If your video is huge and exceeds 2GiB, enable Autosplit Output. This will automatically split the output into multiple files after the first file is 2GiB in size.

Chooses the method of compression and encoding. For a list of all available options see video formats.

Note

Standards

Some containers and codecs are not compatible with each other, so if you are getting errors check that your container and codec are compatible. Like containers and codecs are sometimes not compatible with each other, some codecs do not work with arbitrary dimensions. So, try to stick with common dimensions or research the limitations of the codec you are trying to use.

These are preset Rate.

Presets to change between a fast encode (bigger file size) and more compression (smaller file size).

The number of pictures per Group of Pictures. Set to 0 for \xe2\x80\x9cintra_only\xe2\x80\x9d, which disables inter-frame video. A higher number generally leads to a smaller file but needs a higher-powered device to replay it.

Enables the use of B \xe2\x80\x9frames.

The maximum number of B \xe2\x80\x9frames between non-B-frames.

Sets the average bit rate (quality), which is the count of binary digits per frame. See also: `FFmpeg -b:v`.

Video files can use what is called variable bit rate (VBR). This is used to give some segments of the video less compressing to frames that need more data and less to frames with less data. This can be controlled by the Minimum and Maximum values.

The decoder bitstream buffer size.

Maximum bit rate of the multiplexed stream. Multiplexing is the process of combining separate video and audio streams into a single file, similar to packing a video file and MP3 audio file in a zip-file.

Reduces data fragmentation or muxer overhead depending on the source.

These settings change how sound is exported while rendering. To control how sound is played back from within Blender, see the audio settings in the Preferences.

Audio format to use. For a list of all available options, see video formats.

The number of audio source locations to encode.
Output a single audio channel.

Output two audio channels; typically a left and right channel.

Output a four audio channels.

Output a five audio channels with one LFE channel.

Output a seven audio channels with one LFE channel.

Sets the audio sampling rate.

For each codec, you can control the bit rate (quality) of the sound in the movie. Higher bit rates are bigger files that stream worse but sound better. Use powers of 2 for compatibility.

Sets the output volume of the audio.

Tip

The choice of video format depends on what you are planning to do.

It's not recommended to render directly to a video format in the first instance. If a problem occurs while rendering, the file might become unplayable and you will have to re-render all frames from the beginning. If you first render out a set of static images such as the default PNG format or the higher-quality OpenEXR (which can retain HDR pixel data), you can combine them as an Image Strip in the Video Sequencer. This way, you can easily:

Restart the rendering from the place (the frame) where any problem occurred.

Try out different video encoding options in seconds, rather than minutes or hours as encoding is usually much faster than rendering the 3D scene. Enjoy the rest of the features of the Video Sequencer, such as adding Image Strips from previous renders, audio, video clips, etc.

Tip

You shouldn't post-process a lossy-compressed file as the compression artifacts may become visible. Lossy compression should be reserved as a final delivery format.

--- Fin du chapitre: _output ---

--- Début du chapitre: _overlays ---Reference

Overlays

Clicking the icon toggles all overlays in the Video Sequencer. The drop-down button displays a popover with more detailed settings, which are described below.

Shows an outline around the selected images.

Shows the 2D Cursor.

Shows the Frame Overlay for comparing the current frame to a reference frame.

Shows guides indicating the video area where content can be seen across all screens.

See also

Camera Safe Areas .

Shows file metadata.

Shows Annotations.

--- Fin du chapitre: _overlays ---

--- Début du chapitre: `_packed_data` --- Blender has the ability to encapsulate (incorporate) various kinds of data within the blend-file that is normally saved outside of the blend-file. For example, an image texture that is an external image file can be put inside the blend-file. This allows sharing a full project as a single file, instead of e.g. an archive containing the blend-file and all its dependencies.

You know that a data is packed when you see a little gift box icon displayed next to its path.

Warning

Not all external files can be packed

Some typically heavy external files, like videos from the Sequence Editor or Movie Clips, cannot be packed in a blend-file.

Reference

File > External Data > Pack Resources

Mark all eligible external resource files used by the blend-file as packed. Actual packing will happen on the next save of the blend-file.

Reference

File > External Data > Automatically Pack Resources

When enabled, this option will ensure that all eligible external resource files, existing or added later, are systematically marked as packed. As with Pack Resources, the blend-file must be saved to the drive for this to have an effect.

Disabling that option won't unpack anything, but future external files won't be automatically marked as packed anymore.

A single file can be packed by clicking on the little gift box icon to the left of its file-path UI widget.

Reference

File > External Data > Unpack Resources

Unpack all external resource files stored into a blend-file.

Unpacks all files in the same directory as the blend-file, grouping them in proper folders (like

`textures` for instance). However, if the final file exists already, it will use that file, instead of unpacking it.

As with previous option, but if the final file exists already, it will overwrite it.

Unpacks all files in their original location. However, if the final file exists already, it will use that file, instead of unpacking it.

As with previous option, but if the final file exists already, it will overwrite it.

Only deactivates the Automatically Pack Resources option.

A single file can be unpacked by clicking on the little gift box icon to the left of its file-path UI widget.

Just mark the file as unpacked, without actually writing it or reloading it from the drive.

Unpack the file at the proposed path, which is local to the current blend-file.

If the original file path still exists, mark it as unpacked. Note that it won't be automatically reloaded from the drive. (differs) or (identical) show difference status between the packed version and the one on-drive.

If the original file path still exists but differs from the packed version, mark it as unpacked and overwrite the on-drive file with the packed version.

If the original file path does not exist, mark it as unpacked and write it to drive.

Reference

File \xe2\x80\xa3 External Data \xe2\x80\xa3 Pack Linked Libraries

Mark all linked library files in the current blend-file as packed. Actual packing will happen on the next save of the blend-file.

Reference

File \xe2\x80\xa3 External Data \xe2\x80\xa3 Unpack Linked Libraries

Unpack all used linked library files from this blend-file.

--- Fin du chapitre: _packed_data ---

--- Début du chapitre: _pack_uv_islands --- The Pack UV Islands

Nodescales islands of a UV map and moves them so they fill the UV space as much as possible.

See also

The Pack Islands operator performs a similar operation in the UV editor. The UV map to modify.

Faces to consider when packing islands. UVs that are part of any other face will not be affected.

The distance to leave between UV islands.

Allow Rotating islands for best fit.

This node has no properties.

The modified UVs.

--- Fin du chapitre: _pack_uv_islands ---

--- Début du chapitre: _paint ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Paints on the active color attribute. Hold Shift to blur painted colors instead.

Color attribute \xe2\x80\x99s can be managed in the palette pop-over in the middle of the header.

Note

More information in the Painting Introduction .

This settings has a different effect on this brush. Instead of defining the strength of each individual step in the stroke, it determines the overall Opacity of the applied color.

Use the Flowsetting instead for faster increasing of strength.

Note

More info at General brush settings and on Advanced brush settings.

Amount of paint that is applied per stroke sample. Used to create fast/slow accumulation effect.

Amount of paint that is picked from the surface into the brush color. Can achieve the effect of a wet canvas.

Amount of wet paint that stays in the brush after applying paint to the surface.

Ratio between the brush radius and the radius that is going to be used to sample the color to blend in wet paint.

Amount of random elements that are going to be affected by this brush. Use this for a more detailed airbrush effect. This works best on a high resolution.

Scale of the brush tip in the X axis. This is useful for achieving a painting stroke like a marker or paint roller.

--- Fin du chapitre: `_paint` ---

--- Début du chapitre: `_painting` --- Sculpt Mode also allows painting your geometry via Color Attributes such as Vertex Colors. This ensures that the most common actions related to the sculpting workflow are contained in the same mode, to avoid unnecessary mode switching.

Other sculpt mode features such as face sets, masking and filters can also be used with painting tools.

The painting functionality in Sculpt Mode is limited to a Paint and Smear brush, as well as a Color Filter and Mask by Color tool.

Just like any other brush, Shift can be used to smooth. In the case of painting brushes it will blur the colors within the brush radius instead.

Note

Once any painting tool is executed, the viewport color shading is switched to `\x00\x0cAttribute\x00\x0d`. This ensures that color attributes are shown on all objects once painting is needed.

--- Fin du chapitre: `_painting` ---

--- Début du chapitre: `_panel` --- The purpose of 2D stabilization is to smooth out jerky camera handling on existing real-world footage. To activate the 2D stabilizer, you need to set the toggle in the panel, and additionally you need to enable Show Stable in the Clip Display pop-over. Then you `\x00\x99`ll need to set up some tracking points to detect the image movements.

The 2D Stabilization panel is used to define the data used for 2D stabilization of the shot. Several options are available in this panel: you may add a list of tracks to determine lateral image shifts and another list of tracks to determine tilting and zooming movements. Based on the average contribution of these tracks, a compensating movement is calculated and applied to each frame.

When the footage includes panning and traveling movements, the stabilizer tends to push the image out of the visible area. This can be compensated by animating the parameters for the intentional, `\x00\x0c`expected `\x00\x0d` camera movement.

Note

To activate the 2D stabilizer, you need to set the toggle in the panel, and additionally you need to enable Show Stable in the Clip Display pop-over.

2D Stabilization panel. `\xc2\xb6`

Reference point to anchor stabilization: other frames will be adjusted relative to this frame `\x00\x99`s position, orientation and scale. You might want to select a frame number where your main subject is featured in an optimal way.

In addition to location, stabilizes detected rotation around the rotation pivot point, which is the weighted average of all location tracking points.

Compensates any scale changes relative to center of rotation.

List of tracks to be used to compensate for camera jumps, or location movement.

List of tracks to be used to compensate for camera tilts and scale changes.

Finds smallest scale factor which, when applied to the footage, would eliminate all empty black borders near the image boundaries.

Limits the amount of automatic scaling.

Known relative offset of original shot, will be subtracted, e.g. for panning shots.

Rotation present on original shot, will be compensated, e.g. for deliberate tilting.

Explicitly scale resulting frame to compensate zoom of original shot.

The amount of transformation applied to the footage can be controlled. In

some cases it is not necessary to fully compensate camera jumps. The

amount of stabilization applied to the footage can be controlled. In some

cases you may not want to fully compensate some of the

camera's jumps. Please note that these parameters

influence parameters do control only the compensation

movements calculated by the stabilizer, not the deliberate movements added

through the Expected parameters.

The stabilizer calculates compensation movements with sub-pixel

accuracy. Consequently, a resulting image pixel needs to be derived from

several adjacent source footage pixels. Unfortunately, any interpolation

causes some minor degree of softening and loss of image quality.

No interpolation, uses nearest neighboring pixel. This setting basically

retains the original image's sharpness. The downside is we also

retain residual movement below the size of one pixel, and compensation

movements are done in 1 pixel steps, which might be noticeable as

irregular jumps.

Simple linear interpolation between adjacent pixels.

Highest quality interpolation, most expensive to calculate.

--- Fin du chapitre: _panel ---

--- Début du chapitre: _parent --- When modeling a complex object, such as a watch, you may choose to model the different parts as separate objects. To make all the parts move as one (the watch), you can designate one object as the parent of all the other parts. These other parts become its children, and any translation, rotation, or scale of the parent will also affect its children.

Contrary to most biological lifeforms, each object or bone in Blender has at most one parent. If an object already has a parent object and you give it another parent then Blender will remove the previous parent relationship. When the plural parents is used in this chapter, it references the hierarchy of parents, so the parent, the grandparent, great grandparent, and so on, of an object.

Reference

Object Mode

Object Parent

Ctrl- P

To parent objects, select at least two objects (select the child objects first, and select the parent object last), and press Ctrl- P . The Set Parent To menu will pop up allowing you to select from one of several possible different parenting types. Selecting one of the entries in Set Parent To confirms, and the child/children to parent relationship is created. The selected objects will have their

parent set to the active object ,and as a result will be siblings.

The Set Parent Topop-up menu is context-sensitive, which meansthe number of entries it displays can change depending on what objects are selectedwhen the Ctrl- P shortcut is used.

Moving, rotating or scaling the parent will also usually transform the child/children.Yet transforming the child/children of the parent will not affect the parent.In other words, the direction of influence is from parent to child and not child to parent.

Tip

You can move a child object back to its parent byclearing its origin.

Blender supports many different types of parenting, listed below.Besides parenting the selected objects, some types add a Modifier or Constraint to the child objects,with the parent as the target object or activates a parent property i.e. Follow Path.

Object

- Armature Deform

Bone

- Curve Deform

- Follow Path

- Path Constraint

- Lattice Deform

Vertex

Vertex (Triangle)

The object's current world transform (so its absolute location, rotation and scale in the world) is computed.The new parent is set, and then the Parent Inversematrix is computed such that after settingthe new parent the object is still at its previous world transform.

Hint

Use the Outliner

There is another way to see the parent-child relationship in groups and that is to use the Outlinerviewof theOutliner editor.

Blender can assign a parent without moving the child object.This is achieved via a hidden matrix called the Parent Inversematrix,which sits between thetransformof the parent and the child.

When objects are parented withCtrl- P, Parent Inverse matrix is updated.Depending on the choice in the Set Parent menu, the object's local location,rotation, and scale are also updated. For more details, seeObject Parent.

The Parent Inverse matrix can be cleared by usingClear Parent Inverse.

Note

When setting the parent via the Object Properties panel, the Parent Inverse matrix is always reset.This can cause an unexpected jump in the object's position.To avoid this, use Ctrl- P to set the new parent.

Object Parentis the most general form of parenting that Blender supports.It will take selected objects and make the active object the parent object of all the selected objects. Each child object will inheritthe transformations of the parent. The parent object can be of any type.

If the object has a preexisting parent, that is cleared first. This moves the object to its own location, rotation and scale, without its parent's influence.

There are three operators that allow you to set an object parent. They differ in the way they compute the Parent Inverse matrix and the local Transform of the object.

Object Parent with Keep Transform will keep any previous transformations applied to them from the previous parent object.

Assume that we have a scene consisting of three objects, those being two empty objects named EmptyA and EmptyB, and a Monkey object. Fig. Scene with no parenting shows the three objects with no parenting relationships active on them.

Scene with no parenting.

If you select the Monkey object by LMBclick and then Shift-LMBclick EmptyA object and press Ctrl-P and finally select Object from the Set Parent Topop-up menu. This will result in EmptyA object being the parent object of the Monkey object. With only EmptyA selected rotating/scaling/moving it will result in the Monkey object being altered respectively.

Scale the EmptyA object, so that the Monkey becomes smaller and moves to the left a little.

The monkey is the child object of

EmptyA.

If you select only the Monkey object by LMBclick and then Shift-LMBclick EmptyB object and press Ctrl-P and select Object from the Set Parent Topop-up menu. This will result in EmptyB object being the parent object of the Monkey object. Notice that when you change the parent of the Monkey the scale of the Monkey changed.

The monkey is the child object of

EmptyB.

This happens because the Monkey object never had its scale altered directly, the change came about because it was the child of EmptyA which had its scale altered. Changing the Monkey's parent to EmptyB resulted in those indirect changes in scale being removed, because EmptyB has not had its scale altered.

This is often the required behavior, but it is also sometimes useful that if you change your parent object that the child object keep any previous transformations it got from the old parent object; If instead when changing the parent object of the Monkey from EmptyA to EmptyB we had chosen parenting type Object and enable Keep Transform, the Monkey would keep its scale information it obtained from the old parent EmptyA when it is assigned to the new parent EmptyB.

The Object parent with Keep Transform.

If you want to follow along with the above description here is the blend-file:

File: Parent_-_Object_(Keep_Transform)_(Demo_File).blend.

Bone parenting allows you to make a certain bone in an armature the parent object of another object. This means that when transforming an

armature the child object will only move if the specific bone is the child object of moves.

Three pictures of armatures with four bones. \xc2\xbf

In Fig. Three pictures of armatures with four bones. with the 2nd bone being the bone parent of the child object cube. The cube is only transformed if the 1st or 2nd bones are. Notice altering the 3rd and 4th bones has no effect on the cube.

To use bone parenting, you must first select all the child objects you wish to parent to a specific armature bone, then Shift-LMB select the armature object and switch it into Pose Mode and then select the specific bone you wish to be the parent bone by LMB selecting it. Once done press Ctrl-P and select bone from the Set Parent Topop-up menu.

Now transforming that bone in Pose Mode will result in the child objects also transforming.

Bone relative parenting is an option you can toggle for each bone. This works in the same way as bone parenting with one difference.

With bone parenting if you have parented a bone to some child objects and you select that bone and switch it into Edit Mode and then move that bone; When you switch back into Pose Mode on that bone, the child object which is parented to that bone will snap back to the location of the bone in Pose Mode.

Single armature bone which has a child object cube parented to it using bone parenting. \xc2\xbf

In Fig. Single armature bone which has a child object cube parented to it using bone parenting. the 1st picture shows the position of the cube and armature before the bone is moved in Edit Mode. 2nd picture shows the position of the cube and armature after the bone was selected in Edit Mode, moved and switched back into Pose Mode. Notice that the child object moves to the new location of the pose bone.

Bone relative parenting works differently; If you move a parent bone in Edit Mode, when you switch back to Pose Mode, the child objects will not move to the new location of the Pose Bone.

Single bone with bone relative parent to a cube. \xc2\xbf

In Fig. Single bone with bone relative parent to a cube. the 1st picture shows the position of the cube and armature before the bone is moved in Edit Mode. 2nd picture shows the position of the cube and armature after the bone was selected in Edit Mode, moved and switched back into Pose Mode. Notice that the child object does not move to the new location of the pose bone.

Note

When using Ctrl-P to set parents, choosing \xe2\x80\x9cBone\xe2\x80\x9d or \xe2\x80\x9cBone Relative\xe2\x80\x9d will respectively clear and set the bone \xe2\x80\x99s \xe2\x80\x9cRelative Parenting\xe2\x80\x9d option. Since \xe2\x80\x9cRelative Parenting\xe2\x80\x9d is an option that is set per bone, this influences all child objects of that bone at once. For objects of type curve, surface, mesh and lattice, there is the possibility to use one of its vertices or points as the parent of other objects. You can parent an object to a single vertex or a group of three vertices as well; that way the child/children will move when the parent mesh is deformed.

In Object Mode, select the child/children and then the parent object. Tab into Edit Mode and on the parent object select either one vertex that defines a single point, or select three vertices that define an area (the

three vertices do not have to form a complete face; they can be any three vertices of the parent object), and then press Ctrl- P and confirm.

At this point, if a single vertex was selected, a relationship/parenting line will be drawn from the vertex to the child/children. If three vertices were selected then a relationship/parenting line is drawn from the averaged center of the three points (of the parent object) to the child/children. Now, as the parent mesh deforms and the chosen parent vertex/vertices move, the child/children will move as well.

Vertex parenting can be performed from Object Mode, this is done like regular object parenting, press Ctrl- P in Object Mode and select Vertex or Vertex (Triangle).

The nearest vertices will be used from each object which is typically what you would want.

The small cubes can each be automatically parented to a triad of nearby vertices on the icosphere using the Vertex

(Triangle) in the set parent context menu.

Reshaping the object in Edit Mode then means each of the cubes follows their vertex parent separately.

Scaling the parent icosphere in Object Mode means the child cubes are also scaled as expected.

The parent context menu item means users can rapidly set up a large number of vertex parent relationships, and avoid the tedious effort of establishing each parent-child vertex relationship separately.

Note

It is in fact a sort of reversed hook .

Reference

Object Mode

Object Parent Make Parent without Inverse

This sets the parent, and then resets the Parent Inverse matrix and the object's local location. As a result, the object will move to the location of the parent, but keep its rotation and scale.

The object's current world transform (so its absolute location, rotation and scale in the world) is computed. The new parent is set, and then the local transform values are set in such a way that after setting the new parent the object is still at its previous world transform.

Reference

Object Mode

Object Parent

Alt- P

You can remove a parent-child relationship via Alt- P .

If the parent in the group is selected, nothing is done. If a child or children are selected, they are disassociated from the parent, or freed, and they return to their original location, rotation, and size.

Frees the children from the parent, and keeps the location, rotation, and size given to them by the parent.

See Non-Uniform Scale which may apply here.

Instead of removing the hierarchical parent-child relationship, this clears the Parent Inverse matrix from the selected objects. With an empty matrix, the location, rotation and scale properties of the children are interpreted in the coordinate space of the parent.

A parent with non-uniform scale and rotation in relation to its child may cause a shear effect.

While this is supported by parenting, the shear will be lost when the parent is cleared since it can't be represented by location, scale and rotation.

If Clear and Keep Transformation moves the object, non-uniform scale is the most likely cause.

--- Fin du chapitre: _parent ---

--- Début du chapitre: _parenting ---Reference

Object Mode and Pose Mode

Object/Pose \xe2\x80\xa3 Parent \xe2\x80\xa3 Armature Deform

Ctrl- P

Armature Deform Parenting is a way of creating and setting up an Armature Modifier.

To use Armature Deform Parenting you must first select all the child objects that will be influenced by the armature and then lastly, select the armature object itself. Once all the child objects and the armature are selected, press Ctrl- P and select Armature Deform in the Set Parent Topop-up menu.

The armature will be the parent object of all the other child objects and each child object will have an Armature Modifier with the armature associated (Object field).

Bone associated with Mesh Object.\xc2\xb6

When parenting it will create empty vertex groups on the child objects (if they do not already exist) for and named after each deforming bone in the armature. The newly created vertex groups will be empty. This means they will not have any weights assigned. Vertex groups will only be created for bones which are setup as deforming (Properties \xe2\x80\xa3 Bone \xe2\x80\xa3 Deform Panel).

You can then manually select the vertices and assign them to a particular vertex group of your choosing to have bones in the armature influence them.

Choose this option if you have already created (and weighted) all the vertex groups the mesh requires.

For example, if you have an armature which consists of three bones named \xe2\x80\x93BoneA\xe2\x80\x9d, \xe2\x80\x93BoneB\xe2\x80\x9d and \xe2\x80\x93BoneC\xe2\x80\x9d and cube mesh called \xe2\x80\x93Cube\xe2\x80\x9d. If you parent the cube to the armature, the cube will get three new vertex groups created on it called \xe2\x80\x93BoneA\xe2\x80\x9d, \xe2\x80\x93BoneB\xe2\x80\x9d and \xe2\x80\x93BoneC\xe2\x80\x9d. Notice that each vertex group is empty.

Cube in Edit Mode using Armature Deform with empty groups.\xc2\xb6

With Automatic Weight parenting works similar to With Empty Groups, but it will not leave the vertex groups empty. It calculates how much influence a particular bone would have on vertices based on the distance from those vertices to a particular bone (\xe2\x80\x93bone heat\xe2\x80\x9d algorithm). This influence will be assigned as weights in the vertex groups.

This method of parenting is certainly easier to setup, but it can often lead to armatures which do not deform child objects in ways you would want. Overlaps can occur when it comes to determining which bones should influence certain vertices when calculating influences for more complex armatures and child objects. Symptoms of this confusion are that when transforming the armature in Pose Mode, parts of the child objects do not deform as you expect; If Blender does not give you the results you

require, you will have to manually alter the weights of vertices in relation to the vertex groups they belong to and have influence in. Works in a similar way to With Automatic Weights. The difference is that the influences are calculated based on the Bone Envelopes settings. It will assign a weight to each vertex group the vertices that is inside its bone's influence volume, depending on their distance to this bone.

This means newly included/excluded vertices or new envelope settings will not be taken into account. You will have to apply Armature Deform With Envelope Weights parenting again.

Tip

If you want the envelope setting to be used instantly, bind the Armature Modifier to Bone Envelopes.

Two sets of armatures, each with three bones. \xc2\xbb6

Warning

If you had defined vertex groups using same names as skinned bones, their content will be completely overridden by both Automatic and Envelope Weights. In this case With Empty Groups could be used instead.

See also

Vertex Groups for Bones.

--- Fin du chapitre: _parenting ---

--- Début du chapitre: _particles --- Create spray particles during the secondary particle simulation. Spray particles are those that appear to fly through the air above the liquid surface when there is a bigger splash.

Create foam particles during the secondary particle simulation. Foam particles are those that solely move on the liquid surface.

Create bubble particles during the secondary particle simulation. Bubble particles are those that move below the liquid surface.

Note

Enabling a secondary particle type will also create a particle system for that type of particles. Disabling a particle type will delete this particle system including its settings.

Select particle types that should go into the same particle system. This option has no effect on the outcome of the simulation. It only changes the way particle systems are allocated in the particle settings.

Factor by which to enhance the resolution of the particle simulation.

The scaling factor is coupled to the Resolution Divisions (i.e. the particle simulation is this times bigger than the base simulation).

Upper clamping threshold for marking fluid cells as wave crests. A higher value results in less marked cells.

Lower clamping threshold for marking fluid cells as wave crests. A lower value results in more marked cells.

Upper clamping threshold for marking fluid cells where air is trapped. A higher value results in less marked cells.

Lower clamping threshold for marking fluid cells where air is trapped. A lower value results in more marked cells.

Upper clamping threshold that indicates the fluid speed where cells start to emit particles. A higher value results in generally less particles.

Lower clamping threshold that indicates the fluid speed where cells start to emit particles. A lower value results in generally more particles.

Radius to compute potential for each cell. Higher values are slower but create smoother potential grids.

Radius to compute position update for each particle. Higher values are slower but particles move less chaotic.

Maximum number of particles generated per wave crest cell per frame.

Maximum number of particles generated per trapped air cell per frame.

Highest possible particle lifetime.

Lowest possible particle lifetime.

Amount of buoyancy force that rises bubbles. A high value results in bubble movement mainly upwards.

Amount of drag force that moves bubbles along with the fluid. A high value results in bubble movement mainly along with the fluid.

Delete secondary particles that are inside obstacles or left the domain.

Push secondary particles that left the domain back into the domain.

This option is only available when using the Modular cache type.

The progress will be displayed in the status bar. Pressing Esc will pause the simulation.

Once the simulation has been baked, the cache can be deleted by pressing Free Particles. It is possible to pause or resume a Bake Particles process.

--- Fin du chapitre: _particles ---

--- Début du chapitre: _particle_info --- Cycles Only

The Particle Info node can be used in the material node tree for objects that are used as the instancing objects, when you use Objector Collection Render mode of a particle system.

This node gives access to the data of the particle that spawned the object instance. It can be useful to give some variation to a single material assigned to multiple instances of instancing object.

Note

This node currently only supports parent particles. Info from child particles is not available.

This node has no inputs.

This node has no properties.

Index number of the particle (from 0 to number of particles).

A random per-particle value in the range from 0 to 1. It can for example be used in combination with a color ramp, to randomize the particle color.

Age of the particle in frames.

Total lifespan of the particle in frames.

Location of the particle.

Size of the particle.

Velocity of the particle.

Angular velocity of the particle.

--- Fin du chapitre: _particle_info ---

--- Début du chapitre: _particle_instance --- When a Particle Instance modifier is added to an object, the mesh of this object will be duplicated at the location of the particles of the selected particle system from another target object. This means that to use this modifier, you must have at least one other object that has a Particles System on it.

Because of the correlation in which the Particle Instance modifier is influenced by the underlying particle systems on other objects, some of the apparent effects generated by the modifier can look and act vastly different, depending on the underlying settings of the particle systems it

is associated with. This is worth taking account of, when it seems that the Particle Instance modifier settings do not return the expected results.

The Particle Instance modifier. \xc2\xba6

The target object which has a particle system associated with it. Which particle system from the target Object to apply this modifier to. When enabled, the modifier will use the regular (parents) particle to duplicate the mesh of the modified object.

When enabled, the modifier will use the children particle to duplicate the mesh of the modified object.

Scale the instanced copies of the mesh by the particle size attribute. When this is disabled, all the copies appear the same size as the origin.

See the particle system's Render and Children panels for particle's size options.

When enabled, the modifier will use the unborn particle to duplicate the mesh of the modified object.

When enabled, the modifier will use the alive particle to duplicate the mesh of the modified object.

When enabled, the modifier will use the dead particle to duplicate the mesh of the modified object.

The proportion of particles to be used. Allows you to randomly skip particles to adjust the amount of instances.

Warning

The random algorithm used currently only ensures that relative amount to be respected statistically. The actual amount of instances generated will differ from the theoretical one, depending on the Seed value of the target particle system (and the Offset value described below, too).

That deviation is not significant with high number of particles, but it will be highly noticeable with low numbers (e.g. with 100 particles in the target system, and an Amount value of 0.1, it can generate either up to 15 or 5 instances, instead of the 10 expected).

A relative offset in the range of particles used for instantiation. Allows you to avoid overlapping of the used particles, when the same particle system is used in multiple modifier instances.

Tip

If you want to fully avoid overlaps, your Offset value must be at least as high as your Amount value.

Use World Space, or Local Space of the target object (that the particle system is assigned to).

World space means that the locations of the copies of the modified mesh will depend on the location of the modified object and of the target object.

Local space means that the locations of the copies of the modified mesh will depend only on the location of the modified object.

Specify which axis of the modified object to use as pole axis to apply the rotation from the instantiated particles.

By default, the instances are placed depending on the particles position in the current frame. By enabling Create Along Paths, the instance of the modified object follows/deforms its shape along the particle path (or the hair strand). This allows you to select the position along the particles path regardless of the current frame.

Tip

You can adjust the particles' path (using the Path visualization type) on the Render panel of the Particle System tab.

Note

The particle system must be baked, except for Hair type or Keyed physics.

Specify what percentage of the path that the instance fills, or the position on the path if the Keep Shape option is enabled.

Adds randomness to the Position value of each instance.

Specifies the rotation around the path.

Adds randomness to the Rotation value of each instance.

Enabling this prevents the instance from being deformed, and places it on the path according to the Position value.

With these fields you can select the Color Attribute, which will be filled with colors based on the particles information. These Color Attributes can be used, for example, in a shader to add variance to a material.

A Color Attribute for values based on the particles index.

A Color Attribute for random per-particle values.

Particle Instance modifier example.

The render above shows a single plane mesh object assigned to two different vertex groups and each of those vertex groups is assigned to a separate and independent particle system, with each particle system being assigned to a different Particle Instance modifier. In the case shown the Particle Instance modifiers are added to a sphere and a cube. See example blend-file.

Create Along Path example.

In this example, a single Keyed particle travels through four points (green planes), on an elliptical path. The Particle Instance modifier is added to a cylinder object and then associated with that Keyed particle system.

When the Create Along Path is activated, instead of the cylinder location just following the position of the particle, the cylinder mesh is fitted to the shape of the path followed by the particle. The mesh geometry of the object which is deformed can have an impact on how well the deformation is carried out. In the case of the cylinder, it has many loop cuts along its length so that it can bend at those points to deform along the particle path.

The Particle Instance modifier Create Along Path option works for hair (strand) particles as well as with keyed particles. In this case, the mesh of the modifier will follow the length and profile of the hair strands paths.

Note

Strands, when they are generated, instantly die when created, so for the Create Along Path checkbox to be of any use, you must also have the Dead checkbox enabled.

--- Fin du chapitre: _particle_instance ---

--- Début du chapitre: _particle_system --- The Particle System modifier is a container for Particle Systems.

Note

By default the Particle System modifier does not take into account the modifier stack. Make sure to enable Use Modifier Stack in the Particle properties if you want Particle System modifier to take other modifiers into account.

As the modifier is only a container its actual options are configured in the Particle Properties tab. See the Particle Systems Properties for more information.

Creates a new object of each instanced object or collection. See Make Instances Real for more information.

Converts path particles to mesh objects. See Convert for more information.

Fur made from particles. \xc2\xbb

--- Fin du chapitre: _particle_system ---

--- D but du chapitre: _particle_system_panel ---Reference

Particle System \xe2\x80\xa3 Particle System

Particle System panel. \xc2\xbb

These are the basic settings.

The List View of the objects Particle Modifier(s).

Copies the active particle system to all selected objects.

Copies all particle systems from the active object to all selected objects.

Duplicates the particle system within the active object. The Duplicate Settings option (in the Adjust Last Operation panel) will duplicate settings as well, so the new particle system uses its own settings.

The Data-Block menu for settings.

Main selector of the system type.

In such a system, particles are emitted from the object.

Use Hair type, rendered as strands.

Regrows the hair for each frame. This is useful when you are animating properties.

Enables advanced settings which reflect the same ones as working in Emitter mode.

Note

This manual assumes that this option is enabled.

Controls the number of parts a hair is made of. Increasing this value will improve the quality of animations.

The process for working with standard particles is:

Create the mesh which will emit the particles.

Create one or more Particle Systems to emit from the mesh. Many times, multiple particle systems interact or merge with each other to achieve the overall desired effect.

Tailor each Particle System \xe2\x80\x99s settings to achieve the desired effect.

Animate the base mesh and other particle meshes involved in the scene.

Define and shape the path and flow of the particles.

For Hair particle systems: Sculpt the emitter \xe2\x80\x99s flow (cut the hair to length and comb it for example).

Make final render and do physics simulation(s), and tweak as needed.

To add a new particle system to an object, go to the Particle tab of the Properties editor and click the small + button. An object can have many Particle Systems.

Each particle system has separate settings attached to it. These settings can be shared among different particle systems, so one does not have to copy every setting manually and can use the same effect on multiple objects.

After you have created a particle system, the Properties fills with many panels and buttons. But do not panic! There are two different types of

particle systems, and you can change between these two with the
Typeselector:Emitter and Hair.
The settings in the Particle System tab are partially different for each
system type.

--- Fin du chapitre: _particle_system_panel ---

--- Début du chapitre: _parts --- All nodes in Blender are based on a
similar construction. This applies to any type of node. These parts include
the title, sockets, properties and more.

The title shows the name/type of the node; it can be overridden by
changing the node's Label. On the left side of the title is
the collapse toggle which can be used to collapse the node. This can also
be done with H.

How a node appears when collapsed.

Previews are an overlay that shows a small image above the node
displaying the node result. Not all nodes support previews, but the ones
that do can be toggled using the / icons in the top right-hand corner of
the node next to the title.

Previews can be disabled for the whole node tree by using Preview overlay
toggle.

Sockets are input and output values for the node. They appear as little
colored circles on either side of the node. Unused sockets can be hidden
with Ctrl-H.

Each socket is color-coded depending on what type of data it handles.

Built-in

Used for shaders in Cycles and Eevee.

Used in Geometry Nodes.

Data

Used to pass a true or false value.

Indicates that the socket accepts/produces color information. The colors
may or may not have an alpha component depending on the node tree type.

Indicates that the socket accepts/produces floating-point numbers. It can
either be a single value or a so-called color value

map. (You can think of a value map as a grayscale image where
the brightness of a pixel represents its value.) If a single value is used
as an input for a color map socket, all points of
the map are set to this same value.

Used to pass an integer value (a number without a fractional component).

Used to pass a text value.

Indicates vector, coordinate and normal information.

Indicates a rotation/quaternion.

Indicates a 4x4 matrix of float values, it is often used to
represent a Transformation Matrix.

Data-Blocks

Used to pass a collection data-block.

Used to pass an object data-block.

Used to pass a material data-block.

Used to pass a texture data-block.

Used to pass an image data-block.

The inputs are located on the bottom left side of the node, and provide
the data the node needs to perform its function. Each input socket, except
for the green shader input, when disconnected, has a default value which
can be edited via a color, numeric, or vector interface input. In the

screenshot of the node above, the second color option is set by a color interface input.

Some nodes have special sockets that can accept multiple inputs. These sockets will have an ellipsis shape rather than a circle to indicate their special behavior.

The outputs are located on the top right side of the node, and can be connected to the input of nodes further down the node tree.

Some socket types can be converted to others either implicitly or explicitly. Implicit conversion happens automatically without the need of a conversion node. For example, Float sockets and Color sockets can be linked to each other.

Once a socket conversion is made, data may be lost and cannot be retrieved later down the node tree. Implicit socket conversion can sometimes change the data units as well. When plugging a Value input node into an angle socket, it will default to use radians regardless of the scene's Units. This happens because the Value node has no unit while the angle input does.

Valid conversions:

Between color and vector \xe2\x80\x93 mapping between color channels and vector components.

Between color and float \xe2\x80\x93 the color data is converted to its grayscale equivalent.

Color/float/vector to Shader \xe2\x80\x93 implicitly converts to color and gives the result of using an Emission node.

Between float and integer \xe2\x80\x93 integers simply become floats, floats are truncated.

Between float and vector \xe2\x80\x93 when a float becomes a vector the value is used for each component. When a vector becomes a float the average of the components is taken.

Between float and boolean \xe2\x80\x93 values greater than 0 are true, true maps to 1, and false maps to 0.

Between rotations and matrices.

Explicit conversion requires the use of a conversion node such as the Shader To RGB node or the RGB to BW Node. The Math Node also contains some functions to convert between degrees and radians.

Many nodes have settings which can affect the way they interact with inputs and outputs. Node settings are located below the outputs and above any inputs.

An example of the controls on the Chroma Key node. \xc2\xbb6

--- Fin du chapitre: _parts ---

--- Début du chapitre: _passes ---Reference

Scene \xe2\x80\xa3 View Layers \xe2\x80\xa3 Passes

Passes can be used to split rendered images into colors, direct and indirect light to edit them individually, and also to extract data such as depth or normals.

The final combination of render passes with everything included.

Distance to any visible surfaces.

Note

The Z pass only uses one sample. When depth values need to be blended in case of motion blur or Depth of Field, use the mist pass.

Distance to visible surfaces, mapped to the 0.0 - 1.0 range. When enabled, settings are in World tab. This pass can be used in compositing to fade out objects that are farther away.

World position of objects in the scene.

Surface normal used for shading.

Motion vectors for the Vector Blur node. The four components consist of 2D vectors giving the motion towards the next and previous frame position in pixel space.

Mapped UV coordinates, used to represent where on a mesh a texture gets mapped too. This is represented through the red and green channels of the image. The blue channel is encoded with a constant value of 1 but does not hold any information.

Includes Denoising Albedo, Denoising Normal, and a render pass of the original combined pass before denoising.

Note

The Z, Position, Object Index, and Material Index passes are not anti-aliased.

Creates a mask of the object that can be later read by the ID Mask Node in the Compositor.

Creates a mask of the material that can be later read by the ID Mask Node in the Compositor.

Number of samples per pixel taken, divided by the maximum number of samples. To analyze adaptive sampling.

Z, Index, normal, UV and vector passes are only affected by surfaces with alpha transparency equal to or higher than this threshold. With value 0.0 the first surface hit will always write to these passes, regardless of transparency. With higher values surfaces that are mostly transparent can be skipped until an opaque surface is encountered.

The final combination of render passes with everything included.

Distance to any visible surfaces.

Distance to visible surfaces, mapped to the 0.0 - 1.0 range.

Surface normal used for shading.

World position of objects in the scene.

Motion vectors for the Vector Blur node. The four components consist of 2D vectors giving the motion towards the next and previous frame position in pixel space.

Direct lighting from diffuse and subsurface BSDFs. We define direct lighting as coming from lights, emitting surfaces, the background, or ambient occlusion after a single reflection or transmission off a surface. BSDF color is not included in this pass.

Indirect lighting from diffuse and subsurface BSDFs. We define indirect lighting as coming from lights, emitting surfaces or the background after more than one reflection or transmission off a surface. BSDF color is not included in this pass.

Color weights of diffuse and subsurface BSDFs. These weights are the color input socket for BSDF nodes, modified by any Mix and Add Shader nodes.

Same as above, but for glossy BSDFs.

Same as above, but for transmission BSDFs.

Same as above, but for volumetric BSDFs.

Emission from directly visible surfaces.

Emission from the directly visible background. When the film is set to transparent, this can be used to get the environment color and composite it back in.

Ambient occlusion from directly visible surfaces. BSDF color or AO factor is not included; i.e. it gives a normalized value between 0 and 1.

Extra indirect light information collected by objects with the Shadow Catcher option enabled. Multiply this pass with existing footage using the Mix Node in the Compositor to add the indirect lighting information to the footage.

Note

Transparent BSDFs are given special treatment. A fully transparent surface is treated as if there is no surface there at all; a partially transparent surface is treated as if only part of the light rays can pass through. This means it is not included in the Transmission passes; for that a glass BSDF with index of refraction 1.0 can be used.

Direct lighting from diffuse BSDFs. We define lighting as coming from lights, the background, or ambient occlusion off a surface. BSDF color is not included in this pass.

Color weights of diffuse BSDFs. These weights are the color input socket for BSDF nodes, modified by any Mix and Add Shader nodes.

Same as above, but for specular BSDFs.

The scattering pass from volume objects or world.

Emission from directly visible surfaces.

Emission from the directly visible background. When the film is set to transparent, this can be used to get the environment color and composite it back in.

Shadows from light objects. Mostly useful for compositing objects with shadow into existing footage.

Ambient occlusion from directly visible surfaces. BSDF color or AO factor is not included; i.e. it gives a normalized value between 0 and 1.

Contain Blended surfaces, so they can be adjusted in the compositor and later mixed with opaque passes.

This pass only supports monochromatic opacity. Colored opacity will show differently than in combined pass.

Distance of object that contributes to the ambient occlusion effect.

Cryptomatte is a standard to efficiently create mattes for compositing. Cycles outputs the required render passes, which can then be used in the Blender Compositor or another compositor with Cryptomatte support to create masks for specified objects.

Unlike the Material and Object Index passes, the objects to isolate are selected in compositing. The mattes will be anti-aliased and take into account effects like motion blur and transparency.

Render cryptomatte object pass, for isolating objects in compositing.

Render cryptomatte material pass, for isolating materials in compositing.

Render cryptomatte asset pass, for isolating groups of objects with the same parent in compositing.

Sets how many unique objects can be distinguished per pixel.

Enable Cryptomatte Object render pass in the Passes panel, and render.

In the compositing nodes, create a Cryptomatte node and link the Render Layer matching Image and Cryptomatte passes to it.

Attach a Viewer node to the Pick output of the Cryptomatte node.

Use the Cryptomatte Add/Remove button to sample objects in the Pick Viewer node.

Use the Matte output of the Cryptomatte node to get the alpha mask.

See also

Cryptomatte Node .

Shader AOVs (Arbitrary Output Variables) provide custom render passes for any shader node components. As an artist this can be a good way to fix or

tweak fine details of a scene in post-processing. To use Shader AOVs create the pass in the Shader AOV panel then reference this pass with the AOV Output shading node. Shader AOVs can be added or removed in the Shader AOV panel. In this panel is a list of all AOV passes; each AOV in the list consists of a Name and Data Type.

The name of the render pass; this is the Name that is referenced in the AOV Output node. Any names can be used for these passes, as long as they do not conflict with built-in passes that are enabled.

Shader AOVs can either express a Color or a Value output. The Color type as the name suggest can be used for a color but also for normals. A Value type can be used for any single numerical value.

Cycles only

Light Groups provide render passes that only contains information from the lights within that group. Light Groups can be used to easily tweak the lighting color and intensity of specific lights without having to re-render the scene.

The name of the render pass. This is the name that is used when adding World Shaders, Lights and Objects to Light Groups.

These operators are available in the menu to the right of the Light Group list.

Adds all Light Groups to the View Layer that have been created elsewhere and have lights assigned to them.

Deletes all Light Groups that do not have any lights assigned to them.

All these lighting passes can be combined to produce the final image as follows:

The passes can be combined to produce the final image as follows:

Blended materials are not rendered in render passes except the combined pass. Use the Dithered as Render Method to render transparent materials in render passes.

Depth of field is not rendered in render passes except the combined pass. It is possible to add the depth of field back in the Compositor using the Defocus node.

EEVEE render passes exclude parts of the BSDF equation. Shader to RGB is not supported as it needs the full BSDF equation.

EEVEE supports up to 16 color and 16 value AOV render passes.

--- Fin du chapitre: _passes ---

--- Début du chapitre: _path_animation --- The Path Animation settings can be used to determine how child objects move along a certain path.

Note

This feature is deprecated, but still available. A more future-proof method is the Follow Path Constraint.

Path Animation panel.

The number of frames that are needed to traverse the path, defining the maximum value for the Evaluation Time setting.

Parametric position along the length of the curve that object following it should be at (the position is evaluated by dividing by the Frames value). By default, it is linked to the global frame number, but could be keyframed to give more control over the path animation.

Clamp the curve path children so they can't travel past the start/end point of the curve.

Make the curve path children rotate along the curvature of the path.

This example shows you how setup a Path Animation.

Add an object you want to animate and a path along which this object will move. In this example it's the Monkey and the Bézier Circle.

To parent the monkey to the Bézier circle, first select the monkey then the curve (so that the curve is the active object), press Ctrl- P and select Follow Path. It will automatically animate Evaluation Time and activate Follow option in the Path Animation panel.

Select the monkey and Clear Origin to reset its offset.

You can change the orientation of the monkey by changing the Tracking Axis

. Monkey parented to the Bézier Circle.

The final result.

--- Fin du chapitre: _path_animation ---

--- Début du chapitre: _pen ---

Reference

Edit Mode

Toolbar Curve Pen

The Curve Pen tool allows you to construct and edit curves rapidly.

Curve Pen Preferences

The following preferences can be configured from: Preferences

Keymap 3D View Curve 3D View Tool: Edit Curve, Curve Pen.

LMB click to add a new point connected to an existing point.

The handle type of the extruded points. Can be either Vector or

Auto. However, the handle type switches to Align when handles are moved

(See Move Point).

Ctrl- LMB click on an existing point to delete it.

Ctrl- LMB click on a Curve Segment to insert a new control point between the two adjacent control points. Ctrl- LMB click and drag to control the handles of the inserted points.

LMB drag on a segment in between two control points to adjust the handles, changing the shape of the curve without affecting the location of any control points.

LMB click to select a single point or handle at a time.

LMB drag to move existing points or handles. With an endpoint of a spline selected, click and drag on empty space to Extrude Point and move the handle at the same time.

Make the spline Cyclic by clicking the endpoints consecutively.

The condition for Close Spline to activate.

Turn off the Close Spline functionality.

Close the spline on mouse down. With this option, you may click and drag to adjust the handles of the endpoint.

Activate on mouse release. With this option, the Close

Spline functionality will not be triggered on click and drag.

Double LMB click on a handle to switch handle between Vector and Auto handle types. Can be used to easily switch between sharp corners and smooth curves.

Double LMB click on the control point to cycle through all handle types.

Hold Left Shift while dragging a handle to switch between Free and Align handle types. Can be used to create sharp corners along the curve.

Hold Left Ctrl while dragging a handle to move the closer handle of the adjacent control point. Can be helpful to make adjustments to newly created curve segments.

Hold Spacebar while dragging a handle to move the entire point.

Press RightCtrl while dragging a handle to mirror its movement on the opposite handle of the same point.
Hold LeftAlt while dragging a handle to limit the movement of the handle to its current direction, so only its length can be adjusted.

--- Fin du chapitre: _pen ---

--- Début du chapitre: _performance ---Reference

Properties \xe2\x80\xa3 Render \xe2\x80\xa3 Performance

Uses higher precision normals and tangents which can improve visual quality for dense meshes with high frequency textures at the cost of memory.

--- Fin du chapitre: _performance ---

--- Début du chapitre: _perlin_noise_1d --- The Perlin Noise 1D modifier adds one-dimensional Perlin noise to the stroke. The curvilinear abscissa (value between 0 and 1 determined by a point \xe2\x80\x99s position relative to the first and last point of a stroke) is used as the input to the noise function to generate noisy displacements.

This means that this modifier will give an identical result for two strokes with the same length and sampling interval.

How dense the noise is (kind of a scale factor along the stroke).

How much the noise distorts the stroke in the Angledirection.

The seed of the random generator (the same seed over a stroke will always give the same result).

The \xe2\x80\x99level of detail \xe2\x80\x9d of the noise.

In which direction the noise is applied (0.0 is fully horizontal).

--- Fin du chapitre: _perlin_noise_1d ---

--- Début du chapitre: _perlin_noise_2d --- The Perlin Noise 2D modifier adds one-dimensional Perlin noise to the stroke. The modifier generates noisy displacements using 2D coordinates of stroke vertices as the input of the noise generator.

How dense the noise is (kind of a scale factor along the stroke).

How much the noise distorts the stroke in the Angledirection.

The seed of the random generator (the same seed over a stroke will always give the same result).

The \xe2\x80\x99level of detail \xe2\x80\x9d of the noise.

In which direction the noise is applied (0.0 is fully horizontal).

--- Fin du chapitre: _perlin_noise_2d ---

--- Début du chapitre: _physical_properties ---Reference

Physics \xe2\x80\xa3 Cloth \xe2\x80\xa3 Physical Properties

The mass of the cloth material.

Air has some thickness which slows falling things down.

Cloth model with linear bending springs (old).

Cloth model with angular bending springs.

How much the material resists stretching.

How much the material resists compression.

Overall stiffness of the cloth (only in linear bending model).

How much the material resists shearing.

Wrinkle coefficient. Higher creates more large folds.

Amount of damping in stretching behavior.

Amount of damping in compression behavior.

Amount of damping in stretching behavior (only in linear bending model).

Amount of damping in shearing behavior.

Amount of damping in bending behavior.

As stated in the introduction, cloth physics are simulated through Springs connecting vertices on the surface of a mesh. But these springs only interact on the surface and only apply to 2D surfaces. 3D or Internal Springs can be used to make a mesh behave similarly to a Soft Body. Internal springs can be enabled by toggling the checkbox in the Internal Springs panel header.

The maximum length an internal spring can have during creation. If the distance between internal points is greater than this, no internal spring will be created between these points. A length of zero means that there is no length limit.

The maximum angle that is allowed to use to connect the internal points can diverge from the vertex normal.

Requires the points the internal springs connect to have opposite normal directions.

How much the material resists stretching.

How much the material resists compression.

The Tension and Compression of internal springs can be controlled via a Vertex Group to specify which the portions of the mesh have internal springs or the spring strength.

Maximum tension stiffness value.

Maximum Compression stiffness value.

Cloth pressure allows the simulation of soft-shelled objects such as balloons or balls that are filled with a type of fluid. This fluid is modeled as a gas; to emulate an incompressible liquid set Pressure Scale as high as possible without breaking the simulation. Cloth pressure can be enabled by toggling the checkbox in the Pressure panel header.

Note

Non-manifold meshes will work with cloth pressure however, pressure will escape out of the mesh holes and cause drifting or propulsion forces. One way to get around this is by using the Vertex Group to exclude the non-manifold portions of the mesh.

The uniform pressure that is constantly applied to the mesh. This value is specified in units of Pressure Scale, and can be negative to simulate implosions or any other case where an object has outside pressure pushing inwards.

Use the Target Volume parameter as the initial volume for the cloth, instead of computing it from the mesh itself.

The mesh volume where the inner/outer pressure will be the same. If set to zero, changes in the volume of the object will not affect pressure.

Ambient pressure (in kPa) that exists both inside and outside the object, balancing out when the volume matches the target. Increase the value to make the object resist changes in volume more strongly.

Specifies the density of the fluid contained inside the object (in kg/liter = 1000 kg/m³, use 1 for water), used to generate a hydrostatic pressure gradient that simulates the weight of the fluid. If the value is negative, it instead models buoyancy from a surrounding fluid.

The fluid is not actually simulated, so while the setting helps to achieve a more plausible object shapes at rest, it cannot create realistic fluid dynamics effects. It can also be used to give more weight to a soft body like object with heavy and sufficiently flexible filling, even if it is not a fluid by itself.

The volume of the object is not preserved. If that is desired it should be used together with Pressure Scale. Fluid density $\times$ object size $\times$ 50 is a good start value for Scale to make sure that no more than 10% volume change if the object does not experience higher acceleration than standard gravity.

Cloth pressure can be controlled via a Vertex Group to specify which the portions of the mesh to apply pressure. Zero weight means no pressure while a weight of one means full pressure.

Note, faces with a vertex that has zero weight will be excluded from the Target Volume calculation.

--- Fin du chapitre: `_physical_properties` ---

--- Début du chapitre: `_pinch` ---

Sculpt Mode

Sidebar `\xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3`

Advanced `\xe2\x80\xa3 Brush Type`

Pulls geometry towards the center of the brush. When inverted via Ctrl it will Magnify geometry by pushing them away from the center of the brush.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: `_pinch` ---

--- Début du chapitre: `_pinch_curves` --- Converges adjacent curves to the location at the center of the cursor.

The pinch brush can be inverted by holding Ctrl.

--- Fin du chapitre: `_pinch_curves` ---

--- Début du chapitre: `_piston` ---

Reference
Physics `\xe2\x80\xa3 Rigid Body Constraint`

Piston

A piston permits translation along the X axis of the constraint object. It also allows rotation around the X axis of the constraint object. It is like a combination of the freedoms of a slider with the freedoms of a hinge (neither of which is very free alone).

Enables/disables limit translation around X axis.

Lower limit of X axis translation.

Upper limit of X axis translation.

Enables/disables limit rotation around X axis.

Lower limit of X axis rotation.

Upper limit of X axis rotation.

--- Fin du chapitre: `_piston` ---

--- Début du chapitre: `_pivot` --- The Pivot constraint allows the owner to rotate around a target object. It was originally intended for pivot joints found in human *e.g.* fingers, feet, elbows, etc.

Pivot panel. `\xc2\xb6`

Data ID for the selection of the object to be used as a pivot point. See common constraint properties for more information.

Offset will be an absolute point in space instead of relative to the target.

Offset of pivot from target.

Rotation range on which pivoting should occur.

Use the pivot point in every rotation.

Use the pivot point in the corresponding direction around the corresponding axis.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_pivot` ---

--- Début du chapitre: `_pivot_point` ---Reference

Pivot Point

Period

The Pivot Point is the point around which images are rotated and scaled. It's indicated by the position of the selected tool's gizmo.

See also

The Pivot Point of the 3D Viewport

Use the center of the rectangle that's wrapped as tightly as possible around the selected images' origin points.

Use the averaged-out position of the selected images' origin points.

Use the location of the 2D Cursor, for when you want to specify the pivot point by hand.

Rotate/scale each image around its own origin, rather than rotating/scaling all of them around the same single point like the other options do.

--- Fin du chapitre: `_pivot_point` ---

--- Début du chapitre: `_pixelate` --- The Pixelate node reduces the detail in an image by making individual pixels more prominent. It results in a blocky or mosaic-like appearance, where the fine details of the image are obscured, and the overall image is represented using larger, more noticeable pixels.

Standard color input.

The size of the pixels in the output image, measured in pixels.

Standard color output.

--- Fin du chapitre: `_pixelate` ---

--- Début du chapitre: `_plane` --- A light probe plane records the light incoming from a single direction for all visible points on a plane. The specular reflection direction is the only one currently available.

This type of light probe is suited to smooth planar surfaces.

Each visible planar light probe increases the render time as the scene needs to be rendered for each of them.

Light probe planes only work when the ray tracing method is set to Screen Trace. When enabled, they accelerate the tracing process and complete the missing data from the screen space ray tracing.

Note

Reflections and volumetrics are not supported inside Light probe planes.

If Backface Culling is not enabled, snapping the light probe plane to the planar surface will effectively capture the underside of the surface.

You can manually move the light probe plane above the surface enough for it to not appear in the capture. Alternatively you can disable the light probe visibility in the object visibility panel.

Reference

Object Data › Probe

A probe object only influences the lighting of surfaces inside its influence zone. This influence zone is defined by the distance parameter and the object's scale.

For light probe planes, the influence distance is the distance from the plane. Only surfaces whose normals are aligned with the Reflection Plane will receive the captured reflection.

Define how far below the plane the near clip is when capturing the scene. Increasing this can fix reflection contact problems.

Reference

Object Data \xa3 Viewport Display

Size of the arrow showing the reflection plane normal.

Show the captured reflected image onto a fully reflective plane in the 3D Viewport.

Show the influence bounds in the 3D Viewport.

--- Fin du chapitre: _plane ---

--- Début du chapitre: _plane_track --- Plane Track panel.

Its properties are shown only when a plane track is selected.

The name of the selected plane track is shown. It can also be changed from here.

Toggles the auto-keyframing for corners of the plane track. With this enabled, keyframes will automatically get inserted when any corner is moved.

Field to select or create an image which will be displayed inside the plane track. This image is for preview purposes in the Movie Clip editor only. To include it in your final render, see Plane Track Deform node.

Creates an image from the pixels of the movieclip that the plane marker sees at the current frame. This allows to create an unwarped texture of any flat surface in the footage. The resulting image can then be used for editing and retouching, for example to paint out certain parts of the footage.

Updates the pixels of the active Plane Track's image.

Used to set the opacity of this image. Again, this is for display purposes only, and will not affect your final render.

--- Fin du chapitre: _plane_track ---

--- Début du chapitre: _plane_track_deform --- The Plane Track Deform

Node is used to replace flat planes in footage by another image, using plane tracks from motion tracking.

Before using this node, plane track for the footage should be made in the Movie Clip Editor.

Image to put in place of the plane track, and thus, override that area in the movie clip.

Used to select the movie clip whose plane track to use. For controls see Data-Block Menu .

Used to select the object to which the plane track is linked.

Used to select the plane track to use.

Specify whether to use blur caused by motion of the plane track or not.

Set the number of samples to take for each frame. The higher this number, the smoother the blur effect, but the longer the render, as each virtual intermediate frame has to be rendered.

Note

Samples are taken only from the nextframe, not the previous one. Therefore the blurred object will appear to be slightly ahead of how it would look without motion blur.

Time (in frames) the shutter is open. If you are rendering at 24 fps, and the Shutter is set to 0.5, the time in between frames is 41.67 ms, so the shutter is open for half that, 20.83 ms.

The output by perspective wrapping the image to that plane track.

Produces a black-and-white mask of the plane track.

This can simply be achieved by using the Alpha Over node.

Image output.

This can be achieved by mixing the movie clip and the image by using the plane output as the factor.

Plane output.

Using Image output scales, moves, and skews the input image according to the track while using the original image and mixing it with the movie clip using Plane output as factor will display the part of the image that lies inside that mask. This image shows the difference:

Comparison between image output and original image (see Viewer nodes carefully).

--- Fin du chapitre: _plane_track_deform ---

--- Début du chapitre: _ply ---Reference

Import-Export

File Import/Export Stanford (.ply)

Use the operator to import ASCII or binary PLY-files, you can select multiple files at once. For exporting, you can choose to enable or disable the modifiers during the export and you can choose which data you want to export (UV textures, Color Attributes).

Value by which to scale the imported objects in relation to the world's origin.

Apply current scene's unit (as defined by unit scale) to imported data.

Since many applications use a different axis for Up, these are axis conversion for these settings, Forward and Up axes. By mapping these to different axes you can convert rotations between applications default up and forward axes.

Blender uses Y Forward, Z Up (since the front view looks along the +Y direction). For example, it's common for applications to use Y as the up axis, in that case -Z Forward, Y Up is needed.

Attempts to combine co-located vertices where possible.

The color space that the color data in the ply-file was saved in.

Does not import vertex color data.

Vertex colors in the file are in sRGB Color Space

Vertex colors in the file are in Linear Color Space

Formats the file using the simple ASCII format. This option might be helpful if the program that will later import the file does not support the binary file format.

Only selected objects are exported. Instanced objects, for example collections that are instanced in the scene, are considered selected when their instancer is selected.

Value by which to scale the exported objects in relation to the world's origin.

Since many applications use a different axis for
Up, these are axis conversion for these
settings, Forward and Up axes. By mapping these to different
axes you can convert rotations between applications default up and forward
axes.

Blender uses Y Forward, Z Up (since the front view looks along the +Y
direction). For example, it's common for applications to use Y
as the up axis, in that case -Z Forward, Y Up is needed.

Write out the active UV layers coordinates from Blender.

Write out Blender's face and vertex normals (depending on the
faces smooth setting).

Mostly this isn't needed since most applications will
calculate their own normals but to match Blender's normal map
textures you will need to write these too.

The color space that the color data in the ply-file was saved in.

Does not import vertex color data.

Vertex colors in the file are in sRGBColor Space

Vertex colors in the file are in LinearColor Space

All N-gons with four or more vertices will be triangulated. Meshes in the
scene will not be affected. Behaves like Triangulate Modifier with the
following settings:

N-gon-method: Beauty

Quad-method: Shortest Diagonal

Min vertices: 4

Export objects using the evaluated mesh, meaning the resulting mesh
after all modifiers have been calculated.

--- Fin du chapitre: _ply ---

--- Début du chapitre: _point --- Reference

Physics Rigid Body Constraint
Point

The objects are linked by a point bearing allowing any kind of rotation
around the location of the constraint object, but no relative translation
is permitted. The physics engine will do its best to make sure that the
two points designated by the constraint object on the two constrained
objects are coincident.

--- Fin du chapitre: _point ---

--- Début du chapitre: _points --- The Points node generate a point cloud
with positions and radii defined by fields.

The number of points to create.

The position of each generated point.

The radius of each point.

Note

Since the point cloud is created from scratch, the Position and
Radius inputs can only depend on the index node. Regular input nodes
like the position won't work.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _points ---

--- Début du chapitre: _points_of_curve --- The Points of Curve node
retrieves indices of specific control points in a curve.

The index of the input curve.

Note

By default this uses the index from the field context, which makes it important that the node is evaluated on the curve domain. Values used to sort the curve's control points. By default the control points are sorted by index, so the control points with the smallest indices come first.

Which of the sorted control points to use for the Point Index output. If the value is larger than the total number of control points, it will wrap around to the beginning.

This node has no properties.

The index of one of the curve's control points, chosen by the Sort Index input.

The number of control points in the curve.

--- Fin du chapitre: `_points_of_curve` ---

--- Début du chapitre: `_points_to_curves` --- The Points to Curves node generates a Curves geometry by taking all points and inserting them to new curves. All Attributes from points are propagated to Curve Points. Built-in curves attributes stored in points will be ignored.

Tip

To simplify thinking about points, attributes and their positions in each curve, The weight of each point in curve can be associated with a point attributes value. The sorting and grouping will be reflected on the attributes as like on the Weight and Group ID.

The Point Cloud geometry component.

All points with the same Group ID value will be joined in the same curve. The value of Group ID can be any value (negative, zero, or infinity, etc.). All created curves must have at least a single point. The order of curves depends both on Group ID value and on the order of Group ID values in the Point Cloud.

If the curve contains more than one Point, the Weight of each Point is used to define the order of all points in curve via sorting. The goal of sorting is to have points with the minimal Weight value at the start of curve and the maximum Weight at the end of curve.

Note

If points of curve have the same Weight value, the order will be the same as its original relative location. Without any Weight and Group ID inputs, each point will have the same indices in the curve.

This node has no properties.

The curves with all copied points from the Point Cloud, but joined in curves. All other components aren't saved. The resulting curves are always non-cyclic.

The above example creates a curve Array with connections between curves. This is created by duplicating the Arc primitive curve with the Duplicate Elements Node. Each curve is shifted in a top direction based on its index value. All the curves are converted to the Point Cloud by the Curve to Points Node. Finally, the points are converted to curves by the Points to Curves node.

All the Points of the resulting Curves geometry have the same attributes as points on the initial Arc primitive.

--- Fin du chapitre: `_points_to_curves` ---

--- Début du chapitre: `_points_to_vertices` --- The Points to Vertices node generate a mesh vertex in the output geometry foreach point cloud point in the input geometry.
Standard geometry input.
Boolean field used to determine if each point will be converted to a vertex.
This node has no properties.
Standard geometry output.
--- Fin du chapitre: `_points_to_vertices` ---

--- Début du chapitre: `_points_to_volume` --- The Points to Volume node generates a fog volume sphere around every point in the input geometry. The new volume grid is named `\density`.
It usually makes sense to combine this node with the Volume to Mesh Node.
Warning
This node expects that point positions are not extremely large. For position values of many billions, the behavior isn't guaranteed, and it may be unstable.
Standard geometry input.
Value of voxels inside the generated fog volume.
Specify the approximate number of voxels along the diagonal.
Specify the voxel side length.
Specify the radius of the sphere generated at each point.
How the voxel size is specified.
Specify the approximate number of voxels along the diagonal.
Specify the voxel side length. It is recommended to be careful when tweaking this value, because small changes can have a large effect on the processing time.
Standard geometry output.
--- Fin du chapitre: `_points_to_volume` ---

--- Début du chapitre: `_point_density` --- The Point Density node is available in volume shaders, to render volumetric points for each particle or vertex of another object.
Texture coordinate to sample texture at; defaults to global position (Position output of Geometry node) if the socket is left unconnected.
Where to get points from.
Use each particle position from the specified particle system.
Use each vertex position from the specified object.
Which object's vertices or particle system will be used.
Particle positions from this system will be used.
The coordinate system for mapping points.
Map each point exactly where the source particle or vertex is.
Fit the points from the source particles/vertices inside the bounding box of the object with the point density texture.
Size of the points.
Texel filtering type.
No interpolation, use nearest texel. Produces blocky looking points.
Interpolate linearly between texels, producing soft, round points.
Use cubic falloff, producing very soft points. Useful when points are very densely packed.
The dimensions of the texture holding the point data.
Which attribute of the particle system or mesh is used to color the output.

Lifetime mapped as (0.0 - 1.0) intensity.

Particle speed (absolute magnitude of velocity) mapped as (0.0 - 1.0) intensity.

XYZ velocity mapped to RGB colors.

Use a Color Attribute for coloring the point density texture.

Note

Color Attributes are defined per face corner. A single vertex can have as many different colors as faces it is part of. The actual color of the point density texture is averaged from all vertex corners.

Use weights from a vertex group as intensity values.

Use object-space vertex normals as RGB values.

Texture color output.

Density of volume.

Domain object with Point Density texture using vertices from ball as points. \xc2\xba6

--- Fin du chapitre: _point_density ---

--- Début du chapitre: _point_info --- Cycles Only

The Point Infonode can be used in the material node tree for point cloud objects and gives access to the data of individual points. It can be useful to give some variation to a single material assigned a point cloud object.

This node has no inputs.

This node has no properties.

Location of the particle.

Size of the particle.

A random per-point value in the range from 0 to 1. It can for example be used in combination with a color ramp, to randomize the point color.

--- Fin du chapitre: _point_info ---

--- Début du chapitre: _point_menu --- Reference

Edit Mode

Point \xe2\x80\xa3 Extrude

Toolbar \xe2\x80\xa3 Extrude

E

Extrudes points by duplicating the selected points, which then can be moved. The new points stay connected with the original points of the edit line.

Note

Since Grease Pencil strokes can only have one start and end point, a new stroke will be created when extrude intermediate points in the strokes.

Reference

Edit Mode

Point \xe2\x80\xa3 Smooth

Softens strokes by reducing the differences in the locations of the points along the line, while trying to maintain similar values that make the line fluid and smoother.

The number of times to repeat the procedure.

The amount of the smoothness to apply.

Smooths the stroke \xe2\x80\x99s endpoints.

Preserves the strokes shape.

When enabled, the operator affects the points location.

When enabled, the operator affects the points thickness.

When enabled, the operator affects the points strength (alpha).

Operators for working with vertex groups.

Reference

Edit Mode

Point \xe2\x80\xa3 Set Handle Type

V

Sets the handle type for the points on the B\xe3\x9zier curve that are in the selection.

The handle type to switch to.

The handles are independent of each other.

This handle has a completely automatic length and direction which is set by Blender to ensure the smoothest result. These handles convert to Aligned handles when moved.

Both parts of a handle always point to the previous handle or the next handle which allows you to create curves or sections thereof made of straight lines or with sharp corners. Vector handles convert to Free handles when moved.

These handles always lie in a straight line, and give a continuous curve without sharp angles.

--- Fin du chapitre: _point_menu ---

--- D\xe9but du chapitre: _poke_faces ---Reference

Edit Mode

Face \xe2\x80\xa3 Poke Faces

Splits each selected faces into a triangle fan, creating a new center vertex and triangles between the original face edges and new center vertex. The Offset can be used to make spikes or depressions.

Offset the new center vertex along the face normal.

Multiply the Offset by the average length from the center to the face vertices.

Computes the center of a face.

Using the mean average weighted by edge length.

Using the mean average.

Uses center of bounding box.

--- Fin du chapitre: _poke_faces ---

--- D\xe9but du chapitre: _polygonization --- The Polygonization modifier simplifies strokes as much as possible (in other words, it transforms smooth strokes into jagged polylines).

The maximum distance allowed between the new simplified stroke and the original one (the larger this value is, the more jagged/approximated the resulting polylines are).

--- Fin du chapitre: _polygonization ---

--- D\xe9but du chapitre: _polyline ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Polyline

The Polyline tool creates multiple straight lines using any of the Drawtype brushes.

You can configure the brush main settings exposed on the Tool Settings for convenience. For the draw brushes configuration and settings see: Draw Brush .

The number of stroke points between each stroke edge.

Use a curve widget to define the stroke thickness from the start (left) to end (right) of the stroke.

When enabled, the stroke use a curve profile to control the thickness along the line.

Picks the brush asset used by the tool.

SeeBrush Assetfor more information.

SeeDraw Brushesfor a detailed list of all draw brushes and their options.

Parameters to control to look of the stroke.

SeeDraw Brushesfor details.

Settings to determine the color of strokes.

SeeColor

In the Tool Settings select the brush, material and color type to use with the tool.The Line tool uses Draw Brushtypes.SeeBrush Settingsfor more information.

Click (LMBor the Pentip) and drag the start point.

Release on the desired end point.

Click multiple times on different locations to create multiple connected lines.

Then confirm (Return/ MMB) or cancel (Esc/ RMB).

While dragging you can use Shiftto snapping the line to horizontal, vertical or 45\xc2\xbd angle.

NumpadPlusand NumpadMinusor using the mouse Wheelwill increase or decrease the amount of points in the final line.

click and dragging the start point. \xc2\xbd

Click multiple times to create multiple connected lines. \xc2\xbd

The polyline after confirming. \xc2\xbd

--- Fin du chapitre: _polyline ---

--- Début du chapitre: _poly_build ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Poly Build

Poly Buildcombines several mesh editing tools into one, letting you work more quickly.It\xe2\x80\x99s especially useful for retopology.

When creating a new triangle that shares an edge with an existing one,automatically dissolves this edge so you\xe2\x80\x99re left with a quad.

Creates a new vertex at the mouse cursor, then creates a triangle using this new vertexand the nearest existing edge. If the existing edge already has two neighboring faces,instead creates a new edge using the new vertex and the nearest existing vertex.Holding Ctrlwill preview the result in blue.

Dissolves the vertex/deletes the face under the mouse cursor.Holding Shiftwill highlight the target element in red.

You can move a vertex by dragging it.

You can extrude an edge into a quad by dragging it.

Tip

It is useful to enableSnappingandAuto Mergewhile tweaking vertices to combine them.

--- Fin du chapitre: _poly_build ---

--- Début du chapitre: _pose ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Deform a model simulating armature-like workflow. This can either be useful for posing a model without a rig, adjusting the proportions of a mesh or other fast deformations.

The brush will automatically determine an origin point, indicated with a white line on the brush cursor.

If the Deformation Target is changed, the brush can also be used for cloth sculpting.

Only Radius and Auto-Masking has an impact on the brush behavior for this brush.

Note

More info at General brush settings and on Advanced brush settings.

Deformation type that is used by the brush.

Rotates the mesh around the pose origin. When pressing Ctrl, the brush applies a twist rotation instead (and disables any IK segments that are used).

Scale the mesh based on the pose origin. While holding Ctrl the brush moves the mesh.

Works similar to Scale/Translate however, it applies different scale values along different axes to achieve the stretching effect.

Method to set the rotation origin for the pose origin or individual IK segments.

Sets the rotation origin automatically using the topology and shape of the mesh.

Creates a pose segment per Face Set, starting from the active face set. This can lead to the most accurate and desirable results.

Simulates a Forward Kinematics deformation using the Face Set under the cursor as control.

Offset of the pose origin in relation to the brush radius. This is useful to manipulate areas with a lot of complex shapes like fingers.

Controls the smoothness of the falloff of the deformation.

Controls how many IK segments are going to be created for posing. This can be seen by a divided white line on the cursor. This is also useful for making curved deformations with the pose brush, like hair clumps and tails.

When using Scale/Translate Deformation, do not rotate the segment; only scaling is applied.

Keeps the position of the last segment in the IK chain fixed. If this is disabled, the mesh can be dragged around more freely, creating snake like shapes.

The brush will only affect topologically connected elements. Disabling this will allow deforming multiple disconnected meshes at the same time, for example characters with clothing & shoes.

Disabling this setting can have a big impact on performance, as neighboring elements will be merged internally. Keeping the Max Element Distance as low as possible will help counteract the performance impact.

Maximum distance to search for disconnected loose parts in the mesh.

--- Fin du chapitre: _pose ---

--- Début du chapitre: _pose_library --- This section describes the pose library, which is based on the Asset Browser. For an overview of the asset system, see the Asset Libraries section. The pose library is meant to be used in Pose Mode. In other words, it only works when posing an armature, and not for general object animation.

Note

The pose library is implemented as an add-on. This add-on is enabled by default; disabling it will remove the pose library from Blender's user interface.

The building blocks of the pose library are actually implemented in Blender itself. The add-on only contains the user interface and the logic that determines what is stored in a pose asset. This was intentionally put into an add-on, so that artists or studios who want to change the behavior can do so with an add-on of their own.

A pose asset is an action that has been marked as asset, and that contains exactly one frame of animation data. Usually these are created via the Create Pose Asset button (see below), but any action that is keyed on exactly one frame can be seen as pose asset.

Each pose in the library is stored in its own action data-block. This means that it can get its own name, its own preview image, and can be organized in Asset Catalogs.

A pose library file is typically a blend-file that is dedicated to poses. It can link in a character, props, etc., which can then not only be used to create the poses, but also for rendering previews.

Example pose library of the Sprite Fright character Ellie.

To create a pose in the library from the Action Editor, pose the character, select the relevant bones, and click the Create Pose Asset button. This will create the new pose Action, which will contain keys for the current value of each bone's location, rotation, scale, and Bendy Bone properties.

To create a new pose asset, use the Create Pose Asset button in the Action editor.

The created Action is now assigned to the armature. This makes it possible to inspect which bones are included and to tweak anything. In that respect, it's an Action like any other, and you can add or remove keys as usual. Just make sure that the keys are all on the same frame, in order to keep this a pose instead of an animation snippet; the latter isn't supported at the moment.

True to its name, the Create Pose Asset button automatically marks the Action as Asset. Not only does this make it available in the pose library, it will also act as a fake user to ensure the Action isn't lost after you unassign it from the armature.

The pose asset can be renamed in the Asset Browser. There you can also right-click on the thumbnail, then choose Assign Action to assign the Action to the active Object (see description above).

Note

The Create Pose Asset button creates a new Action. To make sure that this is actually visible in the user interface, so that you know that something happened, it tries to make sure that the Asset Browser shows the newly created pose asset. Because of this, it requires that there is an Asset Browser visible, and that it's set to show the Current File asset library.

This is especially relevant to pose assets, compared to other assets. You cannot mark an object as asset multiple times, but you can create ten pose assets from the same character pose.

Animators eat and breathe time, so there is a fair chance that you already have some poses lined up on the timeline. Creating a pose asset from existing animation is pretty much the same as described above, with a few subtle differences:

Go to the frame with the pose you want to turn into an asset.

Select the relevant bones and click the Create Pose Asset button in the Action Editor.

This creates an Action as before, but this time it also includes any bone property that was keyed on the current frame. In other words: any bone property (regular and custom) that's displayed in yellow in the user interface will be included in the pose asset. This makes it possible to also include properties that control IK/FK switching, for example. As with the pose, the current value is copied into the pose asset, and not the keyed value.

Blender saves which action was previously assigned to the armature.

The new pose Action is assigned to the armature so you can give it a name and inspect/adjust its contents.

Click the Restore Previous Action button (back arrow icon) that appeared next to the Create Pose Asset button. This reassigns the previous Action, so that you're back at the animation you had before.

As described in Design Limitations, Blender only writes data to the currently open blend-file. To copy a pose from some other file into a pose library file, see the following steps:

Pose the character and select the relevant bones.

Click the Copy Pose as Asset button, which is available in the Action Editor. This will create the pose asset (including its thumbnail) and store it in a temporary file somewhere.

Choose an existing pose asset, and open its context menu. Click the Open Blend File option.

A new Blender process will start, and automatically open the asset library file that contains the chosen pose. By the way, this works for all assets, not just poses!

In the Asset Browser, click the Paste as New Asset button. This will load that temporary file, and load all the assets it can find in there. In our case, it will only find a single pose, but future versions of Blender may extend this for other asset types. This is why the button is named so generically \xe2\x80\x93 it is not pose-specific.

Give the pose a name, and click on the \xe2\x80\x93 refresh \xe2\x80\x93 button in the preview image panel to render a new preview if you want.

Save the file and quit Blender.

The original Blender is still running in the background and notices that the new Blender has quit. It automatically refreshes the Asset Browser to show the newly added pose.

When you create a pose asset, Blender may automatically assign it to an asset catalog. This only works if there is an Asset Browser visible; Blender then assigns the pose asset to its active asset catalog. If there are multiple Asset Browsers open, it performs the following steps:

If the current window has one Asset Browser, it uses that one.

If the current window has multiple Asset Browsers, it uses the biggest one.

Otherwise Blender goes over the other windows (if there are any), and do a similar search. The first window it sees that has an Asset Browser wins.

The pose library preview images are rendered with the active Scene camera. This approach was preferred over rendering a specific 3D Viewport for two main reasons:

There is only one scene camera active at any time, making it predictable which camera is used.

The camera, as well as the rest of the scene, can be set up specifically for rendering the thumbnails. Pose library files are intended for that purpose: to contain the poses and render their preview images.

The preview images are rendered using the Workbench Engine. Switch the scene to use that as render engine, and you'll see various options to influence the look. Select a pose asset and press the Generate Preview button to re-render the preview image with the current settings.

You can also animate settings such as MatCap rendering, light positions, and intensities, etc. Use this to your advantage!

Sometimes it's handy to have a few different background colors or camera angles for your poses. Many facial poses are made with a specific camera angle in mind.

Background color can be animated by placing a plane behind the character and animating its material. In this case just for fun, but for more serious applications this could be used to indicate a certain character, or a mood, or anything else.

The active camera can be switched by using camera markers.

Both make it possible to choose a specific frame to pick the background color and camera angle. Pose the character, click the Create Pose Asset button, and the pose action will be keyed on the current frame. This means it's easy to edit the pose and refresh its preview image, because you know exactly which frame it was originally created on. The pose library can be used to pose a character in a few different ways. In short, you can fully apply a pose or blend it into the character's current pose interactively. How exactly these operations work depends on where you use them. This section will explain the use from both the Asset Browser and the 3D Viewport.

The pose library can be used directly from the Asset Browser. The Pose Library panels will appear when the active object is an armature and in Pose Mode. The catalog system and the filter bar at the top can be used to search for specific poses.

The following operators can be accessed by RMB on a pose:

Applies the pose to the character. If there are any bones selected, the pose will be applied only to those bones. This makes it possible to create a "finger guns" pose by applying a fist pose to the hand, and then an "open hand" pose for only the index finger and thumb. Double-clicking a pose will also apply it.

Will mirror the pose from left to right and vice versa. This makes it possible, for example, to apply a left-hand pose to the right hand, reducing the number of poses you have to put into the library. This can of course also be applied for asymmetrical facial expressions that depend on the camera angle. While blending (see below), keep Ctrl pressed to blend the flipped pose.

Allows you to gradually blend a pose from the library into the character's pose. Click the button, then move the mouse left/right to determine the desired blend. A pose asset can be subtracted while blending. Drag to the right to blend as usual, drag to the left to subtract the pose. While blending, you can use Tab to toggle between the original and the blended pose. As usual in Blender, LMB or press Return to confirm; RMB or press Esc to cancel the operator. Blending can also exaggerate a pose, by pressing E (for Extrapolate) and applying a pose for more than 100%.

Select or deselect the bones that are used in the pose. This can be used to create a selection set, or simply show what was part of the pose and what wasn't.

The pose library in use from the Asset Shelf.

Note

The pose library previously lived in the Sidebar within the Pose Library panel. The panel still exists, but now contains a button to open the asset shelf.

In the 3D viewport, poses can be quickly applied from the Asset Shelf. Contrary to the Asset Browser, the shelf allows you to apply poses quicker.

Click on a pose to apply it. A single click is enough. You can also select and apply a pose via the cursor keys. This allows for fast exploration of the poses, to directly see the result on the active character.

Drag the pose thumbnail left to right to blend it into the character's current pose. Just release the mouse button to confirm.

In Blender 3.0, the Asset Browser based pose library, described above, replaced its predecessor pose library system. This section describes how to convert poses from the old pose library to the current system.

Old-style pose libraries can be converted to pose assets in the following way:

In the Action Editor, select the Action containing the pose library you want to convert.

Make sure the scene camera is set up correctly for rendering preview images.

In the Action Editor's Pose Library panel, click the 'Convert Old-Style Pose Library' button.

Open the Asset Browser, and see the poses have been converted.

If you're happy with the result, remove the old pose library Action.

Save the blend-file.

As usual, the blend-file should be saved to a directory marked as asset library in order to use the pose assets from other blend-files.

Note

This conversion does not assign the poses to any catalog, and so they will appear in the 'Unassigned' section of the 'Current File' asset library.

--- Fin du chapitre: _pose_library ---

--- Début du chapitre: _position --- The Position node outputs a vector of each point of the geometry the node is connected to.

The node can work on geometry domains besides points. In that case, the position data will be automatically interpolated to the new domain. For example, when used as part of the input to the Split Edges Node, the position for each edge will be the average position of the edge's two vertices.

For instances themselves, the output is the origin of each instance. However, if the node is for a geometry node that adjusts data inside instances, the position output of this node will be in the local space of each instance. See the Instance Processing page for more details.

This node has no inputs.

This node has no properties.

Vector that indicates the location of each element of the geometry.

--- Fin du chapitre: `_position` ---

--- Début du chapitre: `_posterize` --- The Posterize Node reduces the number of colors in image, converting smooth gradients into sharp transitions. This node is useful for generating masks in particular for rotoscoping.

Standard color input.

The number of colors per channel; A value of 8 will result in $8^3 = 512$ total colors.

This node has no properties.

Standard color output.

--- Fin du chapitre: `_posterize` ---

--- Début du chapitre: `_post_processing` --- Reference

Properties `Output` Post Processing

The Post Processing panel is used to control different options used to process your image after rendering.

Post Processing panel.

Todo.

Renders the output from the compositing node setup, and then applies the Composite node tree on all images, displaying the image inputted in the Composite Output node.

Renders the output of the Video Sequence editor, instead of the view from the 3D scene's active camera. If the sequence contains Scene strips, these will also be rendered as part of the pipeline. If Compositing is also enabled, the Scene strip will be the output of the Compositor.

Dithering is a technique for blurring pixels to prevent banding that is seen in areas of gradients, where stair-stepping appears between colors. Banding artifacts are more noticeable when gradients are longer, or less steep. Dithering was developed for graphics with low bit depths, meaning they had a limited range of possible colors.

Dithering works by taking pixel values and comparing them with a threshold and neighboring pixels then does calculations to generate the appropriate color. Dithering creates the perceived effect of a larger color palette by creating a sort of visual color mixing. For example, if you take a grid and distribute red and yellow pixels evenly across it, the image would appear to be orange.

--- Fin du chapitre: `_post_processing` ---

--- Début du chapitre: `_precision` --- Reference

Object and Edit Modes

Ctrl and/or Shift

Holding Ctrl during a transform operation (such as move, rotate or scale) will toggle Transform Snapping. When using Increment Snap this allows the transformation to be performed in discrete amounts.

Holding Shift during a transform operation will transform the object at 1/10th the speed, allowing much finer control.

The magnitude of the transformation can be viewed in the 3D Viewport header. Releasing Ctrl or Shift during the transformation will cause the movement to revert back to its normal mode of operation.

Note

The snapping behaviors described on this page only apply when Increment Snap is selected.

Tip

It is possible to enable both snapping and precision mode, simply hold Ctrl and Shift. This has the following effects:

Changes in 0.1 unit increments, regardless of zoom level.

Changes in 1 unit increments.

Changes in 0.01 unit increments.

Press G, R or S and then hold either Ctrl, Shift or Shift-Ctrl.

Select the gizmo handle then while moving the mouse hold Ctrl, Shift or Shift-Ctrl to activate precision control or snapping.

See also

Read more about the Transform Gizmo .

Tip

Combining with Other Controls

All of the precision controls detailed on the page can be combined with the Axis Locking controls and used with the different Pivot Points .

One unit (default zoom level).

Snapping while moving objects changes the object location in 1 unit increments. While in an aligned view, the increment amount is changed based on the zoom level. For example, at a base zoom level objects are moved in increments of 1 unit (i.e. between the two light gray lines). Zooming in enough to see the next set of gray lines will snap in increments of 1/10 of a unit. Zooming in further until will snap in increments of 1/100 of a unit and so on until the zoom limit is reached. Zooming out will have the opposite effect and cause movement to happen by increments of 10, 100 units, etc.

Holding Ctrl will cause rotations of 5 degrees.

Holding Ctrl will cause size changes in increments of 0.1 units.

Note

Snapping Modes

Note that when you are Snapping To something other than Increment, holding Ctrl will cause the selection to snap to that nearest element.

Read more about snapping .

Holding Shift during transformations allows for very fine control that does not rely on fixed increments. Rather, large movements of the mouse across the screen only result in small transformations of the selection. In rotation mode the selected element will be rotated in 0.01 degree increments.

--- Fin du chapitre: _precision ---

--- Début du chapitre: _preview --- The Preview panel gives a quick visualization of the active material applied in a simple scene.

Preview the material on a Plane, Sphere, Cube, Hair, Shader Ball, Cloth or Fluid object. This shape is also used for previews when linking and appending materials.

Use the world from the current scene for lighting in the material preview.

Preview shapes.

--- Fin du chapitre: _preview ---

--- Début du chapitre: _previews --- A blend-file can store previews, both for itself, and for some of its data-blocks. You can disable writing any previews when saving a blend-file using the Save Preview Images setting from the Save & Load section of the Preferences.

Blender saves by default a small preview of current scene in the blend-file. This will show in the Thumbnailview of the File Browser . During its installation, Blender also adds a small tool to your OS, that will allow your system file browser to show those previews as file thumbnails as well.

Blender will automatically generate previews for some type of data, mainly the ones related to shading (like images, textures, materials, lights and world shaders).

It can also store previews for scenes, collections and objects, but those need to be generated manually.

These previews can then be used by the Thumbnailview of the File Browser, when linking or appending data-blocks.

Reference

File \xe2\x80\xa3 Data Previews \xe2\x80\xa3 Refresh Data-blocks Previews

Refresh all data-block previews that can be automatically generated by Blender (shading-related ones), in the current blend-file. You still need to save the file if you want to write them to the drive.

Reference

File \xe2\x80\xa3 Data Previews \xe2\x80\xa3 Batch Generate Previews

Generate some data-block types \xe2\x80\x99 previews (you can choose which in its options), in one or more blend-files on your drive. You should not use this operator on the file currently opened in Blender.

This is currently the only way to generate and store in blend-files previews for scenes, collections and objects. Note that since this involves a lot of rendering, even of small sizes, the process may take some time to complete.

Generate previews of scenes and their collections.

Generate previews of collections of objects.

Generate previews of objects.

Generates previews for materials, textures, images, and other internal data.

When enabled, Python scripts and drivers that may be included in the file will be run automatically. Enable this only if you created the file yourself, or you trust that the person who gave it to you did not include any malicious code with it. See [Python Security](#) to configure default trust options.

Keep a backup version (blend1-file) of the files when saving with generated previews.

Reference

File \xe2\x80\xa3 Data Previews \xe2\x80\xa3 Clear Data-blocks Previews

Clear all, a generic type of, or a specific data-block type of previews in the current blend-file. You still need to save the file if you want to clear them from the drive.

Reference

File \xe2\x80\xa3 Data Previews \xe2\x80\xa3 Batch Clear Previews

Clear some data-block types \xe2\x80\x99 previews (you can choose which in its options), in one or more blend-files on your drive. You should not use this operator on the file currently opened in Blender.

Clear previews of scenes and their collections.

Clear previews of collections of objects.

Clear previews of objects.

Clear previews for materials, textures, images, and other internal data.

When enabled, Python scripts and drivers that may be included in the file will be run automatically. Enable this only if you created the file yourself, or you trust that the person who gave it to you did not include any malicious code with it. See [Python Security](#) to configure default trust options.

Keep a backup version (blend1-file) of the files when saving with cleared previews.

--- Fin du chapitre: _previews ---

--- Début du chapitre: _primitives --- Reference

Object Mode and Edit Mode

Add \xe2\x80\xa3 Grease Pencil

Shift- A

In Object Mode, the Add menu provides three different Grease Pencil primitives with preset materials and 2D layers:

Grease Pencil primitives. \xc2\xb6

Adds a Grease Pencil object without any stroke.

Adds a Grease Pencil object with a simple stroke as a reference.

It creates a 2D monkey head. The Monkey's name is \xe2\x80\x9cSuzanne\xe2\x80\x9d and is Blender's mascot. 2D Suzanne is very useful as a standard test.

Sets up a Line Art Modifier for the active scene by creating an \xe2\x80\x9cempty\xe2\x80\x9d Grease Pencil object with a Line Art modifier referencing each object in the scene.

Sets up a Line Art Modifier for the active collection by creating an \xe2\x80\x9cempty\xe2\x80\x9d Grease Pencil object with a Line Art modifier referencing each object in the collection.

Sets up a Line Art Modifier for the active object by creating an \xe2\x80\x9cempty\xe2\x80\x9d Grease Pencil object with a Line Art modifier referencing the active object.

--- Fin du chapitre: _primitives ---

--- Début du chapitre: _principled --- The Principled BSDF that combines multiple layers into a single easy to use node. It can model a wide variety of materials.

It is based on the OpenPBR Surface shading model, and provides parameters compatible with similar PBR shaders found in other software, such as the Disney and Standard Surface models. Image textures painted or baked from software like Substance Painter may be directly linked to the corresponding input in this shader.

The base layer is a mix between metal, diffuse, subsurface, and transmission components. Most materials will use one of these components, though it is possible to smoothly mix between them.

The metal component is opaque and only reflect lights. Diffuse is fully opaque, while subsurface also involves light scattering just below the surface. Both diffuse and subsurface sit below a specular layer. The transmission component includes both specular reflection and refraction. On top of all base layers there is an optional glossy coat. And finally the sheen layer sits on top of all other layers, to add fuzz or dust. Light emission can also be added. Light emits from below the coat and sheen layers, to model for example emissive displays with a coat or dust. Overall color of the material used for diffuse, subsurface, metal and transmission.

Same base color for multiple materials types \xc2\xb6

Specifies microfacet roughness of the surface for specular reflection and transmission. A value of 0.0 gives a perfectly sharp reflection, while 1.0 gives a diffuse reflection.

Roughness from 0.0 to 1.0 \xc2\xbb6

Blends between a dielectric and metallic material model. At 0.0 the material consists of a diffuse or transmissive base layer, with a specular reflection layer on top. A value of 1.0 gives a fully specular reflection tinted with the base color, without diffuse reflection or transmission.

Metallic from 0.0 to 1.0 \xc2\xbb6

Index of refraction (IOR) for specular reflection and transmission. For most materials, the IOR is between 1.0 (vacuum and air) and 4.0 (germanium). The default value of 1.5 is a good approximation for glass.

IOR from 1.0 to 2.0 \xc2\xbb6

Controls the transparency of the surface, with 1.0 fully opaque. Usually linked to the Alpha output of an Image Texture node.

Alpha from 0.0 to 1.0 \xc2\xbb6

Controls the normals of the base layers.

Surface roughness; 0.0 gives standard Lambertian reflection, higher values activate the Oren-Nayar BSDF.

Subsurface scattering is used to render materials such as skin, milk and wax. Light scatters below the surface to create a soft appearance.

Rendering method to simulate subsurface scattering.

An approximation to physically-based volume scattering. This method is less accurate than Random Walk, however, in some situations this method will resolve noise faster.

Cycles Only Provides accurate results for thin and curved objects. Random Walk uses true volumetric scattering inside the mesh, which means that it works best for closed meshes. Overlapping faces and holes in the mesh can cause problems.

Cycles Only Random walk method optimized for skin rendering. The radius is automatically adjusted based on the color texture, and the subsurface entry direction uses a mix of diffuse and specular transmission with custom IOR. This tends to retain greater surface detail and color and matches measured skin more closely.

Blend between diffuse surface and subsurface scattering. Typically should be zero or one (either fully diffuse or subsurface).

Weight from 0.0 to 1.0 \xc2\xbb6

Average distance that light scatters below the surface. Higher radius gives a softer appearance, as light bleeds into shadows and through the object. The scattering distance is specified separately for the RGB channels, to render materials such as skin where red light scatters deeper. The X, Y and Z values are mapped to the R, G and B values, respectively.

Radius from white to red \xc2\xbb6

Scale applied to the radius.

Scale from 0 cm to 50 cm \xc2\xbb6

Index of refraction (IOR) used for rays that enter the subsurface component. This may be set to a different value than the global IOR to simulate different layers of skin.

IOR from 1.0 to 2.0 \xc2\xbb6

Directionality of volume scattering within the subsurface medium. Zero scatters uniformly in all directions, with higher values scattering more

strongly forward. For example, skin has been measured to have an anisotropy of 0.8.

Anisotropy from 0.0 to 1.0

Controls for both the metallic component and specular layer on top of diffuse and subsurface.

Microfacet distribution to use.

A method that is faster than Multiple-scattering GGX but is less physically accurate.

Takes multiple scattering events between microfacets into account. This gives more energy conserving results, which would otherwise be visible as excessive darkening.

Adjustment to the IOR to increase or decrease intensity of the specular layer. 0.5 means no adjustment, 0 removes all reflections, 1 doubles them at normal incidence.

This input is designed for conveniently texturing the IOR and amount of specular reflection.

IOR level from 0.0 to 1.0

Color tint for specular and metallic reflection.

For non-metallic tints provides artistic control over the color specular reflections at normal incidence, while grazing reflections remain white. In reality non-metallic specular reflection is fully white.

For metallic materials tints the edges to simulate complex IOR as found in materials such as gold or copper.

Tint from white to orange

Amount of anisotropy for specular reflection. Higher values give elongated highlights along the tangent direction; negative values give highlights shaped perpendicular to the tangent direction.

Anisotropic from 0.0 to 1.0

Rotates the direction of anisotropy, with 1.0 going full circle.

Compared to the Glossy BSDF node, the direction of highlight elongation is rotated by 90. Add 0.25 to the value to correct.

Anisotropic rotation from 0.0 to 1.0

Controls the tangent direction for anisotropy.

Transmission is used to render materials like glass and liquids, where the surface both reflects light and transmits it into the interior of the object

Mix between fully opaque surface at zero and fully transmissive at one.

Weight from 0.0 to 1.0

Coat on top of the materials, to simulate for example a clearcoat, lacquer or car paint.

Controls the intensity of the coat layer, both the reflection and the tinting. Typically should be zero or one for physically-based materials, but may be textured to vary the amount of coating across the surface.

Weight from 0.0 to 1.0

Roughness of the coat layer.

Roughness from 0.0 to 1.0

Index of refraction (IOR) of the coat layer. Affects its reflectivity as well as the falloff of coat tinting.

IOR from 1.0 to 2.0

Adds a colored tint to the coat layer by modeling absorption in the layer. Saturation increases at shallower angles, as the light travels farther through the medium, depending on the IOR.

Tint from white to blue

Controls the normals of the Coatlayer, for example to add a smooth coating on a rough surface.

Sheen simulates very small fibers on the surface. For cloth this adds a soft velvet like reflection near edges. It can also be used to simulate dust on arbitrary materials.

Controls in the intensity of the sheen layer.

Weight from 0.0 to 1.0

Roughness of the sheen reflection.

Roughness from 0.0 to 1.0

The color of the sheen reflection.

Tint from white to green.

Light emission from the surface.

Color of light emission from the surface.

Emission color variations

Strength of the emitted light. A value of 1.0 ensures that the object in the image has the exact same color as the Emission Color, i.e. make it shadeless.

Strength from 0.0 to 10.0

Thin Film simulates the effect of interference in a thin film sitting on top of the material. This causes the specular reflection to be colored in a way which strongly depends on the view angle as well as the film thickness and the index of refraction (IOR) of the film and the material itself.

This effect is commonly seen on e.g. oil films, soap bubbles or glass coatings. While its influence is more obvious in specular highlights, it also affects transmission.

Note

Thin-film interference is currently only applied to dielectric materials. Support for thin films on top of Metallic is planned in the future.

The thickness of the film in nanometers. A value of 0 disables the simulation. The interference effect is strongest between roughly 100 and 1000 nanometers, since this is near the wavelengths of visible light.

Index of refraction (IOR) of the thin film. The common range for this value is between 1.0 (vacuum and air) and roughly 2.0, though some materials can reach higher values. The default value of 1.33 is a good approximation for water. Note that when the value is set to 1.0 or to the main IOR of the material, the thin film effect disappears since the film optically blends into the air or the material.

Standard shader output.

--- Fin du chapitre: principled ---

--- Début du chapitre: projections ---

Reference
All modes

View Perspective/Orthographic

Numpad5

This operator changes the projection of the viewport camera. Each 3D Viewport supports two different types of projection. These are demonstrated in the Fig. below.

Orthographic projection.

Perspective projection.

Our eyes are used to perspective viewing where distant objects appear smaller. Orthographic projection often seems a bit odd at first, because objects stay the same size regardless of their distance. It is like viewing the scene from an infinitely distant point. Nevertheless,

orthographic viewing can be very useful, because it provides a more technical insight into the scene, making it easier to model and judge proportions.

To toggle between the two projections for the 3D Viewport, select View → Perspective/Orthographic or use the shortcut Numpad5.

Changing the projection for a 3D Viewport does not affect the way the scene will be rendered. Rendering is in perspective by default. If you need to create an orthographic rendering, select the camera, go to the Camera tab in the Properties editor, and set the Type in the Lens panel to Orthographic.

See also

Auto-Perspective Preference

Camera Lens Type

Camera Projections

--- Fin du chapitre: _projections ---

--- Début du chapitre: _project_point --- Applies a projection matrix to a point. Specifically, this node turns the given Euclidean vector (X, Y, Z) into the homogeneous vector (X, Y, Z, 1), multiplies the given projection matrix by it, and turns the resulting homogeneous vector back into a Euclidean one by dividing it by the absolute value of its W component. This last step is also known as perspective division.

The position vector to project.

The projection matrix.

This node has no properties.

The projected position vector.

--- Fin du chapitre: _project_point ---

--- Début du chapitre: _propagate --- Reference

Pose Mode

Pose → Propagate

Alt- P

The Propagate tool copies the pose of the selected bones on the current frame over to the keyframes delimited by the Termination Mode. It automates the process of copying and pasting.

Modes which determine how it decides when to stop overwriting keyframes.

Simply copies the pose to the first keyframe after (but not including any keyframe on) the current frame.

Will simply replace the last keyframe (i.e. making action cyclic).

To all keyframes between current frame and the End frame option. This option is best suited for use from scripts due to the difficulties in setting this frame value, though it is possible to set this manually via the Adjust Last Operation panel if necessary.

To all keyframes from current frame until no more are found.

Will apply the pose of the selected bones to all selected keyframes.

To all keyframes occurring on frames with Scene Markers after the current frame.

Defines the upper-bound for the frame range within which keyframes will be affected (with the lower bound being the current frame).

--- Fin du chapitre: _propagate ---

--- Début du chapitre: _propagate_shapes --- Reference

Edit Mode

Vertex → Propagate to Shapes

Apply selected vertex locations to all othershape keys.

--- Fin du chapitre: `_propagate_shapes` ---

--- Début du chapitre: `_properties` ---Reference

Edit Mode

Armature \xe2\x80\xa3 Bone Settings \xe2\x80\xa3 \xe2\x80\xa6

Shift- W , Shift- Ctrl- W , Alt- W

Most bones\xe2\x80\x99 properties (except the transform ones) are regrouped in each bone\xe2\x80\x99s panels,in the Bonestab in Edit Mode. Let us detail them.

Note that some of them are also available in the 3D Viewport,through the three pop-up menus within the same entry:

Toggle Setting:Shift- Wor Armature \xe2\x80\xa3 Bone Settings

\xe2\x80\xa3 Toggle a Setting

Enable Setting:Shift- Ctrl- Wor Armature \xe2\x80\xa3 Bone Settings

\xe2\x80\xa3 Enable a Setting

Disable Setting:Alt- Wor Armature \xe2\x80\xa3 Bone Settings

\xe2\x80\xa3 Disable a Setting

Always display the bone as wireframe.

(alsoShift- W\xe2\x80\xa3 (Deform, \xe2\x80\xa6)).

(alsoShift- W\xe2\x80\xa3 (Multiply Vertex Group by Envelope, \xe2\x80\xa6)).

These settings control how the bone influences its geometry, along with the bones\xe2\x80\x99 joints radius.This will be detailed in theskinning part.

The bone automatically rotates together with its parent in Pose Mode. For more details,see therelations page.

(alsoShift- W\xe2\x80\xa3 (Locked, \xe2\x80\xa6))This will prevent all editing of the bone in Edit Mode;seebone locking.

--- Fin du chapitre: `_properties` ---

--- Début du chapitre: `_properties_editor` --- The Properties, with Objectproperties shown.\xc2\xb6

The Properties editor shows and allows editing of many active data, including the active scene and object.

The Properties editor has several categories, which can be chosen via tabs (the icons column to its left).Each tab regroups properties and settings of a data type, and is documented in its own manual sections, linked below.

This first tab contains settings for the activetool(in the 3D Viewport)and the currentworkspace.

These tabs contain settings for the active scene.

Render:EEVEE,CyclesorWorkbenchsettings

Output

View Layer

Scene

World

This tabs contain settings for the activeCollection.

These tabs are used to add features, and to change properties for the active object.Dependent on the type of the active object, some of those will be hidden.

Object

Modifiers(orGrease Pencil Modifiers)

Object Visual Effects

Particles

Physics

Object Constraints

The main tab of that category (often the only one) always has the same name, Object Data, but its icon will change based on the actual type of the active object.

Geometry Objects:

Mesh

Curve

Surface

Text

Metaball

Grease Pencil

Rigging and Deformation Objects:

Armature

Bone

Bone Constraints

Lattice

Other Types of Objects:

Empty

Speaker

Camera

Light

Light Probe

Depending on the type of the active object, some of those will be hidden.

Material

Texture

The header of the Properties. \xc2\xbb6

Allows you to search for a property by its name. The results will be highlighted with their corresponding panel also highlighted and expanded. The search also works across multiple Tabs; graying out tabs with no search results. You can start a search using Ctrl- F or clear a search with Alt- F. This filter is useful if you do not remember where a property is within the Properties.

Just below the header is a list of icons and text items which show the owner of the properties being edited, together with some dependency context if needed. In the example above, the material

\xe2\x80\x9cMaterial\xe2\x80\x9d is used by the active object

\xe2\x80\x9cCube\xe2\x80\x9d.

By toggling on the pin icon to the right, Blender will display only the currently shown data-block\xe2\x80\x99s properties in that editor, disregarding further selection changes. Toggle off that pin to switch back to default behavior, showing active data-block\xe2\x80\x99s properties.

Sync tabs with the Outliner data-block selection. See Properties Editor Sync for more information.

Always, when possible, change Property editor tabs when clicking an icon in an Outliner.

Never change Property editor tabs when clicking an icon in an Outliner.

Change tabs only when the Property editor shares a border with an Outliner.

--- Fin du chapitre: _properties_editor ---

--- Début du chapitre: _property_weights ---Reference

Physics \xe2\x80\xa3 Cloth \xe2\x80\xa3 Property Weights

This panel is used to constrain certain cloth properties to a certain vertex group. The properties that they control can be found in a combination of the Physical Properties and Shape panels.

Defines a vertex group to control over structural stiffness.

Maximum tension stiffness value.

Maximum Compression stiffness value.

Vertex group for fine control over shear stiffness.

Maximum shear scaling value.

Vertex group for fine control over bending stiffness.

Maximum bending stiffness value.

Vertex group for shrinking cloth.

Max amount to shrink cloth by, specifying a negative value controls the max amount for the cloth to grow.

--- Fin du chapitre: _property_weights ---

--- Début du chapitre: _proportional_editing ---Reference

Object and Edit Mode

Proportional Editing

O

Proportional Editing popover.\xc2\xbb

Proportional Editing is a way of transforming selected elements while also affecting the nearby unselected elements. The farther away an unselected element is, the less it will be affected (hence the \xe2\x80\x93proportional\xe2\x80\x9d). This feature is very useful for smoothly deforming dense meshes.

Note

Blender also has a Sculpting workflow that contains brushes and tools for proportionally editing a mesh without seeing the individual vertices. Proportional Editing is off, only selected vertices will be affected. Vertices other than the selected vertex are affected, within a defined radius.

You can increase or decrease the radius of the tool\xe2\x80\x99s influence during a transform operation from the Proportional Editing popover or with WheelUp/ WheelDown, or PageUp/ PageDown respectively. As you change the radius, the points surrounding your selection will adjust their positions accordingly.

Influence circle.\xc2\xbb

While editing, you can change the curve profile by either clicking the Falloff icon in the header or pressing Shift-O to get a pie menu.

Constant, No Falloff. \xc2\xbb

Random Falloff. \xc2\xbb

Linear Falloff. \xc2\xbb

Sharp Falloff. \xc2\xbb

Root Falloff. \xc2\xbb

Sphere Falloff. \xc2\xbb

Smooth Falloff. \xc2\xbb

Inverse Square Falloff. \xc2\xbb

Proportional Editing is typically used in Edit Mode, but it can also be used in Object Mode. The tool then works on entire objects rather than individual mesh components.

In the image below, the leftmost cylinder is being scaled up vertically, which also affects the cylinders near it.

Proportional Editing in Object Mode. \xc2\xbb

When working with dense geometry, it can become difficult to make subtle adjustments without causing visible lumps and creases in the model's surface. When you face situations like this, Proportional Editing can help.

Proportional Editing in Edit Mode. 

Rather than using a radius only, the proportional falloff spreads via connected geometry. This means that you can proportionally edit the vertices in a finger of a hand without affecting the other fingers. While the other vertices are physically close (in 3D space), they are far away following the topological edge connections of the mesh. The icon will have a blue center when Connected is active. This mode is only available in Edit Mode.

Depth along the view is ignored when applying the radius.

The difference between having  Projected from View  disabled (left) and enabled (right). 

The image below shows the final render of a low-poly landscape obtained by moving up the vertices of a triangulated grid with Proportional Editing enabled.

A landscape obtained via Proportional Editing. 

--- Fin du chapitre: proportional_editing ---

--- Début du chapitre: proxy --- As projects involve increasingly high-resolution footage, the performance of the video preview can decrease drastically. To combat this, Blender can generate proxies  copies of the original footage stored at a lower quality and/or resolution  to maintain a smooth editing experience without compromising visual fidelity in the end result.

The quickest way to set up proxies for videos is to simply select a Proxy Render Size in the Viewtab (visible when the editor is in Preview or Sequencer & Preview mode). This will automatically enable the selected proxy resolution in all the strips and start generating the downscaled video files.

You can use the Proxytab if you want to configure proxies in more detail (or create proxies for image sequences).

Reference

Sidebar region  Proxy  Proxy Settings
Contains scene-wide proxy settings.

How proxies are stored for the project.

Each strip can specify where to store its proxies (see below).

All proxies are stored in one directory.

The location to store the proxies for the project.

Shows a pop-over that lets you choose the resolution(s) to generate and whether to overwrite existing proxy files. Once you confirm with the Set button, your choices are applied to the selected strips. You can view and tweak these settings for individual strips in the Strip Proxy & Timecode panel (see below).

In the Preview mode, where the Proxytab is not available, this is instead done through the menu View  Proxy  Setup.

Generates proxies and time indices for the selected strips.

In the Preview mode, where the Proxytab is not available, this is instead done through the menu View  Proxy  Rebuild.

Reference

Sidebar region  Proxy & Timecode  Strip Proxy & Timecode

Contains strip-specific proxy settings. The checkbox in the header can be used to enable/disable proxy generation.

By default, all generated proxy videos are stored to the folder `<path to original footage>/BL_proxy/<clipname>`, but this can be changed to a custom directory using this option.

Allows you to use preexisting proxies.

The resolution(s) of the proxy videos to generate; multiple sizes can be selected.

Whether to overwrite existing proxy files or keep them.

The video/image quality for proxies.

When you are working with footage directly copied from a camera without preprocessing it, there might be numerous artifacts, mostly due to seeking to a given frame in the sequence. This happens because such footage usually does not have correct frame rate values in the file header. This issue can still arise when the source clip has the same frame rate as the scene settings. In order for Blender to correctly calculate the frames and frame rate there are two possible solutions:

Preprocess your video with e.g. MEncoder to repair the file header and insert the correct keyframes.

Use the Timecode Index option in Blender.

Ignore generated timecodes, seek in movie stream based on calculated timestamp.

Seek based on timestamps read from movie stream, giving the best match between scene and movie times.

Effectively convert movie to an image sequence, ignoring incomplete or dropped frames, and changes in frame rate.

Note

Record Run is the Timecode Index which usually is best to use, but if the source file is totally damaged, Record Run No Gaps will be the only chance of getting an acceptable result.

--- Fin du chapitre: _proxy ---

--- Début du chapitre: _puff_curves --- Makes the curves stand up. The brush aligns the curve with the surface normal and makes sure that points don't move closer to the root point.

--- Fin du chapitre: _puff_curves ---

--- Début du chapitre: _pull_requests --- This page describes the tools used for code contribution and review.

Reviews are a key measure to assure changes are of a good quality. They help preventing bugs, design inconsistencies, or potential maintenance problems. And having your work reviewed also generally keeps you on your toes.

Note

Writers who have been given commit access can commit to the main repository without needing to fork the repository.

See Commit Guidelines if this applies to you.

This assumes you have the Blender manual repository already checked out on your computer, following the install instructions.

Go to Blender repository and click the Fork button.

Confirm the fork with the default settings.

Now you will have to add your personal fork as a remote in your local git repository. Click SSH to see the correct URL, and then add it like this:

Note

In order to push to the fork repository, you need an SSH key. If you don't already have the file `~/.ssh/id_rsa.pub`, there's a simple command to generate such keys which works on Linux, macOS, and in Git Bash on Windows:

This command will generate a private key `id_rsa` and a public key `id_rsa.pub` in `~/.ssh`. The private key must never be shown or sent to anyone else to avoid compromising your account, but the public key is safe to share.

The contents of `~/.ssh/id_rsa.pub` can be copied and pasted into the account settings on projects.blender.org, after clicking "Add Key". Any name for the SSH key is ok.

The workflow for working with pull requests can be found in the Blender Developer's Documentation.

Note, some text in the above guideline is focused on the main Blender repository, however, the workflow is the same for any Blender project. The pull request text should be usable as the git commit message (see the guidelines for details).

Be explicit when some changes are to be addressed before committing, without the need for a review iteration.

If the pull request is not approved the author is expected to make another iteration.

If the change needs agreement on the design task first, put the pull request on hold by adding a WIP: prefix in the title, indicating the author considers the pull request not ready to be merged. No review is expected unless the author specifically asks for it.

Writers are expected to reply to pull requests in 3 working days.

Add relevant modules/projects to the labels.

Encourage new writers to do review, it's a good way to learn and important to grow the project.

To get the patch file, add `.patch` to the end of the URL of the pull request. Example:

Checkout a pull request into a detached head (not leaving behind a branch). Example:

```
--- Fin du chapitre: _pull_requests ---
```

```
--- Début du chapitre: _push_pull ---Reference
```

Object and Edit Modes

- Toolbar Shrink/Fatten Push/Pull

- Object/Mesh Transform Push/Pull

- Push/Pull distance.

Moves the selected elements closer to (Push) or further away from (Pull) the pivot point, all by the same distance. You can control this distance by moving the mouse up or down, typing a number, or using the slider in the Adjust Last Operation panel.

- Equidistant objects being pushed together.

- Random objects being pushed together.

- Push (middle) vertices around the 3D cursor compared to Scale (right).

```
--- Fin du chapitre: _push_pull ---
```

```
--- Début du chapitre: _python ---While not common practice, Python add-ons can be distributed with their own precompiled libraries.Unlike
```

regular Python scripts, these are not portable between different platforms.

It is possible the library is incompatible with your Blender installation (attempting to load a library built for a different version of Python, or loading a 32-bit library on a 64-bit system).

If the add-on contains .pyd or .so files, check that the distribution is compatible with your operating system.

If Python is raising errors or you have an add-on that just fails when enabled with an error, e.g: ...isnotavalidWin32application. .

A Python traceback. \xc2\xbf

This may be caused by some inconsistency in the Python libraries. While Blender comes with its own bundled Python interpreter, duplicate, incompatible libraries can cause problems.

To find out which Python Library caused the Problem check the error message.

This is normally reported somewhere around the bottom line of the traceback. With the error above you see the problem is caused while trying to import _socket . This corresponds to either a file named _socket.py or _socket.pyd .

To help troubleshoot this problem, the following script can be pasted into the Text editor and run to check for duplicate libraries in your search path. (The output will show in Command Line Window .)

--- Fin du chapitre: _python ---

--- Début du chapitre: _python_console --- The Python Console offers a quick way to test code snippets and explore Blender's API. It executes whatever you type on its >>> prompt and has command history and auto-complete.

Python Console. \xc2\xbf

Increases/decreases the font size.

Moves the cursor to the beginning of the previous word. If the cursor is in the middle of a word, the cursor is moved to the beginning of the current word.

Moves the cursor to the end of the next word. If the cursor is in the middle of a word, the cursor is moved to the end of the current word.

Moves the cursor to the start of the current line.

Shift- Home : Selects all text between the cursor and the start of the current line.

Moves the cursor to the end of the current line.

Shift- End : Selects all text between the cursor and the end of the current line.

Refreshes the console, giving the view a fresh start. Note that command history is not cleared.

Removes everything from the prompt line.

Deletes everything between the cursor and the beginning of the previous word (separated by periods). If the cursor is in the middle of a word, deletes everything to the beginning of the current word.

Deletes everything between the cursor and the end of the next word. If the cursor is in the middle of a word, deletes everything to the end of the current word.

Copies the full history buffer to the clipboard. This can be pasted into a text file to be used as a Python script.

Copies the selected text into the clipboard and deletes it.

Copies the selected text into the clipboard.

Pastes into the command line.
Inserts a tab character at the cursor.
Unindents the selection.
Changes the current command to the previous one from the command history.
Changes the current command to the next one from the command history.
See Auto Completion.

Key Bindings

LMB\xe2\x80\x93 Moves the cursor along the input line.
Left/ Right\xe2\x80\x93 Moves the cursor by one character.
Ctrl- Left/Ctrl- Right\xe2\x80\x93 Moves the cursor by one word.
Shift- Left/Shift- Right\xe2\x80\x93 Selects characters to the left/right.
Shift- Ctrl- Left/Shift- Ctrl- Right\xe2\x80\x93 Selects words to the left/right.
Ctrl- ASelects all text and text history.
Backspace/ Delete\xe2\x80\x93 Erase characters.
Ctrl- Backspace/Ctrl- Delete\xe2\x80\x93 Erase words.
Return\xe2\x80\x93 Execute command.
Shift- Return\xe2\x80\x93 Add to command history without executing.
Some variables and modules are available for convenience:
C: Quick access to bpy.context.
D: Quick access to bpy.data.
bpy: Top level Blender Python API module.
To see the list of global functions and variables, typedir() and press Return to execute it.

The Console can preview the available members of a module or variable. As an example, type bpy and press Tab:

The submodules are listed in green. Attributes and methods will be listed in the same way, with methods being indicated by a trailing (.).

This module gives you access to the current scene, the currently selected objects, the current object mode, and so on.

Note

For the commands below to show the proper output, make sure you have selected object(s) in the 3D Viewport.

Get the current 3D Viewport mode (Object, Edit, Sculpt, etc.):

Get the active object:

Change the active object\xe2\x80\x93s X coordinate to 1:

Move the active object by 0.5 along the X axis:

Change all three location coordinates in one go:

Change only the X and Y coordinates:

Get the selected objects:

Get the selected objects excluding the active one:

Gives you access to all the data in the blend-file, regardless of whether it\xe2\x80\x93s currently active or selected.

\xe2\x80\x93 Operators\xe2\x80\x93 are actions that are normally triggered from a button or menu item but can also be called programmatically. See the bpy.ops API documentation for a list of all operators.

--- Fin du chapitre: _python_console ---

--- D but du chapitre: _python_wheels --- Python wheels (*.whl) are the standard way of distributing Python modules. They are supported in Blender to make self-contained Python Extensions.

By convention, always locate the files under ./wheels/ .

Wheels must be bundled unmodified from Python's package index. Wheels must include their dependencies. Wheels filenames must match Python's binary distribution specification: see docs. Wheels downloaded from Python's package index will follow this convention.

Use forward slashes as path separators when listing them on the manifest

.

Python wheels(*.whl) can be bundled using the following steps.

Download the wheel to the directory ./wheels/ .

For wheels that are platform independent this example downloads jsmin :

For wheels that contain binary compiled files, wheels for all supported platforms should be included:

This example downloads pillow - the popular image manipulation module.

The available platform identifiers are listed on pillow's download page.

In blender_manifest.toml include the wheels as a list of paths, e.g.

Now installing the package will extract the wheel into the extensions own site-packages directory.

Once the extension has been installed you can check the module is being loaded by importing it in the Python console and printing its location:

Wheels can severely impact the size of an extension. To mitigate this, it is possible to build different extension zip files for each unique required platform.

For this you need to use the --split-platforms option from the build command.

Manifest file excerpt:

Generated files from--split-platforms:

my_addon_with_wheels-1.0.0-windows_x64.zip

my_addon_with_wheels-1.0.0-macos_x64.zip

Note

Even though there is a Linux-only wheel present, no Linux zip file is generated. This happens because the platforms field only has Mac and Windows.

--- Fin du chapitre: _python_wheels ---

--- Début du chapitre: _quadratic_bezier --- The Quadratic

Bezier node generates a poly spline curve from the given control points. The generated shape is a parabola.

The number of edges on the curve.

Positions of the three control points. The generated curve passes through the two end points, and is tangent to the lines between the middle point and the two end points.

This node has no properties.

Poly spline generated from the inputs.

--- Fin du chapitre: _quadratic_bezier ---

--- Début du chapitre: _quadrilateral --- The Quadrilateral node generates a polygon with four points, with different modes.

The X axis size of the shape.

The Y axis size of the shape.

The distance between the bottom or top point and the X axis, in Kite mode.

In Parallelogrammode, the relative X difference between the top and bottom edge. In Trapezoidmode, the amount to move the top edge in the positive X axis.
Input vectors for the Pointsmode.
Generate a rectangle-shaped curve with a width and a height.
Generate a rectangle with an offset for the different X values of the top and bottom edges.
Generate a trapezoid-shaped curve with a height, a width for the top and bottom, and an offset.
Generate a kite shape with a width, and the top and bottom points distance from the X axis.
Generate a four point cyclic poly spline by inputting the position vectors directly.
Poly spline generated from the inputs.
--- Fin du chapitre: _quadrilateral ---

--- Début du chapitre: _quaternion_to_rotation --- The Quaternion to Rotationnode converts aquaternion rotationtoa standard rotation.
The W value of the quaternion.
The X value of the quaternion.
The Y value of the quaternion.
The Z value of the quaternion.
Standard rotation value.
--- Fin du chapitre: _quaternion_to_rotation ---

--- Début du chapitre: _radius --- The Radiusnode outputs the radius value at each point on the evaluated geometry. For curves, this value is used for things like determining the size of the mesh created intheCurve to Meshnode. For point clouds, the value is used for the display size of the point in the viewport.
This node has no inputs.
This node has no properties.
Float value indicating radius at each point on the geometry.
--- Fin du chapitre: _radius ---

--- Début du chapitre: _random ---Reference
Object Mode
Select \xe2\x80\xa3 Select Random
Randomly selects unselected vertices, edges, or faces based on percentage probability.
The likelihood of an unselected mesh element being selected. Note that this is not the percentage amount of elements that will be selected.
--- Fin du chapitre: _random ---

--- Début du chapitre: _randomize ---Reference
Edit Mode
Mesh \xe2\x80\xa3 Transform \xe2\x80\xa3 Randomize
The Randomizetool in Edit Mode allows you to displace the vertices of a meshalong their normal.
Distance of the displacement.
Adds a random offset of the amount.
Adds a random offset to the displacement normal.
The random seed is an offset to the random transformation. A different seed will produce a new result.

See also

Object Mode Randomize Transform

--- Fin du chapitre: `_randomize` ---

--- Début du chapitre: `_random_value` --- The Random Value node outputs a white noise like value as a Float, Integer, Vector, or Boolean field. The minimum value of the range where random values are sampled from. This input is only available for Float, Integer, and Vector types. The maximum value of the range where random values are sampled from. This input is only available for Float, Integer, and Vector types. The probability ratio for the random Boolean output to be True. This input is only available for Boolean types. An ID to drive the random number generator seed. By default, this input uses the same value as of the ID Node, which is the id attribute of the context geometry if it exists, and otherwise the index.

Tip

Single Random Value

By default, the random value node generates a value for each unique index. If a single random value is desired, connect a single value (such as an Integer Node) to the ID input.

A field to Seed the random number generator. This can be used to generate a different set of random values, even for two nodes with the same ID input.

The output will be a Float field.

The output will be an Integer field.

The output will be a Vector field.

The output will be a Boolean field.

Random values as a field.

--- Fin du chapitre: `_random_value` ---

--- Début du chapitre: `_raycast` --- The Raycast node intersects rays from one geometry onto another. The source geometry is defined by the context of the node that the Raycast node is connected to. Each ray computes hit points on the target mesh and outputs normals, distances and any surface attribute specified.

Geometry that rays are tested against.

An optional field input evaluated on the Target Geometry that will be interpolated at the hit points. The resulting values are outputted with the Attribute output.

The position from where to start each ray. By default, this is the same as if the Position Node was connected.

Direction of each ray from the starting position. The field is evaluated on the geometry from the context of the field evaluation, not the Target Geometry.

Maximum distance a ray can travel before being considered `no hit`.

How attributes of the target mesh are mapped to the attribute values on the result geometry.

Vertex and corner attributes are interpolated smoothly, with a bilinear function.

Choose the value of the closest vertex without interpolating.

Boolean output that is true for each ray which has hit the Target Geometry.

The location of the intersection point with the target mesh.

The surface Normal vector at the hit location.

The distance from the Source Position to the Hit Position. If the ray does not hit, the Ray Length is returned.

Interpolated values of the Attribute input sampled at the Hit Position.

--- Fin du chapitre: _raycast ---

--- Début du chapitre: _raytracing ---Reference

Render \xe2\x80\xa3 Raytracing

The ray-tracing pipeline goal is to increase the accuracy of surface indirect lighting. This is done by generating ray from each BSDF and finding their intersection with the scene individually.

When disabled, it is replaced by a faster pipeline that uses pre-filtered light-probes. This fallback mode offers a more visually stable and optimized alternative when visual fidelity is not the primary goal. See also

Limitations .

Determine the tracing method used to find scene-ray intersections and indirect lighting.

Use light-probe spheres and planes to find scene intersection. This option has the lowest tracing cost but relies on manually placed light-probes.

Trace ray against the screen depth buffer. Fallback to light-probes if ray exits the view.

Resolution at which the ray-tracing is performed. Lower options will be faster and use less memory but will produce blurrier results.

These settings control the behavior of the screen space ray-tracing. They are only visible if Screen-Trace is the active tracing Method.

Higher values increase precision of the screen space ray-tracing but lower the maximum trace distance. Increased precision also increases performance cost.

How thick to consider the pixels of the depth buffer during the tracing. Higher values will stretch the reflections and add flickering. Lower values may make the ray miss surfaces.

Denoising can be enabled to reduce the amount of noise from the raw ray-traced output. This can help image stability but will also over-blur the final ray-traced output.

Reuse the rays from neighbor pixels. Can introduce some light leaks across surfaces.

Accumulate samples by re-projecting the last ray tracing results. This removes Fireflies but also introduces color bias. Useful for viewport temporal stability or making renders converge faster.

Blur the resolved ray-traced output using a bilateral filter.

Fast GI Approximation is a fallback to the ray-tracing pipeline for BSDF with high roughness. It produces a less noisy output and captures bounce lighting more efficiently than individually traced rays.

This is currently implemented as a screen space effect and will inherit all associated limitations .

Maximum roughness a BSDF can have to use ray-tracing. BSDFs with higher roughness will progressively use the Fast GI Approximation. A value of 1 will raytrace every surfaces and disable the Fast GI.

Determine the method used to compute the fast GI approximation.

Use scene intersections to shadow the distant lighting from light-probes. This is the fastest option.

Compute global illumination taking into account light bouncing off surrounding objects.

Resolution at which the fast GI is computed. Lower options will be faster and use less memory but will produce blurrier results.

Number of GI rays per pixel at the specified Resolution. Higher values will reduce noise.

Number of screen samples per GI ray. Higher values will reduce the noise amount and increase the quality.

Tip

With a higher step count, there is less chance to miss other surfaces that could reflect or block the light. This means that the Fast GI Thickness parameters can be tweaked to lower values without losing too much light bounce energy.

Higher values increase the precision of the scene intersections with the GI rays. Increased precision also increases performance cost.

If non-zero, the maximum distance at which other surfaces will contribute to the fast GI approximation.

Geometric thickness of the surfaces when computing fast GI and ambient occlusion. Reduces light leaking and missing contact occlusion. The effectiveness decreases proportionally to the distance from the shading point, following the inverse square law.

Angular thickness of the surfaces when computing fast GI and ambient occlusion. Reduces energy loss and missing occlusion of far geometry. Higher values will make the very thin objects block or reflect too much light.

Bias the shading normal to reduce self intersection artifacts.

--- Fin du chapitre: _raytracing ---

--- Début du chapitre: _ray_portal --- Cycles Only

The Ray Portal BSDF node transports rays that enter to another location in the scene. It can be used to render portals for visual effects, and other production rendering tricks.

It acts much like a Transparent BSDF: render passes are passed through, and it is affected by light path max transparent bounces.

Note

The Ray Portal BSDF only allows rays to pass through it in one direction. Add a second portal at the target location to make rays go in the other direction as well.

Light sampling does not work efficiently through portals. This can lead to increased noise from lights on the other side of portals. Particularly small lights may be very noisy, or not pass through at all.

Tint rays passing through the portal.

Ray start position at new location. Defaults to the current position, matching the Position output of the Geometry node.

Ray direction at the new location. Defaults to the current view direction, which is the same as the negation of the Incoming output of the Geometry node.

This node has no properties.

Standard shader output.

One use case for the Ray Portal BSDF is to connect two spaces together to create effects like a portal to an alternative dimension, or impossible spaces where something is bigger or smaller on the inside than expected.

To set up a Ray Portal BSDF for a technique like this, augment the Position and Incoming outputs of the Geometry node to set the exit point and direction of the ray through the portal. Here are some examples:

This simple node setup offsets the ray position. In this example, the ray is offset 0 units along the X axis, 4 units along the Y axis, and 5 units along the Z axis.

In this example, the Location of Portal Target and Rotation of Portal Target vectors are obtained from a target portal object using Drivers.

Along with augmenting rays, the ray position and ray direction can be replaced, for effects like a camera feed on a screen.

Using the Ray Portal BSDF to replicate the effect of a camera feed on a screen.

Node setup for replicating a camera feed like effect on a screen.

--- Fin du chapitre: _ray_portal ---

--- Début du chapitre: _realize_instances --- The Realize Instances node makes any instances (efficient duplicates of the same geometry) into real geometry data. This makes it possible to affect each instance individually, whereas without this node, the exact same changes are applied to every instance of the same geometry. However, performance can become much worse when the input contains many instances of complex geometry, which is a fundamental limitation when procedurally processing geometry.

Note

If the input contains multiple volume instances, only the first volume component is moved to the output.

When merging attributes from multiple geometry inputs, the highest complexity data type is chosen for the output attribute. In other words, if a weight attribute has a Boolean type on one geometry input and a vector data type on another geometry, the weight attribute on the output geometry will have a vector data type.

Named and anonymous attributes are propagated from the instance domain to the realized geometry. If the same attribute exists on the geometry and on an instance, the attribute values from the geometry have precedence over the values on the instances.

In order to avoid creating duplicate values, the id attribute has special handling. The id values or indices of each instance are combined with id values from the points on geometry data.

Warning

Like other geometry nodes, this node always outputs generic typed attributes. So instead of a Vertex Group attribute, it will create a Float attribute on the result, and it will create a generic 2D vector attribute instead of a special UV Map attribute. Some other areas of Blender don't properly handle generic attributes in version 3.0.

Custom face corner normals are also not transferred currently. Standard geometry input.

Which top-level instances to realize.

Realize all levels of nested instances for each top-level instances (overrides the value of the Depth input).

Number of levels of nested instances to realize for each top-level instance.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_realize_instances` ---

--- Début du chapitre: `_realtime_compositor` --- The new GPU accelerated compositor introduced in Blender 3.5 and is currently used for viewport compositing.

Compositing nodes operate on data that is either an image or a dimensionless single value. For instance, the `Levels` node outputs a single value, while the `Render Layers` node outputs an image. Node inputs that expect a single value assume a default value if an image is given and ignore the image completely, for instance, the `Transform` node expects single values for its inputs and will assume default values if images were given to those inputs. The default values are those that are considered identity and thus have no effect on the output, so for the `Transform` node, the `X`, `Y`, and `Angle` inputs will have a default value of zero, while the `Scale` input will have a default value of one. On the other hand, if node inputs that expect an image are given a single value, the single value will be assumed to cover the whole compositing space. For instance, the `Filter` node expects its `Factor` input to be an image, but if a single value is given, it will be assumed to be the same for all pixels.

Three types of data exist, all of which are stored in half precision formats:

A signed floating-point number. Integer data is also stored as floats because no integer type exists.

A 4D vector. While it is 4D, it can have different interpretations depending on the node that uses it. It can be treated as a 2D vector with the last two components ignored, for instance, the `Vector` input of the `Displace` node is treated as a 2D vector. It can be treated as a 3D vector with the last component ignored, for instance, the `Vector` input of the `Separate XYZ` node is treated as a 3D vector. It can be treated as two consecutive 2D vectors. For instance the `Velocity Passes` expected by the `Vector Blur` node is assumed to have the 2D Previous Velocity in the `X` and `Y` components of the vector and the 2D Next Velocity in the `Z` and `W` components of the vector.

A 4D vector storing the Red, Green, Blue, and Alpha of the color. The color is free form and does not conform to a specific color space or alpha storage model, instead, appropriate nodes will have settings to control the representation of their output and nodes exist to convert between the different representations.

In case a node input is given data of type other than its own type, the following implicit conversions are performed:

```
Source
Target
Conversion
Float
Vector
f => Vector(f, f, f, 0)
Float
Color
f => Color(f, f, f, 1)
Vector
Float
(x, y, z, w) => Average(x, y, z)
```

```

Vector
Color
(x, y, z, w) => Color(x, y, z, 1)
Color
Float
(r, g, b, a) => Average(r, g, b)
Color
Vector
(r, g, b, a) => Vector(r, g, b, 0)

```

The following example demonstrates implicit conversion between a color type and a float type, since the Math node expects float inputs.

An example that demonstrates implicit conversion between a color type and a float type, since the Math node expects float inputs. \xc2\xba6

The compositor is designed in such a way as to allow compositing in an infinite compositing space. Consequently, images are not only represented by their size, but also by their transformation in that space, much like 3D objects have transformations. An identity transformation represents an image that is centered in space. The rectangular area occupied by an image in that space as defined by its transformation and size is called the Domain of the image. The figure below demonstrates the domains of two example images.

The domains of two example images are illustrated on the compositing space. One of the images is centered in space and the other is scaled down and translated such that it lies in the upper right quadrant of the space. Notice that both images have similar sizes in pixels, yet their apparent sizes are different. \xc2\xba6

Images can be transformed using nodes like the Transform, Translate, and Rotate nodes.

Compositor Nodes operate on a specific rectangular area of the compositing space called the Operation Domain. The nodes only consider the area of the input images that overlap the operation domain and ignore the rest of the images. If an input image doesn't completely overlap the operation domain, the rest of the operation domain for that input will be assumed to be a zero value, a zero vector, or a transparent zero color depending on the type.

For instance, the figure below illustrates a case where the operation domain of a node is the large blue area and the domain of an input image is the small red area. In that case, the input image doesn't completely overlap the operation domain, so the rest of the blue area for that input image is assumed to be zero.

An example case where the operation domain of a node is shown in blue and the domain of an input image is shown in red. Since the input image doesn't completely cover the operation domain of the node, the rest of the blue area for that input image is assumed to be zero. \xc2\xba6

The previous illustration is a representation of a real world example where one uses the Alpha Over node to overlay a small logo on an image, as shown in the figure below. In that case, the operation domain covers the entirety of the viewport as will later be demonstrated, but the logo covers only a small area of it, so the rest of the area is assumed to be a zero transparent color, which is convenient for the use case.

A real world example where the Alpha Over node is used to overlay a small logo on an image. The logo only covers a small area of the operation

domain, which is the whole viewport in this case, so the rest of the area is assumed to be a zero transparent color.

If an input image to a node is not perfectly aligned with the operation domain of the node or have a different size in pixels, the node would typically need to do a process called Interpolation, where the input image is read at the exact positions of the pixels of the operation domain. This can be done using different interpolation methods, including Nearest-Neighbor, Bilinear, and Bicubic interpolations. Those interpolation methods are demonstrated in the following Wikipedia gallery. Transformation nodes like the Transform and Rotate nodes include an interpolation option to set how they prefer their output image to be read and interpolated.

The question remains on how nodes determine their operation domain. Different node types can have different mechanisms for determining their operation domain. But generally, three classes of nodes exist when it comes to the mechanism of determining the operation domain, each of which is presented in one of the following sections.

The operation domain of input nodes like the Image node is a domain with an identity transformation and the same size as their outputs, so for the Image node, the operation domain will be the domain whose size is the size of the image and whose transformation is an identity one.

The operation domain of output nodes like the Viewer node is a domain with an identity transformation and the same size as the final compositor output. For viewport compositing, that size would be the viewport size, and for final render compositing, that size would be the scene render size.

Unless stated otherwise in their respective documentation pages, all other nodes use the following mechanism. One of the inputs of the nodes is designated as the Domain Input of the node, and the operation domain of the node is identical to the domain of that designated input. For many nodes, the domain input can be intuitively identified as the main input of the node, for instance, the domain input for the Filter node is the Image input. But there are some caveats to note, which requires a deeper understanding of the mechanism.

Each input in the node has the so called Domain Priority property, the operation domain of the node is the domain of the non single value input with the highest domain priority. So for instance, the Filter node has two inputs, the domain priority of the Image input is higher than that of the Factor input, and there are four possible configurations:

Both the Image and factor inputs are connected to images. In this case, the Image input is the domain input because it has the highest priority and is connected to an image.

The Image input is connected to an image, but the Factor input isn't. In this case, the Image input is the domain input because it is the only input connected to an image regardless of its priority.

The Image input is not connected to an image but the Factor input is. In this case, the Factor input is the domain input because it is the only input connected to an image regardless of its priority.

Neither the Image nor the Factor inputs are connected to images, in this case, there isn't a domain input because the node is evaluated on single values.

The aforementioned mechanism for determining the operation domain has a number of consequences that needs to be considered as they might be undesirable, each of which is presented in one of the following sections.

The output of nodes will be intuitively clipped to the operation domain, or rather, the domain of the domain input. For instance, if the Foregroundinput is bigger than the Backgroundinput in the Alpha Over node, the output will be clipped to the Backgroundinput because it is the domain input, as shown in the following figure.

The Foregroundinput is bigger than the Backgroundinput in the Alpha Over node, so the output is intuitively clipped to the Backgroundinput because it is the domain input.

The Alpha Over node currently does not support changing the domain priority for its inputs, so as a workaround, one can use a Mix node to achieve the desired behavior, noting that the first Imageinput in the Mix node has the highest domain priority, as shown in the following figure.

Working around the clipping behavior of the Alpha Over node using a Mix node, noting that the first Imageinput in the Mix node has the highest domain priority.

The GPU compositor only supports a single active output target, that is, only one of the Composite nodes or Viewer nodes in the node tree will be considered active and the rest will be ignored. In particular, the compositor searches the Active Node Tree Context and falls back to the Root Node Tree Context if no active output was found in the active node tree context. The active node tree context is the node tree of an expanded node group, that is, when the user selects a node group node and edits its underlying tree, while the root node tree context is the top level node tree without any expanded node groups. The compositor searches for the active Composite node, if none was found, it searches for the active Viewernode, be it a Viewer or a Split Viewernode, if none was found, the compositor doesn't run altogether. Consequently, note that adding a Viewernode will have no effect if there is a Composite node, since the priority is given to Composite nodes.

--- Fin du chapitre: _realtime_compositor ---

--- Début du chapitre: _reconstruction --- Scene orientation tools can be used for orienting object to bundles.

Reference

Tracking

Reconstruction \xe2\x80\xa3 Set Origin

Transform camera in a way which makes active track to be moved to a scene origin. Only translation is applied to the camera.

Reference

Tracking

Reconstruction \xe2\x80\xa3 Set Floor

Use selected three markers to define a floor. Camera will be transformed in a way which makes the selected markers to be flat (have Z = 0).

Reference

Tracking

Reconstruction \xe2\x80\xa3 Set Wall

Similar to the floor orientation, but defines a wall (selected tracks are placed onto the XZ plane).

Reference

Tracking

Reconstruction \xe2\x80\xa3 Set X/Y Axis

Transform camera in a way which makes active track to become on X or Y axis.No translation is applied, meaning scene origin which was specified before will be preserved.

Reference

Tracking

Reconstruction \xe2\x80\xa3 Set Scale

Scale camera or tracking object in a way which makes distancebetween two selected tracks match the given value in Distance.

Reference

Tracking

Reconstruction \xe2\x80\xa3 Apply Solution Scale

Similar to Set Scale , but actually modifies the tracking data.

Reference

Tracking

Reconstruction \xe2\x80\xa3 Link Empty to Track

Creates new empty in 3D Viewport and appends constraint which parts it to the active track.

Reference

Tracking

Reconstruction \xe2\x80\xa3 3D Markers to Mesh

Creates a mesh which vertices matches positions of reconstructed tracks.It is required to have motion solved first before using this operator.Only tracks from the current tracking object will be used.The intention of this operator is to give a nice starting point for a manual mesh reconstruction.

--- Fin du chapitre: _reconstruction ---

--- Début du chapitre: _recover --- Computer crashes, power outages, or simply forgetting to save can result inthe loss or corruption of your work. You can use Blender\xe2\x80\x99s Auto Savefeatureto reduce the chance of losing files when such events occur.

There are options to save a backup of your files like Auto Savethat saves your file automatically over time, and Save on Quit,which saves your blend-file automatically when you exit Blender.In addition to these functions being enabled by default,the Save on Quitfunctionality cannot be disabled.

Note

For your actions, there are options like Undo, Redoand an Undo History,used to roll back from mistakes under normal operation, or return back to a specific action.See Undo & Redo .

By default Blender keeps an additional backup when saving files.So saving renames the previously saved file with a .blend1 extension instead of overwriting it.

This file can be used to revert to a previous state.

See Save Versions to configure the number of versions kept.

Reference

File \xe2\x80\xa3 Recover \xe2\x80\xa3 Last Session

The Recover Last Sessionwill open thequit.blendfilethat is saved into theTemporary Directorywhen you quit Blender under normal operation (seeBlender Session). Note that files in your temporary directory may be deleted when you rebootyour computer (depending on your system configuration).

Reference

File \xe2\x80\xa3 Recover \xe2\x80\xa3 Auto Save

The Recover Auto Save allows you to open the Auto Saved file. You will have to navigate to your Temporary Directory. The Auto Saved files are named using a random number and have a .blend extension.

See Auto Save Preferences to configure auto-save.

When enabled, Python scripts and drivers that may be included in the file will be run automatically. Enable this only if you created the file yourself, or you trust that the person who gave it to you did not include any malicious code with it. See Python Security to configure default trust options.

Tip

Enable the detailed list view when browsing auto-saved files to show which is the most recent.

File Browser displaying a vertical list.

Warning

When recovering an Auto Saved file, any changes made since the last Auto Save will be lost. Only one Auto Saved file exists for each .blend file, i.e. Blender does not keep older versions. Therefore, you will only be able to restore the most recent Auto Save file.

--- Fin du chapitre: _recover ---

--- Début du chapitre: _redistribute_curve_points --- Redistributes existing control points evenly along each curve.

Curves

Factor to blend overall effect.

Use simple feature awareness to keep feature definition.

This node has no properties.

Curves

--- Fin du chapitre: _redistribute_curve_points ---

--- Début du chapitre: _reducing_noise --- When performing a final render, it is important to reduce noise as much as possible. Here we will discuss a number of tricks that, while breaking the laws of physics, are particularly important when rendering animations within a reasonable time. Click to enlarge the example images to see the noise differences well.

Cycles uses path tracing with next event estimation, which is not good at rendering all types of light effects, like caustics, but has the advantage of being able to render more detailed and larger scenes compared to some other rendering algorithms. This is because we do not need to store, for example, a photon map in memory, and because we can keep rays relatively coherent to use an on-demand image cache, compared to e.g. bidirectional path tracing.

We do the inverse of what reality does, tracing light rays from the camera into the scene and onto lights, rather than from the light sources into the scene and then into the camera. This has the advantage that we do not waste light rays that will not end up in the camera, but also means that it is difficult to find some light paths that may contribute a lot. Light rays will be sent either according to the surface BSDF, or in the direction of known light sources.

See also

For more details, see the Light Paths and Sampling documentation.

To understand where noise can come from, take for example the scene below. When we trace a light ray into the location marked by the white

circle on a red dot, the second image below gives an impression of what the diffuse shader \xe2\x80\x9csees\xe2\x80\x9d.

To find the light that is reflected from this surface, we need to find the average color from all these pixels. Note the glossy highlight on the sphere, and the bright spot the light casts on the nearby wall. These hot-spots are much brighter than other parts of the image and will contribute significantly to the lighting of this pixel.

The scene.\xc2\xbb

Irradiance at the shading point.\xc2\xbb

The detected highlights.\xc2\xbb

The light is a known light source, so its location is already known, but the glossy highlight(s) that it causes are a different matter. The best we can do with path tracing is to distribute light rays randomly over the hemisphere, hoping to find all the important bright spots. If for some pixels we miss some bright spot, but we do find it for another, that results in noise. The more samples we take, the higher the probability that we cover all the important sources of light.

With some tricks we can reduce this noise. If we blur the bright spots, they become bigger and less intense, making them easier to find and less noisy. This will not give the same exact result, but often it\xe2\x80\x99s close enough when viewed through a diffuse or soft glossy reflection. Below is an example of using Glossy Filter and Light Falloff .

Using Glossy Filter & Light Falloff.\xc2\xbb

Irradiance at the shading point.\xc2\xbb

The detected highlights.\xc2\xbb

In reality light will bounce a huge number of times due to the speed of light being very high. In practice more bounces will introduce more noise, and it might be good to use something like the Limited Global Illumination preset in the Light Paths Section that uses fewer bounces for different shader types. Diffuse surfaces typically can get away with fewer bounces, while glossy surfaces need a few more, and transmission shaders such as glass usually need the most.

No bounces.\xc2\xbb

Two bounces at max.\xc2\xbb

Four bounces at max.\xc2\xbb

Also important is to use shader colors that do not have components of value 1.0 or values near that; try to keep the maximum value to 0.8 or less and make your lights brighter. In reality, surfaces are rarely perfectly reflecting all light, but there are of course exceptions; usually glass will let most light through, which is why we need more bounces there. High values for the color components tend to introduce noise because light intensity then does not decrease much as it bounces off each surface.

Caustics are a well-known source of noise, causing Fireflies . They happen because the renderer has difficulty finding specular highlights viewed through a soft glossy or diffuse reflection. There is a No Caustics option to disable glossy behind a diffuse reflection entirely. Many renderers will typically disable caustics by default.

Default settings.\xc2\xbb

Caustics disabled.\xc2\xbb

Filter Glossy greater than zero.\xc2\xbb

However, using No Caustics will result in missing light, and it still does not cover the case where a sharp glossy reflection is viewed through a soft glossy reflection. There is a Filter Glossy option to reduce the noise

from such cases at the cost of accuracy. This will blur the sharp glossy reflection to make it easier to find, by increasing the shader Roughness. The above images show default settings, no caustics, and filter glossy set to 1.0.

In reality light in a vacuum will always fall off at a rate of $1/(\text{distance}^2)$. However, as distance goes to zero, this value goes to infinity and we can get very bright spots in the image. These are mostly a problem for indirect lighting, where the probability of hitting such a small but extremely bright spot is low and so happens only rarely. This is a typical recipe for Fireflies .

Hard Falloff.

Soft Falloff.

To reduce this problem, the Light Falloff node has a Smooth factor, that can be used to reduce the maximum intensity a light can contribute to nearby surfaces. The images above show default falloff and smooth value 1.0.

Materials with emission shaders can be configured to use Multiple Importance Sampling (Material Settings). This means that they will get rays sent directly towards them, rather than ending up there based on rays randomly bouncing around. For very bright mesh light sources, this can reduce noise significantly. However, when the emission is not particularly bright, this will take samples away from other brighter light sources for which it is important to find them this way.

The optimal setting here is difficult to guess; it may be a matter of trial and error, but often it is clear that a somewhat glowing object may be only contributing light locally, while a mesh light used as a light would need this option enabled. Here is an example where the emissive spheres contribute little to the lighting, and the image renders with slightly less noise by disabling Multiple Importance on them.

Multiple Importance off.

Multiple Importance on.

The world background also has a Multiple Importance(Settings) option. This is mostly useful for environment maps that have small bright spots in them, rather than being smooth. This option will then, in a preprocess, determine the bright spots, and send light rays directly towards them. Again, enabling this option may take samples away from more important light sources if it is not needed.

With caustics disabled, glass shadows may appear too dark, and with filter glossy the caustics might be too soft. We can make a glass shader that will use a Glass BSDF when viewed directly, and a Transparent BSDF when viewed indirectly. The Transparent BSDF can be used for transparent shadows to find light sources straight through surfaces, and will give properly-colored shadows, but without the caustics. The Light Path node is used to determine when to use which of the two shaders.

Optimized glass shader.

Above we can see the node setup used for the glass transparency trick; on the left the render has dark shadows due to missing caustics, and on the right the render with the trick.

Default Glass BSDF.

Optimized Glass Shader.

When rendering a daylight indoor scene where most of the light is coming in through a window or door opening, it is difficult for the integrator to find its way to them. To fix this, use Light Portals . You then will need

to modify its shape to match that of the opening that you are trying to fill.

Even with all the settings described above there will always end up being some render noise no matter how many samples you use. To fix this there is a post-processing technique to cleanup the final bit of noise. To use this enable Denoising in the Rendertab of the Properties.

Below is an example render by The Pixelary.

Example render before denoising.\xc2\xba

Example render after denoising.\xc2\xba

Ideally with all the previous tricks, Fireflies would be eliminated, but they could still happen. For that, the intensity that any individual light ray sample will contribute to a pixel can be clamped to a maximum value with the integrator Clamp setting .

If set too low this can cause missing highlights in the image, which might be useful to preserve for camera effects such as bloom or glare. To mitigate this conundrum it's often useful to clamp only indirect bounces, leaving highlights directly visible to the camera untouched.

No Clamp (0).\xc2\xba

With Clamp set to 4.\xc2\xba

--- Fin du chapitre: _reducing_noise ---

--- Début du chapitre: _refraction --- The RefractionBSDF is used to add glossy refraction with sharp or microfacet distribution, used for materials that transmit light. For best results this node should be considered as a building block and not be used on its own, but rather mixed with a glossy node using a Fresnel factor. Otherwise it will give quite dark results at the edges for glossy refraction.

Color of the surface, or physically speaking, the probability that light is refracted for each wavelength.

Influences sharpness of the refraction; perfectly sharp at 0.0 and smoother with higher values.

Normal used for shading; if nothing is connected the default shading normal is used.

Microfacet distribution to use.

GGX microfacet distribution.

Cycles Only Beckmann microfacet distribution.

Standard shader output.

Refraction Shader. \xc2\xba

--- Fin du chapitre: _refraction ---

--- Début du chapitre: _regions --- Reference

All modes

View \xe2\x80\xa3 View Regions \xe2\x80\xa3 Clipping Region \xe2\x80\xa6
Alt- B

Allows you to define a clipping region to limit the 3D Viewport display to a portion of 3D space. It can assist in the process of working with complex models and scenes.

Once activated, you have to draw a rectangle with the mouse. It becomes a clipping volume of four planes:

A right-angled parallelepiped (of infinite length) if your view is orthographic.

A rectangular-based pyramid (of infinite height) if your view is in perspective.

Once clipping is used, you will only see what's inside the volume you defined. Tools such as paint, sculpt, selection, transform snapping, etc. will also ignore geometry outside the clipping bounds. To delete this clipping, press Alt- B again.

Selecting a region. ¶

Region selected. ¶

View rotated. ¶

The Region/Volume clipping image shows an example of using the clipping tool with a cube. Start by activating the tool with Alt- B. This will generate a dashed cross-hair cursor. Click with the LMB and drag out a rectangular region. Now clipping is applied against that region in 3D space. Use the MMB to rotate the view and you will see that only what is inside the clipping volume is visible. All the editing tools still function as normal, but only within the clipping volume.

The dark gray area is the clipping volume itself. Once clipping is deactivated with another Alt- B, all of 3D space will become visible again.

Reference

All modes

View ¶ View Regions ¶ Render

Region ¶ View ¶ View Regions ¶ Clear
Render Region

Mark: Ctrl- B Clear: Ctrl- Alt- B

Allows you to limit rendering to a 2D rectangular area. If you're busy tweaking just a small part of the scene, it can be quite wasteful to have the whole viewport in Rendered shading mode or make full-frame renders, so this feature lets you save time.

You can define Render Regions in two different contexts:

If you define one while in Camera View, it will apply not just to the viewport, but also to the final render. If you want to temporarily disable this region rather than clearing it entirely, you can do so in the Output tab of the Properties editor.

If you define one while not in Camera View, it will only apply to the viewport. If you want to temporarily disable this region rather than clearing it entirely, you can do so in the Sidebar.

Both Render Regions can exist at the same time.

Note

Render regions only apply to the viewport when using Cycles, not when using Eevee. However, they always affect the final render.

See also

Zoom Region.

--- Fin du chapitre: _regions ---

--- Début du chapitre: _relations ---Reference

All Modes

Bone ¶ Relations

Bone Relations panel. ¶

In this panel you can manage the relationship of this bone with its parent bone. It also shows the bone collections the bone is assigned to. A Data ID to select the bone to set as a parent.

Changes how transformation of the bone is applied to its child Objects. The Connected checkbox set the head of the bone to be connected with its parent root.

Bones relationships have effects on transformations behavior.

By default, children bones inherit:

Their parent position, with their own offset of course.

Their parent rotation (i.e. they keep a constant rotation relatively to their parent).

Their parent scale, here again with their own offset.

The armature in its rest position. \xc2\xbb

Rotation of a root bone. \xc2\xbb

Scaling of a root bone. \xc2\xbb

Exactly like standard children objects. You can modify this behavior on a per-bone basis, using the Relations panel in the Bonestab:

When disabled, the location transform property is evaluated in the parent bone's local space, rather than using the bone's own rest pose local space orientation.

When disabled, this will break the rotation relationship to the bone's parent. This means that the child will keep its rotation in the armature object space when its parent is rotated.

Specifies which effects of parent scaling the bone inherits:

These inheriting behaviors propagate along the bones hierarchy. So when you scale down a bone, all its descendants are by default scaled down accordingly. However, if you disable one bone's Inherit Scale or Inherit Rotation property in this family, this will break the scaling propagation, i.e. this bone and all its descendants will no longer be affected when you scale one of its ancestors.

The bone inherits all effects of parent scaling and shear.

Full parent effects are applied to the rest state of the child, after which any shear is removed in a way that preserves the bone direction, length and volume, and minimally affects roll on average. The result is combined with the local transformation of the child.

If the inherited scale is non-uniform, this does not prevent shear from reappearing due to local rotation of the child bone, or of its children.

Parent scaling is inherited as if the child was oriented the same as the parent, always applying parent X scale over child X scale, and so on.

Inherits a uniform scaling factor that is the total change in the volume of the parent.

Ignores all scaling and shear of the parent.

Ignores all scaling, provided the parent is not sheared. If it is, there are no guarantees.

This choice replicates the behavior of the old Inherit Scale checkbox, and may be removed in a future release.

Tip

The various Inherit Scale options are provided as tools in avoiding shear that is caused by non-uniform scaling combined with parenting and rotation. There is no obvious best way to achieve that, so different options are useful for different situations.

None Useful for gaining full control over the scaling of the child in order to e.g. manually overwrite it with constraints.

Average Useful to block squash and stretch propagation between sub-rigs, while allowing uniform changes in the size and volume to pass through.

Aligned Can be used within bone chains, e.g. tentacles, in order to propagate lengthwise scaling as lengthwise, and sideways as sideways, no matter how the tentacle bends. Similar to using Nonuniform Copy Scale from parent.

Fix Shear\xe2\x80\x93May be useful at the base of an appendage in order to reallocate squash and stretch between axes based on the difference in rest pose orientations of the parent and child. It behaves closest to Full while suppressing shear.

The yellow outlined Inherit Rotation disabled bone in the armature.
\xc2\xb6

Rotation of a bone with an Inherit Rotation disabled bone among its descendants. \xc2\xb6

Scaling of a bone with an Inherit Rotation disabled bone among its descendants. \xc2\xb6

Connected bones have another specificity: they cannot be moved. Indeed, as their root must be at their parent's tip, if you do not move the parent, you cannot move the child's root, but only its tip, which leads to a child rotation. This is exactly what happens, when you press G with a connected bone selected, Blender automatically switches to rotation operation.

Bones relationships also have important consequences on how selections of multiple bones behave when transformed. There are many different situations which may not be included on this list, however, this should give a good idea of the problem:

Non-related selected bones are transformed independently, as usual.

When several bones of the same family\xe2\x80\x9d are selected, only the most parent\xe2\x80\x9d ones are really transformed\xe2\x80\x93the descendants are just handled through the parent relationship process, as if they were not selected (see Fig. Scaling bones, some of them related. the third tip bone, outlined in yellow, was only scaled down through the parent relationship, exactly as the unselected ones, even though it is selected and active. Otherwise, it should have been twice smaller!)

Scaling bones, some of them related. \xc2\xb6

When connected and unconnected bones are selected, and you start a move operation, only the unconnected bones are affected.

When a child connected hinge bone is in the selection, and the most parent\xe2\x80\x9d selected one is connected, when you press G, nothing happens, because Blender remains in move operation, which of course has no effect on a connected bone.

So, when posing a chain of bones, you should always edit its elements from the root bone to the tip bone. This process is known as Forward Kinematics (FK). We will see in a later page that Blender features another pose method, called Inverse Kinematics (IK), which allows you to pose a whole chain just by moving its tip.

Note

This feature is somewhat extended/completed by the pose library .

This list shows the bone collections the bone is assigned to. Press the eye icon to show or hide the entire bone collection. Press the star icon to show only this bone collection, and others also marked as solo\xe2\x80\x93Press the X icon to remove the bone from that particular collection.

To assign the bone to other bone collections, either use the Move or Shift-M shortcuts (see Moving Bones Between Collections) or go to the Armature properties panel .

--- Fin du chapitre: _relations ---

--- Début du chapitre: _relax ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Relax

The Relax tool can be used to distribute UVs more evenly. It works by pulling vertices along UV edges to bring the UV unwrap into balance.

The Relax tool can be compared with the Minimize Stretch tool which works directly on faces to reduce texture stretching and shearing. You may find that sometimes minimize stretch works better, sometimes the unwrap tool and other times the Relax tool.

First using Unwrap, then Minimize Stretch and touching up with the Relax tool often gives the best results. Remember, you can use \xe2\x80\x9cUndo\xe2\x80\x9d at any time to return to an earlier state. This option controls the radius of the brush, measured in pixels. Allows you to change the brush size interactively by dragging the mouse and then LMB. Typing a number then enter while using Allows you to enter the size numerically.

Controls how much each application of the brush affects the UVs. You can change the strength interactively by pressing Shift- F in the 3D Viewport and then moving the brush and then LMB. You can enter the size numerically also while in Shift- F sizing.

The Falloff allows you to control the Strength falloff of the brush. The falloff is mapped from the center of the brush (left part of the curve) towards its borders (right part of the curve). Changing the shape of the curve will make the brush softer or harder. Read more about using the Curve Widget .

You can choose how the strength of the falloff is determined from the center of the brush to the borders by manually manipulating the control points within the curve widget. There are also a couple of preset custom curves displayed at the bottom of the curve widget that can be used on their own or as a starting point for tweaking.

Smooth.\xc2\xbb

Sphere.\xc2\xbb

Root.\xc2\xbb

Sharp.\xc2\xbb

Linear.\xc2\xbb

Constant.\xc2\xbb

The center strength, the border strength, and the falloff transition between them are evenly distributed.

Similar to Smooth but produces a wider center point of the brush before tapering off.

The strength of the brush is predominately at its strongest point with a steep falloff near the border of the brush.

Similar to a Sphere but the center is a more concentrated point.

The center of the brush is the strongest point then exponentially tapers off to a lower strength, creating a fine point.

With the center being the strongest, the strength will consistently weaken as it reaches the border of the brush.

Similar to Sharp but the center point is more condensed.

A hybrid between Smooth and Sphere.

The strength of the brush remains unified across the entire brush. This will create a sharp edge at the border of the brush.

Locks the boundary of UV islands from being affected by the brush. This is useful to preserve the shape of UV islands.

To edit all islands and not only the island nearest to the brush center when the sculpt stroke was started.

How to determine the edge weighting:

The classic discrete laplace operator applied to the UV graph. Each edge has equal weighting, resulting in triangles which resemble a honeycomb shape, or quads aligned into square grid.

Similar to Laplacian, the HC method uses equal weighting while trying to preserve a gradient between dense regions of the mesh and regions with fewer edges.

Note, this method uses the Humphrey's

Classes operator as described in the paper:

Improved Laplacian Smoothing of Noisy Surface

Meshes.

Edges are weighted according to the discrete Laplace operator (cotangent formula) applied to the 3D geometry. This tries to bring the relative lengths of edges in UV closer to the relative lengths of edges in 3D, resulting in a UV unwrap with less distortion across edge boundaries.

--- Fin du chapitre: _relax ---

--- Début du chapitre: _remesh ---Reference

Sculpt Mode

Tool Settings Remesh

Sidebar Tool Remesh

Ctrl- R

For a general explanation to remeshing, visit the Introduction.

The resolution or the amount of detail the remeshed mesh will have. The value is used to define the size, in object space, of the Voxel. These voxels are assembled around the mesh and are used to determine the new geometry. For example a value of 0.5m will create topological patches that are about 0.5m (assuming Preserve Volume is enabled). Lower values preserve finer details but will result in a mesh with a much more dense topology.

The voxel size also be adjusted from the 3D Viewport using R. Using the shortcut displays an interactive grid overlay showing the resulting voxel size. Moving the mouse closer to center of the grid decreases the voxel size while moving away from the center increase the voxel size. Holding Shift increases the precision; adjusting the voxel size in small increments.

Used to adjust the Voxel Size by picking an area of the mesh to match the denseness of polygons after the remesh operation.

Reduces the final face count by simplifying geometry where detail is not needed. This introduces triangulation to faces that do not need as much detail. Note, an Adaptivity value greater than zero disables Fix Poles.

Tries to produce less poles at the cost of some performance to produce a better topological flow.

Tells the algorithm to try to preserve the original volume of the mesh. Enabling this could make the operator slower depending on the complexity of the mesh.

Reprojects the paint mask onto the new mesh.

Reprojects Face Sets onto the new mesh.

Reprojects the Color Attributes onto the new mesh.

Performs the remeshing operation to create a new manifold mesh based on the volume of the current mesh. Performing this will lose all mesh object data layers associated with the original mesh.

See also

Remesh modifier

Remeshing only works on the original mesh data and ignores generated geometry from modifiers, shape keys, rigging, etc.
Remeshing will not work with the Multiresolution Modifier .

--- Fin du chapitre: _remesh ---

--- Début du chapitre: _remove_named_attribute --- The Remove Named Attribute node deletes an attribute with a certain name from its geometry input. Any attribute that exists on geometry data will be automatically propagated when the geometry storing it is changed, which can be an expensive operation, so using this node can be a simple way to optimize the performance of a geometry node tree or even to lower the memory usage of the entire scene.

Almost all named attributes can be removed. For certain Built-In Attributes, removing it will mean that a default value will be used instead. For example, removing the cyclic attribute on curves means that all curves will be non-cyclic.

Standard geometry input.

The name of the attribute to remove.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _remove_named_attribute ---

--- Début du chapitre: _rename --- Reference

Edit \xe2\x80\xa3 Rename Active Item

F2

The Rename Active Item operator renames the active Bone , Node , Object and Sequence Strip .

When the operator is executed, a pop-up dialog appears. The text field shows the name of the current item and can be overwritten to rename the item. Return confirms the name while Esc cancels the operator.

Reference

Edit \xe2\x80\xa3 Batch Rename

Ctrl- F2

The Batch Rename operator can rename many data-block names at once. This uses a pop-up dialog with operations and their options to change the name. These actions are applied in order, from first to last.

Where to look for the data-blocks that are intended to be renamed.

Operates on the currently selected objects.

Operates on all data in the blend file.

The data-block type to perform the batch rename operations on.

The Batch Rename has several sub Operations to change the data names. The default operation is Find/Replace however, other operations can be added to change the data names further. Below all the operations gives a message in the status bar on how many data-blocks were renamed.

Find/Replace searches for a particular text in the names and optionally replaces it with a new text. Regular Expressions can be used as a powerful way to tailor the Find/ Replace texts and can be enabled using the icon to the right of the text fields.

The text to search for in names.

The text to replace for in matching names found from the Find text.

Search results must exactly match the case of the Find text.

Set Names works the most similar to Rename Active Item by renaming the current data-block without having to do a find and replace operation.

Disregards the current name replacing it with the
\xe2\x80\x9cnew\xe2\x80\x9d name.

Adds text to the beginning of the current name. This is useful for tools
that look for special text in the prefix of a data-block name.

Adds text to the end of the current name. This is useful for tools that
look for special text in the suffix of a data-block name.

Defines the new name or the text to add as a prefix/suffix.

Strip Characters cleans up names by removing certain character types from
either the beginning or the end of the name.

Strips any space characters from the name, e.g. \xe2\x80\x9cLiving Room
\xe2\x80\x9c becomes \xe2\x80\x9cLiving Room\xe2\x80\x9d.

Strips any numerical characters from the name, e.g. cube.001 becomes
cube. .

Strips any punctuation characters (, . ? ! : ; etc.) from the name, e.g.
cube? becomes cube .

Tip

Multiple character types can be removed at once by Shift- LMB on the
types.

Strips any leading characters in the name.

Strips any trailing characters in the name.

Change Case modifies the case of names to be one of the following:

Changes all text to be in upper case, e.g. cube.001 becomes CUBE.001 .

Changes all text to be in lower case, e.g. CUBE.001 becomes cube.001 .

Changes all text to be in title case, e.g. livingroom becomes LivingRoom

.

--- Fin du chapitre: _rename ---

--- Début du chapitre: _render --- In some situations we want to increase
the render speed, access Blender remotely to render something or build
scripts that use the command line.

One advantage of using the command line is that we do not need a
graphical display (no need for X server on Linux for example) and
consequently we can render via a remote shell (typically SSH).

See Command Line Arguments for a full list of arguments (for example to
specify which scene to render, the end frame number, etc.), or simply
run:

See Command Line Launching for specific instructions on launching Blender
from the command line.

Note

Arguments are executed in the order they are given!

The following command will not work, since the output and extension are
set after Blender is told to render:

The following command will behave as expected:

Always position -f or -a as the last arguments.

Render in the background (without UI).

Path to the blend-file to render.

Render only the 10th frame.

Path of where to save the rendered image, using five padded zeros for the
frame number.

Override the image format specified in the blend-file and save to an
OpenEXR image.

Render only the second last frame.

Warning

Arguments are case sensitive! -F and -f are not the same.

Render the whole animation using all the settings saved in the blend-file.

Use the Cycles Render engine. For a list of available render engines, run `blender-Ehelp`.

Set the start frame to 10 and the end frame to 500.

Use only two threads.

In addition to the options above, which apply to all render engines, Cycles has additional options to further control its behavior. See Cycles Render Options

--- Fin du chapitre: `_render` ---

--- Début du chapitre: `_render_layers` --- Read render layers and passes from a scene into the compositing node graph.

This node has no input sockets.

Select the scene within your blend-file. The scene information taken is the raw footage (pre-compositing and pre-sequencing).

Hint

To use composited footage from another scene, it has to be rendered into a multi-layer frameset (e.g. OpenEXR) as an intermediate file store and then imported with Image input node again.

A list of available Render Layers. The render button is a shorthand to re-render the active scene.

Rendered image.

Alpha channel.

Render Passes Sockets

Depending on the Render passes that are enabled, other sockets are available. See render passes.

Note

In the viewport compositor, Render Passes are only supported in EEVEE. For other engines, the passes return a zero value, a zero vector, or a transparent color depending on their type.

--- Fin du chapitre: `_render_layers` ---

--- Début du chapitre: `_repeat_zone` --- Repeat zones allow running nodes many times in a loop. Compared with simply duplicating a node, they support executing a node an arbitrary number of times, possibly determined when the node group is evaluated. For example, the nodes could be repeated based on the number of stories in a building generator.

Repeat zone used to repeat a node group a few times

When adding a repeat zone, two nodes are added, with the zone defined between them. The inputs connected to the Repeat Input node are read at the beginning, before starting the repetitions. Then they are passed to the inside of the zone where they can be changed, and passed to the next iteration. This process is repeated the specified number of times.

Other nodes can be connected as inputs to the inside of the repeat zone from the outside. Those are constant throughout every iteration based on their value at the current frame. However, outputs of the zone must be connected through the Repeat Output node.

Number of times to repeat the execution of the zone. The current iteration is available with the Iteration input on the inside of the zone. Standard geometry input, which is available by default to input geometry into the repeat zone. More bake items can be added by dragging sockets

into the blank socket or in the Bake Itemspanel. Items can be renamed by Ctrl- LMB on the socket name or in the nodes Propertiespanel. Iteration number that is used by inspection features like the Viewer Node or socket inspection .

--- Fin du chapitre: _repeat_zone ---

--- Début du chapitre: _replace_material --- The Replace Materialnode swaps one material with another. Replacing a material with this node is more efficient than creating a selection of all faces with the old material with the Material Selection Node and then using the Set Material Node.

Note

Currently this node only adjusts mesh data.

Standard geometry input.

Material that is going to be replaced.

Material that is replacing the old material.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _replace_material ---

--- Début du chapitre: _replace_string --- The Replace Stringnode replaces a string segment with another.

Standard string input.

The substring to find in String to be replaced.

A string segment which replaces occurrences of the Find substring.

This node has no properties.

Standard string output.

Using the node to add the newline character to a string. \xc2\xbf

--- Fin du chapitre: _replace_string ---

--- Début du chapitre: _reroute --- A node used primarily for organization. Reroute looks and behaves much like a socket on other nodes in that it supports one input connection while allowing multiple output connections.

To quickly add a Reroute node into an existing connection, hold Shift and RMB while sweeping across the link to add a Reroute node.

Input value used for unconnected sockets.

--- Fin du chapitre: _reroute ---

--- Début du chapitre: _resample_curve --- The Resample Curve node creates a poly spline for each input spline. In the Count and Length modes, the control points of the new polysplines will have uniform spacing.

Tip

Use a field as an input to have a different count/length for each spline.

Standard geometry input.

Whether or not to resample each spline. True values mean spline will be resampled to a poly spline, false values mean the spline will be unaffected.

The number of control points on the new splines.

The approximate length between the control points of the new splines.

Tip

A Trim Curve Node can be used with a multiple of the input length to make the distance between each sampled point exact, even when the length of the spline changes.

How to specify the amount of samples.

Sample the specified number of points along each spline.

Calculate the number of samples by splitting each spline into segments with the specified length. The length will be rounded down so that a whole number of samples will fit in each input spline.

Evaluate the spline's points based on the resolution attribute for NURBS and B-spline splines. Changes nothing for poly splines.

Standard geometry output.

--- Fin du chapitre: `_resample_curve` ---

--- Début du chapitre: `_restore_curve_segment_length` --- Restores the length of each curve segment using a previous state after deformation.

Curves

Only affect selected elements.

Factor to blend overall effect.

Reference position before deformation.

Pin each curve at a certain point for the operation.

This node has no properties.

Curves

--- Fin du chapitre: `_restore_curve_segment_length` ---

--- Début du chapitre: `_retopology` --- Retopology is the process of simplifying the topology of a mesh to make it cleaner and easier to work with. Retopology is needed for mangled topology resulting from sculpting or generated topology, for example from a 3D scan. Meshes often need to be retopologized if the mesh is going to be deformed in some way. Deformations can include rigging or physics simulations such as cloth or soft body. Retopology can be done by hand by manipulating geometry in Edit Mode or through automated methods.

Todo 2.81.

Reference

Object Mode, Sculpt Mode

Properties \xe2\x80\xa3 Object Data \xe2\x80\xa3 Remesh

Remeshing is a technique that automatically rebuilds the geometry with a more uniform topology. Remeshing can either add or remove the amount of topology depending on a defined resolution. This technique is especially useful for sculpting, to generate better topology after blocking out the initial shape.

Note

Limitations:

Remeshing only works on the original mesh data and ignores generated geometry from modifiers, shape keys, rigging, etc.

Remeshing will not work with the Multiresolution Modifier.

See also

Remesh modifier

The Voxel Remesher uses OpenVDB to generate a new manifold mesh from the current geometry. It produces a mesh with perfectly even distributed topology and it does not have any performance penalty once the new mesh is calculated. This makes the voxel remesher great for sculpting as it is possible to sculpt at a much higher level of detail than using other features like dyntopo which often adds more performance overhead.

The resolution or the amount of detail the remeshed mesh will have. The value is used to define the size, in object space, of the Voxel. These voxels are assembled around the mesh and are used to determine the new

geometry. For example a value of 0.5 m will create topological patches that are about 0.5 m (assuming Preserve Volume is enabled). Lower values preserve finer details but will result in a mesh with a much more dense topology.

Reduces the final face count by simplifying geometry where detail is not needed. This introduces triangulation to faces that do not need as much detail. Note, an Adaptivity value greater than zero disables Fix Poles.

Tries to produce less Poles at the cost of some performance to produce a better topological flow.

Tells the algorithm to try to preserve the original volume of the mesh. Enabling this could make the operator slower depending on the complexity of the mesh.

Reprojects the Paint Mask onto the new mesh.

Reprojects Face Sets onto the new mesh.

Reprojects the Color Attributes onto the new mesh.

Performs the remeshing operation to create a new manifold mesh based on the volume of the current mesh. Performing this will lose all mesh object data layers associated with the original mesh.

The Quad remesh uses the Quadriflow algorithm to create a Quad-based mesh with few poles and edge loops following the curvature of the surface. This method is relatively slow but generates a higher quality output for final topology.

Warning

Performing Quadriflow Remesh will lose all mesh object data layers associated with the original mesh.

Opens a pop-up used to set parameters for the remesh operation.

Generates a symmetrical mesh using the Mesh Symmetry options.

Tells the algorithm to try to preserve sharp features of the mesh. Enabling this could make the operator slower depending on the complexity of the mesh.

Tells the algorithm to try to preserve the original volume of the mesh. Enabling this could make the operator slower depending on the complexity of the mesh.

Reprojects the Paint Mask onto the new mesh.

Applies the Smooth Normals operator to the resulting mesh.

How to specify the amount of detail for the new mesh.

Specify target number of faces relative to the current mesh.

Input target edge length in the new mesh.

Input target number of faces in the new mesh.

Random Seed to use with the solver; different seeds will cause the remesher to generate different quad layouts on the mesh.

--- Fin du chapitre: _retopology ---

--- Début du chapitre: _reverse_curve --- The Reverse Curve node swaps the start and end of splines. The shape of the splines is not changed.

Tip

When used on the Profile input of the Curve to Mesh Node, this node will flip the normals of the resulting mesh.

Standard geometry input.

Whether or not to change the direction of each spline. True values mean the direction will be changed, false values mean the spline will be unaffected.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_reverse_curve` ---

--- Début du chapitre: `_rgb` --- This node has no input sockets.
The RGB node uses the color picker widget .
A single RGBA color value.

--- Fin du chapitre: `_rgb` ---

--- Début du chapitre: `_rgb_curves` --- The RGB Curves Node performs level adjustments on each color channel.
Controls the amount of influence the node exerts on the image.
Standard color input.
Defines the input color that should be mapped to black.
Defines the input color that should be mapped to white.

Tip

To define the black and white levels, use the eyedropper to select a color sample of a displayed image.

The Combined curve is applied to each channel individually, which may result in a change of hue.

Keeps the hue constant.

The curve to show.

Combined

Red

Green

Blue

A Bézier curve that maps each input level (X axis) to an output level (Y axis). For the curve controls, see Curve widget .

Standard color output.

Below are some common curves you can use to achieve desired effects.

From left to right: 1. Lighten shadows 2. Negative 3. Decrease contrast 4. Posterize. \xc2\xbb

Color correction with curves. \xc2\xbb

In this example, the image has too much red in it, so we run it through an RGB Curves node and reduce the Red channel.

The documentation for the Mix Node has an additional example about fixing overexposure.

Color correction with Black/White Levels. \xc2\xbb

Manually adjusting the RGB curves for color correction can be difficult. Another option for color correction is to use the Black and White Levels instead, which really might be their main purpose.

In this example, the White Level is set to the color of a bright spot of the sand in the background, and the Black Level to the color in the center of the fish's eye. To do this efficiently it is best to bring up the Image Editor showing the original input image. You can then use the levels color picker to easily choose the appropriate colors from the input image, zooming into pixel level if necessary. The result can be fine-tuned with the R, G, and B curves like in the previous example.

The curve for C is used to compensate for the increased contrast that is a side effect of setting Black and White Levels.

Changing colors by inverting the red channel. \xc2\xbb

--- Fin du chapitre: `_rgb_curves` ---

--- Début du chapitre: `_rgb_to_bw` --- The RGB to BW Node makes a color image black-and-white by outputting its luminance.

Note

You can directly connect Color sockets to Value sockets in node graphs, which also converts the image to black-and-white. As such, this node is not always necessary.

Color image input.

This node has no properties.

Grayscale value output.

--- Fin du chapitre: _rgb_to_bw ---

--- Début du chapitre: _rigid_body --- Reference

3D Viewport

Object Mode

Object \xe2\x80\xa3 Rigid Body \xe2\x80\xa3 Calculate Mass

Calculate mass values for rigid body objects based on their volume and density. The volume is calculated automatically, the density needs to be given based on the object you want to simulate.

A list of preset density values for real-world materials, if a material is not given you can research the density and use the Custom preset to input the density manually.

When the Custom Material Preset is selected, this is the input density, in kg/m³, to use.

--- Fin du chapitre: _rigid_body ---

--- Début du chapitre: _rig_features --- After human rig generation a new armature named rig will be added to your scene. This is the character rig you have generated from the human meta-rig and will contain all the features.

The generated rig is accompanied by a script that implements a set of panels that appear in the Item tab of the 3D view sidebar when a bone belonging to the generated rig is active.

This panel is displayed if the armature has an active Action, and is used by operators that apply an operation to multiple keyframes.

When enabled, the operator computes and keyframes its result on every frame that has a key for any of the bones, as opposed to just relevant ones.

When enabled, the operator is limited to a certain frame range.

Specify the frame range to process.

Sets the baking frame range from the scene frame range.

This panel shows properties and operators that are relevant to the selected bones.

This panel contains buttons for toggling visibility of bone collections.

The layout and labels of the buttons are defined in the meta-rig Bone Collection UI panel.

Rigify rigs are built from standardized components called sub-rigs, which are linked together in a parent-child hierarchy. Although the precise behavior of each sub-rig is determined by its implementation, there are certain conventions that are followed by many of them.

Every Rigify rig has a bone called root, which serves as a parent for all bones of the rig. It is assigned to a bone collection called Root. Unless the meta-rig has a custom bone of that name, it is positioned at the origin of the rig object. Its widget looks like a circle with four arrow shaped protrusions.

Many limb-like sub-rigs have a gear-shaped bone at their base.

This bone can in some cases be used to transform the whole sub-rig as a rigid unit, and is also used as a container for its custom properties that are displayed in the Rig Main Properties panel. If you are looking in the Graph Editor for the animated values of the properties, this is most likely the bone to look at.

As an exception, if multiple controls of the sub-rig need their own copy of conceptually the same property, it may be placed on those controls directly instead.

These controls look like blue spheres in the default color scheme, and are the final control layer above the deformation bones themselves.

Tweaks are subordinate to the general IK or FK limb position but can be moved apart, twisted and scaled freely, even reaching virtually impossible limb shapes.

Some sub-rigs provide a slider in their Rig Main Properties when tweaks are selected, which controls the smoothness of the Bendy Bone joint at that position. When zero, the joint deforms with a sharp bend, while setting it to 1 makes the transition smooth for a more rubber hose cartoon like appearance.

Some bones that can be freely moved in space (like IK controls) can be optionally accompanied by a custom pivot control. These controls usually look like a plain axes empty with the axis lines capped with squares or crosses, like the one in the image above. The control can be freely moved to change the location of the pivot, and then rotated or scaled to transform the target bone around the pivot.

A number of rig types provides both IK and FK controls (red for IK and green for FK in the image above), with an ability to switch and snap between them.

Switching is controlled by a slider in Rig Main Properties, usually blending between full IK at 0 and full FK at 1.

Snapping one type of controls to the shape of the other is done via buttons, which form a group of three in their complete set:

The main button will snap on the current frame, and auto-key the result if enabled.

The Action button will bake the change on multiple keyframes, according to Rig Bake Settings.

The Clear button will delete keyframes on the corresponding controls within the bake interval.

Some freely movable controls, e.g. usually the IK controls, can have a mechanism to switch their parent bone between a set of choices, including the root bone, or none at all.

This mechanism is exposed in the Rig Main Properties panel through a row with three controls:

A button that presents a dropdown menu, which allows switching the parent on the current frame while preserving the bone position and orientation in the world space.

A dropdown input field that directly exposes the switch property for keyframing and direct manipulation. Changing the value can cause the bone position to jump.

A button to apply the position preserving parent switch over the bake range of keyframes.

Note

When manually placing a Child Of constraint on the control bone, the built-in parent should be switched to none.

Limbs have a master bone and tweaks. Depending on the user defined meta-rig options, multiple deform bone segments with tweaks will be created. The IK control may have an optional custom pivot, as well as additional predefined pivots.

Rigify's limbs have the following controls in the Sidebar panel:

When set to 1 the FK limb will not rotate with the torso and will retain its rotation relative to the root bone instead.

Controls whether the limb follows IK or FK controls, blending between full IK at 0 and full FK at 1.

Snaps one type of controls to another.

Blends between the limb stretching freely at 1, or having its maximum length constrained at 0.

When the toggle is Off, the IK limb will use the rotational pole vector (the arrow at the base of the limb). Rotating/translating/scaling the arrow will control the IK limb base.

When the toggle is On, the classic pole vector will be displayed and used to orient the IK limb. The arrow will continue to handle the scale and the location of the IK limb base.

Similar to Parent Switching, the row includes buttons to convert the current pose between types, or bake the whole action.

Switches the effective parent of the main IK control.

Switches the effective parent of the classic IK Pole control.

Arms have the simplest control structure: the IK controls consist of the main IK control, the optional custom pivot control, and the optional wrist control (the bent circle), which pivots around the tail rather than the head of the hand bone.

There are no additional controls in the Rig Main Properties panel.

Legs have a more complicated setup, which has:

Two separate IK and FK controls for the toe (this is on by default in the bundled metarigs, and is recommended for stable IK<->FK snapping).

A heel control which can be rotated to command forward or backward roll, sideways rock, or yaw of the heel.

An extra pivot control rotating around the base of the toe.

A custom pivot control.

The properties panel has two additional features:

Standard IK to FK snapping resets the transformations of all IK controls other than the main one. This is not convenient to use in an animation that involves the use of the heel control, because roll and rock would be folded into the transformation of the main control.

This alternative snapping operator tries to deduce the rotation of the heel control so as to keep the main IK control parallel to the ground plane inferred from the current orientation of the IK control. The operator has options to specify which rotational axes to use for the heel control rotation.

If enabled in the sub-rig settings, this slider can be used to control whether the heel rotation (excluding backward roll) is applied at the base or the tip of the toe.

The simplest type of rig for a finger or appendage in general is the simple tentacle sub-rig. It has only basic FK controls and tweaks, with the only automation being the ability to copy certain axes of the local rotation of a FK control to the next one.

For fingers specifically, Rigify has a dedicated fingers sub-rig type, which provides:

A master control (orange), which can be used to rotate the finger as a whole, as well as to bend it via Y scaling.

FK control chain (green) that can also operate as semi-tweaks through allowing translation.

IK control for the tip (red).

Note

IK in this sub-rig is rudimentary and operates as an adjustment for FK. The intended way of use is to pose the finger in FK, and then enable IK after using IK->FK snap if it is necessary to pin the tip of the finger in place.

The properties panel has the following features:

Slider controlling the influence of the IK.

Snaps the IK control to the end of the finger, or adjusts the FK controls to the result of the IK correction.

Has the same effect as Rubber Tweak on limbs, controlling the rubber hose cartoon effect.

Spline Tentacle (Stretch To Fit, Manual Squash & Stretch)

Spline Tentacle (Direct Tip Control)

The spline tentacle is an advanced rig for a flexible appendage (tentacle) based on the Spline IK constraint. The IK control bones manage control points of a Bezier spline curve, which in turn is followed by the IK chain.

The tentacle can be generated in three major modes:

In this simplest mode all bones of the sub-rig deform chain follow the curve and squash & stretch to match its length.

This mode is almost the same, but the chain does not automatically scale to match the curve length. Instead, it tries to cover as much as possible of the curve given its manually scaled length. If the curve is too short, the chain will overhang it and straighten out, but this can result in jitter.

This mode is more similar to the behavior of IK limbs: the final bone of the chain is directly controlled by the tip IK control, while the other bones of the chain stretch and follow the curve to bridge the gap.

The tentacle sub-rig has the following control bones:

The tentacle has the same gear master control as other limbs (seen as a line in the images).

The IK control at the base of the tentacle, which can be used to control the base twist and sideways scale, and is one of the potential switchable parents for other IK controls.

In the Manual Squash & Stretch mode it controls uniform scale of the tentacle in all directions.

Extra start controls, optional and hidden by default. Switchable parents default to the IK Start control. The scale of the control may optionally affect the thickness of the chain via the radius of the curve point.

Controls for the middle of the curve. The switchable parents default to Master, but may be set to IK Start or IK End controls. The scale of the control may optionally affect the thickness of the chain via the radius of the curve point.

Extra end controls, optional and hidden by default. Switchable parents default to the IK End control. The scale of the control may optionally affect the thickness of the chain via the radius of the curve point.

The Direct Tip Control mode adds one more extra end control next to the middle ones that cannot be hidden.

Controls the last control point of the curve, and is one of the potential parents for the other chain controls.

In the Direct Tip Control mode also directly controls the last bone of the chain.

This control is visually attached to the last bone of the chain, and must use Euler rotation.

Stretch To Fit: it controls the twist of the tip of the tentacle, interpolated to nothing at the base.

Manual Squash & Stretch: it also controls the scaling of the tip of the tentacle.

Direct Tip Control: the control does not exist.

If enabled, the rig has an alternative fully FK control chain.

The properties panel has the following features:

If extra controls exist, this property controls how many of them are visible and active.

When a control is disabled, it is snapped to a position extremely close to the corresponding end control point, thus effectively neutralizing its effect. Thus, changing the setting during an animation can cause jumps.

The plus and minus buttons can help with maintaining a continuous transition in an animation by keyframing the change in the property value with Constant interpolation, and also snapping and keyframing the control itself to its "hidden" position.

In the Direct Tip Control mode the twist at the end of the tentacle is deduced from the free form orientation of the tip control, rather than using a separate twist control with constrained Euler rotation. However, for technical reasons, that can only give values within the 180 degrees range of neutral.

A long tentacle can accept more twist than 180 degrees, so a workaround is necessary. This property allows specifying an approximate estimate of the twist value (effectively shifting the neutral position), and the rig then applies the automatic correction within 180 degrees of this value.

If the FK controls are enabled, these provide standard IK-FK switching and snapping.

However, unlike other limbs, for this rig automatic IK to FK snapping can only be approximate and requires manual tuning. For this reason, buttons for baking the snapping over a range of keyframes are not provided.

Switches the parent of the selected IK control.

The spine sub-rig provides a cube shaped torso control with switchable parent, and bent circle shaped hip and chest controls subordinate to it. For low level deformation tweak controls are provided.

The torso control can optionally be accompanied with a custom pivot control. The rig can also optionally provide a full set of FK controls that are subordinate to the normal simplified ones, but above tweaks.

The rig properties panel for the spine controls usually includes options for the head and/or tail as well.

The head sub-rig attaches to the end of the spine, and provides rotational controls for the head and neck, as well as tweaks for fine control of the neck.

If the neck is three or more bones long, an additional tweak-like translational neck bend control is provided (the widget looks like a circle with arrows).

The properties panel contains the following options:

This slider controls the rotations isolation for the neck bones. The neck will follow the orientation of the Torso when set to 0, and the Chest when set to 1.

This slider controls the rotations isolation for the head. The head will follow the orientation of the Torso when set to 0, and the Chest when set to 1.

The tail sub-rig attaches to the start of the spine, and provides FK controls for the tail, as well as a master control that replicates its local rotation around certain axes to all individual bones.

The properties panel contains the following options:

This slider controls the rotations isolation for the tail. The tail will follow the orientation of the Torso when set to 0, and the Hips when set to 1.

Note

This describes the new-style modular face produced by the Upgrade Face operator button.

The foundation of the Rigify face is a network of Bendy Bone chains with controls placed at every bone end. These controls affect all bones that meet at that specific point.

When the controls are merely translated, the B-Bone chains retain the normal automatic bezier handle behavior. Local rotation and/or scaling of the controls are applied on top of that.

In case of certain chains, the transformation of the end and/or middle controls is interpolated to other controls located between them. In such cases the controls often have different colors and/or shapes.

Additionally, certain controls have arbitrary constraints that partially copy transformation from nearby control points.

Certain areas of the face, like eyes or mouth, have additional specialized controllers that apply custom behavior on top of the chains and their controllers within the relevant area.

They have the following controls in addition to the eyelid chains:

This large circular control can be used to transform the whole eye as one unit.

This large control enveloping all individual eye targets has a switchable parent and can be used to specify the point that the eyes should look at.

These small circle controls within the common target control specify the point targeted by each individual eye. Their local scale can also be used to affect the iris or pupil of the eye, depending on how it was weight painted.

The rig properties panel contains the following options:

Controls how much the rotation of the eyeball affects the eyelids.

Depending on the sub-rig generation options, this slider can be split to separately control the horizontal and vertical directions.

If enabled in the sub-rig generation options, this slider can be used to disable the mechanism that forces the eyelids to conform to the sphere of the eye.

Selects the parent for the common target control.

The mouth has the following controls:

Controls rotation of the jaw, directly affecting the main jaw deform bone, as well as chains fully belonging to the jaw. Chains forming the lip loop(s) are adjusted to open the mouth as the jaw rotates or moves.

This control uniformly transforms the lips without moving the jaw.

The rig properties panel contains the following options:

This slider can be changed from 0 to 1 in order to suppress opening of the mouth when the jaw rotates or moves.

--- Fin du chapitre: _rig_features ---

--- Début du chapitre: _rim --- The RimVisual Effect shows a simulated rim light on the object contour.

For simulating the rim light, a masked color silhouette of the object is displaced in horizontal and/or vertical direction.

Many blending modes can be applied to the resulting mask.

Rim Visual Effect. \xc2\xba

Defines the rim light color.

Defines a color to keep unaltered.

The mask blending operation to perform. See Color Blend Modes .

Control the color mask displacement in pixels on the X and Y axis.

Control the blur scale in pixels on the X and Y axis.

Number of blur samples (0 disabled the blur effect).

Original image. \xc2\xba

No blur. \xc2\xba

Blur. \xc2\xba

Mask color: Black. \xc2\xba

--- Fin du chapitre: _rim ---

--- Début du chapitre: _rip ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Rip

V

The Rip tool separates UV components (vertices, edges, faces) from connected components. The components are ripped in the direction of the mouse pointer position.

Before. \xc2\xba

After. \xc2\xba

See also

Mesh editing Rip operation.

--- Fin du chapitre: _rip ---

--- Début du chapitre: _rip_vertices ---Reference

Edit Mode

Vertex \xe2\x80\xa3 Rip Vertices

V

Rip creates a \xe2\x80\x9chole\xe2\x80\x9d in the mesh by making a copy of selected vertices and edges, still linked to the neighboring non-selected vertices, so that the new edges are borders of the faces on one side, and the old ones, borders of the faces on the other side of the rip.

Selected vertex. \xc2\xba

Hole created after using rip on vertex. \xc2\xba

Edges selected. \xc2\xba

Result of rip with edge selection. \xc2\xba

A complex selection of vertices. \xc2\xba

Result of rip operation. \xc2\xba

Rip will only work when edges and/or vertices are selected. Using the tool when a face is selected (explicitly or implicitly), will return an error message \xe2\x80\x9cCannot perform ripping with faces selected this way\xe2\x80\x9d. If your selection includes edges or vertices that are not \xe2\x80\x9cbetween\xe2\x80\x9d two faces Manifold, it will also fail

with the message \xe2\x80\x9cNo proper selection or faces
include\xe2\x80\x9d.

--- Fin du chapitre: _rip_vertices ---

--- Début du chapitre: _rip_vertices_extend ---Reference

Edit Mode

Vertex \xe2\x80\xa3 Rip Vertices and Extend

Alt- D

This tool takes any number of selected vertices and duplicate-drags them along the closest edge to the mouse. When extending an edge loop, it extends the vertices at the endpoints of the loop. The behavior is similar to the Extrude tool, but it creates an n-gon.

It helps to easily add details to existing edges.

--- Fin du chapitre: _rip_vertices_extend ---

--- Début du chapitre: _rip_vertices_fill ---Reference

Edit Mode

Vertex \xe2\x80\xa3 Rip Vertices and Fill

Alt- V

Rip fill works the same as the Rip tool above, but instead of leaving a hole, it fills in the gap with geometry.

Edges selected. \xc2\xb6

Result of rip fill. \xc2\xb6

--- Fin du chapitre: _rip_vertices_fill ---

--- Début du chapitre: _roll_hair_curves --- Rolls up hair curves starting from their tips.

Geometry

Factor to blend overall effect.

Subdivision level applied before deformation.

Level of smoothing on the roll path to include shape variation.

Length of each curve to be rolled.

Radius of the rolls.

Depth offset of the roll.

Taper of the roll.

Offset the roll along the original curve to retain shape.

Axis around which each curve is rolled.

Amount of randomization of the direction of the roll.

Random Seed for the operation.

Preserve each curve\xe2\x80\x99s length during deformation.

This node has no properties.

Geometry

--- Fin du chapitre: _roll_hair_curves ---

--- Début du chapitre: _rotate --- This node rotates an image.

Standard color input.

Rotation angle in degree. Positive values rotate clockwise and negative ones counterclockwise.

Interpolation Methods.

No interpolation, uses nearest neighboring pixel.

Simple interpolation between adjacent pixels.

Highest quality interpolation.

Standard color output.

--- Fin du chapitre: _rotate ---

--- Début du chapitre: _rotate_edge ---Reference

Edit Mode

Edge \xe2\x80\xa3 Rotate Edge CW / Rotate Edge CCW

Rotating an edge clockwise (CW) or counterclockwise (CCW) spins an edge between two faces around their vertices. This is very useful for restructuring a mesh's topology.

The tool operates on selected edges or the shared edge between selected faces.

Two adjacent faces selected. \xc2\xb6

Selected edge rotated. \xc2\xb6

Selected edge. \xc2\xb6

Edge, rotated CW. \xc2\xb6

Warning

To rotate an edge based on faces you must select adjacent face pairs, otherwise Blender notifies you with an error message, \xe2\x80\x9cCould not find any selected edges that can be rotated\x9d. Using either Rotate Edge CW or Rotate Edge CCW will produce exactly the same results as if you had selected the common edge.

--- Fin du chapitre: _rotate_edge ---

--- Début du chapitre: _rotate_euler --- The Rotate Euler node rotates an Euler rotation.

Important

This node is deprecated, use the Rotate Rotation Node instead.

The Euler rotation to rotate.

Specifies how much an Euler rotation is rotated. This input is only available when the rotation type is set to Euler.

The axis to rotate around. This input is only available when the rotation type is set to Axis Angle.

The angle to rotate by around the specified axis. This input is only available when the rotation type is set to Axis Angle.

Use separate axis and angle inputs to control the rotation.

Use an Euler input to control the rotation.

Rotate an Euler rotation in the evaluated object's space.

Rotate an Euler rotation in local space.

The rotated Euler rotation.

--- Fin du chapitre: _rotate_euler ---

--- Début du chapitre: _rotate_hair_curves --- Rotates each hair curve around an axis.

Input Geometry (only curves will be affected).

Factor to influence the rotation angle.

Rotation Axis (Default: Tangent at root).

Angle of rotation.

Random offset to the rotation angle per Curve.

Lock rotation to the axis between the curve ends.

Random Seed for the operation.

This node has no properties.

Geometry

--- Fin du chapitre: _rotate_hair_curves ---

--- Début du chapitre: _rotate_instances --- The Rotate Instances node rotates geometry instances in local or global space.

TheInstancespage contains more information about geometry instances.

Standard geometry input.

Boolean field used to determine if an instance will be rotated.

The Euler rotation to rotate the instances by.

The position around which each instance is rotated. If the Local Spaceinput is true,the location is relative to the initial transform of the instance.

If enabled, the instances are rotated in local space. In other words,they are rotated around the axes described by the initial transform of each instance.When the input is disabled, the pivot point and rotation are specified inthe local space of the modifier object.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _rotate_instances ---

--- Début du chapitre: _rotate_rotation --- The Rotate Rotationnode applies an additional rotation to a given one.

To rotate anEuler Rotation, first use theEuler to Rotation Node.

The starting rotation.

The additional rotation.

Rotate in Global Space .

Rotate in Local Space .

The resulting rotation.

--- Fin du chapitre: _rotate_rotation ---

--- Début du chapitre: _rotate_vector --- The Rotate Vectornode rotates a vector by a given rotation value.

The vector to rotate.

Standard rotation value.

The rotated vector.

--- Fin du chapitre: _rotate_vector ---

--- Début du chapitre: _rotation --- The Rotationinput creates a rotation from euler rotation values.

This node has no inputs.

The amount of rotation about each axes.

Standard rotation output.

--- Fin du chapitre: _rotation ---

--- Début du chapitre: _rotations --- Blender lets you define rotations in several ways. Each one of them has a series of advantages and drawbacks;there is no best rotation mode, as each one is suitable for specific cases.

In all of these modes, positive angle values mean counter-clockwise rotation direction,while negative values define clockwise rotation.

Though you can rotate elements using the global or local transform orientations,these axes are not suitable to define rotations, as the effect of each ofthem cannot be isolated from the other two.

Take, for instance, any three values for X, Y and Z rotation. Perform each one of these using global or local axes.Depending on the order in which you perform these, you will end up with different final orientations.So proper rotation coordinate systems are needed.

The axes system used for performing Euler rotations is the so called Euler gimbal.A gimbal is a particular set of three axes.The special thing

about this is that the axes have a hierarchical relationship between them: one of the axes is at the top of the hierarchy, and has one of the other two axes as its immediate child; at the same time, this child axis is the parent of the remaining axis, the one at the very bottom of the hierarchy.

Which axis is on top, which one in the middle and which at the bottom, depends on the particular Euler gimbal: there are six types of them, as there are six possible combinations: XYZ, XZY, YXZ, YZX, ZXY and ZYX Euler rotation modes. These modes are named using the letters of the axes in order, starting from the axis at the bottom of the hierarchy, and finishing with the one on top.

The main problem of these systems comes when they lose their relative perpendicularity. And this happens when the axis in the middle rotates, causing the axis at the bottom to rotate with it. It keeps getting worse when this bottom axis approaches 90° (or equivalent angles). In that case, it will remain aligned with the axis on top of the hierarchy. In that moment we have just lost one axis of rotation. This can cause discontinuous interpolations when animating. This particular loss of axis is known as the gimbal lock.

Hint

The actual configuration of the gimbal axes can be seen in the 3D Viewport by enabling the Rotate object gizmo and setting it to Gimbal (from the gizmos button in the header). At the same time, rotation mode should be set to any of the Euler modes for the active object.

Now you can perform a rotation around the axis in the middle (e.g. in XYZ Euler mode that is the Y axis), and see how easy it is to end up having a gimbal with just two axes. In the specific case of the XYZ Euler mode with gimbal lock, a rotation around the X axis will have the same effect as rotating around the Z axis, meaning, in practice, that no X axis rotations can be performed.

One advantage of this mode is that animation curves are easy to understand and edit. However, special attention must be done when the middle axis approaches values close to 90° (or equivalent angles). This mode lets us define an axis (X, Y, Z) and a rotation angle (W) around that axis.

If we define the rotation using interactive rotations (with the rotation gizmo), the values of X, Y and Z will not exceed 1.0 in absolute value, and W will be comprised between 0 and 180 degrees.

If you wish to define rotations above 180° (e.g. to define multiple revolutions), you will need to edit the W value directly, but as soon as you perform an interactive rotation, that value will be adjusted again. Same thing goes for axis values.

This system is suitable for elements revolving around a fixed axis, or to animate one of the elements at a time (either the axis or the angle). The problem might come when animating (interpolating) both components at the same time: axis and angle. The resulting effect might not be as expected. The Gimbal gizmo in this rotation mode shows a set of three orthogonal axes in which the Z axis goes along the defined rotation axis, i.e. it points towards the direction defined by the (X, Y, Z) point.

The axis-angle system is free from gimbal lock, but animation curves in this mode are not intuitive at all when animating axis and angle at the same time, in which case they are difficult to understand and edit.

In this mode, rotations are also defined by four values (X, Y, Z and W). X, Y and Z also define an axis, and W an angle, but it does it quite

differently from axis-angle. The important thing here is the relation between all four values.

To describe it in an intuitive way, let's take the effect of the X coordinate: what it does is to rotate the element around the X axis up to 180 degrees. The same goes for Y and Z. The effect of W is to avoid those rotations and leave the element with zero rotation. The final orientation is a combination of these four effects.

As the relation between components is what defines the final orientation, multiplying or dividing all four numbers by a constant value will yield the very same rotation.

This mode is ideal for interpolating between any pair of orientations. It doesn't suffer from gimbal lock or any interpolation undesired effect. The only drawback is that you cannot interpolate between two orientations that are at a distance greater than 180°, as the animation will take the shortest path between them. Thus to animate a revolving element you must set up many intermediate keyframes, 180° from each other at most.

The Gimbal gizmo in this mode is equivalent to the Local one, and doesn't have any special meaning.

The animation curves in this mode are not intuitive, so they are also difficult to understand and edit.

This section is not really useful for 3D artists, but it can be suitable for the curious or the scientist.

Quaternions are a number system extending the complex numbers. They represent a four component vector, whose components are called, in Blender, X, Y, Z and W. When rotating interactively in quaternion mode, the so called norm (length) of the quaternion will remain constant. By definition, the norm of a quaternion equals 1.0 (that's a normalized quaternion). When you select the quaternion mode in Blender, the XYZW components describe a normalized quaternion.

Note

The norm of a quaternion q is defined mathematically as:

However, if one of the quaternion components is locked during the interactive transformation using the proper lock button, the norm will not remain unchanged, as that blocked component will not be able to adjust itself to keep the unit norm.

Hint

Interactive rotations with the gizmo don't change the norm of the current quaternion. Editing a single XYZW component individually you can change the norm. To make the norm 1.0 again you can switch to any rotation mode and back again into quaternion.

The rotation components of a quaternion keep a tight relation with those of axis-angle. To find a correspondence, first of all we must deal with the normalized version of the quaternion, that is, one whose norm equals 1.0. To normalize a quaternion, just divide each one of its components by its norm. As we have seen before, dividing all four values by the same number gives the same orientation.

Once we have calculated the components of the normalized quaternion, the relation with the axis-angle components is as follows:

X, Y and Z mean exactly the same as in axis-angle: they just define an axis around which the rotation takes place.

W can be used to retrieve the actual rotation around the defined angle. The following formula applies (provided that the quaternion is

normalized): $W = \cos\left(\frac{a}{2}\right)$, where a is actually the rotation angle we are looking for. That is: $a = 2 \arccos(W)$. In axis-angle and quaternion modes we can lock rotations in interactive modes in a per component basis, instead of doing it by axis. To do so we can activate this locking ability using the lock buttons next to the corresponding Rotation transform buttons. Regarding rotation animations, all keyframes must be defined in the same rotation mode, which must be the selected rotation mode for the object throughout the entire animation.

--- Fin du chapitre: _rotations ---

--- Début du chapitre: _rotation_to_euler --- The Rotation to Euler node converts a standard rotation socket value to an Euler rotation. Standard rotation socket value. The Euler rotation.

--- Fin du chapitre: _rotation_to_euler ---

--- Début du chapitre: _rotation_to_quaternion --- The Rotation to Quaternion node converts a standard rotation value to a quaternion rotation. Standard rotation value. The W value of the quaternion. The X value of the quaternion. The Y value of the quaternion. The Z value of the quaternion.

--- Fin du chapitre: _rotation_to_quaternion ---

--- Début du chapitre: _sample_curve --- The Sample Curve calculates a point on a curve at a certain distance from the start of the curve, specified by the length or factor inputs. It also outputs data retrieved from that position on the curve. The sampled values are linearly interpolated from the values at the evaluated curve points at each side of the sampled point.

Note

When the curve contains multiple splines, the sample position is found based on the total accumulated length, including the lengths of all previous splines. The order of the splines is the same order as displayed in the Spreadsheet Editor .

Standard geometry input with a curve component.

A field input to evaluate custom attributes. The evaluation is outputted through the Value output.

The portion of the total length used to determine the sample position.

A length in distance units used to determine how far along the curve to travel before sampling.

An index to only evaluate specific splines, these indices can be specified manually or from the Index Node . This input is ignored when the All Curves property is enabled.

The data type used for the evaluated data.

How to find endpoint positions for the trimmed spline. The option acts the same as it does in the Trim Curve Node .

Find the endpoint positions using a factor of each spline's length. The input values should be between 0 or 1.

Find the endpoint positions using a length from the start of each spline. The input values should be between 0 and the length of the splines.

Sample lengths based on the total length of all curves, rather than using a length inside each selected curve.

The value of the input Value at the sample point.

The position at the sample along the spline.

The normalized curve tangent at the sample.

Tip

This output can be combined with the Align Rotation to Vector Node to create a rotation that lines up with direction of the curve. Including the Normal output in a second align node after can align the other rotation axis.

The normalized curve normal at the sample.

Here, the Count mode of the Resample Curve Node is recreated, except a mesh is used for the result instead of a curve. The Z axis of the position can be used as the sample factor because the position is between zero and one for the entire line.

--- Fin du chapitre: _sample_curve ---

--- Début du chapitre: _sample_index --- The Sample Index node retrieves values from a source geometry at a specific index.

Tip

If the Geometry used for the input is the same as the geometry from the field context, this node is equivalent to the Evaluate at Index Node. Using that node is usually preferable since avoiding the geometry socket makes the whole setup easier to use in other situations and share.

Tip

Different components can have same attribute domain (Points). This node simply uses first component that not empty for such domain, checked in the order of: Mesh, Point Cloud, Curve. The Separate Components Node can be used to sample directly from a specific component.

The geometry to retrieve the attribute from.

A field to evaluate on the source Geometry. The values are then retrieved from specific indices for the output.

Which index to use when retrieving the data from the input Value field.

Any index can be connected, resulting in a \xe2\x80\x99cshuffling\xe2\x80\x99d of the values.

The data type to use for the retrieved values.

The attribute domain that the attribute is transferred from, or in other words, the domain used to evaluate the Attribute input. For example, it is possible to transfer data from the faces of one geometry to the points of another.

Clamp the indices to the size of the attribute domain instead of outputting a default value for invalid indices.

The data retrieved from the source Geometry input.

Here the node is used to copy the positions of one object to another. This recreates the behavior of the Transfer Attribute node from Blender versions before 3.4. This works best when their geometries have the same number of points and the same Topology. \xc2\xb6

--- Fin du chapitre: _sample_index ---

--- Début du chapitre: `_sample_nearest` --- The Sample Nearest node retrieves the index of the geometry element in its input geometry that is closest to the input position.

This node is similar to the Geometry Proximity Node, but it outputs the index of the closest element instead of its distance from the current location.

Tip

If you want to find nearest to each point in same geometry, its better to use the Index of Nearest node.

The geometry to sample.

Note

This node only supports point cloud and mesh inputs.

The position to start from when finding the closest location on the target geometry. By default, this is the same as if the Position Node was connected.

The attribute domain to consider the distance from.

The index of the closest geometry element of the chosen domain.

Combining this node with the Sample Index Node gives a setup that can retrieve the closest attribute value from another geometry. This is the same behavior as the Transfer Attribute node in versions of Blender before 3.4.

--- Fin du chapitre: `_sample_nearest` ---

--- Début du chapitre: `_sample_nearest_surface` --- The Sample Nearest Surface node finds values at the closest points on the surface of a source mesh geometry. Non-face attributes are interpolated across the surface.

This node is similar to the Geometry Proximity Node, but it gives the value of any attribute at the closest surface point, not just its position.

Warning

Because the node samples the surface of a mesh rather than its edges or vertices, values from loose points and edges are ignored.

The geometry to retrieve the attribute from.

A field to evaluate on the Source geometry for use with the transfer method.

Is evaluated on the face domain and splits the input mesh into multiple parts, each with its own id.

The position to start from when finding the closest location on the target mesh. By default, this is the same as if the Position Node was connected.

Determines in which group the closest nearest surface is detected.

The data type to use for the retrieved values.

The data retrieved and interpolated from the Source geometry, mapped based on the node's settings and inputs.

Whether the sampling was successful. It can fail when the sampled group is empty.

--- Fin du chapitre: `_sample_nearest_surface` ---

--- Début du chapitre: `_sample_uv_surface` --- The Sample UV Surface node finds values on a mesh's surface at specific UV locations. Internally the process is a reverse UV lookup from a location in 2D space. The node then finds the face that corresponds to each UV coordinate, and the location within that face.

Warning

Because of the node's method of computation, the UV map should not have any overlapping faces. If the UV map is sampled at a location with no faces or overlapping faces, the node will output the default value for the data type, which is zeros for most types.

A geometry containing the mesh with a UV map for sampling.

A field to evaluate on the target Mesh geometry for later sampling at the surface positions.

The mesh UV map to sample, evaluated on the Mesh input. Should not have overlapping faces.

The coordinates to sample within the UV map.

The data type to use for the retrieved values.

The data retrieved and interpolated from the Mesh geometry, mapped based on the node's settings and inputs.

Whether the node could find a single face to sample at the UV coordinate.

--- Fin du chapitre: `_sample_uv_surface` ---

--- Début du chapitre: `_sampling` --- The Sampling modifier changes the definition, precision of the stroke, for the following modifiers.

The smaller this value, the more precise are the strokes. Be careful; too small values will require a huge amount of time and memory during render!

--- Fin du chapitre: `_sampling` ---

--- Début du chapitre: `_save_load` --- Preferences Save/Load section.

Asks for confirmation before closing or opening a new blend-file if the current file has unsaved changes.

Number of versions created (for backup) when saving newer versions of a file.

This option keeps saved versions of your file in the same directory, using extensions: `.blend1`, `.blend2`, etc., with the number increasing to the number of versions you specify.

Older files will be named with a higher number. E.g. with the default setting of 2, you will have three versions of your file:

- *.blend last saved.

- *.blend1 second last saved.

- *.blend2 third last saved.

Number of files displayed in File > Open Recent.

Enables Auto Save. Tells Blender to automatically save a backup copy of your work-in-progress files to the Temporary Directory.

This specifies the number of minutes to wait between each Auto Save. The default value of the Blender installation is 2 minutes. The minimum is 1, and the Maximum is 60 (save every hour).

Select how blend-file preview are generated. These previews are used both in the File Browser and for previews shown in the operating system's file browser.

Do not generate any blend-file previews.

If there is no camera in the 3D Viewport a preview using a screenshot of the active Workspace is generated. If a camera is in the scene, a preview of the viewport from the camera view is used.

Generate a preview by taking a screenshot of the active Workspace.

Generate a preview of a Workbench render from the camera's point of view.

Default value for Relative Paths when loading external files such as images, sounds, and linked libraries. It will be ignored if a path is already set.

Default value for Compress file when saving blend-files.

Default value for Load UI when loading blend-files.

Entering Tab in the Text Editor adds the appropriate number of spaces instead of using characters.

Python scripts (including driver expressions) are not executed by default for security reasons. You may be working on projects where you only load files from trusted sources, making it more convenient to allow scripts to be executed automatically.

Blend-files in these folders will not automatically run Python scripts. This can be used to define where blend-files from untrusted sources are kept.

See also

Python Security .

Hide the Recent panel of the File Browser which displays recently accessed folders.

Hide System Bookmarks in the File Browser.

By activating this, the file region in the File Browser will only show appropriate files (i.e. blend-files when loading a complete Blender setting). The selection of file types may be changed in the file region.

Unhide files and data-blocks with names that start with . in File Browsers and data IDs .

Hint

Data-blocks with names beginning with a. can be selected by typing in a search string that also starts with the. character, even if this setting is disabled.

--- Fin du chapitre: _save_load ---

--- Début du chapitre: _scale --- This node scales the size of an image. Standard color input.

Scale in the axis directions, only available if Space is set to Relative or Absolute.

Coordinate Space to scale relative to.

Percentage values relative to the dimensions of the image input.

Size of an image by using absolute pixel values.

Sizes an image to the size of the final render resolution for the scene. For example, rendering a scene at the standard 1080p resolution but setting the render percentage at 50%, will produce a 1080p image with the scene scaled down 50% and leaving the rest of the image as alpha.

Image dimensions set in the Render panel.

Stretch distorts the image so that it fits into the render size. Fit scales the image until the bigger axis fits into the render size. Crop cuts the image so that it is the same aspect ratio as the render size.

Offset factor for the final scaled image.

Standard color output.

For instance X: 0.5 and Y: 0.5 would produce an image which width and height would be half of what they used to be.

Use this node to match image sizes. Most nodes produce an image that is the same size as the image input into their top image socket. To uniformly combine two images of different size, the second image has to be scaled up to match the resolution of the first.

--- Fin du chapitre: _scale ---

--- Début du chapitre: _scale_cage ---Reference

Object and Edit Modes

Toolbar \xe2\x80\xa3 Scale \xe2\x80\xa3 Scale Cage

The Scale Cagetool is a bounding box around the object(s) which scales objects from a particular point or axis. The tool works by selecting a scale point and dragging inwards or outwards to adjust the scale accordingly. The origin for the scale will be from the point on the cube directly opposite from the point selected. Selecting points on the faces of the cube scales along one axis, selecting points on the edges of the cube scales along two axes, and selecting points on the vertices of the cube scales along all three axes.

Scale Cage tool.\xc2\xb6

Aligns the transformation axes to a specified orientation constraint. See Transform Orientations for more information.

The amount to resize the selection on their respected axis.

Aligns the transformation axes to a specified orientation constraint. See Transform Orientations for more information.

The extruded face will affect nearby geometry. See Proportional Editing for a full reference.

--- Fin du chapitre: _scale_cage ---

--- Début du chapitre: _scale_elements --- Scales the selected faces or edges, letting you specify a scaling factor and pivot point for each one. Connected faces/edges are scaled together using their average factor and pivot point.

Standard geometry input.

Boolean field indicating which elements to scale.

The scaling factor for each element.

The pivot point for each element.

Axis along which to scale each element. This vector is normalized internally, so the length does not matter.

The element type to transform.

Scale faces.

Scale edges.

Scale elements by the same factor in every direction.

Scale elements in a single direction defined by the Axisinput.

Standard geometry output.

The node is useful when combined with the Extrude Mesh Node ,especially in Individualmode where connected faces aren't extruded together.

--- Fin du chapitre: _scale_elements ---

--- Début du chapitre: _scale_instances --- The Scale Instancesnode scales geometry instances in local or global space.

TheInstancespage contains more information about geometry instances.

Standard geometry input.

Boolean field used to determine if an instance will be scaled.

The scale factor to apply to the instance's transform on each axis.

The position from which the instance origins are scaled. Each instance will move away from this location. When the Local Spaceinput is enabled, this location is relative to the initial transform of each instance.

If enabled, the instances are scaled in local space. In other words, they are scaled in the directions described by the initial transform of each instance. When the input is disabled, the Center and Scale inputs are specified in the local space of the modifier object.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_scale_instances` ---

--- Début du chapitre: `_scene` --- Scene strips are a way to insert the render output of another scene into your sequence. Instead of rendering out a video, then inserting the video file, you can insert the scene directly.

The strip length will be determined based on the animation settings in that scene.

Note

Scene strips cannot be used to reference the sequence's own scene; a secondary scene must be used instead.

Existing scene strips can be added from the Add Scene menu. New scenes can also be created directly from the add menu with Add Scene New Scene.

Options

The first frame to start the scene strip.

The channel to place the strip in.

Replace the active strip with the new scene strip.

When creating a new scene you have the following options:

How the new scene is created.

Add new Strip with a new empty Scene with default settings.

Add a new Strip, with an empty scene, and copy settings from the current scene.

Add a Strip and link in the collections from the current scene (shallow copy).

Add a Strip and make a full copy of the current scene.

A Data-Block Menu to select or create the scene to render from.

Input type to use for the Scene strip.

Use the scene's 3D camera as input.

Use the scene's Sequencer timeline as input, allowing one scene to reuse another scene's edit (instead of taking the render output from the scene).

This is similar to how MetaStrips work, with the added advantage of supporting multiple instances of the same data.

This can be used to override the scene's camera with any other object.

It is useful to support switching views within a single scene.

Shows Annotations while in non-render Preview Shading Modes i.e. Solid or Wireframe mode.

Creates a transparent background. This is useful for doing overlays like rendering out Grease Pencil films via the Sequencer.

Volume of the audio taken from the chosen scene.

Scene strips do not render individual Render Passes; only the Combined render pass will be used.

--- Fin du chapitre: `_scene` ---

--- Début du chapitre: `_scene_fbx` --- Reference

Import-Export

File \xe2\x80\xa3 Import/Export \xe2\x80\xa3 FBX (.fbx)

This add-on is enabled by default, in case it is not:

Open Blender and go to Add-ons section of the Preferences .

Search \xe2\x80\x9cFBX\xe2\x80\x9d and check the Enable

Extensioncheckbox.

This format is mainly use for interchanging character animations between applicationsand is supported by applications such as Cinema4D, Maya, Autodesk 3ds Max, Wings3D andengines such as Unity3D, Unreal Engine 3/UDK and Unreal Engine 4.

The exporter can bake mesh modifiers and animation into the FBX so the final result looks the same as in Blender.

Note

Bones would need to get a correction to their orientation(FBX bones seems to be -X aligned, Blender\xe2\x80\x99s are Y aligned),this does not affect skinning or animation, but imported bones in other applications will look wrong.

Animations (FBX AnimStacks, Blender actions) are not linkedto their object,because there is no real way to know which stack to use as \xe2\x80\x98active\xe2\x80\x99 action for a given object, mesh or bone.This may be enhanced to be smarter in the future, but it\xe2\x80\x99s not really considered urgent,so for now you\xe2\x80\x99ll have to link actions to objects manually.

Armature instances are not supported.

Note

Bones\xe2\x80\x99 orientation importing is complex, you may have to play a bit withrelated settings until you get the expected results.

Animation support is minimal currently, we read all curves as if they were \xe2\x80\x98baked\xe2\x80\x99 ones(i.e. a set of close keyframes with linear interpolation).

Imported actions are linked to their related object, bone or shape key, on a \xe2\x80\x98first one wins\xe2\x80\x99 basis.If you export a set of them for a single object, you\xe2\x80\x99ll have to reassign them yourself.

Note

Saving Just Animations

The FBX file format supports files that only contain takes.It is up to you to keep track of which animation belongs to which model.The animation that will be exported is the currently selected action within the Action editor.To reduce the file size, turn off the exporting of any parts you do not want and disable All Actions.For armature animations typically you just leave the armature enabled which is necessary forthat type of animation. Reducing what is output makes the export and future import much faster.Normally each action will have its own name but the current oronly take can be forced to be named \xe2\x80\x9cDefault Take\xe2\x80\x9d. Typically, this option can remain off.

Note

Blender now only supports complex node-based shading. FBX having a fixed pipeline-like support of materials,this add-on converts between them.

TODO.

Todo.

TODO.

TODO.

TODO.

Todo.

TODO.

TODO.

Since many applications use a different axis for
Up, these are axis conversion for these
settings, Forward and Up axes By mapping these to different
axes you can convert rotations between applications default up and forward
axes.

Blender uses Y Forward, Z Up (since the front view looks along the +Y
direction). For example, its common for applications to use Y as the up
axis, in that case -Z Forward, Y Up is needed.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

When referencing paths in exported files you may want some control as to
the method used since absolute paths may only be correct on your own
system. Relative paths, on the other hand, are more portable but mean that
you have to keep your files grouped when moving about on your local file
system. In some cases, the path doesn't matter since the target
application will search a set of predefined paths anyway so you have the
option to strip the path too.

Uses relative paths for files which are in a subdirectory of the
exported location, absolute for any directories outside that.

Uses full paths.

Uses relative paths in every case (except when on a different drive on
Windows).

Uses relative / absolute paths based on the paths used in Blender.

Only write the filename and omit the path component.

Copy the file on exporting and reference it with a relative path.

TODO.

When enabled, export each group or scene to a file.

Choose whether to batch export groups or scenes to files. Note, when
Group/Scene is enabled, you cannot use the animation option Current
Actions since it uses scene data and groups are not attached to any
scenes. Also note, when Group/Scene is enabled you must include the
armature objects in the group for animated actions to work.

When enabled, each file is exported into its own directory, this is
useful when using the Copy Images option. So each directory contains one
model with all the images it uses. Note, this requires a full Python
installation. If you do not have a full Python installation, this button
will not be shown.

Only export the selected objects. Otherwise export all objects in the
scene. Note, this does not apply when batch exporting.

Todo.

Enable/Disable exporting of respective object types.

TODO.

Scale the exported data by this value. 10 is the default because this fits
best with the scale most applications import FBX to.

TODO.

Since many applications use a different axis for
Up, these are axis conversions for Forward and Up
axes. By mapping these to different axes you can convert
rotations between applications default up and forward axes.

Blender uses Y Forward, Z Up (since the front view looks along the +Y
direction). For example, its common for applications to use Y as the up
axis, in that case -Z Forward, Y Up is needed.

TODO.

TODO.

TODO.

Todo.

Export objects using the evaluated mesh, meaning the resulting mesh after
all Modifiers have been calculated.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

TODO.

Export all actions compatible with the selected armatures start/end times
which are derived from the keyframe range of each action. When disabled
only the currently assigned action is exported.

TODO.

TODO.

TODO.

Note that the importer is a new addition and lacks many features the
exporter supports.

- binary FBX files only.

- Version 7.1 or newer.

Mesh: shape keys.

NURBS surfaces, text3D and metaballs are converted to meshes at export
time.

Some of the following features are missing because they are not supported
by the FBX format, others may be added later.

Object instancing. Exported objects do not share
data, instanced objects will each be written with their own data.

Material textures

Vertex shape keys. FBX supports them but this exporter does
not write them yet.

Animated fluid simulation. FBX does not support this kind of
animation. You can however use the OBJ exporter to write a sequence of
files.

Constraints. The result of using constraints is exported as a
keyframe animation however the constraints themselves are not saved in the
FBX.

Instanced objects. At the moment instanced objects are only
written in static scenes (when animation is disabled).

--- Fin du chapitre: _scene_fbx ---

--- Début du chapitre: _scene_gltf2 ---

Reference
Import-Export

File > Import/Export > glTF 2.0 (.glb, .gltf)

This add-on is enabled by default, in case it is not:

Open Blender and go to Add-ons section of the Preferences .

Search glTF 2.0 and check the Enable Extensioncheckbox.

glTF (GL Transmission Format) is used for transmission and loading of 3D models in web and native applications. glTF reduces the size of 3D models and the runtime processing needed to unpack and render those models. This format is commonly used on the web, and has support in various 3D engines such as Unity3D, Unreal Engine 4, and Godot.

This importer/exporter supports the following glTF 2.0 features:

Meshes

Materials (Principled BSDF) and Shadeless (Unlit)

Textures

Cameras

Punctual lights (point, spot, and directional)

Extensions (listed below)

Extras (custom properties)

Animation (keyframe, shape key, and skinning)

glTF's internal structure mimics the memory buffers commonly used by graphics chips when rendering in real-time, such that assets can be delivered to desktop, web, or mobile clients and be promptly displayed with minimal processing. As a result, quads and n-gons are automatically converted to triangles when exporting to glTF. Discontinuous UVs and flat-shaded edges may result in moderately higher vertex counts in glTF compared to Blender, as such vertices are separated for export. Likewise, curves and other non-mesh data are not preserved, and must be converted to meshes prior to export.

When the option is enabled in Exporter, instances are exported using the EXT_mesh_gpu_instancing extension. There are some limitations, at export:

Instances must be meshes, and don't have any children themselves

Instances must all be children of the same object.

This extension doesn't manage material variation. That means that the generated file may include all instances with same materials.

Instances detected are objects sharing the same mesh data.

At import, instances are created by creating objects sharing the same mesh data.

The core material system in glTF supports a metal/rough PBR workflow with the following channels of information:

Base Color

Metallic

Roughness

Baked Ambient Occlusion

Normal Map (tangent space, +Y up)

Emissive

Some additional material properties or types of materials can be expressed using glTF extensions. The complete list can be found in Extensions part of this documentation.

An example of the various image maps available in the glTF 2.0 core format. This is the water bottle sample model shown alongside slices of its various image maps.

The glTF material system is different from Blender's own materials. When a glTF file is imported, the add-on will construct a set of Blender nodes to replicate each glTF material as closely as possible.

The importer supports Metal/Rough PBR (core glTF), Spec/Gloss PBR (KHR_materials_pbrSpecularGlossiness) and some extension materials. The complete list can be found in Extensions part of this documentation.

Tip

Examining the result of the material import process is a good way to see examples of the types of material nodes and settings that can be exported to glTF.

The exporter supports Metal/Rough PBR (core glTF) and Shadeless (KHR_materials_unlit) materials. It will construct a glTF material based on the nodes it recognizes in the Blender material. The material export process handles the settings described below.

Note

When image textures are used by materials, glTF requires that images be in PNG or JPEG format. The add-on will automatically convert images from other formats, increasing export time.

The glTF base color is determined by looking for a Base Color input on a Principled BSDF node. If the input is unconnected, the input's default color (the color field next to the unconnected socket) is used as the Base Color for the glTF material.

A solid base color can be specified directly on the node. If an Image Texture node is found to be connected to the Base Color input, that image will be used as the glTF base color.

An image is used as the glTF base color.

These values are read from the Principled BSDF node. If both of these inputs are unconnected, the node will display sliders to control their respective values between 0.0 and 1.0, and these values will be copied into the glTF.

When using an image, glTF expects the metallic values to be encoded in the blue (B) channel, and roughness to be encoded in the green (G) channel of the same image. If images are connected to the Blender node in a manner that does not follow this convention, the add-on may attempt to adapt the image to the correct form during exporting (with an increased export time).

In the Blender node tree, it is recommended to use a Separate RGB node to separate the channels from an Image Texture node, and connect the green (G) channel to Roughness, and blue (B) to Metallic. The glTF exporter will recognize this arrangement as matching the glTF standard, and that will allow it to simply copy the image texture into the glTF file during export.

The Image Texture node for this should have its Color Space set to Non-Color.

A metallic/roughness image connected in a manner consistent with the glTF standard, allowing it to be used verbatim inside an exported glTF file.

glTF is capable of storing a baked ambient occlusion map. Currently there is no arrangement of nodes that causes Blender to use such a map in exactly the same way as intended in glTF. However, if the exporter finds a custom node group by the name of glTFMaterialOutput, and finds an input named Occlusion on that node group, it will look for an Image Texture attached there to use as the occlusion map in glTF. The effect need not be shown in Blender, as Blender has other ways of showing ambient

occlusion, but this method will allow the exporter to write an occlusion image to the glTF. This can be useful to real-time glTF viewers, particularly on platforms where there may not be spare power for computing such things at render time.

A pre-baked ambient occlusion map, connected to a node that doesn't render but will export to glTF.

Tip

If you enable Shader Editor Add-ons in preferences, you will be able to add this custom node group from Menu: Add > Output > glTF Material Output. glTF stores occlusion in the red (R) channel, allowing it to optionally share the same image with the roughness and metallic channels.

This combination of nodes mimics the way glTF packs occlusion, roughness, and metallic values into a single image.

Tip

The Cycles render engine has a Bake panel that can be used to bake ambient occlusion maps. The resulting image can be saved and connected directly to the glTF Material Output node.

To use a normal map in glTF, connect an Image Texture node's color output to a Normal Map node's color input, and then connect the Normal Map normal output to the Principled BSDF node's normal input. The Image Texture node for this should have its Color Space property set to Non-Color.

The Normal Map node must remain on its default property of Tangent Space as this is the only type of normal map currently supported by glTF. The strength of the normal map can be adjusted on this node. The exporter is not exporting these nodes directly, but will use them to locate the correct image and will copy the strength setting into the glTF.

A normal map image connected such that the exporter will find it and copy it to the glTF file.

Tip

The Cycles render engine has a Bake panel that can be used to bake tangent-space normal maps from almost any other arrangement of normal vector nodes. Switch the Bake type to Normal. Keep the default space settings (space: Tangent, R: +X, G: +Y, B: +Z) when using this bake panel for glTF. The resulting baked image can be saved and plugged into a new material using the Normal Map node as described above, allowing it to export correctly.

See: Cycles Render Baking

An Image Texture node can be connected to the Emission input on the Principled BSDF node to include an emissive map with the glTF material. Alternatively, the Image Texture node can be connected to an Emission shader node, and optionally combined with properties from a Principled BSDF node by way of an Add Shader node.

If the emissive map is alone in the material, it is best to set the Base Color default to black, and the Roughness default to 1.0. This minimizes the influence of the other channels if they are not needed.

This arrangement is supported for backwards compatibility. It is simpler to use the Principled BSDF node directly.

If any component of emissiveFactor is > 1.0,

KHR_materials_emissive_strength extension will be used.

When the Clearcoat input on the Principled BSDF node has a nonzero default value or Image Texture node connected, the KHR_materials_clearcoat glTF extension will be included in the export. This extension will also include a value or Image Texture from the Clearcoat Roughness input if available.

If Image Textures are used, glTF requires that the clearcoat values be written to the red (R) channel, and Clearcoat Roughness to the green (G) channel. If monochrome images are connected, the exporter will remap them to these color channels.

The Clearcoat Normal input accepts the same kinds of inputs as the base Normal input, specifically a tangent-space normal map with +Y up, and a user-defined strength. This input can reuse the same normal map that the base material is using, or can be assigned its own normal map, or can be left disconnected for a smooth coating.

All Image Texture nodes used for clearcoat shading should have their Color Space set to Non-Color.

An example of a complex clearcoat application that will export correctly to glTF. A much simpler, smooth coating can be applied from just the Principled BSDF node alone.

If a Sheen Roughness Texture is used, glTF requires the values be written to the alpha (A) channel.

Tip

Sheen BSDF node is only available on Cycles render engine. You may have to temporarily switch to Cycles to add this node, and get back to EEVEE. When the Specular IOR Level or Specular Tint input of Principled BSDF node have a non default value or Image Texture node connected, the KHR_materials_specular glTF extension will be included in the export. Anisotropic textures and data need to be converted at export, and at import.

At import, some nodes are created to manage this conversion

At export, this exact same nodes are detected, and used to export data.

At export, you can also plug some grayscale textures for Anisotropic and Anisotropic Rotation sockets. Then, exporter will convert these texture into a glTF compatible texture.

Note that the tangent socket must be linked to a tangent node, with UVMat. The chosen UVMat must be the UVMat of the Normal Map.

When the Transmission input on the Principled BSDF node has a non zero default value or Image Texture node connected, the

KHR_materials_transmission glTF extension will be included in the export.

When a texture is used, glTF stores the values in the red (R)

channel. The Color Space should be set to Non-Color.

Transmission is different from alpha blending, because transmission allows full-strength specular reflections. In glTF, alpha blending is intended to represent physical materials that are partially missing from the specified geometry, such as medical gauze wrap. Transmission is intended to represent physical materials that are solid but allow non-specularly-reflected light to transmit through the material, like glass.

Tip

The material's base roughness can be used to blur the transmission, like frosted glass.

Tip

Typically the alpha blend mode of a transmissive material should remain Opaque, the default setting, unless the material only partially covers the specified geometry.

Note

In real-time engines where transmission is supported, various technical limitations in the engine may determine which parts of the scene are visible through the transmissive surface. In particular, transmissive materials may not be visible behind other transmissive materials. These

limitations affect physically-based transmission, but not alpha-blended non-transmissive materials.

Note

If you want to enable refraction on your model, `KHR_materials_transmission` must also be used in addition with `KHR_materials_volume`. See the dedicated Volume part of the documentation.

Warning

Transmission is complex for real-time rendering engines to implement, and support for the `KHR_materials_transmission` glTF extension is not yet widespread.

At import, there are two different situations:

if `KHR_materials_ior` is not set, IOR value of Principled BSDF node is set to 1.5, that is the glTF default value of IOR.

If set, the `KHR_materials_ior` is used to set the IOR value of Principled BSDF.

At export, IOR is included in the export only if one of these extensions are also used:

- `KHR_materials_transmission`

- `KHR_materials_volume`

- `KHR_materials_specular`

IOR of 1.5 are not included in the export, because this is the default glTF IOR value.

Volume can be exported using a Volume Absorption node, linked to Volume socket of Output node. Data will be exported using the

`KHR_materials_volume` extension.

For volume to be exported, some transmission must be set on Principled BSDF node.

Color of Volume Absorption node is used as glTF attenuation color. No texture is allowed for this property.

Density of Volume Absorption node is used as inverse of glTF attenuation distance.

Thickness can be plugged into the Thickness socket of custom group node `glTFMaterialOutput`.

If a texture is used for thickness, it must be plugged on (G) Green channel of the image.

Note

For a full Variants experience, you have to enable UI in Add-on preferences

There are two locations to manage glTF Variants in Blender

In 3D View, on glTF Variants tab

For advanced settings, in Mesh Material Properties (see Advanced glTF Variant checks)

The main concept to understand for using Variants, is that each material slot will be used as equivalent of a glTF primitive.

After importing a glTF file including `KHR_materials_variants` extension, all variants can be displayed.

You can switch Variant, by selecting the variant you want to display, then clicking on Display Variant.

You can switch to default materials (when no Variant are used), by clicking on Reset to default.

You can add a new Variant by clicking the + at right of the Variant list. Then you can change the name by double-clicking.

After changing Materials in Material Slots, you can assign current materials to the active Variant using Assign to Variant. You can also set default materials using Assign as Original. These materials will be exported as default material in glTF. These are materials that will be displayed by any viewer that doesn't manage KHR_materials_variants extension.

If you want to check primitive by primitive, what are Variants used, you can go to Mesh Material Properties.

The glTF Material Variantstab refers to the active material Slot and Material used by this slot. You can see every Variants that are using this material for the given Slot/Primitive.

You can also assign material to Variants from this tab, but recommendation is to perform it from 3D View tab.

For materials where only the front faces will be visible, turn on Backface Culling in the Settings panel of an EEVEE material. When using other engines (Cycles, Workbench) you can temporarily switch to EEVEE to configure this setting, then switch back.

Leave this box unchecked for double-sided materials.

The inverse of this setting controls glTF's DoubleSided flag.

glTF has three alpha modes, depending on whether the alpha value is always 1, always 0 or 1, or can be between 0 and 1. The exporter determines the alpha mode automatically from the nodes connected to the Alpha socket.

In opaque mode, the material alpha is always 1.

In mask mode, the material alpha is always 0 or 1. This creates a cutout transparency, where there is a hard edge between opaque and transparent regions, and can be used for things like leaves or cloth with holes. To enable this mode, use a Math node to round the alpha value to either 0 or 1.

Rounding snaps alpha values that are 0.5 or greater up to 1, and ones below 0.5 down to 0. It is also possible to use a cutoff value different than 0.5 by using Math nodes to do $1 - (\alpha < \text{cutoff})$.

Mask mode is essentially the same as EEVEE's Alpha Clip blend mode, but is done with shader nodes so it works in other render engines.

Materials that use neither of these will use blend mode. Blend mode allows partially transparent surfaces that tint the objects seen through them, like layers of colored film. However, partial transparency is complex to render, and glTF viewers may show visual artifacts in non-trivial scenes that use blend mode.

To avoid artifacts, it may be a good idea to separate out parts of a model that can use opaque or mask mode, and use blend mode only on the parts where it is necessary, or to use only a single layer of transparent polygons in front of opaque objects.

Control over UV map selection and transformations is available by connecting a UV Map node and a Mapping node to any Image Texture node. Settings from the Mapping node are exported using a glTF extension named KHR_texture_transform. There is a mapping type selector across the top. Point is the recommended type for export. Texture and Vector are also supported. The supported offsets are:

Location- X and Y

Rotation- Z only

Scale- X and Y

For the Texturetype, ScaleX and Y must be equal (uniform scaling).

A deliberate choice of UV mapping. \xc2\xba6

Tip

These nodes are optional. Not all glTF readers support multiple UV maps or texture transforms.

Any Image Texture nodes may optionally be multiplied with a constant color or scalar. These will be written as factors in the glTF file, which are numbers that are multiplied with the specified image textures. These are not common.

Use Math node (multiply) for scalar factors. Use second value as factor

Use Mix node (color / multiply) for color factors. Set Factor to 1, and use Color2 (B) as factors

A single material may use all of the above at the same time, if desired.

This figure shows a typical node structure when several of the above options are applied at once:

A Principled BSDF material with an emissive texture. \xc2\xba6

UDIM is a way to store multiple textures in a single image file. The UDIM system is supported by Blender, but is not supported by glTF. When exporting a model that uses UDIM, the add-on will automatically split the image into multiple images, one for each tile, and will update the material nodes to use the new images. All UDIM texture must use the same UV map to be exported.

To export an unlit material, mix in a camera ray, and avoid using the Principled BSDF node.

One of several similar node arrangements that will export

KHR_materials_unlit and render shadeless in Blender. \xc2\xba6

The core glTF 2.0 format can be extended with extra information, using glTF extensions. This allows the file format to hold details that were not considered universal at the time of first publication. Not all glTF readers support all extensions, but some are fairly common.

Certain Blender features can only be exported to glTF via these extensions. The following glTF 2.0 extensions are supported directly by this add-on:

Import

KHR_materials_pbrSpecularGlossiness

KHR_materials_clearcoat

KHR_materials_transmission

KHR_materials_unlit

KHR_materials_emissive_strength

KHR_materials_volume

KHR_materials_sheen

KHR_materials_specular

KHR_materials_anisotropy

KHR_materials_ior

KHR_materials_variants

KHR_lights_punctual

KHR_texture_transform

KHR_mesh_quantization

EXT_mesh_gpu_instancing

Export

KHR_draco_mesh_compression

KHR_lights_punctual

KHR_materials_clearcoat

KHR_materials_transmission

KHR_materials_unlit
KHR_materials_emissive_strength
KHR_materials_volume
KHR_materials_sheen
KHR_materials_specular
KHR_materials_anisotropy
KHR_materials_ior
KHR_materials_variants
KHR_texture_transform
EXT_mesh_gpu_instancing

It is possible for Python developers to add Blender support for additional glTF extensions by writing their own third-party add-on, without modifying this glTF add-on. For more information, see the example on GitHub and if needed, register an extension prefix.

Custom properties are always imported, and will be exported from most objects if the Include Custom Properties option is selected before export. These are stored in the extras field on the corresponding object in the glTF file.

Unlike glTF extensions, custom properties (extras) have no defined namespace, and may be used for any user-specific or application-specific purposes.

A glTF animation changes the transforms of objects or pose bones, or the values of shape keys. One animation can affect multiple objects, and there can be multiple animations in a glTF file.

Imported models are set up so that the first animation in the file is playing automatically. Scrub the Timeline to see it play.

When the file contains multiple animations, the rest will be organized using the Nonlinear Animation editor. Each animation becomes an action stashed to an NLA track. The track name is the name of the glTF animation. To make the animation within that track visible, click Solo (star icon) next to the track you want to play.

This is the fox sample model showing its Run animation.

If an animation affects multiple objects, it will be broken up into multiple parts. The part of the animation that affects one object becomes an action stashed on that object. Use the track names to tell which actions are part of the same animation. To play the whole animation, you need to enable Solo (star icon) for all its tracks.

Note

There is currently no way to see the non-animated pose of a model that had animations.

You can also use the animation switcher that can be found in Dope Sheet editor.

Note

You have to enable UI in Add-on preferences for seeing the animation switcher

You can switch all animation imported. It automatically enables Solo (star icon) for all needed tracks. It also resets non-animated object to Rest transformation.

You can export animations using different ways. How glTF animations are made from actions / NLA is controlled by the Animation export option.

An action will be exported if it is the active action on an object, or it is stashed to an NLA track (e.g. with the Store or Push Down buttons in the

Action Editor). Actions which are not associated with an object in one of these ways are not exported. If you have multiple actions you want to export, make sure they are stashed!

A glTF animation can have a name, which is the action name by default. You can override it by renaming its NLA track from NLA Track / [ActionStash] to the name you want to use. For example, the Fig. fox model will export with three animations, \xe2\x80\x9cSurvey\xe2\x80\x9d, \xe2\x80\x9cWalk\xe2\x80\x9d, and \xe2\x80\x9cRun\xe2\x80\x9d. If you rename two tracks on two different objects to the same name, they will become part of the same glTF animation and will play together. The importer organizes actions so they will be exported correctly with this mode.

This mode is useful if you are exporting for game engine, with an animation library of a character. Each action must be on its own NLA track.

In this mode, the NLA organization is not used, and only one animation is exported using the active actions on all objects.

In this mode, each NLA Track will be exported as an independent glTF animation. This mode is useful if you are using Strip modifiers, or if you get multiple action on a same Track.

If you rename two tracks on two different objects to the same name, they will become part of the same glTF animation and will play together.

Using Scene option, animations will be exported as you can see them in viewport. You can choose to export a single glTF animation, or each object separately.

Note

Remember only certain types of animation are supported:

Object transform (location, rotation, scale)

Pose bones

Shape key values

Animation of other properties, like physics, lights, or materials, will be ignored.

Note

In order to sample shape key animations controlled by drivers using bone transformations, they must be on a mesh object that is a direct child of the bones \xe2\x80\x99 armature.

Note

Only Actions (default) and Active Actions merged mode can handle not sampled animations.

The glTF specification identifies different ways the data can be stored. The importer handles all of these ways. The exporter will ask the user to select one of the following forms:

This produces a single .glb file with all mesh data, image textures, and related information packed into a single binary file.

Tip

Using a single file makes it easy to share or copy the model to other systems and services.

This produces a JSON text-based .gltf file describing the overall structure, along with a .bin file containing mesh and vector data, and optionally a number of .png or .jpg files containing image textures referenced by the .gltf file.

Tip

Having an assortment of separate files makes it much easier for a user to go back and edit any JSON or images after the export has completed.

Note

Be aware that sharing this format requires sharing all of these separate files together as a group.

This produces a JSON text-based glTF file, with all mesh data and image data encoded (using Base64) within the file. This form is useful if the asset must be shared over a plain-text-only connection.

Warning

This is the least efficient of the available forms, and should only be used when required. Available only when you activated it in add-on preferences.

Pack all images into the blend-file.

The glTF format requires discontinuous normals, UVs, and other vertex attributes to be stored as separate vertices, as required for rendering on typical graphics hardware. This option attempts to combine co-located vertices where possible. Currently cannot combine verts with different normals.

How normals are computed during import.

Determines the pose for bones (and consequently, skinned meshes) in Edit Mode. When on, attempts to guess the pose that was used to compute the inverse bind matrices.

Changes the heuristic the importer uses to decide where to place bone tips. Note that the Fortune setting may cause inaccuracies in models that use non-uniform scaling. Otherwise this is purely aesthetic. The default value will not change axis, and is best for re-exporting from Blender. This default option will change display mode (adding shape and changing relationship line) to have a better view, even if original bones axis are not the most accurate (esthetically speaking)

Optional backwards compatibility for non-standard render engines. Applies to lights. Standard: Physically-based glTF lighting units (cd, lx, nt). Unitless: Non-physical, unitless lighting. Useful when exposure controls are not available. Raw (Deprecated): Blender lighting strengths with no conversion

If a texture exists in WebP format, loads the WebP texture instead of the fallback png/jpg one.

See: File Format Variations.

For glTF Separate file format only. Keep original textures files if possible. Warning: if you use more than one texture, where PBR standard requires only one, only one texture will be used. This can lead to unexpected results

For glTF Separate file format only. Folder to place texture files in. Relative to the glTF-file.

Legal rights and conditions for the model.

Store export settings in the blend-file, so they will be recalled next time the file is opened.

Export selected objects only.

Export visible objects only.

Export renderable objects only.

Export objects from active collection only.

Only when Active Collection is On. When On, export recursively objects on nested active collections.

Export active scene only.

Export custom properties as glTF extras.

Export cameras.

Export directional, point, and spot lights. Uses the KHR_lights_punctual glTF extension.

Export using glTF convention, +Y up.

Export Geometry nodes instances. This feature is experimental.

Export using EXT_mesh_gpu_instancing extensions.

Useful in case of non-decomposable TRS matrix. Only skinned meshes will stay children of armature.

Export collections as empty, keeping full hierarchy. If an object is in multiple collections, it will be exported only once, in the first collection it is found.

Export objects using the evaluated mesh, meaning the resulting mesh after all Modifiers have been calculated.

Export UVs (texture coordinates) with meshes.

Export vertex normals with meshes.

Export vertex tangents with meshes.

Export Attributes with meshes, when the name starts with underscore.

Export loose edges as lines, using the material from the first material slot.

Export loose points as glTF points, using the material from the first material slot.

For triangles, use shared accessor for indices. This is more efficient (smaller files when you have lots of materials).

Export vertex color when used in material node tree as Base Color multiplier. This is the default, and the most accurate regarding glTF specification.

Export active vertex colors, even if not used in material node tree. A fully compliant glTF viewer should display this VC as Base Color multiplier.

Do not export vertex color.

Export all vertex colors, additional VC will be COLOR_1, COLOR_2, etc.

Export active vertex color when no material is assigned to the object.

Export full materials, only placeholders (all primitives but without materials), or does not export materials. (In that last case, primitives are merged, losing material slot information).

Output format for images. PNG is lossless and generally preferred, but JPEG might be preferable for web applications due to the smaller file size. If WebP is chosen, all textures will be saved as WebP, without any png/jpg fallback. If None is chosen, materials are exported without textures.

When exporting jpeg or WebP files, the quality of the exported file. Creates WebP textures for every textures, in addition to the existing texture. For already WebP textures, nothing happens.

For all WebP textures, create a png fallback texture.

Export images that are not used in any material.

Export texture info (sampler, image, texcoord) that are not used in any material.

Export shape keys (morph targets).

Export vertex normals with shape keys (morph targets).

Export vertex tangents with shape keys (morph targets).

Sparse Accessor will be used if it saves space (if the exported file is smaller)

If data is empty, omit to export Sparse Accessor. Not all viewers manage it correctly, so this option is Off by default

Export Armatures using rest position as joint rest pose. When Off, the current frame pose is used as rest pose.

Export Deformation bones only, not other bones. Animation for deformation bones are baked.

Remove Armature Objects if possible. If some armature(s) have multiple root bones, we can't remove them.

Useful in case of non-decomposable TRS matrix.

Export skinning data

How many joint vertex influences will be exported. Models may appear incorrectly in many viewers with value different to 4 or 8.

Export all joint vertex influences. Models may appear incorrectly in many viewers.

Optional backwards compatibility for non-standard render engines. Applies to lights. Standard: Physically-based glTF lighting units (cd, lx, nt). Unitless: Non-physical, unitless lighting. Useful when exposure controls are not available Raw (Deprecated): Blender lighting strengths with no conversion

Compress meshes using Google Draco.

Higher compression results in slower encoding and decoding.

Higher values result in better compression rates.

Higher values result in better compression rates.

Higher values result in better compression rates.

Higher values result in better compression rates.

Higher values result in better compression rates.

Animation mode used for export (See Animations)

Export Shape Keys Animation. Need Shape Keys to be exported (See Data - Shape Keys)

Useful when some objects are constrained without being animated themselves.

Export the scene in the current animation frame. When off, frame 0 is used as rest transformation for objects.

Clips animations to selected playback range.

Set all glTF Animation starting at 0. Can be useful for looping animation

When some frames are in negative range, slide or crop the animation.

Export all actions, bound to a single armature. Warning: Option does not support exports including multiple armatures.

Reset pose bones between each action exported. This is needed when some bones are not keyed on some animations.

Apply sampling to all animations. Do not sample animation can lead to wrong animation export.

How often to evaluate animated values (in frames).

Reduce exported file size by removing duplicate keyframes when all identical.

if all keyframes are identical in a rig, force keeping the minimal animation.

if all keyframes are identical for object transformations, force keeping the minimal animation.

When exporting animations, disable viewport for other objects, for performance reasons, when possible.

Restrict actions to be exported to the ones matching the filter.

This importer/exporter is developed through the glTF-Blender-IO repository, where you can file bug reports, submit feature requests, or contribute code.

Discussion and development of the glTF 2.0 format itself takes place on the Khronos Group glTF GitHub repository, and feedback there is welcome.
--- Fin du chapitre: _scene_gltf2 ---

--- Début du chapitre: _scene_settings --- Defines the resolution of every light probe sphere in the scene.
--- Fin du chapitre: _scene_settings ---

--- Début du chapitre: _scene_time --- The Scene Time node outputs the current time in the scene's animation in units of seconds or frames.
This node has no inputs.
This node has no properties.
Current scene time in seconds.
Current scene frame. As an input in geometry nodes, this output may also output non-round numbers, in order to support higher quality motion blur .
--- Fin du chapitre: _scene_time ---

--- Début du chapitre: _scrape --- Reference
Sculpt Mode
Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3
Advanced \xe2\x80\xa3 Brush Type
Similar to the Flatten brush, but only pushes surfaces downward to the medium height.
Although Ctrl can be held to invert the effect to a Fill brush, if Invert to Fill is enabled. When disabled, the inverted direction will push surfaces away.
Note
More info at General brush settings and on Advanced brush settings.
When enabled, holding Ctrl while sculpting changes the brush behavior to be the same as the Fill brush. When disabled, holding Ctrl while sculpting will push vertices above the cursor up away from the cursor.
--- Fin du chapitre: _scrape ---

--- Début du chapitre: _screw --- The Screw modifier is similar to the Screw tool in the Toolbar, in that it takes a profile object, a mesh or a curve, to create a helix-like shape.
Properly aligning the profile object is important. \xc2\xb6
The profile should be properly aligned to the cardinal direction of the object rather than to the screw axis.
The Screw modifier. \xc2\xb6
Degrees for a single helix revolution.
The height of one helix iteration.
Number of revolutions.
The axis along which the helix will be built.
The name of an object to define the axis direction.
Use the distance from the Axis Object to define the height of one helix iteration.
Number of steps used for a single revolution displayed in the 3D Viewport.
As above, but used during render time. Increase to improve quality.
Merge vertices that lie on the axis of rotation. Use this to close off end points with a triangle fan.
Vertices under this distance to the axis are merged.

Stretch the UV coordinates from (0.0 to 1.0) when UVs are present.

Output faces with smooth shading rather than flat shading. The smooth/flat shading of the input geometry is not preserved.

Order of edges is calculated to avoid problems with normals and shading. Only needed for meshes, not curves.

Flip normals direction.

--- Fin du chapitre: _screw ---

--- Début du chapitre: _sculpt --- This page details the general hotkey operators and menu operators in sculpt mode.

Change the position of the object.

Change the orientation of the mesh.

Increase/decrease the size of the mesh.

Morph the mesh to a spherical shape.

See also

Transform Tools.

Reference

Sculpt Mode

Sculpt

Some very common hotkey operators to control the visibility based on face sets. These are not part of any menu and have to be used via the shortcuts. More visibility operators can be found in the Face Sets Menu and the Pie Menu shortcut W. (Since visibility is often toggled via face sets.)

Draw a box to hide faces of a mesh.

Draw a box to reveal hidden faces. This works similar to the Box Select tool.

Hide all face sets except the active one (under the cursor). If face sets are already hidden, then this operator will show everything.

Hide the face set under the cursor. Press Shift- H afterwards to show everything.

Reveal all hidden geometry.

Hides all visible geometry and makes all hidden geometry visible.

Hides all masked vertices.

Grows or shrinks the visible area of the mesh along its surface.

See also

For a more general introduction see Visibility, Masking & Face Sets.

These operators smooths geometry patches based of a Face set .

See also

Edit Face Set Tool

Creates a perfectly flat and smooth geometry patch from the face set. This is the ideal way to trim parts of your mesh if the vertex count is too high for other operations, or the vertex IDs must not be altered (Like when using Multires sculpting).

Creates a smooth as possible geometry patch from the face set by minimizing changes in vertex tangents . This is ideal for creating smooth curved surfaces on complex topology, where just using the smooth brush will not lead to desired results

The trimming operators add or remove geometry from the mesh based on a gesture input. These operators are especially useful for sketching an early base mesh for further sculpting with the voxel remesher .

Flattens the geometry along a plane determined by the camera view and a drawn line. The region of the mesh being flattened is visualized by the side of the line that is shaded.

Removes geometry based on a box selection .
Removes geometry based on a lasso selection .
Adds geometry based on a box selection .
Adds geometry based on a lasso selection .
Applies a deformation to all vertices in the mesh at the same time.Masking, auto-masking and visibility will be taken into account.
To use these operators, click and drag away from left to right or from right to left for a negative effect.

See also

Mesh Filter Tool

Smooths the positions of the vertices to either polish surfaces or remove volume from larger shapes.Especially useful to fix most of the artifacts of the voxel remesher.This filter works similar to the Smooth brush.

Eliminates irregularities of the mesh by making the positions of the vertices more uniform while preserving the volume of the object.This filter works similar to the Surface deformation type of the Smooth brush.
Displaces vertices uniformly along their normal.This filter works similar to the Inflate brush.

Tries to create an even distribution of quads without deforming the volume of the mesh.This filter works the same as holding Shift with the Slide Relax brush.

This will remove the jagged lines visible after drawing or creating a face set.This filter works the same as holding Shift with the Draw Face Set brush.

Sharpens and smooths the mesh based on its curvature,resulting in pinching hard edges and polishing flat surfaces.Especially useful when sculpting hard surfaces and stylized models with creasing and flattening brushes.

Increases the high frequency surface details of the mesh by intensifying the difference between creases and valleys.This filter works similar to the inverted direction of the Smooth brush.

Deletes displacement information of the Multires Modifier ,resetting the mesh to a regular subdivision surface result.This can be used to reset parts of the sculpt or to fix reprojection artifacts after applying a Shrinkwrap Modifier .

Negative strokes will intensify the displacement details,this method works similar to Enhance Details and can give better results in some circumstances.

Randomly moves vertices along the vertex normal.This filter works similar to the Randomize Transform .

Reference

Sculpt Mode

Sculpt \xe2\x80\xa3 Sample Color

Shift- X

Adjust the brush color of the Paint tool to the color under the mouse cursor.

Reference

Sculpt Mode

Sculpt \xe2\x80\xa3 Set Pivot

Like Object and Edit Mode, Sculpt Mode also has a Pivot Point .This is because the basic move, rotate and scale transforms are also supported in Sculpt Mode.But the pivot point in Sculpt Mode is unique. It always moves together with the transformed mesh and can be both manually & automatically placed.

Sets the pivot to the origin of the sculpt.
Sets the pivot position to the average position of the unmasked vertices.
Sets the pivot position to the center of the mask's border. This operation will automatically happen when using Expand .
Sets the pivot position to the active vertex position.
Sets the pivot position to the surface under the cursor.

Tip

For more convenient placement of the pivot point it's recommended to use the shortcut assigned to Surface.

See also

For a more general introduction see Transforming.

Reference

Sculpt Mode

Sculpt > Rebuild BVH

Recalculates the BVH used by Dyntopo to improve performance, which might degrade over time while using Dyntopo.

See also

For a more general introduction see Adaptive Resolution.

Toggles Dyntopo .

Reference

Sculpt Mode

Sculpt > Transfer Sculpt Mode

Alt- Q

Switches Sculpt Mode from the Active object to the object under the mouse. See Switching Objects for more information.

See also

For a more general introduction see Working with Multiple Objects.

--- Fin du chapitre: _sculpt ---

--- Début du chapitre: _scurve --- The curve type used for creating mask splines is almost a Bézier curve, but with some differences. Smooth edges of the mask are defined by feathering. The curve needed to support feathering in a way that stuck to the curve as you edited it, for ease of editing an animation. These are called S-curves.

Besides the handles, every control point also has points that define the feather between the current point and the next point on the spline. Each feather point is stored in UW space, where U means position across spline segment, and W (weight) means distance between main spline and feather points.

S-Curve explained.

This allows for deforming the main spline in almost any way, and the feather will be updated automatically to reflect that change.

For example if there is just rotation of the spline, feather would stay completely unchanged. If one point's feather is moved, the other feathers will be automatically stretched uniformly along that segment and the overall shape will be almost the same as artists would want it to be.

Reference

Mask Mode

Add

Shift- A

There are two primitives available: a Bézier Circle and a Square with vector handles.

--- Fin du chapitre: _scurve ---

--- Début du chapitre: _seams --- For many cases, using the Unwrap calculations of Cube, Cylinder, Sphere, or the regular \xe2\x80\x9cUnwrap\xe2\x80\x9d operators will produce a good UV layout. But for more complex meshes, especially those with lots of indentations, you may want to define a seam to limit and guide the Unwrap operator.

Just like in sewing, a seam is where the ends of the image/cloth are sewn together. In unwrapping, the UV map is discontinuous at the seams. Think of this method as peeling an orange or skinning an animal. You make a series of cuts in the skin, then peel it off. You could then flatten it out, applying some amount of stretching. These cuts are the same as seams.

Simple seam on a cylinder. \xc2\xbb6

When using this method, you need to be aware of how much stretching there is. The more seams there are, the less stretching there is, but this is often an issue for the texturing process. It is a good idea to have as few seams as possible while having the least amount of stretching. Try to hide seams where they will not be seen. In productions where 3D paint is used, this becomes less of an issue, as projection painting can easily deal with seams, as opposed to 2D texturing, where it is difficult to match the edges of different UV islands.

The workflow is the following:

Mark seams.

Unwrap.

Adjust seams and repeat.

Manually adjust UVs.

Reference

3D Viewport

Edit Mode

UV \xe2\x80\xa3 Mark/Clear Seam

Edge \xe2\x80\xa3 Mark/Clear Seam

Reference

UV Editor

View mode

UV \xe2\x80\xa3 Mark/Clear Seam

Seamed Suzanne. \xc2\xbb6

To add an edge to a seam, simply select the edge and press Ctrl- E to Mark Seam, or to remove it, use Ctrl- E to Clear Seam.

In the example to the right, the back-most edge of the cylinder was selected as the seam (to hide the seam), and the default unwrap calculation was used. In the UV Editor, you can see that all the faces are nicely unwrapped, just as if you cut the seam with a scissors and spread out the fabric.

When marking seams, you can use Select Linked in Face Select Mode to check your work. This menu option selects all faces connected to the selected one, up to a seam. If faces outside your intended seam are selected, you know that your seam is not continuous. You do not need continuous seams, however, as long as they resolve regions that may stretch.

Just as there are many ways to skin a cat, there are many ways to go about deciding where seams should go. In general though, you should think as if you were holding the object in one hand, and a pair of sharp scissors in the other, and you want to cut it apart and spread it on the

table with as little tearing as possible. Note that we seamed the outside edges of her ears, to separate the front from the back. Her eyes are disconnected sub-meshes, so they are automatically unwrapped by themselves. A seam runs along the back of her head vertically, so that each side of her head is flattened out.

Another use for seams is to limit the faces unwrapped. For example, when texturing a head, you do not really need to texture the scalp on the top and back of the head since it will be covered in hair. So define a seam at the hairline. Then, when you select a frontal face, and then select linked faces before unwrapping, the select will only go up to the hairline seam, and the scalp will not be unwrapped.

When unwrapping anything that is bilateral, like a head or a body, seam it along the mirror axis. For example, cleave a head or a whole body right down the middle in front view. When you unwrap, you will be able to overlay both halves onto the same Texture Space, so that the image pixels for the right hand will be shared with the left; the right side of the face will match the left, etc.

Note

You do not have to come up with a cone unwrapping that works perfectly for everything everywhere. As we will discuss later, you can easily have multiple UV unwrappings, using different approaches in different areas of your mesh.

Reference

View mode

UV Seams from Islands

Adds seams at the boundaries of existing UV islands. This is useful when modifying the UVs of already unwrapped meshes.

--- Fin du chapitre: _seams ---

--- Début du chapitre: _security --- The ability to include Python scripts within blend-files is valuable for advanced tasks such as rigging and automation. However, it poses a security risk since Python does not restrict what a script can do. Therefore, you should only run scripts from sources you know and trust. Automatic execution is disabled by default; however, some blend-files need this to function properly.

When a blend-file tries to execute a script and is not allowed, a dialog will appear. In it, you can choose to Allow Execution or to Ignore the scripts.

An Auto-run warning in the Info editor's header.

Here are the different ways blend-files may automatically run scripts.

A text data-block can have its Register option enabled which means it will load on start.

Python expressions can be used to drive values and are often used in more advanced rigs and animations.

There are other ways scripts in a blend-file may execute that require user interaction (therefore will run even when auto execution is off), but you should be aware that this is the case since it is not necessarily obvious.

Running a script in the Text editor.

Rendering with Freestyle, because Freestyle uses scripts to control line styles.

Blender provides a number of ways to control whether scripts from a blend-file are allowed to automatically execute.

First, the File Browser has the option Trusted Source which you can use on a case-by-case basis to control auto execution. Since you may forget to set this, or may open a file without going through the File Browser, you can change the default (described next).

In the Preferences, there is the toggle to Auto Run Python Scripts. This means the Trusted Source option in the File Browser will be enabled by default, and scripts can run when blend-files are loaded without using the File Browser. Once enabled you have the option to exclude certain directories; a typical configuration would be to trust all paths except for the download directory.

The Auto Run Python Scripts checkbox.

You may want to perform batch rendering or some other task from the command line, running Blender without an interface. In this case, the Preferences are still used but you may want to override them:

Enable with `-y` or `--enable-autoexec`

Disable with `-Y` or `--disable-autoexec`

To render an animation in background mode, allowing drivers and other scripts to run:

Note

These command-line arguments can be used to start a regular Blender instance and will still override the Preferences.

--- Fin du chapitre: `_security` ---

--- Début du chapitre: `_segments` --- Reference

Edit Mode

Surface `\xe2\x80\xa3 Segments \xe2\x80\xa3 Subdivide`

The Subdivision operator simply subdivides all selected segments by adding one or more control points between the selected segments. Selected grids will be split into four smaller ones.

If used on a 1D surface (a `\xe2\x80\x93 surface curve \xe2\x80\x9d`), this tool works exactly as with curves.

The number of subdivisions to perform.

Reference

Edit Mode

Surface `\xe2\x80\xa3 Segments \xe2\x80\xa3 Switch Direction`

This tool will `\xe2\x80\x93 reverse \xe2\x80\x9d` the direction of any curve with at least one selected element (i.e. the start point will become the end one, and vice versa). Mainly useful when using a curve as path, or the bevel and taper options `\xe2\x80\xa6`

--- Fin du chapitre: `_segments` ---

--- Début du chapitre: `_selecting` --- The active sequence strip is displayed with a light outline. The entire strip could be selected by clicking LMB in the middle of the strip.

The Select menu lets you select strips in different ways.

Selects all the strips in the timeline.

Deselects all the strips in the timeline.

Inverts the current selection.

See Box Select.

Works like Box Select, but also selects any strip handles inside the box.

If a strip has only one handle selected, dragging it will change the strip's length. (If both handles are selected, the complete strip moves instead.)

Select strips that are similar to the active strip. By default, unsimilar strips are deselected, but this can be changed in the Adjust Last Operation region.

Select strips that have the same specific type as the active strip. For example, if the active strip is a Movie strip, this selects all Movie strips.

Select strips that have the same general type (graphics or audio) as the active strip.

If the active strip is an effect strip, selects all effect strips.

Otherwise, selects all non-effect strips. (Despite the name, this operator does not check the effect type.)

Select strips that use the same source (file, scene, movie clip or mask) as the active strip.

Find the effect types that are applied to the active strip, and select all strips that have any of the same effect types applied to them. For example, if the active strip has a Gaussian Blur effect on it, this will select all other strips that are also blurred.

Select strips that are on a lower channel than a selected strip and overlap it in time; then, effect strips linked to the selected content strips; and finally, content strips linked to the selected effect strips. Select strips that partially or completely overlap the active strip in time.

Add/remove neighboring strips to/from the selection.

Select the strips that lie completely to the left or right of the current frame.

Select the strips that intersect the current frame.

Select the left, right, or both handles of the selected strips.

Select the handle of the neighboring strip to the left, right, or on both sides of the selected strips.

Select all the strips that are in the same channels as the currently selected strips.

--- Fin du chapitre: `_selecting` ---

--- Début du chapitre: `_selection` --- The Selection node outputs true for geometry that is selected, and false elsewhere.

The corresponding data flow node is the Set Selection Node.

Note

This node can only be used in the Tool context .

This node has no inputs.

This node has no properties.

Boolean field set to true for geometry that is selected in edit mode.

--- Fin du chapitre: `_selection` ---

--- Début du chapitre: `_selection_paint` --- Paint curves or control points to use as masks for the other tools. The selection visibility can be controlled by the Selection Opacity option in the Viewport Overlays.

By default the selection sets a new selection. The selection paint can be extended by holding Shift and it can be subtracted by holding Ctrl while painting.

Determines whether to set a new selection or remove from it. It can be toggled holding Ctrl while painting.

--- Fin du chapitre: `_selection_paint` ---

--- Début du chapitre: `_selection_sets` --- Reference

Pose Mode

Armature \xe2\x80\xa3 Selection Sets

Selection Sets are a feature that allows the definition of sets of bones for easy selection while animating. The sets can be created in local and linked armature overrides.

A List View listing all selection sets for the selected armature. Here, selection sets can be renamed by double clicking on the name.

To the right of the name is a check box to include that selection set when copying to the clipboard.

Specials

Removes all selection sets from the list.

Removes the selected bones from all selection sets.

Copies the selected set to Blender\xe2\x80\x99s clipboard.

Pastes a selection set from Blender\xe2\x80\x99s clipboard.

Assigns the selected bones to the active selection set.

Removes the selected bones to the active selection set.

Selects all the bones in the active selection set.

Deselects all the bones in the active selection set.

--- Fin du chapitre: _selection_sets ---

--- Début du chapitre: _selection_visibility --- If you have a complex mesh, it is sometimes not easy to paint on the intended vertices. Suppose you only want to paint on a small area of the Mesh and keep the rest untouched. This is where \xe2\x80\x93selection masking\xe2\x80\x93 comes into play. When this mode is enabled, a brush will only paint on the selected vertices or faces. The option is available from the header of the 3D Viewport (see icons surrounded by the yellow frame):

You can choose between Face Selection masking (left button), Vertex selection masking (middle button), and Bone selection (right button). The latter is only available when the mesh has an Armature modifier.

\xc2\xb6

Selection masking has some advantages over the default paint mode:

The original mesh edges are shown, even when modifiers are active.

You can select and deselect faces instead without the need to switch to Edit Mode.

The following standard selection operations are supported:

Alt- LMB\xe2\x80\x93 Single faces

Shift- Alt- LMB\xe2\x80\x93 Select more or remove them from the selection.

A\xe2\x80\x93 All faces, A to deselect.

B\xe2\x80\x93 Box selection.

C\xe2\x80\x93 Circle select with brush.

Ctrl- I\xe2\x80\x93 Invert selection.

L\xe2\x80\x93 Pick linked (under the mouse cursor).

Ctrl- L\xe2\x80\x93 Select linked.

Ctrl- Numpad Plus\xe2\x80\x93 Extend Selection

Ctrl- Numpad Minus\xe2\x80\x93 Shrink Selection

The following only work for face selection and with the selection tool active:

Alt- LMB\xe2\x80\x93 Loop Select

Reference

Vertex and Weight Paint Modes

Vertex Selection

In this mode you can select one or more vertices and then paint only on the selection. All unselected vertices are protected from unintentional changes.

Vertex Selection masking.\xc2\xba

Reference

Texture, Vertex, and Weight Paint Modes

Paint Mask

The Face Selection masking allows you to select faces and limit the paint tool to those faces, very similar to Vertex selection masking.

Face Selection masking.\xc2\xba

Hidden faces.\xc2\xba

You also can hide selected faces as in Edit Mode with the keyboard Shortcut H, then paint on the remaining visible faces and finally unhide the hidden faces again by using Alt- H.

You cannot specifically hide only selected faces in vertex mask selection mode. However, the selection is converted when switching selection modes. So a common trick is to:

Switch to Face selection mask mode to have the selection converted to faces.

Refine your selection next or just hide the faces.

Switch back to Vertex Selection mask mode.

Hide faces will make sure that vertices that belong to visible faces remain visible.

To constrain the paint area further you can use the Clipping Region. Press Alt- B, drag a rectangular area. The selected area will be cut out as the area of interest. The rest of the 3D Viewport gets hidden.

The Clipping Region is used to select interesting parts for local painting.\xc2\xba

You make the entire mesh visible again by pressing Alt- B a second time.

All paint tools that use the view respect this clipping, including box select, and of course brush strokes.

There are two helpful reminders that a Clipping Region is used:

The clipping region is drawn as a grey box in the 3D Viewport

The Text Info overlay will state that the perspective is
\xe2\x80\x9cClipped\xe2\x80\x9d

Reference

Edit Mode

Select \xe2\x80\x93 Select Linked \xe2\x80\x93 Linked

Ctrl- L, Shift- L

Select geometry connected to already selected elements. This is often useful when a mesh has disconnected, overlapping parts, where isolating it any other way would be tedious. Pressing Shift- L will deselect linked any linked elements.

With L you can also select connected geometry directly under the cursor.

--- Fin du chapitre: _selection_visibility ---

--- Début du chapitre: _self_collision ---Reference

Physics \xe2\x80\x93 Soft Body \xe2\x80\x93 Self Collision

Note

Self-Collision is working only if you have activated Use Edges.

When enabled, allows you to control how Blender will prevent the soft body from intersecting with itself. Every vertex is surrounded with an elastic virtual ball. Vertices may not penetrate the balls of other

vertices. If you want a good result you may have to adjust the size of these balls. Normally it works pretty well with the default options.

The Ball Size directly sets the ball size.

The average length of all edges attached to the vertex is calculated and then multiplied with the Ball Size setting. Works well with evenly distributed vertices.

The ball size is as large as the smallest/largest spring length of the vertex multiplied with the Ball Size.

Size = ((Min + Max)/2) * Ball Size.

Fraction of the length of attached edges. The edge length is computed based on the chosen algorithm. This setting is the factor that is multiplied by the spring length. It is a spherical distance (radius) within which, if another vertex of the same mesh enters, the vertex starts to deflect in order to avoid a self-collision.

Set this value to the fractional distance between vertices that you want them to have their own space. Too high of a value will include too many vertices at all times and slow down the calculation. Too low of a level will let other vertices get too close and thus possibly intersect because there will not be enough time to slow them down.

How elastic that ball of personal space is. A high stiffness means that the vertex reacts immediately to another vertex enters their space.

How the vertex reacts. A low value just slows down the vertex as it gets too close. A high value repulses it.

Note

Collisions with other objects are set in the (other) Collision panel. To collide with another object they have to share at least one common layer.
--- Fin du chapitre: _self_collision ---

--- Début du chapitre: _self_object --- The Self Object node outputs the object that contains the geometry nodes modifier currently being executed. This can be used to retrieve the original transforms.

When evaluated in the Tool context, this node returns the Active object.

Note

The geometry cannot be retrieved from this object with the Object Info Node, since its final geometry is still being evaluated.

This node has no inputs.

This node has no properties.

The object currently being evaluated.

--- Fin du chapitre: _self_object ---

--- Début du chapitre: _separate --- Reference

Edit Mode

Mesh > Separate

P

At some point, you will come to a time when you need to cut parts away from a mesh to be separate.

To separate an object, the vertices (or faces) must be selected and then separated, though there are several different ways to do this.

Suzanne dissected neatly.

Separates the selected elements.

Separates fragments based on the materials assigned to the different faces.

Creates one object for every independent (disconnected) fragment of the original mesh.

See also

Joining objects .

--- Fin du chapitre: _separate ---

--- Début du chapitre: _separate_bones ---Reference

Edit Mode

Armature \xe2\x80\xa3 Separate Bones

P

You can, as with meshes, separate the selected bones in a new armature object Armature \xe2\x80\xa3 Separate, Ctrl- Alt- P and of course, in Object Mode, you can join all selected armatures in one Object \xe2\x80\xa3 Join Objects, Ctrl- J.

--- Fin du chapitre: _separate_bones ---

--- Début du chapitre: _separate_color --- Splits an image into its channels, based on a particular Color Model.

Standard color input.

The color model to output.

Red, Green, Blue.

Hue, Saturation, Value.

Hue, Saturation, Lightness.

The outputs of this node depend on the Mode property (see above).

The opacity value.

--- Fin du chapitre: _separate_color ---

--- Début du chapitre: _separate_components --- The Separate Components node splits a geometry into a separate output for each type of data in the geometry.

Standard geometry input.

This node has no properties.

Mesh component of the input geometry.

Curve component of the input geometry.

Point cloud component of the input geometry.

Volume component of the input geometry.

Instances component of the input geometry. Even if the instances contain geometry data with one of the other types, all instances will be added to this output. A Realize Instances Node can be added to move the data from geometry instances to their corresponding outputs.

--- Fin du chapitre: _separate_components ---

--- Début du chapitre: _separate_geometry --- The Separate Geometry node produces two geometry outputs. Based on the Selection input, the input geometry is split between the two outputs.

Tip

This node can be combined with the Compare Node for a more precise control of which parts are separated to a given output geometry.

Standard Geometry input.

Boolean field used to calculate which output each part of the geometry will go to. Parts in the selection will move to the Selection output. Parts not in the selection will move to the Inverted output.

The domain on which the selection field is evaluated.

The selection is on the points, control points, and vertices of the geometry.

The selection is on the edges of the mesh component. The other components are not modified.

The selection is on the faces of the mesh component. The other components are not modified.

The selection is on the splines in the curve component. For each spline, it is either entirely in the selection or not at all. The other components are not modified.

Note

When selecting a domain that doesn't modify all components, the unmodified components will appear in both outputs.

The parts of the geometry in the selection.

The parts of the geometry not in the selection.

--- Fin du chapitre: _separate_geometry ---

--- Début du chapitre: _separate_matrix --- The Separate Matrix node splits a 4x4 matrix into its individual values.

The matrix to split into individual values.

This node has no properties.

The outputs of this node are split into panels for each column of the matrix. Each panel, has four value outputs for the four rows of the matrix.

--- Fin du chapitre: _separate_matrix ---

--- Début du chapitre: _separate_transform --- The Separate Transform node separates a Transformation Matrix into a translation vector, a rotation vector, and a scale vector.

The transformation matrix to separate.

This node has no properties.

The translation vector.

The rotation vector.

The scale vector.

--- Fin du chapitre: _separate_transform ---

--- Début du chapitre: _separate_xyz --- The Separate XYZ Node splits a vector into its individual components.

Standard vector input.

This node has no properties.

X

Y

Z

--- Fin du chapitre: _separate_xyz ---

--- Début du chapitre: _sequencer_preview --- This view type shows both the preview and the sequencer inside one editor.

Figure 1: Combined Sequencer & Preview

In general, it's better to avoid this view type and instead have two editors, one serving as the Preview and the other as the Sequencer. Reasons for this include:

Most of the Preview tools, such as Move and Rotate, are not available in Sequencer & Preview.

You can't add a small editor (such as a File Browser) on the side that only takes up the height of the preview.

You can maximize the preview on another screen.

One way of getting two separate editors is to simply open the default Video Editing workspace. (You may need to click the icon to the right of the tabs to find it.)

--- Fin du chapitre: `_sequencer_preview` ---

--- Début du chapitre: `_settings` ---Reference

Material Settings

While shading nodes control the appearance, these settings control the quality and algorithms that each renderer uses to render the material.

EEVEE specific settings

Cycles specific settings

Index number for the Material Index render pass. This can be used to give a mask to a material and then be read with the ID Mask Node in the Compositor.

Note

Volume Objects are not supported.

These settings control the 3D Viewport display in solid shading. They provide a faster alternative to full shader nodes, which may be too heavy or distracting for tasks like modeling, layout or sculpting.

Diffuse or metal surface color.

Blends between a non-metallic and metallic material model. A value of 1.0 gives a fully specular reflection tinted with the base color, without diffuse reflection or transmission. At 0.0 the material consists of a diffuse or transmissive base layer, with a specular reflection layer on top.

Specifies microfacet roughness of the surface for metal and specular reflection.

--- Fin du chapitre: `_settings` ---

--- Début du chapitre: `_set_alpha` --- The Set Alpha Node adds an alpha channel to an image.

Standard color input.

The amount of Alpha can be set for the whole image by using the input field or per pixel by connecting to the socket.

Method to mix the alpha with the input Image.

Multiply the input images RGBA channels by the Alpha input value. In this cases, the output is using Premultiplied Alpha.

Replace the inputs alpha channel with the Alpha input value. In this cases, the output is using Straight Alpha.

Standard color output.

Note

This is not, and is not intended to be, a general-purpose solution to the problem of compositing an image that does not contain alpha information. You might wish to use chroma keying or difference keying (as discussed elsewhere) if you can. This node is most often used (with a suitable input being provided by means of the socket) in those troublesome cases when you cannot, for some reason, use those techniques directly.

To transition the audience from one scene or shot to another, a common technique is to fade to black. As its name implies, the scene fades to a black screen. You can also fade to white or whatever color you wish, but black is a good neutral color that is easy on the eyes and intellectually

\xe2\x80\x9cresets\xe2\x80\x9d the viewer\xe2\x80\x99s mind. The node tree below shows how to do this using the Set Alpha node.

Fade to black.\xc2\xbf

In the example above, the alpha channel of the swirl image is ignored. Instead, a Time Curve node introduces a factor from 0.0 to 1.0 over 60 frames, or about 2 seconds, to the Set Alpha node. Note that the time curve is exponentially-shaped, so that the overall blackness will fade in slowly and then accelerate toward the end. The Set Alpha node does not need an input image; instead, the flat (shadeless) black color is used. The Set Alpha Node uses the input factor and color to create a black image that has an alpha set which goes from 0.0 to 1.0 over 60 frames, or completely transparent to completely opaque. Think of alpha as a multiplier for how vivid you can see that pixel. These two images are combined by the Alpha Over node completely (a Factor of 1.0) to produce the composite image. The Set Alpha node will thus, depending on the frame being rendered, produce a black image that has some amount of transparency. Setup and animate, and you have an image sequence that fades to black over a two-second period.

Note

No Scene Information Used

This example node tree does not use the Render Layer node. To produce this 2-second animation, no Blender scene information was used. This is an example of using Blender\xe2\x80\x99s powerful compositing abilities separate from its modeling and animation capabilities. (A Render Layer could be substituted for the Image layer, and the \xe2\x80\x9cfade-network\xe2\x80\x9d effect will still produce the same effect.)

To introduce your animation, you will want to present the title of your animation over a background. You can have the title fly in, or fade it in. To fade it in, use the Set Alpha node with the Time Curve node as shown below.

Using Set Alpha to fade in a title.\xc2\xbf

In the above example, a Time curve provides the Alpha value to the input socket. The current Render Layer node, which has the title in view, provides the image. As before, the Alpha Over node mixes (using the alpha values) the background swirl and the alpha title to produce the composite image.

Using Set Alpha to colorize an image.\xc2\xbf

In the example above, notice how the blue tinge of the render input colors the swirl. You can use the Set Alpha node\xe2\x80\x99s color field with this kind of node tree to add a consistent color to a BW image.

In the example tree to the right, use the Alpha value of the Set Alpha node to give a desired degree of colorization. Thread the input image and the Set Alpha node into an Alpha Over node to colorize any black-and-white image in this manner.

--- Fin du chapitre: _set_alpha ---

--- D but du chapitre: _set_attribute ---Reference

Edit Mode

Mesh \xe2\x80\xa3 Set Attribute

Sets the value of the Active Attribute for the selected element.\xe2\x80\x9cActive attribute\xe2\x80\x9d here means the attribute that is currently selected in the Attributes list.

When the operator is executed a pop-up window will display the attribute's name and current value. From here, the value field can be adjusted to change the attribute's value.

See also

Attribute values can be viewed in the Spreadsheet editor.

--- Fin du chapitre: `_set_attribute` ---

--- Début du chapitre: `_set_curve_normal` --- The Set Curve Normal controls the method used to calculate curve normals for every curve.

The node doesn't set the normals directly, those are calculated later as necessary. Combined with the tilt attribute value at each control point, this will define the final normals accessible with the Normal Node.

Internally this node adjusts the values of the `normal_mode` attribute on each curve.

Standard geometry input, containing curves.

Whether or not to change the value on each curve.

Input for the custom normal attribute (`custom_normal`) when using Free mode.

The method for evaluation of the curve's normals

The final normals are calculated to have the smallest twist around the curve tangent across the whole curve.

The final normals are calculated so that they are perpendicular to the Z axis and the tangent. If a series of points is vertical, the X axis is used.

Use the stored custom normal attribute (`custom_normal`) as the final normals.

This mode adds a Normal input that can be used to set the value of the custom normal.

Note

Custom normals are not rotation invariant, meaning normals must be set after any rotation transformations; i.e. at the end of the node tree or at the bottom of the modifier stack.

Standard geometry output.

--- Fin du chapitre: `_set_curve_normal` ---

--- Début du chapitre: `_set_curve_radius` --- The Set Curve Radius controls the radius of the curve, used for operations like the size of the profile in the Curve to Mesh node. The value is set for every control point, and is then interpolated to each evaluated point in between the control points.

The input node for this data is the Radius node.

Standard geometry input, containing curves.

Whether or not to change the value on each control point. True values mean the value will be changed, false values mean the value will remain the same.

The radius value for each control point.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_curve_radius` ---

--- Début du chapitre: `_set_curve_tilt` --- The Set Curve Tilt controls the tilt angle at each curve control point. That angle rotates normal vector

which is generated at each point when evaluating the curve. The normal then can be retrieved with the Normal Node.

The rotation of the normal vector is an Axis Anglerotation. It is the same as the Vector Rotate Node operation with the tangent vector as the axis, the raw evaluated normal is used as the original vector, and the tilt as the rotation angle.

The input node for this data is the Curve Tilt Node.

Standard geometry input, containing curves.

Whether or not to change the value on each control point. True values mean the value will be changed, false values mean the value will remain the same.

The tilt angle for each control point.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _set_curve_tilt ---

--- Début du chapitre: _set_face_set --- The Set Face Set node controls which face set that faces are in.

The input node for this data is the Face Set Node.

Note

This node can only be used in the Tool context .

Standard geometry input.

Boolean field that controls which faces will have the Face Set value applied.

Integer field for specifying which face set each selected face should be moved to. Ignored for faces where the value of Selection is false.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _set_face_set ---

--- Début du chapitre: _set_geometry_name --- The Set Geometry Name node stores a custom name on the geometry, overriding the name which might come from the Object Info Node or a Grease Pencil to Curves Node. The name is displayed in the spreadsheet and can be helpful for debugging purposes.

Standard geometry input.

The new name for the geometry.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _set_geometry_name ---

--- Début du chapitre: _set_hair_curve_profile --- Sets the radius attribute of hair curves according to a profile shape.

Geometry

Replace the original radius.

Base radius to be set if Replace Radius is enabled.

Shape of the radius along the curve.

Factor of the radius at the minimum.

Factor of the radius at the maximum.

This node has no properties.

Geometry

--- Fin du chapitre: _set_hair_curve_profile ---

--- Début du chapitre: _set_handle_positions --- The Set Handle Positions node sets the positions for the handles of a B-spline curve.

curves. They can be used to alter the generated shape of the curve. The input node for this data is the Curve Handle Positions Node. See the Bézier curves page for more details.

Note

When the position is changed, Autohandle types will be converted to Aligned, and Vectorhandle types will be converted to Free.

Note

The left and right handles cannot be changed at the same time with this node. That is because it would break the alignment for left and right handles at the same control point.

Standard geometry input, containing curves.

Whether or not to change the handle position on each control point. True values mean the resolution will be changed, false values mean the resolution will remain the same.

The new handle position.

Note

The handle positions are the global position of the handle, they are not relative to the position of the corresponding control point.

An optional translation for each handle. This is evaluated at the same time as the Position input, meaning that fields evaluated for it will not reflect the changed position.

Whether to set the handle position of the left or right handle. The Lefthandle is closer to the start of the spline, and the Righthandle is closer to the end.

Standard geometry output.

Here, the handles are adjusted to the same position as the control points, but offset down in the Z direction slightly. With the Set Spline Type Node, the curve can be a poly spline in Edit Mode, to make adjustment simpler for the user of the node group.

--- Fin du chapitre: _set_handle_positions ---

--- Début du chapitre: _set_handle_type --- Sets the handle type for the points on the Bézier curve that are in the selection.

A selection for a certain handle type can be retrieved with the Handle Type Selection Node.

Bézier curve.

The points whose handle types will be changed.

Whether to update left or right or both handles. When both are selected, both the left and the right handles will be updated.

Update the left handles.

Update the right handles.

The handle type to switch to. See the Bézier curves page for more details on the different handle types.

Standard curve output.

--- Fin du chapitre: _set_handle_type ---

--- Début du chapitre: _set_id --- The Set ID node fills the id attribute on the input geometry. If the attribute does not exist yet, it will be created with a default value of zero. The ID is also created by the Distribute Points on Faces, and it is used in the Random Value Node and other nodes if it exists.

The input node for this data is the ID Node.

Standard geometry input.

Whether or not to change the value on each point or instance. True values mean the value will be changed, false values mean the value will remain the same.

The value for each element. By default, this input uses the index, which is useful when stable IDs are desired when deleting a dynamic number of instances.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_id` ---

--- Début du chapitre: `_set_instance_transform` --- The Set Instance Transform node transforms geometry instances using a Transformation Matrix.

The Instances page contains more information about geometry instances.

Standard geometry input.

Boolean field used to determine if an instance will be rotated.

The transformation matrix to translate, rotate, and scale individual instances.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_instance_transform` ---

--- Début du chapitre: `_set_material` --- The Set Material changes the material assignment in the specified selection, by adjusting the material_index attribute. If the material is already used on the geometry, the existing material index will be reused.

Note

This node adjusts mesh, point clouds, and volume data; other data types do not support materials.

Standard geometry input containing a mesh.

The material to apply to the geometry.

Whether to change the material of each face. True values mean the material will be changed, false values mean it will remain the same.

Note, volumes and point clouds only support a single material, in these cases a field input will be ignored.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_material` ---

--- Début du chapitre: `_set_material_index` --- The Set Material Index node sets the material index for a geometry.

The node to get this data is the Material Index node.

Standard geometry input.

Whether to change the material index for each face. True values mean the material index will be changed, false values mean it will remain the same.

The new material index.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_material_index` ---

--- Début du chapitre: `_set_point_radius` --- The Set Point Radius node controls the size each selected point cloud point should display with in the viewport.

The input node for this data is the Radius Node.

Standard geometry input.

Float value indicating the radius of the point geometry at each point.

Boolean input for selecting which points will have the radius value applied.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_point_radius` ---

--- Début du chapitre: `_set_position` --- The Set Position node controls the location of each point, the same way as controlling the position attribute. If the input geometry contains instances, this node will affect the location of the origin of each instance.

The input node for this data is the Position Node.

Standard geometry input.

Whether or not to change the position of each point or instance. True values mean the position will be changed, false values mean it will remain the same.

The new position for selected elements. By default, this is the same as if the Position Node was connected, meaning the node will do nothing.

An optional translation for each point. This is evaluated at the same time as the Position input, meaning that fields evaluated for it will not reflect the changed position.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_position` ---

--- Début du chapitre: `_set_selection` --- The Set Selection node controls which geometry is selected.

The input node for this data is the Selection Node.

Note

This node can only be used in the Tool context .

Standard geometry input.

Boolean field for specifying which elements should be selected in the output geometry. Elements for which this field evaluates to false are implicitly de-selected.

Which domain to set the selection on.

Standard geometry output.

--- Fin du chapitre: `_set_selection` ---

--- Début du chapitre: `_set_shade_smooth` --- The Set Shade Smooth node controls whether the mesh's faces look smooth in the viewport and renders. The smooth status of both edges and faces can be controlled, corresponding to the sharp_edge and sharp_face attributes. The input node for this data is the Is Face Smooth Node.

Standard geometry input.

When true, the selected faces will be marked to render smooth shaded. Otherwise the faces will be rendered flat shaded.

Boolean input for selecting which faces will have the Shade Smooth value applied.

Whether to write smoothness of mesh faces or edges.

Standard geometry output.

--- Fin du chapitre: `_set_shade_smooth` ---

--- Début du chapitre: `_set_spline_cyclic` --- The Set Spline Cyclic controls whether each spline will loop back on itself. Each spline has the same number of control points whether or not it is set as cyclic. But when displaying in the viewport or for operations with other nodes, a connection will be made between the first and last control points.

The input node for this data is the Is Spline Cyclic Node.

Standard geometry input.

Whether or not to change the cyclic value on each spline. True values mean the value will be changed, false values mean the value will remain the same.

Whether or not to join the first and last points of each spline.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_spline_cyclic` ---

--- Début du chapitre: `_set_spline_resolution` --- The Set Spline Resolution node sets the value for how many evaluated points should be generated on the curve for every control point. It only has an effect on NURBS, B-spline, and Catmull Rom splines. In case of B-spline splines, the resolution does not have an effect on segments between vector handles.

The evaluated points are displayed in the viewport, used in the Curve to Mesh Node, and optionally used in the Resample Curve Node.

The input node for this data is the Spline Resolution Node.

Standard geometry input.

Whether or not to change the resolution value on each spline. True values mean the value will be changed, false values mean the resolution will remain the same.

The number of evaluated points generated for each control point in NURBS and B-spline splines. It controls the accuracy of operations like trimming or sampling a curve. Higher resolutions are more accurate, but slower.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_set_spline_resolution` ---

--- Début du chapitre: `_set_spline_type` --- Sets the spline type for the splines in the curve component that are in the selection.

Standard geometry input with a curve component.

The splines whose spline types will be changed.

The type to convert the splines in the selection to. Read the Spline Types page for more detail on the different spline types.

Convert to a B-spline. A spline converted from a poly spline gets vector handles, while one converted from NURBS or Catmull Rom spline gets auto handles.

Note

When converting from a NURBS spline to a B-spline, at least six points are needed. When the number of points is not a multiple of three, a full conversion is not possible and the spline has to be truncated.

Convert to a NURBS spline.

Convert to a poly spline.

Convert to a Catmull Rom spline.

Standard curve output.

--- Fin du chapitre: `_set_spline_type` ---

--- Début du chapitre: `_shader_editor` --- The Shader Editor is used to edit materials which are used for rendering. Materials used by Cycles and EEVEE are defined using a node tree. Therefore, the main window of the Shader editor is a node editor.

Shader Editor with the default material node tree.

A list of all shader nodes is available in the rendering section.

The Use Nodes setting is mostly a legacy setting and should always be checked for materials.

The Slot menu can be used to select the active material slot on the active object. The material selector to the right of it can change the material that is in the selected slot.

The pin button will keep the current material selection fixed. When a material is pinned, it will remain visible in the Shader editor even when another object or material is selected elsewhere.

The Options panel in the Sidebar region contains the same settings that are also available in the Material tab in the Properties. They differ depending on the selected render engine. The settings are duplicated to make it possible to edit the entire material from the Shader editor.

--- Fin du chapitre: `_shader_editor` ---

--- Début du chapitre: `_shader_to_rgb` --- EEVEE Only

The Shader to RGB node is typically used for non-photorealistic rendering, to apply additional effects on the output of BSDFs. For example, a color ramp on the output of a diffuse BSDF can be used to create a flexible toon shader.

Using this conversion breaks the PBR pipeline and thus makes the result unpredictable when used in combination with effects such as ambient occlusion, contact shadows, soft shadows and screen space refraction.

Some effects require multiple samples to converge, and applying arbitrary changes to noisy input may not convert to a smooth result.

Warning

If a Shader to RGB node is used, any upstream BSDF will be invisible to the following effects:

Screen Space Reflection

Subsurface Scattering

Alpha Clip and Alpha Hashed blend modes

Shader to RGB node doesn't give expected results in render passes.

Any shader such as a BSDF or Emission node can be linked here.

This node has no properties.

Surface color computed from BSDFs and lighting.

Alpha transparency from any Transparent BSDFs in the input.

Simple toon shading with Shader to RGB and Freestyle.

--- Fin du chapitre: `_shader_to_rgb` ---

--- Début du chapitre: `_shading` --- The appearance of the mesh edges are determined to be evened out or well defined within the 3D Viewport and render. In Edit Mode, individual faces can be selected to determine which faces are smoothed or flattened.

Note

Both Shade Smooth and Flat are also available in Object Mode and function the same way.

Reference

Edit Mode

Face \xe2\x80\xa3 Shade Smooth

Using interpolated vertex normals, the mesh faces will blur at the edges and appear smooth.

Reference

Edit Mode

Face \xe2\x80\xa3 Shade Flat

Face normals are displayed evenly, because of this all the edges of the selected mesh will be easily visible.

Tip

Use the Edge Split modifier and Smooth by Angle Smooth to balance between smooth surfaces and sharp edges.

--- Fin du chapitre: _shading ---

--- Début du chapitre: _shadow --- The Shadow Visual Effect shows a simulated shadow casting by the object.

For simulating the shadow a color silhouette of the object is displaced in horizontal and/or vertical direction on the back of the object.

Shadow Visual Effect. \xc2\xb6

Defines the shadow color.

Control the shadow displacement in pixels on the X and Y axis.

Control the size of the shadow on the X and Y axis.

Sets the shadow rotation around the Grease Pencil object center or another object when Use Object As Pivot is enabled.

When enabled, an Object is used by the shadow as the center of rotation.

Control the blur scale in pixels on the X and Z axis.

Number of blur samples (0 disabled the blur effect).

When enabled, apply a wave distortion to the shadow.

Sets horizontal or vertical direction for the waves.

Controls the strength and the depth of the wave.

Controls the wave period. The time it takes to complete one cycle.

Shifts the wave pattern over the shadow.

Simple Shadow. \xc2\xb6

Blurred Shadow. \xc2\xb6

Stretched shadow with an empty as center of rotation. \xc2\xb6

--- Fin du chapitre: _shadow ---

--- Début du chapitre: _shape --- Reference

Particle System \xe2\x80\xa3 Hair Shape

These settings control the shape of hair curves for rendering.

A shape parameter that controls the transition in thickness between the root and tip. Negative values make the primitive rounded more towards the top, the value of zero makes the primitive linear, and positive values make the primitive rounded more towards the bottom.

Multiplier of the hair width at the root.

Multiplier of the hair width at the tip.

Multiplier for the Root and Tip values. This can be used to change the thickness of the hair.

Sets the thickness at the tip to zero, even when using a nonzero tip multiplier.

--- Fin du chapitre: _shape ---

--- Début du chapitre: _shape_key --- This mode shows the shape keys of the active object and lets you create/adjust keyframes for their values.

The Shape Key Editor.

--- Fin du chapitre: _shape_key ---

--- Début du chapitre: _shape_keys_panel ---Reference

Properties

All modes

Object Data \xe2\x80\xa3 Shape Keys

Shape Keys panel.

The Shape Keys panel is used for authoring shape keys.

A List View.

In Relative mode: Value is the current influence of the shape key used for blending between the shape (value=1.0) and its reference key (value=0.0). The reference key is usually the Basis shape. The weight of the blend can be extrapolated above 1.0 and below 0.0.

In Absolute mode: Value is the Evaluation Time at which the shape will have maximum influence.

If unchecked, the shape key will not be taken into consideration when mixing the shape key stack into the result visible in the 3D Viewport.

Shape keys can be locked to protect them from accidental modification due to inadvertently selecting the wrong key for editing in the list. Most common sculpt and edit mode operators and tools that move vertices abort with an error if the active shape key is locked.

Note

Operators that always modify all shape keys in exactly the same way, like Apply Object Transforms, don't check shape key locks. Neither currently do most edit mode operators that modify topology, because the topology is expected to usually be finalized before shape keys are created.

Add a new shape key with the current deformed shape of the object. This differs from the + button of the list, as that one always copies the Basis shape independently of the current mix.

If your mesh is symmetrical, in Object Mode, you can mirror the shape keys on the X axis. This will not work unless the mesh vertices are perfectly symmetrical. Use the Mesh \xe2\x80\xa3 Symmetrize tool in Edit Mode.

Same as Mirror Shape Key though it detects the mirrored vertices based on the topology of the mesh. The mesh vertices do not have to be perfectly symmetrical for this action to work.

Transfer the current resulting shape from a different object.

Select the object to copy, then the object to copy into. Use this action and a new shape key will be added to the active object with the current mix of the first object.

Transfer the active shape key from a different object regardless of its current influence.

Select the object to copy, then the object to copy into. Use this action and a new shape key will be added to the active object with the active shape of the first object.

Removes all Shape Keys and any effect that they had on the mesh.

Saves the current visible shape to the mesh data and deletes all Shape Keys.

Set the shape keys to Relative or Absolute. See Relative or Absolute Shape Keys.

Show the active shape in the 3D Viewport without blending. Shape Key Lock gets automatically enabled while the object is in Edit Mode.

If enabled, when entering Edit Mode the active shape key will not take maximum influence as is default. Instead, the current blend of shape keys will be visible and can be edited from that state.

Creates an Attribute in the vertex domain called `rest_position` which is a copy of the `position` attribute before shape keys and modifiers are evaluated. Only mesh objects support this option.

Relative Shape Keys options. \xc2\xbb

See Relative or Absolute Shape Keys .

With relative shape keys, the value shown for each shape in the list represents the current weight or influence of that shape in the current Mix.

Set all influence values, or weights, to zero. Useful to quickly guarantee that the result shown in the 3D Viewport is not affected by shapes.

The weight of the blend between the shape key and its reference key (usually the Basis shape).

A value of 0.0 denotes 100% influence of the reference key and 1.0 of the shape key.

Minimum and maximum range for the influence value of the active shape key. Blender can extrapolate results when the Value goes lower than 0.0 or above 1.0.

Limit the active shape key deformation to a vertex group. Useful to break down a complex shape into components by assigning temporary vertex groups to the complex shape and copying the result into new simpler shapes.

Select the shape key to deform from. This is called the Reference Key for that shape.

Note

Rather than storing offsets directly, internally relative keys are stored as snapshots of the mesh shape. The relative deformation offsets are computed by subtracting Reference Key from that snapshot.

Therefore, replacing the Reference Key has the effect of subtracting the difference between the new and old reference from the relative deformation of the current key.

Absolute Shape Keys options. \xc2\xbb

See Relative or Absolute Shape Keys .

With absolute shape keys, the value shown for each shape in the list represents the Evaluation Time at which that shape key will be active.

Absolute shape keys are timed, by order in the list, at a constant interval. This button resets the timing for the keys. Useful if keys were removed or re-ordered.

Controls the interpolation between shape keys.

Linear, Cardinal, Catmull-Rom, B-Spline

Different types of interpolation. \xc2\xbb

The red line represents interpolated values between keys (black dots).

Controls the shape key influence. Scrub to see the effect of the current configuration. Typically, this property is keyed for animation or rigged with a driver.

--- Fin du chapitre: `_shape_keys_panel` ---

--- Début du chapitre: `_sharp_edges` ---

Reference

Edit Mode

Select \xe2\x80\xa3 Select Sharp Edges

Selects edges whose faces intersect at an acute angle. This can be useful to find hard edges in a mesh.

Determines how sharp of an angle at an edge needs to be in order for it to be selected.

--- Fin du chapitre: _sharp_edges ---

--- Début du chapitre: _shear ---Reference

Object and Edit Mode

Toolbar \xe2\x80\xa3 Shear

Object/Mesh/Curve/Surface \xe2\x80\xa3 Transform \xe2\x80\xa3 Shear

Shift- Ctrl- Alt- S

Shearing is a form of movement where parallel surfaces move past one another. During this transform, movement of the selected elements will occur along the horizontal axis of the current view. The axis location will be defined by the Pivot Point. Everything that is above this axis will move (shear) in the same direction as your mouse pointer (but always parallel to the horizontal axis). Everything that is below the horizontal axis will move in the opposite direction.

Shear Offset Factor. \xc2\xb6

How far items are shifted from their original location.

Defines one axis of the imaginary shearing plane.

Defines the other axis of the imaginary shearing plane.

See Transform Orientations .

See Proportional Editing .

Warning

The Axis and Axis Orthographic cannot be the same axis, else the imaginary plane is dimensionless and the objects will disappear into a point.

See below for the result of using Shear on a number of different elements.

The effects of a Shear transform with different Pivot Points. \xc2\xb6

The three frames of the image above show the effects of shearing on the selected vertices when the pivot point is altered. In middle frame, the Pivot Point is set to Median Point and the mouse was moved to the left during the transform. In right frame, the Pivot Point is set to the 3D cursor which is located above the mesh. When the mouse is moved to the left during a Shear transform, the selected vertices are moved to the right as they are below the horizontal axis.

Tip

Shear Transform Magnitude

The magnitude of the Shear transform applied to the selected elements is directly proportional to the distance from the horizontal axis. i.e. the further it is away from the axis, the greater the movement.

The effects of a Shear transform on objects with different Pivot Points.

\xc2\xb6

--- Fin du chapitre: _shear ---

--- Début du chapitre: _sheen --- Cycles Only

The SheenBSDF is used to add reflection to materials that have micro surface details such as cloth or dust. This shader is intended to be layered on top of other shaders such as dielectric or metallic shader setups.

Color of the surface, or physically speaking, the probability that light is reflected for each wavelength.

Controls the amount of color that is reflected back to the camera, higher values reflect more color and can give a dusty appearance, while lower values look fuzzy and darker.

Normal used for shading; if nothing is connected the default shading normal is used.

Sheen shading model.

Classic Ashikhmin velvet, used in Blender versions prior to 4.0

Microflake-based model of multiple scattering between normal-oriented fibers.

Standard shader output.

The Sheen shader example.\xc2\xbb

The Sheen shader behavior.\xc2\xbb

--- Fin du chapitre: _sheen ---

--- Début du chapitre: _shortest_edge_paths --- The Shortest Edge Paths node finds paths along mesh edges to a selection of end vertices. The cost used to define \xe2\x80\x9cshortest\xe2\x80\x9d can be set to anything. By default there is a constant cost for every edge, but a typical input would be the length of each edge.

The output is encoded with vertex indices, and is meant to be used on the vertex domain. For each vertex, the Next Vertex Index output gives the index of the following vertex in the path to the \xe2\x80\x9cclosest\xe2\x80\x9d endpoint.

The node is implemented with Dijkstra\xe2\x80\x99s algorithm.

Tip

The edge length is a natural input to the Edge Cost. It can be implemented with the Edge Vertices Node and the Vector Math Node set to the Distance operation.

See also

This node can be used with the Edge Paths to Selection Node or the Edge Paths to Curves Node to generate new geometry based on the paths.

A selection of the goal vertices that terminate the edge paths.

The weight for each edge, used to determine the meaning of

\xe2\x80\x9cshortest\xe2\x80\x9d

This node has no properties.

The following vertex on the shortest path from every vertex to the closest endpoint (as defined by the cost input).

The remaining cost before an end vertex is reached.

--- Fin du chapitre: _shortest_edge_paths ---

--- Début du chapitre: _show_hide --- Reference

All Modes

Properties \xe2\x80\xa3 Bone \xe2\x80\xa3 Viewport Display

\xe2\x80\xa6 \xe2\x80\xa3 Show/Hide

You do not have to use bone layers to show/hide some bones. As with objects, vertices or control points, you can use H:

H will hide the selected bone(s).

Shift- H will hide all bones but the selected one(s).

Alt- H will show all hidden bones.

You can also use the Hide checkbox of the Bone tab \xe2\x80\xa3 Viewport Display panel.

Note that hidden bones are specific to a mode, i.e. you can hide some bones in Edit Mode, they will still be visible in Pose Mode, and vice versa. Hidden bones in Pose Mode are also invisible in Object Mode. And in Edit Mode, the bone to hide must be fully selected, not just its root or tip.

--- Fin du chapitre: _show_hide ---

--- Début du chapitre: _shrink-fatten ---Reference

Edit Mode

Toolbar \xe2\x80\xa3 Shrink/Fatten

Mesh \xe2\x80\xa3 Transform \xe2\x80\xa3 Shrink/Fatten

Alt- S

Moves the selected vertices \xe2\x80\x9cinwards\xe2\x80\x9d or \xe2\x80\x9coutwards\xe2\x80\x9d along their normal, all by the same distance. You can control this distance by moving the mouse up or down, typing a number, or using the slider in the Adjust Last Operation panel.

Applies a larger offset to vertices that are part of a sharp corner, for a more uniform result. You can toggle this option by pressing S, holding Alt, or clicking the checkbox in the Adjust Last Operation panel.

Mesh before shrink/fatten. \xc2\xb6

Inflated using a positive value. \xc2\xb6

Shrunk using a negative value. \xc2\xb6

--- Fin du chapitre: _shrink-fatten ---

--- Début du chapitre: _shrinkwrap --- The Shrinkwrap constraint is the \xe2\x80\x9cobject counterpart\xe2\x80\x9d of the Shrinkwrap Modifier. It moves the owner origin and therefore the owner object \xe2\x80\x99s location to the surface of its target. This implies that the target must have a surface; thus, you can only use meshes as targets.

Shrinkwrap panel. \xc2\xb6

Data ID used to select the constraint \xe2\x80\x99s target, which must be a mesh object, and is not functional (red state) when it has none. See common constraint properties for more information.

This number field controls the offset of the owner from the shrunk computed position on the target \xe2\x80\x99s surface.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

This selector allows you to select which method to use to compute the point on the target \xe2\x80\x99s surface to which to move the owner \xe2\x80\x99s origin. You have these options:

The chosen target \xe2\x80\x99s surface \xe2\x80\x99s point will be the nearest one to the original owner \xe2\x80\x99s location. This is the default and most commonly useful option.

The target \xe2\x80\x99s surface point is determined by projecting the owner \xe2\x80\x99s origin along a given axis.

This axis is controlled by the radio buttons that show up when you select this type. This means the projection axis can only be aligned with one of the three axes, or their opposites. When the projection of the owner \xe2\x80\x99s origin along the selected direction does not hit the target \xe2\x80\x99s surface, the owner \xe2\x80\x99s location is left unchanged.

+X, +Y, +Z, -X, -Y, -Z

Coordinate space in which the axis direction is specified.

Distance cutoff after which projection is assumed to have failed, leaving the location unchanged.

In addition to the selected projection axis, project in the opposite direction and choose the closest hit.

This radio button allows you to prevent any projection over the \xe2\x80\x9cfront side\xe2\x80\x9d (respectively the \xe2\x80\x9cback side\xe2\x80\x9d) of the target\xe2\x80\x99s faces. The \xe2\x80\x9cside\xe2\x80\x9d of a face is determined by its normal (front being the side \xe2\x80\x9cfrom where\xe2\x80\x9d the normal \xe2\x80\x9coriginates\xe2\x80\x9d).

Off, Front, Back

When used with Project Opposite and Face Culling, it inverts the Front or Back cull choice for the opposite direction.

This method is very similar to the Nearest Surface Point one, except that the owner\xe2\x80\x99s possible shrink locations are limited to the target\xe2\x80\x99s vertices.

This method doesn\xe2\x80\x99t support the Snap Mode setting described below.

This method is similar to Nearest Surface Point, but produces a much smoother projection in return for being significantly slower.

Instead of finding the closest point, it searches for the nearest point that has its interpolated smooth normal pointing towards or away from the original owner position. Non-manifold boundary edges are specially handled as infinitely thin cylinders that emit normals in all perpendicular directions; ignores flat shading.

Most Shrinkwrap types support an additional setting to control how the owner is moved to the target point selected by the methods described above. Some of the choices only differ if Distance is not zero.

The owner location is always changed. The offset is applied along the projection line connecting the original owner location and selected target point towards the original position.

Like On Surface, but the offset is always applied towards the outside of the target.

Like On Surface, but the offset is applied along the smooth normal of the target.

The owner is not moved if it is already inside the target. Offset shrinks the allowed volume towards the inside along the projection line.

The owner is not moved if it is already outside the target. Offset expands the exclusion volume towards the outside along the projection line.

The Inside and Outside options can be used for very crude collision detection. The inside vs outside determination is done based on the target normal and is not always stable near 90 degree and sharper angles in the target mesh.

Whenever Snap Mode is available, it is also possible to align the specified local axis of the object to the smooth normal of the target at the selected point. The axis is selected via radio buttons.

The alignment is performed via smallest rotation, like in Damped Track constraint.

--- Fin du chapitre: `_shrinkwrap` ---

--- Début du chapitre: `_shrinkwrap_hair_curves` --- Shrinkwraps hair curves to a mesh surface from below and optionally from above.

Input Geometry (only curves will be affected).

Surface geometry used for shrinkwrap. This input takes priority over the corresponding object input, if used.

Surface object used for shrinkwrap.

Factor

Distance from the surface used for shrinkwrap.

Blend shrinkwrap for points above the surface.

Amount of steps of smoothing applied after shrinkwrap.

Lock the position of root points.

This node has no properties.

Geometry

--- Fin du chapitre: _shrinkwrap_hair_curves ---

--- Début du chapitre: _sidebar ---Set the frame range of the mask for Sequencer.

Mask Layer panel. \xc2\xba

Mask layers consists of one or several splines and used to

\xe2\x80\x9cgrouped\xe2\x80\x9d operation on splines.Layers can be used to create complex shapes and to define how the splines interact with each other.Splines belonging to the same layer can be animated together, for example by an itemfrom motion tracker footage.Example of such tools might be parenting the whole set of splines to single motion tracking data orsimple to transform all of them together.

Used to set the opacity of the mask layer.

Inverts the values (colors) in the mask layer.

The layer blending operation to perform. See Color Blend Modes .

Modes Merge Addand Merge Subtractgive better results when using a Featheron overlapping masksthan straightforward mathematical addition and subtraction.

Type of the Featherfalloff, controls the shape of the transition between black and white.

Fills the self-intersecting areas.

Overlapping splines from the same layer will generate holes in the mask.

The Overlap option example.\xc2\xba

The Holes option example.\xc2\xba

The purpose of mask layers can be explained with an example.Suppose there are two unwanted people in the footage, and one of them goes from left to right, andthe other in the opposite direction. Two mask layers can then be used to mask them separately byusing a single mask data-block. At the point of intersection of these shapes they will be added together rather thancreating a hole, as would happen if they were on the same layer. If the motion is simple enough,a single motion tracked point can be used to drive the location of the entire mask layer.

Active Spline panel. \xc2\xba

The method used for calculating the offset of the mask spline feather.

Preserves the thickness of the feather, but can give undesirable loops of the feather curve.

Gives a nicer and smoother shape,but can also give an undesirable sharp feather when a curve segment forms an S-shape.

The type of weight (thickness of feather) interpolation between points.

Linearor Ease(i.e. changes occur slowly at the beginning and at the end).

If the spline is closed or not.

Creates splines with filled areas.If disabled, Blender will create curves with a thickness to mask out thin objects such as wires or hair.

Prevent the feather (not the curve itself) from intersecting with itself.

Active Point panel. \xc2\xba6

This panel is shown when both a tracking marker and mask is selected.

In the Movie Clip Editor it is possible to link the whole mask or its points to motion tracks. This way the mask or points will follow the tracks.

Data ID to which the mask or spline is parented to in case of parenting to movie tracking data set to Movie Clip data-block.

Point Track, Plane Track

Object to parent to.

Name of individual tracks.

--- Fin du chapitre: _sidebar ---

--- Début du chapitre: _side_of_active ---Reference

Edit Mode

Select \xe2\x80\xa3 Side of Active

With an active vertex, this tool will select all vertices in a specified direction. It is similarly to the Loop Inner-Region tool in that it will fill select faces within its perimeters, however, it is determined by direction and not by a closed loop.

Determines the behavior of the selection. More information on this can be found in Transform Orientations.

Depending on which Axis is chosen, the selection will encompass the positive or negative axis starting from the active vertex outward.

Where Positive and Negative Axis select all vertices in a given direction, Aligned Axis will only select the vertices that are in-line with the active vertex.

While following along the X axis: (from left to right) active vertex, Aligned, Positive, and Negative. \xc2\xba6

Chooses the direction of the selection.

The amount of influence the selection has outside the original perimeters. The higher the Threshold the more vertices will be selected.

--- Fin du chapitre: _side_of_active ---

--- Début du chapitre: _similar ---Reference

Edit Mode

Select \xe2\x80\xa3 Similar

Shift- G

Select geometry that has similar certain properties to the ones selected, based on a threshold that can be set in tool properties after activating the tool. Tool options change depending on the selection mode:

Selects all vertices that have normals pointing in similar directions to those currently selected.

Selects all vertices that have the same number of faces connected to them.

Selects all vertices in the same vertex group.

Selects all vertices that have the same number of edges connected to them.

Selects all edges that have a similar length as those already selected.

Selects all edges that have a similar direction (angle) as those already selected.

Selects all edges that belong to the same number of faces.

Selects all edges that are between two faces forming a similar angle, as with those already selected.

Selects all edges that have a similarCreasevalue as those already selected.

Selects all edges that have the same Bevel Weightas those already selected.

Selects all edges that have the same Seamstate as those already selected. Seamis a mark used inUV texturing.

Selects all edges that have the same Sharpstate as those already selected. Sharpis a mark used by theEdge Split Modifier.

Selects all faces that use the same material as those already selected.

Selects all faces that have a similar area as those already selected.

Selects all faces that have the same number of edges.

Selects all faces that have a similar perimeter (added values of its edge lengths).

Selects all faces that have a similar normal as those selected.This is a way to select faces that have the same orientation (angle).

Selects all faces that are (nearly) in the same plane as those selected.

Selects all faces with similarface shading.

Selects all faces with similarFreestyle Face Marks.

For quantitative properties, this property selects the type of comparison to between the two numerical values.

Select items with the same value as the active item\xe2\x80\x99s chosen property.

Select items with a larger value as the active item\xe2\x80\x99s chosen property.

Select items with a smaller value as the active item\xe2\x80\x99s chosen property.

For quantitative properties, this property controls howclose the property\xe2\x80\x99s values have to be in the comparison.

Reference

Edit Mode

Select \xe2\x80\xa3 Similar \xe2\x80\xa3 Face Regions

Select matching features on a mesh that has multiple similar areas based on the topology.

--- Fin du chapitre: _similar ---

--- Début du chapitre: _simple_deform --- The Simple Deformmodifier allows the application of a simple deformation toan object. Meshes, lattices, curves, surfaces and texts are supported objects.The deformation is either a rotation (Twist, Bend) or a scaling (Taper, Stretch).The amount of deformation is specified by the Deform Angle (rotation) or Deform Factor (scaling).

The Simple Deform modifier.\xc2\xb6

The deformation is calculated in the localcoordinate space.Be aware that the local axes of an object can differ from the global ones.In the figure above, the global Z axis points up and the local Z axis points at 45\xc2\xb0.The deformation is applied along a Deform axis, which can be set by selection from a list (X, Y or Z).By using the Limitsfield, the influence of the modifier can be restricted to a subset of the Deform axis.All distances are measured from the origin of the object.The vertices that are furthest away from the origin on the Deform axisrepresent the upper and lower limits.The origin of the object and the orientation of the local axes can be defined by an external Deform object (most of the time, an empty).

Defines the kind of deformation which will be applied. The figure below shows the four modes, applied to a text object. The origin of the object is at the very left of the text.

Twist around X axis (180°).\xc2\xba6

Bend around Z axis (180°).\xc2\xba6

Taper along X axis (factor = 2).\xc2\xba6

Stretch along X axis (factor = 0.3).\xc2\xba6

Rotates the mesh around the specified Axis. Each vertex along the Deform axis is rotated around the object's origin. If the origin is inside the object, this results in a twisted appearance. Below the origin, there is a negative rotation and above the origin, the rotation is positive or clockwise. Vertices in the same plane as the origin are not rotated.

The total amount of rotation is specified by the angle and the rotation at each vertex is thus weighted by the distance of the vertex to the origin of the object. Vertices that are furthest away from the object origin have max rotation, positive or negative.

Bends the mesh over the specified Axis. The Bend mode is more complex and less intuitive. The picture below shows the same plane but with different Deform and Bending axes.

Deform axis X or Y (a).\xc2\xba6

Deform axis Z (b).\xc2\xba6

Deform axis X. Local axis Y points down (c).\xc2\xba6

Deform axis X. Local axis Y points up (d).\xc2\xba6

Using a mesh plane and setting the Deform axis to X or Y will not result in any deformation (Fig a). You expect that the bending should be something like Figure (c) or (d). In a 3D world however, selecting the X axis introduces an ambiguity because bending along the X axis could result in Figure (c) or (d). The following pairs describe the selected deform axis vs. the desired bending axis: X and Z, Y and Z, Z and X.

In Fig (a), because of the Deform axis X, the Bending is along the Z. All vertices however have the same Z coordinate equal to the local origin. So, no deformation occurs. In Fig (d) the local axes are rotated around the Deform axis X, so that the Bending axis Z points to the left. So, all vertices are bent for their Z coordinate. The further away from the local origin, the more bending.

This explains also the unexpected result of Fig (b). The Deform axis is set here to Z (pointing up). So, according to the pairs above, the Bending axis defaults to X. All vertices of the plane are bent in their X coordinate. The further away, the more rotation occurs. Negative X coordinates are rotated counterclockwise.

Linearly scales along the specified Axis. The scaling factor is weighted by the distance from the origin of the object in the deform axis. No scaling occurs in the plane of the origin of the object. The maximum scaling occurs at the vertices that are furthest away from the local origin. This can be a positive or negative scaling, depending on the location of the origin. If the local origin is within the object, the deformed object appears tapered.

Stretches the object along the specified Axis. If the local origin is within the object, the deformed mesh looks stretched like pulling a rubber from both sides. With a positive factor, the mesh gets longer in the deformed axis, wider at the borders and thinner at the origin than the original mesh. If the factor is negative, then the mesh is squashed in the deformed axis, thicker at the origin and thinner at the borders.

The total amount of deformation. Can be negative to reverse the deformation.

The name of an object that defines the origin and axis of deformation (usually an empty). This object can be:

Rotated to control the axis (its local Axis is now used as the deformation one).

Moved to control the origin of the deformation.

Scaled to change the deformation factor.

You can set the lower and upper limits of the deformation. The upper limit cannot be lower than the lower one. These limits are mapped on the Deform axis.

These controls whether the coordinates along the two other axes are allowed to change or not. E.g. if you Stretch your object along its Z axis, it is possible to squash along the X axis only, by locking the Y one.

The name of the vertex group that indicates whether and how much each vertex is influenced by the deformation. The amount of influence is determined by the weight in the Weight Paint map.

--- Fin du chapitre: _simple_deform ---

--- Début du chapitre: _simplification --- The Simplification modifier merges stroke vertices that lie close to one another, like the Decimate modifier for meshes.

Measure for how close points have to be to each other to be merged. A higher tolerance means more vertices are merged.

--- Fin du chapitre: _simplification ---

--- Début du chapitre: _simplify --- Reference

Render \xe2\x80\xa3 Simplify

Common Settings

Maximum number of subdivision by the Subdivision Surface modifiers.

Show only a subset of all child hairs and particles.

Automatically scales textures down so that they are no larger than the values chosen. This can help reduce computer memory resources when rendering large scenes with huge textures.

See Common Settings above.

Resolution percentage of volume objects in the viewport. This mostly affects memory usage rather than computation times.

Skip computing custom normals and face corner normals for displaying meshes in the viewport.

See Common Settings above.

Automatically culls objects based on the camera frustum defined by the Margin.

Automatically culls objects based on their distance from the active camera. This is set via the Distance property.

Activates the simplification process only during animation playback.

Shows the fill component in Grease Pencil materials.

Shows Grease Pencil modifiers .

Shows Grease Pencil visual effects .

Shows layers tint overrides.

Use Anti-Aliasing to smooth stroke edges. The amount of anti-aliasing can be adjusted by the Anti-Aliasing Threshold .

--- Fin du chapitre: _simplify ---

--- Début du chapitre: `_simulation` --- You can control the internal timing of the soft body system with this value. It sets the correlation between frame rate and tempo of the simulation. A free falling body should cover a distance of about five meters after one second and travel at a speed of ten meters per seconds.

You can adjust the scale of your scene and simulation with this correlation. If you render with 25 frames per second, you will have to set Speed to 1.3.

--- Fin du chapitre: `_simulation` ---

--- Début du chapitre: `_simulation_nodes` --- Through the use of Simulation Zones, Geometry Nodes can be used to create custom physic simulations through nodes. Simulation zones allow the result of one frame to influence the next one. That way even a set of simple rules can lead to complex results, with the passing of time. The most common type of them is physics simulation, with specific solvers for physical phenomena.

See also

Read more about Simulation Zones

The simulation is automatically cached during playback. The valid cache can be seen as a strong yellow line in the timeline editor. This allows for animators to quickly inspect all the previous frames of a simulation.

Cached frames in the Timeline. \xc2\xbf

When the result is ready to be sent to a render-farm, it can be baked to disk. This allows for the simulation to be rendered in a non-sequential order.

Simulation and Physics, Geometry Nodes user interface. \xc2\xbf

Note

Baking the simulation will bake all the simulations in all modifiers for the selected objects.

Calculate simulations in geometry nodes modifiers from the start to current frame.

Bake simulations in geometry nodes modifiers. In order to bake the simulation, the blend-file must be saved to your computer. The location the file is saved determines where the baked data is also saved. The directory the baked data is saved to can be changed per modifier in the Internal Dependencies .

Delete cached/baked simulations in geometry nodes modifiers

For the cases where the current frame is the only one relevant, users can opt-out of caching the results to save memory.

Combined with the Index of Nearest , this can be used for a number of sphere-based simulations.

Index of Nearest sample file CC-BY Sean Christofferson. \xc2\xbf

--- Fin du chapitre: `_simulation_nodes` ---

--- Début du chapitre: `_simulation_zone` --- Simulation zones allow the result of one frame to influence the next one. That way even a set of simple rules can lead to complex results, with the passing of time. The most common type of them is physics simulation, with specific solvers for physical phenomena.

Initial simulation nodes and simulation zone. \xc2\xbf

When adding a simulation, two nodes are added, defining between them a \xe2\x80\x9cSimulation Zone\xe2\x80\x9d.

The inputs that are connected to the Simulation Input node are evaluated only once, at the beginning of the simulation, passed to the next

simulation state and eventually outputted. Other nodes can be linked inside the simulation region from the outside. Those are re-evaluated every step based on their value at the given frame.

It is not possible to have any link going towards outside. The result of the simulation can only be accessed via the Simulation Output node. This also allows sub-frame interpolation for motion blur.

Note

This node cannot be used in the Tool context \xe2\x80\x94only in the Modifier context.

The simulation is tied to the animation system, with support for sub-steps. It will only be evaluated while the animation frame changes, and is cached like the existing physics simulations in Blender.

In the Node Editor the inputs can be renamed, shuffled and removed. This is also the place where sub-steps can be defined for a simulation.

Standard geometry input, which is available by default to input geometry into the simulation zone. More bake items can be added by dragging sockets into the blank socket or in the Simulation State panel. Items can be renamed by Ctrl-LMB on the socket name or in the nodes Properties panel.

The time in seconds between frames. Essentially this is the inverse of the renderFrame Rate.

This delta is used to drive the simulation by connecting it to node setups that depend on a rate. This will keep the simulation playback consistent when the frame rate changes.

Forward the output of the simulation input node directly to the output node and ignore the nodes in the simulation zone.

The simulation is automatically cached during playback. The valid cache can be seen as a strong yellow line in the timeline editor. This allows for animators to quickly inspect all the previous frames of a simulation.

Cached frames in the Timeline. \xc2\xba6

For the cases where the current frame is the only one relevant, users can opt-out of \xe2\x80\x94Cache\xe2\x80\x94 to save memory.

When the result is ready to be sent to a render-farm, it can be baked to disk. This allows for the simulation to be rendered in a non-sequential order.

Simulation and Physics, Simulation Nodes user interface. \xc2\xba6

Note

Baking the simulation will bake all the simulations in all modifiers for the selected objects.

Combined with the Index of Nearest, this can be used for a number of sphere-based simulations.

Index of Nearest sample file CC-BY Sean Christofferson. \xc2\xba6

--- Fin du chapitre: _simulation_zone ---

--- Début du chapitre: _sinus_displacement --- The Sinus

Displacement modifier adds a sinusoidal displacement to the stroke.

How wide the undulations are along the stroke.

How high the undulations are across the stroke.

Allows \xe2\x80\x94offsetting\xe2\x80\x94

(\xe2\x80\x94moving\xe2\x80\x94) the undulations along the stroke.

Tip

The undulations this modifier produces look exactly the same at a Phase of 0 and any positive or negative multiple of the Wavelength set on the modifier. This can be used for rendering short video sequences with wavy

linesthat can then be seamlessly looped without any visual jumps in the undulations along the line.

Sinus Displacement modifier demo by T.K.(blend-file).\xc2\xbb6

--- Fin du chapitre: _sinus_displacement ---

--- Début du chapitre: _skin --- These rigs implement a flexible system for rigging skin using multiple interacting B-Bone chains.This is developed as the base for a new modular Rigify face rig.These are the main ideas of the system:

One core idea of the system is that most of the deformation should be implemented using a standard powerful B-Bone chain rig. These chains support advanced behavior by interacting with other rig components. This is in contrast to having multiple domain-specific rigs that each generate their own deform chains.

The implementation provides two versions of the chain rig: skin.basic_chain merely attaches B-Bones to the controls with no automation added to the controls themselves. The skin.stretchy_chain rig in addition interpolates motion of the end (and an optional middle) controls to the other controls of the chain.

The deformation part of the system consists of chains of one or more B-Bones connecting control points (nodes). Whenever controls for two chains would completely overlap, they are automatically merged.

For each merged control, one of the chains is selected as the owner, based on heuristic factors like parent depth from root, presence of.T/.B.L/.Rsymmetry markers, and even alphabetical order as the last resort. This can be overridden by an explicit priority setting in cases when it guesses wrong.

The owner and its parents determine additional automation that is placed on the control. As a special case, if a control is merged with its.T/.B.L/.Rsymmetry counterparts (detected purely by naming), the automation from all of the symmetry siblings of the owner is averaged.

Rather than simply using the parent meta-rig bone (ORG) as parent for controls and chain mechanisms, the new system includes an interface for parent rigs. It explicitly provide parent bones and generate controlparent automation mechanisms for their child chain controls by inheriting from the appropriate base and overriding methods.

This allows implementing rigs that integrate and manage their child chains in intelligent ways in order to add extra automation specific to certain areas. The base skin system includes one simple example skin.transform.basicrig, which translates its child control points according to its control bone transformation.

Finally, the new system provides ways to integrate with custom automation directly included in the meta-rig via two extra rig components.

The skin.anchorrig generates a single control with inherited constraints etc., similar to basic.super_copy. However, it also integrates into the skin system as a zero length chain with highest priority. This allows overriding the normal behavior by providing a control point under full control of the user, which other chains would automatically attach to.

The skin.gluerig on the other hand will attach itself to the control that is generated at its position (it is an error if there is none). It can be used to read the position of the control from custom rigging in the meta-rig, or inject constraints into the control bone. It is possible to also detect the control at the tail of the glue bone and use it as

target in the constraints, thus copying transformation between the controls.

This is the basic chain rig, which bridges controls with B-Bones but does not add any automation to the controls themselves.

When controls are merely moved, the chains behave as if using standard automatic handles, but rotating and optionally scaling the controls will adjust the result.

Specifies the number of segments to use. Setting this to 1 disables all advanced behavior and merely bridges the points with a Stretch To bone. This can be enabled to let the chain respond to rotation and scale induced by parents of controls owned by other chains that this chain's control merged into.

Enables using control scale to drive scale and/or easing of the B-Bone. Specifies whether the ends of the chain should smoothly connect when merging controls with its .T / .B.L / .R symmetry counterpart. The relevant option must be enabled on both chains to work.

Specifies whether the end of the chain should connect to the opposite end of a different chain when merging controls. Thus forming a continuous smooth chain in the same direction. The relevant options must be enabled on both chains.

Specifies whether the rig should generate a mechanism to form a sharp corner at the relevant connected end, depending on the angle formed by adjacent control locations. When the control angle becomes sharper than the specified value, ease starts reducing from 1 to 0.

Specifies that the controls should be oriented the same as the selected bone, rather than being aligned to the chain.

Copy to selected rigs that have the same option. Thus allowing to indiscriminately select bones without assigning unnecessary values. Allows overriding the heuristic used to select the primary owner when merging controls.

This rig extends the basic chain with automation that propagates movement of the start and end, and an optional middle control, to other controls. This results in stretching the whole chain when moving one of the ends, rather than just the immediately adjacent B-Bones.

Specifies the position of the middle control within the chain; disabled when zero.

Specifies the influence falloff curves of the start, middle and end controls. Zero results in linear falloff, increasing widens the influence, and -10 disables the influence propagation from that control completely.

Toggle buttons to change the shape of the falloff curve from a power curve that at falloff 1 forms a parabola $((1 - x^{2f}))$ to a curve forming a circle $((1 - x^{2f})^{2{-f}})$.

Computes the falloff curve along the length of the chain, instead of projecting on the straight line connecting its start and end points.

Specifies whether twist of the chain should be propagated to control points between main controls.

Specifies whether perpendicular scaling of the chain should be propagated to control points between main controls.

Allows other chains to see propagated twist and scale via Merge Parent Rotation and Scale when their controls are merged into this chain, instead of it being completely local to this chain.

Optionally specifies bone collections for the end controls.

Optionally specifies bone collections for the middle control, falling back to Primary Control Layers if not set.

The main controls with active falloff have the effect of Merge Parent Rotation and Scale automatically enabled just for them.

This rig effectively acts as a zero-length chain with highest priority, ensuring that it becomes the owner when merging controls with other chains. And also allowing one to input custom automation influence into the skin system.

All constraints on the meta-rig bone are moved to the created control.

Creates a deformation bone parented to the control.

Makes the control a hidden mechanism bone to hide it from the user.

Selects which widget to generate for the control.

Operates the same as in basic.raw_copy , except all constraints are moved from ORG to the control bone.

Specifies the bone used to orient the control, like for other chains.

This rig is in concept similar to skin.anchor, but instead of overriding controls, it is used to read or adjust the state of controls generated by other rigs. The head of the bone must overlap a control of another skin rig.

The rig sets up its ORG bone to read the state of the control, while moving all constraints that were originally on the bone to the control.

Specifies how the ORG bone is connected to the skin control.

Makes the ORG bone a child of the control bone.

Makes the ORG bone a sibling of the control with a Copy Transforms constraint from the control. The resulting local space transformation is the same as control's local space.

Parents the ORG bone to the parent automation a control owned by the glue rig would have had, while making it follow the actual control. This includes both direct and parent-induced motion of the control into the local space transformation of the bone.

Other than adding glue constraints to the control, the rig acts as a one segment basic deform chain. This is convenient when a pair of controls need to be bridged both with glue and a deform bone.

Operates the same as in basic.raw_copy , except all constraints are moved from ORG to the control bone.

Relinks TARGET or any constraints with an empty target bone and no relink specification to reference the control located at the tail of the glue bone.

Switches the tail target to operate similarly to Mirror With Parents.

Allows to add a typical glue constraints with specific Influence, as if it were at the start of the ORG bone constraint stack.

This rig provides a simplistic parent controller , which uses regular translation, rotation, or scale to modify locations but not orientations or scale of its child chain controls.

Specifies whether to generate a visible control, or use the transformation of the ORG bone as a part of more complex and specific rig setup.

--- Fin du chapitre: _skin ---

--- Début du chapitre: _skin_resize ---Reference

Edit Mode

Mesh Transform Skin Resize
Ctrl- A

This tool is used to set a skin radius per vertex when using a Skin Modifier. Non-uniform scaling of the X and Y axes is accessible by the usual axis locking with X or Y. The radius can also be adjusted in the

Transformpanel ofthe Sidebar, in the 3D Viewport. The mean radius of all vertices can be previewedin the 3D Viewport as a dashed circle around a vertex.

Simple creature, made with only the Skin and Subdivision Surface modifiers.\xc2\xba6

--- Fin du chapitre: _skin_resize ---

--- Début du chapitre: _sky --- The Sky Texturenode generates a procedural sky. It\xe2\x80\x99s typically used in combinationwith theWorld Output Node.

Texture coordinate to sample texture at;defaults to Generated texture coordinates if the socket is left unconnected.

Sky model to use.

Based on the 1999 paperby Preetham et al.

Based on the 2012 paperby Hosek and Wilkie.

Improved version of the 1993 modelby Nishita et al.

Note that this sky type is quite bright and makes the image look overexposed with the default scene settings.You can reduce the Exposure setting in Properties \xe2\x80\xa3 Render \xe2\x80\xa3 Filmto fix this.

Sun direction vector.

Atmospheric turbidity.

2: Arctic like

3: clear sky

6: warm/moist day

10: hazy day

Amount of light reflected from the planet surface back into the atmosphere.

Enable/Disable sun disc lighting.

Angular diameter of the sun disc (in degrees).

Multiplier for sun disc lighting.

Rotation of the sun from the horizon (in degrees).

Rotation of the sun around the zenith (in degrees).

The distance from sea level to the location of the camera.For example, if the camera is placed on a beach then a value of 0 should be used.However, if the camera is in the cockpit of a flying airplane then a value of 10 km will be more suitable.Note, this is limited to 60 km because the mathematical model only accountsfor the first two layers of the earth\xe2\x80\x99s atmosphere (which ends around 60 km).

Density of air molecules.

0 no air

1 clear day atmosphere

2 highly polluted day

Density of dust and water droplets.

0 no dust

1 clear day atmosphere

5 city like atmosphere

10 hazy day

Density of ozone molecules;useful to make the sky appear bluer.

0 no ozone

1 clear day atmosphere

2 city like atmosphere

Texture color output.

Example of Sky Texture. \xc2\xba6

--- Fin du chapitre: _sky ---

--- Début du chapitre: `_slice_string` --- The Slice Stringnode extracts a string segment from a larger string.
String value to be sliced.
Integer value used to determine the starting point of the new string within the input string.The first letter of the string is at index 0.
Integer value used to determine how many characters are extracted from the input string.
This node has no properties.
String value of the extracted substring.
--- Fin du chapitre: `_slice_string` ---

--- Début du chapitre: `_slider` ---Reference
Physics \xe2\x80\xa3 Rigid Body Constraint
Slider
The Slider constraint allows relative translation along the X axis of the constraint object,but permits no relative rotation, or relative translation along other axes.
Enables/disables limit translation around X axis.
Lower limit of X axis translation.
Upper limit of X axis translation.
--- Fin du chapitre: `_slider` ---

--- Début du chapitre: `_slide_curves` --- Slides the curves along the surface mesh. This tool requires the curve to have aSurface.Each curve is also rotated by the change in the surface normal.
--- Fin du chapitre: `_slide_curves` ---

--- Début du chapitre: `_slide_relax` ---Reference
Sculpt Mode
Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3 Advanced \xe2\x80\xa3 Brush Type
This brush deforms the topology of the meshwhile minimizing changes to the geometrical shape of the mesh.By default it will drag geometry, but this can be changed in the Deformationsettings.
This brush is especially useful for redistributing topology to areas that require more detail,or sliding geometry to somewhere where they should be.
Holding Shiftchanges the brush effect to Relax geometry,creating an even distribution of topology.
Note
More info at General brush settingsand on Advanced brush settings.
Deformation type that is used by the brush.
Slides the topology of the mesh in the direction of the stroke.
Slides the topology of the mesh towards the center of the stroke.
Slides the topology of the mesh away from the center of the stroke.
--- Fin du chapitre: `_slide_relax` ---

--- Début du chapitre: `_slide_vertices` ---Reference
Edit Mode
Vertex \xe2\x80\xa3 Slide Vertices
Shift- V , or Gtwice
Vertex Slide will transform a vertex along one of its adjacent edges.UseShift- Vto activate tool.The nearest selected vertex to the

mouse cursor will be the control one. Move the mouse along the direction of the desired edge to specify the vertex position. Then press LMB to confirm the transformation.

By default, the offset value of the vertices is a percentage of the edges length along which they move. When Even mode is active, the vertices are shifted by an absolute value.

When Flipped is active, vertices move the same distance from adjacent vertices, instead of moving from their original position.

Toggle clamping the slide within the edge extents.

Selected vertex. \xc2\xba

Positioning vertex interactively. \xc2\xba

Repositioned vertex. \xc2\xba

--- Fin du chapitre: _slide_vertices ---

--- Début du chapitre: _smear ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Smears the painted colors of the active color attribute. It takes the colors under the cursor, and blends them in the direction of your stroke.

Note

More info at General brush settings and on Advanced brush settings.

Smear colors along the direction of the stroke.

Smear colors inwards towards your brush center.

Smear colors outwards away from your brush center.

--- Fin du chapitre: _smear ---

--- Début du chapitre: _smooth --- The SmoothModifier changes the value of one or more stroke/points properties like: location, strength, thickness or UV texture position trying to maintain similar values that make the line fluid and smoother.

Smooth Modifier. \xc2\xba

Combination of stroke/points properties that will be affected by the smooth factor.

Smooth affect the point location.

Smooth affect the point strength (opacity).

Smooth affect the point thickness.

Smooth affect the point UV rotation.

Strength of the smooth effect.

The number of smoothing iterations, equivalent to executing the Smooth tool multiple times. High values can reduce the animation performance (FPS).

When enabled, the smoothing algorithm will try to maintain as close as possible the overall shape of the original stroke.

Smooth the ends of the strokes.

See Influence Filters.

--- Fin du chapitre: _smooth ---

--- Début du chapitre: _smooth_by_angle --- Set the sharpness of mesh edges based on the angle between the neighboring faces.

Note

This is a node group asset that is included in the bundled

\xe2\x80\x9cEssentials\xe2\x80\x9d asset library .

Standard geometry input.

Maximum angle between face normals that will be considered as smooth.
Smooth all edges, even if they have been marked as sharp.
This node has no properties.
Standard geometry output.

--- Fin du chapitre: `_smooth_by_angle` ---

--- Début du chapitre: `_smooth_curves` --- This brush makes curve segments close to one another more parallel.

--- Fin du chapitre: `_smooth_curves` ---

--- Début du chapitre: `_smooth_hair_curves` --- Smooths the shape of hair curves.

Input Geometry (only curves will be affected).

Amount of smoothing. Negative values will result in crumpling the curves.

Shape of the influence along curves (0=constant, 0.5=linear).

Amount of smoothing steps.

Weight used for smoothing.

Lock tip position when smoothing.

Preserve each curve's length during deformation.

This node has no properties.

Geometry

--- Fin du chapitre: `_smooth_hair_curves` ---

--- Début du chapitre: `_smooth_vertices` ---Reference

Edit Mode

Vertex `Smooth Vertices`, Context Menu `Smooth Vertices`

This operator smooths the selected vertices by averaging the angles between the faces.

Smoothing factor.

The number of smoothing iterations.

Limit the effect to certain axes.

Mesh before smoothing. `Mesh before smoothing`

Mesh after one smoothing iteration. `Mesh after one smoothing iteration`

Mesh after ten smoothing iterations. `Mesh after ten smoothing iterations`

Tip

Using the Smooth tool after subdividing can help create a more organic shape.

See also

Smooth Modifier

The Smooth Modifier, which can be limited to a Vertex Group, is a non-destructive alternative to the Smooth operator.

--- Fin du chapitre: `_smooth_vertices` ---

--- Début du chapitre: `_snake_hook` ---Reference

Sculpt Mode

Sidebar `Tool` `Brush Settings` `Brush Type`

Pulls vertices along with the movement of the brush to create long, snake-like forms. During the stroke, geometry will be dynamically picked up & let go.

When the Rakesetting is used, the brush can also be used to rotate geometry via dragging.

Note

More info at General brush settings and on Advanced brush settings.

Pulled geometry tends to lose volume along the stroke. With Magnify value greater than 0.5 this is prevented. More info at Pinch/Magnify

Rotates geometry along the direction of the stroke.

Deformation type that is used by the brush.

Applies the brush falloff to the tip of the brush.

Modifies the entire mesh using an Elastic deformation. More info in the Elastic Deform brush.

--- Fin du chapitre: _snake_hook ---

--- Début du chapitre: _snake_hook_curves --- Extend existing curves in a specific direction of the brush strokes. A curve can only be extended by its tips (the control point opposite to the root).

Note

No new control points are created for the curves. Instead, existing points are sampled along the new curve shape.

--- Fin du chapitre: _snake_hook_curves ---

--- Début du chapitre: _snap --- Reference

Object, Edit, and Pose Mode

Object/Object type \xe2\x80\xa3 Snap

Shift- S

The Snap menu (also available from the 3D header in both Object Mode and Edit Mode Object \xe2\x80\xa3 Snap and Mesh \xe2\x80\xa3 Snap). This menu provides a number of options to move the cursor or your selection to a defined point (the cursor, selection or the grid).

Snaps the currently selected object(s) to the nearest grid point.

Moves each one of the currently selected object(s) to the cursor location.

Places the selection at the position of the 3D cursor. If there are multiple objects selected, they are not moved individually at the cursor position; instead, they are centered around the 3D cursor, maintaining their relative distances.

Moves the selection to the origin of the active object.

Places the cursor to the center of the current selection, unless see below.

Places the cursor to the origin of the world (location 0, 0, 0).

Places the cursor to the nearest grid point.

Places the cursor to the origin of the active (last selected) object.

The Cursor to Selected option is also affected by the current Pivot Point.

For example:

With the Bounding Box Center pivot point active, the Cursor to Selected option will snap the 3D cursor to the center of the bounding box surrounding the objects \xe2\x80\x99 origins.

When the Median Point pivot point is selected, Cursor to Selected will snap the 3D cursor to the median of the object origins.

--- Fin du chapitre: _snap ---

--- Début du chapitre: _snapping --- The icon toggles snapping; you can also do this temporarily by holding Ctrl after starting to transform an image.

Images have multiple snap points; they can snap along their edges, corners, or center.

The drop-down arrow offers the following options:

Snap images to the edges of the render region.
Snap images to the horizontal and vertical center lines of the render region.

Snap images to the snap points of other images.

--- Fin du chapitre: _snapping ---

--- Début du chapitre: _snap_symmetry ---Reference

Edit Mode

Mesh \xe2\x80\xa3 Snap to Symmetry

The Snap to Symmetry tool lets you snap a mesh vertices to their mirrored neighbors.

Useful when dealing with meshes which are mostly symmetrical, but have vertices which have been moved enough that Blender does not detect them as mirrored (when X Mirror option is enabled for example).

This can be caused by accident when editing without X Mirror enabled. Sometimes models imported from other applications are asymmetrical enough that mirror fails too.

Specify the axis and direction to snap. Can be any of the three axes, and either positive to negative, or negative to positive.

Specify the search radius to use when finding matching vertices.

Support for blending mirrored locations from one side to the other (0.5 is an equal mix of both).

Snap vertices along the center axis to zero.

Before Snap to Symmetry. \xc2\xbb

After Snap to Symmetry. \xc2\xbb

--- Fin du chapitre: _snap_symmetry ---

--- Début du chapitre: _soft_body --- The Soft Body modifier is a container for a Soft Body Physics simulation.

As the modifier is only a container its actual options are configured in the Physics Properties tab. See the Soft Body Physics Properties for more information.

The wind cone is a soft body, as the suspension. \xc2\xbb

--- Fin du chapitre: _soft_body ---

--- Début du chapitre: _solidify --- The Solidify modifier takes the surface of any mesh and adds depth, thickness to it.

The Solidify modifier in simple mode. \xc2\xbb

The Solidify modifier in complex mode. \xc2\xbb

This is the default solidify algorithm, which simply extrudes the geometry. This algorithm does not work on geometry where edges have more than two adjacent faces.

Important

If the normals of adjacent faces don't point into the same general direction, simple mode will not be able to solidify the boundary between those. This happens if the normals are not recalculated or for example on one-sided surfaces like a Möbius strip.

This is a solidify algorithm which can handle every geometric situation to guarantee a manifold output geometry. This algorithm is able to solidify shapes like Möbius strips, Klein bottles, architectural wall layouts and many more which the Simple Mode isn't able to do. If the special cases are not present it is recommended to choose Simple because the extra logic makes this algorithm much slower.

Note

There are no options for crease in the Modifier tab because crease is handled in a dynamic way. The modifier will transfer the creases of the original mesh in a smart way to the output mesh to work with the Subdivision Surface modifier.

Choose the kind of thickness handling (thickness solver).

Different thickness options on a non-manifold mesh.  This is similar to Simple Mode without Even Thickness. The new vertices are always in a fixed distance to the old ones.

This is similar to Simple Mode with Even Thickness and High Quality Normals. It adjusts for sharp corners, but may not always work when more than three faces come together.

This is a more advanced model to try to always get the optimal thickness everywhere. For up to three faces it is always guaranteed to find an optimal solution.

Choose the kind of boundary that suits the model the most.

Different boundary options with a matCap. 

No boundary fix is applied. Results are stable.

Adjusts the boundary for an opening to face inwards (like a hole in an egg).

Adjusts the boundary of a planar opening to be a flat (like a cut sphere).

The depth to be solidified.

Important

The modifier thickness is calculated using local vertex coordinates. If the object has a non-uniform scale, the thickness will vary on different sides of the object.

To fix this, either Apply or Clear the scale.

A value between (-1 to 1) to locate the solidified output inside or outside the original mesh. The inside and outside is determined by the face normals. Set to 0.0, the solidified output will be centered on the original mesh.

Maintain thickness by adjusting for sharp corners. Sometimes improves quality but also increases computation time.

Distance within which degenerated geometry is merged.

Fills the gap between the inner and outer edges.

In Simple Mode: Will not extrude surfaces parallel to the original one, but instead will only add the perpendicular rim.

In Complex Mode: Will only leave the generated perpendicular rim.

Note

Fill and Only Rim only make a difference on non-manifold objects, since the rims are generated from the borders of the original geometry.

The weights of the selected vertex group are multiplied onto the Thickness, so vertices with lower weights will be less thick. The vertices which are not part of the vertex group will be used as if their weight was zero.

Reverses the vertex group weights, so that the used weight is one minus the actual weight.

How much the vertex weights are taken into account.

On 0.0, vertices with zero weight will have no thickness at all.

On 0.5, vertices with zero weight will be half as thick as those with full weight.

On 1.0, the weights are ignored and the Thickness value is used for every vertex.

Use the minimal vertex weight assigned to the vertices of a face to make sure that new faces stay parallel to their original ones. This is slow, so disable it when it is not needed.

Note

If the final thickness of a vertex is zero, it will still be solidified. Therefore creating duplicate geometry, which sometimes needs extra care.

Reverse the normals of all geometry (both the inner and outer surfaces).

Normals are calculated to produce a more even thickness. Sometimes improves quality but also increases computation time.

Choose a different material slot index to use for the new geometry. This is applied as an offset from the original material of the face from which it was solidified.

A value of 0 means it will use the same material.

A value of 1 means it will use the material immediately below the original material.

A value of -2 means the material two positions above the original material will be used.

These are clamped to the top-most and bottom-most material slots.

Similarly, you can give another material to the rim faces.

Set `creaseTo` to the inner edges.

Set `creaseTo` to the outer edges.

Set `creaseTo` to the rim.

Edge bevel weight to be added to outside edges.

Edges which will get creases marked.

A value between (0 to 2) to clamp offsets to avoid self-intersection. The amount is determined by the length of the shortest adjacent edge.

Clamp Offset.

If enabled clamping will also consider angles in the geometry, not only lengths.

Vertex group that the generated shell geometry will be weighted to. This allows you to use other modifiers to only affect the shell geometry by using a that modifier's vertex group influence control.

Same as Shell Vertex Group, but for the generated rim geometry.

Solidify thickness is an approximation. While Even Thickness and High Quality Normal should yield good results, the final wall thickness is not guaranteed and may vary depending on the mesh topology. Especially for vertices with more than three adjacent faces.

In order to maintain a precise wall thickness in every case, we would need to add/remove faces on the offset shell, something this modifier does not do since this would add a lot of complexity. The best option to preserve wall thickness is complex mode with constraints thickness mode, but it is also not guaranteed to work perfect in every case.

--- Fin du chapitre: `_solidify` ---

--- Début du chapitre: `_solidify_faces` ---Reference

Edit Mode

Face `Solidify Faces`

This takes a selection of faces and solidifies them by extruding them uniformly to give volume to a non-manifold surface. This is also available as a modifier. After using the tool, you can set the offset distance in the Adjust Last Operation panel.

Amount to offset the newly created surface. Positive values offset the surface inward relative to the normals direction. Negative values offset outward.

Mesh before solidify operation. \xc2\xba6

Solidify with a positive thickness. \xc2\xba6

Solidify with a negative thickness. \xc2\xba6

--- Fin du chapitre: _solidify_faces ---

--- Début du chapitre: _solve ---See Create Plane Track .

Tripod Motion can be used for footage where the camera does not move and only rotates. Such footage can't be tracked with a generic solver approach, and it is impossible to determine the actual feature points in space due to a lack of information. So this solver will solve only the relative camera rotation and then reproject the feature points into a sphere, with the same distance between feature and camera for all feature points.

Note

This is special type of camera solver and it behaves different from regular solver. It means using more tracks doesn't imply more accurate solution. Having 5-10 tracks on frame is likely what shall be commonly used for this kind of solver.

Automatically select keyframes for initial reconstruction. This option enables complex algorithms which tries to find a keyframe pair with minimal reconstruction error and best scene scale guess.

Start (A) and End (B) frame of the range used for reconstruction.

Specifies which parameters should be refined during solve. Such refining is useful when you are not sure about some camera intrinsics, and solver should try to find the best parameter for those intrinsics. But you still have to know approximate initial values \xe2\x80\x93it will fail to find correct values if they were set completely incorrectly initially.

Refine the camera's Focal Length .

Refine the camera's Optical Center .

Refine the camera's Radial Distortion Parameters .

Refine the camera's Tangential Distortion Parameters .

See Solve Solution .

This panel contains operators and their settings which are needed to clean up bad tracks: tracks which are not tracked long enough or which failed to reconstruct accurately.

Tracks or tracked segments shorter than this number of frames will be removed.

Tracks which has reprojection error higher than this value will be removed.

Several actions can be performed for bad tracks:

They can simply be selected.

The whole track can be deleted.

Bad segments of tracked sequence can be removed.

See Clean Tracks .

See Filter Tracks .

See 3D Markers to Mesh .

See Link Empty to Track .

Scene orientation tools can be used for orienting object to bundles.

See Set Origin .

See Set Floor .

See Set Floor .

See Set X/Y Axis .

See Set Scale .

See Apply Solution Scale .

Distance in active scene units which is used by Set/Apply scale.

See Set as Background .

See Setup Tracking Scene .

--- Fin du chapitre: _solve ---

--- Début du chapitre: _solver ---Reference

Physics \xe2\x80\xa3 Soft Body \xe2\x80\xa3 Solver

The settings in the Soft Body Solverpanel determine the accuracy of the simulation.

Minimum simulation steps per frame. Increase this value, if the soft body misses fast-moving collision objects.

Maximum simulation steps per frame. Normally the number of simulation steps is set dynamically (with the Error Limit) but you have probably a good reason to change it.

Use velocities for automatic step sizes. Helps the Solver figure out how much work it needs to do based on how fast things are moving.

Rules the overall quality of the solution delivered. The most critical setting that defines how precise the solver should check for collisions. Start with a value that is half the average edge length. If there are visible errors, jitter, or over-exaggerated responses, decrease the value. The solver keeps track of how \xe2\x80\x9cbad\xe2\x80\x9d it is doing and the Error Limit causes the solver to do some \xe2\x80\x9cadaptive step sizing\xe2\x80\x9d.

Prints on the console how the solver is doing.

Estimate matrix, split to COM , ROT , SCALE .

These settings control how the soft body will react (deform) once it either gets close to or actually intersects (cuts into) another collision object on the same layer.

Calms down (reduces the exit velocity of) a vertex or edge once it penetrates a collision mesh.

Fuzziness while on collision, high values make collision handling faster but less stable. Simulation is faster, but less accurate.

--- Fin du chapitre: _solver ---

--- Début du chapitre: _sort_elements --- The Sort Elements node rearranges geometry elements by changing their indices.

Standard geometry input.

The selection of elements to sort, if left blank, all elements are sorted. Non selected elements will keep their current indices.

Elements with the same group ID are sorted together. If this is not a field, the node has no affect.

The sorted values used to do the reordering. If this is not a field, the node has no affect.

The domain on which the selection and group ID fields are evaluated.

The fields are evaluated on points, control points, and vertices.

The fields are evaluated on the edges of the mesh component.

The fields are evaluated on the faces of the mesh component.

The fields are evaluated on the splines in the curve component.

The fields are evaluated on the top-level instances. Realized instances are ignored.

Standard geometry output.

--- Fin du chapitre: `_sort_elements` ---

--- Début du chapitre: `_sound` --- As well as images and movies the Video Sequencer can also edit audio tracks. You can add Waveform Audio format WAV, mp3 and other audio formats files from your drive, or from sound encoded within a movie, and mix them using an F-Curve as a volume control.

Example of sound editing.

A Sound strip is just like any other strip in the Video Sequencer. You can select and move it, adjust its starting offset using LMB over the strip handles, and Kcut it into pieces. A useful example is cutting out the

silence and dead voice time. You can have as many Sound strips as you wish and the result will be the mixing of all of them. You can give each strip its own name and volume via the Sidebar region.

Overlapping strips are automatically mixed down during the rendering process. For example, you can have the announcer on channel 5, background music on channel 6, and Foley sound effects on channel 7.

See also

In the Playback Popover menu of the Timeline you will find some options concerning audio playback behavior.

The waveform of the audio is shown depending on two options:

The Sequencer Overlay menu has options to show all strip waveforms, none of them, or to use the per-strip options described below.

Each strip has an option Display Waveform. It is only visible when the above overlay option is set to Use Strip Option.

Clipping audio, i.e. values over 100% amplitude, will be shown in red in the waveform.

More strip options are documented in Sound Sidebar Panel .

To animate Sound strips simply hit I over any of its values. Examples of animating an audio strip are to fade in/out background music or to adjust volume levels. Layered/crossed Sound strips are added together; the lower channel does not override and cut out higher channels (unlike image and video strips). This makes Blender an audio mixer. By adding audio tracks and using the curves to adjust each tracks sound level, you have an automated dynamic multi-track audio mixer!

See also

Sounds can be cross-faded by adding a Sound Crossfade effect.

There are two ways to render out your audio. You can either have it encoded with a video file or in its own audio file. Read more on how to select a proper audio format and how to start rendering .

Reference

Add `Sound`

Store the location of the image file relative to the blend-file.

The Start Frame to place the left handle of the strip.

The Channel to place the strip.

Replaces the currently selected strips with the new strip.

Cache the sound in memory, enables Caching in the Source properties.

Merge all sound channels into one channel, enables Mono in the Sound properties.

--- Fin du chapitre: `_sound` ---

--- Début du chapitre: `_sound_crossfade` --- The Sound Crossfade transition works by animating the Volume of two overlapping Sound strips to evenly

fade between them. Because this simply animates a value it does not create a strip like other effects or transitions.

--- Fin du chapitre: `_sound_crossfade` ---

--- Début du chapitre: `_spatial_noise` --- The Spatial Noisemodifier adds some spatial noise to the stroke. Spatial noise displacements are added in the normal direction (i.e. the direction perpendicular to the tangent line) evaluated at each stroke vertex.

How much the noise distorts the stroke.

How wide the noise is along the stroke.

The level of detail of the noise.

When enabled, apply some smoothing over the generated noise.

When disabled, the next generated random value depends on the previous one; otherwise they are completely independent. Disabling this setting gives a more consistent noise along a stroke.

--- Fin du chapitre: `_spatial_noise` ---

--- Début du chapitre: `_speaker` --- Speaker objects.

The speaker object is used to give sound in the 3D Viewport. After adding the object, the various settings can be changed in the Properties.

The Data-Block Menu for loading audio files. There are two properties you can check when loading a sound:

This means the whole sound will be decoded and the raw audio data will be buffered in memory, which results in faster playback, but uses quite a lot of memory. So this should be used for short sound effects that get played more often, but not for longer audio files such as music.

For any 3D audio or panning effects the sound source has to be single channel, otherwise it's assumed that the 3D audio and panning information is already present in the multichannel file. Enable this if you want to use those effects for a file with multiple channels.

Toggles whether or not the sound can be heard.

Adjust the loudness of the sound.

Can be used to bend the pitch of the sound to be either deeper or higher. This basically changes the playback speed of the sound which also results in a pitch change.

There is no setting to choose the start time when the speaker should start playing, because you might want a single speaker to play multiple times. Therefore, you have to open the NLA Editor where you can add Sound strips that define when the sound should start (nothing else, so any other properties of the strips, like length don't matter). When you add a speaker object such a strip will be added at the current frame.

Speaker properties.

Distance attenuation relevant settings.

No matter how far away/close the object is, the distance-based volume won't be lower/higher than this value.

How strong the distance affects the volume. This factor sets the strength of the distance-based volume change, depending on the chosen distance model (see scene settings).

If the object is farther away than this distance, this distance is used to calculate the distance-based volume. Influence of this value also depends on the distance model.

The distance at which the volume is full (1.0). Set this value to the distance used for recording the sound. Usually sound effects recordings should be made exactly 1 m away from sound to get an accurate volume.

Directionality relevant settings.

Imagine a cone with the top at the original of the speaker object and the main axis of it facing in the same direction as the speaker. There are two cones an inner and an outer cone. The angles represent their opening angles, so 360° mean the cone is fully open and there is no directionality anymore. Inside the inner cone the volume is full (1.0), outside the outer cone the volume is, whatever one sets for the outer cone volume and the volume between those two cones, linearly interpolated between this two volumes.

Angle of the outer cone in degrees. Outside this cone, the volume is equal to the Outer volume.

Angle of the inner cone in degrees. Inside the cone, the volume is full.

Volume outside the outer cone.

--- Fin du chapitre: _speaker ---

--- Début du chapitre: _special_characters --- The Special Characters node is used to output string characters that can't be typed directly with the keyboard.

Tip

This node can be used to create a multi-line string for the String to Curves Node, when combined with the Join Strings Node or the Replace String Node.

This node has no inputs.

This node has no properties.

A new line character (escape character \n).

A tab character used to add an indentation in the output.

--- Fin du chapitre: _special_characters ---

--- Début du chapitre: _specular_bsdf --- EEVEE Only

The SpecularBSDF combines multiple layers into a single easy to use node.

It is similar to the Principled BSDF node but uses the specular workflow instead of the metallic. It has far fewer parameters and supports less features. Both might be merged into one node in the future.

The specular workflow functions by specifying the facing (along normal) reflection color. The result may not be physically plausible because there is no energy conservation.

Diffuse surface color. For conductor materials (metals) it should be black.

Amount of specular reflection. Specifies facing (along normal) reflectivity. Conductor materials (metals) can have colored specular reflection.

Hint

To compute this value for a realistic material with a known index of refraction, you may use this special case of the Fresnel formula:

$$f_s = \left(\frac{ior - 1}{ior + 1} \right)^2$$

For example:

water: $ior = 1.33$, $specular = 0.02$

glass: $ior = 1.5$, $specular = 0.04$

diamond: $ior = 2.417$, $specular = 0.17$

Specifies microfacet roughness of the surface for diffuse and specular reflection.

Hint

When converting from the older Glossy BSDF node, use the square root of the original value.

Color of the emitted light. This light is added to the BSDF result.
 Transparency factor. This is the inverse of the alpha channel (1 - alpha) you find in an image. Use an Invert node to convert alpha to transparency. This will only have an effect if the material uses a blend mode other than opaque.
 Controls the normals of the base layers.
 Extra white specular layer on top of others. This is useful for materials like car paint and the like.
 Roughness of clear coat specular.
 Controls the normals of the Clear Coat layer.
 Amount of occlusion to apply to indirect lighting. Usually a baked ambient occlusion map. The final occlusion factor is the minimum of this input and the runtime ambient occlusion effect.
 This node has no properties.
 Standard shader output.
 --- Fin du chapitre: _specular_bsdf ---

--- Début du chapitre: _speed_control --- Speed Control time-warps the strip, making it play faster or slower than it normally would. Playing faster means that some frames are skipped, and the strip will run out of frames before the end frame. When the strip runs out of frames to display, it will just keep repeating the last one action that will appear to be frozen. To avoid this, position the next strip under the original at a point where you want the motion to continue.
 The method used to adjust the speed of the strip.
 Automatically calculates the speed effect based on the length of the input strip. If you scale a strip to 1/2 the original size the sequence will play back at 2 times the speed.
 Multiplies the current speed of the sequence by a Multiply Factor. Thus a value of 0.5 will make the sequence half as fast while 2 would make the sequence twice as fast. Negative values will reverse the input while also adjusting the speed, so a value of negative two will play in reverse and twice as fast as normal.

Note

You will have to manually re-adjust the length of the strip accordingly. Specifies a frame to remap the current frame to, for example, setting the Frame Number value to 50 displays the 50th frame. This can then be manually keyframed to recreate the animation.

Maps the frame range on a percentage scale. For example, using this and a value of 50% will select the frame halfway through the sequence.

Crossfades between frames to reduce screen tearing when the speed is slower than the original frame rate.

Suppose you want to slow your strip down. You need to affect the speed of the video clip without affecting the overall frame rate. Select the clip and Add \xe2\x80\xa3 Effect \xe2\x80\xa3 Speed Control effect strip.

Choose the Multiply option in the Effect Strip panel in the Sidebar. Set the Multiply Factor to be the factor by which you want to adjust the speed. To cut the displayed speed by 50%, enter 0.5. Now, a 275-frame clip will play at half speed, and thus display only the first 137 frames.

If you want the remaining frames to show in slow motion after the first set is displayed, double the Length of the source strip (since effects strip bounds are controlled by their source strips). If you are using a speed factor other than 0.5 then use the formula:

$$\text{new_length} = \text{real_length} / \text{speed_factor}$$

That is it, set your render to animate (in this example) all 550 frames. Keyframing the Frame number.

To get even finer control over your clip timing, you can use curves! While it is possible to keyframe the Multiply factor, usually you want to keyframe the Frame number directly.

Choose the Frame Number option. You now have a Frame number field which you can keyframe. If you want the strip to animate at all you will have to insert some keyframes, otherwise it will look like a still. In most cases you will want to use the Graph editor view to set the curve interpolation to Linear since the default Bezier will rarely be what you want.

Tip

If you choose to keyframe the Speed factor instead, remember to Refresh All or the changes will not take effect.

You can use the speed control to change the frame rate in frames per second (fps) of a video. If you are rendering your video to a sequence set, you can effectively increase or decrease the number of individual image files created, by using a Multiply value less than or greater than one, respectively.

For example, if you captured a five-minute video at 30 fps and wanted to transfer that to film, which runs at 24 fps, you would enter a Multiply Factor of $30/24$, or 1.25 (and enable Interpolation frame blending to create a film blur effect). Instead of producing 5 60 30 = 9000 frames, Blender would produce $9000 / 1.25 = 7200 = 5 \times 60 \times 24$ frames. In this case, you set a start = 1 and end = 7200, set your Format output to jpeg30fps, and image files 0001.jpg through 7200.jpg would be rendered out, but those images cover the entire 9000 frames. The image file 7200.jpg is the same at frame 9000. When you read those images back into your film blend-file at 24 fps, the strip will last exactly 5 minutes.

--- Fin du chapitre: _speed_control ---

--- Début du chapitre: _sphere --- A light probe sphere records the light incoming from many directions at a single location.

They are used for smooth and semi-rough reflections. Sphere probes smoothly blend to light probe volume lighting for completely diffuse reflections.

If Raytracing is turned on, they are used as a fallback if a ray misses.

Note

In both usages, the light probe spheres are shadowed by light probe volume. This is done in order to reduce light leaking in shadowed areas and reduce the need to set up more light probe spheres.

Adjusting their resolution is done inside the Scene datapanel.

The world also has an internal light probe sphere with a resolution that can be adjusted in the World datapanel.

Reference

Object Data > Probe

Shape of the influence volume. Can be set to Sphere or Box.

A probe object only influences the lighting of nearby surfaces. This influence zone is defined by the size parameter and object scaling.

Percentage of the influence distance in which the influence of a probe fades linearly.

Note

In the viewport, capture only happens if an update is detected on the light probe data or position. For renders, the capture happens at the start of each frame.

Define the near and far clip distances when capturing the scene.

Reference

Object Data \xe2\x80\xa3 Custom Parallax

By default, the influence volume is also the parallax volume. The parallax volume is a volume on which the recorded light is projected. It should roughly fit its surrounding area. In some cases it may be better to adjust the parallax volume without touching the influence parameters. In this case, enable the Custom Parallax and change the shape and radius of the parallax volume independently.

Show the captured light using a reflective sphere of the given size.

Show the clipping distance in the 3D Viewport.

Show the influence bounds in the 3D Viewport. The inner sphere is where the falloff starts.

Show the Custom Parallax shape in the 3D Viewport.

--- Fin du chapitre: _sphere ---

--- Début du chapitre: _spin ---Reference

Edit Mode

Mesh \xe2\x80\xa3 Extrude \xe2\x80\xa3 Spin

Toolbar \xe2\x80\xa3 Spin

The Spintool extrudes (or duplicates it if the selection is manifold) the selected elements, rotating around a specific point and axis.

Use the tool to create the sort of objects that you would produce on a lathe (this tool is often called \xe2\x80\x99clathe\xe2\x80\x9d tool or \xe2\x80\x99csweep\xe2\x80\x9d tool in the literature, for this reason). In fact, it does a sort of circular extrusion of your selected elements, centered on the 3D cursor, and around the axis perpendicular to the working view\xe2\x80\xa6

The point of view will determine around which axis the extrusion spins\xe2\x80\xa6

The position of the 3D cursor will be the center of the rotation. Specifies how many copies will be extruded along the \xe2\x80\x99csweep\xe2\x80\x9d.

When enabled, will keep the original selected elements as separated islands in the mesh (i.e. unlinked to the result of the spin extrusion).

Specifies the axis to use as the pivot of the spin operation.

Specifies how many copies will be extruded along the \xe2\x80\x99csweep\xe2\x80\x9d.

Specifies the angle \xe2\x80\x99cswept\xe2\x80\x9d by this tool, in degrees (e.g. set it to 180 for half a turn).

Automatically merges the first and last duplicates, if they make a full revolution which results in overlapping geometry.

Reverses the Normal\xe2\x80\x99s direction for any resulting geometry.

Specifies the center of the spin. By default it uses the cursor position.

Specify the spin axis as a vector. By default it uses the view axis (viewport).

Glass profile. \xc2\xb6

First, create a mesh representing the profile of your object. If you are modeling a hollow object, it is a good idea to thicken the outline. Fig. Glass profile. shows the profile for a wine glass we will model as a demonstration.

We will be rotating the object around the cursor in the top view, so switch to the top view with Numpad7.

Glass profile, top view in Edit Mode, just before spinning. \xc2\xba6 Place the cursor along the center of the profile by entering Edit Mode and selecting one of the vertices along the center, and snapping the 3D cursor to that location with Mesh \xe2\x80\xa3 Snap \xe2\x80\xa3 Cursor to Selection. (Fig. Glass profile, top view in Edit Mode, just before spinning.) shows the wine glass profile from top view, with the cursor correctly positioned.

Select all the vertices with A and select the Spin tool from the Toolbar and use the Gizmo to spin the vertices. Fig. Spun profile. shows the result of a successful spin.

Spun profile using an angle of 360. \xc2\xba6

Spun profile using an angle of 120. \xc2\xba6

Result of spin operation. \xc2\xba6

Result of Duplicate enabled. \xc2\xba6

Duplicate vertices. \xc2\xba6

The spin operation leaves duplicate vertices along the profile. You can select all vertices at the seam with Box select B (shown in Fig. Duplicate vertices.) and perform a Merge by Distance operation.

Notice the selected vertex count before and after the Merge by Distance operation. Vertex count after removing doubles. If all goes well, the final vertex count (38 in this example) should match the number of the original profile noted in Mesh data \xe2\x80\xa3 Vertex and face numbers. If not, some vertices were missed and you will need to weld them manually. Or, worse, too many vertices will have been merged.

Note

Merging Two Vertices into One

To merge (weld) two vertices together, select both of them by Shift- LMB clicking on them. Press S to start scaling and hold down Ctrl while scaling to scale the points down to 0 units in the X, Y and Z axis. LMB to complete the scaling operation and click Mesh \xe2\x80\xa3 Merge \xe2\x80\xa3 By Distance to merge the vertices. Alternatively, you can use Context Menu \xe2\x80\xa3 Merge Vertices (or M). Then, in the new pop-up menu, choose to merge By Distance.

All that remains now is to recalculate the normals to the outside by selecting all vertices, pressing Alt- N and validating Recalculate Normals Outside in the pop-up menu.

--- Fin du chapitre: _spin ---

--- Début du chapitre: _spines --- These rigs are used to generate spine structures, including the head and tail.

Will create a complete bendy and stretchy b-bones spine system based on bone numbers of your bone chain and user defined options.

This is a composite wrapper of `spines.basic_spine`, `spines.super_head` and `spines.basic_tail`. Note that for the tail, the direction of the bones is reversed compared to the separate rig.

Requirement: A chain of at least three connected bones (base system).

Spine required bones. \xc2\xba6

Defines the pivot position for torso and hips.

When checked neck and head systems will be added to your spine rig.

Defines the bone where the neck system starts. The last bone will always be the head system. If neck position is the last bone of the chain, then only the head system will be created ignoring the neck.

When checked tail system will be added to your spine rig.
 Defines the bone where the tail system starts. The next bone will always be the hips system.
 When generating a tail, specifies which local axis rotations should be replicated along the chain.
 If enabled, allows placing the Tweak controls in different bone collections from the IK bones.
 Spine default bones. \xc2\xbf
 Spine with tail bones. \xc2\xbf
 Defines a bendy and stretchy b-bones spine.
 Defines the pivot position for torso and hips.
 If enabled, allows placing the Tweak controls in different bone collections from the IK bones.
 Specifies whether to generate an FK control chain.
 If enabled, allows placing the FK chain in different bone collections from the IK bones.
 Defines a bendy and stretchy b-bones tail.
 Specifies which local axis rotations should be replicated along the chain from each controlbone to the following one.
 If enabled, allows placing the Tweak controls in different bone collections from the IK bones.
 Defines a head rig with follow torso controls.
 If enabled, allows placing the Tweak controls in different bone collections from the IK bones.
 --- Fin du chapitre: _spines ---

--- Début du chapitre: _splash --- When starting Blender, the splash screen appears in the center of the window. It contains options to create new projects or open recent ones. A more detailed description can be found below.
 Blender Splash Screen. \xc2\xbf
 To close the splash screen and start a new project, click anywhere outside the splash screen (but inside the Blender window) or press Esc. The splash screen will disappear revealing the default screen. To reopen the splash screen, click on the Blender icon in the Topbar and select Splash Screen.
 Note
 When starting Blender for the first time or updating to a new version, the \xe2\x80\x9cinteractive region\xe2\x80\x9d contains a Quick Set Up Process .
 The upper part of the splash screen contains the splash image with the Blender version in the top right.
 The interactive region is the bottom half of the splash screen.
 Start a new project based on a template.
 Your most recently opened blend-files. This gives quick and easy access to your recent projects.
 Allows opening an existing blend-file.
 Blender will try to recover the last session based on temporary files.
 See Recovering Data .
 Open Blender\xe2\x80\x99s Development Fundwebsite.
 Open the latest release notes.
 --- Fin du chapitre: _splash ---

--- Début du chapitre: `_spline_ik` --- The Spline IKconstraint aligns a chain of bones along a curve. By leveraging the ease and flexibility of achieving aesthetically pleasing shapes offered by curves and the predictability and well-integrated control offered by bones, Spline IK is an invaluable tool in the riggers' toolbox. It is particularly well suited for rigging flexible body parts such as tails, tentacles, and spines, as well as inorganic items such as ropes.

To set up Spline IK, it is necessary to have a chain of connected bones and a curve to constrain these bones to:

With the last bone in the chain selected, add a Spline IKconstraint from the Bone Constraintstab in the Properties.

Set the `Chain Length` setting to the number of bones in the chain (starting from and including the selected bone) that should be influenced by the curve.

Finally, set Target to the curve that should control the curve.

Note

The IK constraints are special in that they modify multiple bones. For this reason, they ignore their position in the stack and always run after all other constraints on the affected bones. To apply constraints after IK, it is necessary to first copy the final transformation to a new bone chain, e.g. using Copy Transforms .

Spline IK panel. `Target`

Data ID used to select the target curve. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

How many bones are included in the chain.

Ignore the relative length of the bones when fitting to the curve.

Retain the offset of the root joint from the start point of the curve.

Average radius of the endpoints is used to tweak the X and Z scaling of the bones, on top of the X and Z scale mode.

Specifies how the length of a bone is scaled when it is fitted to the curve, in addition to the effects of target curve object scale and curvature.

The bone is reset to its rest pose length.

The bones are stretched to cover the entire length of the curve.

The original Y axis scale of the bone is used.

Scaling that a bone undergoes in the other two directions.

Do not scale the X and Z axes.

Use the original scaling of the bones.

Scale of the X and Z axes is the inverse of the Y scale.

Similar to the Stretch To constraint.

Specifies that Inverse Scale or Volume Preservation should be applied on top of the original scaling of the bones, like in the Stretch To constraint.

See also

This subject is seen in-depth in the Armature Posing section .

--- Fin du chapitre: `_spline_ik` ---

--- Début du chapitre: `_spline_length` --- The Spline Length node outputs the total length of each spline, as a distance, or a number of points. This is different than the Curve Length node, which adds up the total length for all of the curve's splines.

The output values correspond to the spline domain, but the node can be used to output a value for every curve control point as well.

This node has no inputs.

This node has no properties.

The length of each spline.

The number of control points in each spline.

--- Fin du chapitre: `_spline_length` ---

--- Début du chapitre: `_spline_parameter` --- The Spline Parameter node outputs how far along each spline a control point is. The Factor output is different from dividing the index by the total number of control points, because the control points might not be equally spaced along the curve.

The first value is zero, so the output corresponds to the length at the control point rather than including the length of the following segment.

When used on the spline domain, the node outputs the portion of the total length of the curve (including all splines) has been traversed at the start of each spline. The order of the curve's splines is visible in the Spreadsheet Editor.

Warning

For NURBS and Bézier spline curves, keep in mind that the value retrieved from this node is the value at every control point, which may not correspond to the visible evaluated points. For NURBS splines the difference may be even more pronounced and the result may not be as expected. A Resample Curve Node can be used to create a poly spline, where there is a control point for every evaluated point.

Note

When the Length is zero, the Factor is arbitrary. In this case the result is exceptionally calculated dividing the index by the total number of control points or curves.

This node has no inputs.

This node has no properties.

When the node is used on the point domain, the value is the portion of the spline's total length at each control point. On the spline domain it is the portion of the curve's total length at the start of the spline.

When the node is used on the point domain, the value is the distance along the spline to each control point. On the spline domain it is the length along the entire curve at the start of the spline.

Each control point's index on its spline. This is different from the output of the index node, which also counts the control points in all previous splines.

The parameter used to control the radius of the curve. The beginning of the spline has a radius of 0, the end has a radius of 1.

--- Fin du chapitre: `_spline_parameter` ---

--- Début du chapitre: `_spline_resolution` --- The Spline Resolution outputs the number of evaluated curve points that will be generated for every control point on the spline. This node works for NURBS, Bézier, and Catmull Rom splines.

For poly splines, there is a one-to-one correspondence between original points and evaluated points, so the resolution does not have an effect.

On Bézier splines, the resolution does not have an effect on segments between vector handles, since there are no extra evaluated points between the neighboring control points.

The node to set this data is the Set Spline Resolution Node.

This node has no inputs.

This node has no properties.

The integer resolution value for each spline.

--- Fin du chapitre: _spline_resolution ---

--- Début du chapitre: _split --- The Split node combines two images for side-by-side display. Typically used in combination with the Viewer Node.

Shown on the right or top half set by the axis.

And respectively the left or bottom half.

X or Y used as the split axis.

Percentage factor setting the space distribution between the two images.

This node has no output sockets.

Hint

This node could be used to plan scene transitions by comparing the end frame of one scene with the start frame of another to make sure they align.

Example of a Split Viewer node.

--- Fin du chapitre: _split ---

--- Début du chapitre: _split_edges --- Like the Edge Split Modifier, the

Split Edges node splits and duplicates edges within a mesh, breaking

links between faces around those split edges.

Standard geometry input.

A standard Boolean selection input to determine which edges will be split.

Note

Because of mesh topology requirements, sometimes more or fewer edges than are selected will be split.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _split_edges ---

--- Début du chapitre: _split_to_instances --- Splits a selection of geometry elements (such as faces) into groups, then turns each group into an instance.

Standard geometry input.

Boolean field indicating which geometry elements to include.

Integer field indicating which group each element belongs to. Elements with the same ID will be moved into the same output instance.

The type of geometry to extract and split. This is also the domain on which the Selection and Group ID fields are evaluated.

Points, spline control points, and vertices.

Mesh edges.

Mesh faces.

Curve splines.

Top-level instances. Realized instances are ignored.

Grease Pencil layers.

Note

Geometry that doesn't match the selected domain will be removed. For example, if you choose Edge, any faces, splines, and instances in the input geometry will be lost.

The instances containing the grouped geometry elements.

Group ID of each instance.

In the example above, we start with a grid of 1000x1000 square faces serving as pixels. Then, we group these faces into patches by assigning them a group ID sampled from a Voronoi texture, and move each resulting instance by a random amount along the Z axis.

Note that, because the texture outputs floating point values between 0 and 1 while the group ID is an integer, all the values would be rounded to 0 or 1 and we would only get two groups. To get more variation, we multiply the texture value by 1000.

--- Fin du chapitre: _split_to_instances ---

--- Début du chapitre: _spreadsheet --- The Spreadsheet editor is used to inspect the geometry attributes of the active object, typically in order to debug geometry nodes.

The Spreadsheet editor.

The state for which to display data.

Display data from the object with all modifiers applied.

Display data from the original object without any modifiers applied.

Display the data that's fed into the active Viewer Node.

You can also toggle between Evaluated and Viewer Node by clicking the eye icon in the Viewer node's header.

Shows the name of the active object, and (if Object Evaluation State is set to Viewer Node) the name of the Geometry Nodes modifier and the active Viewer node.

You can click one of the arrows between the names to hide the modifier.

Click to lock the editor to the currently active object, and have it keep displaying that object's data even if another object becomes active. Click again to unlock.

This option is only available if the object is in Edit Mode. When checked, only data for the selected geometry elements is shown.

Whether to use the filters that are defined in the Sidebar (see below).

The main view shows the actual spreadsheet. Column names and row indices remain visible when scrolling down or to the side.

Note

Tooltips give more detail about the value, depending on the type. For example, Byte Color attributes are displayed as scene linear floats, but the actual integer values are displayed when hovering over the float values, and Matrix attribute values are only displayed in tooltips.

With region on the left, you can choose what to display data for. The top tree lets you pick from the hierarchy of geometries, such as a mesh inside an instance. The bottom tree lets you pick a domain, such as mesh vertices or curve splines.

Each tree item shows the number of elements inside.

In the Sidebar, you can define filters so that only the rows matching these filters are displayed. Click Add Row Filter and set up the properties described below.

Uncheck to temporarily disable the filter.

The name of the column to filter on. If there is no column with the specified name, the filter will be grayed out and ignored.

If you want to filter on an attribute from another domain, you can use the Store Named Attribute Node to create a copy that's converted to the current domain, then filter on that.

For numerical columns, you can select one of the following comparison operators. Other columns only support Equal To.

Only display rows whose value for the column is equal to the filter value (within the specified threshold).

Only display rows whose value for the column is greater than the filter value.

Only display rows whose value for the column is less than the filter value.

The filter value to compare the row value to.

How much the row's value is allowed to deviate from the filter value before it is excluded.

The status bar shows how many rows and columns there are, and how many rows remain after filtering.

--- Fin du chapitre: _spreadsheet ---

--- Début du chapitre: _sss --- The Subsurface Scattering node is used to add simple subsurface multiple scattering, for materials such as skin, wax, marble, milk and others. For these materials, rather than light being reflect directly off the surface, it will penetrate the surface and bounce around internally before getting absorbed or leaving the surface at a nearby point.

How far the color scatters on average can be configured per RGB color channel. For example, for skin, red colors scatter further, which gives distinctive red-colored shadows, and a soft appearance.

Color of the surface, or physically speaking, the probability that light is reflected for each wavelength.

Global scale factor for the scattering radius.

Average distance that light scatters below the surface. Higher radius gives a softer appearance, as light bleeds into shadows and through the object. The scattering distance is specified separately for the RGB channels, to render materials such as skin where red light scatters deeper. The X, Y and Z values are mapped to the R, G and B values, respectively.

Index of refraction for Subsurface Scattering.

Directionality of subsurface scattering. Higher anisotropy scatters deeper into the object.

Roughness of the glossy surface surrounding the subsurface volume.

Normal used for shading; if nothing is connected the default shading normal is used.

Rendering method to simulate subsurface scattering.

Note

EEVEE does not support the Random Walk methods.

An approximation to physically-based volume scattering. This method is less accurate than Random Walk, however, in some situations this method will resolve noise faster.

Provides accurate results for thin and curved objects. Random Walk uses true volumetric scattering inside the mesh, which means that it works best for closed meshes. Overlapping faces and holes in the mesh can cause problems.

Behaves similarly to Random Walk (Fixed Radius) but modulates the Radius based on the Color, Anisotropy, and IOR. This method thereby

attempts to retain greater surface detail and color than Random Walk (Fixed Radius).

BSSRDF shader output.

Random walk subsurface scattering. \xc2\xbb6

--- Fin du chapitre: _sss ---

--- Début du chapitre: _stabilize_2d --- Stabilizes the footage according to the settings set in Movie Clip Editor \xe2\x80\xa3 Properties \xe2\x80\xa3 Stabilization \xe2\x80\xa3 2D Stabilization. For more information, see 2D Stabilization.

Standard color input.

The movie clip whose stabilization to use.

Various methods for the stabilization. Usually, the same as used in Movie Clip Editor \xe2\x80\xa3 Properties \xe2\x80\xa3 Stabilization \xe2\x80\xa3 2D Stabilization \xe2\x80\xa3 Interpolate. For technical details on their difference, see this. But for most purposes, default of Bilinear should suffice.

Invert the stabilization. If the stabilization calculated is to move the movie clip up by 5 units, this will move the movie clip down by 5 units.

Standard color input.

--- Fin du chapitre: _stabilize_2d ---

--- Début du chapitre: _stack --- The combination of all the constraints affecting an object is called the Constraints Stack. The Stack is in the Constraints panel, below the Add Constraint menu.

Constraints in the stack are evaluated from top to bottom. The order of each constraint has a substantial impact on the final outcome of the stack. Changing the order of the constraints can change the behavior of the entire stack.

The constraints in this example stack are evaluated from top to bottom starting with the \xe2\x80\x9cCopy Location\xe2\x80\x9d constraint and ending with the final \xe2\x80\x9cDamped Track\xe2\x80\x9d constraint. \xc2\xbb6

To change the order of a constraint use the up/down arrows in the header.

--- Fin du chapitre: _stack ---

--- Début du chapitre: _star --- The Star node generates a poly spline in a star pattern by connecting alternating points of two circles. The points on the inner circle are offset by a rotation so that they lie in between the points on the outer circle. This offset can be changed with the twist input.

Number of points on each of the circles.

Radii of the two circles on which to place the control points. The inner radius can be larger than the outer radius.

Angle offset of the inner circle. The twist value rotates the points on the circle corresponding with the inner radius counterclockwise by the given angle.

This node has no properties.

Poly spline generated from the inputs.

A boolean attribute field with a selection of the points on the Outer Radius, which is every other point.

--- Fin du chapitre: _star ---

--- Début du chapitre: `_startup` --- There are some common causes for problems when using Blender. If Blender fails to start, there are a few things to check for:

See if your computer meets the minimum requirements.

Confirm that your graphics card is supported and that the drivers are up to date (see [Troubleshooting Graphics Hardware](#)).

Make sure any antivirus software is not preventing Blender from starting.

If you cannot find a solution to your problem here, try asking the community for help.

The Blender Console Window can display many different types of status and error messages. Some messages simply inform the user what Blender is doing, but have no real impact on Blender's ability to function. Other messages can indicate serious errors that will most likely prevent Blender carrying out a particular task and may even make Blender non-responsive or shut down completely. The Blender Console Window messages can also originate internally from within the Blender code or from external sources such as Python scripts.

This message indicates that Blender was able to find the Python library for the Python interpreter embedded within Blender. If this folder is missing or unable to be found, it is likely that an error will occur, and this message will not appear.

The preferences use this path.

--- Fin du chapitre: `_startup` ---

--- Début du chapitre: `_startup_scene` --- After closing the splash screen, the startup scene is displayed in the 3D Viewport (if no other blend-file was loaded). This startup scene can be customized.

The startup scene.

The gray cube in the center of the scene is a mesh object. Its orange outline indicates that it's selected. The orange dot in the center is its Origin, which indicates its precise location.

The set of concentric black circles is a light source illuminating the cube.

The pyramid with a big triangle above it is the camera, which is used as the point of view for rendering.

The 3D cursor, a cross with a red-and-white circle, determines where newly added objects are placed and can also serve as a transformation pivot point.

The gray lines forming a floor mark the zero height of the world. The red and green lines are the axes of the world coordinate system. They meet at the world origin, which is also where the origin of the Cube is located. The Grid Floor settings are in the Viewport Overlays popover.

The top left corner of the viewport shows various bits of information see [Viewport Overlays](#) for details.

--- Fin du chapitre: `_startup_scene` ---

--- Début du chapitre: `_static_repository` --- To host your own extensions and leverage Blender update system all that is required is to host a static JSON file that is generated by Blender.

To generate a valid JSON file use the server generate Blender command-line tool:

This creates an `index.json` listing from all the .zip extensions found in the `repo-dir` location.

For more details, read the generated JSON API.

To test the generated repository, add a new Remote Repository from the Preferences:

Get Extensions \xe2\x86\x92 Repositories \xe2\x86\x92 [+] \xe2\x86\x92 Add Remote Repository

Paste the location of the generated JSON as the URL. So the example/path/to/packages would use the:

file:///path/to/packages/index.json on Linux and macOS.

file:///C:/path/to/packages/index.json on Windows.

file://HOST/share/path/to/packages/index.json for network shares on Windows.

Tip

Open file:/// in a web browser and navigate to the repository location and copy that as the remote repository URL.

The server-generate command can optionally create a simple website using the --html argument.

This creates an index.html file ready to use, listing extensions which can be dropped into Blender for installation.

In order to support drag and drop for installing from a remote repository, there are a few optional ways to prepare the URLs.

The only strict requirement is that the download URL must end in .zip.

You can pass different arguments to the URL to give more clues to Blender about what to do with the dropped URL.

Link to the JSON file to be used to install the repository in Blender. Supports relative URLs.

Comma-separated list of supported platforms. If omitted, the extension will be available in all operating systems.

Minimum supported Blender version, e.g., 4.2.0.

Blender version that the extension does not support, earlier versions are supported.

Tip

The more details you provide, the better the user experience.

With the exception of the repository, all the other parameters can be extracted from the extensions manifest. Those arguments are to be encoded as part of the URL.

{URL}.zip?repository={repository}&blender_version_min={version_min}&blender_max={version_max_exclusive}&platforms={platform1,platform2}

http://my-site.com/my-

addon.zip?repository=%2Findex.json&blender_version_min=4.2.0&platforms=windows-x64

https://extensions.blender.org/download/sha256:57a6a5f39fa2cc834dc086a27b7b2e572c12fd14f8377fb8bd1c7022df3d7ccb/add-on-amaranth-

v1.0.23.zip?repository=%2Fapi%2Fv1%2Fextensions%2F&blender_version_min=4.2.0&platforms=linux-x64%2Cmacos-x64

Note

%2F and %2C are simply the url-encoded equivalent of / and , respectively.

--- Fin du chapitre: _static_repository ---

--- Début du chapitre: _status_bar --- The Status Bar is located at the bottom of the Blender window and displays contextual information such as keyboard shortcuts, messages, and statistical information. The Status Bar can be hidden by disabling Show Status Bar in Window menu or by dragging from the top edge down.

Status Bar.\xc2\xbb

The left side of the Status Bar displays mouse button shortcuts and the keymap of the active tool. In editors with a Toolbar, tapping Alt (or Option on macOS) shows the hotkeys to change to a desired tool.

Tip

This functionality can be disabled with the Alt Click Tool Prompt preference in the Keymap Preferences).

The middle of the Status Bar displays information about in-progress operations.

Shows the progress of the currently running task (such as rendering or baking). Hovering the mouse pointer over the progress bar will display a time estimate. The task can be aborted by clicking the cancel button (). Informational messages or warnings, such as after saving a file. They disappear after a short time. Click them to show the full message in the Info Editor .

The right side of the Status Bar displays information about the Blender instance. Which information is shown can be chosen by RMB on the Status Bar or in the Preferences .

Shows information about the data in the active scene.

Collection: The name of the active Collection .

Active Object: The name of the active selected object.

Geometry: Information about the current scene depending on the mode and object type. This can be the number of vertices, faces, triangles, or bones.

Objects: The number of selected objects and the total count of objects.

Shows the total amount of time of the playback along with the current frame number and total frame count. The format of the duration text is determined by the Timecode Style .

Shows an estimate of Blender's RAM consumption. On a single-instance single-machine scenario, this estimate provides a measurement against the hardware limit of the machine.

Shows the number of extensions with available updates.

Shows the version number of Blender that is currently running.

--- Fin du chapitre: _status_bar ---

--- Début du chapitre: _steam --- Steam is a software distribution platform. Blender can be downloaded and updated using the Steam client by following the steps described below on Linux, macOS, or Windows.

Download the Steam client for your operating system. Once installed, open the client and login to your Steam account, or create one if you haven't already. Once logged in, navigate to the Store tab, search for Blender, and press the green installation button. Blender should now be available in the Library tab of the Steam client.

See also

When installing Blender from Steam on Linux and Windows, the .blend filename extension will not be automatically associated with Blender. To associate blend-files with Blender, see the processes described on the Linux and Windows installation pages.

When an update for Blender is available on Steam, Steam will automatically update Blender for you.

--- Fin du chapitre: _steam ---

--- Début du chapitre: _stl --- Reference

Import-Export

File \xe2\x80\xa3 Import/Export \xe2\x80\xa3 Stl (.stl)

The STL-file format is useful if you intend to import/export the files for CAD software. It is also commonly used for loading into 3D printing software.

Value by which to scale the imported objects in relation to the world\xe2\x80\x99s origin.

Apply current scene\xe2\x80\x99s unit (as defined by unit scale) to imported data.

Since many applications use a different axis for pointing upwards, these are axis conversion for these settings, Forward and up axes \xe2\x80\x93 By mapping these to different axes you can convert rotations between applications default up and forward axes.

Blender uses Y forward, Z up (since the front view looks along the +Y direction). For example, it is common for applications to use Y as the up axis, in that case -Z forward, Y up is needed.

Use (import) facet normals (note that this will still give flat shading).

Check the imported mesh for corrupt data and fix it if necessary. When disabled, erroneous data may cause crashes displaying or editing the meshes. This option will make the importing slower but is recommended, as data errors are not always obvious.

Exports the stl-file in ASCII format rather than as a binary format

Export each object as a separate STL file.

When checked, only selected objects are exported. Instanced objects, for example collections that are instanced in the scene, are considered \xe2\x80\x98selected\xe2\x80\x99 when their instancer is selected.

Value by which to scale the exported objects in relation to the world\xe2\x80\x99s origin.

Apply current scene\xe2\x80\x99s unit (as defined by unit scale) to exported data.

Since many applications use a different axis for \xe2\x80\x98Up\xe2\x80\x99, these are axis conversion for these settings, Forward and Up axes \xe2\x80\x93 By mapping these to different axes you can convert rotations between applications default up and forward axes.

Blender uses Y Forward, Z Up (since the front view looks along the +Y direction). For example, it\xe2\x80\x99s common for applications to use Y as the up axis, in that case -Z Forward, Y Up is needed.

Export objects using the evaluated mesh, meaning the resulting mesh after all Modifiers have been calculated.

--- Fin du chapitre: _stl ---

--- Début du chapitre: _store_named_attribute --- The Store Named Attributenode stores the result of a field on a geometry as an attribute with the specified name. If the attribute already exists, the datatype and domain will be updated to the values chosen in the node. However, keep in mind that the domain and data type of Built-In Attributes cannot be changed.

Compared with the Capture Attribute Node, this node basically does the same thing, but the attribute gets a name instead of an anonymous reference. For reusing the data in the same node tree, the Capture Attributenode might be preferable since it does not create the chance for name conflicts in the input geometry.

Note

If the input geometry contains multiple geometry component types, the attribute will be created on each component that has the chosen domain. Standard geometry input.

A boolean field input for each element indicating whether to store the attribute value for the given index. If the attribute does not exist, unselected parts are filled with zero values; otherwise, only the selected parts are filled.

The input field to evaluate.

The name to give the stored data.

The data type used for the evaluated data.

Which attribute domain to store the evaluated data on.

Standard geometry output.

--- Fin du chapitre: `_store_named_attribute` ---

--- Début du chapitre: `_straighten_hair_curves` --- Straightens hair curves between root and tip.

Input Geometry (only curves will be affected).

Amount of straightening. Negative values will result in crumpling the curves.

Shape of the influence along curves (0=constant, 0.5=linear).

Preserve each curve's length during deformation.

This node has no properties.

Geometry

--- Fin du chapitre: `_straighten_hair_curves` ---

--- Début du chapitre: `_stretch_to` --- The Stretch To constraint causes its owner to rotate and scale its Y axis towards its target. So it has the same tracking behavior as the Track To constraint. However, it assumes that the Y axis will be the tracking and stretching axis, and does not give you the option of using a different one.

It also optionally has some raw volumetric features, so the owner can squash down as the target moves closer, or thin out as the target moves farther away. Note however, that it is not the real volume of the owner which is thus preserved, but rather the virtual one defined by its scale values. Hence, this feature works even with non-volumetric objects, like empties, 2D meshes or surfaces, and curves.

With bones, the volumetric variation scales them along their own local axes (remember that the local Y axis of a bone is aligned with it, from root to tip).

Stretch To panel. 

Data ID used to select the constraints target, and is not functional (red state) when it has none. See common constraint properties for more information.

This number field sets the rest distance between the owner and its target, i.e. the distance at which there is no deformation (stretching) of the owner.

When clicked, this small button will recalculate the Rest Length value, so that it corresponds to the actual distance between the owner and its target (i.e. the distance before this constraint is applied).

This number field controls the amount of volume variation exponentially to the stretching amount. Note that the 0.0 value is not allowed, if you want to disable the volume feature, use the None button (see below).

Limits for the volume preservation to a minimum and maximum scaling each by a Bulgefactor.

Smoothness factor to make limits less visible.

These buttons control which of the X and/or Z axes should be affected (scaled up/down) to preserve the virtual volume while stretching along the Y axis. If you enable the nonebutton, the volumetric features are disabled.

Specifies how the owner should be rotated to track the target with its Y axis.

These buttons are equivalent to the Upones of the Track To constraint : the owner is rotated around two local axes in the specified order.

The constraint uses a single Swing rotation, equivalent to the Damped Track constraint .

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_stretch_to` ---

--- Début du chapitre: `_string` --- The Stringinput node creates a single string. It can be connected to attribute name socket to make using the same attribute name in multiple places more convenient.

This node has no inputs.

String

Standard string output.

--- Fin du chapitre: `_string` ---

--- Début du chapitre: `_string_length` --- The String Length node outputs the number of characters in the input string.

String value to be evaluated.

This node has no properties.

Integer value representing the length of the input string.

--- Fin du chapitre: `_string_length` ---

--- Début du chapitre: `_string_to_curves` --- The String to Curves converts a string to curve instances. Each unique character used in the string is converted to a curve once, and further uses of that character are more instances of the same geometry.

This makes processing the output geometry very efficient, because each unique character only has to be processed once. However, it means that the result will be the same for every instance of the same character. To process each character individually, the Realize Instances Node can be used.

Tip

Socket inspection can be used to see the value of the string input used when the node was evaluated, by holding the mouse over the socket.

Standard string input.

The size of each character. The values of the other inputs are scaled by this value.

A factor by which the space between each character (kerning) is scaled on the X axis.

A factor by which whitespace between words is scaled on the X axis.

The distance between separate lines in the output. Scaled by the Sizeinput.

The maximum width of each line, though individual words will not be wrapped.

The maximum height for all the lines of the text.

Font glyph used to generate the curve.
Wraps the text at the Text Box Width.
Scales the text size to fit the Text Box Width and Text Box Height.
Only outputs text characters that fit within the width and height, based on the Sizeinput. Any part of the string that did not fit is moved to the Remainderoutput.
Aligns the text to the left.
Aligns the text to the center.
Aligns the text to the right.
Aligns the text to the left and right.
Aligns the text to the left and right with equal character spacing.
Aligns the text to the top.
Aligns the text to the top baseline.
Aligns the text to the middle.
Aligns the text to the bottom baseline.
Aligns the text to the bottom.
Controls where on each character the output Pivot Point is placed.
Place the pivot points at the center of each character\ue2\x80\x99s bounds.
Place the pivot points at the top left of each character\ue2\x80\x99s bounds.
Place the pivot points at the middle of the top of each character\ue2\x80\x99s bounds.
Place the pivot points at the top right of each character\ue2\x80\x99s bounds.
Place the pivot points at the bottom left of each character\ue2\x80\x99s bounds.
Place the pivot points at the middle of bottom of each character\ue2\x80\x99s bounds.
Place the pivot points at the bottom right of each character\ue2\x80\x99s bounds.
Curve instances geometry.
The part of the text that did not fit in the box described by the Text Box Height and Text Box Width inputs. Only used in the Truncate overflow mode.
An attribute field containing the line index of each character (on the instance domain).
Outputs the position described by the Pivot Point drop-down in the local space of each instance.
The node can be used to make overflowing text boxes. Here, the text that does not fit into the first node\ue2\x80\x99s fix-sized text box is passed to a separate String to Curves node. And finally added with a Scale to Fit node.

--- Fin du chapitre: _string_to_curves ---

--- Début du chapitre: _strip --- Reference

Nonlinear Animation

Strip \ue2\x80\xa3 Transform

Move the selected strips in time or to a different track.
Lets you quickly move the selected strips that are on a certain side of the Playhead. This is handy if you need to, say, move all the strips after a certain time point to the right to make space for new ones.
To use this operator, first select some or all strips and place your mouse cursor to the left or right of the Playhead. Then, press E, move the

mouse to move (only) the strips on that side of the Playhead, and press LMB to confirm (or RMB to cancel).

If a strip straddles the Playhead, only its starting/ending point will be moved (again depending on the position of the mouse cursor).

Scales the selected strips, using the Playhead as the pivot point.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Transform \xe2\x80\xa3 Swap

Alt- F

Swap the order of the selected strips in their track.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Transform \xe2\x80\xa3 Move Up

PageUp

Move selected strips up a track if there is room.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Transform \xe2\x80\xa3 Move Down

PageDown

Move selected strips down a track if there is room.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Snap

Move the start of the selected strips to the current frame.

Move the start of the selected strips to the nearest full frame.

Move the start of the selected strips to the nearest second.

Move the start of the selected strips to the nearest marker.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Split

Y

Split the selected strips in two at the current frame.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Duplicate

Alt- D

Creates copies of the selected strips, duplicating any actions they reference. Editing the keyframes in a copied strip therefore doesn't affect the original.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Linked Duplicate

Shift- D

Creates copies of the selected strips, reusing any actions they reference. Editing the keyframes in a copied strip therefore also affects the original (and vice versa). Blender warns you about this by highlighting the other strip in red.

Linked duplicated strip being edited. \xc2\xb6

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Delete

Delete, X

Deletes the selected NLA-Strips.

Reference

Nonlinear Animation
 Add \xe2\x80\xa3 Make Meta
 Ctrl- G
 Groups the selected NLA-strips into a meta strip.
 Select two or more strips.\xc2\xb6
 Combine them into a meta strip.\xc2\xb6
 Reference
 Nonlinear Animation
 Strip \xe2\x80\xa3 Remove Meta
 Ctrl- Alt- G
 Ungroups the selected meta strips, replacing them by their contents.
 Reference
 Nonlinear Animation
 Strip \xe2\x80\xa3 Toggle Muting
 H
 Mutes or unmutes the selected strips. Muted strips have a dotted border
 and don't influence the animation.
 Reference
 Nonlinear Animation
 Strip \xe2\x80\xa3 Bake Action
 Reference
 3D Viewport
 Object and Pose Modes
 Header \xe2\x80\xa3 Object \xe2\x80\xa3 Animation \xe2\x80\xa3 Bake
 Action\xe2\x80\xa6
 The final motion of objects and bones depends not only on the keyframed
 animation, but also on F-Curve modifiers , drivers , and constraints .The
 Bake Action operator computes this final motion and creates a
 corresponding keyframe on every scene frame.
 This can be useful for adding deviation to a cyclic action like a Walk
 Cycle , or to create a keyframe animation from drivers or constraints.
 Start frame for baking.
 End frame for baking.
 Number of frames to skip forward while baking each frame.
 Only key selected bones (Pose baking only).
 Keyframe from the final transformations (with constraints applied).
 Remove all constraints from keyed object/bones, and do
 \xe2\x80\x98visual\xe2\x80\x99 keying.
 Bake animation onto the object then clear parents (objects only).
 Bake animation into the current action instead of creating a new
 one (useful for baking only part of bones in an armature).
 After baking curves, remove redundant keys .
 Which data\xe2\x80\x99s transformations to bake.
 Bake bone transformations.
 Bake object transformations.
 Which channels to bake.
 Bake location channels.
 Bake rotation channels.
 Bake scale channels.
 Bake B-Bone channels.
 Bake custom properties.
 Reference
 Nonlinear Animation
 Strip \xe2\x80\xa3 Apply Scale

Ctrl- A

Applies the scale of the selected strips to their referenced actions.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Clear Scale

Alt- S

Resets the scale of the selected strips.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Sync Action Length

Resets the strip\xe2\x80\x99s length to that of its underlying action, ensuring that it (only) plays from the action\xe2\x80\x99s first keyframe to its last.

See also

The Sync Length Now button in theSidebar, which does the same thing.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Make Single User

U

Duplicates actions where necessary so that each selected strip has its own action that\xe2\x80\x99s not used by any others. This way, you can edit the keyframes in the selected strips knowing that you won\xe2\x80\x99t affect any other part of the animation.

Note

This does not recursively go inside meta strips.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Start Editing Stashed Action

Shift- Tab

Enters Tweak Mode for the selected strip\xe2\x80\x99s action, making its keyframes available for editing in e.g. the Graph Editor . In addition, marks the strip\xe2\x80\x99s track as Solo, muting all the other tracks \xe2\x80\x93 this way, they no longer influence the animation and you can focus exclusively on the action you\xe2\x80\x99re editing.

While the menu item refers to stashed (muted) actions, this only reflects the typical use case. It works on unmuted actions as well.

When you\xe2\x80\x99re done editing, click Strip \xe2\x80\xa3 Stop

Editing Stashed Action or press Shift- Tab again.

Strip in NLA mode.\xc2\xbb

Strip in Tweak mode.\xc2\xbb

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Start Tweaking Strips Actions (Full Stack)

Tab

Enters Tweak Mode for the selected strip\xe2\x80\x99s action, making its keyframes available for editing. Leaves all the other tracks enabled so that you can still see their effects while making changes.

When you\xe2\x80\x99re done, click Strip \xe2\x80\xa3 Stop Tweaking Strips Actions or press Tab again.

Note

For transitions above the tweaked strip, keyframe remapping will fail for channel values that are affected by the transition. A workaround is to tweak the active strip without evaluating the upper NLA stack.

Reference

Nonlinear Animation

Strip \xe2\x80\xa3 Start Tweaking Strips Actions (Lower Stack)

Enters Tweak Mode for the selected strip\xe2\x80\x99s action, making its keyframes available for editing. Mutes any tracks above the current one so that they don\xe2\x80\x99t influence the animation while making changes. When you\xe2\x80\x99re done, click Strip \xe2\x80\xa3 Stop Tweaking Strips Actions or press Tab.

--- Fin du chapitre: _strip ---

--- Début du chapitre: _strips --- A strip tells the animation when something happens and for how long. There are a few different types which are described below.

An action strip plays the keyframes inside an action. You can create one using Add \xe2\x80\xa3 Action. Another way is to click Push Down Action in the NLA\xe2\x80\x99s Action Track \xe2\x80\x93 this will create a strip based on the object\xe2\x80\x99s active action.

Multiple strips can reference the same action, so that you can potentially change multiple parts of the animation by editing a single set of keyframes.

A strip can be shorter than its underlying action, be it through cropping, speeding up, or both. It can also be longer than its underlying action, be it through extending, slowing down, or both. See the Sidebar for details.

A transition strip interpolates between two neighboring action strips. Select them and click Add \xe2\x80\xa3 Transition.

Transition Strip. \xc2\xb6

These strips control when a Speaker plays its sound clip.

A meta strip groups other strips together, letting you move, scale, and copy them as one combined unit.

--- Fin du chapitre: _strips ---

--- Début du chapitre: _stroke --- The stroke settings define the behavior of the sculpted/painted stroke. Any other brush behavior and effect is applied on top of the stroke.

Stroke panel. \xc2\xb6

Defines the way brush strokes are applied to the canvas.

Apply paint on each mouse move step. This is regardless of their distance to each other, and instead depends on the stroke speed. This means that a slower stroke will have more accumulative strength applied.

Leaves only one dab on the canvas which can be placed by dragging.

Creates brush stroke as a series of dots, whose distance (spacing) is determined by the Spacing setting.

Limits brush application to the distance specified by the percentage of the brush radius.

Brush Spacing can be affected by enabling the pressure sensitivity icon, if you are using a Graphics Tablet.

Flow of the brush continues as long as the mouse click is held (spray), determined by the Rate setting.

Interval for how frequent the brush is applied during the stroke.

Creates a single dab at the brush location. Clicking and dragging will resize the dab diameter.

The brush location and orientation are determined by a two point circle, where the first click is one point, and dragging places the second point, opposite from the first.

Clicking and dragging lets you define a line in screen space. The line dabs are separated by Spacing, similar to space strokes. With Alt the line stroke is constrained to 45 degree increments.

Defines the stroke curve with a Bézier curve (dabs are separated according to Spacing). This Bézier curve is stored in Blender as a Paint Curve data-block.

Use Ctrl-RMB to create the initial control point of the curve.

Paint Curves are reusable and can be stored and selected by using the Data-Block Menu.

You can define additional curve control points by using Ctrl-RMB. The handles can be defined by dragging the mouse. The stroke flows in the direction of the first control point to the second control point, and so on.

The control points and handles can be dragged with RMB (In right click select with LMB). To make sure that the handles of a control point are symmetrical, drag them using Shift-RMB. A few transform operators are supported such as moving (G), rotating (R) and scaling (S).

The handles can be selected individually by using LMB (In right click select with RMB), extend the selection by Shift-LMB and deselect/select all by using A.

To delete a curve point, use X.

To confirm and execute the curved stroke, press Return or use the Draw Curve button.

Method used to calculate the distance to generate a new brush step.

Calculates the brush spacing relative to the view.

Calculates the brush spacing relative to all three dimensions of the scene using the stroke location. This avoids artifacts when sculpting across curved surfaces and keeps the spacing much more consistent.

Keep the brush strength consistent, even if the spacing changes. Available for the Space, Line, and Curve stroke methods.

Ratio of samples in a cycle that the brush is enabled. This is useful to create dashed lines in texture paint or stitches in Sculpt Mode. Available for the Space, Line, and Curve stroke methods.

Length of a dash cycle measured in stroke samples. This is useful to create dashed lines in texture paint or stitches in Sculpt Mode. Available for the Space, Line, and Curve stroke methods.

Jitter the position of each step in the brush stroke.

Brush Jitter can be affected by enabling the pressure sensitivity icon, if you are using a Graphics Tablet.

Controls how the brush Jitter is measured.

The Jitter is relative to the view direction i.e. screen space.

The Jitter is measured relative to all three dimensions of the scene. The unit type and scaling can be configured in the Scene Units.

Recent mouse locations (input samples) are averaged together to smooth brush strokes.

Use the same brush Input Samples across all brushes.

Stabilize Stroke makes the stroke lag behind the cursor and creates a smoothed curve to the path of the cursor. This can be enabled pressing Shift-S or by clicking the checkbox found in the header.

Minimum distance from the last point before the stroke continues.

A smooth factor, where higher values result in smoother strokes but the drawing sensation feels like as if you were pulling the stroke.

--- Fin du chapitre: _stroke ---

--- Début du chapitre: `_strokes` --- Strokes are the final rendered lines. Yet you can tweak them, for example, by removing the ones longer/shorter than some threshold, chaining lines into a single stroke or breaking a stroke into several ones based on angles, dashed pattern, etc.

Line Style: Strokes.\xc2\xba

You can choose between three types of line caps:

Flat cap, exactly at the point the line ends.

A half circle centered on the end point of the line.

A square centered on the end point of the line (hence, like the circle, the drawn end of the line is slightly extended compared to its computed value).

Line caps example.\xc2\xba

By default all retrieved lines from the line set are chained together. There are two basic chaining methods:

The default chaining method; it creates simple chains.

This chaining option allows for generating chains of feature edges with sketchy multiple strokes. Basically, it generates Roundstrokes instead of a single one. It is only really useful if you use some random-driven modifiers in the line style!

It specifies the number of rounds in sketchy strokes.

If true, only feature edges of the same object are joined.

Chaining can also be turned off to render each line separately, which can be useful for line styles which depend on accurate representation of the line set.

You can split up chains of Freestyle lines by enabling one of the following:

Splits chains of feature edges when they make a 2D angle above (or below) a minimum (or maximum) threshold.

Splits chains when they are longer than the given value.

Splits chains of feature edges if they cross from one material to another.

Splits the chains using the given dashed pattern (see also Dashed Line).

Length of the specified dash for splitting.

Length of the specified gap for splitting.

You can sort the order of your strokes, allowing the lines to stack in the order given.

A sort key is used to determine the stacking order of lines.

Lines closer to the camera lie on top of further lines.

Longer lines lie on top of shorter lines.

Sort by the projected X or Y value in the image coordinate system.

Use in tandem with the Sort Key to determine the range for sorting. Since the distance of a line from the camera may vary over vertices, this option computes the sort key for a line from the values computed at individual vertices. The value computed for the line is:

The mean of the values obtained for the vertices.

The minimum of the values obtained for the vertices.

The maximum of the values obtained for the vertices.

The value obtained for the first vertex.

The value obtained for the last vertex.

With the given result you can choose to `\xe2\x80\x9cReverse\xe2\x80\x9d` the sort order.

You can also choose to only render selected chains.

Chains longer and/or shorter than 2D Length.

Allows the selection of first N chains.

By enabling the Dashed Linecheckbox, you can specify three pairs of dash and gap lengths. Dash values define the lengths of dash strokes, while gap values specify intervals between two dashes.

If a zero gap is specified, then the corresponding dash is ignored even if it has a nonzero value.

Dashes are treated as separate strokes, meaning that you can apply line caps, as well as color, alpha and thickness modifiers.

Length of the specified dash for dashed lines.

Length of the specified gap for dashed lines.

--- Fin du chapitre: _strokes ---

--- Début du chapitre: _stroke_menu --- This page covers many of the tools in the Strokes menu. These are tools that work primarily on strokes, however, some also work with point selections.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Subdivide

Subdivides the strokes by inserting points between the selected points. The number of subdivisions to perform.

When enabled, limits the effect to only the selected points within the stroke.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Subdivide and Smooth

Subdivides and smooths the strokes by inserting points between the selected points.

The number of subdivisions to perform.

When enabled, limits the effect to only the selected points within the stroke.

Number of times to repeat the procedure.

The amount of the smoothness on subdivided points.

Smooths the stroke \xe2\x80\x99s endpoints.

Preserves the strokes shape.

When enabled, the operator affect the points location.

When enabled, the operator affect the points thickness.

When enabled, the operator affect the points strength (alpha).

Reference

Edit Mode

Stroke \xe2\x80\xa3 Simplify

Uses the RDP algorithm (Ramer-Douglas-Peucker algorithm) for points deletion. The algorithm tries to obtain a similar line shape with fewer points.

Controls the amount of recursively simplifications applied by the algorithm.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Trim

Trims selected stroke to first loop or intersection.

Original stroke. \xc2\xb6

Result of trim operation. \xc2\xb6

Reference

Edit Mode

Stroke \xe2\x80\xa3 Join \xe2\x80\xa3 Join,

Join two or more strokes into a single one.
Ctrl- J Join selected strokes by connecting points.
Join selected strokes by connecting points in a new stroke.
When enabled, do not use geometry to connect the strokes.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Join \xe2\x80\xa3 Join and Copy

Shift- Ctrl- J

Same as Join but Typedefaults to Join and Copy.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Move to Layer

M

A pop-up menu to move the stroke to a different layer. You can choose the layer to move the selected strokes to from a list of layers of the current Grease Pencil object. You can also add a new layer to move the selected stroke to. When creating a new layer, there is another pop-up to type in the name of the new layer.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Assign Material

Changes the material linked to the selected stroke. You can choose the name of the material to be used by the selected stroke from a list of materials of the current Grease Pencil object.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Set as Active Material

Sets the active object material based on the selected stroke material.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Arrange

Change the drawing order of the strokes in the 2D layer.

Moves to the top the selected points/strokes.

Moves the selected points/strokes upper the next one in the drawing order.

Moves the selected points/strokes below the previous one in the drawing order.

Moves to the bottom the selected points/strokes.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Close

F

Close or open strokes by connecting the last and first point.

Close all open selected strokes.

Open all closed selected strokes.

Close or Open selected strokes as required.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Toggle Cyclic

Toggles between an open stroke and closed stroke (cyclic).

Close all open selected strokes.

Open all closed selected strokes.

Close or Open selected strokes as required.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Set Caps

Toggle ending cap styles of the stroke.

Sets stroke start and end points to rounded (default).

Toggle stroke start and end points caps to flat or rounded.

Toggle stroke start point cap to flat or rounded.

Toggle stroke end point cap to flat or rounded.

Stroke ending with rounded caps.\xc2\xb6

Stroke ending with flat caps.\xc2\xb6

Stroke ending with combined caps.\xc2\xb6

Reference

Edit Mode

Stroke \xe2\x80\xa3 Switch Direction

Reverse the direction of the points in the selected strokes(i.e. the start point will become the end one, and vice versa).

Reference

Edit Mode

Stroke \xe2\x80\xa3 Set Uniform Thickness

Makes the thickness equal for the entire stroke.

Thickness value to use on all points of the stroke.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Set Uniform Opacity

Makes the opacity equal for the entire stroke.

Opacity value to use on all points of the stroke.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Scale Thickness

When enabled, scales the stroke thickness during scale transformations.

Reference

Edit Mode

Stroke \xe2\x80\xa3 Set Curve Type

Sets the spline type for the splines in the stroke component that are in the selection.

The type to convert the splines in the selection to. Read the Spline Types page for more detail on the different spline types.

Convert to a B\xc3\xa9zier spline. A spline converted from a poly spline gets vector handles, while one converted from NURBS or Catmull Rom spline gets auto handles.

Note

When converting from a NURBS spline to a B\xc3\xa9zier spline, at least six points are needed. When the number of points is not a multiple of three a full conversion is not possible and the spline has to be truncated.

Convert to a NURBS spline.

Convert to a poly spline.

Convert to a Catmull Rom spline.

Take handle information into account in the conversion

Reference

Edit Mode

Stroke \xe2\x80\xa3 Set Curve Resolution

Sets the number of points generated along each curve segment (between two handles).

Reference

Edit Mode
Stroke \xe2\x80\xa3 Set Curve Resolution
Reset UV transformation to default values.
--- Fin du chapitre: _stroke_menu ---

--- Début du chapitre: _stroke_placement ---Reference

Draw Mode

Stroke Placement

Stroke Placement pop-over.\xc2\xb6

The Stroke Placement selector helps to select the location in which strokes are drawn.

Note

The Stroke Placement selection only affects new strokes and does not affect the existing ones.

Strokes are placed at Grease Pencil object origin.

Strokes are placed at 3D cursor.

Strokes will stick on mesh surfaces.

Distance from the mesh surface to place the new strokes.

Only project the strokes onto selected objects.

Strokes will stick on other strokes.

All the points of the new stroke sticks to other strokes.

Only the start and end points of the new stroke sticks to other strokes.

Only the start point of the new stroke sticks to other strokes.

Origin. \xc2\xb6

3D Cursor. \xc2\xb6

Surface. \xc2\xb6

Stroke. \xc2\xb6

--- Fin du chapitre: _stroke_placement ---

--- Début du chapitre: _structure --- Example of a very basic armature.\xc2\xb6

Armatures mimic real skeletons. They are made out of bones, which are (by default) rigid elements. But you have more possibilities than with real skeletons: In addition to the \xe2\x80\x9cnatural\xe2\x80\x9d rotation of bones, you can also move and even scale them! And your bones do not have to be connected to each other; they can be completely free if you want. However, the most natural and useful setups imply that some bones are related to others, forming so-called \xe2\x80\x9cchains of bones\xe2\x80\x9d, which create some sort of \xe2\x80\x9climbs\xe2\x80\x9d in your armature, as detailed in Chains of Bones.

The bones inside an armature can be completely independent from each other (i.e. the modification of one bone does not affect the others). But this is not often a useful set up: To create a leg, all bones \xe2\x80\x9cafter\xe2\x80\x9d the thigh bone should move \xe2\x80\x9cwith\xe2\x80\x9d it in a well-coordinated manner. This is exactly what happens in armatures by parenting a bone to the next one in the limb, you create a \xe2\x80\x9cchains of bones\xe2\x80\x9d. These chains can be ramified. For example, five fingers attached to a single \xe2\x80\x9chand\xe2\x80\x9d bone.

An armature with two chains of bones. \xc2\xb6

Bones are chained by linking the tip of the parent to the root of the child. Root and tip can be connected, i.e. they are always exactly at the same point; or they can be free, like in a standard parent-child object relationship.

A given bone can be the parent of several children, and hence be part of several chains at the same time.

The bone at the beginning of a chain is called its root bone, and the last bone of a chain is the tip bone (do not confuse them with similar names of bones\& joints!).

Chains of bones are a particularly important topic in posing (especially with the standard forward kinematics versus automatic\& inverse kinematics posing techniques). You create/edit them in Edit Mode, but except in case of connected bones, their relationships have no effect on bone transformations in this mode (i.e. transforming a parent bone will not affect its children).

The easiest way to manage bones relationships is to use the Relations panel in the Bone tab.

--- Fin du chapitre: _structure ---

--- Début du chapitre: _stucci --- Stucci Texture panels.\&

The Stucci texture is based on noise functions. It is often used for stone, asphalt, or oranges, normally for bump mapping to create grainy surfaces.

Plastic is the standard Stucci, while the \& walls\& is where Stucci gets its name. This is a typical wall structure with holes or bumps.

There are two methods available for working with Noise.

Dimension of the Noise table.

Depth of the Stucci calculations.

--- Fin du chapitre: _stucci ---

--- Début du chapitre: _style_guide --- This page covers conventions concerning the translations.

Note

We expect our readers to use the English version of Blender, not a translated one.

The translations are licensed under the same License as the original.

Can be translated, but only as an addition, not as a replacement. See also Adding Text.

Only translate these, when the localized expression is common! See also Technical Terms.

Simply mark the text as fuzzy and/or add a comment. The next translator might understand it.

You probably will not find the original scene if it is a screenshot of a file and it is too much load on the server (and too much work for you).

We expect our readers to use the English UI.

Do not translate it! It will do more harm than good!

In general, the technical terms used in computer graphics are quite new or even downright neologisms invented for the needs, so they do not always have a translation in your language. Moreover, a large part of Blender users use its English interface.

As a result, unless a term has an evident translation, you should preferably use the English one, putting it in italic. You can then find a translation for it, which you will use from time to time (e.g. to avoid repetitions\&). This is also valid in the other way: even when a term has a straightforward translation, do not hesitate to use its English version from time to time, to get the reader used with it\&

If a term is definitively not translatable, simply use the English one, but make sure its glossary entry is translated. In the glossary, the English term is written first (to maintain alphabetic order) with the translated entry following in parenthesis, when appropriate.

Generally, you should always translate exactly what is in the text, and avoid providing updates or extra information.

But sometimes that is necessary, for example when talking about the manual itself: To a foreign reader it is not clear, that they can contribute English text only, whereas this is obvious to an English reader.

In these (rare) cases you can and should provide extra information.

When the manual is updated, those translations which are outdated will be marked as fuzzy. To keep track with that, you can use a tool we created for that task, see How to install it .

--- Fin du chapitre: `_style_guide` ---

--- Début du chapitre: `_subdivide` ---Reference

Edit Mode

Armature `\xe2\x80\xa3 Subdivide`

You can subdivide bones, to get two or more bones where there was just one bone. The tool will subdivide all selected bones, preserving the existing relationships: the bones created from a subdivision always form a connected chain of bones.

To create an arbitrary number of bones from each selected bone in the Subdivide MultiAdjust Last Operation panel.

Specifies the number of cuts. As in mesh editing, if you set `ncuts`, you will get `n + 1` bones for each selected bone.

An armature with one selected bone, just before multi-subdivision.

`\xc2\xb6`

The selected bone has been `\xe2\x80\x99cut\xe2\x80\x9d` two times, giving three sub-bones. `\xc2\xb6`

--- Fin du chapitre: `_subdivide` ---

--- Début du chapitre: `_subdivide_curve` --- The Subdivide Curve node adds more control points in between existing control points on the curve input. For B-splines and poly splines, the shape of the spline will not be changed at all.

With B-splines curves, this can be used to increase the control on the shape of the curve while still having the higher-level provided by B-splines splines. Unlike the Resample Curve Node, where they are converted to poly splines.

Standard geometry input.

The number of control points to create on the segment following each point. When the input is a field, the number of cuts for a segment is determined by the value of the field when evaluated at the previous point. This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_subdivide_curve` ---

--- Début du chapitre: `_subdivide_edge_ring` ---Reference

Edit Mode

Edge `\xe2\x80\xa3 Subdivide Edge-Ring`

Take an edge ring, and subdivide with interpolation options.

SeeBridge Edge Loops.

--- Fin du chapitre: _subdivide_edge_ring ---

--- Début du chapitre: _subdivide_mesh --- The Subdivide Meshnode adds new faces to mesh geometry using a simple interpolation for deformation. Standard geometry input.

The number of subdivisions to apply to the input geometry.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: _subdivide_mesh ---

--- Début du chapitre: _subdivision ---Reference

Render \xe2\x80\xa3 Subdivision

Note

These settings are only available if Experimental Feature Set is turned on.

These settings are used to controlAdaptive Subdivision.

Size ofMicropolygonsin pixels for the final/viewport render.

Multiplier for dicing rate of geometry outside of the camera view.The dicing rate of objects is gradually increased the further they are outside the camera view.Lower values provide higher quality reflections and shadows for off screen objects,while higher values use less memory.

Stop subdividing when this level is reached even if the dicing rate would produce finerTessellation.

Camera to use as reference point when subdividing geometry,useful to avoid crawling artifacts in animations when the scene camera is moving.

--- Fin du chapitre: _subdivision ---

--- Début du chapitre: _subdivision_surface --- The Subdivision Surfacenode adds new faces to mesh geometry using a Catmull-Clark subdivision method.

Standard geometry input.

The number of subdivisions to apply to the input geometry.

Controls how smooth edges should be with Weighted Edge Creases .

Controls how much the subdivision surface should be pulled towards the vertex.Similar to edge creases, but allows individual vertices to be creased.

Controls how subdivision smoothing is applied to UVs.

UVs remain unchanged.

UV islands are smoothed, but their boundary remain unchanged.

UVs are smoothed, corners on discontinuous boundary and junctions of three or more regions are kept sharp.

UVs are smoothed, corners on discontinuous boundary,junctions of three or more regions and darts and concave corners are kept sharp.

UVs are smoothed, boundaries are kept sharp.

UVs and their boundaries are smoothed.

Controls how open boundaries (and corners) are smoothed.

Smooth boundaries, including corners.

Smooth boundaries, but corners are kept sharp.

Standard geometry output.

--- Fin du chapitre: _subdivision_surface ---

--- Début du chapitre: _subtract --- This effect takes away one strip\xe2\x80\x99s color from the second.

Make a negative of an image using this effect, or switch the order of the strips and just darken the strip. Subtracting a hue of blue from a white image will make it yellow, since red and green make yellow.

This strip has no options.

Subtract Effect. \xc2\xba6

--- Fin du chapitre: _subtract ---

--- Début du chapitre: _sun_beams --- The Sun Beams node provides a cheap way of adding sun beams based on image brightness alone.

Sun Beams is a 2D effect for simulating the effect of bright light getting scattered in a medium (Crepuscular Rays). This phenomenon can be created by renderers, but full volumetric lighting is a rather arduous approach and takes a long time to render.

Standard color input.

Source point of the rays as a factor of the image dimensions.

Length of the rays as a factor of the image size.

Standard color output.

Usually, the first step is to define the area from which rays are cast. Any diffuse reflected light from surfaces is not going to contribute to such scattering in the real world, so should be excluded from the input data. Possible ways to achieve this are:

Entirely separate image as a light source.

Brightness/contrast tweaking to leave only the brightest areas.

Muting shadow and midtone colors, which is a bit more flexible.

Masking for ultimate control.

After generating the sun beams from such a light source image they can then be overlaid on the original image. Usually, a simple

\xe2\x80\x9cAdd\xe2\x80\x9d Mix node is sufficient, and physically correct because the scattered light adds to the final result.

--- Fin du chapitre: _sun_beams ---

--- Début du chapitre: _surface --- The surface shader defines the light interaction at the surface of the mesh. One or more BSDFs specify if incoming light is reflected back, refracted into the mesh, or absorbed.

Emission defines how light is emitted from the surface, allowing any surface to become a light source.

Stands for Bidirectional Scattering Distribution Function. It defines how light is reflected and refracted at a surface.

BSDFs reflect an incoming ray on the same side of the surface.

BSDFs transmit an incoming ray through the surface, leaving on the other side.

BSDFs are a type of Transmission, transmitting an incoming ray and changing its direction as it exits on the other side of the surface.

A major difference from non-physically-based renderers is that direct light reflection from lights and indirect light reflection of other surfaces are not decoupled, but rather handled using a single BSDF. This limits the possibilities a bit, but we believe overall it is helpful in creating consistent-looking renders with fewer parameters to tune.

For the glossy BSDFs, the roughness parameter controls the sharpness of the reflection, from 0.0 (perfectly sharp) to 1.0 (very soft).

--- Fin du chapitre: _surface ---

--- Début du chapitre: _surface_deform --- The Surface Deform modifier allows an arbitrary mesh surface to control the deformation of another,

essentially transferring its motion/deformation. One great use for this is to have a proxy mesh for cloth simulation, which will in turn drive the motion of your final and more detailed mesh, which would otherwise not be suitable for simulation.

The Surface Deform modifier. \xc2\xba6

The object to which to bind (this setting is unavailable after binding).

Warning

Target Mesh Validity

While there are no restrictions with regard to the modified mesh, the target object's mesh has a few constraints, which if not followed, will prevent a successful binding:

It must not contain edges with more than two faces.

It must not contain concave faces.

It must not contain overlapping vertices (doubles).

It must not contain faces with collinear edges.

How much a vertex bound to one face of the target will be affected by the surrounding faces (this setting is unavailable after binding). This essentially controls how smooth the deformations are.

Note

While lower values result in smoother deformations, they may also introduce slight artifacts.

The overall amount of influence the modifier has on deforming the mesh. Allows you to define the influence per vertex.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier. The setting reverses the weight values of the group.

Only record bind data for vertices that have nonzero weights in the group at the time of bind. This is an optimization, but adding new vertices to the group will require a rebind.

Bind the current state of the modified mesh to the current state of the target mesh, such that any later change in the target mesh will deform the modified mesh as well. Note that until the bind has been executed, this modifier will have no effect at all.

Once the mesh is bound, the Bind button changes to Unbind. Executing this frees the modified mesh from the target, and resets it to its original shape.

Note

The meshes are bound with regard to global coordinates, but later transformations on the objects are ignored. This means that one can freely transform the target or modified object after binding, without affecting the modified object. The modified mesh will only pick up changes to the target object's mesh itself.

Note

The further a mesh deviates from the target mesh surface, the more likely it is to get undesirable artifacts. This is an inherent characteristic of surface binding in general, so it is recommended to have reasonably well matching meshes, in order to get a good bind.

Cloth simulation copied to an arbitrary mesh with rings as instancing faces. \xc2\xba6

--- Fin du chapitre: _surface_deform ---

--- Début du chapitre: _swirl --- The Swirl Visual Effect applies a swirling pattern to the object. The effect uses an object as the center of the swirl.

Swirl Visual Effect. \xc2\xba
Sets the object to use as the center of the swirl.
External radius size of the swirl.
Rotation angle of the swirl. A value of 0 shows no swirl.
Angle: 0\xc2\xba.\xc2\xba
Angle: 15\xc2\xba.\xc2\xba
Angle: 45\xc2\xba.\xc2\xba
Angle: 90\xc2\xba.\xc2\xba
--- Fin du chapitre: _swirl ---

--- Début du chapitre: _switch --- The Switchnode outputs one of two inputs depending on a condition.Only the input that is passed through the node is computed.
See also
The Menu Switch Node and Index Switch Node can be used to switch between an arbitrary amount of inputs.
Determines which of the two inputs below will be passed through.
Is passed through when the switch is set to false.
Is passed through when the switch is set to true.
Determines the type of the data that is handled by the node.
One of the two inputs without any modifications.
--- Fin du chapitre: _switch ---

--- Début du chapitre: _switch_direction ---Reference
Edit Mode
Armature \xe2\x80\xa3 Switch Direction
Alt- F
This tool allows you to switch the direction of the selected bones(i.e. their root will become their tip, and vice versa).
Switching the direction of a bone will generally break the chain(s) it belongs to.However, if you switch a whole (part of a) chain, the switched bones will still be parented/connected,but in \xe2\x80\x9creversed order\xe2\x80\x9d. See the Fig.Switching example..
An armature with one selected bone, and one selected chain of three bones, just before switching. \xc2\xba
The selected bones have been switched. Bone.005 is no more connected nor parented to anything.The chain of switched bones still exists, but reversed (now Bone.002 is its root, and Bone is its tip).Bone.003 is now a free bone. \xc2\xba
--- Fin du chapitre: _switch_direction ---

--- Début du chapitre: _switch_stereo_view --- The Switch Viewnode combines the views(left and right) into a single stereo 3D output.This can be useful if for example, you need to treat the view as separate images by combining each of the views.
See also
The multi-view workflow .
Left-eye image input.
Right-eye image input.
This node has no properties.
Stereo 3D image output.
Compositor, Backdrop and Split Viewer Node. \xc2\xba
The views to render are defined in the current scene views,in a similar way as you define the composite output resolution in the current scene

render panel, regardless of the Image nodes resolutions or Render Layers from different scenes.

--- Fin du chapitre: `_switch_stereo_view` ---

--- Début du chapitre: `_symmetrize` ---Reference

Edit Mode

Armature `\xe2\x80\xa3 Symmetrize`

This operator will mirror the selected bones along the X axis based on Blender's bonenaming convention for symmetrical armatures, either from left to right or right to left, depending on the selection.

Note

If the side of the bone cannot be determined, it will be ignored.

Bones with the opposite names that don't yet exist will be created, and already existing ones will be overwritten. If matching bones are selected on both sides, mirroring will happen from right to left.

Symmetrized bone and constraint properties will have the necessary changes to mirror their behaviors. When symmetrizing bones with Action Constraints, the necessary keyframes will be added to the target Action to result in symmetrical movement when the Action is activated.

Note

Note that bone or constraint drivers will not be created or affected in any way.

--- Fin du chapitre: `_symmetrize` ---

--- Début du chapitre: `_symmetry` ---Reference

Vertex Paint Mode

Toolbar `\xe2\x80\xa3 Tool \xe2\x80\xa3 Symmetry`

Use this option for mirrored painting on groups that have symmetrical names, like with suffix `\xe2\x80\x9cR\xe2\x80\x9d/`

`\xe2\x80\x9cL\xe2\x80\x9d` or `\xe2\x80\x9cR\xe2\x80\x9d /`

`\xe2\x80\x9cL\xe2\x80\x9d`. If a group has no mirrored counterpart, it will paint symmetrically on the active group itself. You can read more about the naming convention in [Editing Armatures: Naming conventions](#). The conventions for armatures/bones apply here as well.

Mirror the brush strokes across the selected local axes. Note that if you want to alter the directions the axes point in, you must rotate the model in Edit Mode and not in Object Mode.

These settings allow for radial symmetry in the desired axes. The number determines how many times the stroke will be repeated within 360 degrees around the central axes.

Use topology-based mirroring, for when both sides of a mesh have matching mirrored topology. See [here](#) for more information.

--- Fin du chapitre: `_symmetry` ---

--- Début du chapitre: `_system` --- The System section allows you to set graphics card options, memory limits & sound settings.

If your hardware does not support some of the options described on this page, then they will either not be displayed or be corrected on startup.

Preferences System section. `\xc2\xb6`

Changes the computing device the Cycles render engine uses to render images. Cycles can use either the CPU or certain GPUs to render images, for more information see the GPU Rendering page.

When set to None or when the only option is None: the CPU will be used as the computing device for Cycles.

If the system has a compatible Nvidia CUDA device, it will be available as an option for rendering with Cycles.

If the system has a compatible Nvidia OptiX device, it will be available as an option for rendering with Cycles.

If the system has a compatible AMD HIP device, it will be available as an option for rendering with Cycles.

If the system has a compatible Intel oneAPI device, it will be available as an option for rendering with Cycles.

If the system has a compatible Apple Metal device, it will be available as an option for rendering with Cycles.

Allocates resources across multiple GPUs rather than duplicating data, effectively freeing up space for larger scenes. Note that in order for this option to be available, the GPUs must be connected together with a high bandwidth communication protocol.

Currently only NVLink on Nvidia GPUs is supported.

Enables the use of hardware ray tracing on Intel GPUs, providing better overall performance.

Only supported with oneAPI rendering devices..

Speeds up rendering by enabling AMD hardware ray tracing on RDNA2 and above, with shader fallback on older cards. This feature is experimental and some scenes may render incorrectly.

This feature is only available when using a HIPrender device.

MetalRT for ray tracing uses less memory for scenes which use curves extensively, and can give better performance in specific cases.

Disable MetalRT (uses BVH2 layout for intersection queries).

Enable MetalRT for intersection queries.

Automatically pick the fastest intersection method.

Make this installation your default Blender (MS-Windows & Linux only).

On Linux, if Blender is installed from a package manager such as Snap, file association is handled by the package manager.

Make the currently in use Blender installation the default for generating thumbnails and the default for opening blend-files.

Remove file association & thumbnailer.

Register Blender for all users, requires escalated privileges.

Linux Registration

Files are setup files under: /usr/local for all users, otherwise ~/.local is used.

A desktop file & icon is installed so the application is available in launchers.

A file association for *.blend is setup.

The thumbnailer is installed so blend-file thumbnails will be shown in file managers (For All Users only).

Allow Blender to access the internet.

Add-ons that follow this setting will only connect to the internet if enabled. However, Blender cannot prevent third-party add-ons from violating this rule.

The time (in seconds) that online operations may wait before timing out.

Use the systems default when zero.

The maximum number of simultaneous connections an online operation may make.

Do not limit the number of connections when zero.

Number of Undo steps available.

Maximum memory usage in Mb (0 is unlimited).

This enables Blender to save actions done when you are not in Edit Mode. For example, duplicating objects, changing panel settings or switching between modes.

Warning

While disabling this option does save memory, it stops the Adjust Last Operation panel from functioning, also preventing tool options from being changed in some cases. For typical usage, it's best to keep this enabled.

See also

Read more about Undo and Redo options.

The number of lines, buffered in memory of the console window. Useful for debugging purposes and command-line rendering.

Time since last access of a GL texture in seconds, after which it is freed. Set this to 0 to keep textures allocated.

Number of seconds between each run of the GL texture garbage collector.

Time since last access of a GL vertex buffer object (VBO) in seconds after which it is freed (set to 0 to keep VBO allocated).

Number of seconds between each run of the GL vertex buffer object garbage collector.

Upper limit of the Video Sequencer and Movie Clip Editor memory cache (in megabytes). For an optimal Clip editor and Sequencer performance, high values are recommended.

Writes cached strips to disk which can store a lot more than RAM. To use the Disk Cache, this option must be enabled, the Disk Cache Directory and Disk Cache Limit set, then save or reopen the existing blend-file.

The location on disk to store the cache.

Upper limit of the Video Sequencer's disk cache (in gigabytes), setting to zero disables disk cache.

The level of compression to compress image in the disk cache. This has a trade off between saving disk space and requiring more processing. The more compression used requires faster disk write/read speeds and more CPU usage.

When and how Proxies are created.

Build proxies for added movie and image strips in each preview size.

Set up proxies manually.

See also

Sequencer Cache Properties

This panel contains the sound settings for live playback within Blender and are only available with a device other than None. To control these settings for exporting sound see the Encoding Panel and Audio Panel.

Sets the audio engine to use to process and output audio.

No audio playback support (audio strips can still be loaded and rendered normally).

On macOS, CoreAudio is the native audio API. This is the default setting for macOS users and should be preferred.

PulseAudio is the most commonly used sound server on modern Linux distributions. If PulseAudio is available, this should be the preferred setting on Linux.

On Windows, WASAPI is the native audio API introduced with Windows Vista. This is the default setting for Windows users and should be preferred.

High quality professional audio engine that needs a properly configured server running on your system. Supports accurate synchronization with other professional audio applications using Jack.

Available on all platforms in case the native engines do not work. The played back 3D audio might sound different than when rendered. Uses Simple Direct Media Layer API from libsdl.org which supports all platforms. Might be of lower quality and thus should only be used as backup.

The number of audio source locations to output.

Output a single audio channel.

Output two audio channels; typically a left and right channel.

Output a four audio channels.

Output a five audio channels with one LFE channel.

Output a seven audio channels with one LFE channel.

Sets the number of samples used by the audio mixing buffer. Higher buffer sizes can cause latency issues, but if you hear clicks or other problems, try to increase the size.

Sets the audio sampling rate.

Sets the audio sample format.

--- Fin du chapitre: `_system` ---

--- Début du chapitre: `_tabs_panels` --- Top: Horizontal Tab header in the Topbar. Bottom: Vertical Tab header shows tab icons in the Properties.

`\xc2\xba`

Tabs are used to control overlapping sections in the user interface. The content of only one Tab is visible at a time. Tabs are listed in a Tab header, which can be horizontal or vertical.

Vertical tabs can be switched with `Ctrl-Wheel` from anywhere in the tab. You can also cycle through tabs with `Ctrl-Tab`, or press down `LMB` and move the mouse over the tab header icons. (This does not apply to Workspace tabs; see Workspace controls.)

Panels in Properties. `\xc2\xba`

A panel is highlighted in yellow and a subpanel in red.

The smallest organizational unit in the user interface is a panel. The panel header shows the title of the panel. It is always visible. Some panels also include subpanels.

A panel can either be expanded to show its contents, or collapsed to hide its contents. An expanded panel is indicated by a down-arrow (`\xe2\x96\xbc`) in the panel header, while a collapsed panel is shown with a right-arrow (`\xe2\x96\xba`).

Clicking `LMB` on the panel header expands or collapses it.

Pressing `A` expands/collapses the panel under the mouse pointer.

Clicking `Ctrl-LMB` on the header of a collapsed panel will expand it and collapse all others.

Clicking `Ctrl-LMB` on the header of an expanded panel will expand/collapse all its subpanels.

Dragging with `LMB` over the headers will expand or collapse many at once.

You can change the position of a panel within its region by clicking and dragging the grip widget (`:::`) on the right side of its header.

Sometimes it is desirable to view panels from different tabs at the same time. Like, for instance, having access to a camera's properties, while other objects are selected. This has been solved by making panels pinnable.

A pinned panel remains visible regardless of which tab has been selected. You can pin a panel by clicking on the pin icon in its header. Panels that do not have a pin icon can be pinned by `RMB` on the panel header and selecting `Pin`, or by pressing `Shift-LMB`.

Note

Pinning is not available for all panels. For example, it's available in the Sidebar but not in the Properties editor.

Example Presets menu.

A list of available presets. A selection will override the included properties.

New presets can be added based on the currently applied set of properties, which will be saved for later reuse. A pop-up opens where you can set a name, after which you can select it from the list and in some cases additional settings.

Deletes the selected preset.

--- Fin du chapitre: _tabs_panels ---

--- Début du chapitre: _tags --- A different set of tags is available for the different extensions types. This is the list of the tags currently supported by the Blender Extensions Platform:

3D View
Add Curve
Add Mesh
Animation
Bake
Camera
Compositing
Development
Game Engine
Geometry Nodes
Grease Pencil
Import-Export
Lighting
Material
Modeling
Mesh
Node
Object
Paint
Pipeline
Physics
Render
Rigging
Scene
Sculpt
Sequencer
System
Text Editor
Tracking
User Interface
UV
Dark
Light
Colorful
Inspired By
Print
Accessibility
High Contrast

--- Fin du chapitre: _tags ---

--- Début du chapitre: _tangent --- This modifier bases its effect on the traveling direction of the stroke evaluated at the stroke's vertices.

--- Fin du chapitre: _tangent ---

--- Début du chapitre: _templates --- The following guide provides patterns for interface elements and directories.

Each operator should receive its own heading or page based on the length of the content. At the start should be a reference admonition documenting the context of the operator:

Panels should be documented by their own heading, nested panels should use decreasing heading levels. Each panel could have its own page based on the length of documentation and/or the amount of panels. Expanded menus that toggle what properties are presented to the user should be treated like subpanels:

Properties should be documented using definition lists. Properties that are hidden based on other properties should use nested definitions:

Select menus should be documented using the following syntax:

Nodes should always have three headings inputs, properties and outputs with a note of absence if the node has none. At the end of the page can be an optional example(s) section:

should be generally structured as follows:

directory_name/

index.rst (contains links to internal files)

introduction.rst

section_1.rst

section_2.rst

For example:

rendering/

index.rst

cycles/

index.rst

introduction.rst

materials/

index.rst

introduction.rst

volumes.rst

The idea is to enclose all the content of a section inside of a folder. Ideally every section should have an index.rst (containing the TOC for that section) and an introduction.rst (introducing) to the contents of the section.

By default, a table of contents should show two levels of depth:

--- Fin du chapitre: _templates ---

--- Début du chapitre: _text --- The Text strip allows you to directly display text in the Sequence editor. The strip will display the text inserted in its text field on the final sequence.

Tip

All Text strips in a video sequence can be exported as a SubRip file. This is useful when using Text strips as subtitles.

The actual text displayed.

Wraps the text by the percentage of the frame width, setting this to zero disables word wrapping.

Data-Block Menu to choose which font-file is used to render the text.

Use a bold font face with a strong/thick visual appearance.

Use an italicized font face with a slanted visual appearance.

Size of the text.

The text color.

Creates a shadow of the specified color under the text.

Defines the position of the shadow as an angle, 0 being to the right and 90 being below.

Amount to shift the shadow compared to the normal text.

Amount to blur the shadow.

Creates a line with the defined color enclosing the shape of the text.

The thickness of the outline.

Creates a background for the text to improve the readability and clarity of text in some situations. The color and opacity of the box can be adjusted using the color selector.

The distance the box boundaries extends from the boundaries of the font glyphs. The distance is measured as a factor of the image's width.

Positions the text on the X, Y axis.

Horizontal (X) or vertical (Y) anchor point of the text relative to the location.

Text effect.

--- Fin du chapitre: _text ---

--- Début du chapitre: _texture --- The Texture node makes 3D textures available to the Compositor.

Note

The Texture node is not supported by the viewport compositor.

A vector (XYZ) transforming the origin of the texture.

A vector (XYZ) to scale the texture.

The texture can be selected from a list of textures available in the current blend-file or linked-in textures. The textures themselves can not be edited in the Compositor, but in the Texture Node Editor.

Gray-scale color values.

Color values.

--- Fin du chapitre: _texture ---

--- Début du chapitre: _texture_coordinate --- The Texture Coordinate node is commonly used for the coordinates of textures, typically used as inputs for the Vector input for texture nodes.

This node has no inputs.

Specific object to use for object space coordinates. This only affects the Object output.

If the object is generated by instancing from vertices or faces, use texture coordinates from instancer. This only affects the Generated and UV outputs.

From left to right: Sphere with a UV-mapped texture. Small spheres instanced to the faces of the textured sphere using instancing from faces. Small spheres with From Instancer enabled, using the UV map of the large sphere.

Note

From Instancer only works with the UV output when the object is instanced, either from particles or from faces. Automatically-generated texture coordinates from the vertex positions of the mesh without deformation, keeping them sticking to the surface under animation. Range from 0.0 to 1.0 over the bounding box of the undeformed mesh. See Texture Spaces for more information. Object space normal, for texturing objects with the texture staying fixed on the object as it transformed. The Normal output can be used on Point and Spot lights. The coordinates will take the rotation of the light into account. UV texture coordinates from the active render UV map. See UV Mapping for more information.

Note

In order to select UV map other than the active map you must use the UV Map node.

Uses an object as a source for coordinates. Often used with an empty, this is an easy way to place a small image at a given point on the object. This object can also be animated, to move a texture around or through a surface.

Position coordinate in camera space.

Location of shading point on the screen, ranging from 0.0 to 1.0 from the left to right side and bottom to top of the render. This is well suited for blending two objects.

Uses the direction of the reflection vector as coordinates. This is useful for adding reflection maps. You will need this input when using environment maps.

--- Fin du chapitre: _texture_coordinate ---

--- Début du chapitre: _texture_influence --- Reference

Texture Influence

Defines the settings of a Particle system spatial with a texture.

Affect the emission time of the particles.

Affect the life time of the particles.

Affect the density of the particles.

Affect the particles size.

Affect the particles initial velocity.

Affect the particles velocity damping.

Affect the particles gravity.

Affect the particles force fields.

Affect the child hair length.

Affect the child clumping.

Affect the child kink.

Affect the child roughness.

--- Fin du chapitre: _texture_influence ---

--- Début du chapitre: _texture_mapping --- The Texture Mapping Modifier change the strokes texture UV position.

Texture Mapping

The texture transformation will be applied to the stroke/fill or stroke UVs.

Selects the texture fitting method.

The texture keep a consistent length along the strokes.

The texture is normalized to fit the stroke length.

Moves the texture along the strokes.

Rotates the points of the strokes.

Note

The Rotationoption is limited to a range of -90 to 90 degrees.

Factor for the texture scale.

Sets the texture angle.

Moves the texture origin.

X, Y

Factor for the texture scale.

SeeInfluence Filters.

Rotation: 0\xc2\x00.\xc2\x06

Rotation: 45\xc2\x00.\xc2\x06

Rotation: 90\xc2\x00.\xc2\x06

--- Fin du chapitre: _texture_mapping ---

--- Début du chapitre: _texture_slots --- Texture Slots settings.\xc2\x06

The combination of images associated with UV maps is called

\xe2\x80\x99slots\xe2\x80\x9d.

Selecting a Paint Slotsor Canvas Imagewill also display the corresponding image in the Image Editor.

The slot system includes two painting modes:

This mode tries to detect the slots from the materials of the mesh.

For the Cycles renderer, all textures (Image Texturenode) in the material\xe2\x80\x99s node treeare added in the slots tab.

Alist viewof slots.Activate a certain slot to use it for painting by LMBclick on it.

You can just select an existing image and painting will use the active UV layer for painting.

Allows you to select the image used as a canvas.

Create a new image.

Allows you to select the UV layer for painting.(Same as the currently active UV map in the mesh\xe2\x80\x99s UV Mapspanel.)

Set the interpolation mode of the texture. This can be Linear or Closest.

Repack (or save if external file) all edited images.Same as in theImage Editor.

The Add Simple UVsdoes a simple cube unwrap followed by a pack operation.It\xe2\x80\x99s still recommended to make a custom unwrap.

This operator is available when the object does not already have a UV Map.

--- Fin du chapitre: _texture_slots ---

--- Début du chapitre: _texture_space ---Reference

Object Mode and Edit Mode

Object \xe2\x80\x93 Transform \xe2\x80\x93 Move/Scale Texture Space

The Move/Scale Texture Space tool transforms theTexture Spaceof the object, instead of the object or element itself.

--- Fin du chapitre: _texture_space ---

--- Début du chapitre: _text_editor --- This editor can be used to write Python scripts, Open Shading Language scripts,or just plaintext notes. To open it, you can switch to the Scripting workspaceor pressShift- F1lto replace the current editor.

The newly opened Text editor is empty, with a very simple header.More options become available when a text file is created or opened.

Text header. \xc2\xb6
Text header with a text loaded. \xc2\xb6
The standard editor selection button.
Editor\xe2\x80\x99s menus.
Resolves modified file conflicts when an external text file is updated from another program.
Opens the file from drive again, overriding any local changes.
Converts the external text data-block into an internal one.
Hides the warning message until the external text file is modified externally again.
A data-block menu to select a text or to create a new one.After that the header will change.
Executes the text as a Python script Alt- P . See Template Menu.
Toggles for line numbers, word wrapping, and syntax highlighting.
When an OSL-file is opened, this updates the Shader Script node with new options and sockets from the script.
Show or hide the Sidebar.
Displays the text file\xe2\x80\x99s line numbers on the left of the Main View.
Wraps words that don\xe2\x80\x99t fit into the horizontal space by pushing them to a new \xe2\x80\x9cpseudo line\xe2\x80\x9d.
Colors special words, in the Main View, that are used in the Python programming language.
Emphasizes the active line by altering the color of the background.
Increase/decrease the font size of text in the main view.
Moves the view and cursor to the start of the text file.
Moves the view and cursor to the end of the text file.
Moves the cursor to the start of the current line.
Moves the cursor to the end of the current line.
Moves the cursor to the same position in the line above the current line.
Moves the cursor to the same position in the line below the current line.
Moves the cursor to the beginning of the previous word.If the cursor is in the middle of a word, the cursor is moved to the beginning of the current word.
Moves the cursor to the end of the next word.If the cursor is in the middle of a word, the cursor is moved to the end of the current word.
Creates a new text Data Block.
Loads an external text file that is selected via the File Browser.
Reopens (reloads) the current buffer (all non-saved modifications are lost).
Edit text file in external text editor.The external editor can be configured in the User Preferences.
Saves an already open file.
Saves text as a new text file.A File Browser is opened to select the directory to save the file along with giving the file a name and extension.
Runs the text data-block as a Python script on loading the blend-file.Read more about the registration of Python modules in API documentation.
Runs the Python script each time you make a change.
Executes the text as a Python script. See Running Scripts for more information.

SeeUndo & Redo.
 Cuts out the marked text into the clipboard.
 Copies the marked text into the clipboard.
 Pastes the text from the clipboard at the cursor location in the Text editor.
 Duplicates the current line.
 Swaps the current/selected line(s) with the above.
 Swaps the current/selected line(s) with the below.
 Shows the Find & Replacepanel in the Sidebar.
 Finds the next instance of the selected text.
 Shows a pop-up, which lets you select a line number where to move the cursor to.
 Shows a selectable list of words already used in the text.
 Converts the text file to aText Objecteither as One Objector One Object Per Line.
 Selects the entire text file.
 Selects the entire current line.
 Selects the entire current word.
 Selects everything above the cursor.
 Selects everything below the cursor.
 Selects everything between the beginning of the current line and the cursor.
 Selects everything between the cursor and the end of the current line.
 Selects everything between the cursor and the position of the cursor one line above.
 Selects everything between the cursor and the position of the cursor one line below.
 Selects everything between the cursor and the beginning of the previous word.If the cursor is in the middle of a word, select everything to the beginning of the current word.
 Selects everything between the cursor and the end of the next word.If the cursor is in the middle of a word, select everything to the end of the current word.
 Inserts a tab character at the cursor.
 Unindents the selection.
 Toggles whether the selected line(s) are a Python comment.If no lines are selected, the current line is toggled.
 Converts indentation characters To Spacesor To Tabs.
 Contains a number of templates for both Python and Open Shading Language scripts.
 Typing on the keyboard produces text in the text buffer.
 As usual, pressing, dragging and releasing LMBselects text.Pressing RMBopens the context menu.

Tip

The Text editor is also handy when you want to share your blend-file with others:you can leave a note that explains how the file is structured.Be sure to keep the editor visible when saving so they\xe2\x80\x99ll see it!

Searches for instances of a text that occur after the cursor.Using the eyedropper icon will search for the currently selected textand sets the selection to the match. Find Nextsearches for the next instance of the text.

Searches for the text specified in Find Textand replaces it with the new text.Using the eyedropper icon will set the currently selected text as

the replace text. Replace searches for the next match and replaces it. Replace All searches for the match and replaces all occurrences of the match with the new text.

Search is sensitive to uppercase and lowercase letters.

Search again from the start of the file when reaching the end.

Search in all text data-blocks instead of only the active one.

Shows a vertical margin line to help keep text lines at a reasonable length. The position of this margin line is specified by Margin Column.

The size of the font used to display text.

The number of character spaces to display tab characters with.

Whether to use Tabs or Spaces for indentation.

The Text editor footer displays whether the text is saved internally or externally and if there are unsaved changes to an external file. For external files, this region also displays the file path to the text file.

The most notable keystroke is Alt-P which executes the current text as a Python script. You can access not just the standard Python modules, but also a whole bunch of Blender-specific ones; see Scripting & Extending Blender.

--- Fin du chapitre: _text_editor ---

--- Début du chapitre: _themes --- The Theme section allows you to customize interface appearance and colors.

The colors for each editor can be set separately by simply selecting the editor you wish to change in the multi-choice list at the left, and adjusting colors as required. Notice that changes appear in real-time on your screen. In addition, details such as the dot size in the 3D Viewport or the Graph Editor can also be changed.

Select the Theme from a list of predefined Themes.

You add a custom theme to the preset list by LMB on the Add button + .

You remove a custom theme from the preset list by LMB on the Remove button - .

You save a custom theme in the preset list by LMB on the Save button.

This will save the theme to an XML file in the

./scripts/presets/interface_theme/ subdirectory of one of the configuration directories .

Load and apply a Blender XML theme file and add it to the list of theme presets.

Reset to the default theme colors.

Blender comes bundled with a small selection of themes.

This is an example of the theme Blender Light. \xc2\xbb6

--- Fin du chapitre: _themes ---

--- Début du chapitre: _thickness --- Controls the thickness of the Freestyle strokes.

Line Style: Thickness. \xc2\xbb6

The base thickness for this line style.

Control the position of stroke thickness from the original (backbone) stroke geometry. There are four choices:

The thickness is evenly split to the left and right side of the stroke geometry.

The strokes are drawn within object boundary.

The strokes are drawn outside the object boundary.

Specifies the relative position by a number between 0.0 (inside) and 1.0 (outside), in the Thickness Ratio number field just below.

Note

The thickness position options are applied only to strokes of edge types Silhouette and Border, since these are the only edge types defined in terms of the object boundary. Strokes of other edge types are always drawn using the Center option.

The modifier output can be mixed with the base property using the usual methods (see for example the Mix compositing node).

How much the result of this modifier affects the current property.

Along Stroke

Calligraphy

Crease Angle

Curvature 3D

Distance from Camera

Distance from Object

Material

Noise

Tangent

--- Fin du chapitre: `_thickness` ---

--- Début du chapitre: `_thumb` ---Reference

Sculpt Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Brush Settings \xe2\x80\xa3

Advanced \xe2\x80\xa3 Brush Type

Similar to the Grab brush, but instead only moves the geometry along the surface (using the area plane). This is useful for grabbing surfaces with a very specific direction, or without making too impactful changes to the overall object shape.

Note

More info at General brush settings and on Advanced brush settings.

--- Fin du chapitre: `_thumb` ---

--- Début du chapitre: `_tiling` ---Reference

Image Editor

Paint Mode

Sidebar \xe2\x80\xa3 Tools \xe2\x80\xa3 Tiling

Wraps the stroke to the other side of the image as your brush moves off the opposite side of the canvas. Very handy for making seamless textures.

left/right

top/bottom

--- Fin du chapitre: `_tiling` ---

--- Début du chapitre: `_time` --- The Time Curve node generates a factor value (from 0.0 to 1.0) between the scene start and end time, using a curve mapping.

This node has no inputs.

The Y value defined by the curve is the factor output. For the curve controls, see Curve widget .

Tip

Flipping the curve around reverses the time input, but doing so is easily overlooked in the node setup.

Start frame and End frame of the range of time specifying the value the output should last. This range becomes the X axis of the graph. The time input could be reversed by specifying a start frame greater than the end frame.

The Y value of the curve at the current frame.

Hint

The Map Value node can be used to map the output to a more appropriate value. With some curves, it is possible that the Time Curve node may output a number larger than one or less than zero. To be safe, use the Min/Max clamping function of the Map Value node to limit output.

Time controls from left to right: no effect, slow down, freeze, accelerate, reverse.

--- Fin du chapitre: _time ---

--- Début du chapitre: _timeline --- The Timeline editor is used to jump to different frames, manipulate keyframes, and control animation playback.

The Timeline.

The X axis represents time, with the numbers 0/50/100/... being frame numbers. The blue line is the Playhead indicating the current frame, and the diamond shapes are Keyframes, points where you specified a certain value for a certain property at a certain time.

Panning is done by dragging MMB.

Zooming is done by dragging Ctrl-MMB, rolling the mouse Wheel, or pressing Numpad Minus/ Numpad Plus.

You can also use the scrollbars located at the bottom and the right of the editor.

Playhead.

The Playhead is the blue vertical line showing the current frame number.

It can be moved to a new position by clicking or dragging LMB in the scrubbing area at the top.

You can also move it in single-frame increments by pressing Left or Right, or jump to the beginning or end frame by pressing Shift-Left or Shift-Right.

The Frame Range determines the length of the scene's animation. By default, it's set to start at frame 1 and end at frame 250. You can change this using the Start/End inputs in the Timeline header, or in the Output Properties.

By default, the timeline only shows keyframes for selected items. You can make it show all keyframes by unchecking View Only Show Selected.

You can click a keyframe to select it (and deselect all others), or click it while holding Shift to add it to the selection (or remove it if it was already selected). You can also drag a box to select multiple keyframes in one go.

To move the selected keyframes, simply drag one of them.

Alternatively, you can press G, move the mouse, and click LMB to confirm (or RMB to cancel). You can also press S to scale the keyframes in relation to the Playhead.

See the Markers page for more information.

Popovers for Playback and Keying; transport controls; and frame controls

3D Viewport red FPS.

If animation playback can't keep up with the desired Frame Rate, the actual frame rate (shown in the top left corner of the 3D Viewport) will turn red, and the Sync option determines how the situation should be handled.

Play every frame, even if this results in the animation playing slower than intended.

Drop frames if playback becomes slower than the scene's frame rate.

Drop frames if playback becomes too slow to remain synced with audio.

Play bits of the sound in the animation (if there is any) while you drag the Playhead around.

Uncheck to mute all sound.

Don't allow moving the Playhead outside of the Frame Range using the mouse.

Automatically pan the view to catch up when the Playhead goes off screen.

Which editors to update on each animation frame. If an editor is unchecked, it'll only be updated once playback stops (with some exceptions where it'll update on each frame anyway). When starting playback in either the Graph Editor, Dope Sheet or the NLA Editor, all editors will play back regardless of the settings. This is a feature requested by animators to easily play back all views.

Display and allow changing the current scene subframe.

Set the scene's start/end frame to the current frame. If the Preview Range is active (see Frame Controls), that one is changed instead.

The Keying popover contains options that affect keyframe insertion.

Timeline Keying Sets.

A Keying Set is a named collection of animatable properties. If you select one and then press I while not hovering over any input field, Blender will create keyframes for the properties in that keying set.

If you don't have a keying set selected, you'll get keyframes on a default set of properties instead (e.g.

Location/Rotation/Scale for objects).

There are a number of predefined keying sets, but you can also create your own in the Keying Sets panel.

Insert keyframes on the current frame.

Delete keyframes on the current frame.

The keyframe type for newly created keyframes.

When inserting keyframes into trivially cyclic curves, special handling is applied to preserve the cycle integrity (most useful while tweaking an established cycle):

If a key insertion is attempted outside of the main time range of the cycle, it is remapped back inside the range.

When overwriting one of the end keys, the other one is updated accordingly.

In addition, when adding a new curve into an action with a Manual Frame Range and Cyclic Animation enabled, the curve is automatically made cyclic with the period matching the frame range. For convenience, this check and conversion is also done before adding the second keyframe to such a curve.

Auto Keying button.

When the record button () is enabled, Blender will automatically create keyframes on the current frame whenever you transform an object or bone in the 3D Viewport (or change one of its transform properties in the Properties Editor).

One special use case is to record a camera path as you fly through the scene. See Fly/Walk Navigation.

Note

Auto Keying only works for transform properties (Location, Rotation, Scale). It won't create a keyframe if you change, say, the color of a material. You still have to do that manually.

Add or replace keyframes as needed.

Only replace existing keyframes.

By default, Auto Keying will create keyframes even for properties that are not in the active keying set. Use this checkbox to change that.

Adds a new NLA Track for every pass made over the animation to allow non-destructive tweaking.

Displays a pop-up panel to alter properties of the last completed operation. See Adjust Last Operation.

Show or hide the Channels region (the tree of objects and animatable properties on the left).

Pans and zooms the view so that all keyframes are visible.

Reset the horizontal view to the current scene frame range, taking the preview range into account if it is active.

Centers the Timeline to the Playhead.

Shows the Markers region (if any markers are defined). When disabled, the Marker Menu is also hidden and marker operators are not available in this editor.

Shows the time on the X axis and the Playhead as timestamps instead of frame numbers. A timestamp such as 01:03+02 means 1 minute, 3 seconds, 2 frames.

Synchronizes the horizontal panning and scale of the editor with other time-based editors that also have this option enabled. That way, they always show the same section of time.

Only show keyframes related to the selected items. This could be objects, bones, nodes, and so on.

Note

If this option is enabled, the Timeline may not show all material keyframes of the selected objects. Instead, it only shows the keyframes belonging to the selected nodes in the Shader Editor.

Only show curves and drivers that are disabled or have errors. Useful for debugging.

Which simulation caches to show on the timeline.

Baked simulations will be shown as fully opaque, cached simulations will be slightly transparent, and invalid caches will be slightly transparent with dark diagonal stripes.

Timeline Cache.

Area controls. See the user interface documentation for more information.

Markers are used to denote frames with key points or significant events within an animation. Like in most animation editors, they're shown at the bottom of the Timeline.

Markers in an animation editor.

For descriptions of the different marker tools, see Editing Markers.

These buttons are used to set the current frame and control playback.

Transport controls.

Sets the Playhead to the start of the frame range.

Moves the Playhead to the previous keyframe.

Starts playing the animation in reverse.

Starts playing the animation.

Moves the Playhead to the next keyframe.

Sets the Playhead to the end of the frame range.

Stops playing the animation.

The number of the frame that's currently being displayed in the 3D Viewport. This is also the location of the Playhead.

The Preview Range is an alternative Frame Range that you can use for focusing on a particular part of the animation. It lets you repeatedly play a short segment without having to manually rewind or change the frame range of the entire scene.

This range only affects the preview in the 3D Viewport; it doesn't affect rendering.

The boundaries of the Preview Range are shown in dark orange. You can quickly configure and enable it by pressing **P** and dragging a box. To disable it, you can press **Alt-P**.

The start/end frame of the scene (or the preview range, if active).

--- Fin du chapitre: `_timeline` ---

--- Début du chapitre: `_time_curve` --- The Time Curve node generates a factor value (from 0.0 to 1.0) between the scene start and end time, using a curve mapping.

This node has no inputs.

The Y value defined by the curve is the factor output. For the curve controls, see Curve widget .

Tip

Flipping the curve around reverses the time input, but doing so is easily overlooked in the node setup.

Start frame and End frame of the range of time specifying the value the output should last. This range becomes the X axis of the graph. The time input could be reversed by specifying a start frame greater than the end frame.

The Y value of the curve at the current frame.

Hint

The Map Value node can be used to map the output to a more appropriate value. With some curves, it is possible that the Time Curve node may output a number larger than one or less than zero. To be safe, use the Min/Max clamping function of the Map Value node to limit output.

Time controls from left to right: no effect, slow down, freeze, accelerate, reverse. \xc2\xbb6

--- Fin du chapitre: `_time_curve` ---

--- Début du chapitre: `_time_offset` --- The Time Offset Modifier applies a temporal offset to Grease Pencil keyframes on your timeline. If you have duplicated a Grease Pencil object you can use the Time Offset Modifier on the copies to desynchronize their animation. This can give more natural looking results.

Using the Time Offset Modifier it's possible to have Grease Pencil frame ranges play back as repeating loops. Traditionally, 2D animation that uses looped drawings includes characters walking, rising smoke, and falling rain. In Fixed Frame mode the Time Offset Modifier can display drawings on your timeline entirely independently of the playhead position.

This can be handy for displaying drawings that will appear often in your animation. Think of switching between predefined mouth shapes for instance.

Time Offset Modifier. \xc2\xbb6

Offsets keyframes in the default animation playback direction (playhead moving from left to right).

Offsets keyframes in reversed animation playback direction (playhead moving from right to left).

The Frame parameter determines which frame is displayed. This value needs to be animated in order to have the displayed frame change during playback.

The number of the frame to display.

Loop back and forth animation.

It allows to combine the different Modes consecutively.

Number of cycle repeats

The number of frames to offset the original keyframes by.

Controls the speed of the frames playback. 1 is equal to the actual frame rate, could be positive (faster) or negative (slower).

Moves end frame to the animation start to keep animation in a loop.

When enabled, the animation playback is restricted only to a frame range.

Sets the range start and end frames.

See Influence Filters.

--- Fin du chapitre: `_time_offset` ---

--- Début du chapitre: `_tint` ---Reference

Draw Mode

Brush `\xe2\x80\xa3 Tint`

The Tint brush allows you to paint onto strokes point mixing the material base color with a selected color.

Defines how Color Attributes affect to the strokes.

Color Attributes affects both the Stroke and Fill materials.

Color Attributes affects the Stroke material only.

Color Attributes affects the Fill material only.

In the Tool Settings select the brush, color and mode to use with the tool.

You can configure the brush main settings included in the Tool Settings for convenience. For the vertex paint brushes configuration and settings see Vertex Paint Brush.

Ctrl- LMB erase the Color Attribute.

Click and hold LMB or use the pen tip to paint onto the stroke points.

--- Fin du chapitre: `_tint` ---

--- Début du chapitre: `_tips` --- As with all physics-enabled objects, pay close attention to the Animated checkbox in the Rigid Body panel of the Physics tab in the Properties. A common mistake is to use keyframe animation on a Passive physics object without checking the Animated box. The object will move, but the physics engine will behave as if the Passive is still in its starting place, leading to disappointment.

The most common trick is to Keyframe animate the location or rotation of an Active physics object as well as the Animated checkbox. When the curve on the Animated property switches to disabled, the physics engine takes over using the object's last known location, rotation and velocities.

Animating the strengths of various other parameters (a Motor's Target Velocity, a Hinge's limits, etc.) can be used to accomplish a wide variety of interesting results.

Enabling a constraint during the physics simulation often has dramatic results as the physics engine tries to bring into alignment two objects which are often dramatically out of alignment. It is very common for the

affected objects to build up enough kinetic energy to bounce themselves out of camera.

Rigid body dynamics can be baked to normal keyframes with Bake To Keyframes in the Object > Rigid Body menu.

The simplest way of improving simulation stability is to increase the steps per second. However, care has to be taken since making too many steps can cause problems and make the simulation even less stable (if you need more than 1000 steps, you should look at other ways to improve stability).

Increasing the number of solver iterations helps making constraints stronger and also improves object stacking stability.

It is best to avoid small objects, as they are currently unstable. Ideally, objects should be at least 20 cm in diameter. If it is still necessary, setting the collision margin to 0, while generally not recommended, can help making small objects behave more naturally.

When objects are small and/or move very fast, they can pass through each other. Besides what is mentioned above it's also good to avoid using mesh shapes in this case. Mesh shapes consist of individual triangles and therefore do not really have any thickness, so objects can pass through more easily. You can give them some thickness by increasing the collision margin.

Since the rigid body simulation is part of the animation system, it can influence other simulations just like the animation system can.

In order for this to work, the rigid body object needs to have a Collision Modifier. Simply click on Collision in the Physics tab.

Rigid body objects can be scaled, also during the simulation. This works well in most cases, but can sometimes cause problems.

If dynamic scaling is not needed, rigid body objects should have the scale applied by using the Apply Scale tool Ctrl-A.

--- Fin du chapitre: _tips ---

--- Début du chapitre: _tip_remover --- The Tip Remover modifier removes a piece of the stroke at its beginning and end.

Length of stroke to remove at both of its tips.

--- Fin du chapitre: _tip_remover ---

--- Début du chapitre: _todo_list --- This page provides a list of changes that need to be made to the manual. This is a great place for new contributors to start but also check the documentation workboard.

--- Fin du chapitre: _todo_list ---

--- Début du chapitre: _tone_map --- Tone mapping is used to map high dynamic range colors into a more limited dynamic range supported by the display, while preserving the appearance as much as possible.

This is a legacy node. It is recommended to use view transforms in the color management settings instead, and output linear high dynamic range images from the compositor instead of low dynamic range.

HDR image.

The value the average luminance is mapped to.

Normally always 1, but can be used as an extra control to alter the brightness curve.

If not used, set to 1.

If less than zero, darkens image; otherwise, makes it brighter.

Set to 0 to use estimate from input image.

If 0, global; if 1, based on pixel intensity.
If 0, same for all channels; if 1, each independent.
LDRimage.

--- Fin du chapitre: _tone_map ---

--- Début du chapitre: _toolbar --- Mesh Edit Mode tools:

Select or move.

Select geometry by dragging a box.

Select geometry by dragging a circle.

Select geometry by drawing a lasso.

Change the location of the 3D Cursor.

Translation tool.

Rotation tool.

Scale tool.

Change the scale of an object by controlling its cage.

Tool to adjust the objects translation, rotations and scale.

Draw free-hand annotation.

Draw straight line annotation.

Draw a polygon annotation.

Erase previous drawn annotations.

Measure distances in the scene.

Rotates a bone around its local Y axis.

Todo.

Todo.

Creates a new bone connected to the last selected joint.

Creates a new bone between the last selected joint and the mouse position.

Todo.

--- Fin du chapitre: _toolbar ---

--- Début du chapitre: _tools --- Geometry Nodes can be used to expand the core functionality of Blender via node-group-defined tools. They can be shared as any regular node group assets.

Node group tools integrated in the Selection menu.

The way to create Node-based tools is by switching the Geometry Nodes editor context to Tool.

New node groups created in the tool context will be enabled as Tool by default, although users still need to set them as Assets if they want to share them (see below).

Note

The Inspection features are not supported in the Tool context: Viewer Node and Socket Inspection.

For a node group to be shared as a tool, it has to be an Asset first. The asset catalog is used to determine on which menus the node group will show, similar to the regular node group assets. If the catalog name matches an existing menu, the tool will be added to the end of it. Assets that have no catalog assigned to them, or local tools are exposed in the \xe2\x80\x9cUnassigned\xe2\x80\x9d menu.

The asset options need to be set on the Asset Browser .

The node group inputs will be exposed as in the Adjust Last Operation panel.

Node groups must specify which modes and object types they support. This helps to determine where the tool is available in the user

interface. These properties can be configured in popover menus in the Geometry Node editor when in the Toolcontext. Currently only Object, Edit, and Sculpting modes are supported, and only for the Mesh and Hair Curves object types.

Important

For mesh objects, Shape Keys are not supported. Operating a node tool on a mesh with shape keys will remove the shape key data.

The following nodes are only supported in the Toolcontext:

- 3D Cursor Node
- Mouse Position Node
- Viewport Transform Node
- Active Element Node
- Selection Node
- Set Selection Node
- Face Set Node
- Set Face Set Node
- Note

The Self Object node returns the Active object when inside a Tool node group.

These nodes are only supported in the Modifiercontext:

- Simulation Zone
- Viewer Node

--- Fin du chapitre: _tools ---

--- Début du chapitre: _tools_operators ----- Fin du chapitre:
_tools_operators ---

--- Début du chapitre: _tools_settings --- Information on brush settings for every mode can be found in these pages:

See general and advanced Brush here.

See the global brush settings for Stroke settings.

See the global brush settings for Falloff settings.

See the global brush settings for Cursor settings.

--- Fin du chapitre: _tools_settings ---

--- Début du chapitre: _tool_settings ---Reference

Pose Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Pose Options \xe2\x80\xa3 Auto IK
Automatic IK is a tool for quick posing, when enabled, translating a bone will activate inverse kinematics and rotate the parent bone, and the parent \xe2\x80\x99s parent, and so on, to follow the selected bone. The IK chain can only extend from a child to a parent bone if the child is connected to it.

While moving bones, the length of the chain (the number of affected bones) can be increased or decreased using keyboard hotkeys. Pressing PageUp will increase in chain length by one and PageDown decreases the length by one. The chain length can also be controlled with WheelUp or WheelDown.

The initial chain length is 0, which effectively means follow the connections to parent bones as far as possible, with no length limit. So pressing increasing the chain length the first time sets the length to 1 (move only the selected bone), and at this point, decreasing the length point sets it back to 0 (unlimited) again. Thus, you have to increase the

chain length more than once from the initial state to set a finite chain length greater than 1.

This is a more limited feature than using an IK constraint, which can be configured, but it can be useful for quick posing.

Reference

Edit and Pose Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Options \xe2\x80\xa3 X-Axis

Mirror

This option enables automatic mirroring of editing actions along the X axis. You can enable this option in the Tool tab \xe2\x80\xa3 Options panel, while the armature is selected in Edit Mode. When you have pairs of bones of the same name with just a different \xe2\x80\x9cside suffix\xe2\x80\x9d (e.g.

\xe2\x80\x9cR\xe2\x80\x9d/\xe2\x80\x9dL\xe2\x80\x9d, or

\xe2\x80\x9c_right\xe2\x80\x9d/\xe2\x80\x9d_left\xe2\x80\x9d

\xe2\x80\xa6), once this option is enabled, each time you transform (move, rotate, scale\xe2\x80\xa6) a bone, its \xe2\x80\x9cother side\xe2\x80\x9d counterpart will be transformed accordingly, through a symmetry along the armature local X axis. As most rigs have at least one axis of symmetry (animals, humans, \xe2\x80\xa6), it is an easy way to keep the model symmetrical.

Reference

Edit and Pose Mode

Sidebar \xe2\x80\xa3 Tool \xe2\x80\xa3 Options \xe2\x80\xa3 Relative

Mirror

Accounts for any relative transformations when using X-Axis Mirror.

See also

Naming bones .

Relative Mirror is not supported with Auto IK enabled.

--- Fin du chapitre: _tool_settings ---

--- Début du chapitre: _tool_system --- Tools are accessed from the Toolbar.

This is a general introduction to tools. Individual tools have their own documentation.

There can only be one active tool per workspace and mode. This tool is remembered: if you \xe2\x80\x9c are in Edit Mode and have the Extrude tool selected, then switch to Object Mode (which has no Extrude tool) and back to Edit Mode, the Extrude tool will still be active.

Most tools are controlled using just LMB, though some also have modifier keys (shown in the Status Bar while using the tool). This can all be customized in the Keymap Preferences.

Some tools define gizmos (Shear and Spin for example) to help control them.

Reference

T

Expanded tool group. \xc2\xb6

The Toolbar contains buttons for the various tools. Buttons with a small triangle in their bottom right corner are tool groups which can be opened by holding LMB on them for a moment (or dragging LMB to open them instantly).

Hovering your cursor over a tool for a short time will show its name, while hovering longer will show the full tooltip.

Resizing the Toolbar horizontally will display the icons with two columns. Expanding it further will display the icon and its text.

Reference

Shift- Spacebar

Pressing Shift- Spacebar will pop up a small toolbar right at your cursor for faster access. The shortcuts for selecting the tools are displayed on the right.

Alternatively, you can map this action to Spacebar in the Keymap Preferences. Then you'll be able to use Spacebar like a modifier key (similar to holding Ctrl or Shift). For example, you can press Spacebar T for Transform, Spacebar D for Annotate, Spacebar M for Measure and so on. See Spacebar Action .

Reference

Q

The Quick Favorites menu gathers your favorite tools. Any tool or menu item can be added to this pop-up menu via its context menu.

If you have Alt Click Tool Prompt enabled in the Keymap Preferences, tapping Alt will display a tool prompt in the Status Bar. You can then press a key to select the corresponding tool, or tap Alt again to cancel the prompt.

The fallback tool is the one that's selected by default (so the one at the top of the Toolbar). You can change it by either holding LMB on the toolbar button or pressing Alt- W to get a pie menu.

If you bind a key to a tool which is part of a group, you can enable the Cycle option in the keymap editor. Successive presses will then cycle through the tools in that group.

This is enabled by default for the selection tools in the 3D Viewport, for example: pressing W will cycle between Select Box, Select Circle and so on.

Tools can have their own settings, which are available from multiple places:

The Tool's Active Toolpanel in the Sidebar N.

The Active Tooltab in the Properties editor .

The Tool Settings region below the area header.

--- Fin du chapitre: _tool_system ---

--- Début du chapitre: _toon --- Cycles Only

The Toon BSDF is used to create Diffuse and Glossy materials with cartoon light effects.

Color of the surface, or physically speaking, the probability that light is reflected for each wavelength.

Parameter between 0.0 and 1.0 that gives an angle of reflection between 0° and 90°.

This value specifies an angle over which a smooth transition from full to no reflection happens.

Normal used for shading; if nothing is connected the default shading normal is used.

The material component to base the toon effect.

Use shading based on the Diffuse BSDF.

Use shading based on the Glossy BSDF for specular reflection.

Standard shader output.

Example of Toon Shader. \b6

--- Fin du chapitre: _toon ---

--- Début du chapitre: _topbar --- Open the Splash Screen.

Opens a menu displaying the following information about Blender:

Version: The Blender version.

Date: Date when Blender was compiled.

Hash: The Git Hash of the build. This can be useful to give to support personnel when diagnosing a problem.

Branch: Optional branch name.

Windowing Environment: On Linux, this will show either Wayland or X11 depending on the windowing environment that Blender is running on.

Donate: Open Blender's Development Fund website.

What's New: Open the latest release notes.

Credits: Open the credits webpage.

License: Open the license webpage.

Blender Store: Open the Blender Store website.

Blender Website: Open main Blender website.

Install a new application template.

The options to manage files are:

Clears the current scene and loads the selected application template.

Open a blend-file.

Displays a list of the most recently opened blend-files. Hovering over items will show a preview, and information about the blend-file. Select any of the file names in the list to open that blend-file.

Removes items from the recent files list.

Reopens the current file to its last saved version.

This will load a blend-file that Blender automatically saves just before exiting. So this option enables you to recover your last work session, e.g. if you closed Blender by accident.

This will open an automatically saved file to recover it.

Save the current blend-file.

Opens the File Browser to specify file name and location of save.

Saves a copy of the current file.

Save the current Blender file with a numerically incremented name that does not overwrite any existing files.

Links data from an external blend-file (library) to the current one. The editing of that data is only possible in the external library. Link and Append are used to load in only selected parts from another file. See Linked Libraries.

Appends data from an external blend-file to the current one. The new data is copied from the external file, and completely unlinked from it.

Tools for managing data-block previews.

Blender can use information stored in a variety of other format files which are created by other graphics programs. See Import/Export.

Normally you save your work in a blend-file, but you can export some or all of your work to a format that can be processed by other graphics programs. See Import/Export.

Invokes all configured exporters for all collection.

External data, like texture images and other resources, can be stored inside the blend-file (packed) or as separate files (unpacked). Blender keeps track of all unpacked resources via a relative or absolute path. See pack or unpack external data.

Pack all currently used external files into the blend-file and automatically pack any files that are added later. Unchecking this option will only stop the automatic packing for new files; it won't unpack existing ones.

Pack all used external files into the blend-file. After running this operator and saving the blend-file, the external files will no longer be used \xe2\x80\x93 any changes in them will no longer be reflected in the blend-file, and you are free to move or delete them.

Export previously packed files back to external ones. You can choose whether to reuse existing external files or overwrite them.

Pack data-blocks that are linked from an external blend-file into the current one.

Export previously packed data-blocks back to external blend-files. Existing blend-files are overwritten.

Make all paths to external files relative to the current blend-file.

Make all paths to external files absolute (= full path from the system \xe2\x80\x93s root).

This option is useful to check if there are links to unpacked files that no longer exist. After selecting this option, a warning message will appear in the Info editor \xe2\x80\x93s header. If no warning is shown, there are no missing external files.

In case you have broken links in a blend-file, this can help you to fix the problem. A File Browser will show up. Select the desired directory (or a file within that directory), and a search will be performed in it, recursively in all contained directories. Every missing file found in the search will be recovered. Those recoveries will be done as absolute paths, so if you want to have relative paths you will need to select Make Paths Relative.

Note

Recovered files might need to be reloaded. You can do that one by one, or you can save the blend-file and reload it again, so that all external files are reloaded at once.

Opens a dialog to remove unused data-blocks from both the current blend-file or any Linked Data (cannot be undone). See the Outliner for more information.

Opens a pop-up window of the Outliner in Unused Data mode which lists data-blocks and other data that are unused and/or will be lost when the file is reloaded. It includes data-blocks which have only a fake user. You can add/remove the Fake User by clicking on cross/tick icon on the right side of the Outliner.

This menu manages the startup file which is used to store the default scene, workspace, and interface displayed when creating a new file.

Initially this contains the startup scene included with Blender. This can be replaced by your own customized setup.

Saves the current blend-file as the startup file.

Restores the default startup file and preferences.

When an Application Template is in use the following operators are shown:

Loads the default settings to the original Blender settings without the changes made from the current application template.

Loads the default settings to the original application template.

See also

Managing Preferences .

Closes Blender. The current scene is saved to a file called \xe2\x80\x93quit.blend \xe2\x80\x93d in Blender \xe2\x80\x93s temporary directory (which can be found on the \xe2\x80\x93File Paths \xe2\x80\x93d tab of the Preferences).

See Undo & Redo.

See Undo & Redo.

Find a menu based on its name.
 Execute an operator based on its name (Developer Extra only).
 Rename the active object or node; see Rename tool for more information.
 Renames multiple data types at once; see Batch Rename tool for more information.
 Prevents selecting objects that are in a different mode than the current one.

Note

This option can prevent accidental mode changes, such as when you're trying to select a bone in Pose Mode to animate it, but instead click a piece of background scenery (which would normally select that piece and switch to Object Mode).
 You may want to disable Lock Object Modes for example when weighting rigged objects or sculpting/painting where you intentionally want to switch between objects in different modes.
 Open the Preferences window.
 Render the active scene at the current frame.
 Render the animation of the active scene.

See also

Rendering Animations for details.
 Mix the scene's audio to a sound file.

See also

Rendering audio for details.
 Show the Render window. (Press again to switch back to the main Blender window.)
 Playback rendered animation in a separate player.

See also

Animation player for details.
 Preferences for selecting a different animation player than the default one.
 Lock interface during rendering in favor of giving more memory to the renderer.
 Create a new window by copying the current window.
 Create a new window with its own workspace and scene selection.
 Toggle the current window fullscreen.
 Switch to the next workspace.
 Switch to the previous workspace.
 Choose whether the Status Bar at the bottom of the window should be displayed.
 Capture a picture of the current Blender window. A File Browser will open to choose where the screenshot is saved.
 Capture a picture of the selected Editor. Select the Editor by clicking LMB within its area after running the operator. A File Browser will open to choose where the screenshot is saved.
 See Help Menu.

This set of tabs is used to switch between Workspaces, which are essentially predefined window layouts.
 These data-block menus are used to select the current Scene and View Layer.

Fin du chapitre: _topbar ---

Début du chapitre: _to_sphere ---

Reference
 Object and Edit Modes
 Mesh Transform To Sphere

Shift- Alt- S

The To Spheretransformation will give the selection spherical qualities. The Fig.Monkey with increasing sphericity. below shows the results of applying the To Spheretransformation to the monkey mesh.

Monkey with increasing sphericity. \xc2\xba6

The sequence above shows a monkey mesh with a 0, 0.25 (25%), 0.5 (50%) and 1 (100%) To Sphere transform applied.

To Sphere Factor. \xc2\xba6

As can be seen in the below image, the result will be smoother and more spherical when there are more mesh elements available to work with.

To Sphere applied to cubes with different subdivision levels. \xc2\xba6

In this image sequence, To Sphere was applied to the entire cube at levels of 0, 0.25 (25%), 0.5 (50%) and 1 (100%) respectively.

The To Spheretransform will generate different results depending on the number and arrangement of elements that were selected (as shown by the below image).

To Sphere applied to different selections. \xc2\xba6

--- Fin du chapitre: _to_sphere ---

--- Début du chapitre: _trace_image ---Reference

Object Mode

Object \xe2\x80\xa3 Convert \xe2\x80\xa3 Trace Image to Grease Pencil

The Trace Image to Grease Pencil tool traces a black and white image and generates Grease Pencil strokes. If the image is not black and white, it will be internally converted. For better results, convert the images manually to black and white. Also try to keep the resolution of the image small; high resolutions can produce very dense strokes.

Add an Image Empty to the scene.

Run Trace Image to Grease Pencil.

Determines if the image empty is kept or replaced.

New Object: Creates a new grease pencil object and keeps the image

emptySelected Object: Replaces the image empty with the grease pencil object.

The thickness of the generated Grease Pencil strokes.

Determine the Luminance threshold above which strokes are generated.

Determines how to resolve ambiguities during decomposition of an image into paths.

Prioritizes to connect black (foreground) components.

Prioritizes to connect white (background) components.

Always take a left turn.

Always take a right turn.

Prioritizes to connect the color (black or white) that occurs least frequently in the local neighborhood of the current position.

Prioritizes to connect the color (black or white) that occurs most frequently in the local neighborhood of the current position.

Choose pseudo-randomly.

Determines if the image being traced is a single image or image sequence.

The image empty is a single image or the current frame of an image sequence.

The image empty is an Image Sequence .

When enabled, start the tracing process at the current image frame.

Used to trace only one frame of the image sequence, set to zero to trace all.

--- Fin du chapitre: _trace_image ---

--- Début du chapitre: _track --- Track panel.\xc2\xb6

The track name can be changed with this field. Track names are used for linking tracking data to other areas, like a Follow Track constraint.

This toggle controls if the marker is enabled. If a marker is disabled, its position is not used either by solver nor by constraints.

The toggle controls whether the track is locked. Locked tracks cannot be edited at all. This helps to prevent accidental changes to tracks which are \xe2\x80\x9cfinished\xe2\x80\x9d (tracked accurate along the whole footage).

The widget in this panel is called \xe2\x80\x9cTrack Preview\xe2\x80\x9d and it displays the content of the pattern area. This helps to check how accurately the feature is being tracked (controlling that there is no sliding off original position) and also helps to move the track back to the correct position. The track can be moved directly using this widget by mouse dragging.

If an anchor is used (the position in the image which is tracking is different from the position which is used for parenting), a preview widget will display the area around the anchor position. This configuration helps in masking some things when there is no good feature at position where the mask corner should be placed. Details of this technique will be written later.

There is small area below the preview widget which can be used to enlarge the vertical size of preview widget (the area is highlighted with two horizontal lines).

Tracking happens in gray-scale space, so a high contrast between the feature and its background yields more accurate tracking. In such cases disabling some color channels can help.

Display the preview image as gray-scale even if all channels are enabled. Applies mask defined by an annotation tool in the preview widget.

When several tracks are used for 3D camera reconstruction, it is possible to assign a reduced weight to some tracks to control their influence on the solution result. This parameter can (and often need to) be animated. Altering the weights of problem tracking markers can correct or greatly reduce undesirable jumps as a feature disappears or become difficult to track.

Another use of Track Weights is when you want to reconstruct a scene from your camera solution. In that case you can first carefully track and solve your scene, and once you are done, lock all your markers with Ctrl- L , set the tracker weight in the Extra Settings of the tracker settings to zero and use the feature detection to quickly add lots of markers. Now track them and solve the scene again. Since their weight is zero they will not influence your solution at all, but you will have lots of good reference points in your scene.

While Weight parameter is used for 3D reconstruction, the Stabilization Weight is used to control 2D stabilization.

The preset for the Custom Color.

This setting overrides the default marker color used in the Clip editor and 3D Viewport, and it helps to distinguish different type of features (for example, features in the background vs. foreground and so on). Color also can be used for \xe2\x80\x9cgrouping\xe2\x80\x9d tracks so a whole group of tracks can be selected by color using the Select Grouped operator.

Tip

To select good points for tracking, use points in the middle of the footage timeline and track backwards and forwards from there. This will provide a greater chance of the marker and point staying in the camera shot.

--- Fin du chapitre: `_track` ---

--- Début du chapitre: `_tracks` --- A track plays one or more actions in sequence. You can create multiple tracks to play several actions at the same time.

NLA Tracks and Strips.

The track region has the following properties:

When unchecked, mutes all the tracks except the Action Track.

Double-click to change. (Not possible for the Action Track, as this one simply displays the name of the action.)

When unchecked, the track stops contributing to the animation. Its strips receive a dotted outline to indicate this. Note that you can also mute individual strips.

Prevents changes from being made to this track. This is useful, for example, if you want to move the strips in all the tracks except for a few.

Mutes all other tracks, including the Action Track, so that only this track contributes to the animation. This is useful for inspecting the track without any distractions from others.

The topmost track with the orange header holds the action that is being edited. Normally this is the object's active action, but if you select a strip and press Tab to enter Tweak Mode, you can temporarily make that one editable instead in the Action Editor or the Graph Editor, for example.

The Action Track has one of the following buttons:

Not available in Tweak Mode. Creates a new track below the Action Track and moves the active action into it as a strip, leaving the Action Track empty. (If you create a keyframe after this, Blender will automatically create a new active action to hold it.)

Push Down Action button.

Only available in Tweak Mode. When unchecked, the action's keyframes are shown at their original time points, rather than their new time points resulting from the strip being moved and scaled.

Strip at its original time point.

Strip moved. Notice that the keyframes are now shown to start at frame 20, which is also how the animation will behave. Within the action, however, they still start at frame 1.

After unchecking the Pin icon, the keyframes are shown at their original time points.

--- Fin du chapitre: `_tracks` ---

--- Début du chapitre: `_track_position` --- The Track Position node is used to return information about a tracking marker to the Compositor.

This node has no inputs.

Used to select a Movie Clip data-block to use, for controls see Data-Block Menu.

Camera object to get track information from.

The name of the track to get track information from.

Which marker position to use for output.

Outputs an absolute position of a marker.

Outputs the positions of a marker relative to the first marker of a track.

Outputs the positions of a marker relative to the markers of the given Frame.

Outputs the absolute positions of a marker at the given Frame.

The marker's X and Y location.

The velocity of the marker, measured in pixels per frame. This could be used to fake effects like motion blur by connecting it to the Vector Blur Node.

--- Fin du chapitre: `_track_position` ---

--- Début du chapitre: `_track_to` --- The Track To constraint applies rotations to its owner, so that it always points a given

axis towards its target, with another axis permanently maintained as much aligned with the global Z axis (by default) as possible. This tracking is similar to the billboard tracking in 3D (see note below).

This is the preferred tracking constraint, as it has a more easily controlled constraining mechanism.

This constraint shares a close relationship to the Inverse Kinematics constraint in some ways.

Tip

Billboard Tracking

The term "billboard" has a specific meaning in real-time CG programming (i.e. video games!), where it is used for plane objects always facing the camera (they are indeed "trackers", the camera being their "target"). Their main usage is as support for tree or mist textures: if they were not permanently facing the camera, you would often see your trees squeezing to nothing, or your mist turning into a mille-feuille paste, which would be funny but not so credible.

Track To panel. `axis`

Data ID used to select the constraint's target, and is not functional (red state) when it has none. See common constraint properties for more information.

The tracking local axis, i.e. the owner's axis to point at the target. The negative options force the relevant axis to point away from the target.

The "upward-most" local axis, i.e. the owner's axis to be aligned (as much as possible) with the global Z axis (or target Z axis, when the Target button is enabled).

By default, the owner's Up axis is (as much as possible) aligned with the global Z axis, during the tracking rotations. When this button is enabled, the Up axis will be (as much as possible) aligned with the target's local Z axis?

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

Warning

If you choose the same axis for Tracking Axis and Up, the constraint will not be functional anymore (red state).

--- Fin du chapitre: `_track_to` ---

--- Début du chapitre: _transfer_mesh_data ---Reference

Object Mode

Object \xe2\x80\xa3 Link/Transfer Data \xe2\x80\xa3 Transfer Mesh Data

The Data Transfertool transfers several types of data from one mesh to another. Data types include vertex groups, UV maps, Color Attributes, custom normals\xe2\x80\xa6 Transfer works by generating a mapping between source mesh\xe2\x80\x99s elements (vertices, edges, etc.) and destination ones, either on a one-to-one basis, or mapping several source elements to a single destination one by interpolated mapping.

Transfers data layer(s) from active to selected meshes.

Prevent changes to settings to re-run the operator. This is useful if you are editing several settings at once with heavy geometry.

Which data to transfer.

Data types.\xc2\xb6

Add data layers on destination meshes if needed.

Method used to map source vertices to destination ones. Because the options change depending on the Data Type options are explained in Vertex Mapping below.

The simplest option, expects both meshes to have identical number of elements, and match them by order (indices). Useful e.g. between meshes that were identical copies, and got deformed differently.

Those always select only one source element for each destination one, often based on shortest distance.

Uses source\xe2\x80\x99s nearest vertex.

Uses source\xe2\x80\x99s nearest vertex of source\xe2\x80\x99s nearest edge.

Uses source\xe2\x80\x99s nearest vertex of source\xe2\x80\x99s nearest face.

Uses source\xe2\x80\x99s edge which vertices are nearest from destination edge\xe2\x80\x99s vertices.

Uses source\xe2\x80\x99s nearest edge (using edge\xe2\x80\x99s midpoints).

Uses source\xe2\x80\x99s nearest edge of source\xe2\x80\x99s nearest face (using edge\xe2\x80\x99s midpoints).

A face corner is not a real element by itself, it\xe2\x80\x99s some kind of split vertex attached to a specific face. Hence both vertex (location) and face (normal, \xe2\x80\xa6) aspects are used to match them together.

Uses source\xe2\x80\x99s corner having the most similar split normal with destination one, from those sharing the nearest source\xe2\x80\x99s vertex.

Uses source\xe2\x80\x99s corner having the most similar face normal with destination one, from those sharing the nearest source\xe2\x80\x99s vertex.

Uses source\xe2\x80\x99s nearest corner of source\xe2\x80\x99s nearest face.

Uses source\xe2\x80\x99s nearest face.

Uses source\xe2\x80\x99s face which normal is most similar with destination one.

Those use several source elements for each destination one, interpolating their data during the transfer.

Uses nearest point on nearest source\xe2\x80\x99s edge, interpolates data from both source edge\xe2\x80\x99s vertices.

Uses nearest point on nearest source\xe2\x80\x99s face, interpolates data from all that source face\xe2\x80\x99s vertices.

Uses point of face on source hit by projection of destination vertex along its own normal, interpolates data from all that source face's vertices.

This is a sampling process. Several rays are cast from along the destination's edge (interpolating both edge's vertex normals), and if enough of them hit a source's edge, all hit source edges' data are interpolated into destination one.

A face corner is not a real element by itself, it's some kind of split vertex attached to a specific face. Hence both vertex (location) and face (normal, 's aspects are used to match them together.

Uses nearest point of nearest source's face, interpolates data from all that source face's corners.

Uses point of face on source hit by projection of destination corner along its own normal, interpolates data from all that source face's corners.

This is a sampling process. Several rays are cast from the whole destination's face (along its own normal), and if enough of them hit a source's face, all hit source faces' data are interpolated into destination one.

Automatically computes the transformation to get the best possible match between source and destination meshes.

This allows to match and transfer data between two meshes with similar shape, but transformed differently. Note that you'll get best results with exact copies of the same mesh. Otherwise, you'll likely get better results if you visually make them match in 3D space (and use Object Transform) instead.

Evaluate source and destination meshes in global space.

Source elements must be closer than given distance from destination one.

Maximum allowed distance between source and destination element (for non-topology mappings).

The starting ray radius to use when Ray Casting against vertices or edges.

When transferring data between meshes Blender performs a series of ray casts to generate mappings. Blender starts with a ray with the radius defined here, if that does not detect a hit then the radius is progressively increased until a positive hit or a limit is reached.

This property acts as an accuracy/performance control; using a lower ray radius will be more accurate however, might take longer if Blender has to progressively increase the limit. Lower values will work better for dense meshes with lots of detail while larger values are probably better suited for simple meshes.

How to affect destination elements with source values.

Replaces everything in destination (note that Mix Factor is still used).

Only replaces destination value if it is above given threshold Mix Factor. How that threshold is interpreted depends on data type, note that for Boolean values this option fakes a logical AND.

Only replaces destination value if it is below given threshold Mix Factor. How that threshold is interpreted depends on data type, note that for Boolean values this option fakes a logical OR.

Apply that operation, using mix factor to control how much of source or destination value to use. Only available for a few types (vertex groups, Color Attributes).

How much of the transferred data gets mixed into existing one (not supported by all data types).

--- Fin du chapitre: _transfer_mesh_data ---

--- Début du chapitre: `_transfer_mesh_data_layout` ---Reference
Object Mode
Object `\xe2\x80\xa3 Link/Transfer Data \xe2\x80\xa3 Transfer Mesh Data Layout`
Transfers layout of data layer(s) from active to selected meshes.
Which data to transfer.
Data types.`\xc2\xb6`
Also Delete some data layers from destination if necessary, so that it matches the source exactly.
Which layers to transfer, in case of multi-layer types.
Only transfer the active data layer.
Transfer all data layers.
How to match source and destination layers.
Match target data layers to affect by name.
Match target data layers to affect by order (indices).
See also
Data Transfer Modifier
--- Fin du chapitre: `_transfer_mesh_data_layout` ---

--- Début du chapitre: `_transform` --- Transform is a swiss-army knife of image manipulation.It moves, rotates, and scales the images within a strip.
No interpolation, uses nearest neighboring pixel.
Simple interpolation between adjacent pixels.
Highest quality interpolation.
Control whether the input values are in Percentor Pixels.
Moves the input along the X and Y axis.
Scale the input evenly along the X and Y axis.
Scale the image on the X and Y axis.
Rotates the input two-dimensionally along the Z axis.
Transform Effect. `\xc2\xb6`
--- Fin du chapitre: `_transform` ---

--- Début du chapitre: `_transformation` --- This constraint is more complex and versatile than the other `\xe2\x80\x9ctransform\xe2\x80\x9d` constraints.It lets you set the location, rotation or scale of an object/bone based on thelocation, rotation or scale of another, mixing and matching axes as you see fit.An example could be to set a gear`\xe2\x80\x99s` X rotation based on the Y coordinate ofa rail next to it.
The constraint works with input and output value ranges, one for each axis.It first clamps the input value to the Map Fromrange, then offsets and scales itto the corresponding Map Torange. This lets you, say, map a Y coordinatein the range (-3m, 3m) to an X rotation in the range (0, 180`\xc2\xb0`).
Transformation panel. `\xc2\xb6`
The reference object to read a transformation property from. If you don`\xe2\x80\x99t` select one, the constraint`\xe2\x80\x99s`icon will turn red and it will have no effect.
See common constraint properties for more information.
If Targetis an Armature , you can optionally choose abone here to use the transformation of that bone instead of the armature object as a whole.

By default, the input and output values are clamped to the Min/Max values. When you enable Extrapolate, they're allowed to go beyond these limits. This is illustrated with the graphs below, where the X axis represents the input (Map From set to Min= 1 and Max= 4) and the Y axis represents the output (Map To set to Min= 1 and Max= 2).

Extrapolate disabled: the output values are limited to the Map To range.

Extrapolate enabled: the output values can extend beyond the limits.

Standard conversion between spaces. See common constraint properties for more information.

Controls the percentage of effect the constraint has on the object. See common constraint properties for more information.

The transformation to read from the Target (or Bone).

The type of transformation to read.

The type of rotation to use, including different Euler orders, Quaternion, and other Rotation Channel Modes. Defaults to using the Euler order of the constraint owner.

In the Quaternion mode, the channels are converted to weighted angles in the same way as the swing angles of the Swing and X/Y/Z Twist modes.

The input value range for each axis.

The transformation to apply to the owner.

The type of transformation to apply.

Which Euler order to use. Defaults to the order of the constraint owner.

For each of the three output axes, lets you choose the input axis that it should take its value from. You can select the same input axis multiple times.

The output value range for each axis.

Specifies how the result of the constraint is combined with the existing transformation. The set of available choices varies based on the type of transformation.

The result of the constraint replaces the existing transformation.

The new values are multiplied with the existing axis values.

The new values are added to the existing axis values.

The new rotation is added before the existing rotation, as if it were applied to a parent of the constraint owner.

The new rotation is added after the existing rotation, as if it were applied to a child of the constraint owner.

Note

For historical reasons, the Mix mode defaults to Add for location and rotation, and Replace for scale.

When using the rotation of the target as input, whatever the real values are, the constraint will always take them back into the (-180 to 180) range. E.g. if the target has a rotation of 420 degrees around its X axis, the values used as X input by the constraint will be:

$((420 + 180) \text{ modulo } 360) - 180 = 60 - 180 = -120$

As such, this constraint is not really suited for transforming an object based on a gear's rotation. Rotating a gear based on an object's transformation works fine, however.

Similarly, when using the scale transform properties of the target as input, whatever the real values are, the constraint will always take their absolute values (i.e. invert negative ones).

When a Minvalue is higher than its corresponding Maxone,both are considered equal to the Maxone. This means you cannot create reversed mappings.

In the following example, we add a constraint to a gear that sets its X rotation based on the Y position of a rail:

Target: Rail object

Map From: Location

Y Min: -3m

Y Max: 3m

Map To: Rotation

X Source Axis: Y

X Min: 0

X Max: 180

Before moving the rail.

After moving the rail.

By default, the gear will stop rotating if the rail goes outside the (-3m, 3m) range.You can enable Extrapolateto change this.

--- Fin du chapitre: _transformation ---

--- Début du chapitre: _transforming --- Transform tools to move, rotate and scale are also available in Sculpt Mode, but with an important difference to other modes. Sculpt Mode uses its own pivot point, which can be manually positioned Shift- RMB or automatically positioned with Mask Expand. This ensures that the pivot point can be more freely placed and always moves with the transformed geometry.

Optionally instead of keeping the transform tools active, you can enable the viewport gizmo to have access to the gizmo at all times.

Note

The gizmo can in some cases block areas from being sculpted on. In that case move the pivot point somewhere else to be able to click on the desired surface.

Apart from the transform tools there are also special brushes to move, rotate and scale the topology like Pose, Boundary and Elastic Deform.

--- Fin du chapitre: _transforming ---

--- Début du chapitre: _transforms --- Reference

Sculpt Mode

Toolbar Move

Translation tool.

Reference

Sculpt Mode

Toolbar Rotate

Rotation tool.

Reference

Sculpt Mode

Toolbar Scale

Scale tool.

Reference

Sculpt Mode

Toolbar Transform

Tool to adjust the objects translation, rotations and scale.

Each tool has the following settings to change how the unmasked mesh will be transformed.

How the transformation is going to be applied to the target.

Applies the transformation to all vertices in the mesh.
Applies the transformation while dynamically simulating elasticity. Instead of applying this to all vertices, it uses the radius of the cursor as the area of effect.

--- Fin du chapitre: `_transforms` ---

--- Début du chapitre: `_transform_cache` --- The Transform Cache Constraint is used to stream animations from Alembic or USD made at the transformation matrix level (for example rigid bodies, or camera movements).

When importing an Alembic or USD file, Transform Cache constraints are automatically added to objects with animated transforms. For time-varying meshes (so deforming animations), the Mesh Sequence Cache modifier is used.

Transform Cache Constraint. \xc2\xba6

Data-block menu to select the Alembic or USD file.

Path to the Alembic or USD file.

Whether or not the cache is separated in a series of files.

Whether to use a custom frame for looking up data in the cache file, instead of using the current scene frame.

The Frame value is the time to use for looking up the data in the cache file, or to determine which to use in a file sequence.

Subtracted from the current frame to use for looking up the data in the cache file, or to determine which file to use in a file sequence.

Value by which to enlarge or shrink the object with respect to the world's origin.

The name of the Alembic attribute used for generating motion blur data; by default, this is `.velocities` which is standard for most Alembic files.

Note

The Velocity Attribute option is currently for Alembic files only.

Defines how the velocity vectors are interpreted with regard to time.

The velocity unit was encoded in frames and does not need to be scaled by scene FPS.

The velocity unit was encoded in seconds and needs to be scaled by the scene FPS (1 / FPS).

Note

The Velocity Unit option is currently for Alembic files only.

The path to the Alembic or USD object inside the archive or stage.

Controls the percentage of affect the constraint has on the object. See common constraint properties for more information.

--- Fin du chapitre: `_transform_cache` ---

--- Début du chapitre: `_transform_direction` --- The Transform Direction node multiplies a Transformation Matrix by a vector.

The vector.

The transformation matrix.

This node has no properties.

The vector.

--- Fin du chapitre: `_transform_direction` ---

--- Début du chapitre: `_transform_geometry` --- The Transform Geometry Node allows you to move, rotate or scale the geometry. The transformation is applied to the entire geometry, and not per element. The Set Position Node is used for moving individual points of a geometry. For transforming

instances individually, the `instancetranslate`, `rotate`, or `scales` nodes can be used.

Standard geometry input.

Translation of the entire geometry in the local space of the modified object.

Euler rotation in the local space of the modified object.

Scale for the geometry in the local space of the modified object.

A Transformation Matrix, available when using Matrix mode.

How the transformation is specified.

Provide separate inputs for location, rotation and scale.

Use a transformation matrix.

Standard geometry output.

--- Fin du chapitre: `_transform_geometry` ---

--- Début du chapitre: `_transform_gizmo` --- The Transform Gizmo node provides a compound gizmo that can control a position, rotation and scale.

Special gizmo value socket. Everything that linked into this socket will be modified when the gizmo is modified.

Position of the gizmo in the local space of the object.

Local orientation of the gizmo.

Note

The rotation input is ignored by the 3D viewport if the transform orientation is set to global.

The node has properties in the sidebar which allow disabling parts of the gizmo. This can be useful when e.g. controlling only a translation or only a rotation.

Should be joined into the geometry that is controlled by this gizmo.

--- Fin du chapitre: `_transform_gizmo` ---

--- Début du chapitre: `_transform_panel` --- Reference

Edit Mode

Sidebar `Transform`

When nothing is selected, the panel is empty. When more than one control point is selected, the median values are edited and `Median` is added in front of the labels.

The first controls (X, Y, Z) show the coordinates of the selected point or handle (vertex). The last control (W), defines the weight of the selected control point or the median weight.

The Space radio buttons let you choose if those coordinates are relative to the object origin (local) or the global origin (global).

Global, Local

Controls the `goal weight` of selected control points, which is used when a surface has Soft Body physics, forcing the surface to `stick` to their original positions, based on the weight.

Surface objects do not have a Radius property, this value has no effect.

Surface objects do not have a Radius property, this value has no effect.

--- Fin du chapitre: `_transform_panel` ---

--- Début du chapitre: `_transform_point` --- The Transform Point node applies a Transformation Matrix to a position vector.

The position of a point to transform.

The transformation matrix.

This node has no properties.

The transformed point.

--- Fin du chapitre: `_transform_point` ---

--- Début du chapitre: `_translate` --- The Translate node moves an image.

Could also be used to add a 2D camera shake.

Standard color input.

Used to move the input image horizontally and vertically.

Percentage translation values relative to the input image size.

Repeat pixels on the other side when they extend over the image dimensions, making endless translating possible.

None, X Axis, Y Axis, Both Axis

Interpolation Methods.

No interpolation, uses nearest neighboring pixel.

Simple interpolation between adjacent pixels.

Highest quality interpolation.

Standard color output.

--- Fin du chapitre: `_translate` ---

--- Début du chapitre: `_translate_instances` --- The Translate Instances node moves top-level geometry instances in local or global space.

The Instances page contains more information about geometry instances.

Standard geometry input.

Boolean field used to determine if an instance will be translated.

The vector to translate the instances by.

If enabled, the instances are translated relative to their initial rotation. Otherwise they are translated in the local space of the modifier object.

This node has no properties.

Standard geometry output.

--- Fin du chapitre: `_translate_instances` ---

--- Début du chapitre: `_translucent` --- The Translucent BSDF is used to add Lambertian diffuse transmission.

Color of the surface, or physically speaking, the probability that light is transmitted for each wavelength.

Normal used for shading; if nothing is connected the default shading normal is used.

This node has no properties.

Standard shader output.

Translucent shader example.

Translucent shader behavior.

--- Fin du chapitre: `_translucent` ---

--- Début du chapitre: `_transparent` --- The Transparent BSDF is used to add transparency without refraction, passing straight through the surface, as if there were no geometry there. Useful with alpha maps, for example. This shader affects light paths somewhat differently than other BSDFs. Note that only pure white transparent shaders are completely transparent.

Color of the surface, or physically speaking, the probability for each wavelength that light is blocked or passes straight through the surface.

This node has no properties.

Standard shader output.

Transparent shader (pure white).\xc2\xbb6
Transparent shader behavior.\xc2\xbb6
Transparent shader (gray).\xc2\xbb6
--- Fin du chapitre: _transparent ---

--- Début du chapitre: _transpose_matrix --- The Transpose Matrixnode
flips a matrix over its diagonal.
See also
Transpose on Wikipedia.
The matrix to be transposed.
This node has no properties.
The transposed matrix.
--- Fin du chapitre: _transpose_matrix ---

--- Début du chapitre: _triangles_quads ---Reference
Edit Mode

Face \xe2\x80\xa3 Triangles to Quads
Alt- J
This tool converts the selected triangles into quads by taking adjacent
triangles and removing the shared edge to create a quad, based on a
threshold. This tool can be applied on a selection of multiple triangles.
This means you can select the entire mesh and convert triangles that
already form square shapes \xe2\x80\x93to be converted into quads,
without having to concern yourself with individual faces.
Alternatively you can force this operation selecting a pairs of faces
(see hint below for other ways of joining).
To create a quad, this tool needs at least two adjacent triangles. If you
have an even number of selected triangles, it is also possible not to
obtain only quads. In fact, this tool tries to create most even
rectangular quads from the given triangles, which means some triangles
could remain.
Before converting tris to quads. \xc2\xbb6
After converting tris to quads. \xc2\xbb6
This value, between (0 to 180), controls the threshold for this tool to
work on adjacent triangles. With a threshold of 0.0, it will only join
adjacent triangles that form a perfect rectangle (i.e. right-angled
triangles sharing their hypotenuses). Larger values are required for
triangles with a shared edge that is small, relative to the size of the
other edges of the triangles.
Prevents the union of triangles that are not also adjacent in the active
UV map.
Prevents the union of triangles that do not have matching Color
Attributes.
Prevents the union of triangles that share an edge marked as sharp.
Prevents the union of triangles that do not have the same material
assigned.

Hint
When isolated groups of faces are selected, they can be combined with
Create Face or Dissolve Faces ; this is not limited to quads.
--- Fin du chapitre: _triangles_quads ---

--- Début du chapitre: _triangulate --- The Triangulate node converts all
faces in a mesh (quads and n-gons) to triangular faces. It functions the
same as the Triangulate tool in Edit Mode.

Standard geometry input.

A standard Boolean selection input to determine which faces will be triangulated.

Minimum number of vertices a face must have to be triangulated. For example, setting this value to 5, will prevent triangulation of Quads and only triangulate N-gons .

Split the quads in nice triangles, slower method.

Split the quads on their 1st and 3rd vertices.

Split the quads on their 2nd and 4th vertices.

Split the quads along their shortest diagonal.

Split the quads along their longest diagonal. This is the preferred mode for cloth simulations.

Arrange the new triangles nicely, slower method.

Split n-gons using an ear-clipping algorithm (the same method of tessellation used for viewport display).

Standard geometry output.

Mesh before triangulation.\xc2\xbb

Mesh after triangulation.\xc2\xbb

--- Fin du chapitre: _triangulate ---

--- Début du chapitre: _triangulate_faces ---Reference

Edit Mode

Face \xe2\x80\xa3 Triangulate Faces

Ctrl- T

This tool converts each of the selected faces (whether it be quads or n-gons) to triangular faces. See the Triangulate Modifier.

--- Fin du chapitre: _triangulate_faces ---

--- Début du chapitre: _trim ---Reference

Draw Mode

Toolbar \xe2\x80\xa3 Trim

The Trim tool delete points in between intersecting strokes.

Mark newly created End Caps as Flat.

Determine the threshold for stroke intersections.

Draw a dotted line around the strokes you want to trim. After releasing the mouse button all the points on the selected strokes will be deleted until another intersecting stroke is found.

Original drawing.\xc2\xbb

Lasso Selecting the strokes to be trimmed.\xc2\xbb

Final result.\xc2\xbb

--- Fin du chapitre: _trim ---

--- Début du chapitre: _trim_curve --- The Trim Curve node shortens each spline in the curve by removing sections at the start and end of each spline.

B-splines will still be B-splines in the output, with the first and last control point and its handles moved as necessary to preserve the shape. NURBS splines will be transformed into poly splines in order to be trimmed.

Warning

Currently the Trim Curve node does not support cyclic splines.

Note

Since curve normals are calculated the final curve, this node may change the resulting normals when the Minimum twist method is used, since the

Minimum method considers the entire length of the curve to decide the final normals. In some cases the Capture Attribute Node could be used to avoid this, by saving the original normals to be used later.

Standard geometry input with a curve component.

A boolean field input for each curve indicating whether it is trimmed or not.

The factor or length used to determine where to start each output spline.

Note

If the Start input is larger than the End, then the resulting spline will have a single point, located at the sample location of the Start value.

The factor or length used to determine where to end each output spline.

How to find endpoint positions for the trimmed spline.

Find the endpoint positions using a factor of each spline's length. The input values should be between 0 or 1.

Find the endpoint positions using a length from the start of each spline. The input values should be between 0 and the length of the splines.

Standard geometry output.

--- Fin du chapitre: _trim_curve ---

--- Début du chapitre: _trim_hair_curves --- Trims or scales hair curves to a certain length.

Input Geometry (only curves will be affected).

Scale each curve uniformly to reach the target length.

Multiply the original length by a factor.

Use the length input to fully replace the original length.

Target length for the operation.

Mask to blend overall effect.

Trim hair curves randomly up to a certain amount.

Pin each curve at a certain point for the operation.

Random Seed for the operation.

This node has no properties.

Geometry

--- Fin du chapitre: _trim_hair_curves ---

--- Début du chapitre: _trim_tools --- Reference

Sculpt Mode

Trim gesture tools add or remove geometry based on a selection area. This tool is especially useful for sketching an early base mesh for further sculpting with the voxel remesher.

Using Lasso Trim set to Join

The symmetrized mesh.

Sculpting with voxel remeshing.

New geometry is assigned to a new Face Set. When removing geometry, the new interior geometry along the selection will be assigned a new face set instead.

Note

It is not recommended to use this tool on a mesh above 100k vertices when using Difference or Union as the Trim Mode with the Exact Solver. This tool is using a Boolean operation so it might take a long time to process. For higher resolution meshes it is recommended to instead use the Line Project tool or the Fair Positions mode of the Edit Face Set tool to trim geometry.

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Box Trim

Performs a Boolean operation based on the area defined by a box gesture .

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Lasso Trim

Performs a Boolean operation based on the area defined by a lasso gesture

.

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Line Trim

Performs a Boolean operation based on the area defined by a line gesture

.

Note

The Line Trim tool does not support adding geometry. Only Difference mode is supported.

Reference

Sculpt Mode

Toolbar \xe2\x80\xa3 Polyline Trim

Performs a Boolean operation based on the area defined by a polyline gesture .

Algorithm used to calculate the Boolean intersections.

Uses a mathematically simple solver which offers the best

performance; however, this solver lacks support for overlapping geometry.

Uses a mathematically complex solver which offers the best results and has full support for overlapping geometry; however, this solver is much slower than the Fast Solver.

Geometry can be either added or removed by choosing one of these modes.

Removes geometry, filling any holes that are created.

Creates a geometry and joins any intersections with existing geometry.

Similar to Union but joins the mesh as separate geometry, without

performing any Boolean operations with existing geometry.

The method used to orientate the trimming shape.

Use the view to orientate the trimming shape.

Use the surface normal to orientate the trimming shape.

Aligns new geometry orthogonally for 90 degree angles in depth.

Aligns new geometry with the perspective of the current view for a tapered result.

Use cursor location and radius for the dimensions and position of the trimming shape. If not set, the tool uses the full depth of the object from the camera view.

--- Fin du chapitre: _trim_tools ---

--- Début du chapitre: _troubleshooting --- Some common problems people may run into when using drivers.

A security warning in the Drivers panel. \xc2\xb6

An Auto-run warning in the Info editor \xe2\x80\x99s header. \xc2\xb6

By default Blender will restrict execution of Python scripts.

If using a Scripted Expression Driver Type that doesn't

follow the Simple Expressions subset, you will have to open the file as

Trusted Source, or set Auto Run Python Scripts in Preferences \xe2\x80\xa3

Save & Load \xe2\x80\xa3 Blender Files.

The Trusted Source checkbox in the File Browser. \xc2\xb6

The Auto Run Python Scripts checkbox in the Preferences. \xc2\xb6

Parts of the User Interface may use different units of measurements for angles, rotation. In the Graph Editor, while working with Drivers, all angles are Radians.

--- Fin du chapitre: _troubleshooting ---

--- Début du chapitre: _turbulence ---Reference

Physics \xe2\x80\xa3 Force Fields

Turbulence

A Turbulenceforce field creates a random & chaotic 3D noise effect, similar to jets of water or geysers under the ocean.

Indicates the scale of the noise.

Makes the size and strength of the noise relative to the world, instead of the object it is attached to.

Turbulence force field affecting a particle system. \xc2\xb6

--- Fin du chapitre: _turbulence ---

--- Début du chapitre: _types ---Reference

Object Mode

Add

Shift- A

New objects can be created with the Addmenu in the 3D

Viewport\xe2\x80\x99s header.

Objects composed of vertices, edges and polygonal facesand can be edited extensively with Blender\xe2\x80\x99s mesh editing tools. SeeMesh

Primitives.

Mathematically defined objects which can be manipulated with control handlesor control points (instead of vertices), to edit their length and curvature. SeeCurves Primitives.

Mathematically defined patches that are manipulated with control points. These are useful for simple rounded forms and organic landscapes. SeeSurfaces Primitives.

Objects formed by a mathematical function (with no vertices or control points) defining the 3D volume in which the object exists. Meta objects have a liquid-like qualitywhere when two or more metaballs are brought together, they merge by smoothly rounding out the connection, appearing as one unified object. SeeMeta Primitives.

Create a two-dimensional representation of a text.

Container for OpenVDB files that is generatedby other software or Blender\xe2\x80\x99sFluid Simulator.

Objects created by drawing strokes. SeeGrease Pencil Primitives

Used for rigging 3D models to make them pose-able and animatable.

Non-renderable wireframes commonly used for the deformation of other objectswith help of theLattice Modifier.

Null objects that are simple visual transform nodes that do not render. They are useful for controlling the position or movement of other objects.

Empty objects that display images in the 3D Viewport. These images can be used to aid artists in modeling or animating.

Adds amesh planewith materials and texture from an image file. The dimensions of the plane are calculated to match the aspect of the image file.

Empty objects that emit light and are used for lighting the scene in renders.

Used by the EEVEE render engine to record lighting information for indirect lighting.

This is the virtual camera that is used to determine what appears in the render.

Empty objects that bring a source of sound to the scene.

Empty objects that give simulations external forces, creating movement, and are represented in the 3D Viewport as small control objects.

Lets you select from a list of existing collections. Once selected, an empty object will be created, with an instance of the selected collection (collection instancing active).

You can change the options of the object in the Adjust Last Operation panel just after creating it:

You can change the type of some objects after their creation with a selector.

Sets the starting size.

Rotates the new object so that it is aligned in one of the following manners:

Aligns the object to the global space axes, i.e. the object's front faces the negative Y axis (default).

Aligns the object to the view space axes, i.e. the object's front faces the viewport's point of view.

Aligns the object to match the rotation of the 3D Cursor .

Objects are placed, by default, at the position of the 3D Cursor. These values let you place the object in an other position.

Values let you rotate the object so that default rotation is overridden.

--- Fin du chapitre: _types ---

--- Début du chapitre: _udims --- Using UV maps can have one disadvantage, they consist of one texture for the entire mesh. Most of the time this is sufficient but the disadvantage is that the texture is one resolution for the entire mesh. This causes issues if you have a very large mesh with geometry of different importance. When using a singular texture, the resolution might be too low to cover larger UV islands while being inefficient for smaller, less important islands.

UDIM offers a solution to this by being able to spread UV islands across several different textures. UDIM which stands for U DIMension is based on a tile system where each tile is a different texture in the overall UDIM texture array. Basically each tile consists of its own UV space (0-1, 1-2, 2-3) and have its own image assigned to that tile. Tiles are managed in the UDIM Tiles panel where they can have a generated image assigned to them. Generally, you create several textures of different resolutions; for example, you may have a 4k resolution texture for the major details, and 2k and/or 1k textures for less important details.

The UDIM array consists of one main tile, this tile is given the index number of 1001. The next tile that gets added will be 1002 and will be placed to the right of the main tile. The overall UDIM array is ten tiles wide, so tiles 1001 through 1010 are created on the first row. After ten tiles a new row of tiles is started above the main tile; so 1011 will be placed directly above 1001.

To start using a UDIM workflow, you should unwrap a mesh as you would for any other UV map. After that you should decide how many textures you want to split your UV map into. This will be different for every mesh and workflow but a good minimum is 3: one 4k, one 2k, and one 1k image. Then create the desired textures to match how many textures you want.

After this it is the same process of moving UVs to the appropriate tile and scaling and managing them like any other UV map. See Layout Workflow for information on laying out UVs.

When the UVs are correctly set up across the multiple UV islands it is time to add proper textures to the UDIM array. Currently, existing textures cannot be added to a tile, to fill a tile with an existing texture you first must:

Create the desired tiles.

Save the image.

Replace the saved image file with the desired texture by deleting the file and replacing it with a new image file, keeping the old file name. Or by opening the image in another application and modifying the contents of the image.

Other than using a third-party application to edit the UDIM texture it is possible to paint on UDIM textures. This works for either 2D Painting or 3D Painting.

Substitution tokens are special sequences of characters in a filename that can be replaced with more meaningful and context aware information. In this case, tokens are identified by being text wrapped in angle bracket characters.

This substitution is used while loading or saving an image to automatically identify the tile associated with a particular texture in the UDIM array.

The following tokens are supported:

<UDIM> : A 4-digit notation calculated as $1001 + u\text{-tile} + v\text{-tile} * 10$.

<UVTILE> : A notation defined as $u(u\text{-tile} + 1)_v(v\text{-tile} + 1)$.

Examples:

monster-basecolor.<UDIM>.png will load/save files like monster-basecolor.1021.png etc.

monster-basecolor.<UVTILE>.png will load/save files like monster-basecolor.u1_v3.png etc.

Reference

Image Editor, UV Editor

All Modes

Sidebar \xe2\x80\xa3 Image \xe2\x80\xa3 UDIM Tiles

In this panel UDIM tiles are managed; new tiles can be added, tiles can be removed, or tiles can be filled with a generated texture.

List all UDIM tiles associated with the main index (1000 tile). Double clicking on the tile name allows you to alter the tile's label.

Adds new UDIM tiles to the group.

The starting tile index number. UDIMs must start with the 1001 tile and typically increase in incremental order.

The number of tiles to add.

An optional label can be used instead of the index number. These labels are shown in the 2D Viewport.

Occupies the UDIM tile with a generated image; see Fill Tile below.

Deletes the selected UDIM tile from the group. If this tile is not saved and contains data, that data will be lost.

Occupies the UDIM tile with a Generated Image.

Warning

If a tile is not filled, it will not be saved with the image.

--- Fin du chapitre: _udims ---

--- Début du chapitre: _ui_translations --- Todo

Open Blender and go to Add-ons section of the Preferences .
Search \xe2\x80\x9cManage UI Translations\xe2\x80\x9d and check the
Enable Extensioncheckbox.

See Blender translation guidein the Developer Handbook.

Reference

System

Allows managing UI translations directly from within Blender(update main
po-files, update scripts\xe2\x80\x99 translations, etc.).

Topbar \xe2\x80\xa3 File menu, Text editor, any UI control

ui_translate folder

Bastien Montagne

This add-on is bundled with Blender.

--- Fin du chapitre: _ui_translations ---

--- Début du chapitre: _undo_redo --- The tools listed below will let you
roll back an accidental action,redo your last action, or let you choose
to recover to a specific point,by picking from a list of recent actions
recorded by Blender.

Reference

All Modes

Edit \xe2\x80\xa3 Undo

Ctrl- Z

If you want to undo your last action, just press Ctrl- Z .

See also

Memory & Limits Preferencesto change undo settings.

Reference

All Modes

Edit \xe2\x80\xa3 Redo

Shift- Ctrl- Z

To roll back the Undo action, press Shift- Ctrl- Z .

Reference

All Modes

Edit \xe2\x80\xa3 Adjust Last Operation\xe2\x80\xa6

F9

You can tweak the parameters of an operator after running it.In editors
that support it, there is a \xe2\x80\x9chead-up display\xe2\x80\x9d panel
in the bottom leftbased on the last performed operation.Alternatively,
you can create a pop-up with F9which does the same thing.

For example, if your last operation was a rotation in Object Mode,Blender
will show you the last value changed for the angle(see Fig. Rotation
(Object Mode, 60 degrees). left),where you can change your action back
completely by typing Numpad0in the AngleField.There are other useful
options, based on the operator,and you cannot only Undo actions, but
change them completely using the available options.

If you are in Edit Mode,Blender will also change its contents based on
your last action taken.In the second example (on the right), the last
operation was a Move in Object Mode;but a Scaleon a Face in Edit Mode,
and, as you can see,the contents of Adjust Last Operationare different,
because of the mode (Edit Mode)(See Fig. Scale (Edit Mode, Resize face).
right).

Rotation (Object Mode, 60 degrees).\xc2\xbb6

Scale (Edit Mode, Resize face).\xc2\xbb6

Tip

Some operations produce particularly useful results by using Adjust Last Operation. For example, adding a Circle in the 3D Viewport; if you reduce the Verticesto three, you get a perfect equilateral triangle.

Tip

The Adjust Last Operationregion can be hidden by View \xe2\x80\xa3 Adjust Last Operation.

Reference

All Modes

Edit \xe2\x80\xa3 Undo History

The Undo History menu. \xc2\xb6

There is also an Undo History of the last actions taken, recorded by Blender.

The top of the list corresponds to the most recent actions. A small icon of a dot next to one of the entries indicates the current status. Rolling back actions using the Undo Historyfeature will take you back to the action you choose. Much like how you can alternate between going backward in time with Undoand then forward with Redo, you can hop around on the Undo timeline as much as you want as long as you do not make a new change. Once you do make a new change, the Undo History is truncated at that point. Selecting one of the entries in the list takes the current status to that position.

Reference

All Modes

Edit \xe2\x80\xa3 Repeat Last

Shift- R

The Repeat Last feature will repeat your last action when you press Shift- R .

In the example images below, we duplicated a Monkeymesh and moved it a bit. Using repeat Shift- R , the Monkeywas duplicated and moved a second time.

Suzanne. \xc2\xb6

After a Shift- D and move. \xc2\xb6

After a Shift- R . \xc2\xb6

Reference

All Modes

Edit \xe2\x80\xa3 Repeat History \xe2\x80\xa6

The Repeat History menu. \xc2\xb6

The Repeat Historyfeature will present you a list of the last repeated actions, and you can choose the actions you want to repeat. It works in the same way as the Undo History, explained above, but the list contains only repeated actions.

Important

When you quit Blender, the complete list of user actions will be lost, even if you save your file before quitting.

See also

Troubleshooting section on Recovering your lost work.

--- Fin du chapitre: _undo_redo ---

--- Début du chapitre: _unsubdivide ---Reference

Edit Mode

Edge \xe2\x80\xa3 Un-Subdivide

Un-subdivide functions as the reverse of subdivide by attempting to remove edgesthat were the result of a subdivide operation. If additional

editing has been done after the subdivide operation, unexpected results may occur.

How many subdivisions to remove.

--- Fin du chapitre: _unsubdivide ---

--- Début du chapitre: _usage --- For example, we will take an existing blend-file that was made for monoscopic rendering and transform it to be stereo 3D ready.

Creature Factory 2 by Andy Goralczyk rendered in stereo 3D (anaglyph).

Start opening up your project file, in this case turntable.blend from the Creature Factory 2 Open Movie Workshop series from the Blender Institute by Andy Goralczyk.

Turntable Creature Factory 2.

Go to the Output Properties and enable Stereoscopy for this scene.

Scene render views.

Note

When you turn on Stereoscopy in the scene, you get 3D preview in the viewport, as well as multiple panels that are now accessible all over the user interface.

Viewport with 3D visualization.

To tweak the stereo 3D parameters, select the camera in the Outliner. In the Camera panel go to the Stereoscopy tab and change the Convergence Distance.

The viewport will respond in real-time to those changes allowing you to preview the current depth value of the scene.

Stereo convergence distance.

Before fine-tuning the camera parameters, you can set the convergence plane in the viewport based in your scene depth layout. Go outside the camera view and you will instantly see the convergence plane in front of the camera.

You can toggle this and other display settings in the Stereoscopy panel of the 3D Viewport's Sidebar. In the following image, the camera's frustum volumes are also visible.

Viewport plane and volume stereo preview.

If you have a real 3D display at some point, you can change the 3D display mode in the Window menu, by calling the Stereo 3D operator. Be aware that some modes require a fullscreen editor to work.

Window menu, stereo 3D operator.

Before rendering your scene, you can save a Viewport Preview of the animation for testing in the final display. In the Render Output panel you can choose the output Views Format.

The options include individual files per view, top-bottom, anaglyph among others. Pick the one that fits your display requirements.

Once you are happy with the results, you can render out the final animation. In the Image Editor you can inspect the individual views and the stereo result.

Your final animation can be saved in more robust formats. In this example we saved as cross-eyed side-by-side stereo 3D.

Side-by-side cross-eye format.

As this guide showed, there is more to stereo 3D rendering than just generate two images. The earlier the stereo pipeline is considered the smoother it will get. The following sections are a more in-depth view of the individual components we visited in the workflow.

An essential component of the Stereoscopy pipeline is the ability to display the stereo image in a proper display. Blender supports from high-end 3D displays to simple red-cyan glasses. On top of that, you can set a different display mode for each window.

The display mode can be changed via the Window menu or if you create your own shortcuts for the `wm.set_stereo_3d` operator.

Window menu, stereo 3D operator.

Render two differently filtered colored images for each eye. Anaglyph glasses are required. We support red-cyan, green-magenta and yellow-blue glasses.

Render two images for each eye into one interlaced image. A 3D-ready monitor is required. We support Row, Column and Checkerboard Interleaved. An option to Swap Left/Right helps to adjust the image for the screen. This method works better in fullscreen.

Render alternate eyes. This method is also known as Page Flip. This requires the graphic card to support Quad Buffer and it only works in fullscreen.

Render images for left and right eye side-by-side. There is an option to support Cross-Eye glasses. It works only in fullscreen, and it should be used with the Full Editor operator.

Render images for left and right eye one above another. It works only in fullscreen, and it should be used with the Full Editor operator.

Note

Full Screen Stereo 3D Modes

If you have a 3D display most of the time, you will use it to see in stereo 3D, you will have to go to the fullscreen mode. In fact some modes will only work in the full window mode that hides most of the user interface from the work area. In this case it is recommended to work with two monitors, using the 3D screen for visualizing the stereo result while the other screen can be used for the regular Blender work.

When using the Stereo 3D scene view setup, a stereo pair is created on-the-fly and used for rendering and previsualization. For all the purposes this works as two cameras that share most parameters (focal length, clipping, etc.). The stereo pair, however, is offset, and can have unique rotation and shift between itself.

Stereo 3D camera settings.

Set the distance between the camera pair. Although the convergence of a stereo pair can be changed in post-production, different interocular distances will produce different results due to the parts of the scene being occluded from each point of view.

The converge point for the stereo cameras. This is often the distance between a projector and the projection screen. You can visualize this in the 3D Viewport.

Render every pixel rotating the camera around the middle of the interocular distance.

Fade interocular distance to 0 after the given cutoff angle.

Angle at which interocular distance starts to fade to 0.

Angle at which interocular distance is 0.

The stereo camera pair is separated by the interocular distance, and shifted inwards so it converges in the convergence plane. This is the ideal format since it is the one closest to how the human vision works.

This method produces two parallel cameras that do not converge. Since this method needs to be manually converged it cannot be used for

viewing. This method is common when combining real footage with rendered elements.

A less common approach is to rotate the cameras instead of shifting their frustum. The Toe-in method is rarely used in modern 3D productions.

The stereo pair can be constructed around the active camera with a new camera built for each eye (Center Pivot) or using the existing camera and creating (Left or Right). The latter is what is used when only one eye needs to be rendered for an existing mono 2D project.

When you enable Views in the Render Layer panel, a new area is available in the 3D Viewport Sidebar region. In this panel you can pick whether to see the stereo 3D in the viewport, or which camera to see. It also allows you to see the Cameras, the Plane and the Volume of the stereo cameras.

Viewport stereo 3D settings.

When working with the Stereo 3D Viewports setup, you can inspect what each individual generated camera is looking or the combined result of them. In the Multi-View mode you can see the combined result of the left and right cameras (when available) or the current selected camera.

The convergence plane represents the screen as it is perceived by the audience. Visualizing it in the 3D Viewport allows you to layout your scene based on your depth script outside the camera view.

The intersection of the stereo cameras frustums helps planning the show by avoiding elements being visible by only one camera. The volume is defined by the camera's start and end clipping distances. The areas that are in the frustum of one camera only are known as retinal rivalry areas. They are tolerated in the negative space (the region from the convergence plane into the image) but are to be avoided at all costs in the positive space (the area from the convergence plane to the camera).

Viewport 3D: convergence plane and volume display.

Multi-view images can be saved in special formats according to the production requirements. By default the system saves each view as an individual file, thus generating as many files as views to be rendered. In stereo 3D productions, for the final deployment or even intermediary previews it is convenient to save stereo 3D images, that are ready to use with 3D displays or simple anaglyph glasses. The formats supported match the display modes available for the window.

Some stereo 3D formats represent a considerable loss of data. For example, the Anaglyph format will cap out entire color channels from the original image. The Top-Bottom compressed will discard half of your vertical resolution data. The Interlace will mash your data considerably. Once you export in those formats, you can still import the image back in Blender, for it to be treated as Stereo 3D. You will need to match the window stereo 3D display mode to the image stereo 3D format though.

Some formats will preserve the original data, leading to no problems on exporting and importing the files back in Blender. The Individual option will produce separate images that (if saved in a lossless encoding such as PNG or OpenEXR) can be loaded back in production with no loss of data. For the Stereo 3D formats the only lossless options are Top-Bottom and Side-by-Side without the Squeezed Frame option.

Another option is to use multi-view OpenEXR files. This format can save multiple views in a single file and is backward compatible with old OpenEXR viewers (you see only one view though). Multi-view native support is only available to OpenEXR.

After you render your scene with Stereo 3D you will be able to see the rendered result in the combined stereo 3D or to inspect the individual views. This works for Viewer nodes, render results or opened images.

Stereo 3D and view menu. \xc2\xbb

When you drag and drop an image into the Image Editor, Blender will open it as an individual image at first. If your image was saved with one of the Stereo 3D formats, you can change how Blender should interpret the image by switching the mode to Stereo 3D, turning on Use Multi-View and picking the corresponding stereo method.

Views formats and stereo 3D. \xc2\xbb

The Compositor works smoothly with multi-view images. The compositing of a view is completed before the remaining views start to be composited. The pipeline is the same as the single-view workflow, with the difference that you can use images, movies or image sequences in any of the supported multi-view formats.

Compositor, backdrop and Split Viewer node. \xc2\xbb

The views to render are defined in the current scene views, in a similar way as you define the composite output resolution in the current scene render panel, regardless of the Image nodes resolutions or Render Layers from different scenes.

Note

Single-View Images

If the image from an Image node does not have the view you are trying to render, the image will be treated as a single-view image.

If you need to treat the views separately, you can use the Switch View node to combine the views before an Output node.

Tip

Performance

By default, when compositing and rendering from the user interface all views are rendered and then composited. During test iterations you can disable all but one view from the Scene Views panel, and re-enable it after you get the final look.

--- Fin du chapitre: _usage ---

--- D but du chapitre: _usd --- USD files typically represent the scene as a hierarchy of primitives, or prims. Individual prims contain data to describe scene entities, such as geometry, lights, cameras and transform hierarchies. Blender's USD importer converts USD prims to a hierarchy of Blender objects. Like the USD exporter, the importer does not yet handle more advanced USD concepts, such as layers and references. The following USD data types can be imported as Blender objects:

Cameras

Curves

Lights

Materials

Meshes

Primitive Shapes

Volume

For more information on how the various data types are handled, see the following descriptions of the Import Options.

Note

When importing a USDZ archive, it is important to carefully consider the Import Textures option to determine whether and how to copy texture files from the zip archive.

USD provides an `Xformprim` type, containing transform data, which can be used to represent transform hierarchies and to organize the scene. Such `Xformprims` are imported as Blender empty objects.

USD also supports `Scopeprimitives`, which are entities that do not contain transform data, but which serve to group other elements of the scene. Blender doesn't have an exact counterpart to the concept of a scope, so such primitives are imported as Blender empties located at the origin. This is an imperfect representation, because empty objects have a transform and scopes do not, but this approach nonetheless helps preserve the structure of the scene hierarchy.

USD provides a `UsdGeomPointInstancerprim` type, containing instances that are scattered on a primitive's points.

These are imported into Blender as Point Clouds using a Geometry Nodes Modifier and the Instance on Points Node.

The importer supports two types of animation:

Animating transforms: If a USD primitive has time-varying transform data, a `Transform Cache` constraint will be added to the imported Blender object.

Animating geometry: Animating mesh and curve geometry is supported by adding a `Mesh Sequence Cache` modifier to the imported data. Geometry attribute (USD `Primvar`) animation is currently supported only for Color Attributes and UVs. Note that USD file sequences (i.e. a unique file per frame) are not yet supported.

If a USD mesh or geometry subset has a bound material, the importer will assign to the Blender object a material with the same name as the USD material. If a Blender material with the same name already exists in the scene, the existing material may be used, depending on the `Material Name Collision` option. Otherwise, a new material will be created.

If the USD material has a USD Preview Surface shader source, the Viewport Display color, metallic, and roughness are set to the corresponding USD Preview Surface input values.

There is also an `Import USD Preview` option to convert USD Preview Surface shaders to Blender Principled BSDF shader nodes. This option can be lossy, as it does not yet handle converting all shader settings and types, but it can generate approximate visualizations of the materials.

If the imported USD is Y up, a rotation will be automatically applied to root objects to convert to Blender's Z up orientation.

The following options are available when importing from USD:

- Import only the subset of the USD scene rooted at the given primitive.

- Do not import invisible USD primitives. Only applies to primitives with a non-animated visibility attribute. Primitives with animated visibility will always be imported.

- When disabled this allows importing USD primitives which are not defined, such as those with an override specifier.

- Update the scene's start and end frame to match those of the USD stage.

- Add all imported objects to a new collection.

- Select the file relative to the blend-file.

- Value by which to scale the imported objects in relation to the world's origin.

- Scale for the intensity of imported lights.

- Behavior when importing USD attributes as Custom Properties.

- Does not import USD custom attributes.

Imports USD attributes in the `userProperties` namespace as custom properties. The namespace will be stripped from the property names.

Imports all USD custom attributes as custom properties. Namespaces will be retained in the property names.

Import cameras (perspective and orthographic).

Import curve primitives, including USD basis and NURBS curves. (Note that support for B-spline basis is not yet fully implemented.)

Import lights. Does not currently include USD dome, cylinder or geometry lights.

Import materials.

Import meshes.

Import USD OpenVDB field assets.

Imports USD `GeomPoints` as a `Point Cloud` object.

Imports USD primitive shapes (cubes, spheres, cones, ect) as Blender meshes.

Include primitives with `purposeRender`.

Include primitives with `purposeProxy`.

Include primitives with `purposeGuide`.

Attempt to import materials with the given purpose. If no material with this purpose is bound to the primitive, fall back on loading any other bound material.

Attempt to import all `Purpose` materials.

Attempt to import `preview` materials. Load all `Purpose` materials as a fallback.

Attempt to import `full` materials. Load all `Purpose` or `preview` materials, in that order, as a fallback.

Read mesh UV coordinates.

Convert the USD `meshDisplayColor` values to Blender's Color Attributes.

Read USD `PrimVars` as mesh attributes.

Create Subdivision Surface modifiers based on the `USDSubdivisionScheme` attribute.

Check the imported mesh for corrupt data and fix it if necessary. When disabled, erroneous data may cause crashes displaying or editing the meshes. This option will make the importing slower but is recommended, as data errors are not always obvious.

Imports USD blend shapes as Blender's Shape Keys.

Imports USD skeletons as Blender's Armatures.

Also import materials that are not used by any geometry. Note, when this option is false, materials referenced by geometry will still be imported.

Convert USD Preview Surface shaders to Principled BSDF shader networks.

Converts the first discovered USD dome light to a world background shader.

If the `Import USD Preview` option is enabled, the material blend method will automatically be set based on the `opacity` and `opacityThreshold` shader inputs, allowing for visualization of transparent objects.

Behavior when the name of an imported material conflicts with an existing material.

Import each USD material as a unique Blender material.

If a material with the same name already exists, reference that instead of importing.

When importing a USDZ package, the following options specify whether and how texture asset dependencies of the USD should be copied from the zip archive so they can be loaded into Blender.

Behavior when importing textures from a USDZ archive.

Don't import textures. Note that, with this option, material textures may fail to be resolved in Blender.

Import textures as packed data in the Blender file.

Copy files to the directory specified in the Textures Directory option.

Path to the directory where imported textures will be copied, when the Import Textures mode is Copy.

Note that the default textures directory is the relative path //textures, which requires the Blender file to have been saved before importing, so the relative path can be resolved.

Behavior when the name of an imported texture file conflicts with an existing file.

If a file with the same name already exists, use that instead of copying.

Overwrite existing files.

Import USD scene graph instances as collection instances, otherwise they are imported as copies.

Universal Scene Description (USD) files can contain complex layering, overriding, and references to other files. Blender's USD Exporter takes a much simpler approach. When exporting, all visible, supported objects in the scene are exported, optionally limited by their selection state. Blender does not (yet) support exporting invisible objects, USD layers, variants, etc.

The following objects can be exported to USD:

Meshes (of different kinds, see below).

Cameras (perspective cameras only at the moment, not orthogonal ones).

Light (all types except area lights).

Hair (exported as curves, and limited to parent strands).

Volume (both static and animated volumes).

Armatures

When exporting an animation, the final, evaluated mesh is written to USD. This means that the following meshes can be exported:

Static meshes.

Deforming meshes; here the topology of the mesh does not change, but the locations of the vertices change over time. Examples are animated characters or bouncing (but not cracking) objects.

Arbitrarily animated meshes; here the topology does change. An example is the result of a fluid simulation, where splashes of fluid can break off the main body.

Metaballs are exported as animated meshes.

Note

To export the Blender scene as a USDZ archive, set the file extension of the output file to .usdz. The exported USDZ package will be a zip archive containing the USD and its texture file dependencies.

Shot from Spring exported to USD and opened in USDView.

The following options are available when exporting to USD:

If set, add a transform primitive with the given path to the stage as the parent of all exported data.

When checked, only selected objects are exported. Instanced objects, for example collections that are instanced in the scene, are considered selected when their instancer is selected.

Only exports objects that are not hidden. Invisible parents of exported objects are exported as empty transforms.

When checked, the entire scene frame range is exported. When unchecked, only the current scene frame is exported.

ExportsCustom Properties as USD attributes. The Namespace property is used to determine the namespace that the attributes are written to.

If set, add the given namespace as a prefix to exported custom property names. This only applies to property names that do not already have a prefix (e.g., it would apply to namebar but not foo:bar) and does not apply to Blender object and data names which are always exported in the userProperties:blender namespace.

By default, userProperties namespace is used.

Author USD custom attributes containing the original Blender object and object data names.

Preserves UTF-8 encoded characters when writing USD prim and property names (requires software utilizing USD 24.03 or greater when opening the resulting files).

Use relative paths to reference external files (i.e. textures, volumes) in the exported USD file, otherwise use absolute paths.

Convert orientation axis to a different convention to match other applications. Blender uses Y Forward, Z Up (since the front view looks along the +Y direction). For example, it's common for applications to use Y as the up axis, in that case -Z Forward, Y Up is needed.

By mapping these to different axes you can convert rotations between applications default up and forward axes.

The type of transform operators to use to transform prims.

Export with translate, rotate, and scaleXform operators.

Export with translate, orient quaternion, and scaleXform operators.

Export matrix operator.

Determines whether to use Viewport or Render visibility of collection, modifiers, or any other property that can be set for both the Viewport and Render.

Exports Mesh Objects

Exports Light Objects USD does not directly support spot lights, so those are not exported.

Exports Camera Objects Only perspective cameras are exported.

Exports Curve Objects

Exports Point Cloud Objects as UsdGeomPoints primitives.

Exports Volume Objects

Exports parent hair strands are exported as a curve system. Hair strand colors are not exported.

When checked, includes UV coordinates for exported meshes. The name of the UV map in USD is the same as the name in Blender.

Exports UV maps using the USD default name (st) as opposed to Blender's default name (UVMap).

When checked, includes normals for exported meshes. This includes custom loop normals.

Triangulates the mesh before writing. For more detail on the specific option see the Triangulate modifier.

Export shape keys as USD blend shapes.

Absolute shape keys are not supported.

Export Armatures and meshes with Armature Modifiers as USD skeletons and skinned meshes.

Limitations:

Modifiers in addition to Armature modifiers will not be applied.

Bendy bones are not supported.

Only export deform bones and their parents.

Exports material information of the object. By default the exporter approximates the Principled BSDF node tree by converting it to USD's Preview Surface format.

When a mesh has multiple materials assigned, a geometry subset is created for each material. The first material (if any) is always applied to the mesh itself as well (regardless of the existence of geometry subsets), because the Hydra viewport does not support materials on subsets. See USD issue #542 for more information.

Note

If USD Preview Surface Network and MaterialX Network are disabled, the material is set to the viewport materials of meshes.

Approximates a Principled BSDF node tree to by converting it to USD's Preview Surface format.

Warning

Not all nodes are supported; currently only Diffuse, Principle, Image Textures, and UV Map nodes are supported.

Generates material shading graphs using the MaterialX standard. This standard is designed to support a high amount of interoperability among DCCs <Digital Content Creation>. In Blender, MaterialX supports most of the shader nodes and their functionality but has a few caveats (see below).

Implementation Caveats

When using the Principled BSDF, the resulting graph is very usable. However, when using some of the other BSDFs, some of the generated shading graphs are difficult for other DCCs to understand.

Convert the world material to a USD dome light. Currently works for simple materials, consisting of an environment texture connected to a background shader, with an optional vector multiply of the texture color.

Method for exporting textures.

Use original location of textures.

Preserve file paths of textures from already imported USD files. Export remaining textures to a \textures\ folder next to the USD file.

Export textures to a \textures\ folder next to the USD file.

Allow overwriting existing texture files when exporting textures.

Choose a maximum size for all exported textures.

Keep all current texture sizes.

Resize to a maximum of 256 pixels.

Resize to a maximum of 512 pixels.

Resize to a maximum of 1024 pixels.

Resize to a maximum of 2048 pixels.

Resize to a maximum of 4096 pixels.

Specify a custom size.

The size in pixels of the Custom downsampling.

As this is an experimental option. When unchecked, duplicated objects are exported as real objects, so a particle system with 100 particles that is displayed with 100 meshes will have 100 individual meshes in the exported file. When checked, duplicated objects are exported as a reference to the original object. If the original object is not part of the export, the first duplicate is exported as real object and used as reference.

USD seems to support neither per-material nor per-face-group double-sidedness,so Blender uses the flag from the first material to mark the entire mesh as single/double-sided.If there is no material it defaults to double-sided.

The mesh subdivision scheme in USD is \xe2\x80\x98Catmull-Clark\xe2\x80\x99 by default,but Blender uses \xe2\x80\x98None\xe2\x80\x99 instead, indicating that a polygonal mesh is exported.This is necessary for USD to understand the custom normals;otherwise the mesh is always rendered smooth.

Currently only fluid simulations (not meshes in general) have explicit vertex velocities.This is the most important case for exporting velocities, though,as the baked mesh changes topology all the time, andthus computing the velocities at import time in a post-processing step is hard.

When there are multiple materials, the mesh faces are stored as geometry subsetand each material is assigned to the appropriate subset.If there is only one material this is skipped. Note that the geometry subsets are not time-sampled,so it may break when an animated mesh changes topology.

Only the parent strands are exported, and only with a constant color.No UV coordinates, and no information about the normals.

Only perspective cameras are exported.

USD does not directly support spot lights, so those are not exported.

Particles are only written when they are alive, which means that they are always visible.There is currently no code that deals with marking them as invisible outside their lifespan.

Objects instanced by particle system are exported by suffixing the object name withthe particle\xe2\x80\x99s persistent ID, giving each particle transform a unique name.

This is still an experimental feature that can be enabled when exporting to USD.When enabled, instanced object meshes are written to USD as references to the original mesh.The first copy of the mesh is written for real, and the following copies are referencing the first.Which mesh is considered \xe2\x80\x98the first\xe2\x80\x99 is chosen more or less arbitrarily.

Due to a current limitation in the USD library, UDIM textures cannot be include in the USDZ archive.This limitation will likely be addressed in a future version of USD.(See USD pull request #2133.)

--- Fin du chapitre: _usd ---

--- Début du chapitre: _uv ---Reference

3D Viewport

Edit Mode

Header \xe2\x80\xa3 UV

U

Blender offers several ways of mapping UVs.The simpler projection methods use formulas that map 3D space onto 2D space,by interpolating the position of points toward a point/axis/plane through a surface.The more advanced methods can be used with more complex models, and have more specific uses.

Reference

3D Viewport and UV Editor

Edit Mode

UV \xe2\x80\xa3 Unwrap

U

Flattens the mesh surface by cutting along seams .Useful for organic shapes.

Begin by selecting all the faces you want to unwrap.In the 3D Viewport, select UV > Unwrap and select Unwrap.You can also do this from the UV Editor with UV > Unwrap U.This method will unwrap all faces and reset previous work.The UVs menu will appear in the UV Editor after unwrapping has been performed once.

Result of unwrapping Suzanne.

This operation unwraps the faces of the object to provide the best fit scenario based on how the faces are connected and will fit within the image, and takes into account any seams within the selected faces.If possible, each selected face gets its own different area of the image and is not overlapping any other faces UVs.If all faces of an object are selected, then each face is mapped to a part of the image.

Tip

A face's UV image texture only has to use part of the image, not the whole image.Also, portions of the same image can be shared by multiple faces.A face can be mapped to less and less of the total image.

The Adjust Last Operation panel allows fine control over how a mesh is unwrapped:

Uses Angle Based Flattening (ABF). This method gives a good 2D representation of a mesh.

Uses Least Squares Conformal Mapping (LSCM).This usually results in a less accurate UV mapping than Angle Based, but performs better on simpler objects.

Uses Scalable Locally Injective Mapping (SLIM).This tries to balance minimizing area distortion and minimizing angle distortion.

Activating Fill Holes will prevent overlapping from occurring and better represent any holes in the UV regions.

Map UVs will take the image's aspect ratio into consideration.If an image has already been mapped to the Texture Space that is non-square, the projection will take this into account and distort the mapping to appear correctly.

Map UVs taking vertex position after Subdivision Surface Modifier into account.

The Minimum Stretch method is iterative, where each iteration reduces the distortion more.This option says how many iterations to use before stopping.

When using the Minimum Stretch method this option allows faces to flip, which sometimes results in less distortion when there are pins.

The Minimize Stretch method has a feature that allows a user-specified vertex group to control the relative amount of area used by different parts of the unwrapped map.Vertices with higher weights will mark portions of the mesh whose adjacent UV map faces should be stretched larger than smaller-weight areas.When this option is chosen, there are two additional options to control this:

The name of the vertex group with the weights to be used.

A global factor to multiply all the weights. A bigger number will result in a more exaggerated difference between high-weight and low-weight areas.

The method to use when calculating the empty space between islands.

Use scale of existing UVs to multiply margin.

Simple method, just add the margin.

Precisely specify the fraction of the UV unit square for margin. (Slower than other two methods.)

The scale for the empty space between islands.

Reference

3D Viewport

Edit Mode

UV \xe2\x80\xa3 Smart UV Project

U

Smart UV Project, cuts a mesh based on an angle threshold (angular changes in your mesh). This gives you fine control over how automatic seams are created. It is a good method for simple and complex geometric forms, such as mechanical objects or architecture.

This algorithm examines the shape of your object, the selected faces and their relation to one another, and creates a UV map based on this information and settings that you supply.

In the example below, the Smart Mapper mapped all of the faces of a cube to a neat arrangement of three sides on top, three sides on the bottom, for all six sides of the cube to fit squarely, just like the faces of the cube.

Smart UV project on a cube. \xc2\xb6

For more complex mechanical objects, this operator can quickly and easily create a regular and straightforward UV layout.

The Adjust Last Operation panel allows fine control over how a mesh is unwrapped:

This controls how faces are grouped: a higher limit will lead to many small groups but less distortion, while a lower limit will create fewer groups at the expense of more distortion.

The method to use when calculating the empty space between islands.

Use scale of existing UVs to multiply margin.

Simple method, just add the margin.

Precisely specify the fraction of the UV unit square for margin. (Slower than other two methods.)

Rotated to a minimal rectangle, either vertical or horizontal.

Rotate islands to be aligned horizontally.

Rotate islands to be aligned vertically.

This controls how tightly the UV islands are packed together. A higher number will add more space between islands.

Weight projection \xe2\x80\x99s vector by faces with larger areas.

Map UVs will take the image \xe2\x80\x99s aspect ratio into consideration. If an image has already been mapped to the Texture Space that is non-square, the projection will take this into account and distort the mapping to appear correctly.

If the UV map is larger than the (0 to 1) range, the entire map will be scaled to fit inside.

Reference

3D Viewport

Edit Mode

UV \xe2\x80\xa3 Lightmap Pack

U

Lightmap Pack takes each of a mesh \xe2\x80\x99s faces, or selected faces, and packs them into the UV bounds. Lightmaps are used primarily in realtime rendering, where lighting information is baked onto texture maps, when it is needed to use as much UV space as possible. It has several options that appear in the Adjust Last Operation panel:

Only unwraps the selected faces.

Unwraps the whole mesh.

This is useful if mapping more than one mesh. It attempts to fit all of the objects' faces in the UV bounds without overlapping.

If mapping multiple meshes, this option creates a new UV map for each mesh. See UV Maps.

Pre-packing before the more complex Box packing.

This controls how tightly the UV islands are packed together. A higher number will add more space between islands.

Reference

3D Viewport

Edit Mode

UV > Follow Active Quads

U

Extrapolate UVs based on the active quad by following continuous face loops, even if the mesh face is irregularly-shaped.

Note

For a clean 90-degree unwrap it's typically best to first make sure the quad is a rectangle in UV space.

Otherwise any distortion in the active UV is extended which doesn't result in a useful grid-layout.

Note

The resulting unwrap is not clamped within the UV bounds, you may wish to scale down the active quad's UVs so the result is in a usable range.

Method to space UV edge loops.

Space all UVs evenly, where the shape of the quad in the 3D viewport is ignored.

Each face's UVs are calculated based on the edge length.

While this minimizes distortion, adjacent loops may become disconnected.

Average space UVs edge length of each loop.

This has the benefit of minimizing distortion, while keeping UVs connected.

Reference

3D Viewport

Edit Mode

UV > Cube Projection

U

Cube Projection maps the mesh onto the faces of a cube, which is then unfolded. It projects the mesh onto six separate planes, creating six UV islands. In the UV editor, these will appear overlapped, but can be moved. See Editing UVs.

Set the size of the cube to be projected onto.

Map UVs will take the image's aspect ratio into consideration. If an image has already been mapped to the Texture Space that is non-square, the projection will take this into account and distort the mapping to appear correctly.

Any UVs that lie outside the (0 to 1) range will be clipped to that range by being moved to the UV space border it is closest to.

If the UV map is larger than the (0 to 1) range, the entire map will be scaled to fit inside.

Reference

3D Viewport

Edit Mode

UV \xe2\x80\xa3 Cylinder Projection

U

Normally, to unwrap a cylinder (tube) as if you slit it lengthwise and folded it flat, Blender wants the view to be vertical, with the tube standing \xe2\x80\x99cup\xe2\x80\x9d. Different views will project the tube onto the UV map differently, skewing the image if used. However, you can set the axis on which the calculation is done manually.

Use when viewing from the top (at a pole) by using an axis that is straight down from the view.

Use if view is looking at the equator, by using a vertical axis.

Uses the object\xe2\x80\x99s transform to calculate the axis.

How to determine rotation around the pole.

Polar 0 is on the X axis.

Polar 0 is on the Y axis.

How to handle faces at the poles.

UVs are pinched at the poles.

UVs are fanned at the poles.

Separate projections by islands isolated by seams.

The radius of the cylinder to use.

Map UVs will take the image\xe2\x80\x99s aspect ratio into consideration. If an image has already been mapped to a Texture Space that is non-square, the projection will take this into account and distort the mapping to appear correctly.

Any UVs that lie outside the (0 to 1) range will be clipped to that range by being moved to the UV space border it is closest to.

If the UV map is larger than the (0 to 1) range, the entire map will be scaled to fit inside.

Reference

3D Viewport

Edit Mode

UV \xe2\x80\xa3 Sphere Projection

U

Spherical mapping is similar to cylinder but the difference is that a cylindrical mapping projects the UVs on a plane toward the cylinder shape, while a spherical map takes into account the sphere\xe2\x80\x99s curvature, and each latitude line becomes evenly spaced. Sphere Projection is useful for spherical shapes, like eyes, planets, etc. Recall the opening cartographer\xe2\x80\x99s approach to mapping the world? Well, you can achieve the same here when unwrapping a sphere from different points of view. Normally, to unwrap a sphere, view the sphere with the poles at the top and bottom. After unwrapping, Blender will give you an equirectangular projection; the point at the equator facing you will be in the middle of the image. A polar view will give a very different but common projection map. Using an equirectangular projection map of the earth as the UV image will give a good planet mapping onto the sphere.

Using an equirectangular image with a Sphere Projection. \xc2\xb6

Direction of the sphere.

Use when viewing from the top (at a pole) by using an axis that is straight down from the view.

Use if view is looking at the equator, by using a vertical axis.

Uses the object\xe2\x80\x99s transform to calculate the axis.

Select which axis is up.

Polar 0 is on the X axis.

Polar 0 is on the Y axis.

How to handle faces at the poles.

UVs are pinched at the poles.

UVs are fanned at the poles.

Separate projections by islands isolated by seams.

Map UVs will take the image's aspect ratio into consideration. If an image has already been mapped to a Texture Space that is non-square, the projection will take this into account and distort the mapping to appear correctly.

Any UVs that lie outside the (0 to 1) range will be clipped to that range by being moved to the UV space border it is closest to.

If the UV map is larger than the (0 to 1) range, the entire map will be scaled to fit inside.

Reference

3D Viewport

Edit Mode

UV $\rightarrow$ Project from View

U

Project from View takes the current view in the 3D Viewport and flattens the mesh as it appears. Use this option if you are using a picture of a real object as a UV Texture for an object that you have modeled. You will get stretching in areas where the model recedes away from you.

Apply an orthographic projection.

Map UVs to the camera region taking resolution and aspect into account

Map UVs will take the image's aspect ratio into consideration. If an image has already been mapped to a Texture Space that is non-square, the projection will take this into account and distort the mapping to appear correctly.

Any UVs that lie outside the (0 to 1) range will be clipped to that range by being moved to the UV space border it is closest to.

If the UV map is larger than the (0 to 1) range, the entire map will be scaled to fit inside.

Reference

3D Viewport

Edit Mode

UV $\rightarrow$ Project from View (Bounds)

U

The same as Project from View, but with Scale to Bounds activated by default.

Reference

3D Viewport and UV Editor

Edit Mode

UV $\rightarrow$ Reset

U

Reset UVs maps each face to fill the UV grid, giving each face the same mapping.

If you want to use an image that is tileable, the surface will be covered in a smooth repetition of that image, with the image skewed to fit the shape of each individual face. Use this unwrapping option to reset the map and undo any unwrapping (go back to the start).

--- Fin du chapitre: _uv ---

--- Début du chapitre: `_uv_map` --- The UV Mapnode is used to retrieve specific UV maps. Unlike the Texture Coordinate Node which only provides the active UV map, this node can retrieve any UV map belonging to the object using the material.

This node has no inputs.

See the From Instancer option of the Texture Coordinate Node .

UV map to use.

UV mapping coordinates from the specified UV map.

--- Fin du chapitre: `_uv_map` ---

--- Début du chapitre: `_uv_project` --- Projecting the Blender logo onto Suzanne.

The UV Project modifier acts like a slide projector. It emits a UV map from the negative Z axis of a controller object (such as an empty object), and applies it to the object as the light hits it.

Download an example.

The UV Project modifier.

Which UV map to modify. Defaults to the active rendering layer.

Changes the image's aspect ratio. Only apply when a camera is used as projector object.

Scales the image. Only apply when a camera is used as projector object.

Up to ten projector objects are supported. Each face will choose the closest and aligned projector with its surface normal. Projections emit from the negative Z axis (i.e. straight down a camera or light). If the projector is a camera, the projection will adhere to its perspective/orthographic setting.

Specify the projector object(s).

UV Project is great for making spotlights more diverse, and also for creating decals to break up repetition.

Usually, an Image Texture node mapped to the UV map that the modifier targets is added to the object's material.

When projecting geometry in a perspective view, vertices behind the camera are not properly mapped. You can work around this by subdividing geometry so that faces in front of the camera have correctly mapped UVs.

--- Fin du chapitre: `_uv_project` ---

--- Début du chapitre: `_uv_sphere` --- The UV Spherenode generates a spherical mesh mostly out of quads except for triangles at the top and bottom.

Horizontal resolution of the sphere. If this is smaller than three, no mesh is generated.

Vertical resolution of the sphere. If this is smaller than two, no mesh is generated.

Distance of vertices to the origin.

This node has no properties.

Standard geometry output.

A 2D vector representing the default X/Y coordinates of the UV Map for the primitive's shape. This can be connected to the Store Named Attribute Node, to be used once the Geometry Nodes Modifier gets applied. The UV map must be stored on the face corner in order to be accessed.

--- Fin du chapitre: `_uv_sphere` ---

--- Début du chapitre: _uv_texture_spaces --- Reference

All Modes

Properties \xe2\x80\xa3 Data \xe2\x80\xa3 UV Maps

The UV Maps panel in the Data tab. \xc2\xb6

If you have a mesh object selected, you\xe2\x80\x99ll find its UV maps in the Data tab of the Properties editor . After selecting a map, you can view and edit it in the UV editor .

One mesh can have multiple UV maps (e.g. one map per texture), although it\xe2\x80\x99s also possible to reuse a UV map for multiple textures.

Click the camera icon to make a certain UV map the default one for rendering. This will be the map that\xe2\x80\x99s used by the UV output of the Texture Coordinate Node . You can use the UV Map Node to access any other maps.

Duplicates the selected UV map, or creates a new one if the list is empty.

Removes the selected UV map.

Reference

All Modes

Properties \xe2\x80\xa3 Data \xe2\x80\xa3 Texture Space

This panel lets you configure the object\xe2\x80\x99s Texture Space , which is a 3D box used for generating texture coordinates without the use of a UV map. You can visualize the texture space using the option in the Viewport Display panel.

Calculates the texture space automatically.

Lets you define the texture space manually, relative to the object. Note that you can also edit it in the 3D Viewport \xe2\x80\x93 see Editing below.

Use another mesh for texture indices. The vertices of the two objects must be perfectly aligned or the UV map will be distorted.

Modifies the Location and Size to match the object\xe2\x80\x99s bounding box. This disables Auto Texture Space.

Reference

Object Mode and Edit Mode

Object \xe2\x80\xa3 Transform \xe2\x80\xa3 Move/Scale Texture Space

Click one of these menu items, then move the mouse to adjust the texture space and press LMB to confirm. While transforming, you can use keyboard shortcuts to lock certain axes; see the status bar.

When setting up a material shader, you can use the Generated output of the Texture Coordinate Node to read the 3D coordinate inside the object\xe2\x80\x99s texture space. You can then pass this coordinate to a texture node.

Tip

Texture spaces do not have rotation support. You can use a Mapping Node to manually rotate the coordinate in the material shader instead.

--- Fin du chapitre: _uv_texture_spaces ---

--- Début du chapitre: _uv_unwrap --- The UV Unwrap Node generates a UV map islands based on a selection of seam edges. The node implicitly performs a Pack Islands operation upon completion, because the results may not be generally useful otherwise.

See also

The Unwrap operator performs a similar operation in the UV editor. Unlike the Unwrap operator, the node doesn\xe2\x80\x99t perform aspect ratio correction, because it is trivial to implement with a Vector Math Node .

Faces to participate in the unwrap operation. UVs that are part of any other face will not be affected.

Edges to mark where the mesh is \xe2\x80\x99cut\xe2\x80\x99d for the purposes of unwrapping.

The distance to leave between UV islands.

Virtually fill holes in mesh before unwrapping, to better avoid overlaps and preserve symmetry.

This method gives a good 2D representation of a mesh

Uses LSCM (Least Squares Conformal Mapping). This usually gives a less accurate UV mapping than Angle Based, but works better for simpler objects

The generated UV coordinates between 0 and 1 for each face corner in the selected faces.

Note

In order for Blender to recognize the created attribute as a UV map, it must be created with the Store Named Attribute Node on the Face Corner domain with the 2D Vector data type. This is necessary because there is no 2D Vector socket type.

--- Fin du chapitre: _uv_unwrap ---

--- Début du chapitre: _uv_warp --- The UV Warp modifier transforms an object's UV map based on values or two objects. Its purpose is to give you direct control over the object's UVs in the 3D Viewport, allowing you to directly move, rotate, and scale existing UV coordinates using defined values or a controller object or bone.

Which UV map to modify; if not set it defaults to the active rendering layer.

The center point of the UV map to use when applying scale or rotation. With (0, 0) at the bottom left and (1, 1) at the top right.

The axes to use when mapping the 3D coordinates into 2D.

The two objects used to define the transformation. See Usage below.

The vertex group can be used to scale the influence of the transformation per vertex.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier.

The setting reverses the weight values of the group.

Amount to move the UV map.

Amount to scale the UV map.

Amount to rotate the UV map.

How the UVs are warped is determined by the difference between the transforms (location, rotation and scale) of the from and to objects.

If the to object has the same transforms as the from object, the UVs will not be changed.

Assuming the UV Axis of the modifier is X/Y and the scale of the objects is (1, 1, 1), if the to object is one unit away from the from object on the X axis, the UVs will be transformed on the U axis (horizontally) by one full UV space (the entire width of the image).

--- Fin du chapitre: _uv_warp ---

--- Début du chapitre: _value --- The Value Node is a simple node to input numerical values to other nodes in the tree.

This node has no input sockets.

Single numerical value (floating-point).

The value set in the node properties.

In the following example the Value Node is used to control multiple values at once, this makes the node a useful organizational tool.

Example of the Value node. \xc2\xba

Tip

From this you can also make different values proportional to each other by adding a Math Node in between the different links.

--- Fin du chapitre: _value ---

--- Début du chapitre: _value_to_normal --- Computes a normal map.

The heightmap to compute the normal map from.

Size of derivative offset used for calculating normals.

This node has no properties.

Standard normal output.

--- Fin du chapitre: _value_to_normal ---

--- Début du chapitre: _value_to_string --- The Value to String node generates string representation of the input value.

Floating-point value to be converted.

Integer value used to determine the precision of the output value.

The type of numerical value to convert to a string.

Convert a floating-point value to a string.

Convert a 32-bit integer to a string.

String value representation of the input.

--- Fin du chapitre: _value_to_string ---

--- Début du chapitre: _vector --- The Vector input node creates a single vector.

This node has no inputs.

X

Y

Z

Standard vector output.

--- Fin du chapitre: _vector ---

--- Début du chapitre: _vector_blur --- The Vector Blur node is a fast method for simulating Motion Blur in compositing. It uses the vector speed render pass to blur the image pixels in 2D.

Image input, to be linked to the \xe2\x80\x9cCombined\xe2\x80\x9d render pass.

Z depth, to be linked to the \xe2\x80\x9cDepth\xe2\x80\x9d render pass.

Input for the \xe2\x80\x9cVector\xe2\x80\x9d render pass. See Cycles render passes .

Quality factor.

Scaling factor for the motion vector (actually the \xe2\x80\x9cshutter speed\xe2\x80\x9d in frames).

Motion blurred image output.

Even with a correct compositing setup with Image, Z and Speed nodes all linked to the appropriate passes, there may still be artifacts. The 2D

render passes does not contain 3D information, and so the information what is behind a moving object or outside the camera view is lost.

Better results can be achieved by rendering the scene into multiple

render layers, applying vector blur to each render layer, and then

compositing the results together. Typically an animated character would be

rendered in a separate render layer than the background set. Especially if hair or transparency is involved this is important. For other artifacts it can help to slightly blur the Speed pass or to set a Maximum Speed limit. This helps to smoothen out the motion, but too much blurring leads to its own problems.

The speed vector in this example was created by animating the patterned sphere horizontally and using a frame at the mid-point of the sequence.

Render result, no post-processing.

Composite, with Samples set to 32 and Blur set to 1.0.

--- Fin du chapitre: `_vector_blur` ---

--- Début du chapitre: `_vector_curves` --- The Vector Curves node maps an input vector components to a curve.

Use this curve node to slow things down or speed them up from the original scene.

In the shader context the node also has an additional Factor property. Controls the amount of influence the node exerts on the output vector. Standard vector input.

X, Y, Z

For the curve controls see: Curve widget .

Standard vector output.

--- Fin du chapitre: `_vector_curves` ---

--- Début du chapitre: `_vector_displacement` --- The Vector Displacement node is used to displace the surface along arbitrary directions, unlike the regular Displacement node which only displaces along the surface normal.

It is typically used to apply vector displacement maps created by other sculpting software. Vector displacement maps can fully represent the high resolution detail to be applied on a smooth base mesh, unlike regular displacement maps.

By default, Blender only uses Bump Mapping to render displacement. However with true displacement, the rendered geometry will be physically displaced. To use true displacement the Displacement method must be set accordingly.

Tip

For best results when using true displacement, the mesh must be subdivided finely to bring out the detail in the displacement texture.

See also

Material Displacement for more details on displacement workflows.

Vector specifying the displacement along three axes. This is where a texture node can be connected.

Typically a baked vector displacement image texture is used. For Object Space, RGB colors in the image are interpreted as an XYZ offset in object space. For Tangent Space, R is an offset along the tangent, G along the normal and B along the bitangent.

Neutral displacement value that causes no displacement. With the default 0.0, any lower values will cause the surfaces to be pushed inwards, and any higher values will push them outwards.

Increase or decrease the amount of displacement.

Object Space maps work for static meshes, and will render slightly faster with less memory usage. Tangent Space maps can be used for meshes that will be deformed, like animated characters, so the displacement follows the deformation.

Displacement offset to be connected into the Material Output.

Regular and exaggerated vector displacement on a smooth base mesh.

\xc2\xb6

--- Fin du chapitre: _vector_displacement ---

--- Début du chapitre: _vector_math --- The Vector Mathnode performs the selected math operation on the input vectors.

The inputs of the node are dynamic. Some inputs are only available in certain operations. For instance, the Scale input is only available in the Scale operator.

Input vector $(A = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix})$.

Input vector $(B = \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix})$.

Input Scale (s) .

The vector math operator to be applied on the input vectors.

The sum of A and B. $(\begin{pmatrix} A_x + B_x \\ A_y + B_y \\ A_z + B_z \end{pmatrix})$

The difference between A and B. $(\begin{pmatrix} A_x - B_x \\ A_y - B_y \\ A_z - B_z \end{pmatrix})$

The entrywise product of A and B. $(\begin{pmatrix} A_x \cdot B_x \\ A_y \cdot B_y \\ A_z \cdot B_z \end{pmatrix})$

The entrywise division of A by B. Division by zero results in zero.

$(\begin{pmatrix} A_x / B_x \\ A_y / B_y \\ A_z / B_z \end{pmatrix})$

The entrywise combination of the multiply and addition operations. $(A \cdot B + C)$

The cross product of A and B. $(\begin{pmatrix} A_y \cdot B_z - A_z \cdot B_y \\ A_z \cdot B_x - A_x \cdot B_z \\ A_x \cdot B_y - A_y \cdot B_x \end{pmatrix})$

The projection of A onto B.

The reflection of A around the normal B. B need not be normalized.

For a given incident vector A, surface normal B and ratio of indices of refraction (IOR), refract outputs the refraction vector R.

Orients a vector A to point away from a surface B as defined by its normal C. Computes $((\text{dot}(B, C) < 0) ? A : -A)$.

The dot product of A and B. $(A_x \cdot B_x + A_y \cdot B_y + A_z \cdot B_z)$

The distance between A and B.

The length of A. $(\sqrt{A_x^2 + A_y^2 + A_z^2})$

The result of multiplying A by the scalar input Scale.

$(\begin{pmatrix} s \cdot A_x \\ s \cdot A_y \\ s \cdot A_z \end{pmatrix})$

The result of normalizing A. The result vector points to the same direction as A and has a length of 1. If A is (0, 0, 0), the result is (0, 0, 0) as well.

The entrywise output of a value between Min and Max based on the absolute difference between the input value and the nearest integer multiple of Max less than the value.

The result of rounding A to the largest integer multiple of B less than or equal A.

Rounds the input value entrywise down to the nearest integer.

Rounds the input value entrywise up to the nearest integer.

The entrywise modulo of A by B.

Returns the fractional part of the value entrywise.
 The entrywise absolute value of A.
 The entrywise minimum value from A and B.
 The entrywise maximum value from A and B.
 The entrywise Sine of A.
 The entrywise Cosine of A.
 The entrywise Tangent of A.
 The output of the node is dynamic. It is either a vector or a scalar depending on the operator. For instance, the Length operator has a scalar output while the Add operator has a vector output.
 Output vector.
 Output value.

--- Fin du chapitre: `_vector_math` ---

--- Début du chapitre: `_vector_rotate` --- The Vector Rotate Node provides the ability to rotate a vector around a pivot point (Center).
 Vector to be rotated.
 Point to rotate around.
 Axis to rotate around.
 Angle to rotate the input vector by.
 When Type is set to Euler, rotate the input vector by these angles around the X, Y, then Z axes in that order.
 The type of angle input.
 Rotates the vector around the defined axis and the amount of rotation is defined by the Angle input.
 Rotates the vector around an arbitrary axis defined by the Axis input vector. The amount of rotation is defined by the Angle input.
 Rotates the vector about a center point defined by the Center input vector. The amount of rotation on each axis is defined by the Rotation input vector.
 Inverts the rotation angle.
 The rotated vector.

Vector Rotate node example. `\xc2\xba6`

--- Fin du chapitre: `_vector_rotate` ---

--- Début du chapitre: `_velocity` --- Reference
 Particle System `\xe2\x80\xa3 Velocity`
 The initial velocity of particles can be set through different parameters, based on the type of the particle system. If the particle system type is Emitter or Hair, then the following parameters give the particle an initial velocity.
 The emitter `\xe2\x80\x99s` surface normals (i.e. let the surface normal give the particle a starting speed).
 Let the tangent speed give the particle a starting speed.
 Rotates the surface tangent.
 Give an initial velocity in the X, Y, and Z axes.
 X, Y, Z
 The emitter objects movement (i.e. let the object give the particle a starting speed).
 Gives the starting speed a random variation. You can use a texture to only change the value, see Controlling Emission, Interaction and Time.
 --- Fin du chapitre: `_velocity` ---

--- Début du chapitre: `_vertex_color` --- The Color Attribute node provides access to Color Attributes as well as their alpha value. This node has no inputs.

The target Color Attribute. The listed Color Attributes are those of the mesh of the active object. If the active object has no mesh, a warning will be displayed. If the property is marked in red, it means the Color Attribute is not available in the mesh of the active object, but it may be available in other meshes of objects that share this material!

Standard color output.

Alpha value.

--- Fin du chapitre: `_vertex_color` ---

--- Début du chapitre: `_vertex_groups` --- Reference

Particle System \xe2\x80\xa3 Vertex Groups

The Vertex groups panel allows you to specify vertex groups to use for several child particle settings. You can also negate the effect of each vertex group with the checkboxes. You can affect the following attributes:

Defines the density of the particle distribution.

Defines the length of the hair.

Controls the amount of clumping. The weight of 1.0 gives current Clump value, weight of 0.0 completely removes effect.

Controls the frequency of the children Kink.

Adjusts the Uniform roughness parameter.

Adjusts the Random roughness parameter.

Adjusts the Endpoint roughness parameter.

Vertex group to control the children \xe2\x80\x99s Twist effect. Gives control over the direction of the twist, as well as the amount. The weight of 0.5 is neutral, i.e. there is no twist effect.

--- Fin du chapitre: `_vertex_groups` ---

--- Début du chapitre: `_vertex_neighbors` --- The Vertex Neighbors node outputs topology information relating to each vertex of a mesh.

This node has no inputs.

This node has no properties.

The number of vertices connected to this vertex with an edge, equal to the number of connected edges.

The number of faces that contain the vertex.

--- Fin du chapitre: `_vertex_neighbors` ---

--- Début du chapitre: `_vertex_of_corner` --- Outputs the index of the vertex that a face corner is attached to.

The index of the face corner.

Note

If this input is not connected, it uses the index of the context item, which means it \xe2\x80\x99s important that the node is evaluated in the Face Corner domain.

This node has no properties.

The index of the vertex that the face corner is attached to.

--- Fin du chapitre: `_vertex_of_corner` ---

--- Début du chapitre: `_vertex_weights` --- Reference

Edit and Weight Paint Modes

Sidebar region \xe2\x80\xa3 Vertex Weights

Vertex Weights panel. \xc2\xb6

A vertex group assigns a weight (a number between 0 and 1) to each vertex it contains. A group can have multiple vertices, and each vertex can be part of multiple groups.

The Vertex Weight panel in the 3D Viewport's Sidebar shows the vertex groups for the active vertex, and lets you see and edit the associated weights. It's available in Edit Mode, as well as in Weight Paint Mode when Vertex Selection is enabled in the header. While all vertex groups are technically the same, we can still divide them into two types depending on how they're used: Also sometimes called "weight group" or "weight map," this type of vertex group determines which vertices are affected by a certain bone in the Armature. In other words, it defines which part of the mesh deforms when the bone moves around.

The remaining vertex groups are used with shape keys, modifiers, and other areas.

The deform vertex groups are related to each other: the deformation weights of every vertex typically need to add up to 1. For this reason, you can use the filter buttons at the top of the panel to show only these vertex groups (or to exclude them).

The Weight Table shows all the weights associated with the active vertex, which is the vertex that was selected last (and is highlighted in white). If there is no active vertex, or it isn't part of any vertex group, the panel is not displayed.

You can click the name of a vertex group to make it the active one.

Changing the active vertex group.

Enable display of weights in Edit Mode.

When you are in Edit Mode, you can make the weights of the active group visible on the mesh: open the Mesh Edit Mode Overlay popover and enable the Vertex Group Weight option.

Weights in Edit Mode.

You can change the weight for a vertex group by either clicking the number and typing a new one or by dragging left and right with LMB. You can also click the arrows (only shown when hovering) to change the weight in steps of 0.01.

Changing a weight value.

The Paste Weight to Selected button copies the weight from the active vertex to the other selected vertices. Note that, even though it uses the word "paste," it doesn't interact with the Copy button and in fact doesn't use the clipboard at all.

Copying a weight.

The Delete Weight button removes the active vertex from the vertex group, making the row disappear from the list.

Deleting a weight.

Vertex weight operators.

Recalculates the weights of the active vertex so that they add up to 1.0 while retaining their relative magnitude.

Copies all the weights from the active vertex to the other selected vertices.

Tip

Both tools only work on the vertex groups that match the current filter setting.

Locked vertex group.

If a vertex group is locked, its weights become uneditable, and the buttons for copying and normalizing weights become disabled.

Tip

The Normalize and Copy buttons only become disabled if there's a locked vertex group in the current list. If (for example) only non-deforming vertex groups are locked, you can switch to the Deform filter and normalize the groups that way.

--- Fin du chapitre: _vertex_weights ---

--- Début du chapitre: _verts ---Reference

Object Mode

Properties \xe2\x80\xa3 Object Properties \xe2\x80\xa3 Instancing

Instance Vertices allows you to replicate child objects at the location of every vertex of the parent object.

Note

The relative Object Origin position of the parent and child objects determines offset instanced geometry from parent vertex.

Rotates all instanced objects according to the corresponding vertex normals of the parent mesh.

To change the axis of direction of the instanced objects, select the child object and change the Tracking Axis.

There are actually two approaches to modeling using instanced vertices. They can be used as an arranging tool, allowing you to model geometrical arrangements of objects (e.g. the columns of a Greek temple, the trees in a garden, the desks in a classroom). The object can be of any object type which Blender supports. The second approach is to use them to model an object starting from a single part of it (e.g. the spikes in a club, the thorns of a sea-urchin, the tiles in a wall, the petals in a flower).

Note

Download Example Blend-File

You can download a file with the examples described on this page. In this blend, the first example, a monkey parented to a circle is on layer 1; while a tentacle parented to an icosphere is on layer 2.

All you need is a base object (e.g. the tree or the column) and a pattern mesh with its vertices following the pattern you have in mind. In this section, we will use a simple scene for the following part. We will be using a monkey head located at the origin of the coordinate system as our base object and a circle at the same location as our parent mesh.

A monkey head and a circle. \xc2\xbb

Instanced monkeys on Vertices. \xc2\xbb

First, in Object Mode, select the base object and Shift-LMB to add the circle to the selection (order is very important here), and Ctrl- P or Object \xe2\x80\xa3 Parent \xe2\x80\xa3 Object to parent the base object to the circle. Now, the circle is the parent of the monkey; if you move the circle, the monkey will follow it.

With only the circle selected, enable Instancing Vertices; a monkey head should be placed at every vertex of the circle.

The original monkey head at the center and the parent mesh are still shown in the 3D Viewport but neither will be rendered. If the placement and rotation of your monkey head are odd, you might need to clear its rotation Alt-R, scale Alt-S, location Alt-G, and origin Object \xe2\x80\xa3 Clear \xe2\x80\xa3 Origin.

If you now select the base object and modify it in either Object or Edit Mode, all changes will also affect the shape of all instanced objects. You can also select the parent mesh to modify the arrangement of the instances; adding vertices will also add more base objects. Note that the base objects will inherit changes made to the parent mesh in Object Mode, but not in Edit Mode. So scaling the circle up in Object Mode will enlarge the monkey head, while scaling the circle up in Edit Mode will only increase the distance between the base objects. The orientation of the base objects can be controlled by enabling Align to Vertex Normal in the Instancing panel. This will rotate all base objects according to the vertex normals of the parent mesh. To change the orientation of the instanced objects, select the base object and change the Tracking Axis .

Orientation enabled, orientation +Y.

Negative Y.

Positive X.

Positive Z, up X.

Note

The axes of an object can be made visible in the Properties > Object Properties > Viewport Display panel. To display the vertex normals of the parent mesh, enter Edit Mode and enable this visualization in the Display & Shading > Viewport Overlays > Normals where you can also resize the displayed normals as necessary.

Very interesting models can be made using Instancing Vertices and a standard primitive. In this example, a simple tentacle was made by extruding a cube a couple of times. The tentacle object was then parented to an icosphere. With Align to Vertex Normal enabled for the parent mesh (the icosphere), the orientation of the base object (the tentacle) was adapted to the vertex normals of the parent mesh (in this case the tentacle was rotated -90° about the X axis in Edit Mode).

A simple tentacle set to smooth.

Tentacles instanced onto the parent mesh.

Align to Vertex Normal enabled to align instanced geometry.

As in the previous example, the shape and proportions of the arrangement can now be tweaked.

To turn all instanced geometry into real objects, select the icosphere and Make Instances Real. To make the icosphere and the tentacle a single object, make sure they are all selected and go to Object > Join, Ctrl-J.

See also

Other duplication methods are listed here .

--- Fin du chapitre: _verts ---

--- Début du chapitre: _video_formats --- Blender used FFmpeg to handle video encoding/decoding various video formats. These formats are primarily used for compressing rendered sequences into a playable movie. Video formats are composed of a container, a codec, and sometimes audio which is stored using its own codec. The role of the container is to encapsulate video and audio data that is compressed using a codec.

Codecs compress the channels of a video down to save space and enable continuous playback. Lossy codecs make smaller files at the expense of image quality, while lossless codecs compress as much as possible the video/audio, but without losing any existing data.

Some codecs, like H.264, are great for larger images. Codecs are used to encode and decode the movie, and so must be present on both the encoding machine (Blender) and the target machine. The results of the encoding are stored in a container file.

There are dozens, if not hundreds, of codecs, including Xvid, H.264, DivX, Microsoft, and so on. Each has advantages and disadvantages, and compatibility with different players on different operating systems.

Note

Most codecs can only compress the RGB or YUV colors, but some support the Alpha channel as well. Codecs that support RGBA include:

FFmpeg video codec #1

PNG

QuickTime Animation

WebM/VP9 (although Blender will not import the alpha channel due to a limitation of FFmpeg).

A standard for lossy compression of video and audio. It is designed to compress VHS-quality raw digital video and CD audio down to 1.5 Mbit/s. This container enforces the video codec, you can only define quality parameters, and the audio codec.

File Extensions: .mpg , .mpeg

A standard for the generic coding of moving pictures and associated audio information. It describes a combination of lossy video compression and lossy audio data compression methods which permit storage and transmission of movies using currently available storage media (notably DVDs) and transmission bandwidth. This container enforces the video codec, you can only define quality parameters, and the audio codec.

File Extensions: .dvd , .vob , .mpg , .mpeg

While being a video codec, it is also a real container, in which you can store video and audio streams using various codecs. It is widely supported by many modern software and hardware players.

File Extensions: .mp4 , .mpg , .mpeg

A derivative of the Resource Interchange File Format (RIFF). One of the first and most widely used video container formats.

File Extension: .avi

A multi-tracks format. QuickTime and MP4 container formats can use the same codecs. They are mostly interchangeable in a QuickTime-only environment. MP4, being an international standard, has more support.

File Extension: .mov

An intra-frame video compression scheme, used by many digital camcorders back in the days. It uses the discrete cosine transform (DCT, similar algorithm to JPEG) to compress video on a frame-by-frame basis. Audio is stored uncompressed. This container enforces the video codec, you can only define quality parameters.

File extension: .dv

A free open-standard container format, that can hold an unlimited number of video, audio, picture or subtitle tracks in one file.

File Extensions: .ogg , .ogv

A free open-standard container format, a file format that can hold an unlimited number of video, audio, picture or subtitle tracks in one file.

File Extension: .mkv

A container file format used to deliver video over the Internet using Adobe Flash Player. This container enforces the video codec, you can only define quality parameters.

File Extension: .flv

A free open-standard container format, designed to be used for internet streaming. Note that this container can only hold a VP9 video codec, and Vorbis or Opus audio codecs.

File Extension: .webm

These options are not available with all Containers .

For audio-only encoding.

Intended to be usable as both an intermediate format suitable for use while editing, and as a presentation format. It can be either lossless or lossy.

See Containers .

FFV1 is a lossless intra-frame video codec. It can use either variable length coding or arithmetic coding for entropy coding. The encoder and decoder are part of the free, open-source library libavcodec in FFmpeg. Supports an alpha channel.

See Containers .

A modern variation of the MPEG-4 family, this lossy codec is very commonly used. It offers a very good compression/quality ratio.

Lossless video codec created by Ben Rudiak-Gould which is meant to replace uncompressed YCbCr as a video capture format.

See Containers .

See Containers .

Inherits many of the features of MPEG-1, MPEG-2 and other related standards, but also adds new features.

Lossless, this stores each frame as an independent image in the video stream. Compression will be poor, but as every frame is fully self-contained, scrubbing and editing can be simpler. Supports an alpha channel.

Original format of QuickTime videos. Supports an alpha channel.

A free open-standard lossy codec designed together with the Ogg container .

A free open-standard lossy video compression format. One of the most recent codecs, it is widely used for internet streaming.

A free open-standard lossy video compression format, designed as a successor to VP9. AV1 offers great compression rates and visual quality, AV1 produces video files that are about 30% more space efficient than VP9. For video-only encoding.

Advanced Audio Codec, a standardized, lossy compression and encoding scheme for digital audio. AAC generally achieves better sound quality than MP3 at similar bit rates.

Audio Codec 3, an audio compression technology developed by Dolby Laboratories.

Free Lossless Audio Codec. Digital audio compressed by FLAC's algorithm can typically be reduced to 50-60% of its original size.

A lossy audio compression format.

A lossy audio compression format, widely used as final audio format.

A lossy audio compression format, designed to encode speech or general audio and is intended to replace the Vorbis codec.

Pulse Code Modulation, a method used to digitally represent sampled analog signals. It is the standard form for digital audio in computers and various Blu-ray, Compact Disc and DVD formats, as well as other uses such as digital telephone systems.

An open-standard, highly-compressed format comparable to MP3 or AAC. Vorbis generally achieves better sound quality than MP3 at similar bit rates.

Some codecs impose limitations on output size, H.264, for example requires both the height and width to be divisible by 2.

--- Fin du chapitre: `_video_formats` ---

--- Début du chapitre: `_view` --- Annotation tool strokes can be enabled/disabled with the checkbox in the panel header. It is a standard annotations panel where annotation layers and frames can be controlled. There is one difference in the behavior of the annotation tools from other areas \xe2\x80\x93 when a new layer is created \xe2\x80\x93 on-demand \xe2\x80\x93 (when making a stroke without adding a layer before this) the default color for the layer is set to pink. This heightens the color contrast to make the stroke more noticeable on all kinds of movies.

Determines the data-block type the current annotation layer is stored. Store the current annotation layer with the active Movie Clip data-block. Store the current annotation layer with the active Track data-block.

See also

Annotation Tools for more information on general annotation layers.

--- Fin du chapitre: `_view` ---

--- Début du chapitre: `_viewer` --- The Viewer node allows viewing data from inside a geometry node group in the Spreadsheet Editor and the 3D Viewport.

Any geometry connected can be visualized in the viewport and its attribute values can be read in the spreadsheet.

Note

This node cannot be used in the Tool context \xe2\x80\x93 only in the Modifier context.

Using Shift- Ctrl- LMB on any node or socket connects it to the viewer and makes it active. Using the same shortcut on empty space in the node editor makes deactivate the active viewer. When the active viewer is not visible anymore (e.g. another object is selected, or the current node group is exited), it is deactivated. The icon in the viewer node header can also be used to activate and deactivate it.

In the viewport, the Show Viewer option can turn off the viewer node visualization completely in order to see the final output of the object \xe2\x80\x93s evaluation instead.

When the viewer has a geometry and a separate value input connected, the values can be visualized with a viewport overlays. When possible, the attribute domain is used to visualize the data is determined automatically. Otherwise, the viewer node falls back to the face corner domain on meshes and the point domain on curves. When necessary, the domain can be chosen manually.

The spreadsheet now only shows the \xe2\x80\x93 Viewer \xe2\x80\x93 column for the domain that is selected in the Viewer node.

It can be helpful to pin a specific viewer node in the spreadsheet. When pinned, the spreadsheet still references the viewer node even when it becomes inactive.

Geometry that will be displayed in the Spreadsheet.

Field to be evaluated on the geometry. The type for this value is chosen automatically when the keyboard shortcut to link an output is pressed.

However, if the type must be adjusted manually, it is available in the node editor Sidebar.

The data type used to evaluate the Value input, visible in the node sidebar.

The attribute domain used to evaluate the Value input. The Auto option chooses the domain automatically based on the connected nodes.

This node has no outputs.

--- Fin du chapitre: _viewer ---

--- Début du chapitre: _viewpoint --- The menu View > Viewpoint lets you align the viewing direction to a specific axis. This can also be done using the Navigation Gizmo or the following hotkeys:

Numpad7

Numpad1

Numpad3

Ctrl- Numpad7

Ctrl- Numpad1

Ctrl- Numpad3

The above hotkeys align the view to a global (world) axis. You can also align to a local axis of the selected item by additionally holding Shift. This way, you can for example view any mesh face head-on, no matter how it's oriented. (To get out of this local viewpoint, simply align to a global axis again.)

The view can also be aligned by holding Alt- MMB and dragging the mouse in a certain direction.

--- Fin du chapitre: _viewpoint ---

--- Début du chapitre: _viewport --- Blender Preferences Viewport section.

Display the active Object name and frame number at the top left of the 3D Viewport.

Display the name and type of the current view in the top left corner of the 3D Viewport. For example: User Perspective or Top Orthographic.

Show the frames per second screen refresh rate while an animation is played back. It appears in the top left of the 3D Viewport, displaying red if the frame rate set cannot be reached.

Calculate the FPS displayed in the viewport based on an average of the current and previously displayed frames. A value of zero uses the number of frames in 1.0 second.

More samples represent the average FPS over a longer period of time, however sudden changes to performance result in a more gradual increase/decrease over time.

Fewer samples shows an FPS which more closely matches the actual performance, however the value may jitter - making the FPS difficult to comprehend.

Diameter of the gizmo.

Diameter of the HDRI sphere overlay.

Display the axis as an interactive gizmo. Click sets the viewport to display along this axis and dragging orbits the view.

Display simple, less intrusive axis in the viewport.

How vivid the colors of the simple axis are.

Disables the viewport axis.

Diameter of the 3D Viewport Axis widget.

Enable a fresnel effect on edit mesh overlays. It improves shape readability of very dense meshes, but increases eye fatigue when modeling lower poly.

Control the Anti-Aliasing for higher quality rendering.

Display overlays with smooth wire, without this wires will be rendered aliased. To increase the visibility you can disable this for Edit Mode specificity (see below), since edges do not blend into other shaded regions.

Display smooth wire in Edit Mode, without this wires will be rendered aliased.

Limit the maximum resolution for pictures used in textured display to save memory. The limit options are specified in a square of pixels (e.g: the option 256 means a texture of 256x256 pixels). This is useful for game engineers, whereas the texture limit matches paging blocks of the textures in the target graphic card memory.

Sets the level of anisotropic filtering. This improves the quality of textures that are rendered at the cost of performance.

Clip alpha below this threshold in the 3D Viewport. Note that, the default is set to a low value to prevent issues on some GPUs.

Method to render images; the following options are supported:

Automatically use GLSL which runs on the GPU for performance but falls back to the CPU for large images which might be slow when loaded with the GPU.

Uses CPU for display transform and render images as a 2D texture.

Fastest method using GLSL for display transform and render images as a 2D texture.

This option uses an alternative method of picking which uses depth information to select the front-most elements. It is only used for selecting with the cursor (not box select, lasso, circle select, etc.).

Performance varies depending on your OpenGL hardware and drivers.

Under certain circumstances, the GPU will be used to subdivide a mesh with a Subdivision Surface modifier. This typically results in increased subdivision performance.

This feature is not supported on Qualcomm GPUs on Windows

--- Fin du chapitre: _viewport ---

--- Début du chapitre: _viewport_display --- Factor that scales the thickness of the grid that is currently being displayed.

Interpolation method to use for the visualization of the fluid grid.

Linear interpolation between voxels. Gives good smoothness and speed.

Cubic interpolation between voxels. Gives smoothed high quality interpolation, but is slower.

No interpolation between voxels. Gives raw voxels.

Determines how many slices per voxel should be generated.

Renders only a single 2D section of the domain object.

Adjust slice direction according to the view direction.

Slice along the X/Y/Z axis.

Position of the slice relative to the length of the respective domain side.

Display gridlines to differentiate the underlying cells in the current slice of the fluid domain.

Use a specific color map for the visualization of the simulation field. This comes in handy during debugging or when making more advanced adjustments to the simulation. For instance, if the actual color

of a fire simulation is barely visible in the viewport then changing the color profile can help to see the real size of the flame.
The simulation field used in the display options (e.g. density, fuel, heat).

Slice view of \xe2\x80\x9cfire\xe2\x80\x9d grid without color mapping.\xc2\xbb6

Slice view of \xe2\x80\x9cfire\xe2\x80\x9d grid with color mapping.\xc2\xbb6

Scale the selected simulation field by this value.

Visualization options for the vector fields.

Choose to display the vectors as \xe2\x80\x9cStreamlines\xe2\x80\x9d.

Choose to display the vectors as \xe2\x80\x9cNeedles\xe2\x80\x9d.

Choose to display the vector field as \xe2\x80\x9cMarker-And-Cell Grid\xe2\x80\x9d.

Show an individual X/Y/Z component of the MAC grid.

Scale the display vectors by the magnitude of the vectors they represent.

The vector field represented by the display vectors (e.g. fluid velocity, external forces).

Scale the vectors by this size in the viewport.

Advanced coloring options for gridlines.

Color gridlines with flags.

Highlight the cells with values of the displayed grid within the range. Values between the Lower Bound and Upper Bound (inclusive) are considered to be within the range.

Lower bound of the highlighting range.

Upper bound of the highlighting range.

Color used to highlight the cells.

Choose to highlight only a particular type of cells.

--- Fin du chapitre: _viewport_display ---

--- Début du chapitre: _viewport_render --- Viewport rendering lets you create quick preview renders from the current viewpoint (rather than from the active camera, as would be the case with a regular render).

You can use Viewport Render to render both images and animations.

Below is a comparison between the Viewport render and a final render using the Cycles Render engine.

Viewport render using Solid Mode. \xc2\xbb6

Viewport render using Material Preview Mode. \xc2\xbb6

Full render. \xc2\xbb6

Note

Viewport rendering only works for the Workbench and Eevee render engines. It's not supported for Cycles.

Tip

Disable overlays to get a render without \xe2\x80\x9cclutter\xe2\x80\x9d like rigs, empties and so on.

For the most part, Viewport Render uses the current viewport settings. Some settings are located in the properties of the render engine that is used to render the view.

Solid mode uses the render settings of Workbench; Material Preview mode uses the render settings of Eevee.

Additionally, some output settings are used too:

Resolution

Aspect

Output path

File format

Activating Viewport Render will render from the current active view. This means that if you are not in an active camera view, a virtual camera is used to match the current perspective. To get an image from the camera point of view, enter the active camera view with Numpad 0.

As with a normal render, you can abort it with Esc.

To render a still image, use 3D Viewport \xe2\x80\xa3 View \xe2\x80\xa3 Viewport Render Image.

To render an animation, use 3D Viewport \xe2\x80\xa3 View \xe2\x80\xa3 Viewport Render Animation.

To render an animation, but only those frames that have a keyframe, use 3D Viewport \xe2\x80\xa3 View \xe2\x80\xa3 Viewport Render Keyframes. This only renders those frames for which the selected objects have an animation key. The other frames are still written to the output, but will simply repeat the last-rendered frame.

For example, when a six-frame animation is rendered, and the selected objects have a key on frames 3 and 5, the following frames will be output: The 1st frame is always rendered.

The 1st frame is repeated because there is no key on this frame.

The 3rd frame is rendered.

The 3rd frame is repeated because there is no key on this frame.

The 5th frame is rendered.

The 5th frame is repeated because there is no key on this frame.

Tip

You can limit the viewport render to a particular region with Render Regions.

--- Fin du chapitre: _viewport_render ---

--- Début du chapitre: _viewport_transform --- The Viewport Transform node retrieves the view direction and location of the 3D Viewport.

Note

This node can only be used in the Tool context .

This node has no inputs.

This node has no properties.

The 3D Viewport \xe2\x80\x99s perspective or orthographic projection matrix.

The view direction and location of the 3D viewport.

Whether the viewport is using orthographic projection.

--- Fin du chapitre: _viewport_transform ---

--- Début du chapitre: _views --- By default, the 3D Viewport only shows the scene from one viewpoint. By using Quad Views, you can see it from multiple viewpoints at the same time, which gives more context about the changes you \xe2\x80\x99re making.

Reference

All modes

View \xe2\x80\xa3 Area \xe2\x80\xa3 Toggle Quad View

Ctrl- Alt- Q

Toggling Quad View will split the 3D Viewport into four views: three orthographic side views and one user perspective view.

Note

Quad View is different from splitting the area and aligning the views manually. In Quad View, the four views are still part of a single 3D Viewport, so that they share the same display options.

Quad View. \xc2\xbb6

Reference

All modes

Sidebar \xe2\x80\xa3 View \xe2\x80\xa3 Quad View

When disabled, makes it possible to orbit in the orthographic views as well (turning them into perspective views instead).

Syncs the view position between side views. (Requires Lock Rotation to be enabled.)

Clip objects based on what is visible in the other side views.

--- Fin du chapitre: _views ---

--- Début du chapitre: _view_layer ---Reference

Properties \xe2\x80\xa3 Scene \xe2\x80\xa3 View Layer

View Layer panel (shown here for the EEVEE render engine).\xc2\xbb6

The Layer Panel shows the settings of the active View Layer.

The active view layer will be used during rendering.

Only render the active view layer.

Note

This option is ignored when rendering from the command line.

See also

Additional options shown in this panel are different for each render engine. See Render Passes for the options per render engine.

--- Fin du chapitre: _view_layer ---

--- Début du chapitre: _visibility ---Reference

Object Mode

Properties \xe2\x80\xa3 Object Properties \xe2\x80\xa3 Visibility

The Visibility panel controls how objects are interacted with in the viewport and in the final render. These visibility options can also be set in the Outliner.

The object is able to be selected in the 3D Viewport.

The object will be displayed in the 3D Viewport.

The object is able to be in the final render, note that it will still be visible in rendered shading view.

See also

Cycles has additional Visibility properties and also Grease Pencil objects have additional Visibility properties .

Render objects as a holdout or matte, creating a hole in the image with zero Alpha, to fill out in compositing with real footage or another render.

--- Fin du chapitre: _visibility ---

--- Début du chapitre: _visibility_masking_face_sets ---Parts of the mesh

can be hidden in Sculpt Mode. Because hidden faces cannot be sculpted,

hiding makes it easier to isolate what you want to work on. Hiding

geometry also improves the viewport performance.

Hiding is shared between all modes, except Object Mode (i.e.

hiding/showing of faces in one mode will hide the same faces in other modes too).

Unlike Selection Masking in other painting modes, Sculpt Mode primarily uses Masks and Face Sets to easily control the mesh visibility and which faces can currently be edited. The exception is the Clipping Region , which can be used in any mode.

The most common shortcuts are H to hide the face set under the cursor and Shift-H to isolate the face set under the cursor (or show everything).

Inverting the visibility and showing all is also available in the Alt- W pie menu.

Modifying visibility can also be done via the Hide Gesture Tools .

See also

More information for controlling the visibility at Show & Hide.

A mask is used to control which vertices of the mesh are influenced by sculpting and painting. The mask can for example be created/edited via the Mask Gesture Tools and Mask by Color tool.

Internally, masks are stored using the `sculpt_maskAttributes` .

Creating masks follows a slightly different mental model than selecting in other modes. For example Shift- LMB is used for smoothing instead of adding to a mask.

Masking is also conceptually inverted to selection (i.e. You cannot edit masked vertices. But you can edit selected vertices).

Instead a mask is typically always added to the current mask with LMB and subtracted with Ctrl- LMB. So if you wish to edit the masked surfaces, you'll need to use the Invert operator. In the case of masking everything that is visible, the best workflow is to first Clear and then Invert the mask.

Both these operators can be quickly accessed in the Apie menu.

See also

More information about editing and using masks at the Mask Menu

Face sets are used to group your mesh into differently colored faces, which can then be quickly hidden or shown like mentioned above. They can also be used for fast mask creation via the Mask Expand . Face Set Expand is also useful for creating, editing and joining face sets.

More options can be found in the Alt- W pie menu.

Otherwise Face Sets can be created/edited with the Draw Face Sets brush, Mask Gesture Tools . They can also be edited with the Edit Face Set tool.

See also

More information about editing and using face sets at the Face Sets Menu

Internally, face sets are stored using the `sculpt_face_setAttributes` .

Auto-Masking is also a fast way of only editing specific geometry without having to manually create a new mask or hide geometry. This feature is especially useful in combination with face sets.

The mask and face sets display can be toggled and adjusted in the Display Settings .

Note

When Xray shading is enabled, masks and face sets will not be displayed.

--- Fin du chapitre: `_visibility_masking_face_sets` ---

--- Début du chapitre: `_volume` --- Volume rendering is used to render various effects that cannot be represented by hard surfaces alone.

Smoke, fire or clouds are set up using a volume object or fluid simulation, with only a volume shader.

Meshes can also be used to create such shapes by removing the default surface shader and using a volume shader with the mesh shape defining the volume bounds and textures defining the volume density.

Mist is created with a volume shader for the world, or with a large mesh object encompassing the scene.

Absorption in glass is simulated by combining a glass surface shader with refraction and a volume absorption shader for the interior of the object.

Principled Volume is a physically-based volume shader that can be used to create a wide range of volume materials. It supports scattering, absorption and emission in one easy to use node. Fire can be rendered with blackbody emission.

Smoke and fire rendered with Principled Volume shader.

For more control, volume shading components can be manually combined into a custom shader setup.

Volume Absorption will absorb part of the light as it passes through the volume. This can be used to shade for example black smoke or colored glass objects, or mixed with the Volume Scatter node. This node is similar to the transparent BSDF node, it blocks part of the light and lets other light pass straight through.

Volume Scatter lets light scatter in other directions as it hits particles in the volume. The anisotropy defines in which direction the light is more likely to scatter. A value of 0 will let light scatter evenly in all directions (similar to the diffuse BSDF node), negative values let light scatter mostly backwards, and positive values let light scatter mostly forward. This can be used to shade white smoke or clouds for example.

Emission will emit light from the volume, for example for fire.

Volume Absorption, Scatter and Emission

When rendering smoke and fire, volume attributes are used to define the shape and shading of the volume. The Principled Volume shader will use them by default, while custom volume shaders can use the Attribute node to get attributes such as density, color and temperature.

All volume shaders have a density input. The density defines how much of the light will interact with the volume, getting absorbed or scattered, and how much will pass straight through. For effects such as smoke you would specify a density field to indicate where in the volume there is smoke and how much (density bigger than 0), and where there is no smoke (density equals 0).

Volumes in the real world consist of particles, a higher density means there are more particles per unit volume. More particles means there is a higher chance for light to collide with a particle and get absorbed or scattered, rather than passing straight through.

Meshes used for volume render should be closed and Manifold. That means that there should be no holes in the mesh. Each edge must be connected to exactly two faces such that there are no holes or T-shaped faces where three or more faces are connected to an edge.

Normals must point outside for correct results. The normals are used to determine if a ray enters or exits a volume, and if they point in a wrong direction, or there is a hole in the mesh, then the renderer is unable to decide what is the inside or outside of the volume.

These rules are the same as for rendering glass refraction correctly.

A volume shader can also be applied to the world, filling the entire space.

Currently, this is most useful for night time or other dark scenes, as the world surface shader or sun lights will have no effect if a volume shader is used. This is because the world background is assumed to be infinitely far away, which is accurate enough for the sun for example. However, for modeling effects such as fog or atmospheric scattering, it is not a good assumption that the volume fills the entire space, as most of the distance between the sun and the earth is empty space. For such effects it is better to create a volume object surrounding the scene. The size of this object will determine how much light is scattered or absorbed.

Real-world effects such as scattering in clouds or subsurface scattering require many scattering bounces. However, unbiased rendering of such effects can be noisy, so by default the number of bounces is zero in Cycles, and no support is available in Eevee. The effect you get when rendering with zero volume bounces is what is known as single scattering, the effect from more bounces is multiple scattering.

For rendering materials like skin or milk that require multiple scattering, subsurface scattering is more efficient and easier to control. Particularly the random walk method can accurately render such materials. For materials such as clouds or smoke that do not have a well-defined surface, volume rendering is required. These look best with many scattering bounces, but in practice one might have to limit the number of bounces to keep render times acceptable.

--- Fin du chapitre: _volume ---

--- Début du chapitre: _volumes ---Reference

Render Volumes

Volume Step size is the distance between volume shader samples. Cycles automatically estimates this distance based on voxel size in volume objects and smoke simulations.

Render time can be reduced by increasing the step size, at the cost of potentially losing some volume detail. For procedural volume shaders that add detail, step size can be increased per object, material or world.

Global multiplier on the step size for all volumes in renders. Increase to reduce render time, at the cost of losing detail.

Global multiplier on the step size for all volumes in the viewport. Increase for more responsive viewport rendering.

Maximum number of steps through the volume before giving up, to protect from extremely long render times with big objects or small step sizes.

--- Fin du chapitre: _volumes ---

--- Début du chapitre: _volume_absorption --- The Volume Absorption node allows light to be absorbed as it passes through the volume. Typical usage for this node would be water and colored glass.

Color of the volume.

The density of the absorption effect.

This node has no properties.

The Volume Shader output must be plugged into the Volume Input of the Material or World Output node.

Example of Volume Absorption.

--- Fin du chapitre: _volume_absorption ---

--- Début du chapitre: _volume_cube --- The Volume Cube generates a volume from scratch by evaluating an input field on every single voxel in a rectangular prism. The Density field defines the output volume grid's value at every voxel. The field can only depend on the Position Node.

The value for the new grid at each voxel.

The value of the grid outside the rectangular prism controlled by the Min and Max inputs. The node can generate a more memory-efficient volume when the values of the Density input are the same as the background value. One corner of the rectangular prism in which to fill evaluate the field.

The other corner of the rectangular prism in which to fill evaluate the field.

The number of voxels to evaluate the field in on each axis.

Note

Changing these values can have a significant impact on performance. For example, the default values of 32 mean the input field will be evaluated about 33 thousand times. Increasing the values to 100 will give 1 million evaluations, and 1000 would give 1 billion.

This node has no properties.

Geometry containing the generated volume.

--- Fin du chapitre: _volume_cube ---

--- Début du chapitre: _volume_displace ---Reference

This modifier is only available for Volume Objects .

The Volume Displace modifier displaces existing volume grids based on a 3D texture. It uses the RGB color channels of the texture to displace the volume into the X, Y and Z direction.

The Volume Displace modifier. \xc2\xba6

The texture that is evaluated at every voxel to determine how far and in what direction to displace.

Note

Grayscale textures lead to stretching along one axis. It's best to use a color texture.

Controls how far voxels are displaced.

Smaller values result in better performance, but might cut off the volume outside.

This should be modified if the texture offsets the entire volume in one direction and you want to center it again. For performance reasons, the displaced volume should stay close to its original position.

A volume displaced with various strengths. \xc2\xba6

--- Fin du chapitre: _volume_displace ---

--- Début du chapitre: _volume_info --- The Volume Info node provides information about Smoke Domains.

This node has no inputs.

This node has no properties.

Gives the color of the smoke inside the Fluid Domain . The color and vector outputs are the same. The Factor output is an average of the channels.

Gives a scalar defining the density of any smoke inside the Fluid Domain .

Gives a scalar defining the density of any fire inside the Fluid Domain

. All three outputs are the same.

Gives a scalar defining the temperature of the volume. Values in the range 0 - 1 map to 0 - 1000 kelvin. This may be used to render physically-based fire with the Blackbody or Principled Volume shaders. All three outputs are the same.

Smoke density. \xc2\xba6

Computing the color of fire using the Blackbody node. Since the Blackbody node expects its input in Kelvin, the temperature output has to be remapped first. \xc2\xba6

--- Fin du chapitre: _volume_info ---

--- Début du chapitre: _volume_principled --- The Principled Volume shader combines all volume shading components into a single easy to use node.

Volumes like smoke and fire can be rendered with a single shader node, which includes scattering, absorption and blackbody emission.

- Volume scattering color.
- Volume grid for coloring the volume. Use `\color` for smoke simulations.
- Density of the volume.
- Volume grid to define the density, typically `\density`.
- Backward or forward scattering direction.
- Volume shadow color tint.
- Amount of light to emit.
- Emission color tint.
- Blackbody emission for fire. Set to 1 for physically accurate intensity.
- Color tint for blackbody emission.
- Temperature in kelvin for blackbody emission, higher values emit more.
- Volume grid to define the temperature, typically `\temperature`.
- This node has no properties.
- Standard shader output.

--- Fin du chapitre: `_volume_principled` ---

--- Début du chapitre: `_volume_scatter` --- The Volume Scatter node allows light to be scattered as it passes through the volume. Typical usage would be to add fog to a scene. It can also be used with the Volume Absorption node to create smoke.

- Scattering coefficients per color channel.
- The density of the scatter effect.
- Controls the relative amount of backward and forward scattering.
- Refractive index of the scattering particles relative to water. Common ocean waters range between 1.0 and 1.2, while turbid waters with higher density of particles have higher IORs.
- Fraction of light that is scattered backwards. Most oceanic particles have backscatter values between 0.001 (e.g., very large phytoplankton) and 0.1 (e.g., very small mineral particles), pure water has a backscatter of 0.5. Values taken from Ocean Optics Web Book.
- Blending factor between Henyey-Greenstein ($\alpha = 0$) and Cornette & Shanks ($\alpha = 1$) phase functions.
- Diameter of the scattering particles in μm .
- Volume scattering phase function.
- Simple and widely used phase function, useful for approximating scattering in biological tissues.
- Cycles Only Suitable for modeling the scattering of light in underwater environments.
- Cycles Only Suitable for modeling the scattering of interstellar dust.
- Cycles Only Describes the scattering by particles with a size smaller than the wavelength of light, such as the scattering of sunlight in earth's atmosphere.
- Cycles Only Describes the scattering by particles with a size larger than the wavelength of light, such as cloud and fog.

Tip

These phase functions can be combined using a Mix Shader.

Volume scattering phase as a function of angles between the incoming and the outgoing direction, in logarithmic scale. Light comes from the left side. μ_0

The Volume Shader output must be plugged into the Volume Input of the Material or World Output node.

Example of Volume Scatter. \xc2\xba

--- Fin du chapitre: _volume_scatter ---

--- Début du chapitre: _volume_to_mesh --- The Volume to Mesh node generates a mesh on the \xe2\x80\x9csurface\xe2\x80\x9d of a volume. The surface is defined by a threshold value. All voxels with a larger value than the threshold are considered to be inside.

Standard geometry input.

Specifies the approximate resolution of the final mesh. The voxel size is adapted to the size of the entire volume.

Use a fixed resolution that does not change when the volume changes.

Voxels with a larger value are considered to be inside the mesh. The mesh will be generated on the boundary of inside and outside voxels. This is also called \xe2\x80\x9ciso value\xe2\x80\x9d.

Reduces the final face count by simplifying geometry where detail is not needed. This is similar to decimating the final to reduce resolution where it is not needed.

Mode for how the resolution of the final mesh is controlled.

This makes the resolution dependent on the resolution of the grid that is converted. Higher resolution grids result in a higher resolution mesh. In many cases, that is the most efficient mode.

Specifies the approximate resolution of the final mesh. The voxel size is adapted to the size of the entire volume.

Use a fixed resolution that does not change when the volume changes.

Note

This option applies individually for every grid in the input geometry. Standard geometry output.

--- Fin du chapitre: _volume_to_mesh ---

--- Début du chapitre: _voronoi --- Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

The Voronoi Texture node evaluates a Worley Noise at the input texture coordinates.

The inputs are dynamic, they become available if needed depending on the node properties.

Texture coordinate to evaluate the noise at; defaults to Generated texture coordinates if the socket is left unconnected.

Texture coordinate to evaluate the noise at.

Scale of the noise.

Number of noise octaves. The fractional part of the input is multiplied by the magnitude of the highest octave. Higher number of octaves corresponds to a higher evaluation time.

Blend between a smoother noise pattern, and rougher with sharper peaks.

The difference between the scale of each two consecutive octaves. Larger values corresponds to larger scale for higher octaves.

The smoothness of the noise.

Smoothness: 0.0. \xc2\xba

Smoothness: 0.25. \xc2\xba

Smoothness: 0.5. \xc2\xba

Smoothness: 1.0.\xc2\xba
Smoothness: 0.0.\xc2\xba
Smoothness: 0.25.\xc2\xba
Smoothness: 0.5.\xc2\xba
Smoothness: 1.0.\xc2\xba
Exponent of the Minkowski distance metric.
Exponent: 0.5.\xc2\xba
Exponent: 1.0.\xc2\xba
Exponent: 2.0.\xc2\xba
Exponent: 32.0.\xc2\xba
The randomness of the noise.
Randomness: 1.0.\xc2\xba
Randomness: 0.5.\xc2\xba
Randomness: 0.25.\xc2\xba
Randomness: 0.0.\xc2\xba
The dimensions of the space to evaluate the noise in.
Evaluate the noise in 1D space at the input W.
Evaluate the noise in 2D space at the input Vector. The Z component is ignored.
Evaluate the noise in 3D space at the input Vector.
Evaluate the noise in 4D space at the input Vector and the input W as the fourth dimension.
Higher dimensions corresponds to higher render time,so lower dimensions should be used unless higher dimensions are necessary.
The Voronoi feature that the node will compute.
The distance to the closest feature point as well as its position and color.
Distance.\xc2\xba
Color.\xc2\xba
Position.\xc2\xba
The distance to the second closest feature point as well as its position and color.
Distance.\xc2\xba
Color.\xc2\xba
Position.\xc2\xba
A smooth version of F1.
Distance.\xc2\xba
Color.\xc2\xba
Position.\xc2\xba
The distance to the edges of the Voronoi cells.
Distance.\xc2\xba
Distance smaller than 0.05.\xc2\xba
The radius of the n-sphere inscribed in the Voronoi cells.In other words, it is half the distance between the closest feature point and the feature point closest to it.
The n-sphere radius can be used to create tightly packed n-spheres.\xc2\xba
Node tree for the shader to the left.\xc2\xba
The distance metric used to compute the texture.
Use the Euclidean distance metric.
Use the Manhattan distance metric.
Use the Chebychev distance metric.
Use the Minkowski distance metric.The Minkowski distance is a generalization of the aforementioned metrics with an Exponent as a

parameter.Minkowski with an exponent of one is equivalent to the Manhattan distance metric.Minkowski with an exponent of two is equivalent to the Euclidean distance metric.Minkowski with an infinite exponent is equivalent to the Chebyshev distance metric.

Minkowski Exponent: 0.5 (Minkowski 1/2).

Minkowski Exponent: 1.0 (Manhattan).

Minkowski Exponent: 2.0 (Euclidean).

Minkowski Exponent: 32.0 (approximation of Chebyshev).

If enabled, ensures that the output values stay in the range 0.0 to 1.0.In rare cases, the output value may be outside that range when Feature is F2.

Distance.

Cell color. The color is arbitrary.

Position of feature point.

Position of feature point.

N-Sphere radius.

In some configurations of the node, especially for low values of Randomness,rendering artifacts may occur. This happens due to the same reasons described in the Notes section in the White Noise Texture page and can be fixed in a similar manner as described there.

The difference between F1 and Smooth F1 can be used to create beveled Voronoi cells.

Creating a hammered metal shader using the Voronoi Texture node.

--- Fin du chapitre: _voronoi ---

--- Début du chapitre: _vortex ---Reference

Physics \xe2\x80\xa3 Force Fields

Vortex

Vortex force field visualization.

The Vortex force field gives a spiraling force that twists the direction of points around the force object's local Z axis. This can be useful for making a swirling sink, or tornado, or kinks in particle hair. Inwards component of the vortex force.

--- Fin du chapitre: _vortex ---

--- Début du chapitre: _vr_scene_inspection --- The VR Scene Inspection add-on exposes and extends the native virtual reality features of Blender in the user interface.The feature set is limited to scene inspection use cases.More advanced use cases may be enabled through further development inside of Blender.

VR support in Blender is based on the OpenXR specification and requires some set up steps.These are explained in the Head-Mounted Displays (HMD) section.

Open Blender and go to Add-ons section of the Preferences .

Search \xe2\x80\x9cVR Scene Inspection\xe2\x80\x9d and check the Enable Extension checkbox.

Located in the 3D Viewport \xe2\x80\xa3 Sidebar \xe2\x80\xa3 VR tab.

Try to set up a connection to the OpenXR platform to share the viewport with an HMD.

Only track rotational changes of the head, do not allow the HMD to affect the location of the viewer in virtual space.

Skip eye offsets that are normally added for placing the viewer exactly at landmarks. This allows the tracking origin to be defined independently of the HMD position.

Enable default controller actions for viewport navigation, controller tracking, and haptics.

Set visibility of the ground plane in the VR view.

Set visibility of annotation strokes in the VR view.

Set visibility of selection outlines in the VR view.

Set visibility of VR motion controllers. Requires enabling the Use Controller Actions option.

Set visibility of custom operator drawing (e.g. default teleport beam).

Set visibility of object extras, including empties, lights, and cameras.

Set visibility of objects by type.

Preferred visualization of VR motion controllers.

Clipping values of the VR view, as in the 3D Viewport.

Landmarks are used to store reusable base poses (position and rotation) for the viewer in the virtual space. In addition, a base viewer reference scale can be set for landmarks of types Custom Object and Custom Pose.

A list view.

Defines which landmark settings are shown below the list. Changing the selected landmark does not have an influence on the VR view.

Activates a landmark, making it change the base pose of the VR view.

Create a landmark.

Delete the selected landmark.

Create a landmark from the viewer pose of the running VR session.

Create a new camera and landmark from the viewer pose of the running VR session.

Add a new landmark from the active camera object.

Update the selected landmark from the current VR viewer pose.

Move the 3D Cursor to the selected landmark.

Position the scene camera at the selected landmark.

Create a new camera from the selected landmark.

Follow the scene's active camera to define the base pose of the viewer.

Set an arbitrary object to define the base pose of the viewer.

Manually define a position and rotation to use as the base pose of the viewer.

Use input from a gamepad (Microsoft Xbox Controller) instead of motion controllers for VR actions such as viewport navigation.

Enable additional controller bindings to ensure correct input-to-action mappings. Note that a given extension may not be supported by all VR platforms.

Enable bindings for the HP Reverb G2 controllers.

Enable bindings for the HTC Vive Cosmos controllers.

Enable bindings for the HTC Vive Focus 3 controllers.

Enable bindings for the Huawei controllers.

Draw an indicator of the current VR viewer pose (location and rotation in the virtual space) in the current 3D Viewport.

Draw indicators of tracked VR motion controllers in the current 3D viewport. Requires enabling the Use Controller Actions option.

Draw landmark indicators in the current 3D Viewport.

Make the current 3D Viewport follow the perspective of the VR view.

Reference

3D View

View the viewport with virtual reality glasses (head-mounted displays).

3D Viewport \xe2\x80\xa3 Sidebar \xe2\x80\xa3 VR tab

viewport_vr_preview folder
Julian Eisel, Sebastian Koenig, Peter Kim
Julian Eisel, Peter Kim
GPL
Official
This add-on is bundled with Blender.
--- Fin du chapitre: _vr_scene_inspection ---

--- Début du chapitre: _walk_fly --- The standard navigation controls are sometimes limiting, especially for large environments such as architectural models. In these cases, it may be preferable to use first person controls instead, where you can look around while standing in one place rather than orbiting around a central viewpoint.

Blender offers two such alternative navigation methods: Flying and Walking. You can initiate either method from the View Navigation menu. You can also initiate your preferred one (configured in the Preferences) by pressing Shift- Accent Grave.

View Navigation.

Common use cases for Fly/Walk include:

This can be a quick way to navigate a large scene.

When activated from a camera view NumPad0, the camera will move along with you.

You can record the path you take by entering a camera view, enabling Auto Keying in the Timeline, starting animation playback, and finally activating Fly/Walk navigation. The path will be recorded as camera keyframes which can then be used for rendering.

Animation playback can be controlled while Fly/Walk navigation is active, so when you're done recording, you first need to exit the navigation with LMB before you can stop playback.

Reference

All modes

View Navigation Walk Navigation

This navigation method behaves like a typical first person game. It works with a combination of keyboard keys and mouse movement.

Move the mouse in the direction you want to look and use the keys listed below to walk around the scene.

When you are happy with the new view, press LMB to confirm. In case you want to go back to where you started, press Esc or RMB.

All these keys are also listed in the Status Bar while navigating. Settings like mouse sensitivity and default speed can be adjusted in the Preferences.

W/ Up

Move forward.

S/ Down

Move backward.

A/ Left

Strafe left.

D/ Right

Strafe right.

E

Move up (global) only available if Gravity is off.

Q

Move down (global) only available if Gravity is off.

R
Move up (local) \xe2\x80\x93 only available if Gravity is off.

F
Move down (local) \xe2\x80\x93 only available if Gravity is off.

Spacebar
Teleport to the location at the crosshair (offset by the Camera Height value set in the Preferences).

WheelUp/ NumpadPlus
Increase the movement speed.

WheelDown/ NumpadMinus
Decrease the movement speed.

Shift
Speed up the movement temporarily.

Alt
Slow down the movement temporarily.

V
Jump \xe2\x80\x93 only available if Gravity is on.

Tab
Toggle Gravity.

Z
Correct the Z axis of the view (smoothly roll it to ensure it \xe2\x80\x93s upright, not tilted to a side).

Period
Increases the jump height.

Comma
Decreases the jump height.

Reference
All modes
View \xe2\x80\x93 Navigation \xe2\x80\x93 Fly Navigation
On activation, the cursor is centered inside a rectangle that defines a safe zone. When the cursor is outside this zone, the view will rotate/pan. Move the mouse outside the safe zone in the direction you want to look. Click LMB or press Spacebar to keep the current view and exit Fly navigation. In case you want to go back to where you started, press Esc or RMB.

W/ Up
Accelerate forward.

S/ Down
Accelerate backward.

A/ Left
Accelerate left.

D/ Right
Accelerate right.

E
Accelerate upward.

Q
Accelerate downward.

MMB
Drag to pan the view. Flying will pause while you \xe2\x80\x93re doing this.

WheelUp/ NumpadPlus
Increase the acceleration in the direction of motion. If there is no motion, start accelerating forward.

WheelDown/ NumpadMinus

Decrease the acceleration in the direction of motion.If there is no motion, start accelerating backward.

Alt

Slow down as long as the key is held, until the view eventually comes to a standstill.

Ctrl

Disable rotation while held, the view rotation doesn't influence the flight direction.This allows you to fly past an object, keeping it centered in the view even as you fly away from it.

X

Toggle X axis correction. If enabled, the view will smoothly pitch to look at the horizon when the cursor is in the safe zone.

Z

Toggle Z axis correction. If enabled, the view will smoothly roll to an upright orientation.

--- Fin du chapitre: _walk_fly ---

--- Début du chapitre: _warning --- Outputs a custom message that can be referenced in the Warnings panel of the Geometry Nodes Modifier.

This allows node groups to communicate expectations about input values.

By default, warnings are propagated through all parent node groups.However, this can be controlled using the Warning Propagation setting on each node.

Present the warning in the Warnings panel.

The type of message to display, the type affects the icon displayed.

A passthrough of the Show input.

--- Fin du chapitre: _warning ---

--- Début du chapitre: _warp --- The Warp modifier can be used to warp parts of a mesh to a new location in a very flexible way, by using two objects to select the from and to regions.

A Warp modifier applied to a grid mesh.

This modifier is a bit tricky to understand at first.It requires two points, specified by the two target objects' origins.The from point designates a point in space that is pulled toward the to point.It is akin to using the Proportional Editing in Edit Mode.

The Warp modifier.

The object defining the origin transformation of the warp.

The object defining the destination transformation of the warp.

Enables volume preservation when rotating one of the transforms.

Sets how strong the effect is.

The name of a vertex group which is used to control the influence of the modifier.If left empty, the modifier affects all vertices equally.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be deformed by the modifier.

The setting reverses the weight values of the group.

Sets the way the strength of the warp change as it goes from the center of the transform to the Radius value.See Proportional Editing for descriptions of the falloff types.

Sets the distance from the transforms that can be warped by the transform handles.

You can finely control which vertices are affected by the warp, and to what extent, using a texture.

See common masking options for a complete reference.

The Warp modifier can be awkward to use sometimes, and its use case is rather small, but there are a few still. For example, it can be used to have an interactive Proportional Editing that can be used for animations. Another way to use this modifier is similar to the Deform Modifier. This allows you to deform parts of the mesh without having to make a vertex group.

Warp Modifier with a custom falloff curve. \xc2\xbb6

--- Fin du chapitre: _warp ---

--- Début du chapitre: _wave ---Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

The Wave Texture node adds procedural bands or rings with noise distortion.

Hint

In general, textures can be distorted by mixing their texture coordinates with another texture. The distortion built into the Wave Texture Node uses the Color output of the Noise Texture Node.

To replicate this, center its value range around zero, multiply it by a factor proportional to Distortion/ Scale and add the result onto the texture coordinates. Detail, Detail Scale and Roughness of the Wave Texture Node correspond to the inputs on the Noise Texture Node.

Texture coordinate to sample texture at; defaults to Generated texture coordinates if the socket is left unconnected.

Overall texture scale.

Amount of distortion of the wave.

Hint

In general, textures can be distorted by mixing their texture coordinates with another texture. The distortion built into the Wave Texture Node uses the Color output of the Noise Texture Node.

To replicate this, center its value range around zero, multiply it by a factor proportional to Distortion/ Scale and add the result onto the texture coordinates. Detail, Detail Scale, and Roughness of the Wave Texture Node correspond to the inputs on the Noise Texture Node.

Amount of distortion noise detail.

Scale of distortion noise.

Blend between a smoother noise pattern, and rougher with sharper peaks.

Position of the wave along the Bands Direction. This can be used as an input for more control over the distortion.

Bands or Rings shaped waves.

The axis the bands or rings propagate from i.e. which axis they are perpendicular to. When using Bands a Diagonal axis is an option and when using Rings the rings can propagate outwards from a single point by using Spherical direction.

Controls the look of the wave type.

Uses a sawtooth profile.

Uses the standard sine profile.

Texture color output.

Texture intensity output.

Wave Texture. \xc2\xbb

--- Fin du chapitre: _wave ---

--- Début du chapitre: _wavelength --- The Wavelengthnode converts a wavelength value to an RGB value.This can be used to achieve a specific color on the light spectrum.

The color wavelength from 380 to 780 nanometers.

This node has no properties.

RGB color output.

Example of Wavelength node. \xc2\xbb

--- Fin du chapitre: _wavelength ---

--- Début du chapitre: _wave_distortion --- The Wave DistortionVisual applies a wavy effects to the object.

Wave Distortion Effect. \xc2\xbb

Sets horizontal or vertical direction for the waves.

Controls the strength and the depth of the wave.

Controls the wave period. The time it takes to complete one cycle.

Shifts the wave pattern over the Object.

Amplitude: 10 (horizontal).\xc2\xbb

Amplitude: 30 (horizontal).\xc2\xbb

Amplitude: 10 (vertical).\xc2\xbb

Amplitude: 30 (vertical).\xc2\xbb

--- Fin du chapitre: _wave_distortion ---

--- Début du chapitre: _weighted_normal --- This modifier changes the custom normals of a mesh, using various selectable methods.This can be useful to make some faces appear very flat during shading, among other effects.See Normals for a description of normals and custom normals.

The normals around a vertex will be combined to create a custom (per face corner) normal using various weights for each. The Weighting Mode defines how to compute the weights.

Weight according to the area of the face that the normal originates.A larger area means that the normal from that face will get a higher weight in final result.

Weight according to the angle each face forms at the vertex.This is the method Blender uses by default when combining face normals to compute a vertex one.

Weights are obtained by multiplying the face area and corner angle ones.

Determines how strongly the weights are biased according to the face areas and/or corner angles,a bit like a contrast setting for a picture.

A value of 50 means all faces are weighted uniformly.More than 50 means

faces with higher area or angles are given even more weight (more

\xe2\x80\x9ccontrast\xe2\x80\x9d).Less than 50 means faces with higher area or angles are given lesser weights (less

\xe2\x80\x9ccontrast\xe2\x80\x9d).

A weight-rounding threshold which means that, if two angles or areas differ by less than that threshold,they will get equal weights.

Preserve sharp edges ,though smoothing will still happen if there are multiple faces between any two sharp edges.

Use face weights (weak, medium, or strong) as assigned by the Set Strength tool or by the Set Strength mode of a Bevel modifier.

For example, if three faces meet at a vertex and have the face weights weak, medium, and strong, then only the normal associated with the strong face will be used to set the final result.

If a vertex group is specified, the modifier will only affect those vertices. The  button to its right will invert the selection (only affect the vertices not in the vertex group).

--- Fin du chapitre: `_weighted_normal` ---

--- Début du chapitre: `_weights_menu` ---Reference

Edit Mode

Weights

This page covers many of the tools in the Weights menu.

Reference

Edit Mode

Weights  Normalize All

For each point, this tool makes sure that the sum of the weights across all vertex groups is equal to 1. It normalizes all of the vertex groups, except for locked groups, which keep their weight values untouched.

Keep the values of the active group while normalizing all the others.

Reference

Edit Mode

Weights  Normalize

This tool only works on the active vertex group. All points keep their relative weights, but the entire set of weights is scaled up such that the highest weight value is 1.0.

Reference

Edit Mode

Weights  Invert

Replaces each weight of the selected vertex group by -1.0 weight.

Examples:

Original 1.0 converts to 0.0

Original 0.5 remains 0.5

Original 0.0 converts to 1.0

Reference

Edit Mode

Weights  Smooth

Smooths the weights of the active vertex group.

Reference

Edit Mode

Weights  Sample Weight

Shift- X

Adjust the Weight of the Draw tool to the weight of the vertex under the mouse cursor.

--- Fin du chapitre: `_weights_menu` ---

--- Début du chapitre: `_weight_angle` --- This modifier sets the weights of the given vertex group, based on a predetermined angle.

Warning

This modifier does implicit clamping of weight values in the standard (0.0 to 1.0) range. All values below 0.0 will be set to 0.0, and all values above 1.0 will be set to 1.0.

The Vertex Weight Proximity modifier panel. 

The vertex group to affect.

Inverts the influence of the selected vertex group. The setting reverses the weight values of the group.

Sets the angle for the maximum weights value.

The axis along which the angle affects the weights.

X, Y, Z

Coordinate space to be used.

Minimum value for vertex weight.

Multiply the calculated weights with the existing values in the vertex group.

See Influence Filters.

--- Fin du chapitre: _weight_angle ---

--- Début du chapitre: _weight_edit --- This modifier is intended to edit the weights of a vertex group.

The general process is the following, for each vertex:

(Optional) It does the mapping, either through one of the predefined functions, or a custom mapping curve.

It applies the influence factor, and optionally the vertex group or texture mask (0.0 means original weight, 1.0 means fully mapped weight).

It applies back the weight to the vertex, and/or it might optionally remove the vertex from the group if its weight is below a given threshold, or add it if it is above a given threshold.

Important

This modifier does implicit clamping of weight values in the standard (0.0 to 1.0) range. All values below 0.0 will be set to 0.0, and all values above 1.0 will be set to 1.0.

Note

You can view the modified weights in Weight Paint Mode. This also implies that you will have to disable the Vertex Weight Edit modifier if you want to see the original weights of the vertex group you are editing.

The Vertex Weight Edit modifier panel.

The vertex group to affect.

The default weight to assign to all vertices not in the given vertex group.

Adds vertices with a final weight over Add Threshold to the vertex group.

Removes vertices with a final weight below Remove Threshold from the vertex group.

Scale the weights in the vertex group to keep the relative weight but the lowest and highest values follow the full 0 - 1 range.

Type of mapping.

No mapping.

Allows you to manually define the mapping using a curve.

These are classical mapping functions, from spikiest to roundest.

Uses a random value for each vertex.

Creates binary weights (0.0 or 1.0), with 0.5 as cutting value.

Inverts the falloff.

Those settings are the same for the three Vertex Weight modifiers.

The overall influence of the modifier (0.0 will leave the vertex group's weights untouched, 1.0 is standard influence).

Important

Influence only affects weights, adding/removing of vertices to/from vertex group is not prevented by setting this value to 0.0.

In addition, a per-vertex fine control of the effect is possible using either a vertex group or a texture(both are mutually exclusive). The per-vertex values from those will be multiplied with the Global Influence.

See common masking options for a complete reference.

Here is an example of various effects achieved using Vertex Weight Edit modifier (together with the Vertex Weight Proximity modifier) to generate weights used by the Displace modifier.

Concave-type mapping curve.

No mapping curve (linear).

Convex-type mapping curve.

Vertices with a computed weight below 0.1 removed from the vertex group.

The blend-file, TEST_2 scene.

--- Fin du chapitre: _weight_edit ---

--- Début du chapitre: _weight_mix --- This modifier mixes a second vertex group (or a simple value) into the affected vertex group, using different operations.

Important

This modifier does implicit clamping of weight values in the standard (0.0 to 1.0) range. All values below 0.0 will be set to 0.0, and all values above 1.0 will be set to 1.0.

Note

You can view the modified weights in Weight Paint Mode. This also implies that you will have to disable the Vertex Weight Mix modifier if you want to see the original weights of the vertex group you are editing.

The Vertex Weight Mix modifier panel.

A: The vertex group to affect.

B: The second vertex group to mix into the affected one. Leave it empty if you only want to mix in a simple value.

Invert the influence of the vertex group.

A: The default weight to assign to all vertices not in the given vertex group.

B: The default weight to assign to all vertices not in the given second vertex group.

Choose which vertices will be affected.

Affects all vertices, disregarding the vertex groups content.

Affects only vertices belonging to the affected vertex group.

Affects only vertices belonging to the second vertex group.

Affects only vertices belonging to at least one of the vertex groups.

Affects only vertices belonging to both vertex groups.

Important

When using All vertices, Vertices from group B or Vertices from one group, vertices might be added to the affected vertex group.

How the vertex group weights are affected by the other vertex group's weights.

Replaces affected weights with the second group's weights.

Adds the values of Group B to Group A.

Subtracts the values of Group B from Group A.

Multiplies the values of Group B with Group A.

Divides the values of Group A by Group B.

Subtracts the smaller of the two values from the larger.

Adds the values together, then divides by 2.

Uses the smallest weight value of VGroup A or VGroup B weights.
Uses the largest weight value of VGroup A or VGroup B weights.
Scale the weights in the vertex group to keep the relative weight but the lowest and highest values follow the full 0 - 1 range.
Those settings are the same for the three Vertex Weight modifiers, see the Vertex Weight Edit modifier page.
Here is an example of using a texture and the mapping curve to generate weights used by the Wave modifier.
Using intensity.
Using Red.
Using Saturation.
A customized mapping curve.
Custom Mapping disabled.
Custom Mapping enabled.
The blend-file, TEST_4 scene.
--- Fin du chapitre: _weight_mix ---

--- Début du chapitre: _weight_proximity --- This modifier sets the weights of the given vertex group, based on the distance between the object (or its vertices), and another target object (or its geometry).
Warning
This modifier does implicit clamping of weight values in the standard (0.0 to 1.0) range. All values below 0.0 will be set to 0.0, and all values above 1.0 will be set to 1.0.
See also
This documentation refers to the Vertex Weight Proximity Modifier specific to the Grease Pencil object. For uses with other object types refer to the general Vertex Weight Proximity Modifier.
The Vertex Weight Proximity modifier panel.
The vertex group to affect.
Inverts the influence of the selected vertex group. The setting reverses the weight values of the group.
The object from which to compute distances.
Distance mapping to 0.0 weight.
Distance mapping to 1.0 weight.
Minimum value for vertex weight.
Multiply the calculated weights with the existing values in the vertex group.
See Influence Filters.
--- Fin du chapitre: _weight_proximity ---

--- Début du chapitre: _weld --- The Weld modifier looks for groups of vertices within a threshold and merges them, collapsing the surrounding geometry.
The Weld modifier.
Method for choosing which vertices are merged.
Merge includes all geometry including loose parts.
Merge only includes attached geometry i.e. the modifier will not merge loose parts together.
Maximum distance that the vertices must have each other to be merged.
Only collapse short edges which are not adjacent to any face. This is useful for example to stitch the seams used in cloth simulations.

When the Vertex Group option is selected, only vertices with weight above zero will be affected by the modifier.

Inverts the influence of the selected vertex group, meaning that the group now represents vertices that will not be merged by the modifier. The setting reverses the weight values of the group.

--- Fin du chapitre: _weld ---

--- Début du chapitre: _weld_edges_faces ---Reference

Edit Mode

Face \xe2\x80\xa3 Weld Edges into Faces

A tool to split selected faces by loose wire edges. This can be used in a similar way to the Knife tool, but the edges are manually setup first.

--- Fin du chapitre: _weld_edges_faces ---

--- Début du chapitre: _white_noise ---Note

This node is ported from shader nodes. The manual and images are referencing the shader version of the node. This node accepts field inputs and outputs. When not connected the Vector input has an implicit position attribute value.

The White Noise Texture node returns a random number based on an input seed. The seed can be a number, a 2D vector, a 3D vector, or a 4D vector; depending on the Dimensions property. The output number ranges between zero and one.

The inputs are dynamic, they become available if needed depending on the node properties.

Vector used as seed in 2D, 3D, and 4D dimensions.

Value used as seed in 1D and 4D dimensions.

The dimensions of the space to evaluate the noise in.

The Winput is used as seed.

The X and Y components of the Vector input are used as seed.

The Vector input is used as seed.

Both the Vector input and the Winput are used as seed.

Output random value.

Output random color.

The slightest difference in seed values would result in completely different outputs. Consequently, bad precision may have significant impact on the output. Usually, we can mitigate this issue by:

Eliminating the problematic seed value. If the problematic seed value is constant, it should be eliminated by choosing a lower dimension or multiplying it by zero.

Adding an arbitrary value to the seed. The issue might only happen at certain boundaries, like unit boundaries, so simply adding an arbitrary value might solve the issue.

Taking the absolute value of the seed. In computing, zero may be positive or negative, so taking the absolute values unifies the zero into a single value.

Precision issue due to signed zeros on the Z axis. \xc2\xbb6

Mitigating the issue by eliminating the Z axis. \xc2\xbb6

Mitigating the issue by adding an arbitrary value. \xc2\xbb6

Mitigating the issue by taking the absolute value. \xc2\xbb6

Generating cell noise using the Snap vector operation and the White Noise node. \xc2\xbb6

--- Fin du chapitre: _white_noise ---

--- Début du chapitre: `_wind` ---Reference

Physics \xe2\x80\xa3 Force Fields

Wind

Wind force field visualization.\xc2\xb6

The Windforce field gives a constant force in a single direction, along the force object\xe2\x80\x99s local Z axis.The strength of the force is visualized by the spacing of the circles shown.

--- Fin du chapitre: `_wind` ---

--- Début du chapitre: `_windows` --- This guide covers the following topics:

Installing Python(used to \xe2\x80\x9cconvert\xe2\x80\x9d the source files to HTML)

Installing Git and Downloading the Repository

Setting up the Build Environment

Download the Python installation packagefor Windows.In this guide version 3.9.x is used.

Install Python with the installation wizard.Please make sure that you enable the \xe2\x80\x9cAdd Python to PATH\xe2\x80\x9d option:

The option must be enabled so you can build the manual with the make script. \xc2\xb6

All other settings can remain as set by default.

In this guide, we will use the official Git client, though any Git client will do.

Download and install Gitfor Windows.

Simply check out the Blender Manual\xe2\x80\x99s repository using:

The repository will now be downloaded which may take a few minutes depending on your internet connection.

Note

This process can be completed using a graphical Git client,in that case you will just use the repository address in the URL field provided by your client:

Tip

It is recommended to setup and activate a virtual Python environment where dependencies will be installed:

Repeat the.venv/Scripts/activatecommand to re-activate the virtual environment,whenever you open a new terminal to build the documentation.

Open a Command Prompt. (Run as Administrator)

Enter the blender-manual folder which was just added by gitclone :

Inside that folder is a file called requirements.txt which contains a list of all the dependencies we need.Install all the dependencies using Python\xe2\x80\x99s pip command:

If all goes well, you should see the following message when it is finished:

During the setup, some warnings may be shown, but do not worry about them.However, if any errors occur, they may cause some problems.

Note

Every now and then you may want to make sure your dependencies are up to date using:

--- Fin du chapitre: `_windows` ---

--- Début du chapitre: `_wipe` --- The Wipetransition strip can be used to transition from one strip to the next.The wipe will have no effect if created from a single strip instead of two strips.The duration of the

wipe is the intersection of the two source strips and cannot be adjusted. To adjust the start and end of the wipe you must adjust the temporal bounds of the source strips in a way that alters their intersection.

The type of transition used.

Reveals the next strip by uncovering it in a straight line moving across the image.

Similar to Single, but uses two lines either starting from the middle of the image or the outside. Like the blink of an eye.

Reveals the next strip through an expanding (or contracting) circle. Like the aperture of a camera or pupil of an eye. You can blur the transition, so it looks like ink bleeding through a paper.

Like the hands of an analog clock, it sweeps clockwise or (if Wipe In is enabled) counterclockwise from the 9:00 position. As it sweeps, it reveals the next strip.

Controls whether to fade In or Out.

The width of the blur used to blur the transition.

Controls the angle of the line for Single and Double transition types.

Automatically calculates a linear fade over the length of the strip.

Allows you to manually keyframe a custom fade. This can be used with different easings to fine-tune the fade in/out.

Wipe Effect. \xc2\xba

--- Fin du chapitre: _wipe ---

--- Début du chapitre: _wireframe --- The Wireframe node is used to retrieve the edges of an object as it appears to Cycles. As meshes are triangulated before being processed by Cycles, topology will always appear triangulated when viewed with the Wireframe node.

This node has no inputs.

When enabled, the size of edge lines is set in screen space.

Thickness of the edge lines.

Black-and-white mask showing white lines representing edges according to the object's Topology.

Using the Wireframe node to showcase the topology of a mesh. \xc2\xba

--- Fin du chapitre: _wireframe ---

--- Début du chapitre: _wood --- The wood texture is used to generate wood and ring-shaped patterns.

Wood Texture panels. \xc2\xba

Shape of wave to produce bands.

Sine, Saw, Triangle

Set the bands to either straight or ring-shaped, with or without turbulence.

Bands, Rings, Band Noise, Ring Noise

There are two methods available for the Noise function.

Soft, Hard

Dimension of the Noise table.

Turbulence of the Band Noise and Ring Noise types.

--- Fin du chapitre: _wood ---

--- Début du chapitre: _workflow --- Depending on the original footage's properties, achieving good stabilization results might be simple and easy, or it might require some work, dedication and

careful planning. This section covers some practical considerations to help improving the results.

Whenever the camera is basically fixed, or at least

almost stationary, and the footage is crisp and without motion blur, perfect stabilization is easy to achieve. This might be the case when a tripod was used, but wind or vibrations on the floor (e.g. on a stage) caused some minor shakes. Shoulder camera shots done by an experienced operator also frequently fall into this category. Use as few points as possible. Start with a single point right on the main subject.

Track this single point as accurate as possible. Beware of movements and shape changes of the tracked feature. Proceed in small increments (e.g. 50 frames), zoom in and readjust the target point manually when it drifts away. Another option is to use a larger target area for tracking; since we're tracking only a single point, the slower tracking speed might be acceptable.

After enabling the basic (location) stabilization, consider if you really need rotation stabilization. Often, some minor, slow swinging movements are not really noticeable and do not warrant the additional working time and quality loss caused by rotation and scale stabilization.

For rotation, start with one extra point, well spaced but preferably still attached to the main subject.

Consider to fix some slow residual motion by manually animating the Expected parameters, before you even think of adding more tracking markers. Because doing so is often not worth the effort.

If you need to add more points, the most important goal is to achieve symmetry. Place location tracking points symmetrically above and below the horizon. Place rotation tracking points into diagonally opposed direction, always centered around the main focal area.

The 2D stabilizer can not work miracles; some flaws simply can not be fixed satisfactorily. Notorious issues are motion blur, rolling shutter, pumping autofocus and moving compression artifacts. Especially if you do succeed with basic stabilization, such image flaws become yet the more noticeable and annoying. When on set or on location, it might be tempting to fix matters in post production. Resist that deception, it rarely works out well.

Prefer a short exposure time to avoid motion blur. While motion blur is good to render filmed movements more smooth and natural, it seriously impedes the ability to track features precisely. As a guideline, try to get at least to 1/250 s.

Prefer higher frame rates. The more temporal resolution the stabilizer has to work on, the better the results. If you have the option to choose between progressive and interlaced modes, by all means use interlaced and deinterlace the footage to the doubled frame rate. This can be done with the yadiff filter of FFmpeg: use the mode 1 (send_field).

Beware of the Rolling Shutter effect. Avoid fast lateral movements. If you can, prefer a camera which produces less rolling shutter. Also, using a higher frame rate reduces the amount of rolling shutter; another reason to prefer interlaced over progressive for the purpose at hand.

Switch off autofocus. Better plan your movement beforehand, set a fixed focus and rely on depth of field through using a small aperture. Pumping movements might not be so noticeable to the human observer, but the

feature tracking tends to slide away on defocused image elements; fixing this manually after the fact can cause a huge waste of time. Increase the lighting level, at least use a higher sensitivity. This helps to set a fast shutter speed plus a small aperture. Better lighting and good exposure also help to reduce the impact of compression artifacts. If you can, also select a codec with less data reduction, better color space, etc. Inevitably, we're losing some quality through the interpolation necessary for stabilization. Plus we're losing some quality due to color space conversion.

When the footage builds on elaborate intended movement of the camera, the process of stabilization becomes more involved—especially when there is a shift in the main area of interest within the shot. When working with many tracks and fine-grained animation, it is easy to get into a situation where additional manipulations actually decrease the quality, while it might be hard to spot and locate the root cause of problems. Recommendation is to proceed systematically, starting from the general outline down to tweaking of specific aspects. Understand the nature of the movements in the shot, both the intended and the accidental.

Track some relevant features for location.

Establish the basic location stabilization. This includes the decision, which feature to use for what segment of the shot. Work with the track weights to get an overall consistent movement of the weight center, in accordance with the inherent focus of the shot.

Define the panning movements of the virtual camera (through animation of the Expected Position parameter).

Add tracking for rotation and zoom stabilization.

Fine-tuning pass:

Break down the whole duration of the shot into logical segments to define the intended camera movement. Then refine those segments incrementally step-by-step, until the overall result looks satisfactory. Animating some parameters over duration of the shot is often necessary, at least to get the final touch, including control of the scale factor to hide the dancing black borders. Unfortunately there is a known limitation in the current version: it is not possible to open the generic animation editors (Graph editor and Dope Sheet) for animation data beyond the 3D scene. So, while it is possible to set keyframes right within the UI controls of the stabilizer (either through pressing the I key or with the help of the context menu), it is not possible to manipulate the resulting curves graphically. The only way to readjust or remove a misguided keyframe is to locate the timeline to the very frame and then use the context menu of the animated UI control. (Hint: the color of the UI control changes when you have located at precisely the frame number of the keyframe.)

It might not be possible to track a given feature over the whole duration of the shot. The feature might be blurred or obscured; it might even move out of sight entirely, due to deliberate camera movement. In such a situation, we need another tracked feature to take on its role, and we need some overlap time to get a smooth transition without visible jump.

Irregular Tracks. \xc2\xb6

The stabilizer is able to deal with gaps and partial coverage within the given tracks. However, the basic assumption is that each track covers a single, fixed reference point whenever there is any usable/enabled data. Thus, you must not create a given track to

follow several different points, rather you should disable and thus end one track, when tracking this feature is no longer feasible. You may include gaps, when a tracking point is temporarily disabled or unavailable, but you should start a new track for each distinct new feature to be tracked.

Each track contributes to the overall result by the degree controlled through its Stab Weight parameter. It is evaluated on a per-frame basis, which enables us to control the influence of a track by animating this Stab Weight. You may imagine the overall working of the stabilizer as if each tracking point drags the image through a flexible spring: When you turn down the Stab Weight of a tracking point, you decrease the amount of drag it creates. Sometimes the contribution of different tracks has to work partially counter each other. This effect might be used to cancel out spurious movement, e.g. as caused by perspective. But when, in such a situation, one of the involved tracks suddenly goes away, a jump in image position or rotation might be the result. Thus, whenever we notice a jump at the very frame where some partially covered track starts or ends, we need to soften the transition. We do so by animating the Stab Weight gradually down, so that it reaches zero at the boundary point. In a similar vein, when we plan a handover between several partially covered tracks, we define a cross-fade over the duration where the tracks overlap, again by animating the Stab Weight parameters accordingly. But even with such cross-fade smoothing, some residual movement might remain, which then needs to be corrected with the Expected Position or Expected rotation parameters. It is crucial to avoid covershooting movements in such a situation always strive at setting the animation keyframes onto precisely the same frame number for all the tracks and parameters involved.

--- Fin du chapitre: _workflow ---

--- Début du chapitre: _workflow_examples --- Simple Drivers can be configured from the pop-over that appears when adding a new Driver. When adding multiple Drivers or for more advanced configurations, it is useful to have open the Drivers Editor.

Control a property with an object's transform. In this example, the Y rotation of Object 2 will be driven by the X position of Object 1. Starting from a simple setup with two objects:

Add a Driver to the Rotation Y property of the second object via the context menu or with Ctrl- D .

Open the Drivers Editor and select the Y Euler Rotation property in the channels region.

Open the Sidebar region and select the Driverstab.

Configure the driver to be the Averaged Value of a Transform Channel of the first object.

Experiment with moving the first object and notice how it affects the Y rotation of the second object.

Orbit an object's position around a point with a custom Scripted Expression. The object's position will change when scrubbing the timeline. Using trigonometry, circular motion can be defined in 2D using the sine and cosine functions. (See Unit Circle.) In this example, the current frame is used as the variable that induces the

motion. frame is a Simple Expression that corresponds to `bpy.context.scene.frame_current` .

Add a driver to the X Location property.

Set the Driver Type to Scripted Expression.

Add the expression `0+(sin(frame/8)*4)` , where:

- frame/8 : is the current frame of the animation, divided by 8 to slow the orbit down.
- (sin()*4) : multiplies the result of sin(frame/8) by 4 for a bigger circle.
- 0+ : is used to control the offset to the orbit center point.

Add a driver to the Y Location property with the expression `0+(cos(frame/8)*4)` .

Scrub the timeline to see the effect. Experiment with the variables to control the size and center of the orbit.

Create a custom function to get the square of a value (i.e. value 2). Adding the function to the Driver Namespace allows it to be used from driver expressions. The Driver Namespace has a list of built-in functions for use in driver expressions, as well as constants such as `\xc4\x80` and `e`. These can be inspected via the Python Console:

To add a new function to the Driver Namespace, the function itself needs to be implemented and then added to the `bpy.app.driver_namespace` .

Add the following to the Text Editor inside Blender and press Run Script.

Add a driver with a Scripted Expression such as `square(frame)` .

Observe the effect when scrubbing the timeline.

There are more custom function examples available in Blender\Text Editor Templates \Python \Driver Functions. Since Simple Expressions cannot access custom functions, using them only makes sense for complex computations.

Warning

Trying to replace built-in entries of the driver namespace may result in undefined behavior.

The material Attribute Node in the View Layer mode automatically searches for the attribute property in multiple locations. This, for example, can allow setting a certain value of the custom attribute at the Scene or World level, and then overriding it differently for one View Layer .

Context Properties of drivers don't implement this behavior, so if necessary it has to be manually emulated via fallback values and a conditional expression (conditions are Simple Expressions).

For an attribute named `attr` , the node tries the following six RNA path lookups in order:

- `["attr"]` in the active View Layer (custom property).
- `attr` in the active View Layer (built-in property).
- `["attr"]` in the active Scene.
- `attr` in the active Scene.
- `world["attr"]` in the active Scene.
- `world.attr` in the active Scene.

Depending on the specific property it may be sufficient to check only a subset of these locations. For example, the image on the right shows how to access an attribute that is known to definitely be a custom property with a color value.

Driver variables accessing locations that are not final in the lookup chain should use fallback values that are invalid for the attribute (e.g. negative color values), which can then be checked by the conditional

expression. The final variable should fallback to a valid default value to be used when the property is not set at all.

Fix intersection problems that happen when using armatures and weight painting, especially at joints. Shape keys can also be used to tweak and refine a rig, for example to suggest muscle formations. In this example, a shape key is used to improve the deformation at the elbow of a rudimentary arm.

Left: Skeletal mesh deformation without correction. Right: Corrective shape key applied

Add a mesh (in this example, a cylinder with loop cuts).

Add an armature with a chain of bones.

Skin the mesh to the armature using weight painting.

(Note: to parent the mesh to the armature: select the mesh first, then the armature and use Ctrl-P to parent with auto weights.)

Experiment with posing the armature and observe the deformation at the joint. To fix intersection problems or angles that look unsatisfactory, you can associate a Shape Key with a pose.

Pose the armature such that the problems are visible. Be sure to cover the extreme poses that you want to support for the rig.

With the mesh selected, add a new Shape Key in addition to the Basis key. Properties > Mesh tab > Shape Keys

In order to author the shape key on top of the armature deformation, enable both Edit Mode Display and Cage Editing in the Armature modifier. Properties > Modifiers tab > Armature Modifier > Header

Enter Edit Mode and select the new shape key in the properties panel. Adjust the vertices as desired. Select the Basis key to toggle between the original mesh and your edits. (Note: be careful to apply edits only to your shape and not to the original mesh or other existing keys.)

Once you are satisfied with how the deformation looks for the problematic pose, you'll need to configure a driver to activate the shape smoothly when entering that position.

Add a driver to the Value of the shape key you've created.

Open the Drivers Editor and select the driver.

A simple way to configure the driver is with a direct correspondence of the value of a bone's rotation channel to the shape key activation Value. This method has the disadvantage of relying on a single channel of a bone's rotation which might be insufficient to precisely express the condition under which the shape key should be activated.

In the Drivers tab, select the Averaged Value of the rotation of the bone you are posing.

Understand the rotation axis that you are interested in by enabling axes display in the armature or by observing the bone's transform values in the Properties.

Select the rotation channel and set it to local, meaning, the bone's rotation value relative to its parent bone.

Manually set points in the driver curve by selecting a handle and dragging it or inserting values in the F-Curve tab. The Y axis represents the shape key Value, which should go from 0.0 to 1.0. The X axis is usually the frame, but for this driver it represents the rotation value in radians. You can have more than two points in the curve and tweak the transitions with the handles in the curve view (G to move).

To verify that the driver behaves correctly, deselect the option to only show drivers for selected objects. This way, you can pose the armature and keep an eye on the driver.

This method requires an additional target or corrective bone, but it better expresses the spatial condition in 3D space of the bone that is causing the problem.

In armature Edit Mode, add a new bone extruded from Bone 1, in the position at which Bone 2 should have the shape key active. This type of bones usually follow a naming convention such as `TAR-` (target) or `COR-` (corrective).

In the Drivers tab, select the Averaged Value of the rotational difference between the bone you are rotating and the target bone. A rotational difference is the minimum angle between two objects in World Space. It is therefore important that the bones have the same root, so that the only thing affecting the angle between the bones is the rotation of one of them. When the deformation bone (Bone 2) reaches the target rotation (TAR-Bone 2) the rotational difference will be 0.

Manually adjust the driver curve handles so that the shape key Value (Y axis) is 1.0 when the rotational difference (X axis) is 0. The Value should be 0.0 when the arm is extended, at which point the rotational difference should be around 90 or more (in radians).

See the steps in Method 1 on how to adjust the curve handles and confirm that the functionality is working. Pose the armature to verify that the ranges are correct.

Activate different shape keys in succession. In this example, moving a single bone will activate first Key 1 and then Key 2. See also relative shape keys mix additively.

Add two shape keys to a mesh, besides the Basis.

Basis. 6

Key 1: top faces moved up by 1 m. 6

Key 2: inner top moved up by 1 m. 6

Add an armature with a single bone to control the shape keys. The goal is to activate the keys in succession as this bone moves up.

As shown in the picture above, when the bone is halfway up, both Key 1 and Key 2 have an influence. It is a matter of preference if Key 1 should be at its maximum Value before Key 2 starts to become active, or how much they should overlap. This example shows a seamless blend.

For a seamless blend where there is overlap, Key 1 should have a Value of 0.0 when the bone is at the bottom and increase linearly to 1.0 until the bone is past the midpoint height. Key 2 should have a value of 0.0 before the midpoint height and then increase at the same rate than Key 1 until reaching Value 1.0 when the bone is at maximum height.

Add a driver to the Value of Key 1 and Key 2. In the Driverstab, configure both drivers to be the Averaged Value of a variable with the bone's Z location.

Determine the range of the bone's motion in the World Z axis by moving it up so that it is aligned with the top of the mesh when both keys are active. Here we will use [0.0, 2.5].

Configure the driver functions so that the Value of the shape keys (Y axis) is as desired for the bone's height (X axis).

The driver functions should be linear, therefore, they can be defined analytically with a function of type $y = a + bx$, where a is an offset in y and b is the slope.

In the Modifierstab, add a Generator of type Extended Polynomial for both drivers.

Play with the values of a and b so that the curves go from $[0.0, 1.0]$ in the Y axis and from $[0.0, 2.5]$ in the X axis. The curves should overlap in the mid area of the X axis and they should have the same slope (b).

Possible values are Key 1: $y = 0.0 + 0.6x$ and Key 2: $y = -0.5 + 0.6x$.

Note that the functions go outside the range $[0.0, 1.0]$ for the shape keys Value, but that has no effect because Value is clamped in a Range in the Shape Keys panel.

--- Fin du chapitre: _workflow_examples ---

--- Début du chapitre: _workspaces --- Workspaces are essentially predefined window layouts. Each Workspace consists of a set of Areas containing Editors, and is geared towards a specific task such as modeling, animating, or scripting. You typically switch between multiple Workspaces while working on a project.

Workspaces are located at the Topbar. Click on the tabs to switch between the workspaces. You can also use the keyboard shortcuts Ctrl- PageUp and Ctrl- PageDown. Double-click a tab to rename the workspace.

Click on the Add button to add a new workspace.

The context menu contains options to duplicate, delete and reorder workspaces.

Blender's default startup shows the Layout workspace in the main area. This workspace is a general workspace to preview your scene and contains the following Editors:

- 3D Viewport on top left.

- Outliner on top right.

- Properties on bottom right.

- Timeline on bottom left.

Blender's Layout Workspace with four editors.

- 3D Viewport (yellow), Outliner (green), Properties (blue) and Timeline (red).

Blender also has several other workspaces added by default:

- For modification of geometry by modeling tools.

- For modification of meshes by sculpting tools.

- For mapping of image texture coordinates to 3D surfaces.

- For coloring image textures in the 3D Viewport.

- For specifying material properties for rendering.

- For making properties of objects dependent on time.

- For viewing and analyzing rendering results.

- For combining and post-processing of images and rendering information.

- For procedural modeling using Geometry Nodes.

- For interacting with Blender's Python API and writing scripts.

Blender has a couple additional Workspaces to choose from when adding a new Workspace:

- 2D Animation

- General workspace to work with Grease Pencil.

Similar to 2D Animation but contains a larger canvas.

VFX

For creating 2D masks for compositing or video editing.

For calculating camera motion and stabilizing video footage.

Video Editing

For sequencing together media into one video.

The workspaces are saved in the blend-file. When you open a file, enabling Load UI in the File Browser indicates that Blender should use the file's screen layout rather than the current one.

A custom set of workspaces can be saved as a part of the Defaults .

Reference

Properties

Tool tab Workspace

When enabled, the current workspace will remember the currently selected scene. Then, whenever you activate the workspace, it will automatically switch back to that scene.

Switch to this Mode when activating the workspace.

Determines which add-ons are enabled in the active workspace. When unchecked, the global add-ons will be used. When checked, you can enable individual add-ons in the list below.

--- Fin du chapitre: _workspaces ---

--- Début du chapitre: _world --- The World Output node is used to output light a color information to the scene's World.

The appearance of the environment, usually preceded by a Background shader.

Used to add volumetric effects to the world. See the shaders Volume Absorption and Volume Scatter for more information.

Note

It is not possible to have an HDR and volumetric due to the fact that HDR's are assumed to be an infinite distance from the camera.

Render engine the input shaders are used for. By default shaders are shared between Cycles and Eevee, with multiple output nodes specialized shader setups can be created for each.

This node has no outputs.

--- Fin du chapitre: _world ---

--- Début du chapitre: _world_settings --- Reference

World Mist Pass

Note

The mist pass must be enabled in the View Layer tab of the Properties Editor before the settings below are available in the World tab.

Mist can greatly enhance the illusion of depth in your rendering. To create mist, Blender generates a render layer with a depth map ranging between 0.0 and 1.0 that can be used in the Compositor to generate a mist effect.

The distance from the camera at which the mist starts to fade in.

The distance from Start of the mist, that it fades in over. Objects further from the camera than Start + Depth are completely hidden by the mist.

The curve function that controls the rate of change of the mist's strength further and further into the distance.

Uses the same calculation as light falloff ($\frac{1}{x^2}$) and provides the smoothest transition from transparent (0.0) to opaque (1.0).

Has a steeper start than quadratic ($\frac{1}{x}$).

Has the steepest start ($\frac{1}{\sqrt{x}}$) and approaches 1.0 faster than the other two functions.

Tip

A visualization can be activated in the Camera viewport Displaypanel.

Mist example(blend-file).

Reference

World Ray Visibility

As with other objects, Ray Visibility allows you to control which other shaders can see the environment.

Sometimes it may be useful to have a different background that is directly visible versus one that is indirectly lighting the objects. A simple solution to this is to add a Mix node, with the Blend Factor set to Is Camera Ray. The first input color is then the indirect color, and the second the directly visible color. This is useful when using a high-res image for the background and a low-res image for the actual lighting.

Similarly, adding the Is Camera and Is Glossy rays will mean that the high-res image will also be visible in reflections.

Nodes for the trick above.

Reference

World Settings

Controls the sampling method for the world material. Selecting Auto or Manual enables Multiple Importance Sampling while None disables it. Multiple Importance Sampling is a method to sample the background texture such that lighter parts are favored, creating an importance map. It will produce less noise in the render in trade of artifacts (Fireflies). Enable this when using an image texture with small area lights (like the sun), otherwise noise can take a long time to converge.

Below is a comparison between Multiple Importance Sample off and on. Both images are rendered for 25 seconds (Off: 1,500 samples, On: 1,000 samples).

Multiple Importance Sample off.

Multiple Importance Sample on.

Sets the resolution of the importance map. A higher resolution will better detect small features in the map and give more accurate sampling but conversely will take up more memory and render slightly slower. Higher values also may produce less noise when using high-res images.

Maximal number of bounces the background light will contribute to the render.

See also

See Reducing Noise for more information on how to reduce noise.

Mark the World Shader as a refractive caustic caster. This setting can be used in conjunction with the Cast and Receive caustics object settings to selectively speed up refractive caustic rendering of select objects.

For dense volumes lit from far away Distance sampling is more efficient in most cases. Usually this shouldn't be used for World volumes.

If you have got a light inside or near the volume then equiangular sampling is better.

If you have a combination of both, then the multiple importance sampling will be better.

Interpolation method to use for the volume.

Simple interpolation which gives good results for thin volumes.

Smoothed high-quality interpolation needed for more dense volumes, but slower.

Assume volume has the same density everywhere (not using any textures), for faster rendering. Usually this is automatically determined by the renderer. This settings provides a manual control for cases where it is not detected.

Distance between volume shader samples for world volume shaders. See Volume Render Settings for more information.

Select the Light Group to add the current World Surface Shader to.

If the name input into the Light Group field does not align with an existing Light Group, then pressing this button will create a Light Group with that name and assign this World Shader to it.

--- Fin du chapitre: _world_settings ---

--- Début du chapitre: _writing_guide --- The main goals for this manual are as follows:

The manual is written for people educated in computer graphics, who understand the basics of 3D and/or know other 3D software. While some areas of computer graphics are highly technical, this manual shall be kept understandable by non-technical users.

The manual provides detailed functional description of all features, tools and options in Blender. While there is a canonical source of truth for each of Blender's key areas, this does not mean we have to document every small detail. The manual should provide information on what a feature is, how to use it, and its purpose. More background information should be provided when necessary to give deeper understanding of a 3D pipeline.

Computer graphics is an incredibly interesting field, there are many rules, exceptions to the rules, and interesting details. Expanding into details can add unnecessary content, so keep the text concise and relevant to the topic at hand.

Keep in mind that Blender has frequent releases, so try to write content that will not have to be redone the moment some small change is made. This also helps a small documentation community to maintain the manual.

In order to maintain a consistent writing style within the manual, please keep this page in mind and only deviate from it when you have a good reason to do so.

Rules of thumb:

Spell checking is strongly recommended.

Use American English (e.g: modeling and not modelling, color and not colour) also for formatting numbers (e.g: 2,718.28 and not 2 718,28).

Take care about grammar, appropriate wording and use simple English.

Keep sentences short and clear, resulting in text that is easy to read, objective and to the point.

Including why or how an option might be useful is a good idea.

If you are unsure about how a feature works, ask someone else or find out who developed it and ask them.

To be avoided:

Avoid writing in first person perspective, about yourself or your own opinions.

Avoid weasel words and being unnecessarily vague, e.g:

Avoid including specific details such as:

These details are not useful for users to memorize and they become quickly outdated.

Avoid documenting bugs.

Blender often has 100\% of bugs fixed between releases, so it is not realistic to reference even a fraction of them from the manual, while keeping this list up to date.

Issues that are known to the developers and are not going to be resolved before the next release can be documented as Known Limitations. In some cases, it may be best to include them in the troubleshooting section.

Avoid product placements, i.e. unnecessarily promoting software or hardware brands. Keep content vendor-neutral where possible.

Avoid technical explanations about the mathematical/algorithmic implementation of a feature if there is a simpler way to explain it.

(E.g. explaining how mesh smoothing algorithms work is unnecessary, but the blending types of a Mix node do need a mathematical explanation.)

Avoid repetition of large portions of text. Simply explain it once, and from then on refer to that explanation.

For general terminology, consider defining a :term: in the glossary .

Avoid enumerations of similar options, such as listing every preset or every frame rate in a select menu.

Their contents may be summarized or simply omitted. Such lists are only showing what is already obvious in the interface and end up being a lot of text to read and maintain.

Avoid documenting changes in Blender between releases, that is what the release notes are for. We only need to document the current state of Blender.

Unless the unit a value is measured in is obscure and unpredictable, there is no need to mention it.

Do not simply copy the tooltips from Blender. People will come to the manual to learn more than is provided by the UI.

As a last resort you can add comment (which is not shown in the HTML page, but useful for other editors):

This section is specifically about the Glossary section, where we define common terms in Blender and computer graphics.

Rules of thumb:

Define the term before providing any further information.

Avoid using constructs such as "it is" or "xyz is" before the definition.

Avoid repeating the term immediately or using it in the definition.

Avoid adding terms not found in Blender's interface or the manual.

Avoid overly long entries. If an explanation of a complex term is needed, supplement with external links.

Avoid duplicating documentation; if explaining the term is the primary focus of another section of the manual (e.g. if the term is the name of a tool), either just link to that section, or avoid creating a glossary entry entirely.

URL references are to be added at the end, formatted as follows, e.g:

This entry:

Would be written like this instead, putting a definition first:

This entry:

Would be written more like this, avoiding the immediate repetition of the term:

This entry:

Would be written more like this, avoiding the \xe2\x80\x9d is \xe2\x80\x9d:

--- Fin du chapitre: _writing_guide ---

--- Début du chapitre: _z_combine --- The Z Combine node combines two images based on their Z-depth maps. It overlays the images using the provided Z values to detect which parts of one image are in front of the other.

The background image.

Z depth of the background image.

The foreground image.

Z depth of the foreground image.

The chosen Image pixel alpha channel is also carried over. If a pixel is partially or totally transparent, the result of the Z Combine will also be partially transparent; in which case the background image will show through the foreground (chosen) pixel.

Applies Anti-Aliasing to avoid artifacts at sharp edges or areas with a high contrast.

If both Z values are equal, it will use the foreground image. Whichever Z value is less decides which image pixel is used. See Z-buffer.

The combined Z depth, which allows to thread multiple Z-combines together.

Choosing closest pixels. \xc2\xb6

In the example above, the render output from two scenes are mixed using the Z Combine node, one from a sphere of size 1.3, and the other a cube of size 1.0. The sphere and square are located at the same place. The cube is tipped forward, so the corner in the center is closer to the camera than the sphere surface; so Z Combine chooses to use the cube \xe2\x80\x99s pixels. But the sphere is slightly larger (a size of 1.3 versus 1.0), so it does not fit totally inside the cube. At some point, as the cube \xe2\x80\x99s sides recede back away from the camera, the sphere \xe2\x80\x99s sides are closer. When this happens, Z Combine uses the sphere \xe2\x80\x99s pixels to form the resulting picture.

This node can be used to combine a foreground with a background matte painting. Walt Disney pioneered the use of multi-plane mattes, where three or four partial mattes were painted on glass and placed on the left and right at different Z positions; minimal camera moves to the right created the illusion of depth as Bambi moved through the forest.

Note

Valid Input

Z Input Sockets do not accept fixed values; they must get a vector set (see Map Value node). Image Input Sockets will not accept a color since they do not have UV coordinates.

Mix and match images. \xc2\xb6

The Z Combine can be used to merge two images as well. Using the Z values from the sphere and cube scenes above, but inputting different images, yields the example to the right.

Z Combine in action. \xc2\xb6

In this node setup a render scene is mixed with a flat image. In the side view of the scene, the orange cube is 10 units away from the camera, and the blue ball is 20. The 3D cursor is about 15 units away from the camera. The image is Z-in at a location of 15, thus inserting it in between the cube and the ball. The resulting image appears to have the cube on the green image.

Note

Invisible Man Effect

If a foreground image with a higher Alpha than the background, is then mixed in the Z Combine with a slightly magnified background, the outline of the transparent area will distort the background, enough to make it look like seeing a part of the background through an invisible yet Fresnel-lens object.

--- Fin du chapitre: _z_combine ---